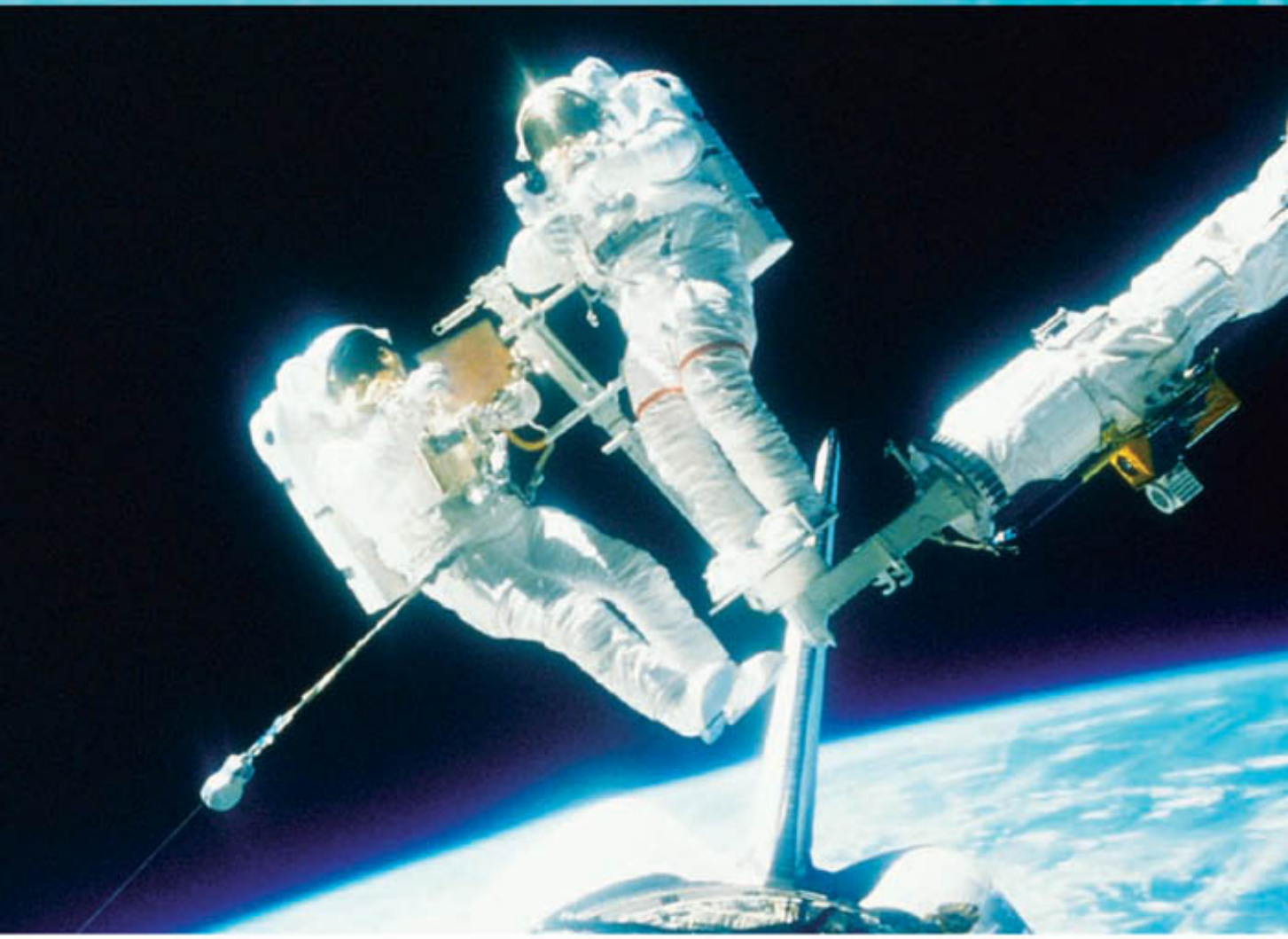
Charles K. Alexander Matthew N.O. Sadiku

Fundamentals of Electric Circuits

Fourth Edition



PRACTICAL APPLICATIONS

Each chapter devotes material to practical applications of the concepts covered in *Fundamentals of Electric Circuits* to help the reader apply the concepts to real-life situations. Here is a sampling of the practical applications found in the text:

- Rechargeable flashlight battery (Problem 1.11)
- Cost of operating toaster (Problem 1.25)
- Potentiometer (Section 2.8)
- Design a lighting system (Problem 2.61)
- Reading a voltmeter (Problem 2.66)
- Controlling speed of a motor (Problem 2.74)
- Electric pencil sharpener (Problem 2.78)
- Calculate voltage of transistor (Problem 3.86)
- Transducer modeling (Problem 4.87)
- Strain gauge (Problem 4.90)
- Wheatstone bridge (Problem 4.91)
- Design a six-bit DAC (Problem 5.83)
- Instrumentation amplifier (Problem 5.88)
- Design an analog computer circuit (Example 6.15)
- Design an op amp circuit (Problem 6.71)
- Design analog computer to solve differential equation (Problem 6.79)
- Electric power plant substation—capacitor bank (Problem 6.83)
- Electronic photo flash unit (Section 7.9)
- Automobile ignition circuit (Section 7.9)
- Welding machine (Problem 7.86)
- Airbag igniter (Problem 8.78)
- Electrical analog to bodily functions—study of convulsions (Problem 8.82)
- Electronic sensing device (Problem 9.87)
- Power transmission system (Problem 9.93)
- Design a Colpitts oscillator (Problem 10.94)
- Stereo amplifier circuit (Problem 13.85)
- Gyrator circuit (Problem 16.69)
- Calculate number of stations allowable in AM broadcast band (Problem 18.63)
- Voice signal—Nyquist rate (Problem 18.65)

COMPUTER TOOLS promote flexibility and meet ABET requirements

- *PSpice* is introduced in Chapter 3 and appears in special sections throughout the text. Appendix D serves as a tutorial on *PSpice* for Windows for readers not familiar with its use. The special sections contain examples and practice problems using *PSpice*. Additional homework problems at the end of each chapter also provide an opportunity to use *PSpice*.
- *MATLAB*® is introduced through a tutorial in Appendix E to show its usage in circuit analysis. A number of examples and practice problems are presented throughout the book in a manner that will allow the student to develop a facility with this powerful tool. A number of end-of-chapter problems will aid in understanding how to effectively use *MATLAB*.
- *KCIDE for Circuits* is a working software environment developed at Cleveland State University. It is designed to help the student work through circuit problems in an organized manner following the process on problem-solving discussed in Section 1.8. Appendix F contains a description of how to use the software. Additional examples can be found at the web site, http://kcide.fennresearch.org/. The actual software package can be downloaded for free from this site. One of the best benefits from using this package is that it automatically generates a Word document and/or a PowerPoint presentation.

CAREERS AND HISTORY of electrical engineering pioneers

Since a course in circuit analysis may be a student's first exposure to electrical engineering, each chapter opens with discussions about how to enhance skills that contribute to successful problem-solving or career-oriented talks on a sub-discipline of electrical engineering. The chapter openers are intended to help students grasp the scope of electrical engineering and give thought to the various careers available to EE graduates. The opening boxes include information on careers in electronics, instrumentation, electromagnetics, control systems, engineering education, and the importance of good communication skills. Historicals throughout the text provide brief biological sketches of such engineering pioneers as Faraday, Ampere, Edison, Henry, Fourier, Volta, and Bell.

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Step 7: The **project manager, editorial team,** and **author team** review the pages for a final accuracy check.

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Fundamentals of

Electric Circuits

fourth edition

Fundamentals of Electric Circuits

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Dedicated to our wives, Kikelomo and Hannah, whose understanding and support have truly made this book possible.

Matthew
and
Chuck

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Preface

You may be wondering why we chose a photo of astronauts working in space on the Space Station for the cover. We actually chose it for several reasons. Obviously, it is very exciting; in fact, space represents the most exciting frontier for the entire world! In addition, much of the station itself consists of all kinds of circuits! One of the most significant circuits within the station is its power distribution system. It is a complete and self contained, modern power generation and distribution system. That is why NASA (especially NASA-Glenn) continues to be at the forefront of both theoretical as well as applied power system research and development. The technology that has gone into the development of space exploration continues to find itself impacting terrestrial technology in many important ways. For some of you, this will be an important career path.

FEATURES

New to This Edition

A course in circuit analysis is perhaps the first exposure students have to electrical engineering. This is also a place where we can enhance some of the skills that they will later need as they learn how to design.

In the fourth edition, we have included a very significant new feature to help students enhance skills that are an important part of the design process. We call this new feature, ***design a problem.***

We know it is not possible to fully develop a student's design skills in a fundamental course like circuits. To fully develop design skills a student needs a design experience normally reserved for their senior year. This does not mean that some of those skills cannot be developed and exercised in a circuits course. The text already included open-ended questions that help students use creativity, which is an important part of learning how to design. We already have some questions that are open desired to add much more into our text in this important area and have developed an approach to do just that. When we develop problems for the student to solve our goal is that in solving the problem the student learn more about the theory and the problem solving process. Why not have the students design problems like we do? That is exactly what we will do in each chapter. Within the normal problem set, we have a set of problems where we ask the student to design a problem. This will have two very important results. The first will be a better understanding of the basic theory and the second will be the enhancement of some of the student's basic design skills.

We are making effective use of the principle of learning by teaching. Essentially we all learn better when we teach a subject. Designing effective problems is a key part of the teaching process. Students

should also be encouraged to develop problems, when appropriate, which have nice numbers and do not necessarily overemphasize complicated mathematical manipulations.

Additionally we have changed almost 40% of the Practice Problems with the idea to better reflect more real component values and to help the student better understand the problem and have added 121 *design a problem* problems. We have also changed and added a total of 357 end-of-chapter problems (this number contains the new *design a problem* problems). This brings up a very important advantage to our textbook, we have a total of 2404 Examples, Practice Problems, Review Questions, and end-of-chapter problems!

Retained from Previous Editions

The main objective of the fourth edition of this book remains the same as the previous editions—to present circuit analysis in a manner that is clearer, more interesting, and easier to understand than other circuit text, and to assist the student in beginning to see the "fun" in engineering. This objective is achieved in the following ways:

- **Chapter Openers and Summaries**
  Each chapter opens with a discussion about how to enhance skills which contribute to successful problem solving as well as successful careers or a career-oriented talk on a sub-discipline of electrical engineering. This is followed by an introduction that links the chapter with the previous chapters and states the chapter objectives. The chapter ends with a summary of key points and formulas.
- **Problem Solving Methodology**
  Chapter 1 introduces a six-step method for solving circuit problems which is used consistently throughout the book and media supplements to promote best-practice problem-solving procedures.
- **Student Friendly Writing Style**
  All principles are presented in a lucid, logical, step-by-step manner. As much as possible, we avoid wordiness and giving too much detail that could hide concepts and impede overall understanding of the material.
- **Boxed Formulas and Key Terms**
  Important formulas are boxed as a means of helping students sort out what is essential from what is not. Also, to ensure that students clearly understand the key elements of the subject matter, key terms are defined and highlighted.
- **Margin Notes**
  Marginal notes are used as a pedagogical aid. They serve multiple uses such as hints, cross-references, more exposition, warnings, reminders not to make some particular common mistakes, and problem-solving insights.
- **Worked Examples**
  Thoroughly worked examples are liberally given at the end of every section. The examples are regarded as a part of the text and are clearly explained without asking the reader to fill in missing steps. Thoroughly worked examples give students a good understanding of the solution process and the confidence to solve problems

themselves. Some of the problems are solved in two or three different ways to facilitate a substantial comprehension of the subject material as well as a comparison of different approaches.

- **Practice Problems**
  To give students practice opportunity, each illustrative example is immediately followed by a practice problem with the answer. The student can follow the example step by step to aid in the solution of the practice problem without flipping pages or looking at the end of the book for answers. The practice problem is also intended to test a student's understanding of the preceding example. It will reinforce their grasp of the material before the student can move on to the next section. Complete solutions to the practice problems are available to students on ARIS.
- **Application Sections**
  The last section in each chapter is devoted to practical application aspects of the concepts covered in the chapter. The material covered in the chapter is applied to at least one or two practical problems or devices. This helps students see how the concepts are applied to real-life situations.
- **Review Questions**
  Ten review questions in the form of multiple-choice objective items are provided at the end of each chapter with answers. The review questions are intended to cover the little "tricks" that the examples and end-of-chapter problems may not cover. They serve as a self-test device and help students determine how well they have mastered the chapter.
- **Computer Tools**
  In recognition of the requirements by ABET® on integrating computer tools, the use of *PSpice, MATLAB, KCIDE for Circuits,* and developing design skills are encouraged in a student-friendly manner. *PSpice* is covered early on in the text so that students can become familiar and use it throughout the text. Appendix D serves as a tutorial on *PSpice* for Windows. *MATLAB* is also introduced early in the book with a tutorial available in Appendix E. *KCIDE for Circuits* is a brand new, state-of-the-art software system designed to help the students maximize their chance of success in problem solving. It is introduced in Appendix F. Finally, *design a problem* problems have been introduced, for the first time. These are meant to help the student develop skills that will be needed in the design process.
- **Historical Tidbits**
  Historical sketches throughout the text provide profiles of important pioneers and events relevant to the study of electrical engineering.
- **Early Op Amp Discussion**
  The operational amplifier (op amp) as a basic element is introduced early in the text.
- **Fourier and Laplace Transforms Coverage**
  To ease the transition between the circuit course and signals and systems courses, Fourier and Laplace transforms are covered lucidly and thoroughly. The chapters are developed in a manner that the interested instructor can go from solutions of first-order

circuits to Chapter 15. This then allows a very natural progression from Laplace to Fourier to AC.

- **Four Color Art Program**
  A completely redesigned interior design and four color art program bring circuit drawings to life and enhance key pedagogical elements throughout the text.
- **Extended Examples**
  Examples worked in detail according to the six-step problem solving method provide a roadmap for students to solve problems in a consistent fashion. At least one example in each chapter is developed in this manner.
- **EC 2000 Chapter Openers**
  Based on ABET's new skill-based CRITERION 3, these chapter openers are devoted to discussions as to how students can acquire the skills that will lead to a significantly enhanced career as an engineer. Because these skills are so very important to the student while in college as well as in their career, we will use the heading, *"Enhancing your Skills and your Career."*
- **Homework Problems**
  There are 358 new or changed end-of-chapter problems which will provide students with plenty of practice as well as reinforce key concepts.
- **Homework Problem Icons**
  Icons are used to highlight problems that relate to engineering design as well as problems that can be solved using *PSpice* or *MATLAB.*
- ***KCIDE for Circuits* Appendix F**
  A new Appendix F provides a tutorial on the Knowledge Capturing Integrated Design Environment (*KCIDE for Circuits*) software, available on ARIS.

Organization

This book was written for a two-semester or three-quarter course in linear circuit analysis. The book may also be used for a one-semester course by a proper selection of chapters and sections by the instructor. It is broadly divided into three parts.

- Part 1, consisting of Chapters 1 to 8, is devoted to dc circuits. It covers the fundamental laws and theorems, circuits techniques, and passive and active elements.
- Part 2, which contains Chapter 9 to 14, deals with ac circuits. It introduces phasors, sinusoidal steady-state analysis, ac power, rms values, three-phase systems, and frequency response.
- Part 3, consisting of Chapters 15 to 19, is devoted to advanced techniques for network analysis. It provides students with a solid introduction to the Laplace transform, Fourier series, Fourier transform, and two-port network analysis.

The material in three parts is more than sufficient for a two-semester course, so the instructor must select which chapters or sections to cover. Sections marked with the dagger sign (†) may be skipped, explained briefly, or assigned as homework. They can be omitted without loss of

continuity. Each chapter has plenty of problems grouped according to the sections of the related material and diverse enough that the instructor can choose some as examples and assign some as homework. As stated earlier, we are using three icons with this edition. We are using ps to denote problems that either require *PSpice* in the solution process, where the circuit complexity is such that *PSpice* would make the solution process easier, and where *PSpice* makes a good check to see if the problem has been solved correctly. We are using ML to denote problems where *MATLAB* is required in the solution process, where *MATLAB* makes sense because of the problem makeup and its complexity, and where *MATLAB* makes a good check to see if the problem has been solved correctly. Finally, we use ed to identify problems that help the student develop skills that are needed for engineering design. More difficult problems are marked with an asterisk (*). Comprehensive problems follow the end-of-chapter problems. They are mostly applications problems that require skills learned from that particular chapter.

Prerequisites

As with most introductory circuit courses, the main prerequisites, for a course using the text, are physics and calculus. Although familiarity with complex numbers is helpful in the later part of the book, it is not required. A very important asset of this text is that ALL the mathematical equations and fundamentals of physics needed by the student, are included in the text.

Supplements

McGraw-Hill's ARIS—Assessment, Review, and Instruction System is a complete, online tutorial, electronic homework, and course management system, designed for greater ease of use than any other system available. Available on adoption, instructors can create and share course materials and assignments with other instructors, edit questions and algorithms, import their own content, and create announcements and due dates for assignments. ARIS has automatic grading and reporting of easy-to-assign algorithmically-generated homework, quizzing, and testing. Once a student is registered in the course, all student activity within McGraw-Hill's ARIS is automatically recorded and available to the instructor through a fully integrated grade book that can be downloaded to Excel. Also included on ARIS are a solutions manual, text image files, transition guides to instructors, and Network Analysis Tutorials, software downloads, complete solutions to text practice problems, FE Exam questions, flashcards, and web links to students. Visit www.mhhe.com/alexander.

Knowledge Capturing Integrated Design Environment for Circuits (*KCIDE for Circuits*) This new software, developed at Cleveland State University and funded by NASA, is designed to help the student work through a circuits problem in an organized manner using the six-step problem-solving methodology in the text. *KCIDE for Circuits* allows students to work a circuit problem in *PSpice* and *MATLAB*, track the

evolution of their solution, and save a record of their process for future reference. In addition, the software automatically generates a Word document and/or a PowerPoint presentation. Appendix F contains a description of how to use the software. Additional examples can be found at the web site, http://kcide.fennresearch.org/, which is linked from ARIS. The software package can be downloaded for free.

Problem Solving Made *Almost* Easy, a companion workbook to *Fundamentals of Electric Circuits*, is available on ARIS for students who wish to practice their problem-solving techniques. The workbook contains a discussion of problem-solving strategies and 150 additional problems with complete solutions provided.

C.O.S.M.O.S This CD, available to instructors only, is a powerful solutions manual tool to help instructors streamline the creation of assignments, quizzes, and tests by using problems and solutions from the textbook, as well as their own custom material. Instructors can edit textbook end-of-chapter problems as well as track which problems have been assigned.

Although the textbook is meant to be self-explanatory and act as a tutor for the student, the personal contact in teaching is not forgotten. It is hoped that the book and supplemental materials supply the instructor with all the pedagogical tools necessary to effectively present the material.

Acknowledgements

We would like to express our appreciation for the loving support we have received from our wives (Hannah and Kikelomo), daughters (Christina, Tamara, Jennifer, Motunrayo, Ann, and Joyce), son (Baixi), and our extended family members.

At McGraw-Hill, we would like to thank the following editorial and production staff: Raghu Srinivasan, publisher and senior sponsoring editor; Lora Kalb-Neyens, developmental editors; Joyce Watters, project manager; Carrie Burger, photo researcher; and Brenda Rolwes, designer. Also, we appreciate the hard work of Tom Hartley at the University of Akron for his very detailed evaluation of various elements of the text and his many valued suggestions for continued improvement of this textbook.

We wish to thank Yongjian Fu and his outstanding team of students, Bramarambha Elka and Saravaran Chinniah, for their efforts in the development of *KCIDE for Circuits*. Their efforts to help us continue to improve this software are also appreciated.

The fourth edition has benefited greatly from the many outstanding reviewers and symposium attendees who contributed to the success of the first three editions! In addition, the following have made important contributions to the fourth edition (in alphabetical order):

Tom Brewer, Georgia Tech
Andy Chan, City University of Hong Kong
Alan Tan Wee Chiat, Multimedia University
Norman Cox, University of Missouri-Rolla
Walter L. Green, University of Tennessee

Dr. Gordon K. Lee, San Diego State University
Gary Perks, Cal Poly State University, San Luis Obispo
Dr. Raghu K. Settaluri, Oregon State University
Ramakant Srivastava, University of Florida
John Watkins, Wichita State University
Yik-Chung Wu, The University of Hong Kong
Xiao-Bang Xu, Clemson University

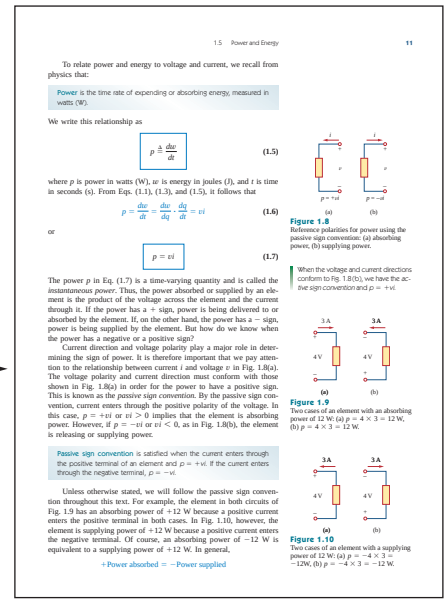
Finally, we appreciate the feedback received from instructors and students who used the previous editions. We want this to continue, so please keep sending us emails or direct them to the publisher. We can be reached at c.alexander@ieee.org for Charles Alexander and sadiku@ieee.org for Matthew Sadiku.

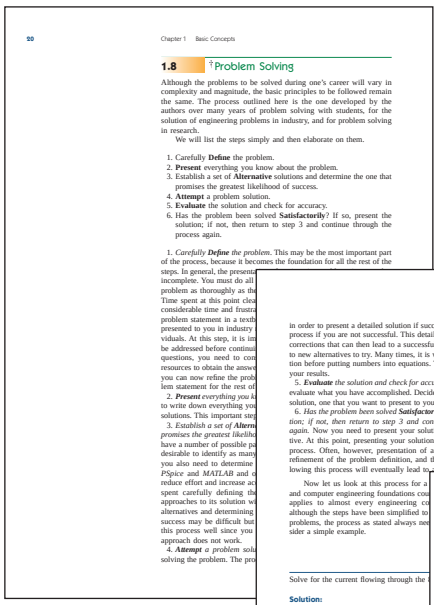
C. K. Alexander and M.N.O. Sadiku

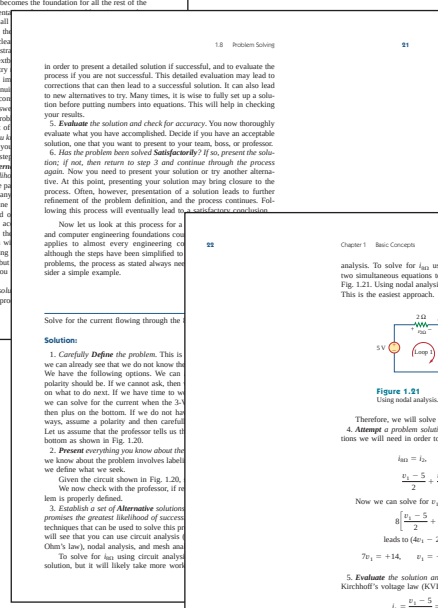
GUIDED TOUR

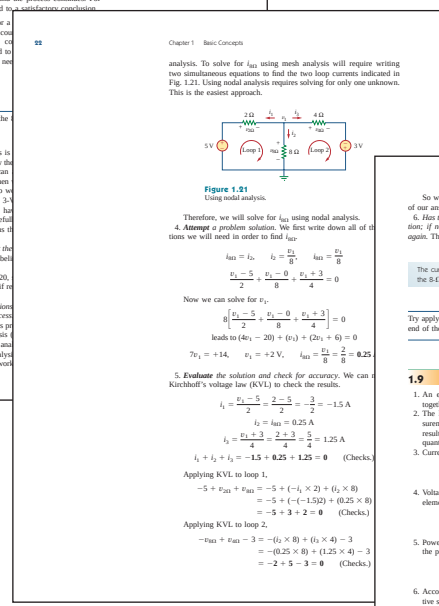
The main objective of this book is to present circuit analysis in a manner that is clearer, more interesting, and easier to understand than other texts. For you, the student, here are some features to help you study and be successful in this course.

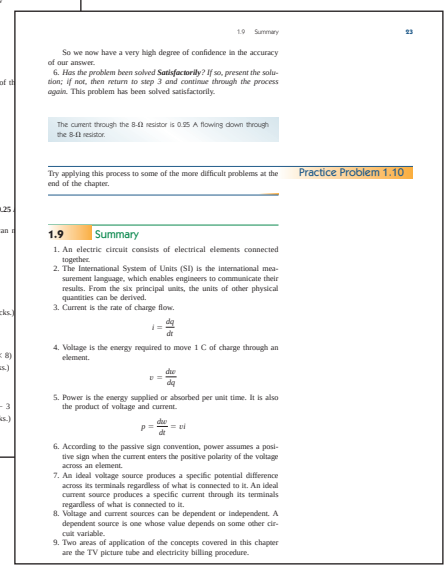
The four color art program brings circuit drawings to life and enhances key concepts throughout the text.



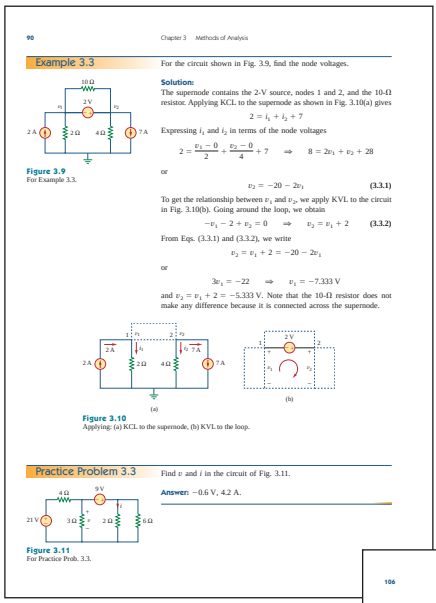








A six-step problem-solving methodology is introduced in Chapter 1 and incorporated into worked examples throughout the text to promote sound, step-by-step problem-solving practices.


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Example 3.3

For the circuit shown in Fig. 3.9, find the node voltages.

Solution:
The supernode contains the 2-V source, nodes 1 and 2, and the 10-Ω resistor. Applying KCL to the supernode as shown in Fig. 3.10(a) gives

$$2 = i_1 + i_2 + 7$$

Expressing i_1 and i_2 in terms of the node voltages

$$2 = \frac{v_1 - 0}{2} + \frac{v_2 - 0}{4} + 7 \quad \Rightarrow \quad 8 = 2v_1 + v_2 + 28$$

or

$$v_2 = -20 - 2v_1 \qquad (3.3.1)$$

To get the relationship between v_1 and v_2, we apply KVL to the circuit in Fig. 3.10(b). Going around the loop, we obtain

$$-v_1 - 2 + v_2 = 0 \quad \Rightarrow \quad v_2 = v_1 + 2 \qquad (3.3.2)$$

From Eqs. (3.3.1) and (3.3.2), we write

$$v_2 = v_1 + 2 = -20 - 2v_1$$

or

$$3v_1 = -22 \quad \Rightarrow \quad v_1 = -7.333 \text{ V}$$

and $v_2 = v_1 + 2 = -5.333$ V. Note that the 10-Ω resistor does not make any difference because it is connected across the supernode.

Figure 3.9
For Example 3.3.

Figure 3.10
Applying: (a) KCL to the supernode, (b) KVL to the loop.

Practice Problem 3.3

Find v and i in the circuit of Fig. 3.11.

Answer: −0.6 V, 4.2 A.

Figure 3.11
For Practice Prob. 3.3.

Each illustrative example is immediately followed by a practice problem and answer to test understanding of the preceding example.

PSpice® for Windows is a student-friendly tool introduced to students early in the text and used throughout, with discussions and examples at the end of each appropriate chapter.

The last section in each chapter is devoted to applications of the concepts covered in the chapter to help students apply the concepts to real-life situations.

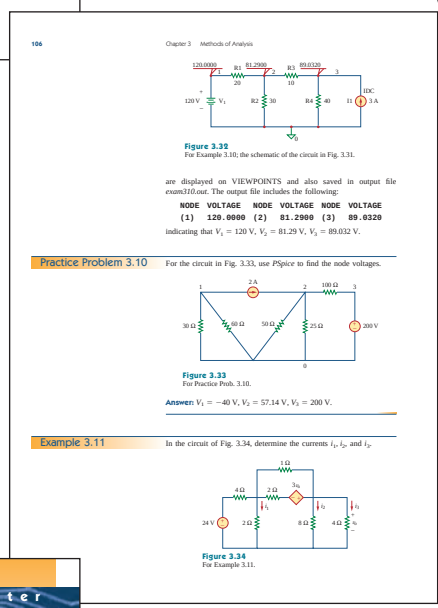

106 Chapter 3 Methods of Analysis

Figure 3.32
For Example 3.10; the schematic of the circuit in Fig. 3.31.

are displayed on VIEWPOINTS and also saved in output file *exam310.out*. The output file includes the following:

```
NODE  VOLTAGE   NODE  VOLTAGE   NODE  VOLTAGE
(1)   120.0000  (2)   81.2900   (3)   89.0320
```

indicating that $V_1 = 120$ V, $V_2 = 81.29$ V, $V_3 = 89.032$ V.

Practice Problem 3.10

For the circuit in Fig. 3.33, use *PSpice* to find the node voltages.

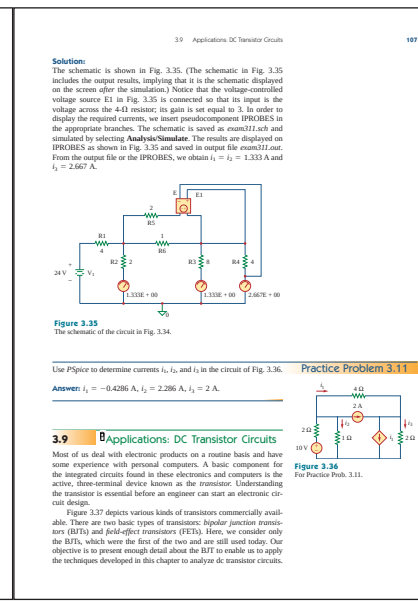
Figure 3.33
For Practice Prob. 3.10.

Answer: $V_1 = -40$ V, $V_2 = 57.14$ V, $V_3 = 200$ V.

Example 3.11

In the circuit of Fig. 3.34, determine the currents i_1, i_2, and i_3.

Figure 3.34
For Example 3.11.


3.9 Applications: DC Transistor Circuits 107

Solution:
The schematic is shown in Fig. 3.35. (The schematic in Fig. 3.35 includes the output results, implying that it is the schematic displayed on the screen *after* the simulation.) Notice that the voltage-controlled voltage source E1 in Fig. 3.35 is connected so that its input is the voltage across the 4-Ω resistor; its gain is set equal to 3. In order to display the required currents, we insert pseudocomponent IPROBES in the appropriate branches. The schematic is saved as *exam311.sch* and simulated by selecting **Analysis/Simulate**. The results are displayed on IPROBES as shown in Fig. 3.35 and saved in output file *exam311.out*. From the output file or the IPROBES, we obtain $i_1 = i_2 = 1.333$ A and $i_3 = 2.667$ A.

Figure 3.35
The schematic of the circuit in Fig. 3.34.

Use *PSpice* to determine currents i_1, i_2, and i_3 in the circuit of Fig. 3.36.

Answer: $i_1 = -0.4286$ A, $i_2 = 2.286$ A, $i_3 = 2$ A.

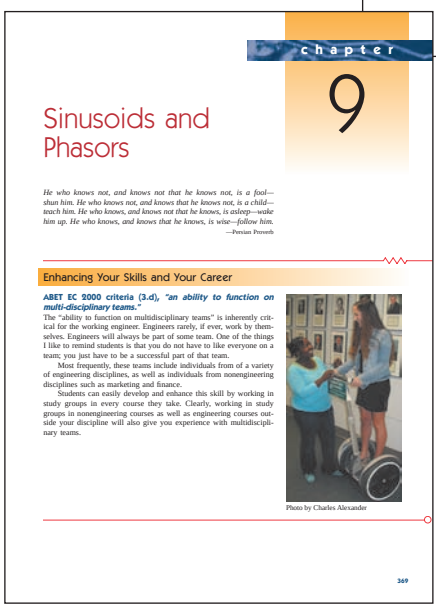
Practice Problem 3.11

Figure 3.36
For Practice Prob. 3.11.

3.9 Applications: DC Transistor Circuits

Most of us deal with electronic products on a routine basis and have some experience with personal computers. A basic component for the integrated circuits found in these electronics and computers is the active, three-terminal device known as the *transistor*. Understanding the transistor is essential before an engineer can start an electronic circuit design.

Figure 3.37 depicts various kinds of transistors commercially available. There are two basic types of transistors: *bipolar junction transistors* (BJTs) and *field-effect transistors* (FETs). Here, we consider only the BJTs, which were the first of the two and are still used today. Our objective is to present enough detail about the BJT to enable us to apply the techniques developed in this chapter to analyze dc transistor circuits.


chapter 9

Sinusoids and Phasors

He who knows not, and knows not that he knows not, is a fool—shun him. He who knows not, and knows that he knows not, is a child—teach him. He who knows, and knows not that he knows, is asleep—wake him up. He who knows, and knows that he knows, is wise—follow him.
—Persian Proverb

Enhancing Your Skills and Your Career

ABET EC 2000 criteria (3.d), *"an ability to function on multi-disciplinary teams."*
The "ability to function on multidisciplinary teams" is inherently critical for the working engineer. Engineers rarely, if ever, work by themselves. Engineers will always be part of some team. One of the things I like to remind students is that you do not have to like everyone on a team; you just have to be a successful part of that team.

Most frequently, these teams include individuals from of a variety of engineering disciplines, as well as individuals from nonengineering disciplines such as marketing and finance.

Students can easily develop and enhance this skill by working in study groups in every course they take. Clearly, working in study groups in nonengineering courses as well as engineering courses outside your discipline will also give you experience with multidisciplinary teams.

Photo by Charles Alexander

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Each chapter opens with a discussion about how to enhance skills that contribute to successful problem solving as well as successful careers or a career-oriented talk on a sub-discipline of electrical engineering to give students some real-world applications of what they are learning.

e2d Icons next to the end-of-chapter homework problems let students know which problems relate to engineering design and which problems can be solved using *PSpice* or *MATLAB*. Appendices on these computer programs provide tutorials for their use.

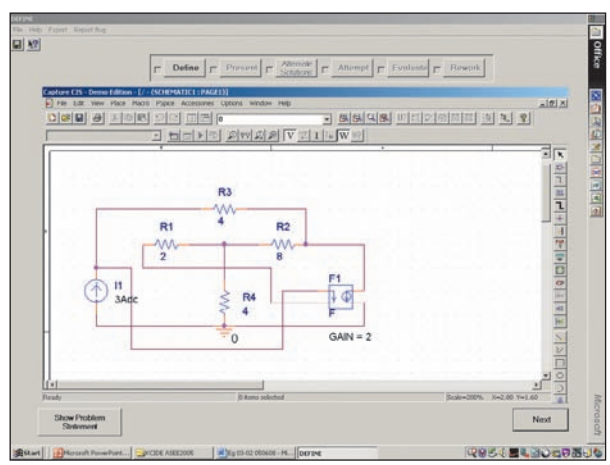
Supplements for Students and Instructors

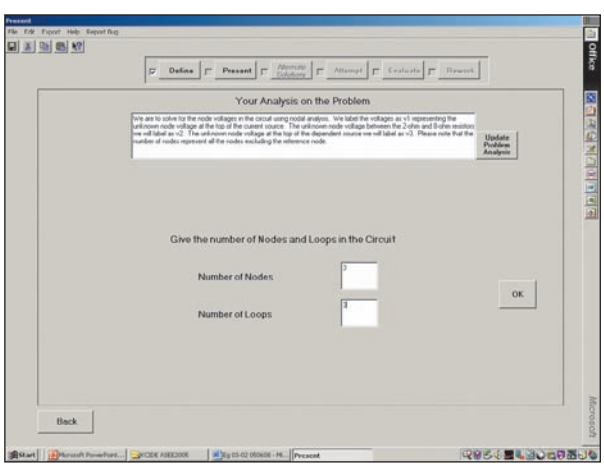


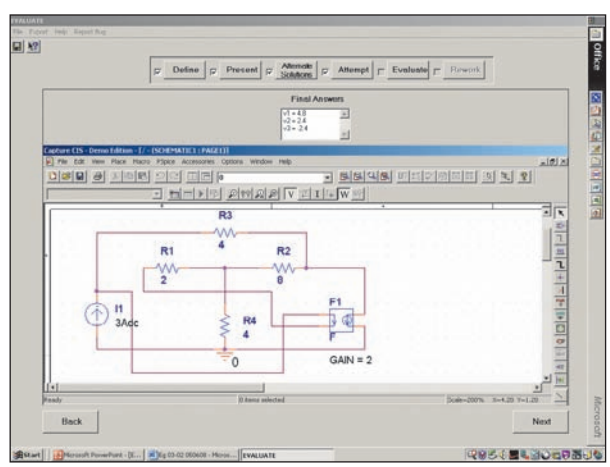
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Knowledge Capturing Integrated Design Environment for Circuits (*KCIDE for Circuits*) software, linked from ARIS, enhances student understanding of the six-step problem-solving methodology in the book. *KCIDE for Circuits* allows students to work a circuit problem in *PSpice* and *MATLAB*, track the evolution of their solution, and save a record of their process for future reference. Appendix F walks the user through this program.







Other resources provided on ARIS.

For Students:

— Network Analysis Tutorials—a series of interactive quizzes to help students practice fundamental concepts in circuits.
— FE Exam Interactive Review Quizzes—chapter based self-quizzes provide hints for solutions and correct solution methods, and help students prepare for the NCEES Fundamentals of Engineering Examination.
— Problem Solving Made *Almost* Easy—a companion workbook to the text, featuring 150 additional problems with complete solutions.
— Complete solutions to Practice Problems in the text
— Flashcards of key terms
— Web links

For Instructors:

— Image Sets—electronic files of text figures for easy integration into your course presentations, exams, and assignments.
— Transition Guides—compare coverage of the third edition to other popular circuits books at the section level to aid transition to teaching from our text.

A Note to the Student

This may be your first course in electrical engineering. Although electrical engineering is an exciting and challenging discipline, the course may intimidate you. This book was written to prevent that. A good textbook and a good professor are an advantage—but you are the one who does the learning. If you keep the following ideas in mind, you will do very well in this course.

- This course is the foundation on which most other courses in the electrical engineering curriculum rest. For this reason, put in as much effort as you can. Study the course regularly.
- Problem solving is an essential part of the learning process. Solve as many problems as you can. Begin by solving the practice problem following each example, and then proceed to the end-of-chapter problems. The best way to learn is to solve a lot of problems. An asterisk in front of a problem indicates a challenging problem.
- *Spice*, a computer circuit analysis program, is used throughout the textbook. *PSpice*, the personal computer version of *Spice*, is the popular standard circuit analysis program at most universities. *PSpice for Windows* is described in Appendix D. Make an effort to learn *PSpice*, because you can check any circuit problem with *PSpice* and be sure you are handing in a correct problem solution.
- *MATLAB* is another software that is very useful in circuit analysis and other courses you will be taking. A brief tutorial on *MATLAB* is given in Appendix E to get you started. The best way to learn *MATLAB* is to start working with it once you know a few commands.
- Each chapter ends with a section on how the material covered in the chapter can be applied to real-life situations. The concepts in this section may be new and advanced to you. No doubt, you will learn more of the details in other courses. We are mainly interested in gaining a general familiarity with these ideas.
- Attempt the review questions at the end of each chapter. They will help you discover some "tricks" not revealed in class or in the textbook.
- Clearly a lot of effort has gone into making the technical details in this book easy to understand. It also contains all the mathematics and physics necessary to understand the theory and will be very useful in your other engineering courses. However, we have also focused on creating a reference for you to use both in school as well as when working in industry or seeking a graduate degree.
- It is very tempting to sell your book after you have completed your classroom experience; however, our advice to you is *DO NOT SELL YOUR ENGINEERING BOOKS!* Books have always been expensive, however, the cost of this book is virtually the same as I paid for my circuits text back in the early 60s in terms of real dollars. In fact, it is actually cheaper. In addition, engineering books of the past are no where near as complete as what is available now.

When I was a student, I did not sell any of my engineering textbooks and was very glad I did not! I found that I needed most of them throughout my career.

A short review on finding determinants is covered in Appendix A, complex numbers in Appendix B, and mathematical formulas in Appendix C. Answers to odd-numbered problems are given in Appendix G.

Have fun!

C. K. A. and M. N. O. S.

About the Authors

Charles K. Alexander is professor of electrical and computer engineering of the Fenn College of Engineering at Cleveland State University, Cleveland, Ohio. He is also the Director of The Center for Research in Electronics and Aerospace Technology (CREATE), and is the Managing Director of the Wright Center for Sensor Systems (WCSSE). From 2002 until 2006 he was Dean of the Fenn College of Engineering. From 2004 until 2007, he was Director of Ohio ICE, a research center in instrumentation, controls, electronics, and sensors (a coalition of CSU, Case, the University of Akron, and a number of Ohio industries). From 1998 until 2002, he was interim director (2000 and 2001) of the Institute for Corrosion and Multiphase Technologies and Stocker Visiting Professor of electrical engineering and computer science at Ohio University. From 1994–1996 he was dean of engineering and computer science at California State University, Northridge.



Charles K. Alexander

From 1989–1994 he was acting dean of the college of engineering at Temple University, and from 1986–1989 he was professor and chairman of the department of electrical engineering at Temple. From 1980–1986 he held the same positions at Tennessee Technological University. He was an associate professor and a professor of electrical engineering at Youngstown State University from 1972–1980, where he was named Distinguished Professor in 1977 in recognition of "outstanding teaching and research." He was assistant professor of electrical engineering at Ohio University in 1971–1972. He received the Ph.D. (1971) and M.S.E.E. (1967) from Ohio University and the B.S.E.E. (1965) from Ohio Northern University.

Dr. Alexander has been a consultant to 23 companies and governmental organizations, including the Air Force and Navy and several law firms. He has received over $85 million in research and development funds for projects ranging from solar energy to software engineering. He has authored 40 publications, including a workbook and a videotape lecture series, and is coauthor of *Fundamentals of Electric Circuits*, *Problem Solving Made* Almost *Easy*, and the fifth edition of the *Standard Handbook of Electronic Engineering*, with McGraw-Hill. He has made more than 500 paper, professional, and technical presentations.

Dr. Alexander is a fellow of the IEEE and served as its president and CEO in 1997. In 1993 and 1994 he was IEEE vice president, professional activities, and chair of the United States Activities Board (USAB). In 1991–1992 he was region 2 director, serving on the Regional Activities Board (RAB) and USAB. He has also been a member of the Educational Activities Board. He served as chair of the USAB Member Activities Council and vice chair of the USAB Professional Activities Council for Engineers, and he chaired the RAB Student Activities Committee and the USAB Student Professional Awareness Committee.

In 1998 he received the Distinguished Engineering Education Achievement Award from the Engineering Council, and in 1996 he received the Distinguished Engineering Education Leadership Award from the same group. When he became a fellow of the IEEE in 1994, the citation read "for leadership in the field of engineering education and the professional development of engineering students." In 1984 he received the IEEE Centennial Medal, and in 1983 he received the IEEE/RAB Innovation Award, given to the IEEE member who best contributes to RAB's goals and objectives.



Matthew N. O. Sadiku

Matthew N. O. Sadiku is presently a professor at Prairie View A&M University. Prior to joining Prairie View, he taught at Florida Atlantic University, Boca Raton, and Temple University, Philadelphia. He has also worked for Lucent/Avaya and Boeing Satellite Systems.

Dr. Sadiku is the author of over 170 professional papers and almost 30 books including Elements of Electromagnetics (Oxford University Press, 3rd ed., 2001), Numerical Techniques in Electromagnetics (2nd ed., CRC Press, 2000), Simulation of Local Area Networks (with M. IIyas, CRC Press, 1994), Metropolitan Area Networks (CRC Press, 1994), and Fundamentals of Electric Circuits (with C. K. Alexander, McGraw-Hill, 3rd ed., 2007). His books are used worldwide, and some of them have been translated into Korean, Chinese, Italian, and Spanish. He was the recipient of the 2000 McGraw-Hill/Jacob Millman Award for outstanding contributions in the field of electrical engineering. He was the IEEE region 2 Student Activities Committee chairman and is an associate editor for IEEE "Transactions on Education." He received his Ph.D. at Tennessee Technological University, Cookeville.

PART TWO

AC Circuits

OUTLINE

9 Sinusoids and Phasors

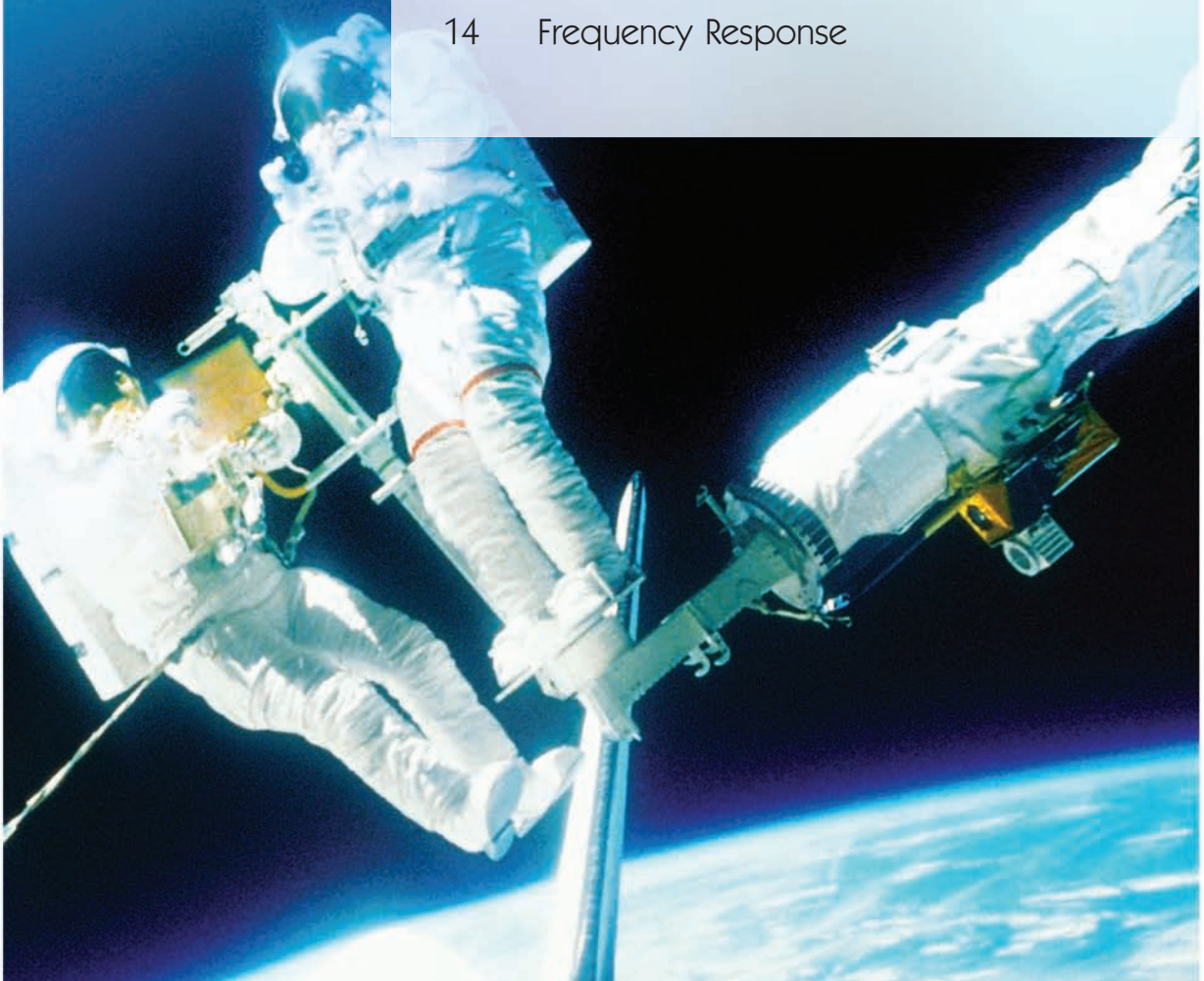
10 Sinusoidal Steady-State Analysis

11 AC Power Analysis

12 Three-Phase Circuits

13 Magnetically Coupled Circuits

14 Frequency Response



chapter

9

Sinusoids and Phasors

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—Persian Proverb

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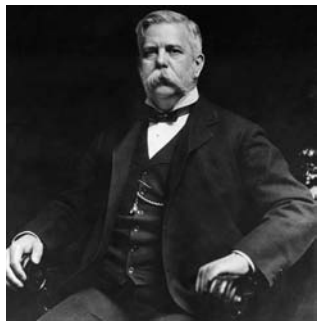
Most frequently, these teams include individuals from of a variety of engineering disciplines, as well as individuals from nonengineering disciplines such as marketing and finance.

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Photo by Charles Alexander

Historical



George Westinghouse. Photo © Bettmann/Corbis

Nikola Tesla (1856–1943) and **George Westinghouse** (1846–1914) helped establish alternating current as the primary mode of electricity transmission and distribution.

Today it is obvious that ac generation is well established as the form of electric power that makes widespread distribution of electric power efficient and economical. However, at the end of the 19th century, which was the better—ac or dc—was hotly debated and had extremely outspoken supporters on both sides. The dc side was lead by Thomas Edison, who had earned a lot of respect for his many contributions. Power generation using ac really began to build after the successful contributions of Tesla. The real commercial success in ac came from George Westinghouse and the outstanding team, including Tesla, he assembled. In addition, two other big names were C. F. Scott and B. G. Lamme.

The most significant contribution to the early success of ac was the patenting of the polyphase ac motor by Tesla in 1888. The induction motor and polyphase generation and distribution systems doomed the use of dc as the prime energy source.

9.1 Introduction

Thus far our analysis has been limited for the most part to dc circuits: those circuits excited by constant or time-invariant sources. We have restricted the forcing function to dc sources for the sake of simplicity, for pedagogic reasons, and also for historic reasons. Historically, dc sources were the main means of providing electric power up until the late 1800s. At the end of that century, the battle of direct current versus alternating current began. Both had their advocates among the electrical engineers of the time. Because ac is more efficient and economical to transmit over long distances, ac systems ended up the winner. Thus, it is in keeping with the historical sequence of events that we considered dc sources first.

We now begin the analysis of circuits in which the source voltage or current is time-varying. In this chapter, we are particularly interested in sinusoidally time-varying excitation, or simply, excitation by a *sinusoid.*

A **sinusoid** is a signal that has the form of the sine or cosine function.

A sinusoidal current is usually referred to as *alternating current* (*ac*). Such a current reverses at regular time intervals and has alternately positive and negative values. Circuits driven by sinusoidal current or voltage sources are called *ac circuits.*

We are interested in sinusoids for a number of reasons. First, nature itself is characteristically sinusoidal. We experience sinusoidal variation in the motion of a pendulum, the vibration of a string, the ripples on the ocean surface, and the natural response of underdamped second-order systems, to mention but a few. Second, a sinusoidal signal is easy to generate and transmit. It is the form of voltage generated throughout

the world and supplied to homes, factories, laboratories, and so on. It is the dominant form of signal in the communications and electric power industries. Third, through Fourier analysis, any practical periodic signal can be represented by a sum of sinusoids. Sinusoids, therefore, play an important role in the analysis of periodic signals. Lastly, a sinusoid is easy to handle mathematically. The derivative and integral of a sinusoid are themselves sinusoids. For these and other reasons, the sinusoid is an extremely important function in circuit analysis.

A sinusoidal forcing function produces both a transient response and a steady-state response, much like the step function, which we studied in Chapters 7 and 8. The transient response dies out with time so that only the steady-state response remains. When the transient response has become negligibly small compared with the steady-state response, we say that the circuit is operating at sinusoidal steady state. It is this *sinusoidal steady-state response* that is of main interest to us in this chapter.

We begin with a basic discussion of sinusoids and phasors. We then introduce the concepts of impedance and admittance. The basic circuit laws, Kirchhoff's and Ohm's, introduced for dc circuits, will be applied to ac circuits. Finally, we consider applications of ac circuits in phase-shifters and bridges.

9.2 Sinusoids

Consider the sinusoidal voltage

$$v(t) = V_m \sin \omega t \tag{9.1}$$

where

V_m = the *amplitude* of the sinusoid
ω = the *angular frequency* in radians/s
ωt = the *argument* of the sinusoid

The sinusoid is shown in Fig. 9.1(a) as a function of its argument and in Fig. 9.1(b) as a function of time. It is evident that the sinusoid repeats itself every T seconds; thus, T is called the *period* of the sinusoid. From the two plots in Fig. 9.1, we observe that $\omega T = 2\pi$,

$$T = \frac{2\pi}{\omega} \tag{9.2}$$

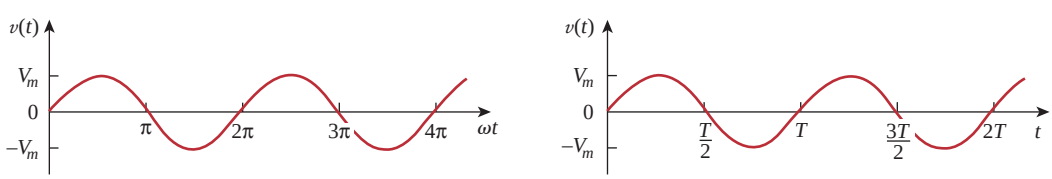


Figure 9.1
A sketch of $V_m \sin \omega t$: (a) as a function of ωt, (b) as a function of t.

Historical



The Burndy Library Collection at The Huntington Library, San Marino, California.

Heinrich Rudorf Hertz (1857–1894), a German experimental physicist, demonstrated that electromagnetic waves obey the same fundamental laws as light. His work confirmed James Clerk Maxwell's celebrated 1864 theory and prediction that such waves existed.

Hertz was born into a prosperous family in Hamburg, Germany. He attended the University of Berlin and did his doctorate under the prominent physicist Hermann von Helmholtz. He became a professor at Karlsruhe, where he began his quest for electromagnetic waves. Hertz successfully generated and detected electromagnetic waves; he was the first to show that light is electromagnetic energy. In 1887, Hertz noted for the first time the photoelectric effect of electrons in a molecular structure. Although Hertz only lived to the age of 37, his discovery of electromagnetic waves paved the way for the practical use of such waves in radio, television, and other communication systems. The unit of frequency, the hertz, bears his name.

The fact that $v(t)$ repeats itself every T seconds is shown by replacing t by $t + T$ in Eq. (9.1). We get

$$\begin{aligned} v(t+T) &= V_m \sin\omega(t+T) = V_m \sin\omega\left(t + \frac{2\pi}{\omega}\right) \\ &= V_m \sin(\omega t + 2\pi) = V_m \sin\omega t = v(t) \end{aligned} \tag{9.3}$$

Hence,

$$\boxed{v(t+T) = v(t)} \tag{9.4}$$

that is, v has the same value at $t + T$ as it does at t and $v(t)$ is said to be *periodic*. In general,

A **periodic function** is one that satisfies $f(t) = f(t + nT)$, for all t and for all integers n.

As mentioned, the *period* T of the periodic function is the time of one complete cycle or the number of seconds per cycle. The reciprocal of this quantity is the number of cycles per second, known as the *cyclic frequency* f of the sinusoid. Thus,

$$\boxed{f = \frac{1}{T}} \tag{9.5}$$

From Eqs. (9.2) and (9.5), it is clear that

$$\omega = 2\pi f \tag{9.6}$$

While ω is in radians per second (rad/s), f is in hertz (Hz).

The unit of f is named after the German physicist Heinrich R. Hertz (1857–1894).

Let us now consider a more general expression for the sinusoid,

$$v(t) = V_m \sin(\omega t + \phi) \tag{9.7}$$

where $(\omega t + \phi)$ is the argument and ϕ is the *phase*. Both argument and phase can be in radians or degrees.

Let us examine the two sinusoids

$$v_1(t) = V_m \sin \omega t \quad \text{and} \quad v_2(t) = V_m \sin(\omega t + \phi) \tag{9.8}$$

shown in Fig. 9.2. The starting point of v_2 in Fig. 9.2 occurs first in time. Therefore, we say that v_2 *leads* v_1 by ϕ or that v_1 *lags* v_2 by ϕ. If $\phi \neq 0$, we also say that v_1 and v_2 are *out of phase*. If $\phi = 0$, then v_1 and v_2 are said to be *in phase*; they reach their minima and maxima at exactly the same time. We can compare v_1 and v_2 in this manner because they operate at the same frequency; they do not need to have the same amplitude.

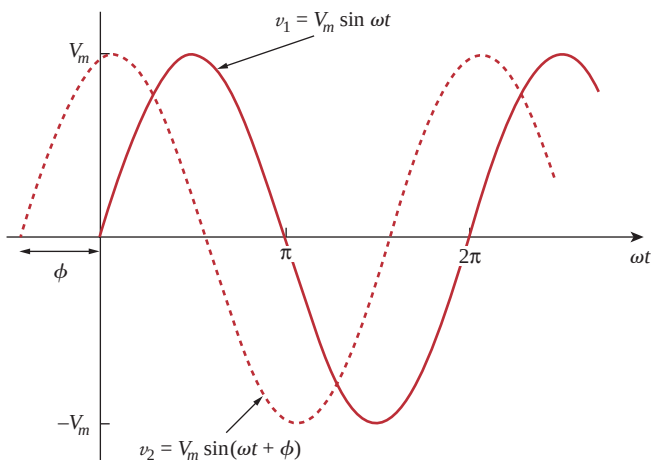


Figure 9.2
Two sinusoids with different phases.

A sinusoid can be expressed in either sine or cosine form. When comparing two sinusoids, it is expedient to express both as either sine or cosine with positive amplitudes. This is achieved by using the following trigonometric identities:

$$\begin{aligned} \sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\ \cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \end{aligned} \tag{9.9}$$

With these identities, it is easy to show that

$$\begin{aligned} \sin(\omega t \pm 180^\circ) &= -\sin \omega t \\ \cos(\omega t \pm 180^\circ) &= -\cos \omega t \\ \sin(\omega t \pm 90^\circ) &= \pm \cos \omega t \\ \cos(\omega t \pm 90^\circ) &= \mp \sin \omega t \end{aligned} \tag{9.10}$$

Using these relationships, we can transform a sinusoid from sine form to cosine form or vice versa.

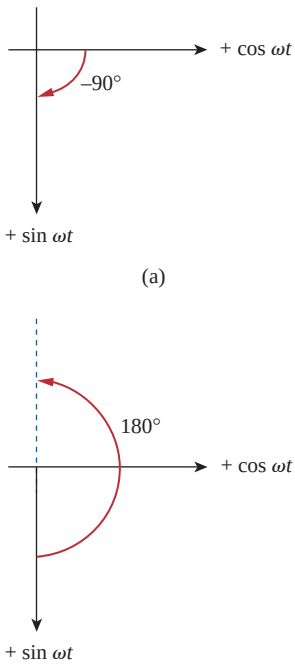


Figure 9.3
A graphical means of relating cosine and sine: (a) $\cos(\omega t - 90^\circ) = \sin\omega t$, (b) $\sin(\omega t + 180^\circ) = -\sin\omega t$.

A graphical approach may be used to relate or compare sinusoids as an alternative to using the trigonometric identities in Eqs. (9.9) and (9.10). Consider the set of axes shown in Fig. 9.3(a). The horizontal axis represents the magnitude of cosine, while the vertical axis (pointing down) denotes the magnitude of sine. Angles are measured positively counterclockwise from the horizontal, as usual in polar coordinates. This graphical technique can be used to relate two sinusoids. For example, we see in Fig. 9.3(a) that subtracting 90° from the argument of $\cos\omega t$ gives $\sin\omega t$, or $\cos(\omega t - 90^\circ) = \sin\omega t$. Similarly, adding 180° to the argument of $\sin\omega t$ gives $-\sin\omega t$, or $\sin(\omega t + 180^\circ) = -\sin\omega t$, as shown in Fig. 9.3(b).

The graphical technique can also be used to add two sinusoids of the same frequency when one is in sine form and the other is in cosine form. To add $A\cos\omega t$ and $B\sin\omega t$, we note that A is the magnitude of $\cos\omega t$ while B is the magnitude of $\sin\omega t$, as shown in Fig. 9.4(a). The magnitude and argument of the resultant sinusoid in cosine form is readily obtained from the triangle. Thus,

$$A\cos\omega t + B\sin\omega t = C\cos(\omega t - \theta) \tag{9.11}$$

where

$$C = \sqrt{A^2 + B^2}, \qquad \theta = \tan^{-1}\frac{B}{A} \tag{9.12}$$

For example, we may add $3\cos\omega t$ and $-4\sin\omega t$ as shown in Fig. 9.4(b) and obtain

$$3\cos\omega t - 4\sin\omega t = 5\cos(\omega t + 53.1^\circ) \tag{9.13}$$

Compared with the trigonometric identities in Eqs. (9.9) and (9.10), the graphical approach eliminates memorization. However, we must not confuse the sine and cosine axes with the axes for complex numbers to be discussed in the next section. Something else to note in Figs. 9.3 and 9.4 is that although the natural tendency is to have the vertical axis point up, the positive direction of the sine function is down in the present case.

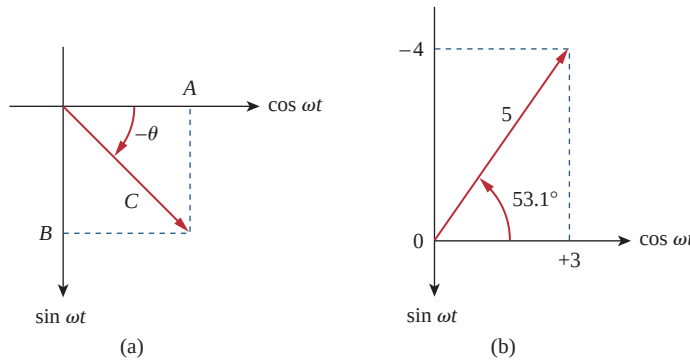


Figure 9.4
(a) Adding $A\cos\omega t$ and $B\sin\omega t$, (b) adding $3\cos\omega t$ and $-4\sin\omega t$.

Example 9.1

Find the amplitude, phase, period, and frequency of the sinusoid

$$v(t) = 12\cos(50t + 10^\circ)$$

Solution:

The amplitude is $V_m = 12$ V.
The phase is $\phi = 10^\circ$.
The angular frequency is $\omega = 50$ rad/s.
The period $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{50} = 0.1257$ s.
The frequency is $f = \dfrac{1}{T} = 7.958$ Hz.

Practice Problem 9.1

Given the sinusoid $5\sin(4\pi t - 60^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Answer: 5, -60°, 12.57 rad/s, 0.5 s, 2 Hz.

Example 9.2

Calculate the phase angle between $v_1 = -10\cos(\omega t + 50^\circ)$ and $v_2 = 12\sin(\omega t - 10^\circ)$. State which sinusoid is leading.

Solution:
Let us calculate the phase in three ways. The first two methods use trigonometric identities, while the third method uses the graphical approach.

■ **METHOD 1** In order to compare v_1 and v_2, we must express them in the same form. If we express them in cosine form with positive amplitudes,

$$v_1 = -10\cos(\omega t + 50^\circ) = 10\cos(\omega t + 50^\circ - 180^\circ)$$
$$v_1 = 10\cos(\omega t - 130^\circ) \quad \text{or} \quad v_1 = 10\cos(\omega t + 230^\circ) \qquad \textbf{(9.2.1)}$$

and

$$v_2 = 12\sin(\omega t - 10^\circ) = 12\cos(\omega t - 10^\circ - 90^\circ)$$
$$v_2 = 12\cos(\omega t - 100^\circ) \qquad \textbf{(9.2.2)}$$

It can be deduced from Eqs. (9.2.1) and (9.2.2) that the phase difference between v_1 and v_2 is 30°. We can write v_2 as

$$v_2 = 12\cos(\omega t - 130^\circ + 30^\circ) \quad \text{or} \quad v_2 = 12\cos(\omega t + 260^\circ) \qquad \textbf{(9.2.3)}$$

Comparing Eqs. (9.2.1) and (9.2.3) shows clearly that v_2 leads v_1 by 30°.

■ **METHOD 2** Alternatively, we may express v_1 in sine form:

$$v_1 = -10\cos(\omega t + 50^\circ) = 10\sin(\omega t + 50^\circ - 90^\circ)$$
$$= 10\sin(\omega t - 40^\circ) = 10\sin(\omega t - 10^\circ - 30^\circ)$$

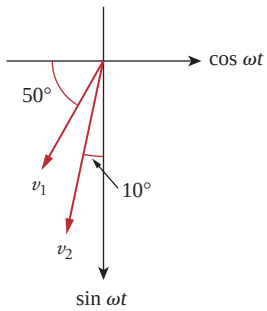


Figure 9.5
For Example 9.2.

But $v_2 = 12\sin(\omega t - 10°)$. Comparing the two shows that v_1 lags v_2 by 30°. This is the same as saying that v_2 leads v_1 by 30°.

■ **METHOD 3** We may regard v_1 as simply $-10\cos\omega t$ with a phase shift of $+50°$. Hence, v_1 is as shown in Fig. 9.5. Similarly, v_2 is $12\sin\omega t$ with a phase shift of $-10°$, as shown in Fig. 9.5. It is easy to see from Fig. 9.5 that v_2 leads v_1 by 30°, that is, $90° - 50° - 10°$.

Practice Problem 9.2

Find the phase angle between

$$i_1 = -4\sin(377t + 25°) \quad \text{and} \quad i_2 = 5\cos(377t - 40°)$$

Does i_1 lead or lag i_2?

Answer: 155°, i_1 leads i_2.

9.3 Phasors

Sinusoids are easily expressed in terms of phasors, which are more convenient to work with than sine and cosine functions.

A **phasor** is a complex number that represents the amplitude and phase of a sinusoid.

Phasors provide a simple means of analyzing linear circuits excited by sinusoidal sources; solutions of such circuits would be intractable otherwise. The notion of solving ac circuits using phasors was first introduced by Charles Steinmetz in 1893. Before we completely define phasors and apply them to circuit analysis, we need to be thoroughly familiar with complex numbers.

Charles Proteus Steinmetz (1865–1923) was a German-Austrian mathematician and electrical engineer.

Appendix B presents a short tutorial on complex numbers.

A complex number z can be written in rectangular form as

$$z = x + jy \tag{9.14a}$$

where $j = \sqrt{-1}$; x is the real part of z; y is the imaginary part of z. In this context, the variables x and y do not represent a location as in two-dimensional vector analysis but rather the real and imaginary parts of z in the complex plane. Nevertheless, we note that there are some resemblances between manipulating complex numbers and manipulating two-dimensional vectors.

The complex number z can also be written in polar or exponential form as

$$z = r\angle\phi = re^{j\phi} \tag{9.14b}$$

Historical



Charles Proteus Steinmetz (1865–1923), a German-Austrian mathematician and engineer, introduced the phasor method (covered in this chapter) in ac circuit analysis. He is also noted for his work on the theory of hysteresis.

Steinmetz was born in Breslau, Germany, and lost his mother at the age of one. As a youth, he was forced to leave Germany because of his political activities just as he was about to complete his doctoral dissertation in mathematics at the University of Breslau. He migrated to Switzerland and later to the United States, where he was employed by General Electric in 1893. That same year, he published a paper in which complex numbers were used to analyze ac circuits for the first time. This led to one of his many textbooks, *Theory and Calculation of ac Phenomena*, published by McGraw-Hill in 1897. In 1901, he became the president of the American Institute of Electrical Engineers, which later became the IEEE.

where r is the magnitude of z, and ϕ is the phase of z. We notice that z can be represented in three ways:

$$\begin{aligned} z &= x + jy && \text{Rectangular form} \\ z &= r\angle\phi && \text{Polar form} \\ z &= re^{j\phi} && \text{Exponential form} \end{aligned} \tag{9.15}$$

The relationship between the rectangular form and the polar form is shown in Fig. 9.6, where the x axis represents the real part and the y axis represents the imaginary part of a complex number. Given x and y, we can get r and ϕ as

$$r = \sqrt{x^2 + y^2}, \qquad \phi = \tan^{-1}\frac{y}{x} \tag{9.16a}$$

On the other hand, if we know r and ϕ, we can obtain x and y as

$$x = r\cos\phi, \qquad y = r\sin\phi \tag{9.16b}$$

Thus, z may be written as

$$\boxed{z = x + jy = r\angle\phi = r(\cos\phi + j\sin\phi)} \tag{9.17}$$

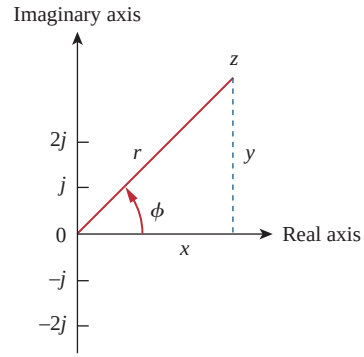


Figure 9.6
Representation of a complex number $z = x + jy = r\angle\phi$.

Addition and subtraction of complex numbers are better performed in rectangular form; multiplication and division are better done in polar form. Given the complex numbers

$$z = x + jy = r\angle\phi, \qquad z_1 = x_1 + jy_1 = r_1\angle\phi_1$$
$$z_2 = x_2 + jy_2 = r_2\angle\phi_2$$

the following operations are important.

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2) \tag{9.18a}$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2) \tag{9.18b}$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2 \tag{9.18c}$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2 \tag{9.18d}$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r} \angle -\phi \tag{9.18e}$$

Square Root:

$$\sqrt{z} = \sqrt{r} \angle \phi/2 \tag{9.18f}$$

Complex Conjugate:

$$z^* = x - jy = r\angle -\phi = re^{-j\phi} \tag{9.18g}$$

Note that from Eq. (9.18e),

$$\frac{1}{j} = -j \tag{9.18h}$$

These are the basic properties of complex numbers we need. Other properties of complex numbers can be found in Appendix B.

The idea of phasor representation is based on Euler's identity. In general,

$$\boxed{e^{\pm j\phi} = \cos\phi \pm j \sin\phi} \tag{9.19}$$

which shows that we may regard $\cos\phi$ and $\sin\phi$ as the real and imaginary parts of $e^{j\phi}$; we may write

$$\cos\phi = \text{Re}(e^{j\phi}) \tag{9.20a}$$

$$\sin\phi = \text{Im}(e^{j\phi}) \tag{9.20b}$$

where Re and Im stand for the *real part of* and the *imaginary part of*. Given a sinusoid $v(t) = V_m \cos(\omega t + \phi)$, we use Eq. (9.20a) to express $v(t)$ as

$$v(t) = V_m \cos(\omega t + \phi) = \text{Re}(V_m e^{j(\omega t + \phi)}) \tag{9.21}$$

or

$$v(t) = \text{Re}(V_m e^{j\phi} e^{j\omega t}) \tag{9.22}$$

Thus,

$$\boxed{v(t) = \text{Re}(\mathbf{V} e^{j\omega t})} \tag{9.23}$$

where

$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi \tag{9.24}$$

V is thus the *phasor representation* of the sinusoid $v(t)$, as we said earlier. In other words, a phasor is a complex representation of the magnitude and phase of a sinusoid. Either Eq. (9.20a) or Eq. (9.20b) can be used to develop the phasor, but the standard convention is to use Eq. (9.20a).

A phasor may be regarded as a mathematical equivalent of a sinusoid with the time dependence dropped.

One way of looking at Eqs. (9.23) and (9.24) is to consider the plot of the *sinor* $\mathbf{V}e^{j\omega t} = V_m e^{j(\omega t+\phi)}$ on the complex plane. As time increases, the sinor rotates on a circle of radius V_m at an angular velocity ω in the counterclockwise direction, as shown in Fig. 9.7(a). We may regard $v(t)$ as the projection of the sinor $\mathbf{V}e^{j\omega t}$ on the real axis, as shown in Fig. 9.7(b). The value of the sinor at time $t = 0$ is the phasor **V** of the sinusoid $v(t)$. The sinor may be regarded as a rotating phasor. Thus, whenever a sinusoid is expressed as a phasor, the term $e^{j\omega t}$ is implicitly present. It is therefore important, when dealing with phasors, to keep in mind the frequency ω of the phasor; otherwise we can make serious mistakes.

If we use sine for the phasor instead of cosine, then $v(t) = V_m \sin(\omega t + \phi) = \text{Im}(V_m e^{j(\omega t+\phi)})$ and the corresponding phasor is the same as that in Eq. (9.24).

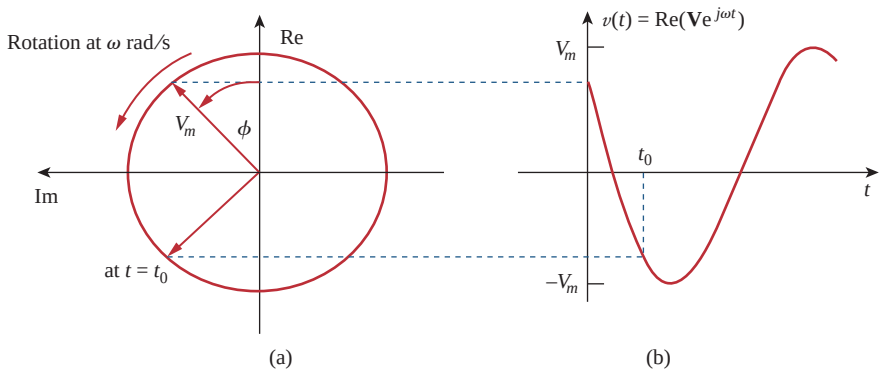


Figure 9.7
Representation of $\mathbf{V}e^{j\omega t}$: (a) sinor rotating counterclockwise, (b) its projection on the real axis, as a function of time.

Equation (9.23) states that to obtain the sinusoid corresponding to a given phasor **V**, multiply the phasor by the time factor $e^{j\omega t}$ and take the real part. As a complex quantity, a phasor may be expressed in rectangular form, polar form, or exponential form. Since a phasor has magnitude and phase ("direction"), it behaves as a vector and is printed in boldface. For example, phasors $\mathbf{V} = V_m\angle\phi$ and $\mathbf{I} = I_m\angle-\theta$ are graphically represented in Fig. 9.8. Such a graphical representation of phasors is known as a *phasor diagram.*

We use lightface italic letters such as z to represent complex numbers but boldface letters such as **V** to represent phasors, because phasors are vector-like quantities.

Equations (9.21) through (9.23) reveal that to get the phasor corresponding to a sinusoid, we first express the sinusoid in the cosine form so that the sinusoid can be written as the real part of a complex number. Then we take out the time factor $e^{j\omega t}$, and whatever is left is the phasor corresponding to the sinusoid. By suppressing the time factor, we transform the sinusoid from the time domain to the phasor domain. This transformation is summarized as follows:

$$v(t) = V_m \cos(\omega t + \phi) \quad \Leftrightarrow \quad \mathbf{V} = V_m\angle\phi \tag{9.25}$$

(Time-domain representation) (Phasor-domain representation)

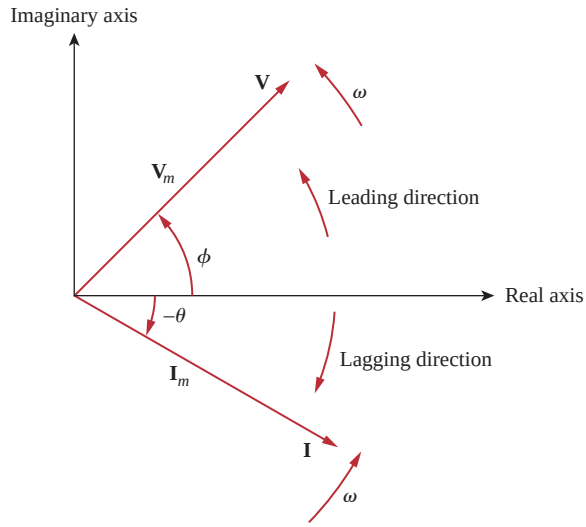


Figure 9.8
A phasor diagram showing $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$.

Given a sinusoid $v(t) = V_m \cos(\omega t + \phi)$, we obtain the corresponding phasor as $\mathbf{V} = V_m \angle \phi$. Equation (9.25) is also demonstrated in Table 9.1, where the sine function is considered in addition to the cosine function. From Eq. (9.25), we see that to get the phasor representation of a sinusoid, we express it in cosine form and take the magnitude and phase. Given a phasor, we obtain the time domain representation as the cosine function with the same magnitude as the phasor and the argument as ωt plus the phase of the phasor. The idea of expressing information in alternate domains is fundamental to all areas of engineering.

TABLE 9.1

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

Note that in Eq. (9.25) the frequency (or time) factor $e^{j\omega t}$ is suppressed, and the frequency is not explicitly shown in the phasor domain representation because ω is constant. However, the response depends on ω. For this reason, the phasor domain is also known as the *frequency domain*.

From Eqs. (9.23) and (9.24), $v(t) = \text{Re}(\mathbf{V}e^{j\omega t}) = V_m \cos(\omega t + \phi)$, so that

$$\begin{aligned} \frac{dv}{dt} &= -\omega V_m \sin(\omega t + \phi) = \omega V_m \cos(\omega t + \phi + 90^\circ) \\ &= \text{Re}(\omega V_m e^{j\omega t} e^{j\phi} e^{j90^\circ}) = \text{Re}(j\omega \mathbf{V} e^{j\omega t}) \end{aligned} \tag{9.26}$$

This shows that the derivative $v(t)$ is transformed to the phasor domain as $j\omega\mathbf{V}$

$$\underbrace{\frac{dv}{dt}}_{\text{(Time domain)}} \quad \Leftrightarrow \quad \underbrace{j\omega\mathbf{V}}_{\text{(Phasor domain)}} \tag{9.27}$$

Differentiating a sinusoid is equivalent to multiplying its corresponding phasor by $j\omega$.

Similarly, the integral of $v(t)$ is transformed to the phasor domain as $\mathbf{V}/j\omega$

$$\underbrace{\int v\,dt}_{\text{(Time domain)}} \quad \Leftrightarrow \quad \underbrace{\frac{\mathbf{V}}{j\omega}}_{\text{(Phasor domain)}} \tag{9.28}$$

Integrating a sinusoid is equivalent to dividing its corresponding phasor by $j\omega$.

Equation (9.27) allows the replacement of a derivative with respect to time with multiplication of $j\omega$ in the phasor domain, whereas Eq. (9.28) allows the replacement of an integral with respect to time with division by $j\omega$ in the phasor domain. Equations (9.27) and (9.28) are useful in finding the steady-state solution, which does not require knowing the initial values of the variable involved. This is one of the important applications of phasors.

Besides time differentiation and integration, another important use of phasors is found in summing sinusoids of the same frequency. This is best illustrated with an example, and Example 9.6 provides one.

Adding sinusoids of the same frequency is equivalent to adding their corresponding phasors.

The differences between $v(t)$ and $\mathbf{V}$ should be emphasized:

1. $v(t)$ is the *instantaneous or time domain* representation, while $\mathbf{V}$ is the *frequency or phasor domain* representation.
2. $v(t)$ is time dependent, while $\mathbf{V}$ is not. (This fact is often forgotten by students.)
3. $v(t)$ is always real with no complex term, while $\mathbf{V}$ is generally complex.

Finally, we should bear in mind that phasor analysis applies only when frequency is constant; it applies in manipulating two or more sinusoidal signals only if they are of the same frequency.

Example 9.3

Evaluate these complex numbers:

(a) $(40\angle 50^\circ + 20\angle -30^\circ)^{1/2}$

(b) $\dfrac{10\angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*}$

Solution:

(a) Using polar to rectangular transformation,

$$40\angle 50^\circ = 40(\cos 50^\circ + j \sin 50^\circ) = 25.71 + j30.64$$

$$20\angle -30^\circ = 20[\cos(-30^\circ) + j \sin(-30^\circ)] = 17.32 - j10$$

Adding them up gives

$$40\angle 50^\circ + 20\angle -30^\circ = 43.03 + j20.64 = 47.72\angle 25.63^\circ$$

Taking the square root of this,

$$(40\angle 50^\circ + 20\angle -30^\circ)^{1/2} = 6.91\angle 12.81^\circ$$

(b) Using polar-rectangular transformation, addition, multiplication, and division,

$$\frac{10\angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*} = \frac{8.66 - j5 + (3 - j4)}{(2 + j4)(3 + j5)}$$

$$= \frac{11.66 - j9}{-14 + j22} = \frac{14.73\angle -37.66^\circ}{26.08\angle 122.47^\circ}$$

$$= 0.565\angle -160.13^\circ$$

Practice Problem 9.3

Evaluate the following complex numbers:

(a) $[(5 + j2)(-1 + j4) - 5\angle 60^\circ]^*$

(b) $\dfrac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5$

Answer: (a) $-15.5 - j13.67$, (b) $8.293 + j7.2$.

Example 9.4

Transform these sinusoids to phasors:

(a) $i = 6\cos(50t - 40^\circ)$ A
(b) $v = -4\sin(30t + 50^\circ)$ V

Solution:

(a) $i = 6\cos(50t - 40^\circ)$ has the phasor

$$\mathbf{I} = 6\angle -40^\circ \text{ A}$$

(b) Since $-\sin A = \cos(A + 90^\circ)$,

$$v = -4\sin(30t + 50^\circ) = 4\cos(30t + 50^\circ + 90^\circ)$$

$$= 4\cos(30t + 140^\circ) \text{ V}$$

The phasor form of v is

$$\mathbf{V} = 4\angle 140^\circ \text{ V}$$

Practice Problem 9.4

Express these sinusoids as phasors:

(a) $v = 7\cos(2t + 40^\circ)$ V
(b) $i = -4\sin(10t + 10^\circ)$ A

Answer: (a) $\mathbf{V} = 7\angle 40^\circ$ V, (b) $\mathbf{I} = 4\angle 100^\circ$ A.

Example 9.5

Find the sinusoids represented by these phasors:

(a) $\mathbf{I} = -3 + j4$ A
(b) $\mathbf{V} = j8e^{-j20^\circ}$ V

Solution:

(a) $\mathbf{I} = -3 + j4 = 5\angle 126.87^\circ$. Transforming this to the time domain gives

$$i(t) = 5\cos(\omega t + 126.87^\circ)\ \text{A}$$

(b) Since $j = 1\angle 90^\circ$,

$$\mathbf{V} = j8\angle -20^\circ = (1\angle 90^\circ)(8\angle -20^\circ)$$
$$= 8\angle 90^\circ - 20^\circ = 8\angle 70^\circ\ \text{V}$$

Converting this to the time domain gives

$$v(t) = 8\cos(\omega t + 70^\circ)\ \text{V}$$

Practice Problem 9.5

Find the sinusoids corresponding to these phasors:

(a) $\mathbf{V} = -10\angle 30^\circ$ V
(b) $\mathbf{I} = j(5 - j12)$ A

Answer: (a) $v(t) = 10\cos(\omega t + 210^\circ)$ V or $10\cos(\omega t - 150^\circ)$ V,
(b) $i(t) = 13\cos(\omega t + 22.62^\circ)$ A.

Example 9.6

Given $i_1(t) = 4\cos(\omega t + 30^\circ)$ A and $i_2(t) = 5\sin(\omega t - 20^\circ)$ A, find their sum.

Solution:

Here is an important use of phasors—for summing sinusoids of the same frequency. Current $i_1(t)$ is in the standard form. Its phasor is

$$\mathbf{I}_1 = 4\angle 30^\circ$$

We need to express $i_2(t)$ in cosine form. The rule for converting sine to cosine is to subtract 90°. Hence,

$$i_2 = 5\cos(\omega t - 20^\circ - 90^\circ) = 5\cos(\omega t - 110^\circ)$$

and its phasor is

$$\mathbf{I}_2 = 5\angle -110^\circ$$

If we let $i = i_1 + i_2$, then

$$\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 = 4\angle 30^\circ + 5\angle -110^\circ$$
$$= 3.464 + j2 - 1.71 - j4.698 = 1.754 - j2.698$$
$$= 3.218\angle -56.97^\circ\ \text{A}$$

Transforming this to the time domain, we get

$$i(t) = 3.218\cos(\omega t - 56.97^\circ)\ \text{A}$$

Of course, we can find $i_1 + i_2$ using Eq. (9.9), but that is the hard way.

Practice Problem 9.6

If $v_1 = -10\sin(\omega t - 30^\circ)$ V and $v_2 = 20\cos(\omega t + 45^\circ)$ V, find $v = v_1 + v_2$.

Answer: $v(t) = 12.158\cos(\omega t + 55.95^\circ)$ V.

Example 9.7

Using the phasor approach, determine the current $i(t)$ in a circuit described by the integrodifferential equation

$$4i + 8\int i\,dt - 3\frac{di}{dt} = 50\cos(2t + 75^\circ)$$

Solution:

We transform each term in the equation from time domain to phasor domain. Keeping Eqs. (9.27) and (9.28) in mind, we obtain the phasor form of the given equation as

$$4\mathbf{I} + \frac{8\mathbf{I}}{j\omega} - 3j\omega\mathbf{I} = 50\angle 75^\circ$$

But $\omega = 2$, so

$$\mathbf{I}(4 - j4 - j6) = 50\angle 75^\circ$$

$$\mathbf{I} = \frac{50\angle 75^\circ}{4 - j10} = \frac{50\angle 75^\circ}{10.77\angle -68.2^\circ} = 4.642\angle 143.2^\circ\ \text{A}$$

Converting this to the time domain,

$$i(t) = 4.642\cos(2t + 143.2^\circ)\ \text{A}$$

Keep in mind that this is only the steady-state solution, and it does not require knowing the initial values.

Practice Problem 9.7

Find the voltage $v(t)$ in a circuit described by the integrodifferential equation

$$2\frac{dv}{dt} + 5v + 10\int v\,dt = 50\cos(5t - 30^\circ)$$

using the phasor approach.

Answer: $v(t) = 5.3\cos(5t - 88^\circ)$ V.

9.4 Phasor Relationships for Circuit Elements

Now that we know how to represent a voltage or current in the phasor or frequency domain, one may legitimately ask how we apply this to circuits involving the passive elements R, L, and C. What we need to do is to transform the voltage-current relationship from the time domain to the frequency domain for each element. Again, we will assume the passive sign convention.

We begin with the resistor. If the current through a resistor R is $i = I_m \cos(\omega t + \phi)$, the voltage across it is given by Ohm's law as

$$v = iR = RI_m \cos(\omega t + \phi) \tag{9.29}$$

The phasor form of this voltage is

$$\mathbf{V} = RI_m \angle \phi \tag{9.30}$$

But the phasor representation of the current is $\mathbf{I} = I_m \angle \phi$. Hence,

$$\mathbf{V} = R\mathbf{I} \tag{9.31}$$

showing that the voltage-current relation for the resistor in the phasor domain continues to be Ohm's law, as in the time domain. Figure 9.9 illustrates the voltage-current relations of a resistor. We should note from Eq. (9.31) that voltage and current are in phase, as illustrated in the phasor diagram in Fig. 9.10.

For the inductor L, assume the current through it is $i = I_m \cos(\omega t + \phi)$. The voltage across the inductor is

$$v = L\frac{di}{dt} = -\omega L I_m \sin(\omega t + \phi) \tag{9.32}$$

Recall from Eq. (9.10) that $-\sin A = \cos(A + 90°)$. We can write the voltage as

$$v = \omega L I_m \cos(\omega t + \phi + 90°) \tag{9.33}$$

which transforms to the phasor

$$\mathbf{V} = \omega L I_m e^{j(\phi + 90°)} = \omega L I_m e^{j\phi} e^{j90°} = \omega L I_m \angle \phi + 90° \tag{9.34}$$

But $I_m \angle \phi = \mathbf{I}$, and from Eq. (9.19), $e^{j90°} = j$. Thus,

$$\mathbf{V} = j\omega L\mathbf{I} \tag{9.35}$$

showing that the voltage has a magnitude of $\omega L I_m$ and a phase of $\phi + 90°$. The voltage and current are 90° out of phase. Specifically, the current lags the voltage by 90°. Figure 9.11 shows the voltage-current relations for the inductor. Figure 9.12 shows the phasor diagram.

For the capacitor C, assume the voltage across it is $v = V_m \cos(\omega t + \phi)$. The current through the capacitor is

$$i = C\frac{dv}{dt} \tag{9.36}$$

By following the same steps as we took for the inductor or by applying Eq. (9.27) on Eq. (9.36), we obtain

$$\mathbf{I} = j\omega C\mathbf{V} \quad \Rightarrow \quad \mathbf{V} = \frac{\mathbf{I}}{j\omega C} \tag{9.37}$$

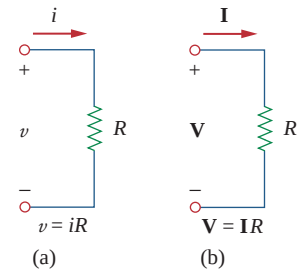


Figure 9.9
Voltage-current relations for a resistor in the: (a) time domain, (b) frequency domain.

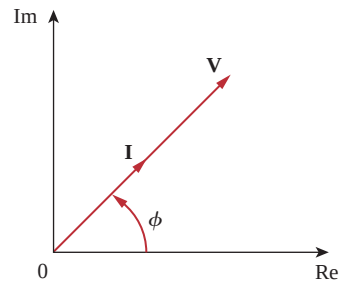


Figure 9.10
Phasor diagram for the resistor.

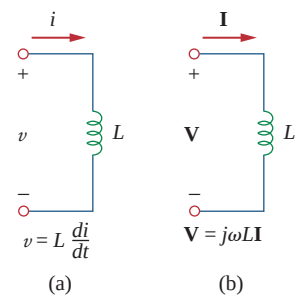


Figure 9.11
Voltage-current relations for an inductor in the: (a) time domain, (b) frequency domain.

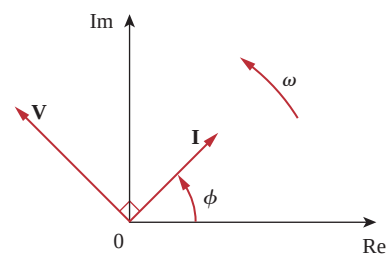


Figure 9.12
Phasor diagram for the inductor; **I** lags **V**.

Although it is equally correct to say that the inductor voltage leads the current by 90°, convention gives the current phase relative to the voltage.

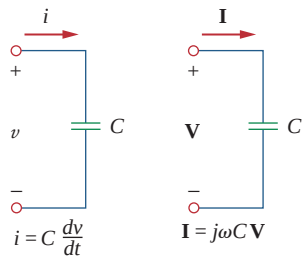


Figure 9.13
Voltage-current relations for a capacitor in the: (a) time domain, (b) frequency domain.

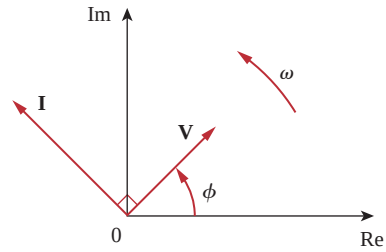


Figure 9.14
Phasor diagram for the capacitor; **I** leads **V**.

showing that the current and voltage are 90° out of phase. To be specific, the current leads the voltage by 90°. Figure 9.13 shows the voltage-current relations for the capacitor; Fig. 9.14 gives the phasor diagram. Table 9.2 summarizes the time domain and phasor domain representations of the circuit elements.

TABLE 9.2

Summary of voltage-current relationships.

Element	Time domain	Frequency domain
R	$v = Ri$	$\mathbf{V} = R\mathbf{I}$
L	$v = L\frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$
C	$i = C\frac{dv}{dt}$	$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$

Example 9.8

The voltage $v = 12\cos(60t + 45°)$ is applied to a 0.1-H inductor. Find the steady-state current through the inductor.

Solution:
For the inductor, $\mathbf{V} = j\omega L\mathbf{I}$, where $\omega = 60$ rad/s and $\mathbf{V} = 12\angle 45°$ V. Hence,

$$\mathbf{I} = \frac{\mathbf{V}}{j\omega L} = \frac{12\angle 45°}{j60 \times 0.1} = \frac{12\angle 45°}{6\angle 90°} = 2\angle -45° \text{ A}$$

Converting this to the time domain,

$$i(t) = 2\cos(60t - 45°) \text{ A}$$

Practice Problem 9.8

If voltage $v = 10\cos(100t + 30°)$ is applied to a 50 μF capacitor, calculate the current through the capacitor.

Answer: $50\cos(100t + 120°)$ mA.

9.5 Impedance and Admittance

In the preceding section, we obtained the voltage-current relations for the three passive elements as

$$\mathbf{V} = R\mathbf{I}, \qquad \mathbf{V} = j\omega L\mathbf{I}, \qquad \mathbf{V} = \frac{\mathbf{I}}{j\omega C} \tag{9.38}$$

These equations may be written in terms of the ratio of the phasor voltage to the phasor current as

$$\frac{\mathbf{V}}{\mathbf{I}} = R, \qquad \frac{\mathbf{V}}{\mathbf{I}} = j\omega L, \qquad \frac{\mathbf{V}}{\mathbf{I}} = \frac{1}{j\omega C} \tag{9.39}$$

From these three expressions, we obtain Ohm's law in phasor form for any type of element as

$$\boxed{\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} \quad \text{or} \quad \mathbf{V} = \mathbf{ZI}} \tag{9.40}$$

where $\mathbf{Z}$ is a frequency-dependent quantity known as *impedance*, measured in ohms.

The impedance $\mathbf{Z}$ of a circuit is the ratio of the phasor voltage $\mathbf{V}$ to the phasor current $\mathbf{I}$, measured in ohms (Ω).

The impedance represents the opposition that the circuit exhibits to the flow of sinusoidal current. Although the impedance is the ratio of two phasors, it is not a phasor, because it does not correspond to a sinusoidally varying quantity.

The impedances of resistors, inductors, and capacitors can be readily obtained from Eq. (9.39). Table 9.3 summarizes their impedances. From the table we notice that $\mathbf{Z}_L = j\omega L$ and $\mathbf{Z}_C = -j/\omega C$. Consider two extreme cases of angular frequency. When $\omega = 0$ (i.e., for dc sources), $\mathbf{Z}_L = 0$ and $\mathbf{Z}_C \to \infty$, confirming what we already know—that the inductor acts like a short circuit, while the capacitor acts like an open circuit. When $\omega \to \infty$ (i.e., for high frequencies), $\mathbf{Z}_L \to \infty$ and $\mathbf{Z}_C = 0$, indicating that the inductor is an open circuit to high frequencies, while the capacitor is a short circuit. Figure 9.15 illustrates this.

As a complex quantity, the impedence may be expressed in rectangular form as

$$\mathbf{Z} = R + jX \tag{9.41}$$

where $R = \text{Re}\,\mathbf{Z}$ is the *resistance* and $X = \text{Im}\,\mathbf{Z}$ is the *reactance*. The reactance X may be positive or negative. We say that the impedance is inductive when X is positive or capacitive when X is negative. Thus, impedance $\mathbf{Z} = R + jX$ is said to be *inductive* or lagging since current lags voltage, while impedance $\mathbf{Z} = R - jX$ is capacitive or leading because current leads voltage. The impedance, resistance, and reactance are all measured in ohms. The impedance may also be expressed in polar form as

$$\mathbf{Z} = |\mathbf{Z}| \angle \theta \tag{9.42}$$

TABLE 9.3

Impedances and admittances of passive elements.

Element	Impedance	Admittance
R	$\mathbf{Z} = R$	$\mathbf{Y} = \frac{1}{R}$
L	$\mathbf{Z} = j\omega L$	$\mathbf{Y} = \frac{1}{j\omega L}$
C	$\mathbf{Z} = \frac{1}{j\omega C}$	$\mathbf{Y} = j\omega C$

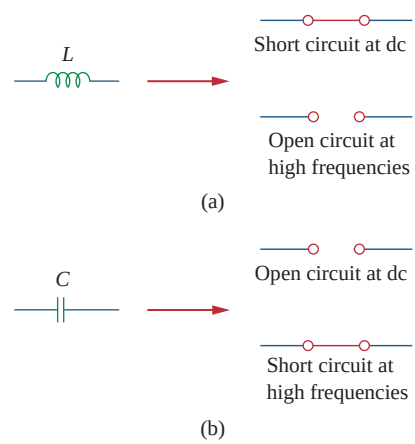


Figure 9.15
Equivalent circuits at dc and high frequencies: (a) inductor, (b) capacitor.

Comparing Eqs. (9.41) and (9.42), we infer that

$$\mathbf{Z} = R + jX = |\mathbf{Z}|\angle\theta \tag{9.43}$$

where

$$|\mathbf{Z}| = \sqrt{R^2 + X^2}, \qquad \theta = \tan^{-1}\frac{X}{R} \tag{9.44}$$

and

$$R = |\mathbf{Z}|\cos\theta, \qquad X = |\mathbf{Z}|\sin\theta \tag{9.45}$$

It is sometimes convenient to work with the reciprocal of impedance, known as *admittance*.

The **admittance** $\mathbf{Y}$ is the reciprocal of impedance, measured in siemens (S).

The admittance $\mathbf{Y}$ of an element (or a circuit) is the ratio of the phasor current through it to the phasor voltage across it, or

$$\mathbf{Y} = \frac{1}{\mathbf{Z}} = \frac{\mathbf{I}}{\mathbf{V}} \tag{9.46}$$

The admittances of resistors, inductors, and capacitors can be obtained from Eq. (9.39). They are also summarized in Table 9.3.

As a complex quantity, we may write $\mathbf{Y}$ as

$$\mathbf{Y} = G + jB \tag{9.47}$$

where $G = \text{Re}\,\mathbf{Y}$ is called the *conductance* and $B = \text{Im}\,\mathbf{Y}$ is called the *susceptance*. Admittance, conductance, and susceptance are all expressed in the unit of siemens (or mhos). From Eqs. (9.41) and (9.47),

$$G + jB = \frac{1}{R + jX} \tag{9.48}$$

By rationalization,

$$G + jB = \frac{1}{R + jX} \cdot \frac{R - jX}{R - jX} = \frac{R - jX}{R^2 + X^2} \tag{9.49}$$

Equating the real and imaginary parts gives

$$G = \frac{R}{R^2 + X^2}, \qquad B = -\frac{X}{R^2 + X^2} \tag{9.50}$$

showing that $G \neq 1/R$ as it is in resistive circuits. Of course, if $X = 0$, then $G = 1/R$.

Example 9.9

Find $v(t)$ and $i(t)$ in the circuit shown in Fig. 9.16.

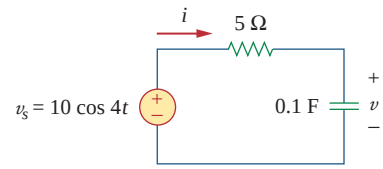


Figure 9.16
For Example 9.9.

Solution:
From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10\angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5\ \Omega$$

Hence the current

$$\mathbf{I} = \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10\angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \tag{9.9.1}$$
$$= 1.6 + j0.8 = 1.789\angle 26.57^\circ \text{ A}$$

The voltage across the capacitor is

$$\mathbf{V} = \mathbf{I}\mathbf{Z}_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789\angle 26.57^\circ}{j4 \times 0.1} \tag{9.9.2}$$
$$= \frac{1.789\angle 26.57^\circ}{0.4\angle 90^\circ} = 4.47\angle -63.43^\circ \text{ V}$$

Converting $\mathbf{I}$ and $\mathbf{V}$ in Eqs. (9.9.1) and (9.9.2) to the time domain, we get

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$
$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

Notice that $i(t)$ leads $v(t)$ by 90° as expected.

Practice Problem 9.9

Refer to Fig. 9.17. Determine $v(t)$ and $i(t)$.

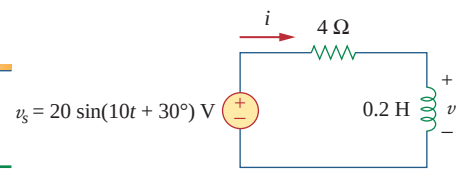


Figure 9.17
For Practice Prob. 9.9.

Answer: $8.944 \sin(10t + 93.43^\circ)$ V, $4.472 \sin(10t + 3.43^\circ)$ A.

9.6 †Kirchhoff's Laws in the Frequency Domain

We cannot do circuit analysis in the frequency domain without Kirchhoff's current and voltage laws. Therefore, we need to express them in the frequency domain.

For KVL, let $v_1, v_2, \ldots, v_n$ be the voltages around a closed loop. Then

$$v_1 + v_2 + \cdots + v_n = 0 \tag{9.51}$$

In the sinusoidal steady state, each voltage may be written in cosine form, so that Eq. (9.51) becomes

$$V_{m1} \cos(\omega t + \theta_1) + V_{m2} \cos(\omega t + \theta_2) + \cdots + V_{mn} \cos(\omega t + \theta_n) = 0 \tag{9.52}$$

This can be written as

$$\text{Re}(V_{m1}e^{j\theta_1}e^{j\omega t}) + \text{Re}(V_{m2}e^{j\theta_2}e^{j\omega t}) + \cdots + \text{Re}(V_{mn}e^{j\theta_n}e^{j\omega t}) = 0$$

or

$$\text{Re}[(V_{m1}e^{j\theta_1} + V_{m2}e^{j\theta_2} + \cdots + V_{mn}e^{j\theta_n})e^{j\omega t}] = 0 \quad (9.53)$$

If we let $\mathbf{V}_k = V_{mk}e^{j\theta_k}$, then

$$\text{Re}[(\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n)e^{j\omega t}] = 0 \quad (9.54)$$

Since $e^{j\omega t} \neq 0$,

$$\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_n = 0 \quad (9.55)$$

indicating that Kirchhoff's voltage law holds for phasors.

By following a similar procedure, we can show that Kirchhoff's current law holds for phasors. If we let $i_1, i_2, \ldots, i_n$ be the current leaving or entering a closed surface in a network at time t, then

$$i_1 + i_2 + \cdots + i_n = 0 \quad (9.56)$$

If $\mathbf{I}_1, \mathbf{I}_2, \ldots, \mathbf{I}_n$ are the phasor forms of the sinusoids $i_1, i_2, \ldots, i_n$, then

$$\mathbf{I}_1 + \mathbf{I}_2 + \cdots + \mathbf{I}_n = 0 \quad (9.57)$$

which is Kirchhoff's current law in the frequency domain.

Once we have shown that both KVL and KCL hold in the frequency domain, it is easy to do many things, such as impedance combination, nodal and mesh analyses, superposition, and source transformation.

9.7 Impedance Combinations

Consider the N series-connected impedances shown in Fig. 9.18. The same current $\mathbf{I}$ flows through the impedances. Applying KVL around the loop gives

$$\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_N = \mathbf{I}(\mathbf{Z}_1 + \mathbf{Z}_2 + \cdots + \mathbf{Z}_N) \quad (9.58)$$

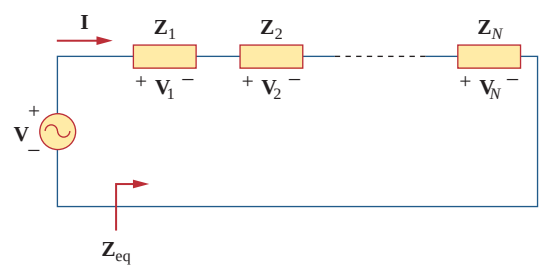


Figure 9.18
N impedances in series.

The equivalent impedance at the input terminals is

$$\mathbf{Z}_{\text{eq}} = \frac{\mathbf{V}}{\mathbf{I}} = \mathbf{Z}_1 + \mathbf{Z}_2 + \cdots + \mathbf{Z}_N$$

or

$$\boxed{\mathbf{Z}_{\text{eq}} = \mathbf{Z}_1 + \mathbf{Z}_2 + \cdots + \mathbf{Z}_N} \quad (9.59)$$

showing that the total or equivalent impedance of series-connected impedances is the sum of the individual impedances. This is similar to the series connection of resistances.

If $N = 2$, as shown in Fig. 9.19, the current through the impedances is

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}_1 + \mathbf{Z}_2} \tag{9.60}$$

Since $\mathbf{V}_1 = \mathbf{Z}_1\mathbf{I}$ and $\mathbf{V}_2 = \mathbf{Z}_2\mathbf{I}$, then

$$\mathbf{V}_1 = \frac{\mathbf{Z}_1}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V}, \qquad \mathbf{V}_2 = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V} \tag{9.61}$$

which is the *voltage-division* relationship.

In the same manner, we can obtain the equivalent impedance or admittance of the N parallel-connected impedances shown in Fig. 9.20. The voltage across each impedance is the same. Applying KCL at the top node,

$$\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 + \cdots + \mathbf{I}_N = \mathbf{V}\left(\frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \cdots + \frac{1}{\mathbf{Z}_N}\right) \tag{9.62}$$

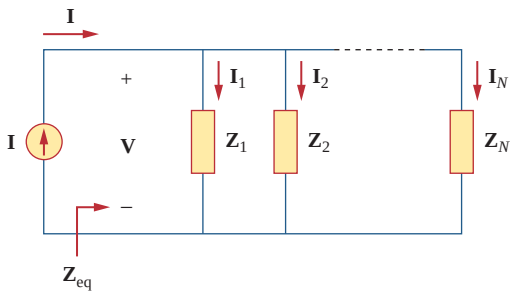


Figure 9.20
N impedances in parallel.

The equivalent impedance is

$$\frac{1}{\mathbf{Z}_{eq}} = \frac{\mathbf{I}}{\mathbf{V}} = \frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2} + \cdots + \frac{1}{\mathbf{Z}_N} \tag{9.63}$$

and the equivalent admittance is

$$\mathbf{Y}_{eq} = \mathbf{Y}_1 + \mathbf{Y}_2 + \cdots + \mathbf{Y}_N \tag{9.64}$$

This indicates that the equivalent admittance of a parallel connection of admittances is the sum of the individual admittances.

When $N = 2$, as shown in Fig. 9.21, the equivalent impedance becomes

$$\mathbf{Z}_{eq} = \frac{1}{\mathbf{Y}_{eq}} = \frac{1}{\mathbf{Y}_1 + \mathbf{Y}_2} = \frac{1}{1/\mathbf{Z}_1 + 1/\mathbf{Z}_2} = \frac{\mathbf{Z}_1\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \tag{9.65}$$

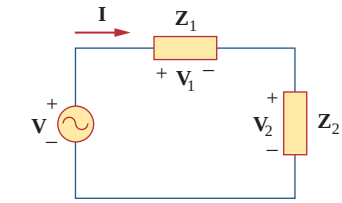


Figure 9.19
Voltage division.

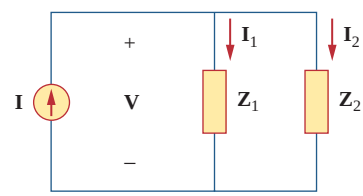


Figure 9.21
Current division.

Also, since

$$\mathbf{V} = \mathbf{I}\mathbf{Z}_{eq} = \mathbf{I}_1\mathbf{Z}_1 = \mathbf{I}_2\mathbf{Z}_2$$

the currents in the impedances are

$$\mathbf{I}_1 = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{I}, \qquad \mathbf{I}_2 = \frac{\mathbf{Z}_1}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{I} \tag{9.66}$$

which is the *current-division* principle.

The delta-to-wye and wye-to-delta transformations that we applied to resistive circuits are also valid for impedances. With reference to Fig. 9.22, the conversion formulas are as follows.

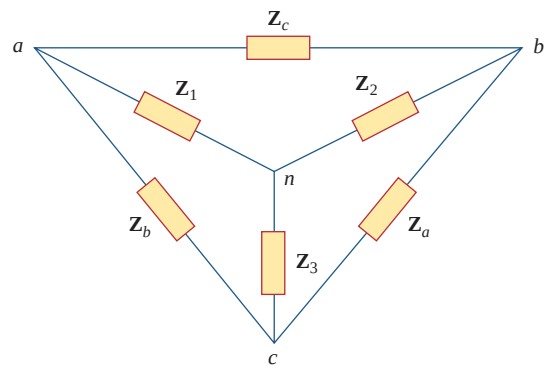


Figure 9.22
Superimposed *Y* and Δ networks.

Y-Δ *Conversion:*

$$\mathbf{Z}_a = \frac{\mathbf{Z}_1\mathbf{Z}_2 + \mathbf{Z}_2\mathbf{Z}_3 + \mathbf{Z}_3\mathbf{Z}_1}{\mathbf{Z}_1}$$
$$\mathbf{Z}_b = \frac{\mathbf{Z}_1\mathbf{Z}_2 + \mathbf{Z}_2\mathbf{Z}_3 + \mathbf{Z}_3\mathbf{Z}_1}{\mathbf{Z}_2} \tag{9.67}$$
$$\mathbf{Z}_c = \frac{\mathbf{Z}_1\mathbf{Z}_2 + \mathbf{Z}_2\mathbf{Z}_3 + \mathbf{Z}_3\mathbf{Z}_1}{\mathbf{Z}_3}$$

Δ-*Y Conversion:*

$$\mathbf{Z}_1 = \frac{\mathbf{Z}_b\mathbf{Z}_c}{\mathbf{Z}_a + \mathbf{Z}_b + \mathbf{Z}_c}$$
$$\mathbf{Z}_2 = \frac{\mathbf{Z}_c\mathbf{Z}_a}{\mathbf{Z}_a + \mathbf{Z}_b + \mathbf{Z}_c} \tag{9.68}$$
$$\mathbf{Z}_3 = \frac{\mathbf{Z}_a\mathbf{Z}_b}{\mathbf{Z}_a + \mathbf{Z}_b + \mathbf{Z}_c}$$

A delta or wye circuit is said to be **balanced** if it has equal impedances in all three branches.

When a Δ-Y circuit is balanced, Eqs. (9.67) and (9.68) become

$$\mathbf{Z}_\Delta = 3\mathbf{Z}_Y \quad \text{or} \quad \mathbf{Z}_Y = \frac{1}{3}\mathbf{Z}_\Delta \tag{9.69}$$

where $\mathbf{Z}_Y = \mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{Z}_3$ and $\mathbf{Z}_\Delta = \mathbf{Z}_a = \mathbf{Z}_b = \mathbf{Z}_c$.

As you see in this section, the principles of voltage division, current division, circuit reduction, impedance equivalence, and Y-Δ transformation all apply to ac circuits. Chapter 10 will show that other circuit techniques—such as superposition, nodal analysis, mesh analysis, source transformation, the Thevenin theorem, and the Norton theorem—are all applied to ac circuits in a manner similar to their application in dc circuits.

Example 9.10

Find the input impedance of the circuit in Fig. 9.23. Assume that the circuit operates at $\omega = 50$ rad/s.

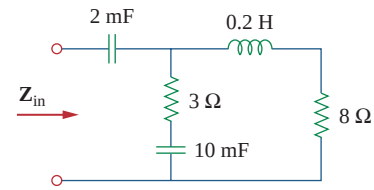


Figure 9.23
For Example 9.10.

Solution:
Let

$\mathbf{Z}_1$ = Impedance of the 2-mF capacitor
$\mathbf{Z}_2$ = Impedance of the 3-Ω resistor in series with the10-mF capacitor
$\mathbf{Z}_3$ = Impedance of the 0.2-H inductor in series with the 8-Ω resistor

Then

$$\mathbf{Z}_1 = \frac{1}{j\omega C} = \frac{1}{j50 \times 2 \times 10^{-3}} = -j10\ \Omega$$

$$\mathbf{Z}_2 = 3 + \frac{1}{j\omega C} = 3 + \frac{1}{j50 \times 10 \times 10^{-3}} = (3 - j2)\ \Omega$$

$$\mathbf{Z}_3 = 8 + j\omega L = 8 + j50 \times 0.2 = (8 + j10)\ \Omega$$

The input impedance is

$$\mathbf{Z}_{in} = \mathbf{Z}_1 + \mathbf{Z}_2 \,\|\, \mathbf{Z}_3 = -j10 + \frac{(3 - j2)(8 + j10)}{11 + j8}$$

$$= -j10 + \frac{(44 + j14)(11 - j8)}{11^2 + 8^2} = -j10 + 3.22 - j1.07\ \Omega$$

Thus,

$$\mathbf{Z}_{in} = 3.22 - j11.07\ \Omega$$

Practice Problem 9.10

Determine the input impedance of the circuit in Fig. 9.24 at $\omega = 10$ rad/s.

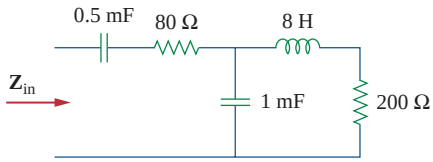


Figure 9.24
For Practice Prob. 9.10.

Answer: $(129.52 - j295)$

Example 9.11

Determine $v_o(t)$ in the circuit of Fig. 9.25.

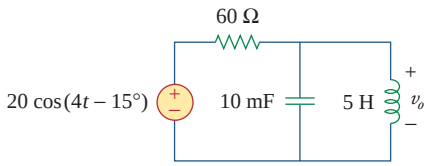


Figure 9.25
For Example 9.11.

Solution:
To do the analysis in the frequency domain, we must first transform the time domain circuit in Fig. 9.25 to the phasor domain equivalent in Fig. 9.26. The transformation produces

$$v_s = 20\cos(4t - 15^\circ) \quad \Rightarrow \quad \mathbf{V}_s = 20\angle -15^\circ \text{ V}, \qquad \omega = 4$$

$$10 \text{ mF} \quad \Rightarrow \quad \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}} = -j25\ \Omega$$

$$5 \text{ H} \quad \Rightarrow \quad j\omega L = j4 \times 5 = j20\ \Omega$$

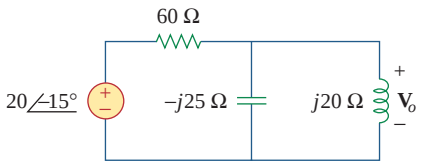


Figure 9.26
The frequency domain equivalent of the circuit in Fig. 9.25.

Let

$\mathbf{Z}_1$ = Impedance of the 60-Ω resistor

$\mathbf{Z}_2$ = Impedance of the parallel combination of the 10-mF capacitor and the 5-H inductor

Then $\mathbf{Z}_1 = 60\ \Omega$ and

$$\mathbf{Z}_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100\ \Omega$$

By the voltage-division principle,

$$\mathbf{V}_o = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V}_s = \frac{j100}{60 + j100}(20\angle -15^\circ)$$

$$= (0.8575\angle 30.96^\circ)(20\angle -15^\circ) = 17.15\angle 15.96^\circ \text{ V}$$

We convert this to the time domain and obtain

$$v_o(t) = 17.15\cos(4t + 15.96^\circ) \text{ V}$$

Practice Problem 9.11

Calculate v_o in the circuit of Fig. 9.27.

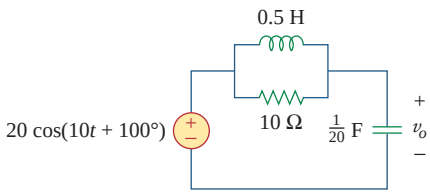


Figure 9.27
For Practice Prob. 9.11.

Answer: $v_o(t) = 14.142\cos(10t - 35^\circ)$ V.

Find current **I** in the circuit of Fig. 9.28.

Example 9.12

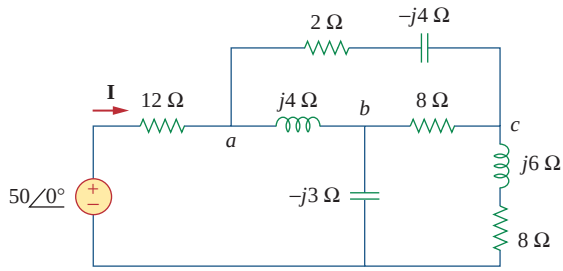


Figure 9.28
For Example 9.12.

Solution:
The delta network connected to nodes *a*, *b*, and *c* can be converted to the *Y* network of Fig. 9.29. We obtain the *Y* impedances as follows using Eq. (9.68):

$$\mathbf{Z}_{an} = \frac{j4(2 - j4)}{j4 + 2 - j4 + 8} = \frac{4(4 + j2)}{10} = (1.6 + j0.8)\ \Omega$$

$$\mathbf{Z}_{bn} = \frac{j4(8)}{10} = j3.2\ \Omega, \qquad \mathbf{Z}_{cn} = \frac{8(2 - j4)}{10} = (1.6 - j3.2)\ \Omega$$

The total impedance at the source terminals is

$$\begin{aligned}\mathbf{Z} &= 12 + \mathbf{Z}_{an} + (\mathbf{Z}_{bn} - j3) \parallel (\mathbf{Z}_{cn} + j6 + 8) \\ &= 12 + 1.6 + j0.8 + (j0.2) \parallel (9.6 + j2.8) \\ &= 13.6 + j0.8 + \frac{j0.2(9.6 + j2.8)}{9.6 + j3} \\ &= 13.6 + j1 = 13.64\angle 4.204^\circ\ \Omega\end{aligned}$$

The desired current is

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{50\angle 0^\circ}{13.64\angle 4.204^\circ} = 3.666\angle -4.204^\circ\ \text{A}$$

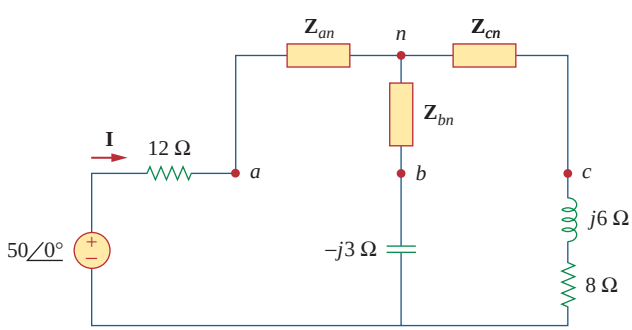


Figure 9.29
The circuit in Fig. 9.28 after delta-to-wye transformation.

Practice Problem 9.12

Find **I** in the circuit of Fig. 9.30.

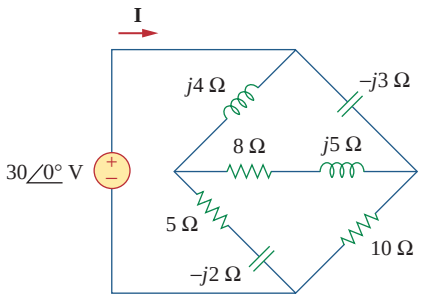


Figure 9.30
For Practice Prob. 9.12.

Answer: $6.364\angle 3.8°$ A.

9.8 † Applications

In Chapters 7 and 8, we saw certain uses of *RC*, *RL*, and *RLC* circuits in dc applications. These circuits also have ac applications; among them are coupling circuits, phase-shifting circuits, filters, resonant circuits, ac bridge circuits, and transformers. This list of applications is inexhaustive. We will consider some of them later. It will suffice here to observe two simple ones: *RC* phase-shifting circuits, and ac bridge circuits.

9.8.1 Phase-Shifters

A phase-shifting circuit is often employed to correct an undesirable phase shift already present in a circuit or to produce special desired effects. An *RC* circuit is suitable for this purpose because its capacitor causes the circuit current to lead the applied voltage. Two commonly used *RC* circuits are shown in Fig. 9.31. (*RL* circuits or any reactive circuits could also serve the same purpose.)

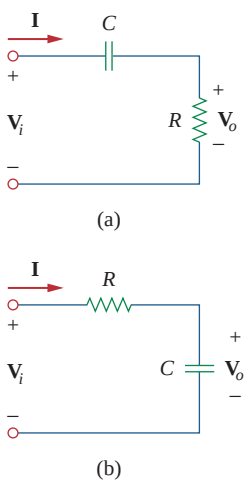


Figure 9.31
Series *RC* shift circuits: (a) leading output, (b) lagging output.

In Fig. 9.31(a), the circuit current **I** leads the applied voltage $\mathbf{V}_i$ by some phase angle θ, where $0 < \theta < 90°$, depending on the values of R and C. If $X_C = -1/\omega C$, then the total impedance is $\mathbf{Z} = R + jX_C$, and the phase shift is given by

$$\theta = \tan^{-1}\frac{X_C}{R} \tag{9.70}$$

This shows that the amount of phase shift depends on the values of R, C, and the operating frequency. Since the output voltage $\mathbf{V}_o$ across the resistor is in phase with the current, $\mathbf{V}_o$ leads (positive phase shift) $\mathbf{V}_i$ as shown in Fig. 9.32(a).

In Fig. 9.31(b), the output is taken across the capacitor. The current **I** leads the input voltage $\mathbf{V}_i$ by θ, but the output voltage $v_o(t)$ across the capacitor lags (negative phase shift) the input voltage $v_i(t)$ as illustrated in Fig. 9.32(b).

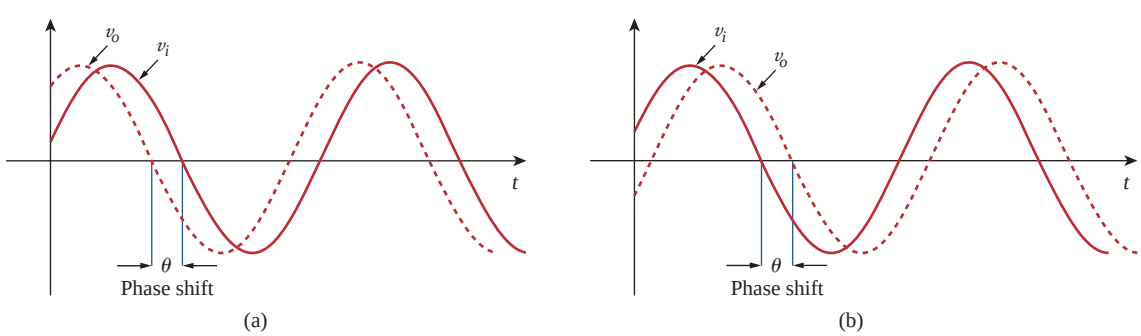


Figure 9.32
Phase shift in *RC* circuits: (a) leading output, (b) lagging output.

We should keep in mind that the simple *RC* circuits in Fig. 9.31 also act as voltage dividers. Therefore, as the phase shift θ approaches 90°, the output voltage $\mathbf{V}_o$ approaches zero. For this reason, these simple *RC* circuits are used only when small amounts of phase shift are required. If it is desired to have phase shifts greater than 60°, simple *RC* networks are cascaded, thereby providing a total phase shift equal to the sum of the individual phase shifts. In practice, the phase shifts due to the stages are not equal, because the succeeding stages load down the earlier stages unless op amps are used to separate the stages.

Example 9.13

Design an *RC* circuit to provide a phase of 90° leading.

Solution:

If we select circuit components of equal ohmic value, say $R = |X_C| = 20\ \Omega$, at a particular frequency, according to Eq. (9.70), the phase shift is exactly 45°. By cascading two similar *RC* circuits in Fig. 9.31(a), we obtain the circuit in Fig. 9.33, providing a positive or leading phase shift of 90°, as we shall soon show. Using the series-parallel combination technique, $\mathbf{Z}$ in Fig. 9.33 is obtained as

$$\mathbf{Z} = 20 \parallel (20 - j20) = \frac{20(20 - j20)}{40 - j20} = 12 - j4\ \Omega \qquad \textbf{(9.13.1)}$$

Using voltage division,

$$\mathbf{V}_1 = \frac{\mathbf{Z}}{\mathbf{Z} - j20}\mathbf{V}_i = \frac{12 - j4}{12 - j24}\mathbf{V}_i = \frac{\sqrt{2}}{3}\angle 45^\circ\, \mathbf{V}_i \qquad \textbf{(9.13.2)}$$

and

$$\mathbf{V}_o = \frac{20}{20 - j20}\mathbf{V}_1 = \frac{\sqrt{2}}{2}\angle 45^\circ\, \mathbf{V}_1 \qquad \textbf{(9.13.3)}$$

Substituting Eq. (9.13.2) into Eq. (9.13.3) yields

$$\mathbf{V}_o = \left(\frac{\sqrt{2}}{2}\angle 45^\circ\right)\left(\frac{\sqrt{2}}{3}\angle 45^\circ\, \mathbf{V}_i\right) = \frac{1}{3}\angle 90^\circ\, \mathbf{V}_i$$

Thus, the output leads the input by 90° but its magnitude is only about 33 percent of the input.

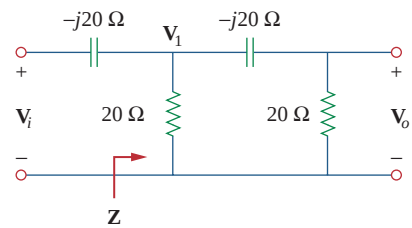


Figure 9.33
An *RC* phase shift circuit with 90° leading phase shift; for Example 9.13.

Practice Problem 9.13

Design an *RC* circuit to provide a 90° lagging phase shift of the output voltage relative to the input voltage. If an ac voltage of 10 V rms is applied, what is the output voltage?

Answer: Figure 9.34 shows a typical design; 3.33 V rms.

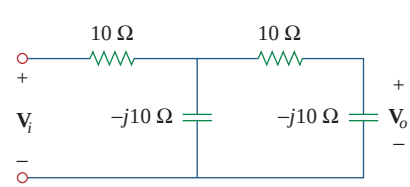


Figure 9.34
For Practice Prob. 9.13.

Example 9.14

For the *RL* circuit shown in Fig. 9.35(a), calculate the amount of phase shift produced at 2 kHz.

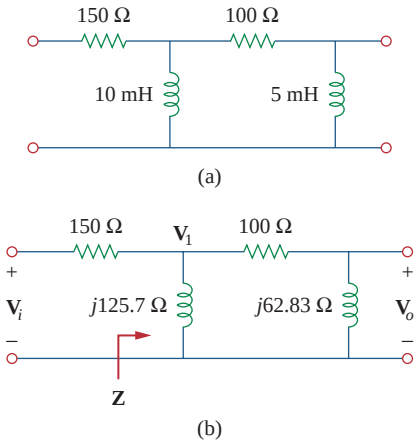


Figure 9.35
For Example 9.14.

Solution:
At 2 kHz, we transform the 10-mH and 5-mH inductances to the corresponding impedances.

$$10\text{ mH} \quad \Rightarrow \quad X_L = \omega L = 2\pi \times 2 \times 10^3 \times 10 \times 10^{-3} = 40\pi = 125.7\ \Omega$$

$$5\text{ mH} \quad \Rightarrow \quad X_L = \omega L = 2\pi \times 2 \times 10^3 \times 5 \times 10^{-3} = 20\pi = 62.83\ \Omega$$

Consider the circuit in Fig. 9.35(b). The impedance $\mathbf{Z}$ is the parallel combination of $j125.7\ \Omega$ and $100 + j62.83\ \Omega$. Hence,

$$\mathbf{Z} = j125.7 \parallel (100 + j62.83) = \frac{j125.7(100 + j62.83)}{100 + j188.5} = 69.56\angle 60.1^\circ\ \Omega \qquad \textbf{(9.14.1)}$$

Using voltage division,

$$\mathbf{V}_1 = \frac{\mathbf{Z}}{\mathbf{Z} + 150}\mathbf{V}_i = \frac{69.56\angle 60.1^\circ}{184.7 + j60.3}\mathbf{V}_i = 0.3582\angle 42.02^\circ\,\mathbf{V}_i \qquad \textbf{(9.14.2)}$$

and

$$\mathbf{V}_o = \frac{j62.832}{100 + j62.832}\mathbf{V}_1 = 0.532\angle 57.86^\circ\,\mathbf{V}_1 \qquad \textbf{(9.14.3)}$$

Combining Eqs. (9.14.2) and (9.14.3),

$$\mathbf{V}_o = (0.532\angle 57.86^\circ)(0.3582\angle 42.02^\circ)\,\mathbf{V}_i = 0.1906\angle 100^\circ\,\mathbf{V}_i$$

showing that the output is about 19 percent of the input in magnitude but leading the input by 100°. If the circuit is terminated by a load, the load will affect the phase shift.

Practice Problem 9.14

Refer to the *RL* circuit in Fig. 9.36. If 1 V is applied, find the magnitude and the phase shift produced at 5 kHz. Specify whether the phase shift is leading or lagging.

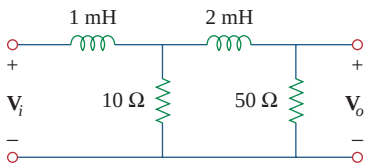


Figure 9.36
For Practice Prob. 9.14.

Answer: 0.172, 120.4°, lagging.

9.8.2 AC Bridges

An ac bridge circuit is used in measuring the inductance L of an inductor or the capacitance C of a capacitor. It is similar in form to the Wheatstone bridge for measuring an unknown resistance (discussed in Section 4.10) and follows the same principle. To measure L and C, however, an ac source is needed as well as an ac meter

instead of the galvanometer. The ac meter may be a sensitive ac ammeter or voltmeter.

Consider the general ac bridge circuit displayed in Fig. 9.37. The bridge is *balanced* when no current flows through the meter. This means that $\mathbf{V}_1 = \mathbf{V}_2$. Applying the voltage division principle,

$$\mathbf{V}_1 = \frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2}\mathbf{V}_s = \mathbf{V}_2 = \frac{\mathbf{Z}_x}{\mathbf{Z}_3 + \mathbf{Z}_x}\mathbf{V}_s \tag{9.71}$$

Thus,

$$\frac{\mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} = \frac{\mathbf{Z}_x}{\mathbf{Z}_3 + \mathbf{Z}_x} \quad \Rightarrow \quad \mathbf{Z}_2\mathbf{Z}_3 = \mathbf{Z}_1\mathbf{Z}_x \tag{9.72}$$

or

$$\boxed{\mathbf{Z}_x = \frac{\mathbf{Z}_3}{\mathbf{Z}_1}\mathbf{Z}_2} \tag{9.73}$$

This is the balanced equation for the ac bridge and is similar to Eq. (4.30) for the resistance bridge except that the R's are replaced by $\mathbf{Z}$'s.

Specific ac bridges for measuring L and C are shown in Fig. 9.38, where L_x and C_x are the unknown inductance and capacitance to be measured while L_s and C_s are a standard inductance and capacitance (the values of which are known to great precision). In each case, two resistors, R_1 and R_2, are varied until the ac meter reads zero. Then the bridge is balanced. From Eq. (9.73), we obtain

$$L_x = \frac{R_2}{R_1}L_s \tag{9.74}$$

and

$$C_x = \frac{R_1}{R_2}C_s \tag{9.75}$$

Notice that the balancing of the ac bridges in Fig. 9.38 does not depend on the frequency f of the ac source, since f does not appear in the relationships in Eqs. (9.74) and (9.75).

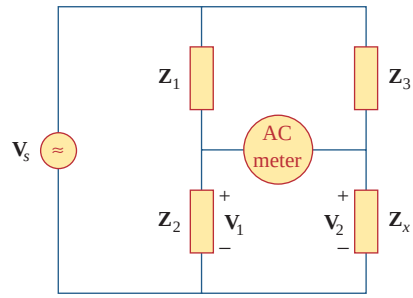
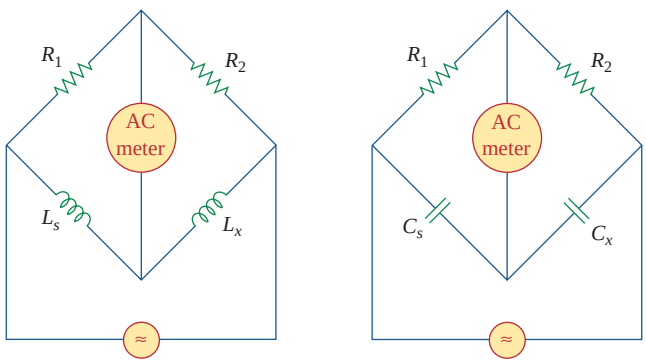


Figure 9.37
A general ac bridge.



(a) (b)

Figure 9.38
Specific ac bridges: (a) for measuring L, (b) for measuring C.

Example 9.15

The ac bridge circuit of Fig. 9.37 balances when $\mathbf{Z}_1$ is a 1-kΩ resistor, $\mathbf{Z}_2$ is a 4.2-kΩ resistor, $\mathbf{Z}_3$ is a parallel combination of a 1.5-MΩ resistor and a 12-pF capacitor, and $f = 2$ kHz. Find: (a) the series components that make up $\mathbf{Z}_x$, and (b) the parallel components that make up $\mathbf{Z}_x$.

Solution:

1. **Define.** The problem is clearly stated.
2. **Present.** We are to determine the unknown components subject to the fact that they balance the given quantities. Since a parallel and series equivalent exists for this circuit, we need to find both.
3. **Alternative.** Although there are alternative techniques that can be used to find the unknown values, a straightforward equality works best. Once we have answers, we can check them by using hand techniques such as nodal analysis or just using *PSpice*.
4. **Attempt.** From Eq. (9.73),

$$\mathbf{Z}_x = \frac{\mathbf{Z}_3}{\mathbf{Z}_1}\mathbf{Z}_2 \tag{9.15.1}$$

where $\mathbf{Z}_x = R_x + jX_x$,

$$\mathbf{Z}_1 = 1000\ \Omega, \qquad \mathbf{Z}_2 = 4200\ \Omega \tag{9.15.2}$$

and

$$\mathbf{Z}_3 = R_3 \,\|\, \frac{1}{j\omega C_3} = \frac{\dfrac{R_3}{j\omega C_3}}{R_3 + 1/j\omega C_3} = \frac{R_3}{1 + j\omega R_3 C_3}$$

Since $R_3 = 1.5\ \text{M}\Omega$ and $C_3 = 12$ pF,

$$\mathbf{Z}_3 = \frac{1.5 \times 10^6}{1 + j2\pi \times 2 \times 10^3 \times 1.5 \times 10^6 \times 12 \times 10^{-12}} = \frac{1.5 \times 10^6}{1 + j0.2262}$$

or

$$\mathbf{Z}_3 = 1.427 - j0.3228\ \text{M}\Omega \tag{9.15.3}$$

(a) Assuming that $\mathbf{Z}_x$ is made up of series components, we substitute Eqs. (9.15.2) and (9.15.3) in Eq. (9.15.1) and obtain

$$\begin{aligned} R_x + jX_x &= \frac{4200}{1000}(1.427 - j0.3228) \times 10^6 \\ &= (5.993 - j1.356)\ \text{M}\Omega \end{aligned} \tag{9.15.4}$$

Equating the real and imaginary parts yields $R_x = 5.993\ \text{M}\Omega$ and a capacitive reactance

$$X_x = \frac{1}{\omega C} = 1.356 \times 10^6$$

or

$$C = \frac{1}{\omega X_x} = \frac{1}{2\pi \times 2 \times 10^3 \times 1.356 \times 10^6} = 58.69\ \text{pF}$$

(b) Z_x remains the same as in Eq. (9.15.4) but R_x and X_x are in parallel. Assuming an *RC* parallel combination,

$$\begin{aligned}\mathbf{Z}_x &= (5.993 - j1.356)\ \text{M}\Omega \\ &= R_x \| \frac{1}{j\omega C_x} = \frac{R_x}{1 + j\omega R_x C_x}\end{aligned}$$

By equating the real and imaginary parts, we obtain

$$R_x = \frac{\text{Real}(\mathbf{Z}_x)^2 + \text{Imag}(\mathbf{Z}_x)^2}{\text{Real}(\mathbf{Z}_x)} = \frac{5.993^2 + 1.356^2}{5.993} = \mathbf{6.3\ M\Omega}$$

$$\begin{aligned}C_x &= -\frac{\text{Imag}(\mathbf{Z}_x)}{\omega[\text{Real}(\mathbf{Z}_x)^2 + \text{Imag}(\mathbf{Z}_x)^2]} \\ &= -\frac{-1.356}{2\pi(2000)(5.917^2 + 1.356^2)} = \mathbf{2.852\ \mu F}\end{aligned}$$

We have assumed a parallel *RC* combination which works in this case.

5. **Evaluate.** Let us now use *PSpice* to see if we indeed have the correct equalities. Running *PSpice* with the equivalent circuits, an open circuit between the "bridge" portion of the circuit, and a 10-volt input voltage yields the following voltages at the ends of the "bridge" relative to a reference at the bottom of the circuit:

```
FREQ        VM($N_0002)  VP($N_0002)
2.000E+03   9.993E+00    -8.634E-03
2.000E+03   9.993E+00    -8.637E-03
```

Since the voltages are essentially the same, then no measurable current can flow through the "bridge" portion of the circuit for any element that connects the two points together and we have a balanced bridge, which is to be expected. This indicates we have properly determined the unknowns.

There is a very important problem with what we have done! Do you know what that is? We have what can be called an ideal, "theoretical" answer, but one that really is not very good in the real world. The difference between the magnitudes of the upper impedances and the lower impedances is much too large and would never be accepted in a real bridge circuit. For greatest accuracy, the overall magnitude of the impedances must at least be within the same relative order. To increase the accuracy of the solution of this problem, I would recommend increasing the magnitude of the top impedances to be in the range of 500 kΩ to 1.5 MΩ. One additional real-world comment: the size of these impedances also creates serious problems in making actual measurements, so the appropriate instruments must be used in order to minimize their loading (which would change the actual voltage readings) on the circuit.

6. **Satisfactory?** Since we solved for the unknown terms and then tested to see if they woked, we validated the results. They can now be presented as a solution to the problem.

Practice Problem 9.15

In the ac bridge circuit of Fig. 9.37, suppose that balance is achieved when $\mathbf{Z}_1$ is a 4.8-kΩ resistor, $\mathbf{Z}_2$ is a 10-Ω resistor in series with a 0.25-μH inductor, $\mathbf{Z}_3$ is a 12-kΩ resistor, and $f = 6$ MHz. Determine the series components that make up $\mathbf{Z}_x$.

Answer: A 25-Ω resistor in series with a 0.625-μH inductor.

9.9 Summary

1. A sinusoid is a signal in the form of the sine or cosine function. It has the general form

$$v(t) = V_m \cos(\omega t + \phi)$$

where V_m is the amplitude, $\omega = 2\pi f$ is the angular frequency, $(\omega t + \phi)$ is the argument, and ϕ is the phase.

2. A phasor is a complex quantity that represents both the magnitude and the phase of a sinusoid. Given the sinusoid $v(t) = V_m \cos(\omega t + \phi)$, its phasor $\mathbf{V}$ is

$$\mathbf{V} = V_m \angle \phi$$

3. In ac circuits, voltage and current phasors always have a fixed relation to one another at any moment of time. If $v(t) = V_m \cos(\omega t + \phi_v)$ represents the voltage through an element and $i(t) = I_m \cos(\omega t + \phi_i)$ represents the current through the element, then $\phi_i = \phi_v$ if the element is a resistor, ϕ_i leads ϕ_v by 90° if the element is a capacitor, and ϕ_i lags ϕ_v by 90° if the element is an inductor.

4. The impedance $\mathbf{Z}$ of a circuit is the ratio of the phasor voltage across it to the phasor current through it:

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = R(\omega) + jX(\omega)$$

The admittance $\mathbf{Y}$ is the reciprocal of impedance:

$$\mathbf{Y} = \frac{1}{\mathbf{Z}} = G(\omega) + jB(\omega)$$

Impedances are combined in series or in parallel the same way as resistances in series or parallel; that is, impedances in series add while admittances in parallel add.

5. For a resistor $\mathbf{Z} = R$, for an inductor $\mathbf{Z} = jX = j\omega L$, and for a capacitor $\mathbf{Z} = -jX = 1/j\omega C$.

6. Basic circuit laws (Ohm's and Kirchhoff's) apply to ac circuits in the same manner as they do for dc circuits; that is,

$$\mathbf{V} = \mathbf{ZI}$$
$$\Sigma\, \mathbf{I}_k = 0 \quad \text{(KCL)}$$
$$\Sigma\, \mathbf{V}_k = 0 \quad \text{(KVL)}$$

7. The techniques of voltage/current division, series/parallel combination of impedance/admittance, circuit reduction, and Y-Δ transformation all apply to ac circuit analysis.
8. AC circuits are applied in phase-shifters and bridges.

Review Questions

9.1 Which of the following is *not* a right way to express the sinusoid $A\cos\omega t$?

(a) $A\cos 2\pi ft$ (b) $A\cos(2\pi t/T)$
(c) $A\cos\omega(t - T)$ (d) $A\sin(\omega t - 90°)$

9.2 A function that repeats itself after fixed intervals is said to be:

(a) a phasor (b) harmonic
(c) periodic (d) reactive

9.3 Which of these frequencies has the shorter period?

(a) 1 krad/s (b) 1 kHz

9.4 If $v_1 = 30\sin(\omega t + 10°)$ and $v_2 = 20\sin(\omega t + 50°)$, which of these statements are true?

(a) v_1 leads v_2 (b) v_2 leads v_1
(c) v_2 lags v_1 (d) v_1 lags v_2
(e) v_1 and v_2 are in phase

9.5 The voltage across an inductor leads the current through it by 90°.

(a) True (b) False

9.6 The imaginary part of impedance is called:

(a) resistance (b) admittance
(c) susceptance (d) conductance
(e) reactance

9.7 The impedance of a capacitor increases with increasing frequency.

(a) True (b) False

9.8 At what frequency will the output voltage $v_o(t)$ in Fig. 9.39 be equal to the input voltage $v(t)$?

(a) 0 rad/s (b) 1 rad/s (c) 4 rad/s
(d) ∞ rad/s (e) none of the above

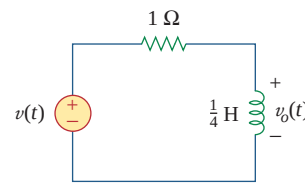


Figure 9.39
For Review Question 9.8.

9.9 A series RC circuit has $|V_R| = 12$ V and $|V_C| = 5$ V. The magnitude of the supply voltage is:

(a) −7 V (b) 7 V (c) 13 V (d) 17 V

9.10 A series RCL circuit has $R = 30\ \Omega$, $X_C = 50\ \Omega$, and $X_L = 90\ \Omega$. The impedance of the circuit is:

(a) $30 + j140\ \Omega$ (b) $30 + j40\ \Omega$
(c) $30 - j40\ \Omega$ (d) $-30 - j40\ \Omega$
(e) $-30 + j40\ \Omega$

Answers: 9.1d, 9.2c, 9.3b, 9.4b,d, 9.5a, 9.6e, 9.7b, 9.8d, 9.9c, 9.10b.

Problems

Section 9.2 Sinusoids

9.1 Given the sinusoidal voltage $v(t) = 50\cos(30t + 10°)$ V, find: (a) the amplitude V_m, (b) the period T, (c) the frequency f, and (d) $v(t)$ at $t = 10$ ms.

9.2 A current source in a linear circuit has

$$i_s = 8\cos(500\pi t - 25°)\ \text{A}$$

(a) What is the amplitude of the current?
(b) What is the angular frequency?
(c) Find the frequency of the current.
(d) Calculate i_s at $t = 2$ ms.

9.3 Express the following functions in cosine form:

(a) $4\sin(\omega t - 30°)$ (b) $-2\sin 6t$
(c) $-10\sin(\omega t + 20°)$

9.4 Design a problem to help other students better understand sinusoids.

9.5 Given $v_1 = 20\sin(\omega t + 60°)$ and $v_2 = 60\cos(\omega t - 10°)$, determine the phase angle between the two sinusoids and which one lags the other.

9.6 For the following pairs of sinusoids, determine which one leads and by how much.

(a) $v(t) = 10\cos(4t - 60°)$ and $i(t) = 4\sin(4t + 50°)$

(b) $v_1(t) = 4\cos(377t + 10°)$ and $v_2(t) = -20\cos 377t$

(c) $x(t) = 13\cos 2t + 5\sin 2t$ and $y(t) = 15\cos(2t - 11.8°)$

Section 9.3 Phasors

9.7 If $f(\phi) = \cos\phi + j\sin\phi$, show that $f(\phi) = e^{j\phi}$.

9.8 Calculate these complex numbers and express your results in rectangular form:

(a) $\dfrac{15\angle 45°}{3 - j4} + j2$

(b) $\dfrac{8\angle -20°}{(2 + j)(3 - j4)} + \dfrac{10}{-5 + j12}$

(c) $10 + (8\angle 50°)(5 - j12)$

9.9 Evaluate the following complex numbers and leave your results in polar form:

(a) $5\angle 30°\left(6 - j8 + \dfrac{3\angle 60°}{2 + j}\right)$

(b) $\dfrac{(10\angle 60°)(35\angle -50°)}{(2 + j6) - (5 + j)}$

9.10 Design a problem to help other students better understand phasors.

9.11 Find the phasors corresponding to the following signals:

(a) $v(t) = 21\cos(4t - 15°)$ V

(b) $i(t) = -8\sin(10t + 70°)$ mA

(c) $v(t) = 120\sin(10t - 50°)$ V

(d) $i(t) = -60\cos(30t + 10°)$ mA

9.12 Let $\mathbf{X} = 8\angle 40°$ and $\mathbf{Y} = 10\angle -30°$. Evaluate the following quantities and express your results in polar form:

(a) $(\mathbf{X} + \mathbf{Y})\mathbf{X}^*$

(b) $(\mathbf{X} - \mathbf{Y})^*$

(c) $(\mathbf{X} + \mathbf{Y})/\mathbf{X}$

9.13 Evaluate the following complex numbers:

(a) $\dfrac{2 + j3}{1 - j6} + \dfrac{7 - j8}{-5 + j11}$

(b) $\dfrac{(5\angle 10°)(10\angle -40°)}{(4\angle -80°)(-6\angle 50°)}$

(c) $\begin{vmatrix} 2 + j3 & -j2 \\ -j2 & 8 - j5 \end{vmatrix}$

9.14 Simplify the following expressions:

(a) $\dfrac{(5 - j6) - (2 + j8)}{(-3 + j4)(5 - j) + (4 - j6)}$

(b) $\dfrac{(240\angle 75° + 160\angle -30°)(60 - j80)}{(67 + j84)(20\angle 32°)}$

(c) $\left(\dfrac{10 + j20}{3 + j4}\right)^2\sqrt{(10 + j5)(16 - j20)}$

9.15 Evaluate these determinants:

(a) $\begin{vmatrix} 10 + j6 & 2 - j3 \\ -5 & -1 + j \end{vmatrix}$

(b) $\begin{vmatrix} 20\angle -30° & -4\angle -10° \\ 16\angle 0° & 3\angle 45° \end{vmatrix}$

(c) $\begin{vmatrix} 1 - j & -j & 0 \\ j & 1 & -j \\ 1 & j & 1 + j \end{vmatrix}$

9.16 Transform the following sinusoids to phasors:

(a) $-10\cos(4t + 75°)$ (b) $5\sin(20t - 10°)$

(c) $4\cos 2t + 3\sin 2t$

9.17 Two voltages v_1 and v_2 appear in series so that their sum is $v = v_1 + v_2$. If $v_1 = 10\cos(50t - \pi/3)$ V and $v_2 = 12\cos(50t + 30°)$ V, find v.

9.18 Obtain the sinusoids corresponding to each of the following phasors:

(a) $\mathbf{V}_1 = 60\angle 15°$ V, $\omega = 1$

(b) $\mathbf{V}_2 = 6 + j8$ V, $\omega = 40$

(c) $\mathbf{I}_1 = 2.8e^{-j\pi/3}$ A, $\omega = 377$

(d) $\mathbf{I}_2 = -0.5 - j1.2$ A, $\omega = 10^3$

9.19 Using phasors, find:

(a) $3\cos(20t + 10°) - 5\cos(20t - 30°)$

(b) $40\sin 50t + 30\cos(50t - 45°)$

(c) $20\sin 400t + 10\cos(400t + 60°) - 5\sin(400t - 20°)$

9.20 A linear network has a current input $4\cos(\omega t + 20°)$ A and a voltage output $10\cos(\omega t + 110°)$ V. Determine the associated impedance.

9.21 Simplify the following:

(a) $f(t) = 5\cos(2t + 15°) - 4\sin(2t - 30°)$

(b) $g(t) = 8\sin t + 4\cos(t + 50°)$

(c) $h(t) = \int_0^t (10\cos 40t + 50\sin 40t)\,dt$

9.22 An alternating voltage is given by $v(t) = 20\cos(5t - 30°)$ V. Use phasors to find

$$10v(t) + 4\frac{dv}{dt} - 2\int_{-\infty}^{t} v(t)\,dt$$

Assume that the value of the integral is zero at $t = -\infty$.

9.23 Apply phasor analysis to evaluate the following.

(a) $v = 50\cos(\omega t + 30°) + 30\cos(\omega t - 90°)$ V

(b) $i = 15\cos(\omega t + 45°) - 10\sin(\omega t + 45°)$ A

9.24 Find $v(t)$ in the following integrodifferential equations using the phasor approach:

(a) $v(t) + \int v\,dt = 5\cos(t + 45°)$ V

(b) $\frac{dv}{dt} + 5v(t) + 4\int v\,dt = 20\sin(4t + 10°)$ V

9.25 Using phasors, determine $i(t)$ in the following equations:

(a) $2\frac{di}{dt} + 3i(t) = 4\cos(2t - 45°)$

(b) $10\int i\,dt + \frac{di}{dt} + 6i(t) = 5\cos(5t + 22°)$ A

9.26 The loop equation for a series *RLC* circuit gives

$$\frac{di}{dt} + 2i + \int_{-\infty}^{t} i\,dt = \cos 2t \text{ A}$$

Assuming that the value of the integral at $t = -\infty$ is zero, find $i(t)$ using the phasor method.

9.27 A parallel *RLC* circuit has the node equation

$$\frac{dv}{dt} + 50v + 100\int v\,dt = 110\cos(377t - 10°) \text{ V}$$

Determine $v(t)$ using the phasor method. You may assume that the value of the integral at $t = -\infty$ is zero.

Section 9.4 Phasor Relationships for Circuit Elements

9.28 Determine the current that flows through an 8-Ω resistor connected to a voltage source $v_s = 110\cos 377t$ V.

9.29 What is the instantaneous voltage across a 2-μF capacitor when the current through it is $i = 4\sin(10^6 t + 25°)$ A?

9.30 A voltage $v(t) = 100\cos(60t + 20°)$ V is applied to a parallel combination of a 40-kΩ resistor and a 50-μF capacitor. Find the steady-state currents through the resistor and the capacitor.

9.31 A series *RLC* circuit has $R = 80\ \Omega$, $L = 240$ mH, and $C = 5$ mF. If the input voltage is $v(t) = 10\cos 2t$, find the currrent flowing through the circuit.

9.32 Using Fig. 9.40, design a problem to help other students better understand phasor relationships for circuit elements.

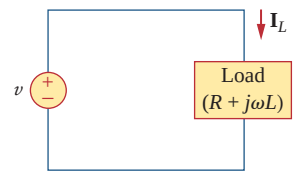


Figure 9.40
For Prob. 9.32.

9.33 A series *RL* circuit is connected to a 110-V ac source. If the voltage across the resistor is 85 V, find the voltage across the inductor.

9.34 What value of ω will cause the forced response v_o in Fig. 9.41 to be zero?

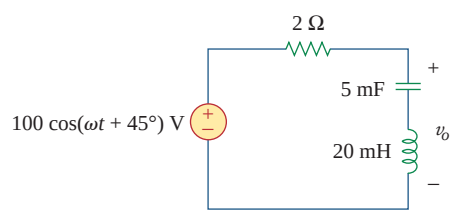


Figure 9.41
For Prob. 9.34.

Section 9.5 Impedance and Admittance

9.35 Find current i in the circuit of Fig. 9.42, when $v_s(t) = 50\cos 200t$ V.

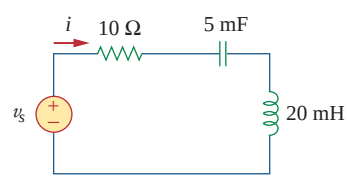


Figure 9.42
For Prob. 9.35.

9.36 Using Fig. 9.43, design a problem to help other students better understand impedance.

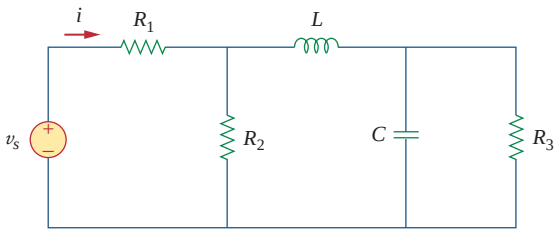


Figure 9.43
For Prob. 9.36.

9.37 Determine the admittance **Y** for the circuit in Fig. 9.44.

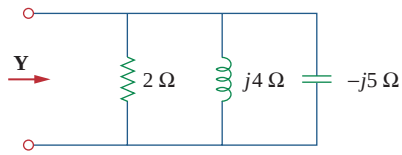
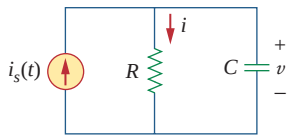
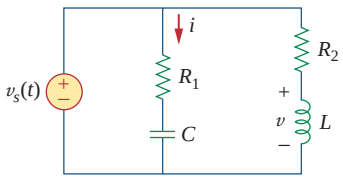


Figure 9.44
For Prob. 9.37.

9.38 Using Fig. 9.45, design a problem to help other students better understand admittance.



(a)



(b)

Figure 9.45
For Prob. 9.38.

9.39 For the circuit shown in Fig. 9.46, find Z_{eq} and use that to find current **I**. Let $\omega = 10$ rad/s.

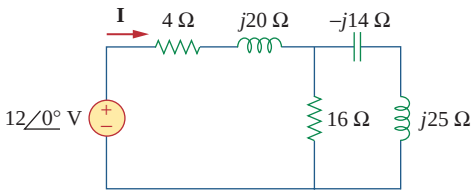


Figure 9.46
For Prob. 9.39.

9.40 In the circuit of Fig. 9.47, find i_o when:

(a) $\omega = 1$ rad/s (b) $\omega = 5$ rad/s

(c) $\omega = 10$ rad/s

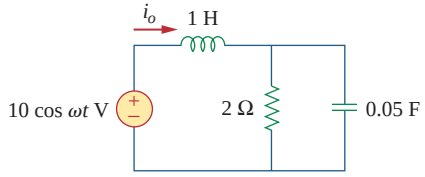


Figure 9.47
For Prob. 9.40.

9.41 Find $v(t)$ in the *RLC* circuit of Fig. 9.48.

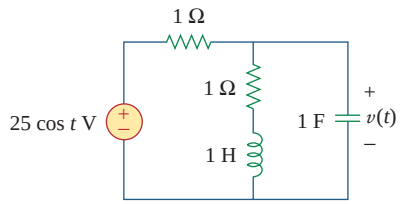


Figure 9.48
For Prob. 9.41.

9.42 Calculate $v_o(t)$ in the circuit of Fig. 9.49.

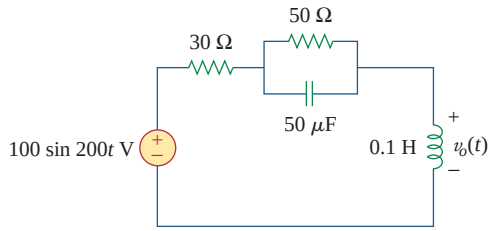


Figure 9.49
For Prob. 9.42.

9.43 Find current $\mathbf{I}_o$ in the circuit shown in Fig. 9.50.

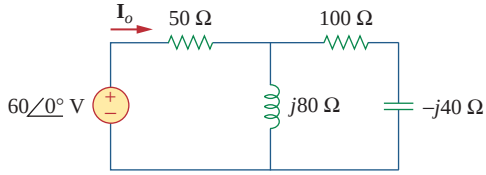


Figure 9.50
For Prob. 9.43.

9.44 Calculate $i(t)$ in the circuit of Fig. 9.51.

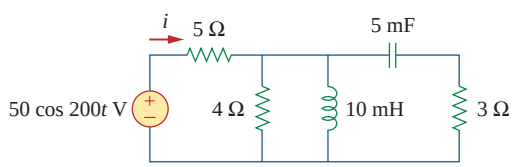


Figure 9.51
For prob. 9.44.

9.45 Find current $\mathbf{I}_o$ in the network of Fig. 9.52.

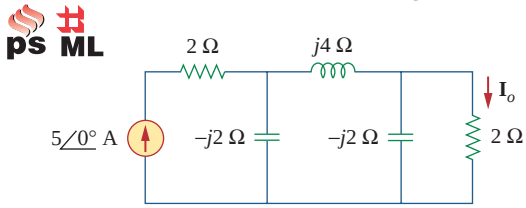


Figure 9.52
For Prob. 9.45.

9.46 If $i_s = 20\cos(10t + 15°)$ A in the circuit of Fig. 9.53, find i_o.

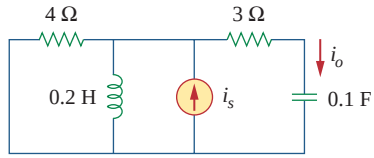


Figure 9.53
For Prob. 9.46.

9.47 In the circuit of Fig. 9.54, determine the value of $i_s(t)$.

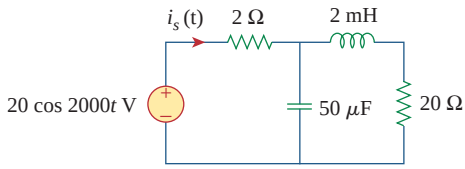


Figure 9.54
For Prob. 9.47.

9.48 Given that $v_s(t) = 20\sin(100t - 40°)$ in Fig. 9.55, determine $i_x(t)$.

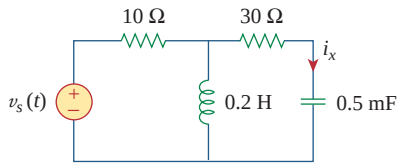


Figure 9.55
For Prob. 9.48.

9.49 Find $v_s(t)$ in the circuit of Fig. 9.56 if the current i_x through the 1-Ω resistor is 0.5 sin 200t A.

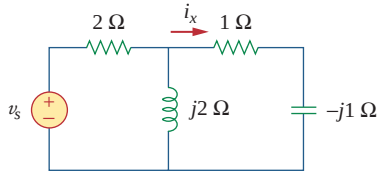


Figure 9.56
For Prob. 9.49.

9.50 Determine v_x in the circuit of Fig. 9.57. Let $i_s(t) = 5\cos(100t + 40°)$ A.

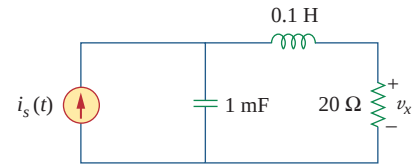


Figure 9.57
For Prob. 9.50.

9.51 If the voltage v_o across the 2-Ω resistor in the circuit of Fig. 9.58 is $-5\cos 2t$ V, obtain i_s.

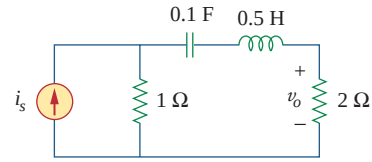


Figure 9.58
For Prob. 9.51.

9.52 If $\mathbf{V}_o = 20\angle 45°$ V in the circuit of Fig. 9.59, find $\mathbf{I}_s$.

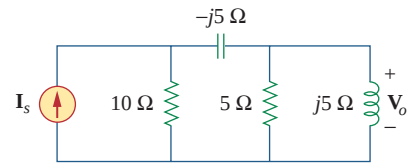


Figure 9.59
For Prob. 9.52.

9.53 Find $\mathbf{I}_o$ in the circuit of Fig. 9.60.

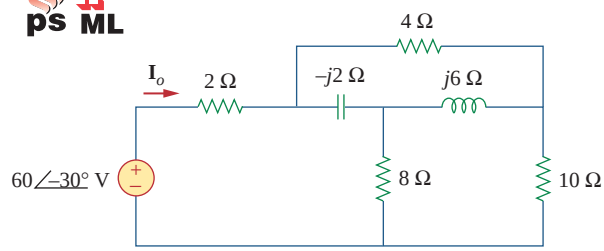


Figure 9.60
For Prob. 9.53.

9.54 In the circuit of Fig. 9.61, find $\mathbf{V}_s$ if $\mathbf{I}_o = 2\angle 0°$ A.

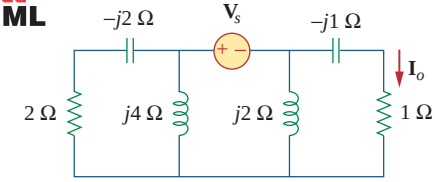


Figure 9.61
For Prob. 9.54.

*9.55 Find **Z** in the network of Fig. 9.62, given that $\mathbf{V}_o = 8\angle 0°$ V.

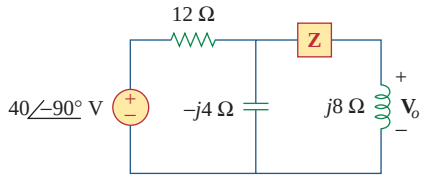


Figure 9.62
For Prob. 9.55.

Section 9.7 Impedance Combinations

9.56 At $\omega = 377$ rad/s, find the input impedance of the circuit shown in Fig. 9.63.

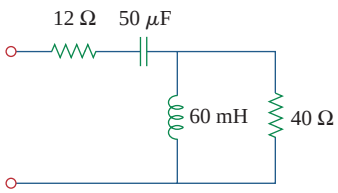


Figure 9.63
For Prob. 9.56.

9.57 At $\omega = 1$ rad/s, obtain the input admittance in the circuit of Fig. 9.64.

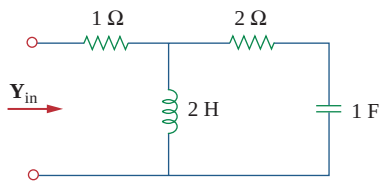


Figure 9.64
For Prob. 9.57.

9.58 Using Fig. 9.65, design a problem to help other students better understand impedance combinations.

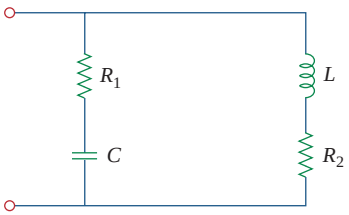


Figure 9.65
For Prob. 9.58.

* An asterisk indicates a challenging problem.

9.59 For the network in Fig. 9.66, find $\mathbf{Z}_{in}$. Let $\omega = 10$ rad/s.

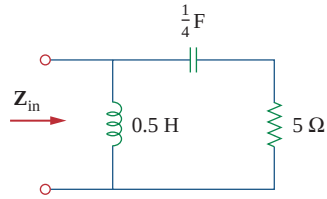


Figure 9.66
For Prob. 9.59.

9.60 Obtain $\mathbf{Z}_{in}$ for the circuit in Fig. 9.67.

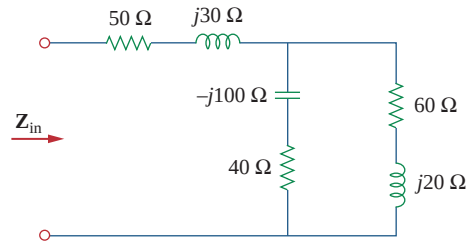


Figure 9.67
For Prob. 9.60.

9.61 Find $\mathbf{Z}_{eq}$ in the circuit of Fig. 9.68.

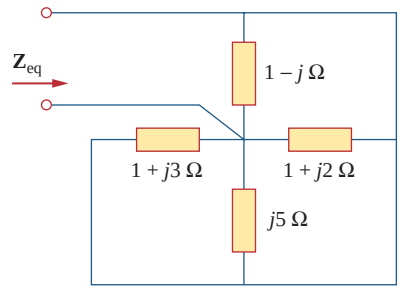


Figure 9.68
For Prob. 9.61.

9.62 For the circuit in Fig. 9.69, find the input impedance $\mathbf{Z}_{in}$ at 10 krad/s.

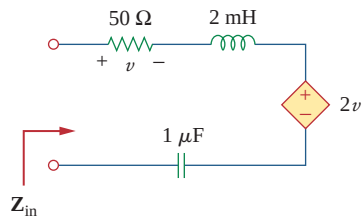


Figure 9.69
For Prob. 9.62.

9.63 For the circuit in Fig. 9.70, find the value of $\mathbf{Z}_T$.

ML

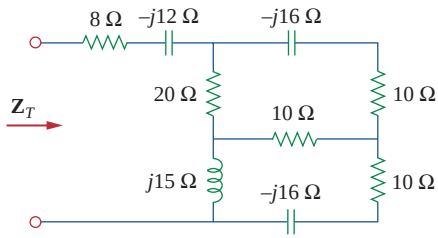


Figure 9.70
For Prob. 9.63.

9.64 Find $\mathbf{Z}_T$ and $\mathbf{I}$ in the circuit of Fig. 9.71.

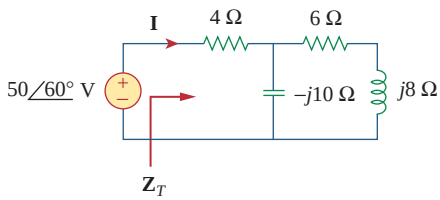


Figure 9.71
For Prob. 9.64.

9.65 Determine $\mathbf{Z}_T$ and $\mathbf{I}$ for the circuit in Fig. 9.72.

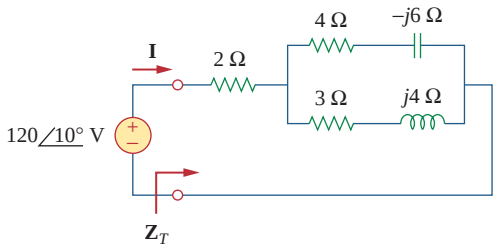


Figure 9.72
For Prob. 9.65.

9.66 For the circuit in Fig. 9.73, calculate $\mathbf{Z}_T$ and $\mathbf{V}_{ab}$.

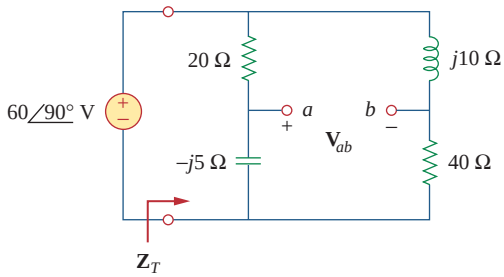


Figure 9.73
For Prob. 9.66.

9.67 At $\omega = 10^3$ rad/s, find the input admittance of each of the circuits in Fig. 9.74.

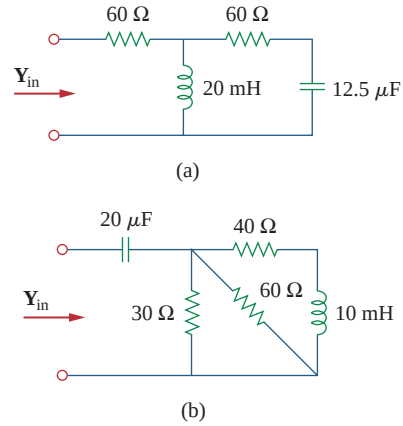


Figure 9.74
For Prob. 9.67.

9.68 Determine $\mathbf{Y}_{eq}$ for the circuit in Fig. 9.75.

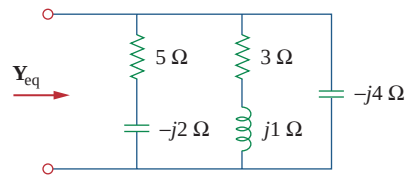


Figure 9.75
For Prob. 9.68.

9.69 Find the equivalent admittance $\mathbf{Y}_{eq}$ of the circuit in Fig. 9.76.

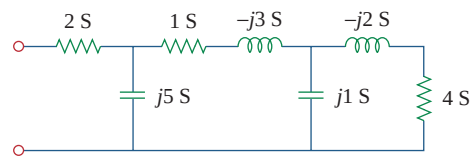


Figure 9.76
For Prob. 9.69.

9.70 Find the equivalent impedance of the circuit in Fig. 9.77.

ML

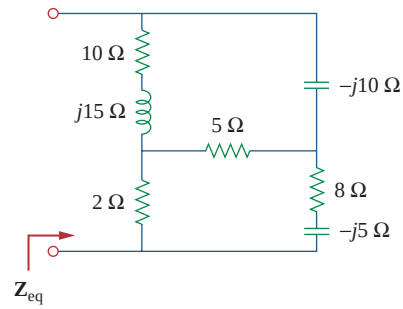


Figure 9.77
For Prob. 9.70.

9.71 Obtain the equivalent impedance of the circuit in Fig. 9.78.

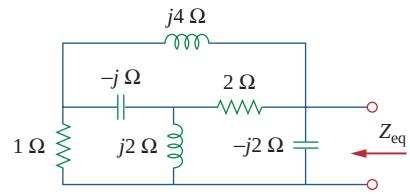


Figure 9.78
For Prob. 9.71.

9.72 Calculate the value of $\mathbf{Z}_{ab}$ in the network of Fig. 9.79.

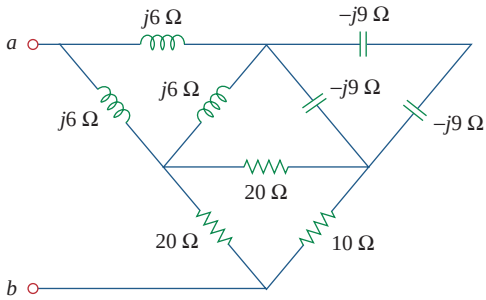


Figure 9.79
For Prob. 9.72.

9.73 Determine the equivalent impedance of the circuit in Fig. 9.80.

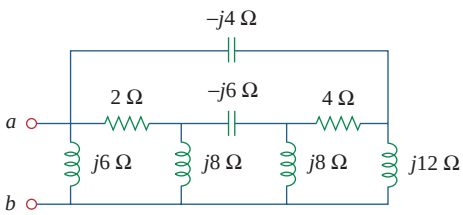


Figure 9.80
For Prob. 9.73.

Section 9.8 Applications

9.74 Design an *RL* circuit to provide a 90° leading phase shift.

9.75 Design a circuit that will transform a sinusoidal voltage input to a cosinusoidal voltage output.

9.76 For the following pairs of signals, determine if v_1 leads or lags v_2 and by how much.

(a) $v_1 = 10\cos(5t - 20°)$, $v_2 = 8\sin 5t$

(b) $v_1 = 19\cos(2t + 90°)$, $v_2 = 6\sin 2t$

(c) $v_1 = -4\cos 10t$, $v_2 = 15\sin 10t$

9.77 Refer to the *RC* circuit in Fig. 9.81.

(a) Calculate the phase shift at 2 MHz.

(b) Find the frequency where the phase shift is 45°.

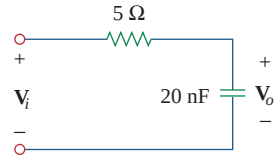


Figure 9.81
For Prob. 9.77.

9.78 A coil with impedance $8 + j6\ \Omega$ is connected in series with a capacitive reactance X. The series combination is connected in parallel with a resistor R. Given that the equivalent impedance of the resulting circuit is $5\angle 0°\ \Omega$, find the value of R and X.

9.79 (a) Calculate the phase shift of the circuit in Fig. 9.82.
(b) State whether the phase shift is leading or lagging (output with respect to input).
(c) Determine the magnitude of the output when the input is 120 V.

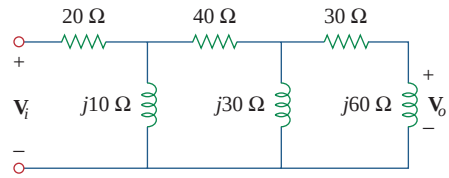


Figure 9.82
For Prob. 9.79.

9.80 Consider the phase-shifting circuit in Fig. 9.83. Let $\mathbf{V}_i = 120$ V operating at 60 Hz. Find:

(a) $\mathbf{V}_o$ when R is maximum

(b) $\mathbf{V}_o$ when R is minimum

(c) the value of R that will produce a phase shift of 45°

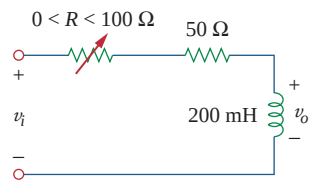


Figure 9.83
For Prob. 9.80.

9.81 The ac bridge in Fig. 9.37 is balanced when $R_1 = 400\ \Omega$, $R_2 = 600\ \Omega$, $R_3 = 1.2\ \text{k}\Omega$, and $C_2 = 0.3\ \mu\text{F}$. Find R_x and C_x. Assume R_2 and C_2 are in series.

9.82 A capacitance bridge balances when $R_1 = 100\ \Omega$, $R_2 = 2\ \text{k}\Omega$, and $C_s = 40\ \mu\text{F}$. What is C_x, the capacitance of the capacitor under test?

9.83 An inductive bridge balances when $R_1 = 1.2\ \text{k}\Omega$, $R_2 = 500\ \Omega$, and $L_s = 250$ mH. What is the value of L_x, the inductance of the inductor under test?

9.84 The ac bridge shown in Fig. 9.84 is known as a *Maxwell bridge* and is used for accurate measurement of inductance and resistance of a coil in terms of a standard capacitance C_s. Show that when the bridge is balanced,

$$L_x = R_2R_3C_s \qquad \text{and} \qquad R_x = \frac{R_2}{R_1}R_3$$

Find L_x and R_x for $R_1 = 40\ \text{k}\Omega$, $R_2 = 1.6\ \text{k}\Omega$, $R_3 = 4\ \text{k}\Omega$, and $C_s = 0.45\ \mu\text{F}$.

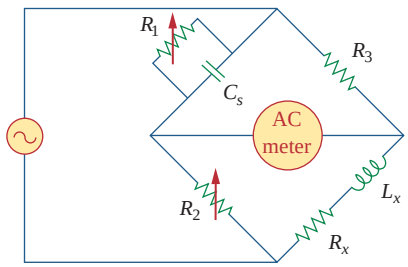


Figure 9.84
Maxwell bridge; For Prob. 9.84.

9.85 The ac bridge circuit of Fig. 9.85 is called a *Wien bridge*. It is used for measuring the frequency of a source. Show that when the bridge is balanced,

$$f = \frac{1}{2\pi\sqrt{R_2R_4C_2C_4}}$$

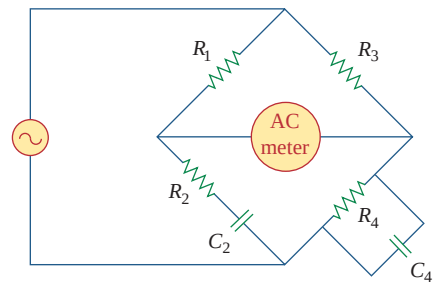


Figure 9.85
Wein bridge; For Prob. 9.85.

Comprehensive Problems

9.86 The circuit shown in Fig. 9.86 is used in a television receiver. What is the total impedance of this circuit?

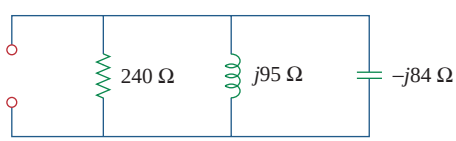


Figure 9.86
For Prob. 9.86.

9.87 The network in Fig. 9.87 is part of the schematic describing an industrial electronic sensing device. What is the total impedance of the circuit at 2 kHz?

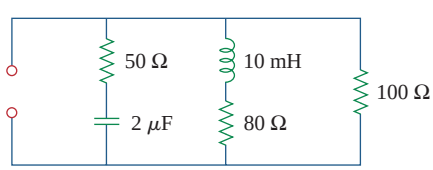


Figure 9.87
For Prob. 9.87.

9.88 A series audio circuit is shown in Fig. 9.88.

(a) What is the impedance of the circuit?

(b) If the frequency were halved, what would be the impedance of the circuit?

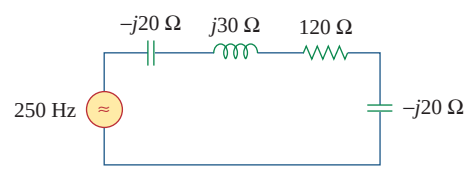


Figure 9.88
For Prob. 9.88.

9.89 An industrial load is modeled as a series combination of a capacitance and a resistance as shown in Fig. 9.89. Calculate the value of an inductance L across the series combination so that the net impedance is resistive at a frequency of 50 kHz.

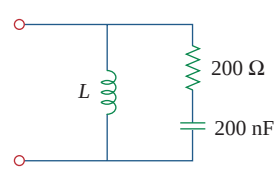


Figure 9.89
For Prob. 9.89.

9.90 An industrial coil is modeled as a series combination of an inductance L and resistance R, as shown in Fig. 9.90. Since an ac voltmeter measures only the magnitude of a sinusoid, the following

measurements are taken at 60 Hz when the circuit operates in the steady state:

$$|\mathbf{V}_s| = 145 \text{ V}, \quad |\mathbf{V}_1| = 50 \text{ V}, \quad |\mathbf{V}_o| = 110 \text{ V}$$

Use these measurements to determine the values of L and R.

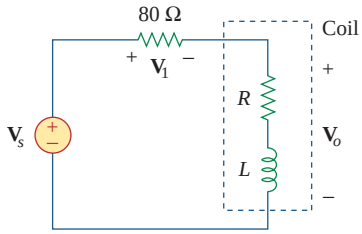


Figure 9.90
For Prob. 9.90.

9.91 Figure 9.91 shows a parallel combination of an inductance and a resistance. If it is desired to connect a capacitor in series with the parallel combination such that the net impedance is resistive at 10 MHz, what is the required value of C?

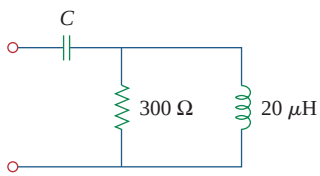


Figure 9.91
For Prob. 9.91.

9.92 A transmission line has a series impedance of $\mathbf{Z} = 100\angle 75^\circ\ \Omega$ and a shunt admittance of $\mathbf{Y} = 450\angle 48^\circ\ \mu\text{S}$. Find: (a) the characteristic impedance $\mathbf{Z}_o = \sqrt{\mathbf{Z}/\mathbf{Y}}$, (b) the propagation constant $\gamma = \sqrt{\mathbf{ZY}}$.

9.93 A power transmission system is modeled as shown in Fig. 9.92. Given the following;

Source voltage $\mathbf{V}_s = 115\angle 0^\circ$ V,
Source impedance $\mathbf{Z}_s = (2 + j)\ \Omega$,
Line impedance $\mathbf{Z}_\ell = (0.8 + j0.6)\ \Omega$,
Load impedance $\mathbf{Z}_L = (46.4 + j37.8)\ \Omega$,
Find the load current $\mathbf{I}_L$.

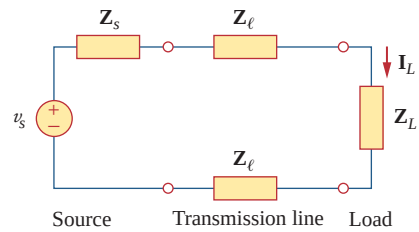


Figure 9.92
For Prob. 9.93.

chapter

10

Sinusoidal Steady-State Analysis

Three men are my friends—he that loves me, he that hates me, he that is indifferent to me. Who loves me, teaches me tenderness; who hates me, teaches me caution; who is indifferent to me, teaches me self-reliance.

—J. E. Dinger

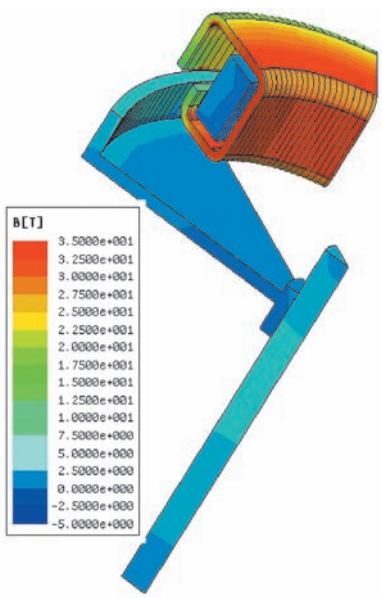
Enhancing Your Career

Career in Software Engineering

Software engineering is that aspect of engineering that deals with the practical application of scientific knowledge in the design, construction, and validation of computer programs and the associated documentation required to develop, operate, and maintain them. It is a branch of electrical engineering that is becoming increasingly important as more and more disciplines require one form of software package or another to perform routine tasks and as programmable microelectronic systems are used in more and more applications.

The role of a software engineer should not be confused with that of a computer scientist; the software engineer is a practitioner, not a theoretician. A software engineer should have good computer-programming skills and be familiar with programming languages, in particular C^{++}, which is becoming increasingly popular. Because hardware and software are closely interlinked, it is essential that a software engineer have a thorough understanding of hardware design. Most important, the software engineer should have some specialized knowledge of the area in which the software development skill is to be applied.

All in all, the field of software engineering offers a great career to those who enjoy programming and developing software packages. The higher rewards will go to those having the best preparation, with the most interesting and challenging opportunities going to those with graduate education.



Output of a modeling software. Courtesy Ansoft

10.1 Introduction

In Chapter 9, we learned that the forced or steady-state response of circuits to sinusoidal inputs can be obtained by using phasors. We also know that Ohm's and Kirchhoff's laws are applicable to ac circuits. In this chapter, we want to see how nodal analysis, mesh analysis, Thevenin's theorem, Norton's theorem, superposition, and source transformations are applied in analyzing ac circuits. Since these techniques were already introduced for dc circuits, our major effort here will be to illustrate with examples.

Analyzing ac circuits usually requires three steps.

Steps to Analyze AC Circuits:

1. Transform the circuit to the phasor or frequency domain.
2. Solve the problem using circuit techniques (nodal analysis, mesh analysis, superposition, etc.).
3. Transform the resulting phasor to the time domain.

Step 1 is not necessary if the problem is specified in the frequency domain. In step 2, the analysis is performed in the same manner as dc circuit analysis except that complex numbers are involved. Having read Chapter 9, we are adept at handling step 3.

Frequency domain analysis of an ac circuit via phasors is much easier than analysis of the circuit in the time domain.

Toward the end of the chapter, we learn how to apply *PSpice* in solving ac circuit problems. We finally apply ac circuit analysis to two practical ac circuits: oscillators and ac transistor circuits.

10.2 Nodal Analysis

The basis of nodal analysis is Kirchhoff's current law. Since KCL is valid for phasors, as demonstrated in Section 9.6, we can analyze ac circuits by nodal analysis. The following examples illustrate this.

Example 10.1

Find i_x in the circuit of Fig. 10.1 using nodal analysis.

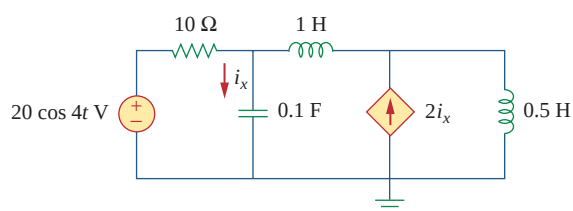


Figure 10.1
For Example 10.1.

Solution:
We first convert the circuit to the frequency domain:

$$\begin{aligned} 20\cos 4t &\Rightarrow 20\angle 0^\circ, \quad \omega = 4 \text{ rad/s} \\ 1 \text{ H} &\Rightarrow j\omega L = j4 \\ 0.5 \text{ H} &\Rightarrow j\omega L = j2 \\ 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j2.5 \end{aligned}$$

Thus, the frequency domain equivalent circuit is as shown in Fig. 10.2.

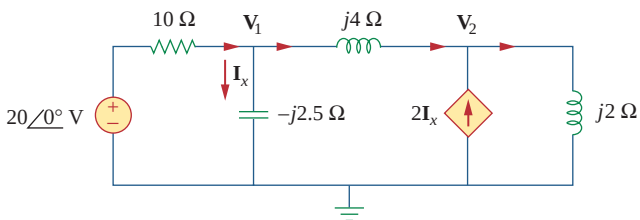


Figure 10.2
Frequency domain equivalent of the circuit in Fig. 10.1.

Applying KCL at node 1,

$$\frac{20 - \mathbf{V}_1}{10} = \frac{\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4}$$

or

$$(1 + j1.5)\mathbf{V}_1 + j2.5\mathbf{V}_2 = 20 \qquad \textbf{(10.1.1)}$$

At node 2,

$$2\mathbf{I}_x + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

But $\mathbf{I}_x = \mathbf{V}_1/-j2.5$. Substituting this gives

$$\frac{2\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

By simplifying, we get

$$11\mathbf{V}_1 + 15\mathbf{V}_2 = 0 \qquad \textbf{(10.1.2)}$$

Equations (10.1.1) and (10.1.2) can be put in matrix form as

$$\begin{bmatrix} 1 + j1.5 & j2.5 \\ 11 & 15 \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

We obtain the determinants as

$$\Delta = \begin{vmatrix} 1 + j1.5 & j2.5 \\ 11 & 15 \end{vmatrix} = 15 - j5$$

$$\Delta_1 = \begin{vmatrix} 20 & j2.5 \\ 0 & 15 \end{vmatrix} = 300, \qquad \Delta_2 = \begin{vmatrix} 1 + j1.5 & 20 \\ 11 & 0 \end{vmatrix} = -220$$

$$\mathbf{V}_1 = \frac{\Delta_1}{\Delta} = \frac{300}{15 - j5} = 18.97\angle 18.43^\circ \text{ V}$$

$$\mathbf{V}_2 = \frac{\Delta_2}{\Delta} = \frac{-220}{15 - j5} = 13.91\angle 198.3^\circ \text{ V}$$

The current $\mathbf{I}_x$ is given by

$$\mathbf{I}_x = \frac{\mathbf{V}_1}{-j2.5} = \frac{18.97\angle 18.43^\circ}{2.5\angle -90^\circ} = 7.59\angle 108.4^\circ \text{ A}$$

Transforming this to the time domain,

$$i_x = 7.59\cos(4t + 108.4^\circ) \text{ A}$$

Practice Problem 10.1

Using nodal analysis, find v_1 and v_2 in the circuit of Fig. 10.3.

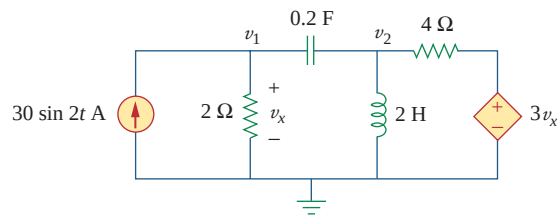


Figure 10.3
For Practice Prob. 10.1.

Answer: $v_1(t) = 33.96\sin(2t + 60.01^\circ)$ V,
$v_2(t) = 99.06\sin(2t + 57.12^\circ)$ V.

Example 10.2

Compute $\mathbf{V}_1$ and $\mathbf{V}_2$ in the circuit of Fig. 10.4.

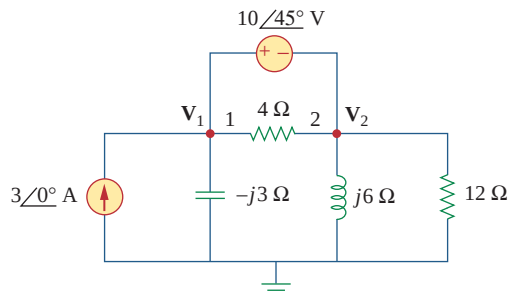


Figure 10.4
For Example 10.2.

Solution:
Nodes 1 and 2 form a supernode as shown in Fig. 10.5. Applying KCL at the supernode gives

$$3 = \frac{\mathbf{V}_1}{-j3} + \frac{\mathbf{V}_2}{j6} + \frac{\mathbf{V}_2}{12}$$

or

$$36 = j4\mathbf{V}_1 + (1 - j2)\mathbf{V}_2 \qquad \textbf{(10.2.1)}$$

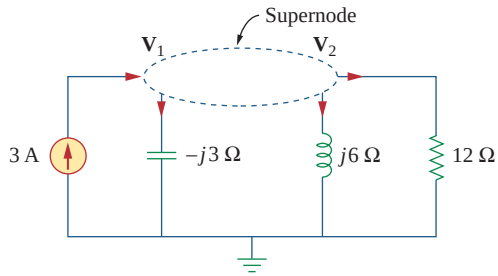


Figure 10.5
A supernode in the circuit of Fig. 10.4.

But a voltage source is connected between nodes 1 and 2, so that

$$\mathbf{V}_1 = \mathbf{V}_2 + 10\angle 45^\circ \qquad \textbf{(10.2.2)}$$

Substituting Eq. (10.2.2) in Eq. (10.2.1) results in

$$36 - 40\angle 135^\circ = (1 + j2)\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{V}_2 = 31.41\angle -87.18^\circ \text{ V}$$

From Eq. (10.2.2),

$$\mathbf{V}_1 = \mathbf{V}_2 + 10\angle 45^\circ = 25.78\angle -70.48^\circ \text{ V}$$

Practice Problem 10.2

Calculate $\mathbf{V}_1$ and $\mathbf{V}_2$ in the circuit shown in Fig. 10.6.

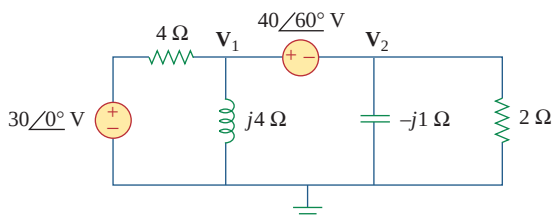


Figure 10.6
For Practice Prob. 10.2.

Answer: $\mathbf{V}_1 = 38.72\angle 69.67^\circ$ V, $\mathbf{V}_2 = 6.752\angle 165.7^\circ$ V.

10.3 Mesh Analysis

Kirchhoff's voltage law (KVL) forms the basis of mesh analysis. The validity of KVL for ac circuits was shown in Section 9.6 and is illustrated in the following examples. Keep in mind that the very nature of using mesh analysis is that it is to be applied to planar circuits.

Example 10.3

Determine current $\mathbf{I}_o$ in the circuit of Fig. 10.7 using mesh analysis.

Solution:
Applying KVL to mesh 1, we obtain

$$(8 + j10 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - j10\mathbf{I}_3 = 0 \qquad \textbf{(10.3.1)}$$

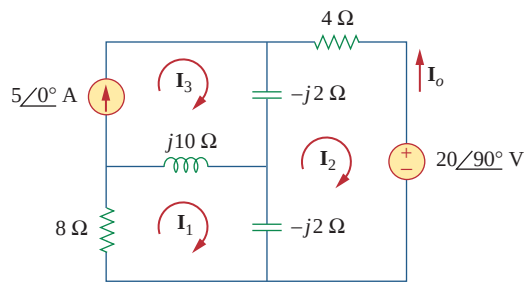


Figure 10.7
For Example 10.3.

For mesh 2,

$$(4 - j2 - j2)\mathbf{I}_2 - (-j2)\mathbf{I}_1 - (-j2)\mathbf{I}_3 + 20\angle 90^\circ = 0 \qquad \textbf{(10.3.2)}$$

For mesh 3, $\mathbf{I}_3 = 5$. Substituting this in Eqs. (10.3.1) and (10.3.2), we get

$$(8 + j8)\mathbf{I}_1 + j2\mathbf{I}_2 = j50 \qquad \textbf{(10.3.3)}$$

$$j2\mathbf{I}_1 + (4 - j4)\mathbf{I}_2 = -j20 - j10 \qquad \textbf{(10.3.4)}$$

Equations (10.3.3) and (10.3.4) can be put in matrix form as

$$\begin{bmatrix} 8 + j8 & j2 \\ j2 & 4 - j4 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} j50 \\ -j30 \end{bmatrix}$$

from which we obtain the determinants

$$\Delta = \begin{vmatrix} 8 + j8 & j2 \\ j2 & 4 - j4 \end{vmatrix} = 32(1 + j)(1 - j) + 4 = 68$$

$$\Delta_2 = \begin{vmatrix} 8 + j8 & j50 \\ j2 & -j30 \end{vmatrix} = 340 - j240 = 416.17\angle -35.22^\circ$$

$$\mathbf{I}_2 = \frac{\Delta_2}{\Delta} = \frac{416.17\angle -35.22^\circ}{68} = 6.12\angle -35.22^\circ \text{ A}$$

The desired current is

$$\mathbf{I}_o = -\mathbf{I}_2 = 6.12\angle 144.78^\circ \text{ A}$$

Practice Problem 10.3

Find $\mathbf{I}_o$ in Fig. 10.8 using mesh analysis.

Answer: $3.582\angle 65.45^\circ$ A.

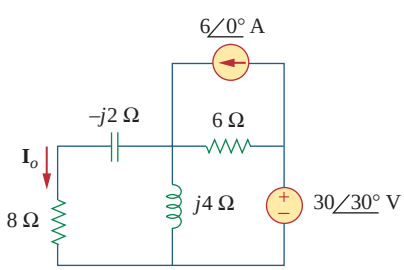


Figure 10.8
For Practice Prob. 10.3.

Solve for $\mathbf{V}_o$ in the circuit of Fig. 10.9 using mesh analysis.

Example 10.4

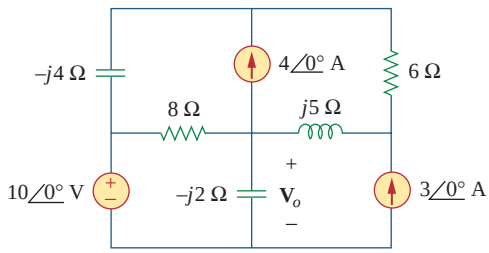


Figure 10.9
For Example 10.4.

Solution:
As shown in Fig. 10.10, meshes 3 and 4 form a supermesh due to the current source between the meshes. For mesh 1, KVL gives

$$-10 + (8 - j2)\mathbf{I}_1 - (-j2)\mathbf{I}_2 - 8\mathbf{I}_3 = 0$$

or

$$(8 - j2)\mathbf{I}_1 + j2\mathbf{I}_2 - 8\mathbf{I}_3 = 10 \qquad \textbf{(10.4.1)}$$

For mesh 2,

$$\mathbf{I}_2 = -3 \qquad \textbf{(10.4.2)}$$

For the supermesh,

$$(8 - j4)\mathbf{I}_3 - 8\mathbf{I}_1 + (6 + j5)\mathbf{I}_4 - j5\mathbf{I}_2 = 0 \qquad \textbf{(10.4.3)}$$

Due to the current source between meshes 3 and 4, at node A,

$$\mathbf{I}_4 = \mathbf{I}_3 + 4 \qquad \textbf{(10.4.4)}$$

■ **METHOD 1** Instead of solving the above four equations, we reduce them to two by elimination.

Combining Eqs. (10.4.1) and (10.4.2),

$$(8 - j2)\mathbf{I}_1 - 8\mathbf{I}_3 = 10 + j6 \qquad \textbf{(10.4.5)}$$

Combining Eqs. (10.4.2) to (10.4.4),

$$-8\mathbf{I}_1 + (14 + j)\mathbf{I}_3 = -24 - j35 \qquad \textbf{(10.4.6)}$$

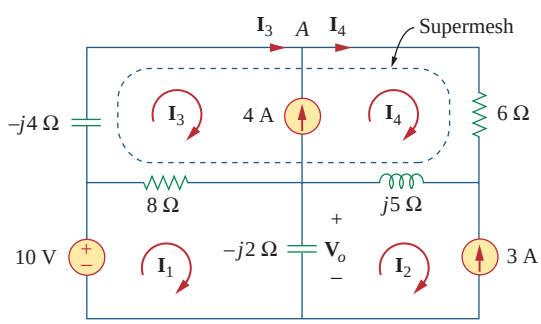


Figure 10.10
Analysis of the circuit in Fig. 10.9.

From Eqs. (10.4.5) and (10.4.6), we obtain the matrix equation

$$\begin{bmatrix} 8 - j2 & -8 \\ -8 & 14 + j \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_3 \end{bmatrix} = \begin{bmatrix} 10 + j6 \\ -24 - j35 \end{bmatrix}$$

We obtain the following determinants

$$\Delta = \begin{vmatrix} 8 - j2 & -8 \\ -8 & 14 + j \end{vmatrix} = 112 + j8 - j28 + 2 - 64 = 50 - j20$$

$$\Delta_1 = \begin{vmatrix} 10 + j6 & -8 \\ -24 - j35 & 14 + j \end{vmatrix} = 140 + j10 + j84 - 6 - 192 - j280$$

$$= -58 - j186$$

Current $\mathbf{I}_1$ is obtained as

$$\mathbf{I}_1 = \frac{\Delta_1}{\Delta} = \frac{-58 - j186}{50 - j20} = 3.618\angle 274.5^\circ \text{ A}$$

The required voltage $\mathbf{V}_0$ is

$$\mathbf{V}_o = -j2(\mathbf{I}_1 - \mathbf{I}_2) = -j2(3.618\angle 274.5^\circ + 3)$$

$$= -7.2134 - j6.568 = 9.756\angle 222.32^\circ \text{ V}$$

■ **METHOD 2** We can use *MATLAB* to solve Eqs. (10.4.1) to (10.4.4). We first cast the equations as

$$\begin{bmatrix} 8 - j2 & j2 & -8 & 0 \\ 0 & 1 & 0 & 0 \\ -8 & -j5 & 8 - j4 & 6 + j5 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \\ \mathbf{I}_3 \\ \mathbf{I}_4 \end{bmatrix} = \begin{bmatrix} 10 \\ -3 \\ 0 \\ 4 \end{bmatrix} \quad \textbf{(10.4.7a)}$$

or

$$\mathbf{AI} = \mathbf{B}$$

By inverting $\mathbf{A}$, we can obtain $\mathbf{I}$ as

$$\mathbf{I} = \mathbf{A}^{-1}\mathbf{B} \quad \textbf{(10.4.7b)}$$

We now apply *MATLAB* as follows:

```
>> A = [(8-j*2)  j*2    -8        0;
        0        1      0         0;
        -8       -j*5   (8-j*4)   (6+j*5);
        0        0      -1        1];
>> B = [10 -3 0 4]';
>> I = inv(A)*B

I =
  0.2828 - 3.6069i
  -3.0000
  -1.8690 - 4.4276i
   2.1310 - 4.4276i
>> Vo = -2*j*(I(1) - I(2))

Vo =
  -7.2138 - 6.5655i
```

as obtained previously.

Calculate current $\mathbf{I}_o$ in the circuit of Fig. 10.11.

Practice Problem 10.4

Answer: $2.538\angle 5.943°$ A.

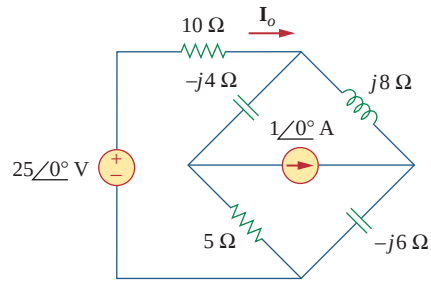


Figure 10.11
For Practice Prob. 10.4.

10.4 Superposition Theorem

Since ac circuits are linear, the superposition theorem applies to ac circuits the same way it applies to dc circuits. The theorem becomes important if the circuit has sources operating at *different* frequencies. In this case, since the impedances depend on frequency, we must have a different frequency domain circuit for each frequency. The total response must be obtained by adding the individual responses in the *time* domain. It is incorrect to try to add the responses in the phasor or frequency domain. Why? Because the exponential factor $e^{j\omega t}$ is implicit in sinusoidal analysis, and that factor would change for every angular frequency ω. It would therefore not make sense to add responses at different frequencies in the phasor domain. Thus, when a circuit has sources operating at different frequencies, one must add the responses due to the individual frequencies in the time domain.

Use the superposition theorem to find $\mathbf{I}_o$ in the circuit in Fig. 10.7.

Example 10.5

Solution:
Let

$$\mathbf{I}_o = \mathbf{I}'_o + \mathbf{I}''_o \tag{10.5.1}$$

where $\mathbf{I}'_o$ and $\mathbf{I}''_o$ are due to the voltage and current sources, respectively. To find $\mathbf{I}'_o$, consider the circuit in Fig. 10.12(a). If we let $\mathbf{Z}$ be the parallel combination of $-j2$ and $8 + j10$, then

$$\mathbf{Z} = \frac{-j2(8 + j10)}{-2j + 8 + j10} = 0.25 - j2.25$$

and current $\mathbf{I}'_o$ is

$$\mathbf{I}'_o = \frac{j20}{4 - j2 + \mathbf{Z}} = \frac{j20}{4.25 - j4.25}$$

or

$$\mathbf{I}'_o = -2.353 + j2.353 \tag{10.5.2}$$

To get $\mathbf{I}''_o$, consider the circuit in Fig. 10.12(b). For mesh 1,

$$(8 + j8)\mathbf{I}_1 - j10\mathbf{I}_3 + j2\mathbf{I}_2 = 0 \tag{10.5.3}$$

For mesh 2,

$$(4 - j4)\mathbf{I}_2 + j2\mathbf{I}_1 + j2\mathbf{I}_3 = 0 \tag{10.5.4}$$

For mesh 3,

$$\mathbf{I}_3 = 5 \tag{10.5.5}$$

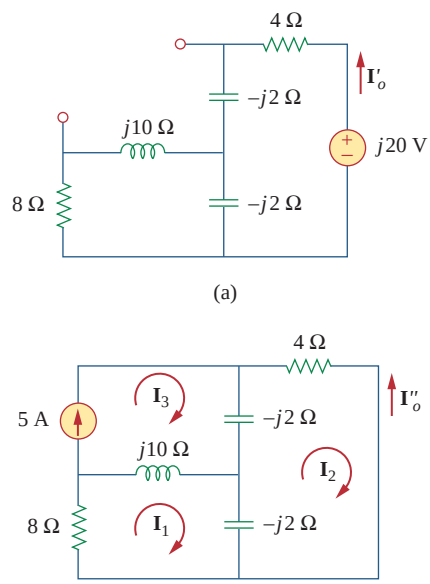


Figure 10.12
Solution of Example 10.5.

From Eqs. (10.5.4) and (10.5.5),

$$(4 - j4)\mathbf{I}_2 + j2\mathbf{I}_1 + j10 = 0$$

Expressing $\mathbf{I}_1$ in terms of $\mathbf{I}_2$ gives

$$\mathbf{I}_1 = (2 + j2)\mathbf{I}_2 - 5 \qquad \textbf{(10.5.6)}$$

Substituting Eqs. (10.5.5) and (10.5.6) into Eq. (10.5.3), we get

$$(8 + j8)[(2 + j2)\mathbf{I}_2 - 5] - j50 + j2\mathbf{I}_2 = 0$$

or

$$\mathbf{I}_2 = \frac{90 - j40}{34} = 2.647 - j1.176$$

Current $\mathbf{I}_o''$ is obtained as

$$\mathbf{I}_o'' = -\mathbf{I}_2 = -2.647 + j1.176 \qquad \textbf{(10.5.7)}$$

From Eqs. (10.5.2) and (10.5.7), we write

$$\mathbf{I}_o = \mathbf{I}_o' + \mathbf{I}_o'' = -5 + j3.529 = 6.12\angle 144.78^\circ \text{ A}$$

which agrees with what we got in Example 10.3. It should be noted that applying the superposition theorem is not the best way to solve this problem. It seems that we have made the problem twice as hard as the original one by using superposition. However, in Example 10.6, superposition is clearly the easiest approach.

Practice Problem 10.5

Find current $\mathbf{I}_o$ in the circuit of Fig. 10.8 using the superposition theorem.

Answer: $3.582\angle 65.45^\circ$ A.

Example 10.6

Find v_o of the circuit of Fig. 10.13 using the superposition theorem.

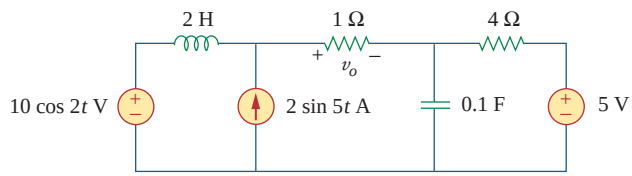


Figure 10.13
For Example 10.6.

Solution:

Since the circuit operates at three different frequencies ($\omega = 0$ for the dc voltage source), one way to obtain a solution is to use superposition, which breaks the problem into single-frequency problems. So we let

$$v_o = v_1 + v_2 + v_3 \qquad \textbf{(10.6.1)}$$

where v_1 is due to the 5-V dc voltage source, v_2 is due to the 10 cos 2t V voltage source, and v_3 is due to the 2 sin 5t A current source.

To find v_1, we set to zero all sources except the 5-V dc source. We recall that at steady state, a capacitor is an open circuit to dc while an inductor is a short circuit to dc. There is an alternative way of looking at this. Since $\omega = 0$, $j\omega L = 0$, $1/j\omega C = \infty$. Either way, the equivalent circuit is as shown in Fig. 10.14(a). By voltage division,

$$-v_1 = \frac{1}{1+4}(5) = 1 \text{ V} \qquad \textbf{(10.6.2)}$$

To find v_2, we set to zero both the 5-V source and the 2 sin 5t current source and transform the circuit to the frequency domain.

$$\begin{aligned} 10\cos 2t &\Rightarrow 10\angle 0^\circ, \quad \omega = 2 \text{ rad/s} \\ 2 \text{ H} &\Rightarrow j\omega L = j4\ \Omega \\ 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j5\ \Omega \end{aligned}$$

The equivalent circuit is now as shown in Fig. 10.14(b). Let

$$\mathbf{Z} = -j5 \parallel 4 = \frac{-j5 \times 4}{4 - j5} = 2.439 - j1.951$$

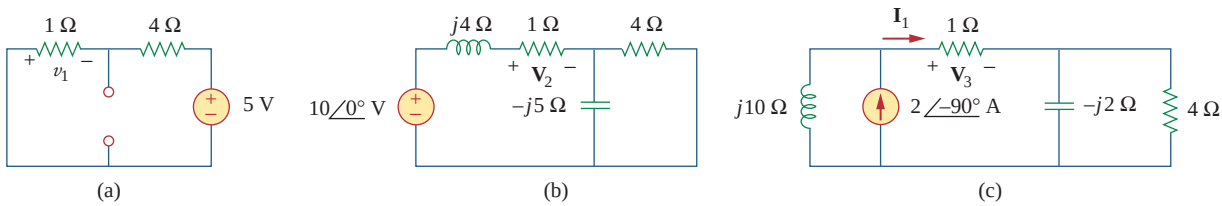


Figure 10.14
Solution of Example 10.6: (a) setting all sources to zero except the 5-V dc source, (b) setting all sources to zero except the ac voltage source, (c) setting all sources to zero except the ac current source.

By voltage division,

$$\mathbf{V}_2 = \frac{1}{1 + j4 + \mathbf{Z}}(10\angle 0^\circ) = \frac{10}{3.439 + j2.049} = 2.498\angle -30.79^\circ$$

In the time domain,

$$v_2 = 2.498\cos(2t - 30.79^\circ) \qquad \textbf{(10.6.3)}$$

To obtain v_3, we set the voltage sources to zero and transform what is left to the frequency domain.

$$\begin{aligned} 2\sin 5t &\Rightarrow 2\angle -90^\circ, \quad \omega = 5 \text{ rad/s} \\ 2 \text{ H} &\Rightarrow j\omega L = j10\ \Omega \\ 0.1 \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j2\ \Omega \end{aligned}$$

The equivalent circuit is in Fig. 10.14(c). Let

$$\mathbf{Z}_1 = -j2 \parallel 4 = \frac{-j2 \times 4}{4 - j2} = 0.8 - j1.6\ \Omega$$

By current division,

$$\mathbf{I}_1 = \frac{j10}{j10 + 1 + \mathbf{Z}_1}(2\angle -90^\circ)\ \text{A}$$

$$\mathbf{V}_3 = \mathbf{I}_1 \times 1 = \frac{j10}{1.8 + j8.4}(-j2) = 2.328\angle -80^\circ\ \text{V}$$

In the time domain,

$$v_3 = 2.33\cos(5t - 80^\circ) = 2.33\sin(5t + 10^\circ)\ \text{V} \qquad \textbf{(10.6.4)}$$

Substituting Eqs. (10.6.2) to (10.6.4) into Eq. (10.6.1), we have

$$v_o(t) = -1 + 2.498\cos(2t - 30.79^\circ) + 2.33\sin(5t + 10^\circ)\ \text{V}$$

Practice Problem 10.6

Calculate v_o in the circuit of Fig. 10.15 using the superposition theorem.

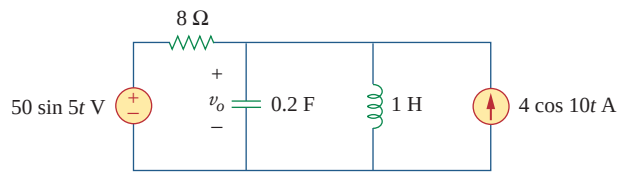


Figure 10.15
For Practice Prob. 10.6.

Answer: $7.718\sin(5t - 81.12^\circ) + 2.102\cos(10t - 86.24^\circ)$ V.

10.5 Source Transformation

As Fig. 10.16 shows, source transformation in the frequency domain involves transforming a voltage source in series with an impedance to a current source in parallel with an impedance, or vice versa. As we go from one source type to another, we must keep the following relationship in mind:

$$\mathbf{V}_s = \mathbf{Z}_s\mathbf{I}_s \quad \Leftrightarrow \quad \mathbf{I}_s = \frac{\mathbf{V}_s}{\mathbf{Z}_s} \qquad \textbf{(10.1)}$$

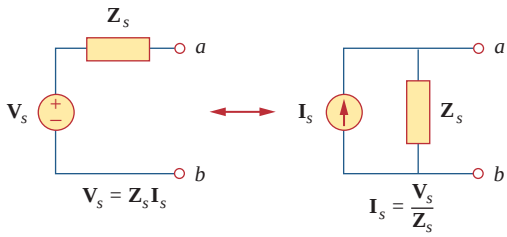


Figure 10.16
Source transformation.

Example 10.7

Calculate $\mathbf{V}_x$ in the circuit of Fig. 10.17 using the method of source transformation.

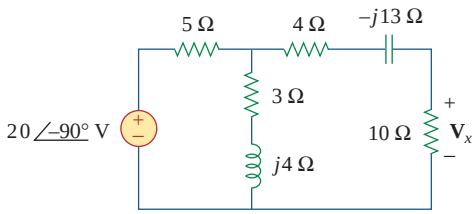


Figure 10.17
For Example 10.7.

Solution:
We transform the voltage source to a current source and obtain the circuit in Fig. 10.18(a), where

$$\mathbf{I}_s = \frac{20\angle -90^\circ}{5} = 4\angle -90^\circ = -j4 \text{ A}$$

The parallel combination of 5-Ω resistance and $(3 + j4)$ impedance gives

$$\mathbf{Z}_1 = \frac{5(3 + j4)}{8 + j4} = 2.5 + j1.25\ \Omega$$

Converting the current source to a voltage source yields the circuit in Fig. 10.18(b), where

$$\mathbf{V}_s = \mathbf{I}_s\mathbf{Z}_1 = -j4(2.5 + j1.25) = 5 - j10 \text{ V}$$

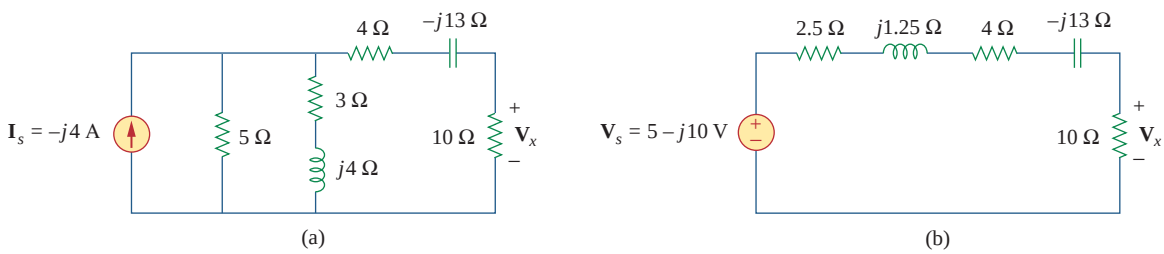


Figure 10.18
Solution of the circuit in Fig. 10.17.

By voltage division,

$$\mathbf{V}_x = \frac{10}{10 + 2.5 + j1.25 + 4 - j13}(5 - j10) = 5.519\angle -28^\circ \text{ V}$$

Practice Problem 10.7

Find $\mathbf{I}_o$ in the circuit of Fig. 10.19 using the concept of source transformation.

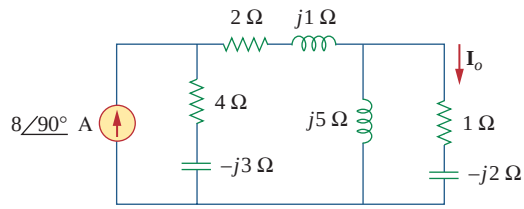


Figure 10.19
For Practice Prob. 10.7.

Answer: $6.576/\underline{99.46°}$ A.

10.6 Thevenin and Norton Equivalent Circuits

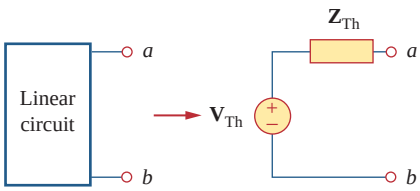


Figure 10.20
Thevenin equivalent.

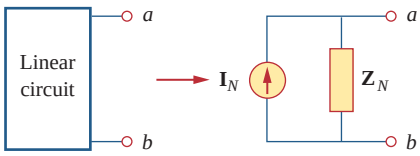


Figure 10.21
Norton equivalent.

Thevenin's and Norton's theorems are applied to ac circuits in the same way as they are to dc circuits. The only additional effort arises from the need to manipulate complex numbers. The frequency domain version of a Thevenin equivalent circuit is depicted in Fig. 10.20, where a linear circuit is replaced by a voltage source in series with an impedance. The Norton equivalent circuit is illustrated in Fig. 10.21, where a linear circuit is replaced by a current source in parallel with an impedance. Keep in mind that the two equivalent circuits are related as

$$\mathbf{V}_{\text{Th}} = \mathbf{Z}_N\mathbf{I}_N, \qquad \mathbf{Z}_{\text{Th}} = \mathbf{Z}_N \tag{10.2}$$

just as in source transformation. $\mathbf{V}_{\text{Th}}$ is the open-circuit voltage while $\mathbf{I}_N$ is the short-circuit current.

If the circuit has sources operating at different frequencies (see Example 10.6, for example), the Thevenin or Norton equivalent circuit must be determined at each frequency. This leads to entirely different equivalent circuits, one for each frequency, not one equivalent circuit with equivalent sources and equivalent impedances.

Example 10.8

Obtain the Thevenin equivalent at terminals *a-b* of the circuit in Fig. 10.22.

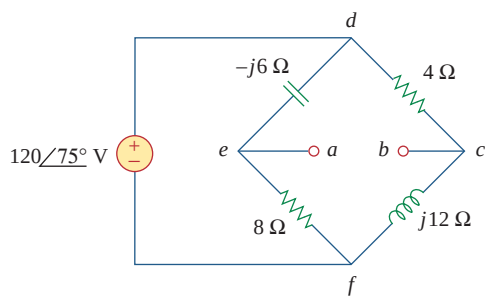


Figure 10.22
For Example 10.8.

Solution:

We find $\mathbf{Z}_{\text{Th}}$ by setting the voltage source to zero. As shown in Fig. 10.23(a), the 8-Ω resistance is now in parallel with the $-j6$ reactance, so that their combination gives

$$\mathbf{Z}_1 = -j6 \parallel 8 = \frac{-j6 \times 8}{8 - j6} = 2.88 - j3.84\ \Omega$$

Similarly, the 4-Ω resistance is in parallel with the $j12$ reactance, and their combination gives

$$\mathbf{Z}_2 = 4 \parallel j12 = \frac{j12 \times 4}{4 + j12} = 3.6 + j1.2\ \Omega$$

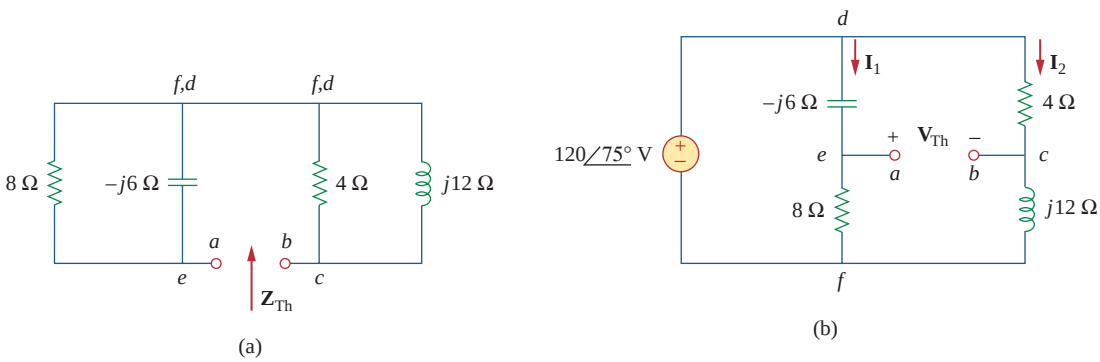


Figure 10.23
Solution of the circuit in Fig. 10.22: (a) finding $\mathbf{Z}_{\text{Th}}$, (b) finding $\mathbf{V}_{\text{Th}}$.

The Thevenin impedance is the series combination of $\mathbf{Z}_1$ and $\mathbf{Z}_2$; that is,

$$\mathbf{Z}_{\text{Th}} = \mathbf{Z}_1 + \mathbf{Z}_2 = 6.48 - j2.64\ \Omega$$

To find $\mathbf{V}_{\text{Th}}$, consider the circuit in Fig. 10.23(b). Currents $\mathbf{I}_1$ and $\mathbf{I}_2$ are obtained as

$$\mathbf{I}_1 = \frac{120\angle 75^\circ}{8 - j6}\ \text{A}, \qquad \mathbf{I}_2 = \frac{120\angle 75^\circ}{4 + j12}\ \text{A}$$

Applying KVL around loop *bcdeab* in Fig. 10.23(b) gives

$$\mathbf{V}_{\text{Th}} - 4\mathbf{I}_2 + (-j6)\mathbf{I}_1 = 0$$

or

$$\begin{aligned}\mathbf{V}_{\text{Th}} = 4\mathbf{I}_2 + j6\mathbf{I}_1 &= \frac{480\angle 75^\circ}{4 + j12} + \frac{720\angle 75^\circ + 90^\circ}{8 - j6} \\ &= 37.95\angle 3.43^\circ + 72\angle 201.87^\circ \\ &= -28.936 - j24.55 = 37.95\angle 220.31^\circ\ \text{V}\end{aligned}$$

Practice Problem 10.8

Find the Thevenin equivalent at terminals *a-b* of the circuit in Fig. 10.24.

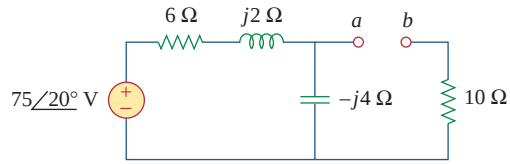


Figure 10.24
For Practice Prob. 10.8.

Answer: $\mathbf{Z}_{\text{Th}} = 12.4 - j3.2\ \Omega$, $\mathbf{V}_{\text{Th}} = 47.42\angle -51.57°$ V.

Example 10.9

Find the Thevenin equivalent of the circuit in Fig. 10.25 as seen from terminals *a-b*.

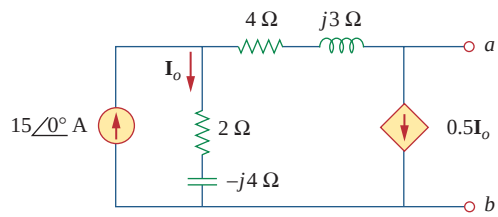


Figure 10.25
For Example 10.9.

Solution:

To find $\mathbf{V}_{\text{Th}}$, we apply KCL at node 1 in Fig. 10.26(a).

$$15 = \mathbf{I}_o + 0.5\mathbf{I}_o \quad \Rightarrow \quad \mathbf{I}_o = 10\ \text{A}$$

Applying KVL to the loop on the right-hand side in Fig. 10.26(a), we obtain

$$-\mathbf{I}_o(2 - j4) + 0.5\mathbf{I}_o(4 + j3) + \mathbf{V}_{\text{Th}} = 0$$

or

$$\mathbf{V}_{\text{Th}} = 10(2 - j4) - 5(4 + j3) = -j55$$

Thus, the Thevenin voltage is

$$\mathbf{V}_{\text{Th}} = 55\angle -90°\ \text{V}$$

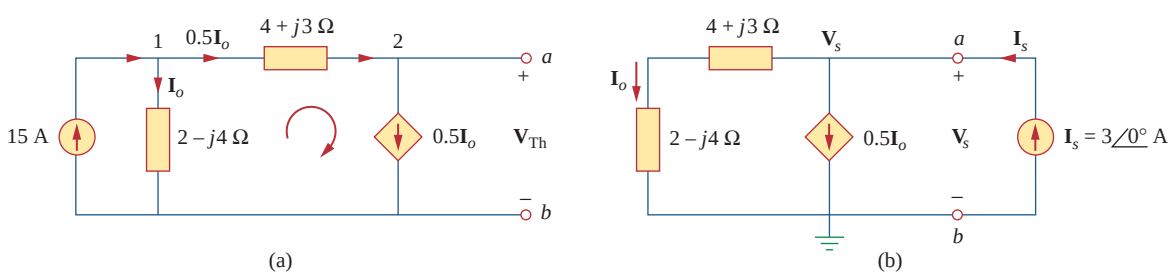


Figure 10.26
Solution of the problem in Fig. 10.25: (a) finding $\mathbf{V}_{\text{Th}}$, (b) finding $\mathbf{Z}_{\text{Th}}$.

To obtain $\mathbf{Z}_{\text{Th}}$, we remove the independent source. Due to the presence of the dependent current source, we connect a 3-A current source (3 is an arbitrary value chosen for convenience here, a number divisible by the sum of currents leaving the node) to terminals *a-b* as shown in Fig. 10.26(b). At the node, KCL gives

$$3 = \mathbf{I}_o + 0.5\mathbf{I}_o \qquad \Rightarrow \qquad \mathbf{I}_o = 2 \text{ A}$$

Applying KVL to the outer loop in Fig. 10.26(b) gives

$$\mathbf{V}_s = \mathbf{I}_o(4 + j3 + 2 - j4) = 2(6 - j)$$

The Thevenin impedance is

$$\mathbf{Z}_{\text{Th}} = \frac{\mathbf{V}_s}{\mathbf{I}_s} = \frac{2(6 - j)}{3} = 4 - j0.6667\ \Omega$$

Practice Problem 10.9

Determine the Thevenin equivalent of the circuit in Fig. 10.27 as seen from the terminals *a-b*.

Answer: $\mathbf{Z}_{\text{Th}} = 4.473\angle -7.64°\ \Omega$, $\mathbf{V}_{\text{Th}} = 29.4\angle 72.9°$ V.

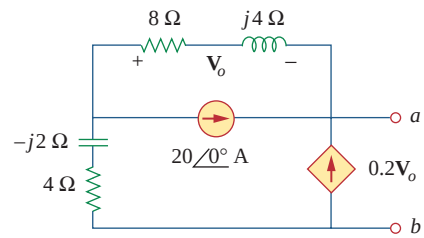


Figure 10.27
For Practice Prob. 10.9.

Example 10.10

Obtain current $\mathbf{I}_o$ in Fig. 10.28 using Norton's theorem.

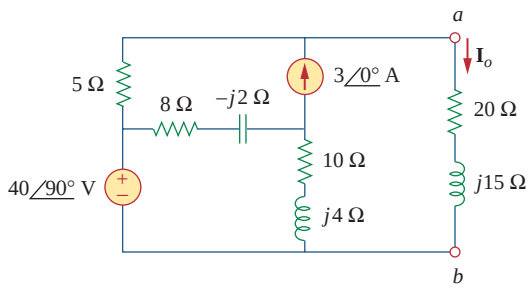


Figure 10.28
For Example 10.10.

Solution:

Our first objective is to find the Norton equivalent at terminals *a-b*. $\mathbf{Z}_N$ is found in the same way as $\mathbf{Z}_{\text{Th}}$. We set the sources to zero as shown in Fig. 10.29(a). As evident from the figure, the $(8 - j2)$ and $(10 + j4)$ impedances are short-circuited, so that

$$\mathbf{Z}_N = 5\ \Omega$$

To get $\mathbf{I}_N$, we short-circuit terminals *a-b* as in Fig. 10.29(b) and apply mesh analysis. Notice that meshes 2 and 3 form a supermesh because of the current source linking them. For mesh 1,

$$-j40 + (18 + j2)\mathbf{I}_1 - (8 - j2)\mathbf{I}_2 - (10 + j4)\mathbf{I}_3 = 0 \qquad \textbf{(10.10.1)}$$

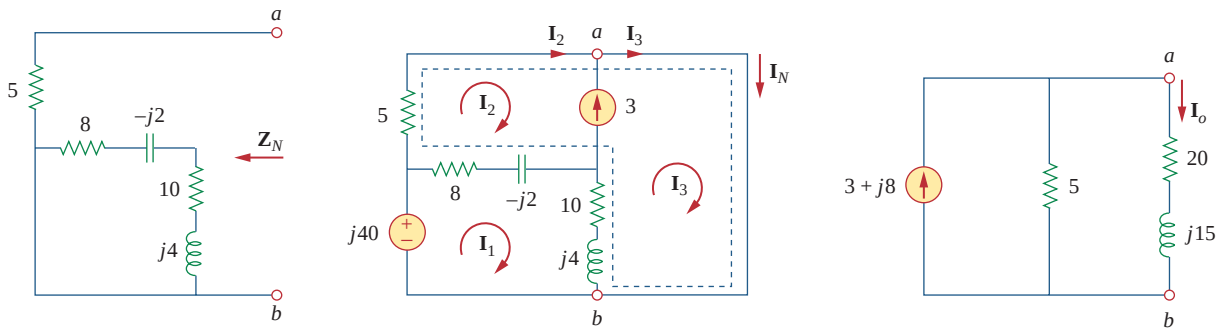


Figure 10.29
Solution of the circuit in Fig. 10.28: (a) finding $\mathbf{Z}_N$, (b) finding $\mathbf{V}_N$, (c) calculating $\mathbf{I}_o$.

For the supermesh,

$$(13 - j2)\mathbf{I}_2 + (10 + j4)\mathbf{I}_3 - (18 + j2)\mathbf{I}_1 = 0 \qquad \textbf{(10.10.2)}$$

At node *a*, due to the current source between meshes 2 and 3,

$$\mathbf{I}_3 = \mathbf{I}_2 + 3 \qquad \textbf{(10.10.3)}$$

Adding Eqs. (10.10.1) and (10.10.2) gives

$$-j40 + 5\mathbf{I}_2 = 0 \quad \Rightarrow \quad \mathbf{I}_2 = j8$$

From Eq. (10.10.3),

$$\mathbf{I}_3 = \mathbf{I}_2 + 3 = 3 + j8$$

The Norton current is

$$\mathbf{I}_N = \mathbf{I}_3 = (3 + j8)\ \text{A}$$

Figure 10.29(c) shows the Norton equivalent circuit along with the impedance at terminals *a*-*b*. By current division,

$$\mathbf{I}_o = \frac{5}{5 + 20 + j15}\mathbf{I}_N = \frac{3 + j8}{5 + j3} = 1.465\angle 38.48^\circ\ \text{A}$$

Practice Problem 10.10

Determine the Norton equivalent of the circuit in Fig. 10.30 as seen from terminals *a*-*b*. Use the equivalent to find $\mathbf{I}_o$.

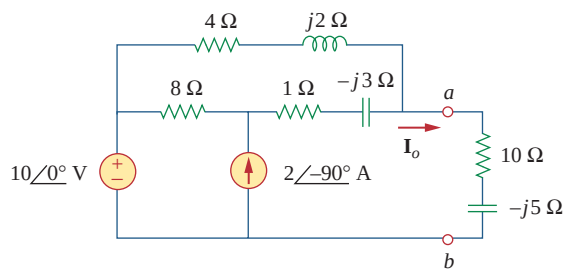


Figure 10.30
For Practice Prob. 10.10.

Answer: $\mathbf{Z}_N = 3.176 + j0.706\ \Omega$, $\mathbf{I}_N = 4.198\angle -32.68^\circ$ A, $\mathbf{I}_o = 985.5\angle -2.101^\circ$ mA.

10.7 Op Amp AC Circuits

The three steps stated in Section 10.1 also apply to op amp circuits, as long as the op amp is operating in the linear region. As usual, we will assume ideal op amps. (See Section 5.2.) As discussed in Chapter 5, the key to analyzing op amp circuits is to keep two important properties of an ideal op amp in mind:

1. No current enters either of its input terminals.
2. The voltage across its input terminals is zero.

The following examples will illustrate these ideas.

Example 10.11

Determine $v_o(t)$ for the op amp circuit in Fig. 10.31(a) if $v_s = 3\cos 1000t$ V.

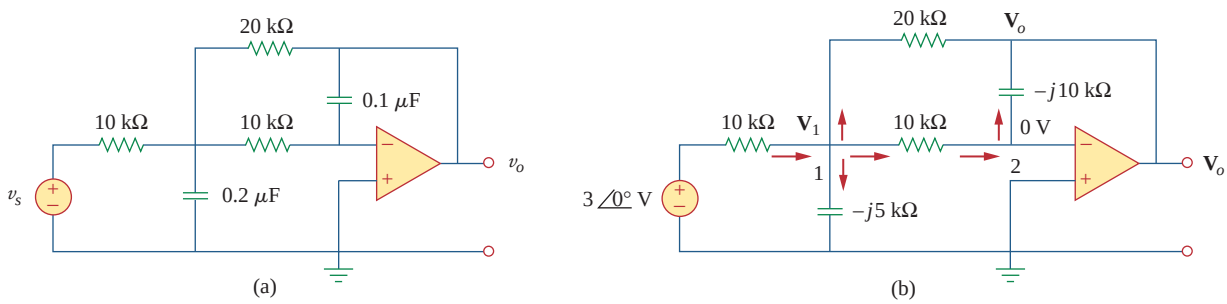


Figure 10.31
For Example 10.11: (a) the original circuit in the time domain, (b) its frequency domain equivalent.

Solution:
We first transform the circuit to the frequency domain, as shown in Fig. 10.31(b), where $\mathbf{V}_s = 3\angle 0°$, $\omega = 1000$ rad/s. Applying KCL at node 1, we obtain

$$\frac{3\angle 0° - \mathbf{V}_1}{10} = \frac{\mathbf{V}_1}{-j5} + \frac{\mathbf{V}_1 - 0}{10} + \frac{\mathbf{V}_1 - \mathbf{V}_o}{20}$$

or

$$6 = (5 + j4)\mathbf{V}_1 - \mathbf{V}_o \qquad \textbf{(10.11.1)}$$

At node 2, KCL gives

$$\frac{\mathbf{V}_1 - 0}{10} = \frac{0 - \mathbf{V}_o}{-j10}$$

which leads to

$$\mathbf{V}_1 = -j\mathbf{V}_o \qquad \textbf{(10.11.2)}$$

Substituting Eq. (10.11.2) into Eq. (10.11.1) yields

$$6 = -j(5 + j4)\mathbf{V}_o - \mathbf{V}_o = (3 - j5)\mathbf{V}_o$$

$$\mathbf{V}_o = \frac{6}{3 - j5} = 1.029\angle 59.04°$$

Hence,

$$v_o(t) = 1.029\cos(1000t + 59.04°)\text{ V}$$

Practice Problem 10.11

Find v_o and i_o in the op amp circuit of Fig. 10.32. Let $v_s = 4 \cos 5000t$ V.

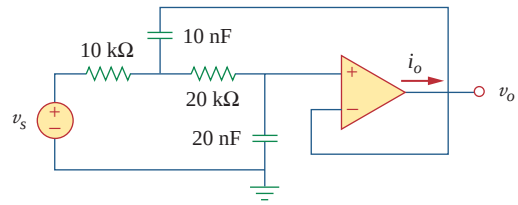


Figure 10.32
For Practice Prob. 10.11.

Answer: 1.3333 sin 5000t V, 133.33 sin 5000t μA.

Example 10.12

Compute the closed-loop gain and phase shift for the circuit in Fig. 10.33. Assume that $R_1 = R_2 = 10$ kΩ, $C_1 = 2$ μF, $C_2 = 1$ μF, and $\omega = 200$ rad/s.

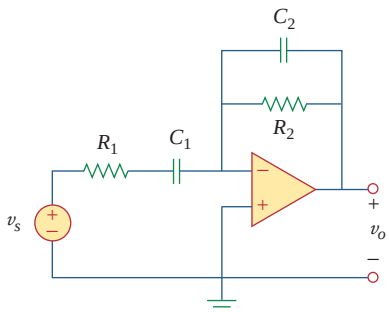


Figure 10.33
For Example 10.12.

Solution:
The feedback and input impedances are calculated as

$$\mathbf{Z}_f = R_2 \,\|\, \frac{1}{j\omega C_2} = \frac{R_2}{1 + j\omega R_2 C_2}$$

$$\mathbf{Z}_i = R_1 + \frac{1}{j\omega C_1} = \frac{1 + j\omega R_1 C_1}{j\omega C_1}$$

Since the circuit in Fig. 10.33 is an inverting amplifier, the closed-loop gain is given by

$$\mathbf{G} = \frac{\mathbf{V}_o}{\mathbf{V}_s} = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i} = \frac{-j\omega C_1 R_2}{(1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2)}$$

Substituting the given values of R_1, R_2, C_1, C_2, and ω, we obtain

$$\mathbf{G} = \frac{-j4}{(1 + j4)(1 + j2)} = 0.434\angle 130.6°$$

Thus, the closed-loop gain is 0.434 and the phase shift is 130.6°.

Practice Problem 10.12

Obtain the closed-loop gain and phase shift for the circuit in Fig. 10.34. Let $R = 10$ kΩ, $C = 1$ μF, and $\omega = 1000$ rad/s.

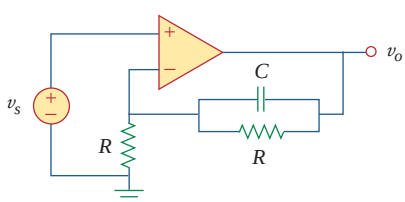


Figure 10.34
For Practice Prob. 10.12.

Answer: 1.015, −5.6°.

10.8 AC Analysis Using *PSpice*

PSpice affords a big relief from the tedious task of manipulating complex numbers in ac circuit analysis. The procedure for using *PSpice* for ac analysis is quite similar to that required for dc analysis. The reader should read Section D.5 in Appendix D for a review of *PSpice* concepts for ac analysis. AC circuit analysis is done in the phasor or frequency domain, and all sources must have the same frequency. Although ac analysis with *PSpice* involves using AC Sweep, our analysis in this chapter requires a single frequency $f = \omega/2\pi$. The output file of *PSpice* contains voltage and current phasors. If necessary, the impedances can be calculated using the voltages and currents in the output file.

Example 10.13

Obtain v_o and i_o in the circuit of Fig. 10.35 using *PSpice*.

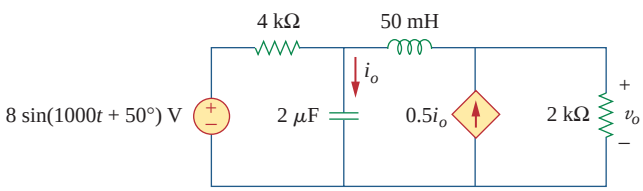


Figure 10.35
For Example 10.13.

Solution:
We first convert the sine function to cosine.

$$8\sin(1000t + 50^\circ) = 8\cos(1000t + 50^\circ - 90^\circ)$$
$$= 8\cos(1000t - 40^\circ)$$

The frequency f is obtained from ω as

$$f = \frac{\omega}{2\pi} = \frac{1000}{2\pi} = 159.155 \text{ Hz}$$

The schematic for the circuit is shown in Fig. 10.36. Notice that the current-controlled current source F1 is connected such that its current flows from node 0 to node 3 in conformity with the original circuit in Fig. 10.35. Since we only want the magnitude and phase of v_o and i_o, we set the attributes of IPRINT and VPRINT1 each to *AC* = *yes*, *MAG* = *yes*, *PHASE* = *yes*. As a single-frequency analysis, we select **Analysis/Setup/AC Sweep** and enter *Total Pts* = 1, *Start Freq* = 159.155, and *Final Freq* = 159.155. After saving the schematic, we simulate it by selecting **Analysis/Simulate.** The output file includes the source frequency in addition to the attributes checked for the pseudocomponents IPRINT and VPRINT1,

```
FREQ          IM(V_PRINT3)    IP(V_PRINT3)
1.592E+02     3.264E-03       -3.743E+01

FREQ          VM(3)           VP(3)
1.592E+02     1.550E+00       -9.518E+01
```

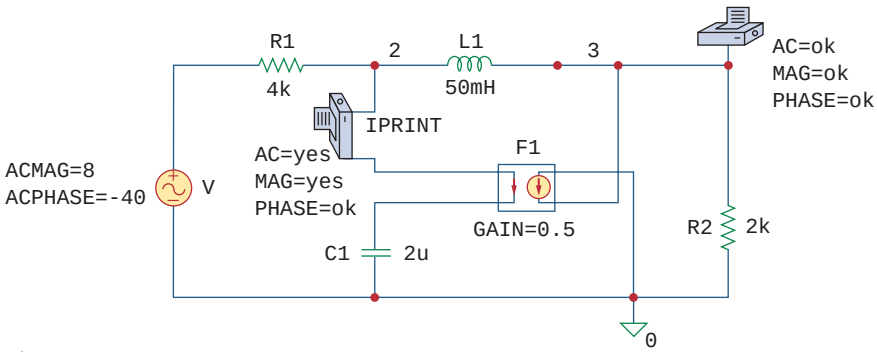


Figure 10.36
The schematic of the circuit in Fig. 10.35.

From this output file, we obtain

$$\mathbf{V}_o = 1.55\angle -95.18^\circ \text{ V}, \qquad \mathbf{I}_o = 3.264\angle -37.43^\circ \text{ mA}$$

which are the phasors for

$$v_o = 1.55\cos(1000t - 95.18^\circ) = 1.55\sin(1000t - 5.18^\circ) \text{ V}$$

and

$$i_o = 3.264\cos(1000t - 37.43^\circ) \text{ mA}$$

Practice Problem 10.13

Use *PSpice* to obtain v_o and i_o in the circuit of Fig. 10.37.

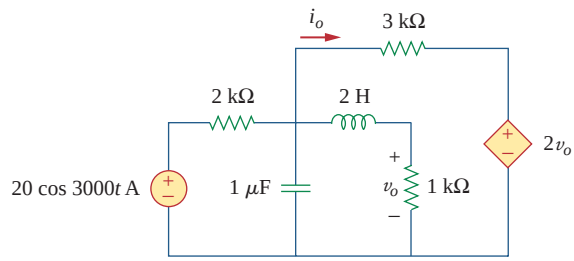


Figure 10.37
For Practice Prob. 10.13.

Answer: 536.4 cos(3000t − 154.6°) mV, 1.088 cos(3000t − 55.12°) mA.

Example 10.14

Find $\mathbf{V}_1$ and $\mathbf{V}_2$ in the circuit of Fig. 10.38.

Solution:

1. **Define.** In its present form, the problem is clearly stated. Again, we must emphasize that time spent here will save lots of time and expense later on! One thing that might have created a problem for you is that, if the reference was missing for this problem, you would then need to ask the individual assigning

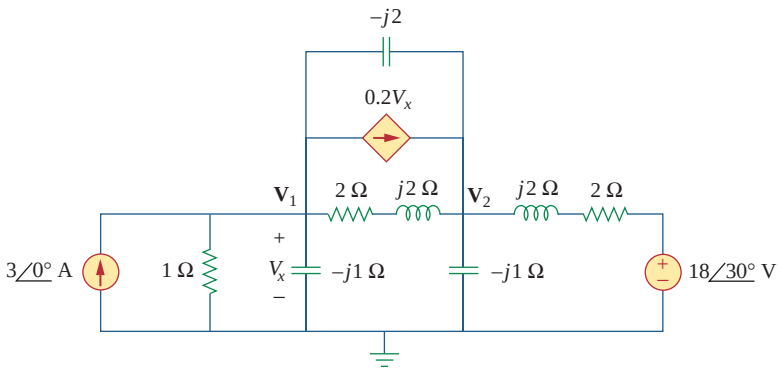


Figure 10.38
For Example 10.14.

the problem where it is to be located. If you could not do that, then you would need to assume where it should be and then clearly state what you did and why you did it.

2. **Present.** The given circuit is a frequency domain circuit and the unknown node voltages $\mathbf{V}_1$ and $\mathbf{V}_2$ are also frequency domain values. Clearly, we need a process to solve for these unknowns in the frequency domain.
3. **Alternative.** We have two direct alternative solution techniques that we can easily use. We can do a straightforward nodal analysis approach or use *PSpice*. Since this example is in a section dedicated to using *PSpice* to solve problems, we will use *PSpice* to find $\mathbf{V}_1$ and $\mathbf{V}_2$. We can then use nodal analysis to check the answer.
4. **Attempt.** The circuit in Fig. 10.35 is in the time domain, whereas the one in Fig. 10.38 is in the frequency domain. Since we are not given a particular frequency and *PSpice* requires one, we select any frequency consistent with the given impedances. For example, if we select $\omega = 1$ rad/s, the corresponding frequency is $f = \omega/2\pi = 0.15916$ Hz. We obtain the values of the capacitance ($C = 1/\omega X_C$) and inductances ($L = X_L/\omega$). Making these changes results in the schematic in Fig. 10.39. To ease wiring, we have

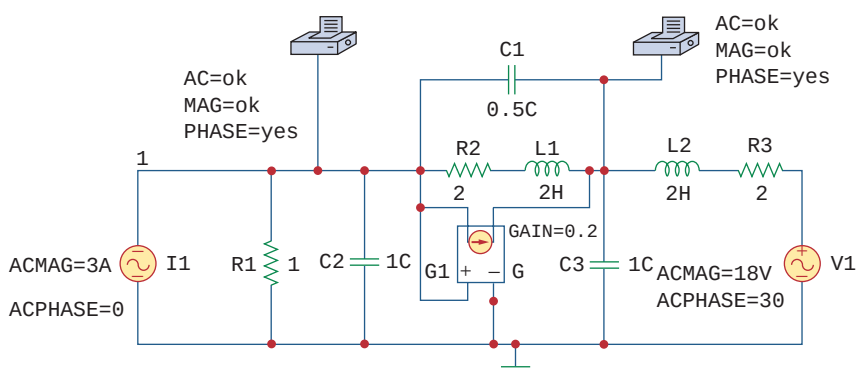


Figure 10.39
Schematic for circuit in the Fig. 10.38.

exchanged the positions of the voltage-controlled current source G1 and the $2 + j2\ \Omega$ impedance. Notice that the current of G1 flows from node 1 to node 3, while the controlling voltage is across the capacitor C2, as required in Fig. 10.38. The attributes of pseudo-components VPRINT1 are set as shown. As a single-frequency analysis, we select **Analysis/Setup/AC Sweep** and enter *Total Pts* = 1, *Start Freq* = 0.15916, and *Final Freq* = 0.15916. After saving the schematic, we select **Analysis/Simulate** to simulate the circuit. When this is done, the output file includes

```
FREQ          VM(1)         VP(1)
1.592E-01     2.708E+00     -5.673E+01

FREQ          VM(3)         VP(3)
1.592E-01     4.468E+00     -1.026E+02
```

from which we obtain,

$$\mathbf{V}_1 = \mathbf{2.708\angle -56.74^\circ\ V} \qquad \text{and} \qquad \mathbf{V}_2 = \mathbf{6.911\angle -80.72^\circ\ V}$$

5. **Evaluate.** One of the most important lessons to be learned is that when using programs such as *PSpice* you still need to validate the answer. There are many opportunities for making a mistake, including coming across an unknown "bug" in *PSpice* that yields incorrect results.

   So, how can we validate this solution? Obviously, we can rework the entire problem with nodal analysis, and perhaps using *MATLAB*, to see if we obtain the same results. There is another way we will use here: write the nodal equations and substitute the answers obtained in the *PSpice* solution, and see if the nodal equations are satisfied.

   The nodal equations for this circuit are given below. Note we have substituted $\mathbf{V}_1 = \mathbf{V}_x$ into the dependent source.

$$-3 + \frac{\mathbf{V}_1 - 0}{1} + \frac{\mathbf{V}_1 - 0}{-j1} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{2 + j2} + 0.2\mathbf{V}_1 + \frac{\mathbf{V}_1 - \mathbf{V}_2}{-j2} = 0$$

$$(1 + j + 0.25 - j0.25 + 0.2 + j0.5)\mathbf{V}_1 - (0.25 - j0.25 + j0.5)\mathbf{V}_2 = 3$$

$$(1.45 + j1.25)\mathbf{V}_1 - (0.25 + j0.25)\mathbf{V}_2 = 3$$

$$1.9144\angle 40.76^\circ\ \mathbf{V}_1 - 0.3536\angle 45^\circ\ \mathbf{V}_2 = 3$$

   Now, to check the answer, we substitute the *PSpice* answers into this.

$$\begin{aligned} 1.9144\angle 40.76^\circ \times 2.708\angle -56.74^\circ - 0.3536\angle 45^\circ \times 6.911\angle -80.72^\circ \\ = 5.184\angle -15.98^\circ - 2.444\angle -35.72^\circ \\ = 4.984 - j1.4272 - 1.9842 + j1.4269 \\ = 3 - j0.0003 \qquad \text{[Answer checks]} \end{aligned}$$

6. **Satisfactory?** Although we used only the equation from node 1 to check the answer, this is more than satisfactory to validate the answer from the *PSpice* solution. We can now present our work as a solution to the problem.

Practice Problem 10.14

Obtain $\mathbf{V}_x$ and $\mathbf{I}_x$ in the circuit depicted in Fig. 10.40.

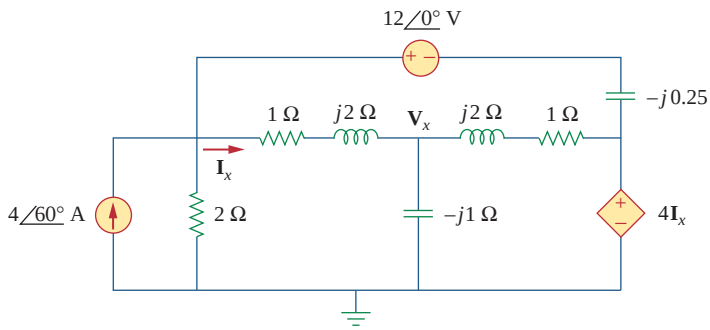


Figure 10.40
For Practice Prob. 10.14.

Answer: $9.842\angle 44.78°$ V, $2.584\angle 158°$ A.

10.9 Applications

The concepts learned in this chapter will be applied in later chapters to calculate electric power and determine frequency response. The concepts are also used in analyzing coupled circuits, three-phase circuits, ac transistor circuits, filters, oscillators, and other ac circuits. In this section, we apply the concepts to develop two practical ac circuits: the capacitance multiplier and the sine wave oscillators.

10.9.1 Capacitance Multiplier

The op amp circuit in Fig. 10.41 is known as a *capacitance multiplier*, for reasons that will become obvious. Such a circuit is used in integrated-circuit technology to produce a multiple of a small physical capacitance C when a large capacitance is needed. The circuit in Fig. 10.41 can be used to multiply capacitance values by a factor up to 1000. For example, a 10-pF capacitor can be made to behave like a 100-nF capacitor.

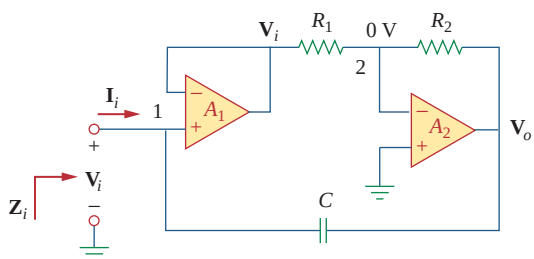


Figure 10.41
Capacitance multiplier.

In Fig. 10.41, the first op amp operates as a voltage follower, while the second one is an inverting amplifier. The voltage follower isolates the capacitance formed by the circuit from the loading imposed by the inverting amplifier. Since no current enters the input terminals of the op amp, the input current $\mathbf{I}_i$ flows through the feedback capacitor. Hence, at node 1,

$$\mathbf{I}_i = \frac{\mathbf{V}_i - \mathbf{V}_o}{1/j\omega C} = j\omega C(\mathbf{V}_i - \mathbf{V}_o) \tag{10.3}$$

Applying KCL at node 2 gives

$$\frac{\mathbf{V}_i - 0}{R_1} = \frac{0 - \mathbf{V}_o}{R_2}$$

or

$$\mathbf{V}_o = -\frac{R_2}{R_1}\mathbf{V}_i \tag{10.4}$$

Substituting Eq. (10.4) into (10.3) gives

$$\mathbf{I}_i = j\omega C\left(1 + \frac{R_2}{R_1}\right)\mathbf{V}_i$$

or

$$\frac{\mathbf{I}_i}{\mathbf{V}_i} = j\omega\left(1 + \frac{R_2}{R_1}\right)C \tag{10.5}$$

The input impedance is

$$\mathbf{Z}_i = \frac{\mathbf{V}_i}{\mathbf{I}_i} = \frac{1}{j\omega C_{\text{eq}}} \tag{10.6}$$

where

$$C_{\text{eq}} = \left(1 + \frac{R_2}{R_1}\right)C \tag{10.7}$$

Thus, by a proper selection of the values of R_1 and R_2, the op amp circuit in Fig. 10.41 can be made to produce an effective capacitance between the input terminal and ground, which is a multiple of the physical capacitance C. The size of the effective capacitance is practically limited by the inverted output voltage limitation. Thus, the larger the capacitance multiplication, the smaller is the allowable input voltage to prevent the op amps from reaching saturation.

A similar op amp circuit can be designed to simulate inductance. (See Prob. 10.89.) There is also an op amp circuit configuration to create a resistance multiplier.

Example 10.15

Calculate C_{eq} in Fig. 10.41 when $R_1 = 10\ \text{k}\Omega$, $R_2 = 1\ \text{M}\Omega$, and $C = 1\ \text{nF}$.

Solution:

From Eq. (10.7)

$$C_{\text{eq}} = \left(1 + \frac{R_2}{R_1}\right)C = \left(1 + \frac{1 \times 10^6}{10 \times 10^3}\right)1\ \text{nF} = 101\ \text{nF}$$

Practice Problem 10.15

Determine the equivalent capacitance of the op amp circuit in Fig. 10.41 if $R_1 = 10\ \text{k}\Omega$, $R_2 = 10\ \text{M}\Omega$, and $C = 10\ \text{nF}$.

Answer: 10 μF.

10.9.2 Oscillators

We know that dc is produced by batteries. But how do we produce ac? One way is using *oscillators*, which are circuits that convert dc to ac.

An **oscillator** is a circuit that produces an ac waveform as output when powered by a dc input.

The only external source an oscillator needs is the dc power supply. Ironically, the dc power supply is usually obtained by converting the ac supplied by the electric utility company to dc. Having gone through the trouble of conversion, one may wonder why we need to use the oscillator to convert the dc to ac again. The problem is that the ac supplied by the utility company operates at a preset frequency of 60 Hz in the United States (50 Hz in some other nations), whereas many applications such as electronic circuits, communication systems, and microwave devices require internally generated frequencies that range from 0 to 10 GHz or higher. Oscillators are used for generating these frequencies.

This corresponds to $\omega = 2\pi f = 377$ rad/s.

In order for sine wave oscillators to sustain oscillations, they must meet the *Barkhausen criteria:*

1. The overall gain of the oscillator must be unity or greater. Therefore, losses must be compensated for by an amplifying device.
2. The overall phase shift (from input to output and back to the input) must be zero.

Three common types of sine wave oscillators are phase-shift, twin *T*, and Wien-bridge oscillators. Here we consider only the Wien-bridge oscillator.

The *Wien-bridge oscillator* is widely used for generating sinusoids in the frequency range below 1 MHz. It is an *RC* op amp circuit with only a few components, easily tunable and easy to design. As shown in Fig. 10.42, the oscillator essentially consists of a noninverting amplifier with two feedback paths: the positive feedback path to the noninverting input creates oscillations, while the negative feedback path to the inverting input controls the gain. If we define the impedances of the *RC* series and parallel combinations as $\mathbf{Z}_s$ and $\mathbf{Z}_p$, then

$$\mathbf{Z}_s = R_1 + \frac{1}{j\omega C_1} = R_1 - \frac{j}{\omega C_1} \tag{10.8}$$

$$\mathbf{Z}_p = R_2 \parallel \frac{1}{j\omega C_2} = \frac{R_2}{1 + j\omega R_2 C_2} \tag{10.9}$$

The feedback ratio is

$$\frac{\mathbf{V}_2}{\mathbf{V}_o} = \frac{\mathbf{Z}_p}{\mathbf{Z}_s + \mathbf{Z}_p} \tag{10.10}$$

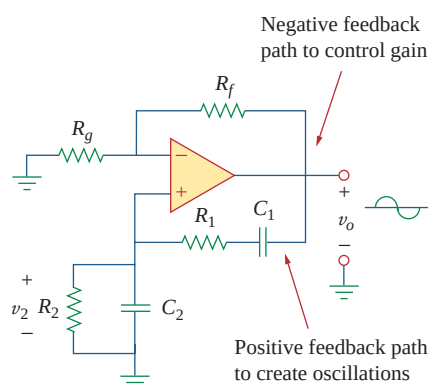


Figure 10.42
Wien-bridge oscillator.

Substituting Eqs. (10.8) and (10.9) into Eq. (10.10) gives

$$\frac{\mathbf{V}_2}{\mathbf{V}_o} = \frac{R_2}{R_2 + \left(R_1 - \dfrac{j}{\omega C_1}\right)(1 + j\omega R_2 C_2)}$$
$$= \frac{\omega R_2 C_1}{\omega(R_2 C_1 + R_1 C_1 + R_2 C_2) + j(\omega^2 R_1 C_1 R_2 C_2 - 1)} \tag{10.11}$$

To satisfy the second Barkhausen criterion, $\mathbf{V}_2$ must be in phase with $\mathbf{V}_o$, which implies that the ratio in Eq. (10.11) must be purely real. Hence, the imaginary part must be zero. Setting the imaginary part equal to zero gives the oscillation frequency ω_o as

$$\omega_o^2 R_1 C_1 R_2 C_2 - 1 = 0$$

or

$$\omega_o = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}} \tag{10.12}$$

In most practical applications, $R_1 = R_2 = R$ and $C_1 = C_2 = C$, so that

$$\omega_o = \frac{1}{RC} = 2\pi f_o \tag{10.13}$$

or

$$\boxed{f_o = \frac{1}{2\pi RC}} \tag{10.14}$$

Substituting Eq. (10.13) and $R_1 = R_2 = R$, $C_1 = C_2 = C$ into Eq. (10.11) yields

$$\frac{\mathbf{V}_2}{\mathbf{V}_o} = \frac{1}{3} \tag{10.15}$$

Thus, in order to satisfy the first Barkhausen criterion, the op amp must compensate by providing a gain of 3 or greater so that the overall gain is at least 1 or unity. We recall that for a noninverting amplifier,

$$\frac{\mathbf{V}_o}{\mathbf{V}_2} = 1 + \frac{R_f}{R_g} = 3 \tag{10.16}$$

or

$$R_f = 2R_g \tag{10.17}$$

Due to the inherent delay caused by the op amp, Wien-bridge oscillators are limited to operating in the frequency range of 1 MHz or less.

Example 10.16

Design a Wien-bridge circuit to oscillate at 100 kHz.

Solution:

Using Eq. (10.14), we obtain the time constant of the circuit as

$$RC = \frac{1}{2\pi f_o} = \frac{1}{2\pi \times 100 \times 10^3} = 1.59 \times 10^{-6} \tag{10.16.1}$$

If we select $R = 10\,\text{k}\Omega$, then we can select $C = 159\,\text{pF}$ to satisfy Eq. (10.16.1). Since the gain must be 3, $R_f/R_g = 2$. We could select $R_f = 20\,\text{k}\Omega$ while $R_g = 10\,\text{k}\Omega$.

Practice Problem 10.16

In the Wien-bridge oscillator circuit in Fig. 10.42, let $R_1 = R_2 = 2.5\ \text{k}\Omega$, $C_1 = C_2 = 1\ \text{nF}$. Determine the frequency f_o of the oscillator.

Answer: 63.66 kHz.

10.10 Summary

1. We apply nodal and mesh analysis to ac circuits by applying KCL and KVL to the phasor form of the circuits.
2. In solving for the steady-state response of a circuit that has independent sources with different frequencies, each independent source *must* be considered separately. The most natural approach to analyzing such circuits is to apply the superposition theorem. A separate phasor circuit for each frequency *must* be solved independently, and the corresponding response should be obtained in the time domain. The overall response is the sum of the time domain responses of all the individual phasor circuits.
3. The concept of source transformation is also applicable in the frequency domain.
4. The Thevenin equivalent of an ac circuit consists of a voltage source $\mathbf{V}_{\text{Th}}$ in series with the Thevenin impedance $\mathbf{Z}_{\text{Th}}$.
5. The Norton equivalent of an ac circuit consists of a current source $\mathbf{I}_N$ in parallel with the Norton impedance $\mathbf{Z}_N\ (=\mathbf{Z}_{\text{Th}})$.
6. *PSpice* is a simple and powerful tool for solving ac circuit problems. It relieves us of the tedious task of working with the complex numbers involved in steady-state analysis.
7. The capacitance multiplier and the ac oscillator provide two typical applications for the concepts presented in this chapter. A capacitance multiplier is an op amp circuit used in producing a multiple of a physical capacitance. An oscillator is a device that uses a dc input to generate an ac output.

Review Questions

10.1 The voltage $\mathbf{V}_o$ across the capacitor in Fig. 10.43 is:

(a) $5\angle 0°$ V (b) $7.071\angle 45°$ V
(c) $7.071\angle -45°$ V (d) $5\angle -45°$ V

10.2 The value of the current $\mathbf{I}_o$ in the circuit of Fig. 10.44 is:

(a) $4\angle 0°$ A (b) $2.4\angle -90°$ A
(c) $0.6\angle 0°$ A (d) -1 A

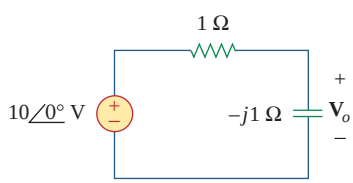


Figure 10.43
For Review Question 10.1.

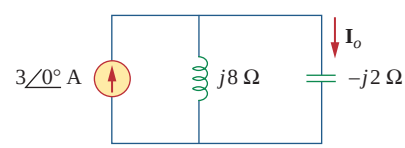


Figure 10.44
For Review Question 10.2.

10.3 Using nodal analysis, the value of $\mathbf{V}_o$ in the circuit of Fig. 10.45 is:

(a) -24 V (b) -8 V
(c) 8 V (d) 24 V

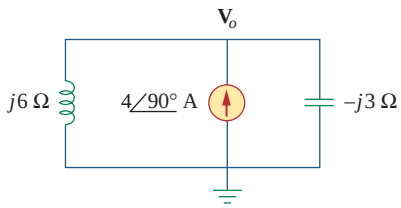


Figure 10.45
For Review Question 10.3.

10.4 In the circuit of Fig. 10.46, current $i(t)$ is:

(a) 10 cos t A (b) 10 sin t A (c) 5 cos t A
(d) 5 sin t A (e) 4.472 cos($t - 63.43°$) A

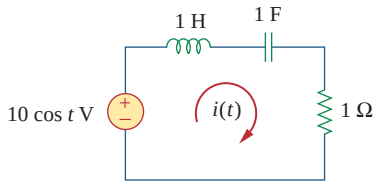


Figure 10.46
For Review Question 10.4.

10.5 Refer to the circuit in Fig. 10.47 and observe that the two sources do not have the same frequency. The current $i_x(t)$ can be obtained by:

(a) source transformation
(b) the superposition theorem
(c) *PSpice*

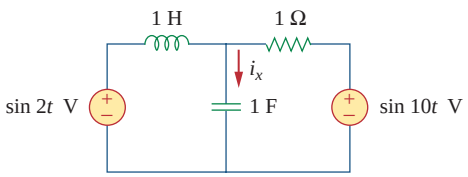


Figure 10.47
For Review Question 10.5.

10.6 For the circuit in Fig. 10.48, the Thevenin impedance at terminals *a-b* is:

(a) 1 Ω (b) $0.5 - j0.5\ \Omega$
(c) $0.5 + j0.5\ \Omega$ (d) $1 + j2\ \Omega$
(e) $1 - j2\ \Omega$

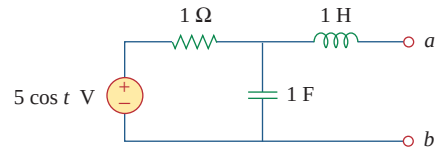


Figure 10.48
For Review Questions 10.6 and 10.7.

10.7 In the circuit of Fig. 10.48, the Thevenin voltage at terminals *a-b* is:

(a) $3.535\angle -45°$ V (b) $3.535\angle 45°$ V
(c) $7.071\angle -45°$ V (d) $7.071\angle 45°$ V

10.8 Refer to the circuit in Fig. 10.49. The Norton equivalent impedance at terminals *a-b* is:

(a) $-j4\ \Omega$ (b) $-j2\ \Omega$
(c) $j2\ \Omega$ (d) $j4\ \Omega$

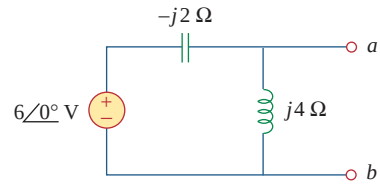


Figure 10.49
For Review Questions 10.8 and 10.9.

10.9 The Norton current at terminals *a-b* in the circuit of Fig. 10.49 is:

(a) $1\angle 0°$ A (b) $1.5\angle -90°$ A
(c) $1.5\angle 90°$ A (d) $3\angle 90°$ A

10.10 *PSpice* can handle a circuit with two independent sources of different frequencies.

(a) True (b) False

Answers: 10.1c, 10.2a, 10.3d, 10.4a, 10.5b, 10.6c, 10.7a, 10.8a, 10.9d, 10.10b.

Problems

Section 10.2 Nodal Analysis

10.1 Determine i in the circuit of Fig. 10.50.

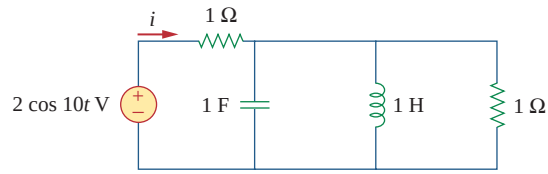


Figure 10.50
For Prob. 10.1.

10.2 Using Fig. 10.51, design a problem to help other students better understand nodal analysis.

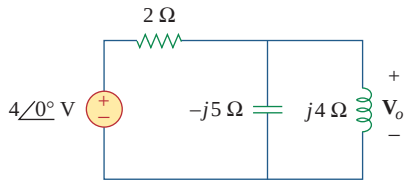


Figure 10.51
For Prob. 10.2.

10.3 Determine v_o in the circuit of Fig. 10.52.

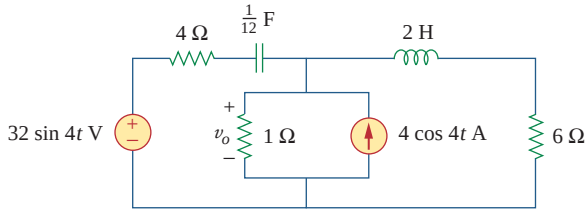


Figure 10.52
For Prob. 10.3.

10.4 Determine i_1 in the circuit of Fig. 10.53.

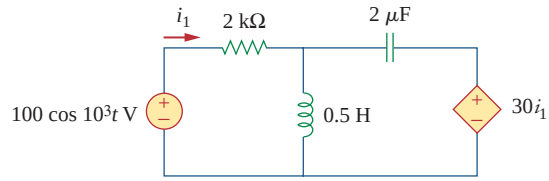


Figure 10.53
For Prob. 10.4.

10.5 Find i_o in the circuit of Fig. 10.54.

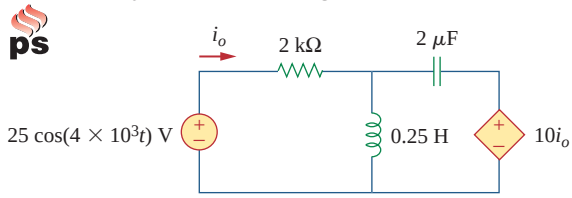


Figure 10.54
For Prob. 10.5.

10.6 Determine $\mathbf{V}_x$ in Fig. 10.55.

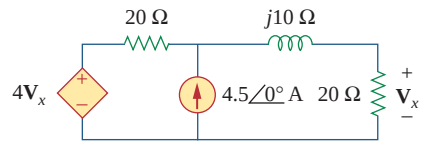


Figure 10.55
For Prob. 10.6.

10.7 Use nodal analysis to find $\mathbf{V}$ in the circuit of Fig. 10.56.

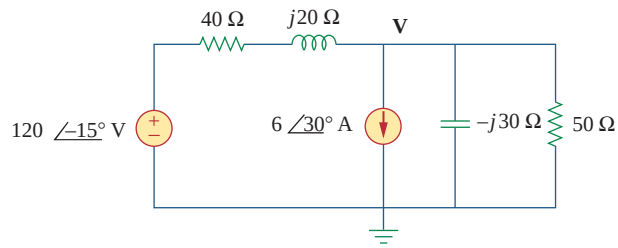


Figure 10.56
For Prob. 10.7.

10.8 Use nodal analysis to find current i_o in the circuit of Fig. 10.57. Let $i_s = 6\cos(200t + 15°)$ A.

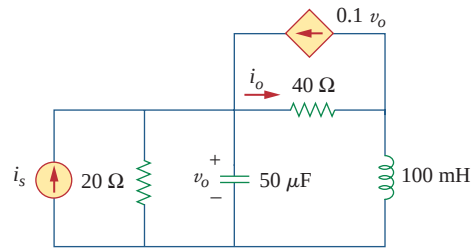


Figure 10.57
For Prob. 10.8.

10.9 Use nodal analysis to find v_o in the circuit of Fig. 10.58.

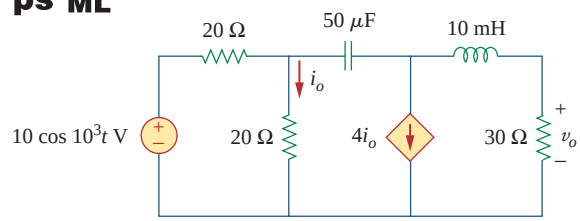


Figure 10.58
For Prob. 10.9.

10.10 Use nodal analysis to find v_o in the circuit of Fig. 10.59. Let $\omega = 2$ krad/s.

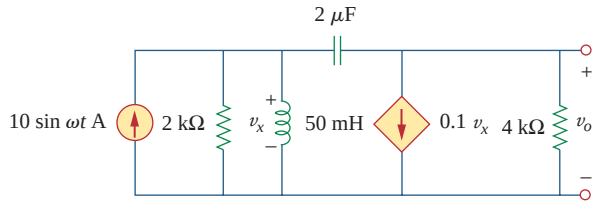


Figure 10.59
For Prob. 10.10.

10.11 Apply nodal analysis to the circuit in Fig. 10.60 and determine $\mathbf{I}_o$.

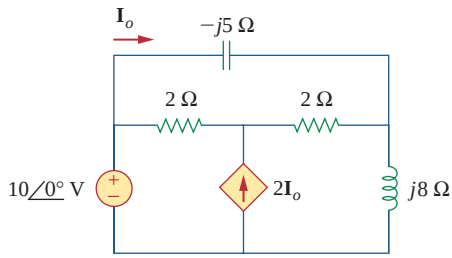


Figure 10.60
For Prob. 10.11.

10.12 Using Fig. 10.61, design a problem to help other students better understand nodal analysis.

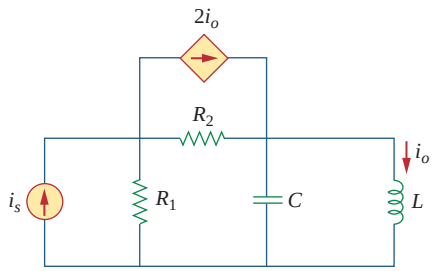


Figure 10.61
For Prob. 10.12.

10.13 Determine $\mathbf{V}_x$ in the circuit of Fig. 10.62 using any method of your choice.

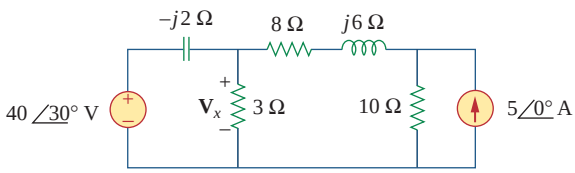


Figure 10.62
For Prob. 10.13.

10.14 Calculate the voltage at nodes 1 and 2 in the circuit of Fig. 10.63 using nodal analysis.

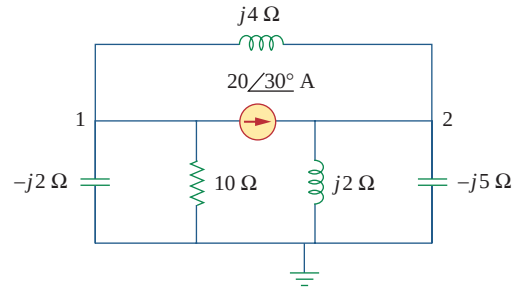


Figure 10.63
For Prob. 10.14.

10.15 Solve for the current $\mathbf{I}$ in the circuit of Fig. 10.64 using nodal analysis.

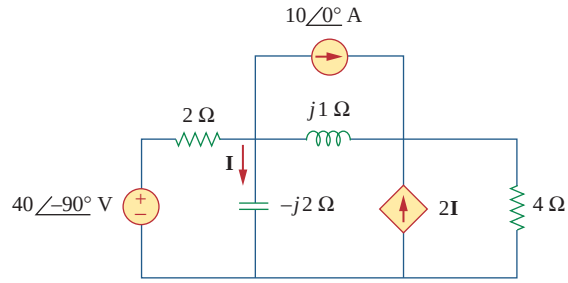


Figure 10.64
For Prob. 10.15.

10.16 Use nodal analysis to find $\mathbf{V}_x$ in the circuit shown in Fig. 10.65.

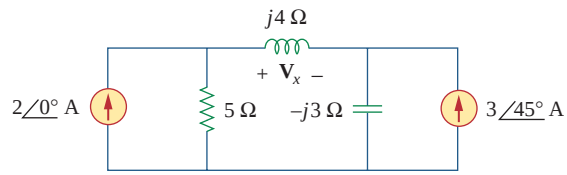


Figure 10.65
For Prob. 10.16.

10.17 By nodal analysis, obtain current $\mathbf{I}_o$ in the circuit of Fig. 10.66.

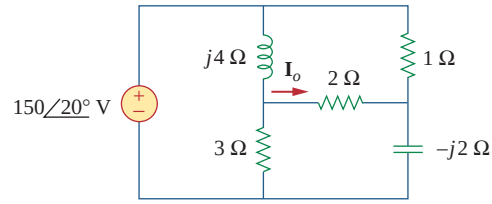


Figure 10.66
For Prob. 10.17.

10.18 Use nodal analysis to obtain $\mathbf{V}_o$ in the circuit of Fig. 10.67 below.

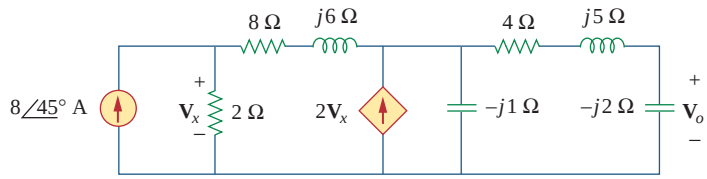


Figure 10.67
For Prob. 10.18.

10.19 Obtain $\mathbf{V}_o$ in Fig. 10.68 using nodal analysis.

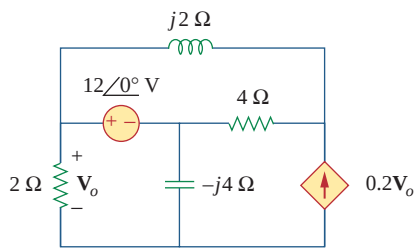


Figure 10.68
For Prob. 10.19.

10.20 Refer to Fig. 10.69. If $v_s(t) = V_m \sin \omega t$ and $v_o(t) = A \sin(\omega t + \phi)$, derive the expressions for A and ϕ.

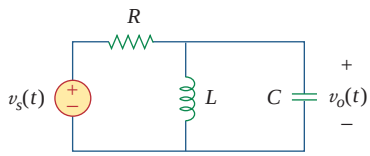


Figure 10.69
For Prob. 10.20.

10.21 For each of the circuits in Fig. 10.70, find $\mathbf{V}_o/\mathbf{V}_i$ for $\omega = 0$, $\omega \to \infty$, and $\omega^2 = 1/LC$.

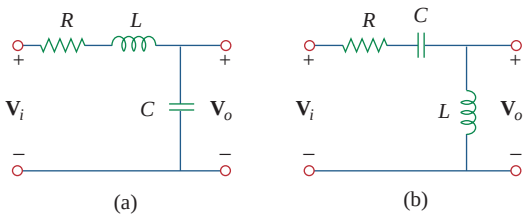


Figure 10.70
For Prob. 10.21.

10.22 For the circuit in Fig. 10.71, determine $\mathbf{V}_o/\mathbf{V}_s$.

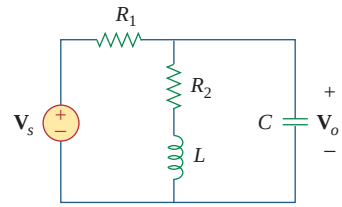


Figure 10.71
For Prob. 10.22.

10.23 Using nodal analysis obtain $\mathbf{V}$ in the circuit of Fig. 10.72.

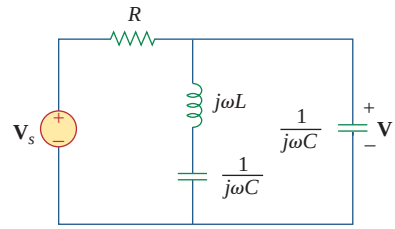


Figure 10.72
For Prob. 10.23.

Section 10.3 Mesh Analysis

10.24 Design a problem to help other students better understand mesh analysis.

10.25 Solve for i_o in Fig. 10.73 using mesh analysis.

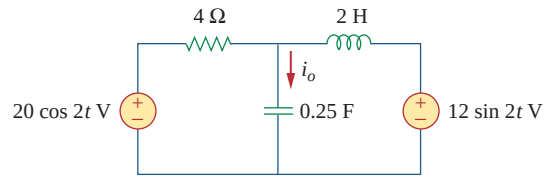


Figure 10.73
For Prob. 10.25.

10.26 Use mesh analysis to find current i_o in the circuit of Fig. 10.74.

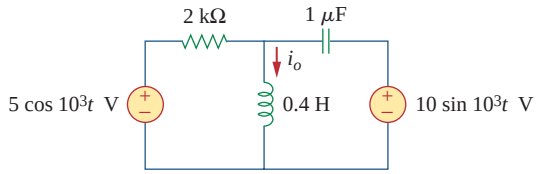


Figure 10.74
For Prob. 10.26.

10.27 Using mesh analysis, find $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit of Fig. 10.75.

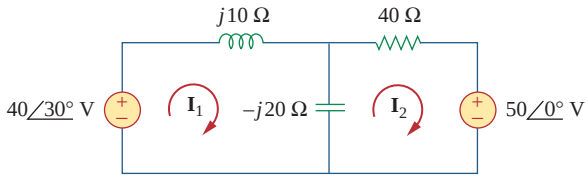


Figure 10.75
For Prob. 10.27.

10.28 In the circuit of Fig. 10.76, determine the mesh currents i_1 and i_2. Let $v_1 = 10 \cos 4t$ **V** and $v_2 = 20 \cos(4t - 30°)$ V.

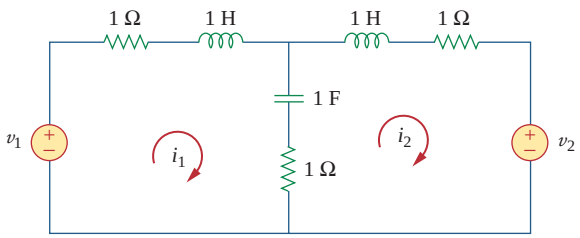


Figure 10.76
For Prob. 10.28.

10.29 Using Fig. 10.77, design a problem to help other students better understand mesh analysis.

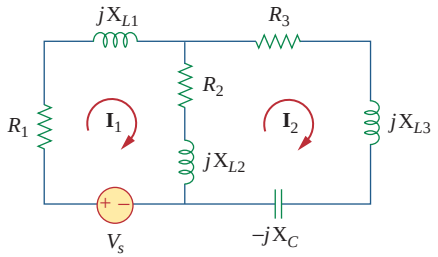


Figure 10.77
For Prob. 10.29.

10.30 Use mesh analysis to find v_o in the circuit of Fig. 10.78. Let $v_{s1} = 240 \cos(100t + 90°)$ V, $v_{s2} = 160 \cos 100t$ V.

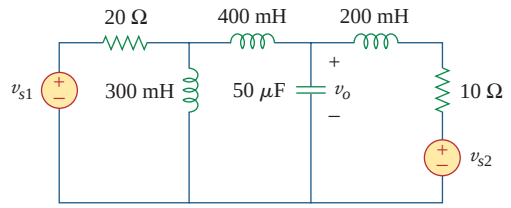


Figure 10.78
For Prob. 10.30.

10.31 Use mesh analysis to determine current $\mathbf{I}_o$ in the circuit of Fig. 10.79 below.

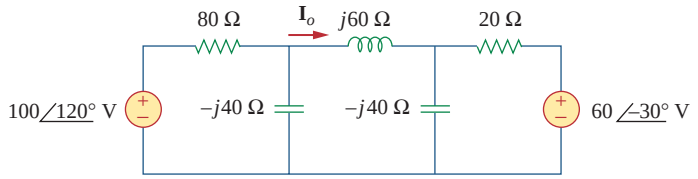


Figure 10.79
For Prob. 10.31.

10.32 Determine $\mathbf{V}_o$ and $\mathbf{I}_o$ in the circuit of Fig. 10.80 using mesh analysis.

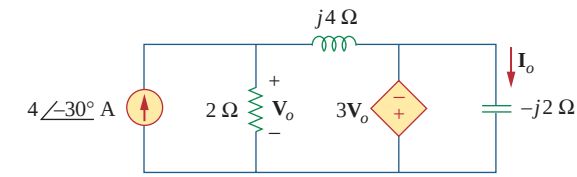


Figure 10.80
For Prob. 10.32.

10.33 Compute $\mathbf{I}$ in Prob. 10.15 using mesh analysis.

10.34 Use mesh analysis to find $\mathbf{I}_o$ in Fig. 10.28 (for Example 10.10).

10.35 Calculate $\mathbf{I}_o$ in Fig. 10.30 (for Practice Prob. 10.10) using mesh analysis.

10.36 Compute $\mathbf{V}_o$ in the circuit of Fig. 10.81 using mesh analysis.

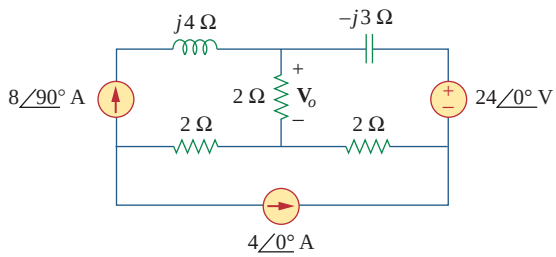


Figure 10.81
For Prob. 10.36.

10.37 Use mesh analysis to find currents $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$ in the circuit of Fig. 10.82.

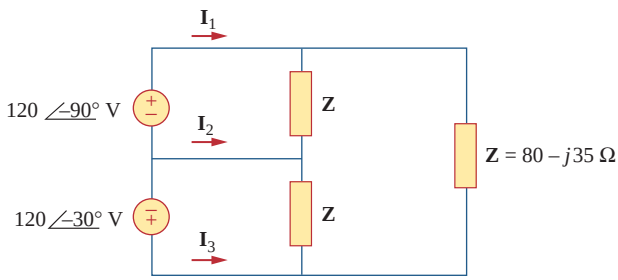


Figure 10.82
For Prob. 10.37.

10.38 Using mesh analysis, obtain $\mathbf{I}_o$ in the circuit shown in Fig. 10.83.

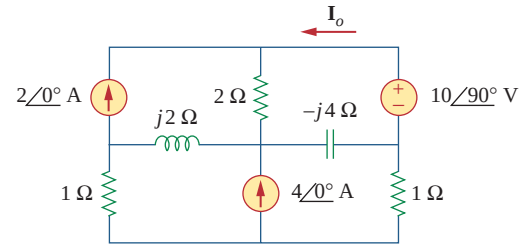


Figure 10.83
For Prob. 10.38.

10.39 Find $\mathbf{I}_1$, $\mathbf{I}_2$, $\mathbf{I}_3$, and $\mathbf{I}_x$ in the circuit of Fig. 10.84.

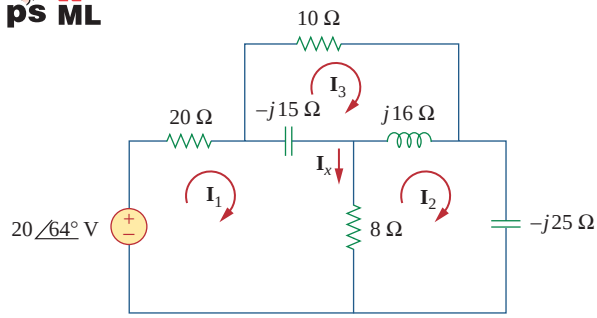


Figure 10.84
For Prob. 10.39.

Section 10.4 Superposition Theorem

10.40 Find i_o in the circuit shown in Fig. 10.85 using superposition.

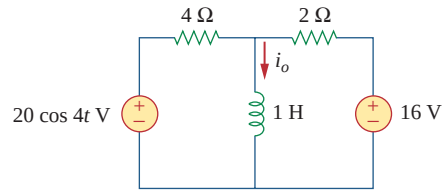


Figure 10.85
For Prob. 10.40.

10.41 Find v_o for the circuit in Fig. 10.86, assuming that $v_s = 3\cos 2t + 8\sin 4t$ V.

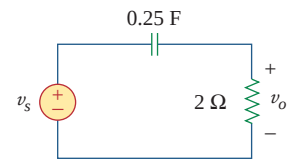


Figure 10.86
For Prob. 10.41.

10.42 Using Fig. 10.87, design a problem to help other students better understand the superposition theorem.

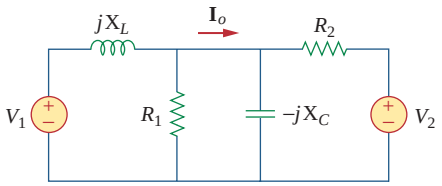


Figure 10.87
For Prob. 10.42.

10.43 Using the superposition principle, find i_x in the circuit of Fig. 10.88.

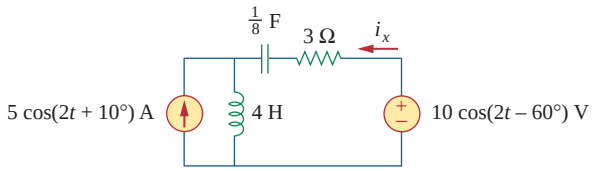


Figure 10.88
For Prob. 10.43.

10.44 Use the superposition principle to obtain v_x in the circuit of Fig. 10.89. Let $v_s = 25 \sin 2t$ V and $i_s = 6 \cos(6t + 10°)$ A.

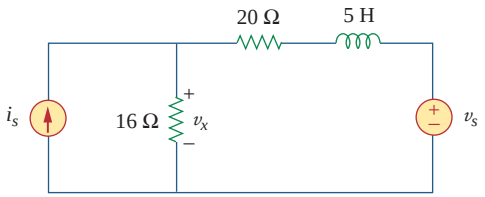


Figure 10.89
For Prob. 10.44.

10.45 Use superposition to find $i(t)$ in the circuit of Fig. 10.90.

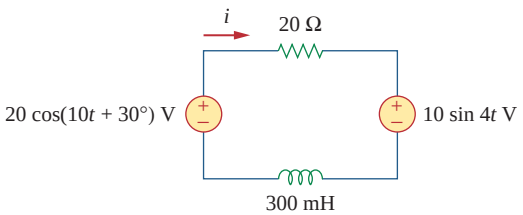


Figure 10.90
For Prob. 10.45.

10.46 Solve for $v_o(t)$ in the circuit of Fig. 10.91 using the superposition principle.

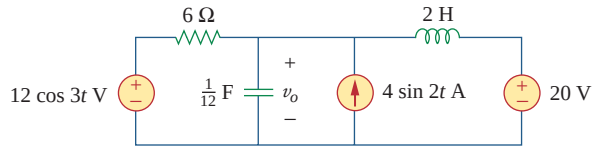


Figure 10.91
For Prob. 10.46.

10.47 Determine i_o in the circuit of Fig. 10.92, using the superposition principle.

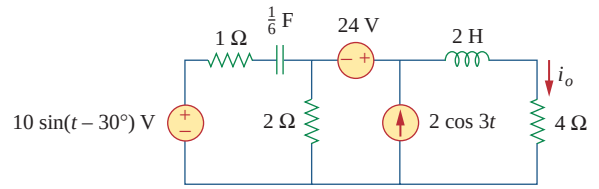


Figure 10.92
For Prob. 10.47.

10.48 Find i_o in the circuit of Fig. 10.93 using superposition.

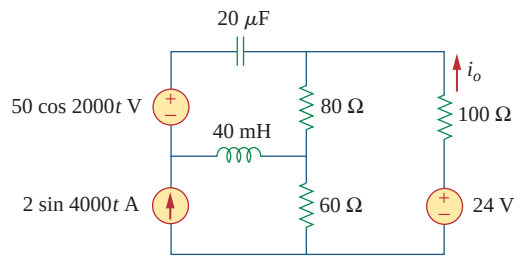


Figure 10.93
For Prob. 10.48.

Section 10.5 Source Transformation

10.49 Using source transformation, find i in the circuit of Fig. 10.94.

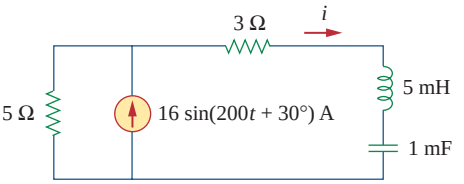


Figure 10.94
For Prob. 10.49.

10.50 Using Fig. 10.95, design a problem to help other students understand source transformation.

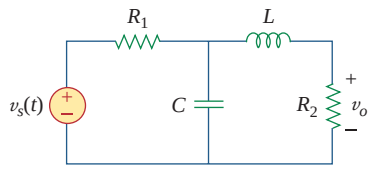


Figure 10.95
For Prob. 10.50.

10.51 Use source transformation to find $\mathbf{I}_o$ in the circuit of Prob. 10.42.

10.52 Use the method of source transformation to find $\mathbf{I}_x$ in the circuit of Fig. 10.96.

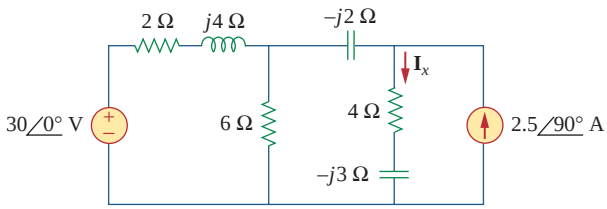


Figure 10.96
For Prob. 10.52.

10.53 Use the concept of source transformation to find $\mathbf{V}_o$ in the circuit of Fig. 10.97.

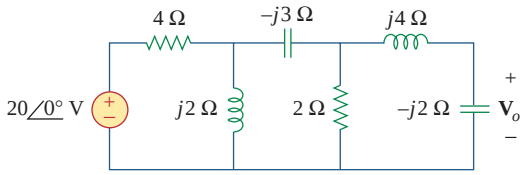
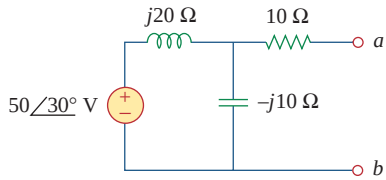


Figure 10.97
For Prob. 10.53.

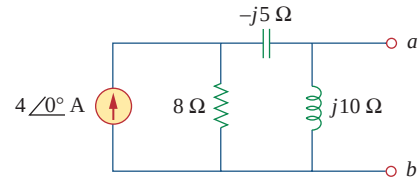
10.54 Rework Prob. 10.7 using source transformation.

Section 10.6 Thevenin and Norton Equivalent Circuits

10.55 Find the Thevenin and Norton equivalent circuits at terminals *a-b* for each of the circuits in Fig. 10.98.



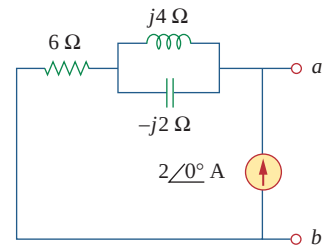
(a)



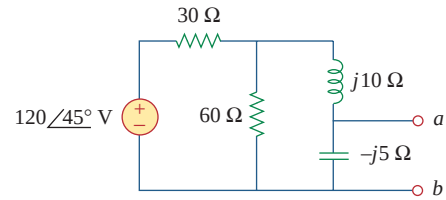
(b)

Figure 10.98
For Prob. 10.55.

10.56 For each of the circuits in Fig. 10.99, obtain Thevenin and Norton equivalent circuits at terminals *a-b*.



(a)



(b)

Figure 10.99
For Prob. 10.56.

10.57 Using Fig. 10.100, design a problem to help other students better understand Thevenin and Norton equivalent circuits.

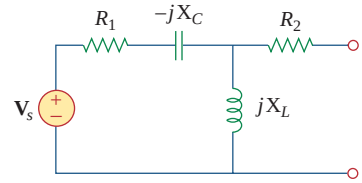


Figure 10.100
For Prob. 10.57.

10.58 For the circuit depicted in Fig. 10.101, find the Thevenin equivalent circuit at terminals *a-b*.

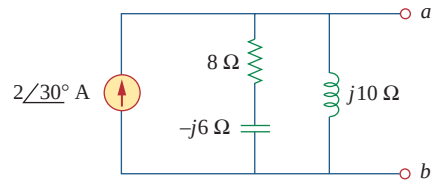


Figure 10.101
For Prob. 10.58.

10.59 Calculate the output impedance of the circuit shown in Fig. 10.102.

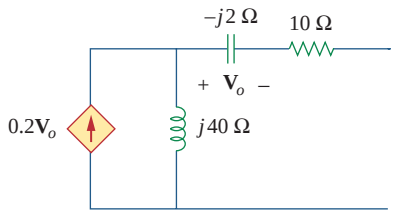


Figure 10.102
For Prob. 10.59.

10.60 Find the Thevenin equivalent of the circuit in Fig. 10.103 as seen from:

(a) terminals *a-b* (b) terminals *c-d*

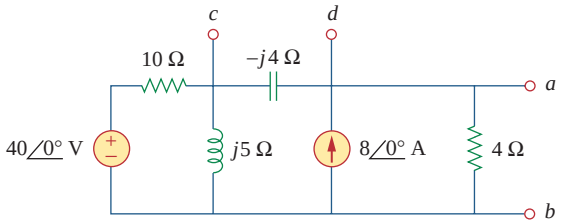


Figure 10.103
For Prob. 10.60.

10.61 Find the Thevenin equivalent at terminals *a-b* of the circuit in Fig. 10.104.

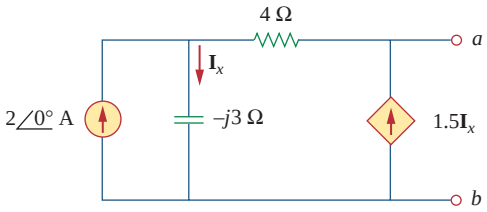


Figure 10.104
For Prob. 10.61.

10.62 Using Thevenin's theorem, find v_o in the circuit of Fig. 10.105.

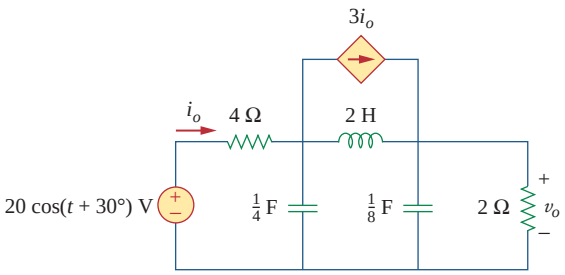


Figure 10.105
For Prob. 10.62.

10.63 Obtain the Norton equivalent of the circuit depicted in Fig. 10.106 at terminals *a-b*.

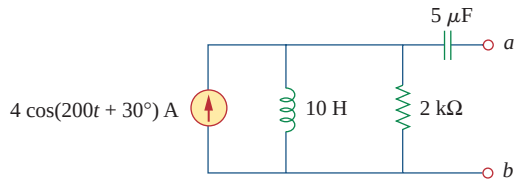


Figure 10.106
For Prob. 10.63.

10.64 For the circuit shown in Fig. 10.107, find the Norton equivalent circuit at terminals *a-b*.

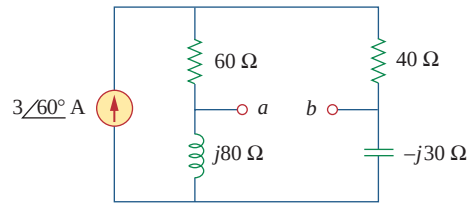


Figure 10.107
For Prob. 10.64.

10.65 Using Fig. 10.108, design a problem to help other students better understand Norton's theorem.

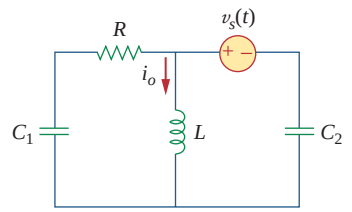


Figure 10.108
For Prob. 10.65.

10.66 At terminals *a-b*, obtain Thevenin and Norton equivalent circuits for the network depicted in Fig. 10.109. Take $\omega = 10$ rad/s.

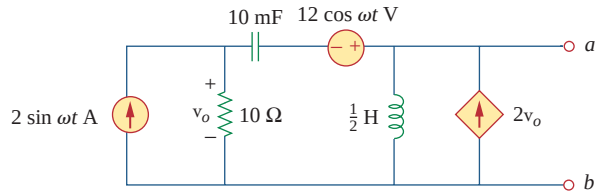


Figure 10.109
For Prob. 10.66.

10.67 Find the Thevenin and Norton equivalent circuits at terminals *a-b* in the circuit of Fig. 10.110.

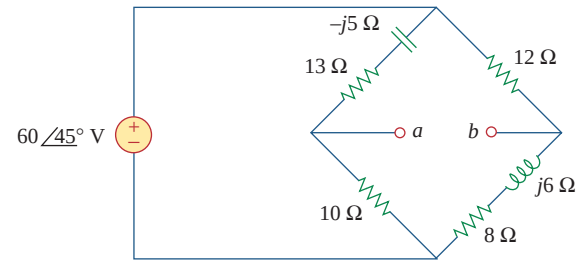


Figure 10.110
For Prob. 10.67.

10.68 Find the Thevenin equivalent at terminals *a-b* in the circuit of Fig. 10.111.

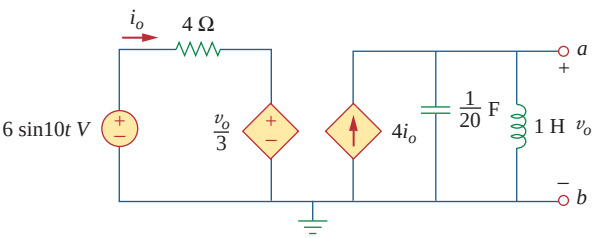


Figure 10.111
For Prob. 10.68.

Section 10.7 Op Amp AC Circuits

10.69 For the differentiator shown in Fig. 10.112, obtain $\mathbf{V}_o/\mathbf{V}_s$. Find $v_o(t)$ when $v_s(t) = \mathbf{V}_m \sin \omega t$ and $\omega = 1/RC$.

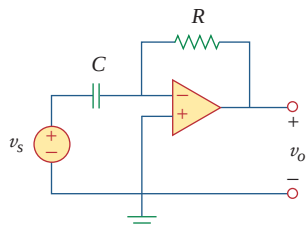


Figure 10.112
For Prob. 10.69.

10.70 Using Fig. 10.113, design a problem to help other students better understand op amps in AC circuits.

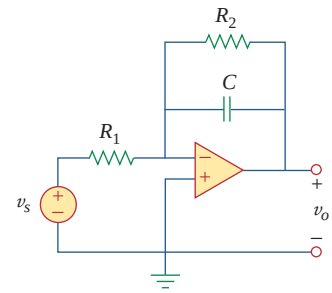


Figure 10.113
For Prob. 10.70.

10.71 Find v_o in the op amp circuit of Fig. 10.114.

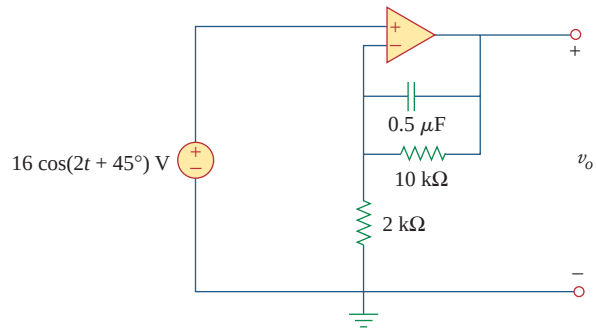


Figure 10.114
For Prob. 10.71.

10.72 Compute $i_o(t)$ in the op amp circuit in Fig. 10.115 if $v_s = 10 \cos(10^4 t + 30°)$ V.

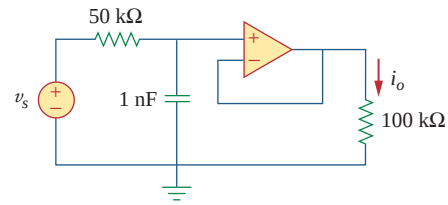


Figure 10.115
For Prob. 10.72.

10.73 If the input impedance is defined as $\mathbf{Z}_{in} = \mathbf{V}_s/\mathbf{I}_s$, find the input impedance of the op amp circuit in Fig. 10.116 when $R_1 = 10$ kΩ, $R_2 = 20$ kΩ, $C_1 = 10$ nF, $C_2 = 20$ nF, and $\omega = 5000$ rad/s.

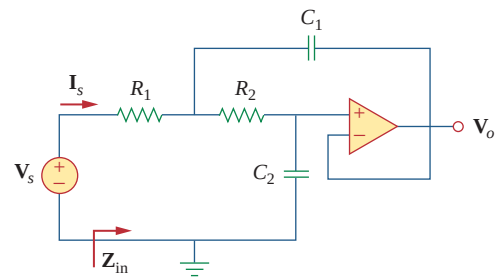


Figure 10.116
For Prob. 10.73.

10.74 Evaluate the voltage gain $\mathbf{A}_v = \mathbf{V}_o/\mathbf{V}_s$ in the op amp circuit of Fig. 10.117. Find $\mathbf{A}_v$ at $\omega = 0$, $\omega \to \infty$, $\omega = 1/R_1C_1$, and $\omega = 1/R_2C_2$.

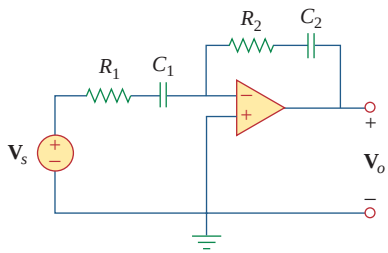


Figure 10.117
For Prob. 10.74.

10.75 In the op amp circuit of Fig. 10.118, find the closed-loop gain and phase shift of the output voltage with respect to the input voltage if $C_1 = C_2 = 1$ nF, $R_1 = R_2 = 100$ kΩ, $R_3 = 20$ kΩ, $R_4 = 40$ kΩ, and $\omega = 2000$ rad/s.

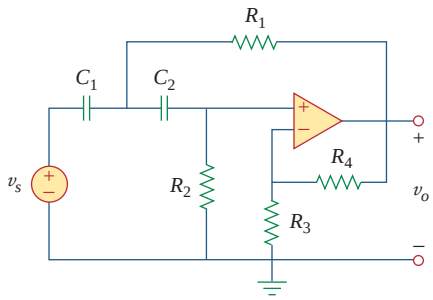


Figure 10.118
For Prob. 10.75.

10.76 Determine $\mathbf{V}_o$ and $\mathbf{I}_o$ in the op amp circuit of Fig. 10.119.

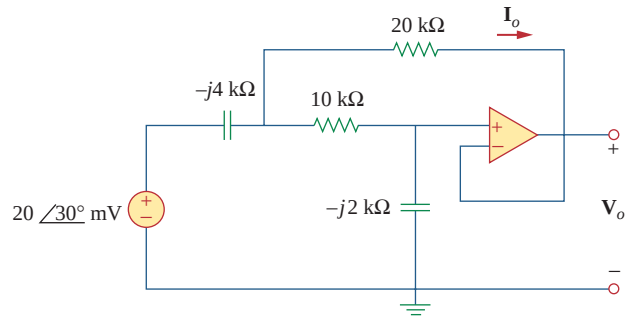


Figure 10.119
For Prob. 10.76.

10.77 Compute the closed-loop gain $\mathbf{V}_o/\mathbf{V}_s$ for the op amp circuit of Fig. 10.120.

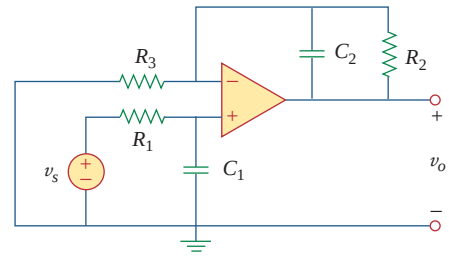


Figure 10.120
For Prob. 10.77.

10.78 Determine $v_o(t)$ in the op amp circuit in Fig. 10.121 below.

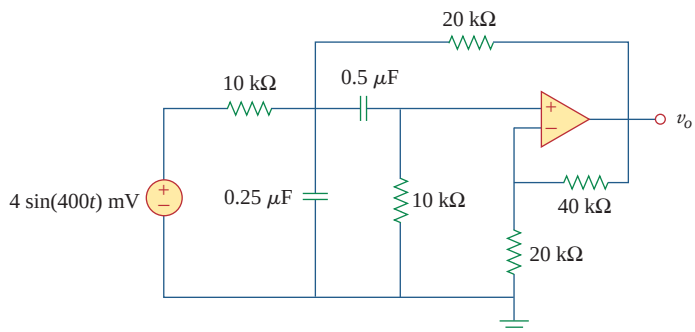


Figure 10.121
For Prob. 10.78.

10.79 For the op amp circuit in Fig. 10.122, obtain $v_o(t)$.

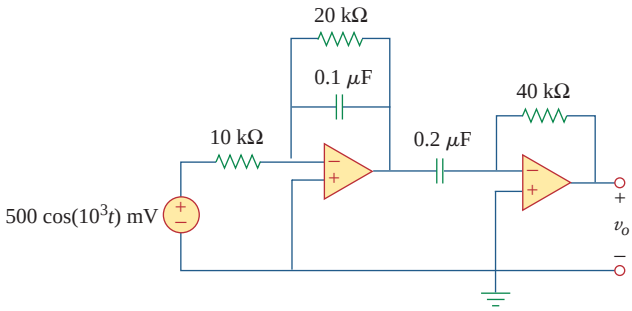


Figure 10.122
For Prob. 10.79.

10.80 Obtain $v_o(t)$ for the op amp circuit in Fig. 10.123 if $v_s = 4\cos(1000t - 60°)$ V.

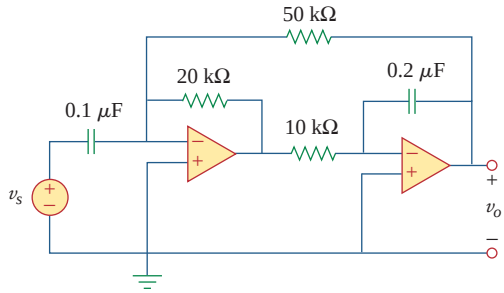


Figure 10.123
For Prob. 10.80.

Section 10.8 AC Analysis Using *PSpice*

10.81 Use *PSpice* to determine $\mathbf{V}_o$ in the circuit of Fig. 10.124. Assume $\omega = 1$ rad/s.

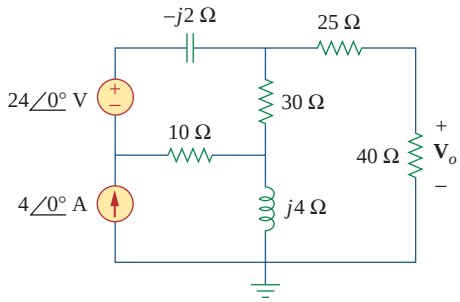


Figure 10.124
For Prob. 10.81.

10.82 Solve Prob. 10.19 using *PSpice*.

10.83 Use *PSpice* to find $v_o(t)$ in the circuit of Fig. 10.125. Let $i_s = 2\cos(10^3 t)$ A.

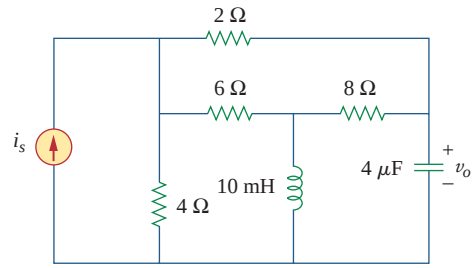


Figure 10.125
For Prob. 10.83.

10.84 Obtain $\mathbf{V}_o$ in the circuit of Fig. 10.126 using *PSpice*.

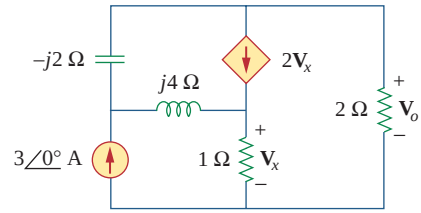


Figure 10.126
For Prob. 10.84.

10.85 Using Fig. 10.127, design a problem to help other students better understand performing AC analysis with *PSpice*.

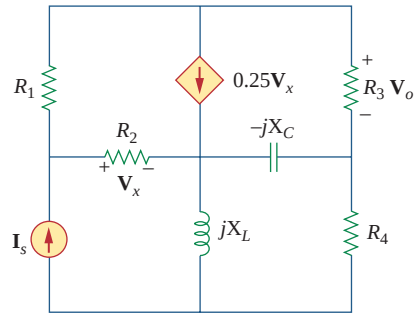


Figure 10.127
For Prob. 10.85.

10.86 Use *PSpice* to find $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ in the network of Fig. 10.128.

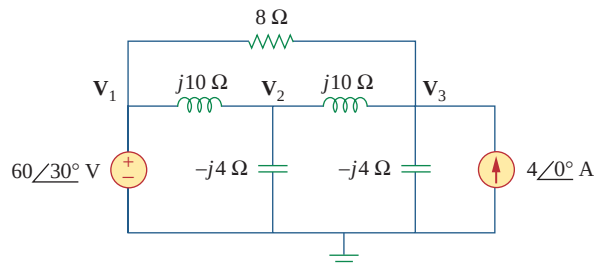


Figure 10.128
For Prob. 10.86.

10.87 Determine $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ in the circuit of Fig. 10.129 using *PSpice*.

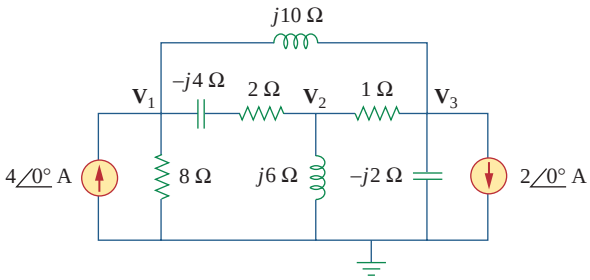


Figure 10.129
For Prob. 10.87.

10.88 Use *PSpice* to find v_o and i_o in the circuit of Fig. 10.130 below.

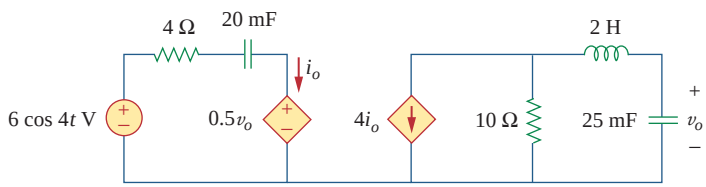


Figure 10.130
For Prob. 10.88.

Section 10.9 Applications

10.89 The op amp circuit in Fig. 10.131 is called an *inductance simulator*. Show that the input impedance is given by

$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{V}_{\text{in}}}{\mathbf{I}_{\text{in}}} = j\omega L_{\text{eq}}$$

where

$$L_{\text{eq}} = \frac{R_1 R_3 R_4}{R_2} C$$

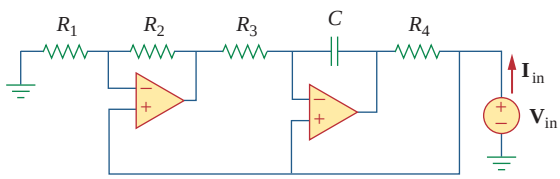


Figure 10.131
For Prob. 10.89.

10.90 Figure 10.132 shows a Wien-bridge network. Show that the frequency at which the phase shift between the input and output signals is zero is $f = \frac{1}{2}\pi RC$, and that the necessary gain is $\mathbf{A}_v = \mathbf{V}_o/\mathbf{V}_i = 3$ at that frequency.

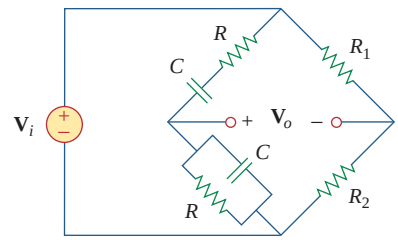


Figure 10.132
For Prob. 10.90.

10.91 Consider the oscillator in Fig. 10.133.

(a) Determine the oscillation frequency.

(b) Obtain the minimum value of R for which oscillation takes place.

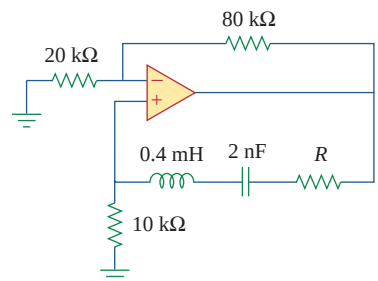


Figure 10.133
For Prob. 10.91.

10.92 The oscillator circuit in Fig. 10.134 uses an ideal op amp.

(a) Calculate the minimum value of R_o that will cause oscillation to occur.

(b) Find the frequency of oscillation.

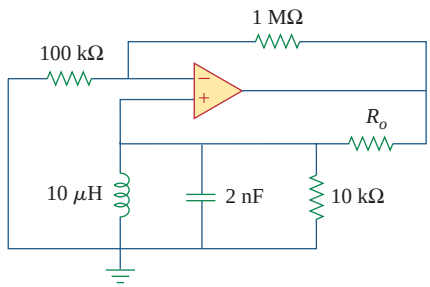


Figure 10.134
For Prob. 10.92.

10.93 Figure 10.135 shows a *Colpitts oscillator*. Show that the oscillation frequency is

$$f_o = \frac{1}{2\pi\sqrt{LC_T}}$$

where $C_T = C_1C_2/(C_1 + C_2)$. Assume $R_i \gg X_{C_2}$.

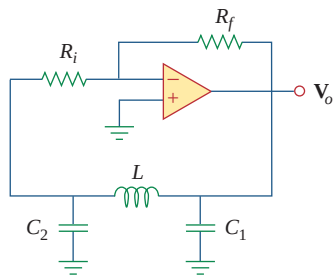


Figure 10.135
A Colpitts oscillator; for Prob. 10.93.

(*Hint:* Set the imaginary part of the impedance in the feedback circuit equal to zero.)

10.94 Design a Colpitts oscillator that will operate at 50 kHz.

10.95 Figure 10.136 shows a *Hartley oscillator*. Show that the frequency of oscillation is

$$f_o = \frac{1}{2\pi\sqrt{C(L_1 + L_2)}}$$

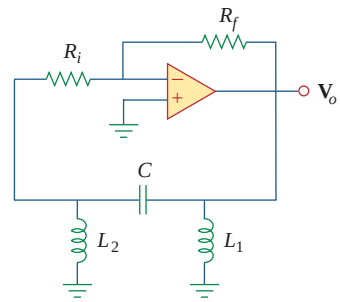


Figure 10.136
A Hartley oscillator; for Prob. 10.95.

10.96 Refer to the oscillator in Fig. 10.137.

(a) Show that

$$\frac{\mathbf{V}_2}{\mathbf{V}_o} = \frac{1}{3 + j(\omega L/R - R/\omega L)}$$

(b) Determine the oscillation frequency f_o.

(c) Obtain the relationship between R_1 and R_2 in order for oscillation to occur.

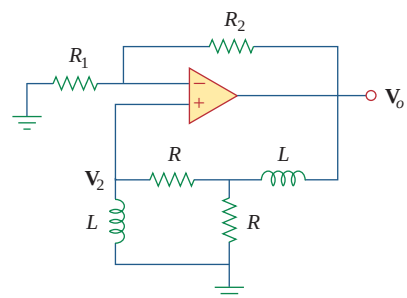


Figure 10.137
For Prob. 10.96.

chapter

11

AC Power Analysis

Four things come not back: the spoken word; the sped arrow; time past; the neglected opportunity.

—Al Halif Omar Ibn

Enhancing Your Career

Career in Power Systems

The discovery of the principle of an ac generator by Michael Faraday in 1831 was a major breakthrough in engineering; it provided a convenient way of generating the electric power that is needed in every electronic, electrical, or electromechanical device we use now.

Electric power is obtained by converting energy from sources such as fossil fuels (gas, oil, and coal), nuclear fuel (uranium), hydro energy (water falling through a head), geothermal energy (hot water, steam), wind energy, tidal energy, and biomass energy (wastes). These various ways of generating electric power are studied in detail in the field of power engineering, which has become an indispensable subdiscipline of electrical engineering. An electrical engineer should be familiar with the analysis, generation, transmission, distribution, and cost of electric power.

The electric power industry is a very large employer of electrical engineers. The industry includes thousands of electric utility systems ranging from large, interconnected systems serving large regional areas to small power companies serving individual communities or factories. Due to the complexity of the power industry, there are numerous electrical engineering jobs in different areas of the industry: power plant (generation), transmission and distribution, maintenance, research, data acquisition and flow control, and management. Since electric power is used everywhere, electric utility companies are everywhere, offering exciting training and steady employment for men and women in thousands of communities throughout the world.



A pole-type transformer with a low-voltage, three-wire distribution system.
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11.1 Introduction

Our effort in ac circuit analysis so far has been focused mainly on calculating voltage and current. Our major concern in this chapter is power analysis.

Power analysis is of paramount importance. Power is the most important quantity in electric utilities, electronic, and communication systems, because such systems involve transmission of power from one point to another. Also, every industrial and household electrical device—every fan, motor, lamp, pressing iron, TV, personal computer—has a power rating that indicates how much power the equipment requires; exceeding the power rating can do permanent damage to an appliance. The most common form of electric power is 50- or 60-Hz ac power. The choice of ac over dc allowed high-voltage power transmission from the power generating plant to the consumer.

We will begin by defining and deriving *instantaneous power* and *average power*. We will then introduce other power concepts. As practical applications of these concepts, we will discuss how power is measured and reconsider how electric utility companies charge their customers.

11.2 Instantaneous and Average Power

As mentioned in Chapter 2, the *instantaneous power* $p(t)$ absorbed by an element is the product of the instantaneous voltage $v(t)$ across the element and the instantaneous current $i(t)$ through it. Assuming the passive sign convention,

$$p(t) = v(t)i(t) \tag{11.1}$$

The **instantaneous power** (in watts) is the power at any instant of time.

We can also think of the instantaneous power as the power absorbed by the element at a specific instant of time. Instantaneous quantities are denoted by lowercase letters.

It is the rate at which an element absorbs energy.

Consider the general case of instantaneous power absorbed by an arbitrary combination of circuit elements under sinusoidal excitation, as shown in Fig. 11.1. Let the voltage and current at the terminals of the circuit be

$$v(t) = V_m \cos(\omega t + \theta_v) \tag{11.2a}$$

$$i(t) = I_m \cos(\omega t + \theta_i) \tag{11.2b}$$

where V_m and I_m are the amplitudes (or peak values), and θ_v and θ_i are the phase angles of the voltage and current, respectively. The instantaneous power absorbed by the circuit is

$$p(t) = v(t)i(t) = V_m I_m \cos(\omega t + \theta_v)\cos(\omega t + \theta_i) \tag{11.3}$$

We apply the trigonometric identity

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)] \tag{11.4}$$

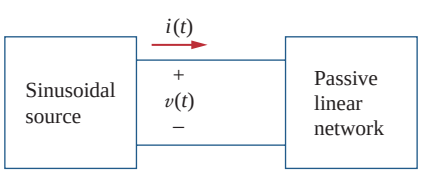


Figure 11.1
Sinusoidal source and passive linear circuit.

and express Eq. (11.3) as

$$p(t) = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2}V_m I_m \cos(2\omega t + \theta_v + \theta_i) \quad \textbf{(11.5)}$$

This shows us that the instantaneous power has two parts. The first part is constant or time independent. Its value depends on the phase difference between the voltage and the current. The second part is a sinusoidal function whose frequency is 2ω, which is twice the angular frequency of the voltage or current.

A sketch of $p(t)$ in Eq. (11.5) is shown in Fig. 11.2, where $T = 2\pi/\omega$ is the period of voltage or current. We observe that $p(t)$ is periodic, $p(t) = p(t + T_0)$, and has a period of $T_0 = T/2$, since its frequency is twice that of voltage or current. We also observe that $p(t)$ is positive for some part of each cycle and negative for the rest of the cycle. When $p(t)$ is positive, power is absorbed by the circuit. When $p(t)$ is negative, power is absorbed by the source; that is, power is transferred from the circuit to the source. This is possible because of the storage elements (capacitors and inductors) in the circuit.

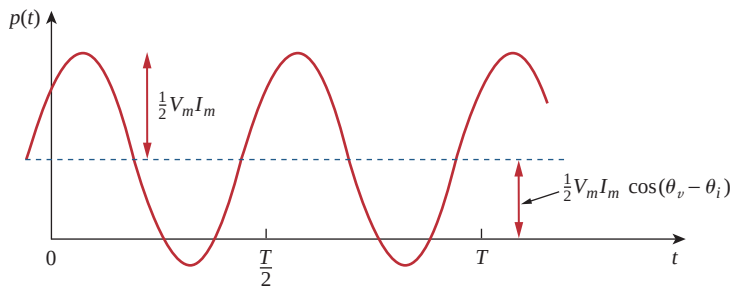


Figure 11.2
The instantaneous power $p(t)$ entering a circuit.

The instantaneous power changes with time and is therefore difficult to measure. The *average* power is more convenient to measure. In fact, the wattmeter, the instrument for measuring power, responds to average power.

The **average power**, in watts, is the average of the instantaneous power over one period.

Thus, the average power is given by

$$P = \frac{1}{T}\int_0^T p(t)\,dt \quad \textbf{(11.6)}$$

Although Eq. (11.6) shows the averaging done over T, we would get the same result if we performed the integration over the actual period of $p(t)$ which is $T_0 = T/2$.

Substituting $p(t)$ in Eq. (11.5) into Eq. (11.6) gives

$$\begin{aligned} P &= \frac{1}{T}\int_0^T \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i)\,dt \\ &\quad + \frac{1}{T}\int_0^T \frac{1}{2}V_m I_m \cos(2\omega t + \theta_v + \theta_i)\,dt \\ &= \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i)\frac{1}{T}\int_0^T dt \\ &\quad + \frac{1}{2}V_m I_m \frac{1}{T}\int_0^T \cos(2\omega t + \theta_v + \theta_i)\,dt \end{aligned} \tag{11.7}$$

The first integrand is constant, and the average of a constant is the same constant. The second integrand is a sinusoid. We know that the average of a sinusoid over its period is zero because the area under the sinusoid during a positive half-cycle is canceled by the area under it during the following negative half-cycle. Thus, the second term in Eq. (11.7) vanishes and the average power becomes

$$P = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) \tag{11.8}$$

Since $\cos(\theta_v - \theta_i) = \cos(\theta_i - \theta_v)$, what is important is the difference in the phases of the voltage and current.

Note that $p(t)$ is time-varying while P does not depend on time. To find the instantaneous power, we must necessarily have $v(t)$ and $i(t)$ in the time domain. But we can find the average power when voltage and current are expressed in the time domain, as in Eq. (11.8), or when they are expressed in the frequency domain. The phasor forms of $v(t)$ and $i(t)$ in Eq. (11.2) are $\mathbf{V} = V_m\angle\theta_v$ and $\mathbf{I} = I_m\angle\theta_i$, respectively. P is calculated using Eq. (11.8) or using phasors $\mathbf{V}$ and $\mathbf{I}$. To use phasors, we notice that

$$\begin{aligned} \frac{1}{2}\mathbf{V}\mathbf{I}^* &= \frac{1}{2}V_m I_m \angle\theta_v - \theta_i \\ &= \frac{1}{2}V_m I_m[\cos(\theta_v - \theta_i) + j\sin(\theta_v - \theta_i)] \end{aligned} \tag{11.9}$$

We recognize the real part of this expression as the average power P according to Eq. (11.8). Thus,

$$P = \frac{1}{2}\text{Re}[\mathbf{V}\mathbf{I}^*] = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) \tag{11.10}$$

Consider two special cases of Eq. (11.10). When $\theta_v = \theta_i$, the voltage and current are in phase. This implies a purely resistive circuit or resistive load R, and

$$P = \frac{1}{2}V_m I_m = \frac{1}{2}I_m^2 R = \frac{1}{2}|\mathbf{I}|^2 R \tag{11.11}$$

where $|\mathbf{I}|^2 = \mathbf{I} \times \mathbf{I}^*$. Equation (11.11) shows that a purely resistive circuit absorbs power at all times. When $\theta_v - \theta_i = \pm 90°$, we have a purely reactive circuit, and

$$P = \frac{1}{2}V_m I_m \cos 90° = 0 \tag{11.12}$$

showing that a purely reactive circuit absorbs no average power. In summary,

> A resistive load (R) absorbs power at all times, while a reactive load (L or C) absorbs zero average power.

Example 11.1

Given that

$$v(t) = 120\cos(377t + 45°)\text{ V} \quad \text{and} \quad i(t) = 10\cos(377t - 10°)\text{ A}$$

find the instantaneous power and the average power absorbed by the passive linear network of Fig. 11.1.

Solution:

The instantaneous power is given by

$$p = vi = 1200\cos(377t + 45°)\cos(377t - 10°)$$

Applying the trigonometric identity

$$\cos A\cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$$

gives

$$p = 600[\cos(754t + 35°) + \cos 55°]$$

or

$$p(t) = 344.2 + 600\cos(754t + 35°)\text{ W}$$

The average power is

$$P = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i) = \frac{1}{2}120(10)\cos[45° - (-10°)]$$
$$= 600\cos 55° = 344.2\text{ W}$$

which is the constant part of $p(t)$ above.

Practice Problem 11.1

Calculate the instantaneous power and average power absorbed by the passive linear network of Fig. 11.1 if

$$v(t) = 165\cos(10t + 20°)\text{ V} \quad \text{and} \quad i(t) = 20\sin(10t + 60°)\text{ A}$$

Answer: $1.0606 + 1.65\cos(20t - 10°)$ kW, 1.0606 kW.

Example 11.2

Calculate the average power absorbed by an impedance $\mathbf{Z} = 30 - j70\ \Omega$ when a voltage $\mathbf{V} = 120\angle 0°$ is applied across it.

Solution:

The current through the impedance is

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{120\angle 0°}{30 - j70} = \frac{120\angle 0°}{76.16\angle -66.8°} = 1.576\angle 66.8°\text{ A}$$

The average power is

$$P = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2}(120)(1.576)\cos(0 - 66.8°) = 37.24 \text{ W}$$

Practice Problem 11.2

A current $\mathbf{I} = 20\angle 30°$ A flows through an impedance $\mathbf{Z} = 40\angle -22°\ \Omega$. Find the average power delivered to the impedance.

Answer: 3.709 kW.

Example 11.3

For the circuit shown in Fig. 11.3, find the average power supplied by the source and the average power absorbed by the resistor.

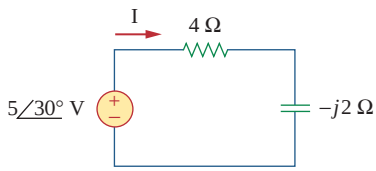


Figure 11.3
For Example 11.3.

Solution:
The current $\mathbf{I}$ is given by

$$\mathbf{I} = \frac{5\angle 30°}{4 - j2} = \frac{5\angle 30°}{4.472\angle -26.57°} = 1.118\angle 56.57° \text{ A}$$

The average power supplied by the voltage source is

$$P = \frac{1}{2}(5)(1.118)\cos(30° - 56.57°) = 2.5 \text{ W}$$

The current through the resistor is

$$\mathbf{I}_R = \mathbf{I} = 1.118\angle 56.57° \text{ A}$$

and the voltage across it is

$$\mathbf{V}_R = 4\mathbf{I}_R = 4.472\angle 56.57° \text{ V}$$

The average power absorbed by the resistor is

$$P = \frac{1}{2}(4.472)(1.118) = 2.5 \text{ W}$$

which is the same as the average power supplied. Zero average power is absorbed by the capacitor.

Practice Problem 11.3

In the circuit of Fig. 11.4, calculate the average power absorbed by the resistor and inductor. Find the average power supplied by the voltage source.

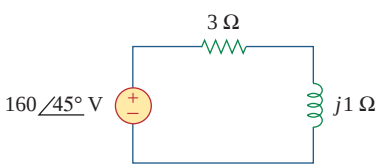


Figure 11.4
For Practice Prob. 11.3.

Answer: 3.84 kW, 0 W, 3.84 kW.

Example 11.4

Determine the average power generated by each source and the average power absorbed by each passive element in the circuit of Fig. 11.5(a).

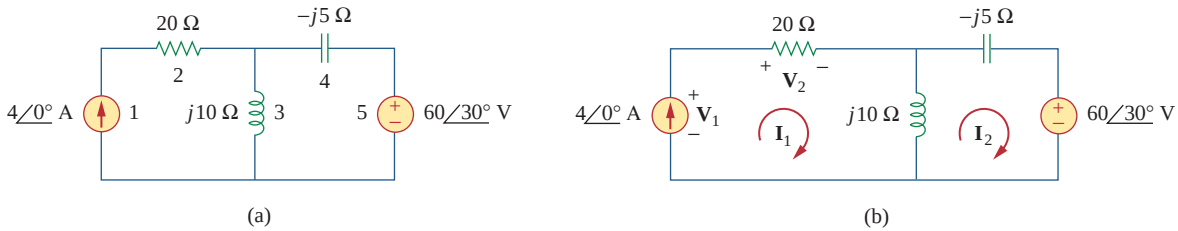


Figure 11.5
For Example 11.4.

Solution:
We apply mesh analysis as shown in Fig. 11.5(b). For mesh 1,

$$\mathbf{I}_1 = 4 \text{ A}$$

For mesh 2,

$$(j10 - j5)\mathbf{I}_2 - j10\mathbf{I}_1 + 60\angle 30^\circ = 0, \qquad \mathbf{I}_1 = 4 \text{ A}$$

or

$$j5\mathbf{I}_2 = -60\angle 30^\circ + j40 \quad \Rightarrow \quad \mathbf{I}_2 = -12\angle -60^\circ + 8 = 10.58\angle 79.1^\circ \text{ A}$$

For the voltage source, the current flowing from it is $\mathbf{I}_2 = 10.58\angle 79.1^\circ$ A and the voltage across it is $60\angle 30^\circ$ V, so that the average power is

$$P_5 = \frac{1}{2}(60)(10.58)\cos(30^\circ - 79.1^\circ) = 207.8 \text{ W}$$

Following the passive sign convention (see Fig. 1.8), this average power is absorbed by the source, in view of the direction of $\mathbf{I}_2$ and the polarity of the voltage source. That is, the circuit is delivering average power to the voltage source.

For the current source, the current through it is $\mathbf{I}_1 = 4\angle 0^\circ$ and the voltage across it is

$$\mathbf{V}_1 = 20\mathbf{I}_1 + j10(\mathbf{I}_1 - \mathbf{I}_2) = 80 + j10(4 - 2 - j10.39) = 183.9 + j20 = 184.984\angle 6.21^\circ \text{ V}$$

The average power supplied by the current source is

$$P_1 = -\frac{1}{2}(184.984)(4)\cos(6.21^\circ - 0) = -367.8 \text{ W}$$

It is negative according to the passive sign convention, meaning that the current source is supplying power to the circuit.

For the resistor, the current through it is $\mathbf{I}_1 = 4\angle 0^\circ$ and the voltage across it is $20\mathbf{I}_1 = 80\angle 0^\circ$, so that the power absorbed by the resistor is

$$P_2 = \frac{1}{2}(80)(4) = 160 \text{ W}$$

For the capacitor, the current through it is $\mathbf{I}_2 = 10.58\angle 79.1°$ and the voltage across it is $-j5\mathbf{I}_2 = (5\angle -90°)(10.58\angle 79.1°) = 52.9\angle 79.1° - 90°$. The average power absorbed by the capacitor is

$$P_4 = \frac{1}{2}(52.9)(10.58)\cos(-90°) = 0$$

For the inductor, the current through it is $\mathbf{I}_1 - \mathbf{I}_2 = 2 - j10.39 = 10.58\angle -79.1°$. The voltage across it is $j10(\mathbf{I}_1 - \mathbf{I}_2) = 10.58\angle -79.1° + 90°$. Hence, the average power absorbed by the inductor is

$$P_3 = \frac{1}{2}(105.8)(10.58)\cos 90° = 0$$

Notice that the inductor and the capacitor absorb zero average power and that the total power supplied by the current source equals the power absorbed by the resistor and the voltage source, or

$$P_1 + P_2 + P_3 + P_4 + P_5 = -367.8 + 160 + 0 + 0 + 207.8 = 0$$

indicating that power is conserved.

Practice Problem 11.4

Calculate the average power absorbed by each of the five elements in the circuit of Fig. 11.6.

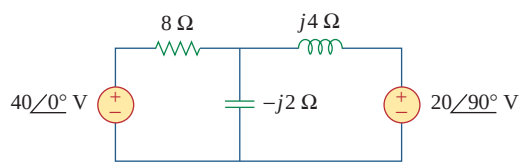


Figure 11.6
For Practice Prob. 11.4.

Answer: 40-V Voltage source: −60 W; *j*20-V Voltage source: −40 W; resistor: 100 W; others: 0 W.

11.3 Maximum Average Power Transfer

In Section 4.8 we solved the problem of maximizing the power delivered by a power-supplying resistive network to a load R_L. Representing the circuit by its Thevenin equivalent, we proved that the maximum power would be delivered to the load if the load resistance is equal to the Thevenin resistance $R_L = R_{\text{Th}}$. We now extend that result to ac circuits.

Consider the circuit in Fig. 11.7, where an ac circuit is connected to a load $\mathbf{Z}_L$ and is represented by its Thevenin equivalent. The load is usually represented by an impedance, which may model an electric

motor, an antenna, a TV, and so forth. In rectangular form, the Thevenin impedance $\mathbf{Z}_{\text{Th}}$ and the load impedance $\mathbf{Z}_L$ are

$$\mathbf{Z}_{\text{Th}} = R_{\text{Th}} + jX_{\text{Th}} \tag{11.13a}$$

$$\mathbf{Z}_L = R_L + jX_L \tag{11.13b}$$

The current through the load is

$$\mathbf{I} = \frac{\mathbf{V}_{\text{Th}}}{\mathbf{Z}_{\text{Th}} + \mathbf{Z}_L} = \frac{\mathbf{V}_{\text{Th}}}{(R_{\text{Th}} + jX_{\text{Th}}) + (R_L + jX_L)} \tag{11.14}$$

From Eq. (11.11), the average power delivered to the load is

$$P = \frac{1}{2}|\mathbf{I}|^2 R_L = \frac{|\mathbf{V}_{\text{Th}}|^2 R_L/2}{(R_{\text{Th}} + R_L)^2 + (X_{\text{Th}} + X_L)^2} \tag{11.15}$$

Our objective is to adjust the load parameters R_L and X_L so that P is maximum. To do this we set $\partial P/\partial R_L$ and $\partial P/\partial X_L$ equal to zero. From Eq. (11.15), we obtain

$$\frac{\partial P}{\partial X_L} = -\frac{|\mathbf{V}_{\text{Th}}|^2 R_L (X_{\text{Th}} + X_L)}{[(R_{\text{Th}} + R_L)^2 + (X_{\text{Th}} + X_L)^2]^2} \tag{11.16a}$$

$$\frac{\partial P}{\partial R_L} = \frac{|\mathbf{V}_{\text{Th}}|^2 [(R_{\text{Th}} + R_L)^2 + (X_{\text{Th}} + X_L)^2 - 2R_L(R_{\text{Th}} + R_L)]}{2[(R_{\text{Th}} + R_L)^2 + (X_{\text{Th}} + X_L)^2]^2} \tag{11.16b}$$

Setting $\partial P/\partial X_L$ to zero gives

$$X_L = -X_{\text{Th}} \tag{11.17}$$

and setting $\partial P/\partial R_L$ to zero results in

$$R_L = \sqrt{R_{\text{Th}}^2 + (X_{\text{Th}} + X_L)^2} \tag{11.18}$$

Combining Eqs. (11.17) and (11.18) leads to the conclusion that for maximum average power transfer, $\mathbf{Z}_L$ must be selected so that $X_L = -X_{\text{Th}}$ and $R_L = R_{\text{Th}}$, i.e.,

$$\boxed{\mathbf{Z}_L = R_L + jX_L = R_{\text{Th}} - jX_{\text{Th}} = \mathbf{Z}_{\text{Th}}^*} \tag{11.19}$$

For **maximum average power transfer**, the load impedance $\mathbf{Z}_L$ must be equal to the complex conjugate of the Thevenin impedance $\mathbf{Z}_{\text{Th}}$.

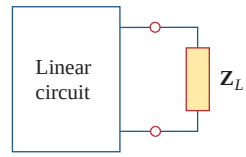
When $\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^*$, we say that the load is matched to the source.

This result is known as the *maximum average power transfer theorem* for the sinusoidal steady state. Setting $R_L = R_{\text{Th}}$ and $X_L = -X_{\text{Th}}$ in Eq. (11.15) gives us the maximum average power as

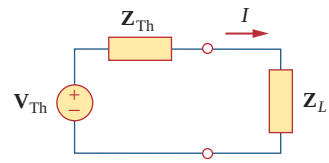
$$\boxed{P_{\max} = \frac{|\mathbf{V}_{\text{Th}}|^2}{8R_{\text{Th}}}} \tag{11.20}$$

In a situation in which the load is purely real, the condition for maximum power transfer is obtained from Eq. (11.18) by setting $X_L = 0$; that is,

$$R_L = \sqrt{R_{\text{Th}}^2 + X_{\text{Th}}^2} = |\mathbf{Z}_{\text{Th}}| \tag{11.21}$$



(a)



(b)

Figure 11.7
Finding the maximum average power transfer: (a) circuit with a load, (b) the Thevenin equivalent.

This means that for maximum average power transfer to a purely resistive load, the load impedance (or resistance) is equal to the magnitude of the Thevenin impedance.

Example 11.5

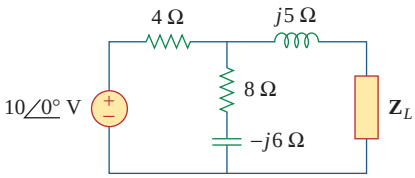


Figure 11.8
For Example 11.5.

Determine the load impedance $\mathbf{Z}_L$ that maximizes the average power drawn from the circuit of Fig. 11.8. What is the maximum average power?

Solution:
First we obtain the Thevenin equivalent at the load terminals. To get $\mathbf{Z}_{\text{Th}}$, consider the circuit shown in Fig. 11.9(a). We find

$$\mathbf{Z}_{\text{Th}} = j5 + 4 \,\|\, (8 - j6) = j5 + \frac{4(8 - j6)}{4 + 8 - j6} = 2.933 + j4.467\ \Omega$$

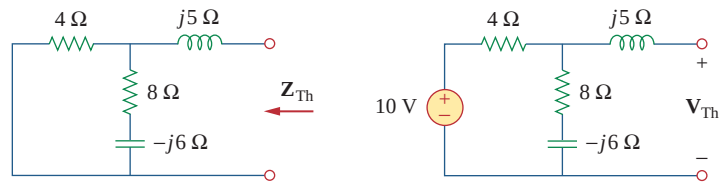


Figure 11.9
Finding the Thevenin equivalent of the circuit in Fig. 11.8.

To find $\mathbf{V}_{\text{Th}}$, consider the circuit in Fig. 11.8(b). By voltage division,

$$\mathbf{V}_{\text{Th}} = \frac{8 - j6}{4 + 8 - j6}(10) = 7.454\angle -10.3^\circ\ \text{V}$$

The load impedance draws the maximum power from the circuit when

$$\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^* = 2.933 - j4.467\ \Omega$$

According to Eq. (11.20), the maximum average power is

$$P_{\max} = \frac{|\mathbf{V}_{\text{Th}}|^2}{8R_{\text{Th}}} = \frac{(7.454)^2}{8(2.933)} = 2.368\ \text{W}$$

Practice Problem 11.5

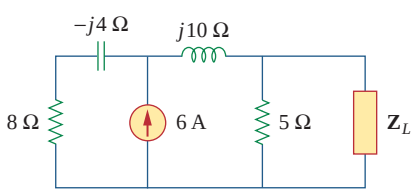


Figure 11.10
For Practice Prob. 11.5.

For the circuit shown in Fig. 11.10, find the load impedance $\mathbf{Z}_L$ that absorbs the maximum average power. Calculate that maximum average power.

Answer: $3.415 - j0.7317\ \Omega$, 12.861 W.

Example 11.6

In the circuit in Fig. 11.11, find the value of R_L that will absorb the maximum average power. Calculate that power.

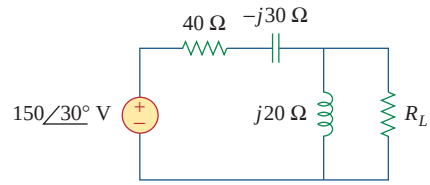


Figure 11.11
For Example 11.6.

Solution:
We first find the Thevenin equivalent at the terminals of R_L.

$$\mathbf{Z}_{\text{Th}} = (40 - j30) \parallel j20 = \frac{j20(40 - j30)}{j20 + 40 - j30} = 9.412 + j22.35\ \Omega$$

By voltage division,

$$\mathbf{V}_{\text{Th}} = \frac{j20}{j20 + 40 - j30}(150\angle 30°) = 72.76\angle 134°\ \text{V}$$

The value of R_L that will absorb the maximum average power is

$$R_L = |\mathbf{Z}_{\text{Th}}| = \sqrt{9.412^2 + 22.35^2} = 24.25\ \Omega$$

The current through the load is

$$\mathbf{I} = \frac{\mathbf{V}_{\text{Th}}}{\mathbf{Z}_{\text{Th}} + R_L} = \frac{72.76\angle 134°}{33.66 + j22.35} = 1.8\angle 100.42°\ \text{A}$$

The maximum average power absorbed by R_L is

$$P_{\max} = \frac{1}{2}|\mathbf{I}|^2 R_L = \frac{1}{2}(1.8)^2(24.25) = 39.29\ \text{W}$$

Practice Problem 11.6

In Fig. 11.12, the resistor R_L is adjusted until it absorbs the maximum average power. Calculate R_L and the maximum average power absorbed by it.

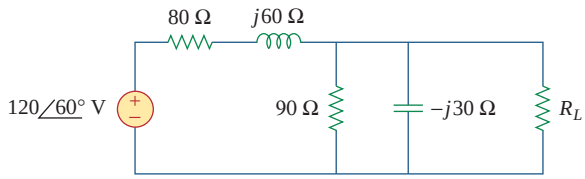


Figure 11.12
For Practice Prob. 11.6.

Answer: 30 Ω, 6.863 W.

11.4 Effective or RMS Value

The idea of *effective value* arises from the need to measure the effectiveness of a voltage or current source in delivering power to a resistive load.

The **effective value** of a periodic current is the dc current that delivers the same average power to a resistor as the periodic current.

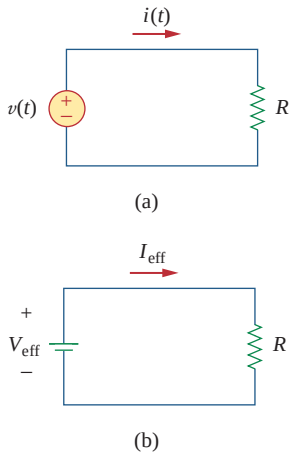


Figure 11.13
Finding the effective current: (a) ac circuit, (b) dc circuit.

In Fig. 11.13, the circuit in (a) is ac while that of (b) is dc. Our objective is to find I_{eff} that will transfer the same power to resistor R as the sinusoid i. The average power absorbed by the resistor in the ac circuit is

$$P = \frac{1}{T}\int_0^T i^2 R\, dt = \frac{R}{T}\int_0^T i^2\, dt \tag{11.22}$$

while the power absorbed by the resistor in the dc circuit is

$$P = I_{\text{eff}}^2 R \tag{11.23}$$

Equating the expressions in Eqs. (11.22) and (11.23) and solving for I_{eff}, we obtain

$$I_{\text{eff}} = \sqrt{\frac{1}{T}\int_0^T i^2\, dt} \tag{11.24}$$

The effective value of the voltage is found in the same way as current; that is,

$$V_{\text{eff}} = \sqrt{\frac{1}{T}\int_0^T v^2\, dt} \tag{11.25}$$

This indicates that the effective value is the (square) *root* of the *mean* (or average) of the *square* of the periodic signal. Thus, the effective value is often known as the *root-mean-square* value, or *rms* value for short; and we write

$$I_{\text{eff}} = I_{\text{rms}}, \qquad V_{\text{eff}} = V_{\text{rms}} \tag{11.26}$$

For any periodic function $x(t)$ in general, the rms value is given by

$$X_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T x^2\, dt} \tag{11.27}$$

The **effective value** of a periodic signal is its root mean square (rms) value.

Equation 11.27 states that to find the rms value of $x(t)$, we first find its *square* x^2 and then find the *mean* of that, or

$$\frac{1}{T}\int_0^T x^2\, dt$$

and then the square *root* ($\sqrt{\quad}$) of that mean. The rms value of a constant is the constant itself. For the sinusoid $i(t) = I_m \cos\omega t$, the effective or rms value is

$$\begin{aligned} I_{\text{rms}} &= \sqrt{\frac{1}{T}\int_0^T I_m^2 \cos^2\omega t\, dt} \\ &= \sqrt{\frac{I_m^2}{T}\int_0^T \frac{1}{2}(1 + \cos 2\omega t)\, dt} = \frac{I_m}{\sqrt{2}} \end{aligned} \tag{11.28}$$

Similarly, for $v(t) = V_m \cos\omega t$,

$$V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \tag{11.29}$$

Keep in mind that Eqs. (11.28) and (11.29) are only valid for sinusoidal signals.

The average power in Eq. (11.8) can be written in terms of the rms values.

$$P = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}}\frac{I_m}{\sqrt{2}}\cos(\theta_v - \theta_i)$$
$$= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) \tag{11.30}$$

Similarly, the average power absorbed by a resistor R in Eq. (11.11) can be written as

$$P = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R} \tag{11.31}$$

When a sinusoidal voltage or current is specified, it is often in terms of its maximum (or peak) value or its rms value, since its average value is zero. The power industries specify phasor magnitudes in terms of their rms values rather than peak values. For instance, the 110 V available at every household is the rms value of the voltage from the power company. It is convenient in power analysis to express voltage and current in their rms values. Also, analog voltmeters and ammeters are designed to read directly the rms value of voltage and current, respectively.

Example 11.7

Determine the rms value of the current waveform in Fig. 11.14. If the current is passed through a 2-Ω resistor, find the average power absorbed by the resistor.

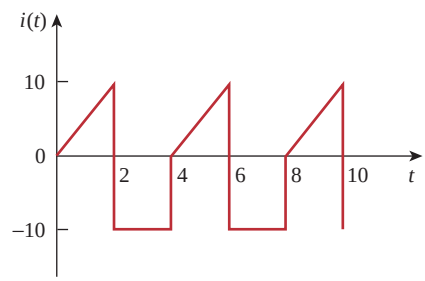


Figure 11.14
For Example 11.7.

Solution:
The period of the waveform is $T = 4$. Over a period, we can write the current waveform as

$$i(t) = \begin{cases} 5t, & 0 < t < 2 \\ -10, & 2 < t < 4 \end{cases}$$

The rms value is

$$I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T i^2\,dt} = \sqrt{\frac{1}{4}\left[\int_0^2 (5t)^2\,dt + \int_2^4 (-10)^2\,dt\right]}$$
$$= \sqrt{\frac{1}{4}\left[25\frac{t^3}{3}\Big|_0^2 + 100t\Big|_2^4\right]} = \sqrt{\frac{1}{4}\left(\frac{200}{3} + 200\right)} = 8.165 \text{ A}$$

The power absorbed by a 2-Ω resistor is

$$P = I_{\text{rms}}^2 R = (8.165)^2(2) = 133.3 \text{ W}$$

Practice Problem 11.7

Find the rms value of the current waveform of Fig. 11.15. If the current flows through a 9-Ω resistor, calculate the average power absorbed by the resistor.

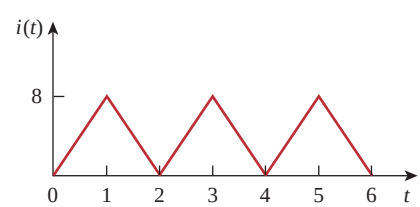


Figure 11.15
For Practice Prob. 11.7.

Answer: 4.318 A, 192 W.

Example 11.8

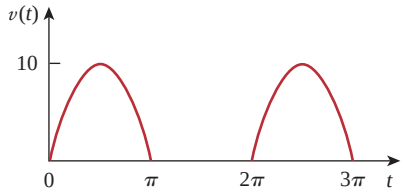


Figure 11.16
For Example 11.8.

The waveform shown in Fig. 11.16 is a half-wave rectified sine wave. Find the rms value and the amount of average power dissipated in a 10-Ω resistor.

Solution:
The period of the voltage waveform is $T = 2\pi$, and

$$v(t) = \begin{cases} 10\sin t, & 0 < t < \pi \\ 0, & \pi < t < 2\pi \end{cases}$$

The rms value is obtained as

$$V_{\text{rms}}^2 = \frac{1}{T}\int_0^T v^2(t)\,dt = \frac{1}{2\pi}\left[\int_0^{\pi}(10\sin t)^2\,dt + \int_{\pi}^{2\pi} 0^2\,dt\right]$$

But $\sin^2 t = \frac{1}{2}(1 - \cos 2t)$. Hence,

$$V_{\text{rms}}^2 = \frac{1}{2\pi}\int_0^{\pi}\frac{100}{2}(1 - \cos 2t)\,dt = \frac{50}{2\pi}\left(t - \frac{\sin 2t}{2}\right)\Big|_0^{\pi}$$

$$= \frac{50}{2\pi}\left(\pi - \frac{1}{2}\sin 2\pi - 0\right) = 25, \qquad V_{\text{rms}} = 5\text{ V}$$

The average power absorbed is

$$P = \frac{V_{\text{rms}}^2}{R} = \frac{5^2}{10} = 2.5\text{ W}$$

Practice Problem 11.8

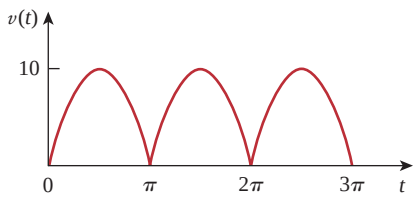


Figure 11.17
For Practice Prob. 11.8.

Find the rms value of the full-wave rectified sine wave in Fig. 11.17. Calculate the average power dissipated in a 6-Ω resistor.

Answer: 7.071 V, 8.333 W.

11.5 Apparent Power and Power Factor

In Section 11.2 we saw that if the voltage and current at the terminals of a circuit are

$$v(t) = V_m\cos(\omega t + \theta_v) \quad \text{and} \quad i(t) = I_m\cos(\omega t + \theta_i) \tag{11.32}$$

or, in phasor form, $\mathbf{V} = V_m\angle\theta_v$ and $\mathbf{I} = I_m\angle\theta_i$, the average power is

$$P = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i) \tag{11.33}$$

In Section 11.4, we saw that

$$P = V_{\text{rms}}I_{\text{rms}}\cos(\theta_v - \theta_i) = S\cos(\theta_v - \theta_i) \tag{11.34}$$

We have added a new term to the equation:

$$\boxed{S = V_{\text{rms}}I_{\text{rms}}} \tag{11.35}$$

The average power is a product of two terms. The product $V_{rms}I_{rms}$ is known as the *apparent power* S. The factor $\cos(\theta_v - \theta_i)$ is called the *power factor* (pf).

The **apparent power** (in VA) is the product of the rms values of voltage and current.

The apparent power is so called because it seems apparent that the power should be the voltage-current product, by analogy with dc resistive circuits. It is measured in volt-amperes or VA to distinguish it from the average or real power, which is measured in watts. The power factor is dimensionless, since it is the ratio of the average power to the apparent power,

$$\text{pf} = \frac{P}{S} = \cos(\theta_v - \theta_i) \tag{11.36}$$

The angle $\theta_v - \theta_i$ is called the *power factor angle*, since it is the angle whose cosine is the power factor. The power factor angle is equal to the angle of the load impedance if $\mathbf{V}$ is the voltage across the load and $\mathbf{I}$ is the current through it. This is evident from the fact that

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{V_m \angle \theta_v}{I_m \angle \theta_i} = \frac{V_m}{I_m} \angle \theta_v - \theta_i \tag{11.37}$$

Alternatively, since

$$\mathbf{V}_{rms} = \frac{\mathbf{V}}{\sqrt{2}} = V_{rms} \angle \theta_v \tag{11.38a}$$

and

$$\mathbf{I}_{rms} = \frac{\mathbf{I}}{\sqrt{2}} = I_{rms} \angle \theta_i \tag{11.38b}$$

the impedance is

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{\mathbf{V}_{rms}}{\mathbf{I}_{rms}} = \frac{V_{rms}}{I_{rms}} \angle \theta_v - \theta_i \tag{11.39}$$

The **power factor** is the cosine of the phase difference between voltage and current. It is also the cosine of the angle of the load impedance.

From Eq. (11.36), the power factor may be seen as that factor by which the apparent power must be multiplied to obtain the real or average power. The value of pf ranges between zero and unity. For a purely resistive load, the voltage and current are in phase, so that $\theta_v - \theta_i = 0$ and pf = 1. This implies that the apparent power is equal to the average power. For a purely reactive load, $\theta_v - \theta_i = \pm 90°$ and pf = 0. In this case the average power is zero. In between these two extreme cases, pf is said to be *leading* or *lagging*. Leading power factor means that current leads voltage, which implies a capacitive load. Lagging power factor means that current lags voltage, implying an inductive

From Eq. (11.36), the power factor may also be regarded as the ratio of the real power dissipated in the load to the apparent power of the load.

load. Power factor affects the electric bills consumers pay the electric utility companies, as we will see in Section 11.9.2.

Example 11.9

A series-connected load draws a current $i(t) = 4\cos(100\pi t + 10°)$ A when the applied voltage is $v(t) = 120\cos(100\pi t - 20°)$ V. Find the apparent power and the power factor of the load. Determine the element values that form the series-connected load.

Solution:
The apparent power is

$$S = V_{\text{rms}} I_{\text{rms}} = \frac{120}{\sqrt{2}} \frac{4}{\sqrt{2}} = 240 \text{ VA}$$

The power factor is

$$\text{pf} = \cos(\theta_v - \theta_i) = \cos(-20° - 10°) = 0.866 \qquad \text{(leading)}$$

The pf is leading because the current leads the voltage. The pf may also be obtained from the load impedance.

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{120\angle -20°}{4\angle 10°} = 30\angle -30° = 25.98 - j15\ \Omega$$

$$\text{pf} = \cos(-30°) = 0.866 \qquad \text{(leading)}$$

The load impedance $\mathbf{Z}$ can be modeled by a 25.98-Ω resistor in series with a capacitor with

$$X_C = -15 = -\frac{1}{\omega C}$$

or

$$C = \frac{1}{15\omega} = \frac{1}{15 \times 100\pi} = 212.2\ \mu\text{F}$$

Practice Problem 11.9

Obtain the power factor and the apparent power of a load whose impedance is $\mathbf{Z} = 60 + j40\ \Omega$ when the applied voltage is $v(t) = 160\cos(377t + 10°)$ V.

Answer: 0.8321 lagging, $177.5\angle 33.69°$ VA.

Example 11.10

Determine the power factor of the entire circuit of Fig. 11.18 as seen by the source. Calculate the average power delivered by the source.

Solution:
The total impedance is

$$\mathbf{Z} = 6 + 4 \,\|\, (-j2) = 6 + \frac{-j2 \times 4}{4 - j2} = 6.8 - j1.6 = 7\angle -13.24°\ \Omega$$

The power factor is

$$\text{pf} = \cos(-13.24) = 0.9734 \qquad \text{(leading)}$$

since the impedance is capacitive. The rms value of the current is

$$\mathbf{I}_{\text{rms}} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{Z}} = \frac{30\angle 0^\circ}{7\angle -13.24^\circ} = 4.286\angle 13.24^\circ \text{ A}$$

The average power supplied by the source is

$$P = V_{\text{rms}} I_{\text{rms}} \text{ pf} = (30)(4.286)0.9734 = 125 \text{ W}$$

or

$$P = I_{\text{rms}}^2 R = (4.286)^2(6.8) = 125 \text{ W}$$

where R is the resistive part of $\mathbf{Z}$.

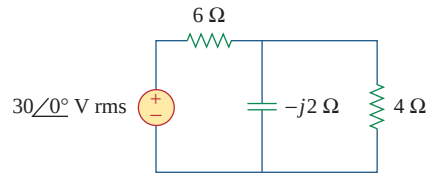


Figure 11.18
For Example 11.10.

Practice Problem 11.10

Calculate the power factor of the entire circuit of Fig. 11.19 as seen by the source. What is the average power supplied by the source?

Answer: 0.936 lagging, 1.062 kW.

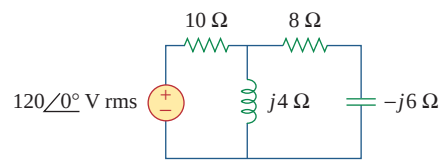


Figure 11.19
For Practice Prob. 11.10.

11.6 Complex Power

Considerable effort has been expended over the years to express power relations as simply as possible. Power engineers have coined the term *complex power*, which they use to find the total effect of parallel loads. Complex power is important in power analysis because it contains *all* the information pertaining to the power absorbed by a given load.

Consider the ac load in Fig. 11.20. Given the phasor form $\mathbf{V} = V_m\angle\theta_v$ and $\mathbf{I} = I_m\angle\theta_i$ of voltage $v(t)$ and current $i(t)$, the *complex power* $\mathbf{S}$ absorbed by the ac load is the product of the voltage and the complex conjugate of the current, or

$$\mathbf{S} = \frac{1}{2}\mathbf{V}\mathbf{I}^* \qquad \textbf{(11.40)}$$

assuming the passive sign convention (see Fig. 11.20). In terms of the rms values,

$$\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^* \qquad \textbf{(11.41)}$$

where

$$\mathbf{V}_{\text{rms}} = \frac{\mathbf{V}}{\sqrt{2}} = V_{\text{rms}}\angle\theta_v \qquad \textbf{(11.42)}$$

and

$$\mathbf{I}_{\text{rms}} = \frac{\mathbf{I}}{\sqrt{2}} = I_{\text{rms}}\angle\theta_i \qquad \textbf{(11.43)}$$

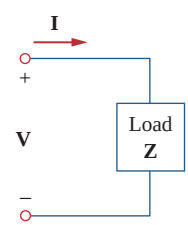


Figure 11.20
The voltage and current phasors associated with a load.

When working with the rms values of currents or voltages, we may drop the subscript rms if no confusion will be caused by doing so.

Thus we may write Eq. (11.41) as

$$\begin{aligned}\mathbf{S} &= V_{\text{rms}} I_{\text{rms}} \angle \theta_v - \theta_i \\ &= V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) + jV_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)\end{aligned} \tag{11.44}$$

This equation can also be obtained from Eq. (11.9). We notice from Eq. (11.44) that the magnitude of the complex power is the apparent power; hence, the complex power is measured in volt-amperes (VA). Also, we notice that the angle of the complex power is the power factor angle.

The complex power may be expressed in terms of the load impedance $\mathbf{Z}$. From Eq. (11.37), the load impedance $\mathbf{Z}$ may be written as

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{I}_{\text{rms}}} = \frac{V_{\text{rms}}}{I_{\text{rms}}} \angle \theta_v - \theta_i \tag{11.45}$$

Thus, $\mathbf{V}_{\text{rms}} = \mathbf{Z}\mathbf{I}_{\text{rms}}$. Substituting this into Eq. (11.41) gives

$$\mathbf{S} = I_{\text{rms}}^2 \mathbf{Z} = \frac{V_{\text{rms}}^2}{\mathbf{Z}^*} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^* \tag{11.46}$$

Since $\mathbf{Z} = R + jX$, Eq. (11.46) becomes

$$\mathbf{S} = I_{\text{rms}}^2 (R + jX) = P + jQ \tag{11.47}$$

where P and Q are the real and imaginary parts of the complex power; that is,

$$P = \text{Re}(\mathbf{S}) = I_{\text{rms}}^2 R \tag{11.48}$$

$$Q = \text{Im}(\mathbf{S}) = I_{\text{rms}}^2 X \tag{11.49}$$

P is the average or real power and it depends on the load's resistance R. Q depends on the load's reactance X and is called the *reactive* (or quadrature) power.

Comparing Eq. (11.44) with Eq. (11.47), we notice that

$$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i), \qquad Q = V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i) \tag{11.50}$$

The real power P is the average power in watts delivered to a load; it is the only useful power. It is the actual power dissipated by the load. The reactive power Q is a measure of the energy exchange between the source and the reactive part of the load. The unit of Q is the *volt-ampere reactive* (VAR) to distinguish it from the real power, whose unit is the watt. We know from Chapter 6 that energy storage elements neither dissipate nor supply power, but exchange power back and forth with the rest of the network. In the same way, the reactive power is being transferred back and forth between the load and the source. It represents a lossless interchange between the load and the source. Notice that:

1. $Q = 0$ for resistive loads (unity pf).
2. $Q < 0$ for capacitive loads (leading pf).
3. $Q > 0$ for inductive loads (lagging pf).

Thus,

Complex power (in VA) is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. As a complex quantity, its real part is real power P and its imaginary part is reactive power Q.

Introducing the complex power enables us to obtain the real and reactive powers directly from voltage and current phasors.

$$\begin{aligned}
\text{Complex Power} &= \mathbf{S} = P + jQ = \mathbf{V}_{\text{rms}}(\mathbf{I}_{\text{rms}})^* \\
&= \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}\angle\theta_v - \theta_i \\
\text{Apparent Power} &= S = |\mathbf{S}| = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}} = \sqrt{P^2 + Q^2} \\
\text{Real Power} &= P = \text{Re}(\mathbf{S}) = S\cos(\theta_v - \theta_i) \\
\text{Reactive Power} &= Q = \text{Im}(\mathbf{S}) = S\sin(\theta_v - \theta_i) \\
\text{Power Factor} &= \frac{P}{S} = \cos(\theta_v - \theta_i)
\end{aligned} \tag{11.51}$$

This shows how the complex power contains *all* the relevant power information in a given load.

S contains *all* power information of a load. The real part of **S** is the real power *P*; its imaginary part is the reactive power *Q*; its magnitude is the apparent power *S*; and the cosine of its phase angle is the power factor pf.

It is a standard practice to represent **S**, *P*, and *Q* in the form of a triangle, known as the *power triangle*, shown in Fig. 11.21(a). This is similar to the impedance triangle showing the relationship between **Z**, *R*, and *X*, illustrated in Fig. 11.21(b). The power triangle has four items—the apparent/complex power, real power, reactive power, and the power factor angle. Given two of these items, the other two can easily be obtained from the triangle. As shown in Fig. 11.22, when **S** lies in the first quadrant, we have an inductive load and a lagging pf. When **S** lies in the fourth quadrant, the load is capacitive and the pf is leading. It is also possible for the complex power to lie in the second or third quadrant. This requires that the load impedance have a negative resistance, which is possible with active circuits.

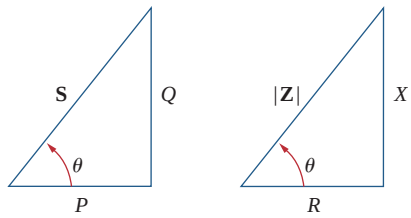


Figure 11.21
(a) Power triangle, (b) impedance triangle.

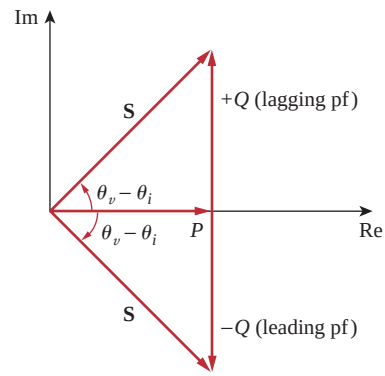


Figure 11.22
Power triangle.

Example 11.11

The voltage across a load is $v(t) = 60\cos(\omega t - 10°)$ V and the current through the element in the direction of the voltage drop is $i(t) = 1.5\cos(\omega t + 50°)$ A. Find: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Solution:

(a) For the rms values of the voltage and current, we write

$$\mathbf{V}_{\text{rms}} = \frac{60}{\sqrt{2}}\angle -10^\circ, \qquad \mathbf{I}_{\text{rms}} = \frac{1.5}{\sqrt{2}}\angle +50^\circ$$

The complex power is

$$\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^* = \left(\frac{60}{\sqrt{2}}\angle -10^\circ\right)\left(\frac{1.5}{\sqrt{2}}\angle -50^\circ\right) = 45\angle -60^\circ \text{ VA}$$

The apparent power is

$$S = |\mathbf{S}| = 45 \text{ VA}$$

(b) We can express the complex power in rectangular form as

$$\mathbf{S} = 45\angle -60^\circ = 45[\cos(-60^\circ) + j\sin(-60^\circ)] = 22.5 - j38.97$$

Since $\mathbf{S} = P + jQ$, the real power is

$$P = 22.5 \text{ W}$$

while the reactive power is

$$Q = -38.97 \text{ VAR}$$

(c) The power factor is

$$\text{pf} = \cos(-60^\circ) = 0.5 \text{ (leading)}$$

It is leading, because the reactive power is negative. The load impedance is

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{60\angle -10^\circ}{1.5\angle +50^\circ} = 40\angle -60^\circ\ \Omega$$

which is a capacitive impedance.

Practice Problem 11.11

For a load, $\mathbf{V}_{\text{rms}} = 110\angle 85^\circ$ V, $\mathbf{I}_{\text{rms}} = 0.4\angle 15^\circ$ A. Determine: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Answer: (a) $44\angle 70^\circ$ VA, 44 VA, (b) 15.05 W, 41.35 VAR, (c) 0.342 lagging, $94.06 + j258.4\ \Omega$.

Example 11.12

A load $\mathbf{Z}$ draws 12 kVA at a power factor of 0.856 lagging from a 120-V rms sinusoidal source. Calculate: (a) the average and reactive powers delivered to the load, (b) the peak current, and (c) the load impedance.

Solution:

(a) Given that $\text{pf} = \cos\theta = 0.856$, we obtain the power angle as $\theta = \cos^{-1} 0.856 = 31.13^\circ$. If the apparent power is $S = 12{,}000$ VA, then the average or real power is

$$P = S\cos\theta = 12{,}000 \times 0.856 = 10.272 \text{ kW}$$

while the reactive power is

$$Q = S\sin\theta = 12{,}000 \times 0.517 = 6.204 \text{ kVA}$$

(b) Since the pf is lagging, the complex power is

$$\mathbf{S} = P + jQ = 10.272 + j6.204 \text{ kVA}$$

From $\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^*$, we obtain

$$\mathbf{I}_{\text{rms}}^* = \frac{\mathbf{S}}{\mathbf{V}_{\text{rms}}} = \frac{10{,}272 + j6204}{120\angle 0^\circ} = 85.6 + j51.7 \text{ A} = 100\angle 31.13^\circ \text{ A}$$

Thus $\mathbf{I}_{\text{rms}} = 100\angle -31.13^\circ$ and the peak current is

$$I_m = \sqrt{2}I_{\text{rms}} = \sqrt{2}(100) = 141.4 \text{ A}$$

(c) The load impedance

$$\mathbf{Z} = \frac{\mathbf{V}_{\text{rms}}}{\mathbf{I}_{\text{rms}}} = \frac{120\angle 0^\circ}{100\angle -31.13^\circ} = 1.2\angle 31.13^\circ\ \Omega$$

which is an inductive impedance.

Practice Problem 11.12

A sinusoidal source supplies 20 kVAR reactive power to load $\mathbf{Z} = 250\angle -75^\circ\ \Omega$. Determine: (a) the power factor, (b) the apparent power delivered to the load, and (c) the rms voltage.

Answer: (a) 0.2588 leading, (b) 20.71 kVA, (c) 2.275 kV.

11.7 †Conservation of AC Power

In fact, we already saw in Examples 11.3 and 11.4 that average power is conserved in ac circuits.

The principle of conservation of power applies to ac circuits as well as to dc circuits (see Section 1.5).

To see this, consider the circuit in Fig. 11.23(a), where two load impedances $\mathbf{Z}_1$ and $\mathbf{Z}_2$ are connected in parallel across an ac source $\mathbf{V}$. KCL gives

$$\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2 \tag{11.52}$$

The complex power supplied by the source is (from now on, unless otherwise specified, all values of voltages and currents will be assumed to be rms values)

$$\mathbf{S} = \mathbf{V}\mathbf{I}^* = \mathbf{V}(\mathbf{I}_1^* + \mathbf{I}_2^*) = \mathbf{V}\mathbf{I}_1^* + \mathbf{V}\mathbf{I}_2^* = \mathbf{S}_1 + \mathbf{S}_2 \tag{11.53}$$

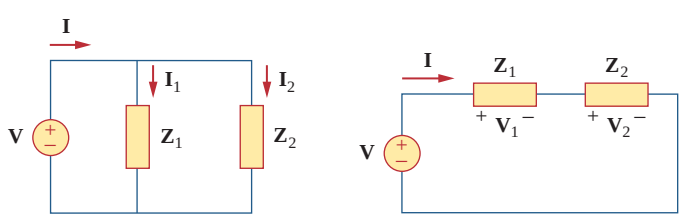


Figure 11.23
An ac voltage source supplied loads connected in: (a) parallel, (b) series.

where $\mathbf{S}_1$ and $\mathbf{S}_2$ denote the complex powers delivered to loads $\mathbf{Z}_1$ and $\mathbf{Z}_2$, respectively.

If the loads are connected in series with the voltage source, as shown in Fig. 11.23(b), KVL yields

$$\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2 \tag{11.54}$$

The complex power supplied by the source is

$$\mathbf{S} = \mathbf{V}\mathbf{I}^* = (\mathbf{V}_1 + \mathbf{V}_2)\mathbf{I}^* = \mathbf{V}_1\mathbf{I}^* + \mathbf{V}_2\mathbf{I}^* = \mathbf{S}_1 + \mathbf{S}_2 \tag{11.55}$$

where $\mathbf{S}_1$ and $\mathbf{S}_2$ denote the complex powers delivered to loads $\mathbf{Z}_1$ and $\mathbf{Z}_2$, respectively.

We conclude from Eqs. (11.53) and (11.55) that whether the loads are connected in series or in parallel (or in general), the total power *supplied* by the source equals the total power *delivered* to the load. Thus, in general, for a source connected to N loads,

$$\boxed{\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 + \cdots + \mathbf{S}_N} \tag{11.56}$$

In fact, all forms of ac power are conserved: instantaneous, real, reactive, and complex.

This means that the total complex power in a network is the sum of the complex powers of the individual components. (This is also true of real power and reactive power, but not true of apparent power.) This expresses the principle of conservation of ac power:

> The complex, real, and reactive powers of the sources equal the respective sums of the complex, real, and reactive powers of the individual loads.

From this we imply that the real (or reactive) power flow from sources in a network equals the real (or reactive) power flow into the other elements in the network.

Example 11.13

Figure 11.24 shows a load being fed by a voltage source through a transmission line. The impedance of the line is represented by the $(4 + j2)\ \Omega$ impedance and a return path. Find the real power and reactive power absorbed by: (a) the source, (b) the line, and (c) the load.

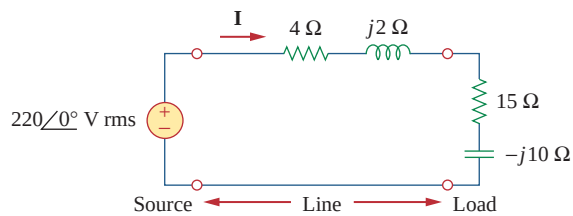


Figure 11.24
For Example 11.13.

Solution:
The total impedance is

$$\mathbf{Z} = (4 + j2) + (15 - j10) = 19 - j8 = 20.62\angle -22.83^\circ\ \Omega$$

The current through the circuit is

$$\mathbf{I} = \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{220\angle 0^\circ}{20.62\angle -22.83^\circ} = 10.67\angle 22.83^\circ \text{ A rms}$$

(a) For the source, the complex power is

$$\begin{aligned}\mathbf{S}_s = \mathbf{V}_s\mathbf{I}^* &= (220\angle 0^\circ)(10.67\angle -22.83^\circ) \\ &= 2347.4\angle -22.83^\circ = (2163.5 - j910.8)\text{ VA}\end{aligned}$$

From this, we obtain the real power as 2163.5 W and the reactive power as 910.8 VAR (leading).

(b) For the line, the voltage is

$$\begin{aligned}\mathbf{V}_{\text{line}} = (4 + j2)\mathbf{I} &= (4.472\angle 26.57^\circ)(10.67\angle 22.83^\circ) \\ &= 47.72\angle 49.4^\circ \text{ V rms}\end{aligned}$$

The complex power absorbed by the line is

$$\begin{aligned}\mathbf{S}_{\text{line}} = \mathbf{V}_{\text{line}}\mathbf{I}^* &= (47.72\angle 49.4^\circ)(10.67\angle -22.83^\circ) \\ &= 509.2\angle 26.57^\circ = 455.4 + j227.7\text{ VA}\end{aligned}$$

or

$$\mathbf{S}_{\text{line}} = |\mathbf{I}|^2\mathbf{Z}_{\text{line}} = (10.67)^2(4 + j2) = 455.4 + j227.7\text{ VA}$$

That is, the real power is 455.4 W and the reactive power is 227.76 VAR (lagging).

(c) For the load, the voltage is

$$\begin{aligned}\mathbf{V}_L = (15 - j10)\mathbf{I} &= (18.03\angle -33.7^\circ)(10.67\angle 22.83^\circ) \\ &= 192.38\angle -10.87^\circ \text{ V rms}\end{aligned}$$

The complex power absorbed by the load is

$$\begin{aligned}\mathbf{S}_L = \mathbf{V}_L\mathbf{I}^* &= (192.38\angle -10.87^\circ)(10.67\angle -22.83^\circ) \\ &= 2053\angle -33.7^\circ = (1708 - j1139)\text{ VA}\end{aligned}$$

The real power is 1708 W and the reactive power is 1139 VAR (leading). Note that $\mathbf{S}_s = \mathbf{S}_{\text{line}} + \mathbf{S}_L$, as expected. We have used the rms values of voltages and currents.

Practice Problem 11.13

In the circuit in Fig. 11.25, the 60-Ω resistor absorbs an average power of 240 W. Find **V** and the complex power of each branch of the circuit. What is the overall complex power of the circuit? (Assume the current through the 60-Ω resistor has no phase shift.)

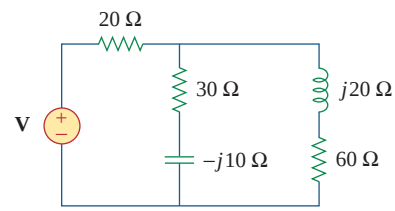


Figure 11.25
For Practice Prob. 11.13.

Answer: $240.67\angle 21.45^\circ$ V (rms); the 20-Ω resistor: 656 VA; the $(30 - j10)\ \Omega$ impedance: $480 - j160$ VA; the $(60 + j20)\ \Omega$ impedance: $240 + j80$ VA; overall: $1376 - j80$ VA.

Example 11.14

In the circuit of Fig. 11.26, $\mathbf{Z}_1 = 60\angle -30^\circ\ \Omega$ and $\mathbf{Z}_2 = 40\angle 45^\circ\ \Omega$. Calculate the total: (a) apparent power, (b) real power, (c) reactive power, and (d) pf, supplied by the source and seen by the source.

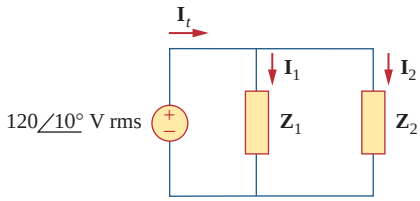


Figure 11.26
For Example 11.14.

Solution:

The current through $\mathbf{Z}_1$ is

$$\mathbf{I}_1 = \frac{\mathbf{V}}{\mathbf{Z}_1} = \frac{120\angle 10^\circ}{60\angle -30^\circ} = 2\angle 40^\circ \text{ A rms}$$

while the current through $\mathbf{Z}_2$ is

$$\mathbf{I}_2 = \frac{\mathbf{V}}{\mathbf{Z}_2} = \frac{120\angle 10^\circ}{40\angle 45^\circ} = 3\angle -35^\circ \text{ A rms}$$

The complex powers absorbed by the impedances are

$$\mathbf{S}_1 = \frac{V_{\text{rms}}^2}{\mathbf{Z}_1^*} = \frac{(120)^2}{60\angle 30^\circ} = 240\angle -30^\circ = 207.85 - j120 \text{ VA}$$

$$\mathbf{S}_2 = \frac{V_{\text{rms}}^2}{\mathbf{Z}_2^*} = \frac{(120)^2}{40\angle -45^\circ} = 360\angle 45^\circ = 254.6 + j254.6 \text{ VA}$$

The total complex power is

$$\mathbf{S}_t = \mathbf{S}_1 + \mathbf{S}_2 = 462.4 + j134.6 \text{ VA}$$

(a) The total apparent power is

$$|\mathbf{S}_t| = \sqrt{462.4^2 + 134.6^2} = 481.6 \text{ VA}.$$

(b) The total real power is

$$P_t = \text{Re}(\mathbf{S}_t) = 462.4 \text{ W or } P_t = P_1 + P_2.$$

(c) The total reactive power is

$$Q_t = \text{Im}(\mathbf{S}_t) = 134.6 \text{ VAR or } Q_t = Q_1 + Q_2.$$

(d) The pf $= P_t/|\mathbf{S}_t| = 462.4/481.6 = 0.96$ (lagging).

We may cross check the result by finding the complex power $\mathbf{S}_s$ supplied by the source.

$$\begin{aligned}\mathbf{I}_t = \mathbf{I}_1 + \mathbf{I}_2 &= (1.532 + j1.286) + (2.457 - j1.721)\\ &= 4 - j0.435 = 4.024\angle -6.21^\circ \text{ A rms}\\ \mathbf{S}_s = \mathbf{V}\mathbf{I}_t^* &= (120\angle 10^\circ)(4.024\angle 6.21^\circ)\\ &= 482.88\angle 16.21^\circ = 463 + j135 \text{ VA}\end{aligned}$$

which is the same as before.

Practice Problem 11.14

Two loads connected in parallel are respectively 2 kW at a pf of 0.75 leading and 4 kW at a pf of 0.95 lagging. Calculate the pf of the two loads. Find the complex power supplied by the source.

Answer: 0.9972 (leading), $6 - j0.4495$ kVA.

11.8 Power Factor Correction

Most domestic loads (such as washing machines, air conditioners, and refrigerators) and industrial loads (such as induction motors) are inductive and operate at a low lagging power factor. Although the inductive nature of the load cannot be changed, we can increase its power factor.

The process of increasing the power factor without altering the voltage or current to the original load is known as **power factor correction.**

Alternatively, power factor correction may be viewed as the addition of a reactive element (usually a capacitor) in parallel with the load in order to make the power factor closer to unity.

An inductive load is modeled as a series combination of an inductor and a resistor.

Since most loads are inductive, as shown in Fig. 11.27(a), a load's power factor is improved or corrected by deliberately installing a capacitor in parallel with the load, as shown in Fig. 11.27(b). The effect of adding the capacitor can be illustrated using either the power triangle or the phasor diagram of the currents involved. Figure 11.28 shows the latter, where it is assumed that the circuit in Fig. 11.27(a) has a power factor of $\cos\theta_1$, while the one in Fig. 11.27(b) has a power factor of $\cos\theta_2$. It is evident from Fig. 11.28 that adding the capacitor has caused the phase angle between the supplied voltage and current to reduce from θ_1 to θ_2, thereby increasing the power factor. We also notice from the magnitudes of the vectors in Fig. 11.28 that with the same supplied voltage, the circuit in Fig. 11.27(a) draws larger current I_L than the current I drawn by the circuit in Fig. 11.27(b). Power companies charge more for larger currents, because they result in increased power losses (by a squared factor, since $P = I_L^2 R$). Therefore, it is beneficial to both the power company and the consumer that every effort is made to minimize current level or keep the power factor as close to unity as possible. By choosing a suitable size for the capacitor, the current can be made to be completely in phase with the voltage, implying unity power factor.

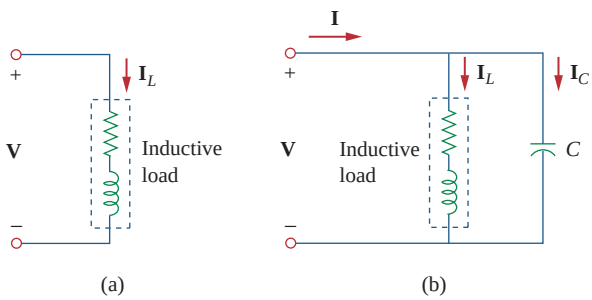


Figure 11.27
Power factor correction: (a) original inductive load, (b) inductive load with improved power factor.

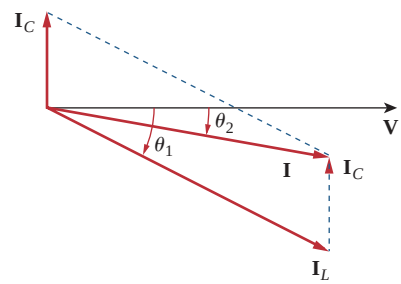


Figure 11.28
Phasor diagram showing the effect of adding a capacitor in parallel with the inductive load.

We can look at the power factor correction from another perspective. Consider the power triangle in Fig. 11.29. If the original inductive load has apparent power S_1, then

$$P = S_1 \cos\theta_1, \qquad Q_1 = S_1 \sin\theta_1 = P \tan\theta_1 \tag{11.57}$$

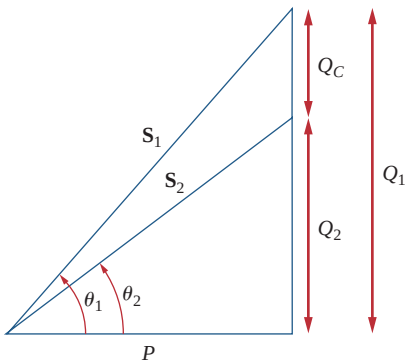


Figure 11.29
Power triangle illustrating power factor correction.

If we desire to increase the power factor from $\cos\theta_1$ to $\cos\theta_2$ without altering the real power (i.e., $P = S_2\cos\theta_2$), then the new reactive power is

$$Q_2 = P\tan\theta_2 \tag{11.58}$$

The reduction in the reactive power is caused by the shunt capacitor; that is,

$$Q_C = Q_1 - Q_2 = P(\tan\theta_1 - \tan\theta_2) \tag{11.59}$$

But from Eq. (11.46), $Q_C = V_{\text{rms}}^2/X_C = \omega C V_{\text{rms}}^2$. The value of the required shunt capacitance C is determined as

$$C = \frac{Q_C}{\omega V_{\text{rms}}^2} = \frac{P(\tan\theta_1 - \tan\theta_2)}{\omega V_{\text{rms}}^2} \tag{11.60}$$

Note that the real power P dissipated by the load is not affected by the power factor correction because the average power due to the capacitance is zero.

Although the most common situation in practice is that of an inductive load, it is also possible that the load is capacitive; that is, the load is operating at a leading power factor. In this case, an inductor should be connected across the load for power factor correction. The required shunt inductance L can be calculated from

$$Q_L = \frac{V_{\text{rms}}^2}{X_L} = \frac{V_{\text{rms}}^2}{\omega L} \quad \Rightarrow \quad L = \frac{V_{\text{rms}}^2}{\omega Q_L} \tag{11.61}$$

where $Q_L = Q_1 - Q_2$, the difference between the new and old reactive powers.

Example 11.15

When connected to a 120-V (rms), 60-Hz power line, a load absorbs 4 kW at a lagging power factor of 0.8. Find the value of capacitance necessary to raise the pf to 0.95.

Solution:
If the pf = 0.8, then

$$\cos\theta_1 = 0.8 \quad \Rightarrow \quad \theta_1 = 36.87^\circ$$

where θ_1 is the phase difference between voltage and current. We obtain the apparent power from the real power and the pf as

$$S_1 = \frac{P}{\cos\theta_1} = \frac{4000}{0.8} = 5000 \text{ VA}$$

The reactive power is

$$Q_1 = S_1\sin\theta = 5000\sin 36.87 = 3000 \text{ VAR}$$

When the pf is raised to 0.95,

$$\cos\theta_2 = 0.95 \quad \Rightarrow \quad \theta_2 = 18.19^\circ$$

The real power P has not changed. But the apparent power has changed; its new value is

$$S_2 = \frac{P}{\cos\theta_2} = \frac{4000}{0.95} = 4210.5 \text{ VA}$$

The new reactive power is

$$Q_2 = S_2 \sin\theta_2 = 1314.4 \text{ VAR}$$

The difference between the new and old reactive powers is due to the parallel addition of the capacitor to the load. The reactive power due to the capacitor is

$$Q_C = Q_1 - Q_2 = 3000 - 1314.4 = 1685.6 \text{ VAR}$$

and

$$C = \frac{Q_C}{\omega V_{\text{rms}}^2} = \frac{1685.6}{2\pi \times 60 \times 120^2} = 310.5\ \mu\text{F}$$

Note: Capacitors are normally purchased for voltages they expect to see. In this case, the maximum voltage this capacitor will see is about 170 V peak. We would suggest purchasing a capacitor with a voltage rating equal to, say, 200 V.

Practice Problem 11.15

Find the value of parallel capacitance needed to correct a load of 140 kVAR at 0.85 lagging pf to unity pf. Assume that the load is supplied by a 110-V (rms), 60-Hz line.

Answer: 30.69 mF.

11.9 †Applications

In this section, we consider two important application areas: how power is measured and how electric utility companies determine the cost of electricity consumption.

11.9.1 Power Measurement

The average power absorbed by a load is measured by an instrument called the *wattmeter*.

The **wattmeter** is the instrument for measuring the average power.

Figure 11.30 shows a wattmeter that consists essentially of two coils: the current coil and the voltage coil. A current coil with very low impedance (ideally zero) is connected in series with the load (Fig. 11.31) and responds to the load current. The voltage coil with very high impedance (ideally infinite) is connected in parallel with the load as shown in Fig. 11.31 and responds to the load voltage. The current coil acts like a short circuit because of its low impedance; the voltage coil

Reactive power is measured by an instrument called the *varmeter*. The varmeter is often connected to the load in the same way as the wattmeter.

Some wattmeters do not have coils; the wattmeter considered here is the electromagnetic type.

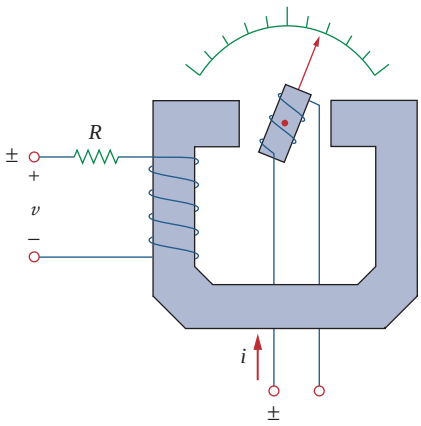


Figure 11.30
A wattmeter.

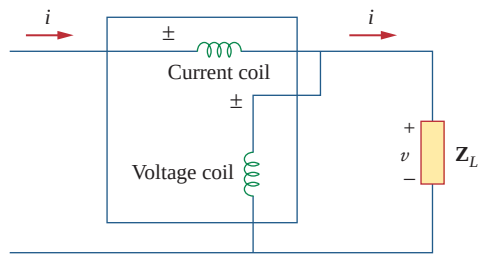


Figure 11.31
The wattmeter connected to the load.

behaves like an open circuit because of its high impedance. As a result, the presence of the wattmeter does not disturb the circuit or have an effect on the power measurement.

When the two coils are energized, the mechanical inertia of the moving system produces a deflection angle that is proportional to the average value of the product $v(t)i(t)$. If the current and voltage of the load are $v(t) = V_m \cos(\omega t + \theta_v)$ and $i(t) = I_m \cos(\omega t + \theta_i)$, their corresponding rms phasors are

$$\mathbf{V}_{\text{rms}} = \frac{V_m}{\sqrt{2}}\angle\theta_v \quad \text{and} \quad \mathbf{I}_{\text{rms}} = \frac{I_m}{\sqrt{2}}\angle\theta_i \qquad \textbf{(11.62)}$$

and the wattmeter measures the average power given by

$$P = |\mathbf{V}_{\text{rms}}||\mathbf{I}_{\text{rms}}| \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) \qquad \textbf{(11.63)}$$

As shown in Fig. 11.31, each wattmeter coil has two terminals with one marked ±. To ensure upscale deflection, the ± terminal of the current coil is toward the source, while the ± terminal of the voltage coil is connected to the same line as the current coil. Reversing both coil connections still results in upscale deflection. However, reversing one coil and not the other results in downscale deflection and no wattmeter reading.

Example 11.16

Find the wattmeter reading of the circuit in Fig. 11.32.

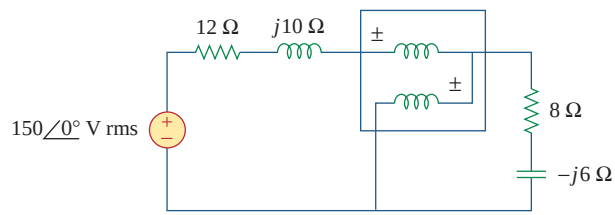


Figure 11.32
For Example 11.16.

Solution:

1. **Define.** The problem is clearly defined. Interestingly, this is a problem where the student could actually validate the results by doing the problem in the laboratory with a real wattmeter.

2. **Present.** This problem consists of finding the average power delivered to a load by an external source with a series impedance.
3. **Alternative.** This is a straightforward circuit problem where all we need to do is find the magnitude and phase of the current through the load and the magnitude and the phase of the voltage across the load. These quantities could also be found by using *PSpice*, which we will use as a check.
4. **Attempt.** In Fig. 11.32, the wattmeter reads the average power absorbed by the $(8 - j6)\ \Omega$ impedance because the current coil is in series with the impedance while the voltage coil is in parallel with it. The current through the circuit is

$$\mathbf{I}_{\text{rms}} = \frac{150\angle 0^\circ}{(12 + j10) + (8 - j6)} = \frac{150}{20 + j4}\ \text{A}$$

The voltage across the $(8 - j6)\ \Omega$ impedance is

$$\mathbf{V}_{\text{rms}} = \mathbf{I}_{\text{rms}}(8 - j6) = \frac{150(8 - j6)}{20 + j4}\ \text{V}$$

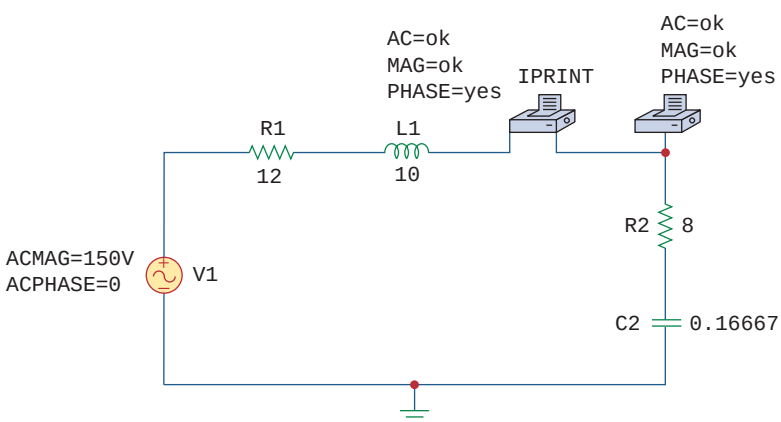
The complex power is

$$\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^* = \frac{150(8 - j6)}{20 + j4} \cdot \frac{150}{20 - j4} = \frac{150^2(8 - j6)}{20^2 + 4^2}$$
$$= 423.7 - j324.6\ \text{VA}$$

The wattmeter reads

$$P = \text{Re}(\mathbf{S}) = \mathbf{432.7\ W}$$

5. **Evaluate.** We can check our results by using *PSpice.*



Simulation yields:

FREQ	IM(V_PRINT2)	IP(V_PRINT2)
1.592E-01	7.354E+00	-1.131E+01

and

FREQ	VM($N_0004)	VP($N_0004)
1.592E-01	7.354E+01	-4.818E+01

To check our answer, all we need is the magnitude of the current (7.354 A) flowing through the load resistor:

$$P = (I_L)^2R = (7.354)^2 8 = \mathbf{432.7\ W}$$

As expected, the answer does check!

6. **Satisfactory?** We have satisfactorily solved the problem and the results can now be presented as a solution to the problem.

Practice Problem 11.16

For the circuit in Fig. 11.33, find the wattmeter reading.

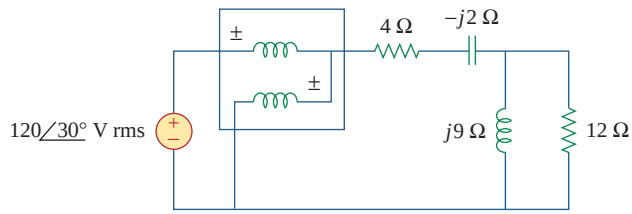


Figure 11.33
For Practice Prob. 11.16.

Answer: 1437 W.

11.9.2 Electricity Consumption Cost

In Section 1.7, we considered a simplified model of the way the cost of electricity consumption is determined. But the concept of power factor was not included in the calculations. Now we consider the importance of power factor in electricity consumption cost.

Loads with low power factors are costly to serve because they require large currents, as explained in Section 11.8. The ideal situation would be to draw minimum current from a supply so that $S = P$, $Q = 0$, and pf $= 1$. A load with nonzero Q means that energy flows back and forth between the load and the source, giving rise to additional power losses. In view of this, power companies often encourage their customers to have power factors as close to unity as possible and penalize some customers who do not improve their load power factors.

Utility companies divide their customers into categories: as residential (domestic), commercial, and industrial, or as small power, medium power, and large power. They have different rate structures for each category. The amount of energy consumed in units of kilowatt-hours (kWh) is measured using a kilowatt-hour meter installed at the customer's premises.

Although utility companies use different methods for charging customers, the tariff or charge to a consumer is often two-part. The first part is fixed and corresponds to the cost of generation, transmission, and distribution of electricity to meet the load requirements of the consumers. This part of the tariff is generally expressed as a certain price

per kW of maximum demand. Or it may be based on kVA of maximum demand, to account for the power factor (pf) of the consumer. A pf penalty charge may be imposed on the consumer whereby a certain percentage of kW or kVA maximum demand is charged for every 0.01 fall in pf below a prescribed value, say 0.85 or 0.9. On the other hand, a pf credit may be given for every 0.01 that the pf exceeds the prescribed value.

The second part is proportional to the energy consumed in kWh; it may be in graded form, for example, the first 100 kWh at 16 cents/kWh, the next 200 kWh at 10 cents/kWh and so forth. Thus, the bill is determined based on the following equation:

$$\text{Total Cost} = \text{Fixed Cost} + \text{Cost of Energy} \tag{11.64}$$

Example 11.17

A manufacturing industry consumes 200 MWh in one month. If the maximum demand is 1600 kW, calculate the electricity bill based on the following two-part rate:

Demand charge: $5.00 per month per kW of billing demand.
Energy charge: 8 cents per kWh for the first 50,000 kWh,
5 cents per kWh for the remaining energy.

Solution:

The demand charge is

$$\$5.00 \times 1600 = \$8000 \tag{11.17.1}$$

The energy charge for the first 50,000 kWh is

$$\$0.08 \times 50{,}000 = \$4000 \tag{11.17.2}$$

The remaining energy is 200,000 kWh − 50,000 kWh = 150,000 kWh, and the corresponding energy charge is

$$\$0.05 \times 150{,}000 = \$7500 \tag{11.17.3}$$

Adding the results of Eqs. (11.17.1) to (11.17.3) gives

$$\text{Total bill for the month} = \$8000 + \$4000 + \$7500 = \$19{,}500$$

It may appear that the cost of electricity is too high. But this is often a small fraction of the overall cost of production of the goods manufactured or the selling price of the finished product.

Practice Problem 11.17

The monthly reading of a paper mill's meter is as follows:

Maximum demand: 32,000 kW
Energy consumed: 500 MWh

Using the two-part rate in Example 11.17, calculate the monthly bill for the paper mill.

Answer: $186,500.

Example 11.18

A 300-kW load supplied at 13 kV (rms) operates 520 hours a month at 80 percent power factor. Calculate the average cost per month based on this simplified tariff:

Energy charge: 6 cents per kWh
Power-factor penalty: 0.1 percent of energy charge for every 0.01 that pf falls below 0.85.
Power-factor credit: 0.1 percent of energy charge for every 0.01 that pf exceeds 0.85.

Solution:
The energy consumed is

$$W = 300 \text{ kW} \times 520 \text{ h} = 156{,}000 \text{ kWh}$$

The operating power factor pf = 80% = 0.8 is 5×0.01 below the prescribed power factor of 0.85. Since there is 0.1 percent energy charge for every 0.01, there is a power-factor penalty charge of 0.5 percent. This amounts to an energy charge of

$$\Delta W = 156{,}000 \times \frac{5 \times 0.1}{100} = 780 \text{ kWh}$$

The total energy is

$$W_t = W + \Delta W = 156{,}000 + 780 = 156{,}780 \text{ kWh}$$

The cost per month is given by

$$\text{Cost} = 6 \text{ cents} \times W_t = \$0.06 \times 156{,}780 = \$9{,}406.80$$

Practice Problem 11.18

An 800-kW induction furnace at 0.88 power factor operates 20 hours per day for 26 days in a month. Determine the electricity bill per month based on the tariff in Example 11.18.

Answer: $24,885.12.

11.10 Summary

1. The instantaneous power absorbed by an element is the product of the element's terminal voltage and the current through the element:

$$p = vi.$$

2. Average or real power P (in watts) is the average of instantaneous power p:

$$P = \frac{1}{T}\int_0^T p\,dt$$

If $v(t) = V_m \cos(\omega t + \theta_v)$ and $i(t) = I_m \cos(\omega t + \theta_i)$, then $V_{\text{rms}} = V_m/\sqrt{2}$, $I_{\text{rms}} = I_m/\sqrt{2}$, and

$$P = \frac{1}{2}V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Inductors and capacitors absorb no average power, while the average power absorbed by a resistor is $(1/2)I_m^2R = I_{\text{rms}}^2R$.

3. Maximum average power is transferred to a load when the load impedance is the complex conjugate of the Thevenin impedance as seen from the load terminals, $\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^*$.
4. The effective value of a periodic signal $x(t)$ is its root-mean-square (rms) value.

$$X_{\text{eff}} = X_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T x^2\,dt}$$

For a sinusoid, the effective or rms value is its amplitude divided by $\sqrt{2}$.

5. The power factor is the cosine of the phase difference between voltage and current:

$$\text{pf} = \cos(\theta_v - \theta_i)$$

It is also the cosine of the angle of the load impedance or the ratio of real power to apparent power. The pf is lagging if the current lags voltage (inductive load) and is leading when the current leads voltage (capacitive load).

6. Apparent power S (in VA) is the product of the rms values of voltage and current:

$$S = V_{\text{rms}}I_{\text{rms}}$$

It is also given by $S = |\mathbf{S}| = \sqrt{P^2 + Q^2}$, where P is the real power and Q is reactive power.

7. Reactive power (in VAR) is:

$$Q = \frac{1}{2}V_mI_m\sin(\theta_v - \theta_i) = V_{\text{rms}}I_{\text{rms}}\sin(\theta_v - \theta_i)$$

8. Complex power $\mathbf{S}$ (in VA) is the product of the rms voltage phasor and the complex conjugate of the rms current phasor. It is also the complex sum of real power P and reactive power Q.

$$\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^* = V_{\text{rms}}I_{\text{rms}}\underline{/\theta_v - \theta_i} = P + jQ$$

Also,

$$\mathbf{S} = I_{\text{rms}}^2\mathbf{Z} = \frac{V_{\text{rms}}^2}{\mathbf{Z}^*}$$

9. The total complex power in a network is the sum of the complex powers of the individual components. Total real power and reactive power are also, respectively, the sums of the individual real powers and the reactive powers, but the total apparent power is not calculated by the process.
10. Power factor correction is necessary for economic reasons; it is the process of improving the power factor of a load by reducing the overall reactive power.
11. The wattmeter is the instrument for measuring the average power. Energy consumed is measured with a kilowatt-hour meter.

Review Questions

11.1 The average power absorbed by an inductor is zero.

(a) True (b) False

11.2 The Thevenin impedance of a network seen from the load terminals is $80 + j55\ \Omega$. For maximum power transfer, the load impedance must be:

(a) $-80 + j55\ \Omega$ (b) $-80 - j55\ \Omega$
(c) $80 - j55\ \Omega$ (d) $80 + j55\ \Omega$

11.3 The amplitude of the voltage available in the 60-Hz, 120-V power outlet in your home is:

(a) 110 V (b) 120 V
(c) 170 V (d) 210 V

11.4 If the load impedance is $20 - j20$, the power factor is

(a) $\angle -45°$ (b) 0 (c) 1
(d) 0.7071 (e) none of these

11.5 A quantity that contains all the power information in a given load is the

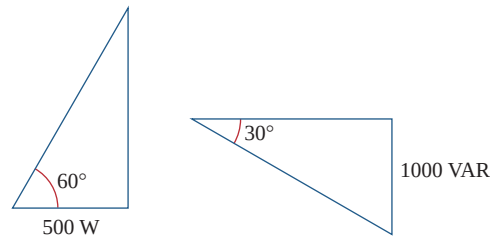
(a) power factor (b) apparent power
(c) average power (d) reactive power
(e) complex power

11.6 Reactive power is measured in:

(a) watts (b) VA
(c) VAR (d) none of these

11.7 In the power triangle shown in Fig. 11.34(a), the reactive power is:

(a) 1000 VAR leading (b) 1000 VAR lagging
(c) 866 VAR leading (d) 866 VAR lagging



(a) (b)

Figure 11.34
For Review Questions 11.7 and 11.8.

11.8 For the power triangle in Fig. 11.34(b), the apparent power is:

(a) 2000 VA (b) 1000 VAR
(c) 866 VAR (d) 500 VAR

11.9 A source is connected to three loads $\mathbf{Z}_1$, $\mathbf{Z}_2$, and $\mathbf{Z}_3$ in parallel. Which of these is not true?

(a) $P = P_1 + P_2 + P_3$ (b) $Q = Q_1 + Q_2 + Q_3$
(c) $S = S_1 + S_2 + S_3$ (d) $\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3$

11.10 The instrument for measuring average power is the:

(a) voltmeter (b) ammeter
(c) wattmeter (d) varmeter
(e) kilowatt-hour meter

Answers: 11.1a, 11.2c, 11.3c, 11.4d, 11.5e, 11.6c, 11.7d, 11.8a, 11.9c, 11.10c.

Problems[1]

Section 11.2 Instantaneous and Average Power

11.1 If $v(t) = 160 \cos 50t$ V and $i(t) = -20 \sin(50t - 30°)$ A, calculate the instantaneous power and the average power.

11.2 Given the circuit in Fig. 11.35, find the average power supplied or absorbed by each element.

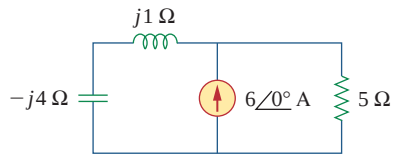


Figure 11.35
For Prob. 11.2.

11.3 A load consists of a 60-Ω resistor in parallel with a 90-μF capacitor. If the load is connected to a voltage source $v_s(t) = 40 \cos 2000t$, find the average power delivered to the load.

11.4 Using Fig. 11.36, design a problem to help other students better understand instantaneous and average power.

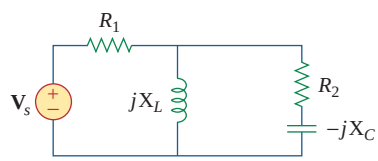


Figure 11.36
For Prob. 11.4.

[1] Starting with problem 11.22, unless otherwise specified, assume that all values of currents and voltages are rms.

11.5 Assuming that $v_s = 16\cos(2t - 40°)$ V in the circuit of Fig. 11.37, find the average power delivered to each of the passive elements.

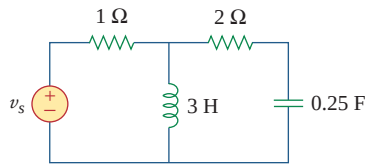


Figure 11.37
For Prob. 11.5.

11.6 For the circuit in Fig. 11.38, $i_s = 3\cos 10^3 t$ A. Find the average power absorbed by the 50-Ω resistor.

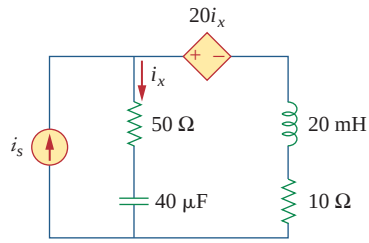


Figure 11.38
For Prob. 11.6.

11.7 Given the circuit of Fig. 11.39, find the average power absorbed by the 10-Ω resistor.

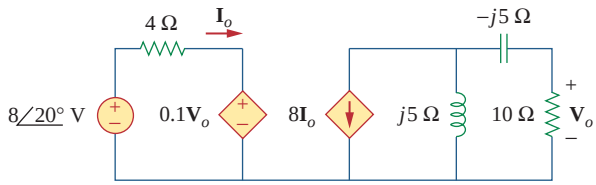


Figure 11.39
For Prob. 11.7.

11.8 In the circuit of Fig. 11.40, determine the average power absorbed by the 40-Ω resistor.

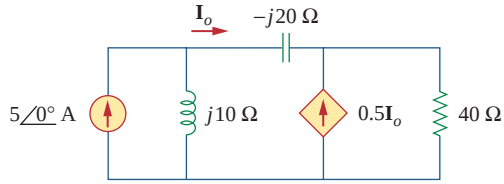


Figure 11.40
For Prob. 11.8.

11.9 For the op amp circuit in Fig. 11.41, $\mathbf{V}_s = 2\angle 30°$ V. Find the average power absorbed by the 20-kΩ resistor.

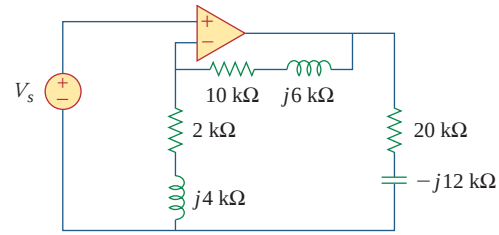


Figure 11.41
For Prob. 11.9.

11.10 In the op amp circuit in Fig. 11.42, find the total average power absorbed by the resistors.

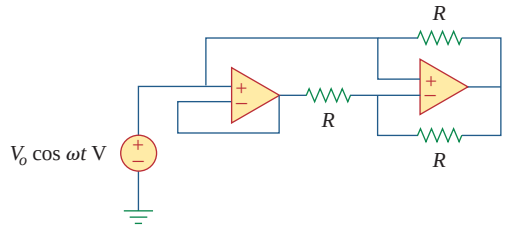


Figure 11.42
For Prob. 11.10.

11.11 For the network in Fig. 11.43, assume that the port impedance is

$$\mathbf{Z}_{ab} = \frac{R}{\sqrt{1 + \omega^2 R^2 C^2}} \angle -\tan^{-1} \omega RC$$

Find the average power consumed by the network when $R = 10$ kΩ, $C = 200$ nF, and $i = 2\sin(377t + 22°)$ mA.

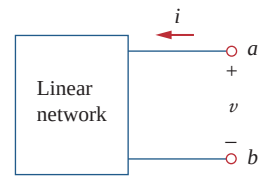


Figure 11.43
For Prob. 11.11.

Section 11.3 Maximum Average Power Transfer

11.12 For the circuit shown in Fig. 11.44, determine the load impedance **Z** for maximum power transfer (to **Z**). Calculate the maximum power absorbed by the load.

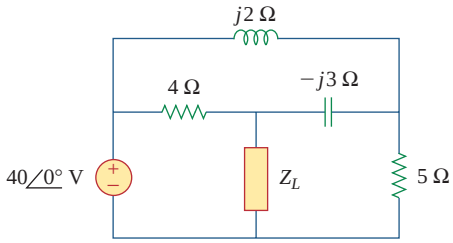


Figure 11.44
For Prob. 11.12.

11.13 The Thevenin impedance of a source is $\mathbf{Z}_{\text{Th}} = 120 + j60\ \Omega$, while the peak Thevenin voltage is $\mathbf{V}_{\text{Th}} = 110 + j0$ V. Determine the maximum available average power from the source.

11.14 Using Fig. 11.45, design a problem to help other students better understand maximum average power transfer.

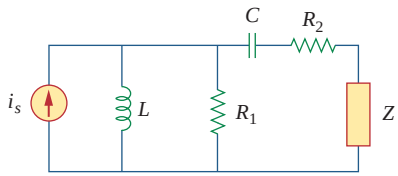


Figure 11.45
For Prob. 11.14.

11.15 In the circuit of Fig. 11.46, find the value of $\mathbf{Z}_L$ that will absorb the maximum power and the value of the maximum power.

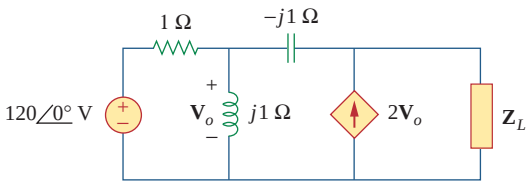


Figure 11.46
For Prob. 11.15.

11.16 For the circuit of Fig. 11.47, find the maximum power delivered to the load $\mathbf{Z}_L$.

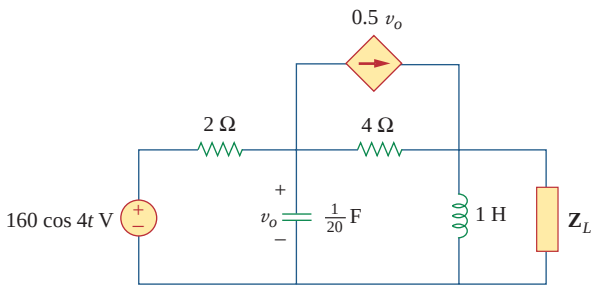


Figure 11.47
For Prob. 11.16.

11.17 Calculate the value of $\mathbf{Z}_L$ in the circuit of Fig. 11.48 in order for $\mathbf{Z}_L$ to receive maximum average power. What is the maximum average power received by $\mathbf{Z}_L$?

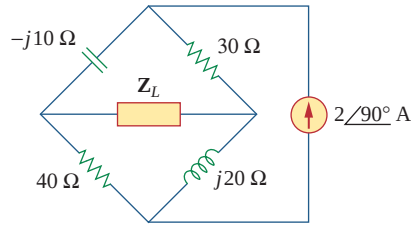


Figure 11.48
For Prob. 11.17.

11.18 Find the value of $\mathbf{Z}_L$ in the circuit of Fig. 11.49 for maximum power transfer.

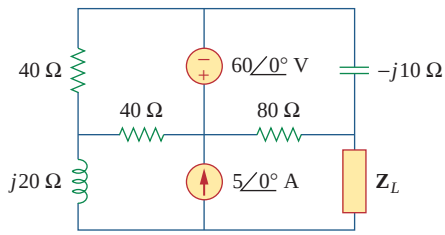


Figure 11.49
For Prob. 11.18.

11.19 The variable resistor R in the circuit of Fig. 11.50 is adjusted until it absorbs the maximum average power. Find R and the maximum average power absorbed.

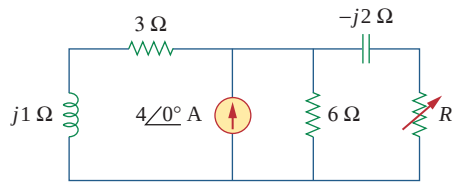


Figure 11.50
For Prob. 11.19.

11.20 The load resistance R_L in Fig. 11.51 is adjusted until it absorbs the maximum average power. Calculate the value of R_L and the maximum average power.

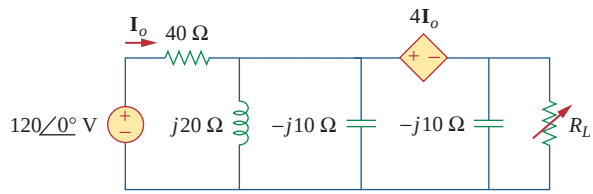


Figure 11.51
For Prob. 11.20.

11.21 Assuming that the load impedance is to be purely resistive, what load should be connected to terminals *a-b* of the circuits in Fig. 11.52 so that the maximum power is transferred to the load?

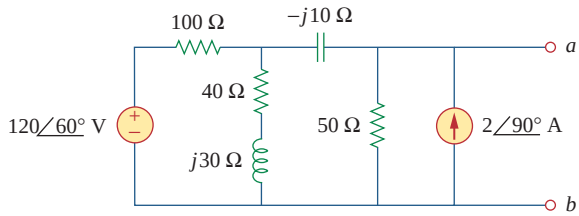


Figure 11.52
For Prob. 11.21.

Section 11.4 Effective or RMS Value

11.22 Find the rms value of the offset sine wave shown in Fig. 11.53.

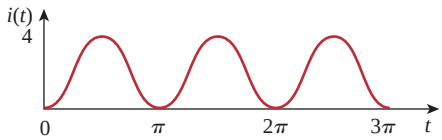


Figure 11.53
For Prob. 11.22.

11.23 Using Fig. 11.54, design a problem to help other students better understand how to find the rms value of a waveshape.

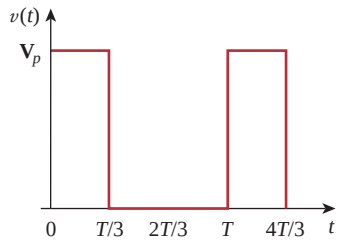


Figure 11.54
For Prob. 11.23.

11.24 Determine the rms value of the waveform in Fig. 11.55.

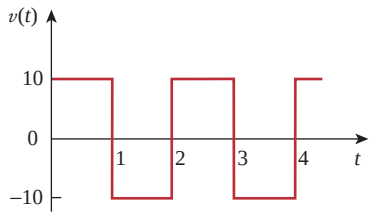


Figure 11.55
For Prob. 11.24.

11.25 Find the rms value of the signal shown in Fig. 11.56.

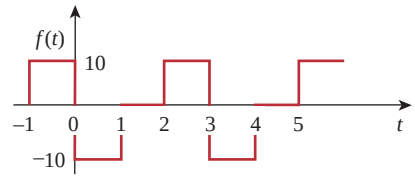


Figure 11.56
For Prob. 11.25.

11.26 Find the effective value of the voltage waveform in Fig. 11.57.

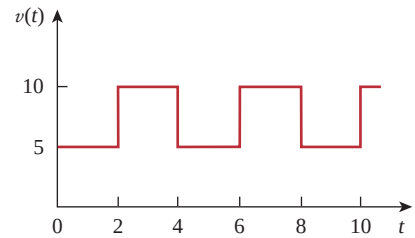


Figure 11.57
For Prob. 11.26.

11.27 Calculate the rms value of the current waveform of Fig. 11.58.

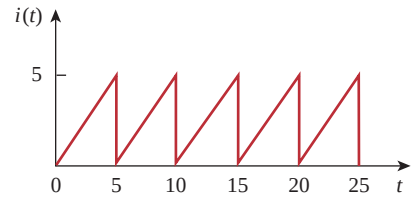


Figure 11.58
For Prob. 11.27.

11.28 Find the rms value of the voltage waveform of Fig. 11.59 as well as the average power absorbed by a 2-Ω resistor when the voltage is applied across the resistor.

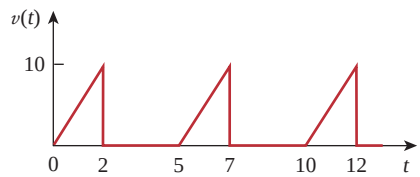


Figure 11.59
For Prob. 11.28.

11.29 Calculate the effective value of the current waveform in Fig. 11.60 and the average power delivered to a 12-Ω resistor when the current runs through the resistor.

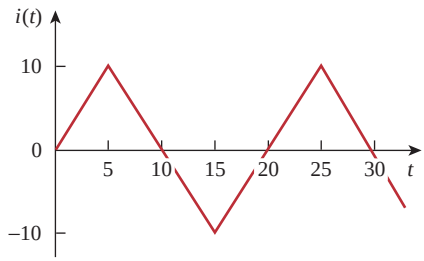


Figure 11.60
For Prob. 11.29.

11.30 Compute the rms value of the waveform depicted in Fig. 11.61.

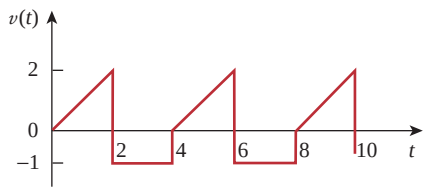


Figure 11.61
For Prob. 11.30.

11.31 Find the rms value of the signal shown in Fig. 11.62.

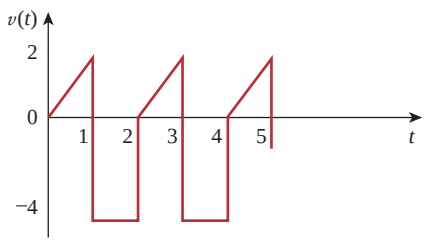


Figure 11.62
For Prob. 11.31.

11.32 Obtain the rms value of the current waveform shown in Fig. 11.63.

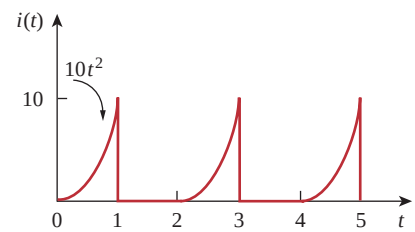


Figure 11.63
For Prob. 11.32.

11.33 Determine the rms value for the waveform in Fig. 11.64.

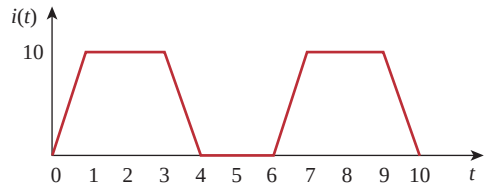


Figure 11.64
For Prob. 11.33.

11.34 Find the effective value of $f(t)$ defined in Fig. 11.65.

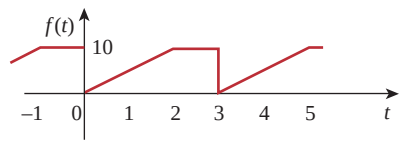


Figure 11.65
For Prob. 11.34.

11.35 One cycle of a periodic voltage waveform is depicted in Fig. 11.66. Find the effective value of the voltage. Note that the cycle starts at $t = 0$ and ends at $t = 6$ s.

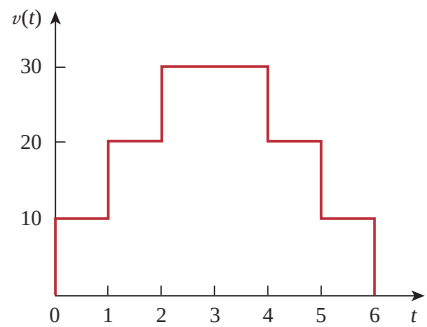


Figure 11.66
For Prob. 11.35.

11.36 Calculate the rms value for each of the following functions:

(a) $i(t) = 10$ A (b) $v(t) = 4 + 3\cos 5t$ V

(c) $i(t) = 8 - 6\sin 2t$ A (d) $v(t) = 5\sin t + 4\cos t$ V

11.37 Design a problem to help other students better understand how to determine the rms value of the sum of multiple currents.

Section 11.5 Apparent Power and Power Factor

11.38 For the power system in Fig. 11.67, find: (a) the average power, (b) the reactive power, (c) the power factor. Note that 220 V is an rms value.

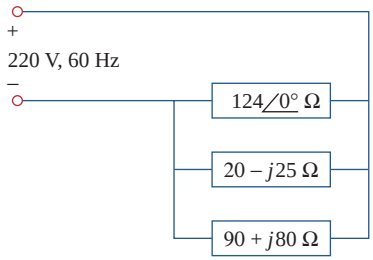


Figure 11.67
For Prob. 11.38.

11.39 An ac motor with impedance $\mathbf{Z}_L = 4.2 + j3.6\ \Omega$ is supplied by a 220-V, 60-Hz source. (a) Find pf, P, and Q. (b) Determine the capacitor required to be connected in parallel with the motor so that the power factor is corrected to unity.

11.40 Design a problem to help other students better understand apparent power and power factor.

11.41 Obtain the power factor for each of the circuits in Fig. 11.68. Specify each power factor as leading or lagging.

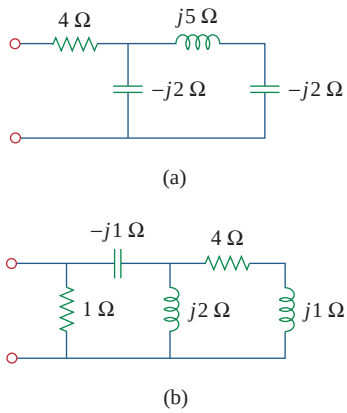


Figure 11.68
For Prob. 11.41.

Section 11.6 Complex Power

11.42 A 110-V rms, 60-Hz source is applied to a load impedance $\mathbf{Z}$. The apparent power entering the load is 120 VA at a power factor of 0.707 lagging.

(a) Calculate the complex power.

(b) Find the rms current supplied to the load.

(c) Determine $\mathbf{Z}$.

(d) Assuming that $\mathbf{Z} = R + j\omega L$, find the values of R and L.

11.43 Design a problem to help other students understand complex power.

11.44 Find the complex power delivered by v_s to the network in Fig. 11.69. Let $v_s = 160 \cos 2000t$ V.

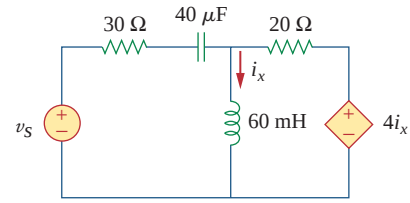


Figure 11.69
For Prob. 11.44.

11.45 The voltage across a load and the current through it are given by

$$v(t) = 20 + 60 \cos 100t \text{ V}$$
$$i(t) = 1 - 0.5 \sin 100t \text{ A}$$

Find:

(a) the rms values of the voltage and of the current

(b) the average power dissipated in the load

11.46 For the following voltage and current phasors, calculate the complex power, apparent power, real power, and reactive power. Specify whether the pf is leading or lagging.

(a) $\mathbf{V} = 220\angle 30°$ V rms, $\mathbf{I} = 0.5\angle 60°$ A rms

(b) $\mathbf{V} = 250\angle -10°$ V rms, $\mathbf{I} = 6.2\angle -25°$ A rms

(c) $\mathbf{V} = 120\angle 0°$ V rms, $\mathbf{I} = 2.4\angle -15°$ A rms

(d) $\mathbf{V} = 160\angle 45°$ V rms, $\mathbf{I} = 8.5\angle 90°$ A rms

11.47 For each of the following cases, find the complex power, the average power, and the reactive power:

(a) $v(t) = 112 \cos(\omega t + 10°)$ V, $i(t) = 4 \cos(\omega t - 50°)$ A

(b) $v(t) = 160 \cos 377t$ V, $i(t) = 4 \cos(377t + 45°)$ A

(c) $\mathbf{V} = 80\angle 60°$ V rms, $\mathbf{Z} = 50\angle 30°\ \Omega$

(d) $\mathbf{I} = 10\angle 60°$ A rms, $\mathbf{Z} = 100\angle 45°\ \Omega$

11.48 Determine the complex power for the following cases:

(a) $P = 269$ W, $Q = 150$ VAR (capacitive)

(b) $Q = 2000$ VAR, pf $= 0.9$ (leading)

(c) $S = 600$ VA, $Q = 450$ VAR (inductive)

(d) $V_{\text{rms}} = 220$ V, $P = 1$ kW, $|\mathbf{Z}| = 40\ \Omega$ (inductive)

11.49 Find the complex power for the following cases:

(a) $P = 4$ kW, pf $= 0.86$ (lagging)

(b) $S = 2$ kVA, $P = 1.6$ kW (capacitive)

(c) $\mathbf{V}_{rms} = 208\angle 20°$ V, $\mathbf{I}_{rms} = 6.5\angle -50°$ A

(d) $\mathbf{V}_{rms} = 120\angle 30°$ V, $\mathbf{Z} = 40 + j60\ \Omega$

11.50 Obtain the overall impedance for the following cases:

(a) $P = 1000$ W, pf $= 0.8$ (leading), $V_{rms} = 220$ V

(b) $P = 1500$ W, $Q = 2000$ VAR (inductive), $I_{rms} = 12$ A

(c) $\mathbf{S} = 4500\angle 60°$ VA, $\mathbf{V} = 120\angle 45°$ V

11.51 For the entire circuit in Fig. 11.70, calculate:

(a) the power factor

(b) the average power delivered by the source

(c) the reactive power

(d) the apparent power

(e) the complex power

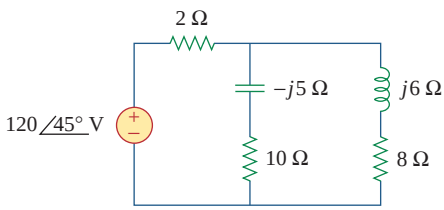


Figure 11.70
For Prob. 11.51.

11.52 In the circuit of Fig. 11.71, device *A* receives 2 kW at 0.8 pf lagging, device *B* receives 3 kVA at 0.4 pf leading, while device *C* is inductive and consumes 1 kW and receives 500 VAR.

(a) Determine the power factor of the entire system.

(b) Find $\mathbf{I}$ given that $\mathbf{V}_s = 240\angle 45°$ V rms.

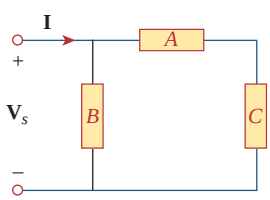


Figure 11.71
For Prob. 11.52.

11.53 In the circuit of Fig. 11.72, load *A* receives 4 kVA at 0.8 pf leading. Load *B* receives 2.4 kVA at 0.6 pf lagging. Box *C* is an inductive load that consumes 1 kW and receives 500 VAR.

(a) Determine $\mathbf{I}$.

(b) Calculate the power factor of the combination.

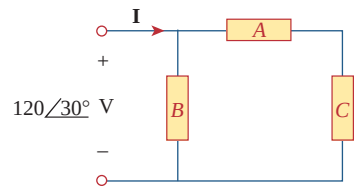


Figure 11.72
For Prob. 11.53.

Section 11.7 Conservation of AC Power

11.54 For the network in Fig. 11.73, find the complex power absorbed by each element.

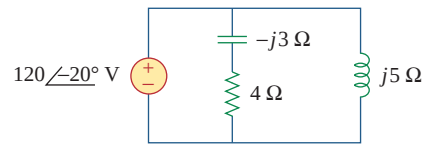


Figure 11.73
For Prob. 11.54.

11.55 Using Fig. 11.74, design a problem to help other students better understand the conservation of AC power.

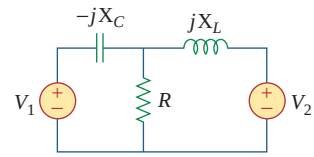


Figure 11.74
For Prob. 11.55.

11.56 Obtain the complex power delivered by the source in the circuit of Fig. 11.75.

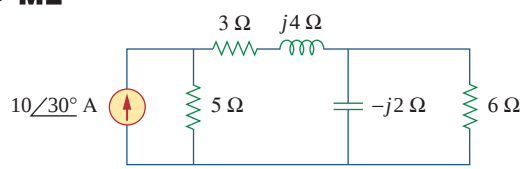


Figure 11.75
For Prob. 11.56.

11.57 For the circuit in Fig. 11.76, find the average, reactive, and complex power delivered by the dependent current source.

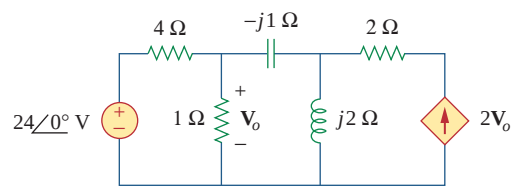


Figure 11.76
For Prob. 11.57.

11.58 Obtain the complex power delivered to the 10-kΩ resistor in Fig. 11.77 below.

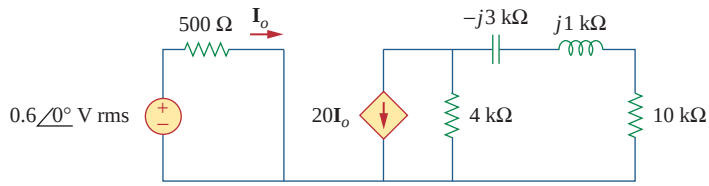


Figure 11.77
For Prob. 11.58.

11.59 Calculate the reactive power in the inductor and capacitor in the circuit of Fig. 11.78.

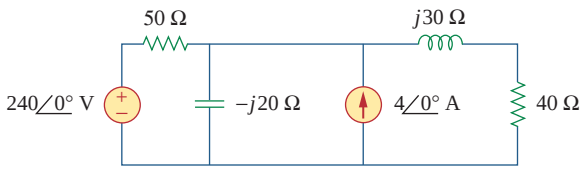


Figure 11.78
For Prob. 11.59.

11.60 For the circuit in Fig. 11.79, find $\mathbf{V}_o$ and the input power factor.

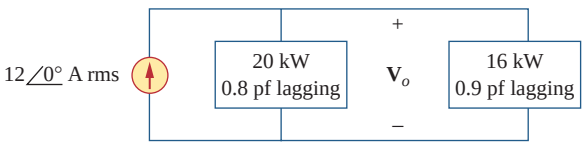


Figure 11.79
For Prob. 11.60.

11.61 Given the circuit in Fig. 11.80, find $\mathbf{I}_o$ and the overall complex power supplied.

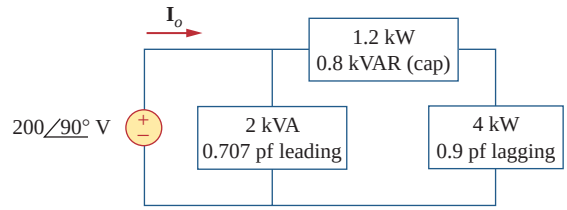


Figure 11.80
For Prob. 11.61.

11.62 For the circuit in Fig. 11.81, find $\mathbf{V}_s$.

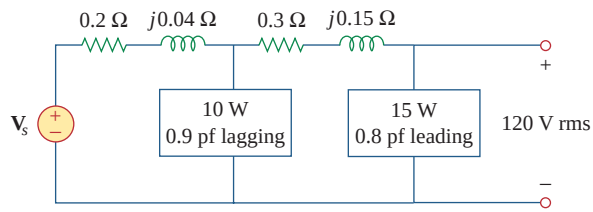


Figure 11.81
For Prob. 11.62.

11.63 Find $\mathbf{I}_o$ in the circuit of Fig. 11.82.

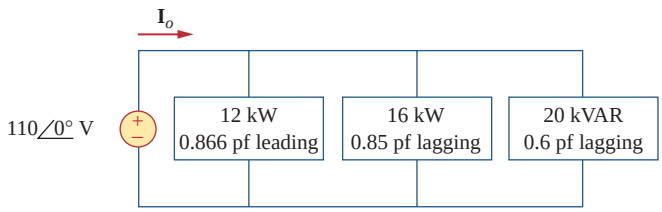


Figure 11.82
For Prob. 11.63.

11.64 Determine $\mathbf{I}_s$ in the circuit of Fig. 11.83, if the voltage source supplies 2.5 kW and 0.4 kVAR (leading).

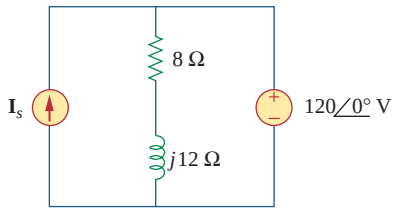


Figure 11.83
For Prob. 11.64.

11.65 In the op amp circuit of Fig. 11.84, $v_s = 4\cos 10^4 t$ V. Find the average power delivered to the 50-kΩ resistor.

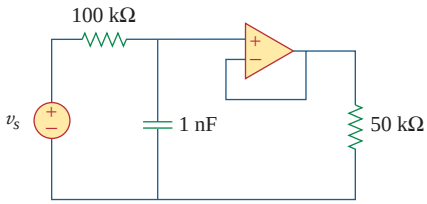


Figure 11.84
For Prob. 11.65.

11.66 Obtain the average power absorbed by the 6-kΩ resistor in the op amp circuit in Fig. 11.85.

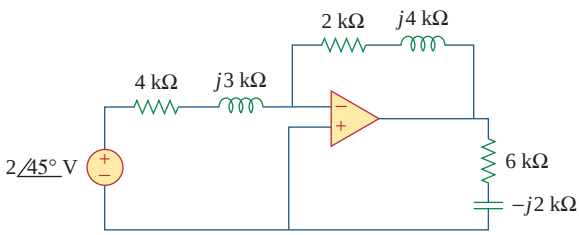


Figure 11.85
For Prob. 11.66.

11.67 For the op amp circuit in Fig. 11.86, calculate:

(a) the complex power delivered by the voltage source

(b) the average power dissipated in the 12-Ω resistor

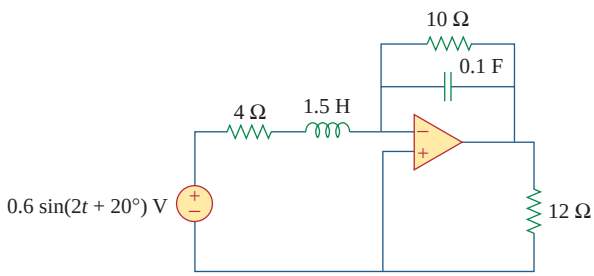


Figure 11.86
For Prob. 11.67.

11.68 Compute the complex power supplied by the current source in the series *RLC* circuit in Fig. 11.87.

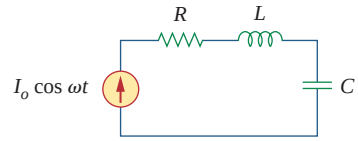


Figure 11.87
For Prob. 11.68.

Section 11.8 Power Factor Correction

11.69 Refer to the circuit shown in Fig. 11.88.

(a) What is the power factor?

(b) What is the average power dissipated?

(c) What is the value of the capacitance that will give a unity power factor when connected to the load?

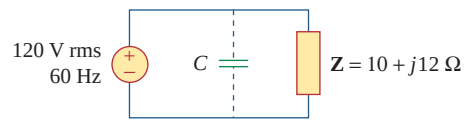


Figure 11.88
For Prob. 11.69.

11.70 Design a problem to help other students better understand power factor correction.

11.71 Three loads are connected in parallel to a $120\angle 0°$ V rms source. Load 1 absorbs 60 kVAR at pf = 0.85 lagging, load 2 absorbs 90 kW and 50 kVAR leading, and load 3 absorbs 100 kW at pf = 1. (a) Find the equivalent impedance. (b) Calculate the power factor of the parallel combination. (c) Determine the current supplied by the source.

11.72 Two loads connected in parallel draw a total of 2.4 kW at 0.8 pf lagging from a 120-V rms, 60-Hz line. One load absorbs 1.5 kW at a 0.707 pf lagging. Determine: (a) the pf of the second load, (b) the parallel element required to correct the pf to 0.9 lagging for the two loads.

11.73 A 240-V rms 60-Hz supply serves a load that is 10 kW (resistive), 15 kVAR (capacitive), and 22 kVAR (inductive). Find:

(a) the apparent power

(b) the current drawn from the supply

(c) the kVAR rating and capacitance required to improve the power factor to 0.96 lagging

(d) the current drawn from the supply under the new power-factor conditions

11.74 A 120-V rms 60-Hz source supplies two loads connected in parallel, as shown in Fig. 11.89.

(a) Find the power factor of the parallel combination.

(b) Calculate the value of the capacitance connected in parallel that will raise the power factor to unity.

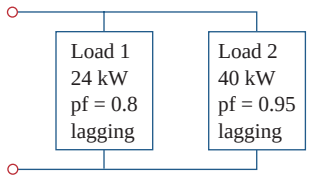


Figure 11.89
For Prob. 11.74.

11.75 Consider the power system shown in Fig. 11.90. Calculate:

(a) the total complex power

(b) the power factor

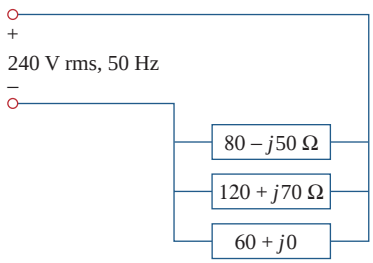


Figure 11.90
For Prob. 11.75.

Section 11.9 Applications

11.76 Obtain the wattmeter reading of the circuit in Fig. 11.91.

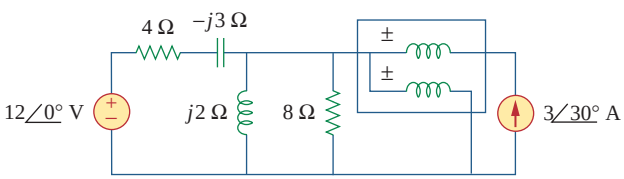


Figure 11.91
For Prob. 11.76.

11.77 What is the reading of the wattmeter in the network of Fig. 11.92?

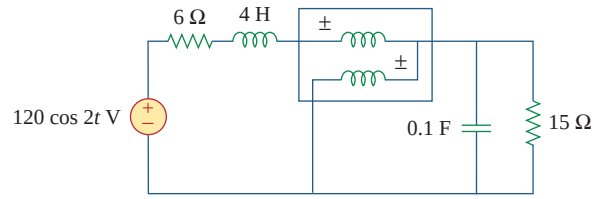


Figure 11.92
For Prob. 11.77.

11.78 Find the wattmeter reading of the circuit shown in Fig. 11.93.

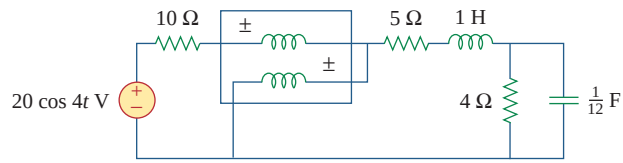


Figure 11.93
For Prob. 11.78.

11.79 Determine the wattmeter reading of the circuit in Fig. 11.94.

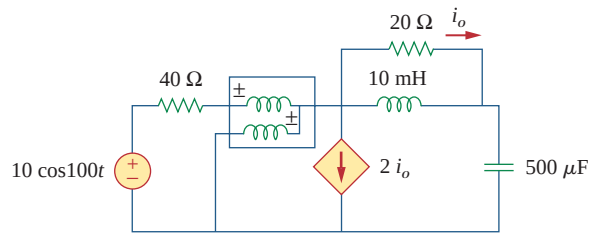


Figure 11.94
For Prob. 11.79.

11.80 The circuit of Fig. 11.95 portrays a wattmeter connected into an ac network.

(a) Find the load current.

(b) Calculate the wattmeter reading.

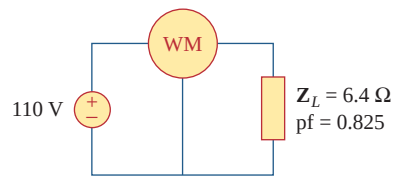


Figure 11.95
For Prob. 11.80.

11.81 Design a problem to help other students better understand how to correct power factor to values other than unity.

11.82 A 240-V rms 60-Hz source supplies a parallel combination of a 5-kW heater and a 30-kVA induction motor whose power factor is 0.82. Determine:

(a) the system apparent power

(b) the system reactive power

(c) the kVA rating of a capacitor required to adjust the system power factor to 0.9 lagging

(d) the value of the capacitor required

11.83 Oscilloscope measurements indicate that the peak voltage across a load and the peak current through it are, respectively, $210\angle 60°$ V and $8\angle 25°$ A. Determine:

(a) the real power

(b) the apparent power

(c) the reactive power

(d) the power factor

11.84 A consumer has an annual consumption of 1200 MWh with a maximum demand of 2.4 MVA. The maximum demand charge is $30 per kVA per annum, and the energy charge per kWh is 4 cents.

(a) Determine the annual cost of energy.

(b) Calculate the charge per kWh with a flat-rate tariff if the revenue to the utility company is to remain the same as for the two-part tariff.

11.85 A regular household system of a single-phase three-wire circuit allows the operation of both 120-V and 240-V, 60-Hz appliances. The household circuit is modeled as shown in Fig. 11.96. Calculate:

(a) the currents $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_n$

(b) the total complex power supplied

(c) the overall power factor of the circuit

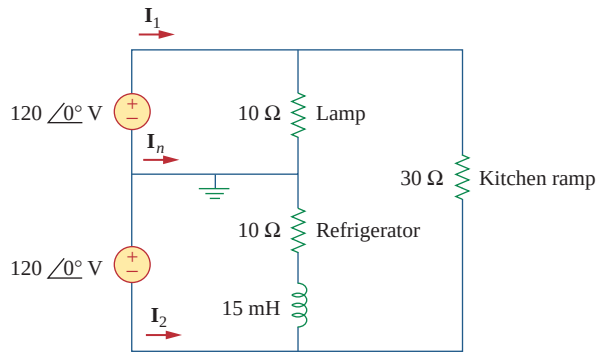


Figure 11.96
For Prob. 11.85.

Comprehensive Problems

11.86 A transmitter delivers maximum power to an antenna when the antenna is adjusted to represent a load of 75-Ω resistance in series with an inductance of 4 μH. If the transmitter operates at 4.12 MHz, find its internal impedance.

11.87 In a TV transmitter, a series circuit has an impedance of 3 kΩ and a total current of 50 mA. If the voltage across the resistor is 80 V, what is the power factor of the circuit?

11.88 A certain electronic circuit is connected to a 110-V ac line. The root-mean-square value of the current drawn is 2 A, with a phase angle of 55°.

(a) Find the true power drawn by the circuit.

(b) Calculate the apparent power.

11.89 An industrial heater has a nameplate that reads: 210 V 60 Hz 12 kVA 0.78 pf lagging
Determine:

(a) the apparent and the complex power

(b) the impedance of the heater

***11.90** A 2000-kW turbine-generator of 0.85 power factor operates at the rated load. An additional load of 300 kW at 0.8 power factor is added.What kVAR of capacitors is required to operate the turbine-generator but keep it from being overloaded?

11.91 The nameplate of an electric motor has the following information:

Line voltage: 220 V rms
Line current: 15 A rms
Line frequency: 60 Hz
Power: 2700 W

Determine the power factor (lagging) of the motor. Find the value of the capacitance C that must be connected across the motor to raise the pf to unity.

11.92 As shown in Fig. 11.97, a 550-V feeder line supplies an industrial plant consisting of a motor drawing 60 kW at 0.75 pf (inductive), a capacitor with a rating of 20 kVAR, and lighting drawing 20 kW.

(a) Calculate the total reactive power and apparent power absorbed by the plant.

(b) Determine the overall pf.

(c) Find the current in the feeder line.

* An asterisk indicates a challenging problem.

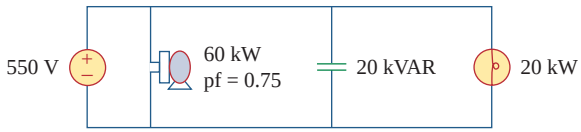


Figure 11.97
For Prob. 11.92.

11.93 A factory has the following four major loads:

- A motor rated at 5 hp, 0.8 pf lagging (1 hp = 0.7457 kW).
- A heater rated at 1.2 kW, 1.0 pf.
- Ten 120-W lightbulbs.
- A synchronous motor rated at 1.6 kVAR, 0.6 pf leading.

(a) Calculate the total real and reactive power.

(b) Find the overall power factor.

11.94 A 1-MVA substation operates at full load at 0.7 power factor. It is desired to improve the power factor to 0.95 by installing capacitors. Assume that new substation and distribution facilities cost $120 per kVA installed, and capacitors cost $30 per kVA installed.

(a) Calculate the cost of capacitors needed.

(b) Find the savings in substation capacity released.

(c) Are capacitors economical for releasing the amount of substation capacity?

11.95 A coupling capacitor is used to block dc current from an amplifier as shown in Fig. 11.98(a). The amplifier and the capacitor act as the source, while the speaker is the load as in Fig. 11.98(b).

(a) At what frequency is maximum power transferred to the speaker?

(b) If $V_s = 4.6$ V rms, how much power is delivered to the speaker at that frequency?

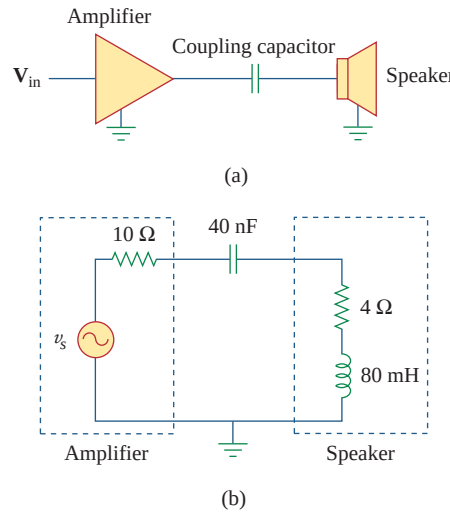


Figure 11.98
For Prob. 11.95.

11.96 A power amplifier has an output impedance of $40 + j8\ \Omega$. It produces a no-load output voltage of 146 V at 300 Hz.

(a) Determine the impedance of the load that achieves maximum power transfer.

(b) Calculate the load power under this matching condition.

11.97 A power transmission system is modeled as shown in Fig. 11.99. If $\mathbf{V}_s = 240\angle 0°$ rms, find the average power absorbed by the load.

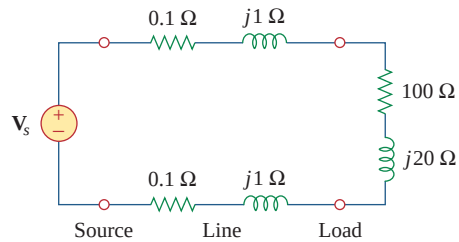


Figure 11.99
For Prob. 11.97.

chapter

12

Three-Phase Circuits

He who cannot forgive others breaks the bridge over which he must pass himself.

—G. Herbert

Enhancing Your Skills and Your Career

ABET EC 2000 criteria (3.e), *"an ability to identify, formulate, and solve engineering problems."*

Developing and enhancing your "ability to identify, formulate, and solve engineering problems" is a primary focus of textbook. Following our six step problem-solving process is the best way to practice this skill. Our recommendation is that you use this process whenever possible. You may be pleased to learn that this process works well for nonengineering courses.

ABET EC 2000 criteria (f), *"an understanding of professional and ethical responsibility."*

"An understanding of professional and ethical responsibility" is required of every engineer. To some extent, this understanding is very personal for each of us. Let us identify some pointers to help you develop this understanding. One of my favorite examples is that an engineer has the responsibility to answer what I call the "unasked question." For instance, assume that you own a car that has a problem with the transmission. In the process of selling that car, the prospective buyer asks you if there is a problem in the right-front wheel bearing. You answer no. However, as an engineer, you are required to inform the buyer that there is a problem with the transmission without being asked.

Your responsibility both professionally and ethically is to perform in a manner that does not harm those around you and to whom you are responsible. Clearly, developing this capability will take time and maturity on your part. I recommend practicing this by looking for professional and ethical components in your day-to-day activities.



Photo by Charles Alexander

12.1 Introduction

So far in this text, we have dealt with single-phase circuits. A single-phase ac power system consists of a generator connected through a pair of wires (a transmission line) to a load. Figure 12.1(a) depicts a single-phase two-wire system, where V_p is the rms magnitude of the source voltage and ϕ is the phase. What is more common in practice is a single-phase three-wire system, shown in Fig. 12.1(b). It contains two identical sources (equal magnitude and the same phase) that are connected to two loads by two outer wires and the neutral. For example, the normal household system is a single-phase three-wire system because the terminal voltages have the same magnitude and the same phase. Such a system allows the connection of both 120-V and 240-V appliances.

Historical note: Thomas Edison invented a *three-wire system*, using three wires instead of four.

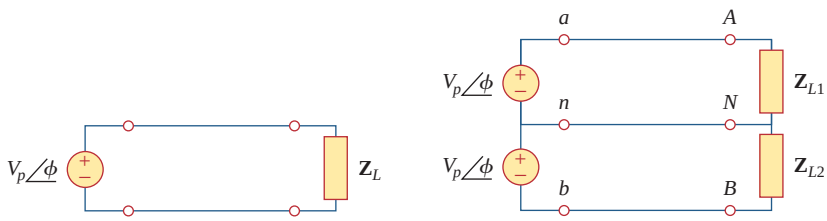


Figure 12.1
Single-phase systems: (a) two-wire type, (b) three-wire type.

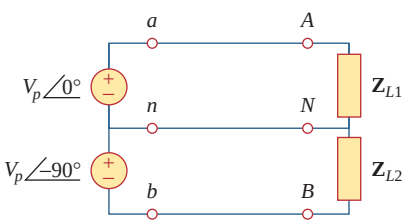


Figure 12.2
Two-phase three-wire system.

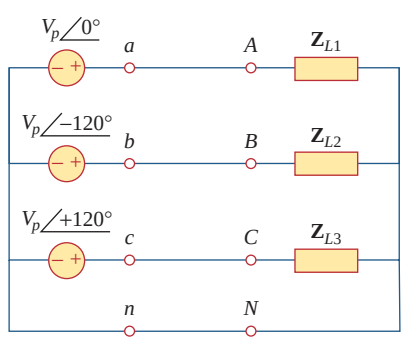
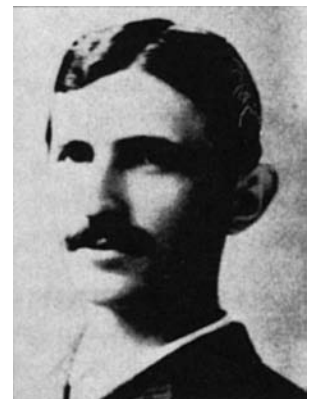


Figure 12.3
Three-phase four-wire system.

Circuits or systems in which the ac sources operate at the same frequency but different phases are known as *polyphase*. Figure 12.2 shows a two-phase three-wire system, and Fig. 12.3 shows a three-phase four-wire system. As distinct from a single-phase system, a two-phase system is produced by a generator consisting of two coils placed perpendicular to each other so that the voltage generated by one lags the other by 90°. By the same token, a three-phase system is produced by a generator consisting of three sources having the same amplitude and frequency but out of phase with each other by 120°. Since the three-phase system is by far the most prevalent and most economical polyphase system, discussion in this chapter is mainly on three-phase systems.

Three-phase systems are important for at least three reasons. First, nearly all electric power is generated and distributed in three-phase, at the operating frequency of 60 Hz (or $\omega = 377$ rad/s) in the United States or 50 Hz (or $\omega = 314$ rad/s) in some other parts of the world. When one-phase or two-phase inputs are required, they are taken from the three-phase system rather than generated independently. Even when more than three phases are needed—such as in the aluminum industry, where 48 phases are required for melting purposes—they can be provided by manipulating the three phases supplied. Second, the instantaneous power in a three-phase system can be constant (not pulsating), as we will see in Section 12.7. This results in uniform power transmission and less vibration of three-phase machines. Third, for the same amount of power, the three-phase system is more economical than the single-phase. The amount of wire required for a three-phase system is less than that required for an equivalent single-phase system.

Historical



Courtesy Smithsonian Institution

Nikola Tesla (1856–1943) was a Croatian-American engineer whose inventions—among them the induction motor and the first polyphase ac power system—greatly influenced the settlement of the ac versus dc debate in favor of ac. He was also responsible for the adoption of 60 Hz as the standard for ac power systems in the United States.

Born in Austria-Hungary (now Croatia), to a clergyman, Tesla had an incredible memory and a keen affinity for mathematics. He moved to the United States in 1884 and first worked for Thomas Edison. At that time, the country was in the "battle of the currents" with George Westinghouse (1846–1914) promoting ac and Thomas Edison rigidly leading the dc forces. Tesla left Edison and joined Westinghouse because of his interest in ac. Through Westinghouse, Tesla gained the reputation and acceptance of his polyphase ac generation, transmission, and distribution system. He held 700 patents in his lifetime. His other inventions include high-voltage apparatus (the tesla coil) and a wireless transmission system. The unit of magnetic flux density, the tesla, was named in honor of him.

We begin with a discussion of balanced three-phase voltages. Then we analyze each of the four possible configurations of balanced three-phase systems. We also discuss the analysis of unbalanced three-phase systems. We learn how to use *PSpice for Windows* to analyze a balanced or unbalanced three-phase system. Finally, we apply the concepts developed in this chapter to three-phase power measurement and residential electrical wiring.

12.2 Balanced Three-Phase Voltages

Three-phase voltages are often produced with a three-phase ac generator (or alternator) whose cross-sectional view is shown in Fig. 12.4. The generator basically consists of a rotating magnet (called the *rotor*) surrounded by a stationary winding (called the *stator*). Three separate

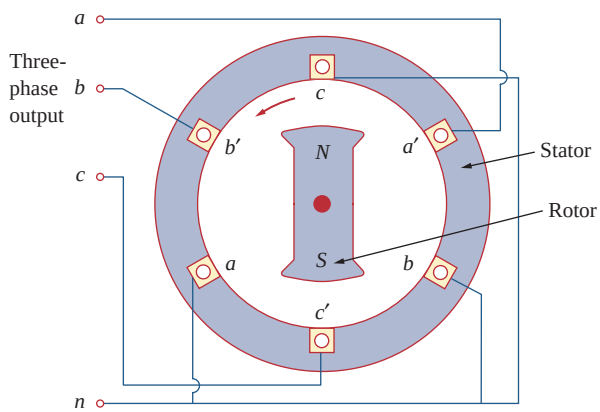


Figure 12.4
A three-phase generator.

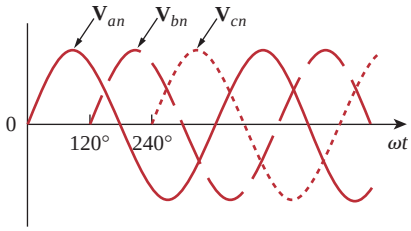


Figure 12.5
The generated voltages are 120° apart from each other.

windings or coils with terminals a-a', b-b', and c-c' are physically placed 120° apart around the stator. Terminals a and a', for example, stand for one of the ends of coils going into and the other end coming out of the page. As the rotor rotates, its magnetic field "cuts" the flux from the three coils and induces voltages in the coils. Because the coils are placed 120° apart, the induced voltages in the coils are equal in magnitude but out of phase by 120° (Fig. 12.5). Since each coil can be regarded as a single-phase generator by itself, the three-phase generator can supply power to both single-phase and three-phase loads.

A typical three-phase system consists of three voltage sources connected to loads by three or four wires (or transmission lines). (Three-phase current sources are very scarce.) A three-phase system is equivalent to three single-phase circuits. The voltage sources can be either wye-connected as shown in Fig. 12.6(a) or delta-connected as in Fig. 12.6(b).

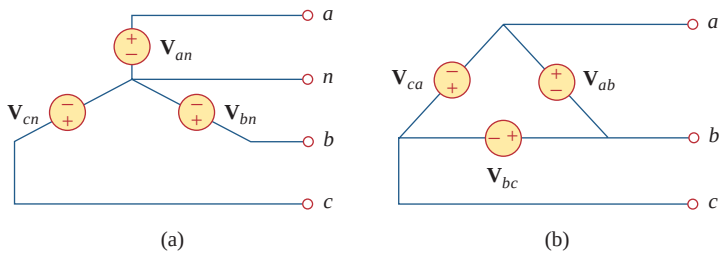


Figure 12.6
Three-phase voltage sources: (a) Y-connected source, (b) Δ-connected source.

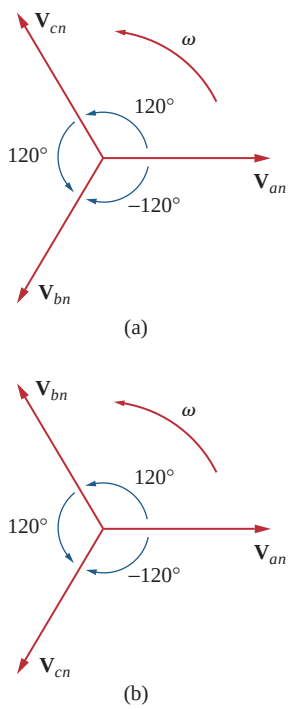


Figure 12.7
Phase sequences: (a) *abc* or positive sequence, (b) *acb* or negative sequence.

Let us consider the wye-connected voltages in Fig. 12.6(a) for now. The voltages $\mathbf{V}_{an}$, $\mathbf{V}_{bn}$, and $\mathbf{V}_{cn}$ are respectively between lines a, b, and c, and the neutral line n. These voltages are called *phase voltages*. If the voltage sources have the same amplitude and frequency ω and are out of phase with each other by 120°, the voltages are said to be *balanced*. This implies that

$$\mathbf{V}_{an} + \mathbf{V}_{bn} + \mathbf{V}_{cn} = 0 \tag{12.1}$$

$$|\mathbf{V}_{an}| = |\mathbf{V}_{bn}| = |\mathbf{V}_{cn}| \tag{12.2}$$

Thus,

Balanced phase voltages are equal in magnitude and are out of phase with each other by 120°.

Since the three-phase voltages are 120° out of phase with each other, there are two possible combinations. One possibility is shown in Fig. 12.7(a) and expressed mathematically as

$$\begin{aligned} \mathbf{V}_{an} &= V_p \angle 0° \\ \mathbf{V}_{bn} &= V_p \angle -120° \\ \mathbf{V}_{cn} &= V_p \angle -240° = V_p \angle +120° \end{aligned} \tag{12.3}$$

where V_p is the effective or rms value of the phase voltages. This is known as the *abc sequence* or *positive sequence*. In this phase sequence, $\mathbf{V}_{an}$ leads $\mathbf{V}_{bn}$, which in turn leads $\mathbf{V}_{cn}$. This sequence is produced when the rotor in Fig. 12.4 rotates counterclockwise. The other possibility is shown in Fig. 12.7(b) and is given by

As a common tradition in power systems, voltage and current in this chapter are in rms values unless otherwise stated.

$$\begin{aligned} \mathbf{V}_{an} &= V_p\angle 0^\circ \\ \mathbf{V}_{cn} &= V_p\angle -120^\circ \\ \mathbf{V}_{bn} &= V_p\angle -240^\circ = V_p\angle +120^\circ \end{aligned} \tag{12.4}$$

This is called the *acb sequence* or *negative sequence*. For this phase sequence, $\mathbf{V}_{an}$ leads $\mathbf{V}_{cn}$, which in turn leads $\mathbf{V}_{bn}$. The *acb* sequence is produced when the rotor in Fig. 12.4 rotates in the clockwise direction. It is easy to show that the voltages in Eqs. (12.3) or (12.4) satisfy Eqs. (12.1) and (12.2). For example, from Eq. (12.3),

$$\begin{aligned} \mathbf{V}_{an} + \mathbf{V}_{bn} + \mathbf{V}_{cn} &= V_p\angle 0^\circ + V_p\angle -120^\circ + V_p\angle +120^\circ \\ &= V_p(1.0 - 0.5 - j0.866 - 0.5 + j0.866) \\ &= 0 \end{aligned} \tag{12.5}$$

The phase sequence may also be regarded as the order in which the phase voltages reach their peak (or maximum) values with respect to time.

The **phase sequence** is the time order in which the voltages pass through their respective maximum values.

The phase sequence is determined by the order in which the phasors pass through a fixed point in the phase diagram.

Reminder: As time increases, each phasor (or sinor) rotates at an angular velocity ω.

In Fig. 12.7(a), as the phasors rotate in the counterclockwise direction with frequency ω, they pass through the horizontal axis in a sequence *abcabca* . . . . Thus, the sequence is *abc* or *bca* or *cab*. Similarly, for the phasors in Fig. 12.7(b), as they rotate in the counterclockwise direction, they pass the horizontal axis in a sequence *acbacba* . . . . This describes the *acb* sequence. The phase sequence is important in three-phase power distribution. It determines the direction of the rotation of a motor connected to the power source, for example.

Like the generator connections, a three-phase load can be either wye-connected or delta-connected, depending on the end application. Figure 12.8(a) shows a wye-connected load, and Fig. 12.8(b) shows a delta-connected load. The neutral line in Fig. 12.8(a) may or may not be there, depending on whether the system is four- or three-wire. (And, of course, a neutral connection is topologically impossible for a delta connection.) A wye- or delta-connected load is said to be *unbalanced* if the phase impedances are not equal in magnitude or phase.

A **balanced load** is one in which the phase impedances are equal in magnitude and in phase.

For a *balanced* wye-connected load,

$$\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{Z}_3 = \mathbf{Z}_Y \tag{12.6}$$

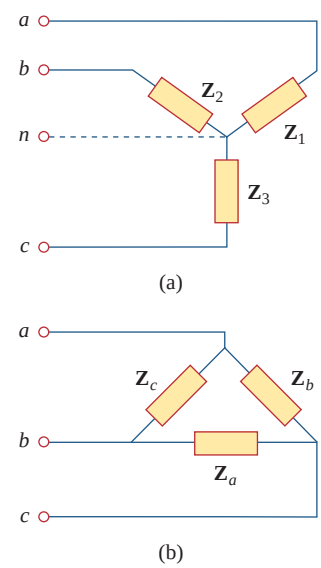


Figure 12.8
Two possible three-phase load configurations: (a) a Y-connected load, (b) a Δ-connected load.

Reminder: A Y-connected load consists of three impedances connected to a neutral node, while a Δ-connected load consists of three impedances connected around a loop. The load is balanced when the three impedances are equal in either case.

where $\mathbf{Z}_Y$ is the load impedance per phase. For a *balanced* delta-connected load,

$$\mathbf{Z}_a = \mathbf{Z}_b = \mathbf{Z}_c = \mathbf{Z}_\Delta \tag{12.7}$$

where $\mathbf{Z}_\Delta$ is the load impedance per phase in this case. We recall from Eq. (9.69) that

$$\mathbf{Z}_\Delta = 3\mathbf{Z}_Y \quad \text{or} \quad \mathbf{Z}_Y = \frac{1}{3}\mathbf{Z}_\Delta \tag{12.8}$$

so we know that a wye-connected load can be transformed into a delta-connected load, or vice versa, using Eq. (12.8).

Since both the three-phase source and the three-phase load can be either wye- or delta-connected, we have four possible connections:

- Y-Y connection (i.e., Y-connected source with a Y-connected load).
- Y-Δ connection.
- Δ-Δ connection.
- Δ-Y connection.

In subsequent sections, we will consider each of these possible configurations.

It is appropriate to mention here that a balanced delta-connected load is more common than a balanced wye-connected load. This is due to the ease with which loads may be added or removed from each phase of a delta-connected load. This is very difficult with a wye-connected load because the neutral may not be accessible. On the other hand, delta-connected sources are not common in practice because of the circulating current that will result in the delta-mesh if the three-phase voltages are slightly unbalanced.

Example 12.1

Determine the phase sequence of the set of voltages

$$v_{an} = 200\cos(\omega t + 10^\circ)$$
$$v_{bn} = 200\cos(\omega t - 230^\circ), \qquad v_{cn} = 200\cos(\omega t - 110^\circ)$$

Solution:

The voltages can be expressed in phasor form as

$$\mathbf{V}_{an} = 200\angle 10^\circ \text{ V}, \qquad \mathbf{V}_{bn} = 200\angle -230^\circ \text{ V}, \qquad \mathbf{V}_{cn} = 200\angle -110^\circ \text{ V}$$

We notice that $\mathbf{V}_{an}$ leads $\mathbf{V}_{cn}$ by 120° and $\mathbf{V}_{cn}$ in turn leads $\mathbf{V}_{bn}$ by 120°. Hence, we have an *acb* sequence.

Practice Problem 12.1

Given that $\mathbf{V}_{bn} = 110\angle 30^\circ$ V, find $\mathbf{V}_{an}$ and $\mathbf{V}_{cn}$, assuming a positive (*abc*) sequence.

Answer: $110\angle 150^\circ$ V, $110\angle -90^\circ$ V.

12.3 Balanced Wye-Wye Connection

We begin with the Y-Y system, because any balanced three-phase system can be reduced to an equivalent Y-Y system. Therefore, analysis of this system should be regarded as the key to solving all balanced three-phase systems.

A **balanced Y-Y system** is a three-phase system with a balanced Y-connected source and a balanced Y-connected load.

Consider the balanced four-wire Y-Y system of Fig. 12.9, where a Y-connected load is connected to a Y-connected source. We assume a balanced load so that load impedances are equal. Although the impedance $\mathbf{Z}_Y$ is the total load impedance per phase, it may also be regarded as the sum of the source impedance $\mathbf{Z}_s$, line impedance $\mathbf{Z}_\ell$, and load impedance $\mathbf{Z}_L$ for each phase, since these impedances are in series. As illustrated in Fig. 12.9, $\mathbf{Z}_s$ denotes the internal impedance of the phase winding of the generator; $\mathbf{Z}_\ell$ is the impedance of the line joining a phase of the source with a phase of the load; $\mathbf{Z}_L$ is the impedance of each phase of the load; and $\mathbf{Z}_n$ is the impedance of the neutral line. Thus, in general

$$\mathbf{Z}_Y = \mathbf{Z}_s + \mathbf{Z}_\ell + \mathbf{Z}_L \tag{12.9}$$

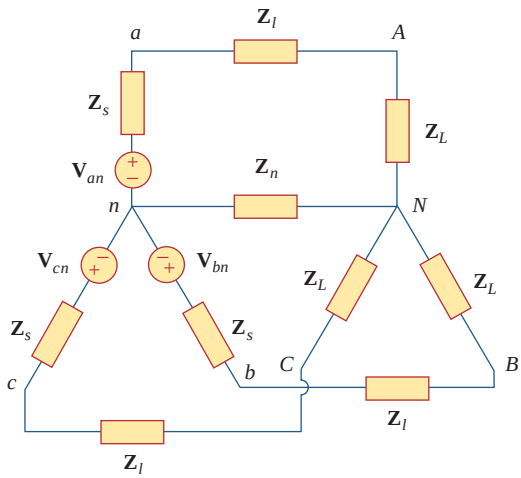


Figure 12.9
A balanced Y-Y system, showing the source, line, and load impedances.

$\mathbf{Z}_s$ and $\mathbf{Z}_\ell$ are often very small compared with $\mathbf{Z}_L$, so one can assume that $\mathbf{Z}_Y = \mathbf{Z}_L$ if no source or line impedance is given. In any event, by lumping the impedances together, the Y-Y system in Fig. 12.9 can be simplified to that shown in Fig. 12.10.

Assuming the positive sequence, the *phase* voltages (or line-to-neutral voltages) are

$$\mathbf{V}_{an} = V_p \angle 0^\circ$$
$$\mathbf{V}_{bn} = V_p \angle -120^\circ, \qquad \mathbf{V}_{cn} = V_p \angle +120^\circ \tag{12.10}$$

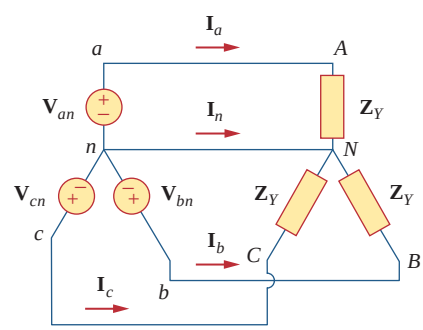


Figure 12.10
Balanced Y-Y connection.

The *line-to-line* voltages or simply *line* voltages $\mathbf{V}_{ab}$, $\mathbf{V}_{bc}$, and $\mathbf{V}_{ca}$ are related to the phase voltages. For example,

$$\begin{aligned}\mathbf{V}_{ab} &= \mathbf{V}_{an} + \mathbf{V}_{nb} = \mathbf{V}_{an} - \mathbf{V}_{bn} = V_p\angle 0^\circ - V_p\angle -120^\circ \\ &= V_p\left(1 + \frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = \sqrt{3}V_p\angle 30^\circ\end{aligned} \tag{12.11a}$$

Similarly, we can obtain

$$\mathbf{V}_{bc} = \mathbf{V}_{bn} - \mathbf{V}_{cn} = \sqrt{3}V_p\angle -90^\circ \tag{12.11b}$$

$$\mathbf{V}_{ca} = \mathbf{V}_{cn} - \mathbf{V}_{an} = \sqrt{3}V_p\angle -210^\circ \tag{12.11c}$$

Thus, the magnitude of the line voltages V_L is $\sqrt{3}$ times the magnitude of the phase voltages V_p, or

$$\boxed{V_L = \sqrt{3}V_p} \tag{12.12}$$

where

$$V_p = |\mathbf{V}_{an}| = |\mathbf{V}_{bn}| = |\mathbf{V}_{cn}| \tag{12.13}$$

and

$$V_L = |\mathbf{V}_{ab}| = |\mathbf{V}_{bc}| = |\mathbf{V}_{ca}| \tag{12.14}$$

Also the line voltages lead their corresponding phase voltages by 30°. Figure 12.11(a) illustrates this. Figure 12.11(a) also shows how to determine $\mathbf{V}_{ab}$ from the phase voltages, while Fig. 12.11(b) shows the same for the three line voltages. Notice that $\mathbf{V}_{ab}$ leads $\mathbf{V}_{bc}$ by 120°, and $\mathbf{V}_{bc}$ leads $\mathbf{V}_{ca}$ by 120°, so that the line voltages sum up to zero as do the phase voltages.

Applying KVL to each phase in Fig. 12.10, we obtain the line currents as

$$\begin{aligned}\mathbf{I}_a &= \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}, \qquad \mathbf{I}_b = \frac{\mathbf{V}_{bn}}{\mathbf{Z}_Y} = \frac{\mathbf{V}_{an}\angle -120^\circ}{\mathbf{Z}_Y} = \mathbf{I}_a\angle -120^\circ \\ \mathbf{I}_c &= \frac{\mathbf{V}_{cn}}{\mathbf{Z}_Y} = \frac{\mathbf{V}_{an}\angle -240^\circ}{\mathbf{Z}_Y} = \mathbf{I}_a\angle -240^\circ\end{aligned} \tag{12.15}$$

We can readily infer that the line currents add up to zero,

$$\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0 \tag{12.16}$$

so that

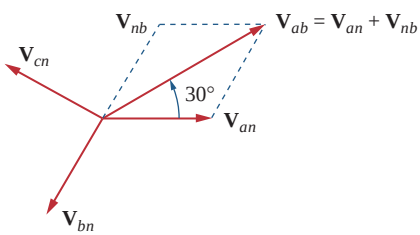
$$\mathbf{I}_n = -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c) = 0 \tag{12.17a}$$

or

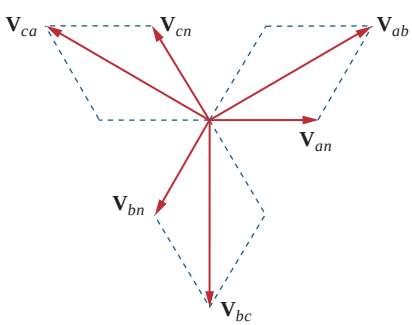
$$\mathbf{V}_{nN} = \mathbf{Z}_n\mathbf{I}_n = 0 \tag{12.17b}$$

that is, the voltage across the neutral wire is zero. The neutral line can thus be removed without affecting the system. In fact, in long distance power transmission, conductors in multiples of three are used with the earth itself acting as the neutral conductor. Power systems designed in this way are well grounded at all critical points to ensure safety.

While the *line* current is the current in each line, the *phase* current is the current in each phase of the source or load. In the Y-Y system, the line current is the same as the phase current. We will use single subscripts



(a)



(b)

Figure 12.11
Phasor diagrams illustrating the relationship between line voltages and phase voltages.

for line currents because it is natural and conventional to assume that line currents flow from the source to the load.

An alternative way of analyzing a balanced Y-Y system is to do so on a "per phase" basis. We look at one phase, say phase *a*, and analyze the single-phase equivalent circuit in Fig. 12.12. The single-phase analysis yields the line current $\mathbf{I}_a$ as

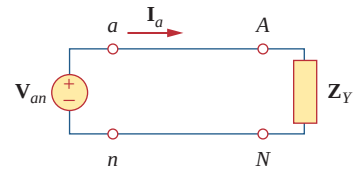


Figure 12.12
A single-phase equivalent circuit.

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y} \tag{12.18}$$

From $\mathbf{I}_a$, we use the phase sequence to obtain other line currents. Thus, as long as the system is balanced, we need only analyze one phase. We may do this even if the neutral line is absent, as in the three-wire system.

Example 12.2

Calculate the line currents in the three-wire Y-Y system of Fig. 12.13.

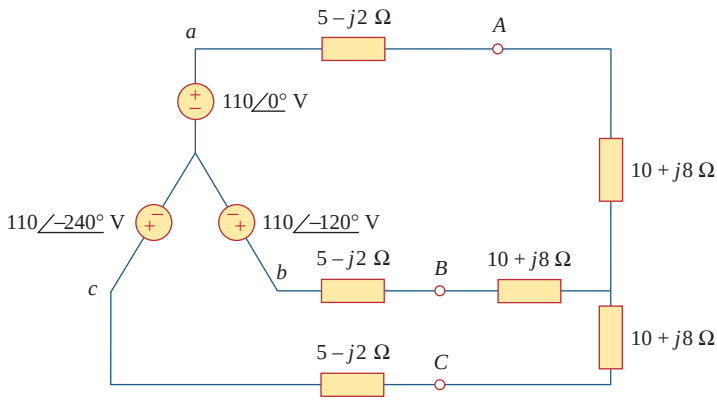


Figure 12.13
Three-wire Y-Y system; for Example 12.2.

Solution:
The three-phase circuit in Fig. 12.13 is balanced; we may replace it with its single-phase equivalent circuit such as in Fig. 12.12. We obtain $\mathbf{I}_a$ from the single-phase analysis as

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}$$

where $\mathbf{Z}_Y = (5 - j2) + (10 + j8) = 15 + j6 = 16.155\angle 21.8°$. Hence,

$$\mathbf{I}_a = \frac{110\angle 0°}{16.155\angle 21.8°} = 6.81\angle -21.8° \text{ A}$$

Since the source voltages in Fig. 12.13 are in positive sequence, the line currents are also in positive sequence:

$$\mathbf{I}_b = \mathbf{I}_a\angle -120° = 6.81\angle -141.8° \text{ A}$$

$$\mathbf{I}_c = \mathbf{I}_a\angle -240° = 6.81\angle -261.8° \text{ A} = 6.81\angle 98.2° \text{ A}$$

Practice Problem 12.2

A Y-connected balanced three-phase generator with an impedance of $0.4 + j0.3\ \Omega$ per phase is connected to a Y-connected balanced load with an impedance of $24 + j19\ \Omega$ per phase. The line joining the generator and the load has an impedance of $0.6 + j0.7\ \Omega$ per phase. Assuming a positive sequence for the source voltages and that $\mathbf{V}_{an} = 120\angle 30^\circ$ V, find: (a) the line voltages, (b) the line currents.

Answer: (a) $207.85\angle 60^\circ$ V, $207.85\angle -60^\circ$ V, $207.85\angle -180^\circ$ V, (b) $3.75\angle -8.66^\circ$ A, $3.75\angle -128.66^\circ$ A, $3.75\angle -111.34^\circ$ A.

12.4 Balanced Wye-Delta Connection

A **balanced Y-Δ system** consists of a balanced Y-connected source feeding a balanced Δ-connected load.

This is perhaps the most practical three-phase system, as the three-phase sources are usually Y-connected while the three-phase loads are usually Δ-connected.

The balanced Y-delta system is shown in Fig. 12.14, where the source is Y-connected and the load is Δ-connected. There is, of course, no neutral connection from source to load for this case. Assuming the positive sequence, the phase voltages are again

$$\mathbf{V}_{an} = V_p\angle 0^\circ$$
$$\mathbf{V}_{bn} = V_p\angle -120^\circ, \qquad \mathbf{V}_{cn} = V_p\angle +120^\circ \tag{12.19}$$

As shown in Section 12.3, the line voltages are

$$\mathbf{V}_{ab} = \sqrt{3}V_p\angle 30^\circ = \mathbf{V}_{AB}, \qquad \mathbf{V}_{bc} = \sqrt{3}V_p\angle -90^\circ = \mathbf{V}_{BC}$$
$$\mathbf{V}_{ca} = \sqrt{3}V_p\angle -150^\circ = \mathbf{V}_{CA} \tag{12.20}$$

showing that the line voltages are equal to the voltages across the load impedances for this system configuration. From these voltages, we can obtain the phase currents as

$$\mathbf{I}_{AB} = \frac{\mathbf{V}_{AB}}{\mathbf{Z}_\Delta}, \qquad \mathbf{I}_{BC} = \frac{\mathbf{V}_{BC}}{\mathbf{Z}_\Delta}, \qquad \mathbf{I}_{CA} = \frac{\mathbf{V}_{CA}}{\mathbf{Z}_\Delta} \tag{12.21}$$

These currents have the same magnitude but are out of phase with each other by 120°.

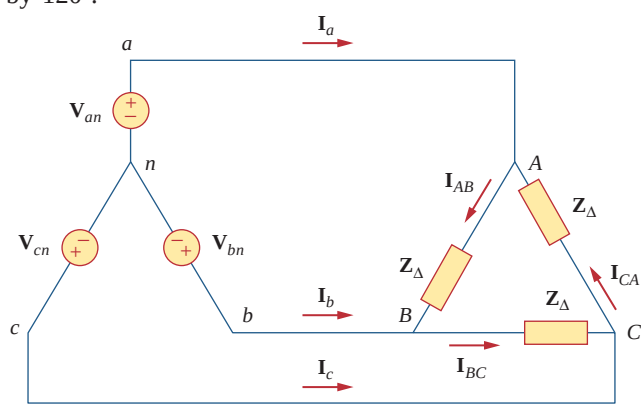


Figure 12.14
Balanced Y-Δ connection.

Another way to get these phase currents is to apply KVL. For example, applying KVL around loop *aABbna* gives

$$-\mathbf{V}_{an} + \mathbf{Z}_\Delta \mathbf{I}_{AB} + \mathbf{V}_{bn} = 0$$

or

$$\mathbf{I}_{AB} = \frac{\mathbf{V}_{an} - \mathbf{V}_{bn}}{\mathbf{Z}_\Delta} = \frac{\mathbf{V}_{ab}}{\mathbf{Z}_\Delta} = \frac{\mathbf{V}_{AB}}{\mathbf{Z}_\Delta} \tag{12.22}$$

which is the same as Eq. (12.21). This is the more general way of finding the phase currents.

The line currents are obtained from the phase currents by applying KCL at nodes *A*, *B*, and *C*. Thus,

$$\mathbf{I}_a = \mathbf{I}_{AB} - \mathbf{I}_{CA}, \qquad \mathbf{I}_b = \mathbf{I}_{BC} - \mathbf{I}_{AB}, \qquad \mathbf{I}_c = \mathbf{I}_{CA} - \mathbf{I}_{BC} \tag{12.23}$$

Since $\mathbf{I}_{CA} = \mathbf{I}_{AB}\angle -240°$,

$$\begin{aligned}\mathbf{I}_a = \mathbf{I}_{AB} - \mathbf{I}_{CA} &= \mathbf{I}_{AB}(1 - 1\angle -240°) \\ &= \mathbf{I}_{AB}(1 + 0.5 - j0.866) = \mathbf{I}_{AB}\sqrt{3}\angle -30°\end{aligned} \tag{12.24}$$

showing that the magnitude I_L of the line current is $\sqrt{3}$ times the magnitude I_p of the phase current, or

$$\boxed{I_L = \sqrt{3} I_p} \tag{12.25}$$

where

$$I_L = |\mathbf{I}_a| = |\mathbf{I}_b| = |\mathbf{I}_c| \tag{12.26}$$

and

$$I_p = |\mathbf{I}_{AB}| = |\mathbf{I}_{BC}| = |\mathbf{I}_{CA}| \tag{12.27}$$

Also, the line currents lag the corresponding phase currents by 30°, assuming the positive sequence. Figure 12.15 is a phasor diagram illustrating the relationship between the phase and line currents.

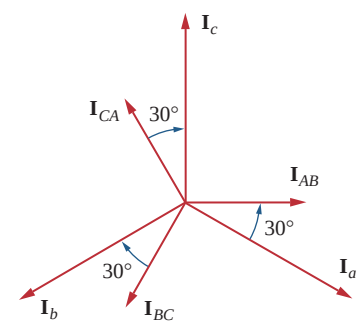


Figure 12.15
Phasor diagram illustrating the relationship between phase and line currents.

An alternative way of analyzing the Y-Δ circuit is to transform the Δ-connected load to an equivalent Y-connected load. Using the Δ-Y transformation formula in Eq. (12.8),

$$\boxed{\mathbf{Z}_Y = \frac{\mathbf{Z}_\Delta}{3}} \tag{12.28}$$

After this transformation, we now have a Y-Y system as in Fig. 12.10. The three-phase Y-Δ system in Fig. 12.14 can be replaced by the single-phase equivalent circuit in Fig. 12.16. This allows us to calculate only the line currents. The phase currents are obtained using Eq. (12.25) and utilizing the fact that each of the phase currents leads the corresponding line current by 30°.

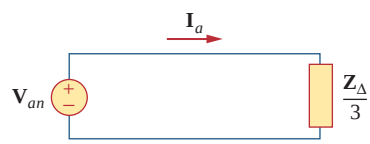


Figure 12.16
A single-phase equivalent circuit of a balanced Y-Δ circuit.

Example 12.3

A balanced *abc*-sequence Y-connected source with $\mathbf{V}_{an} = 100\angle 10°$ V is connected to a Δ-connected balanced load $(8 + j4)\ \Omega$ per phase. Calculate the phase and line currents.

Solution:
This can be solved in two ways.

■ **METHOD 1** The load impedance is

$$\mathbf{Z}_\Delta = 8 + j4 = 8.944\angle 26.57^\circ\ \Omega$$

If the phase voltage $\mathbf{V}_{an} = 100\angle 10^\circ$, then the line voltage is

$$\mathbf{V}_{ab} = \mathbf{V}_{an}\sqrt{3}\angle 30^\circ = 100\sqrt{3}\angle 10^\circ + 30^\circ = \mathbf{V}_{AB}$$

or

$$\mathbf{V}_{AB} = 173.2\angle 40^\circ\ \text{V}$$

The phase currents are

$$\mathbf{I}_{AB} = \frac{\mathbf{V}_{AB}}{\mathbf{Z}_\Delta} = \frac{173.2\angle 40^\circ}{8.944\angle 26.57^\circ} = 19.36\angle 13.43^\circ\ \text{A}$$
$$\mathbf{I}_{BC} = \mathbf{I}_{AB}\angle -120^\circ = 19.36\angle -106.57^\circ\ \text{A}$$
$$\mathbf{I}_{CA} = \mathbf{I}_{AB}\angle +120^\circ = 19.36\angle 133.43^\circ\ \text{A}$$

The line currents are

$$\mathbf{I}_a = \mathbf{I}_{AB}\sqrt{3}\angle -30^\circ = \sqrt{3}(19.36)\angle 13.43^\circ - 30^\circ$$
$$= 33.53\angle -16.57^\circ\ \text{A}$$
$$\mathbf{I}_b = \mathbf{I}_a\angle -120^\circ = 33.53\angle -136.57^\circ\ \text{A}$$
$$\mathbf{I}_c = \mathbf{I}_a\angle +120^\circ = 33.53\angle 103.43^\circ\ \text{A}$$

■ **METHOD 2** Alternatively, using single-phase analysis,

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_\Delta/3} = \frac{100\angle 10^\circ}{2.981\angle 26.57^\circ} = 33.54\angle -16.57^\circ\ \text{A}$$

as above. Other line currents are obtained using the *abc* phase sequence.

Practice Problem 12.3

One line voltage of a balanced Y-connected source is $\mathbf{V}_{AB} = 240\angle -20^\circ$ V. If the source is connected to a Δ-connected load of $20\angle 40^\circ\ \Omega$, find the phase and line currents. Assume the *abc* sequence.

Answer: $12\angle -60^\circ$ A, $12\angle -180^\circ$ A, $12\angle 60^\circ$ A, $20.79\angle -90^\circ$ A, $20.79\angle -150^\circ$ A, $20.79\angle 30^\circ$ A.

12.5 Balanced Delta-Delta Connection

A **balanced Δ-Δ system** is one in which both the balanced source and balanced load are Δ-connected.

The source as well as the load may be delta-connected as shown in Fig. 12.17. Our goal is to obtain the phase and line currents as usual.

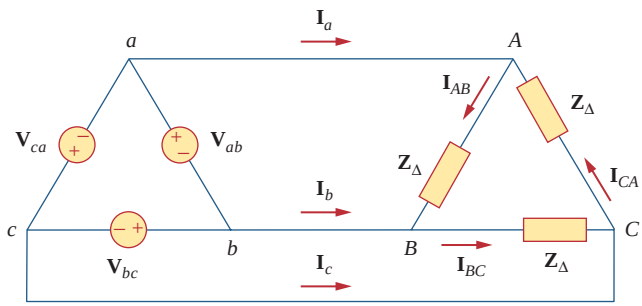


Figure 12.17
A balanced Δ-Δ connection.

Assuming a positive sequence, the phase voltages for a delta-connected source are

$$\mathbf{V}_{ab} = V_p\angle 0^\circ$$
$$\mathbf{V}_{bc} = V_p\angle -120^\circ, \qquad \mathbf{V}_{ca} = V_p\angle +120^\circ \tag{12.29}$$

The line voltages are the same as the phase voltages. From Fig. 12.17, assuming there is no line impedances, the phase voltages of the delta-connected source are equal to the voltages across the impedances; that is,

$$\mathbf{V}_{ab} = \mathbf{V}_{AB}, \qquad \mathbf{V}_{bc} = \mathbf{V}_{BC}, \qquad \mathbf{V}_{ca} = \mathbf{V}_{CA} \tag{12.30}$$

Hence, the phase currents are

$$\mathbf{I}_{AB} = \frac{\mathbf{V}_{AB}}{Z_\Delta} = \frac{\mathbf{V}_{ab}}{Z_\Delta}, \qquad \mathbf{I}_{BC} = \frac{\mathbf{V}_{BC}}{Z_\Delta} = \frac{\mathbf{V}_{bc}}{Z_\Delta}$$
$$\mathbf{I}_{CA} = \frac{\mathbf{V}_{CA}}{Z_\Delta} = \frac{\mathbf{V}_{ca}}{Z_\Delta} \tag{12.31}$$

Since the load is delta-connected just as in the previous section, some of the formulas derived there apply here. The line currents are obtained from the phase currents by applying KCL at nodes *A*, *B*, and *C*, as we did in the previous section:

$$\mathbf{I}_a = \mathbf{I}_{AB} - \mathbf{I}_{CA}, \qquad \mathbf{I}_b = \mathbf{I}_{BC} - \mathbf{I}_{AB}, \qquad \mathbf{I}_c = \mathbf{I}_{CA} - \mathbf{I}_{BC} \tag{12.32}$$

Also, as shown in the last section, each line current lags the corresponding phase current by 30°; the magnitude I_L of the line current is $\sqrt{3}$ times the magnitude I_p of the phase current,

$$I_L = \sqrt{3} I_p \tag{12.33}$$

An alternative way of analyzing the Δ-Δ circuit is to convert both the source and the load to their Y equivalents. We already know that $\mathbf{Z}_Y = \mathbf{Z}_\Delta/3$. To convert a Δ-connected source to a Y-connected source, see the next section.

Example 12.4

A balanced Δ-connected load having an impedance $20 - j15\ \Omega$ is connected to a Δ-connected, positive-sequence generator having $\mathbf{V}_{ab} = 330\angle 0^\circ$ V. Calculate the phase currents of the load and the line currents.

Solution:
The load impedance per phase is

$$\mathbf{Z}_\Delta = 20 - j15 = 25\angle -36.87^\circ\ \Omega$$

Since $\mathbf{V}_{AB} = \mathbf{V}_{ab}$, the phase currents are

$$\mathbf{I}_{AB} = \frac{\mathbf{V}_{AB}}{\mathbf{Z}_\Delta} = \frac{330\angle 0^\circ}{25\angle -36.87} = 13.2\angle 36.87^\circ\ \text{A}$$
$$\mathbf{I}_{BC} = \mathbf{I}_{AB}\angle -120^\circ = 13.2\angle -83.13^\circ\ \text{A}$$
$$\mathbf{I}_{CA} = \mathbf{I}_{AB}\angle +120^\circ = 13.2\angle 156.87^\circ\ \text{A}$$

For a delta load, the line current always lags the corresponding phase current by 30° and has a magnitude $\sqrt{3}$ times that of the phase current. Hence, the line currents are

$$\mathbf{I}_a = \mathbf{I}_{AB}\sqrt{3}\angle -30^\circ = (13.2\angle 36.87^\circ)(\sqrt{3}\angle -30^\circ)$$
$$= 22.86\angle 6.87^\circ\ \text{A}$$
$$\mathbf{I}_b = \mathbf{I}_a\angle -120^\circ = 22.86\angle -113.13^\circ\ \text{A}$$
$$\mathbf{I}_c = \mathbf{I}_a\angle +120^\circ = 22.86\angle 126.87^\circ\ \text{A}$$

Practice Problem 12.4

A positive-sequence, balanced Δ-connected source supplies a balanced Δ-connected load. If the impedance per phase of the load is $18 + j12\ \Omega$ and $\mathbf{I}_a = 19.202\angle 35^\circ$ A, find $\mathbf{I}_{AB}$ and $\mathbf{V}_{AB}$.

Answer: $11.094\angle 65^\circ$ A, $240\angle 98.69^\circ$ V.

12.6 Balanced Delta-Wye Connection

A **balanced Δ-Y system** consists of a balanced Δ-connected source feeding a balanced Y-connected load.

Consider the Δ-Y circuit in Fig. 12.18. Again, assuming the *abc* sequence, the phase voltages of a delta-connected source are

$$\mathbf{V}_{ab} = V_p\angle 0^\circ, \qquad \mathbf{V}_{bc} = V_p\angle -120^\circ$$
$$\mathbf{V}_{ca} = V_p\angle +120^\circ \tag{12.34}$$

These are also the line voltages as well as the phase voltages.

We can obtain the line currents in many ways. One way is to apply KVL to loop *aANBba* in Fig. 12.18, writing

$$-\mathbf{V}_{ab} + \mathbf{Z}_Y\mathbf{I}_a - \mathbf{Z}_Y\mathbf{I}_b = 0$$

or

$$\mathbf{Z}_Y(\mathbf{I}_a - \mathbf{I}_b) = \mathbf{V}_{ab} = V_p\angle 0^\circ$$

Thus,

$$\mathbf{I}_a - \mathbf{I}_b = \frac{V_p\angle 0^\circ}{\mathbf{Z}_Y} \tag{12.35}$$

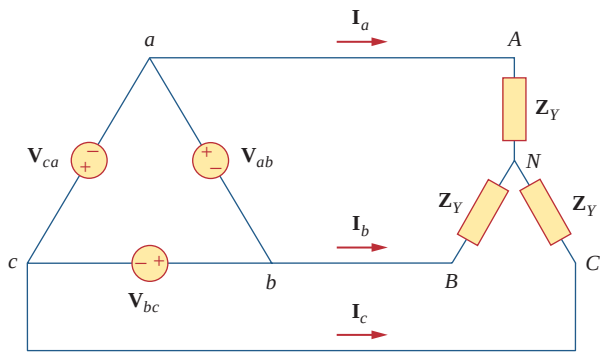


Figure 12.18
A balanced Δ-Y connection.

But $\mathbf{I}_b$ lags $\mathbf{I}_a$ by 120°, since we assumed the *abc* sequence; that is, $\mathbf{I}_b = \mathbf{I}_a/\underline{-120°}$. Hence,

$$\begin{aligned}\mathbf{I}_a - \mathbf{I}_b &= \mathbf{I}_a(1 - 1/\underline{-120°}) \\ &= \mathbf{I}_a\left(1 + \frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = \mathbf{I}_a\sqrt{3}/\underline{30°}\end{aligned} \tag{12.36}$$

Substituting Eq. (12.36) into Eq. (12.35) gives

$$\mathbf{I}_a = \frac{V_p/\sqrt{3}/\underline{-30°}}{\mathbf{Z}_Y} \tag{12.37}$$

From this, we obtain the other line currents $\mathbf{I}_b$ and $\mathbf{I}_c$ using the positive phase sequence, i.e., $\mathbf{I}_b = \mathbf{I}_a/\underline{-120°}$, $\mathbf{I}_c = \mathbf{I}_a/\underline{+120°}$. The phase currents are equal to the line currents.

Another way to obtain the line currents is to replace the delta-connected source with its equivalent wye-connected source, as shown in Fig. 12.19. In Section 12.3, we found that the line-to-line voltages of a wye-connected source lead their corresponding phase voltages by 30°. Therefore, we obtain each phase voltage of the equivalent wye-connected source by dividing the corresponding line voltage of the delta-connected source by $\sqrt{3}$ and shifting its phase by −30°. Thus, the equivalent wye-connected source has the phase voltages

$$\mathbf{V}_{an} = \frac{V_p}{\sqrt{3}}/\underline{-30°}$$

$$\mathbf{V}_{bn} = \frac{V_p}{\sqrt{3}}/\underline{-150°}, \qquad \mathbf{V}_{cn} = \frac{V_p}{\sqrt{3}}/\underline{+90°} \tag{12.38}$$

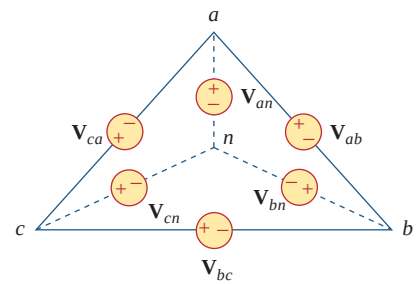


Figure 12.19
Transforming a Δ-connected source to an equivalent Y-connected source.

If the delta-connected source has source impedance $\mathbf{Z}_s$ per phase, the equivalent wye-connected source will have a source impedance of $\mathbf{Z}_s/3$ per phase, according to Eq. (9.69).

Once the source is transformed to wye, the circuit becomes a wye-wye system. Therefore, we can use the equivalent single-phase circuit shown in Fig. 12.20, from which the line current for phase *a* is

$$\mathbf{I}_a = \frac{V_p/\sqrt{3}/\underline{-30°}}{\mathbf{Z}_Y} \tag{12.39}$$

which is the same as Eq. (12.37).

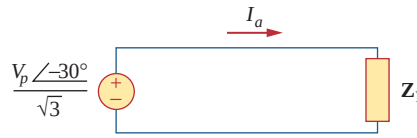


Figure 12.20
The single-phase equivalent circuit.

Alternatively, we may transform the wye-connected load to an equivalent delta-connected load. This results in a delta-delta system, which can be analyzed as in Section 12.5. Note that

$$\mathbf{V}_{AN} = \mathbf{I}_a \mathbf{Z}_Y = \frac{V_p}{\sqrt{3}}\angle -30^\circ \tag{12.40}$$
$$\mathbf{V}_{BN} = \mathbf{V}_{AN}\angle -120^\circ, \qquad \mathbf{V}_{CN} = \mathbf{V}_{AN}\angle +120^\circ$$

As stated earlier, the delta-connected load is more desirable than the wye-connected load. It is easier to alter the loads in any one phase of the delta-connected loads, as the individual loads are connected directly across the lines. However, the delta-connected source is hardly used in practice, because any slight imbalance in the phase voltages will result in unwanted circulating currents.

Table 12.1 presents a summary of the formulas for phase currents and voltages and line currents and voltages for the four connections. Students are advised not to memorize the formulas but to understand how they are derived. The formulas can always be

TABLE 12.1

Summary of phase and line voltages/currents for balanced three-phase systems.[1]

Connection	Phase voltages/currents	Line voltages/currents
Y-Y	$\mathbf{V}_{an} = V_p\angle 0^\circ$ $\mathbf{V}_{bn} = V_p\angle -120^\circ$ $\mathbf{V}_{cn} = V_p\angle +120^\circ$ Same as line currents	$\mathbf{V}_{ab} = \sqrt{3}V_p\angle 30^\circ$ $\mathbf{V}_{bc} = \mathbf{V}_{ab}\angle -120^\circ$ $\mathbf{V}_{ca} = \mathbf{V}_{ab}\angle +120^\circ$ $\mathbf{I}_a = \mathbf{V}_{an}/\mathbf{Z}_Y$ $\mathbf{I}_b = \mathbf{I}_a\angle -120^\circ$ $\mathbf{I}_c = \mathbf{I}_a\angle +120^\circ$
Y-Δ	$\mathbf{V}_{an} = V_p\angle 0^\circ$ $\mathbf{V}_{bn} = V_p\angle -120^\circ$ $\mathbf{V}_{cn} = V_p\angle +120^\circ$ $\mathbf{I}_{AB} = \mathbf{V}_{AB}/\mathbf{Z}_\Delta$ $\mathbf{I}_{BC} = \mathbf{V}_{BC}/\mathbf{Z}_\Delta$ $\mathbf{I}_{CA} = \mathbf{V}_{CA}/\mathbf{Z}_\Delta$	$\mathbf{V}_{ab} = \mathbf{V}_{AB} = \sqrt{3}V_p\angle 30^\circ$ $\mathbf{V}_{bc} = \mathbf{V}_{BC} = \mathbf{V}_{ab}\angle -120^\circ$ $\mathbf{V}_{ca} = \mathbf{V}_{CA} = \mathbf{V}_{ab}\angle +120^\circ$ $\mathbf{I}_a = \mathbf{I}_{AB}\sqrt{3}\angle -30^\circ$ $\mathbf{I}_b = \mathbf{I}_a\angle -120^\circ$ $\mathbf{I}_c = \mathbf{I}_a\angle +120^\circ$
Δ-Δ	$\mathbf{V}_{ab} = V_p\angle 0^\circ$ $\mathbf{V}_{bc} = V_p\angle -120^\circ$ $\mathbf{V}_{ca} = V_p\angle +120^\circ$ $\mathbf{I}_{AB} = \mathbf{V}_{ab}/\mathbf{Z}_\Delta$ $\mathbf{I}_{BC} = \mathbf{V}_{bc}/\mathbf{Z}_\Delta$ $\mathbf{I}_{CA} = \mathbf{V}_{ca}/\mathbf{Z}_\Delta$	Same as phase voltages $\mathbf{I}_a = \mathbf{I}_{AB}\sqrt{3}\angle -30^\circ$ $\mathbf{I}_b = \mathbf{I}_a\angle -120^\circ$ $\mathbf{I}_c = \mathbf{I}_a\angle +120^\circ$
Δ-Y	$\mathbf{V}_{ab} = V_p\angle 0^\circ$ $\mathbf{V}_{bc} = V_p\angle -120^\circ$ $\mathbf{V}_{ca} = V_p\angle +120^\circ$ Same as line currents	Same as phase voltages $\mathbf{I}_a = \dfrac{V_p\angle -30^\circ}{\sqrt{3}\mathbf{Z}_Y}$ $\mathbf{I}_b = \mathbf{I}_a\angle -120^\circ$ $\mathbf{I}_c = \mathbf{I}_a\angle +120^\circ$

[1] Positive or abc sequence is assumed.

obtained by directly applying KCL and KVL to the appropriate three-phase circuits.

Example 12.5

A balanced Y-connected load with a phase impedance of $40 + j25\ \Omega$ is supplied by a balanced, positive sequence Δ-connected source with a line voltage of 210 V. Calculate the phase currents. Use $\mathbf{V}_{ab}$ as reference.

Solution:
The load impedance is

$$\mathbf{Z}_Y = 40 + j25 = 47.17\angle 32^\circ\ \Omega$$

and the source voltage is

$$\mathbf{V}_{ab} = 210\angle 0^\circ\ \text{V}$$

When the Δ-connected source is transformed to a Y-connected source,

$$\mathbf{V}_{an} = \frac{\mathbf{V}_{ab}}{\sqrt{3}}\angle -30^\circ = 121.2\angle -30^\circ\ \text{V}$$

The line currents are

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y} = \frac{121.2\angle -30^\circ}{47.12\angle 32^\circ} = 2.57\angle -62^\circ\ \text{A}$$
$$\mathbf{I}_b = \mathbf{I}_a\angle -120^\circ = 2.57\angle -178^\circ\ \text{A}$$
$$\mathbf{I}_c = \mathbf{I}_a\angle 120^\circ = 2.57\angle 58^\circ\ \text{A}$$

which are the same as the phase currents.

Practice Problem 12.5

In a balanced Δ-Y circuit, $\mathbf{V}_{ab} = 240\angle 15^\circ$ and $\mathbf{Z}_Y = (12 + j15)\ \Omega$. Calculate the line currents.

Answer: $7.21\angle -66.34^\circ$ A, $7.21\angle -173.66^\circ$ A, $7.21\angle 53.66^\circ$ A.

12.7 Power in a Balanced System

Let us now consider the power in a balanced three-phase system. We begin by examining the instantaneous power absorbed by the load. This requires that the analysis be done in the time domain. For a Y-connected load, the phase voltages are

$$v_{AN} = \sqrt{2}V_p \cos\omega t, \qquad v_{BN} = \sqrt{2}V_p \cos(\omega t - 120^\circ)$$
$$v_{CN} = \sqrt{2}V_p \cos(\omega t + 120^\circ) \tag{12.41}$$

where the factor $\sqrt{2}$ is necessary because V_p has been defined as the rms value of the phase voltage. If $\mathbf{Z}_Y = Z\angle\theta$, the phase currents lag behind their corresponding phase voltages by θ. Thus,

$$i_a = \sqrt{2}I_p \cos(\omega t - \theta), \qquad i_b = \sqrt{2}I_p \cos(\omega t - \theta - 120^\circ)$$
$$i_c = \sqrt{2}I_p \cos(\omega t - \theta + 120^\circ) \tag{12.42}$$

where I_p is the rms value of the phase current. The total instantaneous power in the load is the sum of the instantaneous powers in the three phases; that is,

$$\begin{aligned} p &= p_a + p_b + p_c = v_{AN}i_a + v_{BN}i_b + v_{CN}i_c \\ &= 2V_pI_p[\cos\omega t\cos(\omega t - \theta) \\ &\quad + \cos(\omega t - 120^\circ)\cos(\omega t - \theta - 120^\circ) \\ &\quad + \cos(\omega t + 120^\circ)\cos(\omega t - \theta + 120^\circ)] \end{aligned} \tag{12.43}$$

Applying the trigonometric identity

$$\cos A\cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)] \tag{12.44}$$

gives

$$\begin{aligned} p &= V_pI_p[3\cos\theta + \cos(2\omega t - \theta) + \cos(2\omega t - \theta - 240^\circ) \\ &\quad + \cos(2\omega t - \theta + 240^\circ)] \\ &= V_pI_p[3\cos\theta + \cos\alpha + \cos\alpha\cos 240^\circ + \sin\alpha\sin 240^\circ \\ &\quad + \cos\alpha\cos 240^\circ - \sin\alpha\sin 240^\circ] \\ &\quad \text{where } \alpha = 2\omega t - \theta \\ &= V_pI_p\left[3\cos\theta + \cos\alpha + 2\left(-\frac{1}{2}\right)\cos\alpha\right] = 3V_pI_p\cos\theta \end{aligned} \tag{12.45}$$

Thus the total instantaneous power in a balanced three-phase system is constant—it does not change with time as the instantaneous power of each phase does. This result is true whether the load is Y- or Δ-connected. This is one important reason for using a three-phase system to generate and distribute power. We will look into another reason a little later.

Since the total instantaneous power is independent of time, the average power per phase P_p for either the Δ-connected load or the Y-connected load is $p/3$, or

$$P_p = V_pI_p\cos\theta \tag{12.46}$$

and the reactive power per phase is

$$Q_p = V_pI_p\sin\theta \tag{12.47}$$

The apparent power per phase is

$$S_p = V_pI_p \tag{12.48}$$

The complex power per phase is

$$\mathbf{S}_p = P_p + jQ_p = \mathbf{V}_p\mathbf{I}_p^* \tag{12.49}$$

where $\mathbf{V}_p$ and $\mathbf{I}_p$ are the phase voltage and phase current with magnitudes V_p and I_p, respectively. The total average power is the sum of the average powers in the phases:

$$P = P_a + P_b + P_c = 3P_p = 3V_pI_p\cos\theta = \sqrt{3}V_LI_L\cos\theta \tag{12.50}$$

For a Y-connected load, $I_L = I_p$ but $V_L = \sqrt{3}V_p$, whereas for a Δ-connected load, $I_L = \sqrt{3}I_p$ but $V_L = V_p$. Thus, Eq. (12.50) applies for both Y-connected and Δ-connected loads. Similarly, the total reactive power is

$$Q = 3V_pI_p\sin\theta = 3Q_p = \sqrt{3}V_LI_L\sin\theta \tag{12.51}$$

and the total complex power is

$$\mathbf{S} = 3\mathbf{S}_p = 3\mathbf{V}_p\mathbf{I}_p^* = 3I_p^2\mathbf{Z}_p = \frac{3V_p^2}{\mathbf{Z}_p^*} \tag{12.52}$$

where $\mathbf{Z}_p = Z_p\angle\theta$ is the load impedance per phase. ($\mathbf{Z}_p$ could be $\mathbf{Z}_Y$ or $\mathbf{Z}_\Delta$.) Alternatively, we may write Eq. (12.52) as

$$\mathbf{S} = P + jQ = \sqrt{3}V_L I_L\angle\theta \tag{12.53}$$

Remember that V_p, I_p, V_L, and I_L are all rms values and that θ is the angle of the load impedance or the angle between the phase voltage and the phase current.

A second major advantage of three-phase systems for power distribution is that the three-phase system uses a lesser amount of wire than the single-phase system for the same line voltage V_L and the same absorbed power P_L. We will compare these cases and assume in both that the wires are of the same material (e.g., copper with resistivity ρ), of the same length ℓ, and that the loads are resistive (i.e., unity power factor). For the two-wire single-phase system in Fig. 12.21(a), $I_L = P_L/V_L$, so the power loss in the two wires is

$$P_{\text{loss}} = 2I_L^2R = 2R\frac{P_L^2}{V_L^2} \tag{12.54}$$

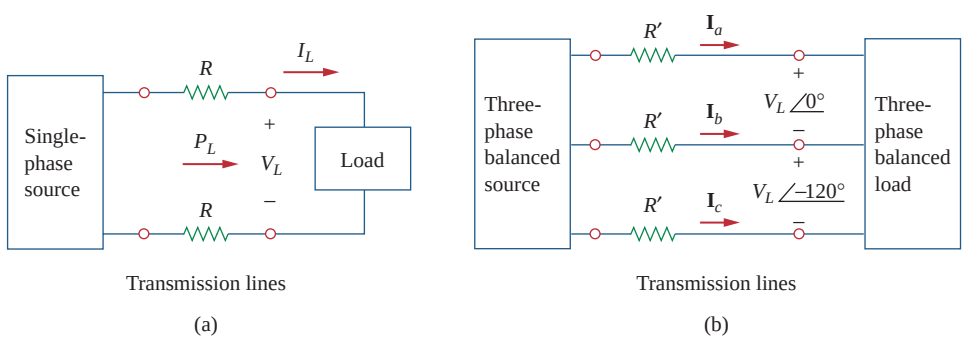


Figure 12.21
Comparing the power loss in (a) a single-phase system, and (b) a three-phase system.

For the three-wire three-phase system in Fig. 12.21(b), $I_L' = |\mathbf{I}_a| = |\mathbf{I}_b| = |\mathbf{I}_c| = P_L/\sqrt{3}V_L$ from Eq. (12.50). The power loss in the three wires is

$$P'_{\text{loss}} = 3(I_L')^2R' = 3R'\frac{P_L^2}{3V_L^2} = R'\frac{P_L^2}{V_L^2} \tag{12.55}$$

Equations (12.54) and (12.55) show that for the same total power delivered P_L and same line voltage V_L,

$$\frac{P_{\text{loss}}}{P'_{\text{loss}}} = \frac{2R}{R'} \tag{12.56}$$

But from Chapter 2, $R = \rho\ell/\pi r^2$ and $R' = \rho\ell/\pi r'^2$, where r and r' are the radii of the wires. Thus,

$$\frac{P_{\text{loss}}}{P'_{\text{loss}}} = \frac{2r'^2}{r^2} \tag{12.57}$$

If the same power loss is tolerated in both systems, then $r^2 = 2r'^2$. The ratio of material required is determined by the number of wires and their volumes, so

$$\begin{aligned}\frac{\text{Material for single-phase}}{\text{Material for three-phase}} &= \frac{2(\pi r^2\ell)}{3(\pi r'^2\ell)} = \frac{2r^2}{3r'^2} \\ &= \frac{2}{3}(2) = 1.333\end{aligned} \tag{12.58}$$

since $r^2 = 2r'^2$. Equation (12.58) shows that the single-phase system uses 33 percent more material than the three-phase system or that the three-phase system uses only 75 percent of the material used in the equivalent single-phase system. In other words, considerably less material is needed to deliver the same power with a three-phase system than is required for a single-phase system.

Example 12.6

Refer to the circuit in Fig. 12.13 (in Example 12.2). Determine the total average power, reactive power, and complex power at the source and at the load.

Solution:

It is sufficient to consider one phase, as the system is balanced. For phase a,

$$\mathbf{V}_p = 110\angle 0^\circ \text{ V} \qquad \text{and} \qquad \mathbf{I}_p = 6.81\angle -21.8^\circ \text{ A}$$

Thus, at the source, the complex power absorbed is

$$\begin{aligned}\mathbf{S}_s = -3\mathbf{V}_p\mathbf{I}_p^* &= -3(110\angle 0^\circ)(6.81\angle 21.8^\circ) \\ &= -2247\angle 21.8^\circ = -(2087 + j834.6) \text{ VA}\end{aligned}$$

The real or average power absorbed is -2087 W and the reactive power is -834.6 VAR.

At the load, the complex power absorbed is

$$\mathbf{S}_L = 3|\mathbf{I}_p|^2\mathbf{Z}_p$$

where $\mathbf{Z}_p = 10 + j8 = 12.81\angle 38.66^\circ$ and $\mathbf{I}_p = \mathbf{I}_a = 6.81\angle -21.8^\circ$. Hence,

$$\begin{aligned}\mathbf{S}_L &= 3(6.81)^2 12.81\angle 38.66^\circ = 1782\angle 38.66 \\ &= (1392 + j1113) \text{ VA}\end{aligned}$$

The real power absorbed is 1391.7 W and the reactive power absorbed is 1113.3 VAR. The difference between the two complex powers is absorbed by the line impedance $(5 - j2)\ \Omega$. To show that this is the case, we find the complex power absorbed by the line as

$$\mathbf{S}_\ell = 3|\mathbf{I}_p|^2\mathbf{Z}_\ell = 3(6.81)^2(5 - j2) = 695.6 - j278.3 \text{ VA}$$

which is the difference between $\mathbf{S}_s$ and $\mathbf{S}_L$; that is, $\mathbf{S}_s + \mathbf{S}_\ell + \mathbf{S}_L = 0$, as expected.

Practice Problem 12.6

For the Y-Y circuit in Practice Prob. 12.2, calculate the complex power at the source and at the load.

Answer: $-(1054 + j843.3)$ VA, $(1012 + j801.6)$ VA.

Example 12.7

A three-phase motor can be regarded as a balanced Y-load. A three-phase motor draws 5.6 kW when the line voltage is 220 V and the line current is 18.2 A. Determine the power factor of the motor.

Solution:
The apparent power is

$$S = \sqrt{3}V_L I_L = \sqrt{3}(220)(18.2) = 6935.13 \text{ VA}$$

Since the real power is

$$P = S\cos\theta = 5600 \text{ W}$$

the power factor is

$$\text{pf} = \cos\theta = \frac{P}{S} = \frac{5600}{6935.13} = 0.8075$$

Practice Problem 12.7

Calculate the line current required for a 30-kW three-phase motor having a power factor of 0.85 lagging if it is connected to a balanced source with a line voltage of 440 V.

Answer: 46.31 A.

Example 12.8

Two balanced loads are connected to a 240-kV rms 60-Hz line, as shown in Fig. 12.22(a). Load 1 draws 30 kW at a power factor of 0.6 lagging, while load 2 draws 45 kVAR at a power factor of 0.8 lagging. Assuming the *abc* sequence, determine: (a) the complex, real, and reactive powers absorbed by the combined load, (b) the line currents, and (c) the kVAR rating of the three capacitors Δ-connected in parallel with the load that will raise the power factor to 0.9 lagging and the capacitance of each capacitor.

Solution:

(a) For load 1, given that $P_1 = 30$ kW and $\cos\theta_1 = 0.6$, then $\sin\theta_1 = 0.8$. Hence,

$$S_1 = \frac{P_1}{\cos\theta_1} = \frac{30 \text{ kW}}{0.6} = 50 \text{ kVA}$$

and $Q_1 = S_1 \sin\theta_1 = 50(0.8) = 40$ kVAR. Thus, the complex power due to load 1 is

$$\mathbf{S}_1 = P_1 + jQ_1 = 30 + j40 \text{ kVA} \qquad \textbf{(12.8.1)}$$

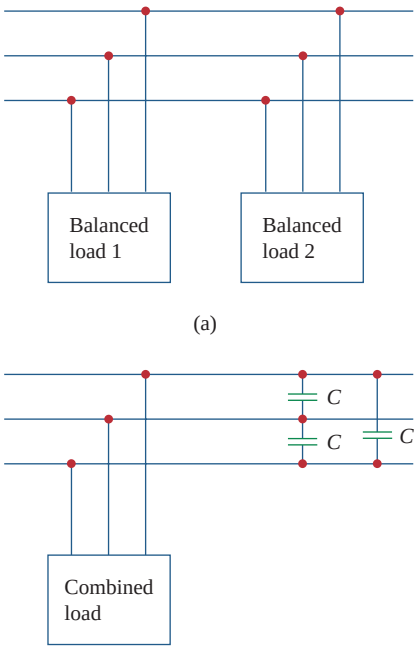


Figure 12.22
For Example 12.8: (a) The original balanced loads, (b) the combined load with improved power factor.

For load 2, if $Q_2 = 45$ kVAR and $\cos\theta_2 = 0.8$, then $\sin\theta_2 = 0.6$. We find

$$S_2 = \frac{Q_2}{\sin\theta_2} = \frac{45 \text{ kVA}}{0.6} = 75 \text{ kVA}$$

and $P_2 = S_2 \cos\theta_2 = 75(0.8) = 60$ kW. Therefore the complex power due to load 2 is

$$\mathbf{S}_2 = P_2 + jQ_2 = 60 + j45 \text{ kVA} \qquad \textbf{(12.8.2)}$$

From Eqs. (12.8.1) and (12.8.2), the total complex power absorbed by the load is

$$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2 = 90 + j85 \text{ kVA} = 123.8\angle 43.36^\circ \text{ kVA} \qquad \textbf{(12.8.3)}$$

which has a power factor of $\cos 43.36^\circ = 0.727$ lagging. The real power is then 90 kW, while the reactive power is 85 kVAR.

(b) Since $S = \sqrt{3} V_L I_L$, the line current is

$$I_L = \frac{S}{\sqrt{3} V_L} \qquad \textbf{(12.8.4)}$$

We apply this to each load, keeping in mind that for both loads, $V_L = 240$ kV. For load 1,

$$I_{L1} = \frac{50{,}000}{\sqrt{3}\, 240{,}000} = 120.28 \text{ mA}$$

Since the power factor is lagging, the line current lags the line voltage by $\theta_1 = \cos^{-1} 0.6 = 53.13^\circ$. Thus,

$$\mathbf{I}_{a1} = 120.28\angle -53.13^\circ$$

For load 2,

$$I_{L2} = \frac{75{,}000}{\sqrt{3}\, 240{,}000} = 180.42 \text{ mA}$$

and the line current lags the line voltage by $\theta_2 = \cos^{-1} 0.8 = 36.87^\circ$. Hence,

$$\mathbf{I}_{a2} = 180.42\angle -36.87^\circ$$

The total line current is

$$\begin{aligned} \mathbf{I}_a = \mathbf{I}_{a1} + \mathbf{I}_{a2} &= 120.28\angle -53.13^\circ + 180.42\angle -36.87^\circ \\ &= (72.168 - j96.224) + (144.336 - j108.252) \\ &= 216.5 - j204.472 = 297.8\angle -43.36^\circ \text{ mA} \end{aligned}$$

Alternatively, we could obtain the current from the total complex power using Eq. (12.8.4) as

$$I_L = \frac{123{,}800}{\sqrt{3}\, 240{,}000} = 297.82 \text{ mA}$$

and

$$\mathbf{I}_a = 297.82\angle -43.36^\circ \text{ mA}$$

which is the same as before. The other line currents, $\mathbf{I}_{b2}$ and $\mathbf{I}_{ca}$, can be obtained according to the *abc* sequence (i.e., $\mathbf{I}_b = 297.82\angle -163.36^\circ$ mA and $\mathbf{I}_c = 297.82\angle 76.64^\circ$ mA).

(c) We can find the reactive power needed to bring the power factor to 0.9 lagging using Eq. (11.59),

$$Q_C = P(\tan\theta_{\text{old}} - \tan\theta_{\text{new}})$$

where $P = 90$ kW, $\theta_{\text{old}} = 43.36°$, and $\theta_{\text{new}} = \cos^{-1} 0.9 = 25.84°$. Hence,

$$Q_C = 90{,}000(\tan 43.36° - \tan 25.84°) = 41.4 \text{ kVAR}$$

This reactive power is for the three capacitors. For each capacitor, the rating $Q'_C = 13.8$ kVAR. From Eq. (11.60), the required capacitance is

$$C = \frac{Q'_C}{\omega V_{\text{rms}}^2}$$

Since the capacitors are Δ-connected as shown in Fig. 12.22(b), V_{rms} in the above formula is the line-to-line or line voltage, which is 240 kV. Thus,

$$C = \frac{13{,}800}{(2\pi 60)(240{,}000)^2} = 635.5 \text{ pF}$$

Practice Problem 12.8

Assume that the two balanced loads in Fig. 12.22(a) are supplied by an 840-V rms 60-Hz line. Load 1 is Y-connected with $30 + j40\ \Omega$ per phase, while load 2 is a balanced three-phase motor drawing 48 kW at a power factor of 0.8 lagging. Assuming the *abc* sequence, calculate: (a) the complex power absorbed by the combined load, (b) the kVAR rating of each of the three capacitors Δ-connected in parallel with the load to raise the power factor to unity, and (c) the current drawn from the supply at unity power factor condition.

Answer: (a) $56.47 + j47.29$ kVA, (b) 15.7 kVAR, (c) 38.813 A.

12.8 †Unbalanced Three-Phase Systems

This chapter would be incomplete without mentioning unbalanced three-phase systems. An unbalanced system is caused by two possible situations: (1) the source voltages are not equal in magnitude and/or differ in phase by angles that are unequal, or (2) load impedances are unequal. Thus,

An **unbalanced system** is due to unbalanced voltage sources or an unbalanced load.

To simplify analysis, we will assume balanced source voltages, but an unbalanced load.

Unbalanced three-phase systems are solved by direct application of mesh and nodal analysis. Figure 12.23 shows an example of an unbalanced three-phase system that consists of balanced source voltages (not shown in the figure) and an unbalanced Y-connected load (shown in the figure). Since the load is unbalanced, $\mathbf{Z}_A$, $\mathbf{Z}_B$, and $\mathbf{Z}_C$ are not equal. The line currents are determined by Ohm's law as

$$\mathbf{I}_a = \frac{\mathbf{V}_{AN}}{\mathbf{Z}_A}, \quad \mathbf{I}_b = \frac{\mathbf{V}_{BN}}{\mathbf{Z}_B}, \quad \mathbf{I}_c = \frac{\mathbf{V}_{CN}}{\mathbf{Z}_C} \tag{12.59}$$

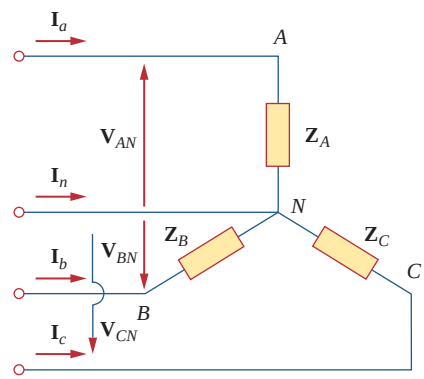


Figure 12.23
Unbalanced three-phase Y-connected load.

A special technique for handling unbalanced three-phase systems is the method of *symmetrical components*, which is beyond the scope of this text.

This set of unbalanced line currents produces current in the neutral line, which is not zero as in a balanced system. Applying KCL at node N gives the neutral line current as

$$\mathbf{I}_n = -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c) \tag{12.60}$$

In a three-wire system where the neutral line is absent, we can still find the line currents $\mathbf{I}_a$, $\mathbf{I}_b$, and $\mathbf{I}_c$ using mesh analysis. At node N, KCL must be satisfied so that $\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0$ in this case. The same could be done for an unbalanced Δ-Y, Y-Δ, or Δ-Δ three-wire system. As mentioned earlier, in long distance power transmission, conductors in multiples of three (multiple three-wire systems) are used, with the earth itself acting as the neutral conductor.

To calculate power in an unbalanced three-phase system requires that we find the power in each phase using Eqs. (12.46) to (12.49). The total power is not simply three times the power in one phase but the sum of the powers in the three phases.

Example 12.9

The unbalanced Y-load of Fig. 12.23 has balanced voltages of 100 V and the *acb* sequence. Calculate the line currents and the neutral current. Take $\mathbf{Z}_A = 15\ \Omega$, $\mathbf{Z}_B = 10 + j5\ \Omega$, $\mathbf{Z}_C = 6 - j8\ \Omega$.

Solution:

Using Eq. (12.59), the line currents are

$$\mathbf{I}_a = \frac{100\angle 0^\circ}{15} = 6.67\angle 0^\circ \text{ A}$$

$$\mathbf{I}_b = \frac{100\angle 120^\circ}{10 + j5} = \frac{100\angle 120^\circ}{11.18\angle 26.56^\circ} = 8.94\angle 93.44^\circ \text{ A}$$

$$\mathbf{I}_c = \frac{100\angle -120^\circ}{6 - j8} = \frac{100\angle -120^\circ}{10\angle -53.13^\circ} = 10\angle -66.87^\circ \text{ A}$$

Using Eq. (12.60), the current in the neutral line is

$$\begin{aligned}\mathbf{I}_n &= -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c) = -(6.67 - 0.54 + j8.92 + 3.93 - j9.2) \\ &= -10.06 + j0.28 = 10.06\angle 178.4^\circ \text{ A}\end{aligned}$$

Practice Problem 12.9

The unbalanced Δ-load of Fig. 12.24 is supplied by balanced line-to-line voltages of 240 V in the positive sequence. Find the line currents. Take $\mathbf{V}_{ab}$ as reference.

Answer: $21.66\angle -41.06^\circ$ A, $34.98\angle -139.8^\circ$ A, $38.24\angle 74.27^\circ$ A.

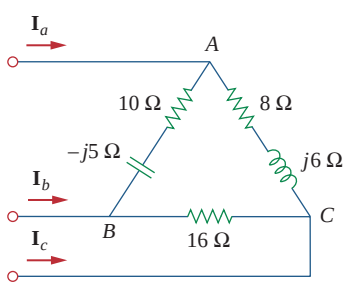


Figure 12.24
Unbalanced Δ-load; for Practice Prob. 12.9.

Example 12.10

For the unbalanced circuit in Fig. 12.25, find: (a) the line currents, (b) the total complex power absorbed by the load, and (c) the total complex power absorbed by the source.

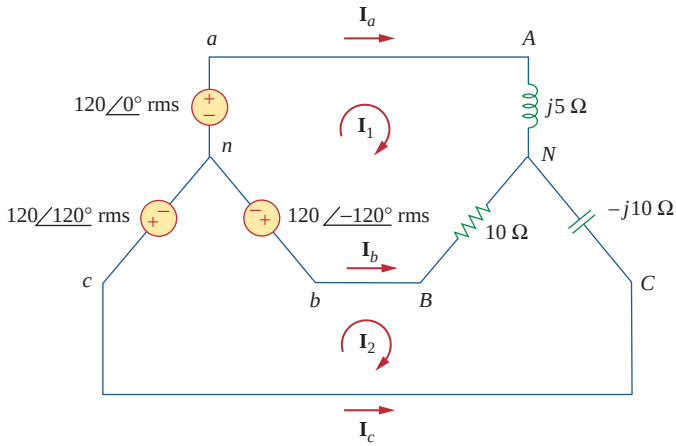


Figure 12.25
For Example 12.10.

Solution:

(a) We use mesh analysis to find the required currents. For mesh 1,

$$120\angle -120^\circ - 120\angle 0^\circ + (10 + j5)\mathbf{I}_1 - 10\mathbf{I}_2 = 0$$

or

$$(10 + j5)\mathbf{I}_1 - 10\mathbf{I}_2 = 120\sqrt{3}\angle 30^\circ \qquad \textbf{(12.10.1)}$$

For mesh 2,

$$120\angle 120^\circ - 120\angle -120^\circ + (10 - j10)\mathbf{I}_2 - 10\mathbf{I}_1 = 0$$

or

$$-10\mathbf{I}_1 + (10 - j10)\mathbf{I}_2 = 120\sqrt{3}\angle -90^\circ \qquad \textbf{(12.10.2)}$$

Equations (12.10.1) and (12.10.2) form a matrix equation:

$$\begin{bmatrix} 10 + j5 & -10 \\ -10 & 10 - j10 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 120\sqrt{3}\angle 30^\circ \\ 120\sqrt{3}\angle -90^\circ \end{bmatrix}$$

The determinants are

$$\Delta = \begin{vmatrix} 10 + j5 & -10 \\ -10 & 10 - j10 \end{vmatrix} = 50 - j50 = 70.71\angle -45^\circ$$

$$\Delta_1 = \begin{vmatrix} 120\sqrt{3}\angle 30^\circ & -10 \\ 120\sqrt{3}\angle -90^\circ & 10 - j10 \end{vmatrix} = 207.85(13.66 - j13.66)$$

$$= 4015\angle -45^\circ$$

$$\Delta_2 = \begin{vmatrix} 10 + j5 & 120\sqrt{3}\angle 30^\circ \\ -10 & 120\sqrt{3}\angle -90^\circ \end{vmatrix} = 207.85(13.66 - j5)$$

$$= 3023.4\angle -20.1^\circ$$

The mesh currents are

$$\mathbf{I}_1 = \frac{\Delta_1}{\Delta} = \frac{4015.23\angle -45^\circ}{70.71\angle -45^\circ} = 56.78\ \text{A}$$

$$\mathbf{I}_2 = \frac{\Delta_2}{\Delta} = \frac{3023.4\angle -20.1^\circ}{70.71\angle -45^\circ} = 42.75\angle 24.9^\circ\ \text{A}$$

The line currents are

$$\mathbf{I}_a = \mathbf{I}_1 = 56.78 \text{ A}, \qquad \mathbf{I}_c = -\mathbf{I}_2 = 42.75\angle -155.1^\circ \text{ A}$$
$$\mathbf{I}_b = \mathbf{I}_2 - \mathbf{I}_1 = 38.78 + j18 - 56.78 = 25.46\angle 135^\circ \text{ A}$$

(b) We can now calculate the complex power absorbed by the load. For phase A,

$$\mathbf{S}_A = |\mathbf{I}_a|^2 \mathbf{Z}_A = (56.78)^2(j5) = j16{,}120 \text{ VA}$$

For phase B,

$$\mathbf{S}_B = |\mathbf{I}_b|^2 \mathbf{Z}_B = (25.46)^2(10) = 6480 \text{ VA}$$

For phase C,

$$\mathbf{S}_C = |\mathbf{I}_c|^2 \mathbf{Z}_C = (42.75)^2(-j10) = -j18{,}276 \text{ VA}$$

The total complex power absorbed by the load is

$$\mathbf{S}_L = \mathbf{S}_A + \mathbf{S}_B + \mathbf{S}_C = 6480 - j2156 \text{ VA}$$

(c) We check the result above by finding the power absorbed by the source. For the voltage source in phase *a*,

$$\mathbf{S}_a = -\mathbf{V}_{an}\mathbf{I}_a^* = -(120\angle 0^\circ)(56.78) = -6813.6 \text{ VA}$$

For the source in phase *b*,

$$\mathbf{S}_b = -\mathbf{V}_{bn}\mathbf{I}_b^* = -(120\angle -120^\circ)(25.46\angle -135^\circ)$$
$$= -3055.2\angle 105^\circ = 790 - j2951.1 \text{ VA}$$

For the source in phase *c*,

$$\mathbf{S}_c = -\mathbf{V}_{bn}\mathbf{I}_c^* = -(120\angle 120^\circ)(42.75\angle 155.1^\circ)$$
$$= -5130\angle 275.1^\circ = -456.03 + j5109.7 \text{ VA}$$

The total complex power absorbed by the three-phase source is

$$\mathbf{S}_s = \mathbf{S}_a + \mathbf{S}_b + \mathbf{S}_c = -6480 + j2156 \text{ VA}$$

showing that $\mathbf{S}_s + \mathbf{S}_L = 0$ and confirming the conservation principle of ac power.

Practice Problem 12.10

Find the line currents in the unbalanced three-phase circuit of Fig. 12.26 and the real power absorbed by the load.

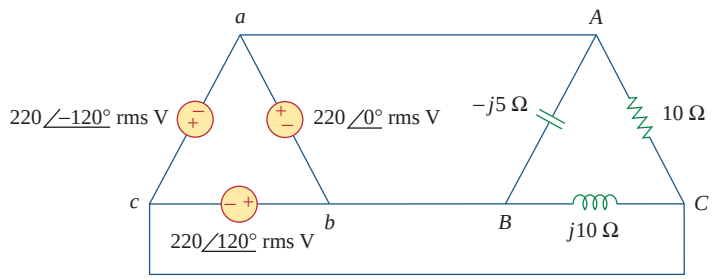


Figure 12.26
For Practice Prob. 12.10.

Answer: $64\angle 80.1^\circ$ A, $38.1\angle -60^\circ$ A, $42.5\angle 225^\circ$ A, 4.84 kW.

12.9 *PSpice* for Three-Phase Circuits

PSpice can be used to analyze three-phase balanced or unbalanced circuits in the same way it is used to analyze single-phase ac circuits. However, a delta-connected source presents two major problems to *PSpice*. First, a delta-connected source is a loop of voltage sources—which *PSpice* does not like. To avoid this problem, we insert a resistor of negligible resistance (say, 1 $\mu\Omega$ per phase) into each phase of the delta-connected source. Second, the delta-connected source does not provide a convenient node for the ground node, which is necessary to run *PSpice*. This problem can be eliminated by inserting balanced wye-connected large resistors (say, 1 MΩ per phase) in the delta-connected source so that the neutral node of the wye-connected resistors serves as the ground node 0. Example 12.12 will illustrate this.

Example 12.11

For the balanced Y-Δ circuit in Fig. 12.27, use *PSpice* to find the line current $\mathbf{I}_{aA}$, the phase voltage $\mathbf{V}_{AB}$, and the phase current $\mathbf{I}_{AC}$. Assume that the source frequency is 60 Hz.

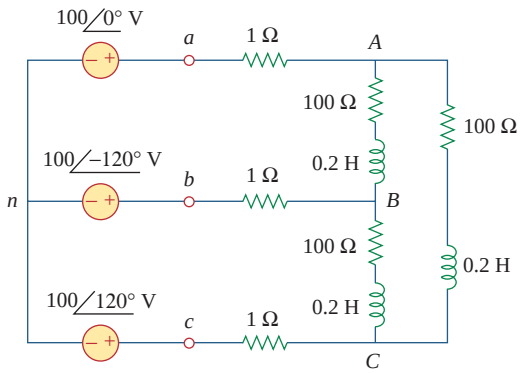


Figure 12.27
For Example 12.11.

Solution:

The schematic is shown in Fig. 12.28. The pseudocomponents IPRINT are inserted in the appropriate lines to obtain $\mathbf{I}_{aA}$ and $\mathbf{I}_{AC}$, while VPRINT2 is inserted between nodes A and B to print differential voltage $\mathbf{V}_{AB}$. We set the attributes of IPRINT and VPRINT2 each to *AC* = *yes*, *MAG* = *yes*, *PHASE* = *yes*, to print only the magnitude and phase of the currents and voltages. As a single-frequency analysis, we select **Analysis/Setup/AC Sweep** and enter *Total Pts* = 1, *Start Freq* = 60, and *Final Freq* = 60. Once the circuit is saved, it is simulated by selecting **Analysis/Simulate**. The output file includes the following:

```
FREQ          V(A,B)          VP(A,B)
6.000E+01     1.699E+02       3.081E+01

FREQ          IM(V_PRINT2)    IP(V_PRINT2)
6.000E+01     2.350E+00       -3.620E+01

FREQ          IM(V_PRINT3)    IP(V_PRINT3)
6.000E+01     1.357E+00       -6.620E+01
```

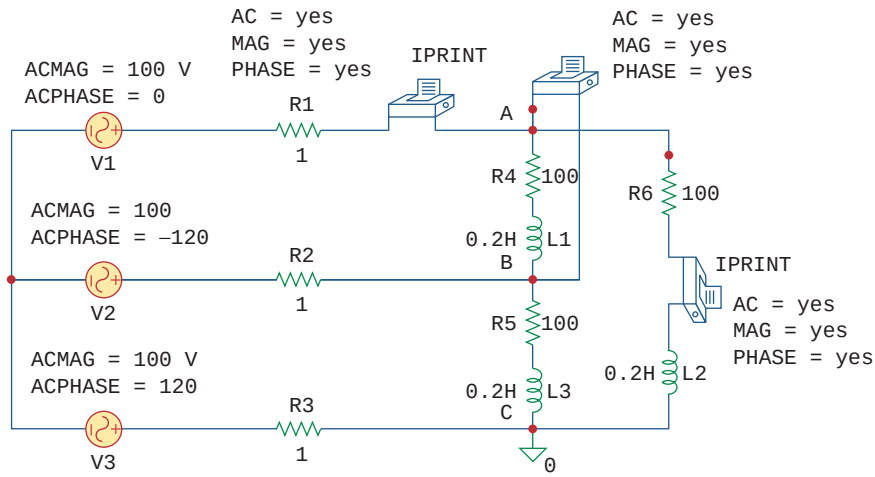


Figure 12.28
Schematic for the circuit in Fig. 12.27.

From this, we obtain

$$\mathbf{I}_{aA} = 2.35\angle -36.2^\circ \text{ A}$$
$$\mathbf{V}_{AB} = 169.9\angle 30.81^\circ \text{ V}, \quad \mathbf{I}_{AC} = 1.357\angle -66.2^\circ \text{ A}$$

Practice Problem 12.11

Refer to the balanced Y-Y circuit of Fig. 12.29. Use *PSpice* to find the line current $\mathbf{I}_{bB}$ and the phase voltage $\mathbf{V}_{AN}$. Take $f = 100$ Hz.

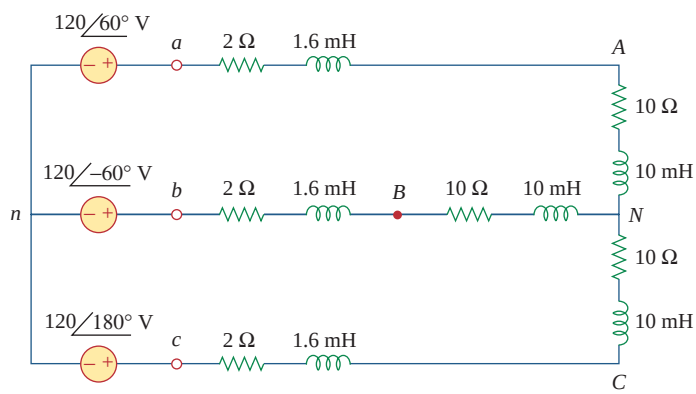


Figure 12.29
For Practice Prob. 12.11.

Answer: $100.9\angle 60.87^\circ$ V, $8.547\angle -91.27^\circ$ A.

Example 12.12

Consider the unbalanced Δ-Δ circuit in Fig. 12.30. Use *PSpice* to find the generator current $\mathbf{I}_{ab}$, the line current $\mathbf{I}_{bB}$, and the phase current $\mathbf{I}_{BC}$.

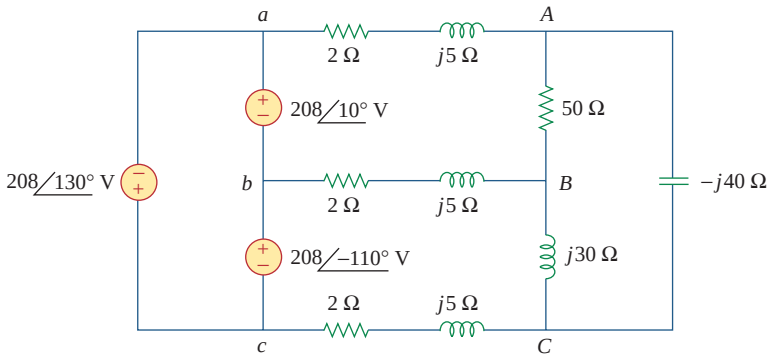


Figure 12.30
For Example 12.12.

Solution:

1. **Define.** The problem and solution process are clearly defined.
2. **Present.** We are to find the generator current flowing from *a* to *b*, the line current flowing from *b* to *B*, and the phase current flowing from *B* to *C*.
3. **Alternative.** Although, there are different approaches to solving this problem, the use of *PSpice* is mandated. Therefore, we will not use another approach.
4. **Attempt.** As mentioned above, we avoid the loop of voltage sources by inserting a 1-$\mu\Omega$ series resistor in the delta-connected source. To provide a ground node 0, we insert balanced wye-connected resistors (1 MΩ per phase) in the delta-connected source, as shown in the schematic in Fig. 12.31. Three IPRINT pseudocomponents with their

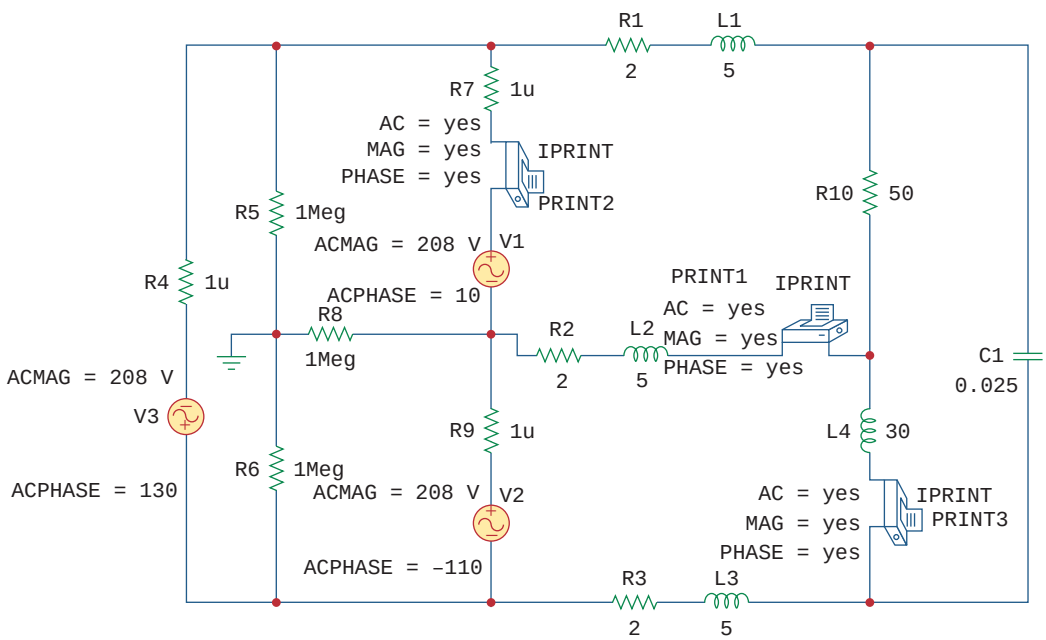


Figure 12.31
Schematic for the circuit in Fig. 12.30.

attributes are inserted to be able to get the required currents $\mathbf{I}_{ab}$, $\mathbf{I}_{bB}$, and $\mathbf{I}_{BC}$. Since the operating frequency is not given and the inductances and capacitances should be specified instead of impedances, we assume $\omega = 1$ rad/s so that $f = 1/2\pi = 0.159155$ Hz. Thus,

$$L = \frac{X_L}{\omega} \qquad \text{and} \qquad C = \frac{1}{\omega X_C}$$

We select **Analysis/Setup/AC Sweep** and enter *Total Pts* = 1, *Start Freq* = 0.159155, and *Final Freq* = 0.159155. Once the schematic is saved, we select **Analysis/Simulate** to simulate the circuit. The output file includes:

```
FREQ          IM(V_PRINT1)   IP(V_PRINT1)
1.592E-01     9.106E+00      1.685E+02

FREQ          IM(V_PRINT2)   IP(V_PRINT2)
1.592E-01     5.959E+00      -1.772E+02

FREQ          IM(V_PRINT3)   IP(V_PRINT3)
1.592E-01     5.500E+00      1.725E+02
```

which yields

$$I_{ab} = \mathbf{5.595}\angle\mathbf{-177.2^\circ}\ \mathbf{A},\ I_{bB} = \mathbf{9.106}\angle\mathbf{168.5^\circ}\ \mathbf{A}, \text{ and}$$
$$I_{BC} = \mathbf{5.5}\angle\mathbf{172.5^\circ}\ \mathbf{A}$$

5. **Evaluate.** We can check our results by using mesh analysis. Let the loop *aABb* be loop 1, the loop *bBCc* be loop 2, and the loop *ACB* be loop 3, with the three loop currents all flowing in the clockwise direction. We then end up with the following loop equations:

Loop 1

$$(54 + j10)I_1 - (2 + j5)I_2 - (50)I_3 = 208\angle 10^\circ = 204.8 + j36.12$$

Loop 2

$$-(2 + j5)I_1 + (4 + j40)I_2 - (j30)I_3 = 208\angle -110^\circ = -71.14 - j195.46$$

Loop 3

$$-(50)I_1 - (j30)I_2 + (50 - j10)I_3 = 0$$

Using MATLAB to solve this we get,

```
>>Z=[(54+10i),(-2-5i),-50;(-2-5i),(4+40i),
-30i;-50,-30i,(50-10i)]

Z=
54.0000+10.0000i-2.0000-5.0000i-50.0000
-2.0000-5.0000i  4.0000+40.0000i  0-30.0000i
-50.0000  0-30.0000i  50.0000-10.0000i
```

```
>>V=[(204.8+36.12i);(-71.14-195.46i);0]

V=
1.0e+002*
2.0480+0.3612i
-0.7114-1.9546i
     0
>>I=inv(Z)*V

I=
8.9317+2.6983i
0.0096+4.5175i
5.4619+3.7964i
```

$$I_{bB} = -I_1 + I_2 = -(8.932 + j2.698) + (0.0096 + j4.518)$$
$$= -8.922 + j1.82 = \mathbf{9.106}\angle\mathbf{168.47}^\circ\ \mathbf{A} \qquad \text{Answer checks}$$
$$I_{BC} = I_2 - I_3 = (0.0096 + j4.518) - (5.462 + j3.796)$$
$$= -5.452 + j0.722 = \mathbf{5.5}\angle\mathbf{172.46}^\circ\ \mathbf{A} \qquad \text{Answer checks}$$

Now to solve for I_{ab}. If we assume a small internal impedance for each source, we can obtain a reasonably good estimate for I_{ab}. Adding in internal resistors of 0.01Ω, and adding a fourth loop around the source circuit, we now get

Loop 1

$$(54.01 + j10)I_1 - (2 + j5)I_2 - (50)I_3 - 0.01I_4 = 208\angle 10^\circ$$
$$= 204.8 + j36.12$$

Loop 2

$$-(2 + j5)I_1 + (4.01 + j40)I_2 - (j30)I_3 - 0.01I_4$$
$$= 208\angle -110^\circ = -71.14 - j195.46$$

Loop 3

$$-(50)I_1 - (j30)I_2 + (50 - j10)I_3 = 0$$

Loop 4

$$-(0.01)I_1 - (0.01)I_2 + (0.03)I_4 = 0$$

```
>>Z=[(54.01+10i),(-2-5i),-50,-0.01;(-2-5i),
(4.01+40i),-30i,-0.01;-50,-30i,(50-10i),
0;-0.01,-0.01,0,0.03]

Z=
54.0100+10.0000i  -2.0000-5.0000i,  -50.0000  -0.0100
-2.0000-5.0000i  4.0100-40.0000i  0-30.0000i  0.0100
-50.0000  0-30.0000i  50.0000-10.0000i  0
-0.0100  -0.0100  0  0.0300

>>V=[(204.8+36.12i);(-71.14-195.46i);0;0]
```

```
V=

1.0e+002*

2.0480+0.3612i
-0.7114-1.9546i
     0
     0
>>I=inv(Z)*V

I=

8.9309+2.6973i
0.0093+4.5159i
5.4623+3.7954i
2.9801+2.4044i
```

$$I_{ab} = -I_1 + I_4 = -(8.931 + j2.697) + (2.98 + j2.404)$$
$$= -5.951 - j0.293 = \mathbf{5.958\angle -177.18^\circ\ A}. \quad \text{Answer checks.}$$

6. **Satisfactory?** We have a satisfactory solution and an adequate check for the solution. We can now present the results as a solution to the problem.

Practice Problem 12.12

For the unbalanced circuit in Fig. 12.32, use *PSpice* to find the generator current $\mathbf{I}_{ca}$, the line current $\mathbf{I}_{cC}$, and the phase current $\mathbf{I}_{AB}$.

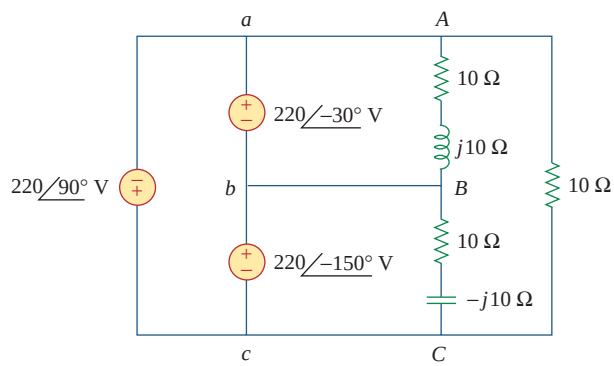


Figure 12.32
For Practice Prob. 12.12.

Answer: $24.68\angle -90^\circ$ A, $37.25\angle 83.8^\circ$ A, $15.556\angle -75^\circ$ A.

12.10 †Applications

Both wye and delta source connections have important practical applications. The wye source connection is used for long distance transmission of electric power, where resistive losses (I^2R) should be

minimal. This is due to the fact that the wye connection gives a line voltage that is $\sqrt{3}$ greater than the delta connection; hence, for the same power, the line current is $\sqrt{3}$ smaller. The delta source connection is used when three single-phase circuits are desired from a three-phase source. This conversion from three-phase to single-phase is required in residential wiring, because household lighting and appliances use single-phase power. Three-phase power is used in industrial wiring where a large power is required. In some applications, it is immaterial whether the load is wye- or delta-connected. For example, both connections are satisfactory with induction motors. In fact, some manufacturers connect a motor in delta for 220 V and in wye for 440 V so that one line of motors can be readily adapted to two different voltages.

Here we consider two practical applications of those concepts covered in this chapter: power measurement in three-phase circuits and residential wiring.

12.10.1 Three-Phase Power Measurement

Section 11.9 presented the wattmeter as the instrument for measuring the average (or real) power in single-phase circuits. A single wattmeter can also measure the average power in a three-phase system that is balanced, so that $P_1 = P_2 = P_3$; the total power is three times the reading of that one wattmeter. However, two or three single-phase wattmeters are necessary to measure power if the system is unbalanced. The *three-wattmeter method* of power measurement, shown in Fig. 12.33, will work regardless of whether the load is balanced or unbalanced, wye- or delta-connected. The three-wattmeter method is well suited for power measurement in a three-phase system where the power factor is constantly changing. The total average power is the algebraic sum of the three wattmeter readings,

$$P_T = P_1 + P_2 + P_3 \tag{12.61}$$

where P_1, P_2, and P_3 correspond to the readings of wattmeters W_1, W_2, and W_3, respectively. Notice that the common or reference point o in Fig. 12.33 is selected arbitrarily. If the load is wye-connected, point o can be connected to the neutral point n. For a delta-connected load, point o can be connected to any point. If point o is connected to point b, for example, the voltage coil in wattmeter W_2 reads zero and $P_2 = 0$, indicating that wattmeter W_2 is not necessary. Thus, two wattmeters are sufficient to measure the total power.

The *two-wattmeter method* is the most commonly used method for three-phase power measurement. The two wattmeters must be properly connected to any two phases, as shown typically in Fig. 12.34. Notice that the current coil of each wattmeter measures the line current, while the respective voltage coil is connected between the line and the third line and measures the line voltage. Also notice that the $\pm$ terminal of the voltage coil is connected to the line to which the corresponding current coil is connected. Although the individual wattmeters no longer read the power taken by any particular phase, the algebraic sum of the two wattmeter readings equals the total average power absorbed by the load, regardless of whether it is wye- or delta-connected, balanced or

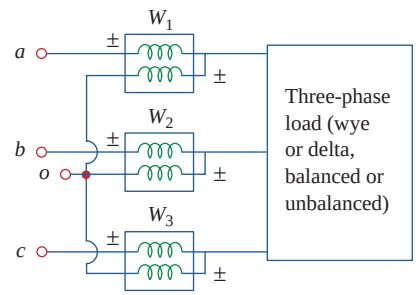


Figure 12.33
Three-wattmeter method for measuring three-phase power.

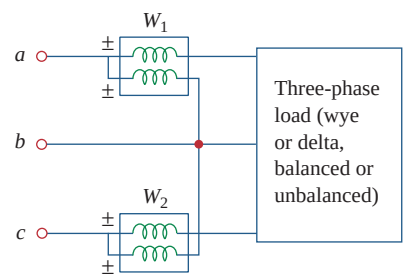


Figure 12.34
Two-wattmeter method for measuring three-phase power.

unbalanced. The total real power is equal to the algebraic sum of the two wattmeter readings,

$$P_T = P_1 + P_2 \tag{12.62}$$

We will show here that the method works for a balanced three-phase system.

Consider the balanced, wye-connected load in Fig. 12.35. Our objective is to apply the two-wattmeter method to find the average power absorbed by the load. Assume the source is in the *abc* sequence and the load impedance $\mathbf{Z}_Y = Z_Y\underline{/\theta}$. Due to the load impedance, each voltage coil leads its current coil by θ, so that the power factor is $\cos\theta$. We recall that each line voltage leads the corresponding phase voltage by 30°. Thus, the total phase difference between the phase current $\mathbf{I}_a$ and line voltage $\mathbf{V}_{ab}$ is $\theta + 30°$, and the average power read by wattmeter W_1 is

$$P_1 = \text{Re}[\mathbf{V}_{ab}\mathbf{I}_a^*] = V_{ab}I_a\cos(\theta + 30°) = V_L I_L\cos(\theta + 30°) \tag{12.63}$$

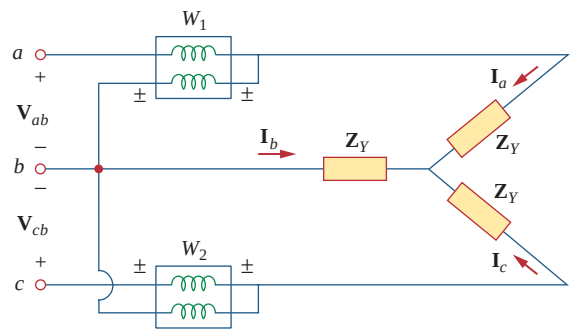


Figure 12.35
Two-wattmeter method applied to a balanced wye load.

Similarly, we can show that the average power read by wattmeter 2 is

$$P_2 = \text{Re}[\mathbf{V}_{cb}\mathbf{I}_c^*] = V_{cb}I_c\cos(\theta - 30°) = V_L I_L\cos(\theta - 30°) \tag{12.64}$$

We now use the trigonometric identities

$$\begin{aligned}\cos(A + B) &= \cos A\cos B - \sin A\sin B\\ \cos(A - B) &= \cos A\cos B + \sin A\sin B\end{aligned} \tag{12.65}$$

to find the sum and the difference of the two wattmeter readings in Eqs. (12.63) and (12.64):

$$\begin{aligned}P_1 + P_2 &= V_L I_L[\cos(\theta + 30°) + \cos(\theta - 30°)]\\ &= V_L I_L(\cos\theta\cos 30° - \sin\theta\sin 30°\\ &\qquad + \cos\theta\cos 30° + \sin\theta\sin 30°)\\ &= V_L I_L 2\cos 30°\cos\theta = \sqrt{3}V_L I_L\cos\theta\end{aligned} \tag{12.66}$$

since $2\cos 30° = \sqrt{3}$. Comparing Eq. (12.66) with Eq. (12.50) shows that the sum of the wattmeter readings gives the total average power,

$$\boxed{P_T = P_1 + P_2} \tag{12.67}$$

Similarly,

$$\begin{aligned} P_1 - P_2 &= V_L I_L[\cos(\theta + 30^\circ) - \cos(\theta - 30^\circ)] \\ &= V_l I_L(\cos\theta \cos 30^\circ - \sin\theta \sin 30^\circ \\ &\qquad - \cos\theta \cos 30^\circ - \sin\theta \sin 30^\circ) \\ &= -V_L I_L 2 \sin 30^\circ \sin\theta \\ P_2 - P_1 &= V_L I_L \sin\theta \end{aligned} \tag{12.68}$$

since $2 \sin 30^\circ = 1$. Comparing Eq. (12.68) with Eq. (12.51) shows that the difference of the wattmeter readings is proportional to the total reactive power, or

$$\boxed{Q_T = \sqrt{3}(P_2 - P_1)} \tag{12.69}$$

From Eqs. (12.67) and (12.69), the total apparent power can be obtained as

$$S_T = \sqrt{P_T^2 + Q_T^2} \tag{12.70}$$

Dividing Eq. (12.69) by Eq. (12.67) gives the tangent of the power factor angle as

$$\tan\theta = \frac{Q_T}{P_T} = \sqrt{3}\frac{P_2 - P_1}{P_2 + P_1} \tag{12.71}$$

from which we can obtain the power factor as $\text{pf} = \cos\theta$. Thus, the two-wattmeter method not only provides the total real and reactive powers, it can also be used to compute the power factor. From Eqs. (12.67), (12.69), and (12.71), we conclude that:

1. If $P_2 = P_1$, the load is resistive.
2. If $P_2 > P_1$, the load is inductive.
3. If $P_2 < P_1$, the load is capacitive.

Although these results are derived from a balanced wye-connected load, they are equally valid for a balanced delta-connected load. However, the two-wattmeter method cannot be used for power measurement in a three-phase four-wire system unless the current through the neutral line is zero. We use the three-wattmeter method to measure the real power in a three-phase four-wire system.

Example 12.13

Three wattmeters W_1, W_2, and W_3 are connected, respectively, to phases *a*, *b*, and *c* to measure the total power absorbed by the unbalanced wye-connected load in Example 12.9 (see Fig. 12.23). (a) Predict the wattmeter readings. (b) Find the total power absorbed.

Solution:

Part of the problem is already solved in Example 12.9. Assume that the wattmeters are properly connected as in Fig. 12.36.

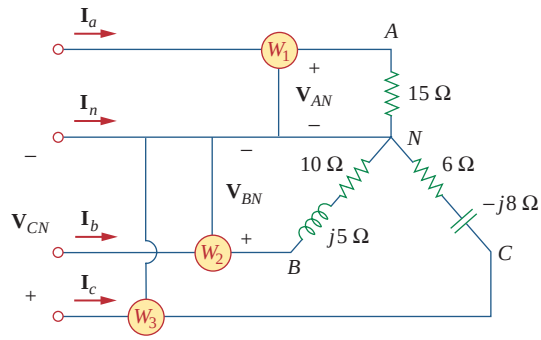


Figure 12.36
For Example 12.13.

(a) From Example 12.9,

$$\mathbf{V}_{AN} = 100\angle 0°, \quad \mathbf{V}_{BN} = 100\angle 120°, \quad \mathbf{V}_{CN} = 100\angle -120° \text{ V}$$

while

$$\mathbf{I}_a = 6.67\angle 0°, \quad \mathbf{I}_b = 8.94\angle 93.44°, \quad \mathbf{I}_c = 10\angle -66.87° \text{ A}$$

We calculate the wattmeter readings as follows:

$$\begin{aligned} P_1 = \text{Re}(\mathbf{V}_{AN}\mathbf{I}_a^*) &= V_{AN}I_a\cos(\theta_{\mathbf{V}_{AN}} - \theta_{\mathbf{I}_a}) \\ &= 100 \times 6.67 \times \cos(0° - 0°) = 667 \text{ W} \\ P_2 = \text{Re}(\mathbf{V}_{BN}\mathbf{I}_b^*) &= V_{BN}I_b\cos(\theta_{\mathbf{V}_{BN}} - \theta_{\mathbf{I}_b}) \\ &= 100 \times 8.94 \times \cos(120° - 93.44°) = 800 \text{ W} \\ P_3 = \text{Re}(\mathbf{V}_{CN}\mathbf{I}_c^*) &= V_{CN}I_c\cos(\theta_{\mathbf{V}_{CN}} - \theta_{\mathbf{I}_c}) \\ &= 100 \times 10 \times \cos(-120° + 66.87°) = 600 \text{ W} \end{aligned}$$

(b) The total power absorbed is

$$P_T = P_1 + P_2 + P_3 = 667 + 800 + 600 = 2067 \text{ W}$$

We can find the power absorbed by the resistors in Fig. 12.36 and use that to check or confirm this result.

$$\begin{aligned} P_T &= |I_a|^2(15) + |I_b|^2(10) + |I_c|^2(6) \\ &= 6.67^2(15) + 8.94^2(10) + 10^2(6) \\ &= 667 + 800 + 600 = 2067 \text{ W} \end{aligned}$$

which is exactly the same thing.

Practice Problem 12.13

Repeat Example 12.13 for the network in Fig. 12.24 (see Practice Prob. 12.9). *Hint:* Connect the reference point *o* in Fig. 12.33 to point *B*.

Answer: (a) 3.92 kW, 0 W, 8.895 kW, (b) 12.815 kW.

Example 12.14

The two-wattmeter method produces wattmeter readings $P_1 = 1560$ W and $P_2 = 2100$ W when connected to a delta-connected load. If the line voltage is 220 V, calculate: (a) the per-phase average power, (b) the per-phase reactive power, (c) the power factor, and (d) the phase impedance.

Solution:
We can apply the given results to the delta-connected load. (a) The total real or average power is

$$P_T = P_1 + P_2 = 1560 + 2100 = 3660 \text{ W}$$

The per-phase average power is then

$$P_p = \frac{1}{3}P_T = 1220 \text{ W}$$

(b) The total reactive power is

$$Q_T = \sqrt{3}(P_2 - P_1) = \sqrt{3}(2100 - 1560) = 935.3 \text{ VAR}$$

so that the per-phase reactive power is

$$Q_p = \frac{1}{3}Q_T = 311.77 \text{ VAR}$$

(c) The power angle is

$$\theta = \tan^{-1}\frac{Q_T}{P_T} = \tan^{-1}\frac{935.3}{3660} = 14.33^\circ$$

Hence, the power factor is

$$\cos\theta = 0.9689 \text{ (lagging)}$$

It is a lagging pf because Q_T is positive or $P_2 > P_1$.
(c) The phase impedance is $\mathbf{Z}_p = Z_p\angle\theta$. We know that θ is the same as the pf angle; that is, $\theta = 14.33^\circ$.

$$Z_p = \frac{V_p}{I_p}$$

We recall that for a delta-connected load, $V_p = V_L = 220$ V. From Eq. (12.46),

$$P_p = V_p I_p \cos\theta \quad \Rightarrow \quad I_p = \frac{1220}{220 \times 0.9689} = 5.723 \text{ A}$$

Hence,

$$Z_p = \frac{V_p}{I_p} = \frac{220}{5.723} = 38.44\ \Omega$$

and

$$\mathbf{Z}_p = 38.44\angle 14.33^\circ\ \Omega$$

Practice Problem 12.14

Let the line voltage $V_L = 208$ V and the wattmeter readings of the balanced system in Fig. 12.35 be $P_1 = -560$ W and $P_2 = 800$ W. Determine:

(a) the total average power
(b) the total reactive power
(c) the power factor
(d) the phase impedance

Is the impedance inductive or capacitive?

Answer: (a) 240 W, (b) 2355.6 VAR, (c) 0.1014, (d) $18.25\angle 84.18^\circ\ \Omega$, inductive.

Example 12.15

The three-phase balanced load in Fig. 12.35 has impedance per phase of $\mathbf{Z}_Y = 8 + j6\ \Omega$. If the load is connected to 208-V lines, predict the readings of the wattmeters W_1 and W_2. Find P_T and Q_T.

Solution:
The impedance per phase is

$$\mathbf{Z}_Y = 8 + j6 = 10\angle 36.87^\circ\ \Omega$$

so that the pf angle is 36.87°. Since the line voltage $V_L = 208$ V, the line current is

$$I_L = \frac{V_p}{|\mathbf{Z}_Y|} = \frac{208/\sqrt{3}}{10} = 12\text{ A}$$

Then

$$\begin{aligned} P_1 &= V_L I_L \cos(\theta + 30^\circ) = 208 \times 12 \times \cos(36.87^\circ + 30^\circ) \\ &= 980.48\text{ W} \\ P_2 &= V_L I_L \cos(\theta - 30^\circ) = 208 \times 12 \times \cos(36.87^\circ - 30^\circ) \\ &= 2478.1\text{ W} \end{aligned}$$

Thus, wattmeter 1 reads 980.48 W, while wattmeter 2 reads 2478.1 W. Since $P_2 > P_1$, the load is inductive. This is evident from the load $\mathbf{Z}_Y$ itself. Next,

$$P_T = P_1 + P_2 = 3.459\text{ kW}$$

and

$$Q_T = \sqrt{3}(P_2 - P_1) = \sqrt{3}(1497.6)\text{ VAR} = 2.594\text{ kVAR}$$

Practice Problem 12.15

If the load in Fig. 12.35 is delta-connected with impedance per phase of $\mathbf{Z}_p = 30 - j40\ \Omega$ and $V_L = 440$ V, predict the readings of the wattmeters W_1 and W_2. Calculate P_T and Q_T.

Answer: 6.166 kW, 0.8021 kW, 6.968 kW, −9.291 kVAR.

12.10.2 Residential Wiring

In the United States, most household lighting and appliances operate on 120-V, 60-Hz, single-phase alternating current. (The electricity may also be supplied at 110, 115, or 117 V, depending on the location.) The local power company supplies the house with a three-wire ac system. Typically, as in Fig. 12.37, the line voltage of, say, 12,000 V is stepped down to 120/240 V with a transformer (more details on transformers in the next chapter). The three wires coming from the transformer are typically colored red (hot), black (hot), and white (neutral). As shown in Fig. 12.38, the two 120-V voltages are opposite in phase and hence add up to zero. That is, $\mathbf{V}_W = 0\angle 0^\circ$, $\mathbf{V}_B = 120\angle 0^\circ$, $\mathbf{V}_R = 120\angle 180^\circ = -\mathbf{V}_B$.

$$\mathbf{V}_{BR} = \mathbf{V}_B - \mathbf{V}_R = \mathbf{V}_B - (-\mathbf{V}_B) = 2\mathbf{V}_B = 240\angle 0^\circ \quad (12.72)$$

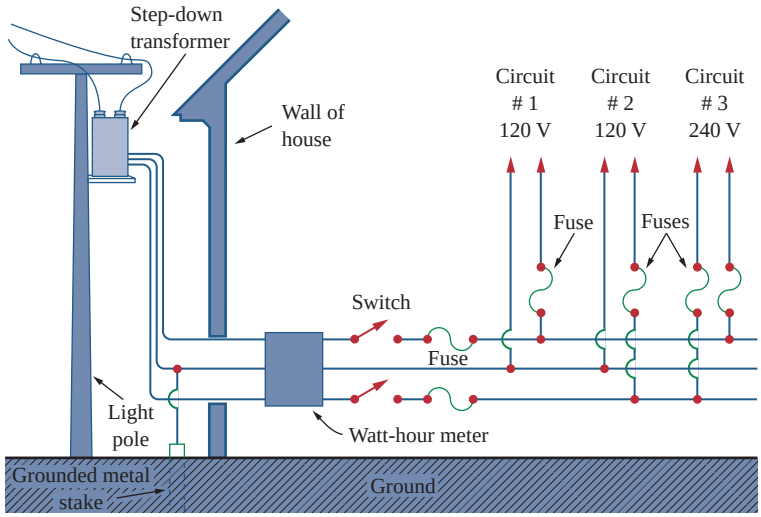


Figure 12.37
A 120/240 household power system.
A. Marcus and C. M. Thomson, *Electricity for Technicians*, 2nd ed. [Englewood Cliffs, NJ: Prentice Hall, 1975], p. 324.

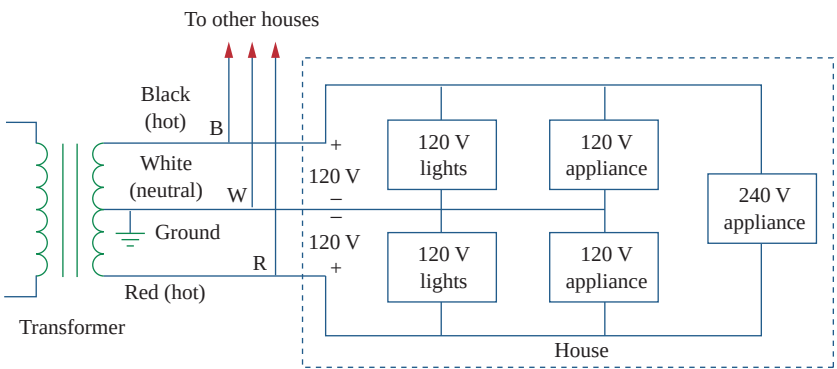


Figure 12.38
Single-phase three-wire residential wiring.

Since most appliances are designed to operate with 120 V, the lighting and appliances are connected to the 120-V lines, as illustrated in Fig. 12.39 for a room. Notice in Fig. 12.37 that all appliances are connected in parallel. Heavy appliances that consume large currents, such as air conditioners, dishwashers, ovens, and laundry machines, are connected to the 240-V power line.

Because of the dangers of electricity, house wiring is carefully regulated by a code drawn by local ordinances and by the National Electrical Code (NEC). To avoid trouble, insulation, grounding, fuses, and circuit breakers are used. Modern wiring codes require a third wire for a separate ground. The ground wire does not carry power like the neutral wire but enables appliances to have a separate ground connection. Figure 12.40 shows the connection of the receptacle to a 120-V rms line and to the ground. As shown in the figure, the neutral line is connected to the ground (the earth) at many critical locations. Although the ground line seems redundant, grounding is important for many reasons. First, it is required by NEC. Second, grounding provides a

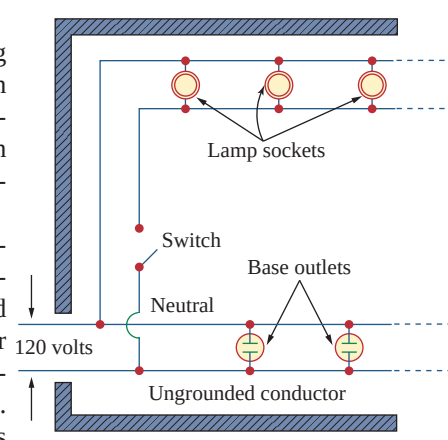


Figure 12.39
A typical wiring diagram of a room.
A. Marcus and C. M. Thomson, *Electricity for Technicians*, 2nd ed. [Englewood Cliffs, NJ: Prentice Hall, 1975], p. 325.

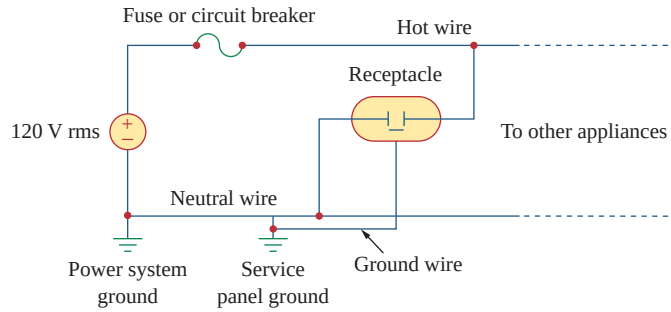


Figure 12.40
Connection of a receptacle to the hot line and to the ground.

convenient path to ground for lightning that strikes the power line. Third, grounds minimize the risk of electric shock. What causes shock is the passage of current from one part of the body to another. The human body is like a big resistor R. If V is the potential difference between the body and the ground, the current through the body is determined by Ohm's law as

$$I = \frac{V}{R} \tag{12.73}$$

The value of R varies from person to person and depends on whether the body is wet or dry. How great or how deadly the shock is depends on the amount of current, the pathway of the current through the body, and the length of time the body is exposed to the current. Currents less than 1 mA may not be harmful to the body, but currents greater than 10 mA can cause severe shock. A modern safety device is the *ground-fault circuit interrupter* (GFCI), used in outdoor circuits and in bathrooms, where the risk of electric shock is greatest. It is essentially a circuit breaker that opens when the sum of the currents i_R, i_W, and i_B through the red, white, and the black lines is not equal to zero, or $i_R + i_W + i_B \neq 0$.

The best way to avoid electric shock is to follow safety guidelines concerning electrical systems and appliances. Here are some of them:

- Never assume that an electrical circuit is dead. Always check to be sure.
- Use safety devices when necessary, and wear suitable clothing (insulated shoes, gloves, etc.).
- Never use two hands when testing high-voltage circuits, since the current through one hand to the other hand has a direct path through your chest and heart.
- Do not touch an electrical appliance when you are wet. Remember that water conducts electricity.
- Be extremely careful when working with electronic appliances such as radio and TV because these appliances have large capacitors in them. The capacitors take time to discharge after the power is disconnected.
- Always have another person present when working on a wiring system, just in case of an accident.

12.11 Summary

1. The phase sequence is the order in which the phase voltages of a three-phase generator occur with respect to time. In an *abc* sequence of balanced source voltages, $\mathbf{V}_{an}$ leads $\mathbf{V}_{bn}$ by 120°, which in turn leads $\mathbf{V}_{cn}$ by 120°. In an *acb* sequence of balanced voltages, $\mathbf{V}_{an}$ leads $\mathbf{V}_{cn}$ by 120°, which in turn leads $\mathbf{V}_{bn}$ by 120°.
2. A balanced wye- or delta-connected load is one in which the three-phase impedances are equal.
3. The easiest way to analyze a balanced three-phase circuit is to transform both the source and the load to a Y-Y system and then analyze the single-phase equivalent circuit. Table 12.1 presents a summary of the formulas for phase currents and voltages and line currents and voltages for the four possible configurations.
4. The line current I_L is the current flowing from the generator to the load in each transmission line in a three-phase system. The line voltage V_L is the voltage between each pair of lines, excluding the neutral line if it exists. The phase current I_p is the current flowing through each phase in a three-phase load. The phase voltage V_p is the voltage of each phase. For a wye-connected load,

$$V_L = \sqrt{3}V_p \quad \text{and} \quad I_L = I_p$$

   For a delta-connected load,

$$V_L = V_p \quad \text{and} \quad I_L = \sqrt{3}I_p$$

5. The total instantaneous power in a balanced three-phase system is constant and equal to the average power.
6. The total complex power absorbed by a balanced three-phase Y-connected or Δ-connected load is

$$\mathbf{S} = P + jQ = \sqrt{3}V_L I_L \underline{/\theta}$$

   where θ is the angle of the load impedances.
7. An unbalanced three-phase system can be analyzed using nodal or mesh analysis.
8. *PSpice* is used to analyze three-phase circuits in the same way as it is used for analyzing single-phase circuits.
9. The total real power is measured in three-phase systems using either the three-wattmeter method or the two-wattmeter method.
10. Residential wiring uses a 120/240-V, single-phase, three-wire system.

Review Questions

12.1 What is the phase sequence of a three-phase motor for which $\mathbf{V}_{AN} = 220\underline{/-100°}$ V and $\mathbf{V}_{BN} = 220\underline{/140°}$ V?

(a) *abc* (b) *acb*

12.2 If in an *acb* phase sequence, $V_{an} = 100\underline{/-20°}$, then $\mathbf{V}_{cn}$ is:

(a) $100\underline{/-140°}$ (b) $100\underline{/100°}$

(c) $100\underline{/-50°}$ (d) $100\underline{/10°}$

12.3 Which of these is not a required condition for a balanced system:

(a) $|\mathbf{V}_{an}| = |\mathbf{V}_{bn}| = |\mathbf{V}_{cn}|$

(b) $\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0$

(c) $V_{an} + V_{bn} + V_{cn} = 0$

(d) Source voltages are 120° out of phase with each other.

(e) Load impedances for the three phases are equal.

12.4 In a Y-connected load, the line current and phase current are equal.

(a) True (b) False

12.5 In a Δ-connected load, the line current and phase current are equal.

(a) True (b) False

12.6 In a Y-Y system, a line voltage of 220 V produces a phase voltage of:

(a) 381 V (b) 311 V (c) 220 V
(d) 156 V (e) 127 V

12.7 In a Δ-Δ system, a phase voltage of 100 V produces a line voltage of:

(a) 58 V (b) 71 V (c) 100 V
(d) 173 V (e) 141 V

12.8 When a Y-connected load is supplied by voltages in *abc* phase sequence, the line voltages lag the corresponding phase voltages by 30°.

(a) True (b) False

12.9 In a balanced three-phase circuit, the total instantaneous power is equal to the average power.

(a) True (b) False

12.10 The total power supplied to a balanced Δ-load is found in the same way as for a balanced Y-load.

(a) True (b) False

Answers: 12.1a, 12.2a, 12.3c, 12.4a, 12.5b, 12.6e, 12.7c, 12.8b, 12.9a, 12.10a.

Problems[1]

Section 12.2 Balanced Three-Phase Voltages

12.1 If $\mathbf{V}_{ab} = 400$ V in a balanced Y-connected three-phase generator, find the phase voltages, assuming the phase sequence is:

(a) *abc* (b) *acb*

12.2 What is the phase sequence of a balanced three-phase circuit for which $\mathbf{V}_{an} = 160\angle 30°$ V and $\mathbf{V}_{cn} = 160\angle -90°$ V? Find $\mathbf{V}_{bn}$.

12.3 Determine the phase sequence of a balanced three-phase circuit in which $\mathbf{V}_{bn} = 208\angle 130°$ V and $\mathbf{V}_{cn} = 208\angle 10°$ V. Obtain $\mathbf{V}_{an}$.

12.4 A three-phase system with *abc* sequence and $\mathbf{V}_L = 200$ V feeds a Y-connected load with $Z_L = 40\angle 30°\ \Omega$. Find the line currents.

12.5 For a Y-connected load, the time-domain expressions for three line-to-neutral voltages at the terminals are:

$$v_{AN} = 150\cos(\omega t + 32°)\text{ V}$$
$$v_{BN} = 150\cos(\omega t - 88°)\text{ V}$$
$$v_{CN} = 150\cos(\omega t + 152°)\text{ V}$$

Write the time-domain expressions for the line-to-line voltages v_{AB}, v_{BC}, and v_{CA}.

Section 12.3 Balanced Wye-Wye Connection

12.6 Using Fig. 12.41, design a problem to help other students better understand balanced wye-wye connected circuits.

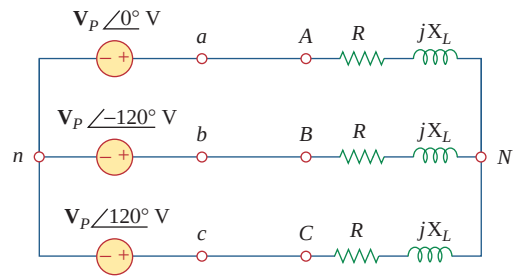


Figure 12.41
For Prob. 12.6.

12.7 Obtain the line currents in the three-phase circuit of Fig. 12.42 on the next page.

12.8 In a balanced three-phase Y-Y system, the source is an *abc* sequence of voltages and $\mathbf{V}_{an} = 220\angle 20°$ V rms. The line impedance per phase is $0.6 + j1.2\ \Omega$, while the per-phase impedance of the load is $10 + j14\ \Omega$. Calculate the line currents and the load voltages.

12.9 A balanced Y-Y four-wire system has phase voltages

$$\mathbf{V}_{an} = 120\angle 0°, \quad \mathbf{V}_{bn} = 120\angle -120°$$
$$\mathbf{V}_{cn} = 120\angle 120°\text{ V}$$

The load impedance per phase is $19 + j13\ \Omega$, and the line impedance per phase is $1 + j2\ \Omega$. Solve for the line currents and neutral current.

[1] Remember that unless stated otherwise, all given voltages and currents are rms values.

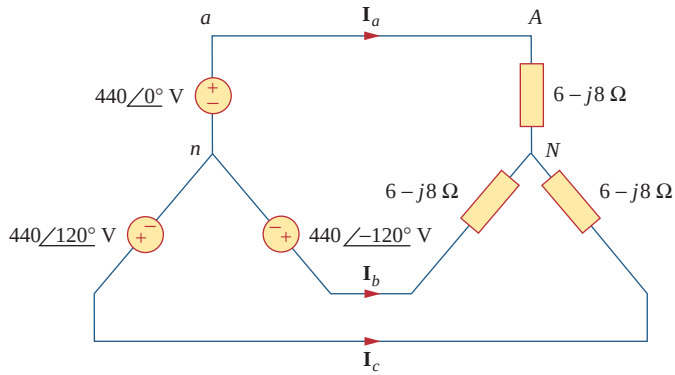


Figure 12.42
For Prob. 12.7.

12.10 For the circuit in Fig. 12.43, determine the current in the neutral line.

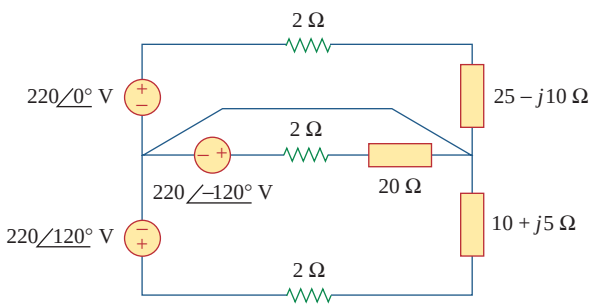


Figure 12.43
For Prob. 12.10.

Section 12.4 Balanced Wye-Delta Connection

12.11 In the Y-Δ system shown in Fig. 12.44, the source is a positive sequence with $\mathbf{V}_{an} = 120\angle 0°$ V and phase impedance $\mathbf{Z}_p = 2 - j3\ \Omega$. Calculate the line voltage $\mathbf{V}_L$ and the line current $\mathbf{I}_L$.

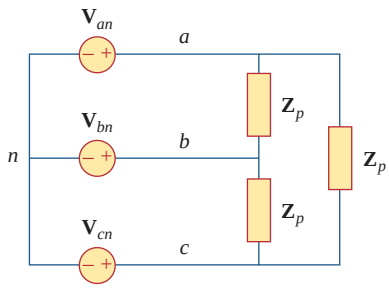


Figure 12.44
For Prob. 12.11.

12.12 Using Fig. 12.45, design a problem to help other students better understand wye-delta connected circuits.

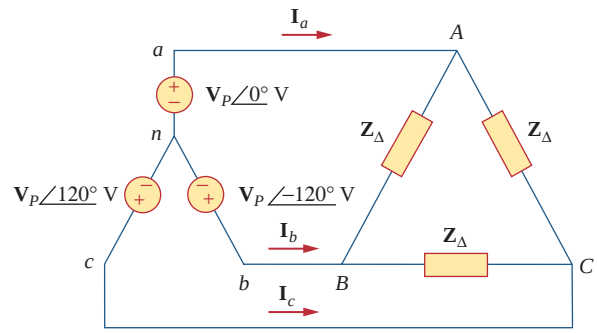


Figure 12.45
For Prob. 12.12.

12.13 In the balanced three-phase Y-Δ system in Fig. 12.46, find the line current I_L and the average power delivered to the load.

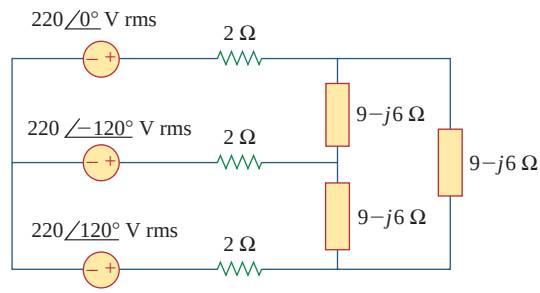


Figure 12.46
For Prob. 12.13.

12.14 Obtain the line currents in the three-phase circuit of Fig. 12.47 on the next page.

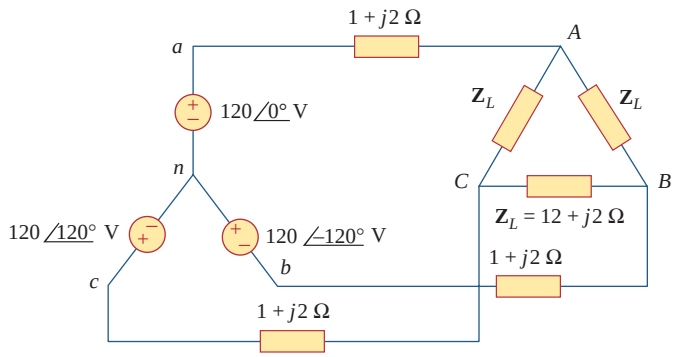


Figure 12.47
For Prob. 12.14.

12.15 The circuit in Fig. 12.48 is excited by a balanced three-phase source with a line voltage of 210 V. If $\mathbf{Z}_l = 1 + j1\ \Omega$, $\mathbf{Z}_\Delta = 24 - j30\ \Omega$, and $\mathbf{Z}_Y = 12 + j5\ \Omega$, determine the magnitude of the line current of the combined loads.

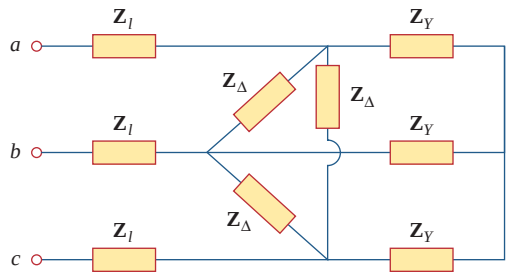


Figure 12.48
For Prob. 12.15.

12.16 A balanced delta-connected load has a phase current $\mathbf{I}_{AC} = 10\angle -30°$ A.

(a) Determine the three line currents assuming that the circuit operates in the positive phase sequence.

(b) Calculate the load impedance if the line voltage is $\mathbf{V}_{AB} = 110\angle 0°$ V.

12.17 A balanced delta-connected load has line current $\mathbf{I}_a = 10\angle -25°$ A. Find the phase currents $\mathbf{I}_{AB}$, $\mathbf{I}_{BC}$, and $\mathbf{I}_{CA}$.

12.18 If $\mathbf{V}_{an} = 440\angle 60°$ V in the network of Fig. 12.49, find the load phase currents $\mathbf{I}_{AB}$, $\mathbf{I}_{BC}$, and $\mathbf{I}_{CA}$.

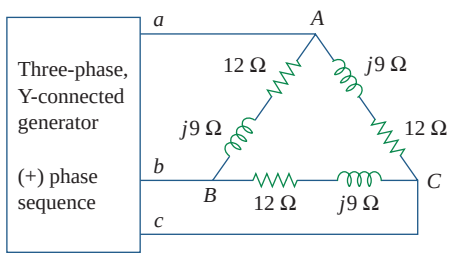


Figure 12.49
For Prob. 12.18.

Section 12.5 Balanced Delta-Delta Connection

12.19 For the Δ-Δ circuit of Fig. 12.50, calculate the phase and line currents.

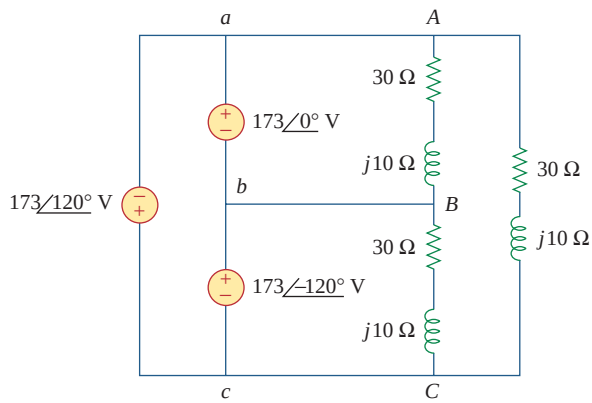


Figure 12.50
For Prob. 12.19.

12.20 Using Fig. 12.51, design a problem to help other students better understand balanced delta-delta connected circuits.

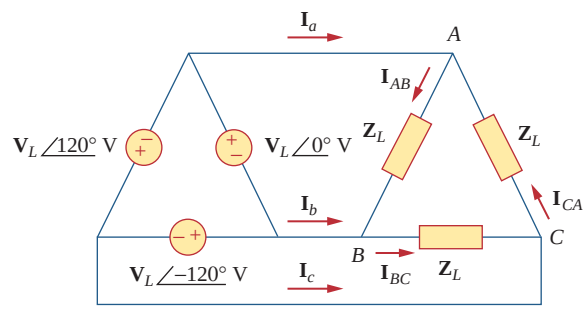


Figure 12.51
For Prob. 12.20.

12.21 Three 440-V generators form a delta-connected source that is connected to a balanced delta-connected load of $\mathbf{Z}_L = 10 + j8\ \Omega$ per phase as shown in Fig. 12.52.

(a) Determine the value of $\mathbf{I}_{AC}$.

(b) What is the value of $\mathbf{I}_{bB}$?

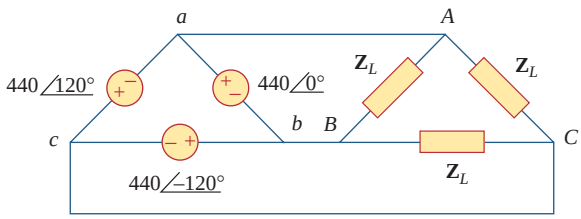


Figure 12.52
For Prob. 12.21.

12.22 Find the line currents $\mathbf{I}_a$, $\mathbf{I}_b$, and $\mathbf{I}_c$ in the three-phase network of Fig. 12.53 below. Take $\mathbf{Z}_\Delta = 12 - j15\ \Omega$, $\mathbf{Z}_Y = 4 + j6\ \Omega$, and $\mathbf{Z}_l = 2\ \Omega$.

12.23 A three-phase balanced system with a line voltage of 202 V rms feeds a delta-connected load with $\mathbf{Z}_p = 25\angle 60°\ \Omega$.

(a) Find the line current.

(b) Determine the total power supplied to the load using two wattmeters connected to the *A* and *C* lines.

12.24 A balanced delta-connected source has phase voltage $\mathbf{V}_{ab} = 440\angle 30°$ V and a positive phase sequence. If this is connected to a balanced delta-connected load, find the line and phase currents. Take the load impedance per phase as $60\angle 30°\ \Omega$ and line impedance per phase as $1 + j1\ \Omega$.

Section 12.6 Balanced Delta-Wye Connection

12.25 In the circuit of Fig. 12.54, if $\mathbf{V}_{ab} = 220\angle 10°$, $\mathbf{V}_{bc} = 220\angle -110°$, $\mathbf{V}_{ca} = 220\angle 130°$ V, find the line currents.

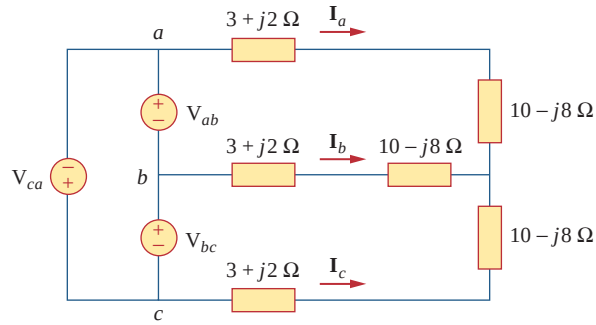


Figure 12.54
For Prob. 12.25.

12.26 Using Fig. 12.55, design a problem to help other students better understand balanced delta connected sources delivering power to balanced wye connected loads.

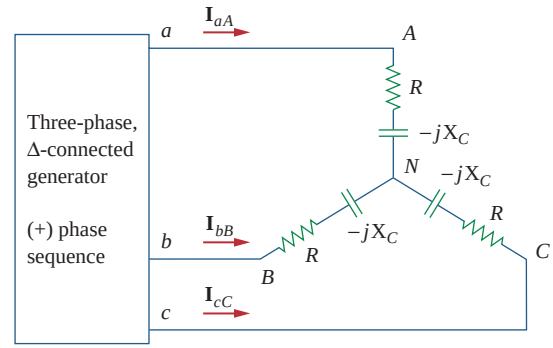


Figure 12.55
For Prob. 12.26.

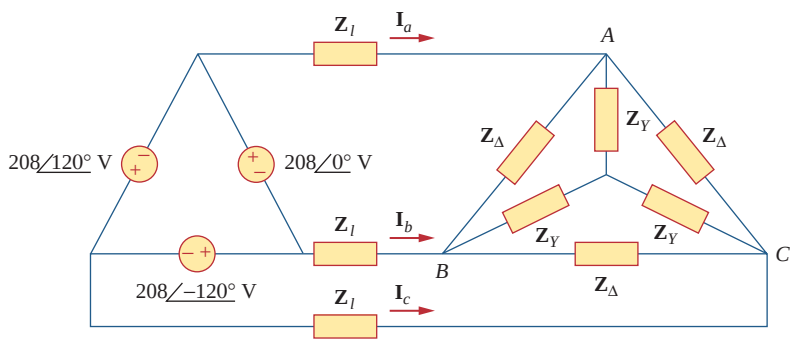


Figure 12.53
For Prob. 12.22.

12.27 A Δ-connected source supplies power to a Y-connected load in a three-phase balanced system. Given that the line impedance is $2 + j1\ \Omega$ per phase while the load impedance is $6 + j4\ \Omega$ per phase, find the magnitude of the line voltage at the load. Assume the source phase voltage $\mathbf{V}_{ab} = 208\angle 0°$ V rms.

12.28 The line-to-line voltages in a Y-load have a magnitude of 220 V and are in the positive sequence at 60 Hz. If the loads are balanced with $Z_1 = Z_2 = Z_3 = 25\angle 30°$, find all line currents and phase voltages.

Section 12.7 Power in a Balanced System

12.29 A balanced three-phase Y-Δ system has $\mathbf{V}_{an} = 120\angle 0°$ V rms and $\mathbf{Z}_\Delta = 51 + j45\ \Omega$. If the line impedance per phase is $0.4 + j1.2\ \Omega$, find the total complex power delivered to the load.

12.30 In Fig. 12.56, the rms value of the line voltage is 208 V. Find the average power delivered to the load.

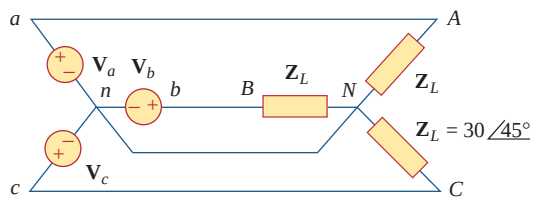


Figure 12.56
For Prob. 12.30.

12.31 A balanced delta-connected load is supplied by a 60-Hz three-phase source with a line voltage of 240 V. Each load phase draws 6 kW at a lagging power factor of 0.8. Find:

(a) the load impedance per phase

(b) the line current

(c) the value of capacitance needed to be connected in parallel with each load phase to minimize the current from the source

12.32 Design a problem to help other students better understand power in a balanced three-phase system.

12.33 A three-phase source delivers 9.6 kVA to a wye-connected load with a phase voltage of 208 V and a power factor of 0.9 lagging. Calculate the source line current and the source line voltage.

12.34 A balanced wye-connected load with a phase impedance of $10 - j16\ \Omega$ is connected to a balanced three-phase generator with a line voltage of 220 V. Determine the line current and the complex power absorbed by the load.

12.35 Three equal impedances, $60 + j30\ \Omega$ each, are delta-connected to a 230-V rms, three-phase circuit. Another three equal impedances, $40 + j10\ \Omega$ each, are wye-connected across the same circuit at the same points. Determine:

(a) the line current

(b) the total complex power supplied to the two loads

(c) the power factor of the two loads combined

12.36 A 4200-V, three-phase transmission line has an impedance of $4 + j\ \Omega$ per phase. If it supplies a load of 1 MVA at 0.75 power factor (lagging), find:

(a) the complex power

(b) the power loss in the line

(c) the voltage at the sending end

12.37 The total power measured in a three-phase system feeding a balanced wye-connected load is 12 kW at a power factor of 0.6 leading. If the line voltage is 208 V, calculate the line current I_L and the load impedance $\mathbf{Z}_Y$.

12.38 Given the circuit in Fig. 12.57 below, find the total complex power absorbed by the load.

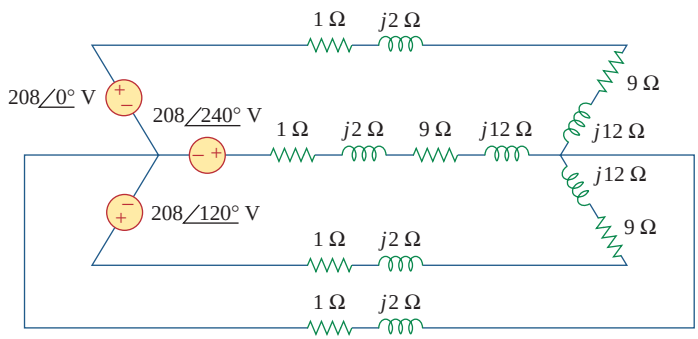


Figure 12.57
For Prob. 12.38.

12.39 Find the real power absorbed by the load in Fig. 12.58.

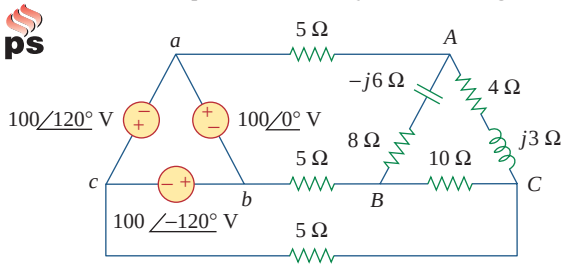


Figure 12.58
For Prob. 12.39.

12.40 For the three-phase circuit in Fig. 12.59, find the average power absorbed by the delta-connected load with $\mathbf{Z}_\Delta = 21 + j24\ \Omega$.

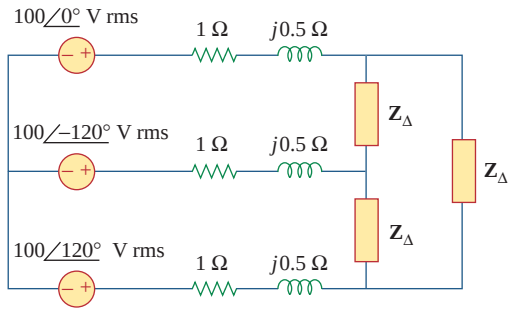


Figure 12.59
For Prob. 12.40.

12.41 A balanced delta-connected load draws 5 kW at a power factor of 0.8 lagging. If the three-phase system has an effective line voltage of 400 V, find the line current.

12.42 A balanced three-phase generator delivers 9.6 kW to a wye-connected load with impedance $30 - j40\ \Omega$ per phase. Find the line current I_L and the line voltage V_L.

12.43 Refer to Fig. 12.48. Obtain the complex power absorbed by the combined loads.

12.44 A three-phase line has an impedance of $1 + j3\ \Omega$ per phase. The line feeds a balanced delta-connected load, which absorbs a total complex power of $12 + j5$ kVA. If the line voltage at the load end has a magnitude of 240 V, calculate the magnitude of the line voltage at the source end and the source power factor.

12.45 A balanced wye-connected load is connected to the generator by a balanced transmission line with an impedance of $0.5 + j2\ \Omega$ per phase. If the load is rated at 450 kW, 0.708 power factor lagging, 440-V line voltage, find the line voltage at the generator.

12.46 A three-phase load consists of three 100-Ω resistors that can be wye- or delta-connected. Determine which connection will absorb the most average power from a three-phase source with a line voltage of 110 V. Assume zero line impedance.

12.47 The following three parallel-connected three-phase loads are fed by a balanced three-phase source:

Load 1: 250 kVA, 0.8 pf lagging
Load 2: 300 kVA, 0.95 pf leading
Load 3: 450 kVA, unity pf

If the line voltage is 13.8 kV, calculate the line current and the power factor of the source. Assume that the line impedance is zero.

12.48 A balanced, positive-sequence wye-connected source has $\mathbf{V}_{an} = 240\angle 0°$ V rms and supplies an unbalanced delta-connected load via a transmission line with impedance $2 + j3\ \Omega$ per phase.

(a) Calculate the line currents if $\mathbf{Z}_{AB} = 40 + j15\ \Omega$, $\mathbf{Z}_{BC} = 60\ \Omega$, $\mathbf{Z}_{CA} = 18 - j12\ \Omega$.

(b) Find the complex power supplied by the source.

12.49 Each phase load consists of a 20-Ω resistor and a 10-Ω inductive reactance. With a line voltage of 220 V rms, calculate the average power taken by the load if:

(a) the three-phase loads are delta-connected

(b) the loads are wye-connected

12.50 A balanced three-phase source with $\mathbf{V}_L = 240$ V rms is supplying 8 kVA at 0.6 power factor lagging to two wye-connected parallel loads. If one load draws 3 kW at unity power factor, calculate the impedance per phase of the second load.

Section 12.8 Unbalanced Three-Phase Systems

12.51 Consider the Δ-Δ system shown in Fig. 12.60. Take $\mathbf{Z}_1 = 8 + j6\ \Omega$, $\mathbf{Z}_2 = 4.2 - j2.2\ \Omega$, $\mathbf{Z}_3 = 10 + j0\ \Omega$.

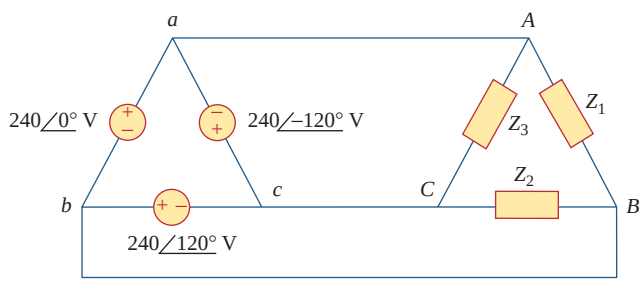


Figure 12.60
For Prob. 12.51.

(a) Find the phase current $\mathbf{I}_{AB}$, $\mathbf{I}_{BC}$, *and* $\mathbf{I}_{CA}$.

(b) Calculate line currents $\mathbf{I}_{aA}$, $\mathbf{I}_{bB}$, and $\mathbf{I}_{cC}$.

12.52 A four-wire wye-wye circuit has

$$\mathbf{V}_{an} = 120\angle 120°, \qquad \mathbf{V}_{bn} = 120\angle 0°$$
$$\mathbf{V}_{cn} = 120\angle -120° \text{ V}$$

If the impedances are

$$\mathbf{Z}_{AN} = 20\angle 60°, \qquad \mathbf{Z}_{BN} = 30\angle 0°$$
$$\mathbf{Z}_{cn} = 40\angle 30° \ \Omega$$

find the current in the neutral line.

12.53 Using Fig. 12.61, design a problem that will help other students better understand unbalanced three-phase systems.

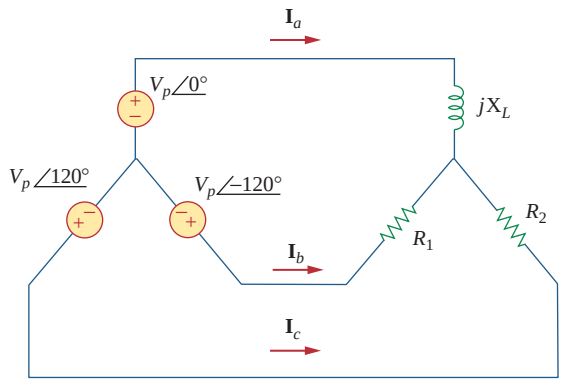


Figure 12.61
For Prob. 12.53.

12.54 A balanced three-phase Y-source with $V_P = 210$ V rms drives a Y-connected three-phase load with phase impedance $\mathbf{Z}_A = 80\ \Omega$, $\mathbf{Z}_B = 60 + j90\ \Omega$, and $\mathbf{Z}_C = j80\ \Omega$. Calculate the line currents and total complex power delivered to the load. Assume that the neutrals are connected.

12.55 A three-phase supply, with the line voltage 240 V rms positively phased, has an unbalanced delta-connected load as shown in Fig. 12.62. Find the phase currents and the total complex power.

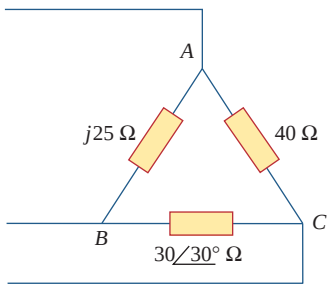


Figure 12.62
For Prob. 12.55.

12.56 Using Fig. 12.63, design a problem to help other students to better understand unbalanced three-phase systems.

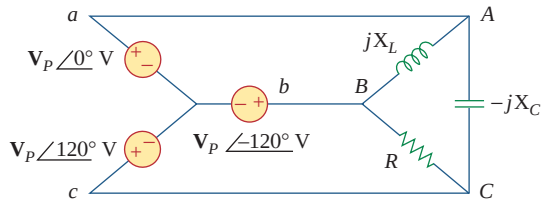


Figure 12.63
For Prob. 12.56.

12.57 Determine the line currents for the three-phase circuit of Fig. 12.64. Let $\mathbf{V}_a = 110\angle 0°$, $\mathbf{V}_b = 110\angle -120°$, $\mathbf{V}_c = 110\angle 120°$ V.

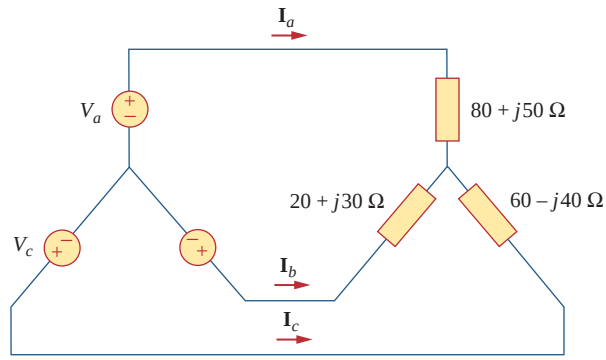


Figure 12.64
For Prob. 12.57.

Section 12.9 *PSpice* for Three-Phase Circuits

12.58 Solve Prob. 12.10 using *PSpice*.

12.59 The source in Fig. 12.65 is balanced and exhibits a positive phase sequence. If $f = 60$ Hz, use *PSpice* to find $\mathbf{V}_{AN}$, $\mathbf{V}_{BN}$, and $\mathbf{V}_{CN}$.

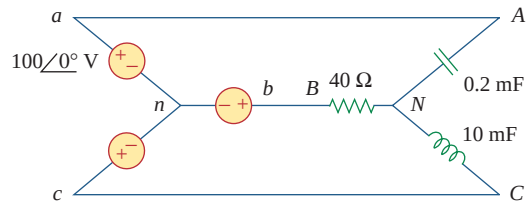


Figure 12.65
For Prob. 12.59.

12.60 Use *PSpice* to determine $\mathbf{I}_o$ in the single-phase, three-wire circuit of Fig. 12.66. Let $\mathbf{Z}_1 = 15 - j10\ \Omega$, $\mathbf{Z}_2 = 30 + j20\ \Omega$, and $\mathbf{Z}_3 = 12 + j5\ \Omega$.

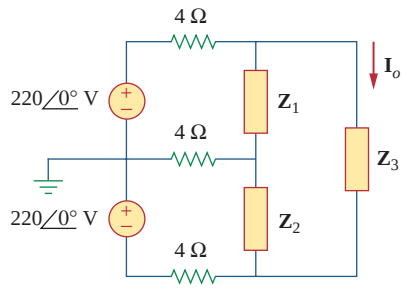


Figure 12.66
For Prob. 12.60.

12.61 Given the circuit in Fig. 12.67, use *PSpice* to determine currents $\mathbf{I}_{aA}$ and voltage $\mathbf{V}_{BN}$.

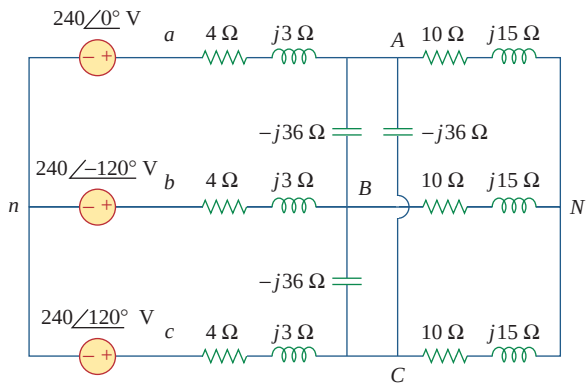


Figure 12.67
For Prob. 12.61.

12.62 Using Fig. 12.68, design a problem to help other students better understand how to use *PSpice* to analyze three-phase circuits.

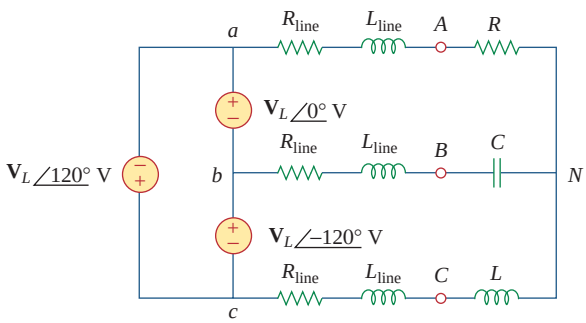


Figure 12.68
For Prob. 12.62.

12.63 Use *PSpice* to find currents $\mathbf{I}_{aA}$ and $\mathbf{I}_{AC}$ in the unbalanced three-phase system shown in Fig. 12.69. Let

$$\mathbf{Z}_l = 2 + j, \qquad \mathbf{Z}_1 = 40 + j20\ \Omega,$$
$$\mathbf{Z}_2 = 50 - j30\ \Omega, \qquad \mathbf{Z}_3 = 25\ \Omega$$

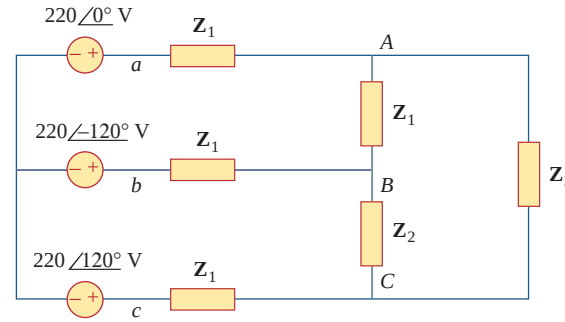


Figure 12.69
For Prob. 12.63.

12.64 For the circuit in Fig. 12.58, use *PSpice* to find the line currents and the phase currents.

12.65 A balanced three-phase circuit is shown in Fig. 12.70 on the next page. Use *PSpice* to find the line currents $\mathbf{I}_{aA}$, $\mathbf{I}_{bB}$, and $\mathbf{I}_{cC}$.

Section 12.10 Applications

12.66 A three-phase, four-wire system operating with a 208-V line voltage is shown in Fig. 12.71. The source voltages are balanced. The power absorbed by the resistive wye-connected load is measured by the three-wattmeter method. Calculate:

(a) the voltage to neutral

(b) the currents $\mathbf{I}_1$, $\mathbf{I}_2$, $\mathbf{I}_3$, and $\mathbf{I}_n$

(c) the readings of the wattmeters

(d) the total power absorbed by the load

***12.67** As shown in Fig. 12.72, a three-phase four-wire line with a phase voltage of 120 V rms and positive phase sequence supplies a balanced motor load at 260 kVA at 0.85 pf lagging. The motor load is connected to the three main lines marked *a*, *b*, and *c*. In addition, incandescent lamps (unity pf) are connected as follows: 24 kW from line *c* to the neutral, 15 kW from line *b* to the neutral, and 9 kW from line *c* to the neutral.

(a) If three wattmeters are arranged to measure the power in each line, calculate the reading of each meter.

(b) Find the magnitude of the current in the neutral line.

12.68 Meter readings for a three-phase wye-connected alternator supplying power to a motor indicate that

* An asterisk indicates a challenging problem.

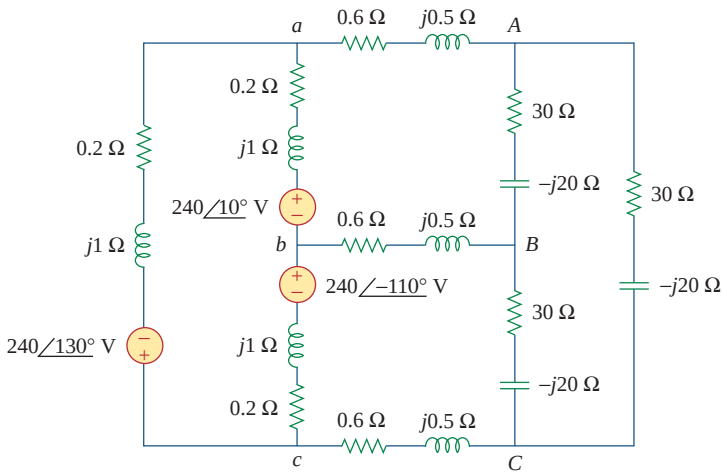


Figure 12.70
For Prob. 12.65.

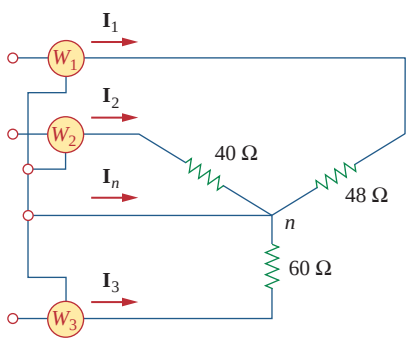


Figure 12.71
For Prob. 12.66.

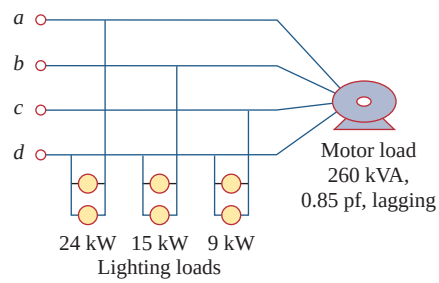


Figure 12.72
For Prob. 12.67.

the line voltages are 330 V, the line currents are 8.4 A, and the total line power is 4.5 kW. Find:

(a) the load in VA

(b) the load pf

(c) the phase current

(d) the phase voltage

12.69 A certain store contains three balanced three-phase loads. The three loads are:

Load 1: 16 kVA at 0.85 pf lagging
Load 2: 12 kVA at 0.6 pf lagging
Load 3: 8 kW at unity pf

The line voltage at the load is 208 V rms at 60 Hz, and the line impedance is $0.4 + j0.8\ \Omega$. Determine the line current and the complex power delivered to the loads.

12.70 The two-wattmeter method gives $P_1 = 1200$ W and $P_2 = -400$ W for a three-phase motor running on a 240-V line. Assume that the motor load is wye-connected and that it draws a line current of 6 A. Calculate the pf of the motor and its phase impedance.

12.71 In Fig. 12.73, two wattmeters are properly connected to the unbalanced load supplied by a balanced source such that $\mathbf{V}_{ab} = 208\angle 0°$ V with positive phase sequence.

(a) Determine the reading of each wattmeter.

(b) Calculate the total apparent power absorbed by the load.

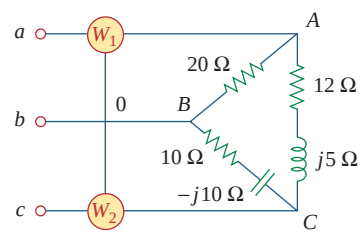


Figure 12.73
For Prob. 12.71.

12.72 If wattmeters W_1 and W_2 are properly connected respectively between lines *a* and *b* and lines *b* and *c* to measure the power absorbed by the delta-connected load in Fig. 12.44, predict their readings.

12.73 For the circuit displayed in Fig. 12.74, find the wattmeter readings.

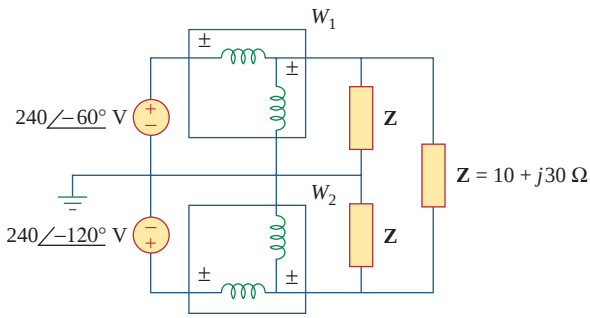


Figure 12.74
For Prob. 12.73.

12.74 Predict the wattmeter readings for the circuit in Fig. 12.75.

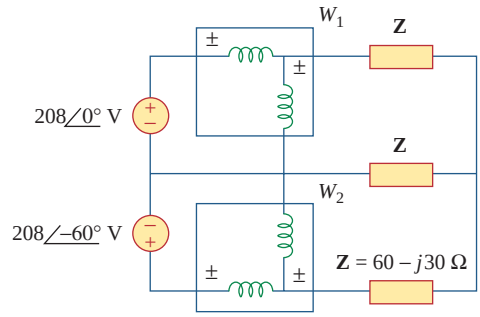


Figure 12.75
For Prob. 12.74.

12.75 A man has a body resistance of 600 Ω. How much current flows through his ungrounded body:

(a) when he touches the terminals of a 12-V autobattery?

(b) when he sticks his finger into a 120-V light socket?

12.76 Show that the I^2R losses will be higher for a 120-V appliance than for a 240-V appliance if both have the same power rating.

Comprehensive Problems

12.77 A three-phase generator supplied 3.6 kVA at a power factor of 0.85 lagging. If 2500 W are delivered to the load and line losses are 80 W per phase, what are the losses in the generator?

12.78 A three-phase 440-V, 51-kW, 60-kVA inductive load operates at 60 Hz and is wye-connected. It is desired to correct the power factor to 0.95 lagging. What value of capacitor should be placed in parallel with each load impedance?

12.79 A balanced three-phase generator has an *abc* phase sequence with phase voltage $\mathbf{V}_{an} = 255\angle 0°$ V. The generator feeds an induction motor which may be represented by a balanced Y-connected load with an impedance of $12 + j5\ \Omega$ per phase. Find the line currents and the load voltages. Assume a line impedance of $2\ \Omega$ per phase.

12.80 A balanced three-phase source furnishes power to the following three loads:

Load 1: 6 kVA at 0.83 pf lagging

Load 2: unknown

Load 3: 8 kW at 0.7071 pf leading

If the line current is 84.6 A rms, the line voltage at the load is 208 V rms, and the combined load has a 0.8 pf lagging, determine the unknown load.

12.81 A professional center is supplied by a balanced three-phase source. The center has four balanced three-phase loads as follows:

Load 1: 150 kVA at 0.8 pf leading

Load 2: 100 kW at unity pf

Load 3: 200 kVA at 0.6 pf lagging

Load 4: 80 kW and 95 kVAR (inductive)

If the line impedance is $0.02 + j0.05\ \Omega$ per phase and the line voltage at the loads is 480 V, find the magnitude of the line voltage at the source.

12.82 A balanced three-phase system has a distribution wire with impedance $2 + j6\ \Omega$ per phase. The system supplies two three-phase loads that are connected in parallel. The first is a balanced wye-connected load that absorbs 400 kVA at a power factor of 0.8 lagging. The second load is a balanced delta-connected load with impedance of $10 + j8\ \Omega$ per phase. If the magnitude of the line voltage at the loads is 2400 V rms, calculate the magnitude of the line voltage at the source and the total complex power supplied to the two loads.

12.83 A commercially available three-phase inductive motor operates at a full load of 120 hp (1 hp = 746 W) at 95 percent efficiency at a lagging power factor of

0.707. The motor is connected in parallel to a 80-kW balanced three-phase heater at unity power factor. If the magnitude of the line voltage is 480 V rms, calculate the line current.

***12.84** Figure 12.76 displays a three-phase delta-connected motor load which is connected to a line voltage of 440 V and draws 4 kVA at a power factor of 72 percent lagging. In addition, a single 1.8 kVAR capacitor is connected between lines *a* and *b*, while a 800-W lighting load is connected between line *c* and neutral. Assuming the *abc* sequence and taking $\mathbf{V}_{an} = V_p\angle 0°$, find the magnitude and phase angle of currents $\mathbf{I}_a$, $\mathbf{I}_b$, $\mathbf{I}_c$, and $\mathbf{I}_n$.

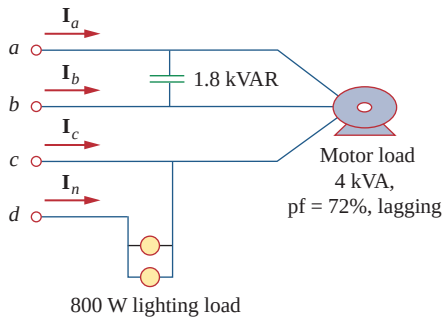


Figure 12.76
For Prob. 12.84.

12.85 Design a three-phase heater with suitable symmetric loads using wye-connected pure resistance. Assume that the heater is supplied by a 240-V line voltage and is to give 27 kW of heat.

12.86 For the single-phase three-wire system in Fig. 12.77, find currents $\mathbf{I}_{aA}$, $\mathbf{I}_{bB}$, and $\mathbf{I}_{nN}$.

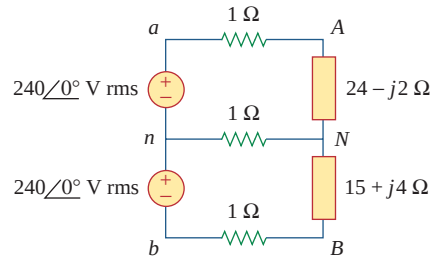


Figure 12.77
For Prob. 12.86.

12.87 Consider the single-phase three-wire system shown in Fig. 12.78. Find the current in the neutral wire and the complex power supplied by each source. Take $\mathbf{V}_s$ as a $115\angle 0°$-V, 60-Hz source.

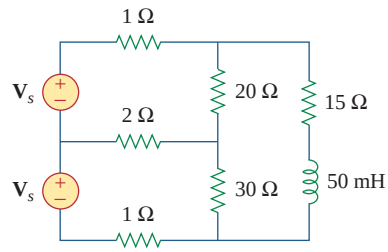


Figure 12.78
For Prob. 12.87.

chapter

13

Magnetically Coupled Circuits

If you would increase your happiness and prolong your life, forget your neighbor's faults . . . . Forget the peculiarities of your friends, and only remember the good points which make you fond of them . . . . Obliterate everything disagreeable from yesterday; write upon today's clean sheet those things lovely and lovable.

—Anonymous

Enhancing Your Career

Career in Electromagnetics

Electromagnetics is the branch of electrical engineering (or physics) that deals with the analysis and application of electric and magnetic fields. In electromagnetics, electric circuit analysis is applied at low frequencies.

The principles of electromagnetics (EM) are applied in various allied disciplines, such as electric machines, electromechanical energy conversion, radar meteorology, remote sensing, satellite communications, bioelectromagnetics, electromagnetic interference and compatibility, plasmas, and fiber optics. EM devices include electric motors and generators, transformers, electromagnets, magnetic levitation, antennas, radars, microwave ovens, microwave dishes, superconductors, and electrocardiograms. The design of these devices requires a thorough knowledge of the laws and principles of EM.

EM is regarded as one of the more difficult disciplines in electrical engineering. One reason is that EM phenomena are rather abstract. But if one enjoys working with mathematics and can visualize the invisible, one should consider being a specialist in EM, since few electrical engineers specialize in this area. Electrical engineers who specialize in EM are needed in microwave industries, radio/TV broadcasting stations, electromagnetic research laboratories, and several communications industries.



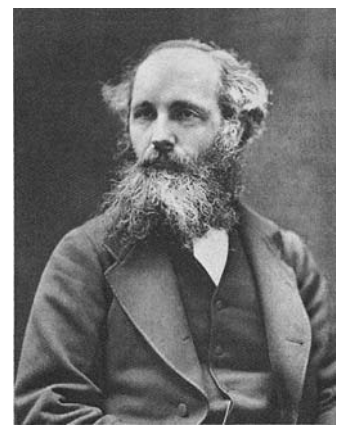
Telemetry receiving station for space satellites. © DV169/Getty Images

Historical

James Clerk Maxwell (1831–1879), a graduate in mathematics from Cambridge University, in 1865 wrote a most remarkable paper in which he mathematically unified the laws of Faraday and Ampere. This relationship between the electric field and magnetic field served as the basis for what was later called electromagnetic fields and waves, a major field of study in electrical engineering. The Institute of Electrical and Electronics Engineers (IEEE) uses a graphical representation of this principle in its logo, in which a straight arrow represents current and a curved arrow represents the electromagnetic field. This relationship is commonly known as *the right-hand rule.* Maxwell was a very active theoretician and scientist. He is best known for the "Maxwell equations." The maxwell, a unit of magnetic flux, was named after him.



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13.1 Introduction

The circuits we have considered so far may be regarded as *conductively coupled*, because one loop affects the neighboring loop through current conduction. When two loops with or without contacts between them affect each other through the magnetic field generated by one of them, they are said to be *magnetically coupled*.

The transformer is an electrical device designed on the basis of the concept of magnetic coupling. It uses magnetically coupled coils to transfer energy from one circuit to another. Transformers are key circuit elements. They are used in power systems for stepping up or stepping down ac voltages or currents. They are used in electronic circuits such as radio and television receivers for such purposes as impedance matching, isolating one part of a circuit from another, and again for stepping up or down ac voltages and currents.

We will begin with the concept of mutual inductance and introduce the dot convention used for determining the voltage polarities of inductively coupled components. Based on the notion of mutual inductance, we then introduce the circuit element known as the *transformer*. We will consider the linear transformer, the ideal transformer, the ideal autotransformer, and the three-phase transformer. Finally, among their

important applications, we look at transformers as isolating and matching devices and their use in power distribution.

13.2 Mutual Inductance

When two inductors (or coils) are in a close proximity to each other, the magnetic flux caused by current in one coil links with the other coil, thereby inducing voltage in the latter. This phenomenon is known as *mutual inductance*.

Let us first consider a single inductor, a coil with N turns. When current i flows through the coil, a magnetic flux ϕ is produced around it (Fig. 13.1). According to Faraday's law, the voltage v induced in the coil is proportional to the number of turns N and the time rate of change of the magnetic flux ϕ; that is,

$$v = N\frac{d\phi}{dt} \tag{13.1}$$

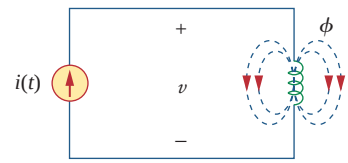


Figure 13.1
Magnetic flux produced by a single coil with N turns.

But the flux ϕ is produced by current i so that any change in ϕ is caused by a change in the current. Hence, Eq. (13.1) can be written as

$$v = N\frac{d\phi}{di}\frac{di}{dt} \tag{13.2}$$

or

$$v = L\frac{di}{dt} \tag{13.3}$$

which is the voltage-current relationship for the inductor. From Eqs. (13.2) and (13.3), the inductance L of the inductor is thus given by

$$L = N\frac{d\phi}{di} \tag{13.4}$$

This inductance is commonly called *self-inductance*, because it relates the voltage induced in a coil by a time-varying current in the same coil.

Now consider two coils with self-inductances L_1 and L_2 that are in close proximity with each other (Fig. 13.2). Coil 1 has N_1 turns, while coil 2 has N_2 turns. For the sake of simplicity, assume that the second inductor carries no current. The magnetic flux ϕ_1 emanating from coil 1 has two components: one component ϕ_{11} links only coil 1, and another component ϕ_{12} links both coils. Hence,

$$\phi_1 = \phi_{11} + \phi_{12} \tag{13.5}$$

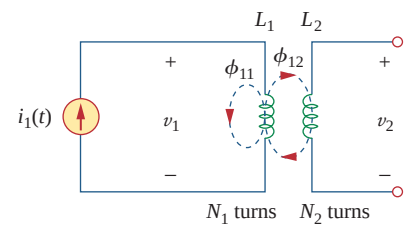


Figure 13.2
Mutual inductance M_{21} of coil 2 with respect to coil 1.

Although the two coils are physically separated, they are said to be *magnetically coupled*. Since the entire flux ϕ_1 links coil 1, the voltage induced in coil 1 is

$$v_1 = N_1\frac{d\phi_1}{dt} \tag{13.6}$$

Only flux ϕ_{12} links coil 2, so the voltage induced in coil 2 is

$$v_2 = N_2\frac{d\phi_{12}}{dt} \tag{13.7}$$

Again, as the fluxes are caused by the current i_1 flowing in coil 1, Eq. (13.6) can be written as

$$v_1 = N_1 \frac{d\phi_1}{di_1}\frac{di_1}{dt} = L_1 \frac{di_1}{dt} \tag{13.8}$$

where $L_1 = N_1\, d\phi_1/di_1$ is the self-inductance of coil 1. Similarly, Eq. (13.7) can be written as

$$v_2 = N_2 \frac{d\phi_{12}}{di_1}\frac{di_1}{dt} = M_{21} \frac{di_1}{dt} \tag{13.9}$$

where

$$M_{21} = N_2 \frac{d\phi_{12}}{di_1} \tag{13.10}$$

M_{21} is known as the *mutual inductance* of coil 2 with respect to coil 1. Subscript 21 indicates that the inductance M_{21} relates the voltage induced in coil 2 to the current in coil 1. Thus, the open-circuit *mutual voltage* (or induced voltage) across coil 2 is

$$\boxed{v_2 = M_{21} \frac{di_1}{dt}} \tag{13.11}$$

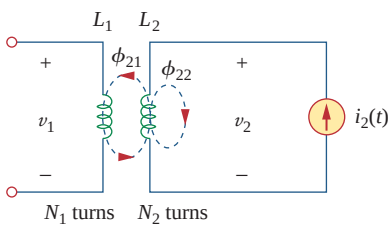


Figure 13.3
Mutual inductance M_{12} of coil 1 with respect to coil 2.

Suppose we now let current i_2 flow in coil 2, while coil 1 carries no current (Fig. 13.3). The magnetic flux ϕ_2 emanating from coil 2 comprises flux ϕ_{22} that links only coil 2 and flux ϕ_{21} that links both coils. Hence,

$$\phi_2 = \phi_{21} + \phi_{22} \tag{13.12}$$

The entire flux ϕ_2 links coil 2, so the voltage induced in coil 2 is

$$v_2 = N_2 \frac{d\phi_2}{dt} = N_2 \frac{d\phi_2}{di_2}\frac{di_2}{dt} = L_2 \frac{di_2}{dt} \tag{13.13}$$

where $L_2 = N_2\, d\phi_2/di_2$ is the self-inductance of coil 2. Since only flux ϕ_{21} links coil 1, the voltage induced in coil 1 is

$$v_1 = N_1 \frac{d\phi_{21}}{dt} = N_1 \frac{d\phi_{21}}{di_2}\frac{di_2}{dt} = M_{12} \frac{di_2}{dt} \tag{13.14}$$

where

$$M_{12} = N_1 \frac{d\phi_{21}}{di_2} \tag{13.15}$$

which is the *mutual inductance* of coil 1 with respect to coil 2. Thus, the open-circuit *mutual voltage* across coil 1 is

$$\boxed{v_1 = M_{12} \frac{di_2}{dt}} \tag{13.16}$$

We will see in the next section that M_{12} and M_{21} are equal; that is,

$$M_{12} = M_{21} = M \tag{13.17}$$

and we refer to M as the mutual inductance between the two coils. Like self-inductance L, mutual inductance M is measured in henrys (H). Keep in mind that mutual coupling only exists when the inductors or coils are in close proximity, and the circuits are driven by time-varying sources. We recall that inductors act like short circuits to dc.

From the two cases in Figs. 13.2 and 13.3, we conclude that mutual inductance results if a voltage is induced by a time-varying current in another circuit. It is the property of an inductor to produce a voltage in reaction to a time-varying current in another inductor near it. Thus,

Mutual inductance is the ability of one inductor to induce a voltage across a neighboring inductor, measured in henrys (H).

Although mutual inductance M is always a positive quantity, the mutual voltage $M\,di/dt$ may be negative or positive, just like the self-induced voltage $L\,di/dt$. However, unlike the self-induced $L\,di/dt$, whose polarity is determined by the reference direction of the current and the reference polarity of the voltage (according to the passive sign convention), the polarity of mutual voltage $M\,di/dt$ is not easy to determine, because four terminals are involved. The choice of the correct polarity for $M\,di/dt$ is made by examining the orientation or particular way in which both coils are physically wound and applying Lenz's law in conjunction with the right-hand rule. Since it is inconvenient to show the construction details of coils on a circuit schematic, we apply the *dot convention* in circuit analysis. By this convention, a dot is placed in the circuit at one end of each of the two magnetically coupled coils to indicate the direction of the magnetic flux if current enters that dotted terminal of the coil. This is illustrated in Fig. 13.4. Given a circuit, the dots are already placed beside the coils so that we need not bother about how to place them. The dots are used along with the dot convention to determine the polarity of the mutual voltage. The dot convention is stated as follows:

If a current **enters** the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is **positive** at the dotted terminal of the second coil.

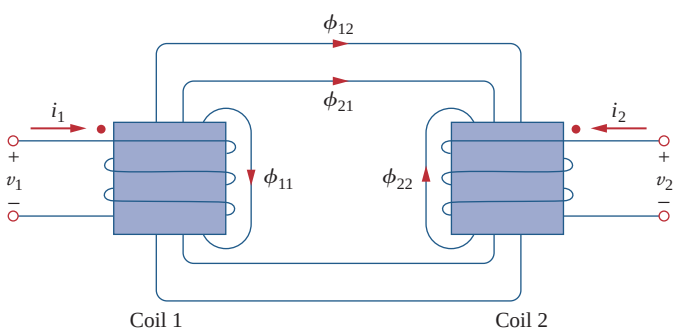


Figure 13.4
Illustration of the dot convention.

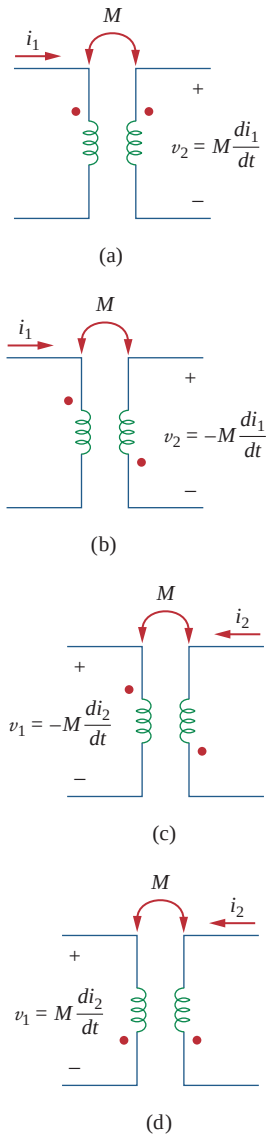


Figure 13.5
Examples illustrating how to apply the dot convention.

Alternatively,

> If a current **leaves** the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is **negative** at the dotted terminal of the second coil.

Thus, the reference polarity of the mutual voltage depends on the reference direction of the inducing current and the dots on the coupled coils. Application of the dot convention is illustrated in the four pairs of mutually coupled coils in Fig. 13.5. For the coupled coils in Fig. 13.5(a), the sign of the mutual voltage v_2 is determined by the reference polarity for v_2 and the direction of i_1. Since i_1 enters the dotted terminal of coil 1 and v_2 is positive at the dotted terminal of coil 2, the mutual voltage is $+M\,di_1/dt$. For the coils in Fig. 13.5(b), the current i_1 enters the dotted terminal of coil 1 and v_2 is negative at the dotted terminal of coil 2. Hence, the mutual voltage is $-M\,di_1/dt$. The same reasoning applies to the coils in Fig. 13.5(c) and 13.5(d).

Figure 13.6 shows the dot convention for coupled coils in series. For the coils in Fig. 13.6(a), the total inductance is

$$L = L_1 + L_2 + 2M \qquad \text{(Series-aiding connection)} \tag{13.18}$$

For the coils in Fig. 13.6(b),

$$L = L_1 + L_2 - 2M \qquad \text{(Series-opposing connection)} \tag{13.19}$$

Now that we know how to determine the polarity of the mutual voltage, we are prepared to analyze circuits involving mutual inductance. As the first example, consider the circuit in Fig. 13.7. Applying KVL to coil 1 gives

$$v_1 = i_1R_1 + L_1\frac{di_1}{dt} + M\frac{di_2}{dt} \tag{13.20a}$$

For coil 2, KVL gives

$$v_2 = i_2R_2 + L_2\frac{di_2}{dt} + M\frac{di_1}{dt} \tag{13.20b}$$

We can write Eq. (13.20) in the frequency domain as

$$\mathbf{V}_1 = (R_1 + j\omega L_1)\mathbf{I}_1 + j\omega M\mathbf{I}_2 \tag{13.21a}$$

$$\mathbf{V}_2 = j\omega M\mathbf{I}_1 + (R_2 + j\omega L_2)\mathbf{I}_2 \tag{13.21b}$$

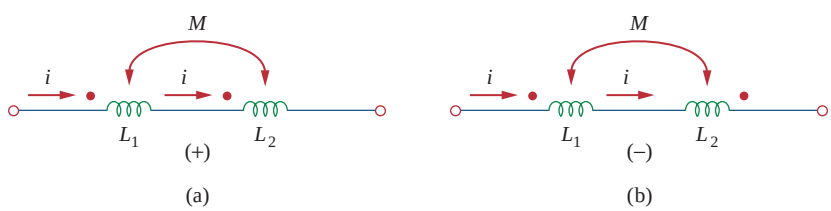


Figure 13.6
Dot convention for coils in series; the sign indicates the polarity of the mutual voltage: (a) series-aiding connection, (b) series-opposing connection.

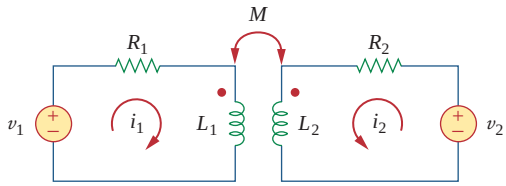


Figure 13.7
Time-domain analysis of a circuit containing coupled coils.

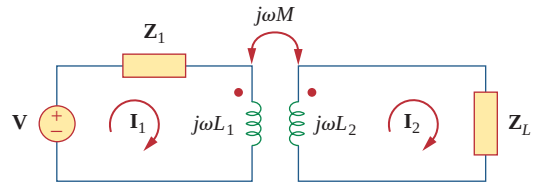


Figure 13.8
Frequency-domain analysis of a circuit containing coupled coils.

As a second example, consider the circuit in Fig. 13.8. We analyze this in the frequency domain. Applying KVL to coil 1, we get

$$\mathbf{V} = (\mathbf{Z}_1 + j\omega L_1)\mathbf{I}_1 - j\omega M\mathbf{I}_2 \qquad \textbf{(13.22a)}$$

For coil 2, KVL yields

$$0 = -j\omega M\mathbf{I}_1 + (\mathbf{Z}_L + j\omega L_2)\mathbf{I}_2 \qquad \textbf{(13.22b)}$$

Equations (13.21) and (13.22) are solved in the usual manner to determine the currents.

At this introductory level we are not concerned with the determination of the mutual inductances of the coils and their dot placements. Like *R*, *L*, and *C*, calculation of *M* would involve applying the theory of electromagnetics to the actual physical properties of the coils. In this text, we assume that the mutual inductance and the placement of the dots are the "givens" of the circuit problem, like the circuit components *R*, *L*, and *C*.

Example 13.1

Calculate the phasor currents $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit of Fig. 13.9.

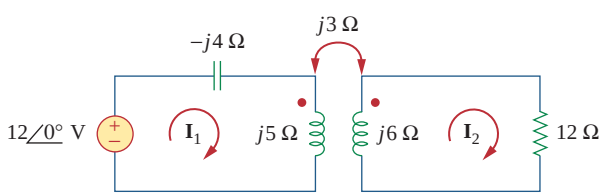


Figure 13.9
For Example 13.1.

Solution:

For coil 1, KVL gives

$$-12 + (-j4 + j5)\mathbf{I}_1 - j3\mathbf{I}_2 = 0$$

or

$$j\mathbf{I}_1 - j3\mathbf{I}_2 = 12 \qquad \textbf{(13.1.1)}$$

For coil 2, KVL gives.

$$-j3\mathbf{I}_1 + (12 + j6)\mathbf{I}_2 = 0$$

or

$$\mathbf{I}_1 = \frac{(12 + j6)\mathbf{I}_2}{j3} = (2 - j4)\mathbf{I}_2 \qquad \textbf{(13.1.2)}$$

Substituting this in Eq. (13.1.1), we get

$$(j2 + 4 - j3)\mathbf{I}_2 = (4 - j)\mathbf{I}_2 = 12$$

or

$$\mathbf{I}_2 = \frac{12}{4 - j} = 2.91\angle 14.04^\circ \text{ A} \tag{13.1.3}$$

From Eqs. (13.1.2) and (13.1.3),

$$\mathbf{I}_1 = (2 - j4)\mathbf{I}_2 = (4.472\angle -63.43^\circ)(2.91\angle 14.04^\circ)$$
$$= 13.01\angle -49.39^\circ \text{ A}$$

Practice Problem 13.1

Determine the voltage $\mathbf{V}_o$ in the circuit of Fig. 13.10.

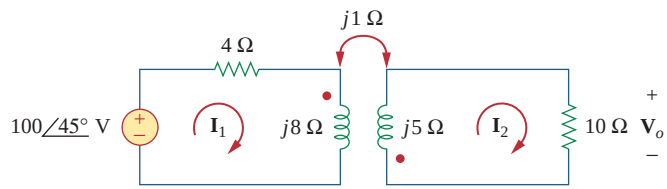


Figure 13.10
For Practice Prob. 13.1.

Answer: $10\angle -135^\circ$ V.

Example 13.2

Calculate the mesh currents in the circuit of Fig. 13.11.

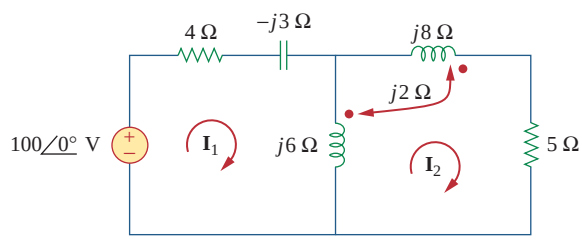


Figure 13.11
For Example 13.2.

Solution:

The key to analyzing a magnetically coupled circuit is knowing the polarity of the mutual voltage. We need to apply the dot rule. In Fig. 13.11, suppose coil 1 is the one whose reactance is 6 Ω, and coil 2 is the one whose reactance is 8 Ω. To figure out the polarity of the mutual voltage in coil 1 due to current $\mathbf{I}_2$, we observe that $\mathbf{I}_2$ leaves the dotted terminal of coil 2. Since we are applying KVL in the clockwise direction, it implies that the mutual voltage is negative, that is, $-j2\mathbf{I}_2$.

Alternatively, it might be best to figure out the mutual voltage by redrawing the relevant portion of the circuit, as shown in Fig. 13.12(a), where it becomes clear that the mutual voltage is $\mathbf{V}_1 = -2j\mathbf{I}_2$.

Thus, for mesh 1 in Fig. 13.11, KVL gives

$$-100 + \mathbf{I}_1(4 - j3 + j6) - j6\mathbf{I}_2 - j2\mathbf{I}_2 = 0$$

or

$$100 = (4 + j3)\mathbf{I}_1 - j8\mathbf{I}_2 \qquad \textbf{(13.2.1)}$$

Similarly, to figure out the mutual voltage in coil 2 due to current $\mathbf{I}_1$, consider the relevant portion of the circuit, as shown in Fig. 13.12(b). Applying the dot convention gives the mutual voltage as $\mathbf{V}_2 = -2j\mathbf{I}_1$. Also, current $\mathbf{I}_2$ sees the two coupled coils in series in Fig. 13.11; since it leaves the dotted terminals in both coils, Eq. (13.18) applies. Therefore, for mesh 2 in Fig. 13.11, KVL gives

$$0 = -2j\mathbf{I}_1 - j6\mathbf{I}_1 + (j6 + j8 + j2 \times 2 + 5)\mathbf{I}_2$$

or

$$0 = -j8\mathbf{I}_1 + (5 + j18)\mathbf{I}_2 \qquad \textbf{(13.2.2)}$$

Putting Eqs. (13.2.1) and (13.2.2) in matrix form, we get

$$\begin{bmatrix} 100 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 + j3 & -j8 \\ -j8 & 5 + j18 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$

The determinants are

$$\Delta = \begin{vmatrix} 4 + j3 & -j8 \\ -j8 & 5 + j18 \end{vmatrix} = 30 + j87$$

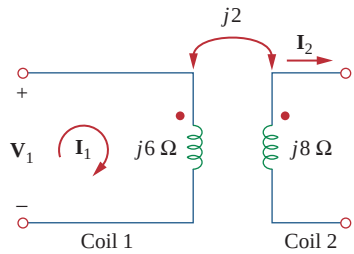
$$\Delta_1 = \begin{vmatrix} 100 & -j8 \\ 0 & 5 + j18 \end{vmatrix} = 100(5 + j18)$$

$$\Delta_2 = \begin{vmatrix} 4 + j3 & 100 \\ -j8 & 0 \end{vmatrix} = j800$$

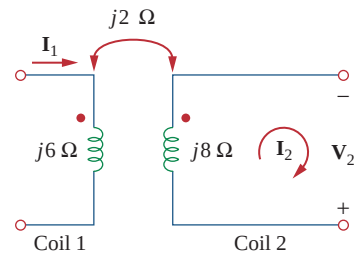
Thus, we obtain the mesh currents as

$$\mathbf{I}_1 = \frac{\Delta_1}{\Delta} = \frac{100(5 + j18)}{30 + j87} = \frac{1{,}868.2\angle 74.5^\circ}{92.03\angle 71^\circ} = 20.3\angle 3.5^\circ \text{ A}$$

$$\mathbf{I}_2 = \frac{\Delta_2}{\Delta} = \frac{j800}{30 + j87} = \frac{800\angle 90^\circ}{92.03\angle 71^\circ} = 8.693\angle 19^\circ \text{ A}$$



(a) $\mathbf{V}_1 = -2j\mathbf{I}_2$



(b) $\mathbf{V}_2 = -2j\mathbf{I}_1$

Figure 13.12
For Example 13.2; redrawing the relevant portion of the circuit in Fig. 13.11 to find mutual voltages by the dot convention.

Practice Problem 13.2

Determine the phasor currents $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit of Fig. 13.13.

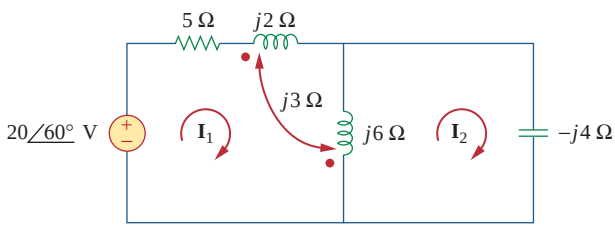


Figure 13.13
For Practice Prob. 13.2.

Answer: $3.583\angle 86.56^\circ$, $5.383\angle 86.56^\circ$ A.

13.3 Energy in a Coupled Circuit

In Chapter 6, we saw that the energy stored in an inductor is given by

$$w = \frac{1}{2}Li^2 \tag{13.23}$$

We now want to determine the energy stored in magnetically coupled coils.

Consider the circuit in Fig. 13.14. We assume that currents i_1 and i_2 are zero initially, so that the energy stored in the coils is zero. If we let i_1 increase from zero to I_1 while maintaining $i_2 = 0$, the power in coil 1 is

$$p_1(t) = v_1 i_1 = i_1 L_1 \frac{di_1}{dt} \tag{13.24}$$

and the energy stored in the circuit is

$$w_1 = \int p_1 dt = L_1 \int_0^{I_1} i_1 di_1 = \frac{1}{2}L_1 I_1^2 \tag{13.25}$$

If we now maintain $i_1 = I_1$ and increase i_2 from zero to I_2, the mutual voltage induced in coil 1 is $M_{12}\, di_2/dt$, while the mutual voltage induced in coil 2 is zero, since i_1 does not change. The power in the coils is now

$$p_2(t) = i_1 M_{12}\frac{di_2}{dt} + i_2 v_2 = I_1 M_{12}\frac{di_2}{dt} + i_2 L_2 \frac{di_2}{dt} \tag{13.26}$$

and the energy stored in the circuit is

$$\begin{aligned} w_2 &= \int p_2 dt = M_{12} I_1 \int_0^{I_2} di_2 + L_2 \int_0^{I_2} i_2 di_2 \\ &= M_{12} I_1 I_2 + \frac{1}{2}L_2 I_2^2 \end{aligned} \tag{13.27}$$

The total energy stored in the coils when both i_1 and i_2 have reached constant values is

$$w = w_1 + w_2 = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 + M_{12} I_1 I_2 \tag{13.28}$$

If we reverse the order by which the currents reach their final values, that is, if we first increase i_2 from zero to I_2 and later increase i_1 from zero to I_1, the total energy stored in the coils is

$$w = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 + M_{21} I_1 I_2 \tag{13.29}$$

Since the total energy stored should be the same regardless of how we reach the final conditions, comparing Eqs. (13.28) and (13.29) leads us to conclude that

$$M_{12} = M_{21} = M \tag{13.30a}$$

and

$$w = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 + M I_1 I_2 \tag{13.30b}$$

This equation was derived based on the assumption that the coil currents both entered the dotted terminals. If one current enters one

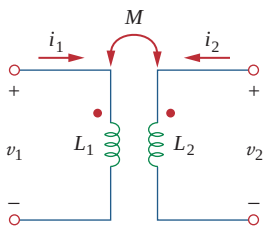


Figure 13.14
The circuit for deriving energy stored in a coupled circuit.

dotted terminal while the other current leaves the other dotted terminal, the mutual voltage is negative, so that the mutual energy MI_1I_2 is also negative. In that case,

$$w = \frac{1}{2}L_1I_1^2 + \frac{1}{2}L_2I_2^2 - MI_1I_2 \tag{13.31}$$

Also, since I_1 and I_2 are arbitrary values, they may be replaced by i_1 and i_2, which gives the instantaneous energy stored in the circuit the general expression

$$\boxed{w = \frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 \pm Mi_1i_2} \tag{13.32}$$

The positive sign is selected for the mutual term if both currents enter or leave the dotted terminals of the coils; the negative sign is selected otherwise.

We will now establish an upper limit for the mutual inductance M. The energy stored in the circuit cannot be negative because the circuit is passive. This means that the quantity $1/2L_1i_1^2 + 1/2L_2i_2^2 - Mi_1i_2$ must be greater than or equal to zero:

$$\frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 - Mi_1i_2 \geq 0 \tag{13.33}$$

To complete the square, we both add and subtract the term $i_1i_2\sqrt{L_1L_2}$ on the right-hand side of Eq. (13.33) and obtain

$$\frac{1}{2}(i_1\sqrt{L_1} - i_2\sqrt{L_2})^2 + i_1i_2(\sqrt{L_1L_2} - M) \geq 0 \tag{13.34}$$

The squared term is never negative; at its least it is zero. Therefore, the second term on the right-hand side of Eq. (13.34) must be greater than zero; that is,

$$\sqrt{L_1L_2} - M \geq 0$$

or

$$M \leq \sqrt{L_1L_2} \tag{13.35}$$

Thus, the mutual inductance cannot be greater than the geometric mean of the self-inductances of the coils. The extent to which the mutual inductance M approaches the upper limit is specified by the *coefficient of coupling* k, given by

$$k = \frac{M}{\sqrt{L_1L_2}} \tag{13.36}$$

or

$$\boxed{M = k\sqrt{L_1L_2}} \tag{13.37}$$

where $0 \leq k \leq 1$ or equivalently $0 \leq M \leq \sqrt{L_1L_2}$. The coupling coefficient is the fraction of the total flux emanating from one coil that links the other coil. For example, in Fig. 13.2,

$$k = \frac{\phi_{12}}{\phi_1} = \frac{\phi_{12}}{\phi_{11} + \phi_{12}} \tag{13.38}$$

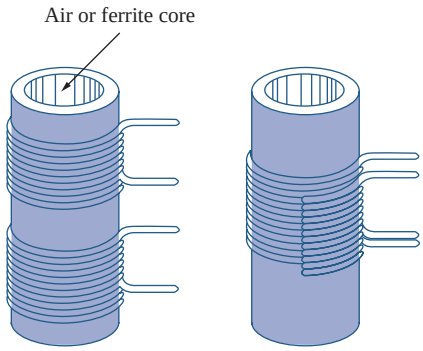


Figure 13.15
Windings: (a) loosely coupled, (b) tightly coupled; cutaway view demonstrates both windings.

and in Fig. 13.3,

$$k = \frac{\phi_{21}}{\phi_2} = \frac{\phi_{21}}{\phi_{21} + \phi_{22}} \tag{13.39}$$

If the entire flux produced by one coil links another coil, then $k = 1$ and we have 100 percent coupling, or the coils are said to be *perfectly coupled*. For $k < 0.5$, coils are said to be *loosely coupled*; and for $k > 0.5$, they are said to be *tightly coupled*. Thus,

The **coupling coefficient** k is a measure of the magnetic coupling between two coils; $0 \le k \le 1$.

We expect k to depend on the closeness of the two coils, their core, their orientation, and their windings. Figure 13.15 shows loosely coupled windings and tightly coupled windings. The air-core transformers used in radio frequency circuits are loosely coupled, whereas iron-core transformers used in power systems are tightly coupled. The linear transformers discussed in Section 3.4 are mostly air-core; the ideal transformers discussed in Sections 13.5 and 13.6 are principally iron-core.

Example 13.3

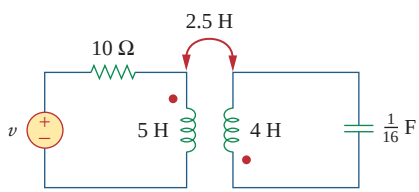


Figure 13.16
For Example 13.3.

Consider the circuit in Fig. 13.16. Determine the coupling coefficient. Calculate the energy stored in the coupled inductors at time $t = 1$ s if $v = 60\cos(4t + 30°)$ V.

Solution:
The coupling coefficient is

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{2.5}{\sqrt{20}} = 0.56$$

indicating that the inductors are tightly coupled. To find the energy stored, we need to calculate the current. To find the current, we need to obtain the frequency-domain equivalent of the circuit.

$$\begin{aligned} 60\cos(4t + 30°) &\Rightarrow 60\angle 30°, \quad \omega = 4 \text{ rad/s} \\ 5 \text{ H} &\Rightarrow j\omega L_1 = j20\ \Omega \\ 2.5 \text{ H} &\Rightarrow j\omega M = j10\ \Omega \\ 4 \text{ H} &\Rightarrow j\omega L_2 = j16\ \Omega \\ \frac{1}{16} \text{ F} &\Rightarrow \frac{1}{j\omega C} = -j4\ \Omega \end{aligned}$$

The frequency-domain equivalent is shown in Fig. 13.17. We now apply mesh analysis. For mesh 1,

$$(10 + j20)\mathbf{I}_1 + j10\mathbf{I}_2 = 60\angle 30° \tag{13.3.1}$$

For mesh 2,

$$j10\mathbf{I}_1 + (j16 - j4)\mathbf{I}_2 = 0$$

or

$$\mathbf{I}_1 = -1.2\mathbf{I}_2 \tag{13.3.2}$$

Substituting this into Eq. (13.3.1) yields

$$\mathbf{I}_2(-12 - j14) = 60\angle 30^\circ \quad \Rightarrow \quad \mathbf{I}_2 = 3.254\angle 160.6^\circ \text{ A}$$

and

$$\mathbf{I}_1 = -1.2\mathbf{I}_2 = 3.905\angle -19.4^\circ \text{ A}$$

In the time-domain,

$$i_1 = 3.905\cos(4t - 19.4^\circ), \qquad i_2 = 3.254\cos(4t + 160.6^\circ)$$

At time $t = 1$ s, $4t = 4$ rad $= 229.2^\circ$, and

$$i_1 = 3.905\cos(229.2^\circ - 19.4^\circ) = -3.389 \text{ A}$$
$$i_2 = 3.254\cos(229.2^\circ + 160.6^\circ) = 2.824 \text{ A}$$

The total energy stored in the coupled inductors is

$$w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 + M i_1 i_2$$
$$= \frac{1}{2}(5)(-3.389)^2 + \frac{1}{2}(4)(2.824)^2 + 2.5(-3.389)(2.824) = 20.73 \text{ J}$$

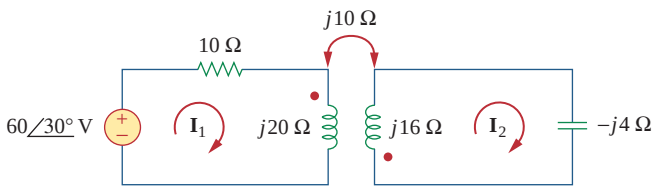


Figure 13.17
Frequency-domain equivalent of the circuit in Fig. 13.16.

Practice Problem 13.3

For the circuit in Fig. 13.18, determine the coupling coefficient and the energy stored in the coupled inductors at $t = 1.5$ s.

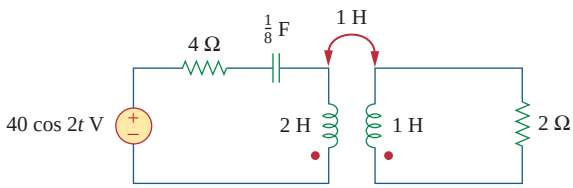


Figure 13.18
For Practice Prob. 13.3.

Answer: 0.7071, 39.4 J.

13.4 Linear Transformers

Here we introduce the transformer as a new circuit element. A transformer is a magnetic device that takes advantage of the phenomenon of mutual inductance.

A **transformer** is generally a four-terminal device comprising two (or more) magnetically coupled coils.

As shown in Fig. 13.19, the coil that is directly connected to the voltage source is called the *primary winding*. The coil connected to the load is called the *secondary winding*. The resistances R_1 and R_2 are included to account for the losses (power dissipation) in the coils. The transformer is said to be *linear* if the coils are wound on a magnetically linear material—a material for which the magnetic permeability is constant. Such materials include air, plastic, Bakelite, and wood. In fact, most materials are magnetically linear. Linear transformers are sometimes called *air-core transformers*, although not all of them are necessarily air-core. They are used in radio and TV sets. Figure 13.20 portrays different types of transformers.

A linear transformer may also be regarded as one whose flux is proportional to the currents in its windings.

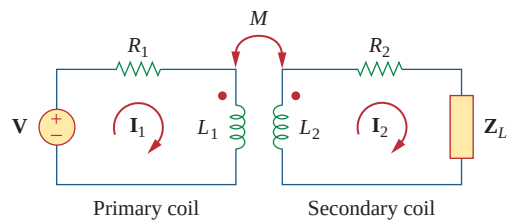
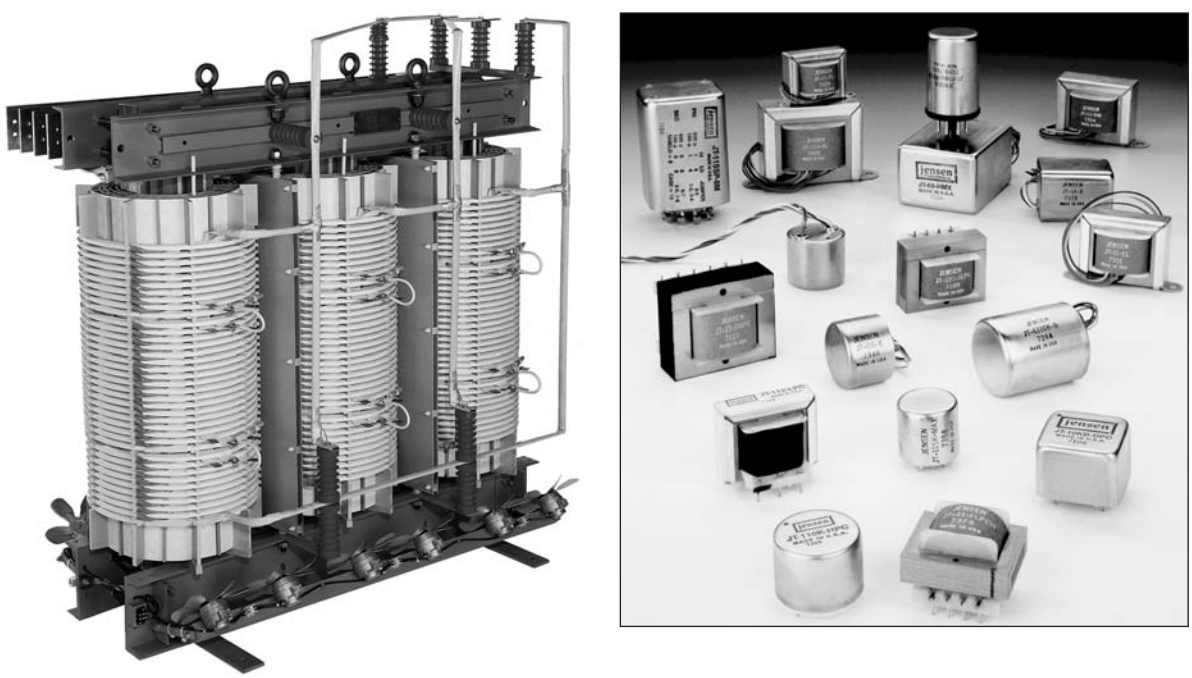


Figure 13.19
A linear transformer.



(a) (b)

Figure 13.20
Different types of transformers: (a) copper wound dry power transformer, (b) audio transformers.
Courtesy of: (a) Electric Service Co., (b) Jensen Transformers.

We would like to obtain the input impedance $\mathbf{Z}_{in}$ as seen from the source, because $\mathbf{Z}_{in}$ governs the behavior of the primary circuit. Applying KVL to the two meshes in Fig. 13.19 gives

$$\mathbf{V} = (R_1 + j\omega L_1)\mathbf{I}_1 - j\omega M\mathbf{I}_2 \tag{13.40a}$$

$$0 = -j\omega M\mathbf{I}_1 + (R_2 + j\omega L_2 + \mathbf{Z}_L)\mathbf{I}_2 \tag{13.40b}$$

In Eq. (13.40b), we express $\mathbf{I}_2$ in terms of $\mathbf{I}_1$ and substitute it into Eq. (13.40a). We get the input impedance as

$$\mathbf{Z}_{in} = \frac{\mathbf{V}}{\mathbf{I}_1} = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + \mathbf{Z}_L} \tag{13.41}$$

Notice that the input impedance comprises two terms. The first term, $(R_1 + j\omega L_1)$, is the primary impedance. The second term is due to the coupling between the primary and secondary windings. It is as though this impedance is reflected to the primary. Thus, it is known as the *reflected impedance* $\mathbf{Z}_R$, and

Some authors call this the *coupled impedance.*

$$\boxed{\mathbf{Z}_R = \frac{\omega^2 M^2}{R_2 + j\omega L_2 + \mathbf{Z}_L}} \tag{13.42}$$

It should be noted that the result in Eq. (13.41) or (13.42) is not affected by the location of the dots on the transformer, because the same result is produced when M is replaced by $-M$.

The little bit of experience gained in Sections 13.2 and 13.3 in analyzing magnetically coupled circuits is enough to convince anyone that analyzing these circuits is not as easy as circuits in previous chapters. For this reason, it is sometimes convenient to replace a magnetically coupled circuit by an equivalent circuit with no magnetic coupling. We want to replace the linear transformer in Fig. 13.21 by an equivalent T or Π circuit, a circuit that would have no mutual inductance.

The voltage-current relationships for the primary and secondary coils give the matrix equation

$$\boxed{\begin{bmatrix}\mathbf{V}_1\\ \mathbf{V}_2\end{bmatrix} = \begin{bmatrix}j\omega L_1 & j\omega M\\ j\omega M & j\omega L_2\end{bmatrix}\begin{bmatrix}\mathbf{I}_1\\ \mathbf{I}_2\end{bmatrix}} \tag{13.43}$$

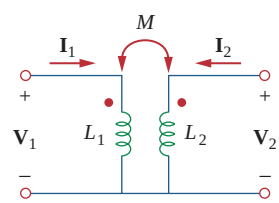


Figure 13.21
Determining the equivalent circuit of a linear transformer.

By matrix inversion, this can be written as

$$\begin{bmatrix}\mathbf{I}_1\\ \mathbf{I}_2\end{bmatrix} = \begin{bmatrix}\dfrac{L_2}{j\omega(L_1L_2 - M^2)} & \dfrac{-M}{j\omega(L_1L_2 - M^2)}\\ \dfrac{-M}{j\omega(L_1L_2 - M^2)} & \dfrac{L_1}{j\omega(L_1L_2 - M^2)}\end{bmatrix}\begin{bmatrix}\mathbf{V}_1\\ \mathbf{V}_2\end{bmatrix} \tag{13.44}$$

Our goal is to match Eqs. (13.43) and (13.44) with the corresponding equations for the T and Π networks.

For the T (or Y) network of Fig. 13.22, mesh analysis provides the terminal equations as

$$\begin{bmatrix}\mathbf{V}_1\\ \mathbf{V}_2\end{bmatrix} = \begin{bmatrix}j\omega(L_a + L_c) & j\omega L_c\\ j\omega L_c & j\omega(L_b + L_c)\end{bmatrix}\begin{bmatrix}\mathbf{I}_1\\ \mathbf{I}_2\end{bmatrix} \tag{13.45}$$

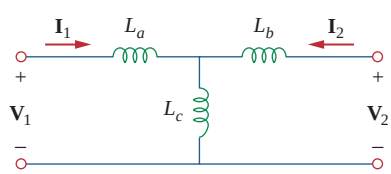


Figure 13.22
An equivalent T circuit.

If the circuits in Figs. 13.21 and 13.22 are equivalents, Eqs. (13.43) and (13.45) must be identical. Equating terms in the impedance matrices of Eqs. (13.43) and (13.45) leads to

$$L_a = L_1 - M, \qquad L_b = L_2 - M, \qquad L_c = M \tag{13.46}$$

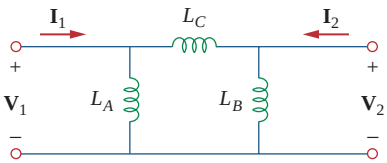


Figure 13.23
An equivalent Π circuit.

For the Π (or Δ) network in Fig. 13.23, nodal analysis gives the terminal equations as

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \dfrac{1}{j\omega L_A} + \dfrac{1}{j\omega L_C} & -\dfrac{1}{j\omega L_C} \\ -\dfrac{1}{j\omega L_C} & \dfrac{1}{j\omega L_B} + \dfrac{1}{j\omega L_C} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} \tag{13.47}$$

Equating terms in admittance matrices of Eqs. (13.44) and (13.47), we obtain

$$L_A = \frac{L_1L_2 - M^2}{L_2 - M}, \qquad L_B = \frac{L_1L_2 - M^2}{L_1 - M}$$
$$L_C = \frac{L_1L_2 - M^2}{M} \tag{13.48}$$

Note that in Figs. 13.22 and 13.23, the inductors are not magnetically coupled. Also note that changing the locations of the dots in Fig. 13.21 can cause M to become $-M$. As Example 13.6 illustrates, a negative value of M is physically unrealizable but the equivalent model is still mathematically valid.

Example 13.4

In the circuit of Fig. 13.24, calculate the input impedance and current $\mathbf{I}_1$. Take $\mathbf{Z}_1 = 60 - j100\ \Omega$, $\mathbf{Z}_2 = 30 + j40\ \Omega$, and $\mathbf{Z}_L = 80 + j60\ \Omega$.

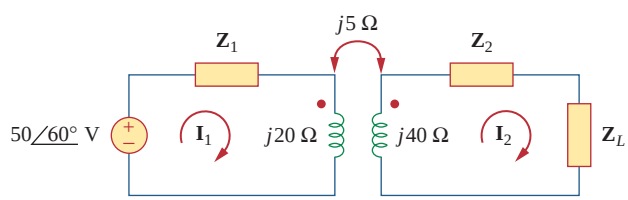


Figure 13.24
For Example 13.4.

Solution:
From Eq. (13.41),

$$\mathbf{Z}_{\text{in}} = \mathbf{Z}_1 + j20 + \frac{(5)^2}{j40 + \mathbf{Z}_2 + \mathbf{Z}_L}$$
$$= 60 - j100 + j20 + \frac{25}{110 + j140}$$
$$= 60 - j80 + 0.14\angle -51.84°$$
$$= 60.09 - j80.11 = 100.14\angle -53.1°\ \Omega$$

Thus,

$$\mathbf{I}_1 = \frac{\mathbf{V}}{\mathbf{Z}_{\text{in}}} = \frac{50\angle 60^\circ}{100.14\angle -53.1^\circ} = 0.5\angle 113.1^\circ \text{ A}$$

Practice Problem 13.4

Find the input impedance of the circuit in Fig. 13.25 and the current from the voltage source.

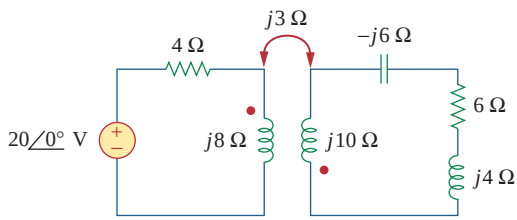


Figure 13.25
For Practice Prob. 13.4.

Answer: $8.58\angle 58.05^\circ\ \Omega$, $2.331\angle -58.05^\circ$ A.

Example 13.5

Determine the T-equivalent circuit of the linear transformer in Fig. 13.26(a).

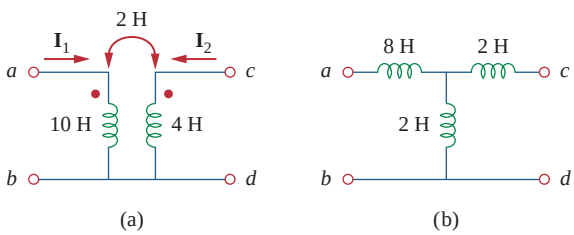


Figure 13.26
For Example 13.5: (a) a linear transformer, (b) its T-equivalent circuit.

Solution:
Given that $L_1 = 10$, $L_2 = 4$, and $M = 2$, the T-equivalent network has the following parameters:

$$L_a = L_1 - M = 10 - 2 = 8 \text{ H}$$
$$L_b = L_2 - M = 4 - 2 = 2 \text{ H}, \qquad L_c = M = 2 \text{ H}$$

The T-equivalent circuit is shown in Fig. 13.26(b). We have assumed that reference directions for currents and voltage polarities in the primary and secondary windings conform to those in Fig. 13.21. Otherwise, we may need to replace M with $-M$, as Example 13.6 illustrates.

Practice Problem 13.5

For the linear transformer in Fig. 13.26(a), find the Π equivalent network.

Answer: $L_A = 18$ H, $L_B = 4.5$ H, $L_C = 18$ H.

Example 13.6

Solve for $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{V}_o$ in Fig. 13.27 (the same circuit as for Practice Prob. 13.1) using the T-equivalent circuit for the linear transformer.

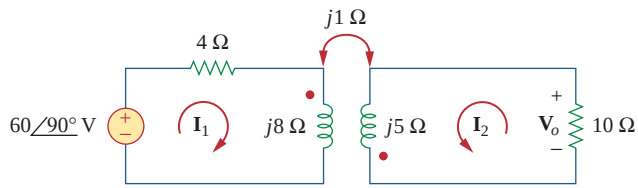


Figure 13.27
For Example 13.6.

Solution:

Notice that the circuit in Fig. 13.27 is the same as that in Fig. 13.10 except that the reference direction for current $\mathbf{I}_2$ has been reversed, just to make the reference directions for the currents for the magnetically coupled coils conform with those in Fig. 13.21.

We need to replace the magnetically coupled coils with the T-equivalent circuit. The relevant portion of the circuit in Fig. 13.27 is shown in Fig. 13.28(a). Comparing Fig. 13.28(a) with Fig. 13.21 shows that there are two differences. First, due to the current reference directions and voltage polarities, we need to replace M by $-M$ to make Fig. 13.28(a) conform with Fig. 13.21. Second, the circuit in Fig. 13.21 is in the time-domain, whereas the circuit in Fig. 13.28(a) is in the frequency-domain. The difference is the factor $j\omega$; that is, L in Fig. 13.21 has been replaced with $j\omega L$ and M with $j\omega M$. Since ω is not specified, we can assume $\omega = 1$ rad/s or any other value; it really does not matter. With these two differences in mind,

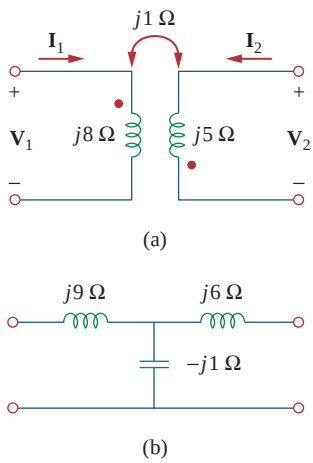


Figure 13.28
For Example 13.6: (a) circuit for coupled coils of Fig. 13.27, (b) T-equivalent circuit.

$$L_a = L_1 - (-M) = 8 + 1 = 9 \text{ H}$$

$$L_b = L_2 - (-M) = 5 + 1 = 6 \text{ H}, \qquad L_c = -M = -1 \text{ H}$$

Thus, the T-equivalent circuit for the coupled coils is as shown in Fig. 13.28(b).

Inserting the T-equivalent circuit in Fig. 13.28(b) to replace the two coils in Fig. 13.27 gives the equivalent circuit in Fig. 13.29, which can be solved using nodal or mesh analysis. Applying mesh analysis, we obtain

$$j6 = \mathbf{I}_1(4 + j9 - j1) + \mathbf{I}_2(-j1) \qquad \textbf{(13.6.1)}$$

and

$$0 = \mathbf{I}_1(-j1) + \mathbf{I}_2(10 + j6 - j1) \qquad \textbf{(13.6.2)}$$

From Eq. (13.6.2),

$$\mathbf{I}_1 = \frac{(10 + j5)}{j}\mathbf{I}_2 = (5 - j10)\mathbf{I}_2 \qquad \textbf{(13.6.3)}$$

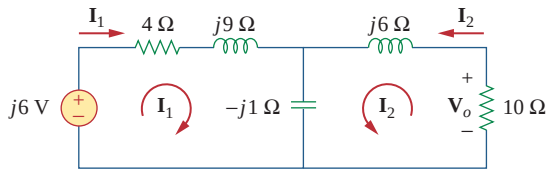


Figure 13.29
For Example 13.6.

Substituting Eq. (13.6.3) into Eq. (13.6.1) gives

$$j6 = (4 + j8)(5 - j10)\mathbf{I}_2 - j\mathbf{I}_2 = (100 - j)\mathbf{I}_2 \simeq 100\mathbf{I}_2$$

Since 100 is very large compared with 1, the imaginary part of $(100 - j)$ can be ignored so that $100 - j \simeq 100$. Hence,

$$\mathbf{I}_2 = \frac{j6}{100} = j0.06 = 0.06\angle 90^\circ \text{ A}$$

From Eq. (13.6.3),

$$\mathbf{I}_1 = (5 - j10)j0.06 = 0.6 + j0.3 \text{ A}$$

and

$$\mathbf{V}_o = -10\mathbf{I}_2 = -j0.6 = 0.6\angle -90^\circ \text{ V}$$

This agrees with the answer to Practice Prob. 13.1. Of course, the direction of $\mathbf{I}_2$ in Fig. 13.10 is opposite to that in Fig. 13.27. This will not affect $\mathbf{V}_o$, but the value of $\mathbf{I}_2$ in this example is the negative of that of $\mathbf{I}_2$ in Practice Prob. 13.1. The advantage of using the T-equivalent model for the magnetically coupled coils is that in Fig. 13.29 we do not need to bother with the dot on the coupled coils.

Practice Problem 13.6

Solve the problem in Example 13.1 (see Fig. 13.9) using the T-equivalent model for the magnetically coupled coils.

Answer: $13\angle -49.4^\circ$ A, $2.91\angle 14.04^\circ$ A.

13.5 Ideal Transformers

An ideal transformer is one with perfect coupling ($k = 1$). It consists of two (or more) coils with a large number of turns wound on a common core of high permeability. Because of this high permeability of the core, the flux links all the turns of both coils, thereby resulting in a perfect coupling.

To see how an ideal transformer is the limiting case of two coupled inductors where the inductances approach infinity and the coupling is perfect, let us reexamine the circuit in Fig. 13.14. In the frequency domain,

$$\mathbf{V}_1 = j\omega L_1\mathbf{I}_1 + j\omega M\mathbf{I}_2 \quad \textbf{(13.49a)}$$

$$\mathbf{V}_2 = j\omega M\mathbf{I}_1 + j\omega L_2\mathbf{I}_2 \quad \textbf{(13.49b)}$$

From Eq. (13.49a), $\mathbf{I}_1 = (\mathbf{V}_1 - j\omega M\mathbf{I}_2)/j\omega L_1$ (we could have also use this equation to develop the current ratios instead of using the conservation of power which we will do shortly). Substituting this in Eq. (13.49b) gives

$$\mathbf{V}_2 = j\omega L_2\mathbf{I}_2 + \frac{M\mathbf{V}_1}{L_1} - \frac{j\omega M^2\mathbf{I}_2}{L_1}$$

But $M = \sqrt{L_1L_2}$ for perfect coupling ($k = 1$). Hence,

$$\mathbf{V}_2 = j\omega L_2\mathbf{I}_2 + \frac{\sqrt{L_1L_2}\mathbf{V}_1}{L_1} - \frac{j\omega L_1L_2\mathbf{I}_2}{L_1} = \sqrt{\frac{L_2}{L_1}}\mathbf{V}_1 = n\mathbf{V}_1$$

where $n = \sqrt{L_2/L_1}$ and is called the *turns ratio*. As $L_1, L_2, M \to \infty$ such that n remains the same, the coupled coils become an ideal transformer. A transformer is said to be ideal if it has the following properties:

1. Coils have very large reactances ($L_1, L_2, M \to \infty$).
2. Coupling coefficient is equal to unity ($k = 1$).
3. Primary and secondary coils are lossless ($R_1 = 0 = R_2$).

An **ideal transformer** is a unity-coupled, lossless transformer in which the primary and secondary coils have infinite self-inductances.

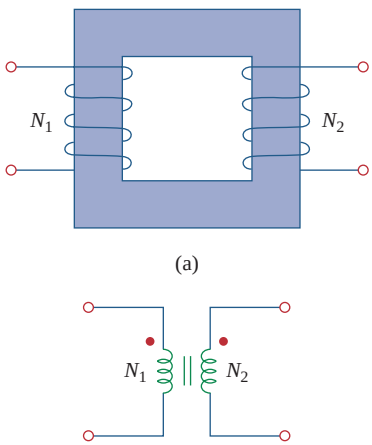


Figure 13.30
(a) Ideal transformer, (b) circuit symbol for ideal transformers.

Iron-core transformers are close approximations to ideal transformers. These are used in power systems and electronics.

Figure 13.30(a) shows a typical ideal transformer; the circuit symbol is in Fig. 13.30(b). The vertical lines between the coils indicate an iron core as distinct from the air core used in linear transformers. The primary winding has N_1 turns; the secondary winding has N_2 turns.

When a sinusoidal voltage is applied to the primary winding as shown in Fig. 13.31, the same magnetic flux ϕ goes through both windings. According to Faraday's law, the voltage across the primary winding is

$$v_1 = N_1\frac{d\phi}{dt} \tag{13.50a}$$

while that across the secondary winding is

$$v_2 = N_2\frac{d\phi}{dt} \tag{13.50b}$$

Dividing Eq. (13.50b) by Eq. (13.50a), we get

$$\frac{v_2}{v_1} = \frac{N_2}{N_1} = n \tag{13.51}$$

where n is, again, the *turns ratio* or *transformation ratio*. We can use the phasor voltages $\mathbf{V}_1$ and $\mathbf{V}_2$ rather than the instantaneous values v_1 and v_2. Thus, Eq. (13.51) may be written as

$$\boxed{\frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{N_2}{N_1} = n} \tag{13.52}$$

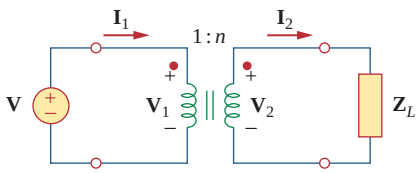


Figure 13.31
Relating primary and secondary quantities in an ideal transformer.

For the reason of power conservation, the energy supplied to the primary must equal the energy absorbed by the secondary, since there are no losses in an ideal transformer. This implies that

$$v_1 i_1 = v_2 i_2 \tag{13.53}$$

In phasor form, Eq. (13.53) in conjunction with Eq. (13.52) becomes

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{\mathbf{V}_2}{\mathbf{V}_1} = n \tag{13.54}$$

showing that the primary and secondary currents are related to the turns ratio in the inverse manner as the voltages. Thus,

$$\boxed{\frac{\mathbf{I}_2}{\mathbf{I}_1} = \frac{N_1}{N_2} = \frac{1}{n}} \tag{13.55}$$

When $n = 1$, we generally call the transformer an *isolation transformer*. The reason will become obvious in Section 13.9.1. If $n > 1$, we have a *step-up transformer*, as the voltage is increased from primary to secondary ($\mathbf{V}_2 > \mathbf{V}_1$). On the other hand, if $n < 1$, the transformer is a *step-down transformer*, since the voltage is decreased from primary to secondary ($\mathbf{V}_2 < \mathbf{V}_1$).

A **step-down transformer** is one whose secondary voltage is less than its primary voltage.

A **step-up transformer** is one whose secondary voltage is greater than its primary voltage.

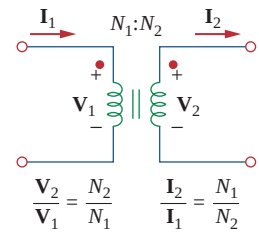
The ratings of transformers are usually specified as V_1/V_2. A transformer with rating 2400/120 V should have 2400 V on the primary and 120 in the secondary (i.e., a step-down transformer). Keep in mind that the voltage ratings are in rms.

Power companies often generate at some convenient voltage and use a step-up transformer to increase the voltage so that the power can be transmitted at very high voltage and low current over transmission lines, resulting in significant cost savings. Near residential consumer premises, step-down transformers are used to bring the voltage down to 120 V. Section 13.9.3 will elaborate on this.

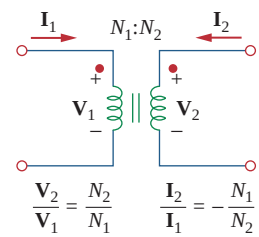
It is important that we know how to get the proper polarity of the voltages and the direction of the currents for the transformer in Fig. 13.31. If the polarity of $\mathbf{V}_1$ or $\mathbf{V}_2$ or the direction of $\mathbf{I}_1$ or $\mathbf{I}_2$ is changed, n in Eqs. (13.51) to (13.55) may need to be replaced by $-n$. The two simple rules to follow are:

1. If $\mathbf{V}_1$ and $\mathbf{V}_2$ are *both* positive or both negative at the dotted terminals, use $+n$ in Eq. (13.52). Otherwise, use $-n$.
2. If $\mathbf{I}_1$ and $\mathbf{I}_2$ *both* enter into or both leave the dotted terminals, use $-n$ in Eq. (13.55). Otherwise, use $+n$.

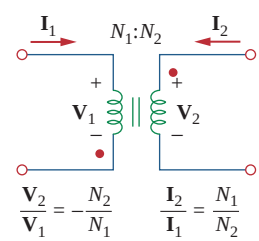
The rules are demonstrated with the four circuits in Fig. 13.32.



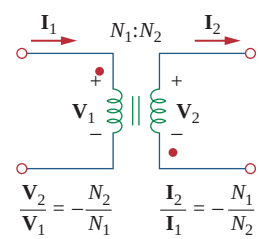
(a)



(b)



(c)



(d)

Figure 13.32
Typical circuits illustrating proper voltage polarities and current directions in an ideal transformer.

Using Eqs. (13.52) and (13.55), we can always express $\mathbf{V}_1$ in terms of $\mathbf{V}_2$ and $\mathbf{I}_1$ in terms of $\mathbf{I}_2$, or vice versa:

$$\mathbf{V}_1 = \frac{\mathbf{V}_2}{n} \quad \text{or} \quad \mathbf{V}_2 = n\mathbf{V}_1 \tag{13.56}$$

$$\mathbf{I}_1 = n\mathbf{I}_2 \quad \text{or} \quad \mathbf{I}_2 = \frac{\mathbf{I}_1}{n} \tag{13.57}$$

The complex power in the primary winding is

$$\mathbf{S}_1 = \mathbf{V}_1\mathbf{I}_1^* = \frac{\mathbf{V}_2}{n}(n\mathbf{I}_2)^* = \mathbf{V}_2\mathbf{I}_2^* = \mathbf{S}_2 \tag{13.58}$$

showing that the complex power supplied to the primary is delivered to the secondary without loss. The transformer absorbs no power. Of course, we should expect this, since the ideal transformer is lossless. The input impedance as seen by the source in Fig. 13.31 is found from Eqs. (13.56) and (13.57) as

$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{V}_1}{\mathbf{I}_1} = \frac{1}{n^2}\frac{\mathbf{V}_2}{\mathbf{I}_2} \tag{13.59}$$

It is evident from Fig. 13.31 that $\mathbf{V}_2/\mathbf{I}_2 = \mathbf{Z}_L$, so that

$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{Z}_L}{n^2} \tag{13.60}$$

Notice that an ideal transformer reflects an impedance as the square of the turns ratio.

The input impedance is also called the *reflected impedance*, since it appears as if the load impedance is reflected to the primary side. This ability of the transformer to transform a given impedance into another impedance provides us a means of *impedance matching* to ensure maximum power transfer. The idea of impedance matching is very useful in practice and will be discussed more in Section 13.9.2.

In analyzing a circuit containing an ideal transformer, it is common practice to eliminate the transformer by reflecting impedances and sources from one side of the transformer to the other. In the circuit of Fig. 13.33, suppose we want to reflect the secondary side of the circuit to the primary side. We find the Thevenin equivalent of the circuit to the right of the terminals *a*-*b*. We obtain $\mathbf{V}_{\text{Th}}$ as the open-circuit voltage at terminals *a*-*b*, as shown in Fig. 13.34(a).

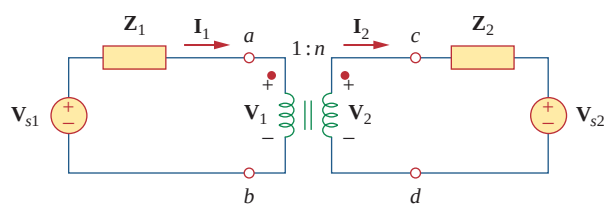


Figure 13.33
Ideal transformer circuit whose equivalent circuits are to be found.

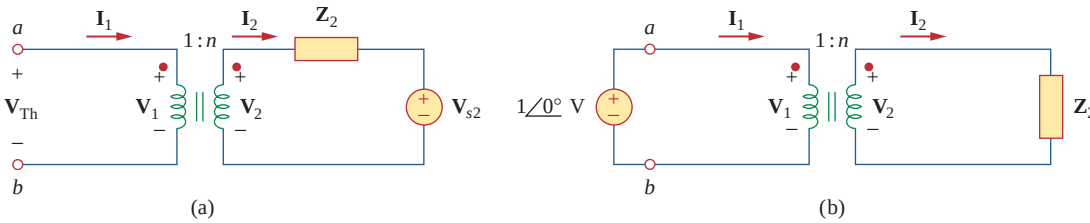


Figure 13.34
(a) Obtaining $\mathbf{V}_{Th}$ for the circuit in Fig. 13.33, (b) obtaining $\mathbf{Z}_{Th}$ for the circuit in Fig. 13.33.

Since terminals *a*-*b* are open, $\mathbf{I}_1 = 0 = \mathbf{I}_2$ so that $\mathbf{V}_2 = \mathbf{V}_{s2}$. Hence, From Eq. (13.56),

$$\mathbf{V}_{Th} = \mathbf{V}_1 = \frac{\mathbf{V}_2}{n} = \frac{\mathbf{V}_{s2}}{n} \tag{13.61}$$

To get $\mathbf{Z}_{Th}$, we remove the voltage source in the secondary winding and insert a unit source at terminals *a*-*b*, as in Fig. 13.34(b). From Eqs. (13.56) and (13.57), $\mathbf{I}_1 = n\mathbf{I}_2$ and $\mathbf{V}_1 = \mathbf{V}_2/n$, so that

$$\mathbf{Z}_{Th} = \frac{\mathbf{V}_1}{\mathbf{I}_1} = \frac{\mathbf{V}_2/n}{n\mathbf{I}_2} = \frac{\mathbf{Z}_2}{n^2}, \qquad \mathbf{V}_2 = \mathbf{Z}_2\mathbf{I}_2 \tag{13.62}$$

which is what we should have expected from Eq. (13.60). Once we have $\mathbf{V}_{Th}$ and $\mathbf{Z}_{Th}$, we add the Thevenin equivalent to the part of the circuit in Fig. 13.33 to the left of terminals *a*-*b*. Figure 13.35 shows the result.

The general rule for eliminating the transformer and reflecting the secondary circuit to the primary side is: divide the secondary impedance by n^2, divide the secondary voltage by n, and multiply the secondary current by n.

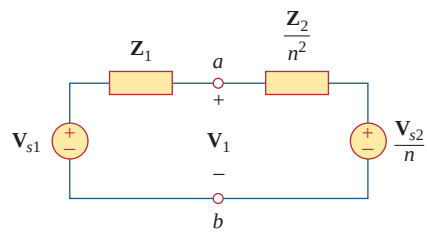


Figure 13.35
Equivalent circuit for Fig. 13.33 obtained by reflecting the secondary circuit to the primary side.

We can also reflect the primary side of the circuit in Fig. 13.33 to the secondary side. Figure 13.36 shows the equivalent circuit.

The rule for eliminating the transformer and reflecting the primary circuit to the secondary side is: multiply the primary impedance by n^2, multiply the primary voltage by n, and divide the primary current by n.

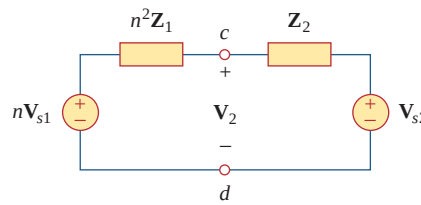


Figure 13.36
Equivalent circuit for Fig. 13.33 obtained by reflecting the primary circuit to the secondary side.

According to Eq. (13.58), the power remains the same, whether calculated on the primary or the secondary side. But realize that this reflection approach only applies if there are no external connections between the primary and secondary windings. When we have external connections between the primary and secondary windings, we simply use regular mesh and nodal analysis. Examples of circuits where there are external connections between the primary and secondary windings are in Figs. 13.39 and 13.40. Also note that if the locations of the dots in Fig. 13.33 are changed, we might have to replace n by $-n$ in order to obey the dot rule, illustrated in Fig. 13.32.

Example 13.7

An ideal transformer is rated at 2400/120 V, 9.6 kVA, and has 50 turns on the secondary side. Calculate: (a) the turns ratio, (b) the number of turns on the primary side, and (c) the current ratings for the primary and secondary windings.

Solution:

(a) This is a step-down transformer, since $V_1 = 2{,}400\ \text{V} > V_2 = 120\ \text{V}$.

$$n = \frac{V_2}{V_1} = \frac{120}{2{,}400} = 0.05$$

(b)

$$n = \frac{N_2}{N_1} \quad \Rightarrow \quad 0.05 = \frac{50}{N_1}$$

or

$$N_1 = \frac{50}{0.05} = 1{,}000 \text{ turns}$$

(c) $S = V_1 I_1 = V_2 I_2 = 9.6$ kVA. Hence,

$$I_1 = \frac{9{,}600}{V_1} = \frac{9{,}600}{2{,}400} = 4\ \text{A}$$

$$I_2 = \frac{9{,}600}{V_2} = \frac{9{,}600}{120} = 80\ \text{A} \quad \text{or} \quad I_2 = \frac{I_1}{n} = \frac{4}{0.05} = 80\ \text{A}$$

Practice Problem 13.7

The primary current to an ideal transformer rated at 3300/110 V is 5 A. Calculate: (a) the turns ratio, (b) the kVA rating, (c) the secondary current.

Answer: (a) 1/30, (b) 16.5 kVA, (c) 150 A.

Example 13.8

For the ideal transformer circuit of Fig. 13.37, find: (a) the source current $\mathbf{I}_1$, (b) the output voltage $\mathbf{V}_o$, and (c) the complex power supplied by the source.

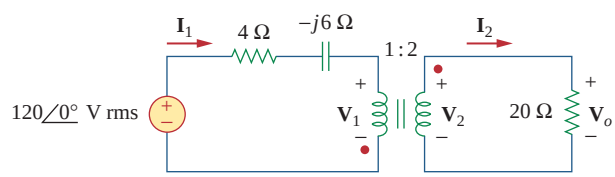


Figure 13.37
For Example 13.8.

Solution:

(a) The 20-Ω impedance can be reflected to the primary side and we get

$$\mathbf{Z}_R = \frac{20}{n^2} = \frac{20}{4} = 5\ \Omega$$

Thus,

$$\mathbf{Z}_{\text{in}} = 4 - j6 + \mathbf{Z}_R = 9 - j6 = 10.82\angle -33.69^\circ\ \Omega$$

$$\mathbf{I}_1 = \frac{120\angle 0^\circ}{\mathbf{Z}_{\text{in}}} = \frac{120\angle 0^\circ}{10.82\angle -33.69^\circ} = 11.09\angle 33.69^\circ\ \text{A}$$

(b) Since both $\mathbf{I}_1$ and $\mathbf{I}_2$ leave the dotted terminals,

$$\mathbf{I}_2 = -\frac{1}{n}\mathbf{I}_1 = -5.545\angle 33.69^\circ\ \text{A}$$

$$\mathbf{V}_o = 20\mathbf{I}_2 = 110.9\angle 213.69^\circ\ \text{V}$$

(c) The complex power supplied is

$$\mathbf{S} = \mathbf{V}_s\mathbf{I}_1^* = (120\angle 0^\circ)(11.09\angle -33.69^\circ) = 1{,}330.8\angle -33.69^\circ\ \text{VA}$$

Practice Problem 13.8

In the ideal transformer circuit of Fig. 13.38, find $\mathbf{V}_o$ and the complex power supplied by the source.

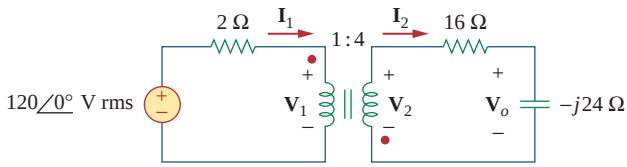


Figure 13.38
For Practice Prob. 13.8.

Answer: $214.7\angle 116.56^\circ$ V, $4.293\angle -26.56^\circ$ kVA.

Example 13.9

Calculate the power supplied to the 10-Ω resistor in the ideal transformer circuit of Fig. 13.39.

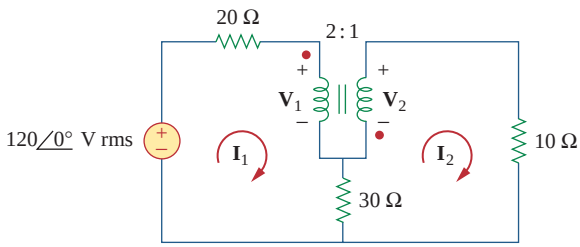


Figure 13.39
For Example 13.9.

Solution:
Reflection to the secondary or primary side cannot be done with this circuit: there is direct connection between the primary and

secondary sides due to the 30-Ω resistor. We apply mesh analysis. For mesh 1,

$$-120 + (20 + 30)\mathbf{I}_1 - 30\mathbf{I}_2 + \mathbf{V}_1 = 0$$

or

$$50\mathbf{I}_1 - 30\mathbf{I}_2 + \mathbf{V}_1 = 120 \tag{13.9.1}$$

For mesh 2,

$$-\mathbf{V}_2 + (10 + 30)\mathbf{I}_2 - 30\mathbf{I}_1 = 0$$

or

$$-30\mathbf{I}_1 + 40\mathbf{I}_2 - \mathbf{V}_2 = 0 \tag{13.9.2}$$

At the transformer terminals,

$$\mathbf{V}_2 = -\frac{1}{2}\mathbf{V}_1 \tag{13.9.3}$$

$$\mathbf{I}_2 = -2\mathbf{I}_1 \tag{13.9.4}$$

(Note that $n = 1/2$.) We now have four equations and four unknowns, but our goal is to get $\mathbf{I}_2$. So we substitute for $\mathbf{V}_1$ and $\mathbf{I}_1$ in terms of $\mathbf{V}_2$ and $\mathbf{I}_2$ in Eqs. (13.9.1) and (13.9.2). Equation (13.9.1) becomes

$$-55\mathbf{I}_2 - 2\mathbf{V}_2 = 120 \tag{13.9.5}$$

and Eq. (13.9.2) becomes

$$15\mathbf{I}_2 + 40\mathbf{I}_2 - \mathbf{V}_2 = 0 \quad \Rightarrow \quad \mathbf{V}_2 = 55\mathbf{I}_2 \tag{13.9.6}$$

Substituting Eq. (13.9.6) in Eq. (13.9.5),

$$-165\mathbf{I}_2 = 120 \quad \Rightarrow \quad \mathbf{I}_2 = -\frac{120}{165} = -0.7272 \text{ A}$$

The power absorbed by the 10-Ω resistor is

$$P = (-0.7272)^2(10) = 5.3 \text{ W}$$

Practice Problem 13.9

Find $\mathbf{V}_o$ in the circuit of Fig. 13.40.

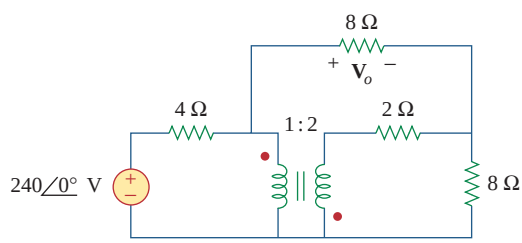


Figure 13.40
For Practice Prob. 13.9.

Answer: 96 V.

13.6 Ideal Autotransformers

Unlike the conventional two-winding transformer we have considered so far, an *autotransformer* has a single continuous winding with a connection point called a *tap* between the primary and secondary sides. The tap is often adjustable so as to provide the desired turns ratio for stepping up or stepping down the voltage. This way, a variable voltage is provided to the load connected to the autotransformer.

An **autotransformer** is a transformer in which both the primary and the secondary are in a single winding.



Figure 13.41
A typical autotransformer.
Courtesy of Todd Systems, Inc.

Figure 13.41 shows a typical autotransformer. As shown in Fig. 13.42, the autotransformer can operate in the step-down or step-up mode. The autotransformer is a type of power transformer. Its major advantage over the two-winding transformer is its ability to transfer larger apparent power. Example 13.10 will demonstrate this. Another advantage is that an autotransformer is smaller and lighter than an equivalent two-winding transformer. However, since both the primary and secondary windings are one winding, *electrical isolation* (no direct electrical connection) is lost. (We will see how the property of electrical isolation in the conventional transformer is practically employed in Section 13.9.1.) The lack of electrical isolation between the primary and secondary windings is a major disadvantage of the autotransformer.

Some of the formulas we derived for ideal transformers apply to ideal autotransformers as well. For the step-down autotransformer circuit of Fig. 13.42(a), Eq. (13.52) gives

$$\frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{N_1 + N_2}{N_2} = 1 + \frac{N_1}{N_2} \tag{13.63}$$

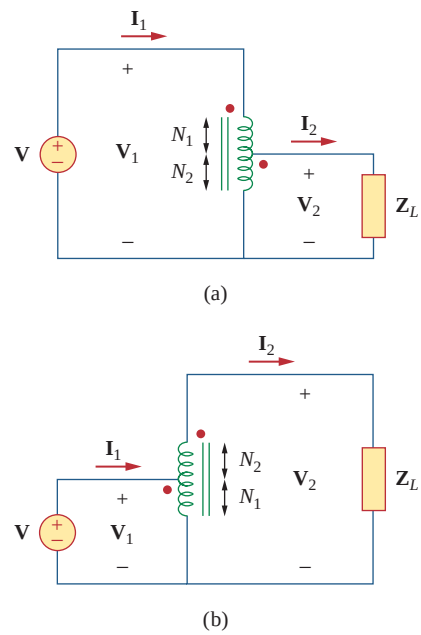


Figure 13.42
(a) Step-down autotransformer, (b) step-up autotransformer.

As an ideal autotransformer, there are no losses, so the complex power remains the same in the primary and secondary windings:

$$\mathbf{S}_1 = \mathbf{V}_1\mathbf{I}_1^* = \mathbf{S}_2 = \mathbf{V}_2\mathbf{I}_2^* \tag{13.64}$$

Equation (13.64) can also be expressed as

$$V_1 I_1 = V_2 I_2$$

or

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} \tag{13.65}$$

Thus, the current relationship is

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{N_2}{N_1 + N_2} \tag{13.66}$$

For the step-up autotransformer circuit of Fig. 13.42(b),

$$\frac{\mathbf{V}_1}{N_1} = \frac{\mathbf{V}_2}{N_1 + N_2}$$

or

$$\frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{N_1}{N_1 + N_2} \tag{13.67}$$

The complex power given by Eq. (13.64) also applies to the step-up autotransformer so that Eq. (13.65) again applies. Hence, the current relationship is

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{N_1 + N_2}{N_1} = 1 + \frac{N_2}{N_1} \tag{13.68}$$

A major difference between conventional transformers and autotransformers is that the primary and secondary sides of the autotransformer are not only coupled magnetically but also coupled conductively. The autotransformer can be used in place of a conventional transformer when electrical isolation is not required.

Example 13.10

Compare the power ratings of the two-winding transformer in Fig. 13.43(a) and the autotransformer in Fig. 13.43(b).

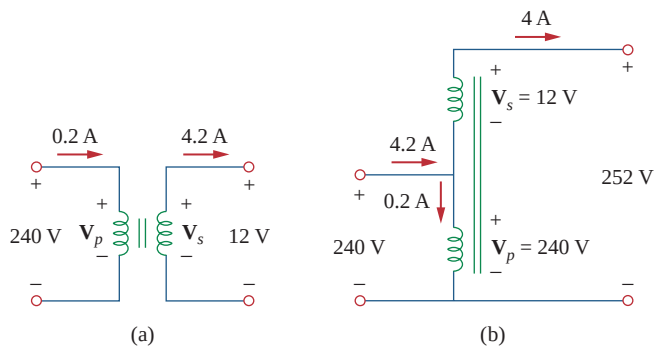


Figure 13.43
For Example 13.10.

Solution:
Although the primary and secondary windings of the autotransformer are together as a continuous winding, they are separated in Fig. 13.43(b) for clarity. We note that the current and voltage of each winding of the autotransformer in Fig. 13.43(b) are the same as those for the two-winding transformer in Fig. 13.43(a). This is the basis of comparing their power ratings.

For the two-winding transformer, the power rating is

$$S_1 = 0.2(240) = 48 \text{ VA} \quad \text{or} \quad S_2 = 4(12) = 48 \text{ VA}$$

For the autotransformer, the power rating is

$$S_1 = 4.2(240) = 1{,}008 \text{ VA} \quad \text{or} \quad S_2 = 4(252) = 1{,}008 \text{ VA}$$

which is 21 times the power rating of the two-winding transformer.

Practice Problem 13.10

Refer to Fig. 13.43. If the two-winding transformer is a 30-VA, 120 V/10 V transformer, what is the power rating of the autotransformer?

Answer: 390 VA.

Example 13.11

Refer to the autotransformer circuit in Fig. 13.44. Calculate: (a) $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_o$ if $\mathbf{Z}_L = 8 + j6\ \Omega$, and (b) the complex power supplied to the load.

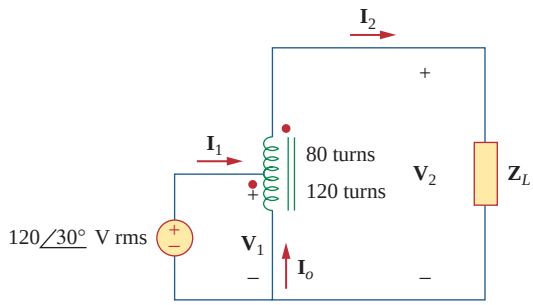


Figure 13.44
For Example 13.11.

Solution:

(a) This is a step-up autotransformer with $N_1 = 80$, $N_2 = 120$, $\mathbf{V}_1 = 120\angle 30°$, so Eq. (13.67) can be used to find $\mathbf{V}_2$ by

$$\frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{N_1}{N_1 + N_2} = \frac{80}{200}$$

or

$$\mathbf{V}_2 = \frac{200}{80}\mathbf{V}_1 = \frac{200}{80}(120\angle 30°) = 300\angle 30°\ \text{V}$$

$$\mathbf{I}_2 = \frac{\mathbf{V}_2}{\mathbf{Z}_L} = \frac{300\angle 30°}{8 + j6} = \frac{300\angle 30°}{10\angle 36.87°} = 30\angle -6.87°\ \text{A}$$

But

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{N_1 + N_2}{N_1} = \frac{200}{80}$$

or

$$\mathbf{I}_1 = \frac{200}{80}\mathbf{I}_2 = \frac{200}{80}(30\angle -6.87°) = 75\angle -6.87°\ \text{A}$$

At the tap, KCL gives

$$\mathbf{I}_1 + \mathbf{I}_o = \mathbf{I}_2$$

or

$$\mathbf{I}_o = \mathbf{I}_2 - \mathbf{I}_1 = 30\angle -6.87° - 75\angle -6.87° = 45\angle 173.13°\ \text{A}$$

(b) The complex power supplied to the load is

$$\mathbf{S}_2 = \mathbf{V}_2\mathbf{I}_2^* = |\mathbf{I}_2|^2\mathbf{Z}_L = (30)^2(10\angle 36.87°) = 9\angle 36.87°\ \text{kVA}$$

Practice Problem 13.11

In the autotransformer circuit of Fig. 13.45, find currents $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_o$. Take $\mathbf{V}_1 = 1{,}250$ V, $\mathbf{V}_2 = 500$ V.

Answer: 12.8 A, 32 A, 19.2 A.

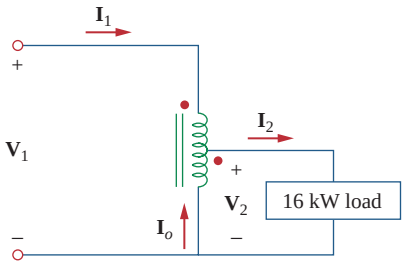


Figure 13.45
For Practice Prob. 13.11.

13.7 †Three-Phase Transformers

To meet the demand for three-phase power transmission, transformer connections compatible with three-phase operations are needed. We can achieve the transformer connections in two ways: by connecting three single-phase transformers, thereby forming a so-called *transformer bank*, or by using a special three-phase transformer. For the same kVA rating, a three-phase transformer is always smaller and cheaper than three single-phase transformers. When single-phase transformers are used, one must ensure that they have the same turns ratio n to achieve a balanced three-phase system. There are four standard ways of connecting three single-phase transformers or a three-phase transformer for three-phase operations: Y-Y, Δ-Δ, Y-Δ, and Δ-Y.

For any of the four connections, the total apparent power S_T, real power P_T, and reactive power Q_T are obtained as

$$S_T = \sqrt{3} V_L I_L \tag{13.69a}$$

$$P_T = S_T \cos\theta = \sqrt{3} V_L I_L \cos\theta \tag{13.69b}$$

$$Q_T = S_T \sin\theta = \sqrt{3} V_L I_L \sin\theta \tag{13.69c}$$

where V_L and I_L are, respectively, equal to the line voltage V_{Lp} and the line current I_{Lp} for the primary side, or the line voltage V_{Ls} and the line current I_{Ls} for the secondary side. Notice From Eq. (13.69) that for each of the four connections, $V_{Ls} I_{Ls} = V_{Lp} I_{Lp}$, since power must be conserved in an ideal transformer.

For the Y-Y connection (Fig. 13.46), the line voltage V_{Lp} at the primary side, the line voltage V_{Ls} on the secondary side, the line current I_{Lp} on the primary side, and the line current I_{Ls} on the secondary side are related to the transformer per phase turns ratio n according to Eqs. (13.52) and (13.55) as

$$V_{Ls} = n V_{Lp} \tag{13.70a}$$

$$I_{Ls} = \frac{I_{Lp}}{n} \tag{13.70b}$$

For the Δ-Δ connection (Fig. 13.47), Eq. (13.70) also applies for the line voltages and line currents. This connection is unique in the

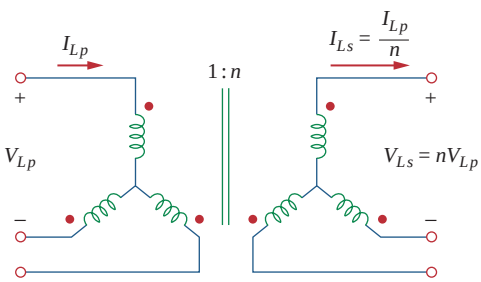


Figure 13.46
Y-Y three-phase transformer connection.

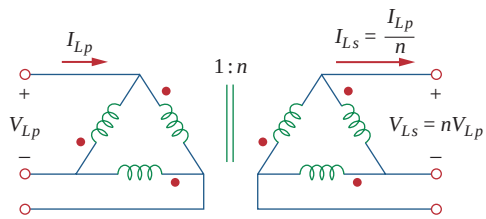


Figure 13.47
Δ-Δ three-phase transformer connection.

sense that if one of the transformers is removed for repair or maintenance, the other two form an *open delta*, which can provide three-phase voltages at a reduced level of the original three-phase transformer.

For the Y-Δ connection (Fig. 13.48), there is a factor of $\sqrt{3}$ arising from the line-phase values in addition to the transformer per phase turns ratio n. Thus,

$$V_{Ls} = \frac{nV_{Lp}}{\sqrt{3}} \tag{13.71a}$$

$$I_{Ls} = \frac{\sqrt{3}I_{Lp}}{n} \tag{13.71b}$$

Similarly, for the Δ-Y connection (Fig. 13.49),

$$V_{Ls} = n\sqrt{3}V_{Lp} \tag{13.72a}$$

$$I_{Ls} = \frac{I_{Lp}}{n\sqrt{3}} \tag{13.72b}$$

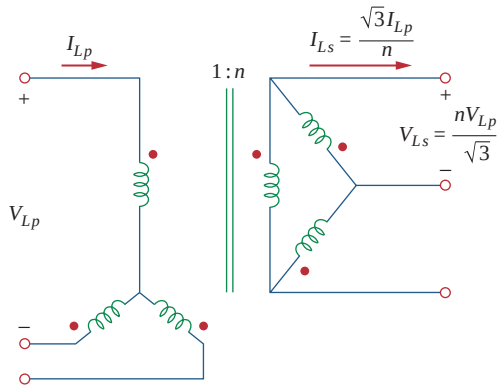


Figure 13.48
Y-Δ three-phase transformer connection.

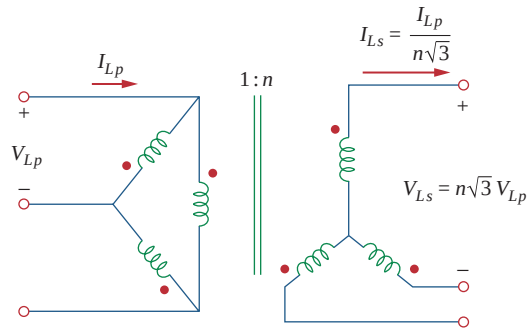


Figure 13.49
Δ-Y three-phase transformer connection.

Example 13.12

The 42-kVA balanced load depicted in Fig. 13.50 is supplied by a three-phase transformer. (a) Determine the type of transformer connections. (b) Find the line voltage and current on the primary side. (c) Determine the kVA rating of each transformer used in the transformer bank. Assume that the transformers are ideal.

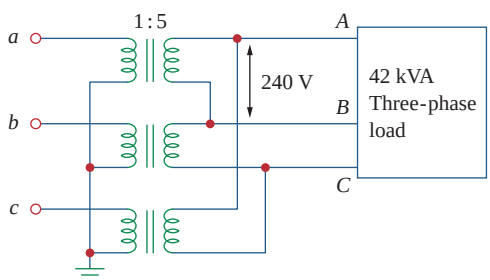


Figure 13.50
For Example 13.12.

Solution:

(a) A careful observation of Fig. 13.50 shows that the primary side is Y-connected, while the secondary side is Δ-connected. Thus, the three-phase transformer is Y-Δ, similar to the one shown in Fig. 13.48.
(b) Given a load with total apparent power $S_T = 42$ kVA, the turns ratio $n = 5$, and the secondary line voltage $V_{Ls} = 240$ V, we can find the secondary line current using Eq. (13.69a), by

$$I_{Ls} = \frac{S_T}{\sqrt{3}V_{Ls}} = \frac{42{,}000}{\sqrt{3}(240)} = 101 \text{ A}$$

From Eq. (13.71),

$$I_{Lp} = \frac{n}{\sqrt{3}}I_{Ls} = \frac{5 \times 101}{\sqrt{3}} = 292 \text{ A}$$

$$V_{Lp} = \frac{\sqrt{3}}{n}V_{Ls} = \frac{\sqrt{3} \times 240}{5} = 83.14 \text{ V}$$

(c) Because the load is balanced, each transformer equally shares the total load and since there are no losses (assuming ideal transformers), the kVA rating of each transformer is $S = S_T/3 = 14$ kVA. Alternatively, the transformer rating can be determined by the product of the phase current and phase voltage of the primary or secondary side. For the primary side, for example, we have a delta connection, so that the phase voltage is the same as the line voltage of 240 V, while the phase current is $I_{Lp}/\sqrt{3} = 58.34$ A. Hence, $S = 240 \times 58.34 = 14$ kVA.

Practice Problem 13.12

A three-phase Δ-Δ transformer is used to step down a line voltage of 625 kV, to supply a plant operating at a line voltage of 12.5 kV. The plant draws 40 MW with a lagging power factor of 85 percent. Find: (a) the current drawn by the plant, (b) the turns ratio, (c) the current on the primary side of the transformer, and (d) the load carried by each transformer.

Answer: (a) 2.1736 kA, (b) 0.02, (c) 43.47 A, (d) 15.69 MVA.

13.8 *PSpice* Analysis of Magnetically Coupled Circuits

PSpice analyzes magnetically coupled circuits just like inductor circuits except that the dot convention must be followed. In *PSpice* Schematic, the dot (not shown) is always next to pin 1, which is the left-hand terminal of the inductor when the inductor with part name L is placed (horizontally) without rotation on a schematic. Thus, the dot or pin 1 will be at the bottom after one 90° counterclockwise rotation, since rotation is always about pin 1. Once the magnetically coupled inductors are arranged with the dot convention in mind and their value

attributes are set in henries, we use the coupling symbol K_LINEAR to define the coupling. For each pair of coupled inductors, take the following steps:

1. Select **Draw/Get New Part** and type K_LINEAR.
2. Hit <enter> or click **OK** and place the K_LINEAR symbol on the schematic, as shown in Fig. 13.51. (Notice that K_LINEAR is not a component and therefore has no pins.)
3. **DCLICKL** on COUPLING and set the value of the coupling coefficient *k*.
4. **DCLICKL** on the boxed K (the coupling symbol) and enter the reference designator names for the coupled inductors as values of Li, i = 1, 2, . . . , 6. For example, if inductors L20 and L23 are coupled, we set L1 = L20 and L2 = L23. L1 and at least one other Li must be assigned values; other Li's may be left blank.

In step 4, up to six coupled inductors with equal coupling can be specified.

For the air-core transformer, the partname is XFRM_LINEAR. It can be inserted in a circuit by selecting **Draw/Get Part Name** and then typing in the part name or by selecting the part name from the analog.slb library. As shown typically in Fig. 13.52(a), the main attributes of the linear transformer are the coupling coefficient *k* and the inductance values L1 and L2 in henries. If the mutual inductance *M* is specified, its value must be used along with L1 and L2 to calculate *k*. Keep in mind that the value of *k* should lie between 0 and 1.

For the ideal transformer, the part name is XFRM_NONLINEAR and is located in the breakout.slb library. Select it by clicking **Draw/Get Part Name** and then typing in the part name. Its attributes are the coupling coefficient and the numbers of turns associated with L1 and L2, as illustrated typically in Fig. 13.52(b). The value of the coefficient of mutual coupling $k = 1$.

PSpice has some additional transformer configurations that we will not discuss here.

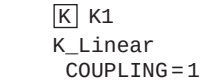
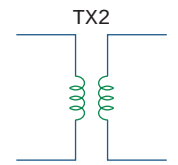
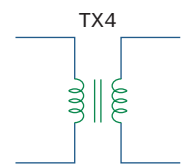


Figure 13.51
K_Linear for defining coupling.



(a)



(b)

Figure 13.52
(a) Linear transformer XFRM_LINEAR, (b) ideal transformer XFRM_NONLINEAR.

Example 13.13

Use *PSpice* to find i_1, i_2, and i_3 in the circuit displayed in Fig. 13.53.

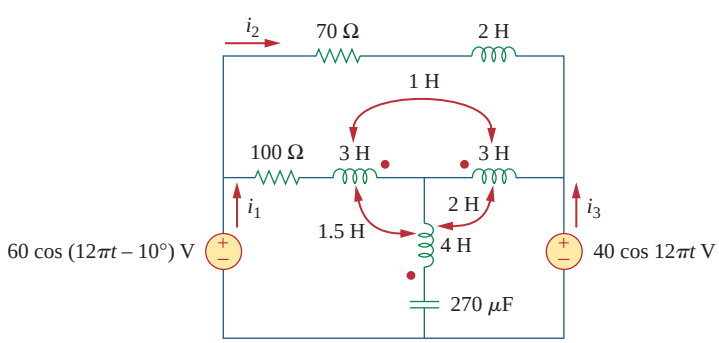


Figure 13.53
For Example 13.13.

Solution:

The coupling coefficients of the three coupled inductors are determined as follows:

$$k_{12} = \frac{M_{12}}{\sqrt{L_1 L_2}} = \frac{1}{\sqrt{3 \times 3}} = 0.3333$$

$$k_{13} = \frac{M_{13}}{\sqrt{L_1 L_3}} = \frac{1.5}{\sqrt{3 \times 4}} = 0.433$$

$$k_{23} = \frac{M_{23}}{\sqrt{L_2 L_3}} = \frac{2}{\sqrt{3 \times 4}} = 0.5774$$

The operating frequency f is obtained from Fig. 13.53 as $\omega = 12\pi = 2\pi f \rightarrow f = 6\text{Hz}$.

The schematic of the circuit is portrayed in Fig. 13.54. Notice how the dot convention is adhered to. For L2, the dot (not shown) is on pin 1 (the left-hand terminal) and is therefore placed without rotation. For L1, in order for the dot to be on the right-hand side of the inductor, the inductor must be rotated through 180°. For L3, the inductor must be rotated through 90° so that the dot will be at the bottom. Note that the 2-H inductor (L_4) is not coupled. To handle the three coupled inductors, we use three K_LINEAR parts provided in the analog library and set the following attributes (by double-clicking on the symbol K in the box):

The right-hand values are the reference designators of the inductors on the schematic.

```
K1  -  K_LINEAR
L1  =  L1
L2  =  L2
COUPLING  =  0.3333

K2  -  K_LINEAR
L1  =  L2
L2  =  L3
COUPLING  =  0.433

K3  -  K_LINEAR
L1  =  L1
L2  =  L3
COUPLING  =  0.5774
```

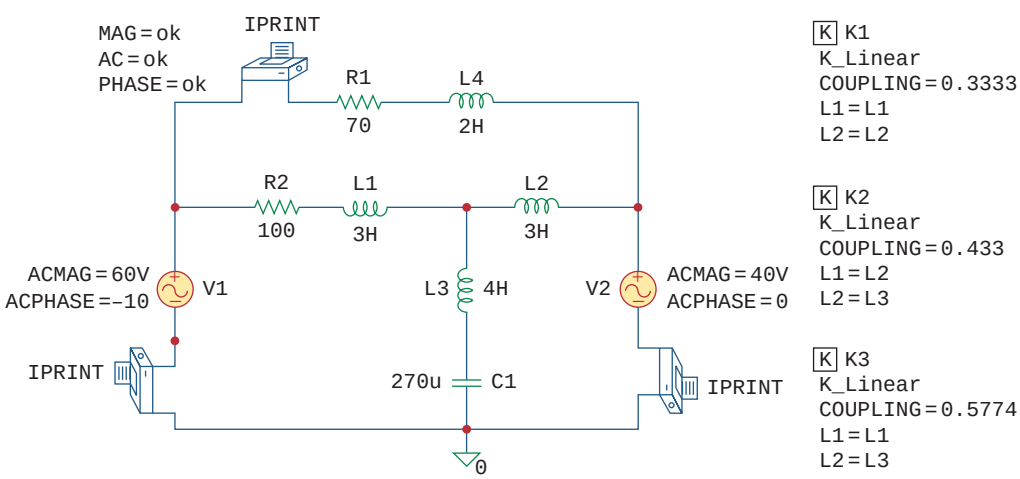


Figure 13.54
Schematic of the circuit of Fig. 13.53.

Three IPRINT pseudocomponents are inserted in the appropriate branches to obtain the required currents i_1, i_2, and i_3. As an AC single-frequency analysis, we select **Analysis/Setup/AC Sweep** and enter *Total Pts* = 1, *Start Freq* = 6, and *Final Freq* = 6. After saving the schematic, we select **Analysis/Simulate** to simulate it. The output file includes:

```
FREQ          IM(V_PRINT2)    IP(V_PRINT2)
6.000E+00     2.114E-01       -7.575E+01
FREQ          IM(V_PRINT1)    IP(V_PRINT1)
6.000E+00     4.654E-01       -7.025E+01
FREQ          IM(V_PRINT3)    IP(V_PRINT3)
6.000E+00     1.095E-01       1.715E+01
```

From this we obtain

$$\mathbf{I}_1 = 0.4654\angle -70.25°$$

$$\mathbf{I}_2 = 0.2114\angle -75.75°, \qquad \mathbf{I}_3 = 0.1095\angle 17.15°$$

Thus,

$$i_1 = 0.4654\cos(12\pi t - 70.25°)\ \text{A}$$
$$i_2 = 0.2114\cos(12\pi t - 75.75°)\ \text{A}$$
$$i_3 = 0.1095\cos(12\pi t + 17.15°)\ \text{A}$$

Practice Problem 13.13

Find i_o in the circuit of Fig. 13.55, using *PSpice*.

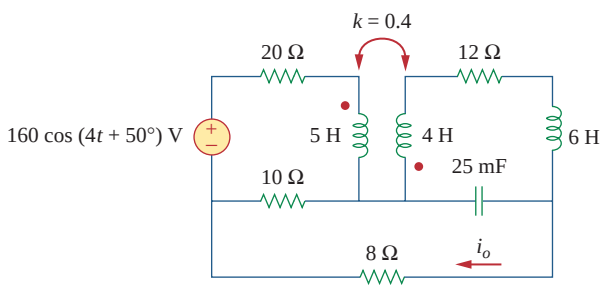


Figure 13.55
For Practice Prob. 13.13.

Answer: 2.012 cos(4*t* + 68.52°) A.

Example 13.14

Find $\mathbf{V}_1$ and $\mathbf{V}_2$ in the ideal transformer circuit of Fig. 13.56 using *PSpice*.

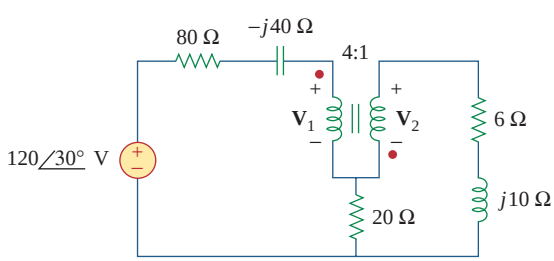


Figure 13.56
For Example 13.14.

Solution:

1. **Define.** The problem is clearly defined and we can proceed to the next step.
2. **Present.** We have an ideal transformer and we are to find the input and the output voltages for that transformer. In addition, we are to use *PSpice* to solve for the voltages.
3. **Alternative.** We are required to use *PSpice*. We can use mesh analysis to perform a check.
4. **Attempt.** As usual, we assume $\omega = 1$ and find the corresponding values of capacitance and inductance of the elements:

$$j10 = j\omega L \quad \Rightarrow \quad L = 10 \text{ H}$$

$$-j40 = \frac{1}{j\omega C} \quad \Rightarrow \quad C = 25 \text{ mF}$$

Reminder: For an ideal transformer, the inductances of both the primary and secondary windings are infinitely large.

Figure 13.57 shows the schematic. For the ideal transformer, we set the coupling factor to 0.99999 and the numbers of turns to 400,000 and 100,000. The two VPRINT2 pseudocomponents are connected across the transformer terminals to obtain $\mathbf{V}_1$ and $\mathbf{V}_2$. As a single-frequency analysis, we select **Analysis/Setup/AC Sweep** and enter *Total Pts* = 1, *Start Freq* = 0.1592, and *Final Freq* = 0.1592. After saving the schematic, we select **Analysis/Simulate** to simulate it. The output file includes:

```
FREQ         VM($N_0003,$N_0006)  VP($N_0003,$N_0006)
1.592E-01    9.112E+01            3.792E+01

FREQ         VM($N_0006,$N_0005)  VP($N_0006,$N_0005)
1.592E-01    2.278E+01            -1.421E+02
```

This can be written as

$$\mathrm{V}_1 = \mathbf{91.12}\angle\mathbf{37.92^\circ}\ \mathbf{V} \quad \text{and} \quad \mathrm{V}_2 = \mathbf{22.78}\angle\mathbf{-142.1^\circ}\ \mathbf{V}$$

5. **Evaluate.** We can check the answer by using mesh analysis as follows:

$$\text{Loop 1} \quad -120\angle 30^\circ + (80 - j40)I_1 + V_1 + 20(I_1 - I_2) = 0$$

$$\text{Loop 2} \quad 20(-I_1 + I_2) - \mathrm{V}_2 + (6 + j10)I_2 = 0$$

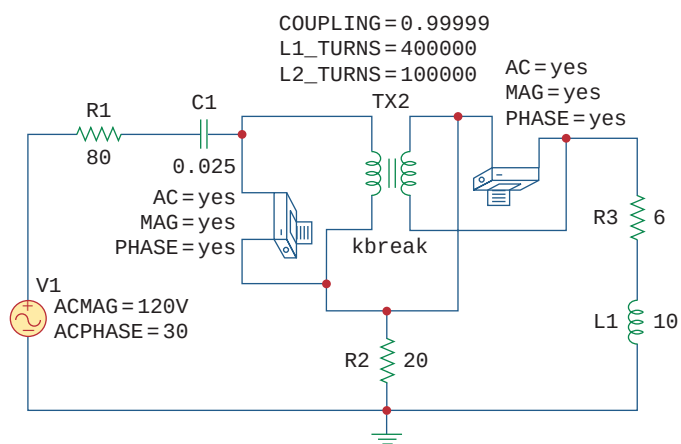


Figure 13.57
The schematic for the circuit in Fig. 13.56.

But $V_2 = -V_1/4$ and $I_2 = -4I_1$. This leads to

$$-120\angle 30^\circ + (80 - j40)I_1 + V_1 + 20(I_1 + 4I_1) = 0$$

$$(180 - j40)I_1 + V_1 = 120\angle 30^\circ$$

$$20(-I_1 - 4I_1) + V_1/4 + (6 + j10)(-4I_1) = 0$$

$$(-124 - j40)I_1 + 0.25V_1 = 0 \quad \text{or} \quad I_1 = V_1/(496 + j160)$$

Substituting this into the first equation yields

$$(180 - j40)V_1/(496 + j160) + V_1 = 120\angle 30^\circ$$

$$(184.39\angle -12.53^\circ/521.2\angle 17.88^\circ)V_1 + V_1$$

$$= (0.3538\angle -30.41^\circ + 1)V_1 = (0.3051 + 1 - j0.17909)V_1 = 120\angle 30^\circ$$

$$V_1 = 120\angle 30^\circ/1.3173\angle -7.81^\circ = \mathbf{91.1\angle 37.81^\circ\ V} \quad \text{and}$$

$$V_2 = \mathbf{22.78\angle -142.19^\circ\ V}$$

Both answers check.

6. **Satisfactory?** We have satisfactorily answered the problem and checked the solution. We can now present the entire solution to the problem.

Practice Problem 13.14

Obtain $\mathbf{V}_1$ and $\mathbf{V}_2$ in the circuit of Fig. 13.58 using *PSpice*.

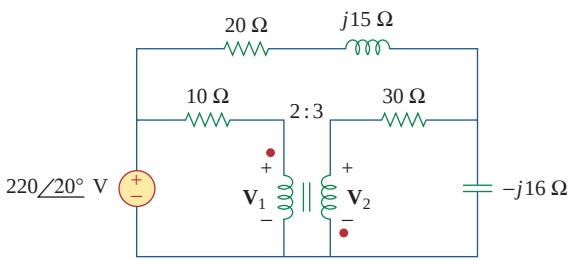


Figure 13.58
For Practice Prob. 13.14.

Answer: $138.82\angle 28.65^\circ$ V, $208.2\angle -151.4^\circ$ V.

13.9 †Applications

Transformers are the largest, the heaviest, and often the costliest of circuit components. Nevertheless, they are indispensable passive devices in electric circuits. They are among the most efficient machines, 95 percent efficiency being common and 99 percent being achievable. They have numerous applications. For example, transformers are used:

- To step up or step down voltage and current, making them useful for power transmission and distribution.
- To isolate one portion of a circuit from another (i.e., to transfer power without any electrical connection).
- As an impedance-matching device for maximum power transfer.
- In frequency-selective circuits whose operation depends on the response of inductances.

Because of these diverse uses, there are many special designs for transformers (only some of which are discussed in this chapter): voltage transformers, current transformers, power transformers, distribution transformers, impedance-matching transformers, audio transformers, single-phase transformers, three-phase transformers, rectifier transformers, inverter transformers, and more. In this section, we consider three important applications: transformer as an isolation device, transformer as a matching device, and power distribution system.

For more information on the many kinds of transformers, a good text is W. M. Flanagan, *Handbook of Transformer Design and Applications*, 2nd ed. (New York: McGraw-Hill, 1993).

13.9.1 Transformer as an Isolation Device

Electrical isolation is said to exist between two devices when there is no physical connection between them. In a transformer, energy is transferred by magnetic coupling, without electrical connection between the primary circuit and secondary circuit. We now consider three simple practical examples of how we take advantage of this property.

First, consider the circuit in Fig. 13.59. A rectifier is an electronic circuit that converts an ac supply to a dc supply. A transformer is often used to couple the ac supply to the rectifier. The transformer serves two purposes. First, it steps up or steps down the voltage. Second, it provides electrical isolation between the ac power supply and the rectifier, thereby reducing the risk of shock hazard in handling the electronic device.

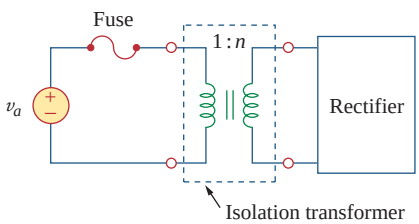


Figure 13.59
A transformer used to isolate an ac supply from a rectifier.

As a second example, a transformer is often used to couple two stages of an amplifier, to prevent any dc voltage in one stage from affecting the dc bias of the next stage. Biasing is the application of a dc voltage to a transistor amplifier or any other electronic device in order to produce a desired mode of operation. Each amplifier stage is biased separately to operate in a particular mode; the desired mode of operation will be compromised without a transformer providing dc isolation. As shown in Fig. 13.60, only the ac signal is coupled through the transformer from one stage to the next. We recall that magnetic coupling does not exist with a dc voltage source. Transformers are used in radio and TV receivers to couple stages of high-frequency amplifiers. When the sole purpose of a transformer is to provide isolation, its turns ratio n is made unity. Thus, an isolation transformer has $n = 1$.

As a third example, consider measuring the voltage across 13.2-kV lines. It is obviously not safe to connect a voltmeter directly to such high-voltage lines. A transformer can be used both to electrically isolate the line power from the voltmeter and to step down the voltage to a safe level, as shown in Fig. 13.61. Once the voltmeter is used to

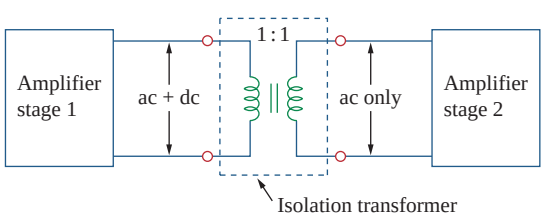


Figure 13.60
A transformer providing dc isolation between two amplifier stages.

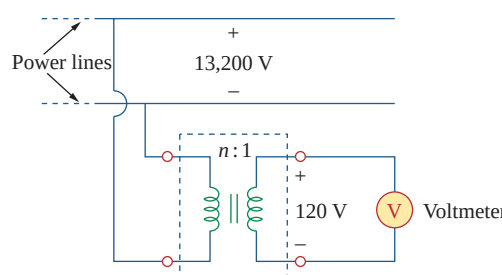


Figure 13.61
A transformer providing isolation between the power lines and the voltmeter.

measure the secondary voltage, the turns ratio is used to determine the line voltage on the primary side.

Example 13.15

Determine the voltage across the load in Fig. 13.62.

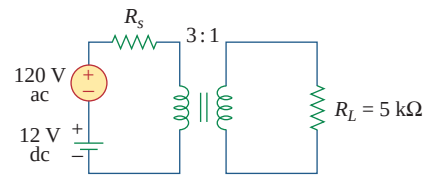


Figure 13.62
For Example 13.15.

Solution:
We can apply the superposition principle to find the load voltage. Let $v_L = v_{L1} + v_{L2}$, where v_{L1} is due to the dc source and v_{L2} is due to the ac source. We consider the dc and ac sources separately, as shown in Fig. 13.63. The load voltage due to the dc source is zero, because a time-varying voltage is necessary in the primary circuit to induce a voltage in the secondary circuit. Thus, $v_{L1} = 0$. For the ac source and a value of R_s so small it can be neglected,

$$\frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{\mathbf{V}_2}{120} = \frac{1}{3} \qquad \text{or} \qquad \mathbf{V}_2 = \frac{120}{3} = 40 \text{ V}$$

Hence, $\mathbf{V}_{L2} = 40$ V ac or $v_{L2} = 40 \cos \omega t$; that is, only the ac voltage is passed to the load by the transformer. This example shows how the transformer provides dc isolation.

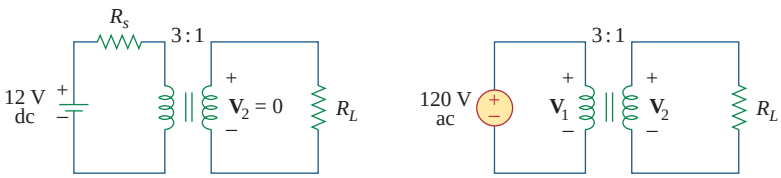


Figure 13.63
For Example 13.15: (a) dc source, (b) ac source.

Practice Problem 13.15

Refer to Fig. 13.61. Calculate the turns ratio required to step down the 13.2-kV line voltage to a safe level of 120 V.

Answer: 110.

13.9.2 Transformer as a Matching Device

We recall that for maximum power transfer, the load resistance R_L must be matched with the source resistance R_s. In most cases, the two resistances are not matched; both are fixed and cannot be altered. However, an iron-core transformer can be used to match the load resistance to the source resistance. This is called *impedance matching*. For example, to connect a loudspeaker to an audio power amplifier requires a transformer, because the speaker's resistance is only a few ohms while the internal resistance of the amplifier is several thousand ohms.

Consider the circuit shown in Fig. 13.64. We recall from Eq. (13.60) that the ideal transformer reflects its load back to the primary with a

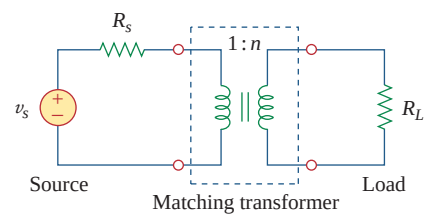


Figure 13.64
Transformer used as a matching device.

scaling factor of n^2. To match this reflected load R_L/n^2 with the source resistance R_s, we set them equal,

$$R_s = \frac{R_L}{n^2} \tag{13.73}$$

Equation (13.73) can be satisfied by proper selection of the turns ratio n. From Eq. (13.73), we notice that a step-down transformer ($n < 1$) is needed as the matching device when $R_s > R_L$, and a step-up ($n > 1$) is required when $R_s < R_L$.

Example 13.16

The ideal transformer in Fig. 13.65 is used to match the amplifier circuit to the loudspeaker to achieve maximum power transfer. The Thevenin (or output) impedance of the amplifier is 192 Ω, and the internal impedance of the speaker is 12 Ω. Determine the required turns ratio.

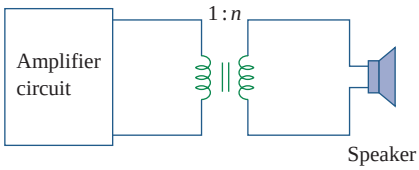


Figure 13.65
Using an ideal transformer to match the speaker to the amplifier; for Example 13.16.

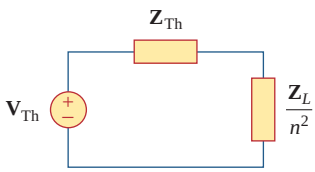


Figure 13.66
Equivalent circuit of the circuit in Fig. 13.65; for Example 13.16.

Solution:

We replace the amplifier circuit with the Thevenin equivalent and reflect the impedance $\mathbf{Z}_L = 12\ \Omega$ of the speaker to the primary side of the ideal transformer. Figure 13.66 shows the result. For maximum power transfer,

$$\mathbf{Z}_{\text{Th}} = \frac{\mathbf{Z}_L}{n^2} \qquad \text{or} \qquad n^2 = \frac{\mathbf{Z}_L}{\mathbf{Z}_{\text{Th}}} = \frac{12}{192} = \frac{1}{16}$$

Thus, the turns ratio is $n = 1/4 = 0.25$.

Using $P = I^2R$, we can show that indeed the power delivered to the speaker is much larger than without the ideal transformer. Without the ideal transformer, the amplifier is directly connected to the speaker. The power delivered to the speaker is

$$P_L = \left(\frac{\mathbf{V}_{\text{Th}}}{\mathbf{Z}_{\text{Th}} + \mathbf{Z}_L}\right)^2 \mathbf{Z}_L = 288\ \mathbf{V}_{\text{Th}}^2\ \mu\text{W}$$

With the transformer in place, the primary and secondary currents are

$$I_p = \frac{\mathbf{V}_{\text{Th}}}{\mathbf{Z}_{\text{Th}} + \mathbf{Z}_L/n^2}, \qquad I_s = \frac{I_p}{n}$$

Hence,

$$P_L = I_s^2 \mathbf{Z}_L = \left(\frac{\mathbf{V}_{\text{Th}}/n}{\mathbf{Z}_{\text{Th}} + \mathbf{Z}_L/n^2}\right)^2 \mathbf{Z}_L$$

$$= \left(\frac{n\mathbf{V}_{\text{Th}}}{n^2\mathbf{Z}_{\text{Th}} + \mathbf{Z}_L}\right)^2 \mathbf{Z}_L = 1{,}302\ \mathbf{V}_{\text{Th}}^2\ \mu\text{W}$$

confirming what was said earlier.

Practice Problem 13.16

Calculate the turns ratio of an ideal transformer required to match a 400-Ω load to a source with internal impedance of 2.5 kΩ. Find the load voltage when the source voltage is 30 V.

Answer: 0.4, 6 V.

13.9.3 Power Distribution

A power system basically consists of three components: generation, transmission, and distribution. The local electric company operates a plant that generates several hundreds of megavolt-amperes (MVA), typically at about 18 kV. As Fig. 13.67 illustrates, three-phase step-up transformers are used to feed the generated power to the transmission line. Why do we need the transformer? Suppose we need to transmit 100,000 VA over a distance of 50 km. Since $S = VI$, using a line voltage of 1,000 V implies that the transmission line must carry 100 A and this requires a transmission line of a large diameter. If, on the other hand, we use a line voltage of 10,000 V, the current is only 10 A. The smaller current reduces the required conductor size, producing considerable savings as well as minimizing transmission line I^2R losses. To minimize losses requires a step-up transformer. Without the transformer, the majority of the power generated would be lost on the transmission line. The ability of the transformer to step up or step down voltage and distribute power economically is one of the major reasons for generating ac rather than dc. Thus, for a given power, the larger the voltage, the better. Today, 1 MV is the largest voltage in use; the level may increase as a result of research and experiments.

One may ask, How would increasing the voltage not increase the current, thereby increasing I^2R losses? Keep in mind that $I = V_\ell/R$, where V_ℓ is the potential difference between the sending and receiving ends of the line. The voltage that is stepped up is the sending end voltage V, not V_ℓ. If the receiving end is V_R, then $V_\ell = V - V_R$. Since V and V_R are close to each other, V_ℓ is small even when V is stepped up.

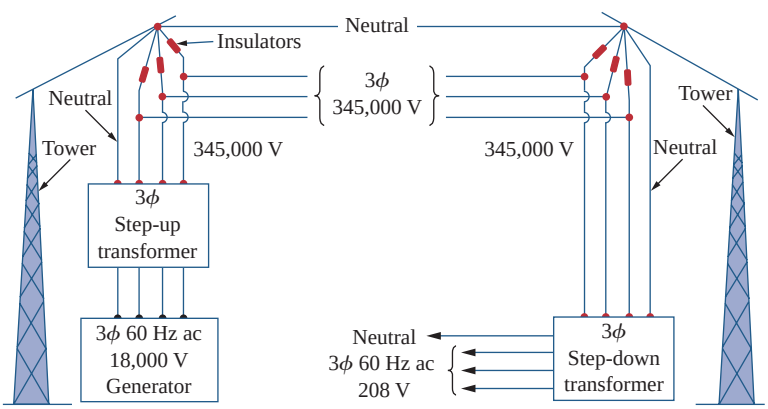


Figure 13.67
A typical power distribution system.
A. Marcus and C. M. Thomson, *Electricity for Technicians, 2nd ed.* [Englewood Cliffs, NJ: Prentice Hall, 1975], p. 337.

Beyond the generation plant, the power is transmitted for hundreds of miles through an electric network called the *power grid.* The three-phase power in the power grid is conveyed by transmission lines hung overhead from steel towers which come in a variety of sizes and shapes. The (aluminum-conductor, steel-reinforced) lines typically have overall diameters up to about 40 mm and can carry current of up to 1,380 A.

At the substations, distribution transformers are used to step down the voltage. The step-down process is usually carried out in stages. Power may be distributed throughout a locality by means of either overhead or underground cables. The substations distribute the power to residential, commercial, and industrial customers. At the receiving end, a residential customer is eventually supplied with 120/240 V, while industrial or commercial customers are fed with higher voltages such

as 460/208 V. Residential customers are usually supplied by distribution transformers often mounted on the poles of the electric utility company. When direct current is needed, the alternating current is converted to dc electronically.

Example 13.17

A distribution transformer is used to supply a household as in Fig. 13.68. The load consists of eight 100-W bulbs, a 350-W TV, and a 15-kW kitchen range. If the secondary side of the transformer has 72 turns, calculate: (a) the number of turns of the primary winding, and (b) the current I_p in the primary winding.

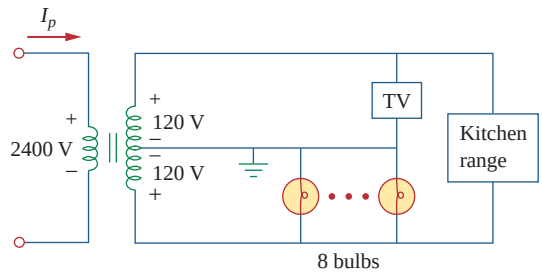


Figure 13.68
For Example 13.17.

Solution:

(a) The dot locations on the winding are not important, since we are only interested in the magnitudes of the variables involved. Since

$$\frac{N_p}{N_s} = \frac{V_p}{V_s}$$

we get

$$N_p = N_s \frac{V_p}{V_s} = 72\frac{2{,}400}{240} = 720 \text{ turns}$$

(b) The total power absorbed by the load is

$$S = 8 \times 100 + 350 + 15{,}000 = 16.15 \text{ kW}$$

But $S = V_p I_p = V_s I_s$, so that

$$I_p = \frac{S}{V_p} = \frac{16{,}150}{2{,}400} = 6.729 \text{ A}$$

Practice Problem 13.17

In Example 13.17, if the eight 100-W bulbs are replaced by twelve 60-W bulbs and the kitchen range is replaced by a 4.5-kW air-conditioner, find: (a) the total power supplied, (b) the current I_p in the primary winding.

Answer: (a) 5.57 kW, (b) 2.321 A.

13.10 Summary

1. Two coils are said to be mutually coupled if the magnetic flux ϕ emanating from one passes through the other. The mutual inductance between the two coils is given by

$$M = k\sqrt{L_1 L_2}$$

where k is the coupling coefficient, $0 < k < 1$.

2. If v_1 and i_1 are the voltage and current in coil 1, while v_2 and i_2 are the voltage and current in coil 2, then

$$v_1 = L_1\frac{di_1}{dt} + M\frac{di_2}{dt} \quad \text{and} \quad v_2 = L_2\frac{di_2}{dt} + M\frac{di_1}{dt}$$

Thus, the voltage induced in a coupled coil consists of self-induced voltage and mutual voltage.

3. The polarity of the mutually-induced voltage is expressed in the schematic by the dot convention.
4. The energy stored in two coupled coils is

$$\frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2$$

5. A transformer is a four-terminal device containing two or more magnetically coupled coils. It is used in changing the current, voltage, or impedance level in a circuit.
6. A linear (or loosely coupled) transformer has its coils wound on a magnetically linear material. It can be replaced by an equivalent T or Π network for the purposes of analysis.
7. An ideal (or iron-core) transformer is a lossless ($R_1 = R_2 = 0$) transformer with unity coupling coefficient ($k = 1$) and infinite inductances ($L_1, L_2, M \to \infty$).
8. For an ideal transformer,

$$\mathbf{V}_2 = n\mathbf{V}_1, \quad \mathbf{I}_2 = \frac{\mathbf{I}_1}{n}, \quad \mathbf{S}_1 = \mathbf{S}_2, \quad \mathbf{Z}_R = \frac{\mathbf{Z}_L}{n^2}$$

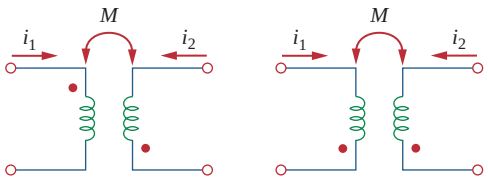
where $n = N_2/N_1$ is the turns ratio. N_1 is the number of turns of the primary winding and N_2 is the number of turns of the secondary winding. The transformer steps up the primary voltage when $n > 1$, steps it down when $n < 1$, or serves as a matching device when $n = 1$.

9. An autotransformer is a transformer with a single winding common to both the primary and the secondary circuits.
10. *PSpice* is a useful tool for analyzing magnetically coupled circuits.
11. Transformers are necessary in all stages of power distribution systems. Three-phase voltages may be stepped up or down by three-phase transformers.
12. Important uses of transformers in electronics applications are as electrical isolation devices and impedance-matching devices.

Review Questions

13.1 Refer to the two magnetically coupled coils of Fig. 13.69(a). The polarity of the mutual voltage is:

(a) Positive (b) Negative



(a) (b)

Figure 13.69
For Review Questions 13.1 and 13.2.

13.2 For the two magnetically coupled coils of Fig. 13.69(b), the polarity of the mutual voltage is:

(a) Positive (b) Negative

13.3 The coefficient of coupling for two coils having $L_1 = 2$ H, $L_2 = 8$ H, $M = 3$ H is:

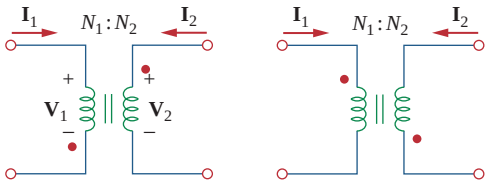
(a) 0.1875 (b) 0.75
(c) 1.333 (d) 5.333

13.4 A transformer is used in stepping down or stepping up:

(a) dc voltages (b) ac voltages
(c) both dc and ac voltages

13.5 The ideal transformer in Fig. 13.70(a) has $N_2/N_1 = 10$. The ratio V_2/V_1 is:

(a) 10 (b) 0.1 (c) −0.1 (d) −10



(a) (b)

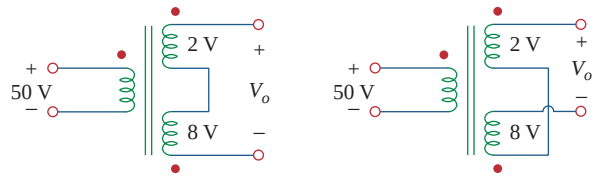
Figure 13.70
For Review Questions 13.5 and 13.6.

13.6 For the ideal transformer in Fig. 13.70(b), $N_2/N_1 = 10$. The ratio I_2/I_1 is:

(a) 10 (b) 0.1 (c) −0.1 (d) −10

13.7 A three-winding transformer is connected as portrayed in Fig. 13.71(a). The value of the output voltage V_o is:

(a) 10 (b) 6 (c) −6 (d) −10



(a) (b)

Figure 13.71
For Review Questions 13.7 and 13.8.

13.8 If the three-winding transformer is connected as in Fig. 13.71(b), the value of the output voltage V_o is:

(a) 10 (b) 6 (c) −6 (d) −10

13.9 In order to match a source with internal impedance of 500 Ω to a 15-Ω load, what is needed is:

(a) step-up linear transformer
(b) step-down linear transformer
(c) step-up ideal transformer
(d) step-down ideal transformer
(e) autotransformer

13.10 Which of these transformers can be used as an isolation device?

(a) linear transformer (b) ideal transformer
(c) autotransformer (d) all of the above

Answers: 13.1b, 13.2a, 13.3b, 13.4b, 13.5d, 13.6b, 13.7c, 13.8a, 13.9d, 13.10b.

Problems[1]

Section 13.2 Mutual Inductance

13.1 For the three coupled coils in Fig. 13.72, calculate the total inductance.

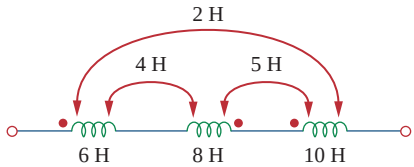


Figure 13.72
For Prob. 13.1.

13.2 Using Fig. 13.73, design a problem to help other students better understand mutual inductance.

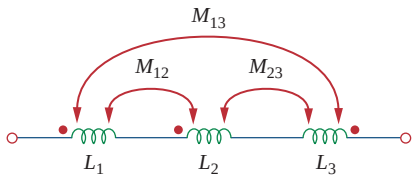


Figure 13.73
For Prob. 13.2.

13.3 Two coils connected in series-aiding fashion have a total inductance of 250 mH. When connected in a series-opposing configuration, the coils have a total inductance of 150 mH. If the inductance of one coil (L_1) is three times the other, find L_1, L_2, and M. What is the coupling coefficient?

13.4 (a) For the coupled coils in Fig. 13.74(a), show that

$$L_{eq} = L_1 + L_2 + 2M$$

(b) For the coupled coils in Fig. 13.74(b), show that

$$L_{eq} = \frac{L_1L_2 - M^2}{L_1 + L_2 - 2M}$$

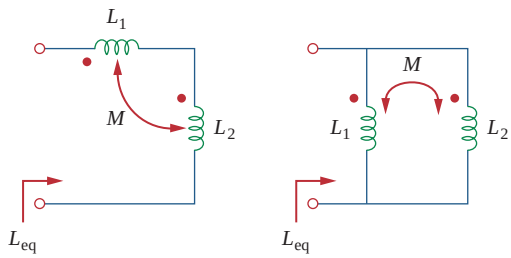


Figure 13.74
For Prob. 13.4.

13.5 Two coils are mutually coupled, with $L_1 = 25$ mH, $L_2 = 60$ mH, and $k = 0.5$. Calculate the maximum possible equivalent inductance if:

(a) the two coils are connected in series

(b) the coils are connected in parallel

13.6 The coils in Fig. 13.75 have $L_1 = 40$ mH, $L_2 = 5$ mH, and coupling coefficient $k = 0.6$. Find $i_1(t)$ and $v_2(t)$, given that $v_1(t) = 10\cos\omega t$ and $i_2(t) = 2\sin\omega t$, $\omega = 2000$ rad/s.

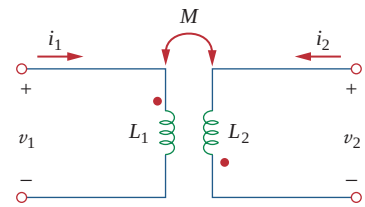


Figure 13.75
For Prob. 13.6.

13.7 For the circuit in Fig. 13.76, find V_o.

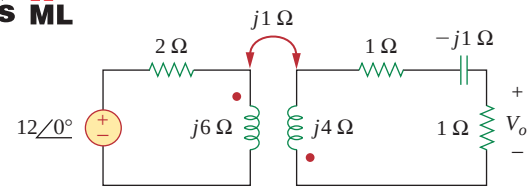


Figure 13.76
For Prob. 13.7.

13.8 Find $v(t)$ for the circuit in Fig. 13.77.

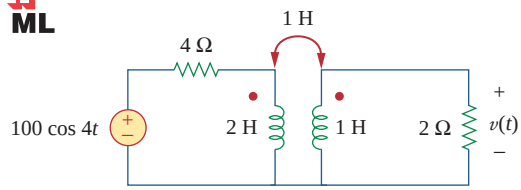


Figure 13.77
For Prob. 13.8.

13.9 Find $\mathbf{V}_x$ in the network shown in Fig. 13.78.

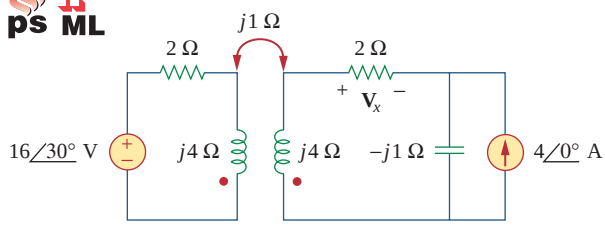


Figure 13.78
For Prob. 13.9.

[1] Remember, unless otherwise specified, assume all values of currents and voltages are rms.

13.10 Find v_o in the circuit of Fig. 13.79.

PS ML

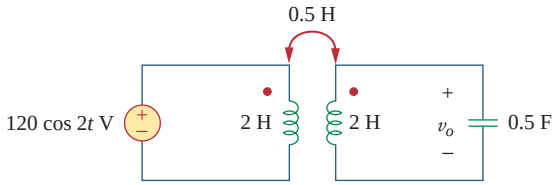


Figure 13.79
For Prob. 13.10.

13.11 Use mesh analysis to find i_x in Fig. 13.80, where

ML

$$i_s = 6\cos(600t)\text{ A} \quad \text{and} \quad v_s = 165\cos(600t + 30°)$$

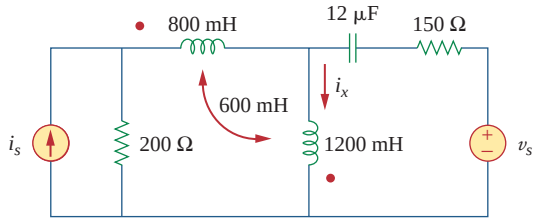


Figure 13.80
For Prob. 13.11.

13.12 Determine the equivalent L_{eq} in the circuit of Fig. 13.81.

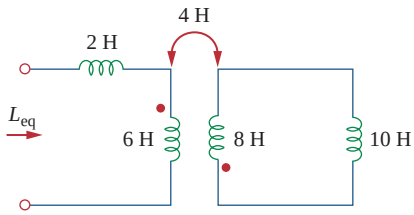


Figure 13.81
For Prob. 13.12.

13.13 For the circuit in Fig. 13.82, determine the impedance seen by the source.

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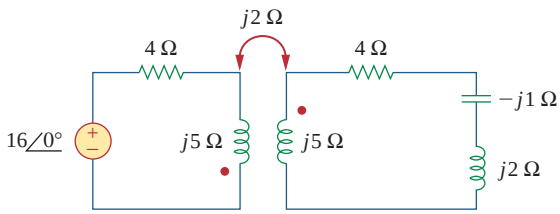


Figure 13.82
For Prob. 13.13.

13.14 Obtain the Thevenin equivalent circuit for the circuit in Fig. 13.83 at terminals *a-b*.

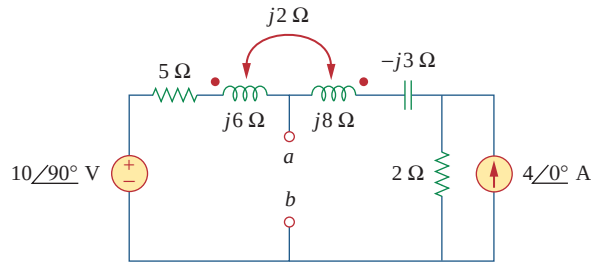


Figure 13.83
For Prob. 13.14.

13.15 Find the Norton equivalent for the circuit in Fig. 13.84 at terminals *a-b*.

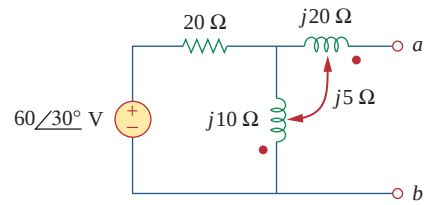


Figure 13.84
For Prob. 13.15.

13.16 Obtain the Norton equivalent at terminals *a-b* of the circuit in Fig. 13.85.

PS ML

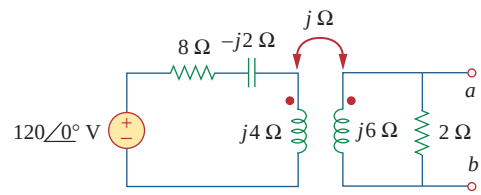


Figure 13.85
For Prob. 13.16.

13.17 In the circuit of Fig. 13.86, Z_L is a 15-mH inductor having an impedance of $j40\ \Omega$. Determine Z_{in} when $k = 0.6$.

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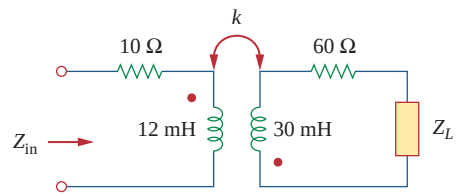


Figure 13.86
For Prob. 13.17.

13.18 Find the Thevenin equivalent to the left of the load **Z** in the circuit of Fig. 13.87.

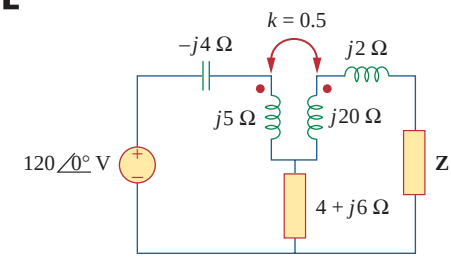


Figure 13.87
For Prob. 13.18.

13.19 Determine an equivalent T-section that can be used to replace the transformer in Fig. 13.88.

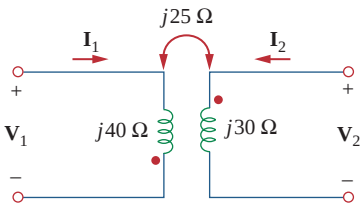


Figure 13.88
For Prob. 13.19.

Section 13.3 Energy in a Coupled Circuit

13.20 Determine currents $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$ in the circuit of Fig. 13.89. Find the energy stored in the coupled coils at $t = 2$ ms. Take $\omega = 1{,}000$ rad/s.

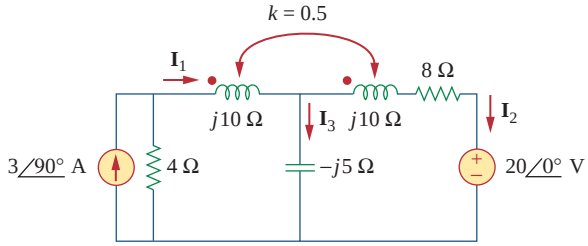


Figure 13.89
For Prob. 13.20.

13.21 Using Fig. 13.90, design a problem to help other students better understand energy in a coupled circuit.

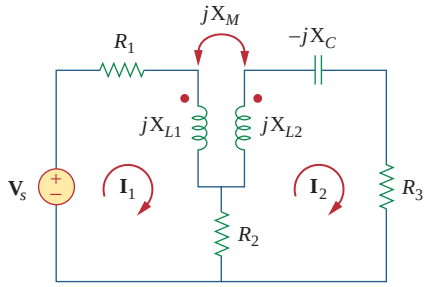


Figure 13.90
For Prob. 13.21.

***13.22** Find current $\mathbf{I}_o$ in the circuit of Fig. 13.91.

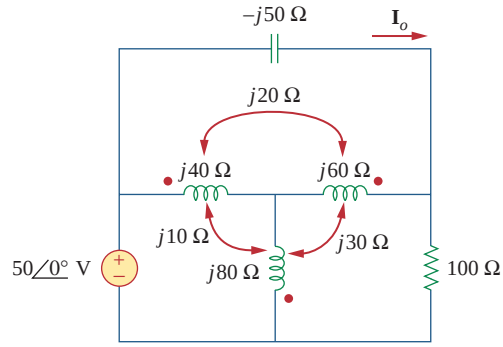


Figure 13.91
For Prob. 13.22.

13.23 If $M = 0.2$ H and $v_s = 120 \cos 10t$ V in the circuit of Fig. 13.92, find i_1 and i_2. Calculate the energy stored in the coupled coils at $t = 15$ ms.

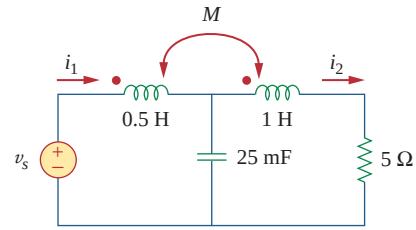


Figure 13.92
For Prob. 13.23.

13.24 In the circuit of Fig. 13.93,

(a) find the coupling coefficient,

(b) calculate v_o,

(c) determine the energy stored in the coupled inductors at $t = 2$ s.

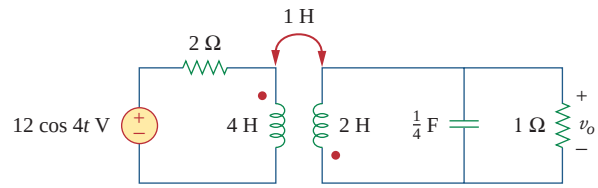


Figure 13.93
For Prob. 13.24.

* An asterisk indicates a challenging problem.

13.25 For the network in Fig. 13.94, find $\mathbf{Z}_{ab}$ and $\mathbf{I}_o$.

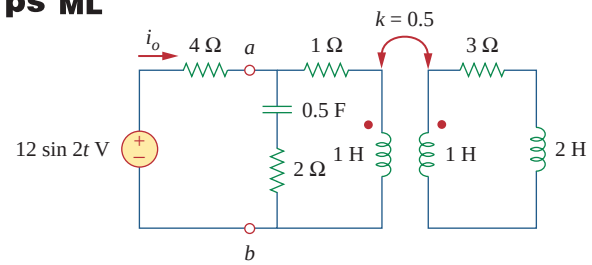


Figure 13.94
For Prob. 13.25.

13.26 Find $\mathbf{I}_o$ in the circuit of Fig. 13.95. Switch the dot on the winding on the right and calculate $\mathbf{I}_o$ again.

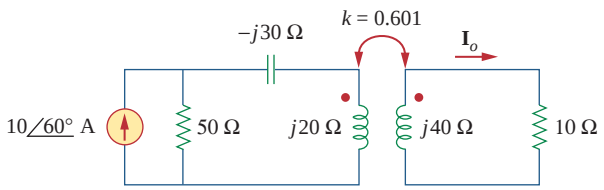


Figure 13.95
For Prob. 13.26.

13.27 Find the average power delivered to the 50-Ω resistor in the circuit of Fig. 13.96.

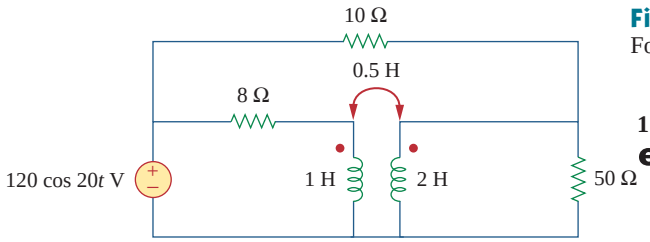


Figure 13.96
For Prob. 13.27.

***13.28** In the circuit of Fig. 13.97, find the value of X that will give maximum power transfer to the 20-Ω load.

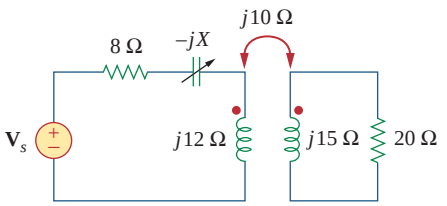


Figure 13.97
For Prob. 13.28.

Section 13.4 Linear Transformers

13.29 In the circuit of Fig. 13.98, find the value of the coupling coefficient k that will make the 10-Ω resistor dissipate 320 W. For this value of k, find the energy stored in the coupled coils at $t = 1.5$ s.

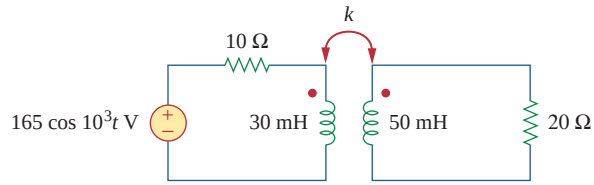


Figure 13.98
For Prob. 13.29.

13.30 (a) Find the input impedance of the circuit in Fig. 13.99 using the concept of reflected impedance.

(b) Obtain the input impedance by replacing the linear transformer by its T equivalent.

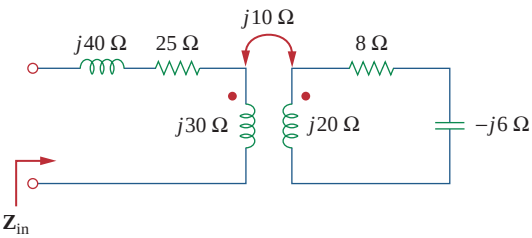


Figure 13.99
For Prob. 13.30.

13.31 Using Fig. 13.100, design a problem to help other students better understand linear transformers and how to find T-equivalent and Π-equivalent circuits.

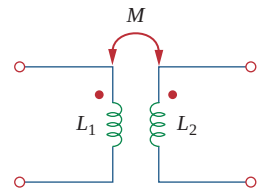


Figure 13.100
For Prob. 13.31.

***13.32** Two linear transformers are cascaded as shown in Fig. 13.101. Show that

$$\mathbf{Z}_{in} = \frac{\omega^2 R(L_a^2 + L_a L_b - M_a^2) + j\omega^3(L_a^2 L_b + L_a L_b^2 - L_a M_b^2 - L_b M_a^2)}{\omega^2(L_a L_b + L_b^2 - M_b^2) - j\omega R(L_a + L_b)}$$

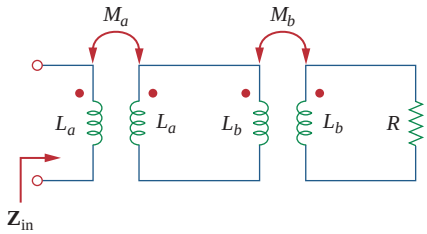


Figure 13.101
For Prob. 13.32.

13.33 Determine the input impedance of the air-core transformer circuit of Fig. 13.102.

ML

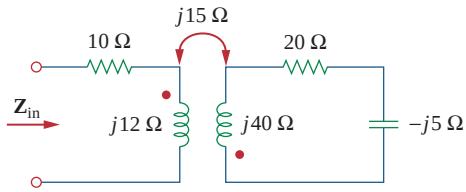


Figure 13.102
For Prob. 13.33.

13.34 Using Fig. 13.103, design a problem to help other students better understand how to find the input impedance of circuits with transformers.

eDd

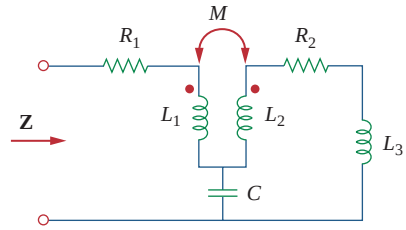


Figure 13.103
For Prob. 13.34.

***13.35** Find currents $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$ in the circuit of Fig. 13.104.

PS ML

Section 13.5 Ideal Transformers

13.36 As done in Fig. 13.32, obtain the relationships between terminal voltages and currents for each of the ideal transformers in Fig. 13.105.

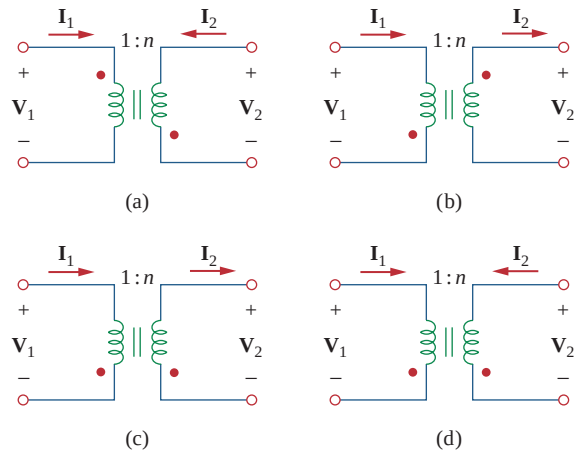


Figure 13.105
For Prob. 13.36.

13.37 A 480/2,400-V rms step-up ideal transformer delivers 50 kW to a resistive load. Calculate:

(a) the turns ratio

(b) the primary current

(c) the secondary current

13.38 Design a problem to help other students better understand ideal transformers.

eDd

13.39 A 1,200/240-V rms transformer has impedance $60\angle -30^\circ\ \Omega$ on the high-voltage side. If the transformer is connected to a $0.8\angle 10^\circ$-Ω load on the low-voltage side, determine the primary and secondary currents when the transformer is connected to 1200 V rms.

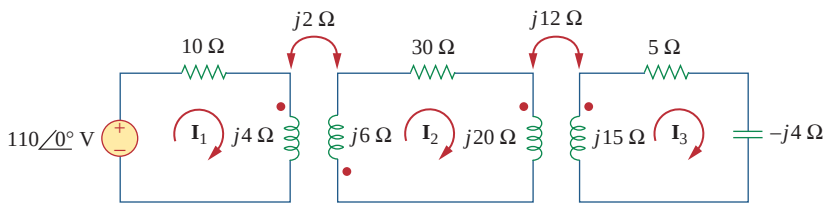


Figure 13.104
For Prob. 13.35.

13.40 The primary of an ideal transformer with a turns ratio of 5 is connected to a voltage source with Thevenin parameters $v_{Th} = 10 \cos 2000t$ V and $R_{Th} = 100\ \Omega$. Determine the average power delivered to a 200-Ω load connected across the secondary winding.

13.41 Determine $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit of Fig. 13.106.

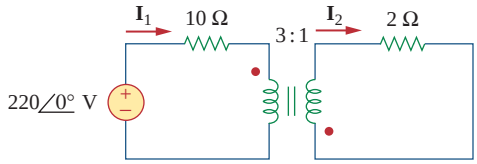


Figure 13.106
For Prob. 13.41.

13.42 For the circuit in Fig. 13.107, determine the power absorbed by the 2-Ω resistor. Assume the 80 V is an rms value.

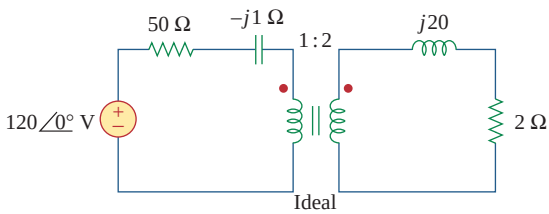


Figure 13.107
For Prob. 13.42.

13.43 Obtain $\mathbf{V}_1$ and $\mathbf{V}_2$ in the ideal transformer circuit of Fig. 13.108.

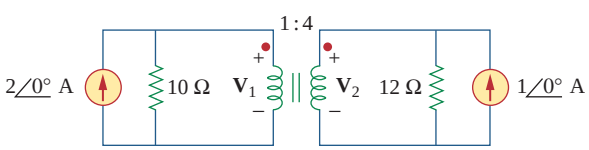


Figure 13.108
For Prob. 13.43.

***13.44** In the ideal transformer circuit of Fig. 13.109, find $i_1(t)$ and $i_2(t)$.

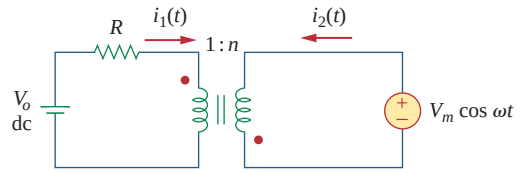


Figure 13.109
For Prob. 13.44.

13.45 For the circuit shown in Fig. 13.110, find the value of the average power absorbed by the 8-Ω resistor.

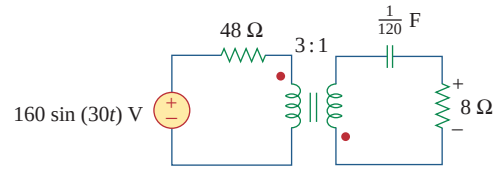


Figure 13.110
For Prob. 13.45.

13.46 (a) Find $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit of Fig. 13.111 below.
(b) Switch the dot on one of the windings. Find $\mathbf{I}_1$ and $\mathbf{I}_2$ again.

13.47 Find $v(t)$ for the circuit in Fig. 13.112.

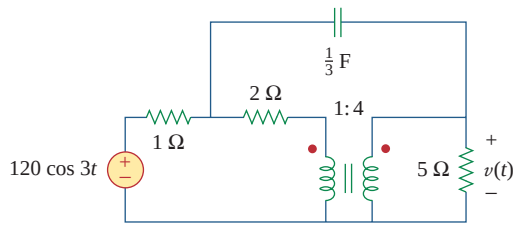


Figure 13.112
For Prob. 13.47.

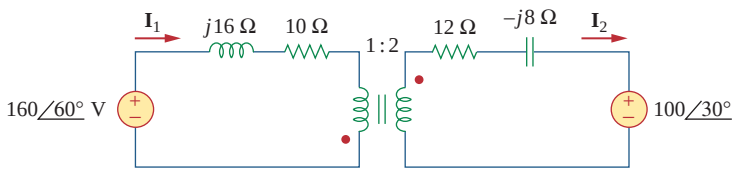


Figure 13.111
For Prob. 13.46.

13.48 Using Fig. 13.113, design a problem to help other students better understand how ideal transformers work.

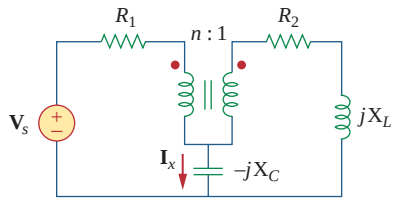


Figure 13.113
For Prob. 13.48.

13.49 Find current i_x in the ideal transformer circuit shown in Fig. 13.114.

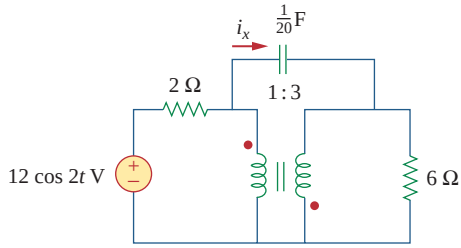


Figure 13.114
For Prob. 13.49.

13.50 Calculate the input impedance for the network in Fig. 13.115.

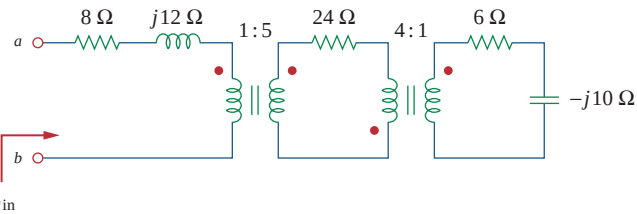


Figure 13.115
For Prob. 13.50.

13.51 Use the concept of reflected impedance to find the input impedance and current $\mathbf{I}_1$ in Fig. 13.116.

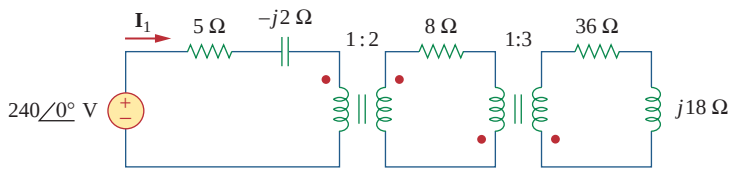


Figure 13.116
For Prob. 13.51.

13.52 For the circuit in Fig. 13.117, determine the turns ratio n that will cause maximum average power transfer to the load. Calculate that maximum average power.

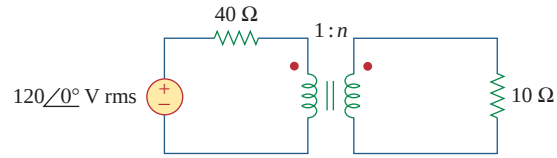


Figure 13.117
For Prob. 13.52.

13.53 Refer to the network in Fig. 13.118.

(a) Find n for maximum power supplied to the 200-Ω load.

(b) Determine the power in the 200-Ω load if $n = 10$.

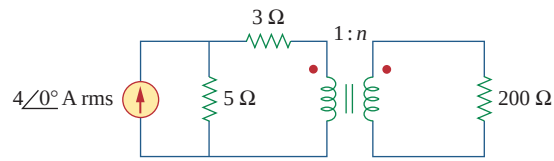


Figure 13.118
For Prob. 13.53.

13.54 A transformer is used to match an amplifier with an 8-Ω load as shown in Fig. 13.119. The Thevenin equivalent of the amplifier is: $V_{Th} = 10$ V, $Z_{Th} = 128\ \Omega$.

(a) Find the required turns ratio for maximum energy power transfer.

(b) Determine the primary and secondary currents.

(c) Calculate the primary and secondary voltages.

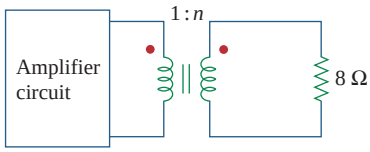


Figure 13.119
For Prob. 13.54.

13.55 For the circuit in Fig. 13.120, calculate the equivalent resistance.

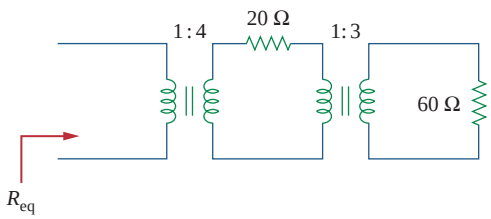


Figure 13.120
For Prob. 13.55.

13.56 Find the power absorbed by the 10-Ω resistor in the ideal transformer circuit of Fig. 13.121.

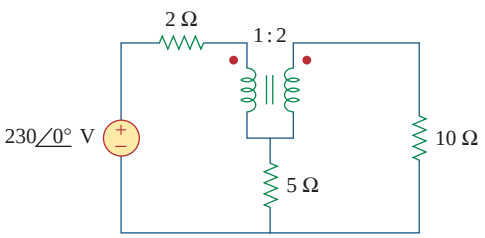


Figure 13.121
For Prob. 13.56.

13.57 For the ideal transformer circuit of Fig. 13.122 below, find:

(a) $\mathbf{I}_1$ and $\mathbf{I}_2$,

(b) $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_o$,

(c) the complex power supplied by the source.

13.58 Determine the average power absorbed by each resistor in the circuit of Fig. 13.123.

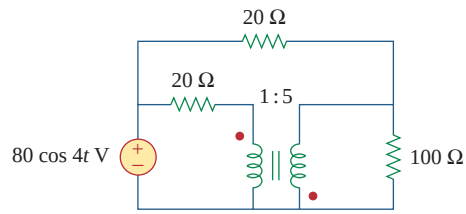


Figure 13.123
For Prob. 13.58.

13.59 In the circuit of Fig. 13.124, let $v_s = 160 \cos 1000t$. Find the average power delivered to each resistor.

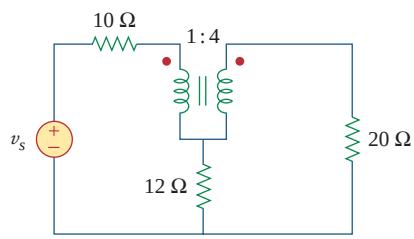


Figure 13.124
For Prob. 13.59.

13.60 Refer to the circuit in Fig. 13.125 on the following page.

(a) Find currents $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$.

(b) Find the power dissipated in the 40-Ω resistor.

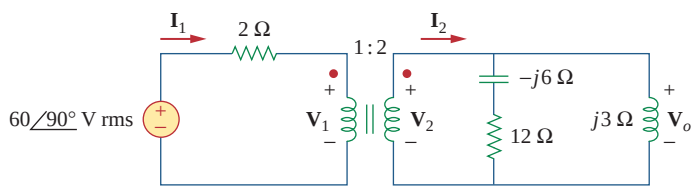


Figure 13.122
For Prob. 13.57.

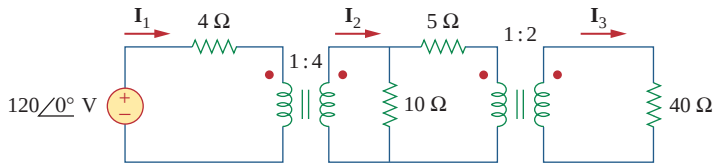


Figure 13.125
For Prob. 13.60.

***13.61** For the circuit in Fig. 13.126, find $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{V}_o$.

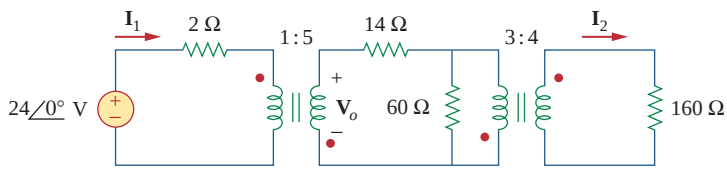


Figure 13.126
For Prob. 13.61.

13.62 For the network in Fig. 13.127, find

(a) the complex power supplied by the source,

(b) the average power delivered to the 18-Ω resistor.

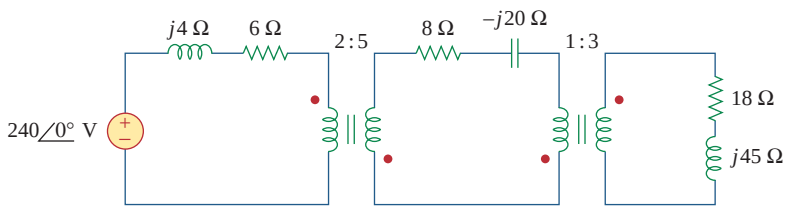


Figure 13.127
For Prob. 13.62.

13.63 Find the mesh currents in the circuit of Fig. 13.128

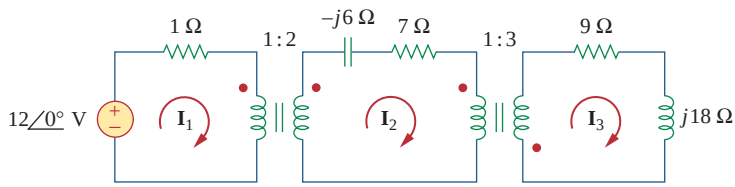


Figure 13.128
For Prob. 13.63.

13.64 For the circuit in Fig. 13.129, find the turns ratio so that the maximum power is delivered to the 30-kΩ resistor.

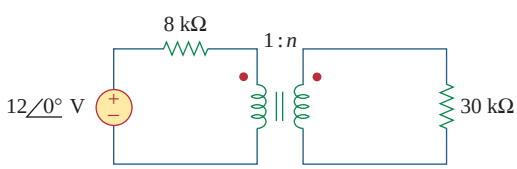


Figure 13.129
For Prob. 13.64.

***13.65** Calculate the average power dissipated by the 20-Ω resistor in Fig. 13.130.

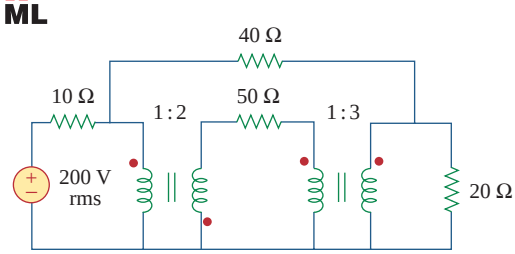


Figure 13.130
For Prob. 13.65.

Section 13.6 Ideal Autotransformers

13.66 Design a problem to help other students better understand how the ideal autotransformer works.

13.67 An autotransformer with a 40 percent tap is supplied by a 400-V, 60-Hz source and is used for step-down operation. A 5-kVA load operating at unity power factor is connected to the secondary terminals. Find:

(a) the secondary voltage

(b) the secondary current

(c) the primary current

13.68 In the ideal autotransformer of Fig. 13.131, calculate $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_o$. Find the average power delivered to the load.

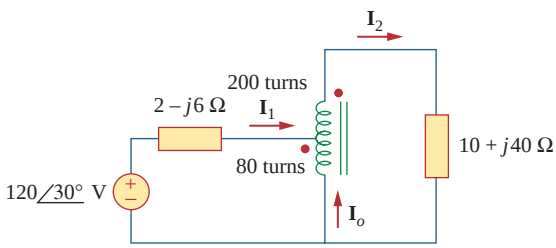


Figure 13.131
For Prob. 13.68.

***13.69** In the circuit of Fig. 13.132, $\mathbf{Z}_L$ is adjusted until maximum average power is delivered to $\mathbf{Z}_L$. Find $\mathbf{Z}_L$ and the maximum average power transferred to it. Take $N_1 = 600$ turns and $N_2 = 200$ turns.

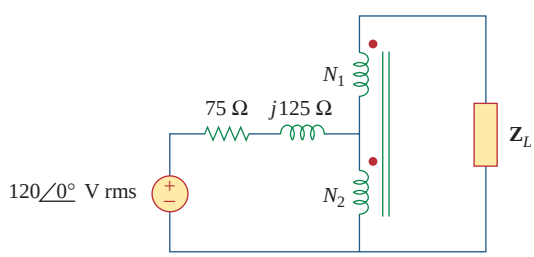


Figure 13.132
For Prob. 13.69.

13.70 In the ideal transformer circuit shown in Fig. 13.133, determine the average power delivered to the load.

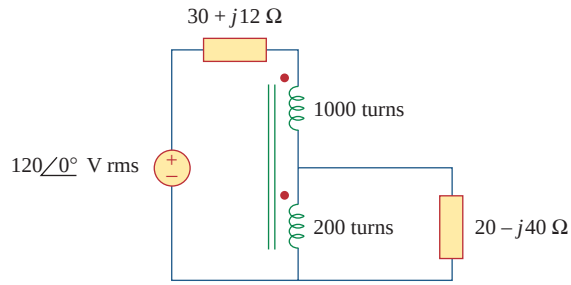


Figure 13.133
For Prob. 13.70.

13.71 In the autotransformer circuit in Fig. 13.134, show that

$$\mathbf{Z}_{\text{in}} = \left(1 + \frac{N_1}{N_2}\right)^2 \mathbf{Z}_L$$

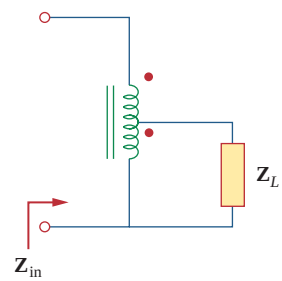


Figure 13.134
For Prob. 13.71.

Section 13.7 Three-Phase Transformers

13.72 In order to meet an emergency, three single-phase transformers with 12,470/7,200 V rms are connected in Δ-Y to form a three-phase transformer which is fed by a 12,470-V transmission line. If the transformer supplies 60 MVA to a load, find:

(a) the turns ratio for each transformer,

(b) the currents in the primary and secondary windings of the transformer,

(c) the incoming and outgoing transmission line currents.

13.73 Figure 13.135 on the following page shows a three-phase transformer that supplies a Y-connected load.

(a) Identify the transformer connection.

(b) Calculate currents $\mathbf{I}_2$ and $\mathbf{I}_c$.

(c) Find the average power absorbed by the load.

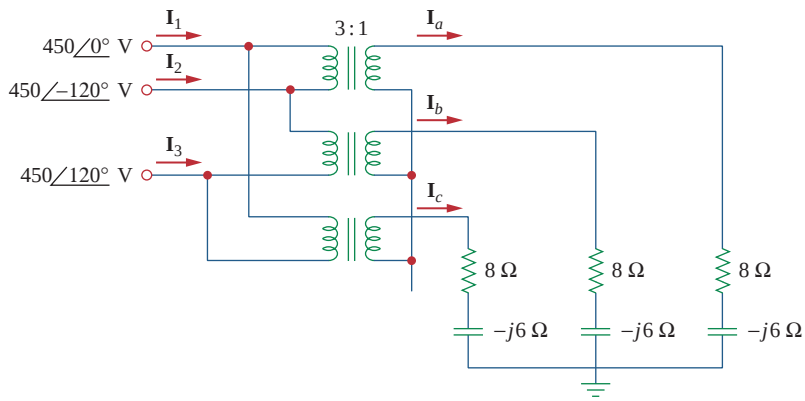


Figure 13.135
For Prob. 13.73.

13.74 Consider the three-phase transformer shown in Fig. 13.136. The primary is fed by a three-phase source with line voltage of 2.4 kV rms, while the secondary supplies a three-phase 120-kW balanced load at pf of 0.8. Determine:

(a) the type of transformer connections,

(b) the values of I_{LS} and I_{PS},

(c) the values of I_{LP} and I_{PP},

(d) the kVA rating of each phase of the transformer.

13.75 A balanced three-phase transformer bank with the Δ-Y connection depicted in Fig. 13.137 is used to step down line voltages from 4,500 V rms to 900 V rms. If the transformer feeds a 120-kVA load, find:

(a) the turns ratio for the transformer,

(b) the line currents at the primary and secondary sides.

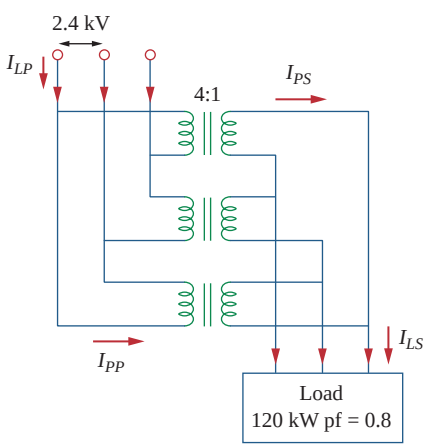


Figure 13.136
For Prob. 13.74.

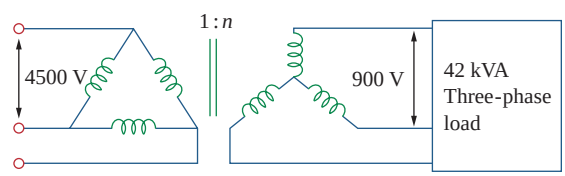


Figure 13.137
For Prob. 13.75.

13.76 Using Fig. 13.138, design a problem to help other students better understand a Y-Δ, three-phase transformer and how they work.

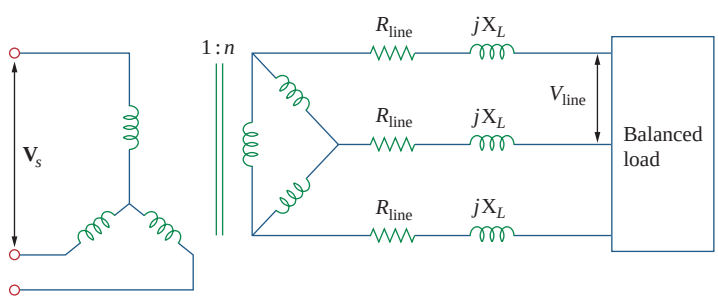


Figure 13.138
For Prob. 13.76.

13.77 The three-phase system of a town distributes power with a line voltage of 13.2 kV. A pole transformer connected to single wire and ground steps down the high-voltage wire to 120 V rms and serves a house as shown in Fig. 13.139.

(a) Calculate the turns ratio of the pole transformer to get 120 V.

(b) Determine how much current a 100-W lamp connected to the 120-V hot line draws from the high-voltage line.

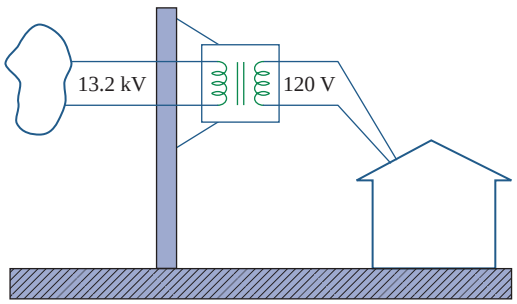


Figure 13.139
For Prob. 13.77.

Section 13.8 *PSpice* Analysis of Magnetically Coupled Circuits

13.78 Use *PSpice* to determine the mesh currents in the circuit of Fig. 13.140. Take $\omega = 1$ rad/s.

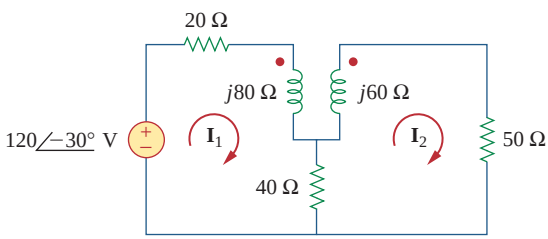


Figure 13.140
For Prob. 13.78.

13.79 Use *PSpice* to find $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$ in the circuit of Fig. 13.141.

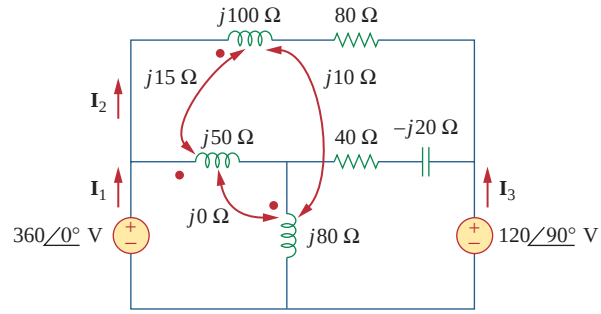


Figure 13.141
For Prob. 13.79.

13.80 Rework Prob. 13.22 using *PSpice*.

13.81 Use *PSpice* to find $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$ in the circuit of Fig. 13.142.

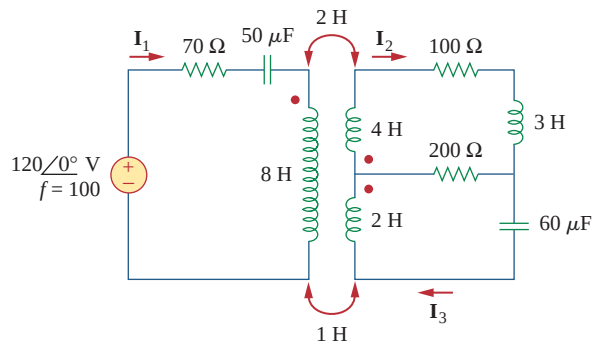


Figure 13.142
For Prob. 13.81.

13.82 Use *PSpice* to find $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{I}_o$ in the circuit of Fig. 13.143.

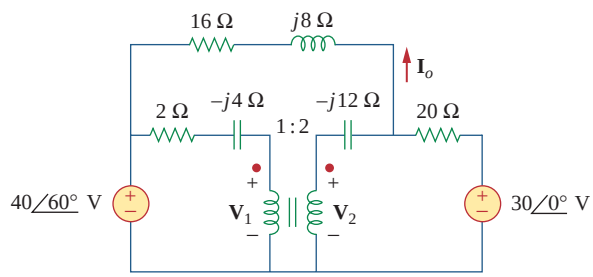


Figure 13.143
For Prob. 13.82.

13.83 Find $\mathbf{I}_x$ and $\mathbf{V}_x$ in the circuit of Fig. 13.144 using *PSpice*.

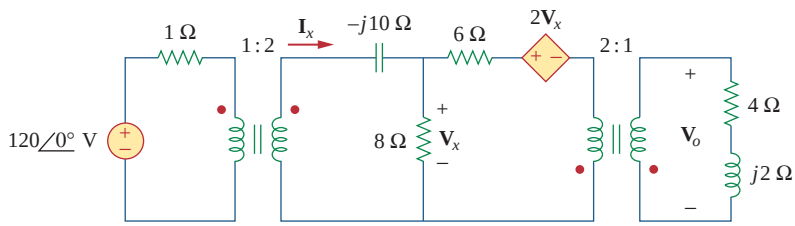


Figure 13.144
For Prob. 13.83.

13.84 Determine $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{I}_3$ in the ideal transformer circuit of Fig. 13.145 using *PSpice*.

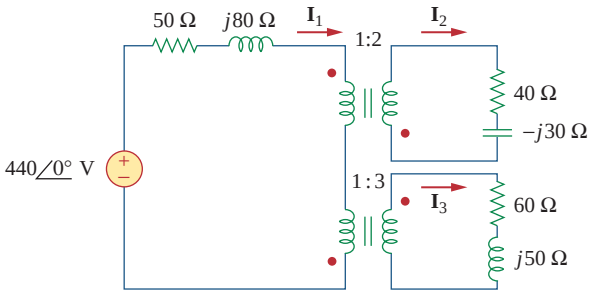


Figure 13.145
For Prob. 13.84.

Section 13.9 Applications

13.85 A stereo amplifier circuit with an output impedance of 7.2 kΩ is to be matched to a speaker with an input impedance of 8 Ω by a transformer whose primary side has 3,000 turns. Calculate the number of turns required on the secondary side.

13.86 A transformer having 2,400 turns on the primary and 48 turns on the secondary is used as an impedance-matching device. What is the reflected value of a 3-Ω load connected to the secondary?

13.87 A radio receiver has an input resistance of 300Ω. When it is connected directly to an antenna system with a characteristic impedance of 75 Ω, an impedance mismatch occurs. By inserting an impedance-matching transformer ahead of the receiver, maximum power can be realized. Calculate the required turns ratio.

13.88 A step-down power transformer with a turns ratio of $n = 0.1$ supplies 12.6 V rms to a resistive load. If the primary current is 2.5 A rms, how much power is delivered to the load?

13.89 A 240/120-V rms power transformer is rated at 10 kVA. Determine the turns ratio, the primary current, and the secondary current.

13.90 A 4-kVA, 2,400/240-V rms transformer has 250 turns on the primary side. Calculate:

(a) the turns ratio,

(b) the number of turns on the secondary side,

(c) the primary and secondary currents.

13.91 A 25,000/240-V rms distribution transformer has a primary current rating of 75 A.

(a) Find the transformer kVA rating.

(b) Calculate the secondary current.

13.92 A 4,800-V rms transmission line feeds a distribution transformer with 1,200 turns on the primary and 28 turns on the secondary. When a 10-Ω load is connected across the secondary, find:

(a) the secondary voltage,

(b) the primary and secondary currents,

(c) the power supplied to the load.

Comprehensive Problems

13.93 A four-winding transformer (Fig. 13.146) is often used in equipment (e.g., PCs, VCRs) that may be operated from either 110 V or 220 V. This makes the equipment suitable for both domestic and foreign use. Show which connections are necessary to provide:

(a) an output of 14 V with an input of 110 V,

(b) an output of 50 V with an input of 220 V.

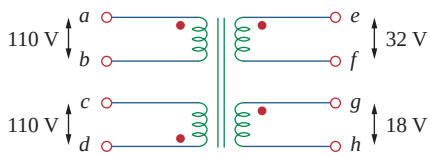


Figure 13.146
For Prob. 13.93.

***13.94** A 440/110-V ideal transformer can be connected to become a 550/440-V ideal autotransformer. There are four possible connections, two of which are wrong. Find the output voltage of:

(a) a wrong connection,

(b) the right connection.

13.95 Ten bulbs in parallel are supplied by a 7,200/120-V transformer as shown in Fig. 13.147, where the bulbs are modeled by the 144-Ω resistors. Find:

(a) the turns ratio n,

(b) the current through the primary winding.

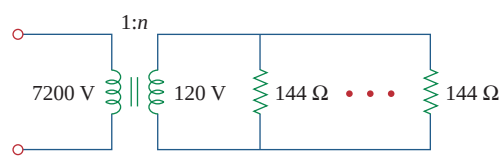


Figure 13.147
For Prob. 13.95.

***13.96** Some modern power transmission systems now have major, high voltage DC transmission segments. There are a lot of good reasons for doing this but we will not go into them here. To go from the AC to DC, power electronics are used. We start with three-phase AC and then rectify it (using a full-wave rectifier). It was found that using a delta to wye and delta combination connected secondary would give us a much smaller ripple after the full-wave rectifier. How is this accomplished? Remember that these are real devices and are wound on common cores.

Hint: using Figs. 13.47 and 13.49, and the fact that each coil of the wye connected secondary and each coil of the delta connected secondary are wound around the same core of each coil of the delta connected primary so the voltage of each of the corresponding coils are in phase. When the output leads of both secondaries are connected through full-wave rectifiers with the same load, you will see that the ripple is now greatly reduced. Please consult the instructor for more help if necessary.

chapter

14

Frequency Response

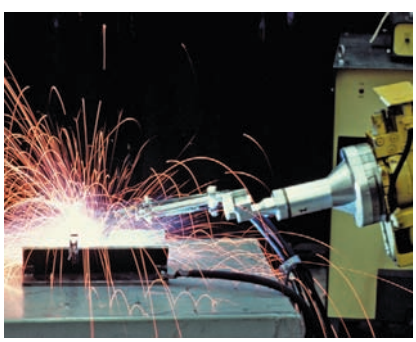
Dost thou love Life? Then do not squander Time; for that is the stuff Life is made.

—Benjamin Franklin

Enhancing Your Career

Career in Control Systems

Control systems are another area of electrical engineering where circuit analysis is used. A control system is designed to regulate the behavior of one or more variables in some desired manner. Control systems play major roles in our everyday life. Household appliances such as heating and air-conditioning systems, switch-controlled thermostats, washers and dryers, cruise controllers in automobiles, elevators, traffic lights, manufacturing plants, navigation systems—all utilize control systems. In the aerospace field, precision guidance of space probes, the wide range of operational modes of the space shuttle, and the ability to maneuver space vehicles remotely from earth all require knowledge of control systems. In the manufacturing sector, repetitive production line operations are increasingly performed by robots, which are programmable control systems designed to operate for many hours without fatigue.



A welding robot. © Vol. 1 PhotoDisc/ Getty Images

Control engineering integrates circuit theory and communication theory. It is not limited to any specific engineering discipline but may involve environmental, chemical, aeronautical, mechanical, civil, and electrical engineering. For example, a typical task for a control system engineer might be to design a speed regulator for a disk drive head.

A thorough understanding of control systems techniques is essential to the electrical engineer and is of great value for designing control systems to perform the desired task.

14.1 Introduction

In our sinusoidal circuit analysis, we have learned how to find voltages and currents in a circuit with a constant frequency source. If we let the amplitude of the sinusoidal source remain constant and vary the frequency, we obtain the circuit's *frequency response*. The frequency response may be regarded as a complete description of the sinusoidal steady-state behavior of a circuit as a function of frequency.

The frequency response of a circuit may also be considered as the variation of the gain and phase with frequency.

The **frequency response** of a circuit is the variation in its behavior with change in signal frequency.

The sinusoidal steady-state frequency responses of circuits are of significance in many applications, especially in communications and control systems. A specific application is in electric filters that block out or eliminate signals with unwanted frequencies and pass signals of the desired frequencies. Filters are used in radio, TV, and telephone systems to separate one broadcast frequency from another.

We begin this chapter by considering the frequency response of simple circuits using their transfer functions. We then consider Bode plots, which are the industry-standard way of presenting frequency response. We also consider series and parallel resonant circuits and encounter important concepts such as resonance, quality factor, cutoff frequency, and bandwidth. We discuss different kinds of filters and network scaling. In the last section, we consider one practical application of resonant circuits and two applications of filters.

14.2 Transfer Function

The transfer function $\mathbf{H}(\omega)$ (also called the *network function*) is a useful analytical tool for finding the frequency response of a circuit. In fact, the frequency response of a circuit is the plot of the circuit's transfer function $\mathbf{H}(\omega)$ versus ω, with ω varying from $\omega = 0$ to $\omega = \infty$.

A transfer function is the frequency-dependent ratio of a forced function to a forcing function (or of an output to an input). The idea of a transfer function was implicit when we used the concepts of impedance and admittance to relate voltage and current. In general, a linear network can be represented by the block diagram shown in Fig. 14.1.

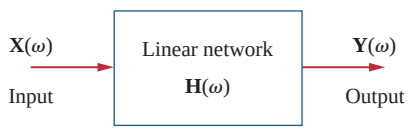


Figure 14.1
A block diagram representation of a linear network.

In this context, $\mathbf{X}(\omega)$ and $\mathbf{Y}(\omega)$ denote the input and output phasors of a network; they should not be confused with the same symbolism used for reactance and admittance. The multiple usage of symbols is conventionally permissible due to lack of enough letters in the English language to express all circuit variables distinctly.

The **transfer function** $\mathbf{H}(\omega)$ of a circuit is the frequency-dependent ratio of a phasor output $\mathbf{Y}(\omega)$ (an element voltage or current) to a phasor input $\mathbf{X}(\omega)$ (source voltage or current).

Thus,

$$\mathbf{H}(\omega) = \frac{\mathbf{Y}(\omega)}{\mathbf{X}(\omega)} \tag{14.1}$$

assuming zero initial conditions. Since the input and output can be either voltage or current at any place in the circuit, there are four possible transfer functions:

$$\mathbf{H}(\omega) = \text{Voltage gain} = \frac{\mathbf{V}_o(\omega)}{\mathbf{V}_i(\omega)} \tag{14.2a}$$

$$\mathbf{H}(\omega) = \text{Current gain} = \frac{\mathbf{I}_o(\omega)}{\mathbf{I}_i(\omega)} \tag{14.2b}$$

$$\mathbf{H}(\omega) = \text{Transfer Impedance} = \frac{\mathbf{V}_o(\omega)}{\mathbf{I}_i(\omega)} \tag{14.2c}$$

$$\mathbf{H}(\omega) = \text{Transfer Admittance} = \frac{\mathbf{I}_o(\omega)}{\mathbf{V}_i(\omega)} \tag{14.2d}$$

Some authors use $\mathbf{H}(j\omega)$ for transfer instead of $\mathbf{H}(\omega)$, since ω and j are an inseparable pair.

where subscripts i and o denote input and output values. Being a complex quantity, $\mathbf{H}(\omega)$ has a magnitude $H(\omega)$ and a phase ϕ; that is, $\mathbf{H}(\omega) = H(\omega)\angle\phi$.

To obtain the transfer function using Eq. (14.2), we first obtain the frequency-domain equivalent of the circuit by replacing resistors, inductors, and capacitors with their impedances R, $j\omega L$, and $1/j\omega C$. We then use any circuit technique(s) to obtain the appropriate quantity in Eq. (14.2). We can obtain the frequency response of the circuit by plotting the magnitude and phase of the transfer function as the frequency varies. A computer is a real time-saver for plotting the transfer function.

The transfer function $\mathbf{H}(\omega)$ can be expressed in terms of its numerator polynomial $\mathbf{N}(\omega)$ and denominator polynomial $\mathbf{D}(\omega)$ as

$$\boxed{\mathbf{H}(\omega) = \frac{\mathbf{N}(\omega)}{\mathbf{D}(\omega)}} \tag{14.3}$$

where $\mathbf{N}(\omega)$ and $\mathbf{D}(\omega)$ are not necessarily the same expressions for the input and output functions, respectively. The representation of $\mathbf{H}(\omega)$ in Eq. (14.3) assumes that common numerator and denominator factors in $\mathbf{H}(\omega)$ have canceled, reducing the ratio to lowest terms. The roots of $\mathbf{N}(\omega) = 0$ are called the *zeros* of $\mathbf{H}(\omega)$ and are usually represented as $j\omega = z_1, z_2, \ldots$. Similarly, the roots of $\mathbf{D}(\omega) = 0$ are the *poles* of $\mathbf{H}(\omega)$ and are represented as $j\omega = p_1, p_2, \ldots$.

A **zero,** as a *root* of the numerator polynomial, is a value that results in a zero value of the function. A **pole,** as a *root* of the denominator polynomial, is a value for which the function is infinite.

A zero may also be regarded as the value of $s = j\omega$ that makes $\mathbf{H}(s)$ zero, and a pole as the value of $s = j\omega$ that makes $\mathbf{H}(s)$ infinite.

To avoid complex algebra, it is expedient to replace $j\omega$ temporarily with s when working with $\mathbf{H}(\omega)$ and replace s with $j\omega$ at the end.

Example 14.1

For the *RC* circuit in Fig. 14.2(a), obtain the transfer function $\mathbf{V}_o/\mathbf{V}_s$ and its frequency response. Let $v_s = V_m \cos\omega t$.

Solution:

The frequency-domain equivalent of the circuit is in Fig. 14.2(b). By voltage division, the transfer function is given by

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}$$

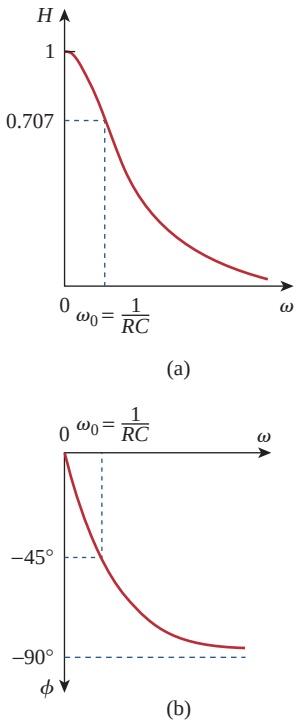


Figure 14.3
Frequency response of the *RC* circuit: (a) amplitude response, (b) phase response.

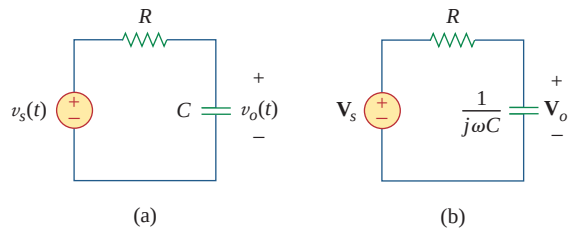


Figure 14.2
For Example 14.1: (a) time-domain *RC* circuit, (b) frequency-domain *RC* circuit.

Comparing this with Eq. (9.18e), we obtain the magnitude and phase of $\mathbf{H}(\omega)$ as

$$H = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}}, \qquad \phi = -\tan^{-1}\frac{\omega}{\omega_0}$$

where $\omega_0 = 1/RC$. To plot H and ϕ for $0 < \omega < \infty$, we obtain their values at some critical points and then sketch.

At $\omega = 0$, $H = 1$ and $\phi = 0$. At $\omega = \infty$, $H = 0$ and $\phi = -90°$. Also, at $\omega = \omega_0$, $H = 1/\sqrt{2}$ and $\phi = -45°$. With these and a few more points as shown in Table 14.1, we find that the frequency response is as shown in Fig. 14.3. Additional features of the frequency response in Fig. 14.3 will be explained in Section 14.6.1 on lowpass filters.

TABLE 14.1
For Example 14.1.

ω/ω_0	H	ϕ	ω/ω_0	H	ϕ
0	1	0	10	0.1	−84°
1	0.71	−45°	20	0.05	−87°
2	0.45	−63°	100	0.01	−89°
3	0.32	−72°	∞	0	−90°

Practice Problem 14.1

Obtain the transfer function $\mathbf{V}_o/\mathbf{V}_s$ of the *RL* circuit in Fig. 14.4, assuming $v_s = V_m \cos\omega t$. Sketch its frequency response.

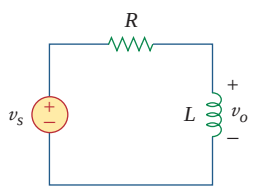


Figure 14.4
RL circuit for Practice Prob. 14.1.

Answer: $j\omega L/(R + j\omega L)$; see Fig. 14.5 for the response.

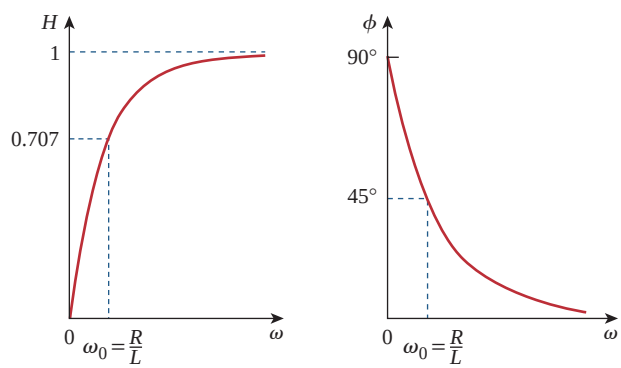


Figure 14.5
Frequency response of the *RL* circuit in Fig. 14.4.

Example 14.2

For the circuit in Fig. 14.6, calculate the gain $\mathbf{I}_o(\omega)/\mathbf{I}_i(\omega)$ and its poles and zeros.

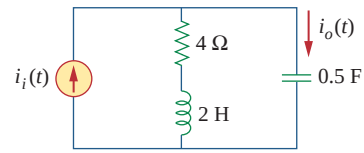


Figure 14.6
For Example 14.2.

Solution:
By current division,

$$\mathbf{I}_o(\omega) = \frac{4 + j2\omega}{4 + j2\omega + 1/j0.5\omega}\mathbf{I}_i(\omega)$$

or

$$\frac{\mathbf{I}_o(\omega)}{\mathbf{I}_i(\omega)} = \frac{j0.5\omega(4 + j2\omega)}{1 + j2\omega + (j\omega)^2} = \frac{s(s + 2)}{s^2 + 2s + 1}, \qquad s = j\omega$$

The zeros are at

$$s(s + 2) = 0 \quad \Rightarrow \quad z_1 = 0, z_2 = -2$$

The poles are at

$$s^2 + 2s + 1 = (s + 1)^2 = 0$$

Thus, there is a repeated pole (or double pole) at $p = -1$.

Practice Problem 14.2

Find the transfer function $\mathbf{V}_o(\omega)/\mathbf{I}_i(\omega)$ for the circuit in Fig. 14.7. Obtain its zeros and poles.

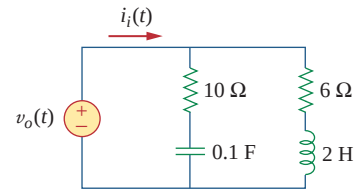


Figure 14.7
For Practice Prob. 14.2.

Answer: $\dfrac{10(s + 1)(s + 3)}{s^2 + 8s + 5}$, $s = j\omega$; zeros: $-1, -3$; poles: $-0.683, -7.317$.

14.3 †The Decibel Scale

It is not always easy to get a quick plot of the magnitude and phase of the transfer function as we did above. A more systematic way of obtaining the frequency response is to use Bode plots. Before we begin to construct Bode plots, we should take care of two important issues: the use of logarithms and decibels in expressing gain.

Since Bode plots are based on logarithms, it is important that we keep the following properties of logarithms in mind:

1. $\log P_1P_2 = \log P_1 + \log P_2$
2. $\log P_1/P_2 = \log P_1 - \log P_2$
3. $\log P^n = n \log P$
4. $\log 1 = 0$

In communications systems, gain is measured in *bels*. Historically, the bel is used to measure the ratio of two levels of power or power gain G; that is,

$$G = \text{Number of bels} = \log_{10}\frac{P_2}{P_1} \qquad \textbf{(14.4)}$$

Historical note: The *bel* is named after Alexander Graham Bell, the inventor of the telephone.

Historical



Alexander Graham Bell (1847–1922) inventor of the telephone, was a Scottish-American scientist.

Bell was born in Edinburgh, Scotland, a son of Alexander Melville Bell, a well-known speech teacher. Alexander the younger also became a speech teacher after graduating from the University of Edinburgh and the University of London. In 1866 he became interested in transmitting speech electrically. After his older brother died of tuberculosis, his father decided to move to Canada. Alexander was asked to come to Boston to work at the School for the Deaf. There he met Thomas A. Watson, who became his assistant in his electromagnetic transmitter experiment. On March 10, 1876, Alexander sent the famous first telephone message: "Watson, come here I want you." The bel, the logarithmic unit introduced in Chapter 14, is named in his honor.

The *decibel* (dB) provides us with a unit of less magnitude. It is 1/10th of a bel and is given by

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} \tag{14.5}$$

When $P_1 = P_2$, there is no change in power and the gain is 0 dB. If $P_2 = 2P_1$, the gain is

$$G_{\text{dB}} = 10 \log_{10} 2 \simeq 3 \text{ dB} \tag{14.6}$$

and when $P_2 = 0.5P_1$, the gain is

$$G_{\text{dB}} = 10 \log_{10} 0.5 \simeq -3 \text{ dB} \tag{14.7}$$

Equations (14.6) and (14.7) show another reason why logarithms are greatly used: The logarithm of the reciprocal of a quantity is simply negative the logarithm of that quantity.

Alternatively, the gain G can be expressed in terms of voltage or current ratio. To do so, consider the network shown in Fig. 14.8. If P_1 is the input power, P_2 is the output (load) power, R_1 is the input resistance, and R_2 is the load resistance, then $P_1 = 0.5V_1^2/R_1$ and $P_2 = 0.5V_2^2/R_2$, and Eq. (14.5) becomes

$$\begin{aligned} G_{\text{dB}} &= 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{V_2^2/R_2}{V_1^2/R_1} \\ &= 10 \log_{10} \left(\frac{V_2}{V_1} \right)^2 + 10 \log_{10} \frac{R_1}{R_2} \end{aligned} \tag{14.8}$$

$$G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1} - 10 \log_{10} \frac{R_2}{R_1} \tag{14.9}$$

For the case when $R_2 = R_1$, a condition that is often assumed when comparing voltage levels, Eq. (14.9) becomes

$$G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1} \tag{14.10}$$

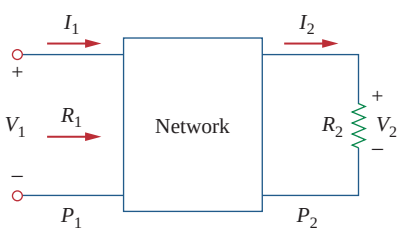


Figure 14.8
Voltage-current relationships for a four-terminal network.

Instead, if $P_1 = I_1^2 R_1$ and $P_2 = I_2^2 R_2$, for $R_1 = R_2$, we obtain

$$G_{\text{dB}} = 20 \log_{10} \frac{I_2}{I_1} \tag{14.11}$$

Three things are important to note from Eqs. (14.5), (14.10), and (14.11):

1. That $10 \log_{10}$ is used for power, while $20 \log_{10}$ is used for voltage or current, because of the square relationship between them ($P = V^2/R = I^2 R$).
2. That the dB value is a logarithmic measurement of the *ratio* of one variable to another *of the same type*. Therefore, it applies in expressing the transfer function H in Eqs. (14.2a) and (14.2b), which are dimensionless quantities, but not in expressing H in Eqs. (14.2c) and (14.2d).
3. It is important to note that we only use voltage and current magnitudes in Eqs. (14.10) and (14.11). Negative signs and angles will be handled independently as we will see in Section 14.4.

With this in mind, we now apply the concepts of logarithms and decibels to construct Bode plots.

14.4 Bode Plots

Obtaining the frequency response from the transfer function as we did in Section 14.2 is an uphill task. The frequency range required in frequency response is often so wide that it is inconvenient to use a linear scale for the frequency axis. Also, there is a more systematic way of locating the important features of the magnitude and phase plots of the transfer function. For these reasons, it has become standard practice to plot the transfer function on a pair of semilogarithmic plots: the magnitude in decibels is plotted against the logarithm of the frequency; on a separate plot, the phase in degrees is plotted against the logarithm of the frequency. Such semilogarithmic plots of the transfer function—known as *Bode plots*—have become the industry standard.

Historical note: Named after Hendrik W. Bode (1905–1982), an engineer with the Bell Telephone Laboratories, for his pioneering work in the 1930s and 1940s.

Bode plots are semilog plots of the magnitude (in decibels) and phase (in degrees) of a transfer function versus frequency.

Bode plots contain the same information as the nonlogarithmic plots discussed in the previous section, but they are much easier to construct, as we shall see shortly.

The transfer function can be written as

$$\mathbf{H} = H \angle \phi = H e^{j\phi} \tag{14.12}$$

Taking the natural logarithm of both sides,

$$\ln \mathbf{H} = \ln H + \ln e^{j\phi} = \ln H + j\phi \tag{14.13}$$

Thus, the real part of $\ln \mathbf{H}$ is a function of the magnitude while the imaginary part is the phase. In a Bode magnitude plot, the gain

$$\boxed{H_{\text{dB}} = 20 \log_{10} H} \tag{14.14}$$

TABLE 14.2

Specific gain and their decibel values.*

Magnitude H	$20 \log_{10} H$ (dB)
0.001	−60
0.01	−40
0.1	−20
0.5	−6
$1/\sqrt{2}$	−3
1	0
$\sqrt{2}$	3
2	6
10	20
20	26
100	40
1000	60

* Some of these values are approximate.

is plotted in decibels (dB) versus frequency. Table 14.2 provides a few values of H with the corresponding values in decibels. In a Bode phase plot, ϕ is plotted in degrees versus frequency. Both magnitude and phase plots are made on semilog graph paper.

A transfer function in the form of Eq. (14.3) may be written in terms of factors that have real and imaginary parts. One such representation might be

$$\mathbf{H}(\omega) = \frac{K(j\omega)^{\pm 1}(1 + j\omega/z_1)[1 + j2\zeta_1\omega/\omega_k + (j\omega/\omega_k)^2]\cdots}{(1 + j\omega/p_1)[1 + j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2]\cdots} \quad \textbf{(14.15)}$$

which is obtained by dividing out the poles and zeros in $\mathbf{H}(\omega)$. The representation of $\mathbf{H}(\omega)$ as in Eq. (14.15) is called the *standard form*. $\mathbf{H}(\omega)$ may include up to seven types of different factors that can appear in various combinations in a transfer function. These are:

1. A gain K
2. A pole $(j\omega)^{-1}$ or zero $(j\omega)$ at the origin
3. A simple pole $1/(1 + j\omega/p_1)$ or zero $(1 + j\omega/z_1)$
4. A quadratic pole $1/[1 + j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2]$ or zero $[1 + j2\zeta_1\omega/\omega_k + (j\omega/\omega_k)^2]$

In constructing a Bode plot, we plot each factor separately and then add them graphically. The factors can be considered one at a time and then combined additively because of the logarithms involved. It is this mathematical convenience of the logarithm that makes Bode plots a powerful engineering tool.

We will now make straight-line plots of the factors listed above. We shall find that these straight-line plots known as Bode plots approximate the actual plots to a reasonable degree of accuracy.

The origin is where $\omega = 1$ or $\log \omega = 0$ and the gain is zero.

Constant term: For the gain K, the magnitude is $20 \log_{10} K$ and the phase is 0°; both are constant with frequency. Thus, the magnitude and phase plots of the gain are shown in Fig. 14.9. If K is negative, the magnitude remains $20 \log_{10} |K|$ but the phase is $\pm 180°$.

A decade is an interval between two frequencies with a ratio of 10; e.g., between ω_0 and $10\omega_0$, or between 10 and 100 Hz. Thus, 20 dB/decade means that the magnitude changes 20 dB whenever the frequency changes tenfold or one decade.

Pole/zero at the origin: For the zero $(j\omega)$ at the origin, the magnitude is $20 \log_{10} \omega$ and the phase is 90°. These are plotted in Fig. 14.10, where we notice that the slope of the magnitude plot is 20 dB/decade, while the phase is constant with frequency.

The Bode plots for the pole $(j\omega)^{-1}$ are similar except that the slope of the magnitude plot is −20 dB/decade while the phase is −90°. In general, for $(j\omega)^N$, where N is an integer, the magnitude plot will have a slope of $20N$ dB/decade, while the phase is $90N$ degrees.

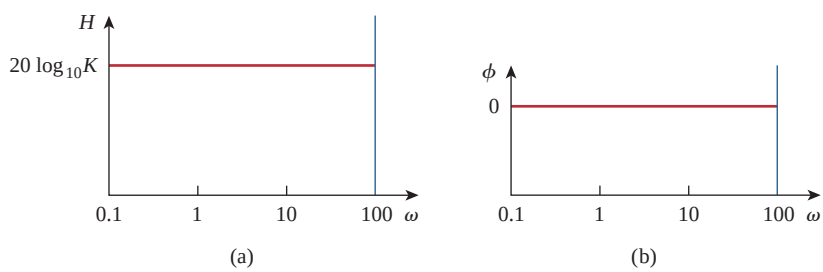


Figure 14.9
Bode plots for gain K: (a) magnitude plot, (b) phase plot.

Simple pole/zero: For the simple zero $(1 + j\omega/z_1)$, the magnitude is $20 \log_{10} |1 + j\omega/z_1|$ and the phase is $\tan^{-1} \omega/z_1$. We notice that

$$H_{\text{dB}} = 20 \log_{10} \left| 1 + \frac{j\omega}{z_1} \right| \quad \Rightarrow \quad 20 \log_{10} 1 = 0 \quad \text{as} \quad \omega \to 0 \tag{14.16}$$

$$H_{\text{dB}} = 20 \log_{10} \left| 1 + \frac{j\omega}{z_1} \right| \quad \Rightarrow \quad 20 \log_{10} \frac{\omega}{z_1} \quad \text{as} \quad \omega \to \infty \tag{14.17}$$

showing that we can approximate the magnitude as zero (a straight line with zero slope) for small values of ω and by a straight line with slope 20 dB/decade for large values of ω. The frequency $\omega = z_1$ where the two asymptotic lines meet is called the *corner frequency* or *break frequency*. Thus the approximate magnitude plot is shown in Fig. 14.11(a), where the actual plot is also shown. Notice that the approximate plot is close to the actual plot except at the break frequency, where $\omega = z_1$ and the deviation is $20 \log_{10} |(1 + j1)| = 20 \log_{10} \sqrt{2} \simeq 3$ dB.

The phase $\tan^{-1}(\omega/z_1)$ can be expressed as

$$\phi = \tan^{-1}\left(\frac{\omega}{z_1}\right) = \begin{cases} 0, & \omega = 0 \\ 45^\circ, & \omega = z_1 \\ 90^\circ, & \omega \to \infty \end{cases} \tag{14.18}$$

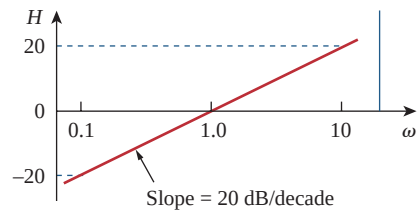
As a straight-line approximation, we let $\phi \simeq 0$ for $\omega \leq z_1/10$, $\phi \simeq 45^\circ$ for $\omega = z_1$, and $\phi \simeq 90^\circ$ for $\omega \geq 10z_1$. As shown in Fig. 14.11(b) along with the actual plot, the straight-line plot has a slope of 45° per decade.

The Bode plots for the pole $1/(1 + j\omega/p_1)$ are similar to those in Fig. 14.11 except that the corner frequency is at $\omega = p_1$, the magnitude has a slope of -20 dB/decade, and the phase has a slope of -45° per decade.

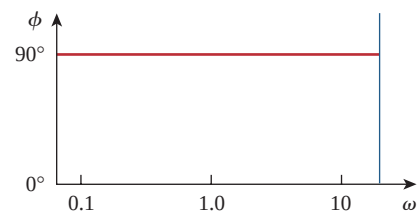
Quadratic pole/zero: The magnitude of the quadratic pole $1/[1 + j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2]$ is $-20 \log_{10} |1 + j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2|$ and the phase is $-\tan^{-1}(2\zeta_2\omega/\omega_n)/(1 - \omega^2/\omega_n^2)$. But

$$H_{\text{dB}} = -20 \log_{10} \left| 1 + \frac{j2\zeta_2\omega}{\omega_n} + \left(\frac{j\omega}{\omega_n}\right)^2 \right| \quad \Rightarrow \quad 0 \quad \text{as} \quad \omega \to 0 \tag{14.19}$$

The special case of dc ($\omega = 0$) does not appear on Bode plots because $\log 0 = -\infty$, implying that zero frequency is infinitely far to the left of the origin of Bode plots.

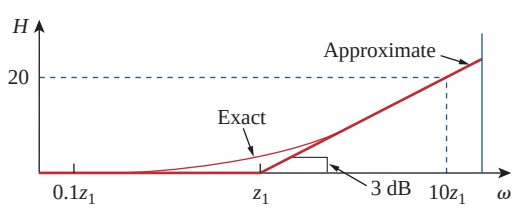


(a)

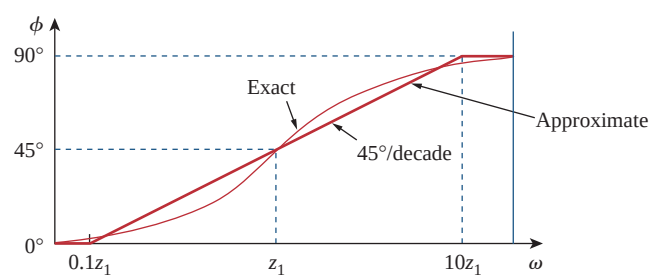


(b)

Figure 14.10
Bode plot for a zero ($j\omega$) at the origin: (a) magnitude plot, (b) phase plot.



(a)



(b)

Figure 14.11
Bode plots of zero $(1 + j\omega/z_1)$: (a) magnitude plot, (b) phase plot.

and

$$H_{\text{dB}} = -20\log_{10}\left|1 + \frac{j2\zeta_2\omega}{\omega_n} + \left(\frac{j\omega}{\omega_n}\right)^2\right| \quad \Rightarrow \quad -40\log_{10}\frac{\omega}{\omega_n} \quad \text{as} \quad \omega \to \infty \tag{14.20}$$

Thus, the amplitude plot consists of two straight asymptotic lines: one with zero slope for $\omega < \omega_n$ and the other with slope -40 dB/decade for $\omega > \omega_n$, with ω_n as the corner frequency. Figure 14.12(a) shows the approximate and actual amplitude plots. Note that the actual plot depends on the damping factor ζ_2 as well as the corner frequency ω_n. The significant peaking in the neighborhood of the corner frequency should be added to the straight-line approximation if a high level of accuracy is desired. However, we will use the straight-line approximation for the sake of simplicity.

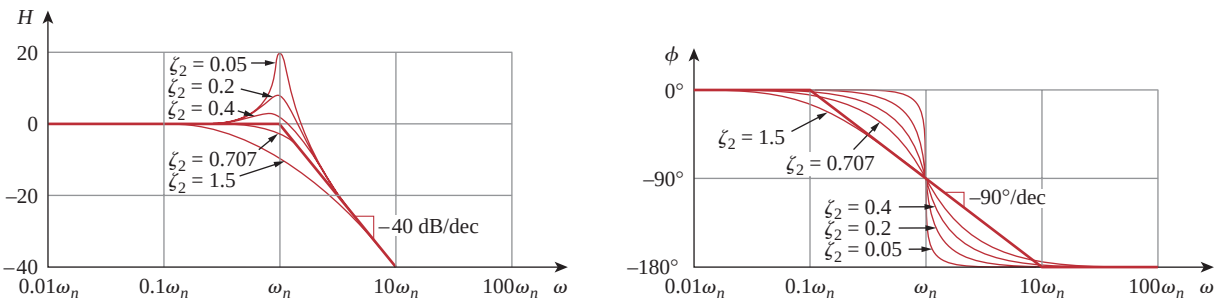


Figure 14.12
Bode plots of quadratic pole $[1 + j2\zeta\omega/\omega_n - \omega^2/\omega_n^2]^{-1}$: (a) magnitude plot, (b) phase plot.

There is another procedure for obtaining Bode plots that is faster and perhaps more efficient than the one we have just discussed. It consists in realizing that zeros cause an increase in slope, while poles cause a decrease. By starting with the low-frequency asymptote of the Bode plot, moving along the frequency axis, and increasing or decreasing the slope at each corner frequency, one can sketch the Bode plot immediately from the transfer function without the effort of making individual plots and adding them. This procedure can be used once you become proficient in the one discussed here.

Digital computers have rendered the procedure discussed here almost obsolete. Several software packages such as *PSpice*, *MATLAB*, Mathcad, and Micro-Cap can be used to generate frequency response plots. We will discuss *PSpice* later in the chapter.

The phase can be expressed as

$$\phi = -\tan^{-1}\frac{2\zeta_2\omega/\omega_n}{1 - \omega^2/\omega_n^2} = \begin{cases} 0, & \omega = 0 \\ -90^\circ, & \omega = \omega_n \\ -180^\circ, & \omega \to \infty \end{cases} \tag{14.21}$$

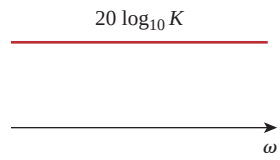
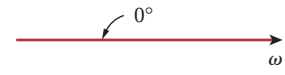
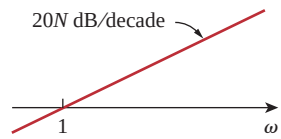
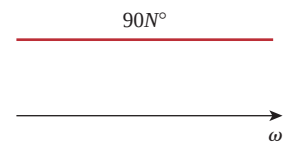
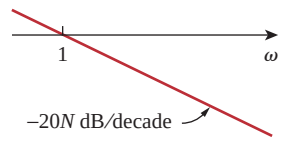
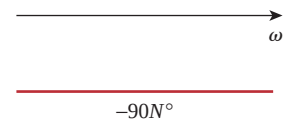
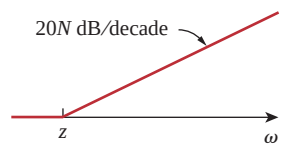
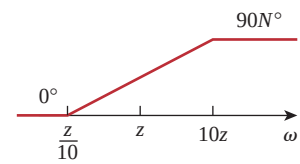
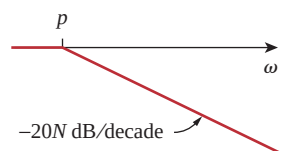
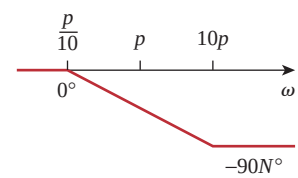
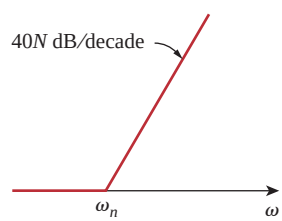
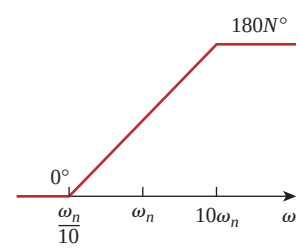
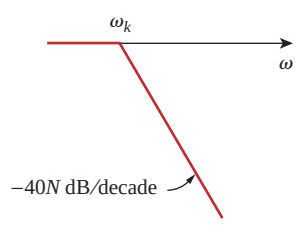
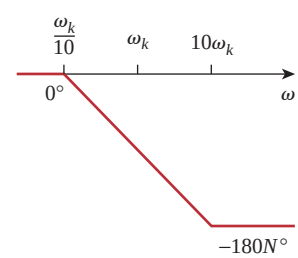
The phase plot is a straight line with a slope of -90° per decade starting at $\omega_n/10$ and ending at $10\omega_n$, as shown in Fig. 14.12(b). We see again that the difference between the actual plot and the straight-line plot is due to the damping factor. Notice that the straight-line approximations for both magnitude and phase plots for the quadratic pole are the same as those for a double pole, i.e. $(1 + j\omega/\omega_n)^{-2}$. We should expect this because the double pole $(1 + j\omega/\omega_n)^{-2}$ equals the quadratic pole $1/[1 + j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2]$ when $\zeta_2 = 1$. Thus, the quadratic pole can be treated as a double pole as far as straight-line approximation is concerned.

For the quadratic zero $[1 + j2\zeta_1\omega/\omega_k + (j\omega/\omega_k)^2]$, the plots in Fig. 14.12 are inverted because the magnitude plot has a slope of 40 dB/decade while the phase plot has a slope of 90° per decade.

Table 14.3 presents a summary of Bode plots for the seven factors. Of course, not every transfer function has all seven factors. To sketch the Bode plots for a function $\mathbf{H}(\omega)$ in the form of Eq. (14.15), for example, we first record the corner frequencies on the semilog graph paper, sketch the factors one at a time as discussed above, and then combine

TABLE 14.3

Summary of Bode straight-line magnitude and phase plots.

Factor	Magnitude	Phase
K		
$(j\omega)^N$		
$\frac{1}{(j\omega)^N}$		
$\left(1+\frac{j\omega}{z}\right)^N$		
$\frac{1}{(1+j\omega/p)^N}$		
$\left[1+\frac{2j\omega\zeta}{\omega_n}+\left(\frac{j\omega}{\omega_n}\right)^2\right]^N$		
$\frac{1}{[1+2j\omega\zeta/\omega_k+(j\omega/\omega_k)^2]^N}$		

additively the graphs of the factors. The combined graph is often drawn from left to right, changing slopes appropriately each time a corner frequency is encountered. The following examples illustrate this procedure.

Example 14.3

Construct the Bode plots for the transfer function

$$\mathbf{H}(\omega) = \frac{200\,j\omega}{(j\omega + 2)(j\omega + 10)}$$

Solution:

We first put $\mathbf{H}(\omega)$ in the standard form by dividing out the poles and zeros. Thus,

$$\mathbf{H}(\omega) = \frac{10\,j\omega}{(1 + j\omega/2)(1 + j\omega/10)}$$

$$= \frac{10|j\omega|}{|1 + j\omega/2||1 + j\omega/10|}\underline{/90^\circ - \tan^{-1}\omega/2 - \tan^{-1}\omega/10}$$

Hence, the magnitude and phase are

$$H_{\text{dB}} = 20\log_{10}10 + 20\log_{10}|j\omega| - 20\log_{10}\left|1 + \frac{j\omega}{2}\right| - 20\log_{10}\left|1 + \frac{j\omega}{10}\right|$$

$$\phi = 90^\circ - \tan^{-1}\frac{\omega}{2} - \tan^{-1}\frac{\omega}{10}$$

We notice that there are two corner frequencies at $\omega = 2, 10$. For both the magnitude and phase plots, we sketch each term as shown by the dotted lines in Fig. 14.13. We add them up graphically to obtain the overall plots shown by the solid curves.

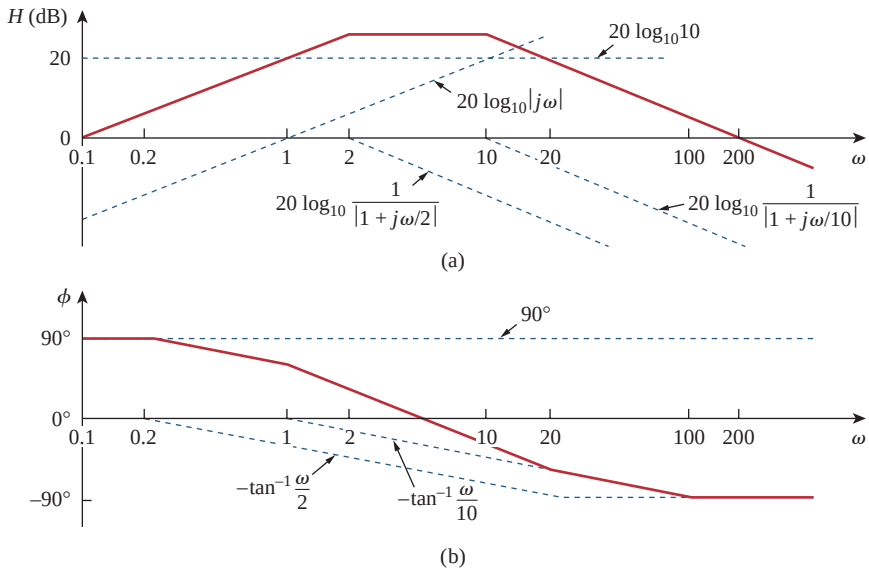


Figure 14.13
For Example 14.3: (a) magnitude plot, (b) phase plot.

Practice Problem 14.3

Draw the Bode plots for the transfer function

$$\mathbf{H}(\omega) = \frac{5(j\omega + 2)}{j\omega(j\omega + 10)}$$

Answer: See Fig. 14.14.

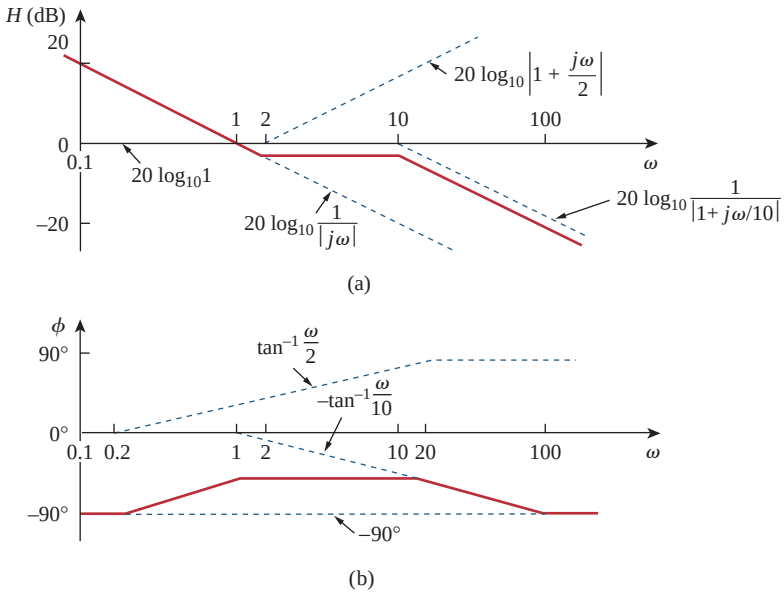


Figure 14.14
For Practice Prob. 14.3: (a) magnitude plot, (b) phase plot.

Example 14.4

Obtain the Bode plots for

$$\mathbf{H}(\omega) = \frac{j\omega + 10}{j\omega(j\omega + 5)^2}$$

Solution:
Putting $\mathbf{H}(\omega)$ in the standard form, we get

$$\mathbf{H}(\omega) = \frac{0.4(1 + j\omega/10)}{j\omega(1 + j\omega/5)^2}$$

From this, we obtain the magnitude and phase as

$$H_{\text{dB}} = 20\log_{10}0.4 + 20\log_{10}\left|1 + \frac{j\omega}{10}\right| - 20\log_{10}|j\omega| - 40\log_{10}\left|1 + \frac{j\omega}{5}\right|$$

$$\phi = 0° + \tan^{-1}\frac{\omega}{10} - 90° - 2\tan^{-1}\frac{\omega}{5}$$

There are two corner frequencies at $\omega = 5, 10$ rad/s. For the pole with corner frequency at $\omega = 5$, the slope of the magnitude plot is -40 dB/decade and that of the phase plot is $-90°$ per decade due to the power of 2. The

magnitude and the phase plots for the individual terms (in dotted lines) and the entire $\mathbf{H}(j\omega)$ (in solid lines) are in Fig. 14.15.

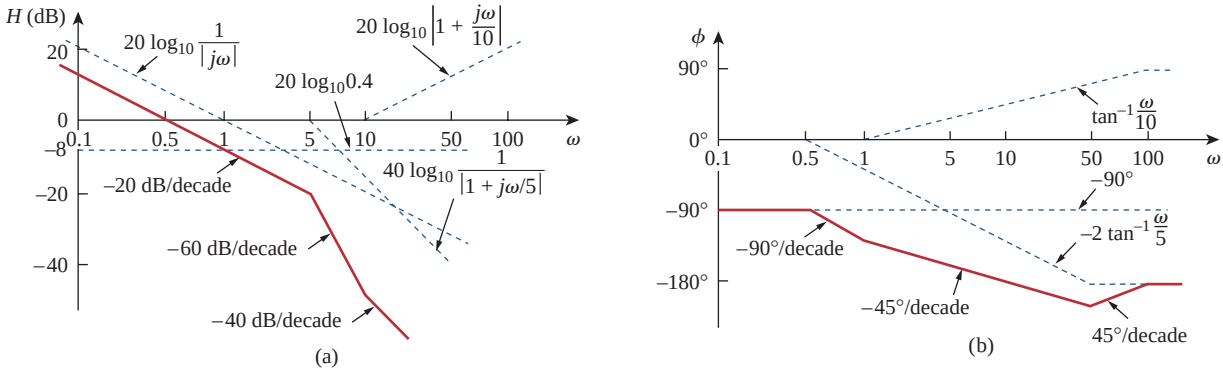


Figure 14.15
Bode plots for Example 14.4: (a) magnitude plot, (b) phase plot.

Practice Problem 14.4

Sketch the Bode plots for

$$\mathbf{H}(\omega) = \frac{50\,j\omega}{(j\omega + 4)(j\omega + 10)^2}$$

Answer: See Fig. 14.16.

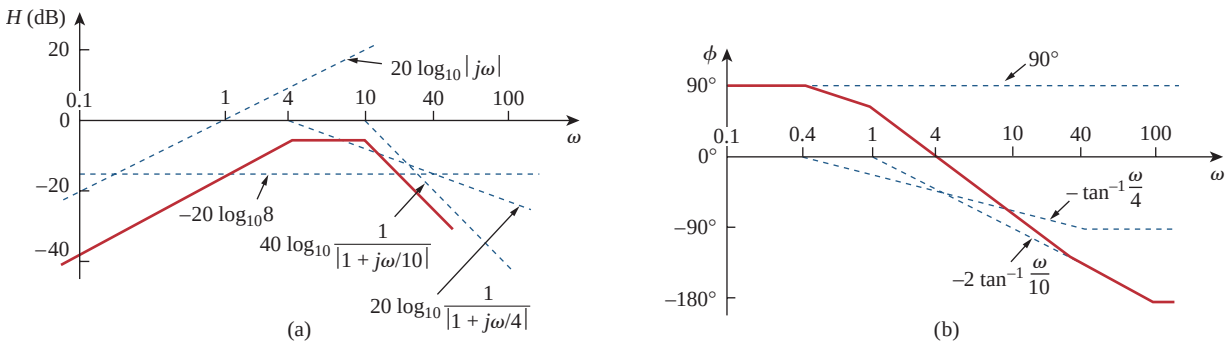


Figure 14.16
For Practice Prob. 14.4: (a) magnitude plot, (b) phase plot.

Example 14.5

Draw the Bode plots for

$$\mathbf{H}(s) = \frac{s + 1}{s^2 + 12s + 100}$$

Solution:

1. **Define.** The problem is clearly stated and we follow the technique outlined in the chapter.
2. **Present.** We are to develop the approximate bode plot for the given function, $\mathbf{H}(s)$.
3. **Alternative.** The two most effective choices would be the approximation technique outlined in the chapter, which we will

use here, and *MATLAB*, which can actually give us the exact Bode plots.

4. **Attempt.** We express $\mathbf{H}(s)$ as

$$\mathbf{H}(\omega) = \frac{1/100(1 + j\omega)}{1 + j\omega 1.2/10 + (j\omega/10)^2}$$

For the quadratic pole, $\omega_n = 10$ rad/s, which serves as the corner frequency. The magnitude and phase are

$$H_{\text{dB}} = -20 \log_{10} 100 + 20 \log_{10} |1 + j\omega| - 20 \log_{10} \left| 1 + \frac{j\omega 1.2}{10} - \frac{\omega^2}{100} \right|$$

$$\phi = 0° + \tan^{-1} \omega - \tan^{-1} \left[\frac{\omega 1.2/10}{1 - \omega^2/100} \right]$$

Figure 14.17 shows the Bode plots. Notice that the quadratic pole is treated as a repeated pole at ω_k, that is, $(1 + j\omega/\omega_k)^2$, which is an approximation.

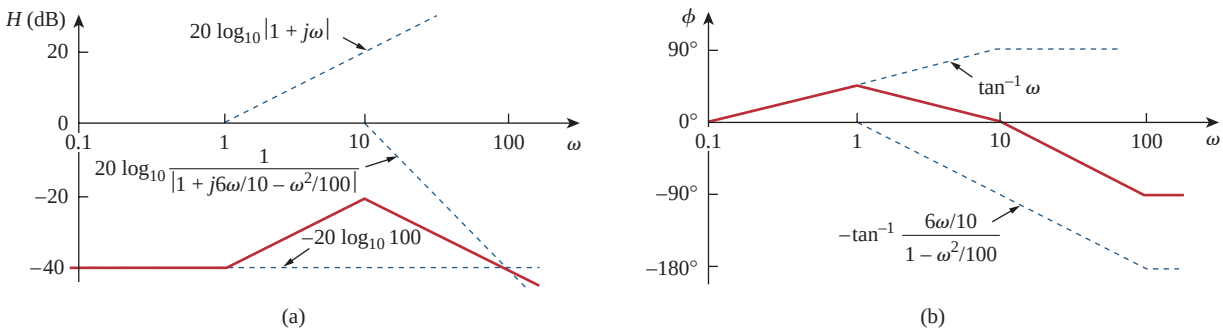


Figure 14.17
Bode plots for Example 14.5: (a) magnitude plot, (b) phase plot.

5. **Evaluate.** Although we could use *MATLAB* to validate the solution, we will use a more straightforward approach. First, we must realize that the denominator assumes that $\zeta = 0$ for the approximation, so we will use the following equation to check our answer:

$$\mathbf{H}(s) \simeq \frac{s + 1}{s^2 + 10^2}$$

We also note that we need to actually solve for H_{dB} and the corresponding phase angle ϕ. First, let $\omega = 0$.

$$H_{\text{dB}} = 20 \log_{10}(1/100) = -40 \qquad \text{and} \qquad \phi = 0°$$

Now try $\omega = 1$.

$$H_{\text{dB}} = 20 \log_{10}(1.4142/99) = -36.9 \text{ dB}$$

which is the expected 3 dB up from the corner frequency.

$$\phi = 45° \qquad \text{from} \qquad \mathbf{H}(j) = \frac{j + 1}{-1 + 100}$$

Now try $\omega = 100$.

$$H_{\text{dB}} = 20\log_{10}(100) - 20\log_{10}(9900) = 39.91 \text{ dB}$$

ϕ is 90° from the numerator minus 180°, which gives −90°. We now have checked three different points and got close agreement, and, since this is an approximation, we can feel confident that we have worked the problem successfully.

You can reasonably ask why did we not check at $\omega = 10$? If we just use the approximate value we used above, we end up with an infinite value, which is to be expected from $\zeta = 0$ (see Fig. 14.12a). If we used the actual value of $\mathbf{H}(j10)$ we will still end up being far from the approximate values, since $\zeta = 0.6$ and Fig. 14.12a shows a significant deviation from the approximation. We could have reworked the problem with $\zeta = 0.707$, which would have gotten us closer to the approximation. However, we really have enough points without doing this.

6. **Satisfactory?** We are satisfied the problem has been worked successfully and we can present the results as a solution to the problem.

Practice Problem 14.5

Construct the Bode plots for

$$H(s) = \frac{10}{s(s^2 + 80s + 400)}$$

Answer: See Fig. 14.18.

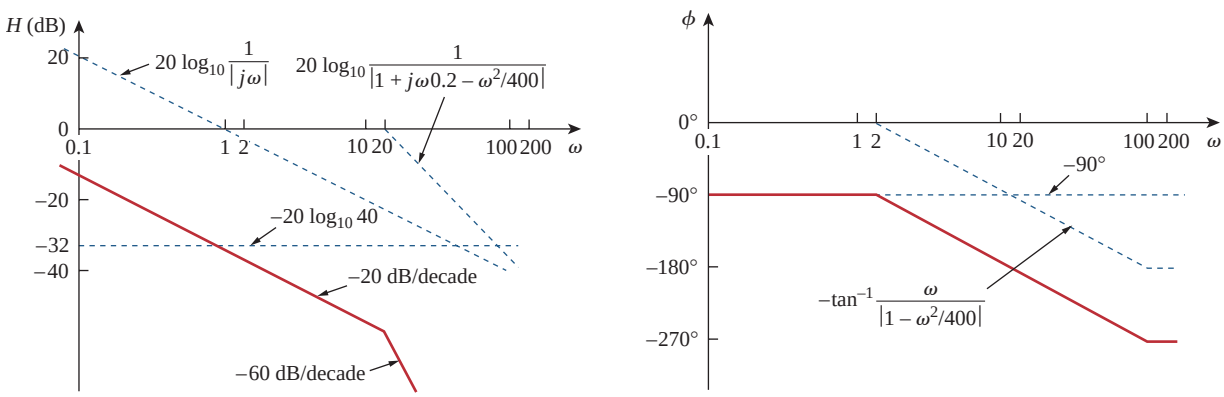


Figure 14.18
For Practice Prob. 14.5: (a) magnitude plot, (b) phase plot.

Example 14.6

Given the Bode plot in Fig. 14.19, obtain the transfer function $\mathbf{H}(\omega)$.

Solution:

To obtain $\mathbf{H}(\omega)$ from the Bode plot, we keep in mind that a zero always causes an upward turn at a corner frequency, while a pole causes a

downward turn. We notice from Fig. 14.19 that there is a zero $j\omega$ at the origin which should have intersected the frequency axis at $\omega = 1$. This is indicated by the straight line with slope +20 dB/decade. The fact that this straight line is shifted by 40 dB indicates that there is a 40-dB gain; that is,

$$40 = 20 \log_{10} K \quad \Rightarrow \quad \log_{10} K = 2$$

or

$$K = 10^2 = 100$$

In addition to the zero $j\omega$ at the origin, we notice that there are three factors with corner frequencies at $\omega = 1$, 5, and 20 rad/s. Thus, we have:

1. A pole at $p = 1$ with slope −20 dB/decade to cause a downward turn and counteract the zero at the origin. The pole at $p = 1$ is determined as $1/(1 + j\omega/1)$.
2. Another pole at $p = 5$ with slope −20 dB/decade causing a downward turn. The pole is $1/(1 + j\omega/5)$.
3. A third pole at $p = 20$ with slope −20 dB/decade causing a further downward turn. The pole is $1/(1 + j\omega/20)$.

Putting all these together gives the corresponding transfer function as

$$\mathbf{H}(\omega) = \frac{100 j\omega}{(1 + j\omega/1)(1 + j\omega/5)(1 + j\omega/20)} = \frac{j\omega 10^4}{(j\omega + 1)(j\omega + 5)(j\omega + 20)}$$

or

$$\mathbf{H}(s) = \frac{10^4 s}{(s + 1)(s + 5)(s + 20)}, \qquad s = j\omega$$

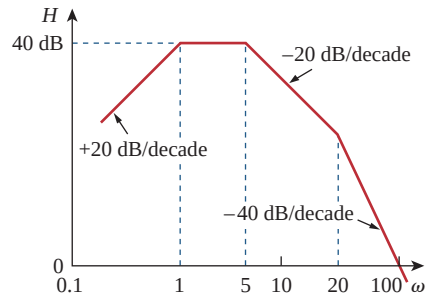


Figure 14.19
For Example 14.6.

Practice Problem 14.6

Obtain the transfer function $\mathbf{H}(\omega)$ corresponding to the Bode plot in Fig. 14.20.

Answer: $\mathbf{H}(\omega) = \dfrac{4{,}000(s + 5)}{(s + 10)(s + 100)^2}$.

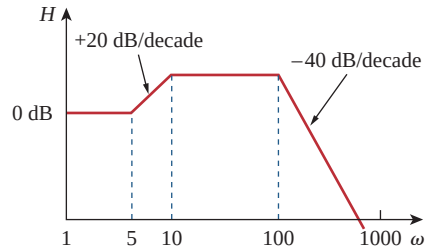


Figure 14.20
For Practice Prob. 14.6.

To see how to use *MATLAB* to produce Bode plots, refer to Section 14.11.

14.5 Series Resonance

The most prominent feature of the frequency response of a circuit may be the sharp peak (or *resonant peak*) exhibited in its amplitude characteristic. The concept of resonance applies in several areas of science and engineering. Resonance occurs in any system that has a complex conjugate pair of poles; it is the cause of oscillations of stored energy from one form to another. It is the phenomenon that allows frequency

discrimination in communications networks. Resonance occurs in any circuit that has at least one inductor and one capacitor.

Resonance is a condition in an *RLC* circuit in which the capacitive and inductive reactances are equal in magnitude, thereby resulting in a purely resistive impedance.

Resonant circuits (series or parallel) are useful for constructing filters, as their transfer functions can be highly frequency selective. They are used in many applications such as selecting the desired stations in radio and TV receivers.

Consider the series *RLC* circuit shown in Fig. 14.21 in the frequency domain. The input impedance is

$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C} \tag{14.22}$$

or

$$\mathbf{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right) \tag{14.23}$$

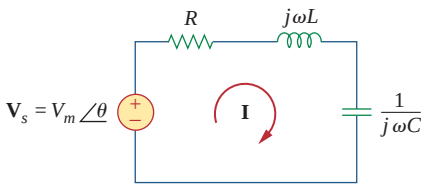


Figure 14.21
The series resonant circuit.

Resonance results when the imaginary part of the transfer function is zero, or

$$\text{Im}(\mathbf{Z}) = \omega L - \frac{1}{\omega C} = 0 \tag{14.24}$$

The value of ω that satisfies this condition is called the *resonant frequency* ω_0. Thus, the resonance condition is

$$\omega_0 L = \frac{1}{\omega_0 C} \tag{14.25}$$

or

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}} \tag{14.26}$$

Since $\omega_0 = 2\pi f_0$,

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz} \tag{14.27}$$

Note that at resonance:

1. The impedance is purely resistive, thus, $\mathbf{Z} = R$. In other words, the *LC* series combination acts like a short circuit, and the entire voltage is across *R*.
2. The voltage $\mathbf{V}_s$ and the current $\mathbf{I}$ are in phase, so that the power factor is unity.
3. The magnitude of the transfer function $\mathbf{H}(\omega) = \mathbf{Z}(\omega)$ is minimum.
4. The inductor voltage and capacitor voltage can be much more than the source voltage.

The frequency response of the circuit's current magnitude

$$I = |\mathbf{I}| = \frac{V_m}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}} \tag{14.28}$$

Note No. 4 becomes evident from the fact that

$$|\mathbf{V}_L| = \frac{V_m}{R}\omega_0 L = QV_m$$

$$|\mathbf{V}_C| = \frac{V_m}{R}\frac{1}{\omega_0 C} = QV_m$$

where *Q* is the quality factor, defined in Eq. (14.38).

is shown in Fig. 14.22; the plot only shows the symmetry illustrated in this graph when the frequency axis is a logarithm. The average power dissipated by the *RLC* circuit is

$$P(\omega) = \frac{1}{2} I^2 R \tag{14.29}$$

The highest power dissipated occurs at resonance, when $I = V_m/R$, so that

$$P(\omega_0) = \frac{1}{2}\frac{V_m^2}{R} \tag{14.30}$$

At certain frequencies $\omega = \omega_1, \omega_2$, the dissipated power is half the maximum value; that is,

$$P(\omega_1) = P(\omega_2) = \frac{(V_m/\sqrt{2})^2}{2R} = \frac{V_m^2}{4R} \tag{14.31}$$

Hence, ω_1 and ω_2 are called the *half-power frequencies.*

The half-power frequencies are obtained by setting Z equal to $\sqrt{2}R$, and writing

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R \tag{14.32}$$

Solving for ω, we obtain

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$
$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \tag{14.33}$$

We can relate the half-power frequencies with the resonant frequency. From Eqs. (14.26) and (14.33),

$$\omega_0 = \sqrt{\omega_1 \omega_2} \tag{14.34}$$

showing that the resonant frequency is the geometric mean of the half-power frequencies. Notice that ω_1 and ω_2 are in general not symmetrical around the resonant frequency ω_0, because the frequency response is not generally symmetrical. However, as will be explained shortly, symmetry of the half-power frequencies around the resonant frequency is often a reasonable approximation.

Although the height of the curve in Fig. 14.22 is determined by R, the width of the curve depends on other factors. The width of the response curve depends on the *bandwidth B*, which is defined as the difference between the two half-power frequencies,

$$B = \omega_2 - \omega_1 \tag{14.35}$$

This definition of bandwidth is just one of several that are commonly used. Strictly speaking, B in Eq. (14.35) is a half-power bandwidth, because it is the width of the frequency band between the half-power frequencies.

The "sharpness" of the resonance in a resonant circuit is measured quantitatively by the *quality factor Q*. At resonance, the reactive energy

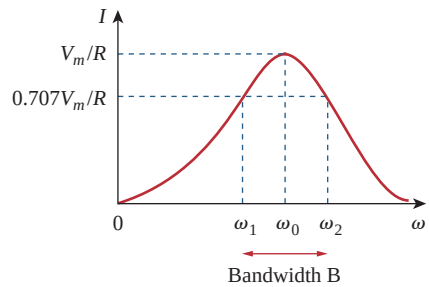


Figure 14.22
The current amplitude versus frequency for the series resonant circuit of Fig. 14.21.

Although the same symbol Q is used for the reactive power, the two are not equal and should not be confused. Q here is dimensionless, whereas reactive power Q is in VAR. This may help distinguish between the two.

in the circuit oscillates between the inductor and the capacitor. The quality factor relates the maximum or peak energy stored to the energy dissipated in the circuit per cycle of oscillation:

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}} \tag{14.36}$$

It is also regarded as a measure of the energy storage property of a circuit in relation to its energy dissipation property. In the series RLC circuit, the peak energy stored is $\frac{1}{2}LI^2$, while the energy dissipated in one period is $\frac{1}{2}(I^2R)(1/f_0)$. Hence,

$$Q = 2\pi \frac{\frac{1}{2}LI^2}{\frac{1}{2}I^2R(1/f_0)} = \frac{2\pi f_0 L}{R} \tag{14.37}$$

or

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR} \tag{14.38}$$

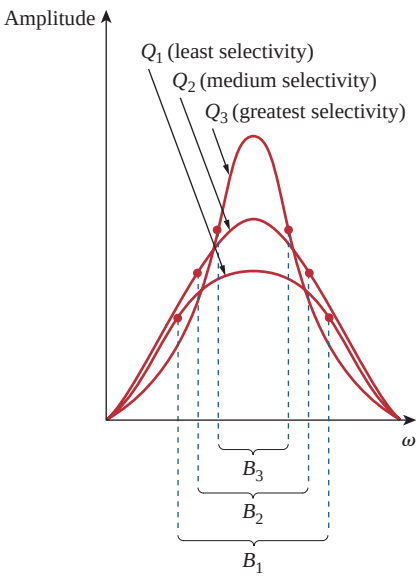


Figure 14.23
The higher the circuit Q, the smaller the bandwidth.

Notice that the quality factor is dimensionless. The relationship between the bandwidth B and the quality factor Q is obtained by substituting Eq. (14.33) into Eq. (14.35) and utilizing Eq. (14.38).

$$B = \frac{R}{L} = \frac{\omega_0}{Q} \tag{14.39}$$

or $B = \omega_0^2 CR$. Thus

The quality factor of a resonant circuit is the ratio of its resonant frequency to its bandwidth.

The quality factor is a measure of the selectivity (or "sharpness" of resonance) of the circuit.

Keep in mind that Eqs. (14.33), (14.38), and (14.39) only apply to a series RLC circuit.

As illustrated in Fig. 14.23, the higher the value of Q, the more selective the circuit is but the smaller the bandwidth. The *selectivity* of an RLC circuit is the ability of the circuit to respond to a certain frequency and discriminate against all other frequencies. If the band of frequencies to be selected or rejected is narrow, the quality factor of the resonant circuit must be high. If the band of frequencies is wide, the quality factor must be low.

A resonant circuit is designed to operate at or near its resonant frequency. It is said to be a *high-Q circuit* when its quality factor is equal to or greater than 10. For high-Q circuits ($Q \geq 10$), the half-power frequencies are, for all practical purposes, symmetrical around the resonant frequency and can be approximated as

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \qquad \omega_2 \simeq \omega_0 + \frac{B}{2} \tag{14.40}$$

High-Q circuits are used often in communications networks.

We see that a resonant circuit is characterized by five related parameters: the two half-power frequencies ω_1 and ω_2, the resonant frequency ω_0, the bandwidth B, and the quality factor Q.

Example 14.7

In the circuit of Fig. 14.24, $R = 2\ \Omega$, $L = 1$ mH, and $C = 0.4\ \mu$F. (a) Find the resonant frequency and the half-power frequencies. (b) Calculate the quality factor and bandwidth. (c) Determine the amplitude of the current at ω_0, ω_1, and ω_2.

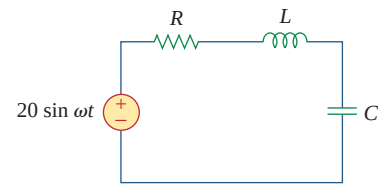


Figure 14.24
For Example 14.7.

Solution:

(a) The resonant frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 0.4 \times 10^{-6}}} = 50 \text{ krad/s}$$

■ **METHOD 1** The lower half-power frequency is

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$= -\frac{2}{2 \times 10^{-3}} + \sqrt{(10^3)^2 + (50 \times 10^3)^2}$$

$$= -1 + \sqrt{1 + 2500} \text{ krad/s} = 49 \text{ krad/s}$$

Similarly, the upper half-power frequency is

$$\omega_2 = 1 + \sqrt{1 + 2500} \text{ krad/s} = 51 \text{ krad/s}$$

(b) The bandwidth is

$$B = \omega_2 - \omega_1 = 2 \text{ krad/s}$$

or

$$B = \frac{R}{L} = \frac{2}{10^{-3}} = 2 \text{ krad/s}$$

The quality factor is

$$Q = \frac{\omega_0}{B} = \frac{50}{2} = 25$$

■ **METHOD 2** Alternatively, we could find

$$Q = \frac{\omega_0 L}{R} = \frac{50 \times 10^3 \times 10^{-3}}{2} = 25$$

From Q, we find

$$B = \frac{\omega_0}{Q} = \frac{50 \times 10^3}{25} = 2 \text{ krad/s}$$

Since $Q > 10$, this is a high-Q circuit and we can obtain the half-power frequencies as

$$\omega_1 = \omega_0 - \frac{B}{2} = 50 - 1 = 49 \text{ krad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 50 + 1 = 51 \text{ krad/s}$$

as obtained earlier.
(c) At $\omega = \omega_0$,

$$I = \frac{V_m}{R} = \frac{20}{2} = 10 \text{ A}$$

At $\omega = \omega_1, \omega_2$,

$$I = \frac{V_m}{\sqrt{2}R} = \frac{10}{\sqrt{2}} = 7.071 \text{ A}$$

Practice Problem 14.7

A series-connected circuit has $R = 4\ \Omega$ and $L = 25$ mH. (a) Calculate the value of C that will produce a quality factor of 50. (b) Find ω_1, ω_2, and B. (c) Determine the average power dissipated at $\omega = \omega_0, \omega_1, \omega_2$. Take $V_m = 100$ V.

Answer: (a) 0.625 μF, (b) 7920 rad/s, 8080 rad/s, 160 rad/s, (c) 1.25 kW, 0.625 kW, 0.625 kW.

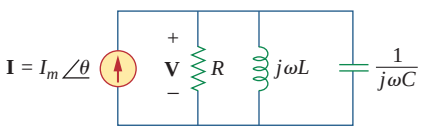


Figure 14.25
The parallel resonant circuit.

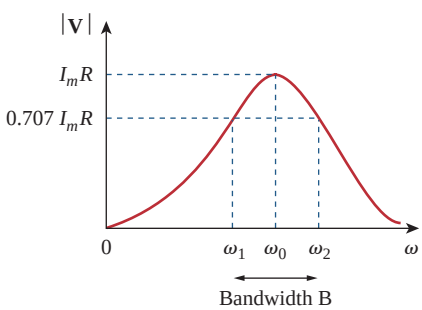


Figure 14.26
The current amplitude versus frequency for the series resonant circuit of Fig. 14.25.

14.6 Parallel Resonance

The parallel *RLC* circuit in Fig. 14.25 is the dual of the series *RLC* circuit. So we will avoid needless repetition. The admittance is

$$\mathbf{Y} = H(\omega) = \frac{\mathbf{I}}{\mathbf{V}} = \frac{1}{R} + j\omega C + \frac{1}{j\omega L} \tag{14.41}$$

or

$$\mathbf{Y} = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right) \tag{14.42}$$

Resonance occurs when the imaginary part of $\mathbf{Y}$ is zero,

$$\omega C - \frac{1}{\omega L} = 0 \tag{14.43}$$

or

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}} \tag{14.44}$$

which is the same as Eq. (14.26) for the series resonant circuit. The voltage $|\mathbf{V}|$ is sketched in Fig. 14.26 as a function of frequency. Notice that at resonance, the parallel *LC* combination acts like an open circuit, so that the entire current flows through *R*. Also, the inductor and capacitor current can be much more than the source current at resonance.

We exploit the duality between Figs. 14.21 and 14.25 by comparing Eq. (14.42) with Eq. (14.23). By replacing *R*, *L*, and *C* in the

We can see this from the fact that

$$|\mathbf{I}_L| = \frac{I_m R}{\omega_0 L} = QI_m$$

$$|\mathbf{I}_C| = \omega_0 C I_m R = QI_m$$

where Q is the quality factor, defined in Eq. (14.47).

expressions for the series circuit with $1/R$, C, and L respectively, we obtain for the parallel circuit

$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$
$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}} \tag{14.45}$$

$$B = \omega_2 - \omega_1 = \frac{1}{RC} \tag{14.46}$$

$$Q = \frac{\omega_0}{B} = \omega_0 RC = \frac{R}{\omega_0 L} \tag{14.47}$$

It should be noted that Eqs. (14.45) to (14.47) apply only to a parallel *RLC* circuit. Using Eqs. (14.45) and (14.47), we can express the half-power frequencies in terms of the quality factor. The result is

$$\omega_1 = \omega_0\sqrt{1 + \left(\frac{1}{2Q}\right)^2} - \frac{\omega_0}{2Q}, \qquad \omega_2 = \omega_0\sqrt{1 + \left(\frac{1}{2Q}\right)^2} + \frac{\omega_0}{2Q} \tag{14.48}$$

Again, for high-Q circuits ($Q \geq 10$)

$$\omega_1 \simeq \omega_0 - \frac{B}{2}, \qquad \omega_2 \simeq \omega_0 + \frac{B}{2} \tag{14.49}$$

Table 14.4 presents a summary of the characteristics of the series and parallel resonant circuits. Besides the series and parallel *RLC* considered here, other resonant circuits exist. Example 14.9 treats a typical example.

TABLE 14.4

Summary of the characteristics of resonant *RLC* circuits.

Characteristic	Series circuit	Parallel circuit
Resonant frequency, ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor, Q	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$
Bandwidth, B	$\frac{\omega_0}{Q}$	$\frac{\omega_0}{Q}$
Half-power frequencies, ω_1, ω_2	$\omega_0\sqrt{1+\left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$	$\omega_0\sqrt{1+\left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$
For $Q \geq 10$, ω_1, ω_2	$\omega_0 \pm \frac{B}{2}$	$\omega_0 \pm \frac{B}{2}$

Example 14.8

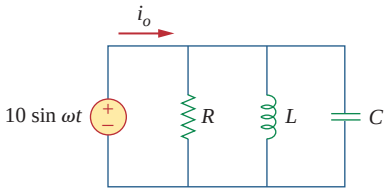


Figure 14.27
For Example 14.8.

In the parallel *RLC* circuit of Fig. 14.27, let $R = 8\ \text{k}\Omega$, $L = 0.2$ mH, and $C = 8\ \mu\text{F}$. (a) Calculate ω_0, Q, and B. (b) Find ω_1 and ω_2. (c) Determine the power dissipated at ω_0, ω_1, and ω_2.

Solution:

(a)

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.2 \times 10^{-3} \times 8 \times 10^{-6}}} = \frac{10^5}{4} = 25\ \text{krad/s}$$

$$Q = \frac{R}{\omega_0 L} = \frac{8 \times 10^3}{25 \times 10^3 \times 0.2 \times 10^{-3}} = 1{,}600$$

$$B = \frac{\omega_0}{Q} = 15.625\ \text{rad/s}$$

(b) Due to the high value of Q, we can regard this as a high-Q circuit, Hence,

$$\omega_1 = \omega_0 - \frac{B}{2} = 25{,}000 - 7.812 = 24{,}992\ \text{rad/s}$$

$$\omega_2 = \omega_0 + \frac{B}{2} = 25{,}000 + 7.812 = 25{,}008\ \text{rad/s}$$

(c) At $\omega = \omega_0$, $\mathbf{Y} = 1/R$ or $\mathbf{Z} = R = 8\ \text{k}\Omega$. Then

$$\mathbf{I}_o = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{10\angle -90^\circ}{8{,}000} = 1.25\angle -90^\circ\ \text{mA}$$

Since the entire current flows through R at resonance, the average power dissipated at $\omega = \omega_0$ is

$$P = \frac{1}{2}|\mathbf{I}_o|^2 R = \frac{1}{2}(1.25 \times 10^{-3})^2 (8 \times 10^3) = 6.25\ \text{mW}$$

or

$$P = \frac{V_m^2}{2R} = \frac{100}{2 \times 8 \times 10^3} = 6.25\ \text{mW}$$

At $\omega = \omega_1, \omega_2$,

$$P = \frac{V_m^2}{4R} = 3.125\ \text{mW}$$

Practice Problem 14.8

A parallel resonant circuit has $R = 100\ \text{k}\Omega$, $L = 20$ mH, and $C = 5$ nF. Calculate ω_0, ω_1, ω_2, Q, and B.

Answer: 100 krad/s, 99 krad/s, 101 krad/s, 50, 2 krad/s.

Example 14.9

Determine the resonant frequency of the circuit in Fig. 14.28.

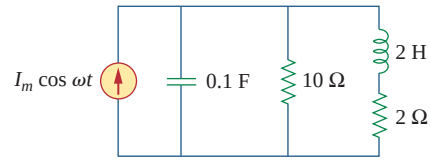


Figure 14.28
For Example 14.9.

Solution:
The input admittance is

$$\mathbf{Y} = j\omega 0.1 + \frac{1}{10} + \frac{1}{2 + j\omega 2} = 0.1 + j\omega 0.1 + \frac{2 - j\omega 2}{4 + 4\omega^2}$$

At resonance, $\text{Im}(\mathbf{Y}) = 0$ and

$$\omega_0 0.1 - \frac{2\omega_0}{4 + 4\omega_0^2} = 0 \quad \Rightarrow \quad \omega_0 = 2 \text{ rad/s}$$

Practice Problem 14.9

Calculate the resonant frequency of the circuit in Fig. 14.29.

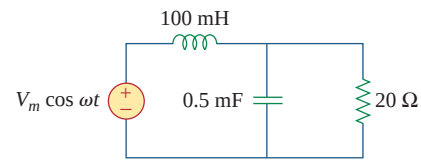


Figure 14.29
For Practice Prob. 14.9

Answer: 100 rad/s.

14.7 Passive Filters

The concept of filters has been an integral part of the evolution of electrical engineering from the beginning. Several technological achievements would not have been possible without electrical filters. Because of this prominent role of filters, much effort has been expended on the theory, design, and construction of filters and many articles and books have been written on them. Our discussion in this chapter should be considered introductory.

A **filter** is a circuit that is designed to pass signals with desired frequencies and reject or attenuate others.

As a frequency-selective device, a filter can be used to limit the frequency spectrum of a signal to some specified band of frequencies. Filters are the circuits used in radio and TV receivers to allow us to select one desired signal out of a multitude of broadcast signals in the environment.

A filter is a *passive filter* if it consists of only passive elements *R*, *L*, and *C*. It is said to be an *active filter* if it consists of active elements (such as transistors and op amps) in addition to passive elements *R*, *L*, and *C*. We consider passive filters in this section and active filters in the next section. *LC* filters have been used in practical applications for more than eight decades. *LC* filter technology feeds related areas such as equalizers, impedance-matching networks, transformers, shaping networks, power dividers, attenuators, and directional couplers, and is continuously providing practicing engineers with oppurtunities to innovate and experiment. Besides the *LC* filters we study in these sections, there are other kinds of filters—such as digital filters, electromechanical filters, and microwave filters—which are beyond the level of this text.

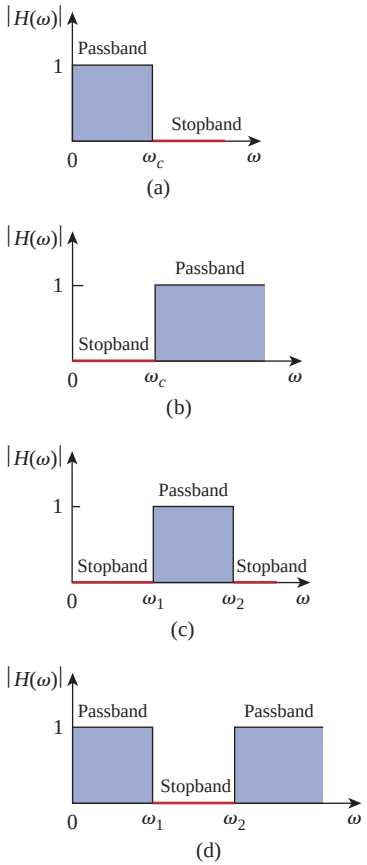


Figure 14.30
Ideal frequency response of four types of filter: (a) lowpass filter, (b) highpass filter, (c) bandpass filter, (d) bandstop filter.

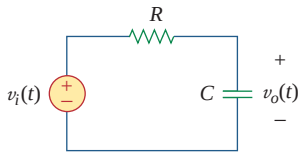


Figure 14.31
A lowpass filter.

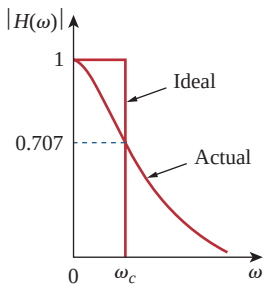


Figure 14.32
Ideal and actual frequency response of a lowpass filter.

As shown in Fig. 14.30, there are four types of filters whether passive or active:

1. A *lowpass filter* passes low frequencies and stops high frequencies, as shown ideally in Fig. 14.30(a).
2. A *highpass filter* passes high frequencies and rejects low frequencies, as shown ideally in Fig. 14.30(b).
3. A *bandpass filter* passes frequencies within a frequency band and blocks or attenuates frequencies outside the band, as shown ideally in Fig. 14.30(c).
4. A *bandstop filter* passes frequencies outside a frequency band and blocks or attenuates frequencies within the band, as shown ideally in Fig. 14.30(d).

Table 14.5 presents a summary of the characteristics of these filters. Be aware that the characteristics in Table 14.5 are only valid for first- or second-order filters—but one should not have the impression that only these kinds of filter exist. We now consider typical circuits for realizing the filters shown in Table 14.5.

TABLE 14.5

Summary of the characteristics of ideal filters.

Type of Filter	$H(0)$	$H(\infty)$	$H(\omega_c)$ or $H(\omega_0)$
Lowpass	1	0	$1/\sqrt{2}$
Highpass	0	1	$1/\sqrt{2}$
Bandpass	0	0	1
Bandstop	1	1	0

ω_c is the cutoff frequency for lowpass and highpass filters; ω_0 is the center frequency for bandpass and bandstop filters.

14.7.1 Lowpass Filter

A typical lowpass filter is formed when the output of an *RC* circuit is taken off the capacitor as shown in Fig. 14.31. The transfer function (see also Example 14.1) is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{1/j\omega C}{R + 1/j\omega C}$$

$$\mathbf{H}(\omega) = \frac{1}{1 + j\omega RC} \tag{14.50}$$

Note that $\mathbf{H}(0) = 1$, $\mathbf{H}(\infty) = 0$. Figure 14.32 shows the plot of $|H(\omega)|$, along with the ideal characteristic. The half-power frequency, which is equivalent to the corner frequency on the Bode plots but in the context of filters is usually known as the *cutoff frequency* ω_c, is obtained by setting the magnitude of $\mathbf{H}(\omega)$ equal to $1/\sqrt{2}$, thus,

$$H(\omega_c) = \frac{1}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}}$$

or

$$\omega_c = \frac{1}{RC} \tag{14.51}$$

The cutoff frequency is also called the *rolloff frequency.*

A **lowpass filter** is designed to pass only frequencies from dc up to the cutoff frequency ω_c.

A lowpass filter can also be formed when the output of an *RL* circuit is taken off the resistor. Of course, there are many other circuits for lowpass filters.

The cutoff frequency is the frequency at which the transfer function **H** drops in magnitude to 70.71% of its maximum value. It is also regarded as the frequency at which the power dissipated in a circuit is half of its maximum value.

14.7.2. Highpass Filter

A highpass filter is formed when the output of an *RC* circuit is taken off the resistor as shown in Fig. 14.33. The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + 1/j\omega C}$$

$$\mathbf{H}(\omega) = \frac{j\omega RC}{1 + j\omega RC} \tag{14.52}$$

Note that $\mathbf{H}(0) = 0$, $\mathbf{H}(\infty) = 1$. Figure 14.34 shows the plot of $|H(\omega)|$. Again, the corner or cutoff frequency is

$$\omega_c = \frac{1}{RC} \tag{14.53}$$

A **highpass filter** is designed to pass all frequencies above its cutoff frequency ω_c.

A highpass filter can also be formed when the output of an *RL* circuit is taken off the inductor.

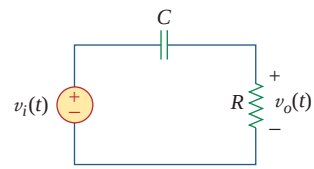


Figure 14.33
A highpass filter.

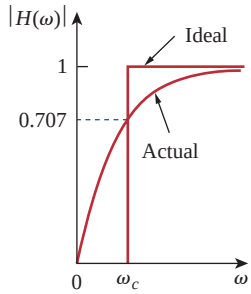


Figure 14.34
Ideal and actual frequency response of a highpass filter.

14.7.3 Bandpass Filter

The *RLC* series resonant circuit provides a bandpass filter when the output is taken off the resistor as shown in Fig. 14.35. The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + j(\omega L - 1/\omega C)} \tag{14.54}$$

We observe that $\mathbf{H}(0) = 0$, $\mathbf{H}(\infty) = 0$. Figure 14.36 shows the plot of $|H(\omega)|$. The bandpass filter passes a band of frequencies ($\omega_1 < \omega < \omega_2$) centered on ω_0, the center frequency, which is given by

$$\omega_0 = \frac{1}{\sqrt{LC}} \tag{14.55}$$

A **bandpass filter** is designed to pass all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

Since the bandpass filter in Fig. 14.35 is a series resonant circuit, the half-power frequencies, the bandwidth, and the quality factor are determined as in Section 14.5. A bandpass filter can also be formed by cascading the lowpass filter (where $\omega_2 = \omega_c$) in Fig. 14.31 with the

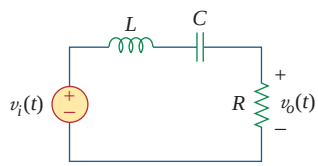


Figure 14.35
A bandpass filter.

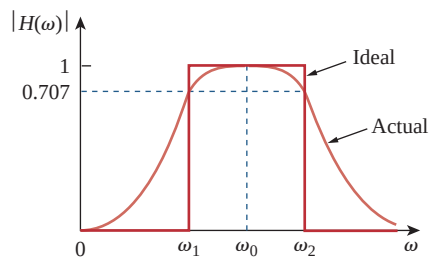


Figure 14.36
Ideal and actual frequency response of a bandpass filter.

highpass filter (where $\omega_1 = \omega_c$) in Fig. 14.33. However, the result would not be the same as just adding the output of the lowpass filter to the input of the highpass filter, because one circuit loads the other and alters the desired transfer function.

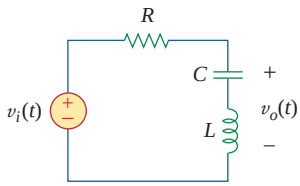


Figure 14.37
A bandstop filter.

14.7.4 Bandstop Filter

A filter that prevents a band of frequencies between two designated values (ω_1 and ω_2) from passing is variably known as a *bandstop*, *band-reject*, or *notch* filter. A bandstop filter is formed when the output *RLC* series resonant circuit is taken off the *LC* series combination as shown in Fig. 14.37. The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)} \tag{14.56}$$

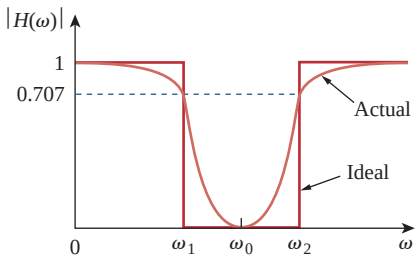


Figure 14.38
Ideal and actual frequency response of a bandstop filter.

Notice that $\mathbf{H}(0) = 1$, $\mathbf{H}(\infty) = 1$. Figure 14.38 shows the plot of $|H(\omega)|$. Again, the center frequency is given by

$$\omega_0 = \frac{1}{\sqrt{LC}} \tag{14.57}$$

while the half-power frequencies, the bandwidth, and the quality factor are calculated using the formulas in Section 14.5 for a series resonant circuit. Here, ω_0 is called the *frequency of rejection*, while the corresponding bandwidth ($B = \omega_2 - \omega_1$) is known as the *bandwidth of rejection*. Thus,

A **bandstop filter** is designed to stop or eliminate all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

Notice that adding the transfer functions of the bandpass and the bandstop gives unity at any frequency for the same values of *R*, *L*, and *C*. Of course, this is not true in general but true for the circuits treated here. This is due to the fact that the characteristic of one is the inverse of the other.

In concluding this section, we should note that:

1. From Eqs. (14.50), (14.52), (14.54), and (14.56), the maximum gain of a passive filter is unity. To generate a gain greater than unity, one should use an active filter as the next section shows.
2. There are other ways to get the types of filters treated in this section.
3. The filters treated here are the simple types. Many other filters have sharper and complex frequency responses.

Example 14.10

Determine what type of filter is shown in Fig. 14.39. Calculate the corner or cutoff frequency. Take $R = 2\ \text{k}\Omega$, $L = 2\ \text{H}$, and $C = 2\ \mu\text{F}$.

Solution:
The transfer function is

$$\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R \parallel 1/sC}{sL + R \parallel 1/sC}, \qquad s = j\omega \tag{14.10.1}$$

But

$$R \parallel \frac{1}{sC} = \frac{R/sC}{R + 1/sC} = \frac{R}{1 + sRC}$$

Substituting this into Eq. (14.10.1) gives

$$\mathbf{H}(s) = \frac{R/(1 + sRC)}{sL + R/(1 + sRC)} = \frac{R}{s^2RLC + sL + R}, \qquad s = j\omega$$

or

$$\mathbf{H}(\omega) = \frac{R}{-\omega^2RLC + j\omega L + R} \tag{14.10.2}$$

Since $\mathbf{H}(0) = 1$ and $\mathbf{H}(\infty) = 0$, we conclude from Table 14.5 that the circuit in Fig. 14.39 is a second-order lowpass filter. The magnitude of $\mathbf{H}$ is

$$H = \frac{R}{\sqrt{(R - \omega^2RLC)^2 + \omega^2L^2}} \tag{14.10.3}$$

The corner frequency is the same as the half-power frequency, i.e., where $\mathbf{H}$ is reduced by a factor of $1/\sqrt{2}$. Since the dc value of $H(\omega)$ is 1, at the corner frequency, Eq. (14.10.3) becomes after squaring

$$H^2 = \frac{1}{2} = \frac{R^2}{(R - \omega_c^2RLC)^2 + \omega_c^2L^2}$$

or

$$2 = (1 - \omega_c^2LC)^2 + \left(\frac{\omega_cL}{R}\right)^2$$

Substituting the values of R, L, and C, we obtain

$$2 = (1 - \omega_c^2\, 4 \times 10^{-6})^2 + (\omega_c\, 10^{-3})^2$$

Assuming that ω_c is in krad/s,

$$2 = (1 - 4\omega_c^2)^2 + \omega_c^2 \qquad \text{or} \qquad 16\omega_c^4 - 7\omega_c^2 - 1 = 0$$

Solving the quadratic equation in ω_c^2, we get $\omega_c^2 = 0.5509$ and -0.1134. Since ω_c is real,

$$\omega_c = 0.742 \text{ krad/s} = 742 \text{ rad/s}$$

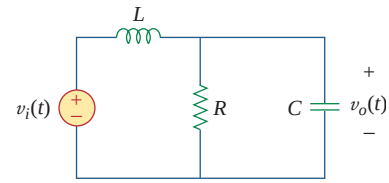


Figure 14.39
For Example 14.10.

Practice Problem 14.10

For the circuit in Fig. 14.40, obtain the transfer function $\mathbf{V}_o(\omega)/\mathbf{V}_i(\omega)$. Identify the type of filter the circuit represents and determine the corner frequency. Take $R_1 = 100\ \Omega = R_2$, $L = 2$ mH.

Answer: $\dfrac{R_2}{R_1 + R_2}\left(\dfrac{j\omega}{j\omega + \omega_c}\right)$, highpass filter

$\omega_c = \dfrac{R_1R_2}{(R_1 + R_2)L} = 25$ krad/s.

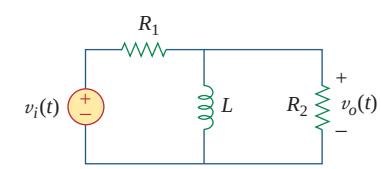


Figure 14.40
For Practice Prob. 14.10.

Example 14.11

If the bandstop filter in Fig. 14.37 is to reject a 200-Hz sinusoid while passing other frequencies, calculate the values of L and C. Take $R = 150\ \Omega$ and the bandwidth as 100 Hz.

Solution:
We use the formulas for a series resonant circuit in Section 14.5.

$$B = 2\pi(100) = 200\pi \text{ rad/s}$$

But

$$B = \frac{R}{L} \quad \Rightarrow \quad L = \frac{R}{B} = \frac{150}{200\pi} = 0.2387 \text{ H}$$

Rejection of the 200-Hz sinusoid means that f_0 is 200 Hz, so that ω_0 in Fig. 14.38 is

$$\omega_0 = 2\pi f_0 = 2\pi(200) = 400\pi$$

Since $\omega_0 = 1/\sqrt{LC}$,

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(400\pi)^2(0.2387)} = 2.653\ \mu\text{F}$$

Practice Problem 14.11

Design a bandpass filter of the form in Fig. 14.35 with a lower cutoff frequency of 20.1 kHz and an upper cutoff frequency of 20.3 kHz. Take $R = 20\ \text{k}\Omega$. Calculate L, C, and Q.

Answer: 15.92 H, 3.9 pF, 101.

14.8 Active Filters

There are three major limitations to the passive filters considered in the previous section. First, they cannot generate gain greater than 1; passive elements cannot add energy to the network. Second, they may require bulky and expensive inductors. Third, they perform poorly at frequencies below the audio frequency range (300 Hz $< f <$ 3,000 Hz). Nevertheless, passive filters are useful at high frequencies.

Active filters consist of combinations of resistors, capacitors, and op amps. They offer some advantages over passive *RLC* filters. First, they are often smaller and less expensive, because they do not require inductors. This makes feasible the integrated circuit realizations of filters. Second, they can provide amplifier gain in addition to providing the same frequency response as *RLC* filters. Third, active filters can be combined with buffer amplifiers (voltage followers) to isolate each stage of the filter from source and load impedance effects. This isolation allows designing the stages independently and then cascading them to realize the desired transfer function. (Bode plots, being logarithmic, may be added when transfer functions are cascaded.) However, active filters are less reliable and less stable. The practical limit of most active

filters is about 100 kHz—most active filters operate well below that frequency.

Filters are often classified according to their order (or number of poles) or their specific design type.

14.8.1 First-Order Lowpass Filter

One type of first-order filter is shown in Fig. 14.41. The components selected for Z_i and Z_f determine whether the filter is lowpass or highpass, but one of the components must be reactive.

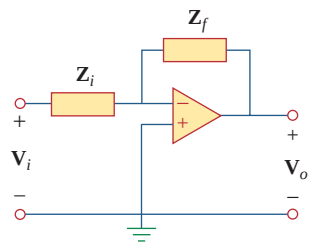


Figure 14.41
A general first-order active filter.

Figure 14.42 shows a typical active lowpass filter. For this filter, the transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i} \tag{14.58}$$

where $\mathbf{Z}_i = R_i$ and

$$\mathbf{Z}_f = R_f \,\|\, \frac{1}{j\omega C_f} = \frac{R_f/j\omega C_f}{R_f + 1/j\omega C_f} = \frac{R_f}{1 + j\omega C_f R_f} \tag{14.59}$$

Therefore,

$$\mathbf{H}(\omega) = -\frac{R_f}{R_i}\frac{1}{1 + j\omega C_f R_f} \tag{14.60}$$

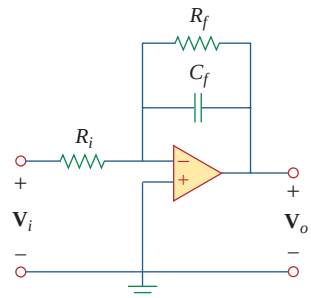


Figure 14.42
Active first-order lowpass filter.

We notice that Eq. (14.60) is similar to Eq. (14.50), except that there is a low frequency ($\omega \to 0$) gain or dc gain of $-R_f/R_i$. Also, the corner frequency is

$$\omega_c = \frac{1}{R_f C_f} \tag{14.61}$$

which does not depend on R_i. This means that several inputs with different R_i could be summed if required, and the corner frequency would remain the same for each input.

14.8.2 First-Order Highpass Filter

Figure 14.43 shows a typical highpass filter. As before,

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i} \tag{14.62}$$

where $\mathbf{Z}_i = R_i + 1/j\omega C_i$ and $\mathbf{Z}_f = R_f$ so that

$$\mathbf{H}(\omega) = -\frac{R_f}{R_i + 1/j\omega C_i} = -\frac{j\omega C_i R_f}{1 + j\omega C_i R_i} \tag{14.63}$$

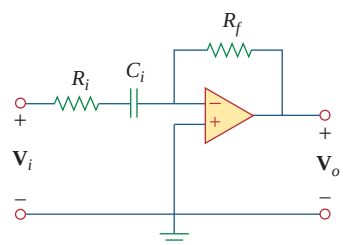


Figure 14.43
Active first-order highpass filter.

This is similar to Eq. (14.52), except that at very high frequencies ($\omega \to \infty$), the gain tends to $-R_f/R_i$. The corner frequency is

$$\omega_c = \frac{1}{R_i C_i} \tag{14.64}$$

14.8.3 Bandpass Filter

The circuit in Fig. 14.42 may be combined with that in Fig. 14.43 to form a bandpass filter that will have a gain K over the required range of frequencies. By cascading a unity-gain lowpass filter, a unity-gain

This way of creating a bandpass filter, not necessarily the best, is perhaps the easiest to understand.

highpass filter, and an inverter with gain $-R_f/R_i$, as shown in the block diagram of Fig. 14.44(a), we can construct a bandpass filter whose frequency response is that in Fig. 14.44(b). The actual construction of the bandpass filter is shown in Fig. 14.45.

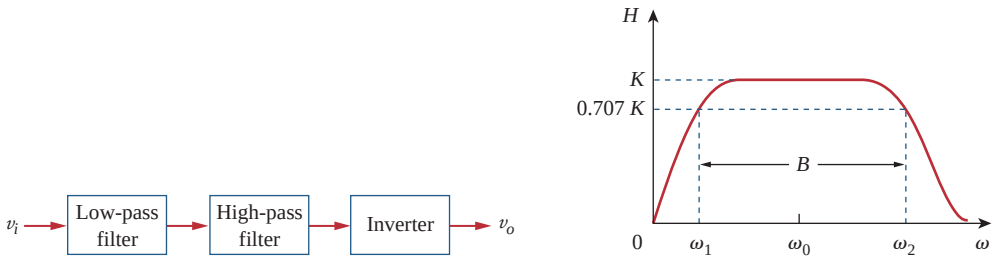


Figure 14.44
Active bandpass filter: (a) block diagram, (b) frequency response.

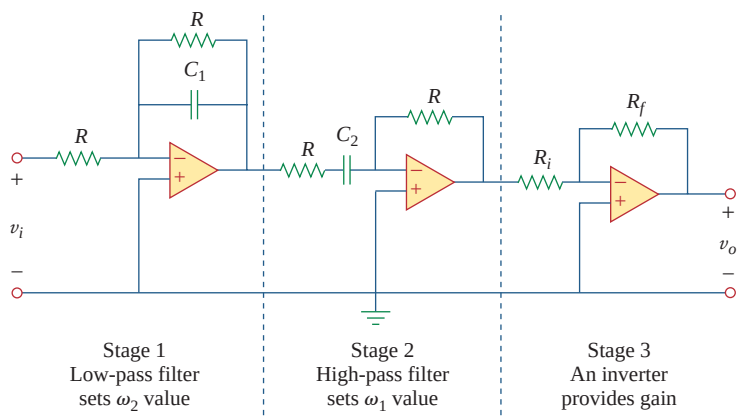


Figure 14.45
Active bandpass filter.

The analysis of the bandpass filter is relatively simple. Its transfer function is obtained by multiplying Eqs. (14.60) and (14.63) with the gain of the inverter; that is,

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \left(-\frac{1}{1 + j\omega C_1 R}\right)\left(-\frac{j\omega C_2 R}{1 + j\omega C_2 R}\right)\left(-\frac{R_f}{R_i}\right)$$
$$= -\frac{R_f}{R_i}\frac{1}{1 + j\omega C_1 R}\frac{j\omega C_2 R}{1 + j\omega C_2 R} \tag{14.65}$$

The lowpass section sets the upper corner frequency as

$$\omega_2 = \frac{1}{RC_1} \tag{14.66}$$

while the highpass section sets the lower corner frequency as

$$\omega_1 = \frac{1}{RC_2} \tag{14.67}$$

With these values of ω_1 and ω_2, the center frequency, bandwidth, and quality factor are found as follows:

$$\omega_0 = \sqrt{\omega_1 \omega_2} \tag{14.68}$$

$$B = \omega_2 - \omega_1 \tag{14.69}$$

$$Q = \frac{\omega_0}{B} \tag{14.70}$$

To find the passband gain K, we write Eq. (14.65) in the standard form of Eq. (14.15),

$$\mathbf{H}(\omega) = -\frac{R_f}{R_i} \frac{j\omega/\omega_1}{(1 + j\omega/\omega_1)(1 + j\omega/\omega_2)} = -\frac{R_f}{R_i} \frac{j\omega\omega_2}{(\omega_1 + j\omega)(\omega_2 + j\omega)} \tag{14.71}$$

At the center frequency $\omega_0 = \sqrt{\omega_1 \omega_2}$, the magnitude of the transfer function is

$$|\mathbf{H}(\omega_0)| = \left| \frac{R_f}{R_i} \frac{j\omega_0\omega_2}{(\omega_1 + j\omega_0)(\omega_2 + j\omega_0)} \right| = \frac{R_f}{R_i} \frac{\omega_2}{\omega_1 + \omega_2} \tag{14.72}$$

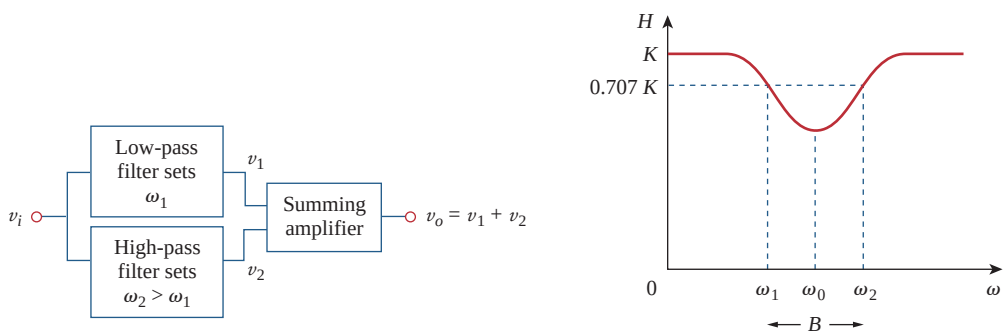
Thus, the passband gain is

$$K = \frac{R_f}{R_i} \frac{\omega_2}{\omega_1 + \omega_2} \tag{14.73}$$

14.8.4 Bandreject (or Notch) Filter

A bandreject filter may be constructed by parallel combination of a lowpass filter and a highpass filter and a summing amplifier, as shown in the block diagram of Fig. 14.46(a). The circuit is designed such that the lower cutoff frequency ω_1 is set by the lowpass filter while the upper cutoff frequency ω_2 is set by the highpass filter. The gap between ω_1 and ω_2 is the bandwidth of the filter. As shown in Fig. 14.46(b), the filter passes frequencies below ω_1 and above ω_2. The block diagram in Fig. 14.46(a) is actually constructed as shown in Fig. 14.47. The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = -\frac{R_f}{R_i}\left(-\frac{1}{1 + j\omega C_1 R} - \frac{j\omega C_2 R}{1 + j\omega C_2 R}\right) \tag{14.74}$$



(a) (b)

Figure 14.46
Active bandreject filter: (a) block diagram, (b) frequency response.

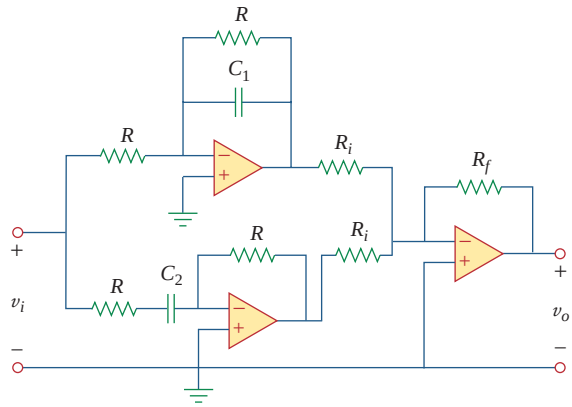


Figure 14.47
Active bandreject filter.

The formulas for calculating the values of ω_1, ω_2, the center frequency, bandwidth, and quality factor are the same as in Eqs. (14.66) to (14.70).

To determine the passband gain K of the filter, we can write Eq. (14.74) in terms of the upper and lower corner frequencies as

$$\begin{aligned}\mathbf{H}(\omega) &= \frac{R_f}{R_i}\left(\frac{1}{1 + j\omega/\omega_2} + \frac{j\omega/\omega_1}{1 + j\omega/\omega_1}\right) \\ &= \frac{R_f}{R_i}\frac{(1 + j2\omega/\omega_1 + (j\omega)^2/\omega_1\omega_1)}{(1 + j\omega/\omega_2)(1 + j\omega/\omega_1)}\end{aligned} \tag{14.75}$$

Comparing this with the standard form in Eq. (14.15) indicates that in the two passbands ($\omega \to 0$ and $\omega \to \infty$) the gain is

$$K = \frac{R_f}{R_i} \tag{14.76}$$

We can also find the gain at the center frequency by finding the magnitude of the transfer function at $\omega_0 = \sqrt{\omega_1\omega_2}$, writing

$$\begin{aligned}H(\omega_0) &= \left|\frac{R_f}{R_i}\frac{(1 + j2\omega_0/\omega_1 + (j\omega_0)^2/\omega_1\omega_1)}{(1 + j\omega_0/\omega_2)(1 + j\omega_0/\omega_1)}\right| \\ &= \frac{R_f}{R_i}\frac{2\omega_1}{\omega_1 + \omega_2}\end{aligned} \tag{14.77}$$

Again, the filters treated in this section are only typical. There are many other active filters that are more complex.

Example 14.12

Design a lowpass active filter with a dc gain of 4 and a corner frequency of 500 Hz.

Solution:
From Eq. (14.61), we find

$$\omega_c = 2\pi f_c = 2\pi(500) = \frac{1}{R_f C_f} \tag{14.12.1}$$

The dc gain is

$$H(0) = -\frac{R_f}{R_i} = -4 \qquad \textbf{(14.12.2)}$$

We have two equations and three unknowns. If we select $C_f = 0.2\ \mu$F, then

$$R_f = \frac{1}{2\pi(500)0.2 \times 10^{-6}} = 1.59\ \text{k}\Omega$$

and

$$R_i = \frac{R_f}{4} = 397.5\ \Omega$$

We use a 1.6-kΩ resistor for R_f and a 400-Ω resistor for R_i. Figure 14.42 shows the filter.

Practice Problem 14.12

Design a highpass filter with a high-frequency gain of 5 and a corner frequency of 2 kHz. Use a 0.1-μF capacitor in your design.

Answer: $R_i = 800\ \Omega$ and $R_f = 4\ \text{k}\Omega$.

Example 14.13

Design a bandpass filter in the form of Fig. 14.45 to pass frequencies between 250 Hz and 3,000 Hz and with $K = 10$. Select $R = 20\ \text{k}\Omega$.

Solution:

1. **Define.** The problem is clearly stated and the circuit to be used in the design is specified.
2. **Present.** We are asked to use the op amp circuit specified in Fig. 14.45 to design a bandpass filter. We are given the value of R to use (20 kΩ). In addition, the frequency range of the signals to be passed is 250 Hz to 3 kHz.
3. **Alternative.** We will use the equations developed in Section 14.8.3 to obtain a solution. We will then use the resulting transfer function to validate the answer.
4. **Attempt.** Since $\omega_1 = 1/RC_2$, we obtain

$$C_2 = \frac{1}{R\omega_1} = \frac{1}{2\pi f_1 R} = \frac{1}{2\pi \times 250 \times 20 \times 10^3} = \textbf{31.83 nF}$$

Similarly, since $\omega_2 = 1/RC_1$,

$$C_1 = \frac{1}{R\omega_2} = \frac{1}{2\pi f_2 R} = \frac{1}{2\pi \times 3{,}000 \times 20 \times 10^3} = \textbf{2.65 nF}$$

From Eq. (14.73),

$$\frac{R_f}{R_i} = K\frac{\omega_1 + \omega_2}{\omega_2} = K\frac{f_1 + f_2}{f_2} = \frac{10(3{,}250)}{3{,}000} = 10.83$$

If we select $R_i = \mathbf{10\ k\Omega}$, then $R_f = 10.83R_i \simeq \mathbf{108.3\ k\Omega}$.

5. **Evaluate.** The output of the first op amp is given by

$$\frac{V_i - 0}{20\ \text{k}\Omega} + \frac{V_1 - 0}{20\ \text{k}\Omega} + \frac{s2.65 \times 10^{-9}(V_1 - 0)}{1}$$

$$= 0 \rightarrow V_1 = -\frac{V_i}{1 + 5.3 \times 10^{-5}s}$$

The output of the second op amp is given by

$$\frac{V_1 - 0}{20\ \text{k}\Omega + \dfrac{1}{s31.83\ \text{nF}}} + \frac{V_2 - 0}{20\ \text{k}\Omega} = 0 \rightarrow$$

$$V_2 = -\frac{6.366 \times 10^{-4}sV_1}{1 + 6.366 \times 10^{-4}s}$$

$$= \frac{6.366 \times 10^{-4}sV_i}{(1 + 6.366 \times 10^{-4}s)(1 + 5.3 \times 10^{-5}s)}$$

The output of the third op amp is given by

$$\frac{V_2 - 0}{10\ \text{k}\Omega} + \frac{V_o - 0}{108.3\ \text{k}\Omega} = 0 \rightarrow V_o = 10.83V_2 \rightarrow j2\pi \times 25°$$

$$V_o = -\frac{6.894 \times 10^{-3}sV_i}{(1 + 6.366 \times 10^{-4}s)(1 + 5.3 \times 10^{-5}s)}$$

Let $j2\pi \times 25°$ and solve for the magnitude of V_o/V_i.

$$\frac{V_o}{V_i} = \frac{-j10.829}{(1 + j1)(1)}$$

$|V_o/V_i| = \mathbf{(0.7071)10.829}$, which is the lower corner frequency point.

Let $s = j2\pi \times 3000 = j18.849\ \text{k}\Omega$. We then get

$$\frac{V_o}{V_i} = \frac{-j129.94}{(1 + j12)(1 + j1)}$$

$$= \frac{129.94\angle -90°}{(12.042\angle 85.24°)(1.4142\angle 45°)} = \mathbf{(0.7071)10.791\angle -18.61°}$$

Clearly this is the upper corner frequency and the answer checks.

6. **Satisfactory?** We have satisfactorily designed the circuit and can present the results as a solution to the problem.

Practice Problem 14.13

Design a notch filter based on Fig. 14.47 for $\omega_0 = 20$ krad/s, $K = 5$, and $Q = 10$. Use $R = R_i = 10\ \text{k}\Omega$.

Answer: $C_1 = 4.762$ nF, $C_2 = 5.263$ nF, and $R_f = 50\ \text{k}\Omega$.

14.9 Scaling

In designing and analyzing filters and resonant circuits or in circuit analysis in general, it is sometimes convenient to work with element values of 1 Ω, 1 H, or 1 F, and then transform the values to realistic

values by *scaling*. We have taken advantage of this idea by not using realistic element values in most of our examples and problems; mastering circuit analysis is made easy by using convenient component values. We have thus eased calculations, knowing that we could use scaling to then make the values realistic.

There are two ways of scaling a circuit: *magnitude* or *impedance scaling*, and *frequency scaling*. Both are useful in scaling responses and circuit elements to values within the practical ranges. While magnitude scaling leaves the frequency response of a circuit unaltered, frequency scaling shifts the frequency response up or down the frequency spectrum.

14.9.1 Magnitude Scaling

Magnitude scaling is the process of increasing all impedances in a network by a factor, the frequency response remaining unchanged.

Recall that impedances of individual elements R, L, and C are given by

$$\mathbf{Z}_R = R, \qquad \mathbf{Z}_L = j\omega L, \qquad \mathbf{Z}_C = \frac{1}{j\omega C} \tag{14.78}$$

In magnitude scaling, we multiply the impedance of each circuit element by a factor K_m and let the frequency remain constant. This gives the new impedances as

$$\mathbf{Z}'_R = K_m\mathbf{Z}_R = K_m R, \qquad \mathbf{Z}'_L = K_m\mathbf{Z}_L = j\omega K_m L$$
$$\mathbf{Z}'_C = K_m\mathbf{Z}_C = \frac{1}{j\omega C/K_m} \tag{14.79}$$

Comparing Eq. (14.79) with Eq. (14.78), we notice the following changes in the element values: $R \to K_m R$, $L \to K_m L$, and $C \to C/K_m$. Thus, in magnitude scaling, the new values of the elements and frequency are

$$\boxed{\begin{aligned} R' &= K_m R, & L' &= K_m L \\ C' &= \frac{C}{K_m}, & \omega' &= \omega \end{aligned}} \tag{14.80}$$

The primed variables are the new values and the unprimed variables are the old values. Consider the series or parallel *RLC* circuit. We now have

$$\omega'_0 = \frac{1}{\sqrt{L'C'}} = \frac{1}{\sqrt{K_m LC/K_m}} = \frac{1}{\sqrt{LC}} = \omega_0 \tag{14.81}$$

showing that the resonant frequency, as expected, has not changed. Similarly, the quality factor and the bandwidth are not affected by magnitude scaling. Also, magnitude scaling does not affect transfer functions in the forms of Eqs. (14.2a) and (14.2b), which are dimensionless quantities.

14.9.2 Frequency Scaling

Frequency scaling is equivalent to relabeling the frequency axis of a frequency response plot. It is needed when translating frequencies such as a resonant frequency, a corner frequency, a bandwidth, etc., to a realistic level. It can be used to bring capacitance and inductance values into a range that is convenient to work with.

Frequency scaling is the process of shifting the frequency response of a network up or down the frequency axis while leaving the impedance the same.

We achieve frequency scaling by multiplying the frequency by a factor K_f while keeping the impedance the same.

From Eq. (14.78), we see that the impedances of L and C are frequency-dependent. If we apply frequency scaling to $\mathbf{Z}_L(\omega)$ and $\mathbf{Z}_C(\omega)$ in Eq. (14.78), we obtain

$$\mathbf{Z}_L = j(\omega K_f)L' = j\omega L \quad \Rightarrow \quad L' = \frac{L}{K_f} \tag{14.82a}$$

$$\mathbf{Z}_C = \frac{1}{j(\omega K_f)C'} = \frac{1}{j\omega C} \quad \Rightarrow \quad C' = \frac{C}{K_f} \tag{14.82b}$$

since the impedance of the inductor and capacitor must remain the same after frequency scaling. We notice the following changes in the element values: $L \to L/K_f$ and $C \to C/K_f$. The value of R is not affected, since its impedance does not depend on frequency. Thus, in frequency scaling, the new values of the elements and frequency are

$$\boxed{\begin{aligned} R' &= R, & L' &= \frac{L}{K_f} \\ C' &= \frac{C}{K_f}, & \omega' &= K_f\,\omega \end{aligned}} \tag{14.83}$$

Again, if we consider the series or parallel RLC circuit, for the resonant frequency

$$\omega_0' = \frac{1}{\sqrt{L'C'}} = \frac{1}{\sqrt{(L/K_f)(C/K_f)}} = \frac{K_f}{\sqrt{LC}} = K_f\,\omega_0 \tag{14.84}$$

and for the bandwidth

$$B' = K_f B \tag{14.85}$$

but the quality factor remains the same ($Q' = Q$).

14.9.3 Magnitude and Frequency Scaling

If a circuit is scaled in magnitude and frequency at the same time, then

$$\boxed{\begin{aligned} R' &= K_m R, & L' &= \frac{K_m}{K_f} L \\ C' &= \frac{1}{K_m K_f} C, & \omega' &= K_f\,\omega \end{aligned}} \tag{14.86}$$

These are more general formulas than those in Eqs. (14.80) and (14.83). We set $K_m = 1$ in Eq. (14.86) when there is no magnitude scaling or $K_f = 1$ when there is no frequency scaling.

Example 14.14

A fourth-order Butterworth lowpass filter is shown in Fig. 14.48(a). The filter is designed such that the cutoff frequency $\omega_c = 1$ rad/s. Scale the circuit for a cutoff frequency of 50 kHz using 10- kΩ resistors.

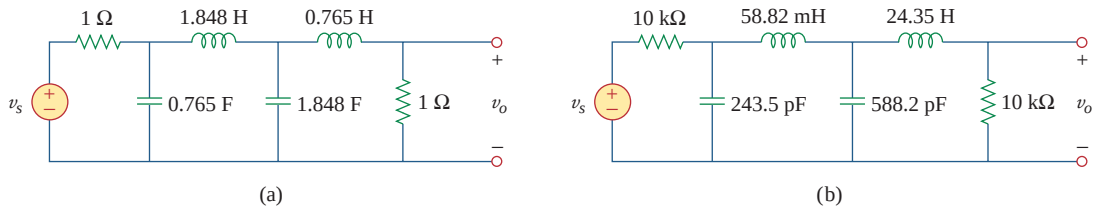


Figure 14.48
For Example 14.14: (a) Normalized Butterworth lowpass filter, (b) scaled version of the same lowpass filter.

Solution:
If the cutoff frequency is to shift from $\omega_c = 1$ rad/s to $\omega_c' = 2\pi(50)$ krad/s, then the frequency scale factor is

$$K_f = \frac{\omega_c'}{\omega_c} = \frac{100\pi \times 10^3}{1} = \pi \times 10^5$$

Also, if each 1-Ω resistor is to be replaced by a 10- kΩ resistor, then the magnitude scale factor must be

$$K_m = \frac{R'}{R} = \frac{10 \times 10^3}{1} = 10^4$$

Using Eq. (14.86),

$$L_1' = \frac{K_m}{K_f} L_1 = \frac{10^4}{\pi \times 10^5}(1.848) = 58.82 \text{ mH}$$

$$L_2' = \frac{K_m}{K_f} L_2 = \frac{10^4}{\pi \times 10^5}(0.765) = 24.35 \text{ mH}$$

$$C_1' = \frac{C_1}{K_m K_f} = \frac{0.765}{\pi \times 10^9} = 243.5 \text{ pF}$$

$$C_2' = \frac{C_2}{K_m K_f} = \frac{1.848}{\pi \times 10^9} = 588.2 \text{ pF}$$

The scaled circuit is shown in Fig. 14.48(b). This circuit uses practical values and will provide the same transfer function as the prototype in Fig. 14.48(a), but shifted in frequency.

Practice Problem 14.14

A third-order Butterworth filter normalized to $\omega_c = 1$ rad/s is shown in Fig. 14.49. Scale the circuit to a cutoff frequency of 10 kHz. Use 15-nF capacitors.

Answer: $R_1' = R_2' = 1.061$ kΩ, $C_1' = C_2' = 15$ nF, $L' = 33.77$ mH.

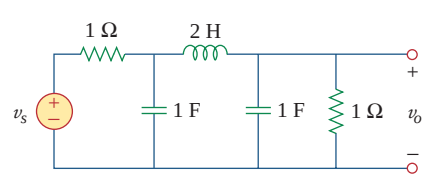


Figure 14.49
For Practice Prob. 14.14.

14.10 Frequency Response Using *PSpice*

PSpice is a useful tool in the hands of the modern circuit designer for obtaining the frequency response of circuits. The frequency response is obtained using the AC Sweep as discussed in Section D.5 (Appendix D). This requires that we specify in the AC Sweep dialog box *Total Pts*, *Start Freq*, *End Freq*, and the sweep type. *Total Pts* is the number of points in the frequency sweep, and *Start Freq* and *End Freq* are, respectively, the starting and final frequencies, in hertz. In order to know what frequencies to select for *Start Freq* and *End Freq*, one must have an idea of the frequency range of interest by making a rough sketch of the frequency response. In a complex circuit where this may not be possible, one may use a trial-and-error approach.

There are three types of sweeps:

Linear: The frequency is varied linearly from *Start Freq* to *End Freq* with *Total Pts* equally spaced points (or responses).

Octave: The frequency is swept logarithmically by octaves from *Start Freq* to *End Freq* with *Total Pts* per octave. An octave is a factor of 2 (e.g., 2 to 4, 4 to 8, 8 to 16).

Decade: The frequency is varied logarithmically by decades from *Start Freq* to *End Freq* with *Total Pts* per decade. A decade is a factor of 10 (e.g., from 2 Hz to 20 Hz, 20 Hz to 200 Hz, 200 Hz to 2 kHz).

It is best to use a linear sweep when displaying a narrow frequency range of interest, as a linear sweep displays the frequency range well in a narrow range. Conversely, it is best to use a logarithmic (octave or decade) sweep for displaying a wide frequency range of interest—if a linear sweep is used for a wide range, all the data will be crowded at the high- or low-frequency end and insufficient data at the other end.

With the above specifications, *PSpice* performs a steady-state sinusoidal analysis of the circuit as the frequency of all the independent sources is varied (or swept) from *Start Freq* to *End Freq*.

The *PSpice* A/D program produces a graphical output. The output data type may be specified in the *Trace Command Box* by adding one of the following suffixes to V or I:

M	Amplitude of the sinusoid.
P	Phase of the sinusoid.
dB	Amplitude of the sinusoid in decibels, i.e., 20 $\log_{10}$ (amplitude).

Example 14.15

Determine the frequency response of the circuit shown in Fig. 14.50.

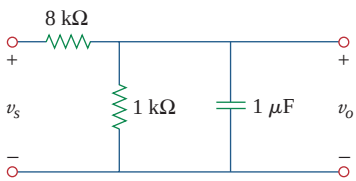


Figure 14.50
For Example 14.15.

Solution:

We let the input voltage v_s be a sinusoid of amplitude 1 V and phase 0°. Figure 14.51 is the schematic for the circuit. The capacitor is rotated 270° counterclockwise to ensure that pin 1 (the positive terminal) is on top. The voltage marker is inserted to the output voltage across the capacitor. To perform a linear sweep for $1 < f < 1000$ Hz with 50 points, we select **Analysis/Setup/AC Sweep**, **DCLICK** *Linear*, type 50 in the *Total Pts* box, type 1 in the *Start Freq* box, and type 1000 in the *End Freq* box. After saving the file, we select **Analysis/Simulate** to simulate the circuit. If there are no errors, the *PSpice* A/D window will

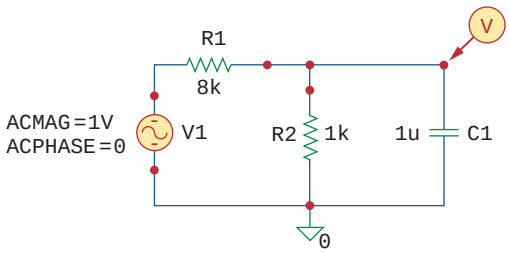


Figure 14.51
The schematic for the circuit in Fig. 14.50.

display the plot of V(C1:1), which is the same as V_o or $H(\omega) = V_o/1$, as shown in Fig. 14.52(a). This is the magnitude plot, since V(C1:1) is the same as VM(C1:1). To obtain the phase plot, select **Trace/Add** in the *PSpice* A/D menu and type VP(C1:1) in the **Trace Command** box. Figure 14.52(b) shows the result. By hand, the transfer function is

$$H(\omega) = \frac{V_o}{V_s} = \frac{1{,}000}{9{,}000 + j\omega 8}$$

or

$$H(\omega) = \frac{1}{9 + j16\pi \times 10^{-3}}$$

showing that the circuit is a lowpass filter as demonstrated in Fig. 14.52. Notice that the plots in Fig. 14.52 are similar to those in Fig. 14.3 (note that the horizontal axis in Fig. 14.52 is logrithic while the horizontal axis in Fig. 14.3 is linear.)

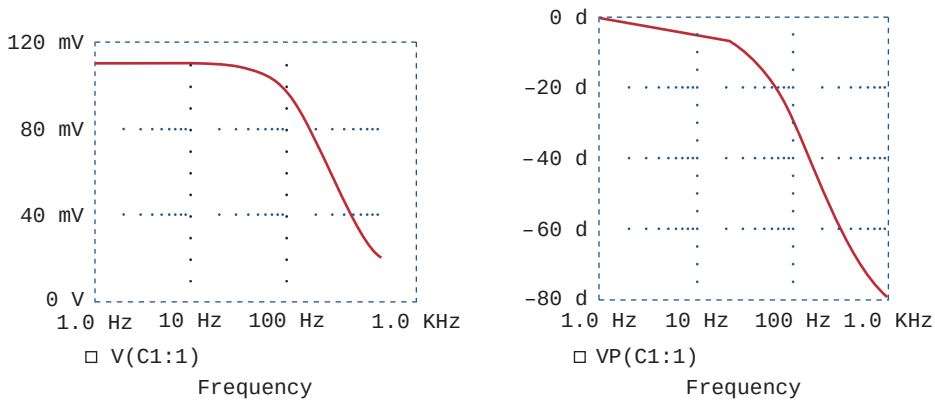


Figure 14.52
For Example 14.15: (a) magnitude plot, (b) phase plot of the frequency response.

Practice Problem 14.15

Obtain the frequency response of the circuit in Fig. 14.53 using *PSpice*. Use a linear frequency sweep and consider $1 < f < 1000$ Hz with 100 points.

Answer: See Fig. 14.54.

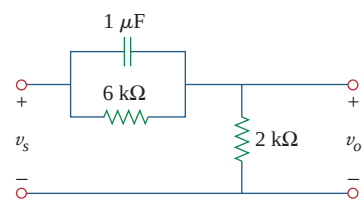


Figure 14.53
For Practice Prob. 14.15.

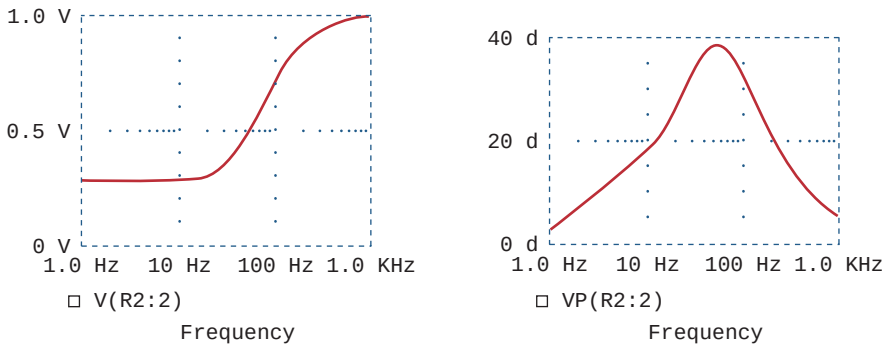


Figure 14.54
For Practice Problem 14.15: (a) magnitude plot, (b) phase plot of the frequency response.

Example 14.16

Use *PSpice* to generate the gain and phase Bode plots of V in the circuit of Fig. 14.55.

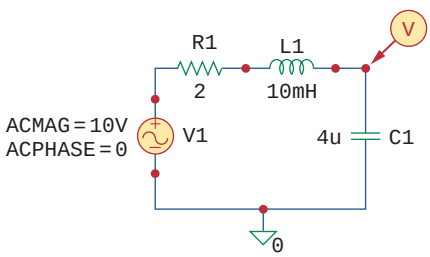


Figure 14.55
For Example 14.16.

Solution:

The circuit treated in Example 14.15 is first-order while the one in this example is second-order. Since we are interested in Bode plots, we use decade frequency sweep for $300 < f < 3{,}000$ Hz with 50 points per decade. We select this range because we know that the resonant frequency of the circuit is within the range. Recall that

$$\omega_0 = \frac{1}{\sqrt{LC}} = 5 \text{ krad/s} \qquad \text{or} \qquad f_0 = \frac{\omega}{2\pi} = 795.8 \text{ Hz}$$

After drawing the circuit as in Fig. 14.55, we select **Analysis/Setup/AC Sweep, DCLICK** *Decade*, enter 50 in the *Total Pts* box, 300 as the *Start Freq*, and 3,000 in the *End Freq* box. Upon saving the file, we simulate it by selecting **Analysis/Simulate**. This will automatically bring up the *PSpice* A/D window and display V(C1:1) if there are no errors. Since we are interested in the Bode plot, we select **Trace/Add** in the *PSpice* A/D menu and type dB(V(C1:1)) in the **Trace Command** box. The result is the Bode magnitude plot in Fig. 14.56(a). For the

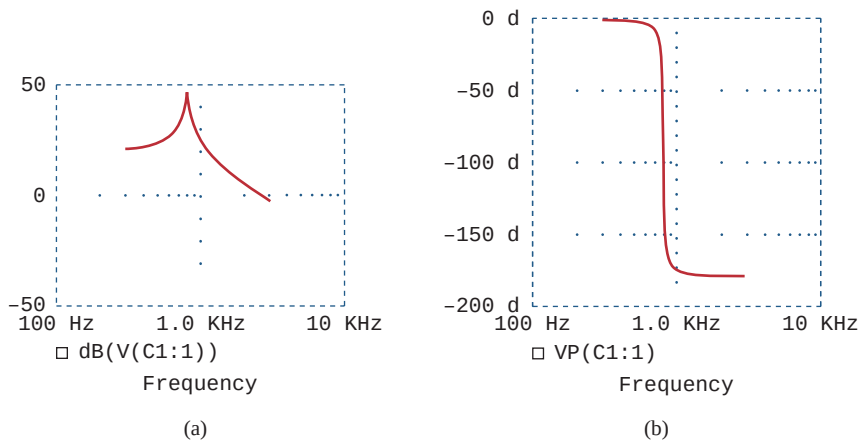


Figure 14.56
For Example 14.16: (a) Bode plot, (b) phase plot of the response.

phase plot, we select **Trace/Add** in the *PSpice* A/D menu and type VP(C1:1) in the **Trace Command** box. The result is the Bode phase plot of Fig. 14.56(b). Notice that the plots confirm the resonant frequency of 795.8 Hz.

Practice Problem 14.16

Consider the network in Fig. 14.57. Use *PSpice* to obtain the Bode plots for V_o over a frequency from 1 kHz to 100 kHz using 20 points per decade.

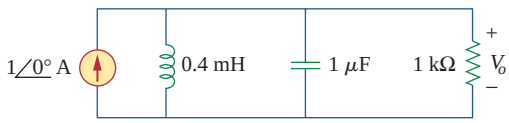


Figure 14.57
For Practice Prob. 14.16.

Answer: See Fig. 14.58.

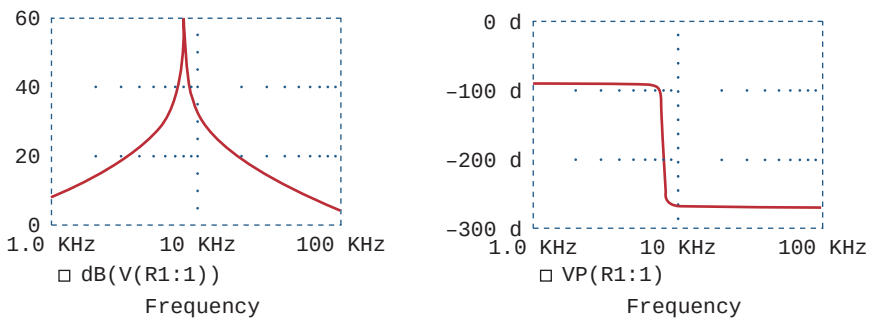


Figure 14.58
For Practice Prob. 14.16: Bode (a) magnitude plot, (b) phase plot.

14.11 Computation Using *MATLAB*

MATLAB is a software package that is widely used for engineering computation and simulation. A review of *MATLAB* is provided in Appendix E for the beginner. This section shows how to use the software to numerically perform most of the operations presented in this chapter and Chapter 15. The key to describing a system in *MATLAB* is to specify the numerator (num) and denominator (den) of the transfer function of the system. Once this is done, we can use several *MATLAB* commands to obtain the system's Bode plots (frequency response) and the system's response to a given input.

The command **bode** produces the Bode plots (both magnitude and phase) of a given transfer function $H(s)$. The format of the command is **bode** (num, den), where num is the numerator of $H(s)$ and den is its denominator. The frequency range and number of points are automatically selected. For example, consider the transfer function in Example 14.3. It is better to first write the numerator and denominator in polynomial forms.

Thus,

$$H(s) = \frac{200\,j\omega}{(j\omega + 2)(j\omega + 10)} = \frac{200s}{s^2 + 12s + 20}, \qquad s = j\omega$$

Using the following commands, the Bode plots are generated as shown in Fig. 14.59. If necessary, the command **logspace** can be included to generate a logarithmically spaced frequency and the command **semilogx** can be used to produce a semilog scale.

```
>> num = [200 0];      % specify the numerator of H(s)
>> den = [1 12 20];    % specify the denominator of H(s)
>> bode(num, den);     % determine and draw Bode plots
```

The step response $y(t)$ of a system is the output when the input $x(t)$ is the unit step function. The command **step** plots the step response of a system given the numerator and denominator of the transfer function of that system. The time range and number of points are automatically selected. For example, consider a second-order system with the transfer function

$$H(s) = \frac{12}{s^2 + 3s + 12}$$

We obtain the step response of the system shown in Fig. 14.60 by using the following commands.

```
>> n = 12;
>> d = [1 3 12];
>> step(n,d);
```

We can verify the plot in Fig. 14.60 by obtaining $y(t) = x(t) * u(t)$ or $Y(s) = X(s)H(s)$.

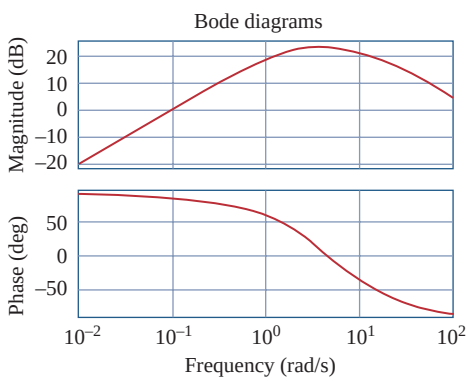


Figure 14.59
Magnitude and phase plots.

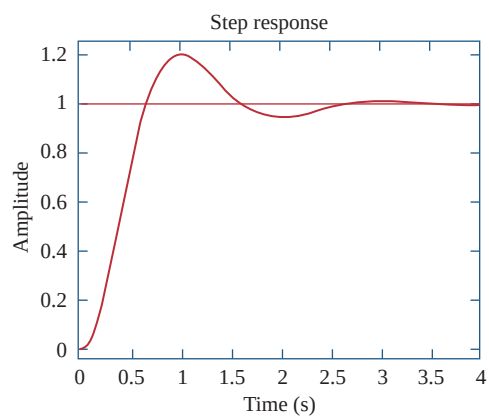


Figure 14.60
The Step response of $H(s) = 12/(s^2 + 3s + 12)$.

The command **lsim** is a more general command than **step**. It calculates the time response of a system to any arbitrary input signal. The format of the command is $y =$ **lsim** (num, den, x, t), where $x(t)$ is the input signal, t is the time vector, and $y(t)$ is the output generated. For example, assume a system is described by the transfer function

$$H(s) = \frac{s + 4}{s^3 + 2s^2 + 5s + 10}$$

To find the response $y(t)$ of the system to input $x(t) = 10e^{-t}u(t)$, we use the following *MATLAB* commands. Both the response $y(t)$ and the input $x(t)$ are plotted in Fig. 14.61.

```
>> t = 0:0.02:5; % time vector 0 < t < 5 with increment
     0.02
>> x = 10*exp(-t);
>> num = [1  4];
>> den = [1  2  5  10];
>> y = lsim(num,den,x,t);
>> plot(t,x,t,y)
```

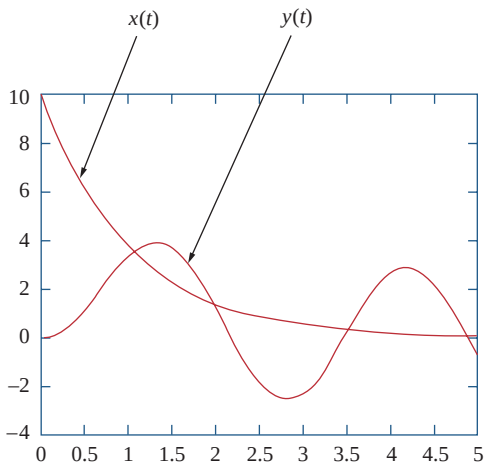


Figure 14.61
The response of the system described by $H(s) = (s + 4)/(s^2 + 2s^2 + 5s + 10)$ to an exponential input.

14.12 † Applications

Resonant circuits and filters are widely used, particularly in electronics, power systems, and communications systems. For example, a Notch filter with a cutoff frequency at 60 Hz may be used to eliminate the 60-Hz power line noise in various communications electronics. Filtering of signals in communications systems is necessary in order to select the desired signal from a host of others in the same range (as in the case of radio receivers discussed next) and also to minimize the effects of noise and interference on the desired signal. In this section, we consider one practical application of resonant circuits and two applications of filters. The focus of each application is not to understand the details of how each device works but to see how the circuits considered in this chapter are applied in the practical devices.

14.12.1 Radio Receiver

Series and parallel resonant circuits are commonly used in radio and TV receivers to tune in stations and to separate the audio signal from the radio-frequency carrier wave. As an example, consider the block diagram

of an AM radio receiver shown in Fig. 14.62. Incoming amplitude-modulated radio waves (thousands of them at different frequencies from different broadcasting stations) are received by the antenna. A resonant circuit (or a bandpass filter) is needed to select just one of the incoming waves. The selected signal is very weak and is amplified in stages in order to generate an audible audio-frequency wave. Thus, we have the radio frequency (RF) amplifier to amplify the selected broadcast signal, the intermediate frequency (IF) amplifier to amplify an internally generated signal based on the RF signal, and the audio amplifier to amplify the audio signal just before it reaches the loudspeaker. It is much easier to amplify the signal at three stages than to build an amplifier to provide the same amplification for the entire band.

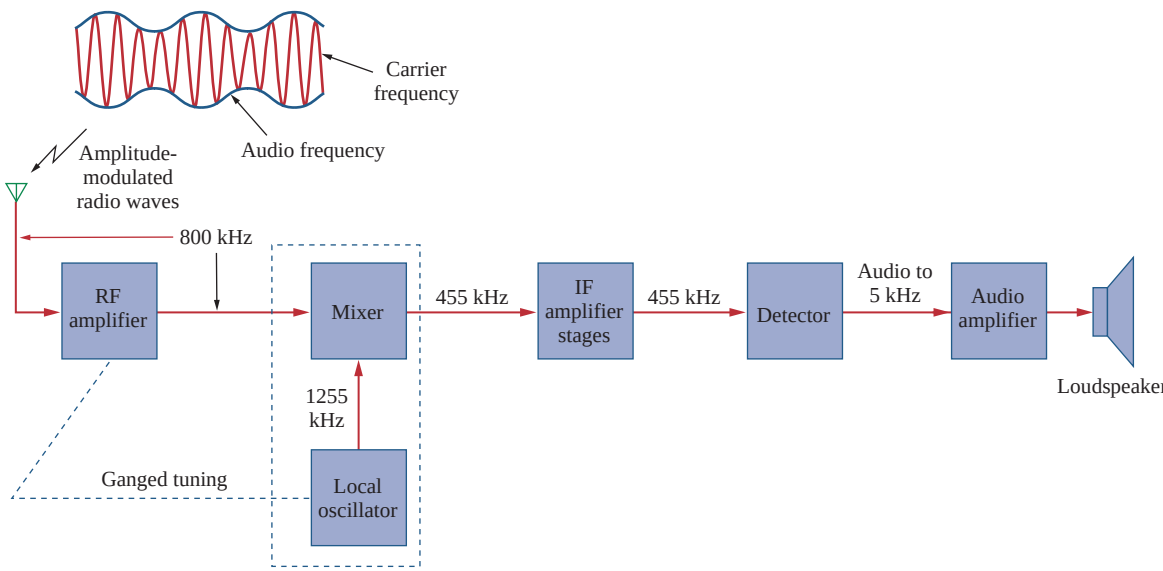


Figure 14.62
A simplified block diagram of a superheterodyne AM radio receiver.

The type of AM receiver shown in Fig. 14.62 is known as the *superheterodyne receiver*. In the early development of radio, each amplification stage had to be tuned to the frequency of the incoming signal. This way, each stage must have several tuned circuits to cover the entire AM band (540 to 1600 kHz). To avoid the problem of having several resonant circuits, modern receivers use a *frequency mixer* or *heterodyne* circuit, which always produces the same IF signal (445 kHz) but retains the audio frequencies carried on the incoming signal. To produce the constant IF frequency, the rotors of two separate variable capacitors are mechanically coupled with one another so that they can be rotated simultaneously with a single control; this is called *ganged tuning*. A *local oscillator* ganged with the RF amplifier produces an RF signal that is combined with the incoming wave by the frequency mixer to produce an output signal that contains the sum and the difference frequencies of the two signals. For example, if the resonant circuit is tuned to receive an 800-kHz incoming signal, the local oscillator must produce a 1,255-kHz signal, so that the sum ($1{,}255 + 800 = 2{,}055$ kHz) and the difference ($1{,}255 - 800 = 455$ kHz) of frequencies are available at the output of

the mixer. However, only the difference, 455 kHz, is used in practice. This is the only frequency to which all the IF amplifier stages are tuned, regardless of the station dialed. The original audio signal (containing the "intelligence") is extracted in the detector stage. The detector basically removes the IF signal, leaving the audio signal. The audio signal is amplified to drive the loudspeaker, which acts as a transducer converting the electrical signal to sound.

Our major concern here is the tuning circuit for the AM radio receiver. The operation of the FM radio receiver is different from that of the AM receiver discussed here, and in a much different range of frequencies, but the tuning is similar.

Example 14.17

The resonant or tuner circuit of an AM radio is portrayed in Fig. 14.63. Given that $L = 1\ \mu\text{H}$, what must be the range of C to have the resonant frequency adjustable from one end of the AM band to another?

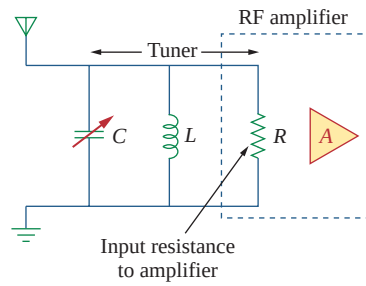


Figure 14.63
The tuner circuit for Example 14.17.

Solution:
The frequency range for AM broadcasting is 540 to 1,600 kHz. We consider the low and high ends of the band. Since the resonant circuit in Fig. 14.63 is a parallel type, we apply the ideas in Section 14.6. From Eq. (14.44),

$$\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

or

$$C = \frac{1}{4\pi^2 f_0^2 L}$$

For the high end of the AM band, $f_0 = 1{,}600$ kHz, and the corresponding C is

$$C_1 = \frac{1}{4\pi^2 \times 1{,}600^2 \times 10^6 \times 10^{-6}} = 9.9\ \text{nF}$$

For the low end of the AM band, $f_0 = 540$ kHz, and the corresponding C is

$$C_2 = \frac{1}{4\pi^2 \times 540^2 \times 10^6 \times 10^{-6}} = 86.9\ \text{nF}$$

Thus, C must be an adjustable (gang) capacitor varying from 9.9 nF to 86.9 nF.

Practice Problem 14.17

For an FM radio receiver, the incoming wave is in the frequency range from 88 to 108 MHz. The tuner circuit is a parallel RLC circuit with a 4-μH coil. Calculate the range of the variable capacitor necessary to cover the entire band.

Answer: From 0.543 pF to 0.818 pF.

14.12.2 Touch-Tone Telephone

A typical application of filtering is the touch-tone telephone set shown in Fig. 14.64. The keypad has 12 buttons arranged in four rows and three columns. The arrangement provides 12 distinct signals by using seven tones divided into two groups: the low-frequency group (697 to 941 Hz) and the high-frequency group (1,209 to 1,477 Hz). Pressing a button generates a sum of two sinusoids corresponding to its unique pair of frequencies. For example, pressing the number 6 button generates sinusoidal tones with frequencies 770 Hz and 1,477 Hz.

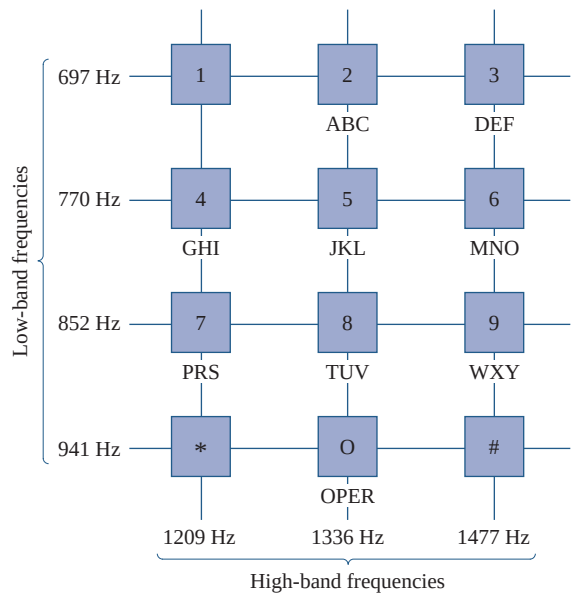


Figure 14.64
Frequency assignments for touch-tone dialing.
Adapted from G. Daryanani, *Principles of Active Network Synthesis and Design* [New York: John Wiley & Sons], 1976, p. 79.

When a caller dials a telephone number, a set of signals is transmitted to the telephone office, where the touch-tone signals are decoded by detecting the frequencies they contain. Figure 14.65 shows the block diagram for the detection scheme. The signals are first amplified and separated into their respective groups by the low-pass (LP) and highpass (HP) filters. The limiters (L) are used to convert the separated tones into square waves. The individual tones are identified using seven bandpass (BP) filters, each filter passing one tone and rejecting other tones. Each filter is followed by a detector (D), which is energized when its input voltage exceeds a certain level. The outputs of the detectors provide the required dc signals needed by the switching system to connect the caller to the party being called.

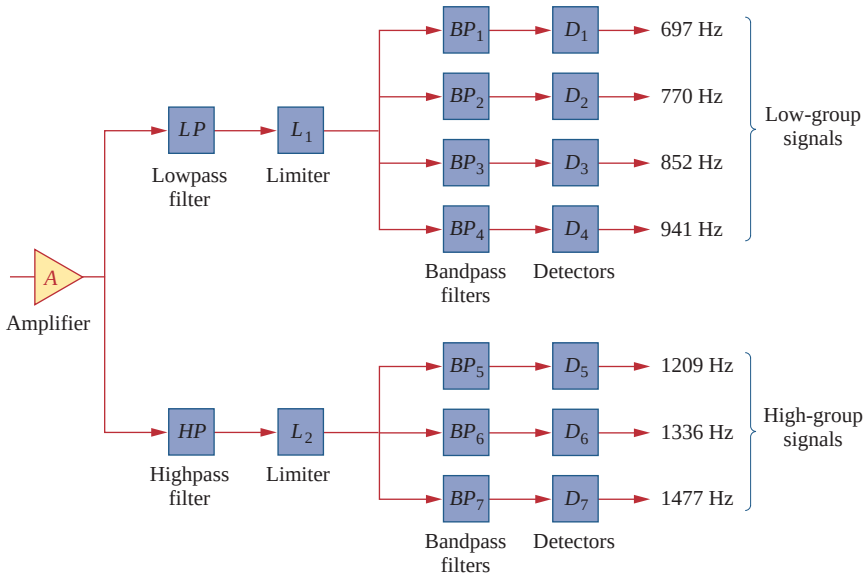


Figure 14.65
Block diagram of detection scheme.
G. Daryanani, *Principles of Active Network Synthesis and Design* [New York: John Wiley & Sons], 1976, p. 79.

Example 14.18

Using the standard 600-Ω resistor used in telephone circuits and a series *RLC* circuit, design the bandpass filter BP_2 in Fig. 14.65.

Solution:
The bandpass filter is the series *RLC* circuit in Fig. 14.35. Since BP_2 passes frequencies 697 Hz to 852 Hz and is centered at $f_0 = 770$ Hz, its bandwidth is

$$B = 2\pi(f_2 - f_1) = 2\pi(852 - 697) = 973.89 \text{ rad/s}$$

From Eq. (14.39),

$$L = \frac{R}{B} = \frac{600}{973.89} = 0.616 \text{ H}$$

From Eq. (14.27) or (14.55),

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{4\pi^2 f_0^2 L} = \frac{1}{4\pi^2 \times 770^2 \times 0.616} = 69.36 \text{ nF}$$

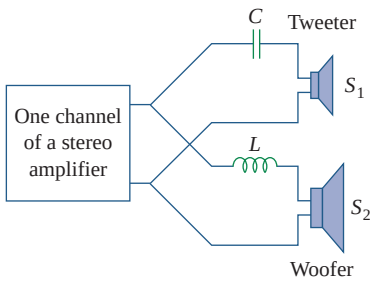
Practice Problem 14.18

Repeat Example 14.18 for bandpass filter BP_6.

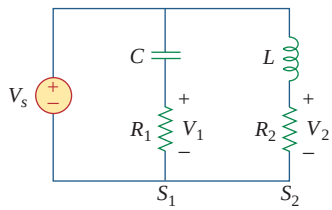
Answer: 0.356 H, 39.83 nF.

14.12.3 Crossover Network

Another typical application of filters is the *crossover network* that couples an audio amplifier to woofer and tweeter speakers, as shown in Fig. 14.66(a). The network basically consists of one highpass *RC*



(a)



(b)

Figure 14.66
(a) A crossover network for two loudspeakers, (b) equivalent model.

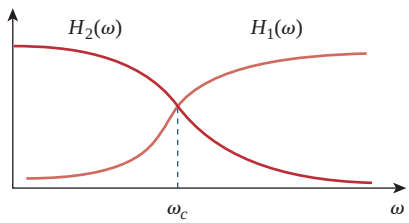


Figure 14.67
Frequency responses of the crossover network in Fig. 14.66.

filter and one lowpass *RL* filter. It routes frequencies higher than a prescribed crossover frequency f_c to the tweeter (high-frequency loudspeaker) and frequencies below f_c into the woofer (low-frequency loudspeaker). These loudspeakers have been designed to accommodate certain frequency responses. A woofer is a low-frequency loudspeaker designed to reproduce the lower part of the frequency range, up to about 3 kHz. A tweeter can reproduce audio frequencies from about 3 kHz to about 20 kHz. The two speaker types can be combined to reproduce the entire audio range of interest and provide the optimum in frequency response.

By replacing the amplifier with a voltage source, the approximate equivalent circuit of the crossover network is shown in Fig. 14.66(b), where the loudspeakers are modeled by resistors. As a highpass filter, the transfer function V_1/V_s is given by

$$H_1(\omega) = \frac{V_1}{V_s} = \frac{j\omega R_1 C}{1 + j\omega R_1 C} \tag{14.87}$$

Similarly, the transfer function of the lowpass filter is given by

$$H_2(\omega) = \frac{V_2}{V_s} = \frac{R_2}{R_2 + j\omega L} \tag{14.88}$$

The values of R_1, R_2, L, and C may be selected such that the two filters have the same cutoff frequency, known as the *crossover frequency*, as shown in Fig. 14.67.

The principle behind the crossover network is also used in the resonant circuit for a TV receiver, where it is necessary to separate the video and audio bands of RF carrier frequencies. The lower-frequency band (picture information in the range from about 30 Hz to about 4 MHz) is channeled into the receiver's video amplifier, while the high-frequency band (sound information around 4.5 MHz) is channeled to the receiver's sound amplifier.

Example 14.19

In the crossover network of Fig. 14.66, suppose each speaker acts as a 6-Ω resistance. Find C and L if the crossover frequency is 2.5 kHz.

Solution:
For the highpass filter,

$$\omega_c = 2\pi f_c = \frac{1}{R_1 C}$$

or

$$C = \frac{1}{2\pi f_c R_1} = \frac{1}{2\pi \times 2.5 \times 10^3 \times 6} = 10.61\ \mu\text{F}$$

For the lowpass filter,

$$\omega_c = 2\pi f_c = \frac{R_2}{L}$$

or

$$L = \frac{R_2}{2\pi f_c} = \frac{6}{2\pi \times 2.5 \times 10^3} = 382\ \mu\text{H}$$

Practice Problem 14.19

If each speaker in Fig. 14.66 has an 8-Ω resistance and $C = 10\ \mu$F, find L and the crossover frequency.

Answer: 0.64 mH, 1.989 kHz.

14.13 Summary

1. The transfer function $\mathbf{H}(\omega)$ is the ratio of the output response $\mathbf{Y}(\omega)$ to the input excitation $\mathbf{X}(\omega)$; that is, $\mathbf{H}(\omega) = \mathbf{Y}(\omega)/\mathbf{X}(\omega)$.
2. The frequency response is the variation of the transfer function with frequency.
3. Zeros of a transfer function $\mathbf{H}(s)$ are the values of $s = j\omega$ that make $H(s) = 0$, while poles are the values of s that make $H(s) \to \infty$.
4. The decibel is the unit of logarithmic gain. For a voltage or current gain G, its decibel equivalent is $G_{\text{dB}} = 20 \log_{10} G$.
5. Bode plots are semilog plots of the magnitude and phase of the transfer function as it varies with frequency. The straight-line approximations of H (in dB) and ϕ (in degrees) are constructed using the corner frequencies defined by the poles and zeros of $\mathbf{H}(\omega)$.
6. The resonant frequency is that frequency at which the imaginary part of a transfer function vanishes. For series and parallel *RLC* circuits.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

7. The half-power frequencies (ω_1, ω_2) are those frequencies at which the power dissipated is one-half of that dissipated at the resonant frequency. The geometric mean between the half-power frequencies is the resonant frequency, or

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

8. The bandwidth is the frequency band between half-power frequencies:

$$B = \omega_2 - \omega_1$$

9. The quality factor is a measure of the sharpness of the resonance peak. It is the ratio of the resonant (angular) frequency to the bandwidth,

$$Q = \frac{\omega_0}{B}$$

10. A filter is a circuit designed to pass a band of frequencies and reject others. Passive filters are constructed with resistors, capacitors, and inductors. Active filters are constructed with resistors, capacitors, and an active device, usually an op amp.
11. Four common types of filters are lowpass, highpass, bandpass, and bandstop. A lowpass filter passes only signals whose frequencies are below the cutoff frequency ω_c. A highpass filter passes only signals whose frequencies are above the cutoff frequency ω_c. A bandpass filter passes only signals whose frequencies are within a prescribed

range ($\omega_1 < \omega < \omega_2$). A bandstop filter passes only signals whose frequencies are outside a prescribed range ($\omega_1 > \omega > \omega_2$).

12. Scaling is the process whereby unrealistic element values are magnitude-scaled by a factor K_m and/or frequency-scaled by a factor K_f to produce realistic values.

$$R' = K_m R, \qquad L' = \frac{K_m}{K_f} L, \qquad C' = \frac{1}{K_m K_f} C$$

13. *PSpice* can be used to obtain the frequency response of a circuit if a frequency range for the response and the desired number of points within the range are specified in the AC Sweep.
14. The radio receiver—one practical application of resonant circuits—employs a bandpass resonant circuit to tune in one frequency among all the broadcast signals picked up by the antenna.
15. The touch-tone telephone and the crossover network are two typical applications of filters. The touch-tone telephone system employs filters to separate tones of different frequencies to activate electronic switches. The crossover network separates signals in different frequency ranges so that they can be delivered to different devices such as tweeters and woofers in a loudspeaker system.

Review Questions

14.1 A zero of the transfer function

$$H(s) = \frac{10(s + 1)}{(s + 2)(s + 3)}$$

is at

(a) 10 (b) −1 (c) −2 (d) −3

14.2 On the Bode magnitude plot, the slope of $1/(5 + j\omega)^2$ for large values of ω is

(a) 20 dB/decade (b) 40 dB/decade
(c) −40 dB/decade (d) −20 dB/decade

14.3 On the Bode phase plot for $0.5 < \omega < 50$, the slope of $[1 + j10\omega - \omega^2/25]^2$ is

(a) 45°/decade (b) 90°/decade
(c) 135°/decade (d) 180°/decade

14.4 How much inductance is needed to resonate at 5 kHz with a capacitance of 12 nF?

(a) 2,652 H (b) 11.844 H
(c) 3.333 H (d) 84.43 mH

14.5 The difference between the half-power frequencies is called the:

(a) quality factor (b) resonant frequency
(c) bandwidth (d) cutoff frequency

14.6 In a series *RLC* circuit, which of these quality factors has the steepest magnitude response curve near resonance?

(a) $Q = 20$ (b) $Q = 12$
(c) $Q = 8$ (d) $Q = 4$

14.7 In a parallel *RLC* circuit, the bandwidth B is directly proportional to R.

(a) True (b) False

14.8 When the elements of an *RLC* circuit are both magnitude-scaled and frequency-scaled, which quality is unaffected?

(a) resistor (b) resonant frequency
(c) bandwidth (d) quality factor

14.9 What kind of filter can be used to select a signal of one particular radio station?

(a) lowpass (b) highpass
(c) bandpass (d) bandstop

14.10 A voltage source supplies a signal of constant amplitude, from 0 to 40 kHz, to an RC lowpass filter. A load resistor, connected in parallel across the capacitor, experiences the maximum voltage at:

(a) dc (b) 10 kHz
(c) 20 kHz (d) 40 kHz

Answers: 14.1b, 14.2c, 14.3d, 14.4d, 14.5c, 14.6a, 14.7b, 14.8d, 14.9c, 14.10a.

Problems

Section 14.2 Transfer Function

14.1 Find the transfer function $\mathbf{V}_o/\mathbf{V}_i$ of the *RC* circuit in Fig. 14.68. Express it using $\omega_0 = 1/RC$.

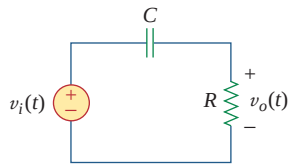


Figure 14.68
For Prob. 14.1.

14.2 Using Fig. 14.69, design a problem to help other students better understand how to determine transfer functions.

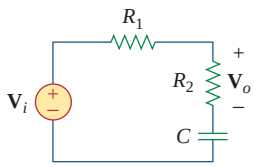


Figure 14.69
For Prob. 14.2.

14.3 Given the circuit in Fig. 14.70, $R_1 = 2\ \Omega$, $R_2 = 5\ \Omega$, $C_1 = 0.1$ F, and $C_2 = 0.2$ F, determine the transfer function $\mathbf{H}(s) = \mathbf{V}_o(s)/\mathbf{V}_i(s)$.

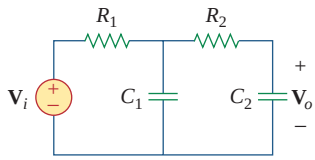
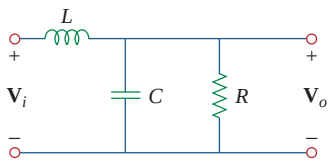
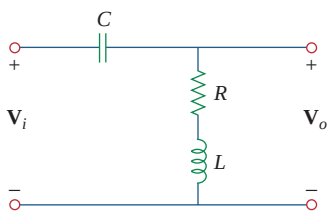


Figure 14.70
For Prob. 14.3.

14.4 Find the transfer function $\mathbf{H}(\omega) = \mathbf{V}_o/\mathbf{V}_i$ of the circuits shown in Fig. 14.71.



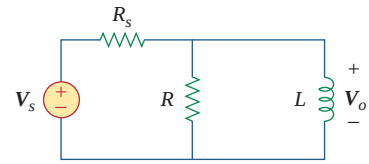
(a)



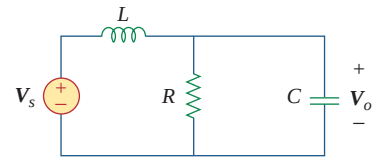
(b)

Figure 14.71
For Prob. 14.4.

14.5 For each of the circuits shown in Fig. 14.72, find $\mathbf{H}(s) = \mathbf{V}_o(s)/\mathbf{V}_s(s)$.



(a)



(b)

Figure 14.72
For Prob. 14.5.

14.6 For the circuit shown in Fig. 14.73, find $\mathbf{H}(s) = \mathbf{I}_o(s)/\mathbf{I}_s(s)$.

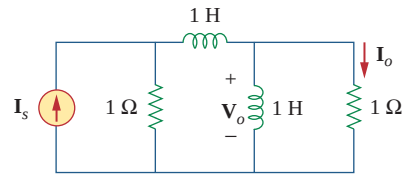


Figure 14.73
For Prob. 14.6.

Section 14.3 The Decibel Scale

14.7 Calculate $|\mathbf{H}(\omega)|$ if H_{dB} equals

(a) 0.05 dB (b) −6.2 dB (c) 104.7 dB

14.8 Design a problem to help other students calculate the magnitude in dB and phase in degrees of a variety of transfer functions at a single value of ω.

Section 14.4 Bode Plots

14.9 A ladder network has a voltage gain of

$$\mathbf{H}(\omega) = \frac{10}{(1 + j\omega)(10 + j\omega)}$$

Sketch the Bode plots for the gain.

14.10 Design a problem to help other students better understand how to determine the Bode magnitude and phase plots of a given transfer function in terms of $j\omega$.

14.11 Sketch the Bode plots for

$$\mathbf{H}(\omega) = \frac{10 + j\omega}{j\omega(2 + j\omega)}$$

14.12 A transfer function is given by

$$T(s) = \frac{s + 1}{s(s + 10)}$$

Sketch the magnitude and phase Bode plots.

14.13 Construct the Bode plots for

$$G(s) = \frac{s + 1}{s^2(s + 10)}, \qquad s = j\omega$$

14.14 Draw the Bode plots for

$$\mathbf{H}(\omega) = \frac{50(j\omega + 1)}{j\omega(-\omega^2 + 10j\omega + 25)}$$

14.15 Construct the Bode magnitude and phase plots for

$$H(s) = \frac{40(s + 1)}{(s + 2)(s + 10)}, \qquad s = j\omega$$

14.16 Sketch Bode magnitude and phase plots for

$$H(s) = \frac{10}{s(s^2 + s + 16)}, \quad s = j\omega$$

14.17 Sketch the Bode plots for

$$G(s) = \frac{s}{(s + 2)^2(s + 1)}, \qquad s = j\omega$$

14.18 A linear network has this transfer function

$$H(s) = \frac{7s^2 + s + 4}{s^3 + 8s^2 + 14s + 5}, \qquad s = j\omega$$

Use *MATLAB* or equivalent to plot the magnitude and phase (in degrees) of the transfer function. Take $0.1 < \omega < 10$ rad/s.

14.19 Sketch the asymptotic Bode plots of the magnitude and phase for

$$H(s) = \frac{100s}{(s + 10)(s + 20)(s + 40)}, \qquad s = j\omega$$

14.20 Design a more complex problem than given in Prob. 14.10, to help other students better understand how to determine the Bode magnitude and phase plots of a given transfer function in terms of $j\omega$. Include at least a second order repeated root.

14.21 Sketch the magnitude Bode plot for

$$H(s) = \frac{s(s + 20)}{(s + 1)(s^2 + 60s + 400)}, \qquad s = j\omega$$

14.22 Find the transfer function $\mathbf{H}(\omega)$ with the Bode magnitude plot shown in Fig. 14.74.

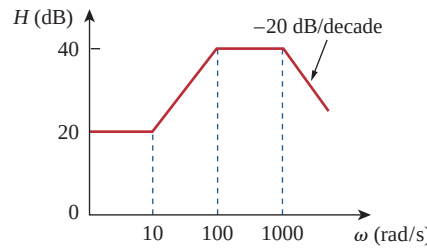


Figure 14.74
For Prob. 14.22.

14.23 The Bode magnitude plot of $\mathbf{H}(\omega)$ is shown in Fig. 14.75. Find $\mathbf{H}(\omega)$.

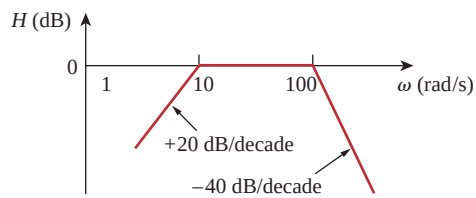


Figure 14.75
For Prob. 14.23.

14.24 The magnitude plot in Fig. 14.76 represents the transfer function of a preamplifier. Find $H(s)$.

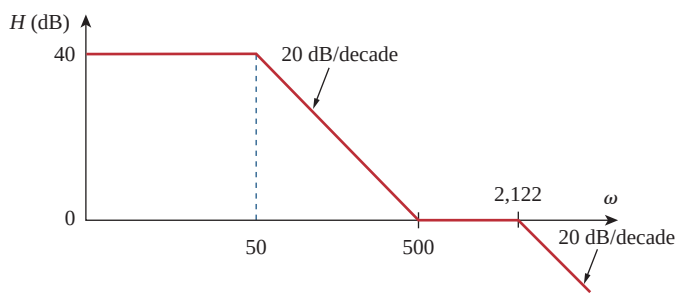


Figure 14.76
For Prob. 14.24.

Section 14.5 Series Resonance

14.25 A series *RLC* network has $R = 2\ \text{k}\Omega$, $L = 40$ mH, and $C = 1\ \mu$F. Calculate the impedance at resonance and at one-fourth, one-half, twice, and four times the resonant frequency.

14.26 Design a problem to help other students better understand ω_0, Q, and B at resonance in series *RLC* circuits.

14.27 Design a series *RLC* resonant circuit with $\omega_0 = 40$ rad/s and $B = 10$ rad/s.

14.28 Design a series *RLC* circuit with $B = 20$ rad/s and $\omega_0 = 1{,}000$ rad/s. Find the circuit's Q. Let $R = 10\ \Omega$.

14.29 Let $v_s = 120\cos(at)$ V in the circuit of Fig. 14.77. Find ω_0, Q, and B, as seen by the capacitor.

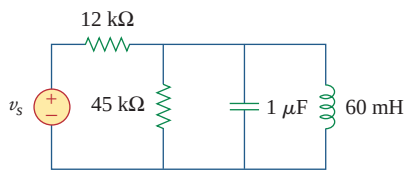


Figure 14.77
For Prob. 14.29.

14.30 A circuit consisting of a coil with inductance 10 mH and resistance 20 Ω is connected in series with a capacitor and a generator with an rms voltage of 120 V. Find:

(a) the value of the capacitance that will cause the circuit to be in resonance at 15 kHz

(b) the current through the coil at resonance

(c) the Q of the circuit

Section 14.6 Parallel Resonance

14.31 Design a parallel resonant *RLC* circuit with $\omega_0 = 10$ rad/s and $Q = 20$. Calculate the bandwidth of the circuit. Let $R = 10\ \Omega$.

14.32 Design a problem to help other students better understand the quality factor, the resonant frequency, and bandwidth of a parallel *RLC* circuit.

14.33 A parallel resonant circuit with quality factor 120 has a resonant frequency of 6×10^6 rad/s. Calculate the bandwidth and half-power frequencies.

14.34 A parallel *RLC* circuit is resonant at 5.6 MHz, has a Q of 80, and has a resistive branch of 40 kΩ. Determine the values of L and C in the other two branches.

14.35 A parallel *RLC* circuit has $R = 5\ \text{k}\Omega$, $L = 8$ mH, and $C = 60\ \mu$F. Determine:

(a) the resonant frequency

(b) the bandwidth

(c) the quality factor

14.36 It is expected that a parallel *RLC* resonant circuit has a midband admittance of 25×10^{-3} S, quality factor of 80, and a resonant frequency of 200 krad/s. Calculate the values of R, L, and C. Find the bandwidth and the half-power frequencies.

14.37 Rework Prob. 14.25 if the elements are connected in parallel.

14.38 Find the resonant frequency of the circuit in Fig. 14.78.

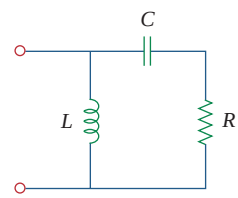


Figure 14.78
For Prob. 14.38.

14.39 For the "tank" circuit in Fig. 14.79, find the resonant frequency.

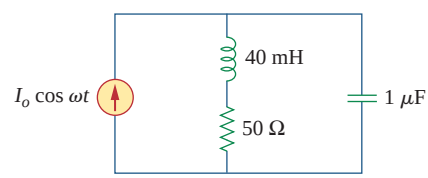


Figure 14.79
For Probs. 14.39 and 14.91.

14.40 A parallel resonance circuit has a resistance of 2 kΩ and half-power frequencies of 86 kHz and 90 kHz. Determine:

(a) the capacitance

(b) the inductance

(c) the resonant frequency

(d) the bandwidth

(e) the quality factor

14.41 Using Fig. 14.80, design a problem to help other students better understand the quality factor, the resonant frequency, and bandwidth of *RLC* circuits.

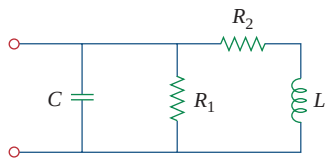


Figure 14.80
For Prob. 14.41.

14.42 For the circuits in Fig. 14.81, find the resonant frequency ω_0, the quality factor Q, and the bandwidth B.

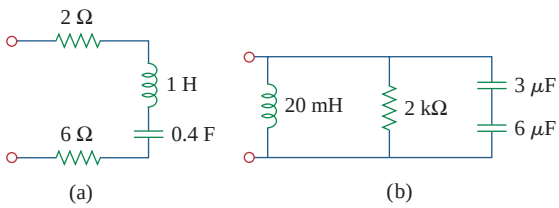


Figure 14.81
For Prob. 14.42.

14.43 Calculate the resonant frequency of each of the circuits in Fig. 14.82.

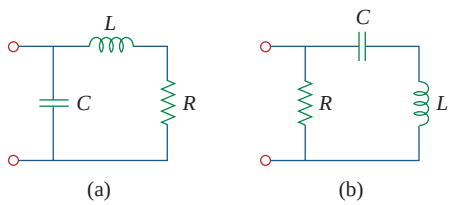


Figure 14.82
For Prob. 14.43.

***14.44** For the circuit in Fig. 14.83, find:

(a) the resonant frequency ω_0

(b) $\mathbf{Z}_{\text{in}}(\omega_0)$

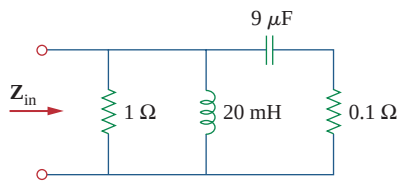


Figure 14.83
For Prob. 14.44.

14.45 For the circuit shown in Fig. 14.84, find ω_0, B, and Q, as seen by the voltage across the inductor.

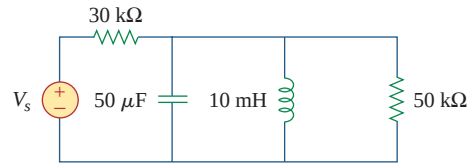


Figure 14.84
For Prob. 14.45.

14.46 For the network illustrated in Fig. 14.85, find

(a) the transfer function $\mathbf{H}(\omega) = \mathbf{V}_o(\omega)/\mathbf{I}(\omega)$,

(b) the magnitude of $\mathbf{H}$ at $\omega_0 = 1$ rad/s.

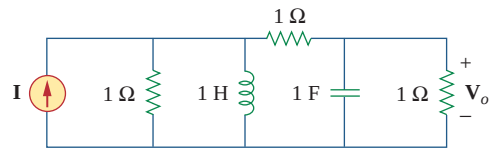


Figure 14.85
For Probs. 14.46, 14.78, and 14.92.

Section 14.7 Passive Filters

14.47 Show that a series LR circuit is a lowpass filter if the output is taken across the resistor. Calculate the corner frequency f_c if $L = 2$ mH and $R = 10$ kΩ.

14.48 Find the transfer function $\mathbf{V}_o/\mathbf{V}_s$ of the circuit in Fig. 14.86. Show that the circuit is a lowpass filter.

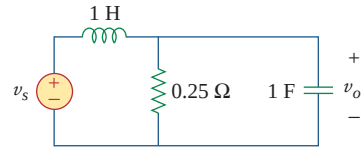


Figure 14.86
For Prob. 14.48.

14.49 Design a problem to help other students better understand lowpass filters described by transfer functions.

14.50 Determine what type of filter is in Fig. 14.87. Calculate the corner frequency f_c.

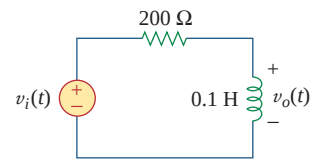


Figure 14.87
For Prob. 14.50.

* An asterisk indicates a challenging problem.

14.51 Design an *RL* lowpass filter that uses a 40-mH coil and has a cutoff frequency of 5 kHz.

14.52 Design a problem to help other students better understand passive highpass filters.

14.53 Design a series *RLC* type bandpass filter with cutoff frequencies of 10 kHz and 11 kHz. Assuming $C = 80$ pF, find R, L, and Q.

14.54 Design a passive bandstop filter with $\omega_0 = 10$ rad/s and $Q = 20$.

14.55 Determine the range of frequencies that will be passed by a series *RLC* bandpass filter with $R = 10\ \Omega$, $L = 25$ mH, and $C = 0.4\ \mu$F. Find the quality factor.

14.56 (a) Show that for a bandpass filter,

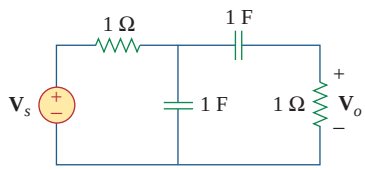
$$\mathbf{H}(s) = \frac{sB}{s^2 + sB + \omega_0^2}, \qquad s = j\omega$$

where B = bandwidth of the filter and ω_0 is the center frequency.

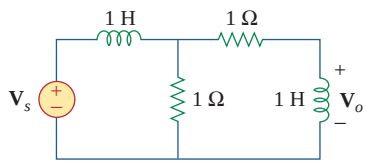
(b) Similarly, show that for a bandstop filter,

$$\mathbf{H}(s) = \frac{s^2 + \omega_0^2}{s^2 + sB + \omega_0^2}, \qquad s = j\omega$$

14.57 Determine the center frequency and bandwidth of the bandpass filters in Fig. 14.88.



(a)



(b)

Figure 14.88
For Prob. 14.57.

14.58 The circuit parameters for a series *RLC* bandstop filter are $R = 2$ kΩ, $L = 0.1$ H, $C = 40$ pF. Calculate:

(a) the center frequency

(b) the half-power frequencies

(c) the quality factor

14.59 Find the bandwidth and center frequency of the bandstop filter of Fig. 14.89.

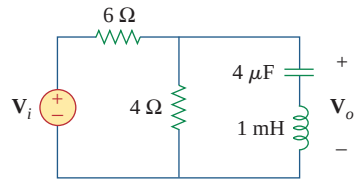
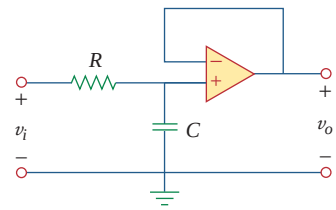


Figure 14.89
For Prob. 14.59.

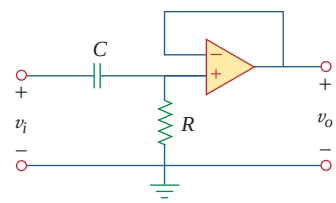
Section 14.8 Active Filters

14.60 Obtain the transfer function of a highpass filter with a passband gain of 10 and a cutoff frequency of 50 rad/s.

14.61 Find the transfer function for each of the active filters in Fig. 14.90.



(a)



(b)

Figure 14.90
For Probs. 14.61 and 14.62.

14.62 The filter in Fig. 14.90(b) has a 3-dB cutoff frequency at 1 kHz. If its input is connected to a 120-mV variable frequency signal, find the output voltage at:

(a) 200 Hz (b) 2 kHz (c) 10 kHz

14.63 Design an active first-order highpass filter with

$$\mathbf{H}(s) = -\frac{100s}{s + 10}, \qquad s = j\omega$$

Use a 1-μF capacitor.

14.64 Obtain the transfer function of the active filter in Fig. 14.91 on the next page. What kind of filter is it?

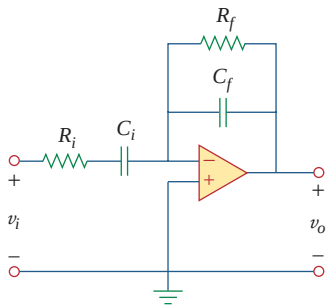


Figure 14.91
For Prob. 14.64.

14.65 A highpass filter is shown in Fig. 14.92. Show that the transfer function is

$$\mathbf{H}(\omega) = \left(1 + \frac{R_f}{R_i}\right)\frac{j\omega RC}{1 + j\omega RC}$$

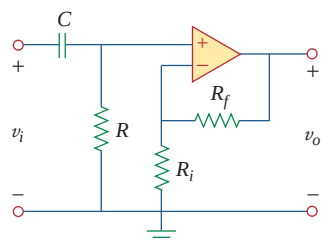


Figure 14.92
For Prob. 14.65.

14.66 A "general" first-order filter is shown in Fig. 14.93.

(a) Show that the transfer function is

$$\mathbf{H}(s) = \frac{R_4}{R_3 + R_4} \times \frac{s + (1/R_1C)[R_1/R_2 - R_3/R_4]}{s + 1/R_2C},$$
$$s = j\omega$$

(b) What condition must be satisfied for the circuit to operate as a highpass filter?

(c) What condition must be satisfied for the circuit to operate as a lowpass filter?

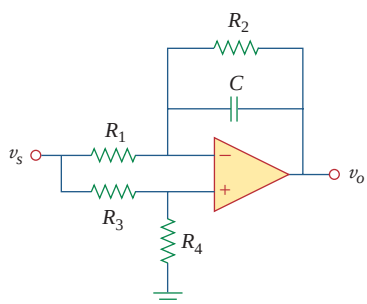


Figure 14.93
For Prob. 14.66.

14.67 Design an active lowpass filter with dc gain of 0.25 and a corner frequency of 500 Hz.

14.68 Design a problem to help other students better understand the design of active highpass filters when specifying a high-frequency gain and a corner frequency.

14.69 Design the filter in Fig. 14.94 to meet the following requirements:

(a) It must attenuate a signal at 2 kHz by 3 dB compared with its value at 10 MHz.

(b) It must provide a steady-state output of $v_o(t) = 10\sin(2\pi \times 10^8 t + 180°)$ V for an input $v_s(t) = 4\sin(2\pi \times 10^8 t)$ V.

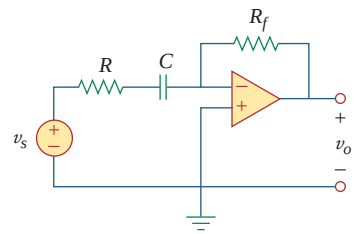


Figure 14.94
For Prob. 14.69.

***14.70** A second-order active filter known as a Butterworth filter is shown in Fig. 14.95.

(a) Find the transfer function $\mathbf{V}_o/\mathbf{V}_i$.

(b) Show that it is a lowpass filter.

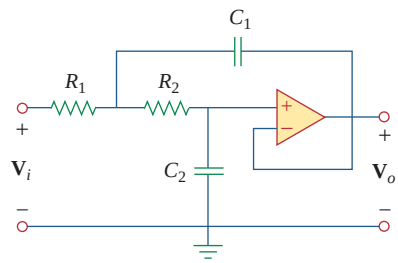


Figure 14.95
For Prob. 14.70.

Section 14.9 Scaling

14.71 Use magnitude and frequency scaling on the circuit of Fig. 14.76 to obtain an equivalent circuit in which the inductor and capacitor have magnitude 1 H and 1 F respectively.

14.72 Design a problem to help other students better understand magnitude and frequency scaling.

14.73 Calculate the values of R, L, and C that will result in $R = 12\ \text{k}\Omega$, $L = 40\ \mu\text{H}$, and $C = 300$ nF respectively when magnitude-scaled by 800 and frequency-scaled by 1000.

14.74 A circuit has $R_1 = 3\ \Omega$, $R_2 = 10\ \Omega$, $L = 2$H, and $C = 1/10$ F. After the circuit is magnitude-scaled by 100 and frequency-scaled by 10^6, find the new values of the circuit elements.

14.75 In an *RLC* circuit, $R = 20\ \Omega$, $L = 4$ H, and $C = 1$ F. The circuit is magnitude-scaled by 10 and frequency-scaled by 10^5. Calculate the new values of the elements.

14.76 Given a parallel *RLC* circuit with $R = 5$ kΩ, $L = 10$ mH, and $C = 20\ \mu$F, if the circuit is magnitude-scaled by $K_m = 500$ and frequency-scaled by $K_f = 10^5$, find the resulting values of R, L, and C.

14.77 A series *RLC* circuit has $R = 10\ \Omega$, $\omega_0 = 40$ rad/s, and $B = 5$ rad/s. Find L and C when the circuit is scaled:

(a) in magnitude by a factor of 600,

(b) in frequency by a factor of 1,000,

(c) in magnitude by a factor of 400 and in frequency by a factor of 10^5.

14.78 Redesign the circuit in Fig. 14.85 so that all resistive elements are scaled by a factor of 1,000 and all frequency-sensitive elements are frequency-scaled by a factor of 10^4.

***14.79** Refer to the network in Fig. 14.96.

(a) Find $\mathbf{Z}_{\text{in}}(s)$.

(b) Scale the elements by $K_m = 10$ and $K_f = 100$. Find $\mathbf{Z}_{\text{in}}(s)$ and ω_0.

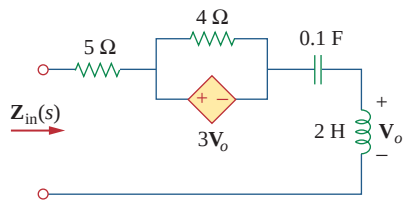


Figure 14.96
For Prob. 14.79.

14.80 (a) For the circuit in Fig. 14.97, draw the new circuit after it has been scaled by $K_m = 200$ and $K_f = 10^4$.

(b) Obtain the Thevenin equivalent impedance at terminals *a*-*b* of the scaled circuit at $\omega = 10^4$ rad/s.

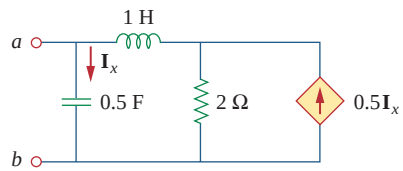


Figure 14.97
For Prob. 14.80.

14.81 The circuit shown in Fig. 14.98 has the impedance

$$Z(s) = \frac{1{,}000(s+1)}{(s+1+j50)(s+1-j50)}, \qquad s = j\omega$$

Find:

(a) the values of R, L, C, and G

(b) the element values that will raise the resonant frequency by a factor of 10^3 by frequency scaling

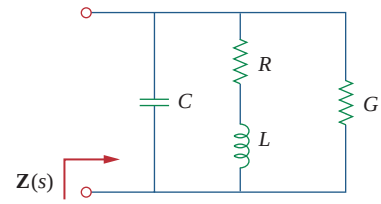


Figure 14.98
For Prob. 14.81.

14.82 Scale the lowpass active filter in Fig. 14.99 so that its corner frequency increases from 1 rad/s to 200 rad/s. Use a 1-μF capacitor.

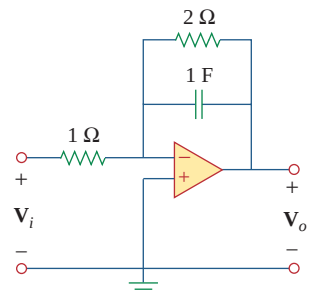


Figure 14.99
For Prob. 14.82.

14.83 The op amp circuit in Fig. 14.100 is to be magnitude-scaled by 100 and frequency-scaled by 10^5. Find the resulting element values.

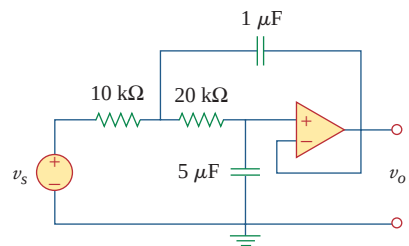


Figure 14.100
For Prob. 14.83.

Section 14.10 Frequency Response Using *PSpice*

14.84 Using *PSpice*, obtain the frequency response of the circuit in Fig. 14.101 on the next page.

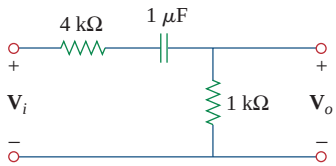


Figure 14.101
For Prob. 14.84.

14.85 Use *PSpice* to obtain the magnitude and phase plots of $\mathbf{V}_o/\mathbf{I}_s$ of the circuit in Fig. 14.102.

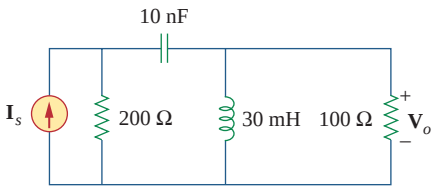


Figure 14.102
For Prob. 14.85.

14.86 Using Fig. 14.103, design a problem to help other students better understand how to use *PSpice* to obtain the frequency response (magnitude and phase of I) in electrical circuits.

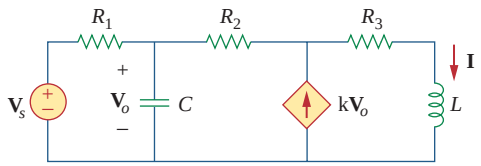


Figure 14.103
For Prob. 14.86.

14.87 In the interval $0.1 < f < 100$ Hz, plot the response of the network in Fig. 14.104. Classify this filter and obtain ω_0.

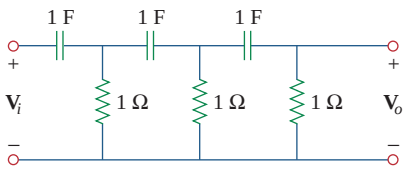


Figure 14.104
For Prob. 14.87.

14.88 Use *PSpice* to generate the magnitude and phase Bode plots of $\mathbf{V}_o$ in the circuit of Fig. 14.105.

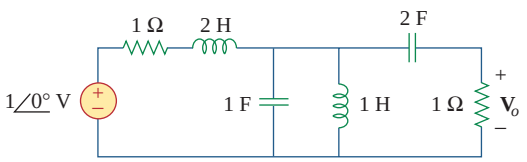


Figure 14.105
For Prob. 14.88.

14.89 Obtain the magnitude plot of the response $\mathbf{V}_o$ in the network of Fig. 14.106 for the frequency interval $100 < f < 1{,}000$ Hz.

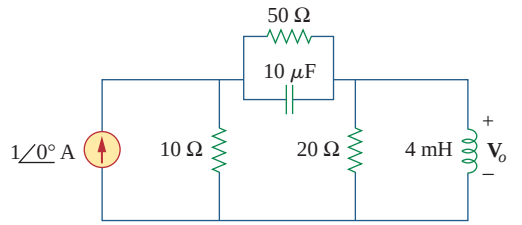


Figure 14.106
For Prob. 14.89.

14.90 Obtain the frequency response of the circuit in Fig. 14.40 (see Practice Problem 14.10). Take $R_1 = R_2 = 100\ \Omega$, $L = 2$ mH. Use $1 < f < 100{,}000$ Hz.

14.91 For the "tank" circuit of Fig. 14.79, obtain the frequency response (voltage across the capacitor) using *PSpice*. Determine the resonant frequency of the circuit.

14.92 Using *PSpice*, plot the magnitude of the frequency response of the circuit in Fig. 14.85.

Section 14.12 Applications

14.93 For the phase shifter circuit shown in Fig. 14.107, find $\mathrm{H} = V_o/V_s$.

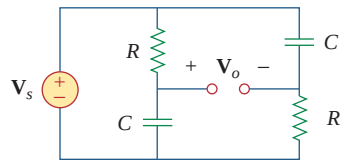


Figure 14.107
For Prob. 14.93.

14.94 For an emergency situation, an engineer needs to make an *RC* highpass filter. He has one 10-pF capacitor, one 30-pF capacitor, one 1.8-kΩ resistor, and one 3.3-kΩ resistor available. Find the greatest cutoff frequency possible using these elements.

14.95 A series-tuned antenna circuit consists of a variable capacitor (40 pF to 360 pF) and a 240-μH antenna coil that has a dc resistance of 12 Ω.

(a) Find the frequency range of radio signals to which the radio is tunable.

(b) Determine the value of Q at each end of the frequency range.

14.96 The crossover circuit in Fig. 14.108 is a lowpass filter that is connected to a woofer. Find the transfer function $\mathbf{H}(\omega) = \mathbf{V}_o(\omega)/\mathbf{V}_i(\omega)$.

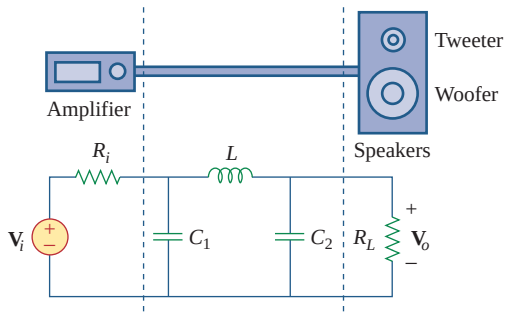


Figure 14.108
For Prob. 14.96.

14.97 The crossover circuit in Fig. 14.109 is a highpass filter that is connected to a tweeter. Determine the transfer function $\mathbf{H}(\omega) = \mathbf{V}_o(\omega)/\mathbf{V}_i(\omega)$.

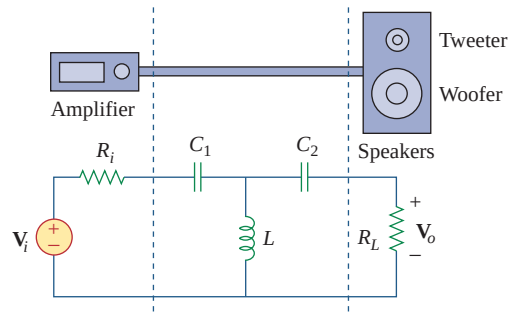


Figure 14.109
For Prob. 14.97.

Comprehensive Problems

14.98 A certain electronic test circuit produced a resonant curve with half-power points at 432 Hz and 454 Hz. If $Q = 20$, what is the resonant frequency of the circuit?

14.99 In an electronic device, a series circuit is employed that has a resistance of 100 Ω, a capacitive reactance of 5 kΩ, and an inductive reactance of 300 Ω when used at 2 MHz. Find the resonant frequency and bandwidth of the circuit.

14.100 In a certain application, a simple *RC* lowpass filter is designed to reduce high frequency noise. If the desired corner frequency is 20 kHz and $C = 0.5\ \mu$F, find the value of R.

14.101 In an amplifier circuit, a simple *RC* highpass filter is needed to block the dc component while passing the time-varying component. If the desired rolloff frequency is 15 Hz and $C = 10\ \mu$F, find the value of R.

14.102 Practical *RC* filter design should allow for source and load resistances as shown in Fig. 14.110. Let $R = 4$ kΩ and $C = 40$-nF. Obtain the cutoff frequency when:

(a) $R_s = 0$, $R_L = \infty$,

(b) $R_s = 1$ kΩ, $R_L = 5$ kΩ.

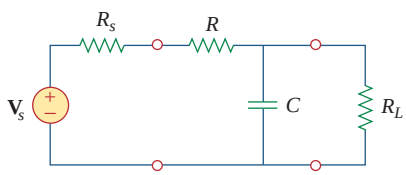


Figure 14.110
For Prob. 14.102.

14.103 The *RC* circuit in Fig. 14.111 is used for a lead compensator in a system design. Obtain the transfer function of the circuit.

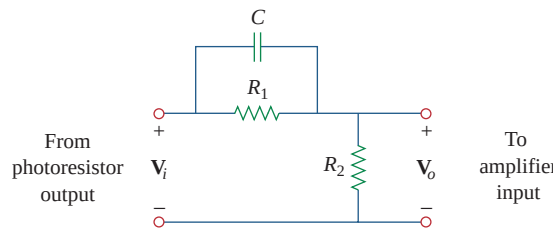


Figure 14.111
For Prob. 14.103.

14.104 A low-quality-factor, double-tuned bandpass filter is shown in Fig. 14.112. Use *PSpice* to generate the magnitude plot of $\mathbf{V}_o(\omega)$.

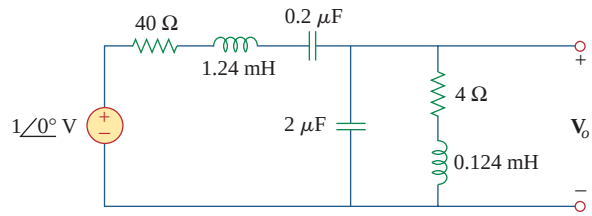


Figure 14.112
For Prob. 14.104.

PART THREE

Advanced Circuit Analysis

OUTLINE

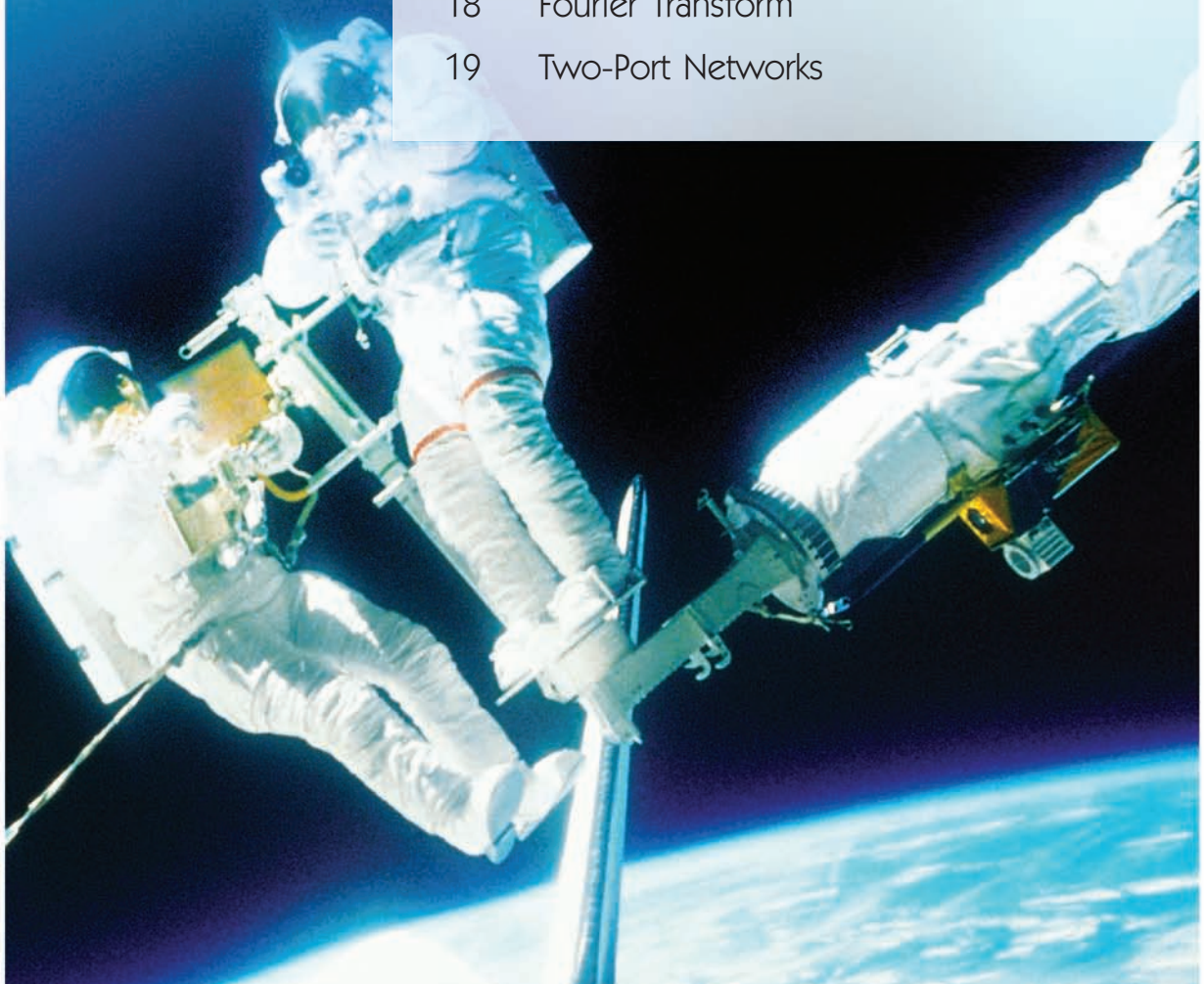
15 Introduction to the Laplace Transform

16 Applications of the Laplace Transform

17 The Fourier Series

18 Fourier Transform

19 Two-Port Networks



chapter

15

Introduction to the Laplace Transform

The important thing about a problem is not its solution, but the strength we gain in finding the solution.

—Anonymous

Enhancing Your Skills and Your Career

ABET EC 2000 criteria (3.h), *"the broad education necessary to understand the impact of engineering solutions in a global and societal context."*

As a student, you must make sure you acquire "the broad education necessary to understand the impact of engineering solutions in a global and societal context." To some extent, if you are already enrolled in an ABET-accredited engineering program, then some of the courses you are required to take must meet this criteria. My recommendation is that even if you are in such a program, you look at all the elective courses you take to make sure that you expand your awareness of global issues and societal concerns. The engineers of the future must fully understand that they and their activities affect all of us in one way or another.



Photo by Charles Alexander

ABET EC 2000 criteria (3.i), *"need for, and an ability to engage in life-long learning."*

You must be fully aware of and recognize the "need for, and an ability to engage in life-long learning." It almost seems absurd that this need and ability must be stated. Yet, you would be surprised at how many engineers do not really understand this concept. The only way to be really able to keep up with the explosion in technology we are facing now and will be facing in the future is through constant learning. This learning must include nontechnical issues as well as the latest technology in your field.

The best way to keep up with the state of the art in your field is through your colleagues and association with individuals you meet through your technical organization or organizations (especially IEEE). Reading state-of-the-art technical articles is the next best way to stay current.

Historical



Pierre Simon Laplace (1749–1827), a French astronomer and mathematician, first presented the transform that bears his name and its applications to differential equations in 1779.

Born of humble origins in Beaumont-en-Auge, Normandy, France, Laplace became a professor of mathematics at the age of 20. His mathematical abilities inspired the famous mathematician Simeon Poisson, who called Laplace the Isaac Newton of France. He made important contributions in potential theory, probability theory, astronomy, and celestial mechanics. He was widely known for his work, *Traite de Mecanique Celeste (Celestial Mechanics)*, which supplemented the work of Newton on astronomy. The Laplace transform, the subject of this chapter, is named after him.

15.1 Introduction

Our goal in this and the following chapters is to develop techniques for analyzing circuits with a wide variety of inputs and responses. Such circuits are modeled by *differential equations* whose solutions describe the total response behavior of the circuits. Mathematical methods have been devised to systematically determine the solutions of differential equations. We now introduce the powerful method of *Laplace transformation*, which involves turning differential equations into *algebraic equations*, thus greatly facilitating the solution process.

The idea of transformation should be familiar by now. When using phasors for the analysis of circuits, we transform the circuit from the time domain to the frequency or phasor domain. Once we obtain the phasor result, we transform it back to the time domain. The Laplace transform method follows the same process: we use the Laplace transformation to transform the circuit from the time domain to the frequency domain, obtain the solution, and apply the inverse Laplace transform to the result to transform it back to the time domain.

The Laplace transform is significant for a number of reasons. First, it can be applied to a wider variety of inputs than phasor analysis. Second, it provides an easy way to solve circuit problems involving initial conditions, because it allows us to work with algebraic equations instead of differential equations. Third, the Laplace transform is capable of providing us, in one single operation, the total response of the circuit comprising both the natural and forced responses.

We begin with the definition of the Laplace transform which gives rise to its most essential properties. By examining these properties, we shall see how and why the method works. This also helps us to better appreciate the idea of mathematical transformations. We also consider some properties of the Laplace transform that are very helpful in circuit analysis. We then consider the inverse Laplace transform, transfer functions, and convolution. In this chapter, we will focus on the mechanics of the Laplace transformation. In Chapter 16 we will examine how

the Laplace transform is applied in circuit analysis, network stability, and network synthesis.

15.2 Definition of the Laplace Transform

Given a function $f(t)$, its Laplace transform, denoted by $F(s)$ or $\mathcal{L}[f(t)]$, is defined by

$$\mathcal{L}[f(t)] = F(s) = \int_{0^-}^{\infty} f(t)e^{-st}\,dt \tag{15.1}$$

where s is a complex variable given by

$$s = \sigma + j\omega \tag{15.2}$$

Since the argument st of the exponent e in Eq. (15.1) must be dimensionless, it follows that s has the dimensions of frequency and units of inverse seconds (s^{-1}) or "frequency." In Eq. (15.1), the lower limit is specified as 0^- to indicate a time just before $t = 0$. We use 0^- as the lower limit to include the origin and capture any discontinuity of $f(t)$ at $t = 0$; this will accommodate functions—such as singularity functions—that may be discontinuous at $t = 0$.

For an ordinary function $f(t)$, the lower limit can be replaced by 0.

It should be noted that the integral in Eq. (15.1) is a definite integral with respect to time. Hence, the result of integration is independent of time and only involves the variable "s."

Equation (15.1) illustrates the general concept of transformation. The function $f(t)$ is transformed into the function $F(s)$. Whereas the former function involves t as its argument, the latter involves s. We say the transformation is from t-domain to s-domain. Given the interpretation of s as frequency, we arrive at the following description of the Laplace transform:

> The **Laplace transform** is an integral transformation of a function $f(t)$ from the time domain into the complex frequency domain, giving $F(s)$.

When the Laplace transform is applied to circuit analysis, the differential equations represent the circuit in the time domain. The terms in the differential equations take the place of $f(t)$. Their Laplace transform, which corresponds to $F(s)$, constitutes algebraic equations representing the circuit in the frequency domain.

We assume in Eq. (15.1) that $f(t)$ is ignored for $t < 0$. To ensure that this is the case, a function is often multiplied by the unit step. Thus, $f(t)$ is written as $f(t)u(t)$ or $f(t)$, $t \geq 0$.

The Laplace transform in Eq. (15.1) is known as the *one-sided* (or *unilateral*) Laplace transform. The *two-sided* (or *bilateral*) Laplace transform is given by

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}\,dt \tag{15.3}$$

The one-sided Laplace transform in Eq. (15.1), being adequate for our purposes, is the only type of Laplace transform that we will treat in this book.

$$|e^{j\omega t}| = \sqrt{\cos^2 \omega t + \sin^2 \omega t} = 1$$

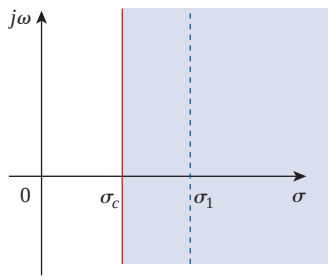


Figure 15.1
Region of convergence for the Laplace transform.

A function $f(t)$ may not have a Laplace transform. In order for $f(t)$ to have a Laplace transform, the integral in Eq. (15.1) must converge to a finite value. Since $|e^{j\omega t}| = 1$ for any value of t, the integral converges when

$$\int_{0^-}^{\infty} e^{-\sigma t} |f(t)|\, dt < \infty \tag{15.4}$$

for some real value $\sigma = \sigma_c$. Thus, the region of convergence for the Laplace transform is $\text{Re}(s) = \sigma > \sigma_c$, as shown in Fig. 15.1. In this region, $|F(s)| < \infty$ and $F(s)$ exists. $F(s)$ is undefined outside the region of convergence. Fortunately, all functions of interest in circuit analysis satisfy the convergence criterion in Eq. (15.4) and have Laplace transforms. Therefore, it is not necessary to specify σ_c in what follows.

A companion to the direct Laplace transform in Eq. (15.1) is the *inverse* Laplace transform given by

$$\mathcal{L}^{-1}[F(s)] = f(t) = \frac{1}{2\pi j} \int_{\sigma_1 - j\infty}^{\sigma_1 + j\infty} F(s) e^{st}\, ds \tag{15.5}$$

where the integration is performed along a straight line ($\sigma_1 + j\omega$, $-\infty < \omega < \infty$) in the region of convergence, $\sigma_1 > \sigma_c$. See Fig. 15.1. The direct application of Eq. (15.5) involves some knowledge about complex analysis beyond the scope of this book. For this reason, we will not use Eq. (15.5) to find the inverse Laplace transform. We will rather use a look-up table, to be developed in Section 15.3. The functions $f(t)$ and $F(s)$ are regarded as a Laplace transform pair where

$$f(t) \quad \Leftrightarrow \quad F(s) \tag{15.6}$$

meaning that there is one-to-one correspondence between $f(t)$ and $F(s)$. The following examples derive the Laplace transforms of some important functions.

Example 15.1

Determine the Laplace transform of each of the following functions: (a) $u(t)$, (b) $e^{-at}u(t)$, $a \geq 0$, and (c) $\delta(t)$.

Solution:

(a) For the unit step function $u(t)$, shown in Fig. 15.2(a), the Laplace transform is

$$\begin{aligned} \mathcal{L}[u(t)] &= \int_{0^-}^{\infty} 1e^{-st}\, dt = -\frac{1}{s} e^{-st} \Big|_0^{\infty} \\ &= -\frac{1}{s}(0) + \frac{1}{s}(1) = \frac{1}{s} \end{aligned} \tag{15.1.1}$$

(b) For the exponential function, shown in Fig. 15.2(b), the Laplace transform is

$$\begin{aligned} \mathcal{L}[e^{-at}u(t)] &= \int_{0^-}^{\infty} e^{-at} e^{-st}\, dt \\ &= -\frac{1}{s+a} e^{-(s+a)t} \Big|_0^{\infty} = \frac{1}{s+a} \end{aligned} \tag{15.1.2}$$

(c) For the unit impulse function, shown in Fig. 15.2(c),

$$\mathcal{L}[\delta(t)] = \int_{0^-}^{\infty} \delta(t)e^{-st}\,dt = e^{-0} = 1 \qquad \textbf{(15.1.3)}$$

since the impulse function $\delta(t)$ is zero everywhere except at $t = 0$. The sifting property in Eq. (7.33) has been applied in Eq. (15.1.3).

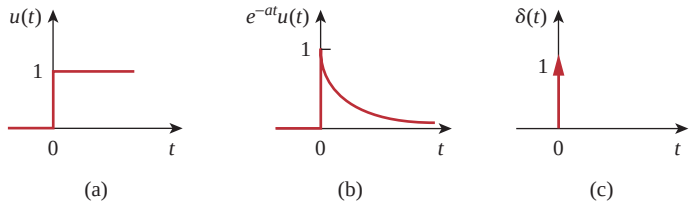


Figure 15.2
For Example 15.1: (a) unit step function, (b) exponential function, (c) unit impulse function.

Practice Problem 15.1

Find the Laplace transforms of these functions: $r(t) = tu(t)$, that is, the ramp function; $e^{-at}u(t)$; and $e^{-j\omega t}u(t)$.

Answer: $1/s^2$, $1/(s + a)$, $1/(s + j\omega)$.

Example 15.2

Determine the Laplace transform of $f(t) = \sin\omega t\,u(t)$.

Solution:
Using Eq. (B.27) in addition to Eq. (15.1), we obtain the Laplace transform of the sine function as

$$F(s) = \mathcal{L}[\sin\omega t] = \int_0^{\infty} (\sin\omega t)e^{-st}\,dt = \int_0^{\infty} \left(\frac{e^{j\omega t} - e^{-j\omega t}}{2j}\right) e^{-st}\,dt$$

$$= \frac{1}{2j}\int_0^{\infty} (e^{-(s-j\omega)t} - e^{-(s+j\omega)t})\,dt$$

$$= \frac{1}{2j}\left(\frac{1}{s - j\omega} - \frac{1}{s + j\omega}\right) = \frac{\omega}{s^2 + \omega^2}$$

Practice Problem 15.2

Find the Laplace transform of $f(t) = 10\cos\omega t\,u(t)$.

Answer: $10s/(s^2 + \omega^2)$.

15.3 Properties of the Laplace Transform

The properties of the Laplace transform help us to obtain transform pairs without directly using Eq. (15.1) as we did in Examples 15.1 and 15.2. As we derive each of these properties, we should keep in mind the definition of the Laplace transform in Eq. (15.1).

Linearity

If $F_1(s)$ and $F_2(s)$ are, respectively, the Laplace transforms of $f_1(t)$ and $f_2(t)$, then

$$\mathcal{L}[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(s) + a_2 F_2(s) \tag{15.7}$$

where a_1 and a_2 are constants. Equation 15.7 expresses the linearity property of the Laplace transform. The proof of Eq. (15.7) follows readily from the definition of the Laplace transform in Eq. (15.1).

For example, by the linearity property in Eq. (15.7), we may write

$$\mathcal{L}[\cos\omega t u(t)] = \mathcal{L}\left[\frac{1}{2}(e^{j\omega t} + e^{-j\omega t})\right] = \frac{1}{2}\mathcal{L}[e^{j\omega t}] + \frac{1}{2}\mathcal{L}[e^{-j\omega t}] \tag{15.8}$$

But from Example 15.1(b), $\mathcal{L}[e^{-at}] = 1/(s + a)$. Hence,

$$\mathcal{L}[\cos\omega t u(t)] = \frac{1}{2}\left(\frac{1}{s - j\omega} + \frac{1}{s + j\omega}\right) = \frac{s}{s^2 + \omega^2} \tag{15.9}$$

Scaling

If $F(s)$ is the Laplace transform of $f(t)$, then

$$\mathcal{L}[f(at)] = \int_{0^-}^{\infty} f(at)e^{-st}\,dt \tag{15.10}$$

where a is a constant and $a > 0$. If we let $x = at$, $dx = a\,dt$, then

$$\mathcal{L}[f(at)] = \int_{0^-}^{\infty} f(x)e^{-x(s/a)}\frac{dx}{a} = \frac{1}{a}\int_{0^-}^{\infty} f(x)e^{-x(s/a)}\,dx \tag{15.11}$$

Comparing this integral with the definition of the Laplace transform in Eq. (15.1) shows that s in Eq. (15.1) must be replaced by s/a while the dummy variable t is replaced by x. Hence, we obtain the scaling property as

$$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right) \tag{15.12}$$

For example, we know from Example 15.2 that

$$\mathcal{L}[\sin\omega t u(t)] = \frac{\omega}{s^2 + \omega^2} \tag{15.13}$$

Using the scaling property in Eq. (15.12),

$$\mathcal{L}[\sin 2\omega t u(t)] = \frac{1}{2}\frac{\omega}{(s/2)^2 + \omega^2} = \frac{2\omega}{s^2 + 4\omega^2} \tag{15.14}$$

which may also be obtained from Eq. (15.13) by replacing ω with 2ω.

Time Shift

If $F(s)$ is the Laplace transform of $f(t)$, then

$$\mathcal{L}[f(t - a)u(t - a)] = \int_{0^-}^{\infty} f(t - a)u(t - a)e^{-st}\,dt \qquad a \geq 0 \tag{15.15}$$

But $u(t-a)=0$ for $t<a$ and $u(t-a)=1$ for $t>a$. Hence,

$$\mathcal{L}[f(t-a)u(t-a)] = \int_a^\infty f(t-a)e^{-st}\,dt \tag{15.16}$$

If we let $x=t-a$, then $dx=dt$ and $t=x+a$. As $t\to a$, $x\to 0$ and as $t\to\infty$, $x\to\infty$. Thus,

$$\begin{aligned}\mathcal{L}[f(t-a)u(t-a)] &= \int_{0^-}^\infty f(x)e^{-s(x+a)}\,dx \\ &= e^{-as}\int_{0^-}^\infty f(x)e^{-sx}\,dx = e^{-as}F(s)\end{aligned}$$

or

$$\boxed{\mathcal{L}[f(t-a)u(t-a)] = e^{-as}F(s)} \tag{15.17}$$

In other words, if a function is delayed in time by a, the result in the s-domain is found by multiplying the Laplace transform of the function (without the delay) by e^{-as}. This is called the *time-delay* or *time-shift property* of the Laplace transform.

As an example, we know from Eq. (15.9) that

$$\mathcal{L}[\cos\omega t\,u(t)] = \frac{s}{s^2+\omega^2}$$

Using the time-shift property in Eq. (15.17),

$$\mathcal{L}[\cos\omega(t-a)u(t-a)] = e^{-as}\frac{s}{s^2+\omega^2} \tag{15.18}$$

Frequency Shift

If $F(s)$ is the Laplace transform of $f(t)$, then

$$\begin{aligned}\mathcal{L}[e^{-at}f(t)u(t)] &= \int_0^\infty e^{-at}f(t)e^{-st}\,dt \\ &= \int_0^\infty f(t)e^{-(s+a)t}\,dt = F(s+a)\end{aligned}$$

or

$$\boxed{\mathcal{L}[e^{-at}f(t)u(t)] = F(s+a)} \tag{15.19}$$

That is, the Laplace transform of $e^{-at}f(t)$ can be obtained from the Laplace transform of $f(t)$ by replacing every s with $s+a$. This is known as *frequency shift* or *frequency translation.*

As an example, we know that

$$\cos\omega t\,u(t) \quad\Leftrightarrow\quad \frac{s}{s^2+\omega^2}$$

and

$$\sin\omega t\,u(t) \quad\Leftrightarrow\quad \frac{\omega}{s^2+\omega^2} \tag{15.20}$$

Using the shift property in Eq. (15.19), we obtain the Laplace transform of the damped sine and damped cosine functions as

$$\mathcal{L}[e^{-at}\cos\omega t u(t)] = \frac{s+a}{(s+a)^2+\omega^2} \tag{15.21a}$$

$$\mathcal{L}[e^{-at}\sin\omega t u(t)] = \frac{\omega}{(s+a)^2+\omega^2} \tag{15.21b}$$

Time Differentiation

Given that $F(s)$ is the Laplace transform of $f(t)$, the Laplace transform of its derivative is

$$\mathcal{L}\left[\frac{df}{dt}u(t)\right] = \int_{0^-}^{\infty}\frac{df}{dt}e^{-st}\,dt \tag{15.22}$$

To integrate this by parts, we let $u = e^{-st}$, $du = -se^{-st}\,dt$, and $dv = (df/dt)\,dt = df(t)$, $v = f(t)$. Then

$$\mathcal{L}\left[\frac{df}{dt}u(t)\right] = f(t)e^{-st}\Big|_{0^-}^{\infty} - \int_{0^-}^{\infty} f(t)[-se^{-st}]\,dt$$

$$= 0 - f(0^-) + s\int_{0^-}^{\infty} f(t)e^{-st}\,dt = sF(s) - f(0^-)$$

or

$$\boxed{\mathcal{L}[f'(t)] = sF(s) - f(0^-)} \tag{15.23}$$

The Laplace transform of the second derivative of $f(t)$ is a repeated application of Eq. (15.23) as

$$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s\mathcal{L}[f'(t)] - f'(0^-) = s[sF(s) - f(0^-)] - f'(0^-)$$

$$= s^2F(s) - sf(0^-) - f'(0^-)$$

or

$$\boxed{\mathcal{L}[f''(t)] = s^2F(s) - sf(0^-) - f'(0^-)} \tag{15.24}$$

Continuing in this manner, we can obtain the Laplace transform of the nth derivative of $f(t)$ as

$$\boxed{\mathcal{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - s^{n-1}f(0^-) - s^{n-2}f'(0^-) - \cdots - s^0f^{(n-1)}(0^-)} \tag{15.25}$$

As an example, we can use Eq. (15.23) to obtain the Laplace transform of the sine from that of the cosine. If we let $f(t) = \cos\omega t u(t)$, then $f(0) = 1$ and $f'(t) = -\omega\sin\omega t u(t)$. Using Eq. (15.23) and the scaling property,

$$\mathcal{L}[\sin\omega t u(t)] = -\frac{1}{\omega}\mathcal{L}[f'(t)] = -\frac{1}{\omega}[sF(s) - f(0^-)]$$

$$= -\frac{1}{\omega}\left(s\frac{s}{s^2+\omega^2} - 1\right) = \frac{\omega}{s^2+\omega^2} \tag{15.26}$$

as expected.

Time Integration

If $F(s)$ is the Laplace transform of $f(t)$, the Laplace transform of its integral is

$$\mathcal{L}\left[\int_0^t f(t)\,dt\right] = \int_{0^-}^{\infty}\left[\int_0^t f(x)\,dx\right]e^{-st}\,dt \tag{15.27}$$

To integrate this by parts, we let

$$u = \int_0^t f(x)\,dx, \qquad du = f(t)\,dt$$

and

$$dv = e^{-st}\,dt, \qquad v = -\frac{1}{s}e^{-st}$$

Then

$$\mathcal{L}\left[\int_0^t f(t)\,dt\right] = \left[\int_0^t f(x)\,dx\right]\left(-\frac{1}{s}e^{-st}\right)\Bigg|_{0^-}^{\infty} - \int_{0^-}^{\infty}\left(-\frac{1}{s}\right)e^{-st}f(t)\,dt$$

For the first term on the right-hand side of the equation, evaluating the term at $t = \infty$ yields zero due to $e^{-s\infty}$ and evaluating it at $t = 0$ gives $\frac{1}{s}\int_0^0 f(x)\,dx = 0$. Thus, the first term is zero, and

$$\mathcal{L}\left[\int_0^t f(t)\,dt\right] = \frac{1}{s}\int_{0^-}^{\infty} f(t)e^{-st}\,dt = \frac{1}{s}F(s)$$

or simply,

$$\boxed{\mathcal{L}\left[\int_0^t f(t)\,dt\right] = \frac{1}{s}F(s)} \tag{15.28}$$

As an example, if we let $f(t) = u(t)$, from Example 15.1(a), $F(s) = 1/s$. Using Eq. (15.28),

$$\mathcal{L}\left[\int_0^t f(t)\,dt\right] = \mathcal{L}[t] = \frac{1}{s}\left(\frac{1}{s}\right)$$

Thus, the Laplace transform of the ramp function is

$$\mathcal{L}[t] = \frac{1}{s^2} \tag{15.29}$$

Applying Eq. (15.28), this gives

$$\mathcal{L}\left[\int_0^t t\,dt\right] = \mathcal{L}\left[\frac{t^2}{2}\right] = \frac{1}{s}\frac{1}{s^2}$$

or

$$\mathcal{L}[t^2] = \frac{2}{s^3} \tag{15.30}$$

Repeated applications of Eq. (15.28) lead to

$$\mathcal{L}[t^n] = \frac{n!}{s^{n+1}} \tag{15.31}$$

Similarly, using integration by parts, we can show that

$$\mathcal{L}\left[\int_{-\infty}^{t} f(t)\,dt\right] = \frac{1}{s}F(s) + \frac{1}{s}f^{-1}(0^-) \tag{15.32}$$

where

$$f^{-1}(0^-) = \int_{-\infty}^{0^-} f(t)\,dt$$

Frequency Differentiation

If $F(s)$ is the Laplace transform of $f(t)$, then

$$F(s) = \int_{0^-}^{\infty} f(t)e^{-st}\,dt$$

Taking the derivative with respect to s,

$$\frac{dF(s)}{ds} = \int_{0^-}^{\infty} f(t)(-te^{-st})\,dt = \int_{0^-}^{\infty} (-tf(t))e^{-st}dt = \mathcal{L}[-tf(t)]$$

and the frequency differentiation property becomes

$$\boxed{\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}} \tag{15.33}$$

Repeated applications of this equation lead to

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n F(s)}{ds^n} \tag{15.34}$$

For example, we know from Example 15.1(b) that $\mathcal{L}[e^{-at}] = 1/(s + a)$. Using the property in Eq. (15.33),

$$\mathcal{L}[te^{-at}u(t)] = -\frac{d}{ds}\left(\frac{1}{s+a}\right) = \frac{1}{(s+a)^2} \tag{15.35}$$

Note that if $a = 0$, we obtain $\mathcal{L}[t] = 1/s^2$ as in Eq. (15.29), and repeated applications of Eq. (15.33) will yield Eq. (15.31).

Time Periodicity

If function $f(t)$ is a periodic function such as shown in Fig. 15.3, it can be represented as the sum of time-shifted functions shown in Fig. 15.4. Thus,

$$\begin{aligned} f(t) &= f_1(t) + f_2(t) + f_3(t) + \cdots \\ &= f_1(t) + f_1(t - T)u(t - T) \\ &\quad + f_1(t - 2T)u(t - 2T) + \cdots \end{aligned} \tag{15.36}$$

where $f_1(t)$ is the same as the function $f(t)$ gated over the interval $0 < t < T$, that is,

$$f_1(t) = f(t)[u(t) - u(t - T)] \tag{15.37a}$$

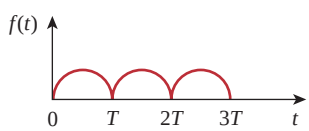


Figure 15.3
A periodic function.

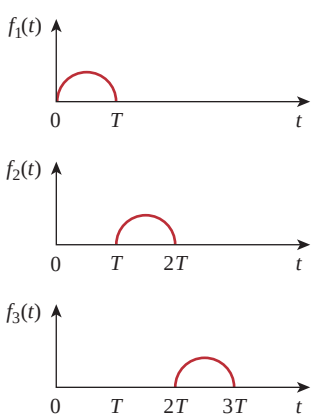


Figure 15.4
Decomposition of the periodic function in Fig. 15.2.

or

$$f_1(t) = \begin{cases} f(t), & 0 < t < T \\ 0, & \text{otherwise} \end{cases} \tag{15.37b}$$

We now transform each term in Eq. (15.36) and apply the time-shift property in Eq. (15.17). We obtain

$$\begin{aligned} F(s) &= F_1(s) + F_1(s)e^{-Ts} + F_1(s)e^{-2Ts} + F_1(s)e^{-3Ts} + \cdots \\ &= F_1(s)[1 + e^{-Ts} + e^{-2Ts} + e^{-3Ts} + \cdots] \end{aligned} \tag{15.38}$$

But

$$1 + x + x^2 + x^3 + \cdots = \frac{1}{1 - x} \tag{15.39}$$

if $|x| < 1$. Hence,

$$\boxed{F(s) = \frac{F_1(s)}{1 - e^{-Ts}}} \tag{15.40}$$

where $F_1(s)$ is the Laplace transform of $f_1(t)$; in other words, $F_1(s)$ is the transform $f(t)$ defined over its first period only. Equation (15.40) shows that the Laplace transform of a periodic function is the transform of the first period of the function divided by $1 - e^{-Ts}$.

Initial and Final Values

The initial-value and final-value properties allow us to find the initial value $f(0)$ and the final value $f(\infty)$ of $f(t)$ directly from its Laplace transform $F(s)$. To obtain these properties, we begin with the differentiation property in Eq. (15.23), namely,

$$sF(s) - f(0) = \mathcal{L}\left[\frac{df}{dt}\right] = \int_{0^-}^{\infty} \frac{df}{dt} e^{-st}\, dt \tag{15.41}$$

If we let $s \to \infty$, the integrand in Eq. (15.41) vanishes due to the damping exponential factor, and Eq. (15.41) becomes

$$\lim_{s\to\infty} [sF(s) - f(0)] = 0$$

Since $f(0)$ is independent of s, we can write

$$\boxed{f(0) = \lim_{s\to\infty} sF(s)} \tag{15.42}$$

This is known as the *initial-value theorem*. For example, we know from Eq. (15.21a) that

$$f(t) = e^{-2t}\cos 10t \quad \Leftrightarrow \quad F(s) = \frac{s + 2}{(s + 2)^2 + 10^2} \tag{15.43}$$

Using the initial-value theorem,

$$\begin{aligned} f(0) = \lim_{s\to\infty} sF(s) &= \lim_{s\to\infty} \frac{s^2 + 2s}{s^2 + 4s + 104} \\ &= \lim_{s\to\infty} \frac{1 + 2/s}{1 + 4/s + 104/s^2} = 1 \end{aligned}$$

which confirms what we would expect from the given $f(t)$.

In Eq. (15.41), we let $s \to 0$; then

$$\lim_{s\to 0}[sF(s) - f(0^-)] = \int_{0^-}^{\infty} \frac{df}{dt} e^{0t}\, dt = \int_{0^-}^{\infty} df = f(\infty) - f(0^-)$$

or

$$\boxed{f(\infty) = \lim_{s\to 0} sF(s)} \tag{15.44}$$

This is referred to as the *final-value theorem*. In order for the final-value theorem to hold, all poles of $F(s)$ must be located in the left half of the s plane (see Fig. 15.1 or Fig. 15.9); that is, the poles must have negative real parts. The only exception to this requirement is the case in which $F(s)$ has a simple pole at $s = 0$, because the effect of $1/s$ will be nullified by $sF(s)$ in Eq. (15.44). For example, from Eq. (15.21b),

$$f(t) = e^{-2t} \sin 5t\, u(t) \quad \Leftrightarrow \quad F(s) = \frac{5}{(s+2)^2 + 5^2} \tag{15.45}$$

Applying the final-value theorem,

$$f(\infty) = \lim_{s\to 0} sF(s) = \lim_{s\to 0} \frac{5s}{s^2 + 4s + 29} = 0$$

as expected from the given $f(t)$. As another example,

$$f(t) = \sin t\, u(t) \quad \Leftrightarrow \quad f(s) = \frac{1}{s^2 + 1} \tag{15.46}$$

so that

$$f(\infty) = \lim_{s\to 0} sF(s) = \lim_{s\to 0} \frac{s}{s^2 + 1} = 0$$

This is incorrect, because $f(t) = \sin t$ oscillates between $+1$ and -1 and does not have a limit as $t \to \infty$. Thus, the final-value theorem cannot be used to find the final value of $f(t) = \sin t$, because $F(s)$ has poles at $s = \pm j$, which are not in the left half of the s plane. In general, the final-value theorem does not apply in finding the final values of sinusoidal functions—these functions oscillate forever and do not have final values.

The initial-value and final-value theorems depict the relationship between the origin and infinity in the time domain and the s-domain. They serve as useful checks on Laplace transforms.

Table 15.1 provides a list of the properties of the Laplace transform. The last property (on convolution) will be proved in Section 15.5. There are other properties, but these are enough for present purposes. Table 15.2 summarizes the Laplace transforms of some common functions. We have omitted the factor $u(t)$ except where it is necessary.

We should mention that many software packages, such as Mathcad, *MATLAB*, Maple, and Mathematica, offer symbolic math. For example, Mathcad has symbolic math for the Laplace, Fourier, and Z transforms as well as the inverse function.

TABLE 15.1

Properties of the Laplace transform.

Property	$f(t)$	$F(s)$
Linearity	$a_1 f_1(t) + a_2 f_2(t)$	$a_1F_1(s) + a_2F_2(s)$
Scaling	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Time shift	$f(t-a)u(t-a)$	$e^{-as}F(s)$
Frequency shift	$e^{-at}f(t)$	$F(s+a)$
Time differentiation	$\frac{df}{dt}$	$sF(s) - f(0^-)$
	$\frac{d^2f}{dt^2}$	$s^2F(s) - sf(0^-) - f'(0^-)$
	$\frac{d^3f}{dt^3}$	$s^3F(s) - s^2f(0^-) - sf'(0^-) - f''(0^-)$
	$\frac{d^nf}{dt^n}$	$s^nF(s) - s^{n-1}f(0^-) - s^{n-2}f'(0^-) - \cdots - f^{(n-1)}(0^-)$
Time integration	$\int_0^t f(t)dt$	$\frac{1}{s}F(s)$
Frequency differentiation	$tf(t)$	$-\frac{d}{ds}F(s)$
Frequency integration	$\frac{f(t)}{t}$	$\int_s^\infty F(s)ds$
Time periodicity	$f(t) = f(t+nT)$	$\frac{F_1(s)}{1-e^{-sT}}$
Initial value	$f(0)$	$\lim_{s\to\infty} sF(s)$
Final value	$f(\infty)$	$\lim_{s\to 0} sF(s)$
Convolution	$f_1(t) * f_2(t)$	$F_1(s)F_2(s)$

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
te^{-at}	$\frac{1}{(s+a)^2}$
t^ne^{-at}	$\frac{n!}{(s+a)^{n+1}}$
$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
$\sin(\omega t + \theta)$	$\frac{s\sin\theta + \omega\cos\theta}{s^2+\omega^2}$
$\cos(\omega t + \theta)$	$\frac{s\cos\theta - \omega\sin\theta}{s^2+\omega^2}$
$e^{-at}\sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$

*Defined for $t \geq 0$; $f(t) = 0$, for $t < 0$.

Example 15.3

Obtain the Laplace transform of $f(t) = \delta(t) + 2u(t) - 3e^{-2t}u(t)$.

Solution:

By the linearity property,

$$F(s) = \mathcal{L}[\delta(t)] + 2\mathcal{L}[u(t)] - 3\mathcal{L}[e^{-2t}u(t)]$$
$$= 1 + 2\frac{1}{s} - 3\frac{1}{s+2} = \frac{s^2+s+4}{s(s+2)}$$

Practice Problem 15.3

Find the Laplace transform of $f(t) = (\cos(3t) + e^{-5t})u(t)$.

Answer: $\dfrac{2s^2+5s+9}{(s+5)(s^2+9)}$.

Example 15.4

Determine the Laplace transform of $f(t) = t^2 \sin 2t\, u(t)$.

Solution:
We know that

$$\mathcal{L}[\sin 2t] = \frac{2}{s^2 + 2^2}$$

Using frequency differentiation in Eq. (15.34),

$$F(s) = \mathcal{L}[t^2 \sin 2t] = (-1)^2 \frac{d^2}{ds^2}\left(\frac{2}{s^2 + 4}\right)$$
$$= \frac{d}{ds}\left(\frac{-4s}{(s^2 + 4)^2}\right) = \frac{12s^2 - 16}{(s^2 + 4)^3}$$

Practice Problem 15.4

Find the Laplace transform of $f(t) = t^2 \cos 3t\, u(t)$.

Answer: $\dfrac{2s(s^2 - 27)}{(s^2 + 9)^3}$.

Example 15.5

Find the Laplace transform of the gate function in Fig. 15.5.

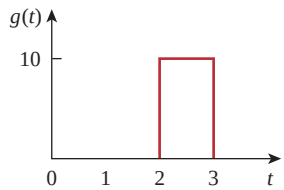


Figure 15.5
The gate function; for Example 15.5.

Solution:
We can express the gate function in Fig. 15.5 as

$$g(t) = 10[u(t - 2) - u(t - 3)]$$

Since we know the Laplace transform of $u(t)$, we apply the time-shift property and obtain

$$G(s) = 10\left(\frac{e^{-2s}}{s} - \frac{e^{-3s}}{s}\right) = \frac{10}{s}(e^{-2s} - e^{-3s})$$

Practice Problem 15.5

Find the Laplace transform of the function $h(t)$ in Fig. 15.6.

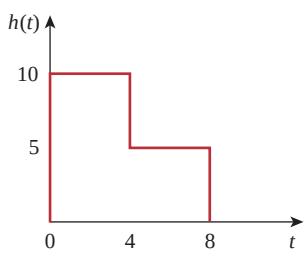


Figure 15.6
For Practice Prob. 15.5.

Answer: $\dfrac{5}{s}(2 - e^{-4s} - e^{-8s})$.

Example 15.6

Calculate the Laplace transform of the periodic function in Fig. 15.7.

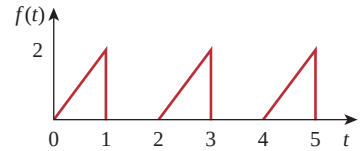


Figure 15.7
For Example 15.6.

Solution:
The period of the function is $T = 2$. To apply Eq. (15.40), we first obtain the transform of the first period of the function.

$$\begin{aligned} f_1(t) &= 2t[u(t) - u(t-1)] = 2tu(t) - 2tu(t-1) \\ &= 2tu(t) - 2(t-1+1)u(t-1) \\ &= 2tu(t) - 2(t-1)u(t-1) - 2u(t-1) \end{aligned}$$

Using the time-shift property,

$$F_1(s) = \frac{2}{s^2} - 2\frac{e^{-s}}{s^2} - \frac{2}{s}e^{-s} = \frac{2}{s^2}(1 - e^{-s} - se^{-s})$$

Thus, the transform of the periodic function in Fig. 15.7 is

$$F(s) = \frac{F_1(s)}{1 - e^{-Ts}} = \frac{2}{s^2(1 - e^{-2s})}(1 - e^{-s} - se^{-s})$$

Practice Problem 15.6

Determine the Laplace transform of the periodic function in Fig. 15.8.

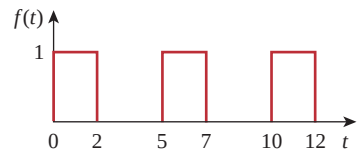


Figure 15.8
For Practice Prob. 15.6.

Answer: $\dfrac{1 - e^{-2s}}{s(1 - e^{-5s})}$.

Example 15.7

Find the initial and final values of the function whose Laplace transform is

$$H(s) = \frac{20}{(s+3)(s^2 + 8s + 25)}$$

Solution:
Applying the initial-value theorem,

$$h(0) = \lim_{s\to\infty} sH(s) = \lim_{s\to\infty} \frac{20s}{(s+3)(s^2+8s+25)}$$

$$= \lim_{s\to\infty} \frac{20/s^2}{(1+3/s)(1+8/s+25/s^2)} = \frac{0}{(1+0)(1+0+0)} = 0$$

To be sure that the final-value theorem is applicable, we check where the poles of $H(s)$ are located. The poles of $H(s)$ are $s = -3, -4 \pm j3$, which all have negative real parts: they are all located on the left half of the s plane (Fig. 15.9). Hence, the final-value theorem applies and

$$h(\infty) = \lim_{s\to 0} sH(s) = \lim_{s\to 0} \frac{20s}{(s+3)(s^2+8s+25)}$$

$$= \frac{0}{(0+3)(0+0+25)} = 0$$

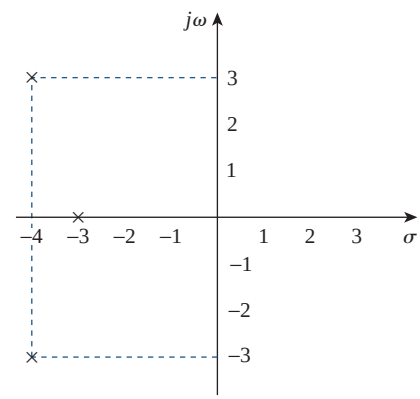


Figure 15.9
For Example 15.7: Poles of $H(s)$.

Both the initial and final values could be determined from $h(t)$ if we knew it. See Example 15.11, where $h(t)$ is given.

Practice Problem 15.7

Obtain the initial and the final values of

$$G(s) = \frac{3s^3 + 2s + 10}{s(s+2)^2(s+3)}$$

Answer: 3, 0.8333.

15.4 The Inverse Laplace Transform

Given $F(s)$, how do we transform it back to the time domain and obtain the corresponding $f(t)$? By matching entries in Table 15.2, we avoid using Eq. (15.5) to find $f(t)$.

Suppose $F(s)$ has the general form of

$$F(s) = \frac{N(s)}{D(s)} \tag{15.47}$$

where $N(s)$ is the numerator polynomial and $D(s)$ is the denominator polynomial. The roots of $N(s) = 0$ are called the *zeros* of $F(s)$, while the roots of $D(s) = 0$ are the *poles* of $F(s)$. Although Eq. (15.47) is similar in form to Eq. (14.3), here $F(s)$ is the Laplace transform of a function, which is not necessarily a transfer function. We use *partial fraction expansion* to break $F(s)$ down into simple terms whose inverse transform we obtain from Table 15.2. Thus, finding the inverse Laplace transform of $F(s)$ involves two steps.

Software packages such as *MATLAB*, Mathcad, and Maple are capable of finding partial fraction expansions quite easily.

Steps to Find the Inverse Laplace Transform:

1. Decompose $F(s)$ into simple terms using partial fraction expansion.
2. Find the inverse of each term by matching entries in Table 15.2.

Let us consider the three possible forms $F(s)$ may take and how to apply the two steps to each form.

15.4.1 Simple Poles

Recall from Chapter 14 that a simple pole is a first-order pole. If $F(s)$ has only simple poles, then $D(s)$ becomes a product of factors, so that

$$F(s) = \frac{N(s)}{(s+p_1)(s+p_2)\cdots(s+p_n)} \tag{15.48}$$

where $s = -p_1, -p_2, \ldots, -p_n$ are the simple poles, and $p_i \neq p_j$ for all $i \neq j$ (i.e., the poles are distinct). Assuming that the degree of $N(s)$ is

Otherwise, we must first apply long division so that $F(s) = N(s)/D(s) = Q(s) + R(s)/D(s)$, where the degree of $R(s)$, the remainder of the long division, is less than the degree of $D(s)$.

less than the degree of $D(s)$, we use partial fraction expansion to decompose $F(s)$ in Eq. (15.48) as

$$F(s) = \frac{k_1}{s + p_1} + \frac{k_2}{s + p_2} + \cdots + \frac{k_n}{s + p_n} \tag{15.49}$$

The expansion coefficients $k_1, k_2, \ldots, k_n$ are known as the *residues* of $F(s)$. There are many ways of finding the expansion coefficients. One way is using the *residue method*. If we multiply both sides of Eq. (15.49) by $(s + p_1)$, we obtain

$$(s + p_1)F(s) = k_1 + \frac{(s + p_1)k_2}{s + p_2} + \cdots + \frac{(s + p_1)k_n}{s + p_n} \tag{15.50}$$

Since $p_i \neq p_j$, setting $s = -p_1$ in Eq. (15.50) leaves only k_1 on the right-hand side of Eq. (15.50). Hence,

$$(s + p_1)F(s)\,|_{s=-p_1} = k_1 \tag{15.51}$$

Thus, in general,

$$\boxed{k_i = (s + p_i)F(s)\,|_{s=-p_i}} \tag{15.52}$$

This is known as *Heaviside's theorem*. Once the values of k_i are known, we proceed to find the inverse of $F(s)$ using Eq. (15.49). Since the inverse transform of each term in Eq. (15.49) is $\mathcal{L}^{-1}[k/(s + a)] = ke^{-at}u(t)$, then, from Table 15.2,

Historical note: Named after Oliver Heaviside (1850–1925), an English engineer, the pioneer of operational calculus.

$$\boxed{f(t) = (k_1e^{-p_1t} + k_2e^{-p_2t} + \cdots + k_ne^{-p_nt})u(t)} \tag{15.53}$$

15.4.2 Repeated Poles

Suppose $F(s)$ has n repeated poles at $s = -p$. Then we may represent $F(s)$ as

$$F(s) = \frac{k_n}{(s + p)^n} + \frac{k_{n-1}}{(s + p)^{n-1}} + \cdots + \frac{k_2}{(s + p)^2} + \frac{k_1}{s + p} + F_1(s) \tag{15.54}$$

where $F_1(s)$ is the remaining part of $F(s)$ that does not have a pole at $s = -p$. We determine the expansion coefficient k_n as

$$k_n = (s + p)^nF(s)\,|_{s=-p} \tag{15.55}$$

as we did above. To determine k_{n-1}, we multiply each term in Eq. (15.54) by $(s + p)^n$ and differentiate to get rid of k_n, then evaluate the result at $s = -p$ to get rid of the other coefficients except k_{n-1}. Thus, we obtain

$$k_{n-1} = \frac{d}{ds}[(s + p)^nF(s)]\,|_{s=-p} \tag{15.56}$$

Repeating this gives

$$k_{n-2} = \frac{1}{2!}\frac{d^2}{ds^2}[(s + p)^nF(s)]\,|_{s=-p} \tag{15.57}$$

The mth term becomes

$$k_{n-m} = \frac{1}{m!}\frac{d^m}{ds^m}[(s+p)^n F(s)]\,|_{s=-p} \tag{15.58}$$

where $m = 1, 2, \ldots, n-1$. One can expect the differentiation to be difficult to handle as m increases. Once we obtain the values of $k_1, k_2, \ldots, k_n$ by partial fraction expansion, we apply the inverse transform

$$\mathcal{L}^{-1}\left[\frac{1}{(s+a)^n}\right] = \frac{t^{n-1}e^{-at}}{(n-1)!}u(t) \tag{15.59}$$

to each term on the right-hand side of Eq. (15.54) and obtain

$$f(t) = \left(k_1 e^{-pt} + k_2 te^{-pt} + \frac{k_3}{2!}t^2 e^{-pt} + \cdots + \frac{k_n}{(n-1)!}t^{n-1}e^{-pt}\right)u(t) + f_1(t) \tag{15.60}$$

15.4.3 Complex Poles

A pair of complex poles is simple if it is not repeated; it is a double or multiple pole if repeated. Simple complex poles may be handled the same way as simple real poles, but because complex algebra is involved the result is always cumbersome. An easier approach is a method known as *completing the square*. The idea is to express each complex pole pair (or quadratic term) in $D(s)$ as a complete square such as $(s+\alpha)^2 + \beta^2$ and then use Table 15.2 to find the inverse of the term.

Since $N(s)$ and $D(s)$ always have real coefficients and we know that the complex roots of polynomials with real coefficients must occur in conjugate pairs, $F(s)$ may have the general form

$$F(s) = \frac{A_1 s + A_2}{s^2 + as + b} + F_1(s) \tag{15.61}$$

where $F_1(s)$ is the remaining part of $F(s)$ that does not have this pair of complex poles. If we complete the square by letting

$$s^2 + as + b = s^2 + 2\alpha s + \alpha^2 + \beta^2 = (s+\alpha)^2 + \beta^2 \tag{15.62}$$

and we also let

$$A_1 s + A_2 = A_1(s+\alpha) + B_1\beta \tag{15.63}$$

then Eq. (15.61) becomes

$$F(s) = \frac{A_1(s+\alpha)}{(s+\alpha)^2 + \beta^2} + \frac{B_1\beta}{(s+\alpha)^2 + \beta^2} + F_1(s) \tag{15.64}$$

From Table 15.2, the inverse transform is

$$f(t) = (A_1 e^{-\alpha t}\cos\beta t + B_1 e^{-\alpha t}\sin\beta t)u(t) + f_1(t) \tag{15.65}$$

The sine and cosine terms can be combined using Eq. (9.11).

Whether the pole is simple, repeated, or complex, a general approach that can always be used in finding the expansion coefficients

is the *method of algebra*, illustrated in Examples 15.9 to 15.11. To apply the method, we first set $F(s) = N(s)/D(s)$ equal to an expansion containing unknown constants. We multiply the result through by a common denominator. Then we determine the unknown constants by equating coefficients (i.e., by algebraically solving a set of simultaneous equations for these coefficients at like powers of s).

Another general approach is to substitute specific, convenient values of s to obtain as many simultaneous equations as the number of unknown coefficients, and then solve for the unknown coefficients. We must make sure that each selected value of s is not one of the poles of $F(s)$. Example 15.11 illustrates this idea.

Example 15.8

Find the inverse Laplace transform of

$$F(s) = \frac{3}{s} - \frac{5}{s+1} + \frac{6}{s^2+4}$$

Solution:

The inverse transform is given by

$$f(t) = \mathcal{L}^{-1}[F(s)] = \mathcal{L}^{-1}\left(\frac{3}{s}\right) - \mathcal{L}^{-1}\left(\frac{5}{s+1}\right) + \mathcal{L}^{-1}\left(\frac{6}{s^2+4}\right)$$
$$= (3 - 5e^{-t} + 3\sin 2t)u(t), \qquad t \geq 0$$

where Table 15.2 has been consulted for the inverse of each term.

Practice Problem 15.8

Determine the inverse Laplace transform of

$$F(s) = 1 + \frac{3}{s+4} - \frac{5s}{s^2+25}$$

Answer: $\delta(t) + (4e^{-4t} - 5\cos(5t))u(t)$.

Example 15.9

Find $f(t)$ given that

$$F(s) = \frac{s^2+12}{s(s+2)(s+3)}$$

Solution:

Unlike in the previous example where the partial fractions have been provided, we first need to determine the partial fractions. Since there are three poles, we let

$$\frac{s^2+12}{s(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3} \qquad \textbf{(15.9.1)}$$

where A, B, and C are the constants to be determined. We can find the constants using two approaches.

METHOD 1 Residue method:

$$A = sF(s)\,|_{s=0} = \frac{s^2 + 12}{(s+2)(s+3)}\bigg|_{s=0} = \frac{12}{(2)(3)} = 2$$

$$B = (s+2)F(s)\,|_{s=-2} = \frac{s^2 + 12}{s(s+3)}\bigg|_{s=-2} = \frac{4+12}{(-2)(1)} = -8$$

$$C = (s+3)F(s)\,|_{s=-3} = \frac{s^2 + 12}{s(s+2)}\bigg|_{s=-3} = \frac{9+12}{(-3)(-1)} = 7$$

METHOD 2 Algebraic method: Multiplying both sides of Eq. (15.9.1) by $s(s+2)(s+3)$ gives

$$s^2 + 12 = A(s+2)(s+3) + Bs(s+3) + Cs(s+2)$$

or

$$s^2 + 12 = A(s^2 + 5s + 6) + B(s^2 + 3s) + C(s^2 + 2s)$$

Equating the coefficients of like powers of s gives

$$\begin{aligned} &\text{Constant:} && 12 = 6A && \Rightarrow && A = 2 \\ &s: && 0 = 5A + 3B + 2C && \Rightarrow && 3B + 2C = -10 \\ &s^2: && 1 = A + B + C && \Rightarrow && B + C = -1 \end{aligned}$$

Thus, $A = 2$, $B = -8$, $C = 7$, and Eq. (15.9.1) becomes

$$F(s) = \frac{2}{s} - \frac{8}{s+2} + \frac{7}{s+3}$$

By finding the inverse transform of each term, we obtain

$$f(t) = (2 - 8e^{-2t} + 7e^{-3t})u(t)$$

Practice Problem 15.9

Find $f(t)$ if

$$F(s) = \frac{6(s+2)}{(s+1)(s+3)(s+4)}$$

Answer: $f(t) = (e^{-t} + 3e^{-3t} - 4e^{-4t})u(t)$.

Example 15.10

Calculate $v(t)$ given that

$$V(s) = \frac{10s^2 + 4}{s(s+1)(s+2)^2}$$

Solution:

While the previous example is on simple roots, this example is on repeated roots. Let

$$V(s) = \frac{10s^2 + 4}{s(s+1)(s+2)^2} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+2)^2} + \frac{D}{s+2} \qquad (15.10.1)$$

METHOD 1 **Residue method:**

$$A = sV(s)\,|_{s=0} = \frac{10s^2 + 4}{(s+1)(s+2)^2}\bigg|_{s=0} = \frac{4}{(1)(2)^2} = 1$$

$$B = (s+1)V(s)\,|_{s=-1} = \frac{10s^2 + 4}{s(s+2)^2}\bigg|_{s=-1} = \frac{14}{(-1)(1)^2} = -14$$

$$C = (s+2)^2V(s)\,|_{s=-2} = \frac{10s^2 + 4}{s(s+1)}\bigg|_{s=-2} = \frac{44}{(-2)(-1)} = 22$$

$$D = \frac{d}{ds}[(s+2)^2V(s)]\bigg|_{s=-2} = \frac{d}{ds}\left(\frac{10s^2+4}{s^2+s}\right)\bigg|_{s=-2}$$

$$= \frac{(s^2+s)(20s) - (10s^2+4)(2s+1)}{(s^2+s)^2}\bigg|_{s=-2} = \frac{52}{4} = 13$$

METHOD 2 **Algebraic method:** Multiplying Eq. (15.10.1) by $s(s+1)(s+2)^2$, we obtain

$$10s^2 + 4 = A(s+1)(s+2)^2 + Bs(s+2)^2 + Cs(s+1) + Ds(s+1)(s+2)$$

or

$$10s^2 + 4 = A(s^3 + 5s^2 + 8s + 4) + B(s^3 + 4s^2 + 4s) + C(s^2 + s) + D(s^3 + 3s^2 + 2s)$$

Equating coefficients,

Constant: $4 = 4A \quad \Rightarrow \quad A = 1$

s: $0 = 8A + 4B + C + 2D \quad \Rightarrow \quad 4B + C + 2D = -8$

s^2: $10 = 5A + 4B + C + 3D \quad \Rightarrow \quad 4B + C + 3D = 5$

s^3: $0 = A + B + D \quad \Rightarrow \quad B + D = -1$

Solving these simultaneous equations gives $A = 1, B = -14, C = 22, D = 13$, so that

$$V(s) = \frac{1}{s} - \frac{14}{s+1} + \frac{13}{s+2} + \frac{22}{(s+2)^2}$$

Taking the inverse transform of each term, we get

$$v(t) = (1 - 14e^{-t} + 13e^{-2t} + 22te^{-2t})u(t)$$

Practice Problem 15.10

Obtain $g(t)$ if

$$G(s) = \frac{s^3 + 2s + 6}{s(s+1)^2(s+3)}$$

Answer: $(2 - 3.25e^{-t} - 1.5te^{-t} + 2.25e^{-3t})u(t)$.

Example 15.11

Find the inverse transform of the frequency-domain function in Example 15.7:

$$H(s) = \frac{20}{(s+3)(s^2+8s+25)}$$

Solution:

In this example, $H(s)$ has a pair of complex poles at $s^2 + 8s + 25 = 0$ or $s = -4 \pm j3$. We let

$$H(s) = \frac{20}{(s+3)(s^2+8s+25)} = \frac{A}{s+3} + \frac{Bs+C}{(s^2+8s+25)} \quad \textbf{(15.11.1)}$$

We now determine the expansion coefficients in two ways.

■ **METHOD 1** **Combination of methods:** We can obtain A using the method of residue,

$$A = (s+3)H(s)\,|_{s=-3} = \frac{20}{s^2+8s+25}\bigg|_{s=-3} = \frac{20}{10} = 2$$

Although B and C can be obtained using the method of residue, we will not do so, to avoid complex algebra. Rather, we can substitute two specific values of s [say $s = 0$, 1, which are not poles of $F(s)$] into Eq. (15.11.1). This will give us two simultaneous equations from which to find B and C. If we let $s = 0$ in Eq. (15.11.1), we obtain

$$\frac{20}{75} = \frac{A}{3} + \frac{C}{25}$$

or

$$20 = 25A + 3C \quad \textbf{(15.11.2)}$$

Since $A = 2$, Eq. (15.11.2) gives $C = -10$. Substituting $s = 1$ into Eq. (15.11.1) gives

$$\frac{20}{(4)(34)} = \frac{A}{4} + \frac{B+C}{34}$$

or

$$20 = 34A + 4B + 4C \quad \textbf{(15.11.3)}$$

But $A = 2$, $C = -10$, so that Eq. (15.11.3) gives $B = -2$.

■ **METHOD 2** **Algebraic method:** Multiplying both sides of Eq. (15.11.1) by $(s+3)(s^2+8s+25)$ yields

$$\begin{aligned} 20 &= A(s^2+8s+25) + (Bs+C)(s+3) \\ &= A(s^2+8s+25) + B(s^2+3s) + C(s+3) \end{aligned} \quad \textbf{(15.11.4)}$$

Equating coefficients gives

s^2: $\quad 0 = A + B \quad \Rightarrow \quad A = -B$

s: $\quad 0 = 8A + 3B + C = 5A + C \quad \Rightarrow \quad C = -5A$

Constant: $\quad 20 = 25A + 3C = 25A - 15A \quad \Rightarrow \quad A = 2$

That is, $B = -2$, $C = -10$. Thus,

$$\begin{aligned} H(s) &= \frac{2}{s+3} - \frac{2s+10}{(s^2+8s+25)} = \frac{2}{s+3} - \frac{2(s+4)+2}{(s+4)^2+9} \\ &= \frac{2}{s+3} - \frac{2(s+4)}{(s+4)^2+9} - \frac{2}{3}\frac{3}{(s+4)^2+9} \end{aligned}$$

Taking the inverse of each term, we obtain

$$h(t) = \left(2e^{-3t} - 2e^{-4t}\cos 3t - \frac{2}{3}e^{-4t}\sin 3t\right)u(t) \quad \textbf{(15.11.5)}$$

It is alright to leave the result this way. However, we can combine the cosine and sine terms as

$$h(t) = (2e^{-3t} - Re^{-4t}\cos(3t - \theta))u(t) \quad \textbf{(15.11.6)}$$

To obtain Eq. (15.11.6) from Eq. (15.11.5), we apply Eq. (9.11). Next, we determine the coefficient R and the phase angle θ:

$$R = \sqrt{2^2 + \left(\tfrac{2}{3}\right)^2} = 2.108, \qquad \theta = \tan^{-1}\frac{\frac{2}{3}}{2} = 18.43°$$

Thus,

$$h(t) = (2e^{-3t} - 2.108e^{-4t}\cos(3t - 18.43°))u(t)$$

Practice Problem 15.11

Find $g(t)$ given that

$$G(s) = \frac{10}{(s+1)(s^2 + 4s + 13)}$$

Answer: $e^{-t} - e^{-2t}\cos 3t - \frac{1}{3}e^{-2t}\sin 3t, \ t \geq 0.$

15.5 The Convolution Integral

The term *convolution* means "folding." Convolution is an invaluable tool to the engineer because it provides a means of viewing and characterizing physical systems. For example, it is used in finding the response $y(t)$ of a system to an excitation $x(t)$, knowing the system impulse response $h(t)$. This is achieved through the *convolution integral*, defined as

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)h(t - \lambda)\, d\lambda \quad \textbf{(15.66)}$$

or simply

$$y(t) = x(t) * h(t) \quad \textbf{(15.67)}$$

where λ is a dummy variable and the asterisk denotes convolution. Equation (15.66) or (15.67) states that the output is equal to the input convolved with the unit impulse response. The convolution process is commutative:

$$y(t) = x(t) * h(t) = h(t) * x(t) \quad \textbf{(15.68a)}$$

or

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)h(t - \lambda)\, d\lambda = \int_{-\infty}^{\infty} h(\lambda)x(t - \lambda)\, d\lambda \quad \textbf{(15.68b)}$$

This implies that the order in which the two functions are convolved is immaterial. We will see shortly how to take advantage of this commutative property when performing graphical computation of the convolution integral.

The **convolution** of two signals consists of time-reversing one of the signals, shifting it, and multiplying it point by point with the second signal, and integrating the product.

The convolution integral in Eq. (15.66) is the general one; it applies to any linear system. However, the convolution integral can be simplified if we assume that a system has two properties. First, if $x(t) = 0$ for $t < 0$, then

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)h(t-\lambda)\,d\lambda = \int_{0}^{\infty} x(\lambda)h(t-\lambda)\,d\lambda \tag{15.69}$$

Second, if the system's impulse response is *causal* (i.e., $h(t) = 0$ for $t < 0$), then $h(t-\lambda) = 0$ for $t - \lambda < 0$ or $\lambda > t$, so that Eq. (15.69) becomes

$$y(t) = h(t) * x(t) = \int_{0}^{t} x(\lambda)h(t-\lambda)\,d\lambda \tag{15.70}$$

Here are some properties of the convolution integral.

1. $x(t) * h(t) = h(t) * x(t)$ (Commutative)
2. $f(t) * [x(t) + y(t)] = f(t) * x(t) + f(t) * y(t)$ (Distributive)
3. $f(t) * [x(t) * y(t)] = [f(t) * x(t)] * y(t)$ (Associative)
4. $f(t) * \delta(t) = \int_{-\infty}^{\infty} f(\lambda)\delta(t-\lambda)\,d\lambda = f(t)$
5. $f(t) * \delta(t - t_o) = f(t - t_o)$
6. $f(t) * \delta'(t) = \int_{-\infty}^{\infty} f(\lambda)\delta'(t-\lambda)\,d\lambda = f'(t)$
7. $f(t) * u(t) = \int_{-\infty}^{\infty} f(\lambda)u(t-\lambda)\,d\lambda = \int_{-\infty}^{t} f(\lambda)\,d\lambda$

Before learning how to evaluate the convolution integral in Eq. (15.70), let us establish the link between the Laplace transform and the convolution integral. Given two functions $f_1(t)$ and $f_2(t)$ with Laplace transforms $F_1(s)$ and $F_2(s)$, respectively, their convolution is

$$f(t) = f_1(t) * f_2(t) = \int_{0}^{t} f_1(\lambda)f_2(t-\lambda)\,d\lambda \tag{15.71}$$

Taking the Laplace transform gives

$$F(s) = \mathcal{L}[f_1(t) * f_2(t)] = F_1(s)F_2(s) \tag{15.72}$$

To prove that Eq. (15.72) is true, we begin with the fact that $F_1(s)$ is defined as

$$F_1(s) = \int_{0^-}^{\infty} f_1(\lambda)e^{-s\lambda}\,d\lambda \tag{15.73}$$

Multiplying this with $F_2(s)$ gives

$$F_1(s)F_2(s) = \int_{0^-}^{\infty} f_1(\lambda)[F_2(s)e^{-s\lambda}]\,d\lambda \tag{15.74}$$

We recall from the time shift property in Eq. (15.17) that the term in brackets can be written as

$$\begin{aligned} F_2(s)e^{-s\lambda} &= \mathcal{L}[\, f_2(t-\lambda)u(t-\lambda)] \\ &= \int_{0^-}^{\infty} f_2(t-\lambda)u(t-\lambda)e^{-s\lambda}\, dt \end{aligned} \tag{15.75}$$

Substituting Eq. (15.75) into Eq. (15.74) gives

$$F_1(s)F_2(s) = \int_{0^-}^{\infty} f_1(\lambda)\left[\int_{0^-}^{\infty} f_2(t-\lambda)u(t-\lambda)e^{-s\lambda}dt\right] d\lambda \tag{15.76}$$

Interchanging the order of integration results in

$$F_1(s)F_2(s) = \int_{0^-}^{\infty}\left[\int_{0^-}^{t} f_1(\lambda)\, f_2(t-\lambda)\, d\lambda\right] e^{-s\lambda}\, d\lambda \tag{15.77}$$

The integral in brackets extends only from 0 to t because the delayed unit step $u(t-\lambda) = 1$ for $\lambda < t$ and $u(t-\lambda) = 0$ for $\lambda > t$. We notice that the integral is the convolution of $f_1(t)$ and $f_2(t)$ as in Eq. (15.71). Hence,

$$\boxed{F_1(s)F_2(s) = \mathcal{L}[\, f_1(t) * f_2(t)]} \tag{15.78}$$

as desired. This indicates that convolution in the time domain is equivalent to multiplication in the s-domain. For example, if $x(t) = 4e^{-t}$ and $h(t) = 5e^{-2t}$, applying the property in Eq. (15.78), we get

$$\begin{aligned} h(t) * x(t) &= \mathcal{L}^{-1}[H(s)X(s)] = \mathcal{L}^{-1}\left[\left(\frac{5}{s+2}\right)\left(\frac{4}{s+1}\right)\right] \\ &= \mathcal{L}^{-1}\left[\frac{20}{s+1} + \frac{-20}{s+2}\right] \\ &= 20(e^{-t} - e^{-2t}), \qquad t \geq 0 \end{aligned} \tag{15.79}$$

Although we can find the convolution of two signals using Eq. (15.78), as we have just done, if the product $F_1(s)F_2(s)$ is very complicated, finding the inverse may be tough. Also, there are situations in which $f_1(t)$ and $f_2(t)$ are available in the form of experimental data and there are no explicit Laplace transforms. In these cases, one must do the convolution in the time domain.

The process of convolving two signals in the time domain is better appreciated from a graphical point of view. The graphical procedure for evaluating the convolution integral in Eq. (15.70) usually involves four steps.

Steps to Evaluate the Convolution Integral:

1. Folding: Take the mirror image of $h(\lambda)$ about the ordinate axis to obtain $h(-\lambda)$.
2. Displacement: Shift or delay $h(-\lambda)$ by t to obtain $h(t-\lambda)$.
3. Multiplication: Find the product of $h(t-\lambda)$ and $x(\lambda)$.
4. Integration: For a given time t, calculate the area under the product $h(t-\lambda)x(\lambda)$ for $0 < \lambda < t$ to get $y(t)$ at t.

The folding operation in step 1 is the reason for the term *convolution.* The function $h(t - \lambda)$ scans or slides over $x(\lambda)$. In view of this superposition procedure, the convolution integral is also known as the *superposition integral.*

To apply the four steps, it is necessary to be able to sketch $x(\lambda)$ and $h(t - \lambda)$. To get $x(\lambda)$ from the original function $x(t)$ involves merely replacing every t with λ. Sketching $h(t - \lambda)$ is the key to the convolution process. It involves reflecting $h(\lambda)$ about the vertical axis and shifting it by t. Analytically, we obtain $h(t - \lambda)$ by replacing every t in $h(t)$ by $t - \lambda$. Since convolution is commutative, it may be more convenient to apply steps 1 and 2 to $x(t)$ instead of $h(t)$. The best way to illustrate the procedure is with some examples.

Example 15.12

Find the convolution of the two signals in Fig. 15.10.

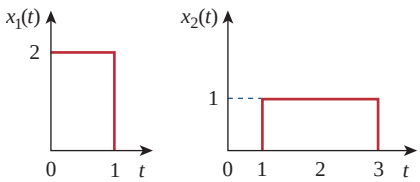


Figure 15.10
For Example 15.12.

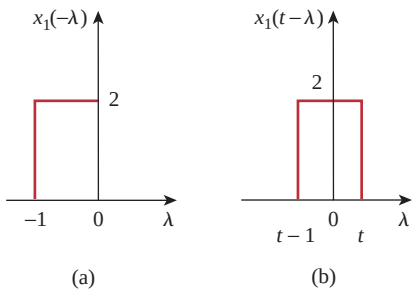


Figure 15.11
(a) Folding $x_1(\lambda)$, (b) shifting $x_1(-\lambda)$ by t.

Solution:

We follow the four steps to get $y(t) = x_1(t) * x_2(t)$. First, we fold $x_1(t)$ as shown in Fig. 15.11(a) and shift it by t as shown in Fig. 15.11(b). For different values of t, we now multiply the two functions and integrate to determine the area of the overlapping region.

For $0 < t < 1$, there is no overlap of the two functions, as shown in Fig. 15.12(a). Hence,

$$y(t) = x_1(t) * x_2(t) = 0, \qquad 0 < t < 1 \tag{15.12.1}$$

For $1 < t < 2$, the two signals overlap between 1 and t, as shown in Fig. 15.12(b).

$$y(t) = \int_1^t (2)(1)\, d\lambda = 2\lambda \Big|_1^t = 2(t - 1), \qquad 1 < t < 2 \tag{15.12.2}$$

For $2 < t < 3$, the two signals completely overlap between $(t - 1)$ and t, as shown in Fig. 15.12(c). It is easy to see that the area under the curve is 2. Or

$$y(t) = \int_{t-1}^t (2)(1)\, d\lambda = 2\lambda \Big|_{t-1}^t = 2, \qquad 2 < t < 3 \tag{15.12.3}$$

For $3 < t < 4$, the two signals overlap between $(t - 1)$ and 3, as shown in Fig. 15.12(d).

$$\begin{aligned} y(t) &= \int_{t-1}^3 (2)(1)\, d\lambda = 2\lambda \Big|_{t-1}^3 \\ &= 2(3 - t + 1) = 8 - 2t, \qquad 3 < t < 4 \end{aligned} \tag{15.12.4}$$

For $t > 4$, the two signals do not overlap [Fig. 15.12(e)], and

$$y(t) = 0, \qquad t > 4 \tag{15.12.5}$$

Combining Eqs. (15.12.1) to (15.12.5), we obtain

$$y(t) = \begin{cases} 0, & 0 \le t \le 1 \\ 2t - 2, & 1 \le t \le 2 \\ 2, & 2 \le t \le 3 \\ 8 - 2t, & 3 \le t \le 4 \\ 0, & t \ge 4 \end{cases} \tag{15.12.6}$$

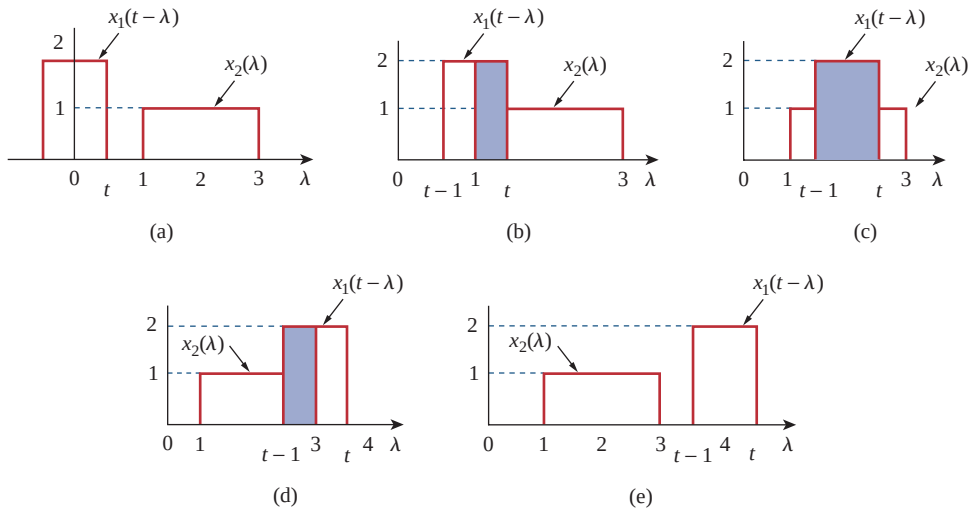


Figure 15.12
Overlapping of $x_1(t - \lambda)$ and $x_2(\lambda)$ for: (a) $0 < t < 1$, (b) $1 < t < 2$, (c) $2 < t < 3$, (d) $3 < t < 4$, (e) $t > 4$.

which is sketched in Fig. 15.13. Notice that $y(t)$ in this equation is continuous. This fact can be used to check the results as we move from one range of t to another. The result in Eq. (15.12.6) can be obtained without using the graphical procedure—by directly using Eq. (15.70) and the properties of step functions. This will be illustrated in Example 15.14.

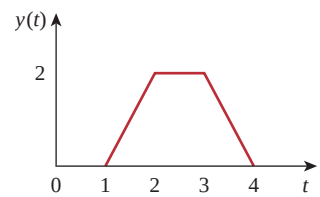


Figure 15.13
Convolution of signals $x_1(t)$ and $x_2(t)$ in Fig. 15.10.

Practice Problem 15.12

Graphically convolve the two functions in Fig. 15.14. To show how powerful working in the s-domain is, verify your answer by performing the equivalent operation in the s-domain.

Answer: The result of the convolution $y(t)$ is shown in Fig. 15.15, where

$$y(t) = \begin{cases} t, & 0 \le t \le 2 \\ 6 - 2t, & 2 \le t \le 3 \\ 0, & \text{otherwise.} \end{cases}$$

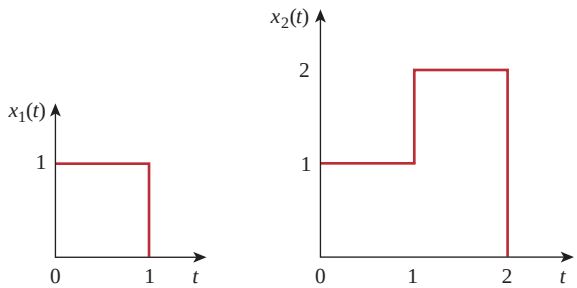


Figure 15.14
For Practice Prob. 15.12.

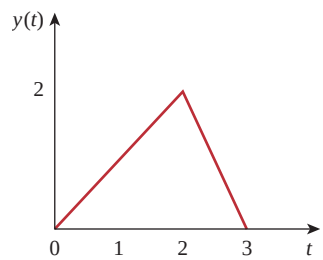


Figure 15.15
Convolution of the signals in Fig. 15.14.

Example 15.13

Graphically convolve $g(t)$ and $u(t)$ shown in Fig. 15.16.

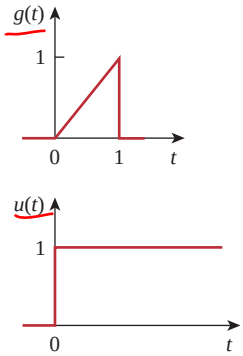


Figure 15.16
For Example 15.13.

Solution:
Let $y(t) = g(t) * u(t)$. We can find $y(t)$ in two ways.

■ **METHOD 1** Suppose we fold $g(t)$, as in Fig. 15.17(a), and shift it by t, as in Fig. 15.17(b). Since $g(t) = t, 0 < t < 1$ originally, we expect that $g(t - \lambda) = t - \lambda, 0 < t - \lambda < 1$ or $t - 1 < \lambda < t$. There is no overlap of the two functions when $t < 0$ so that $y(0) = 0$ for this case.

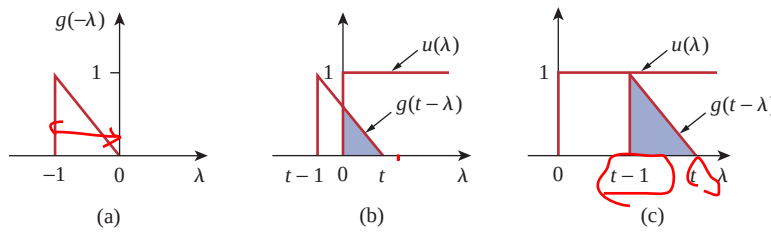


Figure 15.17
Convolution of $g(t)$ and $u(t)$ in Fig. 15.16 with $g(t)$ folded.

For $0 < t < 1$, $g(t - \lambda)$ and $u(\lambda)$ overlap from 0 to t, as evident in Fig. 15.17(b). Therefore,

$$y(t) = \int_0^t (1)(t - \lambda)\, d\lambda = \left(t\lambda - \frac{1}{2}\lambda^2\right)\Big|_0^t$$
$$= t^2 - \frac{t^2}{2} = \frac{t^2}{2}, \qquad 0 \le t \le 1 \tag{15.13.1}$$

For $t > 1$, the two functions overlap completely between $(t - 1)$ and t [see Fig. 15.17(c)]. Hence,

$$y(t) = \int_{t-1}^{t} (1)(t - \lambda)\, d\lambda$$
$$= \left(t\lambda - \frac{1}{2}\lambda^2\right)\Big|_{t-1}^{t} = \frac{1}{2}, \qquad t \ge 1 \tag{15.13.2}$$

Thus, from Eqs. (15.13.1) and (15.13.2),

$$y(t) = \begin{cases} \frac{1}{2}t^2, & 0 \le t \le 1 \\ \frac{1}{2}, & t \ge 1 \end{cases}$$

■ **METHOD 2** Instead of folding g, suppose we fold the unit step function $u(t)$, as in Fig. 15.18(a), and then shift it by t, as in Fig. 15.18(b). Since $u(t) = 1$ for $t > 0$, $u(t - \lambda) = 1$ for $t - \lambda > 0$ or $\lambda < t$, the two functions overlap from 0 to t, so that

$$y(t) = \int_0^t (1)\lambda\, d\lambda = \frac{1}{2}\lambda^2\Big|_0^t = \frac{t^2}{2}, \qquad 0 \le t \le 1 \tag{15.13.3}$$

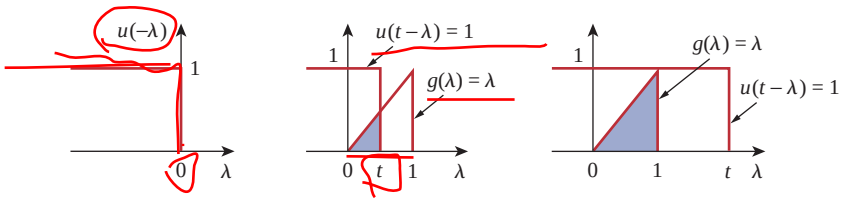


Figure 15.18
Convolution of $g(t)$ and $u(t)$ in Fig. 15.16 with $u(t)$ folded.

For $t > 1$, the two functions overlap between 0 and 1, as shown in Fig. 15.18(c). Hence,

$$y(t) = \int_0^1 (1)\lambda \, d\lambda = \frac{1}{2}\lambda^2 \Big|_0^1 = \frac{1}{2}, \qquad t \geq 1 \qquad \textbf{(15.13.4)}$$

And, from Eqs. (15.13.3) and (15.13.4),

$$y(t) = \begin{cases} \frac{1}{2}t^2, & 0 \leq t \leq 1 \\ \frac{1}{2}, & t \geq 1 \end{cases}$$

Although the two methods give the same result, as expected, notice that it is more convenient to fold the unit step function $u(t)$ than fold $g(t)$ in this example. Figure 15.19 shows $y(t)$.

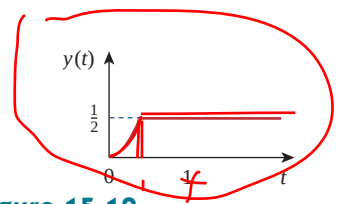


Figure 15.19
Result of Example 15.13.

Practice Problem 15.13

Given $g(t)$ and $f(t)$ in Fig. 15.20, graphically find $y(t) = g(t) * f(t)$.

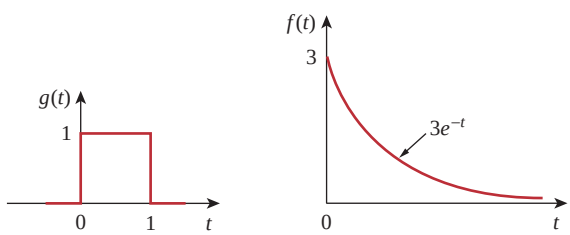


Figure 15.20
For Practice Prob. 15.13.

Answer: $y(t) = \begin{cases} 3(1 - e^{-t}), & 0 \leq t \leq 1 \\ 3(e - 1)e^{-t}, & t \geq 1 \\ 0, & \text{elsewhere.} \end{cases}$

Example 15.14

For the *RL* circuit in Fig. 15.21(a), use the convolution integral to find the response $i_o(t)$ due to the excitation shown in Fig. 15.21(b).

Solution:

1. **Define.** The problem is clearly stated and the method of solution is also specified.

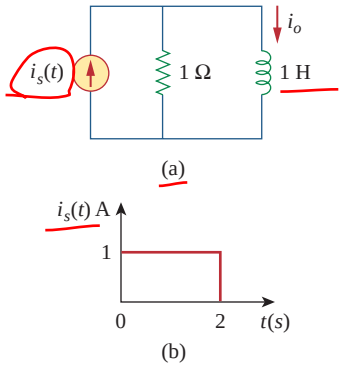


Figure 15.21
For Example 15.14.

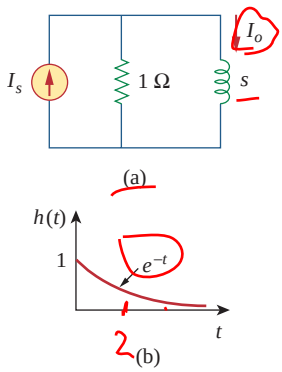


Figure 15.22
For the circuit in Fig. 15.21(a): (a) its *s*-domain equivalent, (b) its impulse response.

2. **Present.** We are to use the convolution integral to solve for the response $i_o(t)$ due to $i_s(t)$ shown in Fig. 15.21(b).
3. **Alternative.** We have learned to do convolution by using the convolution integral and how to do it graphically. In addition, we could always work in the *s*-domain to solve for the current. We will solve for the current using the convolution integral and then check it using the graphical approach.
4. **Attempt.** As we stated, this problem can be solved in two ways: directly using the convolution integral or using the graphical technique. To use either approach, we first need the unit impulse response $h(t)$ of the circuit. In the *s*-domain, applying the current division principle to the circuit in Fig. 15.22(a) gives

$$I_o = \frac{1}{s+1} I_s$$

Hence,

$$H(s) = \frac{I_o}{I_s} = \frac{1}{s+1} \tag{15.14.1}$$

and the inverse Laplace transform of this gives

$$h(t) = e^{-t}u(t) \tag{15.14.2}$$

Figure 15.22(b) shows the impulse response $h(t)$ of the circuit.

To use the convolution integral directly, recall that the response is given in the *s*-domain as

$$I_o(s) = H(s)I_s(s)$$

With the given $i_s(t)$ in Fig. 15.21(b),

$$i_s(t) = u(t) - u(t-2)$$

so that

$$\begin{aligned} i_o(t) = h(t) * i_s(t) &= \int_0^t i_s(\lambda)h(t-\lambda)\,d\lambda \\ &= \int_0^t [u(\lambda) - u(\lambda-2)]e^{-(t-\lambda)}\,d\lambda \end{aligned} \tag{15.14.3}$$

Since $u(\lambda - 2) = 0$ for $0 < \lambda < 2$, the integrand involving $u(\lambda)$ is nonzero for all $\lambda > 0$, whereas the integrand involving $u(\lambda - 2)$ is nonzero only for $\lambda > 2$. The best way to handle the integral is to do the two parts separately. For $0 < t < 2$,

$$\begin{aligned} i_o'(t) &= \int_0^t (1)e^{-(t-\lambda)}\,d\lambda = e^{-t}\int_0^t (1)e^{\lambda}\,d\lambda \\ &= e^{-t}(e^t - 1) = 1 - e^{-t}, \qquad 0 < t < 2 \end{aligned} \tag{15.14.4}$$

For $t > 2$,

$$\begin{aligned} i_o''(t) &= \int_2^t (1)e^{-(t-\lambda)}\,d\lambda = e^{-t}\int_2^t e^{\lambda}\,d\lambda \\ &= e^{-t}(e^t - e^2) = 1 - e^2e^{-t}, \qquad t > 2 \end{aligned} \tag{15.14.5}$$

Substituting Eqs. (15.14.4) and (15.14.5) into Eq. (15.14.3) gives

$$\begin{aligned} i_o(t) &= i_o'(t) - i_o''(t) \\ &= (1 - e^{-t})[u(t-2) - u(t)] - (1 - e^2 e^{-t})u(t-2) \\ &= \begin{cases} 1 - e^{-t}\,\text{A}, & 0 < t < 2 \\ (e^2 - 1)e^{-t}\,\text{A}, & t > 2 \end{cases} \end{aligned} \tag{15.14.6}$$

5. **Evaluate.** To use the graphical technique, we may fold $i_s(t)$ in Fig. 15.21(b) and shift by t, as shown in Fig. 15.23(a). For $0 < t < 2$, the overlap between $i_s(t - \lambda)$ and $h(\lambda)$ is from 0 to t, so that

$$i_o(t) = \int_0^t (1)e^{-\lambda}\,d\lambda = -e^{-\lambda}\Big|_0^t = (1 - e^{-t})\,\text{A}, \qquad 0 \le t \le 2 \tag{15.14.7}$$

For $t > 2$, the two functions overlap between $(t - 2)$ and t, as in Fig. 15.23(b). Hence,

$$\begin{aligned} i_o(t) &= \int_{t-2}^t (1)e^{-\lambda}\,d\lambda = -e^{-\lambda}\Big|_{t-2}^t = -e^{-t} + e^{-(t-2)} \\ &= (e^2 - 1)e^{-t}\,\text{A}, \qquad t \ge 0 \end{aligned} \tag{15.14.8}$$

From Eqs. (15.14.7) and (15.14.8), the response is

$$i_o(t) = \begin{cases} 1 - e^{-t}\,\text{A}, & 0 \le t \le 2 \\ (e^2 - 1)e^{-t}\,\text{A}, & t \ge 2 \end{cases} \tag{15.14.9}$$

which is the same as in Eq. (15.14.6). Thus, the response $i_o(t)$ along the excitation $i_s(t)$ is as shown in Fig. 15.24.

6. **Satisfactory?** We have satisfactorily solved the problem and can present the results as a solution to the problem.

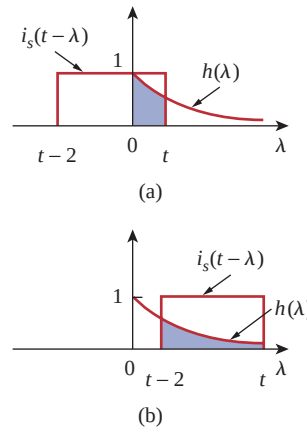


Figure 15.23
For Example 15.14.

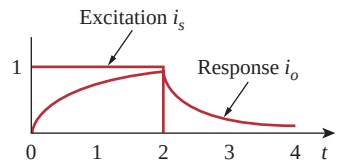


Figure 15.24
For Example 15.14; excitation and response.

Practice Problem 15.14

Use convolution to find $v_o(t)$ in the circuit of Fig. 15.25(a) when the excitation is the signal shown in Fig. 15.25(b). To show how powerful working in the s-domain is, verify your answer by performing the equivalent operation in the s-domain.

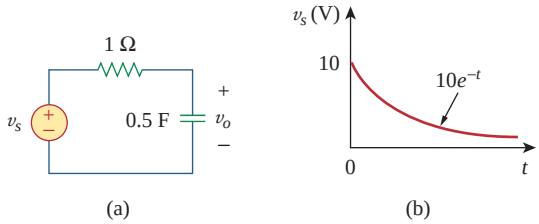


Figure 15.25
For Practice Prob. 15.14.

Answer: $20(e^{-t} - e^{-2t})$ V.

15.6 †Application to Integrodifferential Equations

The Laplace transform is useful in solving linear integrodifferential equations. Using the differentiation and integration properties of Laplace transforms, each term in the integrodifferential equation is transformed.

Initial conditions are automatically taken into account. We solve the resulting algebraic equation in the *s*-domain. We then convert the solution back to the time domain by using the inverse transform. The following examples illustrate the process.

Example 15.15

Use the Laplace transform to solve the differential equation

$$\frac{d^2v(t)}{dt^2} + 6\frac{dv(t)}{dt} + 8v(t) = 2u(t)$$

subject to $v(0) = 1, v'(0) = -2$.

Solution:

We take the Laplace transform of each term in the given differential equation and obtain

$$[s^2V(s) - sv(0) - v'(0)] + 6[sV(s) - v(0)] + 8V(s) = \frac{2}{s}$$

Substituting $v(0) = 1, v'(0) = -2$,

$$s^2V(s) - s + 2 + 6sV(s) - 6 + 8V(s) = \frac{2}{s}$$

or

$$(s^2 + 6s + 8)V(s) = s + 4 + \frac{2}{s} = \frac{s^2 + 4s + 2}{s}$$

Hence,

$$V(s) = \frac{s^2 + 4s + 2}{s(s + 2)(s + 4)} = \frac{A}{s} + \frac{B}{s + 2} + \frac{C}{s + 4}$$

where

$$A = sV(s)\,|_{s=0} = \frac{s^2 + 4s + 2}{(s + 2)(s + 4)}\bigg|_{s=0} = \frac{2}{(2)(4)} = \frac{1}{4}$$

$$B = (s + 2)V(s)\,|_{s=-2} = \frac{s^2 + 4s + 2}{s(s + 4)}\bigg|_{s=-2} = \frac{-2}{(-2)(2)} = \frac{1}{2}$$

$$C = (s + 4)V(s)\,|_{s=-4} = \frac{s^2 + 4s + 2}{s(s + 2)}\bigg|_{s=-4} = \frac{2}{(-4)(-2)} = \frac{1}{4}$$

Hence,

$$V(s) = \frac{\frac{1}{4}}{s} + \frac{\frac{1}{2}}{s + 2} + \frac{\frac{1}{4}}{s + 4}$$

By the inverse Laplace transform,

$$v(t) = \frac{1}{4}(1 + 2e^{-2t} + e^{-4t})u(t)$$

Practice Problem 15.15

Solve the following differential equation using the Laplace transform method.

$$\frac{d^2v(t)}{dt^2} + 4\frac{dv(t)}{dt} + 4v(t) = e^{-t}$$

if $v(0) = v'(0) = 2$.

Answer: $(2e^{-t} + 4te^{-2t})u(t)$.

Example 15.16

Solve for the response $y(t)$ in the following integrodifferential equation.

$$\frac{dy}{dt} + 5y(t) + 6\int_0^t y(\tau)\,d\tau = u(t), \qquad y(0) = 2$$

Solution:
Taking the Laplace transform of each term, we get

$$[sY(s) - y(0)] + 5Y(s) + \frac{6}{s}Y(s) = \frac{1}{s}$$

Substituting $y(0) = 2$ and multiplying through by s,

$$Y(s)(s^2 + 5s + 6) = 1 + 2s$$

or

$$Y(s) = \frac{2s + 1}{(s + 2)(s + 3)} = \frac{A}{s + 2} + \frac{B}{s + 3}$$

where

$$A = (s + 2)Y(s)\,|_{s=-2} = \left.\frac{2s + 1}{s + 3}\right|_{s=-2} = \frac{-3}{1} = -3$$

$$B = (s + 3)Y(s)\,|_{s=-3} = \left.\frac{2s + 1}{s + 2}\right|_{s=-3} = \frac{-5}{-1} = 5$$

Thus,

$$Y(s) = \frac{-3}{s + 2} + \frac{5}{s + 3}$$

Its inverse transform is

$$y(t) = (-3e^{-2t} + 5e^{-3t})u(t)$$

Practice Problem 15.16

Use the Laplace transform to solve the integrodifferential equation

$$\frac{dy}{dt} + 3y(t) + 2\int_0^t y(\tau)\,d\tau = 2e^{-3t}, \qquad y(0) = 0$$

Answer: $(-e^{-t} + 4e^{-2t} - 3e^{-3t})u(t)$.

15.7 Summary

1. The Laplace transform allows a signal represented by a function in the time domain to be analyzed in the s-domain (or complex frequency domain). It is defined as

$$\mathcal{L}[f(t)] = F(s) = \int_0^\infty f(t)e^{-st}\,dt$$

2. Properties of the Laplace transform are listed in Table 15.1, while the Laplace transforms of basic common functions are listed in Table 15.2.
3. The inverse Laplace transform can be found using partial fraction expansions and using the Laplace transform pairs in Table 15.2 as a look-up table. Real poles lead to exponential functions and complex poles to damped sinusoids.
4. The convolution of two signals consists of time-reversing one of the signals, shifting it, multiplying it point by point with the second signal, and integrating the product. The convolution integral relates the convolution of two signals in the time domain to the inverse of the product of their Laplace transforms:

$$\mathcal{L}^{-1}[F_1(s)F_2(s)] = f_1(t) * f_2(t) = \int_0^t f_1(\lambda)f_2(t-\lambda)\,d\lambda$$

5. In the time domain, the output $y(t)$ of the network is the convolution of the impulse response with the input $x(t)$,

$$y(t) = h(t) * x(t)$$

   Convolution may be regarded as the flip-shift-multiply-time-area method.
6. The Laplace transform can be used to solve a linear integrodifferential equation.

Review Questions

15.1 Every function $f(t)$ has a Laplace transform.

(a) True (b) False

15.2 The variable s in the Laplace transform $H(s)$ is called

(a) complex frequency (b) transfer function
(c) zero (d) pole

15.3 The Laplace transform of $u(t-2)$ is:

(a) $\dfrac{1}{s+2}$ (b) $\dfrac{1}{s-2}$
(c) $\dfrac{e^{2s}}{s}$ (d) $\dfrac{e^{-2s}}{s}$

15.4 The zero of the function

$$F(s) = \frac{s+1}{(s+2)(s+3)(s+4)}$$

is at

(a) -4 (b) -3
(c) -2 (d) -1

15.5 The poles of the function

$$F(s) = \frac{s+1}{(s+2)(s+3)(s+4)}$$

are at

(a) -4 (b) -3
(c) -2 (d) -1

15.6 If $F(s) = 1/(s+2)$, then $f(t)$ is

(a) $e^{2t}u(t)$ (b) $e^{-2t}u(t)$
(c) $u(t-2)$ (d) $u(t+2)$

15.7 Given that $F(s) = e^{-2s}/(s+1)$, then $f(t)$ is

(a) $e^{-2(t-1)}u(t-1)$ (b) $e^{-(t-2)}u(t-2)$

(c) $e^{-t}u(t-2)$ (d) $e^{-t}u(t+1)$

(e) $e^{-(t-2)}u(t)$

15.8 The initial value of $f(t)$ with transform

$$F(s) = \frac{s+1}{(s+2)(s+3)}$$

is:

(a) nonexistent (b) ∞ (c) 0

(d) 1 (e) $\frac{1}{6}$

15.9 The inverse Laplace transform of

$$\frac{s+2}{(s+2)^2+1}$$

is:

(a) $e^{-t}\cos 2t$ (b) $e^{-t}\sin 2t$

(c) $e^{-2t}\cos t$ (d) $e^{-2t}\sin 2t$

(e) none of the above

15.10 The result of $u(t) * u(t)$ is:

(a) $u^2(t)$ (b) $tu(t)$

(c) $t^2u(t)$ (d) $\delta(t)$

Answers: 15.1b, 15.2a, 15.3d, 15.4d, 15.5a,b,c, 15.6b, 15.7b, 15.8d, 15.9c, 15.10b.

Problems

Sections 15.2 and 15.3 Definition and Properties of the Laplace Transform

15.1 Find the Laplace transform of:

(a) $\cosh at$ (b) $\sinh at$

[*Hint:* $\cosh x = \frac{1}{2}(e^x + e^{-x})$,

$\sinh x = \frac{1}{2}(e^x - e^{-x})$.]

15.2 Determine the Laplace transform of:

(a) $\cos(\omega t + \theta)$ (b) $\sin(\omega t + \theta)$

15.3 Obtain the Laplace transform of each of the following functions:

(a) $e^{-2t}\cos 3tu(t)$ (b) $e^{-2t}\sin 4tu(t)$

(c) $e^{-3t}\cosh 2tu(t)$ (d) $e^{-4t}\sinh tu(t)$

(e) $te^{-t}\sin 2tu(t)$

15.4 Design a problem to help other students better understand how to find the Laplace transform of different time varying functions.

15.5 Find the Laplace transform of each of the following functions:

(a) $t^2\cos(2t + 30°)u(t)$ (b) $3t^4e^{-2t}u(t)$

(c) $2tu(t) - 4\frac{d}{dt}\delta(t)$ (d) $2e^{-(t-1)}u(t)$

(e) $5u(t/2)$ (f) $6e^{-t/3}u(t)$

(g) $\frac{d^n}{dt^n}\delta(t)$

15.6 Find $F(s)$ given that

$$f(t) = \begin{cases} 2t, & 0 < t < 1 \\ t, & 1 < t < 2 \\ 0, & \text{otherwise} \end{cases}$$

15.7 Find the Laplace transform of the following signals:

(a) $f(t) = (2t + 4)u(t)$

(b) $g(t) = (4 + 3e^{-2t})u(t)$

(c) $h(t) = (6\sin(3t) + 8\cos(3t))u(t)$

(d) $x(t) = (e^{-2t}\cosh(4t))u(t)$

15.8 Find the Laplace transform $F(s)$, given that $f(t)$ is:

(a) $2tu(t-4)$

(b) $5\cos(t)\,\delta(t-2)$

(c) $e^{-t}u(t-t)$

(d) $\sin(2t)u(t-\tau)$

15.9 Determine the Laplace transforms of these functions:

(a) $f(t) = (t-4)u(t-2)$

(b) $g(t) = 2e^{-4t}u(t-1)$

(c) $h(t) = 5\cos(2t-1)u(t)$

(d) $p(t) = 6[u(t-2) - u(t-4)]$

15.10 In two different ways, find the Laplace transform of

$$g(t) = \frac{d}{dt}(2e^{-4t}\cos(2t))$$

15.11 Find $F(s)$ if:

(a) $f(t) = 6e^{-t}\cosh 2t$ (b) $f(t) = 3te^{-2t}\sinh 4t$

(c) $f(t) = 8e^{-3t}\cosh tu(t-2)$

15.12 If $g(t) = e^{-2t}\cos 4t$, find $G(s)$.

15.13 Find the Laplace transform of the following functions:

(a) $t\cos tu(t)$ (b) $e^{-t}t\sin tu(t)$

(c) $\dfrac{\sin\beta t}{t}u(t)$

15.14 Find the Laplace transform of the signal in Fig. 15.26.

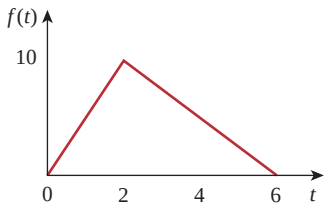


Figure 15.26
For Prob. 15.14.

15.15 Determine the Laplace transform of the function in Fig. 15.27.

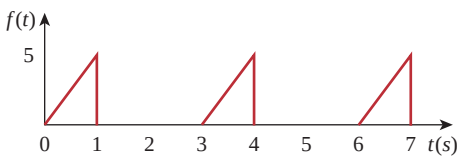


Figure 15.27
For Prob. 15.15.

15.16 Obtain the Laplace transform of $f(t)$ in Fig. 15.28.

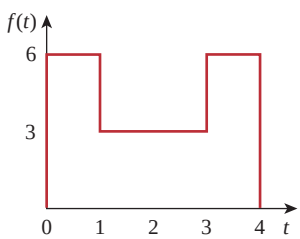


Figure 15.28
For Prob. 15.16.

15.17 Using Fig. 15.29, design a problem to help other students better understand the Laplace transform of a simple, non-periodic waveshape.

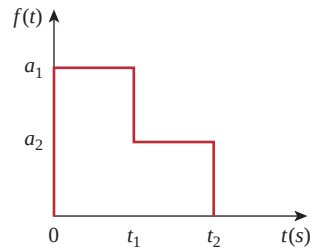
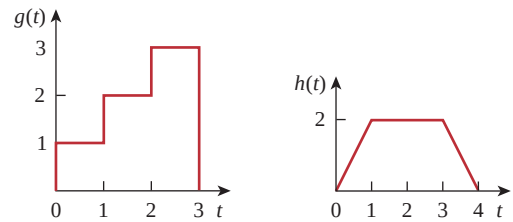


Figure 15.29
For Prob. 15.17.

15.18 Obtain the Laplace transforms of the functions in Fig. 15.30.



(a) (b)

Figure 15.30
For Prob. 15.18.

15.19 Calculate the Laplace transform of the train of unit impulses in Fig. 15.31.

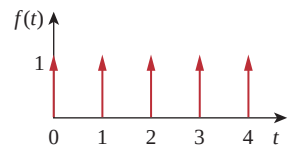


Figure 15.31
For Prob. 15.19.

15.20 Using Fig. 15.32, design a problem to help other students better understand the Laplace transform of a simple, periodic waveshape.

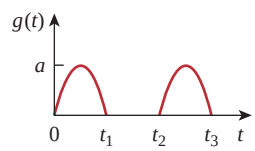


Figure 15.32
For Prob. 15.20.

15.21 Obtain the Laplace transform of the periodic waveform in Fig. 15.33.

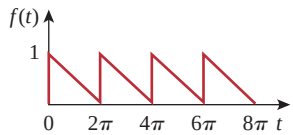
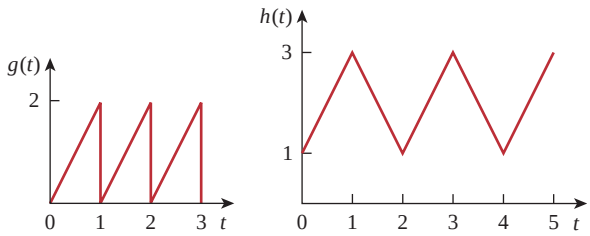


Figure 15.33
For Prob. 15.21.

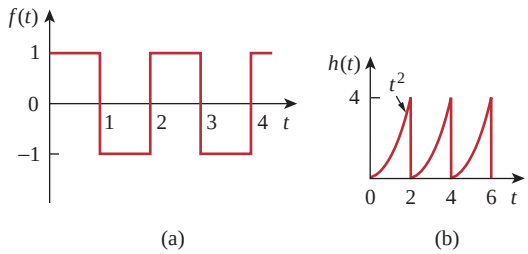
15.22 Find the Laplace transforms of the functions in Fig. 15.34.



(a) (b)

Figure 15.34
For Prob. 15.22.

15.23 Determine the Laplace transforms of the periodic functions in Fig. 15.35.



(a) (b)

Figure 15.35
For Prob. 15.23.

15.24 Design a problem to help other students better understand how to find the initial and final value of a transfer function.

15.25 Let

$$F(s) = \frac{5(s+1)}{(s+2)(s+3)}$$

(a) Use the initial and final value theorems to find $f(0)$ and $f(\infty)$.

(b) Verify your answer in part (a) by finding $f(t)$, using partial fractions.

15.26 Determine the initial and final values of $f(t)$, if they exist, given that:

(a) $F(s) = \dfrac{s^2+3}{s^3+4s^2+6}$

(b) $F(s) = \dfrac{s^2-2s+1}{(s-2)(s^2+2s+4)}$

Section 15.4 The Inverse Laplace Transform

15.27 Determine the inverse Laplace transform of each of the following functions:

(a) $F(s) = \dfrac{1}{s} + \dfrac{2}{s+1}$

(b) $G(s) = \dfrac{3s+1}{s+4}$

(c) $H(s) = \dfrac{4}{(s+1)(s+3)}$

(d) $J(s) = \dfrac{12}{(s+2)^2(s+4)}$

15.28 Design a problem to help other students better understand how to find the inverse Laplace transform.

15.29 Find the inverse Laplace transform of:

$$V(s) = \frac{s+13}{s(s^2+4s+13)}$$

15.30 Find the inverse Laplace transform of:

(a) $F_1(s) = \dfrac{6s^2+8s+3}{s(s^2+2s+5)}$

(b) $F_2(s) = \dfrac{s^2+5s+6}{(s+1)^2(s+4)}$

(c) $F_3(s) = \dfrac{10}{(s+1)(s^2+4s+8)}$

15.31 Find $f(t)$ for each $F(s)$:

(a) $\dfrac{10s}{(s+1)(s+2)(s+3)}$

(b) $\dfrac{2s^2+4s+1}{(s+1)(s+2)^3}$

(c) $\dfrac{s+1}{(s+2)(s^2+2s+5)}$

15.32 Determine the inverse Laplace transform of each of the following functions:

(a) $\dfrac{8(s+1)(s+3)}{s(s+2)(s+4)}$

(b) $\dfrac{s^2-2s+4}{(s+1)(s+2)^2}$

(c) $\dfrac{s^2+1}{(s+3)(s^2+4s+5)}$

15.33 Calculate the inverse Laplace transform of:

(a) $\dfrac{6(s-1)}{s^4-1}$ (b) $\dfrac{e^{-\pi s}}{s^2+1}$

(c) $\dfrac{3}{s(s+1)^3}$

15.34 Find the time functions that have the following Laplace transforms:

(a) $F(s) = 10 + \dfrac{s^2+1}{s^2+4}$

(b) $G(s) = \dfrac{e^{-s}+4e^{-2s}}{s^2+6s+8}$

(c) $H(s) = \dfrac{(s+1)e^{-2s}}{s(s+3)(s+4)}$

15.35 Obtain $f(t)$ for the following transforms:

(a) $F(s) = \dfrac{(s+3)e^{-6s}}{(s+1)(s+2)}$

(b) $F(s) = \dfrac{4-e^{-2s}}{s^2+5s+4}$

(c) $F(s) = \dfrac{se^{-s}}{(s+3)(s^2+4)}$

15.36 Obtain the inverse Laplace transforms of the following functions:

(a) $X(s) = \dfrac{1}{s^2(s+2)(s+3)}$

(b) $Y(s) = \dfrac{1}{s(s+1)^2}$

(c) $Z(s) = \dfrac{1}{s(s+1)(s^2+6s+10)}$

15.37 Find the inverse Laplace transform of:

(a) $H(s) = \dfrac{s+4}{s(s+2)}$

(b) $G(s) = \dfrac{s^2+4s+5}{(s+3)(s^2+2s+2)}$

(c) $F(s) = \dfrac{e^{-4s}}{s+2}$

(d) $D(s) = \dfrac{10s}{(s^2+1)(s^2+4)}$

15.38 Find $f(t)$ given that:

(a) $F(s) = \dfrac{s^2+4s}{s^2+10s+26}$

(b) $F(s) = \dfrac{5s^2+7s+29}{s(s^2+4s+29)}$

***15.39** Determine $f(t)$ if:

(a) $F(s) = \dfrac{2s^3+4s^2+1}{(s^2+2s+17)(s^2+4s+20)}$

(b) $F(s) = \dfrac{s^2+4}{(s^2+9)(s^2+6s+3)}$

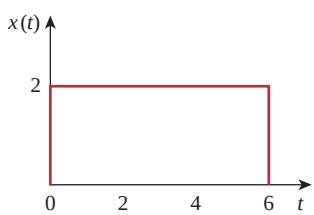
15.40 Show that

$$\mathcal{L}^{-1}\left[\frac{4s^2+7s+13}{(s+2)(s^2+2s+5)}\right] = \left[\sqrt{2}e^{-t}\cos(2t+45°) + 3e^{-2t}\right]u(t)$$

Section 15.5 The Convolution Integral

***15.41** Let $x(t)$ and $y(t)$ be as shown in Fig. 15.36. Find $z(t) = x(t) * y(t)$.

15.42 Design a problem to help other students better understand how to convolve two functions together.



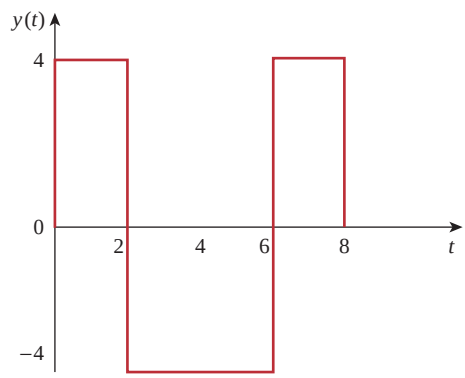


Figure 15.36
For Prob. 15.41.

* An asterisk indicates a challenging problem.

15.43 Find $y(t) = x(t) * h(t)$ for each paired $x(t)$ and $h(t)$ in Fig. 15.37.

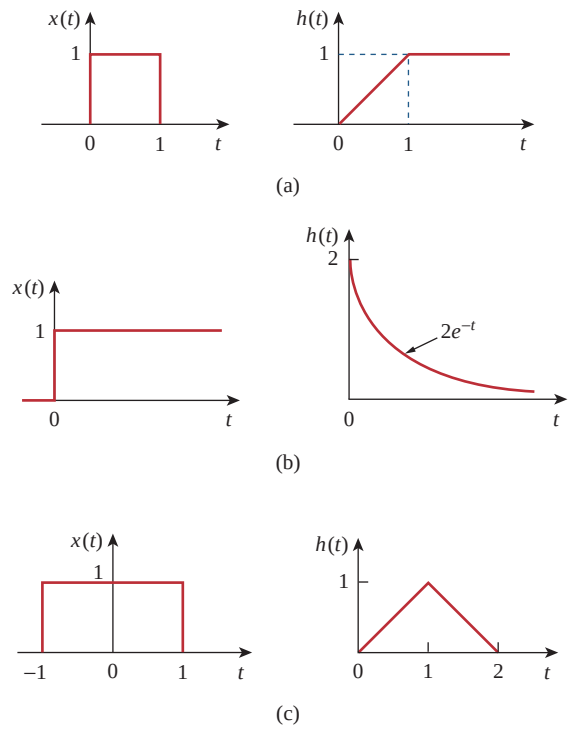


Figure 15.37
For Prob. 15.43.

15.44 Obtain the convolution of the pairs of signals in Fig. 15.38.

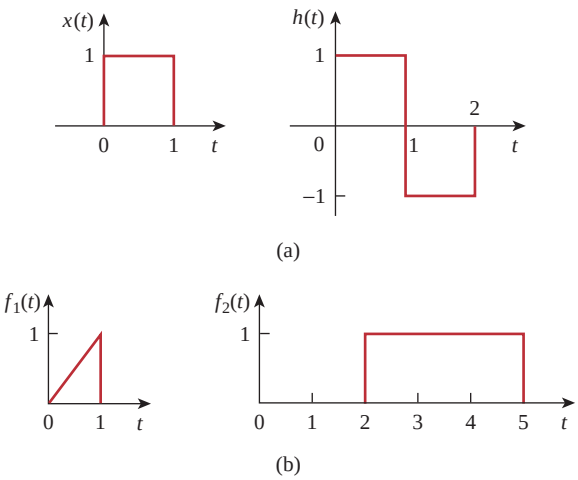


Figure 15.38
For Prob. 15.44.

15.45 Given $h(t) = 4e^{-2t}u(t)$ and $x(t) = \delta(t) - 2e^{-2t}u(t)$, find $y(t) = x(t) * h(t)$.

15.46 Given the following functions

$$x(t) = 2\delta(t), \qquad y(t) = 4u(t), \qquad z(t) = e^{-2t}u(t),$$

evaluate the following convolution operations.

(a) $x(t) * y(t)$
(b) $x(t) * z(t)$
(c) $y(t) * z(t)$
(d) $y(t) * [\,y(t) + z(t)]$

15.47 A system has the transfer function

$$H(s) = \frac{s}{(s+1)(s+2)}$$

(a) Find the impulse response of the system.
(b) Determine the output $y(t)$, given that the input is $x(t) = u(t)$.

15.48 Find $f(t)$ using convolution given that:

(a) $F(s) = \dfrac{4}{(s^2 + 2s + 5)^2}$

(b) $F(s) = \dfrac{2s}{(s+1)(s^2+4)}$

***15.49** Use the convolution integral to find:

(a) $t * e^{at}u(t)$
(b) $\cos(t) * \cos(t)u(t)$

Section 15.6 Application to Integrodifferential Equations

15.50 Use the Laplace transform to solve the differential equation

$$\frac{d^2v(t)}{dt^2} + 2\frac{dv(t)}{dt} + 10v(t) = 3\cos 2t$$

subject to $v(0) = 1$, $dv(0)/dt = -2$.

15.51 Given that $v(0) = 2$ and $dv(0)/dt = 4$, solve

$$\frac{d^2v}{dt^2} + 5\frac{dv}{dt} + 6v = 6e^{-t}u(t)$$

15.52 Use the Laplace transform to find $i(t)$ for $t > 0$ if

$$\frac{d^2i}{dt^2} + 3\frac{di}{dt} + 2i + \delta(t) = 0,$$
$$i(0) = 0, \qquad i'(0) = 3$$

***15.53** Use Laplace transforms to solve for $x(t)$ in

$$x(t) = \cos t + \int_0^t e^{\lambda - t}x(\lambda)\,d\lambda$$

15.54 Design a problem to help other students better understand solving second order differential equations with a time varying input.

15.55 Solve for $y(t)$ in the following differential equation if the initial conditions are zero.

$$\frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 8\frac{dy}{dt} = e^{-t}\cos 2t$$

15.56 Solve for $v(t)$ in the integrodifferential equation

$$4\frac{dv}{dt} + 12\int_{-\infty}^{t} v\,dt = 0$$

given that $v(0) = 2$.

15.57 Design a problem to help other students better understand solving integrodifferential equations with a periodic input, using Laplace transforms.

15.58 Given that

$$\frac{dv}{dt} + 2v + 5\int_0^t v(\lambda)\,d\lambda = 4u(t)$$

with $v(0) = -1$, determine $v(t)$ for $t > 0$.

15.59 Solve the integrodifferential equation

$$\frac{dy}{dt} + 4y + 3\int_0^t y\,dt = 6e^{-2t}, \qquad y(0) = -1$$

15.60 Solve the following integrodifferential equation

$$2\frac{dx}{dt} + 5x + 3\int_0^t x\,dt + 4 = \sin 4t, \qquad x(0) = 1$$

chapter

16

Applications of the Laplace Transform

Communication skills are the most important skills any engineer can have. A very critical element in this tool set is the ability to ask a question and understand the answer, a very simple thing and yet it may make the difference between success and failure!

—James A. Watson

Enhancing Your Skills and Your Career

Asking Questions

In over 30 years of teaching, I have struggled with determining how best to help students learn. Regardless of how much time students spend in studying for a course, the most helpful activity for students is learning how to ask questions in class and then asking those questions. The student, by asking questions, becomes actively involved in the learning process and no longer is merely a passive receptor of information. I think this active involvement contributes so much to the learning process that it is probably the single most important aspect to the development of a modern engineer. In fact, asking questions is the basis of science. As Charles P. Steinmetz rightly said, "No man really becomes a fool until he stops asking questions."



Photo by Charles Alexander

It seems very straightforward and quite easy to ask questions. Have we not been doing that all our lives? The truth is to ask questions in an appropriate manner and to maximize the learning process takes some thought and preparation.

I am sure that there are several models one could effectively use. Let me share what has worked for me. The most important thing to keep in mind is that you do not have to form a perfect question. Since the question-and-answer format allows the question to be developed in an iterative manner, the original question can easily be refined as you go. I frequently tell students that they are most welcome to read their questions in class.

Here are three things you should keep in mind when asking questions. First, prepare your question. If you are like many students who are either shy or have not learned to ask questions in class, you may wish to start with a question you have written down outside of class. Second, wait for an appropriate time to ask the question. Simply use your judgment on that. Third, be prepared to clarify your question by paraphrasing it or saying it in a different way in case you are asked to repeat the question.

One last comment: not all professors like students to ask questions in class even though they may say they do. You need to find out which professors like classroom questions. Good luck in enhancing one of your most important skills as an engineer.

16.1 Introduction

Now that we have introduced the Laplace transform, let us see what we can do with it. Please keep in mind that with the Laplace transform we actually have one of the most powerful mathematical tools for analysis, synthesis, and design. Being able to look at circuits and systems in the *s*-domain can help us to understand how our circuits and systems really function. In this chapter we will take an in-depth look at how easy it is to work with circuits in the *s*-domain. In addition, we will briefly look at physical systems. We are sure you have studied some mechanical systems and may have used the same differential equations to describe them as we use to describe our electric circuits. Actually that is a wonderful thing about the physical universe in which we live; the same differential equations can be used to describe any linear circuit, system, or process. The key is the term *linear*.

A **system** is a mathematical model of a physical process relating the input to the output.

It is entirely appropriate to consider circuits as systems. Historically, circuits have been discussed as a separate topic from systems, so we will actually talk about circuits and systems in this chapter realizing that circuits are nothing more than a class of electrical systems.

The most important thing to remember is that everything we discussed in the last chapter and in this chapter applies to any linear system. In the last chapter, we saw how we can use Laplace transforms to solve linear differential equations and integral equations. In this chapter, we introduce the concept of modeling circuits in the *s*-domain. We can use that principle to help us solve just about any kind of linear circuit. We will take a quick look at how state variables can be used to analyze systems with multiple inputs and multiple outputs. Finally, we examine how the Laplace transform is used in network stability analysis and in network synthesis.

16.2 Circuit Element Models

Having mastered how to obtain the Laplace transform and its inverse, we are now prepared to employ the Laplace transform to analyze circuits. This usually involves three steps.

Steps in Applying the Laplace Transform:

1. Transform the circuit from the time domain to the *s*-domain.
2. Solve the circuit using nodal analysis, mesh analysis, source transformation, superposition, or any circuit analysis technique with which we are familiar.
3. Take the inverse transform of the solution and thus obtain the solution in the time domain.

Only the first step is new and will be discussed here. As we did in phasor analysis, we transform a circuit in the time domain to the frequency or *s*-domain by Laplace transforming each term in the circuit.

As one can infer from step 2, all the circuit analysis techniques applied for dc circuits are applicable to the *s*-domain.

For a resistor, the voltage-current relationship in the time domain is

$$v(t) = Ri(t) \tag{16.1}$$

Taking the Laplace transform, we get

$$\boxed{V(s) = RI(s)} \tag{16.2}$$

For an inductor,

$$v(t) = L\frac{di(t)}{dt} \tag{16.3}$$

Taking the Laplace transform of both sides gives

$$V(s) = L[sI(s) - i(0^-)] = sLI(s) - Li(0^-) \tag{16.4}$$

or

$$\boxed{I(s) = \frac{1}{sL}V(s) + \frac{i(0^-)}{s}} \tag{16.5}$$

The *s*-domain equivalents are shown in Fig. 16.1, where the initial condition is modeled as a voltage or current source.

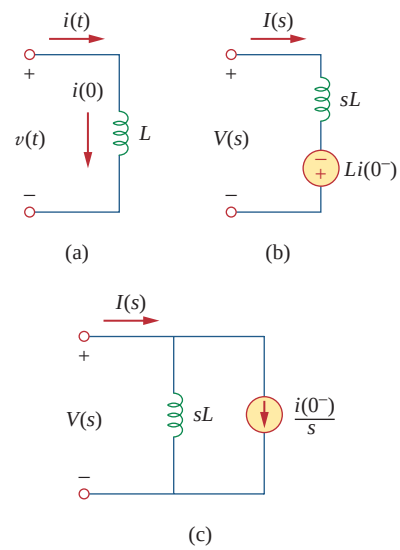


Figure 16.1
Representation of an inductor: (a) time-domain, (b,c) *s*-domain equivalents.

For a capacitor,

$$i(t) = C\frac{dv(t)}{dt} \tag{16.6}$$

which transforms into the *s*-domain as

$$I(s) = C[sV(s) - v(0^-)] = sCV(s) - Cv(0^-) \tag{16.7}$$

or

$$\boxed{V(s) = \frac{1}{sC}I(s) + \frac{v(0^-)}{s}} \tag{16.8}$$

The *s*-domain equivalents are shown in Fig. 16.2. With the *s*-domain equivalents, the Laplace transform can be used readily to solve first- and

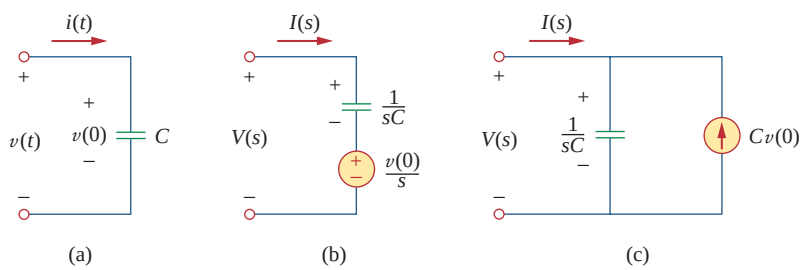


Figure 16.2
Representation of a capacitor: (a) time-domain, (b,c) *s*-domain equivalents.

The elegance of using the Laplace transform in circuit analysis lies in the automatic inclusion of the initial conditions in the transformation process, thus providing a complete (transient and steady-state) solution.

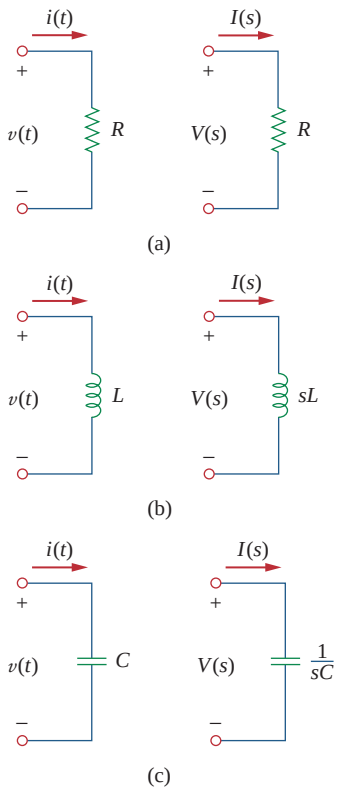


Figure 16.3
Time-domain and *s*-domain representations of passive elements under zero initial conditions.

second-order circuits such as those we considered in Chapters 7 and 8. We should observe from Eqs. (16.3) to (16.8) that the initial conditions are part of the transformation. This is one advantage of using the Laplace transform in circuit analysis. Another advantage is that a complete response—transient and steady state—of a network is obtained. We will illustrate this with Examples 16.2 and 16.3. Also, observe the duality of Eqs. (16.5) and (16.8), confirming what we already know from Chapter 8 (see Table 8.1), namely, that L and C, $I(s)$ and $V(s)$, and $v(0)$ and $i(0)$ are dual pairs.

If we assume zero initial conditions for the inductor and the capacitor, the above equations reduce to:

$$\begin{aligned} &\text{Resistor:} && V(s) = RI(s) \\ &\text{Inductor:} && V(s) = sLI(s) \\ &\text{Capacitor:} && V(s) = \frac{1}{sC} I(s) \end{aligned} \tag{16.9}$$

The *s*-domain equivalents are shown in Fig. 16.3.

We define the impedance in the *s*-domain as the ratio of the voltage transform to the current transform under zero initial conditions; that is,

$$Z(s) = \frac{V(s)}{I(s)} \tag{16.10}$$

Thus, the impedances of the three circuit elements are

$$\begin{aligned} &\text{Resistor:} && Z(s) = R \\ &\text{Inductor:} && Z(s) = sL \\ &\text{Capacitor:} && Z(s) = \frac{1}{sC} \end{aligned} \tag{16.11}$$

Table 16.1 summarizes these. The admittance in the *s*-domain is the reciprocal of the impedance, or

$$Y(s) = \frac{1}{Z(s)} = \frac{I(s)}{V(s)} \tag{16.12}$$

The use of the Laplace transform in circuit analysis facilitates the use of various signal sources such as impulse, step, ramp, exponential, and sinusoidal.

The models for dependent sources and op amps are easy to develop drawing from the simple fact that if the Laplace transform of $f(t)$ is $F(s)$, then the Laplace transform of $af(t)$ is $aF(s)$—the linearity property. The dependent source model is a little easier in that we deal with a single value. The dependent source can have only two controlling values, a constant times either a voltage or a current. Thus,

$$\mathcal{L}[av(t)] = aV(s) \tag{16.13}$$

$$\mathcal{L}[ai(t)] = aI(s) \tag{16.14}$$

The ideal op amp can be treated just like a resistor. Nothing within an op amp, either real or ideal, does anything more than multiply a voltage by a constant. Thus, we only need to write the equations as we always do using the constraint that the input voltage to the op amp has to be zero and the input current has to be zero.

TABLE 16.1

Impedance of an element in the *s*-domain.*

Element	$Z(s) = V(s)/I(s)$
Resistor	R
Inductor	sL
Capacitor	$1/sC$

* Assuming zero initial conditions

Example 16.1

Find $v_o(t)$ in the circuit of Fig. 16.4, assuming zero initial conditions.

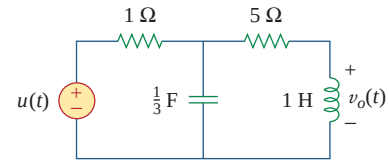


Figure 16.4
For Example 16.1.

Solution:
We first transform the circuit from the time domain to the *s*-domain.

$$u(t) \quad \Rightarrow \quad \frac{1}{s}$$
$$1\text{ H} \quad \Rightarrow \quad sL = s$$
$$\frac{1}{3}\text{F} \quad \Rightarrow \quad \frac{1}{sC} = \frac{3}{s}$$

The resulting *s*-domain circuit is in Fig. 16.5. We now apply mesh analysis. For mesh 1,

$$\frac{1}{s} = \left(1 + \frac{3}{s}\right)I_1 - \frac{3}{s}I_2 \qquad \textbf{(16.1.1)}$$

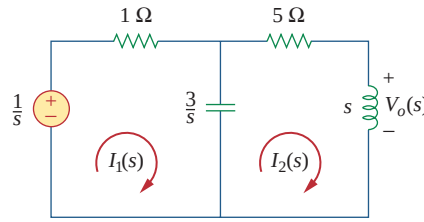


Figure 16.5
Mesh analysis of the frequency-domain equivalent of the same circuit.

For mesh 2,

$$0 = -\frac{3}{s}I_1 + \left(s + 5 + \frac{3}{s}\right)I_2$$

or

$$I_1 = \frac{1}{3}(s^2 + 5s + 3)I_2 \qquad \textbf{(16.1.2)}$$

Substituting this into Eq. (16.1.1),

$$\frac{1}{s} = \left(1 + \frac{3}{s}\right)\frac{1}{3}(s^2 + 5s + 3)I_2 - \frac{3}{s}I_2$$

Multiplying through by $3s$ gives

$$3 = (s^3 + 8s^2 + 18s)I_2 \quad \Rightarrow \quad I_2 = \frac{3}{s^3 + 8s^2 + 18s}$$

$$V_o(s) = sI_2 = \frac{3}{s^2 + 8s + 18} = \frac{3}{\sqrt{2}}\,\frac{\sqrt{2}}{(s+4)^2 + (\sqrt{2})^2}$$

Taking the inverse transform yields

$$v_o(t) = \frac{3}{\sqrt{2}}e^{-4t}\sin\sqrt{2}t\text{ V}, \qquad t \geq 0$$

Practice Problem 16.1

Determine $v_o(t)$ in the circuit of Fig. 16.6, assuming zero initial conditions.

Answer: $20(1 - e^{-2t} - 2te^{-2t})u(t)$ V.

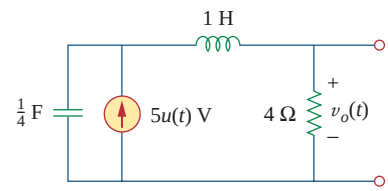


Figure 16.6
For Practice Prob. 16.1.

Example 16.2

Find $v_o(t)$ in the circuit of Fig. 16.7. Assume $v_o(0) = 5$ V.

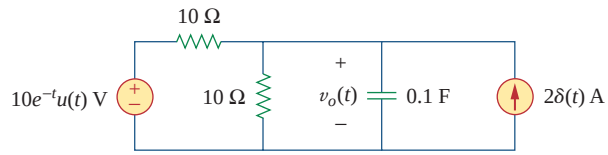


Figure 16.7
For Example 16.2.

Solution:

We transform the circuit to the *s*-domain as shown in Fig. 16.8. The initial condition is included in the form of the current source $Cv_o(0) = 0.1(5) = 0.5$ A. [See Fig. 16.2(c).] We apply nodal analysis. At the top node,

$$\frac{10/(s+1) - V_o}{10} + 2 + 0.5 = \frac{V_o}{10} + \frac{V_o}{10/s}$$

or

$$\frac{1}{s+1} + 2.5 = \frac{2V_o}{10} + \frac{sV_o}{10} = \frac{1}{10}V_o(s+2)$$

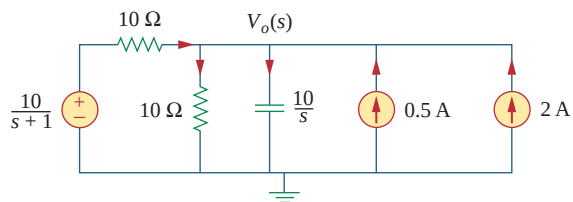


Figure 16.8
Nodal analysis of the equivalent of the circuit in Fig. 16.7.

Multiplying through by 10,

$$\frac{10}{s+1} + 25 = V_o(s+2)$$

or

$$V_o = \frac{25s + 35}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

where

$$A = (s+1)V_o(s)\,|_{s=-1} = \left.\frac{25s+35}{(s+2)}\right|_{s=-1} = \frac{10}{1} = 10$$

$$B = (s+2)V_o(s)\,|_{s=-2} = \left.\frac{25s+35}{(s+1)}\right|_{s=-2} = \frac{-15}{-1} = 15$$

Thus,

$$V_o(s) = \frac{10}{s+1} + \frac{15}{s+2}$$

Taking the inverse Laplace transform, we obtain

$$v_o(t) = (10e^{-t} + 15e^{-2t})u(t)\ \text{V}$$

Practice Problem 16.2

Find $v_o(t)$ in the circuit shown in Fig. 16.9. Note that, since the voltage input is multiplied by $u(t)$, the voltage source is a short for all $t < 0$ and $i_L(0) = 0$.

Answer: $(24e^{-2t} - 4e^{-t/3})u(t)$ V.

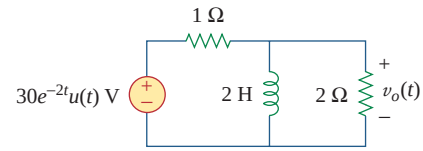
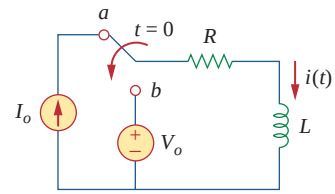


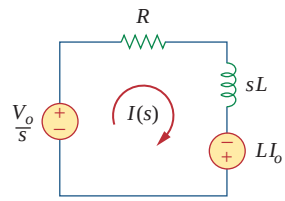
Figure 16.9
For Practice Prob. 16.2.

Example 16.3

In the circuit of Fig. 16.10(a), the switch moves from position a to position b at $t = 0$. Find $i(t)$ for $t > 0$.



(a)



(b)

Figure 16.10
For Example 16.3.

Solution:

The initial current through the inductor is $i(0) = I_o$. For $t > 0$, Fig. 16.10(b) shows the circuit transformed to the s-domain. The initial condition is incorporated in the form of a voltage source as $Li(0) = LI_o$. Using mesh analysis,

$$I(s)(R + sL) - LI_o - \frac{V_o}{s} = 0 \qquad \textbf{(16.3.1)}$$

or

$$I(s) = \frac{LI_o}{R + sL} + \frac{V_o}{s(R + sL)} = \frac{I_o}{s + R/L} + \frac{V_o/L}{s(s + R/L)} \qquad \textbf{(16.3.2)}$$

Applying partial fraction expansion on the second term on the right-hand side of Eq. (16.3.2) yields

$$I(s) = \frac{I_o}{s + R/L} + \frac{V_o/R}{s} - \frac{V_o/R}{(s + R/L)} \qquad \textbf{(16.3.3)}$$

The inverse Laplace transform of this gives

$$i(t) = \left(I_o - \frac{V_o}{R}\right)e^{-t/\tau} + \frac{V_o}{R}, \qquad t \geq 0 \qquad \textbf{(16.3.4)}$$

where $\tau = R/L$. The term in parentheses is the transient response, while the second term is the steady-state response. In other words, the final value is $i(\infty) = V_o/R$, which we could have predicted by applying the final-value theorem on Eq. (16.3.2) or (16.3.3); that is,

$$\lim_{s\to 0} sI(s) = \lim_{s\to 0}\left(\frac{sI_o}{s + R/L} + \frac{V_o/L}{s + R/L}\right) = \frac{V_o}{R} \qquad \textbf{(16.3.5)}$$

Equation (16.3.4) may also be written as

$$i(t) = I_o e^{-t/\tau} + \frac{V_o}{R}(1 - e^{-t/\tau}), \qquad t \geq 0 \qquad \textbf{(16.3.6)}$$

The first term is the natural response, while the second term is the forced response. If the initial condition $I_o = 0$, Eq. (16.3.6) becomes

$$i(t) = \frac{V_o}{R}(1 - e^{-t/\tau}), \qquad t \geq 0 \qquad \textbf{(16.3.7)}$$

which is the step response, since it is due to the step input V_o with no initial energy.

Practice Problem 16.3

The switch in Fig. 16.11 has been in position b for a long time. It is moved to position a at $t = 0$. Determine $v(t)$ for $t > 0$.

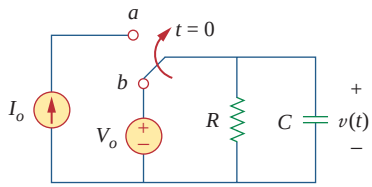


Figure 16.11
For Practice Prob. 16.3.

Answer: $v(t) = (V_o - I_oR)e^{-t/\tau} + I_oR,\ t > 0$, where $\tau = RC$.

16.3 Circuit Analysis

Circuit analysis is again relatively easy to do when we are in the s-domain. We merely need to transform a complicated set of mathematical relationships in the time domain into the s-domain where we convert operators (derivatives and integrals) into simple multipliers of s and $1/s$. This now allows us to use algebra to set up and solve our circuit equations. The exciting thing about this is that *all* of the circuit theorems and relationships we developed for dc circuits are perfectly valid in the s-domain.

Remember, **equivalent circuits,** with capacitors and inductors, only exist in the s-domain; they cannot be transformed back into the time domain.

Example 16.4

Consider the circuit in Fig. 16.12(a). Find the value of the voltage across the capacitor assuming that the value of $v_s(t) = 10u(t)$ V and assume that at $t = 0$, -1 A flows through the inductor and $+5$ V is across the capacitor.

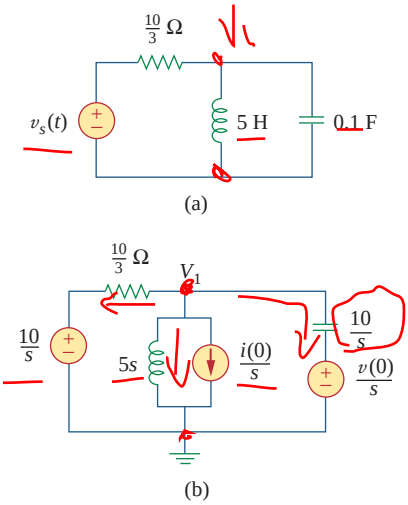


Figure 16.12
For Example 16.4.

Solution:

Figure 16.12(b) represents the entire circuit in the s-domain with the initial conditions incorporated. We now have a straightforward nodal analysis problem. Since the value of V_1 is also the value of the capacitor voltage in the time domain and is the only unknown node voltage, we only need to write one equation.

$$\frac{V_1 - 10/s}{10/3} + \frac{V_1 - 0}{5s} + \frac{i(0)}{s} + \frac{V_1 - [v(0)/s]}{1/(0.1s)} = 0 \qquad \textbf{(16.4.1)}$$

or

$$0.1\left(s + 3 + \frac{2}{s}\right)V_1 = \frac{3}{s} + \frac{1}{s} + 0.5 \qquad \textbf{(16.4.2)}$$

where $v(0) = 5$ V and $i(0) = -1$ A. Simplifying we get

$$(s^2 + 3s + 2)\,V_1 = 40 + 5s$$

or

$$V_1 = \frac{40 + 5s}{(s + 1)(s + 2)} = \frac{35}{s + 1} - \frac{30}{s + 2} \qquad \textbf{(16.4.3)}$$

Taking the inverse Laplace transform yields

$$v_1(t) = (35e^{-t} - 30e^{-2t})u(t)\ \text{V} \qquad \textbf{(16.4.4)}$$

Practice Problem 16.4

For the circuit shown in Fig. 16.12 with the same initial conditions, find the current through the inductor for all time $t > 0$.

Answer: $i(t) = (3 - 7e^{-t} + 3e^{-2t})u(t)$ A.

Example 16.5

For the circuit shown in Fig. 16.12, and the initial conditions used in Example 16.4, use superposition to find the value of the capacitor voltage.

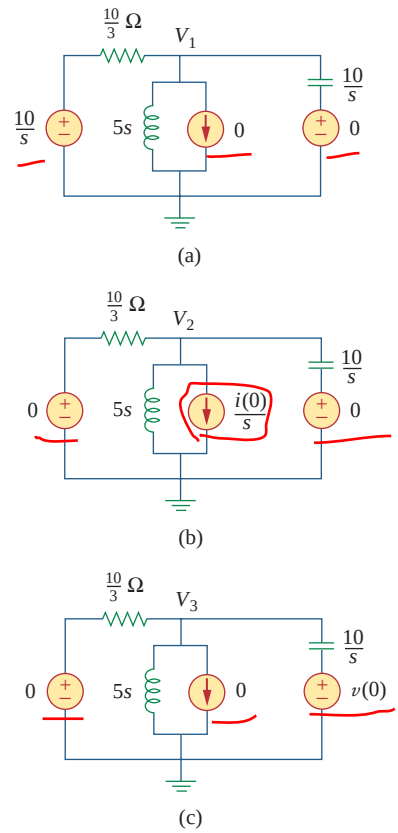


Figure 16.13
For Example 16.5.

Solution:
Since the circuit in the *s*-domain actually has three independent sources, we can look at the solution one source at a time. Figure 16.13 presents the circuits in the *s*-domain considering one source at a time. We now have three nodal analysis problems. First, let us solve for the capacitor voltage in the circuit shown in Fig. 16.13(a).

$$\frac{V_1 - 10/s}{10/3} + \frac{V_1 - 0}{5s} + 0 + \frac{V_1 - 0}{1/(0.1s)} = 0$$

or

$$0.1\left(s + 3 + \frac{2}{s}\right)V_1 = \frac{3}{s}$$

Simplifying we get

$$(s^2 + 3s + 2)V_1 = 30$$

$$V_1 = \frac{30}{(s+1)(s+2)} = \frac{30}{s+1} - \frac{30}{s+2}$$

or

$$v_1(t) = (30e^{-t} - 30e^{-2t})u(t)\ \text{V} \qquad \textbf{(16.5.1)}$$

For Fig. 16.13(b) we get,

$$\frac{V_2 - 0}{10/3} + \frac{V_2 - 0}{5s} - \frac{1}{s} + \frac{V_2 - 0}{1/(0.1s)} = 0$$

or

$$0.1\left(s + 3 + \frac{2}{s}\right)V_2 = \frac{1}{s}$$

This leads to

$$V_2 = \frac{10}{(s+1)(s+2)} = \frac{10}{s+1} - \frac{10}{s+2}$$

Taking the inverse Laplace transform, we get

$$v_2(t) = (10e^{-t} - 10e^{-2t})u(t)\ \text{V} \qquad \textbf{(16.5.2)}$$

For Fig. 16.13(c),

$$\frac{V_3 - 0}{10/3} + \frac{V_3 - 0}{5s} - 0 + \frac{V_3 - 5/s}{1/(0.1s)} = 0$$

or

$$0.1\left(s + 3 + \frac{2}{s}\right)V_3 = 0.5$$

$$V_3 = \frac{5s}{(s+1)(s+2)} = \frac{-5}{s+1} + \frac{10}{s+2}$$

This leads to

$$v_3(t) = (-5e^{-t} + 10e^{-2t})u(t)\ \text{V} \qquad \textbf{(16.5.3)}$$

Now all we need to do is to add Eqs. (16.5.1), (16.5.2), and (16.5.3):

$$\begin{aligned} v(t) &= v_1(t) + v_2(t) + v_3(t) \\ &= \{(30 + 10 - 5)e^{-t} + (-30 + 10 - 10)e^{-2t}\}u(t)\ \text{V} \end{aligned}$$

or

$$v(t) = (35e^{-t} - 30e^{-2t})u(t)\ \text{V}$$

which agrees with our answer in Example 16.4.

Practice Problem 16.5

For the circuit shown in Fig. 16.12, and the same initial conditions in Example 16.4, find the current through the inductor for all time $t > 0$ using superposition.

Answer: $i(t) = (3 - 7e^{-t} + 3e^{-2t})u(t)$ A.

Example 16.6

Assume that there is no initial energy stored in the circuit of Fig. 16.14 at $t = 0$ and that $i_s = 10\,u(t)$ A. (a) Find $V_o(s)$ using Thevenin's theorem. (b) Apply the initial- and final-value theorems to find $v_o(0^+)$ and $v_o(\infty)$. (c) Determine $v_o(t)$.

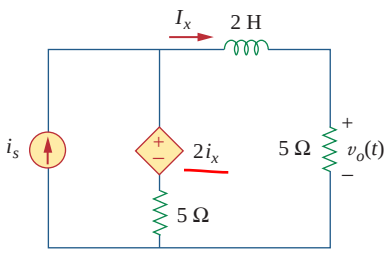


Figure 16.14
For Example 16.6.

Solution:
Since there is no initial energy stored in the circuit, we assume that the initial inductor current and initial capacitor voltage are zero at $t = 0$.

(a) To find the Thevenin equivalent circuit, we remove the 5-Ω resistor and then find V_{oc} (V_{Th}) and I_{sc}. To find V_{Th}, we use the Laplace-transformed circuit in Fig. 16.15(a). Since $I_x = 0$, the dependent voltage source contributes nothing, so

$$V_{oc} = V_{Th} = 5\left(\frac{10}{s}\right) = \frac{50}{s}$$

To find Z_{Th}, we consider the circuit in Fig. 16.15(b), where we first find I_{sc}. We can use nodal analysis to solve for V_1 which then leads to I_{sc} ($I_{sc} = I_x = V_1/2s$).

$$-\frac{10}{s} + \frac{(V_1 - 2I_x) - 0}{5} + \frac{V_1 - 0}{2s} = 0$$

along with

$$I_x = \frac{V_1}{2s}$$

leads to

$$V_1 = \frac{100}{2s + 3}$$

Hence,

$$I_{sc} = \frac{V_1}{2s} = \frac{100/(2s + 3)}{2s} = \frac{50}{s(2s + 3)}$$

and

$$Z_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{50/s}{50/[s(2s + 3)]} = 2s + 3$$

The given circuit is replaced by its Thevenin equivalent at terminals *a*-*b* as shown in Fig. 16.16. From Fig. 16.16,

$$V_o = \frac{5}{5 + Z_{Th}} V_{Th} = \frac{5}{5 + 2s + 3}\left(\frac{50}{s}\right) = \frac{250}{s(2s + 8)} = \frac{125}{s(s + 4)}$$

(b) Using the initial-value theorem we find

$$v_o(0) = \lim_{s\to\infty} sV_o(s) = \lim_{s\to\infty} \frac{125}{s + 4} = \lim_{s\to\infty} \frac{125/s}{1 + 4/s} = \frac{0}{1} = 0$$

Using the final-value theorem we find

$$v_o(\infty) = \lim_{s\to 0} sV_o(s) = \lim_{s\to 0} \frac{125}{s + 4} = \frac{125}{4} = 31.25 \text{ V}$$

(c) By partial fraction,

$$V_o = \frac{125}{s(s + 4)} = \frac{A}{s} + \frac{B}{s + 4}$$

$$A = sV_o(s)\Big|_{s=0} = \frac{125}{s + 4}\Big|_{s=0} = 31.25$$

$$B = (s + 4)V_o(s)\Big|_{s=-4} = \frac{125}{s}\Big|_{s=-4} = -31.25$$

$$V_o = \frac{31.25}{s} - \frac{31.25}{s + 4}$$

Taking the inverse Laplace transform gives

$$v_o(t) = 31.25(1 - e^{-4t})u(t) \text{ V}$$

Notice that the values of $v_o(0)$ and $v_o(\infty)$ obtained in part (b) are confirmed.

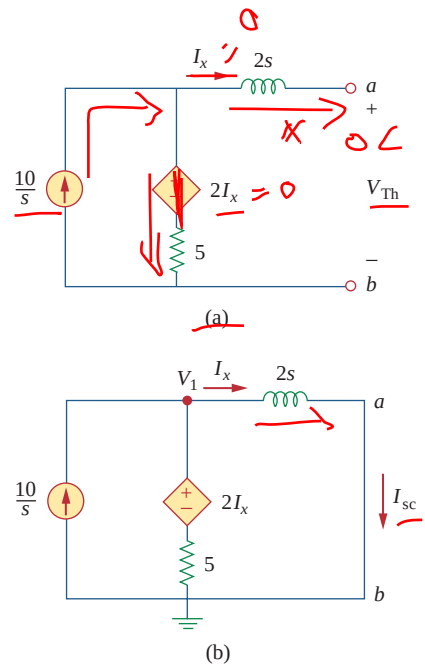


Figure 16.15
For Example 16.6: (a) finding V_{Th}, (b) determining Z_{Th}.

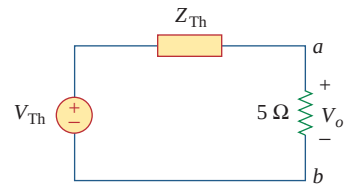


Figure 16.16
The Thevenin equivalent of the circuit in Fig. 16.14 in the *s*-domain.

Practice Problem 16.6

The initial energy in the circuit of Fig. 16.17 is zero at $t = 0$. Assume that $v_s = 15u(t)$ V. (a) Find $V_o(s)$ using the Thevenin theorem. (b) Apply the initial- and final-value theorems to find $v_o(0)$ and $v_o(\infty)$. (c) Obtain $v_o(t)$.

Answer: (a) $V_o(s) = \frac{4(s+0.25)}{s(s+0.3)}$, (b) 4, 3.333 V, (c) $(10 + 2e^{-0.3t})u(t)$ V.

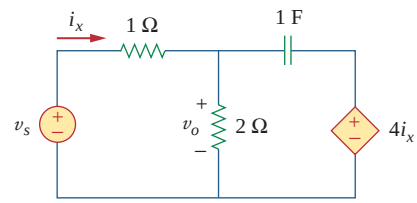


Figure 16.17
For Practice Prob. 16.6.

16.4 Transfer Functions

For electrical networks, the transfer function is also known as the *network function*.

The *transfer function* is a key concept in signal processing because it indicates how a signal is processed as it passes through a network. It is a fitting tool for finding the network response, determining (or designing for) network stability, and network synthesis. The transfer function of a network describes how the output behaves with respect to the input. It specifies the transfer from the input to the output in the s-domain, assuming no initial energy.

The **transfer function** $H(s)$ is the ratio of the output response $Y(s)$ to the input excitation $X(s)$, assuming all initial conditions are zero.

Thus,

$$H(s) = \frac{Y(s)}{X(s)} \tag{16.15}$$

The transfer function depends on what we define as input and output. Since the input and output can be either current or voltage at any place in the circuit, there are four possible transfer functions:

Some authors would not consider Eqs. (16.16c) and (16.16d) transfer functions.

$$H(s) = \text{Voltage gain} = \frac{V_o(s)}{V_i(s)} \tag{16.16a}$$

$$H(s) = \text{Current gain} = \frac{I_o(s)}{I_i(s)} \tag{16.16b}$$

$$H(s) = \text{Impedance} = \frac{V(s)}{I(s)} \tag{16.16c}$$

$$H(s) = \text{Admittance} = \frac{I(s)}{V(s)} \tag{16.16d}$$

Thus, a circuit can have many transfer functions. Note that $H(s)$ is dimensionless in Eqs. (16.16a) and (16.16b).

Each of the transfer functions in Eq. (16.16) can be found in two ways. One way is to assume any convenient input $X(s)$, use any circuit analysis technique (such as current or voltage division, nodal or mesh analysis) to find the output $Y(s)$, and then obtain the ratio of the two. The other approach is to apply the *ladder method*, which involves walking our way through the circuit. By this approach, we assume that the output is 1 V or 1 A as appropriate and use the basic laws of Ohm and Kirchhoff (KCL only) to obtain the input. The transfer function becomes unity divided by the input. This approach may be more convenient to use when the circuit has many loops or nodes so that applying nodal or mesh analysis becomes cumbersome. In the first method, we assume an input and find the output; in the second method, we assume the output and find the input. In both methods, we calculate $H(s)$ as the ratio of output to input transforms. The two methods rely on the linearity property, since we only deal with linear circuits in this book. Example 16.8 illustrates these methods.

Equation (16.15) assumes that both $X(s)$ and $Y(s)$ are known. Sometimes, we know the input $X(s)$ and the transfer function $H(s)$. We find the output $Y(s)$ as

$$Y(s) = H(s)X(s) \tag{16.17}$$

and take the inverse transform to get $y(t)$. A special case is when the input is the unit impulse function, $x(t) = \delta(t)$, so that $X(s) = 1$. For this case,

$$Y(s) = H(s) \quad \text{or} \quad y(t) = h(t) \tag{16.18}$$

where

$$h(t) = \mathcal{L}^{-1}[H(s)] \tag{16.19}$$

The unit impulse response is the output response of a circuit when the input is a unit impulse.

The term $h(t)$ represents the *unit impulse response*—it is the time-domain response of the network to a unit impulse. Thus, Eq. (16.19) provides a new interpretation for the transfer function: $H(s)$ is the Laplace transform of the unit impulse response of the network. Once we know the impulse response $h(t)$ of a network, we can obtain the response of the network to *any* input signal using Eq. (16.17) in the s-domain or using the convolution integral (section 15.5) in the time domain.

Example 16.7

The output of a linear system is $y(t) = 10e^{-t}\cos 4t\,u(t)$ when the input is $x(t) = e^{-t}u(t)$. Find the transfer function of the system and its impulse response.

Solution:

If $x(t) = e^{-t}u(t)$ and $y(t) = 10e^{-t}\cos 4t\,u(t)$, then

$$X(s) = \frac{1}{s+1} \quad \text{and} \quad Y(s) = \frac{10(s+1)}{(s+1)^2 + 4^2}$$

Hence,

$$H(s) = \frac{Y(s)}{X(s)} = \frac{10(s+1)^2}{(s+1)^2 + 16} = \frac{10(s^2 + 2s + 1)}{s^2 + 2s + 17}$$

To find $h(t)$, we write $H(s)$ as

$$H(s) = 10 - 40\frac{4}{(s+1)^2 + 4^2}$$

From Table 15.2, we obtain

$$h(t) = 10\delta(t) - 40e^{-t}\sin 4t\,u(t)$$

Practice Problem 16.7

The transfer function of a linear system is

$$H(s) = \frac{2s}{s+6}$$

Find the output $y(t)$ due to the input $5e^{-3t}u(t)$ and its impulse response.

Answer: $-10e^{-3t} + 20e^{-6t}$, $t \geq 0$, $2\delta(t) - 12e^{-6t}u(t)$.

Example 16.8

Determine the transfer function $H(s) = V_o(s)/I_o(s)$ of the circuit in Fig. 16.18.

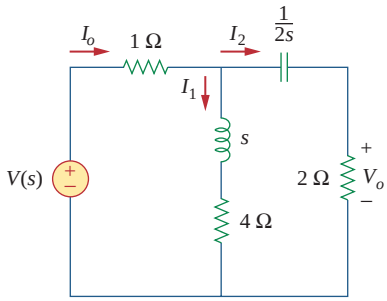


Figure 16.18
For Example 16.8.

Solution:

■ **METHOD 1** By current division,

$$I_2 = \frac{(s + 4)I_o}{s + 4 + 2 + 1/2s}$$

But

$$V_o = 2I_2 = \frac{2(s + 4)I_o}{s + 6 + 1/2s}$$

Hence,

$$H(s) = \frac{V_o(s)}{I_o(s)} = \frac{4s(s + 4)}{2s^2 + 12s + 1}$$

■ **METHOD 2** We can apply the ladder method. We let $V_o = 1$ V. By Ohm's law, $I_2 = V_o/2 = 1/2$ A. The voltage across the $(2 + 1/2s)$ impedance is

$$V_1 = I_2\left(2 + \frac{1}{2s}\right) = 1 + \frac{1}{4s} = \frac{4s + 1}{4s}$$

This is the same as the voltage across the $(s + 4)$ impedance. Hence,

$$I_1 = \frac{V_1}{s + 4} = \frac{4s + 1}{4s(s + 4)}$$

Applying KCL at the top node yields

$$I_o = I_1 + I_2 = \frac{4s + 1}{4s(s + 4)} + \frac{1}{2} = \frac{2s^2 + 12s + 1}{4s(s + 4)}$$

Hence,

$$H(s) = \frac{V_o}{I_o} = \frac{1}{I_o} = \frac{4s(s + 4)}{2s^2 + 12s + 1}$$

as before.

Practice Problem 16.8

Find the transfer function $H(s) = I_1(s)/I_o(s)$ in the circuit of Fig. 16.18.

Answer: $\dfrac{4s + 1}{2s^2 + 12s + 1}$.

Example 16.9

For the *s*-domain circuit in Fig. 16.19, find: (a) the transfer function $H(s) = V_o/V_i$, (b) the impulse response, (c) the response when $v_i(t) = u(t)$ V, (d) the response when $v_i(t) = 8\cos 2t$ V.

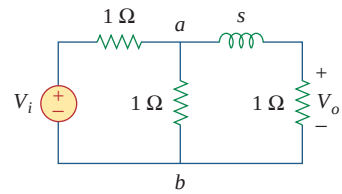


Figure 16.19
For Example 16.9.

Solution:

(a) Using voltage division,

$$V_o = \frac{1}{s+1}V_{ab} \tag{16.9.1}$$

But

$$V_{ab} = \frac{1 \| (s+1)}{1 + 1 \| (s+1)}V_i = \frac{(s+1)/(s+2)}{1 + (s+1)/(s+2)}V_i$$

or

$$V_{ab} = \frac{s+1}{2s+3}V_i \tag{16.9.2}$$

Substituting Eq. (16.9.2) into Eq. (16.9.1) results in

$$V_o = \frac{V_i}{2s+3}$$

Thus, the transfer function is

$$H(s) = \frac{V_o}{V_i} = \frac{1}{2s+3}$$

(b) We may write $H(s)$ as

$$H(s) = \frac{1}{2}\frac{1}{s+\frac{3}{2}}$$

Its inverse Laplace transform is the required impulse response:

$$h(t) = \frac{1}{2}e^{-3t/2}u(t)$$

(c) When $v_i(t) = u(t)$, $V_i(s) = 1/s$, and

$$V_o(s) = H(s)V_i(s) = \frac{1}{2s(s+\frac{3}{2})} = \frac{A}{s} + \frac{B}{s+\frac{3}{2}}$$

where

$$A = sV_o(s)|_{s=0} = \frac{1}{2(s+\frac{3}{2})}\bigg|_{s=0} = \frac{1}{3}$$

$$B = \left(s+\frac{3}{2}\right)V_o(s)\bigg|_{s=-3/2} = \frac{1}{2s}\bigg|_{s=-3/2} = -\frac{1}{3}$$

Hence, for $v_i(t) = u(t)$,

$$V_o(s) = \frac{1}{3}\left(\frac{1}{s} - \frac{1}{s+\frac{3}{2}}\right)$$

and its inverse Laplace transform is

$$v_o(t) = \frac{1}{3}(1 - e^{-3t/2})u(t)\ \text{V}$$

(d) When $v_i(t) = 8\cos 2t$, then $V_i(s) = \dfrac{8s}{s^2 + 4}$, and

$$V_o(s) = H(s)V_i(s) = \frac{4s}{(s + \frac{3}{2})(s^2 + 4)} = \frac{A}{s + \frac{3}{2}} + \frac{Bs + C}{s^2 + 4} \tag{16.9.3}$$

where

$$A = \left(s + \frac{3}{2}\right)V_o(s)\Big|_{s=-3/2} = \frac{4s}{s^2 + 4}\Big|_{s=-3/2} = -\frac{24}{25}$$

To get B and C, we multiply Eq. (16.9.3) by $(s + 3/2)(s^2 + 4)$. We get

$$4s = A(s^2 + 4) + B\left(s^2 + \frac{3}{2}s\right) + C\left(s + \frac{3}{2}\right)$$

Equating coefficients,

$$\text{Constant:} \quad 0 = 4A + \frac{3}{2}C \quad \Rightarrow \quad C = -\frac{8}{3}A$$

$$s: \quad 4 = \frac{3}{2}B + C$$

$$s^2: \quad 0 = A + B \quad \Rightarrow \quad B = -A$$

Solving these gives $A = -24/25$, $B = 24/25$, $C = 64/25$. Hence, for $v_i(t) = 8\cos 2t$ V,

$$V_o(s) = \frac{-\frac{24}{25}}{s + \frac{3}{2}} + \frac{24}{25}\frac{s}{s^2 + 4} + \frac{32}{25}\frac{2}{s^2 + 4}$$

and its inverse is

$$v_o(t) = \frac{24}{25}\left(-e^{-3t/2} + \cos 2t + \frac{4}{3}\sin 2t\right)u(t) \text{ V}$$

Practice Problem 16.9

Rework Example 16.9 for the circuit shown in Fig. 16.20.

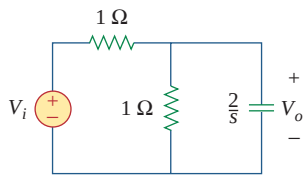


Figure 16.20
For Practice Prob. 16.9.

Answer: (a) $2/(s + 4)$, (b) $2e^{-4t}u(t)$, (c) $\frac{1}{2}(1 - e^{-4t})u(t)$ V, (d) $3.2(-e^{-4t} + \cos 2t + \frac{1}{2}\sin 2t)\,u(t)$ V.

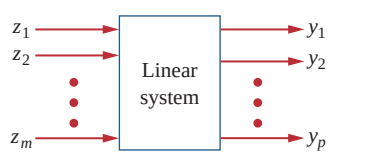


Figure 16.21
A linear system with m inputs and p outputs.

16.5 State Variables

Thus far in this book we have considered techniques for analyzing systems with only one input and only one output. Many engineering systems have many inputs and many outputs, as shown in Fig. 16.21. The state variable method is a very important tool in analyzing systems and understanding such highly complex systems. Thus, the state variable model is more general than the single-input, single-output model, such as a transfer function. Although the topic cannot be adequately covered in one chapter, let alone one section of a chapter, we will cover it briefly at this point.

In the state variable model, we specify a collection of variables that describe the internal behavior of the system. These variables are known as the *state variables* of the system. They are the variables that determine the future behavior of a system when the present state of the system and the input signals are known. In other words, they are those variables which, if known, allow all other system parameters to be determined by using only algebraic equations.

A **state variable** is a physical property that characterizes the state of a system, regardless of how the system got to that state.

Common examples of state variables are the pressure, volume, and temperature. In an electric circuit, the state variables are the inductor current and capacitor voltage since they collectively describe the energy state of the system.

The standard way to represent the state equations is to arrange them as a set of first-order differential equations:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{z} \tag{16.20}$$

where

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} = \text{state vector representing } n \text{ state vectors}$$

and the dot represents the first derivative with respect to time, i.e.,

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \vdots \\ \dot{x}_n(t) \end{bmatrix}$$

and

$$\mathbf{z}(t) = \begin{bmatrix} z_1(t) \\ z_2(t) \\ \vdots \\ z_m(t) \end{bmatrix} = \text{input vector representing } m \text{ inputs}$$

A and **B** are respectively $n \times n$ and $n \times m$ matrices. In addition to the state equation in Eq. (16.20), we need the output equation. The complete state model or state space is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{z} \tag{16.21a}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{z} \tag{16.21b}$$

where

$$\mathbf{y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_p(t) \end{bmatrix} = \text{the output vector representing } p \text{ outputs}$$

C and **D** are, respectively, $p \times n$ and $p \times m$ matrices. For the special case of single-input single-output, $n = m = p = 1$.

Assuming zero initial conditions, the transfer function of the system is found by taking the Laplace transform of Eq. (16.21a); we obtain

$$s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{Z}(s) \quad \rightarrow \quad (s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{B}\mathbf{Z}(s)$$

or

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{Z}(s) \tag{16.22}$$

where **I** is the identity matrix. Taking the Laplace transform of Eq. (16.21b) yields

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{Z}(s) \tag{16.23}$$

Substituting Eq. (16.22) into Eq. (16.23) and dividing by $\mathbf{Z}(s)$ gives the transfer function as

$$\mathbf{H}(s) = \frac{\mathbf{Y}(s)}{\mathbf{Z}(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D} \tag{16.24}$$

where

$\mathbf{A}$ = system matrix
$\mathbf{B}$ = input coupling matrix
$\mathbf{C}$ = output matrix
$\mathbf{D}$ = feedforward matrix

In most cases, $\mathbf{D} = \mathbf{0}$, so the degree of the numerator of $H(s)$ in Eq. (16.24) is less than that of the denominator. Thus,

$$\boxed{\mathbf{H}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}} \tag{16.25}$$

Because of the matrix computation involved, *MATLAB* can be used to find the transfer function.

To apply state variable analysis to a circuit, we follow the following three steps.

Steps to Apply the State Variable Method to Circuit Analysis:

1. Select the inductor current i and capacitor voltage v as the state variables, making sure they are consistent with the passive sign convention.
2. Apply KCL and KVL to the circuit and obtain circuit variables (voltages and currents) in terms of the state variables. This should lead to a set of first-order differential equations necessary and sufficient to determine all state variables.
3. Obtain the output equation and put the final result in state-space representation.

Steps 1 and 3 are usually straightforward; the major task is in step 2. We will illustrate this with examples.

Example 16.10

Find the state-space representation of the circuit in Fig. 16.22. Determine the transfer function of the circuit when v_s is the input and i_x is the output. Take $R = 1\Omega$, $C = 0.25$ F, and $L = 0.5$ H.

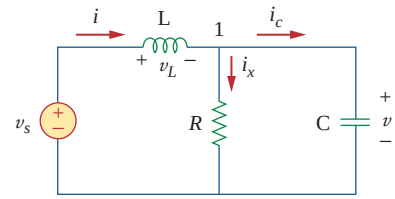


Figure 16.22
For Example 16.10.

Solution:
We select the inductor current i and capacitor voltage v as the state variables.

$$v_L = L\frac{di}{dt} \tag{16.10.1}$$

$$i_C = C\frac{dv}{dt} \tag{16.10.2}$$

Applying KCL at node 1 gives

$$i = i_x + i_C \quad \rightarrow \quad C\frac{dv}{dt} = i - \frac{v}{R}$$

or

$$\dot{v} = -\frac{v}{RC} + \frac{i}{C} \tag{16.10.3}$$

since the same voltage v is across both R and C. Applying KVL around the outer loop yields

$$v_s = v_L + v \quad \rightarrow \quad L\frac{di}{dt} = -v + v_s$$

$$\dot{i} = -\frac{v}{L} + \frac{v_s}{L} \tag{16.10.4}$$

Equations (16.10.3) and (16.10.4) constitute the state equations. If we regard i_x as the output,

$$i_x = \frac{v}{R} \tag{16.10.5}$$

Putting Eqs. (16.10.3), (16.10.4), and (16.10.5) in the standard form leads to

$$\begin{bmatrix} \dot{v} \\ \dot{i} \end{bmatrix} = \begin{bmatrix} \frac{-1}{RC} & \frac{1}{C} \\ \frac{-1}{L} & 0 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} v_s \tag{16.10.6a}$$

$$i_x = \begin{bmatrix} \frac{1}{R} & 0 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} \tag{16.10.6b}$$

If $R = 1$, $C = \frac{1}{4}$, and $L = \frac{1}{2}$, we obtain from Eq. (16.10.6) matrices

$$\mathbf{A} = \begin{bmatrix} \frac{-1}{RC} & \frac{1}{C} \\ \frac{-1}{L} & 0 \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -2 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix},$$

$$\mathbf{C} = \begin{bmatrix} \frac{1}{R} & 0 \end{bmatrix} = [1 \quad 0]$$

$$s\mathbf{I} - \mathbf{A} = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} -4 & 4 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} s+4 & -4 \\ 2 & s \end{bmatrix}$$

Taking the inverse of this gives

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\text{adjoint of } \mathbf{A}}{\text{determinant of } \mathbf{A}} = \frac{\begin{bmatrix} s & 4 \\ -2 & s+4 \end{bmatrix}}{s^2 + 4s + 8}$$

Thus, the transfer function is given by

$$\mathbf{H}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \frac{[1 \quad 0]\begin{bmatrix} s & 4 \\ -2 & s+4 \end{bmatrix}\begin{bmatrix} 0 \\ 2 \end{bmatrix}}{s^2 + 4s + 8} = \frac{[1 \quad 0]\begin{bmatrix} 8 \\ 2s+8 \end{bmatrix}}{s^2 + 4s + 8}$$

$$= \frac{8}{s^2 + 4s + 8}$$

which is the same thing we would get by directly Laplace transforming the circuit and obtaining $\mathbf{H}(s) = I_x(s)/V_s(s)$. The real advantage of the state variable approach comes with multiple inputs and multiple outputs. In this case, we have one input v_s and one output i_x. In the next example, we will have two inputs and two outputs.

Practice Problem 16.10

Obtain the state variable model for the circuit shown in Fig. 16.23. Let $R_1 = 1$, $R_2 = 2$, $C = 0.5$, and $L = 0.2$ and obtain the transfer function.

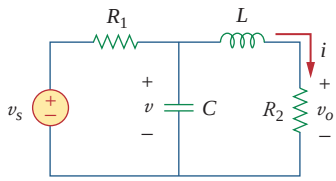


Figure 16.23
For Practice Prob. 16.10.

Answer:

$$\begin{bmatrix} \dot{v} \\ \dot{i} \end{bmatrix} = \begin{bmatrix} \frac{-1}{R_1 C} & \frac{-1}{C} \\ \frac{1}{L} & \frac{-R_2}{L} \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1 C} \\ 0 \end{bmatrix} v_s, \qquad v_o = [0 \quad R_2]\begin{bmatrix} v \\ i \end{bmatrix}$$

$$\mathbf{H}(s) = \frac{20}{s^2 + 12s + 30}$$

Example 16.11

Consider the circuit in Fig. 16.24, which may be regarded as a two-input, two-output system. Determine the state variable model and find the transfer function of the system.

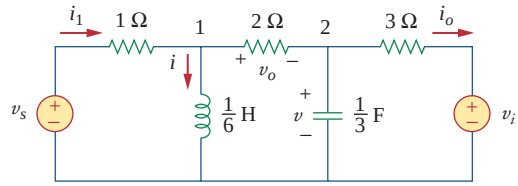


Figure 16.24
For Example 16.11.

Solution:

In this case, we have two inputs v_s and v_i and two outputs v_o and i_o. Again, we select the inductor current i and capacitor voltage v as the state variables. Applying KVL around the left-hand loop gives

$$-v_s + i_1 + \frac{1}{6}\dot{i} = 0 \quad \rightarrow \quad \dot{i} = 6v_s - 6i_1 \qquad \textbf{(16.11.1)}$$

We need to eliminate i_1. Applying KVL around the loop containing v_s, 1-Ω resistor, 2-Ω resistor, and $\frac{1}{3}$-F capacitor yields

$$v_s = i_1 + v_o + v \qquad \textbf{(16.11.2)}$$

But at node 1, KCL gives

$$i_1 = i + \frac{v_o}{2} \quad \rightarrow \quad v_o = 2(i_1 - i) \qquad \textbf{(16.11.3)}$$

Substituting this in Eq. (16.11.2),

$$v_s = 3i_1 + v - 2i \quad \rightarrow \quad i_1 = \frac{2i - v + v_s}{3} \tag{16.11.4}$$

Substituting this in Eq. (16.11.1) gives

$$\dot{i} = 2v - 4i + 4v_s \tag{16.11.5}$$

which is one state equation. To obtain the second one, we apply KCL at node 2.

$$\frac{v_o}{2} = \frac{1}{3}\dot{v} + i_o \quad \rightarrow \quad \dot{v} = \frac{3}{2}v_o - 3i_o \tag{16.11.6}$$

We need to eliminate v_o and i_o. From the right-hand loop, it is evident that

$$i_o = \frac{v - v_i}{3} \tag{16.11.7}$$

Substituting Eq. (16.11.4) into Eq. (16.11.3) gives

$$v_o = 2\left(\frac{2i - v + v_s}{3} - i\right) = -\frac{2}{3}(v + i - v_s) \tag{16.11.8}$$

Substituting Eqs. (16.11.7) and (16.11.8) into Eq. (16.11.6) yields the second state equation as

$$\dot{v} = -2v - i + v_s + v_i \tag{16.11.9}$$

The two output equations are already obtained in Eqs. (16.11.7) and (16.11.8). Putting Eqs. (16.11.5) and (16.11.7) to (16.11.9) together in the standard form leads to the state model for the circuit, namely,

$$\begin{bmatrix} \dot{v} \\ \dot{i} \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} v_s \\ v_i \end{bmatrix} \tag{16.11.10a}$$

$$\begin{bmatrix} v_o \\ i_o \end{bmatrix} = \begin{bmatrix} -\frac{2}{3} & -\frac{2}{3} \\ \frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} \frac{2}{3} & 0 \\ 0 & -\frac{1}{3} \end{bmatrix} \begin{bmatrix} v_s \\ v_i \end{bmatrix} \tag{16.11.10b}$$

Practice Problem 16.11

For the electric circuit in Fig. 16.25, determine the state model. Take v_o and i_o as the output variables.

Answer:

$$\begin{bmatrix} \dot{v} \\ \dot{i} \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 4 & -8 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & -8 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$\begin{bmatrix} v_o \\ i_o \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

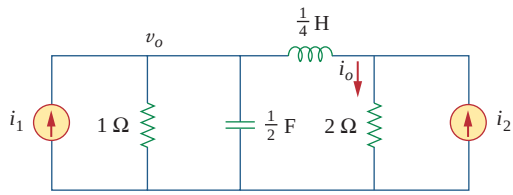


Figure 16.25
For Practice Prob. 16.11.

Example 16.12

Assume we have a system where the output is $y(t)$ and the input is $z(t)$. Let the following differential equation describe the relationship between the input and the output.

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 5z(t) \qquad \textbf{(16.12.1)}$$

Obtain the state model and the transfer function of the system.

Solution:

First, we select the state variables. Let $x_1 = y(t)$, therefore

$$\dot{x}_1 = \dot{y}(t) \qquad \textbf{(16.12.2)}$$

Now let

$$x_2 = \dot{x}_1 = \dot{y}(t) \qquad \textbf{(16.12.3)}$$

Note that at this time we are looking at a second-order system that would normally have two first-order terms in the solution.

Now we have $\dot{x}_2 = \ddot{y}(t)$, where we can find the value $\dot{x}_2$ from Eq. (16.12.1), i.e.,

$$\dot{x}_2 = \ddot{y}(t) = -2y(t) - 3\dot{y}(t) + 5z(t) = -2x_1 - 3x_2 + 5z(t) \qquad \textbf{(16.12.4)}$$

From Eqs. (16.12.2) to (16.12.4), we can now write the following matrix equations:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 5\end{bmatrix}z(t) \qquad \textbf{(16.12.5)}$$

$$\mathbf{y}(t) = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} \qquad \textbf{(16.12.6)}$$

We now obtain the transfer function.

$$s\mathbf{I} - \mathbf{A} = s\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} - \begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix} = \begin{bmatrix}s & -1\\ 2 & s+3\end{bmatrix}$$

The inverse is

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\begin{bmatrix}s+3 & 1\\ -2 & s\end{bmatrix}}{s(s+3)+2}$$

The transfer function is

$$\mathbf{H}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \frac{\begin{pmatrix}1 & 0\end{pmatrix}\begin{bmatrix}s+3 & 1\\ -2 & s\end{bmatrix}\begin{pmatrix}0\\ 5\end{pmatrix}}{s(s+3)+2} = \frac{\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}5\\ 5s\end{pmatrix}}{s(s+3)+2}$$

$$= \frac{5}{(s+1)(s+2)}$$

To check this, we directly apply the Laplace transfer to each term in Eq. (16.12.1). Since initial conditions are zero, we get

$$[s^2 + 3s + 2]Y(s) = 5Z(s) \quad \rightarrow \quad H(s) = \frac{Y(s)}{Z(s)} = \frac{5}{s^2 + 3s + 2}$$

which is in agreement with what we got previously.

Practice Problem 16.12

Develop a set of state variable equations that represent the following differential equation.

$$\frac{d^3y}{dt^3} + 18\frac{d^2y}{dt^2} + 20\frac{dy}{dt} + 5y = z(t)$$

Answer:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & -20 & -18 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = [1 \quad 0 \quad 0].$$

16.6 †Applications

So far we have considered three applications of Laplace's transform: circuit analysis in general, obtaining transfer functions, and solving linear integrodifferential equations. The Laplace transform also finds application in other areas in circuit analysis, signal processing, and control systems. Here we will consider two more important applications: network stability and network synthesis.

16.6.1 Network Stability

A circuit is *stable* if its impulse response $h(t)$ is bounded (i.e., $h(t)$ converges to a finite value) as $t \to \infty$; it is *unstable* if $h(t)$ grows without bound as $t \to \infty$. In mathematical terms, a circuit is stable when

$$\lim_{t\to\infty} |h(t)| = \text{finite} \tag{16.26}$$

Since the transfer function $H(s)$ is the Laplace transform of the impulse response $h(t)$, $H(s)$ must meet certain requirements in order for Eq. (16.26) to hold. Recall that $H(s)$ may be written as

$$H(s) = \frac{N(s)}{D(s)} \tag{16.27}$$

where the roots of $N(s) = 0$ are called the *zeros* of $H(s)$ because they make $H(s) = 0$, while the roots of $D(s) = 0$ are called the *poles* of $H(s)$ since they cause $H(s) \to \infty$. The zeros and poles of $H(s)$ are often located in the s plane as shown in Fig. 16.26(a). Recall from Eqs. (15.47) and (15.48) that $H(s)$ may also be written in terms of its poles as

$$H(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s + p_1)(s + p_2)\cdots(s + p_n)} \tag{16.28}$$

$H(s)$ must meet two requirements for the circuit to be stable. First, the degree of $N(s)$ must be less than the degree of $D(s)$; otherwise, long division would produce

$$H(s) = k_n s^n + k_{n-1}s^{n-1} + \cdots + k_1 s + k_0 + \frac{R(s)}{D(s)} \tag{16.29}$$

where the degree of $R(s)$, the remainder of the long division, is less than the degree of $D(s)$. The inverse of $H(s)$ in Eq. (16.29) does not meet the condition in Eq. (16.26). Second, all the poles of $H(s)$ in

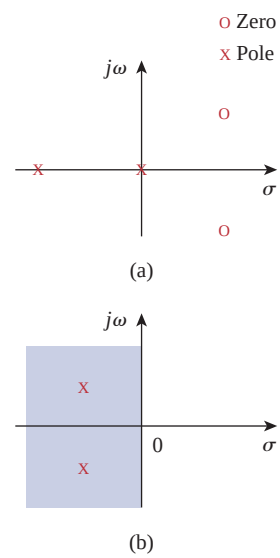


Figure 16.26
The complex s plane: (a) poles and zeros plotted, (b) left-half plane.

Eq. (16.27) (i.e., all the roots of $D(s) = 0$) must have negative real parts; in other words, all the poles must lie in the left half of the s plane, as shown typically in Fig. 16.26(b). The reason for this will be apparent if we take the inverse Laplace transform of $H(s)$ in Eq. (16.27). Since Eq. (16.27) is similar to Eq. (15.48), its partial fraction expansion is similar to the one in Eq. (15.49) so that the inverse of $H(s)$ is similar to that in Eq. (15.53). Hence,

$$h(t) = (k_1 e^{-p_1 t} + k_2 e^{-p_2 t} + \cdots + k_n e^{-p_n t})u(t) \qquad \textbf{(16.30)}$$

We see from this equation that each pole p_i must be positive (i.e., pole $s = -p_i$ in the left-half plane) in order for $e^{-p_i t}$ to decrease with increasing t. Thus,

A circuit is **stable** when all the poles of its transfer function $H(s)$ lie in the left half of the s plane.

An unstable circuit never reaches steady state because the transient response does not decay to zero. Consequently, steady-state analysis is only applicable to stable circuits.

A circuit made up exclusively of passive elements (R, L, and C) and independent sources cannot be unstable, because that would imply that some branch currents or voltages would grow indefinitely with sources set to zero. Passive elements cannot generate such indefinite growth. Passive circuits either are stable or have poles with zero real parts. To show that this is the case, consider the series RLC circuit in Fig. 16.27. The transfer function is given by

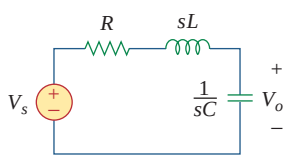


Figure 16.27
A typical RLC circuit.

$$H(s) = \frac{V_o}{V_s} = \frac{1/sC}{R + sL + 1/sC}$$

or

$$H(s) = \frac{1/L}{s^2 + sR/L + 1/LC} \qquad \textbf{(16.31)}$$

Notice that $D(s) = s^2 + sR/L + 1/LC = 0$ is the same as the characteristic equation obtained for the series RLC circuit in Eq. (8.8). The circuit has poles at

$$p_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \qquad \textbf{(16.32)}$$

where

$$\alpha = \frac{R}{2L}, \qquad \omega_0 = \frac{1}{LC}$$

For $R, L, C > 0$, the two poles always lie in the left half of the s plane, implying that the circuit is always stable. However, when $R = 0$, $\alpha = 0$ and the circuit becomes unstable. Although ideally this is possible, it does not really happen, because R is never zero.

On the other hand, active circuits or passive circuits with controlled sources can supply energy, and they can be unstable. In fact, an oscillator is a typical example of a circuit designed to be unstable. An oscillator is designed such that its transfer function is of the form

$$H(s) = \frac{N(s)}{s^2 + \omega_0^2} = \frac{N(s)}{(s + j\omega_0)(s - j\omega_0)} \qquad \textbf{(16.33)}$$

so that its output is sinusoidal.

Example 16.13

Determine the values of k for which the circuit in Fig. 16.28 is stable.

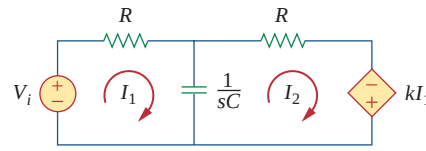


Figure 16.28
For Example 16.13.

Solution:
Applying mesh analysis to the first-order circuit in Fig. 16.28 gives

$$V_i = \left(R + \frac{1}{sC}\right)I_1 - \frac{I_2}{sC} \tag{16.13.1}$$

and

$$0 = -kI_1 + \left(R + \frac{1}{sC}\right)I_2 - \frac{I_1}{sC}$$

or

$$0 = -\left(k + \frac{1}{sC}\right)I_1 + \left(R + \frac{1}{sC}\right)I_2 \tag{16.13.2}$$

We can write Eqs. (16.13.1) and (16.13.2) in matrix form as

$$\begin{bmatrix} V_i \\ 0 \end{bmatrix} = \begin{bmatrix} \left(R + \frac{1}{sC}\right) & -\frac{1}{sC} \\ -\left(k + \frac{1}{sC}\right) & \left(R + \frac{1}{sC}\right) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

The determinant is

$$\Delta = \left(R + \frac{1}{sC}\right)^2 - \frac{k}{sC} - \frac{1}{s^2C^2} = \frac{sR^2C + 2R - k}{sC} \tag{16.13.3}$$

The characteristic equation ($\Delta = 0$) gives the single pole as

$$p = \frac{k - 2R}{R^2C}$$

which is negative when $k < 2R$. Thus, we conclude the circuit is stable when $k < 2R$ and unstable for $k > 2R$.

Practice Problem 16.13

For what value of β is the circuit in Fig. 16.29 stable?

Answer: $\beta > -1/R$.

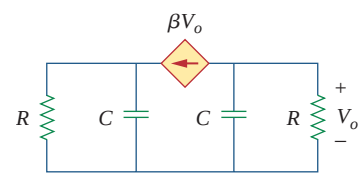


Figure 16.29
For Practice Prob. 16.13.

Example 16.14

An active filter has the transfer function

$$H(s) = \frac{k}{s^2 + s(4 - k) + 1}$$

For what values of k is the filter stable?

Solution:
As a second-order circuit, $H(s)$ may be written as

$$H(s) = \frac{N(s)}{s^2 + bs + c}$$

where $b = 4 - k$, $c = 1$, and $N(s) = k$. This has poles at $p^2 + bp + c = 0$; that is,

$$p_{1,2} = \frac{-b \pm \sqrt{b^2 - 4c}}{2}$$

For the circuit to be stable, the poles must be located in the left half of the s plane. This implies that $b > 0$.

Applying this to the given $H(s)$ means that for the circuit to be stable, $4 - k > 0$ or $k < 4$.

Practice Problem 16.14 A second-order active circuit has the transfer function

$$H(s) = \frac{1}{s^2 + s(25 + \alpha) + 25}$$

Find the range of the values of α for which the circuit is stable. What is the value of α that will cause oscillation?

Answer: $\alpha > -25$, $\alpha = -25$.

16.6.2 Network Synthesis

Network synthesis may be regarded as the process of obtaining an appropriate network to represent a given transfer function. Network synthesis is easier in the s-domain than in the time domain.

In network analysis, we find the transfer function of a given network. In network synthesis, we reverse the approach: given a transfer function, we are required to find a suitable network.

Network synthesis is finding a network that represents a given transfer function.

Keep in mind that in synthesis, there may be many different answers—or possibly no answers—because there are many circuits that can be used to represent the same transfer function; in network analysis, there is only one answer.

Network synthesis is an exciting field of prime engineering importance. Being able to look at a transfer function and come up with the type of circuit it represents is a great asset to a circuit designer. Although network synthesis constitutes a whole course by itself and requires some experience, the following examples are meant to whet your appetite.

Example 16.15

Given the transfer function

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{10}{s^2 + 3s + 10}$$

realize the function using the circuit in Fig. 16.30(a). (a) Select $R = 5\ \Omega$, and find L and C. (b) Select $R = 1\ \Omega$, and find L and C.

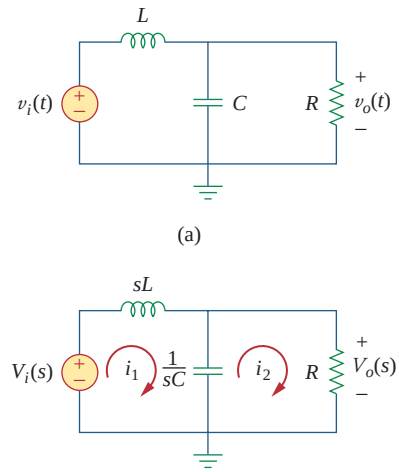


Figure 16.30
For Example 16.15.

Solution:

1. **Define.** The problem is clearly and completely defined. This problem is what we call a synthesis problem: given a transfer function, synthesize a circuit that produces the given transfer function. However, to keep the problem more manageable, we give a circuit that produces the desired transfer function.

   Had one of the variables, R in this case, not been given a value, then the problem would have had an infinite number of answers. An open-ended problem of this kind would require some additional assumptions that would have narrowed the set of solutions.
2. **Present.** A transfer function of the voltage out versus the voltage in is equal to $10/(s^2 + 3s + 10)$. A circuit, Fig. 16.30, is also given that should be able to produce the required transfer function. Two different values of R, 5 Ω and 1 Ω, are to be used to calculate the values of L and C that produce the given transfer function.
3. **Alternative.** All solution paths involve determining the transfer function of Fig. 16.30 and then matching the various terms of the transfer function. Two approaches would be to use mesh analysis or nodal analysis. Since we are looking for a ratio of voltages, nodal analysis makes the most sense.
4. **Attempt.** Using nodal analysis leads to

$$\frac{V_o(s) - V_i(s)}{sL} + \frac{V_o(s) - 0}{1/(sC)} + \frac{V_o(s) - 0}{R} = 0$$

Now multiply through by sLR:

$$RV_o(s) - RV_i(s) + s^2RLCV_o(s) + sLV_o(s) = 0$$

Collecting terms we get

$$(s^2RLC + sL + R)V_o(s) = RV_i(s)$$

or

$$\frac{V_o(s)}{V_i(s)} = \frac{1/(LC)}{s^2 + [1/(RC)]s + 1/(LC)}$$

Matching the two transfer functions produces two equations with three unknowns.

$$LC = 0.1 \qquad \text{or} \qquad L = \frac{0.1}{C}$$

and

$$RC = \frac{1}{3} \qquad \text{or} \qquad C = \frac{1}{3R}$$

We have a constraint equation, $R = 5\ \Omega$ for (a) and $= 1\ \Omega$ for (b).

(a) $C = 1/(3 \times 5) =$ **66.67 mF** and $L =$ **1.5 H**

(b) $C = 1/(3 \times 1) =$ **333.3 mF** and $L =$ **300 mH**

5. **Evaluate.** There are different ways of checking the answer. Solving for the transfer function by using mesh analysis seems the most straightforward and the approach we could use here. However, it should be pointed out that this is mathematically more complex and will take longer than the original nodal analysis approach. Other approaches also exist. We can assume an input for $v_i(t)$, $v_i(t) = u(t)$ V and, using either nodal analysis or mesh analysis, see if we get the same answer we would get with just using the transfer function. That is the approach we will try using mesh analysis.

Let $v_i(t) = u(t)\,V$ or $V_i(s) = 1/s$. This will produce

$$V_o(s) = 10/(s^3 + 3s^2 + 10s)$$

Based on Fig. 16.30, mesh analysis leads to

(a) For loop 1,

$$-(1/s) + 1.5sI_1 + [1/(0.06667s)](I_1 - I_2) = 0$$

or

$$(1.5s^2 + 15)I_1 - 15I_2 = 1$$

For loop 2,

$$(15/s)(I_2 - I_1) + 5I_2 = 0$$

or

$$-15I_1 + (5s + 15)I_2 = 0 \qquad \text{or} \qquad I_1 = (0.3333s + 1)I_2$$

Substituting into the first equation we get

$$(0.5s^3 + 1.5s^2 + 5s + 15)I_2 - 15I_2 = 1$$

or

$$I_2 = 2/(s^3 + 3s^2 + 10s)$$

but

$$V_o(s) = 5I_2 = 10/(s^3 + 3s^2 + 10s)$$

and the answer checks.

(b) For loop 1,

$$-(1/s) + 0.3sI_1 + [1/(0.3333s)](I_1 - I_2) = 0$$

or

$$(0.3s^2 + 3)I_1 - 3I_2 = 1$$

For loop 2,

$$(3/s)(I_2 - I_1) + I_2 = 0$$

or

$$-3I_1 + (s + 3)I_2 = 0 \qquad \text{or} \qquad I_1 = (0.3333s + 1)I_2$$

Substituting into the first equation we get

$$(0.09999s^3 + 0.3s^2 + s + 3)I_2 - 3I_2 = 1$$

or

$$I_2 = 10/(s^3 + 3s^2 + 10s)$$

but $V_o(s) = 1 \times I_2 = 10/(s^3 + 3s^2 + 10s)$
and the answer checks.

6. **Satisfactory?** We have clearly identified values of L and C for each of the conditions. In addition, we have carefully checked the answers to see if they are correct. The problem has been adequately solved. The results can now be presented as a solution to the problem.

Practice Problem 16.15

Realize the function

$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{4s}{s^2 + 4s + 20}$$

using the circuit in Fig. 16.31. Select $R = 2\ \Omega$, and determine L and C.

Answer: 0.5 H, 0.1 F.

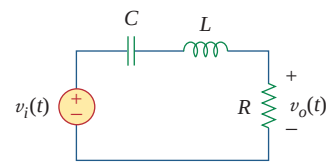


Figure 16.31
For Practice Prob. 16.15.

Example 16.16

Synthesize the function

$$T(s) = \frac{V_o(s)}{V_s(s)} = \frac{10^6}{s^2 + 100s + 10^6}$$

using the topology in Fig. 16.32.

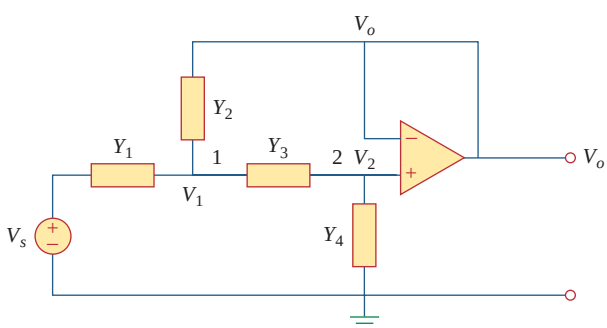


Figure 16.32
For Example 16.16.

Solution:
We apply nodal analysis to nodes 1 and 2. At node 1,

$$(V_s - V_1)Y_1 = (V_1 - V_o)Y_2 + (V_1 - V_2)Y_3 \qquad \textbf{(16.16.1)}$$

At node 2,

$$(V_1 - V_2)Y_3 = (V_2 - 0)Y_4 \qquad \textbf{(16.16.2)}$$

But $V_2 = V_o$, so Eq. (16.16.1) becomes

$$Y_1V_s = (Y_1 + Y_2 + Y_3)V_1 - (Y_2 + Y_3)V_o \qquad \textbf{(16.16.3)}$$

and Eq. (16.16.2) becomes

$$V_1Y_3 = (Y_3 + Y_4)V_o$$

or

$$V_1 = \frac{1}{Y_3}(Y_3 + Y_4)V_o \qquad \textbf{(16.16.4)}$$

Substituting Eq. (16.16.4) into Eq. (16.16.3) gives

$$Y_1V_s = (Y_1 + Y_2 + Y_3)\frac{1}{Y_3}(Y_3 + Y_4)V_o - (Y_2 + Y_3)V_o$$

or

$$Y_1Y_3V_s = [Y_1Y_3 + Y_4(Y_1 + Y_2 + Y_3)]V_o$$

Thus,

$$\frac{V_o}{V_s} = \frac{Y_1Y_3}{Y_1Y_3 + Y_4(Y_1 + Y_2 + Y_3)} \qquad \textbf{(16.16.5)}$$

To synthesize the given transfer function $T(s)$, compare it with the one in Eq. (16.16.5). Notice two things: (1) Y_1Y_3 must not involve s because the numerator of $T(s)$ is constant; (2) the given transfer function is second-order, which implies that we must have two capacitors. Therefore, we must make Y_1 and Y_3 resistive, while Y_2 and Y_4 are capacitive. So we select

$$Y_1 = \frac{1}{R_1}, \qquad Y_2 = sC_1, \qquad Y_3 = \frac{1}{R_2}, \qquad Y_4 = sC_2 \qquad \textbf{(16.16.6)}$$

Substituting Eq. (16.16.6) into Eq. (16.16.5) gives

$$\frac{V_o}{V_s} = \frac{1/(R_1R_2)}{1/(R_1R_2) + sC_2(1/R_1 + 1/R_2 + sC_1)}$$
$$= \frac{1/(R_1R_2C_1C_2)}{s^2 + s(R_1 + R_2)/(R_1R_2C_1) + 1/(R_1R_2C_1C_2)}$$

Comparing this with the given transfer function $T(s)$, we notice that

$$\frac{1}{R_1R_2C_1C_2} = 10^6, \qquad \frac{R_1 + R_2}{R_1R_2\,C_1} = 100$$

If we select $R_1 = R_2 = 10\ \text{k}\Omega$, then

$$C_1 = \frac{R_1 + R_2}{100R_1R_2} = \frac{20 \times 10^3}{100 \times 100 \times 10^6} = 2\ \mu\text{F}$$
$$C_2 = \frac{10^{-6}}{R_1R_2C_1} = \frac{10^{-6}}{100 \times 10^6 \times 2 \times 10^{-6}} = 5\ \text{nF}$$

Thus, the given transfer function is realized using the circuit shown in Fig. 16.33.

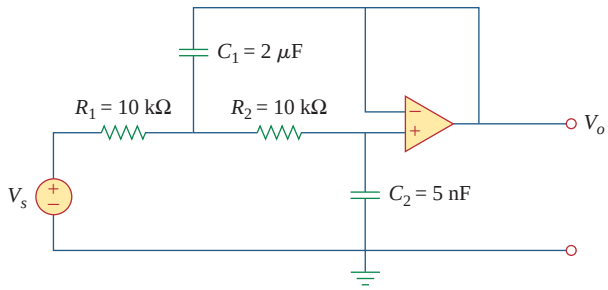


Figure 16.33
For Example 16.16.

Practice Problem 16.16

Synthesize the function

$$\frac{V_o(s)}{V_{\text{in}}} = \frac{-2s}{s^2 + 6s + 10}$$

using the op amp circuit shown in Fig. 16.34. Select

$$Y_1 = \frac{1}{R_1}, \qquad Y_2 = sC_1, \qquad Y_3 = sC_2, \qquad Y_4 = \frac{1}{R_2}$$

Let $R_1 = 1$ kΩ, and determine C_1, C_2, and R_2.

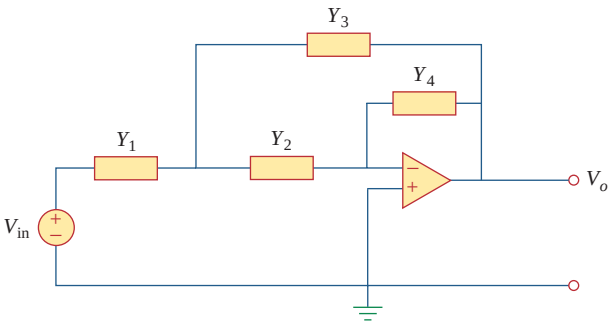


Figure 16.34
For Practice Prob. 16.16.

Answer: 0.1 mF, 0.5 mF, 2 kΩ.

16.7 Summary

1. The Laplace transform can be used to analyze a circuit. We convert each element from the time domain to the *s*-domain, solve the problem using any circuit analysis technique, and convert the result to the time domain using the inverse transform.
2. In the *s*-domain, the circuit elements are replaced with the initial condition at $t = 0$ as follows. (Please note, voltage models are

given below, but the corresponding current models work equally well.):

$$\text{Resistor:} \quad v_R = Ri \quad \rightarrow \quad V_R = RI$$

$$\text{Inductor:} \quad v_L = L\frac{di}{dt} \quad \rightarrow \quad V_L = sLI - Li(0^-)$$

$$\text{Capactior:} \quad v_C = \int i\,dt \quad \rightarrow \quad V_C = \frac{1}{sC} - \frac{v(0^-)}{s}$$

3. Using the Laplace transform to analyze a circuit results in a complete (both transient and steady state) response because the initial conditions are incorporated in the transformation process.
4. The transfer function $H(s)$ of a network is the Laplace transform of the impulse response $h(t)$.
5. In the s-domain, the transfer function $H(s)$ relates the output response $Y(s)$ and an input excitation $X(s)$; that is, $H(s) = Y(s)/X(s)$.
6. The state variable model is a useful tool for analyzing complex systems with several inputs and outputs. State variable analysis is a powerful technique that is most popularly used in circuit theory and control. The state of a system is the smallest set of quantities (known as state variables) that we must know in order to determine its future response to any given input. The state equation in state variable form is

$$\dot{\mathbf{x}} = \mathbf{A}x + \mathbf{B}z$$

while the output equation is

$$\mathbf{y} = \mathbf{C}x + \mathbf{D}z$$

7. For an electric circuit, we first select capacitor voltages and inductor current as state variables. We then apply KCL and KVL to obtain the state equations.
8. Two other areas of applications of the Laplace transform covered in this chapter are circuit stability and synthesis. A circuit is stable when all the poles of its transfer function lie in the left half of the s plane. Network synthesis is the process of obtaining an appropriate network to represent a given transfer function for which analysis in the s-domain is well suited.

Review Questions

16.1 The voltage through a resistor with current $i(t)$ in the s-domain is $sRI(s)$.

(a) True (b) False

16.2 The current through an RL series circuit with input voltage $v(t)$ is given in the s-domain as:

(a) $V(s)\left[R + \frac{1}{sL}\right]$ (b) $V(s)(R + sL)$

(c) $\frac{V(s)}{R + 1/sL}$ (d) $\frac{V(s)}{R + sL}$

16.3 The impedance of a 10-F capacitor is:

(a) $10/s$ (b) $s/10$ (c) $1/10s$ (d) $10s$

16.4 We can usually obtain the Thevenin equivalent in the time domain.

(a) True (b) False

16.5 A transfer function is defined only when all initial conditions are zero.

(a) True (b) False

16.6 If the input to a linear system is $\delta(t)$ and the output is $e^{-2t}u(t)$, the transfer function of the system is:

(a) $\frac{1}{s+2}$ (b) $\frac{1}{s-2}$ (c) $\frac{s}{s+2}$ (d) $\frac{s}{s-2}$

(e) None of the above

16.7 If the transfer function of a system is

$$H(s) = \frac{s^2 + s + 2}{s^3 + 4s^2 + 5s + 1}$$

it follows that the input is $X(s) = s^3 + 4s^2 + 5s + 1$, while the output is $Y(s) = s^2 + s + 2$.

(a) True (b) False

16.8 A network has its transfer function as

$$H(s) = \frac{s+1}{(s-2)(s+3)}$$

The network is stable.

(a) True (b) False

16.9 Which of the following equations is called the state equation?

(a) $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{z}$

(b) $\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{z}$

(c) $\mathbf{H}(s) = \mathbf{Y}(s)/\mathbf{Z}(s)$

(d) $\mathbf{H}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$

16.10 A single-input, single-output system is described by the state model as:

$$\dot{x}_1 = 2x_1 - x_2 + 3z$$
$$\dot{x}_2 = -4x_2 - z$$
$$y = 3x_1 - 2x_2 + z$$

Which of the following matrices is incorrect?

(a) $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 0 & -4 \end{bmatrix}$ (b) $\mathbf{B} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

(c) $\mathbf{C} = [3 \quad -2]$ (d) $\mathbf{D} = \mathbf{0}$

Answers: 16.1b, 16.2d, 16.3c, 16.4b, 16.5b, 16.6a, 16.7b, 16.8b, 16.9a, 16.10d.

Problems

Sections 16.2 and 16.3 Circuit Element Models and Circuit Analysis

16.1 Determine $i(t)$ in the circuit of Fig. 16.35 by means of the Laplace transform.

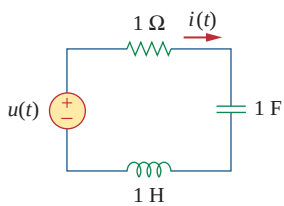


Figure 16.35
For Prob. 16.1.

16.2 Using Fig. 16.36, design a problem to help other students better understand circuit analysis using Laplace transforms.

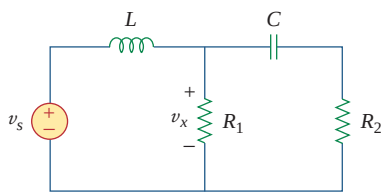


Figure 16.36
For Prob. 16.2.

16.3 Find $i(t)$ for $t > 0$ for the circuit in Fig. 16.37. Assume $i_s = 24u(t) + 12\delta(t)$ mA. (Hint: Can we use superposition to help solve this problem?)

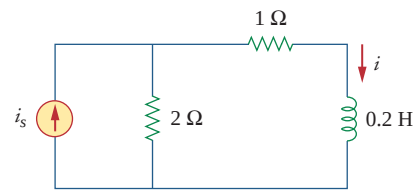


Figure 16.37
For Prob. 16.3.

16.4 The capacitor in the circuit of Fig. 16.38 is initially uncharged. Find $v_o(t)$ for $t > 0$.

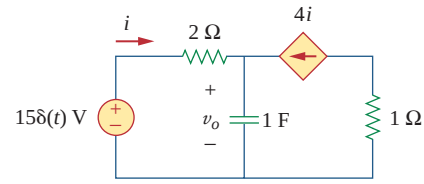


Figure 16.38
For Prob. 16.4.

16.5 If $i_s(t) = e^{-2t}u(t)$ A in the circuit shown in Fig. 16.39, find the value of $i_o(t)$.

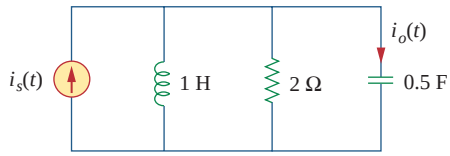


Figure 16.39
For Prob. 16.5.

16.6 Find $v(t)$, $t > 0$ in the circuit of Fig. 16.40. Let $v_s = 20$ V.

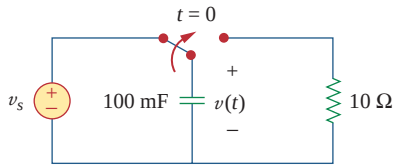


Figure 16.40
For Prob. 16.6.

16.7 Find $v_o(t)$, for all $t > 0$, in the circuit of Fig. 16.41.

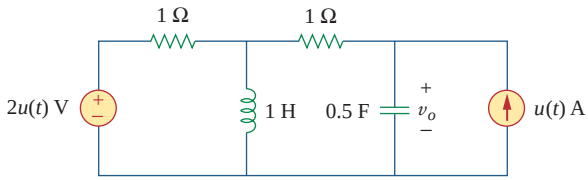


Figure 16.41
For Prob. 16.7.

16.8 If $v_o(0) = -1$ V, obtain $v_o(t)$ in the circuit of Fig. 16.42.

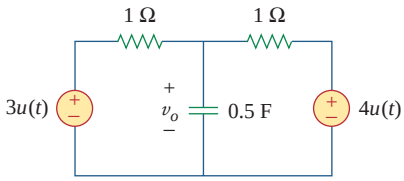


Figure 16.42
For Prob. 16.8.

16.9 Find the input impedance $Z_{in}(s)$ of each of the circuits in Fig. 16.43.

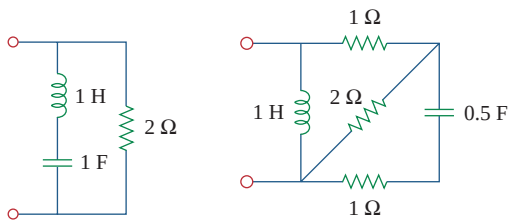


Figure 16.43
For Prob. 16.9.

16.10 Using Fig. 16.44, design a problem to help other students understand how to use Thevenin's theorem (in the *s*-domain) to aid in circuit analysis.

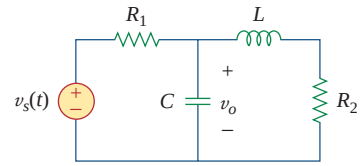


Figure 16.44
For Prob. 16.10.

16.11 Solve for the mesh currents in the circuit of Fig. 16.45. You may leave your results in the *s*-domain.

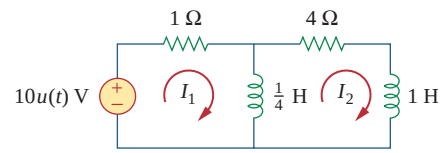


Figure 16.45
For Prob. 16.11.

16.12 Find $v_o(t)$ in the circuit of Fig. 16.46.

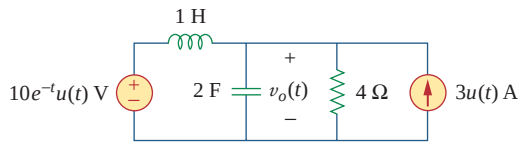


Figure 16.46
For Prob. 16.12.

16.13 Determine $i_o(t)$ in the circuit of Fig. 16.47.

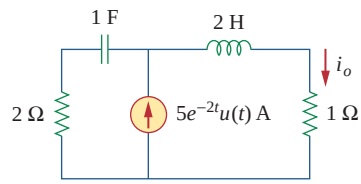


Figure 16.47
For Prob. 16.13.

***16.14** Determine $i_o(t)$ in the network shown in Fig. 16.48.

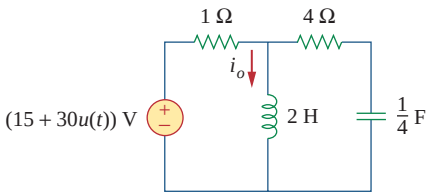


Figure 16.48
For Prob. 16.14.

16.15 Find $V_x(s)$ in the circuit shown in Fig. 16.49.

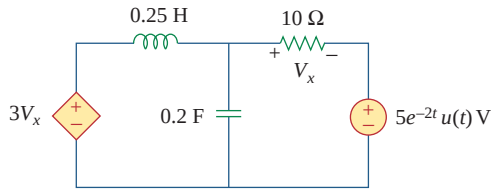


Figure 16.49
For Prob. 16.15.

***16.16** Find $i_o(t)$ for $t > 0$ in the circuit of Fig. 16.50.

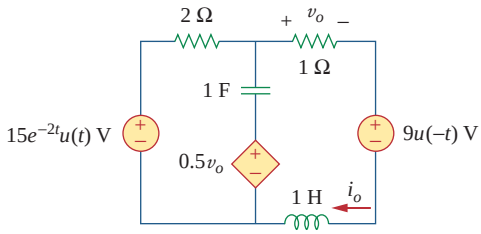


Figure 16.50
For Prob. 16.16.

16.17 Calculate $i_o(t)$ for $t > 0$ in the network of Fig. 16.51.

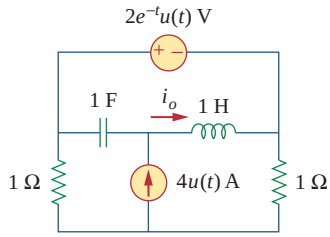


Figure 16.51
For Prob. 16.17.

16.18 (a) Find the Laplace transform of the voltage shown in Fig. 16.52(a). (b) Using that value of $v_s(t)$ in the circuit shown in Fig. 16.52(b), find the value of $v_o(t)$.

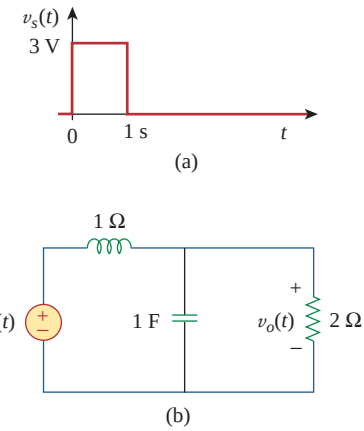


Figure 16.52
For Prob. 16.18.

16.19 Using Fig. 16.53, design a problem to help other students better understand circuit analysis in the *s*-domain with circuits that have dependent sources.

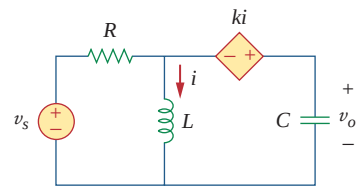


Figure 16.53
For Prob. 16.19.

16.20 Find $v_o(t)$ in the circuit of Fig. 16.54 if $v_x(0) = 2$ V and $i(0) = 1$ A.

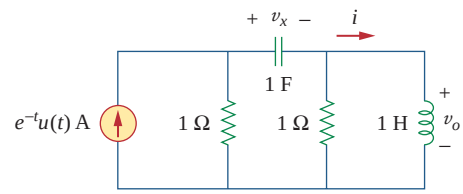


Figure 16.54
For Prob. 16.20.

16.21 Find the voltage $v_o(t)$ in the circuit of Fig. 16.55 by means of the Laplace transform.

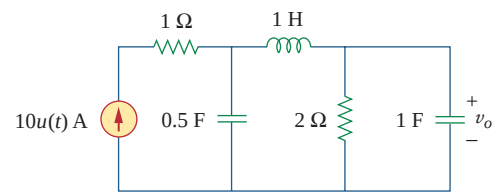


Figure 16.55
For Prob. 16.21.

* An asterisk indicates a challenging problem.

16.22 Using Fig. 16.56, design a problem to help other students better understand solving for node voltages by working in the *s*-domain.

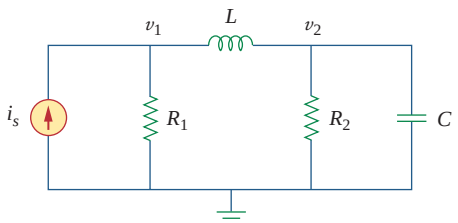


Figure 16.56
For Prob. 16.22.

16.23 Consider the parallel *RLC* circuit of Fig. 16.57. Find $v(t)$ and $i(t)$ given that $v(0) = 5$ and $i(0) = -2$ A.

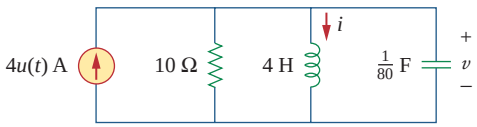


Figure 16.57
For Prob. 16.23.

16.24 The switch in Fig. 16.58 moves from position 1 to position 2 at $t = 0$. Find $v(t)$, for all $t > 0$.

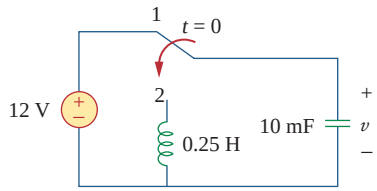


Figure 16.58
For Prob. 16.24.

16.25 For the *RLC* circuit shown in Fig. 16.59, find the complete response if $v(0) = 2$ V when the switch is closed.

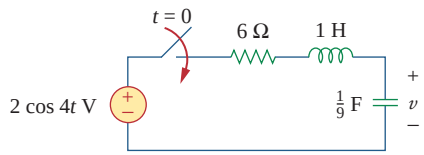


Figure 16.59
For Prob. 16.25.

16.26 For the op amp circuit in Fig. 16.60, find $v_o(t)$ for $t > 0$. Take $v_s = 10e^{-5t}u(t)$ V.

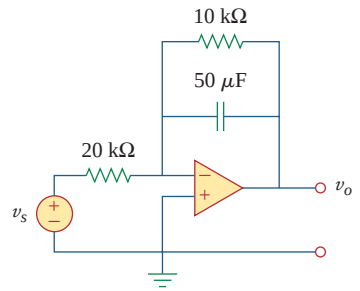


Figure 16.60
For Prob. 16.26.

16.27 Find $I_1(s)$ and $I_2(s)$ in the circuit of Fig. 16.61.

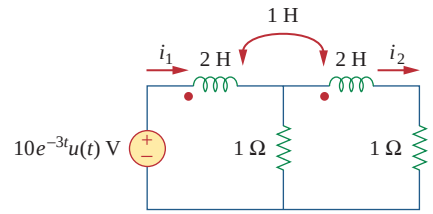


Figure 16.61
For Prob. 16.27.

16.28 Using Fig. 16.62, design a problem to help other students better understand how to do circuit analysis with circuits that have mutually coupled elements by working in the *s*-domain.

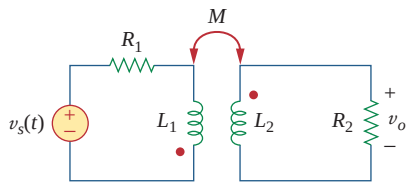


Figure 16.62
For Prob. 16.28.

16.29 For the ideal transformer circuit in Fig. 16.63, determine $i_o(t)$.

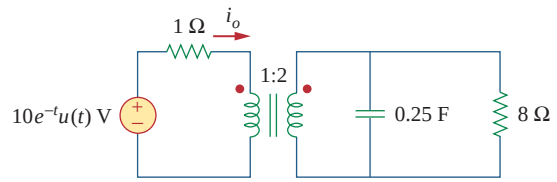


Figure 16.63
For Prob. 16.29.

Section 16.4 Transfer Functions

16.30 The transfer function of a system is

$$H(s) = \frac{s^2}{3s + 1}$$

Find the output when the system has an input of $4e^{-t/3}u(t)$.

16.31 When the input to a system is a unit step function, the response is $10 \cos 2t\,u(t)$. Obtain the transfer function of the system.

16.32 Design a problem to help other students better understand how to find outputs when given a transfer function and an input.

16.33 When a unit step is applied to a system at $t = 0$, its response is

$$y(t) = \left[4 + \frac{1}{2}e^{-3t} - e^{-2t}(2\cos 4t + 3\sin 4t)\right]u(t)$$

What is the transfer function of the system?

16.34 For the circuit in Fig. 16.64, find $H(s) = V_o(s)/V_s(s)$. Assume zero initial conditions.

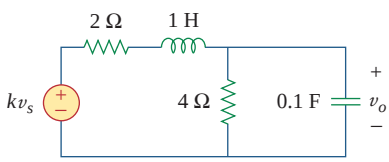


Figure 16.64
For Prob. 16.34.

16.35 Obtain the transfer function $H(s) = V_o/V_s$ for the circuit of Fig. 16.65.

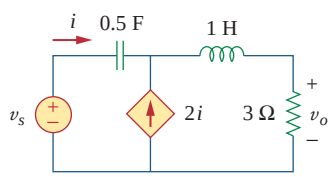


Figure 16.65
For Prob. 16.35.

16.36 The transfer function of a certain circuit is

$$H(s) = \frac{5}{s+1} - \frac{3}{s+2} + \frac{6}{s+4}$$

Find the impulse response of the circuit.

16.37 For the circuit in Fig. 16.66, find:

(a) I_1/V_s (b) I_2/V_x

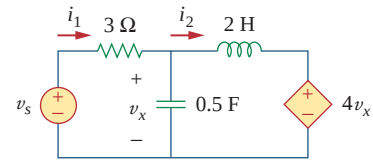


Figure 16.66
For Prob. 16.37.

16.38 Refer to the network in Fig. 16.67. Find the following transfer functions:

(a) $H_1(s) = V_o(s)/V_s(s)$
(b) $H_2(s) = V_o(s)/I_s(s)$
(c) $H_3(s) = I_o(s)/I_s(s)$
(d) $H_4(s) = I_o(s)/V_s(s)$

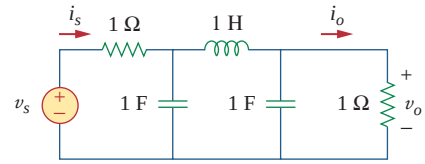


Figure 16.67
For Prob. 16.38.

16.39 Calculate the gain $H(s) = V_o/V_s$ in the op amp circuit of Fig. 16.68.

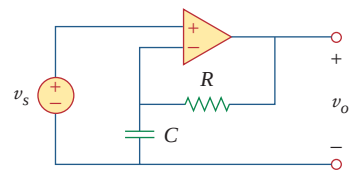


Figure 16.68
For Prob. 16.39.

16.40 Refer to the *RL* circuit in Fig. 16.69. Find:

(a) the impulse response $h(t)$ of the circuit.
(b) the unit step response of the circuit.

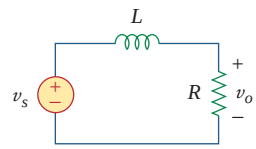


Figure 16.69
For Prob. 16.40.

16.41 A parallel *RL* circuit has $R = 4\ \Omega$ and $L = 1$ H. The input to the circuit is $i_s(t) = 2e^{-t}u(t)$ A. Find the inductor current $i_L(t)$ for all $t > 0$ and assume that $i_L(0) = -2$ A.

16.42 A circuit has a transfer function

$$H(s) = \frac{s + 4}{(s + 1)(s + 2)^2}$$

Find the impulse response.

Section 16.5 State Variables

16.43 Develop the state equations for Prob. 16.1.

16.44 Develop the state equations for the problem you designed in Prob. 16.2.

16.45 Develop the state equations for the circuit shown in Fig. 16.70.

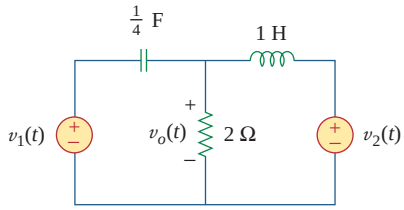


Figure 16.70
For Prob. 16.45.

16.46 Develop the state equations for the circuit shown in Fig. 16.71.

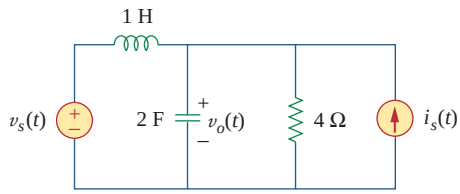


Figure 16.71
For Prob. 16.46.

16.47 Develop the state equations for the circuit shown in Fig. 16.72.

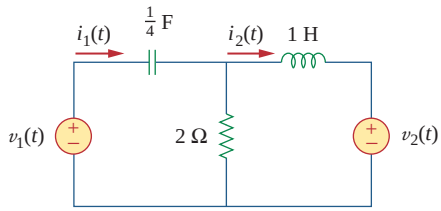


Figure 16.72
For Prob. 16.47.

16.48 Develop the state equations for the following differential equation.

$$\frac{d^2y(t)}{dt^2} + \frac{6\,dy(t)}{dt} + 7y(t) = z(t)$$

***16.49** Develop the state equations for the following differential equation.

$$\frac{d^2y(t)}{dt^2} + \frac{7\,dy(t)}{dt} + 9y(t) = \frac{dz(t)}{dt} + z(t)$$

***16.50** Develop the state equations for the following differential equation.

$$\frac{d^3y(t)}{dt^3} + \frac{6\,d^2y(t)}{dt^2} + \frac{11\,dy(t)}{dt} + 6y(t) = z(t)$$

***16.51** Given the following state equation, solve for $y(t)$:

$$\dot{\mathbf{x}} = \begin{bmatrix} -4 & 4 \\ -2 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u(t)$$

$$\mathbf{y}(t) = [1 \quad 0]x$$

***16.52** Given the following state equation, solve for $y_1(t)$ and $y_2(t)$.

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & -1 \\ 2 & -4 \end{bmatrix} x + \begin{bmatrix} 1 & 1 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} u(t) \\ 2u(t) \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} -2 & -2 \\ 1 & -0 \end{bmatrix} x + \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} u(t) \\ 2u(t) \end{bmatrix}$$

Section 16.6 Applications

16.53 Show that the parallel *RLC* circuit shown in Fig. 16.73 is stable.

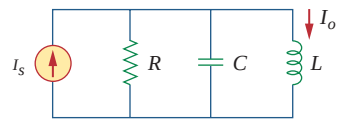


Figure 16.73
For Prob. 16.53.

16.54 A system is formed by cascading two systems as shown in Fig. 16.74. Given that the impulse response of the systems are

$$h_1(t) = 3e^{-t}u(t), \qquad h_2(t) = e^{-4t}u(t)$$

(a) Obtain the impulse response of the overall system.

(b) Check if the overall system is stable.



Figure 16.74
For Prob. 16.54.

16.55 Determine whether the op amp circuit in Fig. 16.75 is stable.

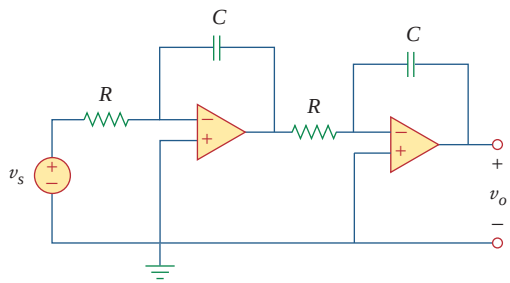


Figure 16.75
For Prob. 16.55.

16.56 It is desired to realize the transfer function

$$\frac{V_2(s)}{V_1(s)} = \frac{2s}{s^2 + 2s + 6}$$

using the circuit in Fig. 16.76. Choose $R = 1\ \text{k}\Omega$ and find L and C.

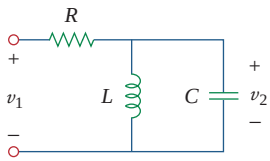


Figure 16.76
For Prob. 16.56.

16.57 Design an op amp circuit, using Fig. 16.77, that will realize the following transfer function:

$$\frac{V_o(s)}{V_i(s)} = -\frac{s + 1000}{2(s + 4000)}$$

Choose $C_1 = 10\ \mu\text{F}$; determine R_1, R_2, and C_2.

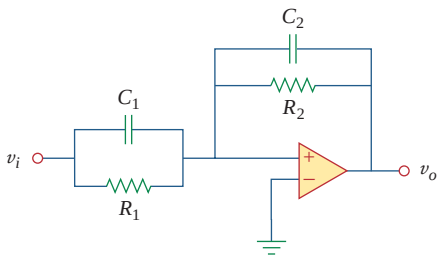


Figure 16.77
For Prob. 16.57.

16.58 Realize the transfer function

$$\frac{V_o(s)}{V_s(s)} = -\frac{s}{s + 10}$$

using the circuit in Fig. 16.78. Let $Y_1 = sC_1$, $Y_2 = 1/R_1$, $Y_3 = sC_2$. Choose $R_1 = 1\ \text{k}\Omega$ and determine C_1 and C_2.

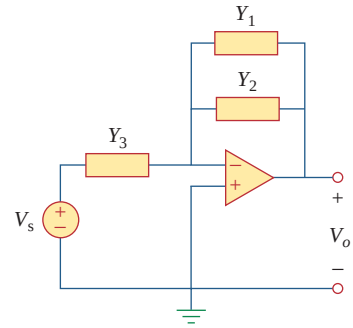


Figure 16.78
For Prob. 16.58.

16.59 Synthesize the transfer function

$$\frac{V_o(s)}{V_{\text{in}}(s)} = \frac{10^6}{s^2 + 100s + 10^6}$$

using the topology of Fig. 16.79. Let $Y_1 = 1/R_1$, $Y_2 = 1/R_2$, $Y_3 = sC_1$, $Y_4 = sC_2$. Choose $R_1 = 1\ \text{k}\Omega$ and determine C_1, C_2, and R_2.

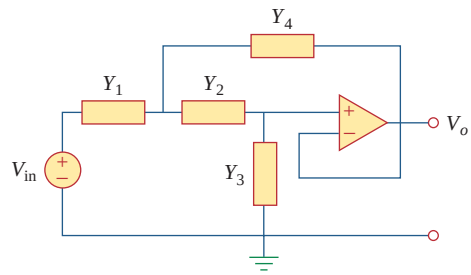


Figure 16.79
For Prob. 16.59.

Comprehensive Problems

16.60 Obtain the transfer function of the op amp circuit in Fig. 16.80 in the form of

$$\frac{V_o(s)}{V_i(s)} = \frac{as}{s^2 + bs + c}$$

where a, b, and c are constants. Determine the constants.

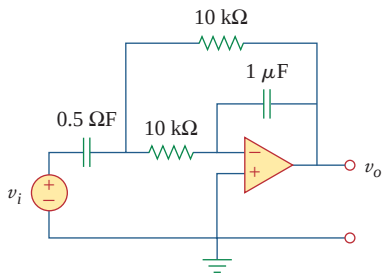


Figure 16.80
For Prob. 16.60.

16.61 A certain network has an input admittance $Y(s)$. The admittance has a pole at $s = -3$, a zero at $s = -1$, and $Y(\infty) = 0.25$ S.

(a) Find $Y(s)$.

(b) An 8-V battery is connected to the network via a switch. If the switch is closed at $t = 0$, find the current $i(t)$ through $Y(s)$ using the Laplace transform.

16.62 A gyrator is a device for simulating an inductor in a network. A basic gyrator circuit is shown in Fig. 16.81. By finding $V_i(s)/I_o(s)$, show that the inductance produced by the gyrator is $L = CR^2$.

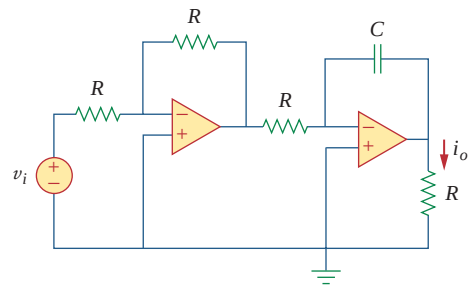


Figure 16.81
For Prob. 16.62.

chapter

17

The Fourier Series

Research is to see what everybody else has seen, and think what nobody has thought.

—Albert Szent Györgyi

Enhancing Your Skills and Your Career

ABET EC 2000 criteria (3.j), *"a knowledge of contemporary issues."*

Engineers must have knowledge of contemporary issues. To have a truly meaningful career in the twenty-first century, you must have knowledge of contemporary issues, especially those that may directly affect your job and/or work. One of the easiest ways to achieve this is to read a lot—newspapers, magazines, and contemporary books. As a student enrolled in an ABET-accredited program, some of the courses you take will be directed toward meeting this criteria.

ABET EC 2000 criteria (3.k), *"an ability to use the techniques, skills, and modern engineering tools necessary for engineering practice."*

The successful engineer must have the "ability to use the techniques, skills, and modern engineering tools necessary for engineering practice." Clearly, a major focus of this textbook is to do just that. Learning to use skillfully the tools that facilitate your working in a modern "knowledge capturing integrated design environment" (KCIDE) is fundamental to your performance as an engineer. The ability to work in a modern KCIDE environment requires a thorough understanding of the tools associated with that environment.

The successful engineer, therefore, must keep abreast of the new design, analysis, and simulation tools. That engineer must also use those tools until he or she is comfortable with using them. The engineer also must make sure software results are consistent with real-world actualities. It is probably in this area that most engineers have the greatest difficulty. Thus, successful use of these tools requires constant learning and relearning the fundamentals of the area in which the engineer is working.



Photo by Charles Alexander

Historical



Jean Baptiste Joseph Fourier (1768–1830), a French mathematician, first presented the series and transform that bear his name. Fourier's results were not enthusiastically received by the scientific world. He could not even get his work published as a paper.

Born in Auxerre, France, Fourier was orphaned at age 8. He attended a local military college run by Benedictine monks, where he demonstrated great proficiency in mathematics. Like most of his contemporaries, Fourier was swept into the politics of the French Revolution. He played an important role in Napoleon's expeditions to Egypt in the later 1790s. Due to his political involvement, he narrowly escaped death twice.

17.1 Introduction

We have spent a considerable amount of time on the analysis of circuits with sinusoidal sources. This chapter is concerned with a means of analyzing circuits with periodic, nonsinusoidal excitations. The notion of periodic functions was introduced in Chapter 9; it was mentioned there that the sinusoid is the most simple and useful periodic function. This chapter introduces the Fourier series, a technique for expressing a periodic function in terms of sinusoids. Once the source function is expressed in terms of sinusoids, we can apply the phasor method to analyze circuits.

The Fourier series is named after Jean Baptiste Joseph Fourier (1768–1830). In 1822, Fourier's genius came up with the insight that any practical periodic function can be represented as a sum of sinusoids. Such a representation, along with the superposition theorem, allows us to find the response of circuits to arbitrary periodic inputs using phasor techniques.

We begin with the trigonometric Fourier series. Later we consider the exponential Fourier series. We then apply Fourier series in circuit analysis. Finally, practical applications of Fourier series in spectrum analyzers and filters are demonstrated.

17.2 Trigonometric Fourier Series

While studying heat flow, Fourier discovered that a nonsinusoidal periodic function can be expressed as an infinite sum of sinusoidal functions. Recall that a periodic function is one that repeats every T seconds. In other words, a periodic function $f(t)$ satisfies

$$f(t) = f(t + nT) \tag{17.1}$$

where n is an integer and T is the period of the function.

According to the *Fourier theorem*, any practical periodic function of frequency ω_0 can be expressed as an infinite sum of sine or cosine functions that are integral multiples of ω_0. Thus, $f(t)$ can be expressed as

$$f(t) = a_0 + a_1 \cos\omega_0 t + b_1 \sin\omega_0 t + a_2 \cos 2\omega_0 t + b_2 \sin 2\omega_0 t + a_3 \cos 3\omega_0 t + b_3 \sin 3\omega_0 t + \cdots \quad \textbf{(17.2)}$$

or

$$f(t) = \underbrace{a_0}_{\text{dc}} + \underbrace{\sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)}_{\text{ac}} \quad \textbf{(17.3)}$$

where $\omega_0 = 2\pi/T$ is called the *fundamental frequency* in radians per second. The sinusoid $\sin n\omega_0 t$ or $\cos n\omega_0 t$ is called the nth harmonic of $f(t)$; it is an odd harmonic if n is odd and an even harmonic if n is even. Equation 17.3 is called the *trigonometric Fourier series* of $f(t)$. The constants a_n and b_n are the *Fourier coefficients*. The coefficient a_0 is the dc component or the average value of $f(t)$. (Recall that sinusoids have zero average values.) The coefficients a_n and b_n (for $n \neq 0$) are the amplitudes of the sinusoids in the ac component. Thus,

The harmonic frequency ω_n is an integral multiple of the fundamental frequency ω_0, i.e., $\omega_n = n\omega_0$.

The **Fourier series** of a periodic function $f(t)$ is a representation that resolves $f(t)$ into a dc component and an ac component comprising an infinite series of harmonic sinusoids.

A function that can be represented by a Fourier series as in Eq. (17.3) must meet certain requirements, because the infinite series in Eq. (17.3) may or may not converge. These conditions on $f(t)$ to yield a convergent Fourier series are as follows:

1. $f(t)$ is single-valued everywhere.
2. $f(t)$ has a finite number of finite discontinuities in any one period.
3. $f(t)$ has a finite number of maxima and minima in any one period.
4. The integral $\int_{t_0}^{t_0+T} |f(t)|\,dt < \infty$ for any t_0.

Historical note: Although Fourier published his theorem in 1822, it was P. G. L. Dirichlet (1805–1859) who later supplied an acceptable proof of the theorem.

These conditions are called *Dirichlet conditions*. Although they are not necessary conditions, they are sufficient conditions for a Fourier series to exist.

A major task in Fourier series is the determination of the Fourier coefficients a_0, a_n, and b_n. The process of determining the

A software package like *Mathcad* or *Maple* can be used to evaluate the Fourier coefficients.

coefficients is called *Fourier analysis*. The following trigonometric integrals are very helpful in Fourier analysis. For any integers m and n,

$$\int_0^T \sin n\omega_0 t\, dt = 0 \tag{17.4a}$$

$$\int_0^T \cos n\omega_0 t\, dt = 0 \tag{17.4b}$$

$$\int_0^T \sin n\omega_0 t \cos m\omega_0 t\, dt = 0 \tag{17.4c}$$

$$\int_0^T \sin n\omega_0 t \sin m\omega_0 t\, dt = 0, \qquad (m \neq n) \tag{17.4d}$$

$$\int_0^T \cos n\omega_0 t \cos m\omega_0 t\, dt = 0, \qquad (m \neq n) \tag{17.4e}$$

$$\int_0^T \sin^2 n\omega_0 t\, dt = \frac{T}{2} \tag{17.4f}$$

$$\int_0^T \cos^2 n\omega_0 t\, dt = \frac{T}{2} \tag{17.4g}$$

Let us use these identities to evaluate the Fourier coefficients.

We begin by finding a_0. We integrate both sides of Eq. (17.3) over one period and obtain

$$\begin{aligned}\int_0^T f(t)\, dt &= \int_0^T \left[a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)\right] dt \\ &= \int_0^T a_0\, dt + \sum_{n=1}^{\infty} \left[\int_0^T a_n \cos n\omega_0 t\, dt \right. \\ &\qquad \left. + \int_0^T b_n \sin n\omega_0 t\, dt\right] dt\end{aligned} \tag{17.5}$$

Invoking the identities of Eqs. (17.4a) and (17.4b), the two integrals involving the ac terms vanish. Hence,

$$\int_0^T f(t)\, dt = \int_0^T a_0\, dt = a_0 T$$

or

$$\boxed{a_0 = \frac{1}{T}\int_0^T f(t)\, dt} \tag{17.6}$$

showing that a_0 is the average value of $f(t)$.

To evaluate a_n, we multiply both sides of Eq. (17.3) by $\cos m\omega_0 t$ and integrate over one period:

$$\begin{aligned}\int_0^T f(t)\cos m\omega_0 t\,dt &= \int_0^T \left[a_0 + \sum_{n=1}^{\infty}(a_n\cos n\omega_0 t + b_n \sin n\omega_0 t)\right]\cos m\omega_0 t\,dt \\ &= \int_0^T a_0\cos m\omega_0 t\,dt + \sum_{n=1}^{\infty}\left[\int_0^T a_n\cos n\omega_0 t\cos m\omega_0 t\,dt \right. \\ &\quad \left. + \int_0^T b_n \sin n\omega_0 t\cos m\omega_0 t\,dt\right]dt \end{aligned} \tag{17.7}$$

The integral containing a_0 is zero in view of Eq. (17.4b), while the integral containing b_n vanishes according to Eq. (17.4c). The integral containing a_n will be zero except when $m = n$, in which case it is $T/2$, according to Eqs. (17.4e) and (17.4g). Thus,

$$\int_0^T f(t)\cos m\omega_0 t\,dt = a_n\frac{T}{2}, \qquad \text{for } m = n$$

or

$$\boxed{a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt} \tag{17.8}$$

In a similar vein, we obtain b_n by multiplying both sides of Eq. (17.3) by $\sin m\omega_0 t$ and integrating over the period. The result is

$$\boxed{b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\,dt} \tag{17.9}$$

Be aware that since $f(t)$ is periodic, it may be more convenient to carry the integrations above from $-T/2$ to $T/2$ or generally from t_0 to $t_0 + T$ instead of 0 to T. The result will be the same.

An alternative form of Eq. (17.3) is the *amplitude-phase* form

$$\boxed{f(t) = a_0 + \sum_{n=1}^{\infty} A_n\cos(n\omega_0 t + \phi_n)} \tag{17.10}$$

We can use Eqs. (9.11) and (9.12) to relate Eq. (17.3) to Eq. (17.10), or we can apply the trigonometric identity

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta \tag{17.11}$$

to the ac terms in Eq. (17.10) so that

$$\begin{aligned} a_0 + \sum_{n=1}^{\infty} A_n\cos(n\omega_0 t + \phi_n) = a_0 &+ \sum_{n=1}^{\infty}(A_n\cos\phi_n)\cos n\omega_0 t \\ &- (A_n\sin\phi_n)\sin n\omega_0 t \end{aligned} \tag{17.12}$$

Equating the coefficients of the series expansions in Eqs. (17.3) and (17.12) shows that

$$a_n = A_n \cos \phi_n, \qquad b_n = -A_n \sin \phi_n \tag{17.13a}$$

or

$$A_n = \sqrt{a_n^2 + b_n^2}, \qquad \phi_n = -\tan^{-1} \frac{b_n}{a_n} \tag{17.13b}$$

To avoid any confusion in determining ϕ_n, it may be better to relate the terms in complex form as

$$A_n \angle \phi_n = a_n - jb_n \tag{17.14}$$

The convenience of this relationship will become evident in Section 17.6. The plot of the amplitude A_n of the harmonics versus $n\omega_0$ is called the *amplitude spectrum* of $f(t)$; the plot of the phase ϕ_n versus $n\omega_0$ is the *phase spectrum* of $f(t)$. Both the amplitude and phase spectra form the *frequency spectrum* of $f(t)$.

The frequency spectrum is also known as the *line spectrum* in view of the discrete frequency components.

The **frequency spectrum** of a signal consists of the plots of the amplitudes and phases of the harmonics versus frequency.

Thus, the Fourier analysis is also a mathematical tool for finding the spectrum of a periodic signal. Section 17.6 will elaborate more on the spectrum of a signal.

To evaluate the Fourier coefficients a_0, a_n, and b_n, we often need to apply the following integrals:

$$\int \cos at \, dt = \frac{1}{a} \sin at \tag{17.15a}$$

$$\int \sin at \, dt = -\frac{1}{a} \cos at \tag{17.15b}$$

$$\int t \cos at \, dt = \frac{1}{a^2} \cos at + \frac{1}{a} t \sin at \tag{17.15c}$$

$$\int t \sin at \, dt = \frac{1}{a^2} \sin at - \frac{1}{a} t \cos at \tag{17.15d}$$

It is also useful to know the values of the cosine, sine, and exponential functions for integral multiples of π. These are given in Table 17.1, where n is an integer.

TABLE 17.1

Values of cosine, sine, and exponential functions for integral multiples of π.

Function	Value
$\cos 2n\pi$	1
$\sin 2n\pi$	0
$\cos n\pi$	$(-1)^n$
$\sin n\pi$	0
$\cos \frac{n\pi}{2}$	$\begin{cases} (-1)^{n/2}, & n = \text{even} \\ 0, & n = \text{odd} \end{cases}$
$\sin \frac{n\pi}{2}$	$\begin{cases} (-1)^{(n-1)/2}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases}$
$e^{j2n\pi}$	1
$e^{jn\pi}$	$(-1)^n$
$e^{jn\pi/2}$	$\begin{cases} (-1)^{n/2}, & n = \text{even} \\ j(-1)^{(n-1)/2}, & n = \text{odd} \end{cases}$

Example 17.1

Determine the Fourier series of the waveform shown in Fig. 17.1. Obtain the amplitude and phase spectra.

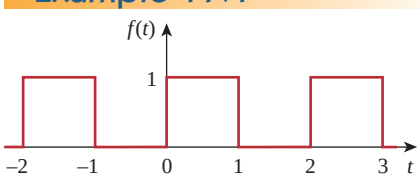


Figure 17.1
For Example 17.1; a square wave.

Solution:
The Fourier series is given by Eq. (17.3), namely,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \tag{17.1.1}$$

Our goal is to obtain the Fourier coefficients a_0, a_n, and b_n using Eqs. (17.6), (17.8), and (17.9). First, we describe the waveform as

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ 0, & 1 < t < 2 \end{cases} \tag{17.1.2}$$

and $f(t) = f(t + T)$. Since $T = 2$, $\omega_0 = 2\pi/T = \pi$. Thus,

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt = \frac{1}{2}\left[\int_0^1 1\,dt + \int_1^2 0\,dt\right] = \frac{1}{2}t\Big|_0^1 = \frac{1}{2} \tag{17.1.3}$$

Using Eq. (17.8) along with Eq. (17.15a),

$$\begin{aligned} a_n &= \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt \\ &= \frac{2}{2}\left[\int_0^1 1\cos n\pi t\,dt + \int_1^2 0\cos n\pi t\,dt\right] \\ &= \frac{1}{n\pi}\sin n\pi t\Big|_0^1 = \frac{1}{n\pi}[\sin n\pi - \sin(0)] = 0 \end{aligned} \tag{17.1.4}$$

From Eq. (17.9) with the aid of Eq. (17.15b),

$$\begin{aligned} b_n &= \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\,dt \\ &= \frac{2}{2}\left[\int_0^1 1\sin n\pi t\,dt + \int_1^2 0\sin n\pi t\,dt\right] \\ &= -\frac{1}{n\pi}\cos n\pi t\Big|_0^1 \\ &= -\frac{1}{n\pi}(\cos n\pi - 1), \qquad \cos n\pi = (-1)^n \\ &= \frac{1}{n\pi}[1 - (-1)^n] = \begin{cases} \dfrac{2}{n\pi}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases} \end{aligned} \tag{17.1.5}$$

Substituting the Fourier coefficients in Eqs. (17.1.3) to (17.1.5) into Eq. (17.1.1) gives the Fourier series as

$$f(t) = \frac{1}{2} + \frac{2}{\pi}\sin\pi t + \frac{2}{3\pi}\sin 3\pi t + \frac{2}{5\pi}\sin 5\pi t + \cdots \tag{17.1.6}$$

Since $f(t)$ contains only the dc component and the sine terms with the fundamental component and odd harmonics, it may be written as

$$f(t) = \frac{1}{2} + \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{1}{n}\sin n\pi t, \qquad n = 2k - 1 \tag{17.1.7}$$

By summing the terms one by one as demonstrated in Fig. 17.2, we notice how superposition of the terms can evolve into the original square. As more and more Fourier components are added, the sum gets closer and closer to the square wave. However, it is not possible in practice to sum the series in Eq. (17.1.6) or (17.1.7) to infinity. Only a partial sum ($n = 1, 2, 3, \ldots, N$, where N is finite) is possible. If we plot the partial sum (or truncated series) over one period for a large N

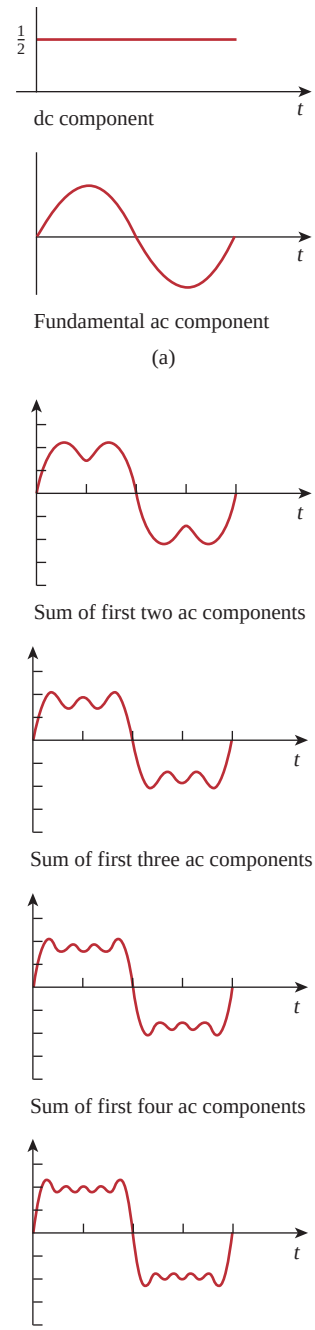
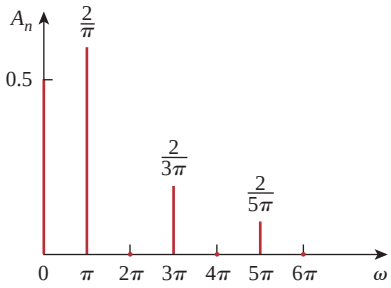


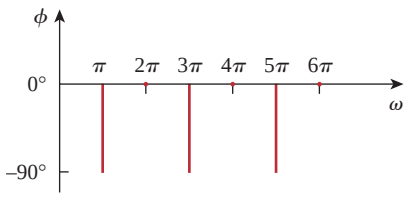
Figure 17.2
Evolution of a square wave from its Fourier components.

Summing the Fourier terms by hand calculation may be tedious. A computer is helpful to compute the terms and plot the sum like those shown in Fig. 17.2.

Historical note: Named after the mathematical physicist Josiah Willard Gibbs, who first observed it in 1899.



(a)



(b)

Figure 17.4
For Example 17.1: (a) amplitude and (b) phase spectrum of the function shown in Fig. 17.1.

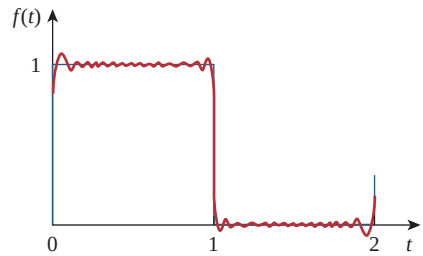


Figure 17.3
Truncating the Fourier series at $N = 11$; Gibbs phenomenon.

as in Fig. 17.3, we notice that the partial sum oscillates above and below the actual value of $f(t)$. At the neighborhood of the points of discontinuity $(x = 0, 1, 2, \dots)$, there is overshoot and damped oscillation. In fact, an overshoot of about 9 percent of the peak value is always present, regardless of the number of terms used to approximate $f(t)$. This is called the *Gibbs phenomenon*.

Finally, let us obtain the amplitude and phase spectra for the signal in Fig. 17.1. Since $a_n = 0$,

$$A_n = \sqrt{a_n^2 + b_n^2} = |b_n| = \begin{cases} \dfrac{2}{n\pi}, & n = \text{odd} \\ 0, & n = \text{even} \end{cases} \tag{17.1.8}$$

and

$$\phi_n = -\tan^{-1}\frac{b_n}{a_n} = \begin{cases} -90°, & n = \text{odd} \\ 0, & n = \text{even} \end{cases} \tag{17.1.9}$$

The plots of A_n and ϕ_n for different values of $n\omega_0 = n\pi$ provide the amplitude and phase spectra in Fig. 17.4. Notice that the amplitudes of the harmonics decay very fast with frequency.

Practice Problem 17.1

Find the Fourier series of the square wave in Fig. 17.5. Plot the amplitude and phase spectra.

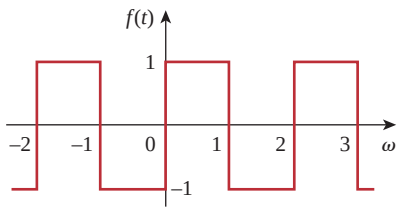
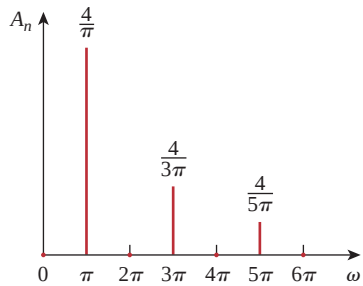
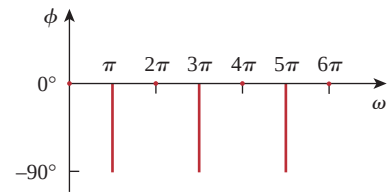


Figure 17.5
For Practice Prob. 17.1.

Answer: $f(t) = \dfrac{4}{\pi}\displaystyle\sum_{k=1}^{\infty}\frac{1}{n}\sin n\pi t$, $n = 2k - 1$. See Fig. 17.6 for the spectra.



(a)



(b)

Figure 17.6
For Practice Prob. 17.1: amplitude and phase spectra for the function shown in Fig. 17.5.

Example 17.2

Obtain the Fourier series for the periodic function in Fig. 17.7 and plot the amplitude and phase spectra.

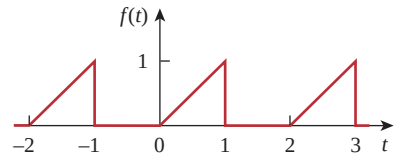


Figure 17.7
For Example 17.2.

Solution:
The function is described as

$$f(t) = \begin{cases} t, & 0 < t < 1 \\ 0, & 1 < t < 2 \end{cases}$$

Since $T = 2$, $\omega_0 = 2\pi/T = \pi$. Then

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt = \frac{1}{2}\left[\int_0^1 t\,dt + \int_1^2 0\,dt\right] = \frac{1}{2}\frac{t^2}{2}\Big|_0^1 = \frac{1}{4} \quad \textbf{(17.2.1)}$$

To evaluate a_n and b_n, we need the integrals in Eq. (17.15):

$$\begin{aligned} a_n &= \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt \\ &= \frac{2}{2}\left[\int_0^1 t\cos n\pi t\,dt + \int_1^2 0\cos n\pi t\,dt\right] \\ &= \left[\frac{1}{n^2\pi^2}\cos n\pi t + \frac{t}{n\pi}\sin n\pi t\right]\Big|_0^1 \\ &= \frac{1}{n^2\pi^2}(\cos n\pi - 1) + 0 = \frac{(-1)^n - 1}{n^2\pi^2} \end{aligned} \quad \textbf{(17.2.2)}$$

since $\cos n\pi = (-1)^n$; and

$$\begin{aligned} b_n &= \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\,dt \\ &= \frac{2}{2}\left[\int_0^1 t\sin n\pi t\,dt + \int_1^2 0\sin n\pi t\,dt\right] \\ &= \left[\frac{1}{n^2\pi^2}\sin n\pi t - \frac{t}{n\pi}\cos n\pi t\right]\Big|_0^1 \\ &= 0 - \frac{\cos n\pi}{n\pi} = \frac{(-1)^{n+1}}{n\pi} \end{aligned} \quad \textbf{(17.2.3)}$$

Substituting the Fourier coefficients just found into Eq. (17.3) yields

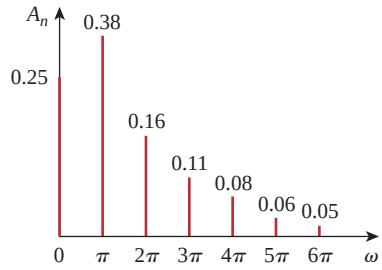
$$f(t) = \frac{1}{4} + \sum_{n=1}^{\infty}\left[\frac{[(-1)^n - 1]}{(n\pi)^2}\cos n\pi t + \frac{(-1)^{n+1}}{n\pi}\sin n\pi t\right]$$

To obtain the amplitude and phase spectra, we notice that, for even harmonics, $a_n = 0$, $b_n = -1/n\pi$, so that

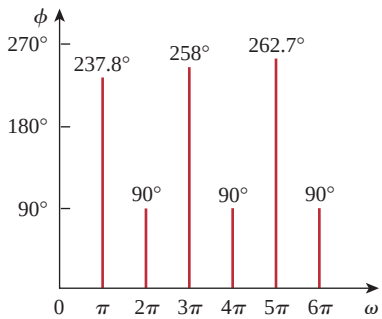
$$A_n\underline{/\phi_n} = a_n - jb_n = 0 + j\frac{1}{n\pi} \quad \textbf{(17.2.4)}$$

Hence,

$$\begin{aligned} A_n &= |b_n| = \frac{1}{n\pi}, \qquad n = 2, 4, \ldots \\ \phi_n &= 90°, \qquad n = 2, 4, \ldots \end{aligned} \quad \textbf{(17.2.5)}$$



(a)



(b)

Figure 17.8
For Example 17.2: (a) amplitude spectrum, (b) phase spectrum.

For odd harmonics, $a_n = -2/(n^2\pi^2)$, $b_n = 1/(n\pi)$ so that

$$A_n\underline{/\phi_n} = a_n - jb_n = -\frac{2}{n^2\pi^2} - j\frac{1}{n\pi} \tag{17.2.6}$$

That is,

$$\begin{aligned} A_n &= \sqrt{a_n^2 + b_n^2} = \sqrt{\frac{4}{n^4\pi^4} + \frac{1}{n^2\pi^2}} \\ &= \frac{1}{n^2\pi^2}\sqrt{4 + n^2\pi^2}, \qquad n = 1, 3, \ldots \end{aligned} \tag{17.2.7}$$

From Eq. (17.2.6), we observe that ϕ lies in the third quadrant, so that

$$\phi_n = 180° + \tan^{-1}\frac{n\pi}{2}, \qquad n = 1, 3, \ldots \tag{17.2.8}$$

From Eqs. (17.2.5), (17.2.7), and (17.2.8), we plot A_n and ϕ_n for different values of $n\omega_0 = n\pi$ to obtain the amplitude spectrum and phase spectrum as shown in Fig. 17.8.

Practice Problem 17.2

Determine the Fourier series of the sawtooth waveform in Fig. 17.9.

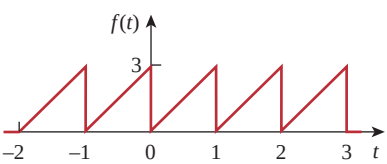


Figure 17.9
For Practice Prob. 17.2.

Answer: $f(t) = \frac{3}{2} - \frac{3}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin 2\pi nt.$

17.3 Symmetry Considerations

We noticed that the Fourier series of Example 17.1 consisted only of the sine terms. One may wonder if a method exists whereby one can know in advance that some Fourier coefficients would be zero and avoid the unnecessary work involved in the tedious process of calculating them. Such a method does exist; it is based on recognizing the existence of symmetry. Here we discuss three types of symmetry: (1) even symmetry, (2) odd symmetry, (3) half-wave symmetry.

17.3.1 Even Symmetry

A function $f(t)$ is *even* if its plot is symmetrical about the vertical axis; that is,

$$f(t) = f(-t) \tag{17.16}$$

Examples of even functions are t^2, t^4, and $\cos t$. Figure 17.10 shows more examples of periodic even functions. Note that each of these examples satisfies Eq. (17.16). A main property of an even function $f_e(t)$ is that:

$$\int_{-T/2}^{T/2} f_e(t)\,dt = 2\int_{0}^{T/2} f_e(t)\,dt \tag{17.17}$$

because integrating from $-T/2$ to 0 is the same as integrating from 0 to $T/2$. Utilizing this property, the Fourier coefficients for an even function become

$$\begin{aligned} a_0 &= \frac{2}{T}\int_0^{T/2} f(t)\,dt \\ a_n &= \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega_0 t\,dt \\ b_n &= 0 \end{aligned} \tag{17.18}$$

Since $b_n = 0$, Eq. (17.3) becomes a *Fourier cosine series*. This makes sense because the cosine function is itself even. It also makes intuitive sense that an even function contains no sine terms since the sine function is odd.

To confirm Eq. (17.18) quantitatively, we apply the property of an even function in Eq. (17.17) in evaluating the Fourier coefficients in Eqs. (17.6), (17.8), and (17.9). It is convenient in each case to integrate over the interval $-T/2 < t < T/2$, which is symmetrical about the origin. Thus,

$$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} f(t)\,dt = \frac{1}{T}\left[\int_{-T/2}^{0} f(t)\,dt + \int_{0}^{T/2} f(t)\,dt\right] \tag{17.19}$$

We change variables for the integral over the interval $-T/2 < t < 0$ by letting $t = -x$, so that $dt = -dx$, $f(t) = f(-t) = f(x)$, since $f(t)$ is an even function, and when $t = -T/2$, $x = T/2$. Then,

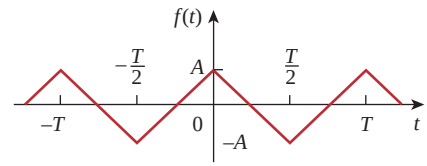
$$\begin{aligned} a_0 &= \frac{1}{T}\left[\int_{T/2}^{0} f(x)(-dx) + \int_{0}^{T/2} f(t)\,dt\right] \\ &= \frac{1}{T}\left[\int_{0}^{T/2} f(x)\,dx + \int_{0}^{T/2} f(t)\,dt\right] \end{aligned} \tag{17.20}$$

showing that the two integrals are identical. Hence,

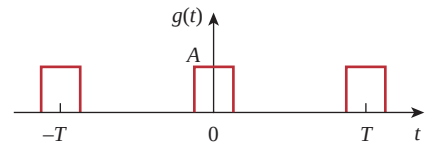
$$a_0 = \frac{2}{T}\int_0^{T/2} f(t)\,dt \tag{17.21}$$

as expected. Similarly, from Eq. (17.8),

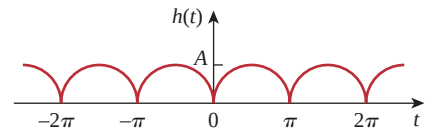
$$a_n = \frac{2}{T}\left[\int_{-T/2}^{0} f(t)\cos n\omega_0 t\,dt + \int_{0}^{T/2} f(t)\cos n\omega_0 t\,dt\right] \tag{17.22}$$



(a)



(b)



(c)

Figure 17.10
Typical examples of even periodic functions.

We make the same change of variables that led to Eq. (17.20) and note that both $f(t)$ and $\cos n\omega_0 t$ are even functions, implying that $f(-t) = f(t)$ and $\cos(-n\omega_0 t) = \cos n\omega_0 t$. Equation (17.22) becomes

$$\begin{aligned} a_n &= \frac{2}{T}\left[\int_{T/2}^{0} f(-x)\cos(-n\omega_0 x)(-dx) + \int_{0}^{T/2} f(t)\cos n\omega_0 t\, dt\right] \\ &= \frac{2}{T}\left[\int_{T/2}^{0} f(x)\cos(n\omega_0 x)(-dx) + \int_{0}^{T/2} f(t)\cos n\omega_0 t\, dt\right] \\ &= \frac{2}{T}\left[\int_{0}^{T/2} f(x)\cos(n\omega_0 x)\, dx + \int_{0}^{T/2} f(t)\cos n\omega_0 t\, dt\right] \end{aligned} \tag{17.23a}$$

or

$$a_n = \frac{4}{T}\int_{0}^{T/2} f(t)\cos n\omega_0 t\, dt \tag{17.23b}$$

as expected. For b_n, we apply Eq. (17.9),

$$b_n = \frac{2}{T}\left[\int_{-T/2}^{0} f(t)\sin n\omega_0 t\, dt + \int_{0}^{T/2} f(t)\sin n\omega_0 t\, dt\right] \tag{17.24}$$

We make the same change of variables but keep in mind that $f(-t) = f(t)$ but $\sin(-n\omega_0 t) = -\sin n\omega_0 t$. Equation (17.24) yields

$$\begin{aligned} b_n &= \frac{2}{T}\left[\int_{T/2}^{0} f(-x)\sin(-n\omega_0 x)(-dx) + \int_{0}^{T/2} f(t)\sin n\omega_0 t\, dt\right] \\ &= \frac{2}{T}\left[\int_{T/2}^{0} f(x)\sin n\omega_0 x\, dx + \int_{0}^{T/2} f(t)\sin n\omega_0 t\, dt\right] \\ &= \frac{2}{T}\left[-\int_{0}^{T/2} f(x)\sin(n\omega_0 x)\, dx + \int_{0}^{T/2} f(t)\sin n\omega_0 t\, dt\right] \\ &= 0 \end{aligned} \tag{17.25}$$

confirming Eq. (17.18).

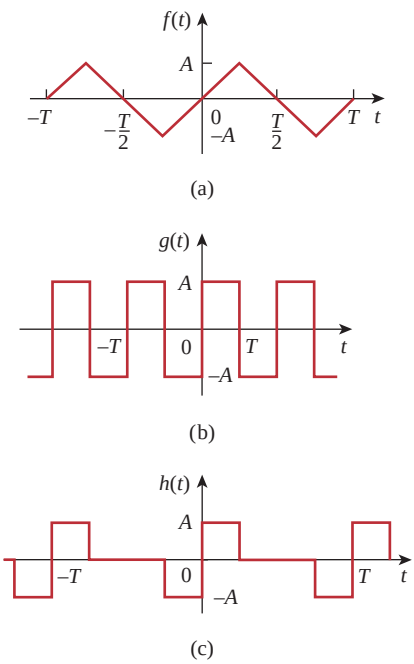


Figure 17.11
Typical examples of odd periodic functions.

17.3.2 Odd Symmetry

A function $f(t)$ is said to be *odd* if its plot is antisymmetrical about the vertical axis:

$$\boxed{f(-t) = -f(t)} \tag{17.26}$$

Examples of odd functions are t, t^3, and $\sin t$. Figure 17.11 shows more examples of periodic odd functions. All these examples satisfy Eq. (17.26). An odd function $f_o(t)$ has this major characteristic:

$$\int_{-T/2}^{T/2} f_o(t)\, dt = 0 \tag{17.27}$$

because integration from $-T/2$ to 0 is the negative of that from 0 to $T/2$. With this property, the Fourier coefficients for an odd function become

$$\boxed{\begin{aligned} a_0 &= 0, \qquad a_n = 0 \\ b_n &= \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega_0 t\, dt \end{aligned}} \tag{17.28}$$

which give us a *Fourier sine series*. Again, this makes sense because the sine function is itself an odd function. Also, note that there is no dc term for the Fourier series expansion of an odd function.

The quantitative proof of Eq. (17.28) follows the same procedure taken to prove Eq. (17.18) except that $f(t)$ is now odd, so that $f(t) = -f(t)$. With this fundamental but simple difference, it is easy to see that $a_0 = 0$ in Eq. (17.20), $a_n = 0$ in Eq. (17.23a), and b_n in Eq. (17.24) becomes

$$\begin{aligned} b_n &= \frac{2}{T}\left[\int_{T/2}^{0} f(-x)\sin(-n\omega_0 x)(-dx) + \int_0^{T/2} f(t)\sin n\omega_0 t\, dt\right] \\ &= \frac{2}{T}\left[-\int_{T/2}^{0} f(x)\sin n\omega_0 x\, dx + \int_0^{T/2} f(t)\sin n\omega_0 t\, dt\right] \\ &= \frac{2}{T}\left[\int_0^{T/2} f(x)\sin(n\omega_0 x)\, dx + \int_0^{T/2} f(t)\sin n\omega_0 t\, dt\right] \\ b_n &= \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega_0 t\, dt \end{aligned} \tag{17.29}$$

as expected.

It is interesting to note that any periodic function $f(t)$ with neither even nor odd symmetry may be decomposed into even and odd parts. Using the properties of even and odd functions from Eqs. (17.16) and (17.26), we can write

$$f(t) = \underbrace{\frac{1}{2}[f(t) + f(-t)]}_{\text{even}} + \underbrace{\frac{1}{2}[f(t) - f(-t)]}_{\text{odd}} = f_e(t) + f_o(t) \tag{17.30}$$

Notice that $f_e(t) = \frac{1}{2}[f(t) + f(-t)]$ satisfies the property of an even function in Eq. (17.16), while $f_o(t) = \frac{1}{2}[f(t) - f(-t)]$ satisfies the property of an odd function in Eq. (17.26). The fact that $f_e(t)$ contains only the dc term and the cosine terms, while $f_o(t)$ has only the sine terms, can be exploited in grouping the Fourier series expansion of $f(t)$ as

$$f(t) = \underbrace{a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t}_{\text{even}} + \underbrace{\sum_{n=1}^{\infty} b_n \sin n\omega_0 t}_{\text{odd}} = f_e(t) + f_o(t) \tag{17.31}$$

It follows readily from Eq. (17.31) that when $f(t)$ is even, $b_n = 0$, and when $f(t)$ is odd, $a_0 = 0 = a_n$.

Also, note the following properties of odd and even functions:

1. The product of two even functions is also an even function.
2. The product of two odd functions is an even function.
3. The product of an even function and an odd function is an odd function.
4. The sum (or difference) of two even functions is also an even function.
5. The sum (or difference) of two odd functions is an odd function.
6. The sum (or difference) of an even function and an odd function is neither even nor odd.

Each of these properties can be proved using Eqs. (17.16) and (17.26).

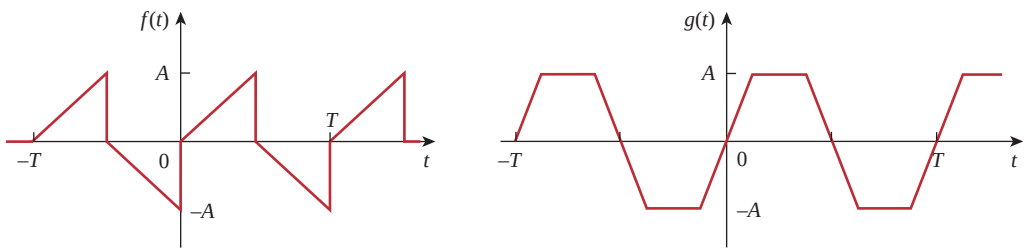
17.3.3 Half-Wave Symmetry

A function is half-wave (odd) symmetric if

$$f\left(t - \frac{T}{2}\right) = -f(t) \tag{17.32}$$

which means that each half-cycle is the mirror image of the next half-cycle. Notice that functions $\cos n\omega_0 t$ and $\sin n\omega_0 t$ satisfy Eq. (17.32) for odd values of n and therefore possess half-wave symmetry when n is odd. Figure 17.12 shows other examples of half-wave symmetric functions. The functions in Figs. 17.11(a) and 17.11(b) are also half-wave symmetric. Notice that for each function, one half-cycle is the inverted version of the adjacent half-cycle. The Fourier coefficients become

$$\begin{aligned} a_0 &= 0 \\ a_n &= \begin{cases} \dfrac{4}{T}\displaystyle\int_0^{T/2} f(t)\cos n\omega_0 t\,dt, & \text{for } n \text{ odd} \\ 0, & \text{for } n \text{ even} \end{cases} \\ b_n &= \begin{cases} \dfrac{4}{T}\displaystyle\int_0^{T/2} f(t)\sin n\omega_0 t\,dt, & \text{for } n \text{ odd} \\ 0, & \text{for } n \text{ even} \end{cases} \end{aligned} \tag{17.33}$$



(a) (b)

Figure 17.12
Typical examples of half-wave odd symmetric functions.

showing that the Fourier series of a half-wave symmetric function contains only odd harmonics.

To derive Eq. (17.33), we apply the property of half-wave symmetric functions in Eq. (17.32) in evaluating the Fourier coefficients in Eqs. (17.6), (17.8), and (17.9). Thus,

$$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} f(t)\,dt = \frac{1}{T}\left[\int_{-T/2}^{0} f(t)\,dt + \int_{0}^{T/2} f(t)\,dt\right] \tag{17.34}$$

We change variables for the integral over the interval $-T/2 < t < 0$ by letting $x = t + T/2$, so that $dx = dt$; when $t = -T/2, x = 0$; and when $t = 0, x = T/2$. Also, we keep Eq. (17.32) in mind; that is, $f(x - T/2) = -f(x)$. Then,

$$\begin{aligned} a_0 &= \frac{1}{T}\left[\int_{0}^{T/2} f\left(x - \frac{T}{2}\right)dx + \int_{0}^{T/2} f(t)\,dt\right] \\ &= \frac{1}{T}\left[-\int_{0}^{T/2} f(x)\,dx + \int_{0}^{T/2} f(t)\,dt\right] = 0 \end{aligned} \tag{17.35}$$

confirming the expression for a_0 in Eq. (17.33). Similarly,

$$a_n = \frac{2}{T}\left[\int_{-T/2}^{0} f(t)\cos n\omega_0 t\,dt + \int_{0}^{T/2} f(t)\cos n\omega_0 t\,dt\right] \tag{17.36}$$

We make the same change of variables that led to Eq. (17.35) so that Eq. (17.36) becomes

$$\begin{aligned} a_n = \frac{2}{T}\left[\int_{0}^{T/2} f\left(x - \frac{T}{2}\right)\cos n\omega_0\left(x - \frac{T}{2}\right)dx \right. \\ \left. + \int_{0}^{T/2} f(t)\cos n\omega_0 t\,dt\right] \end{aligned} \tag{17.37}$$

Since $f(x - T/2) = -f(x)$ and

$$\begin{aligned} \cos n\omega_0\left(x - \frac{T}{2}\right) &= \cos(n\omega_0 t - n\pi) \\ &= \cos n\omega_0 t \cos n\pi + \sin n\omega_0 t \sin n\pi \\ &= (-1)^n \cos n\omega_0 t \end{aligned} \tag{17.38}$$

substituting these in Eq. (17.37) leads to

$$\begin{aligned} a_n &= \frac{2}{T}[1 - (-1)^n]\int_{0}^{T/2} f(t)\cos n\omega_0 t\,dt \\ &= \begin{cases} \dfrac{4}{T}\displaystyle\int_{0}^{T/2} f(t)\cos n\omega_0 t\,dt, & \text{for } n \text{ odd} \\ 0, & \text{for } n \text{ even} \end{cases} \end{aligned} \tag{17.39}$$

confirming Eq. (17.33). By following a similar procedure, we can derive b_n as in Eq. (17.33).

Table 17.2 summarizes the effects of these symmetries on the Fourier coefficients. Table 17.3 provides the Fourier series of some common periodic functions.

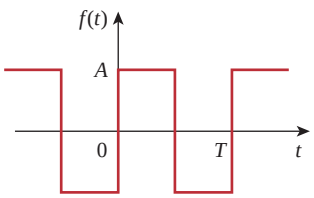
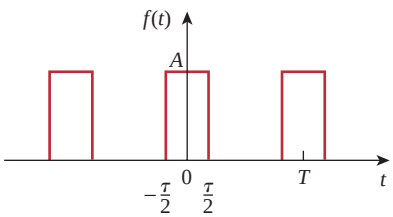
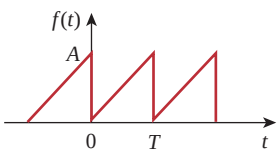
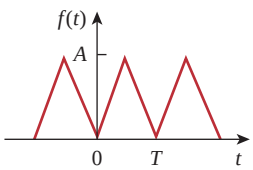
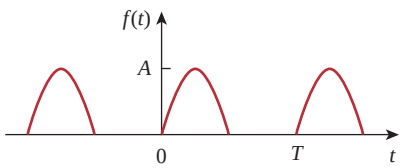
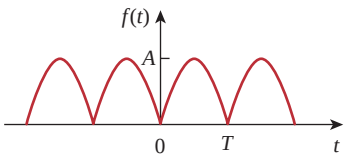
TABLE 17.2

Effects of symmetry on Fourier coefficients.

Symmetry	a_0	a_n	b_n	Remarks
Even	$a_0 \neq 0$	$a_n \neq 0$	$b_n = 0$	Integrate over $T/2$ and multiply by 2 to get the coefficients.
Odd	$a_0 = 0$	$a_n = 0$	$b_n \neq 0$	Integrate over $T/2$ and multiply by 2 to get the coefficients.
Half-wave	$a_0 = 0$	$a_{2n} = 0$	$b_{2n} = 0$	Integrate over $T/2$ and multiply by 2 to get
		$a_{2n+1} \neq 0$	$b_{2n+1} \neq 0$	the coefficients.

TABLE 17.3

The Fourier series of common functions.

Function	Fourier series
1. Square wave 	$f(t) = \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin(2n-1)\omega_0 t$
2. Rectangular pulse train 	$f(t) = \frac{A\tau}{T} + \frac{2A}{T} \sum_{n=1}^{\infty} \frac{1}{n} \sin\frac{n\pi\tau}{T} \cos n\omega_0 t$
3. Sawtooth wave 	$f(t) = \frac{A}{2} - \frac{A}{\pi} \sum_{n=1}^{\infty} \frac{\sin n\omega_0 t}{n}$
4. Triangular wave 	$f(t) = \frac{A}{2} - \frac{4A}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)\omega_0 t$
5. Half-wave rectified sine 	$f(t) = \frac{A}{\pi} + \frac{A}{2} \sin\omega_0 t - \frac{2A}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos 2n\omega_0 t$
6. Full-wave rectified sine 	$f(t) = \frac{2A}{\pi} - \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos n\omega_0 t$

Example 17.3

Find the Fourier series expansion of $f(t)$ given in Fig. 17.13.

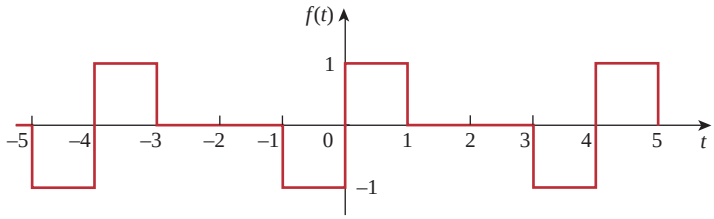


Figure 17.13
For Example 17.3.

Solution:
The function $f(t)$ is an odd function. Hence $a_0 = 0 = a_n$. The period is $T = 4$, and $\omega_0 = 2\pi/T = \pi/2$, so that

$$b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega_0 t\,dt$$

$$= \frac{4}{4}\left[\int_0^1 1\sin\frac{n\pi}{2}t\,dt + \int_1^2 0\sin\frac{n\pi}{2}t\,dt\right]$$

$$= -\frac{2}{n\pi}\cos\frac{n\pi t}{2}\Big|_0^1 = \frac{2}{n\pi}\left(1 - \cos\frac{n\pi}{2}\right)$$

Hence,

$$f(t) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\left(1 - \cos\frac{n\pi}{2}\right)\sin\frac{n\pi}{2}t$$

which is a Fourier sine series.

Practice Problem 17.3

Find the Fourier series of the function $f(t)$ in Fig. 17.14.

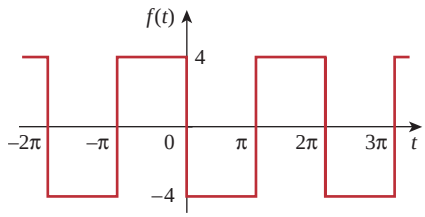


Figure 17.14
For Practice Prob. 17.3.

Answer: $f(t) = -\frac{16}{\pi}\sum_{k=1}^{\infty}\frac{1}{n}\sin nt,\ n = 2k - 1.$

Example 17.4

Determine the Fourier series for the half-wave rectified cosine function shown in Fig. 17.15.

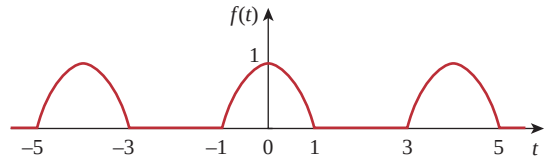


Figure 17.15
A half-wave rectified cosine function; for Example 17.4.

Solution:
This is an even function so that $b_n = 0$. Also, $T = 4$, $\omega_0 = 2\pi/T = \pi/2$. Over a period,

$$f(t) = \begin{cases} 0, & -2 < t < -1 \\ \cos\dfrac{\pi}{2}t, & -1 < t < 1 \\ 0, & 1 < t < 2 \end{cases}$$

$$a_0 = \frac{2}{T}\int_0^{T/2} f(t)\,dt = \frac{2}{4}\left[\int_0^1 \cos\frac{\pi}{2}t\,dt + \int_1^2 0\,dt\right]$$

$$= \frac{1}{2}\frac{2}{\pi}\sin\frac{\pi}{2}t\,\Big|_0^1 = \frac{1}{\pi}$$

$$a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos n\omega_0 t\,dt = \frac{4}{4}\left[\int_0^1 \cos\frac{\pi}{2}t\cos\frac{n\pi t}{2}\,dt + 0\right]$$

But $\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$. Then

$$a_n = \frac{1}{2}\int_0^1 \left[\cos\frac{\pi}{2}(n+1)t + \cos\frac{\pi}{2}(n-1)t\right]dt$$

For $n = 1$,

$$a_1 = \frac{1}{2}\int_0^1 [\cos\pi t + 1]\,dt = \frac{1}{2}\left[\frac{\sin\pi t}{\pi} + t\right]\Big|_0^1 = \frac{1}{2}$$

For $n > 1$,

$$a_n = \frac{1}{\pi(n+1)}\sin\frac{\pi}{2}(n+1) + \frac{1}{\pi(n-1)}\sin\frac{\pi}{2}(n-1)$$

For n = odd ($n = 1, 3, 5, \ldots$), $(n+1)$ and $(n-1)$ are both even, so

$$\sin\frac{\pi}{2}(n+1) = 0 = \sin\frac{\pi}{2}(n-1), \qquad n = \text{ odd}$$

For n = even ($n = 2, 4, 6, \ldots$), $(n+1)$ and $(n-1)$ are both odd. Also,

$$\sin\frac{\pi}{2}(n+1) = -\sin\frac{\pi}{2}(n-1) = \cos\frac{n\pi}{2} = (-1)^{n/2}, \qquad n = \text{even}$$

Hence,

$$a_n = \frac{(-1)^{n/2}}{\pi(n+1)} + \frac{-(-1)^{n/2}}{\pi(n-1)} = \frac{-2(-1)^{n/2}}{\pi(n^2-1)}, \qquad n = \text{ even}$$

Thus,

$$f(t) = \frac{1}{\pi} + \frac{1}{2}\cos\frac{\pi}{2}t - \frac{2}{\pi}\sum_{n=\text{even}}^{\infty}\frac{(-1)^{n/2}}{(n^2-1)}\cos\frac{n\pi}{2}t$$

To avoid using $n = 2, 4, 6, \ldots$ and also to ease computation, we can replace n by $2k$, where $k = 1, 2, 3, \ldots$ and obtain

$$f(t) = \frac{1}{\pi} + \frac{1}{2}\cos\frac{\pi}{2}t - \frac{2}{\pi}\sum_{k=1}^{\infty}\frac{(-1)^{k}}{(4k^2-1)}\cos k\pi t$$

which is a Fourier cosine series.

Practice Problem 17.4

Find the Fourier series expansion of the function in Fig. 17.16.

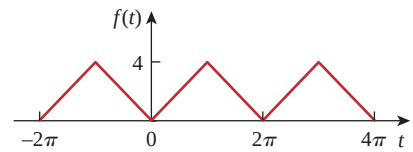


Figure 17.16
For Practice Prob. 17.4.

Answer: $f(t) = 2 - \frac{16}{\pi^2}\sum_{k=1}^{\infty}\frac{1}{n^2}\cos nt,\ n = 2k-1.$

Example 17.5

Calculate the Fourier series for the function in Fig. 17.17.

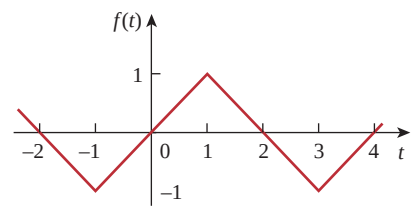


Figure 17.17
For Example 17.5.

Solution:
The function in Fig. 17.17 is half-wave odd symmetric, so that $a_0 = 0 = a_n$. It is described over half the period as

$$f(t) = t, \qquad -1 < t < 1$$

$T = 4$, $\omega_0 = 2\pi/T = \pi/2$. Hence,

$$b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin n\omega_0 t\, dt$$

Instead of integrating $f(t)$ from 0 to 2, it is more convenient to integrate from -1 to 1. Applying Eq. (17.15d),

$$b_n = \frac{4}{4}\int_{-1}^{1} t\sin\frac{n\pi t}{2}dt = \left[\frac{\sin n\pi t/2}{n^2\pi^2/4} - \frac{t\cos n\pi t/2}{n\pi/2}\right]\Bigg|_{-1}^{1}$$

$$= \frac{4}{n^2\pi^2}\left[\sin\frac{n\pi}{2} - \sin\left(-\frac{n\pi}{2}\right)\right] - \frac{2}{n\pi}\left[\cos\frac{n\pi}{2} - \cos\left(-\frac{n\pi}{2}\right)\right]$$

$$= \frac{8}{n^2\pi^2}\sin\frac{n\pi}{2}$$

since $\sin(-x) = -\sin x$ is an odd function, while $\cos(-x) = \cos x$ is an even function. Using the identities for $\sin n\pi/2$ in Table 17.1,

$$b_n = \frac{8}{n^2\pi^2}(-1)^{(n-1)/2}, \quad n = \text{odd} = 1, 3, 5, \ldots$$

Thus,

$$f(t) = \sum_{n=1,3,5}^{\infty} b_n \sin\frac{n\pi}{2}t$$

where b_n is given above.

Practice Problem 17.5 Determine the Fourier series of the function in Fig. 17.12(a). Take $A = 2$ and $T = 2\pi$.

Answer: $f(t) = \frac{4}{\pi}\sum_{k=1}^{\infty}\left(\frac{-2}{n^2\pi}\cos nt + \frac{1}{n}\sin nt\right)$, $n = 2k - 1$.

17.4 Circuit Applications

We find that in practice, many circuits are driven by nonsinusoidal periodic functions. To find the steady-state response of a circuit to a nonsinusoidal periodic excitation requires the application of a Fourier series, ac phasor analysis, and the superposition principle. The procedure usually involves four steps.

Steps for Applying Fourier Series:

1. Express the excitation as a Fourier series.
2. Transform the circuit from the time domain to the frequency domain.
3. Find the response of the dc and ac components in the Fourier series.
4. Add the individual dc and ac responses using the superposition principle.

The first step is to determine the Fourier series expansion of the excitation. For the periodic voltage source shown in Fig. 17.18(a), for example, the Fourier series is expressed as

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t + \theta_n) \tag{17.40}$$

(The same could be done for a periodic current source.) Equation (17.40) shows that $v(t)$ consists of two parts: the dc component V_0 and the ac component $\mathbf{V}_n = V_n\angle\theta_n$ with several harmonics. This Fourier series representation may be regarded as a set of series-connected sinusoidal sources, with each source having its own amplitude and frequency, as shown in Fig. 17.18(b).

The third step is finding the response to each term in the Fourier series. The response to the dc component can be determined in the

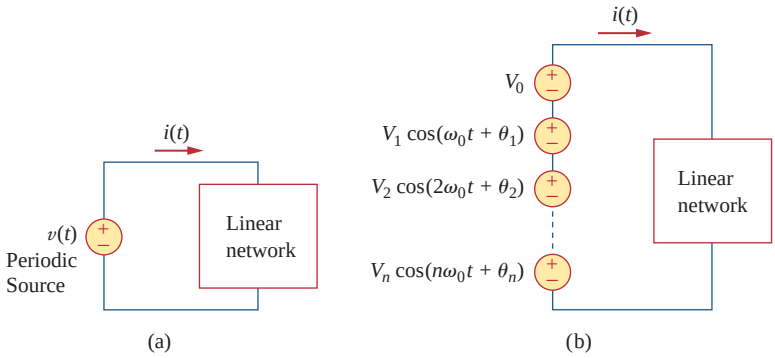


Figure 17.18
(a) Linear network excited by a periodic voltage source, (b) Fourier series representation (time-domain).

frequency domain by setting $n = 0$ or $\omega = 0$ as in Fig. 17.19(a), or in the time domain by replacing all inductors with short circuits and all capacitors with open circuits. The response to the ac component is obtained by applying the phasor techniques covered in Chapter 9, as shown in Fig. 17.19(b). The network is represented by its impedance $\mathbf{Z}(n\omega_0)$ or admittance $\mathbf{Y}(n\omega_0)$. $\mathbf{Z}(n\omega_0)$ is the input impedance at the source when ω is everywhere replaced by $n\omega_0$, and $\mathbf{Y}(n\omega_0)$ is the reciprocal of $\mathbf{Z}(n\omega_0)$.

Finally, following the principle of superposition, we add all the individual responses. For the case shown in Fig. 17.19,

$$\begin{aligned} i(t) &= i_0(t) + i_1(t) + i_2(t) + \cdots \\ &= \mathbf{I}_0 + \sum_{n=1}^{\infty} |\mathbf{I}_n| \cos(n\omega_0 t + \psi_n) \end{aligned} \tag{17.41}$$

where each component $\mathbf{I}_n$ with frequency $n\omega_0$ has been transformed to the time domain to get $i_n(t)$, and ψ_n is the argument of $\mathbf{I}_n$.

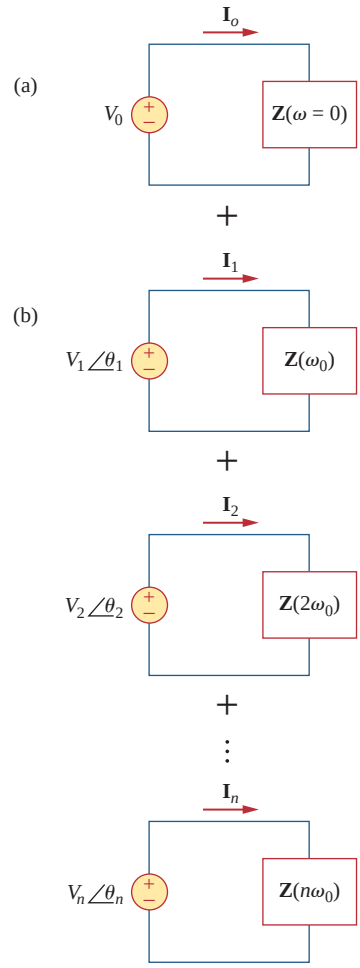


Figure 17.19
Steady-state responses: (a) dc component, (b) ac component (frequency domain).

Example 17.6

Let the function $f(t)$ in Example 17.1 be the voltage source $v_s(t)$ in the circuit of Fig. 17.20. Find the response $v_o(t)$ of the circuit.

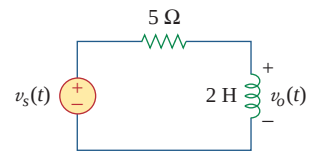


Figure 17.20
For Example 17.6.

Solution:
From Example 17.1,

$$v_s(t) = \frac{1}{2} + \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{1}{n} \sin n\pi t, \qquad n = 2k - 1$$

where $\omega_n = n\omega_0 = n\pi$ rad/s. Using phasors, we obtain the response $\mathbf{V}_o$ in the circuit of Fig. 17.20 by voltage division:

$$\mathbf{V}_o = \frac{j\omega_n L}{R + j\omega_n L} \mathbf{V}_s = \frac{j2n\pi}{5 + j2n\pi} \mathbf{V}_s$$

For the dc component ($\omega_n = 0$ or $n = 0$)

$$\mathbf{V}_s = \frac{1}{2} \quad \Rightarrow \quad \mathbf{V}_o = 0$$

This is expected, since the inductor is a short circuit to dc. For the nth harmonic,

$$\mathbf{V}_s = \frac{2}{n\pi}\angle -90^\circ \tag{17.6.1}$$

and the corresponding response is

$$\begin{aligned}\mathbf{V}_o &= \frac{2n\pi\angle 90^\circ}{\sqrt{25 + 4n^2\pi^2}\angle \tan^{-1} 2n\pi/5}\left(\frac{2}{n\pi}\angle -90^\circ\right) \\ &= \frac{4\angle -\tan^{-1} 2n\pi/5}{\sqrt{25 + 4n^2\pi^2}}\end{aligned} \tag{17.6.2}$$

In the time domain,

$$v_o(t) = \sum_{k=1}^{\infty} \frac{4}{\sqrt{25 + 4n^2\pi^2}} \cos\left(n\pi t - \tan^{-1}\frac{2n\pi}{5}\right), \qquad n = 2k - 1$$

The first three terms ($k = 1, 2, 3$ or $n = 1, 3, 5$) of the odd harmonics in the summation give us

$$\begin{aligned}v_o(t) = {} & 0.4981\cos(\pi t - 51.49^\circ) + 0.2051\cos(3\pi t - 75.14^\circ) \\ & + 0.1257\cos(5\pi t - 80.96^\circ) + \cdots \text{V}\end{aligned}$$

Figure 17.21 shows the amplitude spectrum for output voltage $v_o(t)$, while that of the input voltage $v_s(t)$ is in Fig. 17.4(a). Notice that the two spectra are close. Why? We observe that the circuit in Fig. 17.20 is a highpass filter with the corner frequency $\omega_c = R/L = 2.5$ rad/s, which is less than the fundamental frequency $\omega_0 = \pi$ rad/s. The dc component is not passed and the first harmonic is slightly attenuated, but higher harmonics are passed. In fact, from Eqs. (17.6.1) and (17.6.2), $\mathbf{V}_o$ is identical to $\mathbf{V}_s$ for large n, which is characteristic of a highpass filter.

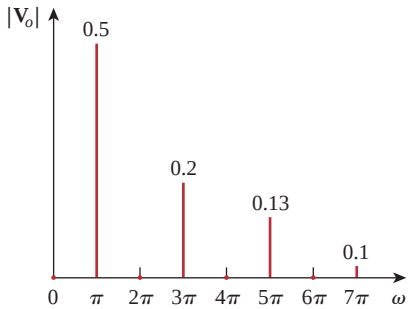


Figure 17.21
For Example 17.6: Amplitude spectrum of the output voltage.

Practice Problem 17.6

If the sawtooth waveform in Fig. 17.9 (see Practice Prob. 17.2) is the voltage source $v_s(t)$ in the circuit of Fig. 17.22, find the response $v_o(t)$.

Answer: $v_o(t) = \frac{1}{2} - \frac{1}{\pi}\sum_{n=1}^{\infty} \frac{\sin(2\pi nt - \tan^{-1} 4n\pi)}{n\sqrt{1 + 16n^2\pi^2}}$ V.

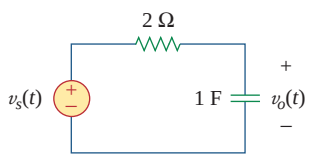


Figure 17.22
For Practice Prob. 17.6.

Example 17.7

Find the response $i_o(t)$ of the circuit of Fig. 17.23 if the input voltage $v(t)$ has the Fourier series expansion

$$v(t) = 1 + \sum_{n=1}^{\infty} \frac{2(-1)^n}{1 + n^2}(\cos nt - n\sin nt)$$

Solution:

Using Eq. (17.13), we can express the input voltage as

$$v(t) = 1 + \sum_{n=1}^{\infty} \frac{2(-1)^n}{\sqrt{1+n^2}} \cos(nt + \tan^{-1} n)$$
$$= 1 - 1.414\cos(t + 45°) + 0.8944\cos(2t + 63.45°)$$
$$- 0.6345\cos(3t + 71.56°) - 0.4851\cos(4t + 78.7°) + \cdots$$

We notice that $\omega_0 = 1$, $\omega_n = n$ rad/s. The impedance at the source is

$$\mathbf{Z} = 4 + j\omega_n 2 \,\|\, 4 = 4 + \frac{j\omega_n 8}{4 + j\omega_n 2} = \frac{8 + j\omega_n 8}{2 + j\omega_n}$$

The input current is

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} = \frac{2 + j\omega_n}{8 + j\omega_n 8}\mathbf{V}$$

where $\mathbf{V}$ is the phasor form of the source voltage $v(t)$. By current division,

$$\mathbf{I}_o = \frac{4}{4 + j\omega_n 2}\mathbf{I} = \frac{\mathbf{V}}{4 + j\omega_n 4}$$

Since $\omega_n = n$, $\mathbf{I}_o$ can be expressed as

$$\mathbf{I}_o = \frac{\mathbf{V}}{4\sqrt{1+n^2}\underline{/\tan^{-1} n}}$$

For the dc component ($\omega_n = 0$ or $n = 0$)

$$\mathbf{V} = 1 \quad \Rightarrow \quad \mathbf{I}_o = \frac{\mathbf{V}}{4} = \frac{1}{4}$$

For the nth harmonic,

$$\mathbf{V} = \frac{2(-1)^n}{\sqrt{1+n^2}}\underline{/\tan^{-1} n}$$

so that

$$\mathbf{I}_o = \frac{1}{4\sqrt{1+n^2}\underline{/\tan^{-1} n}} \frac{2(-1)^n}{\sqrt{1+n^2}}\underline{/\tan^{-1} n} = \frac{(-1)^n}{2(1+n^2)}$$

In the time domain,

$$i_o(t) = \frac{1}{4} + \sum_{n=1}^{\infty} \frac{(-1)^n}{2(1+n^2)} \cos nt \text{ A}$$

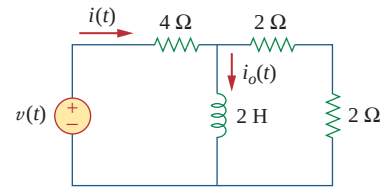


Figure 17.23
For Example 17.7.

Practice Problem 17.7

If the input voltage in the circuit of Fig. 17.24 is

$$v(t) = \frac{1}{3} + \frac{1}{\pi^2}\sum_{n=1}^{\infty}\left(\frac{1}{n^2}\cos nt - \frac{\pi}{n}\sin nt\right)\text{V}$$

determine the response $i_o(t)$.

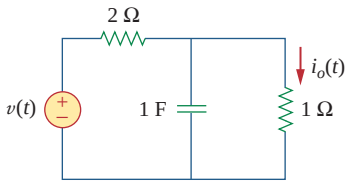


Figure 17.24
For Practice Prob. 17.7.

Answer: $\frac{1}{9} + \sum_{n=1}^{\infty} \frac{\sqrt{1+n^2\pi^2}}{n^2\pi^2\sqrt{9+4n^2}} \cos\left(nt - \tan^{-1}\frac{2n}{3} + \tan^{-1} n\pi\right)$ A.

17.5 Average Power and RMS Values

Recall the concepts of average power and rms value of a periodic signal that we discussed in Chapter 11. To find the average power absorbed by a circuit due to a periodic excitation, we write the voltage and current in amplitude-phase form [see Eq. (17.10)] as

$$v(t) = V_{dc} + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t - \theta_n) \tag{17.42}$$

$$i(t) = I_{dc} + \sum_{m=1}^{\infty} I_m \cos(m\omega_0 t - \phi_m) \tag{17.43}$$

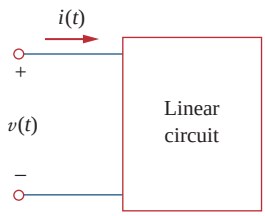


Figure 17.25
The voltage polarity reference and current reference direction.

Following the passive sign convention (Fig. 17.25), the average power is

$$P = \frac{1}{T}\int_0^T vi\, dt \tag{17.44}$$

Substituting Eqs. (17.42) and (17.43) into Eq. (17.44) gives

$$\begin{aligned} P = {} & \frac{1}{T}\int_0^T V_{dc} I_{dc}\, dt + \sum_{m=1}^{\infty} \frac{I_m V_{dc}}{T}\int_0^T \cos(m\omega_0 t - \phi_m)\, dt \\ & + \sum_{n=1}^{\infty} \frac{V_n I_{dc}}{T}\int_0^T \cos(n\omega_0 t - \theta_n)\, dt \\ & + \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \frac{V_n I_m}{T}\int_0^T \cos(n\omega_0 t - \theta_n)\cos(m\omega_0 t - \phi_m)\, dt \end{aligned} \tag{17.45}$$

The second and third integrals vanish, since we are integrating the cosine over its period. According to Eq. (17.4e), all terms in the fourth integral are zero when $m \neq n$. By evaluating the first integral and applying Eq. (17.4g) to the fourth integral for the case $m = n$, we obtain

$$\boxed{P = V_{dc} I_{dc} + \frac{1}{2}\sum_{n=1}^{\infty} V_n I_n \cos(\theta_n - \phi_n)} \tag{17.46}$$

This shows that in average-power calculation involving periodic voltage and current, the total average power is the sum of the average powers in each harmonically related voltage and current.

Given a periodic function $f(t)$, its rms value (or the effective value) is given by

$$F_{rms} = \sqrt{\frac{1}{T}\int_0^T f^2(t)\, dt} \tag{17.47}$$

Substituting $f(t)$ in Eq. (17.10) into Eq. (17.47) and noting that $(a + b)^2 = a^2 + 2ab + b^2$, we obtain

$$\begin{aligned} F_{\text{rms}}^2 &= \frac{1}{T}\int_0^T \left[a_0^2 + 2\sum_{n=1}^{\infty} a_0 A_n \cos(n\omega_0 t + \phi_n) \right. \\ &\qquad \left. + \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} A_n A_m \cos(n\omega_0 t + \phi_n)\cos(m\omega_0 t + \phi_m) \right] dt \\ &= \frac{1}{T}\int_0^T a_0^2\, dt + 2\sum_{n=1}^{\infty} a_0 A_n \frac{1}{T}\int_0^T \cos(n\omega_0 t + \phi_n)\, dt \\ &\quad + \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} A_n A_m \frac{1}{T}\int_0^T \cos(n\omega_0 t + \phi_n)\cos(m\omega_0 t + \phi_m)\, dt \end{aligned} \tag{17.48}$$

Distinct integers n and m have been introduced to handle the product of the two series summations. Using the same reasoning as above, we get

$$F_{\text{rms}}^2 = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} A_n^2$$

or

$$F_{\text{rms}} = \sqrt{a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} A_n^2} \tag{17.49}$$

In terms of Fourier coefficients a_n and b_n, Eq. (17.49) may be written as

$$F_{\text{rms}} = \sqrt{a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} (a_n^2 + b_n^2)} \tag{17.50}$$

If $f(t)$ is the current through a resistor R, then the power dissipated in the resistor is

$$P = RF_{\text{rms}}^2 \tag{17.51}$$

Or if $f(t)$ is the voltage across a resistor R, the power dissipated in the resistor is

$$P = \frac{F_{\text{rms}}^2}{R} \tag{17.52}$$

One can avoid specifying the nature of the signal by choosing a 1-Ω resistance. The power dissipated by the 1-Ω resistance is

$$\boxed{P_{1\Omega} = F_{\text{rms}}^2 = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} (a_n^2 + b_n^2)} \tag{17.53}$$

This result is known as *Parseval's theorem*. Notice that a_0^2 is the power in the dc component, while $\frac{1}{2}(a_n^2 + b_n^2)$ is the ac power in the nth harmonic. Thus, Parseval's theorem states that the average power in a periodic signal is the sum of the average power in its dc component and the average powers in its harmonics.

Historical note: Named after the French mathematician Marc-Antoine Parseval Deschemes (1755–1836).

Example 17.8

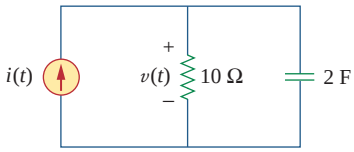


Figure 17.26
For Example 17.8.

Determine the average power supplied to the circuit in Fig. 17.26 if $i(t) = 2 + 10\cos(t + 10°) + 6\cos(3t + 35°)$ A.

Solution:
The input impedance of the network is

$$\mathbf{Z} = 10 \parallel \frac{1}{j2\omega} = \frac{10(1/j2\omega)}{10 + 1/j2\omega} = \frac{10}{1 + j20\omega}$$

Hence,

$$\mathbf{V} = \mathbf{IZ} = \frac{10\mathbf{I}}{\sqrt{1 + 400\omega^2}\angle \tan^{-1} 20\omega}$$

For the dc component, $\omega = 0$,

$$\mathbf{I} = 2 \text{ A} \quad \Rightarrow \quad \mathbf{V} = 10(2) = 20 \text{ V}$$

This is expected, because the capacitor is an open circuit to dc and the entire 2-A current flows through the resistor. For $\omega = 1$ rad/s,

$$\mathbf{I} = 10\angle 10° \quad \Rightarrow \quad \mathbf{V} = \frac{10(10\angle 10°)}{\sqrt{1 + 400}\angle \tan^{-1} 20} = 5\angle -77.14°$$

For $\omega = 3$ rad/s,

$$\mathbf{I} = 6\angle 35° \quad \Rightarrow \quad \mathbf{V} = \frac{10(6\angle 35°)}{\sqrt{1 + 3600}\angle \tan^{-1} 60} = 1\angle -54.04°$$

Thus, in the time domain,

$$v(t) = 20 + 5\cos(t - 77.14°) + 1\cos(3t - 54.04°) \text{ V}$$

We obtain the average power supplied to the circuit by applying Eq. (17.46), as

$$P = V_{\text{dc}} I_{\text{dc}} + \frac{1}{2}\sum_{n=1}^{\infty} V_n I_n \cos(\theta_n - \phi_n)$$

To get the proper signs of θ_n and ϕ_n, we have to compare v and i in this example with Eqs. (17.42) and (17.43). Thus,

$$P = 20(2) + \frac{1}{2}(5)(10)\cos[77.14° - (-10°)] + \frac{1}{2}(1)(6)\cos[54.04° - (-35°)]$$
$$= 40 + 1.247 + 0.05 = 41.5 \text{ W}$$

Alternatively, we can find the average power absorbed by the resistor as

$$P = \frac{V_{\text{dc}}^2}{R} + \frac{1}{2}\sum_{n=1}^{\infty}\frac{|V_n|^2}{R} = \frac{20^2}{10} + \frac{1}{2}\cdot\frac{5^2}{10} + \frac{1}{2}\cdot\frac{1^2}{10}$$
$$= 40 + 1.25 + 0.05 = 41.5 \text{ W}$$

which is the same as the power supplied, since the capacitor absorbs no average power.

Practice Problem 17.8

The voltage and current at the terminals of a circuit are

$$v(t) = 128 + 192\cos 120\pi t + 96\cos(360\pi t - 30°)$$
$$i(t) = 8\cos(120\pi t - 10°) + 3.2\cos(360\pi t - 60°)$$

Find the average power absorbed by the circuit.

Answer: 22.23 kW.

Example 17.9

Find an estimate for the rms value of the voltage in Example 17.7.

Solution:

From Example 17.7, $v(t)$ is expressed as

$$v(t) = 1 - 1.414\cos(t + 45°) + 0.8944\cos(2t + 63.45°) - 0.6345\cos(3t + 71.56°) - 0.4851\cos(4t + 78.7°) + \cdots \text{V}$$

Using Eq. (17.49), we find

$$V_{\text{rms}} = \sqrt{a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} A_n^2}$$
$$= \sqrt{1^2 + \frac{1}{2}\left[(-1.414)^2 + (0.8944)^2 + (-0.6345)^2 + (-0.4851)^2 + \cdots\right]}$$
$$= \sqrt{2.7186} = 1.649 \text{ V}$$

This is only an estimate, as we have not taken enough terms of the series. The actual function represented by the Fourier series is

$$v(t) = \frac{\pi e^t}{\sinh \pi}, \qquad -\pi < t < \pi$$

with $v(t) = v(t + T)$. The exact rms value of this is 1.776 V.

Practice Problem 17.9

Find the rms value of the periodic current

$$i(t) = 8 + 30\cos 2t - 20\sin 2t + 15\cos 4t - 10\sin 4t \text{ A}$$

Answer: 29.61 A.

17.6 Exponential Fourier Series

A compact way of expressing the Fourier series in Eq. (17.3) is to put it in exponential form. This requires that we represent the sine and cosine functions in the exponential form using Euler's identity:

$$\cos n\omega_0 t = \frac{1}{2}[e^{jn\omega_0 t} + e^{-jn\omega_0 t}] \tag{17.54a}$$

$$\sin n\omega_0 t = \frac{1}{2j}[e^{jn\omega_0 t} - e^{-jn\omega_0 t}] \tag{17.54b}$$

Substituting Eq. (17.54) into Eq. (17.3) and collecting terms, we obtain

$$f(t) = a_0 + \frac{1}{2}\sum_{n=1}^{\infty}[(a_n - jb_n)e^{jn\omega_0 t} + (a_n + jb_n)e^{-jn\omega_0 t}] \quad (17.55)$$

If we define a new coefficient c_n so that

$$c_0 = a_0, \qquad c_n = \frac{(a_n - jb_n)}{2}, \qquad c_{-n} = c_n^* = \frac{(a_n + jb_n)}{2} \quad (17.56)$$

then $f(t)$ becomes

$$f(t) = c_0 + \sum_{n=1}^{\infty}(c_n e^{jn\omega_0 t} + c_{-n}e^{-jn\omega_0 t}) \quad (17.57)$$

or

$$\boxed{f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}} \quad (17.58)$$

This is the *complex* or *exponential Fourier series* representation of $f(t)$. Note that this exponential form is more compact than the sine-cosine form in Eq. (17.3). Although the exponential Fourier series coefficients c_n can also be obtained from a_n and b_n using Eq. (17.56), they can also be obtained directly from $f(t)$ as

$$\boxed{c_n = \frac{1}{T}\int_0^T f(t)e^{-jn\omega_0 t}\,dt} \quad (17.59)$$

where $\omega_0 = 2\pi/T$, as usual. The plots of the magnitude and phase of c_n versus $n\omega_0$ are called the *complex amplitude spectrum* and *complex phase spectrum* of $f(t)$, respectively. The two spectra form the complex frequency spectrum of $f(t)$.

The **exponential Fourier series** of a periodic function $f(t)$ describes the spectrum of $f(t)$ in terms of the amplitude and phase angle of ac components at positive and negative harmonic frequencies.

The coefficients of the three forms of Fourier series (sine-cosine form, amplitude-phase form, and exponential form) are related by

$$\boxed{A_n\angle\phi_n = a_n - jb_n = 2c_n} \quad (17.60)$$

or

$$c_n = |c_n|\angle\theta_n = \frac{\sqrt{a_n^2 + b_n^2}}{2}\angle -\tan^{-1}b_n/a_n \quad (17.61)$$

if only $a_n > 0$. Note that the phase θ_n of c_n is equal to ϕ_n.

In terms of the Fourier complex coefficients c_n, the rms value of a periodic signal $f(t)$ can be found as

$$\begin{aligned} F_{\text{rms}}^2 &= \frac{1}{T}\int_0^T f^2(t)\,dt = \frac{1}{T}\int_0^T f(t)\left[\sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}\right]dt \\ &= \sum_{n=-\infty}^{\infty} c_n\left[\frac{1}{T}\int_0^T f(t)e^{jn\omega_0 t}\,dt\right] \\ &= \sum_{n=-\infty}^{\infty} c_n c_n^* = \sum_{n=-\infty}^{\infty} |c_n|^2 \end{aligned} \tag{17.62}$$

or

$$F_{\text{rms}} = \sqrt{\sum_{n=-\infty}^{\infty} |c_n|^2} \tag{17.63}$$

Equation (17.62) can be written as

$$F_{\text{rms}}^2 = |c_0|^2 + 2\sum_{n=1}^{\infty} |c_n|^2 \tag{17.64}$$

Again, the power dissipated by a 1-Ω resistance is

$$P_{1\Omega} = F_{\text{rms}}^2 = \sum_{n=-\infty}^{\infty} |c_n|^2 \tag{17.65}$$

which is a restatement of Parseval's theorem. The *power spectrum* of the signal $f(t)$ is the plot of $|c_n|^2$ versus $n\omega_0$. If $f(t)$ is the voltage across a resistor R, the average power absorbed by the resistor is F_{rms}^2/R; if $f(t)$ is the current through R, the power is $F_{\text{rms}}^2 R$.

As an illustration, consider the periodic pulse train of Fig. 17.27. Our goal is to obtain its amplitude and phase spectra. The period of the pulse train is $T = 10$, so that $\omega_0 = 2\pi/T = \pi/5$. Using Eq. (17.59),

$$\begin{aligned} c_n &= \frac{1}{T}\int_{-T/2}^{T/2} f(t)e^{-jn\omega_0 t}\,dt = \frac{1}{10}\int_{-1}^{1} 10e^{-jn\omega_0 t}\,dt \\ &= \frac{1}{-jn\omega_0}e^{-jn\omega_0 t}\Big|_{-1}^{1} = \frac{1}{-jn\omega_0}(e^{-jn\omega_0} - e^{jn\omega_0}) \\ &= \frac{2}{n\omega_0}\frac{e^{jn\omega_0} - e^{-jn\omega_0}}{2j} = 2\frac{\sin n\omega_0}{n\omega_0}, \qquad \omega_0 = \frac{\pi}{5} \\ &= 2\frac{\sin n\pi/5}{n\pi/5} \end{aligned} \tag{17.66}$$

and

$$f(t) = 2\sum_{n=-\infty}^{\infty} \frac{\sin n\pi/5}{n\pi/5} e^{jn\pi t/5} \tag{17.67}$$

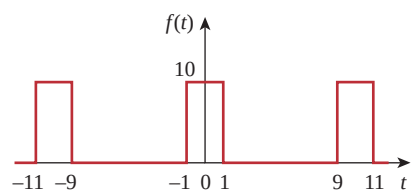


Figure 17.27
The periodic pulse train.

Notice from Eq. (17.66) that c_n is the product of 2 and a function of the form $\sin x/x$. This function is known as the *sinc function*; we write it as

$$\text{sinc}(x) = \frac{\sin x}{x} \tag{17.68}$$

The sinc function is called the *sampling function* in communication theory, where it is very useful.

Some properties of the sinc function are important here. For zero argument, the value of the sinc function is unity,

$$\text{sinc}(0) = 1 \tag{17.69}$$

This is obtained by applying L'Hopital's rule to Eq. (17.68). For an integral multiple of π, the value of the sinc function is zero,

$$\text{sinc}(n\pi) = 0, \qquad n = 1, 2, 3, \ldots \tag{17.70}$$

Also, the sinc function shows even symmetry. With all this in mind, we can obtain the amplitude and phase spectra of $f(t)$. From Eq. (17.66), the magnitude is

$$|c_n| = 2\left|\frac{\sin n\pi/5}{n\pi/5}\right| \tag{17.71}$$

while the phase is

$$\theta_n = \begin{cases} 0°, & \sin\dfrac{n\pi}{5} > 0 \\ 180°, & \sin\dfrac{n\pi}{5} < 0 \end{cases} \tag{17.72}$$

Examining the input and output spectra allows visualization of the effect of a circuit on a periodic signal.

Figure 17.28 shows the plot of $|c_n|$ versus n for n varying from -10 to 10, where $n = \omega/\omega_0$ is the normalized frequency. Figure 17.29 shows the plot of θ_n versus n. Both the amplitude spectrum and phase spectrum are called *line spectra*, because the values of $|c_n|$ and θ_n occur only at discrete values of frequencies. The spacing between the lines is ω_0. The power spectrum, which is the plot of $|c_n|^2$ versus $n\omega_0$, can also be plotted. Notice that the sinc function forms the envelope of the amplitude spectrum.

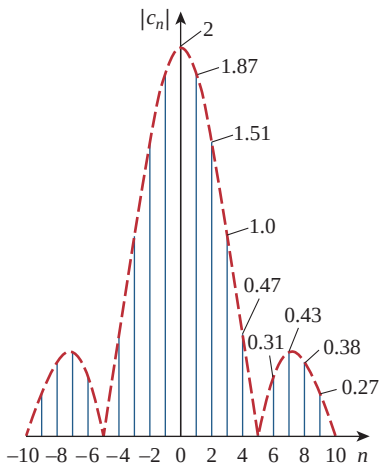


Figure 17.28
The amplitude of a periodic pulse train.

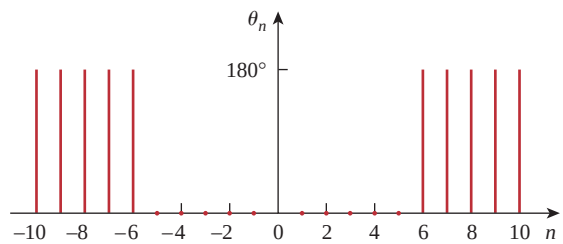


Figure 17.29
The phase spectrum of a periodic pulse train.

Example 17.10

Find the exponential Fourier series expansion of the periodic function $f(t) = e^t$, $0 < t < 2\pi$ with $f(t + 2\pi) = f(t)$.

Solution:
Since $T = 2\pi$, $\omega_0 = 2\pi/T = 1$. Hence,

$$c_n = \frac{1}{T}\int_0^T f(t)e^{-jn\omega_0 t}\,dt = \frac{1}{2\pi}\int_0^{2\pi} e^t e^{-jnt}\,dt$$

$$= \frac{1}{2\pi}\frac{1}{1-jn}e^{(1-jn)t}\Big|_0^{2\pi} = \frac{1}{2\pi(1-jn)}[e^{2\pi}e^{-j2\pi n} - 1]$$

But by Euler's identity,

$$e^{-j2\pi n} = \cos 2\pi n - j\sin 2\pi n = 1 - j0 = 1$$

Thus,

$$c_n = \frac{1}{2\pi(1 - jn)}[e^{2\pi} - 1] = \frac{85}{1 - jn}$$

The complex Fourier series is

$$f(t) = \sum_{n=-\infty}^{\infty} \frac{85}{1 - jn} e^{jnt}$$

We may want to plot the complex frequency spectrum of $f(t)$. If we let $c_n = |c_n| \angle \theta_n$, then

$$|c_n| = \frac{85}{\sqrt{1 + n^2}}, \qquad \theta_n = \tan^{-1} n$$

By inserting in negative and positive values of n, we obtain the amplitude and the phase plots of c_n versus $n\omega_0 = n$, as in Fig. 17.30.

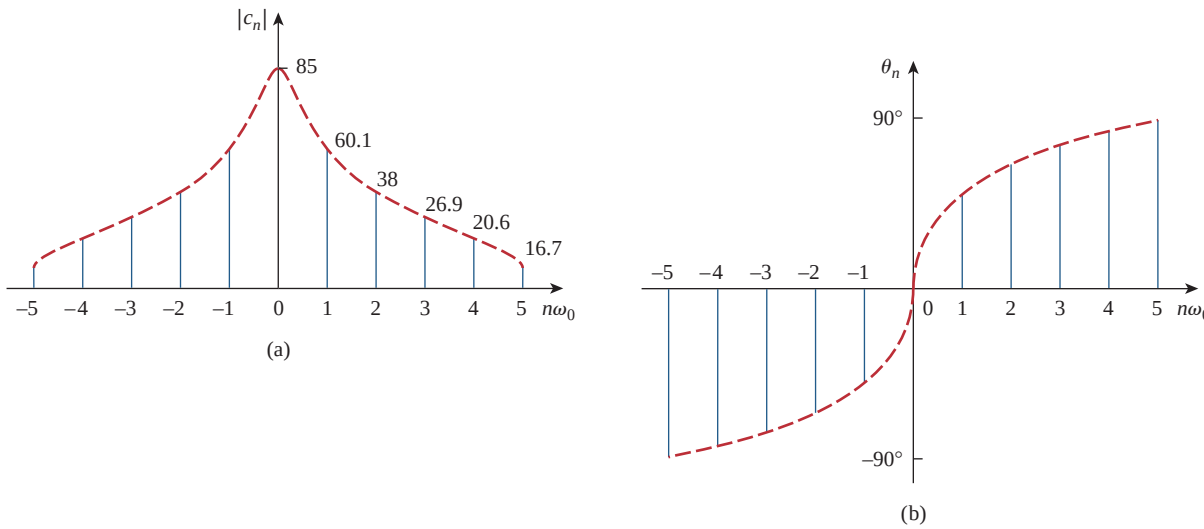


Figure 17.30
The complex frequency spectrum of the function in Example 17.10: (a) amplitude spectrum, (b) phase spectrum.

Practice Problem 17.10

Obtain the complex Fourier series of the function in Fig. 17.1.

Answer: $f(t) = \frac{1}{2} - \sum_{\substack{n=-\infty \\ n\neq 0 \\ n=\text{odd}}}^{\infty} \frac{j}{n\pi} e^{jn\pi t}$.

Example 17.11

Find the complex Fourier series of the sawtooth wave in Fig. 17.9. Plot the amplitude and the phase spectra.

Solution:
From Fig. 17.9, $f(t) = t$, $0 < t < 1$, $T = 1$ so that $\omega_0 = 2\pi/T = 2\pi$. Hence,

$$c_n = \frac{1}{T}\int_0^T f(t)e^{-jn\omega_0 t}\,dt = \frac{1}{1}\int_0^1 te^{-j2n\pi t}\,dt \qquad \textbf{(17.11.1)}$$

But

$$\int te^{at}\,dt = \frac{e^{at}}{a^2}(ax - 1) + C$$

Applying this to Eq. (17.11.1) gives

$$\begin{aligned} c_n &= \frac{e^{-j2n\pi t}}{(-j2n\pi)^2}(-j2n\pi t - 1)\Big|_0^1 \\ &= \frac{e^{-j2n\pi}(-j2n\pi - 1) + 1}{-4n^2\pi^2} \end{aligned} \tag{17.11.2}$$

Again,

$$e^{-j2\pi n} = \cos 2\pi n - j \sin 2\pi n = 1 - j0 = 1$$

so that Eq. (17.11.2) becomes

$$c_n = \frac{-j2n\pi}{-4n^2\pi^2} = \frac{j}{2n\pi} \tag{17.11.3}$$

This does not include the case when $n = 0$. When $n = 0$,

$$c_0 = \frac{1}{T}\int_0^T f(t)\,dt = \frac{1}{1}\int_0^1 t\,dt = \frac{t^2}{2}\Big|_1^0 = 0.5 \tag{17.11.4}$$

Hence,

$$f(t) = 0.5 + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{j}{2n\pi} e^{j2n\pi t} \tag{17.11.5}$$

and

$$|c_n| = \begin{cases} \dfrac{1}{2|n|\pi}, & n \neq 0 \\ 0.5, & n = 0 \end{cases}, \qquad \theta_n = 90°, \qquad n \neq 0 \tag{17.11.6}$$

By plotting $|c_n|$ and θ_n for different n, we obtain the amplitude spectrum and the phase spectrum shown in Fig. 17.31.

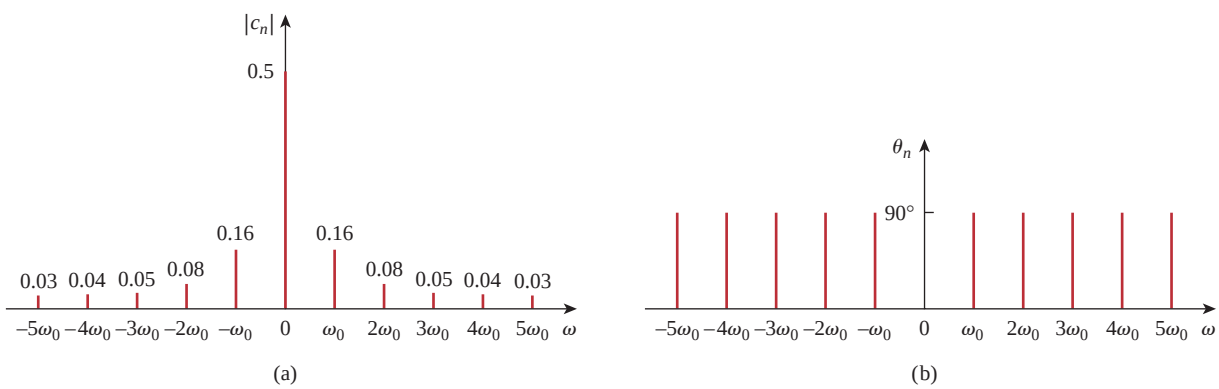


Figure 17.31
For Example 17.11: (a) amplitude spectrum, (b) phase spectrum.

Practice Problem 17.11

Obtain the complex Fourier series expansion of $f(t)$ in Fig. 17.17. Show the amplitude and phase spectra.

Answer: $f(t) = -\sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{j(-1)^n}{n\pi} e^{jn\pi t}$. See Fig. 17.32 for the spectra.

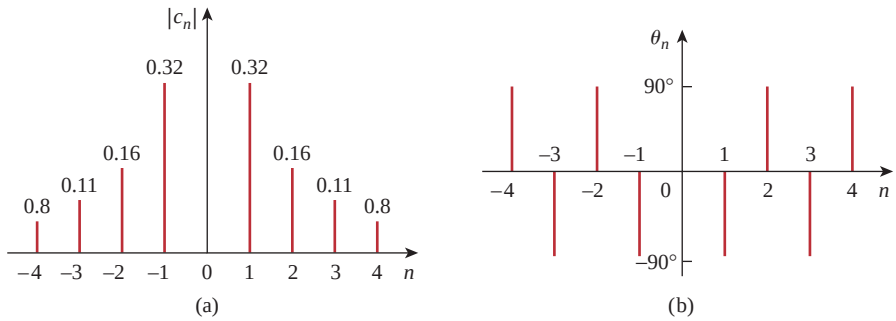


Figure 17.32
For Practice Prob. 17.11: (a) amplitude spectrum, (b) phase spectrum.

17.7 Fourier Analysis with *PSpice*

Fourier analysis is usually performed with *PSpice* in conjunction with transient analysis. Therefore, we must do a transient analysis in order to perform a Fourier analysis.

To perform the Fourier analysis of a waveform, we need a circuit whose input is the waveform and whose output is the Fourier decomposition. A suitable circuit is a current (or voltage) source in series with a 1-Ω resistor as shown in Fig. 17.33. The waveform is inputted as $v_s(t)$ using VPULSE for a pulse or VSIN for a sinusoid, and the attributes of the waveform are set over its period T. The output V(1) from node 1 is the dc level (a_0) and the first nine harmonics (A_n) with their corresponding phases ψ_n; that is,

$$v_o(t) = a_0 + \sum_{n=1}^{9} A_n \sin(n\omega_0 t + \psi_n) \qquad \textbf{(17.73)}$$

where

$$A_n = \sqrt{a_n^2 + b_n^2}, \qquad \psi_n = \phi_n - \frac{\pi}{2}, \qquad \phi_n = \tan^{-1}\frac{b_n}{a_n} \qquad \textbf{(17.74)}$$

Notice in Eq. (17.74) that the *PSpice* output is in the sine and angle form rather than the cosine and angle form in Eq. (17.10). The *PSpice* output also includes the normalized Fourier coefficients. Each coefficient a_n is normalized by dividing it by the magnitude of the fundamental a_1, so that the normalized component is a_n/a_1. The corresponding phase ψ_n is normalized by subtracting from it the phase ψ_1 of the fundamental, so that the normalized phase is $\psi_n - \psi_1$.

There are two types of Fourier analyses offered by *PSpice for Windows: Discrete Fourier Transform* (DFT) performed by the *PSpice*

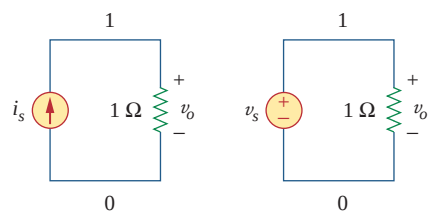


Figure 17.33
Fourier analysis with *PSpice* using: (a) a current source, (b) a voltage source.

program and *Fast Fourier Transform* (FFT) performed by the *PSpice A/D* program. While DFT is an approximation of the exponential Fourier series, FTT is an algorithm for rapid efficient numerical computation of DFT. A full discussion of DFT and FTT is beyond the scope of this book.

17.7.1 Discrete Fourier Transform

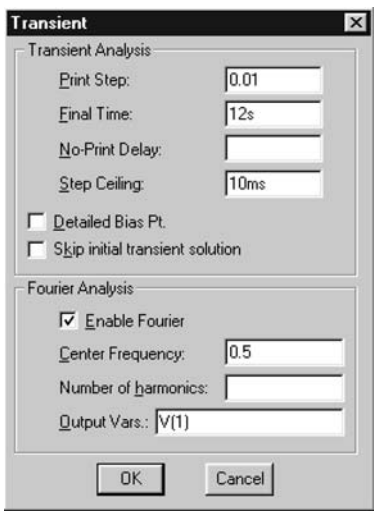


Figure 17.34
Transient dialog box.

A discrete Fourier transform (DFT) is performed by the *PSpice* program, which tabulates the harmonics in an output file. To enable a Fourier analysis, we select **Analysis/Setup/Transient** and bring up the Transient dialog box, shown in Fig. 17.34. The *Print Step* should be a small fraction of the period *T*, while the *Final Time* could be 6T. The *Center Frequency* is the fundamental frequency $f_0 = 1/T$. The particular variable whose DFT is desired, V(1) in Fig. 17.34, is entered in the **Output Vars** command box. In addition to filling in the Transient dialog box, **DCLICK** *Enable Fourier*. With the Fourier analysis enabled and the schematic saved, run *PSpice* by selecting **Analysis/Simulate** as usual. The program executes a harmonic decomposition into Fourier components of the result of the transient analysis. The results are sent to an output file which can be retrieved by selecting **Analysis/Examine Output**. The output file includes the dc value and the first nine harmonics by default, although you can specify more in the *Number of harmonics* box (see Fig. 17.34).

17.7.2 Fast Fourier Transform

A fast Fourier transform (FFT) is performed by the *PSpice A/D* program and displays as a *PSpice A/D* plot the complete spectrum of a transient expression. As explained above, we first construct the schematic in Fig. 17.33(b) and enter the attributes of the waveform. We also need to enter the *Print Step* and the *Final Time* in the Transient dialog box. Once this is done, we can obtain the FFT of the waveform in two ways.

One way is to insert a voltage marker at node 1 in the schematic of the circuit in Fig. 17.33(b). After saving the schematic and selecting **Analysis/Simulate**, the waveform V(1) will be displayed in the *PSpice A/D* window. Double clicking the FFT icon in the *PSpice A/D* menu will automatically replace the waveform with its FFT. From the FFT-generated graph, we can obtain the harmonics. In case the FFT-generated graph is crowded, we can use the *User Defined* data range (see Fig. 17.35) to specify a smaller range.

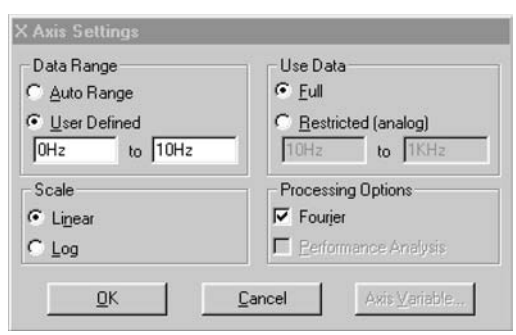


Figure 17.35
X axis settings dialog box.

Another way of obtaining the FFT of V(1) is to not insert a voltage marker at node 1 in the schematic. After selecting **Analysis/Simulate**, the *PSpice A/D* window will come up with no graph on it. We select **Trace/Add** and type V(1) in the **Trace Command** box and **DCLICKL OK**. We now select **Plot/X-Axis Settings** to bring up the *X-Axis Setting* dialog box shown in Fig. 17.35 and then select **Fourier/OK**. This will cause the FFT of the selected trace (or traces) to be displayed. This second approach is useful for obtaining the FFT of any trace associated with the circuit.

A major advantage of the FFT method is that it provides graphical output. But its major disadvantage is that some of the harmonics may be too small to see.

In both DFT and FFT, we should let the simulation run for a large number of cycles and use a small value of *Step Ceiling* (in the Transient dialog box) to ensure accurate results. The *Final Time* in the Transient dialog box should be at least five times the period of the signal to allow the simulation to reach steady state.

Example 17.12

Use *PSpice* to determine the Fourier coefficients of the signal in Fig. 17.1.

Solution:

Figure 17.36 shows the schematic for obtaining the Fourier coefficients. With the signal in Fig. 17.1 in mind, we enter the attributes of the voltage source VPULSE as shown in Fig. 17.36. We will solve this example using both the DFT and FFT approaches.

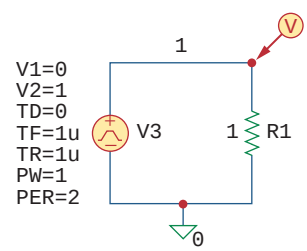


Figure 17.36
Schematic for Example 17.12.

■ **METHOD 1** **DFT Approach:** (The voltage marker in Fig. 17.36 is not needed for this method.) From Fig. 17.1, it is evident that $T = 2$ s,

$$f_0 = \frac{1}{T} = \frac{1}{2} = 0.5 \text{ Hz}$$

So, in the transient dialog box, we select the *Final Time* as $6T = 12$ s, the *Print Step* as 0.01 s, the *Step Ceiling* as 10 ms, the *Center Frequency* as 0.5 Hz, and the output variable as V(1). (In fact, Fig. 17.34 is for this particular example.) When *PSpice* is run, the output file contains the following result:

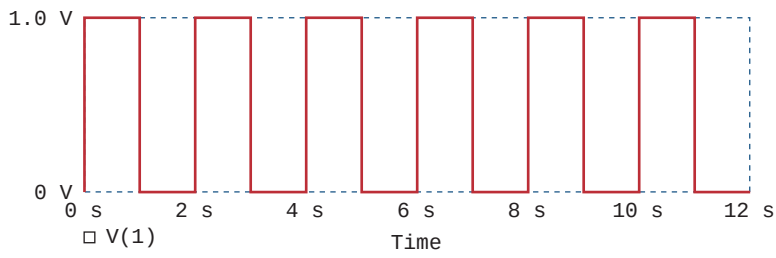
FOURIER COEFFICIENTS OF TRANSIENT RESPONSE V(1)

DC COMPONENT = 4.989950E-01

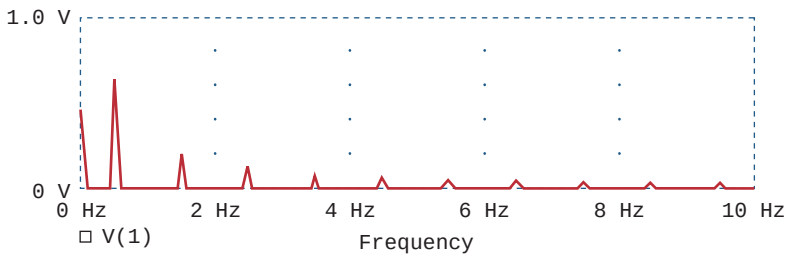
HARMONIC NO	FREQUENCY (HZ)	FOURIER COMPONENT	NORMALIZED COMPONENT	PHASE (DEG)	NORMALIZED PHASE (DEG)
1	5.000E-01	6.366E-01	1.000E+00	-1.809E-01	0.000E+00
2	1.000E+00	2.012E-03	3.160E-03	-9.226E+01	-9.208E+01
3	1.500E+00	2.122E-01	3.333E-01	-5.427E-01	-3.619E-01
4	2.000E+00	2.016E-03	3.167E-03	-9.451E+01	-9.433E+01
5	2.500E+00	1.273E-01	1.999E-01	-9.048E-01	-7.239E-01
6	3.000E+00	2.024E-03	3.180E-03	-9.676E+01	-9.658E+01
7	3.500E+00	9.088E-02	1.427E-01	-1.267E+00	-1.086E+00
8	4.000E+00	2.035E-03	3.197E-03	-9.898E+01	-9.880E+01
9	4.500E+00	7.065E-02	1.110E-01	-1.630E+00	-1.449E+00

Comparing the result with that in Eq. (17.1.7) (see Example 17.1) or with the spectra in Fig. 17.4 shows a close agreement. From Eq. (17.1.7), the dc component is 0.5 while *PSpice* gives 0.498995. Also, the signal has only odd harmonics with phase $\psi_n = -90°$, whereas *PSpice* seems to indicate that the signal has even harmonics although the magnitudes of the even harmonics are small.

■ **METHOD 2** **FFT Approach:** With voltage marker in Fig. 17.36 in place, we run *PSpice* and obtain the waveform V(1) shown in Fig. 17.37(a) on the *PSpice A/D* window. By double clicking the FFT icon in the *PSpice A/D* menu and changing the X-axis setting to 0 to 10 Hz, we obtain the FFT of V(1) as shown in Fig. 17.37(b). The FFT-generated graph contains the dc and harmonic components within the selected frequency range. Notice that the magnitudes and frequencies of the harmonics agree with the DFT-generated tabulated values.



(a)



(b)

Figure 17.37
(a) Original waveform of Fig. 17.1, (b) FFT of the waveform.

Practice Problem 17.12 Obtain the Fourier coefficients of the function in Fig. 17.7 using *PSpice*.

Answer:

```
FOURIER COEFFICIENTS OF TRANSIENT RESPONSE V(1)

DC COMPONENT = 4.950000E-01

HARMONIC    FREQUENCY     FOURIER      NORMALIZED      PHASE       NORMALIZED
   NO         (HZ)       COMPONENT     COMPONENT      (DEG)     PHASE (DEG)

    1      1.000E+00     3.184E-01     1.000E+00   -1.782E+02    0.000E+00
    2      2.000E+00     1.593E-01     5.002E-01   -1.764E+02    1.800E+00
    3      3.000E+00     1.063E-01     3.338E-01   -1.746E+02    3.600E+00
```

(continued)

(continued)

4	4.000E+00	7.979E-02	2.506E-03	-1.728E+02	5.400E+00
5	5.000E+00	6.392E-01	2.008E-01	-1.710E+02	7.200E+00
6	6.000E+00	5.337E-02	1.676E-03	-1.692E+02	9.000E+00
7	7.000E+00	4.584E-02	1.440E-01	-1.674E+02	1.080E+01
8	8.000E+00	4.021E-02	1.263E-01	-1.656E+02	1.260E+01
9	9.000E+00	3.584E-02	1.126E-01	-1.638E+02	1.440E+01

Example 17.13

If $v_s = 12\sin(200\pi t)u(t)$ V in the circuit of Fig. 17.38, find $i(t)$.

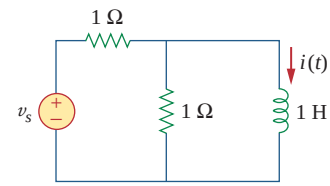


Figure 17.38
For Example 17.13.

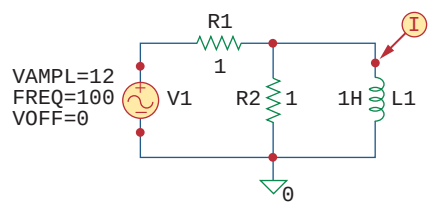


Figure 17.39
Schematic of the circuit in Fig. 17.38.

Solution:

1. **Define.** Although the problem appears to be clearly stated, it might be advisable to check with the individual who assigned the problem to make sure he or she wants the transient response rather than the steady-state response; in the latter case the problem becomes trivial.
2. **Present.** We are to determine the response $i(t)$ given the input $v_s(t)$, using *PSpice* and Fourier analysis.
3. **Alternative.** We will use DFT to perform the initial analysis. We will then check using the FFT approach.
4. **Attempt.** The schematic is shown in Fig. 17.39. We may use the DFT approach to obtain the Fourier coefficents of $i(t)$. Since the period of the input waveform is $T = 1/100 = 10$ ms, in the Transient dialog box we select *Print Step:* 0.1 ms, *Final Time:* 100 ms, *Center Frequency:* 100 Hz, *Number of harmonics:* 4, and *Output Vars:* I(L1). When the circuit is simulated, the output file includes the following:

FOURIER COEFFICIENTS OF TRANSIENT RESPONSE I(VD)

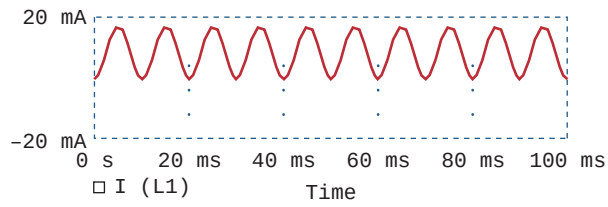
DC COMPONENT = 8.583269E-03

HARMONIC NO	FREQUENCY (HZ)	FOURIER COMPONENT	NORMALIZED COMPONENT	PHASE (DEG)	NORMALIZED PHASE (DEG)
1	1.000E+02	8.730E-03	1.000E+00	-8.984E+01	0.000E+00
2	2.000E+02	1.017E-04	1.165E-02	-8.306E+01	6.783E+00
3	3.000E+02	6.811E-05	7.802E-03	-8.235E+01	7.490E+00
4	4.000E+02	4.403E-05	5.044E-03	-8.943E+01	4.054E+00

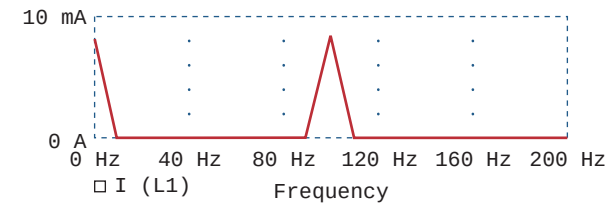
With the Fourier coefficients, the Fourier series describing the current $i(t)$ can be obtained using Eq. (17.73); that is,

$$\begin{aligned} i(t) = {} & 8.5833 + 8.73\sin(2\pi \cdot 100t - 89.84^\circ) \\ & + 0.1017\sin(2\pi \cdot 200t - 83.06^\circ) \\ & + 0.068\sin(2\pi \cdot 300t - 82.35^\circ) + \cdots \text{mA} \end{aligned}$$

5. **Evaluate.** We can also use the FFT approach to cross-check our result. The current marker is inserted at pin 1 of the inductor as shown in Fig. 17.39. Running *PSpice* will automatically produce the plot of I(L1) in the *PSpice A/D* window, as shown in



(a)



(b)

Figure 17.40
For Example 17.13: (a) plot of $i(t)$, (b) the FFT of $i(t)$.

Fig. 17.40(a). By double clicking the FFT icon and setting the range of the X-axis from 0 to 200 Hz, we generate the FFT of I(L1) shown in Fig. 17.40(b). It is clear from the FFT-generated plot that only the dc component and the first harmonic are visible. Higher harmonics are negligibly small.

One final observation, does the answer make sense? Let us look at the actual transient response, $i(t) = (9.549e^{-0.5t} - 9.549)\cos(200\pi t)u(t)$ mA. The period of the cosine wave is 10 ms while the time constant of the exponential is 2000 ms (2 seconds). So, the answer we obtained by Fourier techniques does agree.

6. **Satisfactory?** Clearly, we have solved the problem satisfactorily using the specified approach. We can now present our results as a solution to the problem.

Practice Problem 17.13

A sinusoidal current source of amplitude 4 A and frequency 2 kHz is applied to the circuit in Fig. 17.41. Use *PSpice* to find $v(t)$.

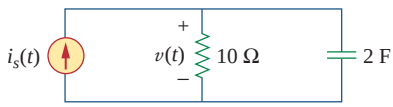


Figure 17.41
For Practice Prob. 17.13.

Answer: $v(t) = -150.72 + 145.5\sin(4\pi \cdot 10^3 t + 90°) + \cdots \mu$V. The Fourier components are shown below:

```
FOURIER COEFFICIENTS OF TRANSIENT RESPONSE V(R1:1)

DC COMPONENT = -1.507169E-04

HARMONIC     FREQUENCY     FOURIER      NORMALIZED      PHASE      NORMALIZED
   NO          (HZ)       COMPONENT     COMPONENT       (DEG)     PHASE (DEG)

    1       2.000E+03     1.455E-04     1.000E+00     9.006E+01      0.000E+00
    2       4.000E+03     1.851E-06     1.273E-02     9.597E+01      5.910E+00
    3       6.000E+03     1.406E-06     9.662E-03     9.323E+01      3.167E+00
    4       8.000E+03     1.010E-06     6.946E-02     8.077E+01     -9.292E+00
```

17.8 †Applications

We demonstrated in Section 17.4 that the Fourier series expansion permits the application of the phasor techniques used in ac analysis to circuits containing nonsinusoidal periodic excitations. The Fourier series has many other practical applications, particularly in communications and signal processing. Typical applications include spectrum analysis, filtering, rectification, and harmonic distortion. We will consider two of these: spectrum analyzers and filters.

17.8.1 Spectrum Analyzers

The Fourier series provides the spectrum of a signal. As we have seen, the spectrum consists of the amplitudes and phases of the harmonics versus frequency. By providing the spectrum of a signal $f(t)$, the Fourier series helps us identify the pertinent features of the signal. It demonstrates which frequencies are playing an important role in the shape of the output and which ones are not. For example, audible sounds have significant components in the frequency range of 20 Hz to 15 kHz, while visible light signals range from 10^5 GHz to 10^6 GHz. Table 17.4 presents some other signals and the frequency ranges of their components. A periodic function is said to be *band-limited* if its amplitude spectrum contains only a finite number of coefficients A_n or c_n. In this case, the Fourier series becomes

$$f(t) = \sum_{n=-N}^{N} c_n e^{jn\omega_0 t} = a_0 + \sum_{n=1}^{N} A_n \cos(n\omega_0 t + \phi_n) \quad \textbf{(17.75)}$$

This shows that we need only $2N + 1$ terms (namely, $a_0, A_1, A_2, \ldots, A_N, \phi_1, \phi_2, \ldots, \phi_N$) to completely specify $f(t)$ if ω_0 is known. This leads to the *sampling theorem:* a band-limited periodic function whose Fourier series contains N harmonics is uniquely specified by its values at $2N + 1$ instants in one period.

A *spectrum analyzer* is an instrument that displays the amplitude of the components of a signal versus frequency. It shows the various frequency components (spectral lines) that indicate the amount of energy at each frequency.

It is unlike an oscilloscope, which displays the entire signal (all components) versus time. An oscilloscope shows the signal in the time domain, while the spectrum analyzer shows the signal in the frequency domain. There is perhaps no instrument more useful to a circuit analyst than the spectrum analyzer. An analyzer can conduct noise and spurious signal analysis, phase checks, electromagnetic interference and filter examinations, vibration measurements, radar measurements, and more. Spectrum analyzers are commercially available in various sizes and shapes. Figure 17.42 displays a typical one.

TABLE 17.4

Frequency ranges of typical signals.

Signal	Frequency Range
Audible sounds	20 Hz to 15 kHz
AM radio	540–1600 kHz
Short-wave radio	3–36 MHz
Video signals (U.S. standards)	dc to 4.2 MHz
VHF television, FM radio	54–216 MHz
UHF television	470–806 MHz
Cellular telephone	824–891.5 MHz
Microwaves	2.4–300 GHz
Visible light	10^5–10^6 GHz
X-rays	10^8–10^9 GHz

17.8.2 Filters

Filters are an important component of electronics and communications systems. Chapter 14 presented a full discussion on passive and active filters. Here, we investigate how to design filters to select the fundamental component (or any desired harmonic) of the input signal and reject other harmonics. This filtering process cannot be accomplished

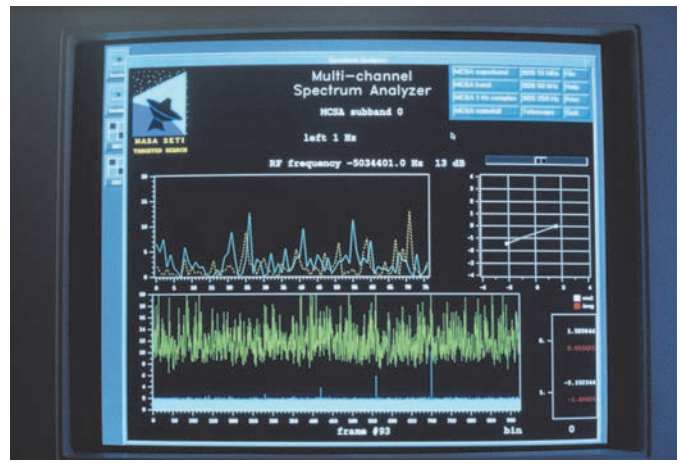


Figure 17.42
A typical spectrum analyzer. © SETI Institute/SPL/Photo Researchers, Inc.

without the Fourier series expansion of the input signal. For the purpose of illustration, we will consider two cases, a lowpass filter and a bandpass filter. In Example 17.6, we already looked at a highpass *RL* filter.

The output of a lowpass filter depends on the input signal, the transfer function $H(\omega)$ of the filter, and the corner or half-power frequency ω_c. We recall that $\omega_c = 1/RC$ for an *RC* passive filter. As shown in Fig. 17.43(a), the lowpass filter passes the dc and low-frequency components, while blocking the high-frequency components. By making ω_c sufficiently large ($\omega_c \gg \omega_0$, e.g., making C small), a large number of the harmonics can be passed. On the other hand, by making ω_c sufficiently small ($\omega_c \ll \omega_0$), we can block out all the ac components and pass only dc, as shown typically in Fig. 17.43(b). (See Fig. 17.2(a) for the Fourier series expansion of the square wave.)

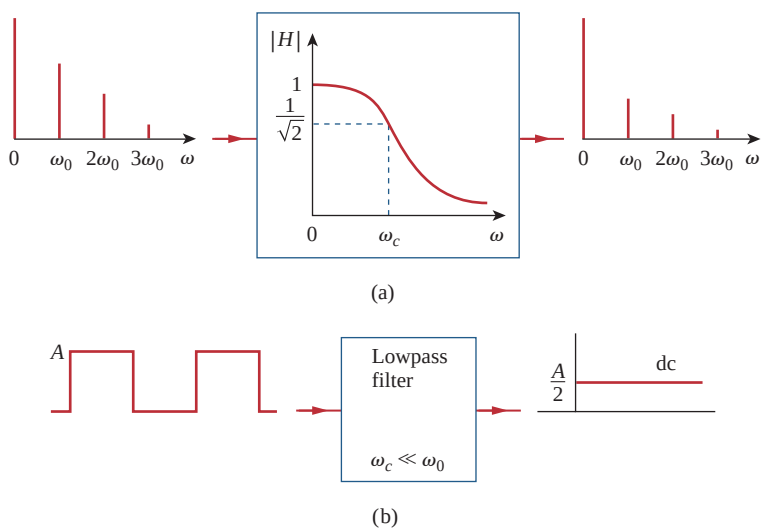


Figure 17.43
(a) Input and output spectra of a lowpass filter, (b) the lowpass filter passes only the dc component when $\omega_c \ll \omega_0$.

Similarly, the output of a bandpass filter depends on the input signal, the transfer function of the filter $H(\omega)$, its bandwidth B, and its center frequency ω_c. As illustrated in Fig. 17.44(a), the filter passes all the harmonics of the input signal within a band of frequencies ($\omega_1 < \omega < \omega_2$) centered around ω_c. We have assumed that ω_0, $2\omega_0$, and $3\omega_0$ are within that band. If the filter is made highly selective ($B \ll \omega_0$) and $\omega_c = \omega_0$, where ω_0 is the fundamental frequency of the input signal, the filter passes only the fundamental component ($n = 1$) of the input and blocks out all higher harmonics. As shown in Fig. 17.44(b), with a square wave as input, we obtain a sine wave of the same frequency as the output. (Again, refer to Fig. 17.2(a).)

In this section, we have used ω_c for the center frequency of the bandpass filter instead of ω_0 as in Chapter 14, to avoid confusing ω_0 with the fundamental frequency of the input signal.

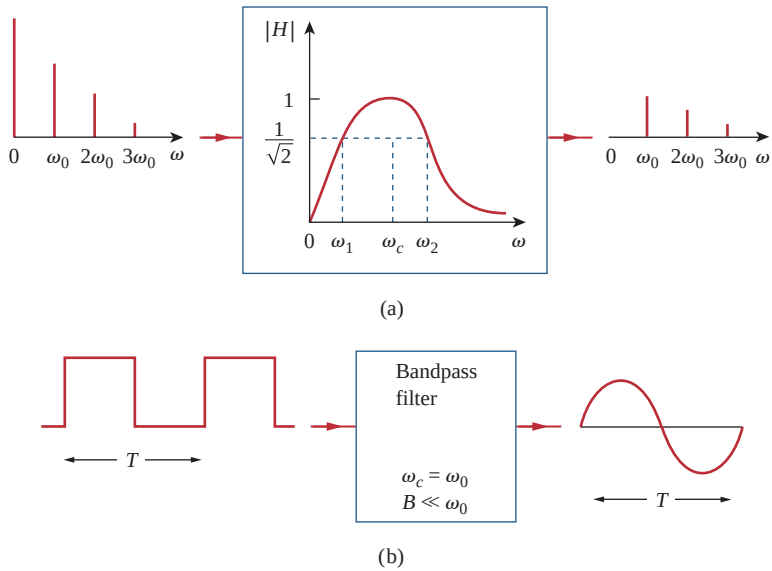


Figure 17.44
(a) Input and output spectra of a bandpass filter, (b) the bandpass filter passes only the fundamental component when $B \ll \omega_0$.

Example 17.14

If the sawtooth waveform in Fig. 17.45(a) is applied to an ideal lowpass filter with the transfer function shown in Fig. 17.45(b), determine the output.

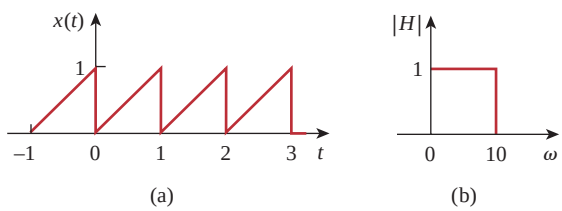


Figure 17.45
For example 17.14.

Solution:
The input signal in Fig. 17.45(a) is the same as the signal in Fig. 17.9. From Practice Prob. 17.2, we know that the Fourier series expansion is

$$x(t) = \frac{1}{2} - \frac{1}{\pi}\sin\omega_0 t - \frac{1}{2\pi}\sin 2\omega_0 t - \frac{1}{3\pi}\sin 3\omega_0 t - \cdots$$

where the period is $T = 1$ s and the fundamental frequency is $\omega_0 = 2\pi$ rad/s. Since the corner frequency of the filter is $\omega_c = 10$ rad/s, only the dc component and harmonics with $n\omega_0 < 10$ will be passed. For $n = 2$, $n\omega_0 = 4\pi = 12.566$ rad/s, which is higher than 10 rad/s, meaning that second and higher harmonics will be rejected. Thus, only the dc and fundamental components will be passed. Hence, the output of the filter is

$$y(t) = \frac{1}{2} - \frac{1}{\pi}\sin 2\pi t$$

Practice Problem 17.14

Rework Example 17.14 if the lowpass filter is replaced by the ideal bandpass filter shown in Fig. 17.46.

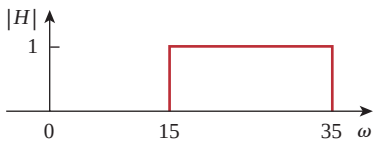


Figure 17.46
For Practice Prob. 17.14.

Answer: $y(t) = -\frac{1}{3\pi}\sin 3\omega_0 t - \frac{1}{4\pi}\sin 4\omega_0 t - \frac{1}{5\pi}\sin 5\omega_0 t.$

17.9 Summary

1. A periodic function is one that repeats itself every T seconds; that is, $f(t \pm nT) = f(t)$, $n = 1, 2, 3, \ldots$.
2. Any nonsinusoidal periodic function $f(t)$ that we encounter in electrical engineering can be expressed in terms of sinusoids using Fourier series:

$$f(t) = \underbrace{a_0}_{\text{dc}} + \underbrace{\sum_{n=1}^{\infty}(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)}_{\text{ac}}$$

where $\omega_0 = 2\pi/T$ is the fundamental frequency. The Fourier series resolves the function into the dc component a_0 and an ac component containing infinitely many harmonically related sinusoids. The Fourier coefficients are determined as

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt, \qquad a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t\,dt$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t\,dt$$

If $f(t)$ is an even function, $b_n = 0$, and when $f(t)$ is odd, $a_0 = 0$ and $a_n = 0$. If $f(t)$ is half-wave symmetric, $a_0 = a_n = b_n = 0$ for even values of n.

3. An alternative to the trigonometric (or sine-cosine) Fourier series is the amplitude-phase form

$$f(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n)$$

where

$$A_n = \sqrt{a_n^2 + b_n^2}, \qquad \phi_n = -\tan^{-1}\frac{b_n}{a_n}$$

4. Fourier series representation allows us to apply the phasor method in analyzing circuits when the source function is a nonsinusoidal periodic function. We use phasor technique to determine the response of each harmonic in the series, transform the responses to the time domain, and add them up.
5. The average-power of periodic voltage and current is

$$P = V_{\text{dc}} I_{\text{dc}} + \frac{1}{2}\sum_{n=1}^{\infty} V_n I_n \cos(\theta_n - \phi_n)$$

In other words, the total average power is the sum of the average powers in each harmonically related voltage and current.

6. A periodic function can also be represented in terms of an exponential (or complex) Fourier series as

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

where

$$c_n = \frac{1}{T}\int_0^T f(t) e^{-jn\omega_0 t}\, dt$$

and $\omega_0 = 2\pi/T$. The exponential form describes the spectrum of $f(t)$ in terms of the amplitude and phase of ac components at positive and negative harmonic frequencies. Thus, there are three basic forms of Fourier series representation: the trigonometric form, the amplitude-phase form, and the exponential form.

7. The frequency (or line) spectrum is the plot of A_n and ϕ_n or $|c_n|$ and $\theta_n$versus frequency.
8. The rms value of a periodic function is given by

$$F_{\text{rms}} = \sqrt{a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty} A_n^2}$$

The power dissipated by a 1-Ω resistance is

$$P_{1\Omega} = F_{\text{rms}}^2 = a_0^2 + \frac{1}{2}\sum_{n=1}^{\infty}(a_n^2 + b_n^2) = \sum_{n=-\infty}^{\infty} |c_n|^2$$

This relationship is known as *Parseval's theorem*.

9. Using *PSpice*, a Fourier analysis of a circuit can be performed in conjunction with the transient analysis.
10. Fourier series find application in spectrum analyzers and filters. The spectrum analyzer is an instrument that displays the discrete Fourier spectra of an input signal, so that an analyst can determine the frequencies and relative energies of the signal's components. Because the Fourier spectra are discrete spectra, filters can be designed for great effectiveness in blocking frequency components of a signal that are outside a desired range.

Review Questions

17.1 Which of the following cannot be a Fourier series?

(a) $t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \frac{t^5}{5}$

(b) $5 \sin t + 3 \sin 2t - 2 \sin 3t + \sin 4t$

(c) $\sin t - 2 \cos 3t + 4 \sin 4t + \cos 4t$

(d) $\sin t + 3 \sin 2.7t - \cos \pi t + 2 \tan \pi t$

(e) $1 + e^{-j\pi t} + \frac{e^{-j2\pi t}}{2} + \frac{e^{-j3\pi t}}{3}$

17.2 If $f(t) = t, 0 < t < \pi, f(t + n\pi) = f(t)$, the value of ω_0 is

(a) 1 (b) 2 (c) π (d) 2π

17.3 Which of the following are even functions?

(a) $t + t^2$ (b) $t^2 \cos t$ (c) e^{t^2}

(d) $t^2 + t^4$ (e) $\sinh t$

17.4 Which of the following are odd functions?

(a) $\sin t + \cos t$ (b) $t \sin t$

(c) $t \ln t$ (d) $t^3 \cos t$

(e) $\sinh t$

17.5 If $f(t) = 10 + 8 \cos t + 4 \cos 3t + 2 \cos 5t + \cdots$, the magnitude of the dc component is:

(a) 10 (b) 8 (c) 4

(d) 2 (e) 0

17.6 If $f(t) = 10 + 8 \cos t + 4 \cos 3t + 2 \cos 5t + \cdots$, the angular frequency of the 6th harmonic is

(a) 12 (b) 11 (c) 9

(d) 6 (e) 1

17.7 The function in Fig. 17.14 is half-wave symmetric.

(a) True (b) False

17.8 The plot of $|c_n|$ versus $n\omega_0$ is called:

(a) complex frequency spectrum

(b) complex amplitude spectrum

(c) complex phase spectrum

17.9 When the periodic voltage $2 + 6 \sin \omega_0 t$ is applied to a 1-Ω resistor, the integer closest to the power (in watts) dissipated in the resistor is:

(a) 5 (b) 8 (c) 20

(d) 22 (e) 40

17.10 The instrument for displaying the spectrum of a signal is known as:

(a) oscilloscope (b) spectrogram

(c) spectrum analyzer (d) Fourier spectrometer

Answers: 17.1a,d, 17.2b, 17.3b,c,d, 17.4d,e, 17.5a, 17.6d, 17.7a, 17.8b, 17.9d, 17.10c.

Problems

Section 17.2 Trigonometric Fourier Series

17.1 Evaluate each of the following functions and see if it is periodic. If periodic, find its period.

(a) $f(t) = \cos \pi t + 2 \cos 3\pi t + 3 \cos 5\pi t$

(b) $y(t) = \sin t + 4 \cos 2\pi t$

(c) $g(t) = \sin 3t \cos 4t$

(d) $h(t) = \cos^2 t$

(e) $z(t) = 4.2 \sin(0.4\pi t + 10°) + 0.8 \sin(0.6\pi t + 50°)$

(f) $p(t) = 10$

(g) $q(t) = e^{-\pi t}$

17.2 Using MATLAB, synthesize the periodic waveform for which the Fourier trigonometric Fourier series is

$$f(t) = \frac{1}{2} - \frac{4}{\pi^2}\left(\cos t + \frac{1}{9}\cos 3t + \frac{1}{25}\cos 5t + \cdots\right)$$

17.3 Give the Fourier coefficients a_0, a_n, and b_n of the waveform in Fig. 17.47. Plot the amplitude and phase spectra.

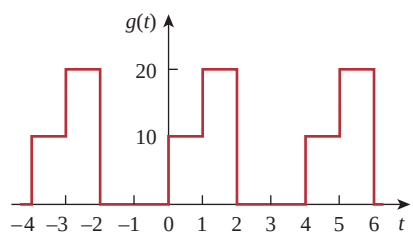


Figure 17.47
For Prob. 17.3.

17.4 Find the Fourier series expansion of the backward sawtooth waveform of Fig. 17.48. Obtain the amplitude and phase spectra.

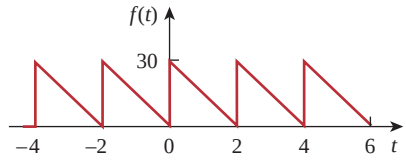
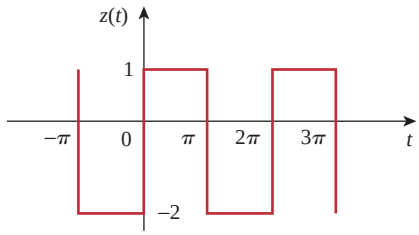


Figure 17.48
For Probs. 17.4 and 17.66.

17.5 Obtain the Fourier series expansion for the waveform shown in Fig. 17.49.



(a)

Figure 17.49
For Prob. 17.5.

17.6 Find the trigonometric Fourier series for

$$f(t) = \begin{cases} 20, & 0 < t < \pi \\ 40, & \pi < t < 2\pi \end{cases} \quad \text{and} \quad f(t + 2\pi) = f(t).$$

***17.7** Determine the Fourier series of the periodic function in Fig. 17.50.

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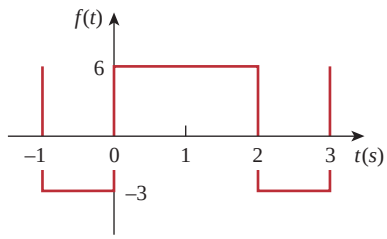


Figure 17.50
For Prob. 17.7.

17.8 Using Fig. 17.51, design a problem to help other students better understand how to determine the exponention Fourier series from a periodic wave shape.

ezd

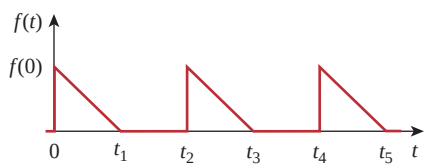


Figure 17.51
For Prob. 17.8.

* An asterisk indicates a challenging problem.

17.9 Determine the Fourier coefficients a_n and b_n of the first three harmonic terms of the rectified cosine wave in Fig. 17.52.

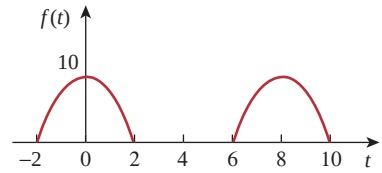


Figure 17.52
For Prob. 17.9.

17.10 Find the exponential Fourier series for the waveform in Fig. 17.53.

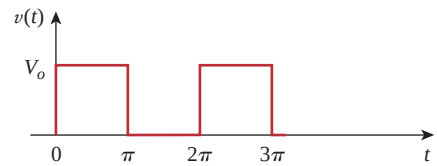


Figure 17.53
For Prob. 17.10.

17.11 Obtain the exponential Fourier series for the signal in Fig. 17.54.

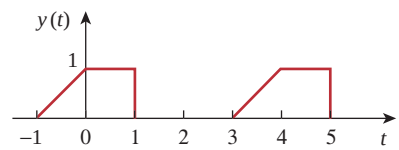


Figure 17.54
For Prob. 17.11.

***17.12** A voltage source has a periodic waveform defined over its period as

$$v(t) = t(2\pi - t)\ \text{V}, \qquad 0 < t < 2\pi$$

Find the Fourier series for this voltage.

17.13 Design a problem to help other students better understand obtaining the Fourier series from a periodic function.

ezd

17.14 Find the quadrature (cosine and sine) form of the Fourier series

$$f(t) = 2 + \sum_{n=1}^{\infty} \frac{10}{n^3 + 1} \cos\left(2nt + \frac{n\pi}{4}\right)$$

17.15 Express the Fourier series

$$f(t) = 10 + \sum_{n=1}^{\infty} \frac{4}{n^2 + 1} \cos 10nt + \frac{1}{n^3} \sin 10nt$$

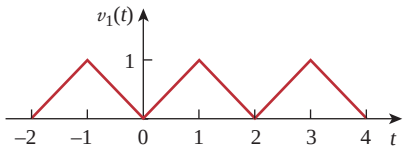
(a) in a cosine and angle form,

(b) in a sine and angle form.

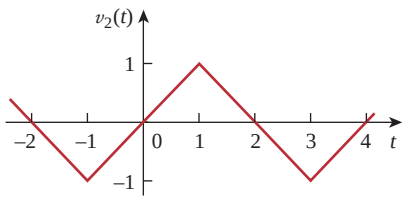
17.16 The waveform in Fig. 17.55(a) has the following Fourier series:

$$v_1(t) = \frac{1}{2} - \frac{4}{\pi^2}\left(\cos \pi t + \frac{1}{9}\cos 3\pi t + \frac{1}{25}\cos 5\pi t + \cdots\right) \text{V}$$

Obtain the Fourier series of $v_2(t)$ in Fig. 17.55(b).



(a)



(b)

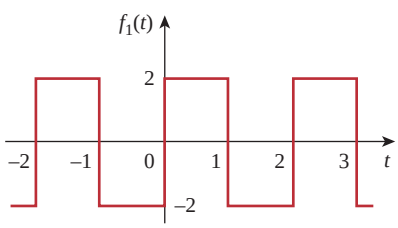
Figure 17.55
For Probs. 17.16 and 17.69.

Section 17.3 Symmetry Considerations

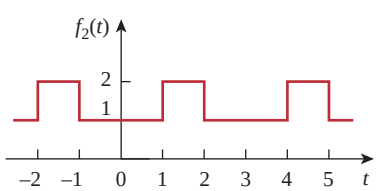
17.17 Determine if these functions are even, odd, or neither.

(a) $1 + t$ (b) $t^2 - 1$ (c) $\cos n\pi t \sin n\pi t$
(d) $\sin^2 \pi t$ (e) e^{-t}

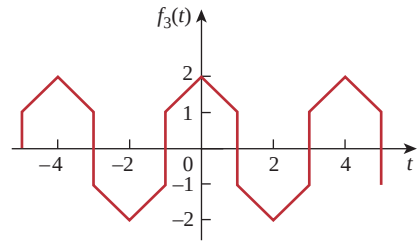
17.18 Determine the fundamental frequency and specify the type of symmetry present in the functions in Fig. 17.56.



(a)



(b)



(c)

Figure 17.56
For Probs. 17.18 and 17.63.

17.19 Obtain the Fourier series for the periodic waveform in Fig. 17.57.

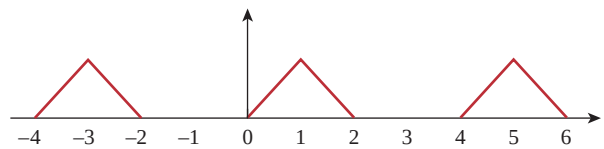


Figure 17.57
For Prob. 17.19.

17.20 Find the Fourier series for the signal in Fig. 17.58. Evaluate $f(t)$ at $t = 2$ using the first three nonzero harmonics.

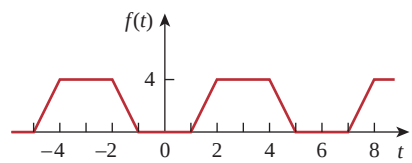


Figure 17.58
For Probs. 17.20 and 17.67.

17.21 Determine the trigonometric Fourier series of the signal in Fig. 17.59.

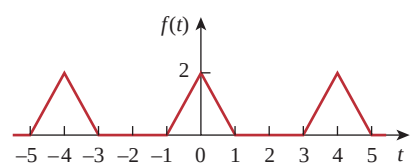


Figure 17.59
For Prob. 17.21.

17.22 Calculate the Fourier coefficients for the function in Fig. 17.60.

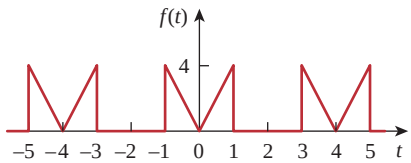


Figure 17.60
For Prob. 17.22.

17.23 Using Fig. 17.61, design a problem to help other students better understand finding the Fourier series of a periodic wave shape.

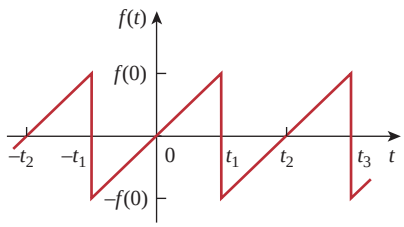


Figure 17.61
For Prob. 17.23.

17.24 In the periodic function of Fig. 17.62,

(a) find the trigonometric Fourier series coefficients a_2 and b_2,

(b) calculate the magnitude and phase of the component of $f(t)$ that has $\omega_n = 10$ rad/s,

(c) use the first four nonzero terms to estimate $f(\pi/2)$,

(d) show that

$$\frac{\pi}{4} = \frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \cdots$$

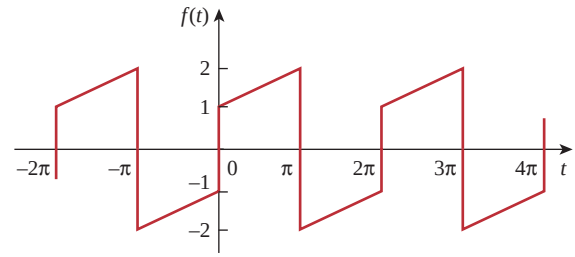


Figure 17.62
For Probs. 17.24 and 17.60.

17.25 Determine the Fourier series representation of the function in Fig. 17.63.

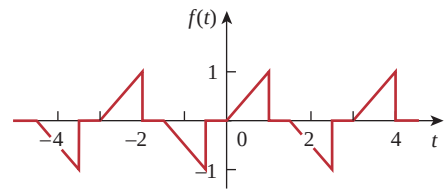


Figure 17.63
For Prob. 17.25.

17.26 Find the Fourier series representation of the signal shown in Fig. 17.64.

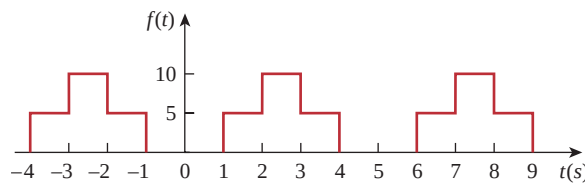


Figure 17.64
For Prob. 17.26.

17.27 For the waveform shown in Fig. 17.65 below,

(a) specify the type of symmetry it has,

(b) calculate a_3 and b_3,

(c) find the rms value using the first five nonzero harmonics.

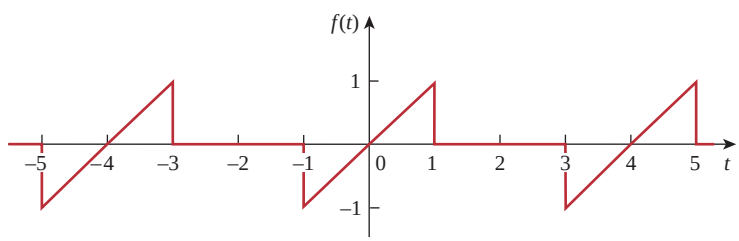


Figure 17.65
For Prob. 17.27.

17.28 Obtain the trigonometric Fourier series for the voltage waveform shown in Fig. 17.66.

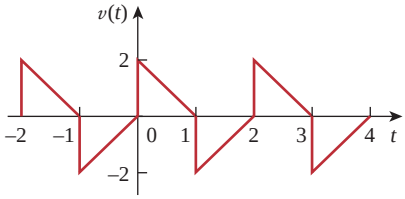


Figure 17.66
For Prob. 17.28.

17.29 Determine the Fourier series expansion of the sawtooth function in Fig. 17.67.

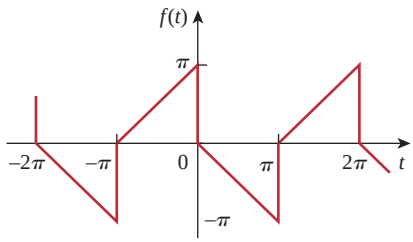


Figure 17.67
For Prob. 17.29.

17.30 (a) If $f(t)$ is an even function, show that

$$c_n = \frac{2}{T}\int_0^{T/2} f(t)\cos n\omega_o t\, dt$$

(b) If $f(t)$ is an odd function, show that

$$c_n = -\frac{j2}{T}\int_0^{T/2} f(t)\sin n\omega_o t\, dt$$

17.31 Let a_n and b_n be the Fourier series coefficients of $f(t)$ and let ω_o be its fundamental frequency. Suppose $f(t)$ is time-scaled to give $h(t) = f(\alpha t)$. Express the a_n' and b_n', and ω_o', of $h(t)$ in terms of a_n, b_n, and ω_o of $f(t)$.

Section 17.4 Circuit Applications

17.32 Find $i(t)$ in the circuit of Fig. 17.68 given that

$$i_s(t) = 1 + \sum_{n=1}^{\infty}\frac{1}{n^2}\cos 3nt\ \text{A}$$

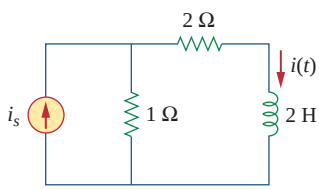


Figure 17.68
For Prob. 17.32.

17.33 In the circuit shown in Fig. 17.69, the Fourier series expansion of $v_s(t)$ is

$$v_s(t) = 3 + \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin(n\pi t)$$

Find $v_o(t)$.

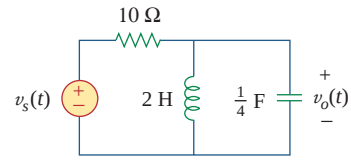


Figure 17.69
For Prob. 17.33.

17.34 Using Fig. 17.70, design a problem to help other students better understand circuit responses to a Fourier series.

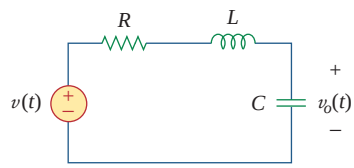


Figure 17.70
For Prob. 17.34.

17.35 If v_s in the circuit of Fig. 17.71 is the same as function $f_2(t)$ in Fig. 17.56(b), determine the dc component and the first three nonzero harmonics of $v_o(t)$.

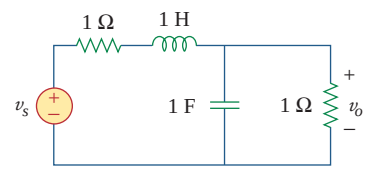
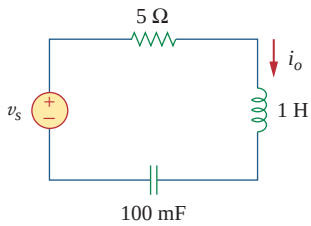
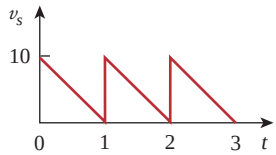


Figure 17.71
For Prob. 17.35.

***17.36** Find the response i_o for the circuit in Fig. 17.72(a), where $v_s(t)$ is shown in Fig. 17.72(b).



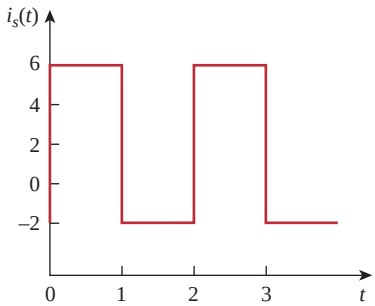
(a)



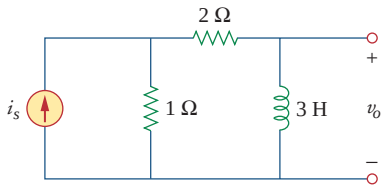
(b)

Figure 17.72
For Prob. 17.36.

17.37 If the periodic current waveform in Fig. 17.73(a) is applied to the circuit in Fig. 17.73(b), find v_o.



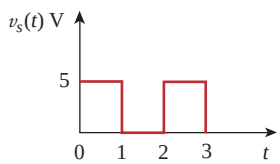
(a)



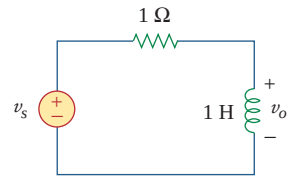
(b)

Figure 17.73
For Prob. 17.37.

17.38 If the square wave shown in Fig. 17.74(a) is applied to the circuit in Fig. 17.74(b), find the Fourier series for $v_o(t)$.



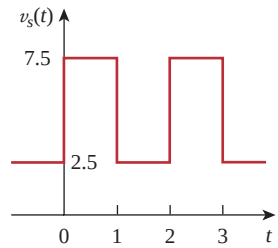
(a)



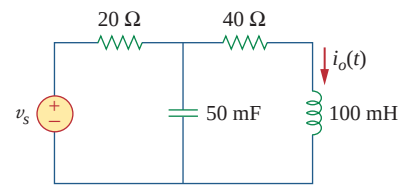
(b)

Figure 17.74
For Prob. 17.38.

17.39 If the periodic voltage in Fig. 17.75(a) is applied to the circuit in Fig. 17.75(b), find $i_o(t)$.



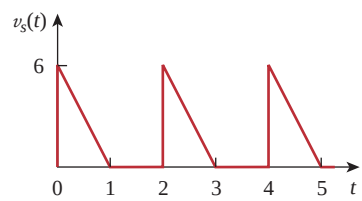
(a)



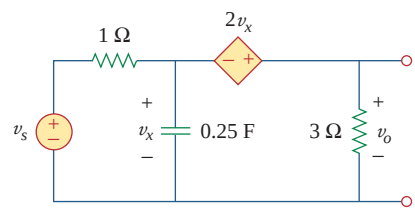
(b)

Figure 17.75
For Prob. 17.39.

***17.40** The signal in Fig. 17.76(a) is applied to the circuit in Fig. 17.76(b). Find $v_o(t)$.



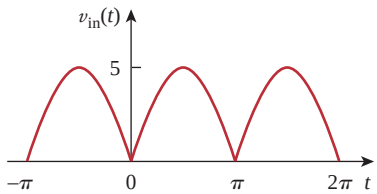
(a)



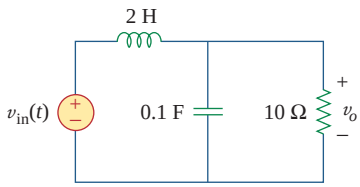
(b)

Figure 17.76
For Prob. 17.40.

17.41 The full-wave rectified sinusoidal voltage in Fig. 17.77(a) is applied to the lowpass filter in Fig. 17.77(b). Obtain the output voltage $v_o(t)$ of the filter.



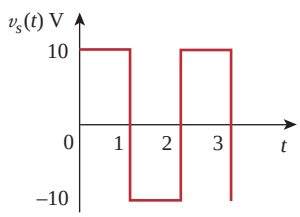
(a)



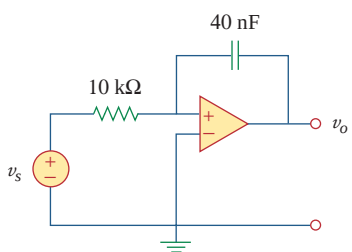
(b)

Figure 17.77
For Prob. 17.41.

17.42 The square wave in Fig. 17.78(a) is applied to the circuit in Fig. 17.78(b). Find the Fourier series of $v_o(t)$.



(a)



(b)

Figure 17.78
For Prob. 17.42.

Section 17.5 Average Power and RMS Values

17.43 The voltage across the terminals of a circuit is

$$v(t) = 30 + 20\cos(60\pi t + 45^\circ) + 10\cos(60\pi t - 45^\circ)\ \text{V}$$

If the current entering the terminal at higher potential is

$$i(t) = 6 + 4\cos(60\pi t + 10^\circ) - 2\cos(120\pi t - 60^\circ)\ \text{A}$$

find:

(a) the rms value of the voltage,

(b) the rms value of the current,

(c) the average power absorbed by the circuit.

***17.44** Design a problem to help other students better understand how to find the rms voltage across and the rms current through an electrical element given a Fourier series for both the current and the voltage. In addition, have them calculate the average power delivered to the element and the power spectrum.

17.45 A series *RLC* circuit has $R = 10\ \Omega$, $L = 2$ mH, and $C = 40\ \mu$F. Determine the effective current and average power absorbed when the applied voltage is

$$v(t) = 100\cos 1000t + 50\cos 2000t + 25\cos 3000t\ \text{V}$$

17.46 Use *MATLAB* to plot the following sinusoids for $0 < t < 5$:

(a) $5\cos 3t - 2\cos(3t - \pi/3)$

(b) $8\sin(\pi t + \pi/4) + 10\cos(\pi t - \pi/8)$

17.47 The periodic current waveform in Fig. 17.79 is applied across a 2-kΩ resistor. Find the percentage of the total average power dissipation caused by the dc component.

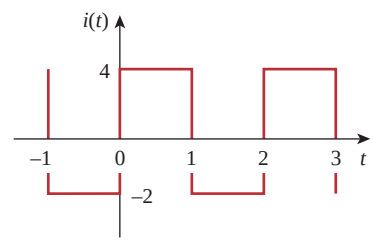


Figure 17.79
For Prob. 17.47.

17.48 For the circuit in Fig. 17.80,

$$i(t) = 20 + 16\cos(10t + 45^\circ) + 12\cos(20t - 60^\circ)\ \text{mA}$$

(a) find $v(t)$, and

(b) calculate the average power dissipated in the resistor.

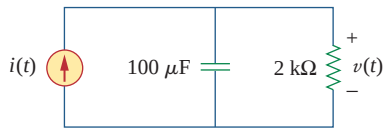


Figure 17.80
For Prob. 17.48.

17.49 (a) For the periodic waveform in Prob. 17.5, find the rms value.

(b) Use the first five harmonic terms of the Fourier series in Prob. 17.5 to determine the effective value of the signal.

(c) Calculate the percentage error in the estimated rms value of $z(t)$ if

$$\%\ \text{error} = \left(\frac{\text{estimated value}}{\text{exact value}} - 1\right) \times 100$$

Section 17.6 Exponential Fourier Series

17.50 Obtain the exponential Fourier series for $f(t) = 2t$, $-1 < t < 1$, with $f(t + 2n) = f(t)$ for all integer values of n.

17.51 Design a problem to help other students better understand how to find the exponential Fourier series of a given periodic function.

17.52 Calculate the complex Fourier series for $f(t) = e^t$, $-\pi < t < \pi$, with $f(t + 2\pi n) = f(t)$ for all integer values of n.

17.53 Find the complex Fourier series for $f(t) = e^{-t}$, $0 < t < 1$, with $f(t + n) = f(t)$ for all integer values of n.

17.54 Find the exponential Fourier series for the function in Fig. 17.81.

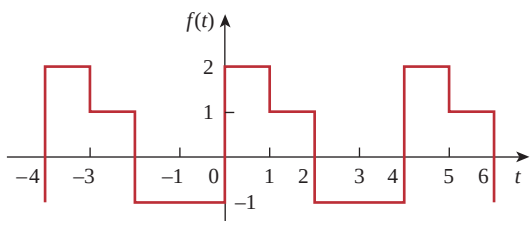


Figure 17.81
For Prob. 17.54.

17.55 Obtain the exponential Fourier series expansion of the half-wave rectified sinusoidal current of Fig. 17.82.

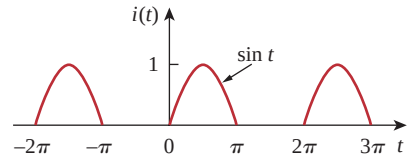


Figure 17.82
For Prob. 17.55.

17.56 The Fourier series trigonometric representation of a periodic function is

$$f(t) = 10 + \sum_{n=1}^{\infty}\left(\frac{1}{n^2+1}\cos n\pi t + \frac{n}{n^2+1}\sin n\pi t\right)$$

Find the exponential Fourier series representation of $f(t)$.

17.57 The coefficients of the trigonometric Fourier series representation of a function are:

$$b_n = 0, \qquad a_n = \frac{6}{n^3 - 2}, \qquad n = 0, 1, 2, \ldots$$

If $\omega_n = 50n$, find the exponential Fourier series for the function.

17.58 Find the exponential Fourier series of a function that has the following trigonometric Fourier series coefficients:

$$a_0 = \frac{\pi}{4}, \qquad b_n = \frac{(-1)^n}{n}, \qquad a_n = \frac{(-1)^n - 1}{\pi n^2}$$

Take $T = 2\pi$.

17.59 The complex Fourier series of the function in Fig. 17.83(a) is

$$f(t) = \frac{1}{2} - \sum_{n=-\infty}^{\infty}\frac{je^{-j(2n+1)t}}{(2n+1)\pi}$$

Find the complex Fourier series of the function $h(t)$ in Fig. 17.83(b).

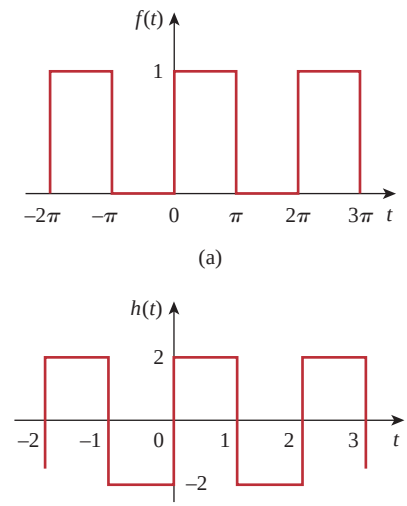


Figure 17.83
For Prob.17.59.

17.60 Obtain the complex Fourier coefficients of the signal in Fig. 17.62.

17.61 The spectra of the Fourier series of a function are shown in Fig. 17.84. (a) Obtain the trigonometric Fourier series. (b) Calculate the rms value of the function.

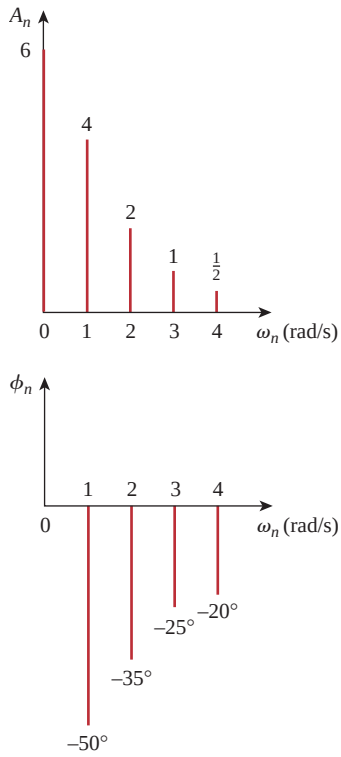
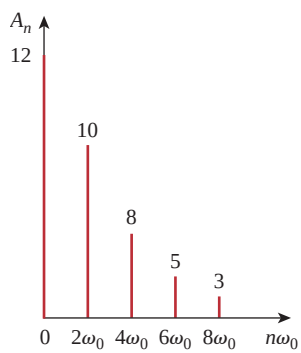


Figure 17.84
For Prob. 17.61.

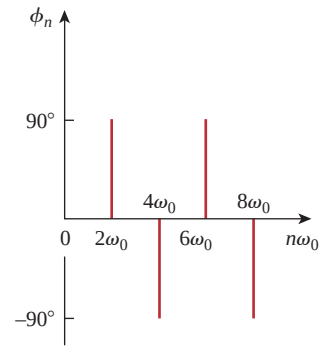
17.62 The amplitude and phase spectra of a truncated Fourier series are shown in Fig. 17.85.

(a) Find an expression for the periodic voltage using the amplitude-phase form. See Eq. (17.10).

(b) Is the voltage an odd or even function of t?



(a)



(b)

Figure 17.85
For Prob. 17.62.

17.63 Plot the amplitude spectrum for the signal $f_2(t)$ in Fig. 17.56(b). Consider the first five terms.

17.64 Design a problem to help other students better understand the amplitude and phase spectra of a given Fourier series.

17.65 Given that

$$f(t) = \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \left(\frac{20}{n^2\pi^2} \cos 2nt - \frac{3}{n\pi} \sin 2nt \right)$$

plot the first five terms of the amplitude and phase spectra for the function.

Section 17.7 Fourier Analysis with *PSpice*

17.66 Determine the Fourier coefficients for the waveform in Fig. 17.48 using *PSpice*.

17.67 Calculate the Fourier coefficients of the signal in Fig. 17.58 using *PSpice*.

17.68 Use *PSpice* to find the Fourier components of the signal in Prob. 17.7.

17.69 Use *PSpice* to obtain the Fourier coefficients of the waveform in Fig. 17.55(a).

17.70 Design a problem to help other students better understand how to use *PSpice* to solve circuit problems with periodic inputs.

17.71 Use *PSpice* to solve Prob. 17.39.

Section 17.8 Applications

17.72 The signal displayed by a medical device can be approximated by the waveform shown in Fig. 17.86. Find the Fourier series representation of the signal.

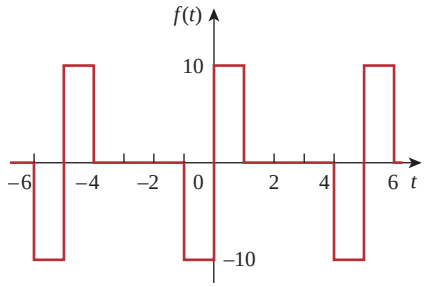


Figure 17.86
For Prob. 17.72.

17.73 A spectrum analyzer indicates that a signal is made up of three components only: 640 kHz at 2 V, 644 kHz at 1 V, 636 kHz at 1 V. If the signal is applied across a 10-Ω resistor, what is the average power absorbed by the resistor?

17.74 A certain band-limited periodic current has only three frequencies in its Fourier series representation: dc, 50 Hz, and 100 Hz. The current may be represented as

$$i(t) = 4 + 6\sin 100\pi t + 8\cos 100\pi t - 3\sin 200\pi t - 4\cos 200\pi t \text{ A}$$

(a) Express $i(t)$ in amplitude-phase form.

(b) If $i(t)$ flows through a 2-Ω resistor, how many watts of average power will be dissipated?

17.75 Design a lowpass *RC* filter with a resistance $R = 2\ k\Omega$. The input to the filter is a periodic rectangular pulse train (see Table 17.3) with $A = 1\ V$, $T = 10$ ms, and $\tau = 1$ ms. Select C such that the dc component of the output is 50 times greater than the fundamental component of the output.

17.76 A periodic signal given by $v_s(t) = 10$ V for $0 < t < 1$ and 0 V for $1 < t < 2$ is applied to the highpass filter in Fig. 17.87. Determine the value of R such that the output signal $v_o(t)$ has an average power of at least 70 percent of the average power of the input signal.

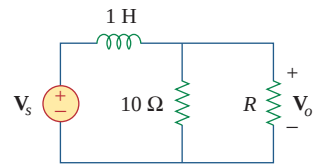


Figure 17.87
For Prob. 17.76.

Comprehensive Problems

17.77 The voltage across a device is given by

$$v(t) = -2 + 10\cos 4t + 8\cos 6t + 6\cos 8t - 5\sin 4t - 3\sin 6t - \sin 8t \text{ V}$$

Find:

(a) the period of $v(t)$,

(b) the average value of $v(t)$,

(c) the effective value of $v(t)$.

17.78 A certain band-limited periodic voltage has only three harmonics in its Fourier series representation. The harmonics have the following rms values: fundamental 40 V, third harmonic 20 V, fifth harmonic 10 V.

(a) If the voltage is applied across a 5-Ω resistor, find the average power dissipated by the resistor.

(b) If a dc component is added to the periodic voltage and the measured power dissipated increases by 5 percent, determine the value of the dc component added.

17.79 Write a program to compute the Fourier coefficients (up to the 10th harmonic) of the square wave in Table 17.3 with $A = 10$ and $T = 2$.

17.80 Write a computer program to calculate the exponential Fourier series of the half-wave rectified sinusoidal current of Fig. 17.82. Consider terms up to the 10th harmonic.

17.81 Consider the full-wave rectified sinusoidal current in Table 17.3. Assume that the current is passed through a 1-Ω resistor.

(a) Find the average power absorbed by the resistor.

(b) Obtain c_n for $n = 1, 2, 3$, and 4.

(c) What fraction of the total power is carried by the dc component?

(d) What fraction of the total power is carried by the second harmonic ($n = 2$)?

17.82 A band-limited voltage signal is found to have the complex Fourier coefficients presented in the table below. Calculate the average power that the signal would supply a 4-Ω resistor.

$n\omega_0$	$\lvert c_n\rvert$	θ_n
0	10.0	0°
ω	8.5	15°
2ω	4.2	30°
3ω	2.1	45°
4ω	0.5	60°
5ω	0.2	75°

chapter

18

Fourier Transform

Planning is doing today to make us better tomorrow because the future belongs to those who make the hard decisions today.

—Business Week

Enhancing Your Skills and Your Career

Career in Communications Systems

Communications systems apply the principles of circuit analysis. A communication system is designed to convey information from a source (the transmitter) to a destination (the receiver) via a channel (the propagation medium). Communications engineers design systems for transmitting and receiving information. The information can be in the form of voice, data, or video.

We live in the information age—news, weather, sports, shopping, financial, business inventory, and other sources make information available to us almost instantly via communications systems. Some obvious examples of communications systems are the telephone network, mobile cellular telephones, radio, cable TV, satellite TV, fax, and radar. Mobile radio, used by police and fire departments, aircraft, and various businesses is another example.

The field of communications is perhaps the fastest growing area in electrical engineering. The merging of the communications field with computer technology in recent years has led to digital data communications networks such as local area networks, metropolitan area networks, and broadband integrated services digital networks. For example, the Internet (the "information superhighway") allows educators, business people, and others to send electronic mail from their computers worldwide, log onto remote databases, and transfer files. The Internet has hit the world like a tidal wave and is drastically changing the way people do business, communicate, and get information. This trend will continue.

A communications engineer designs systems that provide high-quality information services. The systems include hardware for generating, transmitting, and receiving information signals. Communications engineers are employed in numerous communications industries and places where communications systems are routinely used. More and more government agencies, academic departments, and businesses are demanding faster and more accurate transmission of information. To meet these needs, communications engineers are in high demand. Therefore, the future is in communications and every electrical engineer must prepare accordingly.



Photo by Charles Alexander

18.1 Introduction

Fourier series enable us to represent a periodic function as a sum of sinusoids and to obtain the frequency spectrum from the series. The Fourier transform allows us to extend the concept of a frequency spectrum to nonperiodic functions. The transform assumes that a nonperiodic function is a periodic function with an infinite period. Thus, the Fourier transform is an integral representation of a nonperiodic function that is analogous to a Fourier series representation of a periodic function.

The Fourier transform is an *integral transform* like the Laplace transform. It transforms a function in the time domain into the frequency domain. The Fourier transform is very useful in communications systems and digital signal processing, in situations where the Laplace transform does not apply. While the Laplace transform can only handle circuits with inputs for $t > 0$ with initial conditions, the Fourier transform can handle circuits with inputs for $t < 0$ as well as those for $t > 0$.

We begin by using a Fourier series as a stepping stone in defining the Fourier transform. Then we develop some of the properties of the Fourier transform. Next, we apply the Fourier transform in analyzing circuits. We discuss Parseval's theorem, compare the Laplace and Fourier transforms, and see how the Fourier transform is applied in amplitude modulation and sampling.

18.2 Definition of the Fourier Transform

We saw in the previous chapter that a nonsinusoidal periodic function can be represented by a Fourier series, provided that it satisfies the Dirichlet conditions. What happens if a function is not periodic? Unfortunately, there are many important nonperiodic functions—such as a unit step or an exponential function—that we cannot represent by a Fourier series. As we shall see, the Fourier transform allows a transformation from the time to the frequency domain, even if the function is not periodic.

Suppose we want to find the Fourier transform of a nonperiodic function $p(t)$, shown in Fig. 18.1(a). We consider a periodic function $f(t)$ whose shape over one period is the same as $p(t)$, as shown in Fig. 18.1(b). If we let the period $T \to \infty$, only a single pulse of width τ [the desired nonperiodic function in Fig. 18.1(a)] remains, because the adjacent pulses have been moved to infinity. Thus, the function $f(t)$ is no longer periodic. In other words, $f(t) = p(t)$ as $T \to \infty$. It is interesting to consider the spectrum of $f(t)$ for $A = 10$ and $\tau = 0.2$ (see Section 17.6). Figure 18.2 shows the effect of increasing T on the spectrum. First, we notice that the general shape of the spectrum remains the same, and the frequency at which the envelope first becomes zero remains the same. However, the amplitude of the spectrum and the spacing between adjacent components both decrease, while the number of harmonics increases. Thus, over a range of frequencies, the sum of the amplitudes of the harmonics remains almost constant. As the total "strength" or energy of the

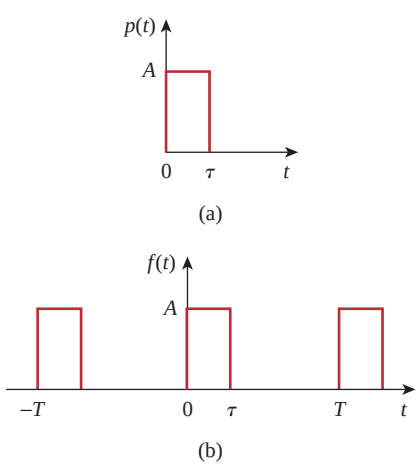


Figure 18.1
(a) A nonperiodic function, (b) increasing T to infinity makes $f(t)$ become the nonperiodic function in (a).

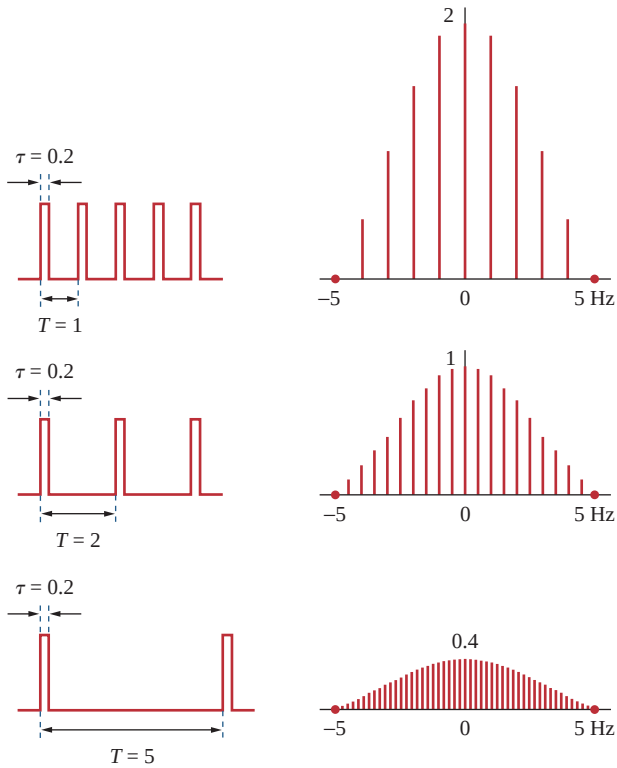


Figure 18.2
Effect of increasing T on the spectrum of the periodic pulse trains in Fig. 18.1(b).
L. Balmer, *Signals and Systems: An Introduction* [London: Prentice-Hall, 1991], p. 229.

components within a band must remain unchanged, the amplitudes of the harmonics must decrease as T increases. Since $f = 1/T$, as T increases, f or ω decreases, so that the discrete spectrum ultimately becomes continuous.

To further understand this connection between a nonperiodic function and its periodic counterpart, consider the exponential form of a Fourier series in Eq. (17.58), namely,

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \tag{18.1}$$

where

$$c_n = \frac{1}{T}\int_{-T/2}^{T/2} f(t)e^{-jn\omega_0 t}\,dt \tag{18.2}$$

The fundamental frequency is

$$\omega_0 = \frac{2\pi}{T} \tag{18.3}$$

and the spacing between adjacent harmonics is

$$\Delta\omega = (n+1)\omega_0 - n\omega_0 = \omega_0 = \frac{2\pi}{T} \tag{18.4}$$

Substituting Eq. (18.2) into Eq. (18.1) gives

$$\begin{aligned} f(t) &= \sum_{n=-\infty}^{\infty} \left[\frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t}\, dt \right] e^{jn\omega_0 t} \\ &= \sum_{n=-\infty}^{\infty} \left[\frac{\Delta\omega}{2\pi} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t}\, dt \right] e^{jn\omega_0 t} \\ &= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \left[\int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t}\, dt \right] \Delta\omega e^{jn\omega_0 t} \end{aligned} \tag{18.5}$$

If we let $T \to \infty$, the summation becomes integration, the incremental spacing $\Delta\omega$ becomes the differential separation $d\omega$, and the discrete harmonic frequency $n\omega_0$ becomes a continuous frequency ω. Thus, as $T \to \infty$,

$$\begin{aligned} \sum_{n=-\infty}^{\infty} &\Rightarrow \int_{-\infty}^{\infty} \\ \Delta\omega &\Rightarrow d\omega \\ n\omega_0 &\Rightarrow \omega \end{aligned} \tag{18.6}$$

so that Eq. (18.5) becomes

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) e^{-j\omega t}\, dt \right] e^{j\omega t}\, d\omega \tag{18.7}$$

Some authors use $F(j\omega)$ instead of $F(\omega)$ to represent the Fourier transform.

The term in the brackets is known as the *Fourier transform* of $f(t)$ and is represented by $F(\omega)$. Thus,

$$F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t}\, dt \tag{18.8}$$

where $\mathcal{F}$ is the Fourier transform operator. It is evident from Eq. (18.8) that:

The **Fourier transform** is an integral transformation of $f(t)$ from the time domain to the frequency domain.

In general, $F(\omega)$ is a complex function; its magnitude is called the *amplitude spectrum*, while its phase is called the *phase spectrum*. Thus $F(\omega)$ is the *spectrum*.

Equation (18.7) can be written in terms of $F(\omega)$, and we obtain the *inverse Fourier transform* as

$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t}\, d\omega \tag{18.9}$$

The function $f(t)$ and its transform $F(\omega)$ form the Fourier transform pairs:

$$f(t) \quad \Leftrightarrow \quad F(\omega) \tag{18.10}$$

since one can be derived from the other.

The Fourier transform $F(\omega)$ exists when the Fourier integral in Eq. (18.8) converges. A sufficient but not necessary condition that $f(t)$ has a Fourier transform is that it be completely integrable in the sense that

$$\int_{-\infty}^{\infty} |f(t)|\,dt < \infty \tag{18.11}$$

For example, the Fourier transform of the unit ramp function $tu(t)$ does not exist, because the function does not satisfy the condition above.

To avoid the complex algebra that explicitly appears in the Fourier transform, it is sometimes expedient to temporarily replace $j\omega$ with s and then replace s with $j\omega$ at the end.

Example 18.1

Find the Fourier transform of the following functions: (a) $\delta(t - t_0)$, (b) $e^{j\omega_0 t}$, (c) $\cos\omega_0 t$.

Solution:

(a) For the impulse function,

$$F(\omega) = \mathcal{F}[\delta(t - t_0)] = \int_{-\infty}^{\infty} \delta(t - t_0)e^{-j\omega t}\,dt = e^{-j\omega t_0} \tag{18.1.1}$$

where the sifting property of the impulse function in Eq. (7.32) has been applied. For the special case $t_0 = 0$, we obtain

$$\mathcal{F}[\delta(t)] = 1 \tag{18.1.2}$$

This shows that the magnitude of the spectrum of the impulse function is constant; that is, all frequencies are equally represented in the impulse function.

(b) We can find the Fourier transform of $e^{j\omega_0 t}$ in two ways. If we let

$$F(\omega) = \delta(\omega - \omega_0)$$

then we can find $f(t)$ using Eq. (18.9), writing

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(\omega - \omega_0)e^{j\omega t}\,d\omega$$

Using the sifting property of the impulse function gives

$$f(t) = \frac{1}{2\pi}e^{j\omega_0 t}$$

Since $F(\omega)$ and $f(t)$ constitute a Fourier transform pair, so too must $2\pi\delta(\omega - \omega_0)$ and $e^{j\omega_0 t}$,

$$\mathcal{F}[e^{j\omega_0 t}] = 2\pi\delta(\omega - \omega_0) \tag{18.1.3}$$

Alternatively, from Eq. (18.1.2),

$$\delta(t) = \mathcal{F}^{-1}[1]$$

Using the inverse Fourier transform formula in Eq. (18.9),

$$\delta(t) = \mathcal{F}^{-1}[1] = \frac{1}{2\pi}\int_{-\infty}^{\infty} 1e^{j\omega t}\,d\omega$$

or

$$\int_{-\infty}^{\infty} e^{j\omega t}\, d\omega = 2\pi\delta(t) \tag{18.1.4}$$

Interchanging variables t and ω results in

$$\int_{-\infty}^{\infty} e^{j\omega t}\, dt = 2\pi\delta(\omega) \tag{18.1.5}$$

Using this result, the Fourier transform of the given function is

$$\mathcal{F}[e^{j\omega_0 t}] = \int_{-\infty}^{\infty} e^{j\omega_0 t} e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} e^{j(\omega_0-\omega)}\, dt = 2\pi\delta(\omega_0 - \omega)$$

Since the impulse function is an even function, with $\delta(\omega_0 - \omega) = \delta(\omega - \omega_0)$,

$$\mathcal{F}[e^{j\omega_0 t}] = 2\pi\delta(\omega - \omega_0) \tag{18.1.6}$$

By simply changing the sign of ω_0, we readily obtain

$$\mathcal{F}[e^{-j\omega_0 t}] = 2\pi\delta(\omega + \omega_0) \tag{18.1.7}$$

Also, by setting $\omega_0 = 0$,

$$\mathcal{F}[1] = 2\pi\delta(\omega) \tag{18.1.8}$$

(c) By using the result in Eqs. (18.1.6) and (18.1.7), we get

$$\begin{aligned} \mathcal{F}[\cos\omega_0 t] &= \mathcal{F}\left[\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}\right] \\ &= \frac{1}{2}\mathcal{F}[e^{j\omega_0 t}] + \frac{1}{2}\mathcal{F}[e^{-j\omega_0 t}] \\ &= \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0) \end{aligned} \tag{18.1.9}$$

The Fourier transform of the cosine signal is shown in Fig. 18.3.

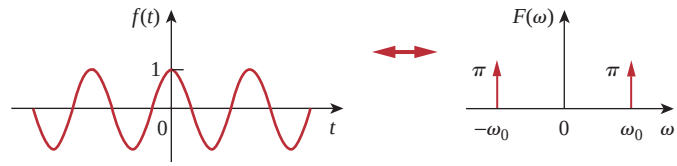


Figure 18.3
Fourier transform of $f(t) = \cos\omega_0 t$.

Practice Problem 18.1

Determine the Fourier transforms of the following functions: (a) gate function $g(t) = 4u(t-1) - 8u(t-2)$, (b) $4\delta(t+2)$, (c) $10\sin\omega_0 t$.

Answer: (a) $(4e^{-j\omega} - 8e^{-j2\omega})/j\omega$, (b) $4e^{j2\omega}$,
(c) $j10\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$.

Example 18.2

Derive the Fourier transform of a single rectangular pulse of width τ and height A, shown in Fig. 18.4.

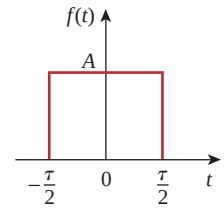


Figure 18.4
A rectangular pulse; for Example 18.2.

Solution:

$$F(\omega) = \int_{-\tau/2}^{\tau/2} Ae^{-j\omega t}\,dt = -\frac{A}{j\omega}e^{-j\omega t}\Big|_{-\tau/2}^{\tau/2}$$

$$= \frac{2A}{\omega}\left(\frac{e^{j\omega\tau/2} - e^{-j\omega\tau/2}}{2j}\right)$$

$$= A\tau\frac{\sin\omega\tau/2}{\omega\tau/2} = A\tau\,\text{sinc}\,\frac{\omega\tau}{2}$$

If we make $A = 10$ and $\tau = 2$ as in Fig. 17.27 (like in Section 17.6), then

$$F(\omega) = 20\,\text{sinc}\,\omega$$

whose amplitude spectrum is shown in Fig. 18.5. Comparing Fig. 18.4 with the frequency spectrum of the rectangular pulses in Fig. 17.28, we notice that the spectrum in Fig. 17.28 is discrete and its envelope has the same shape as the Fourier transform of a single rectangular pulse.

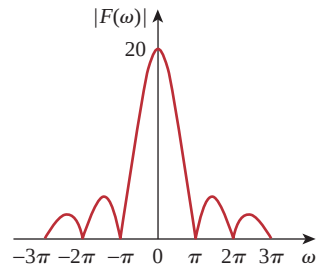


Figure 18.5
Amplitude spectrum of the rectangular pulse in Fig. 18.4: for Example 18.2.

Practice Problem 18.2

Obtain the Fourier transform of the function in Fig. 18.6.

Answer: $\dfrac{10(\cos\omega - 1)}{j\omega}$.

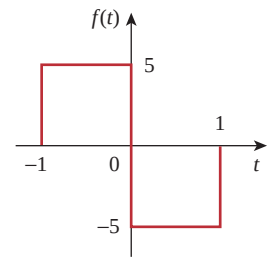


Figure 18.6
For Practice Prob. 18.2.

Example 18.3

Obtain the Fourier transform of the "switched-on" exponential function shown in Fig. 18.7.

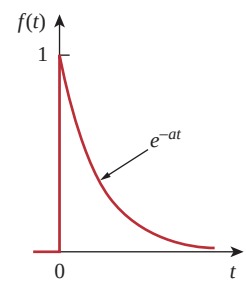


Figure 18.7
For Example 18.3.

Solution:
From Fig. 18.7,

$$f(t) = e^{-at}u(t) = \begin{cases} e^{-at}, & t > 0 \\ 0, & t < 0 \end{cases}$$

Hence,

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt = \int_{0}^{\infty} e^{-at}\,e^{-j\omega t}\,dt = \int_{0}^{\infty} e^{-(a+j\omega)t}\,dt$$

$$= \frac{-1}{a + j\omega}e^{-(a+j\omega)t}\Big|_{0}^{\infty} = \frac{1}{a + j\omega}$$

Practice Problem 18.3

Determine the Fourier transform of the "switched-off" exponential function in Fig. 18.8.

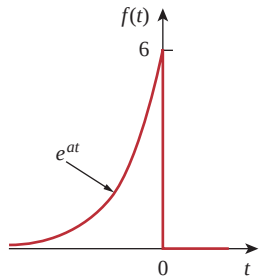


Figure 18.8
For Practice Prob. 18.3.

Answer: $\dfrac{6}{a - j\omega}$.

18.3 Properties of the Fourier Transform

We now develop some properties of the Fourier transform that are useful in finding the transforms of complicated functions from the transforms of simple functions. For each property, we will first state and derive it, and then illustrate it with some examples.

Linearity

If $F_1(\omega)$ and $F_2(\omega)$ are the Fourier transforms of $f_1(t)$ and $f_2(t)$, respectively, then

$$\mathcal{F}[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(\omega) + a_2 F_2(\omega) \tag{18.12}$$

where a_1 and a_2 are constants. This property simply states that the Fourier transform of a linear combination of functions is the same as the linear combination of the transforms of the individual functions. The proof of the linearity property in Eq. (18.12) is straightforward. By definition,

$$\begin{aligned}\mathcal{F}[a_1 f_1(t) + a_2 f_2(t)] &= \int_{-\infty}^{\infty} [a_1 f_1(t) + a_2 f_2(t)] e^{-j\omega t}\, dt \\ &= \int_{-\infty}^{\infty} a_1 f_1(t) e^{-j\omega t}\, dt + \int_{-\infty}^{\infty} a_2 f_2(t) e^{-j\omega t}\, dt \\ &= a_1 F_1(\omega) + a_2 F_2(\omega)\end{aligned} \tag{18.13}$$

For example, $\sin\omega_0 t = \frac{1}{2j}(e^{j\omega_0 t} - e^{-j\omega_0 t})$. Using the linearity property,

$$\begin{aligned}F[\sin\omega_0 t] &= \frac{1}{2j}[\mathcal{F}(e^{j\omega_0 t}) - \mathcal{F}(e^{-j\omega_0 t})] \\ &= \frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] \\ &= j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]\end{aligned} \tag{18.14}$$

Time Scaling

If $F(\omega) = \mathcal{F}[f(t)]$, then

$$\mathcal{F}[f(at)] = \frac{1}{|a|} F\left(\frac{\omega}{a}\right) \tag{18.15}$$

where a is a constant. Equation (18.15) shows that time expansion ($|a| > 1$) corresponds to frequency compression, or conversely, time

compression ($|a| < 1$) implies frequency expansion. The proof of the time-scaling property proceeds as follows.

$$\mathcal{F}[f(at)] = \int_{-\infty}^{\infty} f(at)e^{-j\omega t}\,dt \tag{18.16}$$

If we let $x = at$, so that $dx = a\,dt$, then

$$\mathcal{F}[f(at)] = \int_{-\infty}^{\infty} f(x)e^{-j\omega x/a}\frac{dx}{a} = \frac{1}{a}F\left(\frac{\omega}{a}\right) \tag{18.17}$$

For example, for the rectangular pulse $p(t)$ in Example 18.2,

$$\mathcal{F}[p(t)] = A\tau\,\text{sinc}\,\frac{\omega\tau}{2} \tag{18.18a}$$

Using Eq. (18.15),

$$\mathcal{F}[p(2t)] = \frac{A\tau}{2}\,\text{sinc}\,\frac{\omega\tau}{4} \tag{18.18b}$$

It may be helpful to plot $p(t)$ and $p(2t)$ and their Fourier transforms. Since

$$p(t) = \begin{cases} A, & -\dfrac{\tau}{2} < t < \dfrac{\tau}{2} \\ 0, & \text{otherwise} \end{cases} \tag{18.19a}$$

then replacing every t with $2t$ gives

$$p(2t) = \begin{cases} A, & -\dfrac{\tau}{2} < 2t < \dfrac{\tau}{2} \\ 0, & \text{otherwise} \end{cases} = \begin{cases} A, & -\dfrac{\tau}{4} < t < \dfrac{\tau}{4} \\ 0, & \text{otherwise} \end{cases} \tag{18.19b}$$

showing that $p(2t)$ is time compressed, as shown in Fig. 18.9(b). To plot both Fourier transforms in Eq. (18.18), we recall that the sinc function has zeros when its argument is $n\pi$, where n is an integer. Hence, for the transform of $p(t)$ in Eq. (18.18a), $\omega\tau/2 = 2\pi f\tau/2 = n\pi \rightarrow f = n/\tau$, and for the transform of $p(2t)$ in Eq. (18.18b), $\omega\tau/4 = 2\pi f\tau/4 = n\pi \rightarrow f = 2n/\tau$. The plots of the Fourier transforms are shown in Fig. 18.9, which shows that time compression corresponds with frequency expansion. We should expect this intuitively, because when the signal is squashed in time, we expect it to change more rapidly, thereby causing higher-frequency components to exist.

Time Shifting

If $F(\omega) = \mathcal{F}[f(t)]$, then

$$\boxed{\mathcal{F}[f(t - t_0)] = e^{-j\omega t_0}F(\omega)} \tag{18.20}$$

that is, a delay in the time domain corresponds to a phase shift in the frequency domain. To derive the time shifting property, we note that

$$\mathcal{F}[f(t - t_0)] = \int_{-\infty}^{\infty} f(t - t_0)e^{-j\omega t}\,dt \tag{18.21}$$

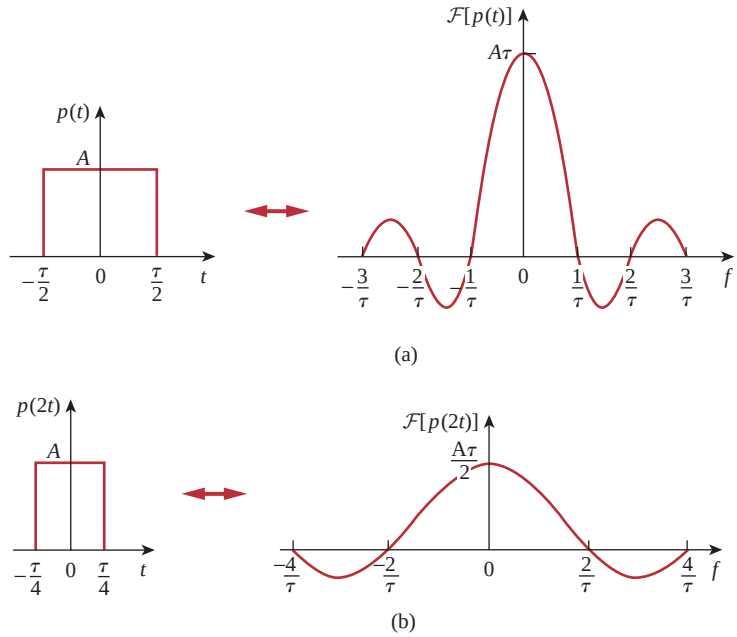


Figure 18.9
The effect of time scaling: (a) transform of the pulse, (b) time compression of the pulse causes frequency expansion.

If we let $x = t - t_0$ so that $dx = dt$ and $t = x + t_0$, then

$$\begin{aligned}\mathcal{F}[\,f(t - t_0)] &= \int_{-\infty}^{\infty} f(x)e^{-j\omega(x+t_0)}\,dx \\ &= e^{-j\omega t_0}\int_{-\infty}^{\infty} f(x)e^{-j\omega x}\,dx = e^{-j\omega t_0}\,F(\omega)\end{aligned} \tag{18.22}$$

Similarly, $\mathcal{F}[\,f(t + t_0)] = e^{j\omega t_0}\,F(\omega)$.

For example, from Example 18.3,

$$\mathcal{F}[e^{-at}u(t)] = \frac{1}{a + j\omega} \tag{18.23}$$

The transform of $f(t) = e^{-(t-2)}u(t - 2)$ is

$$F(\omega) = \mathcal{F}\left[e^{-(t-2)}u(t - 2)\right] = \frac{e^{-j2\omega}}{1 + j\omega} \tag{18.24}$$

Frequency Shifting (or Amplitude Modulation)

This property states that if $F(\omega) = \mathcal{F}[\,f(t)]$, then

$$\boxed{\mathcal{F}\left[\,f(t)e^{j\omega_0 t}\right] = F(\omega - \omega_0)} \tag{18.25}$$

meaning, a frequency shift in the frequency domain adds a phase shift to the time function. By definition,

$$\begin{aligned}\mathcal{F}[\,f(t)e^{j\omega_0 t}] &= \int_{-\infty}^{\infty} f(t)e^{j\omega_0 t}\,e^{-j\omega t}\,dt \\ &= \int_{-\infty}^{\infty} f(t)e^{-j(\omega-\omega_0)t}\,dt = F(\omega - \omega_0)\end{aligned} \tag{18.26}$$

For example, $\cos\omega_0 t = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})$. Using the property in Eq. (18.25),

$$\begin{aligned}\mathcal{F}[\,f(t)\cos\omega_0 t] &= \frac{1}{2}\mathcal{F}[\,f(t)e^{j\omega_0 t}] + \frac{1}{2}\mathcal{F}[\,f(t)e^{-j\omega_0 t}] \\ &= \frac{1}{2}F(\omega - \omega_0) + \frac{1}{2}F(\omega + \omega_0)\end{aligned} \tag{18.27}$$

This is an important result in modulation where frequency components of a signal are shifted. If, for example, the amplitude spectrum of $f(t)$ is as shown in Fig. 18.10(a), then the amplitude spectrum of $f(t)\cos\omega_0 t$ will be as shown in Fig. 18.10(b). We will elaborate on amplitude modulation in Section 18.7.1.

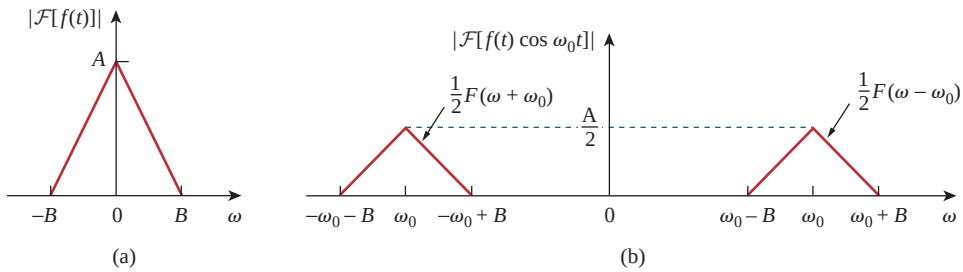


Figure 18.10
Amplitude spectra of: (a) signal $f(t)$, (b) modulated signal $f(t)\cos\omega_0 t$.

Time Differentiation

Given that $F(\omega) = \mathcal{F}[\,f(t)]$, then

$$\boxed{\mathcal{F}[\,f'(t)] = j\omega F(\omega)} \tag{18.28}$$

In other words, the transform of the derivative of $f(t)$ is obtained by multiplying the transform of $f(t)$ by $j\omega$. By definition,

$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\,d\omega \tag{18.29}$$

Taking the derivative of both sides with respect to t gives

$$f'(t) = \frac{j\omega}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega = j\omega\mathcal{F}^{-1}[F(\omega)]$$

or

$$\mathcal{F}[\,f'(t)] = j\omega F(\omega) \tag{18.30}$$

Repeated applications of Eq. (18.30) give

$$\boxed{\mathcal{F}\big[\,f^{(n)}(t)\big] = (j\omega)^n F(\omega)} \tag{18.31}$$

For example, if $f(t) = e^{-at}u(t)$, then

$$f'(t) = -ae^{-at}u(t) + e^{-at}\delta(t) = -af(t) + e^{-at}\delta(t) \quad \textbf{(18.32)}$$

Taking the Fourier transforms of the first and last terms, we obtain

$$j\omega F(\omega) = -aF(\omega) + 1 \quad \Rightarrow \quad F(\omega) = \frac{1}{a + j\omega} \quad \textbf{(18.33)}$$

which agrees with the result in Example 18.3.

Time Integration

Given that $F(\omega) = \mathcal{F}[f(t)]$, then

$$\mathcal{F}\left[\int_{-\infty}^{t} f(t)\,dt\right] = \frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega) \quad \textbf{(18.34)}$$

that is, the transform of the integral of $f(t)$ is obtained by dividing the transform of $f(t)$ by $j\omega$ and adding the result to the impulse term that reflects the dc component $F(0)$. Someone might ask, "How do we know that when we take the Fourier transform for time integration, we should integrate over the interval $[-\infty, t]$ and not $[-\infty, \infty]$?" When we integrate over $[-\infty, \infty]$, the result does not depend on time anymore, and the Fourier transform of a constant is what we will eventually get. But when we integrate over $[-\infty, t]$, we get the integral of the function from the past to time t, so that the result depends on t and we can take the Fourier transform of that.

If ω is replaced by 0 in Eq. (18.8),

$$F(0) = \int_{-\infty}^{\infty} f(t)\,dt \quad \textbf{(18.35)}$$

indicating that the dc component is zero when the integral of $f(t)$ over all time vanishes. The proof of the time integration in Eq. (18.34) will be given later when we consider the convolution property.

For example, we know that $\mathcal{F}[\delta(t)] = 1$ and that integrating the impulse function gives the unit step function [see Eq. (7.39a)]. By applying the property in Eq. (18.34), we obtain the Fourier transform of the unit step function as

$$\mathcal{F}[u(t)] = \mathcal{F}\left[\int_{-\infty}^{t} \delta(t)\,dt\right] = \frac{1}{j\omega} + \pi\delta(\omega) \quad \textbf{(18.36)}$$

Reversal

If $F(\omega) = \mathcal{F}[f(t)]$, then

$$\mathcal{F}[f(-t)] = F(-\omega) = F^*(\omega) \quad \textbf{(18.37)}$$

where the asterisk denotes the complex conjugate. This property states that reversing $f(t)$ about the time axis reverses $F(\omega)$ about the frequency axis. This may be regarded as a special case of time scaling for which $a = -1$ in Eq. (18.15).

For example, $1 = u(t) + u(-t)$. Hence,

$$\begin{aligned}\mathcal{F}[1] &= \mathcal{F}[u(t)] + \mathcal{F}[u(-t)] \\ &= \frac{1}{j\omega} + \pi\delta(\omega) \\ &\quad -\frac{1}{j\omega} + \pi\delta(-\omega) \\ &= 2\pi\delta(\omega)\end{aligned}$$

Duality

This property states that if $F(\omega)$ is the Fourier transform of $f(t)$, then the Fourier transform of $F(t)$ is $2\pi f(-\omega)$; we write

$$\boxed{\mathcal{F}[f(t)] = F(\omega) \quad \Rightarrow \quad \mathcal{F}[F(t)] = 2\pi f(-\omega)} \tag{18.38}$$

This expresses the symmetry property of the Fourier transform. To derive this property, we recall that

$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\,d\omega$$

or

$$2\pi f(t) = \int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega \tag{18.39}$$

Replacing t by $-t$ gives

$$2\pi f(-t) = \int_{-\infty}^{\infty} F(\omega)e^{-j\omega t}\,d\omega$$

If we interchange t and ω, we obtain

$$2\pi f(-\omega) = \int_{-\infty}^{\infty} F(t)e^{-j\omega t}\,dt = \mathcal{F}[F(t)] \tag{18.40}$$

as expected.

For example, if $f(t) = e^{-|t|}$, then

$$F(\omega) = \frac{2}{\omega^2 + 1} \tag{18.41}$$

By the duality property, the Fourier transform of $F(t) = 2/(t^2 + 1)$ is

$$2\pi f(\omega) = 2\pi e^{-|\omega|} \tag{18.42}$$

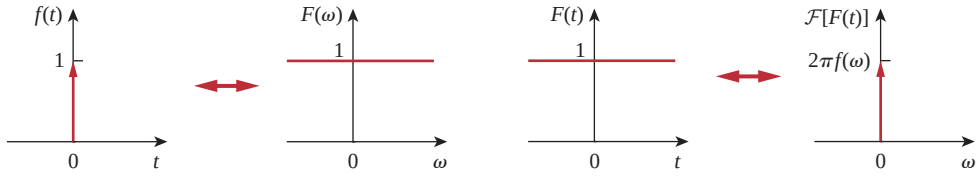
Figure 18.11 shows another example of the duality property. It illustrates the fact that if $f(t) = \delta(t)$ so that $F(\omega) = 1$, as in Fig. 18.11(a), then the Fourier transform of $F(t) = 1$ is $2\pi f(\omega) = 2\pi\delta(\omega)$ as shown in Fig. 18.11(b).

Since $f(t)$ is the sum of the signals in Figs. 18.7 and 18.8, $F(\omega)$ is the sum of the results in Example 18.3 and Practice Prob. 18.3.

Convolution

Recall from Chapter 15 that if $x(t)$ is the input excitation to a circuit with an impulse function of $h(t)$, then the output response $y(t)$ is given by the convolution integral

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\lambda)x(t-\lambda)\,d\lambda \tag{18.43}$$



(a) (b)

Figure 18.11
A typical illustration of the duality property of the Fourier transform: (a) transform of impulse, (b) transform of unit dc level.

If $X(\omega)$, $H(\omega)$, and $Y(\omega)$ are the Fourier transforms of $x(t)$, $h(t)$, and $y(t)$, respectively, then

$$\boxed{Y(\omega) = \mathcal{F}[h(t) * x(t)] = H(\omega)X(\omega)} \tag{18.44}$$

which indicates that convolution in the time domain corresponds with multiplication in the frequency domain.

To derive the convolution property, we take the Fourier transform of both sides of Eq. (18.43) to get

$$Y(\omega) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(\lambda)x(t - \lambda)\, d\lambda \right] e^{-j\omega t}\, dt \tag{18.45}$$

Exchanging the order of integration and factoring $h(\lambda)$, which does not depend on t, we have

$$Y(\omega) = \int_{-\infty}^{\infty} h(\lambda) \left[\int_{-\infty}^{\infty} x(t - \lambda)e^{-j\omega t}\, dt \right] d\lambda$$

For the integral within the brackets, let $\tau = t - \lambda$ so that $t = \tau + \lambda$ and $dt = d\tau$. Then,

$$\begin{aligned} Y(\omega) &= \int_{-\infty}^{\infty} h(\lambda) \left[\int_{-\infty}^{\infty} x(\tau)e^{-j\omega(\tau+\lambda)}\, d\tau \right] d\lambda \\ &= \int_{-\infty}^{\infty} h(\lambda)e^{-j\omega\lambda}\, d\lambda \int_{-\infty}^{\infty} x(\tau)e^{-j\omega\tau}\, d\tau = H(\omega)X(\omega) \end{aligned} \tag{18.46}$$

The important relationship in Eq. (18.46) is the key reason for using the Fourier transform in the analysis of linear systems.

as expected. This result expands the phasor method beyond what was done with the Fourier series in the previous chapter.

To illustrate the convolution property, suppose both $h(t)$ and $x(t)$ are identical rectangular pulses, as shown in Fig. 18.12(a) and 18.12(b). We recall from Example 18.2 and Fig. 18.5 that the Fourier transforms of the rectangular pulses are sinc functions, as shown in Fig. 18.12(c) and 18.12(d). According to the convolution property, the product of the sinc functions should give us the convolution of the rectangular pulses in the time domain. Thus, the convolution of the pulses in Fig. 18.12(e) and the product of the sinc functions in Fig. 18.12(f) form a Fourier pair.

In view of the duality property, we expect that if convolution in the time domain corresponds with multiplication in the frequency

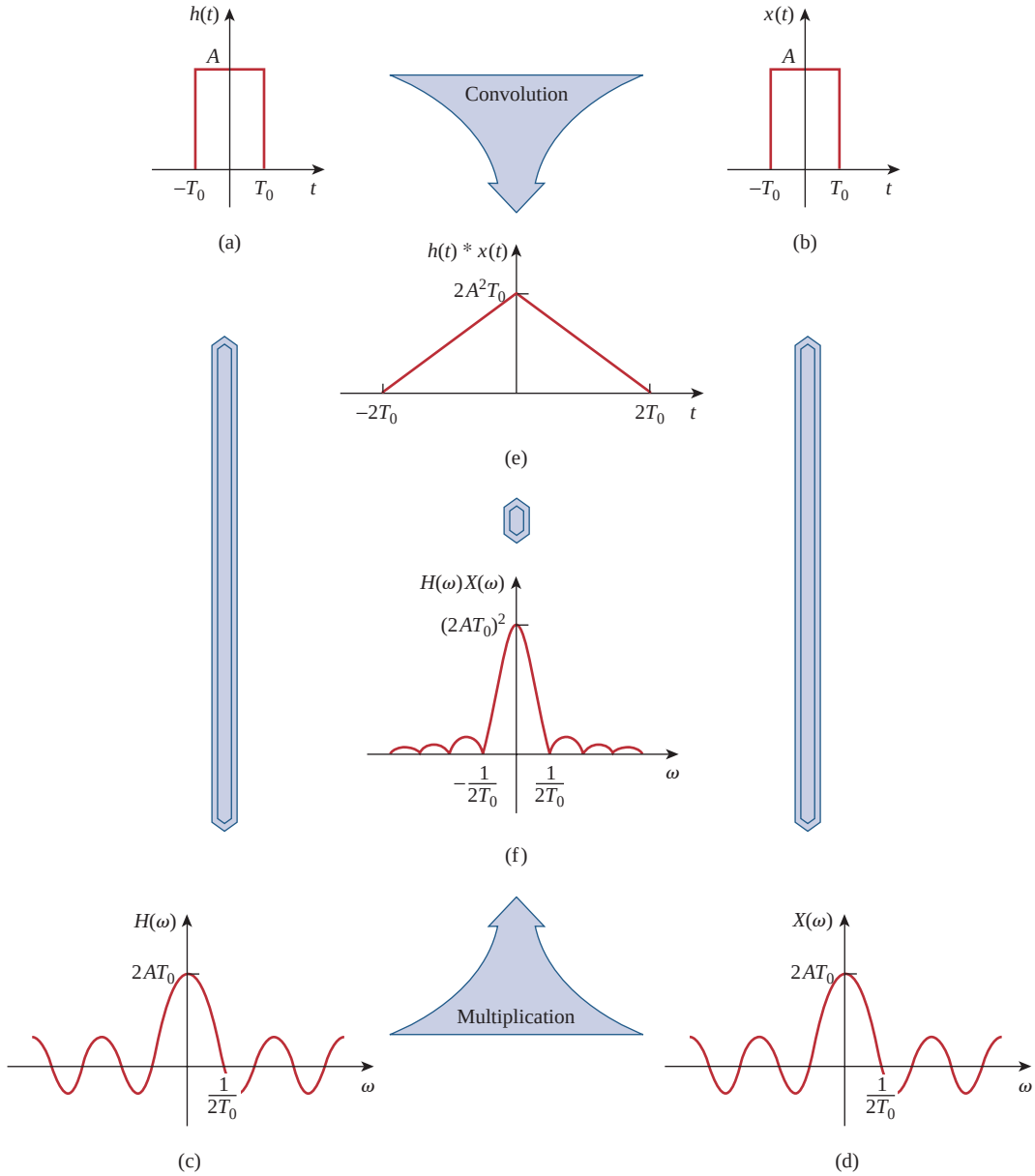


Figure 18.12
Graphical illustration of the convolution property.
E. O. Brigham, *The Fast Fourier Transform* [Englewood Cliffs, NJ: Prentice Hall, 1974], p. 60.

domain, then multiplication in the time domain should have a correspondence in the frequency domain. This happens to be the case. If $f(t) = f_1(t)\, f_2(t)$, then

$$F(\omega) = \mathcal{F}[\, f_1(t)\, f_2(t)] = \frac{1}{2\pi} F_1(\omega) * F_2(\omega) \qquad \textbf{(18.47)}$$

or

$$F(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F_1(\lambda) F_2(\omega - \lambda)\, d\lambda \qquad \textbf{(18.48)}$$

which is convolution in the frequency domain. The proof of Eq. (18.48) readily follows from the duality property in Eq. (18.38).

Let us now derive the time integration property in Eq. (18.34). If we replace $x(t)$ with the unit step function $u(t)$ and $h(t)$ with $f(t)$ in Eq. (18.43), then

$$\int_{-\infty}^{\infty} f(\lambda)u(t-\lambda)\,d\lambda = f(t) * u(t) \tag{18.49}$$

But by the definition of the unit step function,

$$u(t-\lambda) = \begin{cases} 1, & t-\lambda > 0 \\ 0, & t-\lambda > 0 \end{cases}$$

We can write this as

$$u(t-\lambda) = \begin{cases} 1, & \lambda < t \\ 0, & \lambda > t \end{cases}$$

Substituting this into Eq. (18.49) makes the interval of integration change from $[-\infty, \infty]$ to $[-\infty, t]$, and thus Eq. (18.49) becomes

$$\int_{-\infty}^{t} f(\lambda)\,d\lambda = u(t) * f(t)$$

Taking the Fourier transform of both sides yields

$$\mathcal{F}\left[\int_{-\infty}^{t} f(\lambda)\,d\lambda\right] = U(\omega)F(\omega) \tag{18.50}$$

But from Eq. (18.36), the Fourier transform of the unit step function is

$$U(\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

Substituting this into Eq. (18.50) gives

$$\begin{aligned} \mathcal{F}\left[\int_{-\infty}^{t} f(\lambda)\,d\lambda\right] &= \left(\frac{1}{j\omega} + \pi\delta(\omega)\right)F(\omega) \\ &= \frac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega) \end{aligned} \tag{18.51}$$

which is the time integration property of Eq. (18.34). Note that in Eq. (18.51), $F(\omega)\delta(\omega) = F(0)\delta(\omega)$, since $\delta(\omega)$ is only nonzero at $\omega = 0$.

Table 18.1 lists these properties of the Fourier transform. Table 18.2 presents the transform pairs of some common functions. Note the similarities between these tables and Tables 15.1 and 15.2.

TABLE 18.1

Properties of the Fourier transform.

Property	$f(t)$	$F(\omega)$
Linearity	$a_1 f_1(t) + a_2 f_2(t)$	$a_1F_1(\omega) + a_2F_2(\omega)$
Scaling	$f(at)$	$\frac{1}{\lvert a\rvert}F\left(\frac{\omega}{a}\right)$
Time shift	$f(t-a)$	$e^{-j\omega a}F(\omega)$
Frequency shift	$e^{j\omega_0 t}f(t)$	$F(\omega-\omega_0)$

TABLE 18.1 *(continued)*

Property	$f(t)$	$F(\omega)$
Modulation	$\cos(\omega_0 t) f(t)$	$\frac{1}{2}[F(\omega + \omega_0) + F(\omega - \omega_0)]$
Time differentiation	$\frac{df}{dt}$	$j\omega F(\omega)$
	$\frac{d^n f}{dt^n}$	$(j\omega)^n F(\omega)$
Time integration	$\int_{-\infty}^{t} f(t)\, dt$	$\frac{F(\omega)}{j\omega} + \pi F(0)\,\delta(\omega)$
Frequency differentiation	$t^n f(t)$	$(j)^n \frac{d^n}{d\omega^n} F(\omega)$
Reversal	$f(-t)$	$F(-\omega)$ or $F^*(\omega)$
Duality	$F(t)$	$2\pi f(-\omega)$
Convolution in t	$f_1(t) * f_2(t)$	$F_1(\omega)F_2(\omega)$
Convolution in ω	$f_1(t) f_2(t)$	$\frac{1}{2\pi} F_1(\omega) * F_2(\omega)$

TABLE 18.2

Fourier transform pairs.

$f(t)$	$F(\omega)$
$\delta(t)$	1
1	$2\pi\,\delta(\omega)$
$u(t)$	$\pi\,\delta(\omega) + \frac{1}{j\omega}$
$u(t + \tau) - u(t - \tau)$	$2\frac{\sin\omega\tau}{\omega}$
$\lvert t\rvert$	$\frac{-2}{\omega^2}$
$\operatorname{sgn}(t)$	$\frac{2}{j\omega}$
$e^{-at}u(t)$	$\frac{1}{a + j\omega}$
$e^{at}u(-t)$	$\frac{1}{a - j\omega}$
$t^n e^{-at}u(t)$	$\frac{n!}{(a + j\omega)^{n+1}}$
$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2 + \omega^2}$
$e^{j\omega_0 t}$	$2\pi\,\delta(\omega - \omega_0)$
$\sin\omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$\cos\omega_0 t$	$\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
$e^{-at}\sin\omega_0 t u(t)$	$\frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$
$e^{-at}\cos\omega_0 t u(t)$	$\frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$

Example 18.4

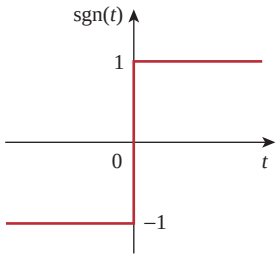


Figure 18.13
The signum function of Example 18.4.

Find the Fourier transforms of the following functions: (a) signum function sgn(t), shown in Fig. 18.13, (b) the double-sided exponential $e^{-a|t|}$, and (c) the sinc function $(\sin t)/t$.

Solution:

(a) We can obtain the Fourier transform of the *signum* function in three ways.

■ **METHOD 1** We can write the signum function in terms of the unit step function as

$$\text{sgn}(t) = f(t) = u(t) - u(-t)$$

But from Eq. (18.36),

$$U(\omega) = \mathcal{F}[u(t)] = \pi\delta(\omega) + \frac{1}{j\omega}$$

Applying this and the reversal property, we obtain

$$\begin{aligned}\mathcal{F}[\text{sgn}(t)] &= U(\omega) - U(-\omega) \\ &= \left(\pi\delta(\omega) + \frac{1}{j\omega}\right) - \left(\pi\delta(-\omega) + \frac{1}{-j\omega}\right) = \frac{2}{j\omega}\end{aligned}$$

■ **METHOD 2** since $\delta(\omega) = \delta(-\omega)$, another way of writing the signum function in terms of the unit step function is

$$f(t) = \text{sgn}(t) = -1 + 2u(t)$$

Taking the Fourier transform of each term gives

$$F(\omega) = -2\pi\delta(\omega) + 2\left(\pi\delta(\omega) + \frac{1}{j\omega}\right) = \frac{2}{j\omega}$$

■ **METHOD 3** We can take the derivative of the signum function in Fig. 18.13 and obtain

$$f'(t) = 2\delta(t)$$

Taking the transform of this,

$$j\omega F(\omega) = 2 \quad \Rightarrow \quad F(\omega) = \frac{2}{j\omega}$$

as obtained previously.

(b) The double-sided exponential can be expressed as

$$f(t) = e^{-a|t|} = e^{-at}u(t) + e^{at}u(-t) = y(t) + y(-t)$$

where $y(t) = e^{-at}u(t)$ so that $Y(\omega) = 1/(a + j\omega)$. Applying the reversal property,

$$\mathcal{F}\left[e^{-a|t|}\right] = Y(\omega) + Y(-\omega) = \left(\frac{1}{a + j\omega} + \frac{1}{a - j\omega}\right) = \frac{2a}{a^2 + \omega^2}$$

(c) From Example 18.2,

$$\mathcal{F}\left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right)\right] = \tau\frac{\sin(\omega\tau/2)}{\omega\tau/2} = \tau\,\text{sinc}\,\frac{\omega\tau}{2}$$

Setting $\tau/2 = 1$ gives

$$\mathcal{F}[u(t+1) - u(t-1)] = 2\frac{\sin\omega}{\omega}$$

Applying the duality property yields

$$\mathcal{F}\left[2\frac{\sin t}{t}\right] = 2\pi[U(\omega+1) - U(\omega-1)]$$

or

$$\mathcal{F}\left[\frac{\sin t}{t}\right] = \pi[U(\omega+1) - U(\omega-1)]$$

Practice Problem 18.4

Determine the Fourier transforms of these functions: (a) gate function $g(t) = u(t) - u(t-1)$, (b) $f(t) = te^{-2t}u(t)$, and (c) sawtooth pulse $p(t) = 10t[u(t) - u(t-2)]$.

Answer: (a) $(1 - e^{-j\omega})\left[\pi\delta(\omega) + \frac{1}{j\omega}\right]$, (b) $\frac{1}{(2+j\omega)^2}$, (c) $\frac{10(e^{-j2\omega} - 1)}{\omega^2} + \frac{20j}{\omega}e^{-j2\omega}$.

Example 18.5

Find the Fourier transform of the function in Fig. 18.14.

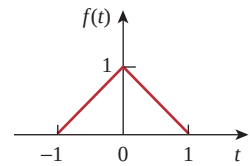


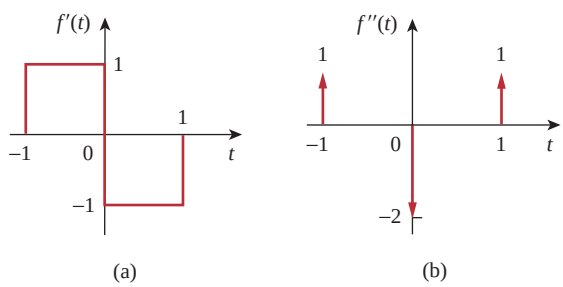
Figure 18.14
For Example 18.5.

Solution:
The Fourier transform can be found directly using Eq. (18.8), but it is much easier to find it using the derivative property. We can express the function as

$$f(t) = \begin{cases} 1 + t, & -1 < t < 0 \\ 1 - t, & 0 < t < 1 \end{cases}$$

Its first derivative is shown in Fig. 18.15(a) and is given by

$$f'(t) = \begin{cases} 1, & -1 < t < 0 \\ -1, & 0 < t < 1 \end{cases}$$



(a) (b)

Figure 18.15
First and second derivatives of $f(t)$ in Fig. 18.14; for Example 18.5.

Its second derivative is in Fig. 18.15(b) and is given by

$$f''(t) = \delta(t+1) - 2\delta(t) + \delta(t-1)$$

Taking the Fourier transform of both sides,

$$(j\omega)^2 F(\omega) = e^{j\omega} - 2 + e^{-j\omega} = -2 + 2\cos\omega$$

or

$$F(\omega) = \frac{2(1-\cos\omega)}{\omega^2}$$

Practice Problem 18.5

Determine the Fourier transform of the function in Fig. 18.16.

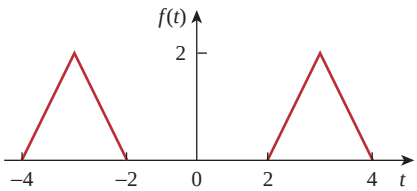


Figure 18.16
For Practice Prob. 18.5.

Answer: $(8\cos 3\omega - 4\cos 4\omega - 4\cos 2\omega)/\omega^2$.

Example 18.6

Obtain the inverse Fourier transform of:

(a) $F(\omega) = \dfrac{10j\omega + 4}{(j\omega)^2 + 6j\omega + 8}$ (b) $G(\omega) = \dfrac{\omega^2 + 21}{\omega^2 + 9}$

Solution:

(a) To avoid complex algebra, we can replace $j\omega$ with s for the moment. Using partial fraction expansion,

$$F(s) = \frac{10s+4}{s^2+6s+8} = \frac{10s+4}{(s+4)(s+2)} = \frac{A}{s+4} + \frac{B}{s+2}$$

where

$$A = (s+4)F(s)\big|_{s=-4} = \left.\frac{10s+4}{(s+2)}\right|_{s=-4} = \frac{-36}{-2} = 18$$

$$B = (s+2)F(s)\big|_{s=-2} = \left.\frac{10s+4}{(s+4)}\right|_{s=-2} = \frac{-16}{2} = -8$$

Substituting $A = 18$ and $B = -8$ in $F(s)$ and s with $j\omega$ gives

$$F(j\omega) = \frac{18}{j\omega+4} + \frac{-8}{j\omega+2}$$

With the aid of Table 18.2, we obtain the inverse transform as

$$f(t) = (18e^{-4t} - 8e^{-2t})u(t)$$

(b) We simplify $G(\omega)$ as

$$G(\omega) = \frac{\omega^2+21}{\omega^2+9} = 1 + \frac{12}{\omega^2+9}$$

With the aid of Table 18.2, the inverse transform is obtained as

$$g(t) = \delta(t) + 2e^{-3|t|}$$

Practice Problem 18.6

Find the inverse Fourier transform of:

(a) $H(\omega) = \dfrac{6(3 + j2\omega)}{(1 + j\omega)(4 + j\omega)(2 + j\omega)}$

(b) $Y(\omega) = \pi\delta(\omega) + \dfrac{1}{j\omega} + \dfrac{2(1 + j\omega)}{(1 + j\omega)^2 + 16}$

Answer: (a) $h(t) = (2e^{-t} + 3e^{-2t} - 5e^{-4t})u(t)$,
(b) $y(t) = (1 + 2e^{-t}\cos 4t)u(t)$.

18.4 Circuit Applications

The Fourier transform generalizes the phasor technique to nonperiodic functions. Therefore, we apply Fourier transforms to circuits with nonsinusoidal excitations in exactly the same way we apply phasor techniques to circuits with sinusoidal excitations. Thus, Ohm's law is still valid:

$$V(\omega) = Z(\omega)I(\omega) \tag{18.52}$$

where $V(\omega)$ and $I(\omega)$ are the Fourier transforms of the voltage and current and $Z(\omega)$ is the impedance. We get the same expressions for the impedances of resistors, inductors, and capacitors as in phasor analysis, namely,

$$\begin{array}{lcl} R & \Rightarrow & R \\ L & \Rightarrow & j\omega L \\ C & \Rightarrow & \dfrac{1}{j\omega C} \end{array} \tag{18.53}$$

Once we transform the functions for the circuit elements into the frequency domain and take the Fourier transforms of the excitations, we can use circuit techniques such as voltage division, source transformation, mesh analysis, node analysis, or Thevenin's theorem, to find the unknown response (current or voltage). Finally, we take the inverse Fourier transform to obtain the response in the time domain.

Although the Fourier transform method produces a response that exists for $-\infty < t < \infty$, Fourier analysis cannot handle circuits with initial conditions.

The transfer function is again defined as the ratio of the output response $Y(\omega)$ to the input excitation $X(\omega)$; that is,

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \tag{18.54}$$

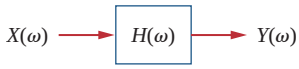


Figure 18.17
Input-output relationship of a circuit in the frequency domain.

or

$$Y(\omega) = H(\omega)X(\omega) \tag{18.55}$$

The frequency domain input-output relationship is portrayed in Fig. 18.17. Equation (18.55) shows that if we know the transfer function and the input, we can readily find the output. The relationship in Eq. (18.54) is the principal reason for using the Fourier transform in circuit analysis. Notice that $H(\omega)$ is identical to $H(s)$ with $s = j\omega$. Also, if the input is an impulse function [i.e., $x(t) = \delta(t)$], then $X(\omega) = 1$, so that the response is

$$Y(\omega) = H(\omega) = \mathcal{F}[h(t)] \tag{18.56}$$

indicating that $H(\omega)$ is the Fourier transform of the impulse response $h(t)$.

Example 18.7

Find $v_o(t)$ in the circuit of Fig. 18.18 for $v_i(t) = 2e^{-3t}u(t)$.

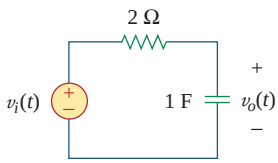


Figure 18.18
For Example 18.7.

Solution:
The Fourier transform of the input voltage is

$$V_i(\omega) = \frac{2}{3 + j\omega}$$

and the transfer function obtained by voltage division is

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = \frac{1/j\omega}{2 + 1/j\omega} = \frac{1}{1 + j2\omega}$$

Hence,

$$V_o(\omega) = V_i(\omega)H(\omega) = \frac{2}{(3 + j\omega)(1 + j2\omega)}$$

or

$$V_o(\omega) = \frac{1}{(3 + j\omega)(0.5 + j\omega)}$$

By partial fractions,

$$V_o(\omega) = \frac{-0.4}{3 + j\omega} + \frac{0.4}{0.5 + j\omega}$$

Taking the inverse Fourier transform yields

$$v_o(t) = 0.4(e^{-0.5t} - e^{-3t})u(t)$$

Practice Problem 18.7

Determine $v_o(t)$ in Fig. 18.19 if $v_i(t) = 5\text{sgn}(t) = (-5 + 10u(t))$ V.

Answer: $-5 + 10(1 - e^{-4t})u(t)$ V.

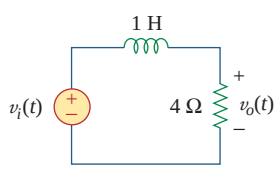


Figure 18.19
For Practice Prob. 18.7.

Example 18.8

Using the Fourier transform method, find $i_o(t)$ in Fig. 18.20 when $i_s(t) = 10 \sin 2t$ A.

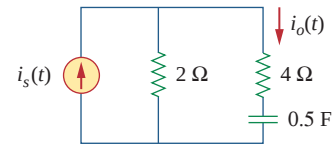


Figure 18.20
For Example 18.8.

Solution:
By current division,

$$H(\omega) = \frac{I_o(\omega)}{I_s(\omega)} = \frac{2}{2 + 4 + 2/j\omega} = \frac{j\omega}{1 + j\omega 3}$$

If $i_s(t) = 10 \sin 2t$, then

$$I_s(\omega) = j\pi 10[\delta(\omega + 2) - \delta(\omega - 2)]$$

Hence,

$$I_o(\omega) = H(\omega)I_s(\omega) = \frac{10\pi\omega[\delta(\omega - 2) - \delta(\omega + 2)]}{1 + j\omega 3}$$

The inverse Fourier transform of $I_o(\omega)$ cannot be found using Table 18.2. We resort to the inverse Fourier transform formula in Eq. (18.9) and write

$$i_o(t) = \mathcal{F}^{-1}[I_o(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{10\pi\omega[\delta(\omega - 2) - \delta(\omega + 2)]}{1 + j\omega 3} e^{j\omega t}\, d\omega$$

We apply the sifting property of the impulse function, namely,

$$\delta(\omega - \omega_0)\, f(\omega) = f(\omega_0)$$

or

$$\int_{-\infty}^{\infty} \delta(\omega - \omega_0) f(\omega)\, d\omega = f(\omega_0)$$

and obtain

$$\begin{aligned} i_o(t) &= \frac{10\pi}{2\pi}\left[\frac{2}{1 + j6}\, e^{j2t} - \frac{-2}{1 - j6}\, e^{-j2t}\right] \\ &= 10\left[\frac{e^{j2t}}{6.082e^{j80.54^\circ}} + \frac{e^{-j2t}}{6.082\, e^{-j80.54^\circ}}\right] \\ &= 1.644\left[e^{j(2t - 80.54^\circ)} + e^{-j(2t - 80.54^\circ)}\right] \\ &= 3.288 \cos(2t - 80.54^\circ) \text{ A} \end{aligned}$$

Practice Problem 18.8

Find the current $i_o(t)$ in the circuit in Fig. 18.21, given that $i_s(t) = 5 \cos 4t$ A.

Answer: $2.95 \cos(4t + 26.57^\circ)$ A.

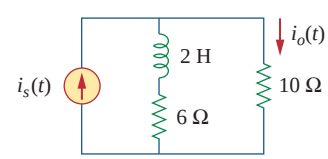


Figure 18.21
For Practice Prob. 18.8.

18.5 Parseval's Theorem

Parseval's theorem demonstrates one practical use of the Fourier transform. It relates the energy carried by a signal to the Fourier transform of the signal. If $p(t)$ is the power associated with the signal, the energy carried by the signal is

$$W = \int_{-\infty}^{\infty} p(t)\,dt \tag{18.57}$$

In order to be able to compare the energy content of current and voltage signals, it is convenient to use a 1-Ω resistor as the base for energy calculation. For a 1-Ω resistor, $p(t) = v^2(t) = i^2(t) = f^2(t)$, where $f(t)$ stands for either voltage or current. The energy delivered to the 1-Ω resistor is

$$W_{1\Omega} = \int_{-\infty}^{\infty} f^2(t)\,dt \tag{18.58}$$

Parseval's theorem states that this same energy can be calculated in the frequency domain as

$$W_{1\Omega} = \int_{-\infty}^{\infty} f^2(t)\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\,d\omega \tag{18.59}$$

Parseval's theorem states that the total energy delivered to a 1-Ω resistor equals the total area under the square of $f(t)$ or $1/2\pi$ times the total area under the square of the magnitude of the Fourier transform of $f(t)$.

In fact, $|F(\omega)|^2$ is sometimes known as the energy spectral density of signal $f(t)$.

Parseval's theorem relates energy associated with a signal to its Fourier transform. It provides the physical significance of $F(\omega)$, namely, that $|F(\omega)|^2$ is a measure of the energy density (in joules per hertz) corresponding to $f(t)$.

To derive Eq. (18.59), we begin with Eq. (18.58) and substitute Eq. (18.9) for one of the $f(t)$'s. We obtain

$$W_{1\Omega} = \int_{-\infty}^{\infty} f^2(t)\,dt = \int_{-\infty}^{\infty} f(t)\left[\frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\,d\omega\right]dt \tag{18.60}$$

The function $f(t)$ can be moved inside the integral within the brackets, since the integral does not involve time:

$$W_{1\Omega} = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t)F(\omega)e^{j\omega t}\,d\omega\,dt \tag{18.61}$$

Reversing the order of integration,

$$\begin{aligned} W_{1\Omega} &= \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\left[\int_{-\infty}^{\infty} f(t)e^{-j(-\omega)t}\,dt\right]d\omega \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)F(-\omega)\,d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)F^*(\omega)\,d\omega \end{aligned} \tag{18.62}$$

But if $z = x + jy$, $zz^* = (x + jy)(x - jy) = x^2 + y^2 = |z|^2$. Hence,

$$W_{1\Omega} = \int_{-\infty}^{\infty} f^2(t)\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\,d\omega \qquad \textbf{(18.63)}$$

as expected. Equation (18.63) indicates that the energy carried by a signal can be found by integrating either the square of $f(t)$ in the time domain or $1/2\pi$ times the square of $F(\omega)$ in the frequency domain.

Since $|F(\omega)|^2$ is an even function, we may integrate from 0 to ∞ and double the result; that is,

$$W_{1\Omega} = \int_{-\infty}^{\infty} f^2(t)\,dt = \frac{1}{\pi}\int_{0}^{\infty} |F(\omega)|^2\,d\omega \qquad \textbf{(18.64)}$$

We may also calculate the energy in any frequency band $\omega_1 < \omega < \omega_2$ as

$$W_{1\Omega} = \frac{1}{\pi}\int_{\omega_1}^{\omega_2} |F(\omega)|^2\,d\omega \qquad \textbf{(18.65)}$$

Notice that Parseval's theorem as stated here applies to nonperiodic functions. Parseval's theorem for periodic functions was presented in Sections 17.5 and 17.6. As evident in Eq. (18.63), Parseval's theorem shows that the energy associated with a nonperiodic signal is spread over the entire frequency spectrum, whereas the energy of a periodic signal is concentrated at the frequencies of its harmonic components.

Example 18.9

The voltage across a 10-Ω resistor is $v(t) = 5e^{-3t}u(t)$ V. Find the total energy dissipated in the resistor.

Solution:

1. **Define.** The problem is well defined and clearly stated.
2. **Present.** We are given the voltage across the resistor for all time and are asked to find the energy dissipated by the resistor. We note that the voltage is zero for all time less than zero. Thus, we only need to consider the time from zero to infinity.
3. **Alternative.** There are basically two ways to find this answer. The first would be to find the answer in the time domain. We will use the second approach to find the answer using Fourier analysis.
4. **Attempt.** In the time domain,

$$W_{10\Omega} = 0.1\int_{-\infty}^{\infty} f^2(t)\,dt = 0.1\int_{0}^{\infty} 25e^{-6t}\,dt$$

$$= 2.5\frac{e^{-6t}}{-6}\bigg|_0^{\infty} = \frac{2.5}{6} = \textbf{416.7 mJ}$$

5. **Evaluate.** In the frequency domain,

$$F(\omega) = V(\omega) = \frac{5}{3 + j\omega}$$

so that

$$|F(\omega)|^2 = F(\omega)F(\omega)^* = \frac{25}{9 + \omega^2}$$

Hence, the energy dissipated is

$$W_{10\Omega} = \frac{0.1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\, d\omega = \frac{0.1}{\pi}\int_{0}^{\infty} \frac{25}{9 + \omega^2}\, d\omega$$

$$= \frac{2.5}{\pi}\left(\frac{1}{3}\tan^{-1}\frac{\omega}{3}\right)\Bigg|_0^{\infty} = \frac{2.5}{\pi}\left(\frac{1}{3}\right)\left(\frac{\pi}{2}\right) = \frac{2.5}{6} = \mathbf{416.7\ mJ}$$

6. **Satisfactory?** We have satisfactorily solved the problem and can present the results as a solution to the problem.

Practice Problem 18.9

(a) Calculate the total energy absorbed by a 1-Ω resistor with $i(t) = 5e^{-2|t|}$ A in the time domain. (b) Repeat (a) in the frequency domain.

Answer: (a) 12.5 J, (b) 12.5 J.

Example 18.10

Calculate the fraction of the total energy dissipated by a 1-Ω resistor in the frequency band $-10 < \omega < 10$ rad/s when the voltage across it is $v(t) = e^{-2t}u(t)$.

Solution:

Given that $f(t) = v(t) = e^{-2t}u(t)$, then

$$F(\omega) = \frac{1}{2 + j\omega} \quad \Rightarrow \quad |F(\omega)|^2 = \frac{1}{4 + \omega^2}$$

The total energy dissipated by the resistor is

$$W_{1\Omega} = \frac{1}{\pi}\int_{0}^{\infty} |F(\omega)|^2\, d\omega = \frac{1}{\pi}\int_{0}^{\infty} \frac{d\omega}{4 + \omega^2}$$

$$= \frac{1}{\pi}\left(\frac{1}{2}\tan^{-1}\frac{\omega}{2}\Bigg|_0^{\infty}\right) = \frac{1}{\pi}\left(\frac{1}{2}\right)\frac{\pi}{2} = 0.25\ \text{J}$$

The energy in the frequencies $-10 < \omega < 10$ rad/s is

$$W = \frac{1}{\pi}\int_{0}^{10} |F(\omega)|^2\, d\omega = \frac{1}{\pi}\int_{0}^{10} \frac{d\omega}{4 + \omega^2} = \frac{1}{\pi}\left(\frac{1}{2}\tan^{1}\frac{\omega}{2}\Bigg|_0^{10}\right)$$

$$= \frac{1}{2\pi}\tan^{-1} 5 = \frac{1}{2\pi}\left(\frac{78.69^\circ}{180^\circ}\pi\right) = 0.218\ \text{J}$$

Its percentage of the total energy is

$$\frac{W}{W_{1\Omega}} = \frac{0.218}{0.25} = 87.4\ \%$$

Practice Problem 18.10

A 2-Ω resistor has $i(t) = 2e^{-t}u(t)$ A. What percentage of the total energy is in the frequency band $-4 < \omega < 4$ rad/s?

Answer: 84.4 percent.

18.6 Comparing the Fourier and Laplace Transforms

It is worthwhile to take some moments to compare the Laplace and Fourier transforms. The following similarities and differences should be noted:

1. The Laplace transform defined in Chapter 15 is one-sided in that the integral is over $0 < t < \infty$, making it only useful for positive-time functions, $f(t)$, $t > 0$. The Fourier transform is applicable to functions defined for all time.
2. For a function $f(t)$ that is nonzero for positive time only (i.e., $f(t) = 0, t < 0$) and $\int_0^\infty |f(t)|\,dt < \infty$, the two transforms are related by

$$F(\omega) = F(s)\,|_{s=j\omega} \tag{18.66}$$

   This equation also shows that the Fourier transform can be regarded as a special case of the Laplace transform with $s = j\omega$. Recall that $s = \sigma + j\omega$. Therefore, Eq. (18.66) shows that the Laplace transform is related to the entire s plane, whereas the Fourier transform is restricted to the $j\omega$ axis. See Fig. 15.1.
3. The Laplace transform is applicable to a wider range of functions than the Fourier transform. For example, the function $tu(t)$ has a Laplace transform but no Fourier transform. But Fourier transforms exist for signals that are not physically realizable and have no Laplace transforms.
4. The Laplace transform is better suited for the analysis of transient problems involving initial conditions, since it permits the inclusion of the initial conditions, whereas the Fourier transform does not. The Fourier transform is especially useful for problems in the steady state.
5. The Fourier transform provides greater insight into the frequency characteristics of signals than does the Laplace transform.

In other words, if all the poles of $F(s)$ lie in the left-hand side of the s plane, then one can obtain the Fourier transform $F(\omega)$ from the corresponding Laplace transform $F(s)$ by merely replacing s by $j\omega$. Note that this is not the case, for example, for $u(t)$ or $\cos atu(t)$.

Some of the similarities and differences can be observed by comparing Tables 15.1 and 15.2 with Tables 18.1 and 18.2.

18.7 †Applications

Besides its usefulness for circuit analysis, the Fourier transform is used extensively in a variety of fields such as optics, spectroscopy, acoustics, computer science, and electrical engineering. In electrical engineering, it is applied in communications systems and signal processing, where frequency response and frequency spectra are vital. Here we consider two simple applications: amplitude modulation (AM) and sampling.

18.7.1 Amplitude Modulation

Electromagnetic radiation or transmission of information through space has become an indispensable part of a modern technological society. However, transmission through space is only efficient and economical at high frequencies (above 20 kHz). To transmit intelligent signals—such as for speech and music—contained in the low-frequency range of 50 Hz to 20 kHz is expensive; it requires a huge amount of power and large antennas. A common method of transmitting low-frequency audio information is to transmit a high-frequency signal, called a *carrier*, which is controlled in some way to correspond to the audio information. Three characteristics (amplitude, frequency, or phase) of a carrier can be controlled so as to allow it to carry the intelligent signal, called the *modulating signal*. Here we will only consider the control of the carrier's amplitude. This is known as *amplitude modulation.*

Amplitude modulation (AM) is a process whereby the amplitude of the carrier is controlled by the modulating signal.

AM is used in ordinary commercial radio bands and the video portion of commercial television.

Suppose the audio information, such as voice or music (or the modulating signal in general) to be transmitted is $m(t) = V_m \cos \omega_m t$, while the high-frequency carrier is $c(t) = V_c \cos \omega_c t$, where $\omega_c \gg \omega_m$. Then an AM signal $f(t)$ is given by

$$f(t) = V_c[1 + m(t)] \cos \omega_c t \tag{18.67}$$

Figure 18.22 illustrates the modulating signal $m(t)$, the carrier $c(t)$, and the AM signal $f(t)$. We can use the result in Eq. (18.27) together with the Fourier transform of the cosine function (see Example 18.1 or Table 18.1) to determine the spectrum of the AM signal:

$$\begin{aligned} F(\omega) &= \mathcal{F}[V_c \cos \omega_c t] + \mathcal{F}[V_c m(t) \cos \omega_c t] \\ &= V_c \pi[\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] \\ &\quad + \frac{V_c}{2}[M(\omega - \omega_c) + M(\omega + \omega_c)] \end{aligned} \tag{18.68}$$

where $M(\omega)$ is the Fourier transform of the modulating signal $m(t)$. Shown in Fig. 18.23 is the frequency spectrum of the AM signal. Figure 18.23 indicates that the AM signal consists of the carrier and two other sinusoids. The sinusoid with frequency $\omega_c - \omega_m$ is known as the *lower sideband*, while the one with frequency $\omega_c + \omega_m$ is known as the *upper sideband*.

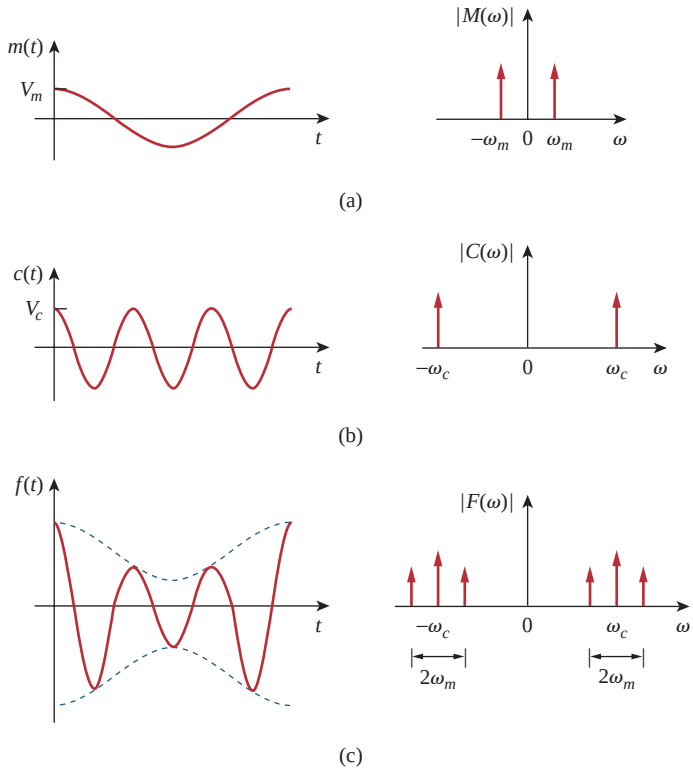


Figure 18.22
Time domain and frequency display of: (a) modulating signal, (b) carrier signal, (c) AM signal.

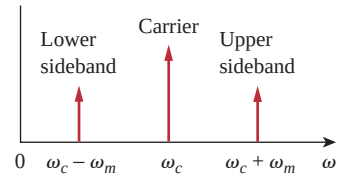


Figure 18.23
Frequency spectrum of AM signal.

Notice that we have assumed that the modulating signal is sinusoidal to make the analysis easy. In real life, $m(t)$ is a nonsinusoidal, band-limited signal—its frequency spectrum is within the range between 0 and $\omega_u = 2\pi f_u$ (i.e., the signal has an upper frequency limit). Typically, $f_u = 5$ kHz for AM radio. If the frequency spectrum of the modulating signal is as shown in Fig. 18.24(a), then the frequency spectrum of the AM signal is shown in Fig. 18.24(b). Thus, to avoid any interference, carriers for AM radio stations are spaced 10 kHz apart.

At the receiving end of the transmission, the audio information is recovered from the modulated carrier by a process known as *demodulation*.

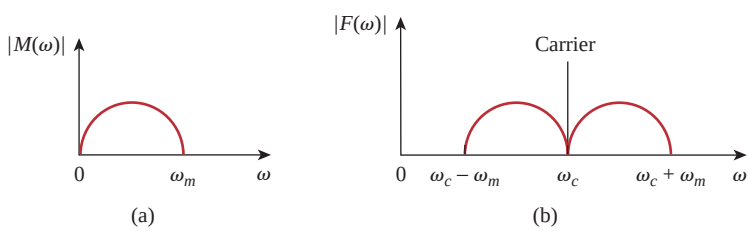


Figure 18.24
Frequency spectrum of: (a) modulating signal, (b) AM signal.

Example 18.11

A music signal has frequency components from 15 Hz to 30 kHz. If this signal could be used to amplitude modulate a 1.2-MHz carrier, find the range of frequencies for the lower and upper sidebands.

Solution:
The lower sideband is the difference of the carrier and modulating frequencies. It will include the frequencies from

$$1{,}200{,}000 - 30{,}000 \text{ Hz} = 1{,}170{,}000 \text{ Hz}$$

to

$$1{,}200{,}000 - 15 \text{ Hz} = 1{,}199{,}985 \text{ Hz}$$

The upper sideband is the sum of the carrier and modulating frequencies. It will include the frequencies from

$$1{,}200{,}000 + 15 \text{ Hz} = 1{,}200{,}015 \text{ Hz}$$

to

$$1{,}200{,}000 + 30{,}000 \text{ Hz} = 1{,}230{,}000 \text{ Hz}$$

Practice Problem 18.11

If a 2-MHz carrier is modulated by a 4-kHz intelligent signal, determine the frequencies of the three components of the AM signal that results.

Answer: 2,004,000 Hz, 2,000,000 Hz, 1,996,000 Hz.

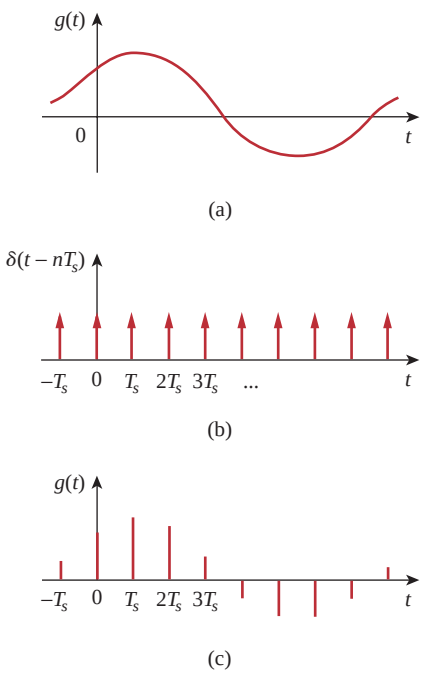


Figure 18.25
(a) Continuous (analog) signal to be sampled, (b) train of impulses, (c) sampled (digital) signal.

18.7.2 Sampling

In analog systems, signals are processed in their entirety. However, in modern digital systems, only samples of signals are required for processing. This is possible as a result of the sampling theorem given in Section 17.8.1. The sampling can be done by using a train of pulses or impulses. We will use impulse sampling here.

Consider the continuous signal $g(t)$ shown in Fig. 18.25(a). This can be multiplied by a train of impulses $\delta(t - nT_s)$ shown in Fig. 18.25(b), where T_s is the *sampling interval* and $f_s = 1/T_s$ is the *sampling frequency* or the *sampling rate*. The sampled signal $g_s(t)$ is therefore

$$g_s(t) = g(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} g(nT_s)\,\delta(t - nT_s) \quad \textbf{(18.69)}$$

The Fourier transform of this is

$$G_s(\omega) = \sum_{n=-\infty}^{\infty} g(nT_s)\mathcal{F}[\delta(t - nT_s)] = \sum_{n=-\infty}^{\infty} g(nT_s)e^{-jn\omega T_s} \quad \textbf{(18.70)}$$

It can be shown that

$$\sum_{n=-\infty}^{\infty} g(nT_s)e^{-jn\omega T_s} = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} G(\omega + n\omega_s) \quad \textbf{(18.71)}$$

where $\omega_s = 2\pi/T_s$. Thus, Eq. (18.70) becomes

$$G_s(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} G(\omega + n\omega_s) \tag{18.72}$$

This shows that the Fourier transform $G_s(\omega)$ of the sampled signal is a sum of translates of the Fourier transform of the original signal at a rate of $1/T_s$.

In order to ensure optimum recovery of the original signal, what must be the sampling interval? This fundamental question in sampling is answered by an equivalent part of the sampling theorem:

> A band-limited signal, with no frequency component higher than W hertz, may be completely recovered from its samples taken at a frequency at least twice as high as $2W$ samples per second.

In other words, for a signal with bandwidth W hertz, there is no loss of information or overlapping if the sampling frequency is at least twice the highest frequency in the modulating signal. Thus,

$$\frac{1}{T_s} = f_s \geq 2W \tag{18.73}$$

The sampling frequency $f_s = 2W$ is known as the *Nyquist frequency* or rate, and $1/f_s$ is the *Nyquist interval.*

Example 18.12

A telephone signal with a cutoff frequency of 5 kHz is sampled at a rate 60 percent higher than the minimum allowed rate. Find the sampling rate.

Solution:
The minimum sample rate is the Nyquist rate $= 2W = 2 \times 5 = 10$ kHz. Hence,

$$f_s = 1.60 \times 2W = 16 \text{ kHz}$$

Practice Problem 18.12

An audio signal that is band-limited to 12.5 kHz is digitized into 8-bit samples. What is the maximum sampling interval that must be used to ensure complete recovery?

Answer: 40 μs.

18.8 Summary

1. The Fourier transform converts a nonperiodic function $f(t)$ into a transform $F(\omega)$, where

$$F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt$$

2. The inverse Fourier transform of $F(\omega)$ is

$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)e^{j\omega t}\,d\omega$$

3. Important Fourier transform properties and pairs are summarized in Tables 18.1 and 18.2, respectively.
4. Using the Fourier transform method to analyze a circuit involves finding the Fourier transform of the excitation, transforming the circuit element into the frequency domain, solving for the unknown response, and transforming the response to the time domain using the inverse Fourier transform.
5. If $H(\omega)$ is the transfer function of a network, then $H(\omega)$ is the Fourier transform of the network's impulse response; that is,

$$H(\omega) = \mathcal{F}[h(t)]$$

The output $V_o(\omega)$ of the network can be obtained from the input $V_i(\omega)$ using

$$V_o(\omega) = H(\omega)V_i(\omega)$$

6. Parseval's theorem gives the energy relationship between a function $f(t)$ and its Fourier transform $F(\omega)$. The 1-Ω energy is

$$W_{1\Omega} = \int_{-\infty}^{\infty} f^2(t)\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\,d\omega$$

The theorem is useful in calculating energy carried by a signal either in the time domain or in the frequency domain.
7. Typical applications of the Fourier transform are found in amplitude modulation (AM) and sampling. For AM application, a way of determining the sidebands in an amplitude-modulated wave is derived from the modulation property of the Fourier transform. For sampling application, we found that no information is lost in sampling (required for digital transmission) if the sampling frequency is equal to at least twice the Nyquist rate.

Review Questions

18.1 Which of these functions does not have a Fourier transform?

(a) $e^{t}u(-t)$ (b) $te^{-3t}u(t)$
(c) $1/t$ (d) $|t|u(t)$

18.2 The Fourier transform of e^{j2t} is:

(a) $\dfrac{1}{2+j\omega}$ (b) $\dfrac{1}{-2+j\omega}$
(c) $2\pi\delta(\omega-2)$ (d) $2\pi\delta(\omega+2)$

18.3 The inverse Fourier transform of $\dfrac{e^{-j\omega}}{2+j\omega}$ is

(a) e^{-2t} (b) $e^{-2t}u(t-1)$
(c) $e^{-2(t-1)}$ (d) $e^{-2(t-1)}u(t-1)$

18.4 The inverse Fourier transform of $\delta(\omega)$ is:

(a) $\delta(t)$ (b) $u(t)$ (c) 1 (d) $1/2\pi$

18.5 The inverse Fourier transform of $j\omega$ is:

(a) $\delta'(t)$ (b) $u'(t)$
(c) $1/t$ (d) undefined

18.6 Evaluating the integral $\displaystyle\int_{-\infty}^{\infty}\frac{10\delta(\omega)}{4+\omega^2}\,d\omega$ results in:

(a) 0 (b) 2 (c) 2.5 (d) ∞

18.7 The integral $\displaystyle\int_{-\infty}^{\infty}\frac{10\delta(\omega-1)}{4+\omega^2}\,d\omega$ gives:

(a) 0 (b) 2 (c) 2.5 (d) ∞

18.8 The current through an initially uncharged 1-F capacitor is $\delta(t)$ A. The voltage across the capacitor is:

(a) $u(t)$ V (b) $-1/2 + u(t)$ V
(c) $e^{-t}u(t)$ V (d) $\delta(t)$ V

18.9 A unit step current is applied through a 1-H inductor. The voltage across the inductor is:

(a) $u(t)$ V (b) sgn(t) V
(c) $e^{-t}u(t)$ V (d) $\delta(t)$ V

18.10 Parseval's theorem is only for nonperiodic functions.

(a) True (b) False

Answers: 18.1c, 18.2c, 18.3d, 18.4d, 18.5a, 18.6c, 18.7b, 18.8a, 18.9d, 18.10b

Problems

†Sections 18.2 and 18.3 Fourier Transform and its Properties

18.1 Obtain the Fourier transform of the function in Fig. 18.26.

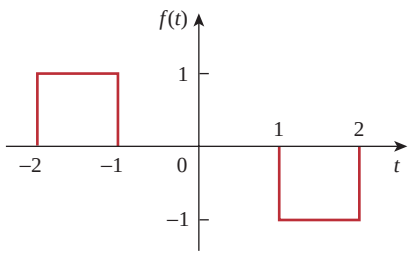


Figure 18.26
For Prob. 18.1.

18.2 Using Fig. 18.27, design a problem to help other students better understand the Fourier transform given a wave shape.

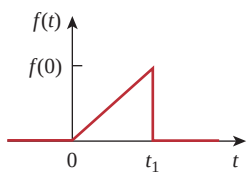


Figure 18.27
For Prob. 18.2.

18.3 Calculate the Fourier transform of the signal in Fig. 18.28.

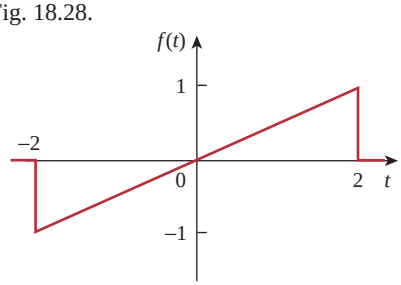


Figure 18.28
For Prob. 18.3.

18.4 Find the Fourier transform of the waveform shown in Fig. 18.29.

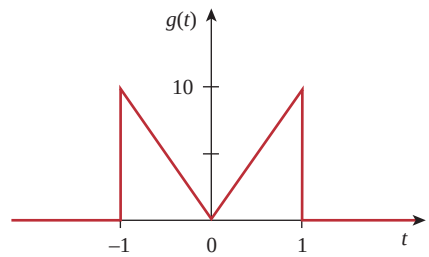


Figure 18.29
For Prob. 18.4.

18.5 Obtain the Fourier transform of the signal shown in Fig. 18.30.

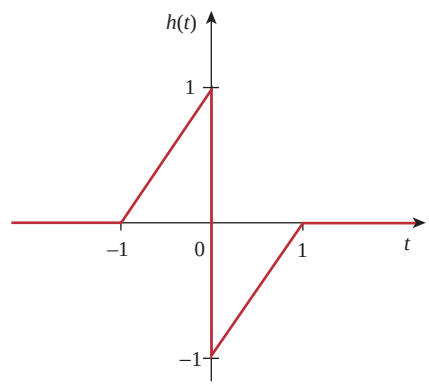


Figure 18.30
For Prob. 18.5.

18.6 Find the Fourier transforms of both functions in Fig. 18.31 on the following page.

† We have marked (with the *MATLAB* icon) the problems where we are asking the student to find the Fourier transform of a wave shape. We do this because you can use *MATLAB* to plot the results as a check.

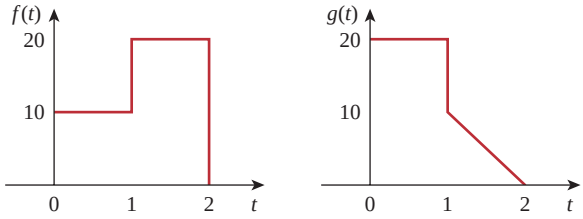


Figure 18.31
For Prob. 18.6.

18.7 Find the Fourier transforms of the signals in Fig. 18.32.

ML

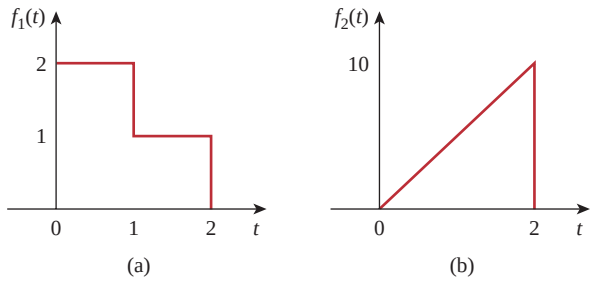
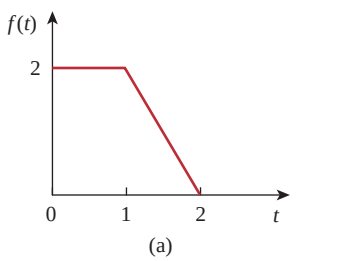


Figure 18.32
For Prob. 18.7.

18.8 Obtain the Fourier transforms of the signals shown in Fig. 18.33.

ML



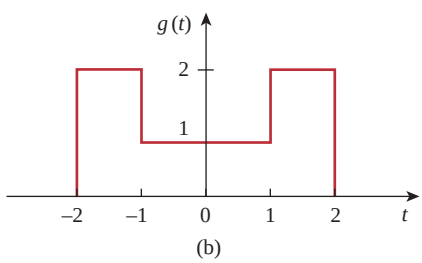
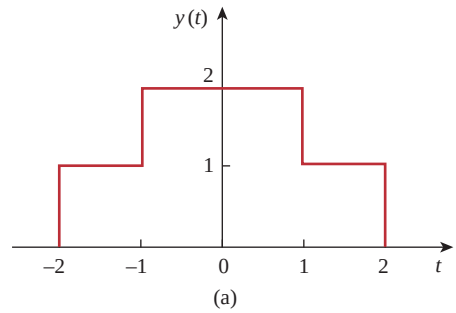


Figure 18.33
For Prob. 18.8.

18.9 Determine the Fourier transforms of the signals in Fig. 18.34.

ML



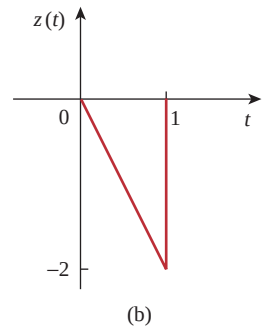


Figure 18.34
For Prob. 18.9.

18.10 Obtain the Fourier transforms of the signals shown in Fig. 18.35.

ML

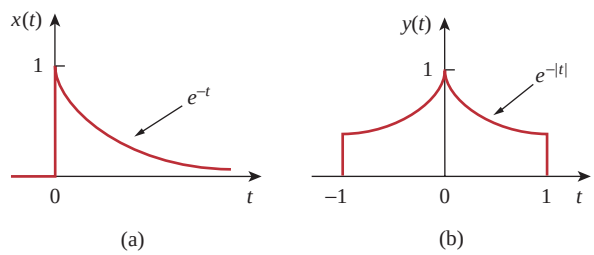


Figure 18.35
For Prob. 18.10.

18.11 Find the Fourier transform of the "sine-wave pulse" shown in Fig. 18.36.

ML

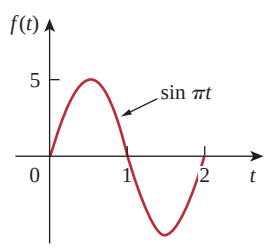


Figure 18.36
For Prob. 18.11.

18.12 Find the Fourier transform of the following signals.

(a) $f_1(t) = e^{-3t}\sin(10t)u(t)$

(b) $f_2(t) = e^{-4t}\cos(10t)u(t)$

18.13 Find the Fourier transform of the following signals:

(a) $f(t) = \cos(at - \pi/3), \quad -\infty < t < \infty$

(b) $g(t) = u(t+1)\sin \pi t, \quad -\infty < t < \infty$

(c) $h(t) = (1 + A\sin at)\cos bt, \quad -\infty < t < \infty,$ where A, a, and b are constants

(d) $i(t) = 1 - t, \quad 0 < t < 4$

18.14 Design a problem to help other students better understand finding the Fourier transform of a variety of time varying functions (do at least three).

18.15 Find the Fourier transforms of the following functions:

(a) $f(t) = \delta(t+3) - \delta(t-3)$

(b) $f(t) = \int_{-\infty}^{\infty} 2\delta(t-1)\,dt$

(c) $f(t) = \delta(3t) - \delta'(2t)$

***18.16** Determine the Fourier transforms of these functions:

(a) $f(t) = 10/t^2$

(b) $g(t) = 20/(4 + t^2)$

18.17 Find the Fourier transforms of:

(a) $\cos 2t u(t)$

(b) $\sin 10t u(t)$

18.18 Given that $F(\omega) = \mathcal{F}[f(t)]$, prove the following results, using the definition of Fourier transform:

(a) $\mathcal{F}[f(t - t_0)] = e^{-j\omega t_0}F(\omega)$

(b) $\mathcal{F}\left[\dfrac{df(t)}{dt}\right] = j\omega F(\omega)$

(c) $\mathcal{F}[f(-t)] = F(-\omega)$

(d) $\mathcal{F}[tf(t)] = j\dfrac{d}{d\omega}F(\omega)$

18.19 Find the Fourier transform of

$$f(t) = \cos 2\pi t[u(t) - u(t-1)]$$

18.20 (a) Show that a periodic signal with exponential Fourier series

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

has the Fourier transform

$$F(\omega) = \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_0)$$

where $\omega_0 = 2\pi/T$.

(b) Find the Fourier transform of the signal in Fig. 18.37.

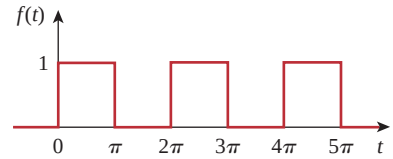


Figure 18.37
For Prob. 18.20(b).

18.21 Show that

$$\int_{-\infty}^{\infty}\left(\frac{\sin a\omega}{a\omega}\right)^2 d\omega = \frac{\pi}{a}$$

Hint: Use the fact that

$$\mathcal{F}[u(t+a) - u(t-a)] = 2a\left(\frac{\sin a\omega}{a\omega}\right).$$

18.22 Prove that if $F(\omega)$ is the Fourier transform of $f(t)$,

$$\mathcal{F}[f(t)\sin\omega_0 t] = \frac{j}{2}[F(\omega + \omega_0) - F(\omega - \omega_0)]$$

18.23 If the Fourier transform of $f(t)$ is

$$F(\omega) = \frac{10}{(2 + j\omega)(5 + j\omega)}$$

determine the transforms of the following:

(a) $f(-3t)$ (b) $f(2t-1)$ (c) $f(t)\cos 2t$

(d) $\dfrac{d}{dt}f(t)$ (e) $\int_{-\infty}^{t} f(t)\,dt$

18.24 Given that $\mathcal{F}[f(t)] = (j/\omega)(e^{-j\omega} - 1)$, find the Fourier transforms of:

(a) $x(t) = f(t) + 3$ (b) $y(t) = f(t-2)$

(c) $h(t) = f'(t)$

(d) $g(t) = 4f\left(\dfrac{2}{3}t\right) + 10f\left(\dfrac{5}{3}t\right)$

18.25 Obtain the inverse Fourier transform of the following signals.

(a) $G(\omega) = \dfrac{10}{j\omega - 2}$

(b) $H(\omega) = \dfrac{6}{\omega^2 + 4}$

(c) $X(\omega) = \dfrac{5}{(j\omega - 1)(j\omega - 2)}$

18.26 Determine the inverse Fourier transforms of the following:

(a) $F(\omega) = \dfrac{e^{-j2\omega}}{1 + j\omega}$

(b) $H(\omega) = \dfrac{1}{(j\omega + 4)^2}$

(c) $G(\omega) = 2u(\omega + 1) - 2u(\omega - 1)$

* An asterisk indicates a challenging problem.

18.27 Find the inverse Fourier transforms of the following functions:

(a) $F(\omega) = \dfrac{100}{j\omega(j\omega + 10)}$

(b) $G(\omega) = \dfrac{10j\omega}{(-j\omega + 2)(j\omega + 3)}$

(c) $H(\omega) = \dfrac{60}{-\omega^2 + j40\omega + 1300}$

(d) $Y(\omega) = \dfrac{\delta(\omega)}{(j\omega + 1)(j\omega + 2)}$

18.28 Find the inverse Fourier transforms of:

(a) $\dfrac{\pi\delta(\omega)}{(5 + j\omega)(2 + j\omega)}$

(b) $\dfrac{10\delta(\omega + 2)}{j\omega(j\omega + 1)}$

(c) $\dfrac{20\delta(\omega - 1)}{(2 + j\omega)(3 + j\omega)}$

(d) $\dfrac{5\pi\delta(\omega)}{5 + j\omega} + \dfrac{5}{j\omega(5 + j\omega)}$

***18.29** Determine the inverse Fourier transforms of:

(a) $F(\omega) = 4\delta(\omega + 3) + \delta(\omega) + 4\delta(\omega - 3)$

(b) $G(\omega) = 4u(\omega + 2) - 4u(\omega - 2)$

(c) $H(\omega) = 6\cos 2\omega$

18.30 For a linear system with input $x(t)$ and output $y(t)$, find the impulse response for the following cases:

(a) $x(t) = e^{-at}u(t)$, $\quad y(t) = u(t) - u(-t)$

(b) $x(t) = e^{-t}u(t)$, $\quad y(t) = e^{-2t}u(t)$

(c) $x(t) = \delta(t)$, $\quad y(t) = e^{-at}\sin btu(t)$

18.31 Given a linear system with output $y(t)$ and impulse response $h(t)$, find the corresponding input $x(t)$ for the following cases:

(a) $y(t) = te^{-at}u(t)$, $\quad h(t) = e^{-at}u(t)$

(b) $y(t) = u(t + 1) - u(t - 1)$, $\quad h(t) = \delta(t)$

(c) $y(t) = e^{-at}u(t)$, $\quad h(t) = \text{sgn}(t)$

***18.32** Determine the functions corresponding to the following Fourier transforms:

(a) $F_1(\omega) = \dfrac{e^{j\omega}}{-j\omega + 1}$ $\quad$ (b) $F_2(\omega) = 2e^{|\omega|}$

(c) $F_3(\omega) = \dfrac{1}{(1 + \omega^2)^2}$ $\quad$ (d) $F_4(\omega) = \dfrac{\delta(\omega)}{1 + j2\omega}$

***18.33** Find $f(t)$ if:

(a) $F(\omega) = 2\sin\pi\omega[u(\omega + 1) - u(\omega - 1)]$

(b) $F(\omega) = \dfrac{1}{\omega}(\sin 2\omega - \sin\omega) + \dfrac{j}{\omega}(\cos 2\omega - \cos\omega)$

18.34 Determine the signal $f(t)$ whose Fourier transform is shown in Fig. 18.38. (*Hint:* Use the duality property.)

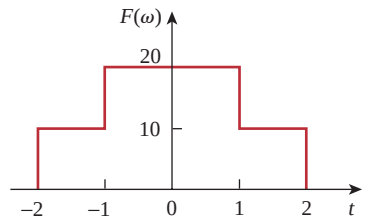


Figure 18.38
For Prob. 18.34.

18.35 A signal $f(t)$ has Fourier transform

$$F(\omega) = \frac{1}{2 + j\omega}$$

Determine the Fourier transform of the following signals:

(a) $x(t) = f(3t - 1)$

(b) $y(t) = f(t)\cos 5t$

(c) $z(t) = \dfrac{d}{dt}f(t)$

(d) $h(t) = f(t) * f(t)$

(e) $i(t) = tf(t)$

Section 18.4 Circuit Applications

18.36 The transfer function of a circuit is

$$H(\omega) = \frac{2}{j\omega + 2}$$

If the input signal to the circuit is $v_s(t) = e^{-4t}u(t)$ V, find the output signal. Assume all initial conditions are zero.

18.37 Find the transfer function $I_o(\omega)/I_s(\omega)$ for the circuit in Fig. 18.39.

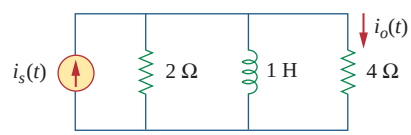


Figure 18.39
For Prob. 18.37.

18.38 Using Fig. 18.40, design a problem to help other students better understand using Fourier transforms to do circuit analysis.

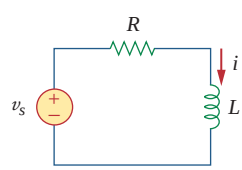


Figure 18.40
For Prob. 18.38.

18.39 Given the circuit in Fig. 18.41, with its excitation, determine the Fourier transform of $i(t)$.

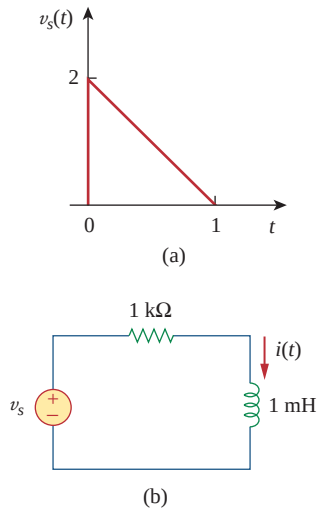


Figure 18.41
For Prob. 18.39.

18.40 Determine the current $i(t)$ in the circuit of Fig. 18.42(b), given the voltage source shown in Fig. 18.42(a).

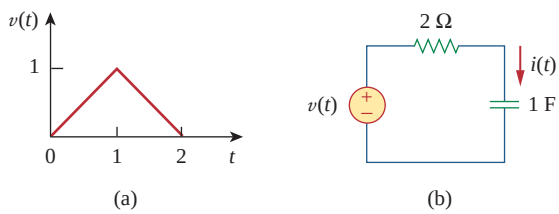


Figure 18.42
For Prob. 18.40.

18.41 Determine the Fourier transform of $v(t)$ in the circuit shown in Fig. 18.43.

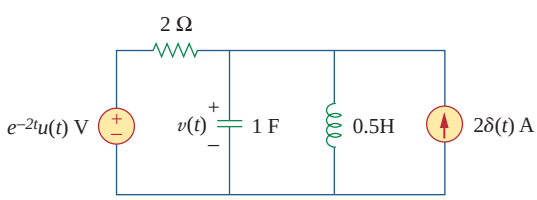


Figure 18.43
For Prob. 18.41.

18.42 Obtain the current $i_o(t)$ in the circuit of Fig. 18.44.

(a) Let $i(t) = \text{sgn}(t)$ A.

(b) Let $i(t) = 4[u(t) - u(t-1)]$ A.

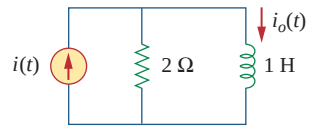


Figure 18.44
For Prob. 18.42.

18.43 Find $v_o(t)$ in the circuit of Fig. 18.45, where $i_s = 5e^{-t}u(t)$ A.

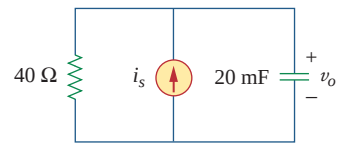


Figure 18.45
For Prob. 18.43.

18.44 If the rectangular pulse in Fig. 18.46(a) is applied to the circuit in Fig. 18.46(b), find v_o at $t = 1$ s.

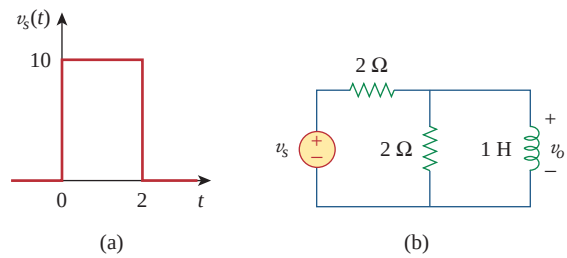


Figure 18.46
For Prob. 18.44.

18.45 Use the Fourier transform to find i(t) in the circuit of Fig. 18.47 if $v_s(t) = 10e^{-2t}u(t)$.

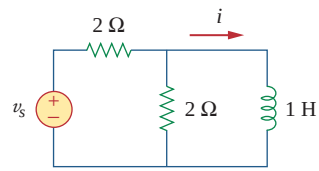


Figure 18.47
For Prob. 18.45.

18.46 Determine the Fourier transform of $i_o(t)$ in the circuit of Fig. 18.48.

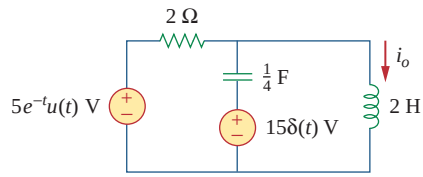


Figure 18.48
For Prob. 18.46.

18.47 Find the voltage $v_o(t)$ in the circuit of Fig. 18.49. Let $i_s(t) = 8e^{-t}u(t)$ A.

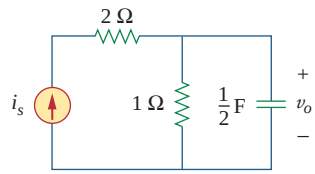


Figure 18.49
For Prob. 18.47.

18.48 Find $i_o(t)$ in the op amp circuit of Fig. 18.50.

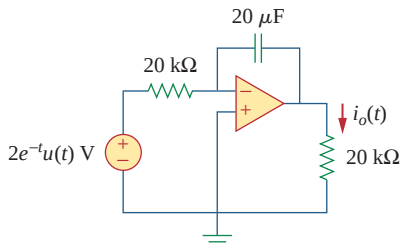


Figure 18.50
For Prob. 18.48.

18.49 Use the Fourier transform method to obtain $v_o(t)$ in the circuit of Fig. 18.51.

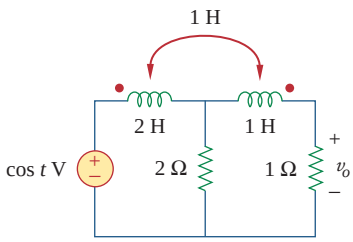


Figure 18.51
For Prob. 18.49.

18.50 Determine $v_o(t)$ in the transformer circuit of Fig. 18.52.

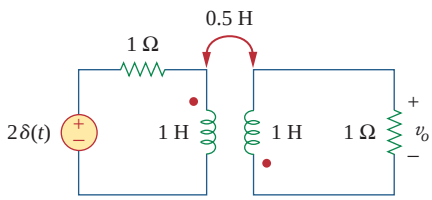


Figure 18.52
For Prob. 18.50.

18.51 Find the energy dissipated by the resistor in the circuit of Fig. 18.53.

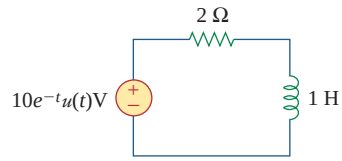


Figure 18.53
For Prob. 18.51.

Section 18.5 Parseval's Theorem

18.52 For $F(\omega) = \dfrac{1}{3 + j\omega}$, find $J = \displaystyle\int_{-\infty}^{\infty} f^2(t)\,dt$.

18.53 If $f(t) = e^{-2|t|}$, find $J = \displaystyle\int_{-\infty}^{\infty} |F(\omega)|^2\,d\omega$.

18.54 Design a problem to help other students better understand finding the total energy in a given signal.

18.55 Let $f(t) = 5e^{-(t-2)}u(t)$. Find $F(\omega)$ and use it to find the total energy in $f(t)$.

18.56 The voltage across a 1-Ω resistor is $v(t) = te^{-2t}u(t)$ V. (a) What is the total energy absorbed by the resistor? (b) What fraction of this energy absorbed is in the frequency band $-2 \le \omega \le 2$?

18.57 Let $i(t) = 5e^{t}u(-t)$ A. Find the total energy carried by $i(t)$ and the percentage of the 1-Ω energy in the frequency range of $-5 < \omega < 5$ rad/s.

Section 18.6 Applications

18.58 An AM signal is specified by

$$f(t) = 10(1 + 4\cos 200\pi t)\cos \pi \times 10^4 t$$

Determine the following:

(a) the carrier frequency,

(b) the lower sideband frequency,

(c) the upper sideband frequency.

18.59 For the linear system in Fig. 18.54, when the input voltage is $v_i(t) = 2\delta(t)$ V, the output is $v_o(t) = 10e^{-2t} - 6e^{-4t}$ V. Find the output when the input is $v_i(t) = 4e^{-t}u(t)$ V.

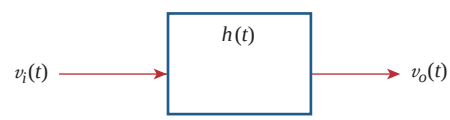


Figure 18.54
For Prob. 18.9.

18.60 A band-limited signal has the following Fourier series representation:

$$i_s(t) = 10 + 8\cos(2\pi t + 30°) + 5\cos(4\pi t - 150°)\text{mA}$$

If the signal is applied to the circuit in Fig. 18.55, find $v(t)$.

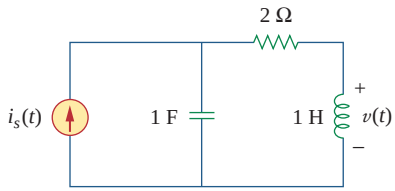


Figure 18.55
For Prob. 18.60.

18.61 In a system, the input signal $x(t)$ is amplitude-modulated by $m(t) = 2 + \cos\omega_0 t$. The response $y(t) = m(t)x(t)$. Find $Y(\omega)$ in terms of $X(\omega)$.

18.62 A voice signal occupying the frequency band of 0.4 to 3.5 kHz is used to amplitude-modulate a 10-MHz carrier. Determine the range of frequencies for the lower and upper sidebands.

18.63 For a given locality, calculate the number of stations allowable in the AM broadcasting band (540 to 1600 kHz) without interference with one another.

18.64 Repeat the previous problem for the FM broadcasting band (88 to 108 MHz), assuming that the carrier frequencies are spaced 200 kHz apart.

18.65 The highest-frequency component of a voice signal is 3.4 kHz. What is the Nyquist rate of the sampler of the voice signal?

18.66 A TV signal is band-limited to 4.5 MHz. If samples are to be reconstructed at a distant point, what is the maximum sampling interval allowable?

***18.67** Given a signal $g(t) = \text{sinc}(200\pi t)$, find the Nyquist rate and the Nyquist interval for the signal.

Comprehensive Problems

18.68 The voltage signal at the input of a filter is $v(t) = 50e^{-2|t|}$ V. What percentage of the total 1-Ω energy content lies in the frequency range of $1 < \omega < 5$ rad/s?

18.69 A signal with Fourier transform

$$F(\omega) = \frac{20}{4 + j\omega}$$

is passed through a filter whose cutoff frequency is 2 rad/s (i.e., $0 < \omega < 2$). What fraction of the energy in the input signal is contained in the output signal?

chapter

19

Two-Port Networks

Never put off till tomorrow what you can do today.
Never trouble another for what you can do yourself.
Never spend your money before you have it.
Never buy what you do not want because it is cheap.
Pride costs us more than hunger, thirst, and cold.
We seldom repent having eaten too little.
Nothing is troublesome that we do willingly.
How much pain the evils have cost us that have never happened!
Take things always by the smooth handle.
When angry, count ten before you speak; if very angry, a hundred.

—Thomas Jefferson

Enhancing Your Career

Career in Education

While two thirds of all engineers work in private industry, some work in academia and prepare students for engineering careers. The course on circuit analysis you are studying is an important part of the preparation process. If you enjoy teaching others, you may want to consider becoming an engineering educator.

Engineering professors work on state-of-the-art research projects, teach courses at graduate and undergraduate levels, and provide services to their professional societies and the community at large. They are expected to make original contributions in their areas of specialty. This requires a broad-based education in the fundamentals of electrical engineering and a mastery of the skills necessary for communicating their efforts to others.

If you like to do research, to work at the frontiers of engineering, to make contributions to technological advancement, to invent, consult, and/or teach, consider a career in engineering education. The best way to start is by talking with your professors and benefiting from their experience.

A solid understanding of mathematics and physics at the undergraduate level is vital to your success as an engineering professor. If you are having difficulty in solving your engineering textbook problems, start correcting any weaknesses you have in your mathematics and physics fundamentals.

Most universities these days require that engineering professors have a doctor's degree. In addition, some universities require that they be actively involved in research leading to publications in reputable journals.



Photo by James Watson

To prepare yourself for a career in engineering education, get as broad an education as possible, because electrical engineering is changing rapidly and becoming interdisciplinary. Without doubt, engineering education is a rewarding career. Professors get a sense of satisfaction and fulfillment as they see their students graduate, become leaders in their profession, and contribute significantly to the betterment of humanity.

19.1 Introduction

A pair of terminals through which a current may enter or leave a network is known as a *port*. Two-terminal devices or elements (such as resistors, capacitors, and inductors) result in one-port networks. Most of the circuits we have dealt with so far are two-terminal or one-port circuits, represented in Fig. 19.1(a). We have considered the voltage across or current through a single pair of terminals—such as the two terminals of a resistor, a capacitor, or an inductor. We have also studied four-terminal or two-port circuits involving op amps, transistors, and transformers, as shown in Fig. 19.1(b). In general, a network may have *n* ports. A port is an access to the network and consists of a pair of terminals; the current entering one terminal leaves through the other terminal so that the net current entering the port equals zero.

In this chapter, we are mainly concerned with *two-port* networks (or, simply, *two-ports*).

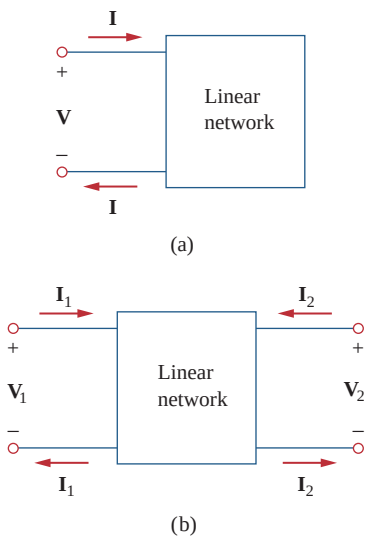


Figure 19.1
(a) One-port network, (b) two-port network.

A **two-port network** is an electrical network with two separate ports for input and output.

Thus, a two-port network has two terminal pairs acting as access points. As shown in Fig. 19.1(b), the current entering one terminal of a pair leaves the other terminal in the pair. Three-terminal devices such as transistors can be configured into two-port networks.

Our study of two-port networks is for at least two reasons. First, such networks are useful in communications, control systems, power systems, and electronics. For example, they are used in electronics to model transistors and to facilitate cascaded design. Second, knowing the parameters of a two-port network enables us to treat it as a "black box" when embedded within a larger network.

To characterize a two-port network requires that we relate the terminal quantities $\mathbf{V}_1$, $\mathbf{V}_2$, $\mathbf{I}_1$, and $\mathbf{I}_2$ in Fig. 19.1(b), out of which two are independent. The various terms that relate these voltages and currents are called *parameters*. Our goal in this chapter is to derive six sets of these parameters. We will show the relationship between these parameters and how two-port networks can be connected in series, parallel, or cascade. As with op amps, we are only interested in the terminal behavior of the circuits. And we will assume that the two-port circuits contain no independent sources, although they can contain dependent sources. Finally, we will apply some of the concepts developed in this chapter to the analysis of transistor circuits and synthesis of ladder networks.

19.2 Impedance Parameters

Impedance and admittance parameters are commonly used in the synthesis of filters. They are also useful in the design and analysis of impedance-matching networks and power distribution networks. We

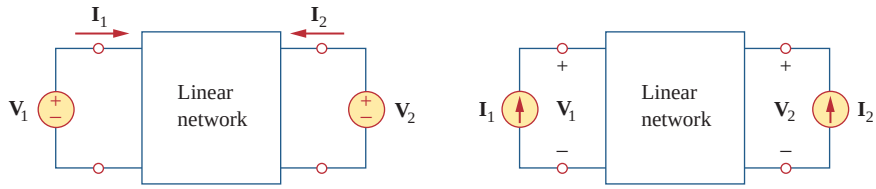


Figure 19.2
The linear two-port network: (a) driven by voltage sources, (b) driven by current sources.

discuss impedance parameters in this section and admittance parameters in the next section.

A two-port network may be voltage-driven as in Fig. 19.2(a) or current-driven as in Fig. 19.2(b). From either Fig. 19.2(a) or (b), the terminal voltages can be related to the terminal currents as

$$\begin{aligned} \mathbf{V}_1 &= \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2 \\ \mathbf{V}_2 &= \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2 \end{aligned} \tag{19.1}$$

Reminder: Only two of the four variables ($\mathbf{V}_1$, $\mathbf{V}_2$, $\mathbf{I}_1$, and $\mathbf{I}_2$) are independent. The other two can be found using Eq. (19.1).

or in matrix form as

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = [\mathbf{z}] \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} \tag{19.2}$$

where the $\mathbf{z}$ terms are called the *impedance parameters*, or simply *z parameters*, and have units of ohms.

The values of the parameters can be evaluated by setting $\mathbf{I}_1 = 0$ (input port open-circuited) or $\mathbf{I}_2 = 0$ (output port open-circuited). Thus,

$$\begin{aligned} \mathbf{z}_{11} &= \left.\frac{\mathbf{V}_1}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0}, & \mathbf{z}_{12} &= \left.\frac{\mathbf{V}_1}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0} \\ \mathbf{z}_{21} &= \left.\frac{\mathbf{V}_2}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0}, & \mathbf{z}_{22} &= \left.\frac{\mathbf{V}_2}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0} \end{aligned} \tag{19.3}$$

Since the *z* parameters are obtained by open-circuiting the input or output port, they are also called the *open-circuit impedance parameters*. Specifically,

$$\begin{aligned} \mathbf{z}_{11} &= \text{Open-circuit input impedance} \\ \mathbf{z}_{12} &= \text{Open-circuit transfer impedance from port 1 to port 2} \\ \mathbf{z}_{21} &= \text{Open-circuit transfer impedance from port 2 to port 1} \\ \mathbf{z}_{22} &= \text{Open-circuit output impedance} \end{aligned} \tag{19.4}$$

According to Eq. (19.3), we obtain $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$ by connecting a voltage $\mathbf{V}_1$ (or a current source $\mathbf{I}_1$) to port 1 with port 2 open-circuited as in Fig. 19.3(a) and finding $\mathbf{I}_1$ and $\mathbf{V}_2$; we then get

$$\mathbf{z}_{11} = \frac{\mathbf{V}_1}{\mathbf{I}_1}, \qquad \mathbf{z}_{21} = \frac{\mathbf{V}_2}{\mathbf{I}_1} \tag{19.5}$$

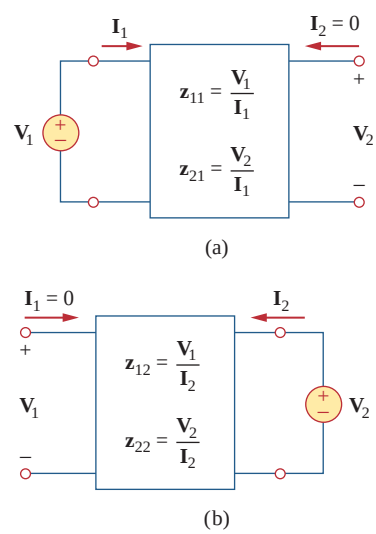


Figure 19.3
Determination of the *z* parameters: (a) finding $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$, (b) finding $\mathbf{z}_{12}$ and $\mathbf{z}_{22}$.

Similarly, we obtain $\mathbf{z}_{12}$ and $\mathbf{z}_{22}$ by connecting a voltage $\mathbf{V}_2$ (or a current source $\mathbf{I}_2$) to port 2 with port 1 open-circuited as in Fig. 19.3(b) and finding $\mathbf{I}_2$ and $\mathbf{V}_1$; we then get

$$\mathbf{z}_{12} = \frac{\mathbf{V}_1}{\mathbf{I}_2}, \qquad \mathbf{z}_{22} = \frac{\mathbf{V}_2}{\mathbf{I}_2} \tag{19.6}$$

The above procedure provides us with a means of calculating or measuring the *z* parameters.

Sometimes $\mathbf{z}_{11}$ and $\mathbf{z}_{22}$ are called *driving-point impedances*, while $\mathbf{z}_{21}$ and $\mathbf{z}_{12}$ are called *transfer impedances*. A driving-point impedance is the input impedance of a two-terminal (one-port) device. Thus, $\mathbf{z}_{11}$ is the input driving-point impedance with the output port open-circuited, while $\mathbf{z}_{22}$ is the output driving-point impedance with the input port open-circuited.

When $\mathbf{z}_{11} = \mathbf{z}_{22}$, the two-port network is said to be *symmetrical*. This implies that the network has mirrorlike symmetry about some center line; that is, a line can be found that divides the network into two similar halves.

When the two-port network is linear and has no dependent sources, the transfer impedances are equal ($\mathbf{z}_{12} = \mathbf{z}_{21}$), and the two-port is said to be *reciprocal*. This means that if the points of excitation and response are interchanged, the transfer impedances remain the same. As illustrated in Fig. 19.4, a two-port is reciprocal if interchanging an ideal voltage source at one port with an ideal ammeter at the other port gives the same ammeter reading. The reciprocal network yields $\mathbf{V} = \mathbf{z}_{12}\mathbf{I}$ according to Eq. (19.1) when connected as in Fig. 19.4(a), but yields $\mathbf{V} = \mathbf{z}_{21}\mathbf{I}$ when connected as in Fig. 19.4(b). This is possible only if $\mathbf{z}_{12} = \mathbf{z}_{21}$. Any two-port that is made entirely of resistors, capacitors, and inductors must be reciprocal. A reciprocal network can be replaced by the T-equivalent circuit in Fig. 19.5(a). If the network is not reciprocal, a more general equivalent network is shown in Fig. 19.5(b); notice that this figure follows directly from Eq. (19.1).

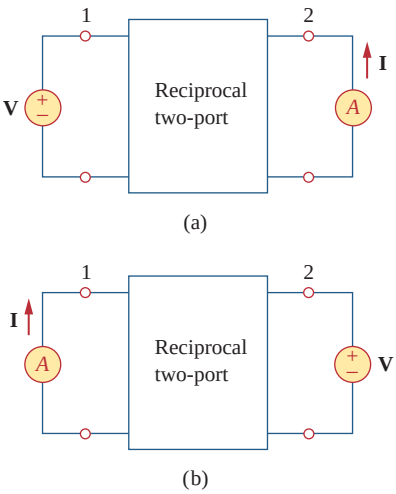


Figure 19.4
Interchanging a voltage source at one port with an ideal ammeter at the other port produces the same reading in a reciprocal two-port.

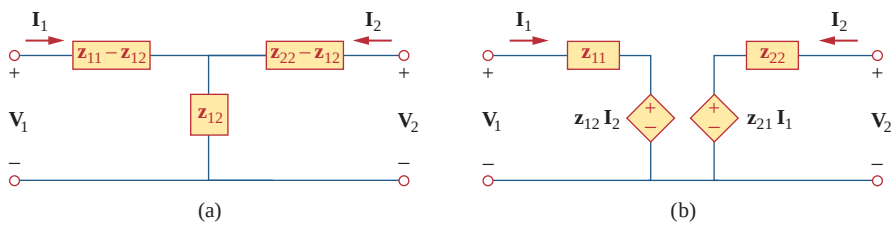


Figure 19.5
(a) T-equivalent circuit (for reciprocal case only), (b) general equivalent circuit.

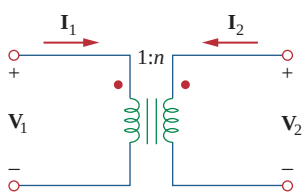


Figure 19.6
An ideal transformer has no *z* parameters.

It should be mentioned that for some two-port networks, the *z* parameters do not exist because they cannot be described by Eq. (19.1). As an example, consider the ideal transformer of Fig. 19.6. The defining equations for the two-port network are:

$$\mathbf{V}_1 = \frac{1}{n}\mathbf{V}_2, \qquad \mathbf{I}_1 = -n\mathbf{I}_2 \tag{19.7}$$

Observe that it is impossible to express the voltages in terms of the currents, and vice versa, as Eq. (19.1) requires. Thus, the ideal transformer has no *z* parameters. However, it does have hybrid parameters, as we shall see in Section 19.4.

Example 19.1

Determine the *z* parameters for the circuit in Fig. 19.7.

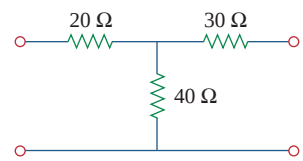


Figure 19.7
For Example 19.1.

Solution:

■ **METHOD 1** To determine $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$, we apply a voltage source $\mathbf{V}_1$ to the input port and leave the output port open as in Fig. 19.8(a). Then,

$$\mathbf{z}_{11} = \frac{\mathbf{V}_1}{\mathbf{I}_1} = \frac{(20 + 40)\mathbf{I}_1}{\mathbf{I}_1} = 60\ \Omega$$

that is, $\mathbf{z}_{11}$ is the input impedance at port 1.

$$\mathbf{z}_{21} = \frac{\mathbf{V}_2}{\mathbf{I}_1} = \frac{40\mathbf{I}_1}{\mathbf{I}_1} = 40\ \Omega$$

To find $\mathbf{z}_{12}$ and $\mathbf{z}_{22}$, we apply a voltage source $\mathbf{V}_2$ to the output port and leave the input port open as in Fig. 19.8(b). Then,

$$\mathbf{z}_{12} = \frac{\mathbf{V}_1}{\mathbf{I}_2} = \frac{40\mathbf{I}_2}{\mathbf{I}_2} = 40\ \Omega, \qquad \mathbf{z}_{22} = \frac{\mathbf{V}_2}{\mathbf{I}_2} = \frac{(30 + 40)\mathbf{I}_2}{\mathbf{I}_2} = 70\ \Omega$$

Thus,

$$[\mathbf{z}] = \begin{bmatrix} 60\ \Omega & 40\ \Omega \\ 40\ \Omega & 70\ \Omega \end{bmatrix}$$

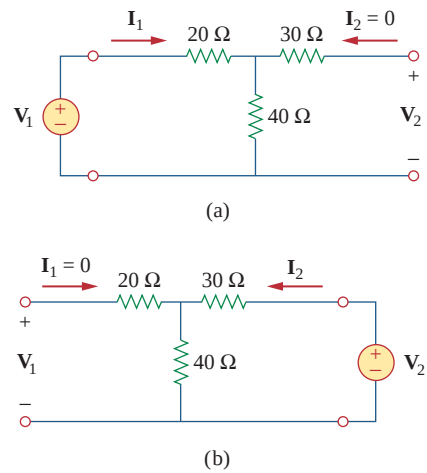


Figure 19.8
For Example 19.1: (a) finding $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$, (b) finding $\mathbf{z}_{12}$ and $\mathbf{z}_{22}$.

■ **METHOD 2** Alternatively, since there is no dependent source in the given circuit, $\mathbf{z}_{12} = \mathbf{z}_{21}$ and we can use Fig. 19.5(a). Comparing Fig. 19.7 with Fig. 19.5(a), we get

$$\mathbf{z}_{12} = 40\ \Omega = \mathbf{z}_{21}$$
$$\mathbf{z}_{11} - \mathbf{z}_{12} = 20 \quad \Rightarrow \quad \mathbf{z}_{11} = 20 + \mathbf{z}_{12} = 60\ \Omega$$
$$\mathbf{z}_{22} - \mathbf{z}_{12} = 30 \quad \Rightarrow \quad \mathbf{z}_{22} = 30 + \mathbf{z}_{12} = 70\ \Omega$$

Practice Problem 19.1

Find the *z* parameters of the two-port network in Fig. 19.9.

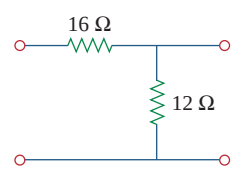


Figure 19.9
For Practice Prob. 19.1.

Answer: $\mathbf{z}_{11} = 28\ \Omega$, $\mathbf{z}_{12} = \mathbf{z}_{21} = \mathbf{z}_{22} = 12\ \Omega$.

Example 19.2

Find $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit in Fig. 19.10.

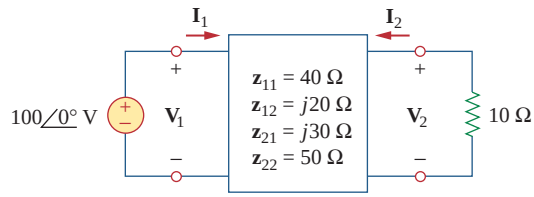


Figure 19.10
For Example 19.2.

Solution:
This is not a reciprocal network. We may use the equivalent circuit in Fig. 19.5(b) but we can also use Eq. (19.1) directly. Substituting the given z parameters into Eq. (19.1),

$$\mathbf{V}_1 = 40\mathbf{I}_1 + j20\mathbf{I}_2 \tag{19.2.1}$$

$$\mathbf{V}_2 = j30\mathbf{I}_1 + 50\mathbf{I}_2 \tag{19.2.2}$$

Since we are looking for $\mathbf{I}_1$ and $\mathbf{I}_2$, we substitute

$$\mathbf{V}_1 = 100\angle 0°, \qquad \mathbf{V}_2 = -10\mathbf{I}_2$$

into Eqs. (19.2.1) and (19.2.2), which become

$$100 = 40\mathbf{I}_1 + j20\mathbf{I}_2 \tag{19.2.3}$$

$$-10\mathbf{I}_2 = j30\mathbf{I}_1 + 50\mathbf{I}_2 \quad \Rightarrow \quad \mathbf{I}_1 = j2\mathbf{I}_2 \tag{19.2.4}$$

Substituting Eq. (19.2.4) into Eq. (19.2.3) gives

$$100 = j80\mathbf{I}_2 + j20\mathbf{I}_2 \quad \Rightarrow \quad \mathbf{I}_2 = \frac{100}{j100} = -j$$

From Eq. (19.2.4), $\mathbf{I}_1 = j2(-j) = 2$. Thus,

$$\mathbf{I}_1 = 2\angle 0° \text{ A}, \qquad \mathbf{I}_2 = 1\angle -90° \text{ A}$$

Practice Problem 19.2

Calculate $\mathbf{I}_1$ and $\mathbf{I}_2$ in the two-port of Fig. 19.11.

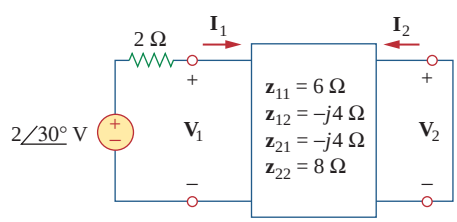


Figure 19.11
For Practice Prob. 19.2.

Answer: $200\angle 30°$ mA, $100\angle 120°$ mA.

19.3 Admittance Parameters

In the previous section we saw that impedance parameters may not exist for a two-port network. So there is a need for an alternative means of describing such a network. This need may be met by the second set of parameters, which we obtain by expressing the terminal currents in terms of the terminal voltages. In either Fig. 19.12(a) or (b), the terminal currents can be expressed in terms of the terminal voltages as

$$\begin{aligned} \mathbf{I}_1 &= \mathbf{y}_{11}\mathbf{V}_1 + \mathbf{y}_{12}\mathbf{V}_2 \\ \mathbf{I}_2 &= \mathbf{y}_{21}\mathbf{V}_1 + \mathbf{y}_{22}\mathbf{V}_2 \end{aligned} \tag{19.8}$$

or in matrix form as

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = [\mathbf{y}] \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} \tag{19.9}$$

The **y** terms are known as the *admittance parameters* (or, simply, *y parameters*) and have units of siemens.

The values of the parameters can be determined by setting $\mathbf{V}_1 = 0$ (input port short-circuited) or $\mathbf{V}_2 = 0$ (output port short-circuited). Thus,

$$\begin{aligned} \mathbf{y}_{11} &= \left.\frac{\mathbf{I}_1}{\mathbf{V}_1}\right|_{\mathbf{V}_2=0}, & \mathbf{y}_{12} &= \left.\frac{\mathbf{I}_1}{\mathbf{V}_2}\right|_{\mathbf{V}_1=0} \\ \mathbf{y}_{21} &= \left.\frac{\mathbf{I}_2}{\mathbf{V}_1}\right|_{\mathbf{V}_2=0}, & \mathbf{y}_{22} &= \left.\frac{\mathbf{I}_2}{\mathbf{V}_2}\right|_{\mathbf{V}_1=0} \end{aligned} \tag{19.10}$$

Since the *y* parameters are obtained by short-circuiting the input or output port, they are also called the *short-circuit admittance parameters*. Specifically,

$$\begin{aligned} \mathbf{y}_{11} &= \text{Short-circuit input admittance} \\ \mathbf{y}_{12} &= \text{Short-circuit transfer admittance from port 2 to port 1} \\ \mathbf{y}_{21} &= \text{Short-circuit transfer admittance from port 1 to port 2} \\ \mathbf{y}_{22} &= \text{Short-circuit output admittance} \end{aligned} \tag{19.11}$$

Following Eq. (19.10), we obtain $\mathbf{y}_{11}$ and $\mathbf{y}_{21}$ by connecting a current $\mathbf{I}_1$ to port 1 and short-circuiting port 2 as in Fig. 19.12(a), finding $\mathbf{V}_1$ and $\mathbf{I}_2$, and then calculating

$$\mathbf{y}_{11} = \frac{\mathbf{I}_1}{\mathbf{V}_1}, \qquad \mathbf{y}_{21} = \frac{\mathbf{I}_2}{\mathbf{V}_1} \tag{19.12}$$

Similarly, we obtain $\mathbf{y}_{12}$ and $\mathbf{y}_{22}$ by connecting a current source $\mathbf{I}_2$ to port 2 and short-circuiting port 1 as in Fig. 19.12(b), finding $\mathbf{I}_1$ and $\mathbf{V}_2$, and then getting

$$\mathbf{y}_{12} = \frac{\mathbf{I}_1}{\mathbf{V}_2}, \qquad \mathbf{y}_{22} = \frac{\mathbf{I}_2}{\mathbf{V}_2} \tag{19.13}$$

This procedure provides us with a means of calculating or measuring the *y* parameters. The impedance and admittance parameters are collectively referred to as *immittance* parameters.

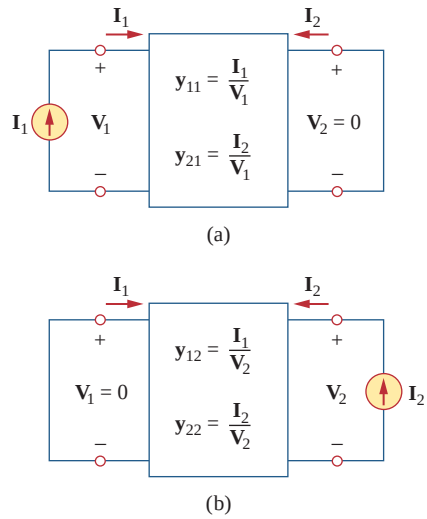


Figure 19.12
Determination of the *y* parameters: (a) finding y_{11} and y_{21}, (b) finding y_{12} and y_{22}.

For a two-port network that is linear and has no dependent sources, the transfer admittances are equal ($\mathbf{y}_{12} = \mathbf{y}_{21}$). This can be proved in the same way as for the *z* parameters. A reciprocal network ($\mathbf{y}_{12} = \mathbf{y}_{21}$) can be modeled by the Π-equivalent circuit in Fig. 19.13(a). If the network is not reciprocal, a more general equivalent network is shown in Fig. 19.13(b).

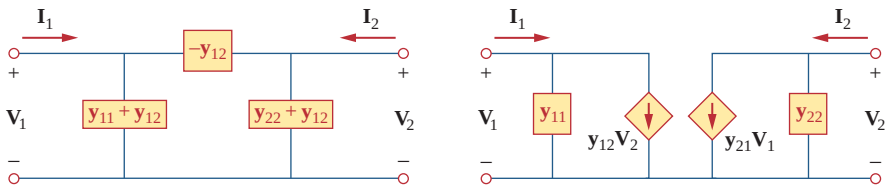


Figure 19.13
(a) Π-equivalent circuit (for reciprocal case only), (b) general equivalent circuit.

Example 19.3

Obtain the *y* parameters for the Π network shown in Fig. 19.14.

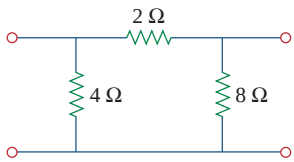


Figure 19.14
For Example 19.3.

Solution:

■ **METHOD 1** To find $\mathbf{y}_{11}$ and $\mathbf{y}_{21}$, short-circuit the output port and connect a current source $\mathbf{I}_1$ to the input port as in Fig. 19.15(a). Since the 8-Ω resistor is short-circuited, the 2-Ω resistor is in parallel with the 4-Ω resistor. Hence,

$$\mathbf{V}_1 = \mathbf{I}_1(4 \parallel 2) = \frac{4}{3}\mathbf{I}_1, \qquad \mathbf{y}_{11} = \frac{\mathbf{I}_1}{\mathbf{V}_1} = \frac{\mathbf{I}_1}{\frac{4}{3}\mathbf{I}_1} = 0.75 \text{ S}$$

By current division,

$$-\mathbf{I}_2 = \frac{4}{4+2}\mathbf{I}_1 = \frac{2}{3}\mathbf{I}_1, \qquad \mathbf{y}_{21} = \frac{\mathbf{I}_2}{\mathbf{V}_1} = \frac{-\frac{2}{3}\mathbf{I}_1}{\frac{4}{3}\mathbf{I}_1} = -0.5 \text{ S}$$

To get $\mathbf{y}_{12}$ and $\mathbf{y}_{22}$, short-circuit the input port and connect a current source $\mathbf{I}_2$ to the output port as in Fig. 19.15(b). The 4-Ω resistor is short-circuited so that the 2-Ω and 8-Ω resistors are in parallel.

$$\mathbf{V}_2 = \mathbf{I}_2(8 \parallel 2) = \frac{8}{5}\mathbf{I}_2, \qquad \mathbf{y}_{22} = \frac{\mathbf{I}_2}{\mathbf{V}_2} = \frac{\mathbf{I}_2}{\frac{8}{5}\mathbf{I}_2} = \frac{5}{8} = 0.625 \text{ S}$$

By current division,

$$-\mathbf{I}_1 = \frac{8}{8+2}\mathbf{I}_2 = \frac{4}{5}\mathbf{I}_2, \qquad \mathbf{y}_{12} = \frac{\mathbf{I}_1}{\mathbf{V}_2} = \frac{-\frac{4}{5}\mathbf{I}_2}{\frac{8}{5}\mathbf{I}_2} = -0.5 \text{ S}$$

■ **METHOD 2** Alternatively, comparing Fig. 19.14 with Fig. 19.13(a),

$$\mathbf{y}_{12} = -\frac{1}{2}\text{ S} = \mathbf{y}_{21}$$

$$\mathbf{y}_{11} + \mathbf{y}_{12} = \frac{1}{4} \quad \Rightarrow \quad \mathbf{y}_{11} = \frac{1}{4} - \mathbf{y}_{12} = 0.75 \text{ S}$$

$$\mathbf{y}_{22} + \mathbf{y}_{12} = \frac{1}{8} \quad \Rightarrow \quad \mathbf{y}_{22} = \frac{1}{8} - \mathbf{y}_{12} = 0.625 \text{ S}$$

as obtained previously.

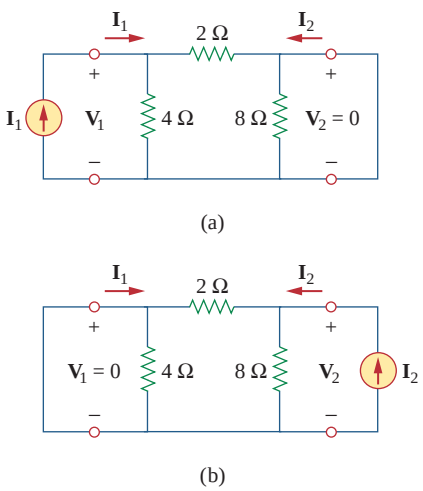


Figure 19.15
For Example 19.3: (a) finding $\mathbf{y}_{11}$ and $\mathbf{y}_{21}$, (b) finding $\mathbf{y}_{12}$ and $\mathbf{y}_{22}$.

Practice Problem 19.3

Obtain the *y* parameters for the *T* network shown in Fig. 19.16.

Answer: $\mathbf{y}_{11} = 75.77$ mS, $\mathbf{y}_{12} = \mathbf{y}_{21} = -30.3$ mS, $\mathbf{y}_{22} = 45.47$ mS.

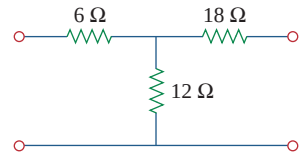


Figure 19.16
For Practice Prob. 19.3.

Example 19.4

Determine the *y* parameters for the two-port shown in Fig. 19.17.

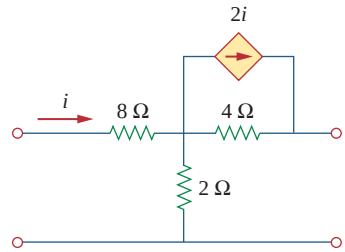


Figure 19.17
For Example 19.4.

Solution:
We follow the same procedure as in the previous example. To get $\mathbf{y}_{11}$ and $\mathbf{y}_{21}$, we use the circuit in Fig. 19.18(a), in which port 2 is short-circuited and a current source is applied to port 1. At node 1,

$$\frac{\mathbf{V}_1 - \mathbf{V}_o}{8} = 2\mathbf{I}_1 + \frac{\mathbf{V}_o}{2} + \frac{\mathbf{V}_o - 0}{4}$$

But $\mathbf{I}_1 = \dfrac{\mathbf{V}_1 - \mathbf{V}_o}{8}$; therefore,

$$0 = \frac{\mathbf{V}_1 - \mathbf{V}_o}{8} + \frac{3\mathbf{V}_o}{4}$$

$$0 = \mathbf{V}_1 - \mathbf{V}_o + 6\mathbf{V}_o \quad \Rightarrow \quad \mathbf{V}_1 = -5\mathbf{V}_o$$

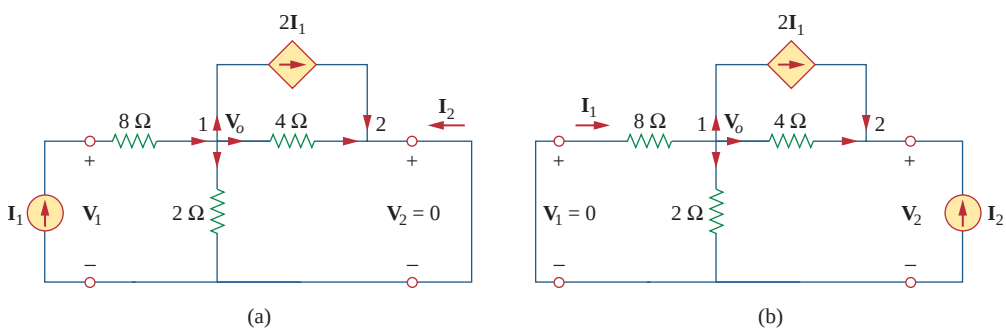


Figure 19.18
Solution of Example 19.4: (a) finding $\mathbf{y}_{11}$ and $\mathbf{y}_{21}$, (b) finding $\mathbf{y}_{12}$ and $\mathbf{y}_{22}$.

Hence,

$$\mathbf{I}_1 = \frac{-5\mathbf{V}_o - \mathbf{V}_o}{8} = -0.75\mathbf{V}_o$$

and

$$\mathbf{y}_{11} = \frac{\mathbf{I}_1}{\mathbf{V}_1} = \frac{-0.75\mathbf{V}_o}{-5\mathbf{V}_o} = 0.15 \text{ S}$$

At node 2,

$$\frac{\mathbf{V}_o - 0}{4} + 2\mathbf{I}_1 + \mathbf{I}_2 = 0$$

or

$$-\mathbf{I}_2 = 0.25\mathbf{V}_o - 1.5\mathbf{V}_o = -1.25\mathbf{V}_o$$

Hence,

$$\mathbf{y}_{21} = \frac{\mathbf{I}_2}{\mathbf{V}_1} = \frac{1.25\mathbf{V}_o}{-5\mathbf{V}_o} = -0.25 \text{ S}$$

Similarly, we get $\mathbf{y}_{12}$ and $\mathbf{y}_{22}$ using Fig. 19.18(b). At node 1,

$$\frac{0 - \mathbf{V}_o}{8} = 2\mathbf{I}_1 + \frac{\mathbf{V}_o}{2} + \frac{\mathbf{V}_o - \mathbf{V}_2}{4}$$

But $\mathbf{I}_1 = \dfrac{0 - \mathbf{V}_o}{8}$; therefore,

$$0 = -\frac{\mathbf{V}_o}{8} + \frac{\mathbf{V}_o}{2} + \frac{\mathbf{V}_o - \mathbf{V}_2}{4}$$

or

$$0 = -\mathbf{V}_o + 4\mathbf{V}_o + 2\mathbf{V}_o - 2\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{V}_2 = 2.5\mathbf{V}_o$$

Hence,

$$\mathbf{y}_{12} = \frac{\mathbf{I}_1}{\mathbf{V}_2} = \frac{-\mathbf{V}_o/8}{2.5\mathbf{V}_o} = -0.05 \text{ S}$$

At node 2,

$$\frac{\mathbf{V}_o - \mathbf{V}_2}{4} + 2\mathbf{I}_1 + \mathbf{I}_2 = 0$$

or

$$-\mathbf{I}_2 = 0.25\mathbf{V}_o - \frac{1}{4}(2.5\mathbf{V}_o) - \frac{2\mathbf{V}_o}{8} = -0.625\mathbf{V}_o$$

Thus,

$$\mathbf{y}_{22} = \frac{\mathbf{I}_2}{\mathbf{V}_2} = \frac{0.625\mathbf{V}_o}{2.5\mathbf{V}_o} = 0.25 \text{ S}$$

Notice that $\mathbf{y}_{12} \neq \mathbf{y}_{21}$ in this case, since the network is not reciprocal.

Practice Problem 19.4

Obtain the *y* parameters for the circuit in Fig. 19.19.

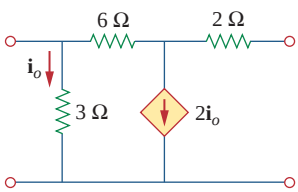


Figure 19.19
For Practice Prob. 19.4.

Answer: $\mathbf{y}_{11} = 0.625$ S, $\mathbf{y}_{12} = -0.125$ S, $\mathbf{y}_{21} = 0.375$ S, $\mathbf{y}_{22} = 0.125$ S.

19.4 Hybrid Parameters

The *z* and *y* parameters of a two-port network do not always exist. So there is a need for developing another set of parameters. This third set of parameters is based on making $\mathbf{V}_1$ and $\mathbf{I}_2$ the dependent variables. Thus, we obtain

$$\begin{aligned} \mathbf{V}_1 &= \mathbf{h}_{11}\mathbf{I}_1 + \mathbf{h}_{12}\mathbf{V}_2 \\ \mathbf{I}_2 &= \mathbf{h}_{21}\mathbf{I}_1 + \mathbf{h}_{22}\mathbf{V}_2 \end{aligned} \tag{19.14}$$

or in matrix form,

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{h}_{11} & \mathbf{h}_{12} \\ \mathbf{h}_{21} & \mathbf{h}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{V}_2 \end{bmatrix} = [\mathbf{h}] \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{V}_2 \end{bmatrix} \tag{19.15}$$

The **h** terms are known as the *hybrid parameters* (or, simply, *h parameters*) because they are a hybrid combination of ratios. They are very useful for describing electronic devices such as transistors (see Section 19.9); it is much easier to measure experimentally the *h* parameters of such devices than to measure their *z* or *y* parameters. In fact, we have seen that the ideal transformer in Fig. 19.6, described by Eq. (19.7), does not have *z* parameters. The ideal transformer can be described by the hybrid parameters, because Eq. (19.7) conforms with Eq. (19.14).

The values of the parameters are determined as

$$\begin{aligned} \mathbf{h}_{11} &= \left.\frac{\mathbf{V}_1}{\mathbf{I}_1}\right|_{\mathbf{V}_2=0}, & \mathbf{h}_{12} &= \left.\frac{\mathbf{V}_1}{\mathbf{V}_2}\right|_{\mathbf{I}_1=0} \\ \mathbf{h}_{21} &= \left.\frac{\mathbf{I}_2}{\mathbf{I}_1}\right|_{\mathbf{V}_2=0}, & \mathbf{h}_{22} &= \left.\frac{\mathbf{I}_2}{\mathbf{V}_2}\right|_{\mathbf{I}_1=0} \end{aligned} \tag{19.16}$$

It is evident from Eq. (19.16) that the parameters $\mathbf{h}_{11}$, $\mathbf{h}_{12}$, $\mathbf{h}_{21}$, and $\mathbf{h}_{22}$ represent an impedance, a voltage gain, a current gain, and an admittance, respectively. This is why they are called the hybrid parameters. To be specific,

$$\begin{aligned} \mathbf{h}_{11} &= \text{Short-circuit input impedance} \\ \mathbf{h}_{12} &= \text{Open-circuit reverse voltage gain} \\ \mathbf{h}_{21} &= \text{Short-circuit forward current gain} \\ \mathbf{h}_{22} &= \text{Open-circuit output admittance} \end{aligned} \tag{19.17}$$

The procedure for calculating the *h* parameters is similar to that used for the *z* or *y* parameters. We apply a voltage or current source to the appropriate port, short-circuit or open-circuit the other port, depending on the parameter of interest, and perform regular circuit analysis. For reciprocal networks, $\mathbf{h}_{12} = -\mathbf{h}_{21}$. This can be proved in the same way as we proved that $\mathbf{z}_{12} = \mathbf{z}_{21}$. Figure 19.20 shows the hybrid model of a two-port network.

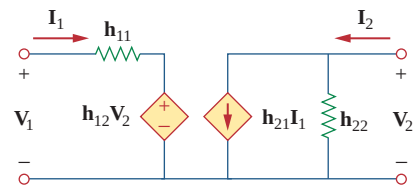


Figure 19.20
The *h*-parameter equivalent network of a two-port network.

A set of parameters closely related to the *h* parameters are the *g parameters* or *inverse hybrid parameters*. These are used to describe the terminal currents and voltages as

$$\begin{aligned} \mathbf{I}_1 &= \mathbf{g}_{11}\mathbf{V}_1 + \mathbf{g}_{12}\mathbf{I}_2 \\ \mathbf{V}_2 &= \mathbf{g}_{21}\mathbf{V}_1 + \mathbf{g}_{22}\mathbf{I}_2 \end{aligned} \tag{19.18}$$

or

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{g}_{11} & \mathbf{g}_{12} \\ \mathbf{g}_{21} & \mathbf{g}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_2 \end{bmatrix} = [\mathbf{g}] \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_2 \end{bmatrix} \tag{19.19}$$

The values of the g parameters are determined as

$$\mathbf{g}_{11} = \frac{\mathbf{I}_1}{\mathbf{V}_1}\bigg|_{\mathbf{I}_2=0}, \qquad \mathbf{g}_{12} = \frac{\mathbf{I}_1}{\mathbf{I}_2}\bigg|_{\mathbf{V}_1=0}$$
$$\mathbf{g}_{21} = \frac{\mathbf{V}_2}{\mathbf{V}_1}\bigg|_{\mathbf{I}_2=0}, \qquad \mathbf{g}_{22} = \frac{\mathbf{V}_2}{\mathbf{I}_2}\bigg|_{\mathbf{V}_1=0} \tag{19.20}$$

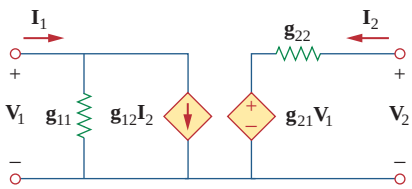


Figure 19.21
The g-parameter model of a two-port network.

Thus, the inverse hybrid parameters are specifically called

$\mathbf{g}_{11}$ = Open-circuit input admittance
$\mathbf{g}_{12}$ = Short-circuit reverse current gain
$\mathbf{g}_{21}$ = Open-circuit forward voltage gain
$\mathbf{g}_{22}$ = Short-circuit output impedance (19.21)

Figure 19.21 shows the inverse hybrid model of a two-port network. The g parameters are frequently used to model field-effect transistors.

Example 19.5

Find the hybrid parameters for the two-port network of Fig. 19.22.

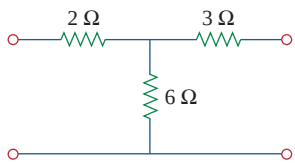


Figure 19.22
For Example 19.5.

Solution:
To find $\mathbf{h}_{11}$ and $\mathbf{h}_{21}$, we short-circuit the output port and connect a current source $\mathbf{I}_1$ to the input port as shown in Fig. 19.23(a). From Fig. 19.23(a),

$$\mathbf{V}_1 = \mathbf{I}_1(2 + 3 \parallel 6) = 4\mathbf{I}_1$$

Hence,

$$\mathbf{h}_{11} = \frac{\mathbf{V}_1}{\mathbf{I}_1} = 4\ \Omega$$

Also, from Fig. 19.23(a) we obtain, by current division,

$$-\mathbf{I}_2 = \frac{6}{6+3}\mathbf{I}_1 = \frac{2}{3}\mathbf{I}_1$$

Hence,

$$\mathbf{h}_{21} = \frac{\mathbf{I}_2}{\mathbf{I}_1} = -\frac{2}{3}$$

To obtain $\mathbf{h}_{12}$ and $\mathbf{h}_{22}$, we open-circuit the input port and connect a voltage source $\mathbf{V}_2$ to the output port as in Fig. 19.23(b). By voltage division,

$$\mathbf{V}_1 = \frac{6}{6+3}\mathbf{V}_2 = \frac{2}{3}\mathbf{V}_2$$

Hence,

$$\mathbf{h}_{12} = \frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{2}{3}$$

Also,

$$\mathbf{V}_2 = (3 + 6)\mathbf{I}_2 = 9\mathbf{I}_2$$

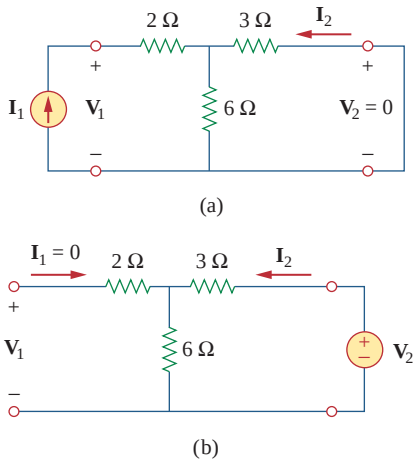


Figure 19.23
For Example 19.5: (a) computing $\mathbf{h}_{11}$ and $\mathbf{h}_{21}$, (b) computing $\mathbf{h}_{12}$ and $\mathbf{h}_{22}$.

Thus,

$$\mathbf{h}_{22} = \frac{\mathbf{I}_2}{\mathbf{V}_2} = \frac{1}{9}\,\text{S}$$

Practice Problem 19.5

Determine the *h* parameters for the circuit in Fig. 19.24.

Answer: $\mathbf{h}_{11} = 2.4\ \Omega$, $\mathbf{h}_{12} = 0.4$, $\mathbf{h}_{21} = -0.4$, $\mathbf{h}_{22} = 200$ mS.

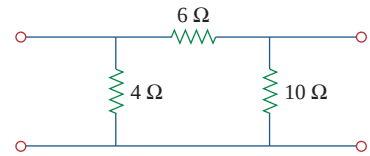


Figure 19.24
For Practice Prob. 19.5.

Example 19.6

Determine the Thevenin equivalent at the output port of the circuit in Fig. 19.25.

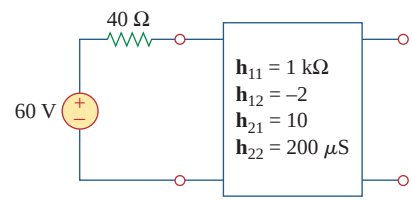


Figure 19.25
For Example 19.6.

Solution:

To find $\mathbf{Z}_{\text{Th}}$ and $\mathbf{V}_{\text{Th}}$, we apply the normal procedure, keeping in mind the formulas relating the input and output ports of the *h* model. To obtain $\mathbf{Z}_{\text{Th}}$, remove the 60-V voltage source at the input port and apply a 1-V voltage source at the output port, as shown in Fig. 19.26(a). From Eq. (19.14),

$$\mathbf{V}_1 = \mathbf{h}_{11}\mathbf{I}_1 + \mathbf{h}_{12}\mathbf{V}_2 \qquad \textbf{(19.6.1)}$$

$$\mathbf{I}_2 = \mathbf{h}_{21}\mathbf{I}_1 + \mathbf{h}_{22}\mathbf{V}_2 \qquad \textbf{(19.6.2)}$$

But $\mathbf{V}_2 = 1$, and $\mathbf{V}_1 = -40\mathbf{I}_1$. Substituting these into Eqs. (19.6.1) and (19.6.2), we get

$$-40\mathbf{I}_1 = \mathbf{h}_{11}\mathbf{I}_1 + \mathbf{h}_{12} \quad \Rightarrow \quad \mathbf{I}_1 = -\frac{\mathbf{h}_{12}}{40 + \mathbf{h}_{11}} \qquad \textbf{(19.6.3)}$$

$$\mathbf{I}_2 = \mathbf{h}_{21}\mathbf{I}_1 + \mathbf{h}_{22} \qquad \textbf{(19.6.4)}$$

Substituting Eq. (19.6.3) into Eq. (19.6.4) gives

$$\mathbf{I}_2 = \mathbf{h}_{22} - \frac{\mathbf{h}_{21}\mathbf{h}_{12}}{\mathbf{h}_{11} + 40} = \frac{\mathbf{h}_{11}\mathbf{h}_{22} - \mathbf{h}_{21}\mathbf{h}_{12} + \mathbf{h}_{22}40}{\mathbf{h}_{11} + 40}$$

Therefore,

$$\mathbf{Z}_{\text{Th}} = \frac{\mathbf{V}_2}{\mathbf{I}_2} = \frac{1}{\mathbf{I}_2} = \frac{\mathbf{h}_{11} + 40}{\mathbf{h}_{11}\mathbf{h}_{22} - \mathbf{h}_{21}\mathbf{h}_{12} + \mathbf{h}_{22}40}$$

Substituting the values of the *h* parameters,

$$\mathbf{Z}_{\text{Th}} = \frac{1000 + 40}{10^3 \times 200 \times 10^{-6} + 20 + 40 \times 200 \times 10^{-6}}$$

$$= \frac{1040}{20.21} = 51.46\ \Omega$$

To get $\mathbf{V}_{\text{Th}}$, we find the open-circuit voltage $\mathbf{V}_2$ in Fig. 19.26(b). At the input port,

$$-60 + 40\mathbf{I}_1 + \mathbf{V}_1 = 0 \quad \Rightarrow \quad \mathbf{V}_1 = 60 - 40\mathbf{I}_1 \qquad \textbf{(19.6.5)}$$

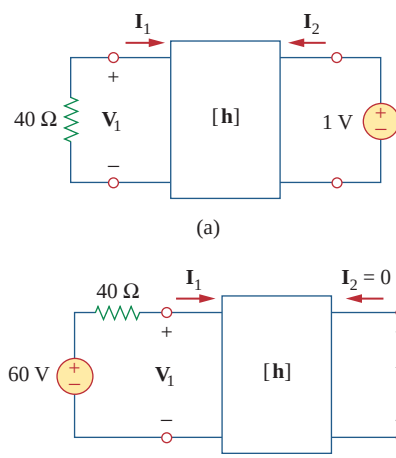


Figure 19.26
For Example 19.6: (a) finding $\mathbf{Z}_{\text{Th}}$, (b) finding $\mathbf{V}_{\text{Th}}$.

At the output,

$$\mathbf{I}_2 = 0 \tag{19.6.6}$$

Substituting Eqs. (19.6.5) and (19.6.6) into Eqs. (19.6.1) and (19.6.2), we obtain

$$60 - 40\mathbf{I}_1 = \mathbf{h}_{11}\mathbf{I}_1 + \mathbf{h}_{12}\mathbf{V}_2$$

or

$$60 = (\mathbf{h}_{11} + 40)\mathbf{I}_1 + \mathbf{h}_{12}\mathbf{V}_2 \tag{19.6.7}$$

and

$$0 = \mathbf{h}_{21}\mathbf{I}_1 + \mathbf{h}_{22}\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{I}_1 = -\frac{\mathbf{h}_{22}}{\mathbf{h}_{21}}\mathbf{V}_2 \tag{19.6.8}$$

Now substituting Eq. (19.6.8) into Eq. (19.6.7) gives

$$60 = \left[-(\mathbf{h}_{11} + 40)\frac{\mathbf{h}_{22}}{\mathbf{h}_{21}} + \mathbf{h}_{12}\right]\mathbf{V}_2$$

or

$$\mathbf{V}_{\text{Th}} = \mathbf{V}_2 = \frac{60}{-(\mathbf{h}_{11} + 40)\mathbf{h}_{22}/\mathbf{h}_{21} + \mathbf{h}_{12}} = \frac{60\mathbf{h}_{21}}{\mathbf{h}_{12}\mathbf{h}_{21} - \mathbf{h}_{11}\mathbf{h}_{22} - 40\mathbf{h}_{22}}$$

Substituting the values of the *h* parameters,

$$\mathbf{V}_{\text{Th}} = \frac{60 \times 10}{-20.21} = -29.69 \text{ V}$$

Practice Problem 19.6

Find the impedance at the input port of the circuit in Fig. 19.27.

Answer: 1.6667 kΩ.

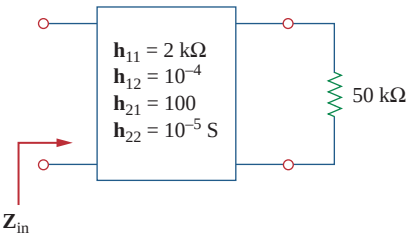


Figure 19.27
For Practice Prob. 19.6.

Example 19.7

Find the *g* parameters as functions of *s* for the circuit in Fig. 19.28.

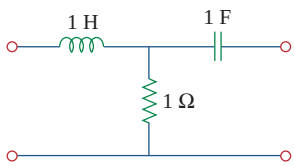


Figure 19.28
For Example 19.7.

Solution:
In the *s* domain,

$$1 \text{ H} \quad \Rightarrow \quad sL = s, \qquad 1 \text{ F} \quad \Rightarrow \quad \frac{1}{sC} = \frac{1}{s}$$

To get $\mathbf{g}_{11}$ and $\mathbf{g}_{21}$, we open-circuit the output port and connect a voltage source $\mathbf{V}_1$ to the input port as in Fig. 19.29(a). From the figure,

$$\mathbf{I}_1 = \frac{\mathbf{V}_1}{s + 1}$$

or

$$\mathbf{g}_{11} = \frac{\mathbf{I}_1}{\mathbf{V}_1} = \frac{1}{s+1}$$

By voltage division,

$$\mathbf{V}_2 = \frac{1}{s+1}\mathbf{V}_1$$

or

$$\mathbf{g}_{21} = \frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{1}{s+1}$$

To obtain $\mathbf{g}_{12}$ and $\mathbf{g}_{22}$, we short-circuit the input port and connect a current source $\mathbf{I}_2$ to the output port as in Fig. 19.29(b). By current division,

$$\mathbf{I}_1 = -\frac{1}{s+1}\mathbf{I}_2$$

or

$$\mathbf{g}_{12} = \frac{\mathbf{I}_1}{\mathbf{I}_2} = -\frac{1}{s+1}$$

Also,

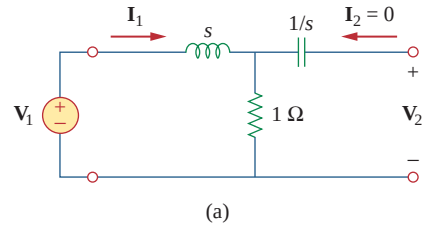
$$\mathbf{V}_2 = \mathbf{I}_2\left(\frac{1}{s} + s \parallel 1\right)$$

or

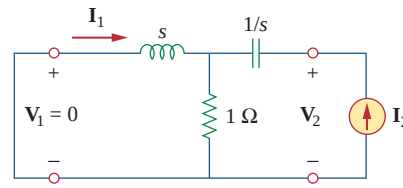
$$\mathbf{g}_{22} = \frac{\mathbf{V}_2}{\mathbf{I}_2} = \frac{1}{s} + \frac{s}{s+1} = \frac{s^2+s+1}{s(s+1)}$$

Thus,

$$[\mathbf{g}] = \begin{bmatrix} \dfrac{1}{s+1} & -\dfrac{1}{s+1} \\ \dfrac{1}{s+1} & \dfrac{s^2+s+1}{s(s+1)} \end{bmatrix}$$



(a)



(b)

Figure 19.29
Determining the *g* parameters in the *s* domain for the circuit in Fig. 19.28.

Practice Problem 19.7

For the ladder network in Fig. 19.30, determine the *g* parameters in the *s* domain.

Answer: $[\mathbf{g}] = \begin{bmatrix} \dfrac{s+2}{s^2+3s+1} & -\dfrac{1}{s^2+3s+1} \\ \dfrac{1}{s^2+3s+1} & \dfrac{s(s+2)}{s^2+3s+1} \end{bmatrix}$.

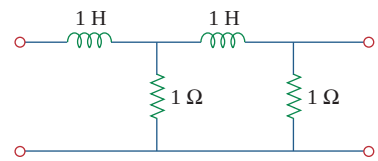


Figure 19.30
For Practice Prob. 19.7.

19.5 Transmission Parameters

Since there are no restrictions on which terminal voltages and currents should be considered independent and which should be dependent variables, we expect to be able to generate many sets of parameters.

Another set of parameters relates the variables at the input port to those at the output port. Thus,

$$\begin{aligned} \mathbf{V}_1 &= \mathbf{A}\mathbf{V}_2 - \mathbf{B}\mathbf{I}_2 \\ \mathbf{I}_1 &= \mathbf{C}\mathbf{V}_2 - \mathbf{D}\mathbf{I}_2 \end{aligned} \tag{19.22}$$

or

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} \begin{bmatrix} \mathbf{V}_2 \\ -\mathbf{I}_2 \end{bmatrix} = [\mathbf{T}] \begin{bmatrix} \mathbf{V}_2 \\ -\mathbf{I}_2 \end{bmatrix} \tag{19.23}$$

Equations (19.22) and (19.23) relate the input variables ($\mathbf{V}_1$ and $\mathbf{I}_1$) to the output variables ($\mathbf{V}_2$ and $-\mathbf{I}_2$). Notice that in computing the transmission parameters, $-\mathbf{I}_2$ is used rather than $\mathbf{I}_2$, because the current is considered to be leaving the network, as shown in Fig. 19.31, as opposed to entering the network as in Fig. 19.1(b). This is done merely for conventional reasons; when you cascade two-ports (output to input), it is most logical to think of $\mathbf{I}_2$ as leaving the two-port. It is also customary in the power industry to consider $\mathbf{I}_2$ as leaving the two-port.

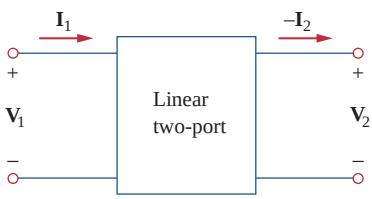


Figure 19.31
Terminal variables used to define the **ADCB** parameters.

The two-port parameters in Eqs. (19.22) and (19.23) provide a measure of how a circuit transmits voltage and current from a source to a load. They are useful in the analysis of transmission lines (such as cable and fiber) because they express sending-end variables ($\mathbf{V}_1$ and $\mathbf{I}_1$) in terms of the receiving-end variables ($\mathbf{V}_2$ and $-\mathbf{I}_2$). For this reason, they are called *transmission parameters*. They are also known as **ABCD** parameters. They are used in the design of telephone systems, microwave networks, and radars.

The transmission parameters are determined as

$$\begin{aligned} \mathbf{A} &= \left.\frac{\mathbf{V}_1}{\mathbf{V}_2}\right|_{\mathbf{I}_2=0}, & \mathbf{B} &= -\left.\frac{\mathbf{V}_1}{\mathbf{I}_2}\right|_{\mathbf{V}_2=0} \\ \mathbf{C} &= \left.\frac{\mathbf{I}_1}{\mathbf{V}_2}\right|_{\mathbf{I}_2=0}, & \mathbf{D} &= -\left.\frac{\mathbf{I}_1}{\mathbf{I}_2}\right|_{\mathbf{V}_2=0} \end{aligned} \tag{19.24}$$

Thus, the transmission parameters are called, specifically,

$$\begin{aligned} \mathbf{A} &= \text{Open-circuit voltage ratio} \\ \mathbf{B} &= \text{Negative short-circuit transfer impedance} \\ \mathbf{C} &= \text{Open-circuit transfer admittance} \\ \mathbf{D} &= \text{Negative short-circuit current ratio} \end{aligned} \tag{19.25}$$

A and **D** are dimensionless, **B** is in ohms, and **C** is in siemens. Since the transmission parameters provide a direct relationship between input and output variables, they are very useful in cascaded networks.

Our last set of parameters may be defined by expressing the variables at the output port in terms of the variables at the input port. We obtain

$$\begin{aligned} \mathbf{V}_2 &= \mathbf{a}\mathbf{V}_1 - \mathbf{b}\mathbf{I}_1 \\ \mathbf{I}_2 &= \mathbf{c}\mathbf{V}_1 - \mathbf{d}\mathbf{I}_1 \end{aligned} \tag{19.26}$$

or

$$\begin{bmatrix} \mathbf{V}_2 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{a} & \mathbf{b} \\ \mathbf{c} & \mathbf{d} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ -\mathbf{I}_1 \end{bmatrix} = [\mathbf{t}] \begin{bmatrix} \mathbf{V}_1 \\ -\mathbf{I}_1 \end{bmatrix} \tag{19.27}$$

The parameters **a**, **b**, **c**, and **d** are called the *inverse transmission*, or *t*, *parameters*. They are determined as follows:

$$\mathbf{a} = \left.\frac{\mathbf{V}_2}{\mathbf{V}_1}\right|_{\mathbf{I}_1=0}, \qquad \mathbf{b} = -\left.\frac{\mathbf{V}_2}{\mathbf{I}_1}\right|_{\mathbf{V}_1=0}$$
$$\mathbf{c} = \left.\frac{\mathbf{I}_2}{\mathbf{V}_1}\right|_{\mathbf{I}_1=0}, \qquad \mathbf{d} = -\left.\frac{\mathbf{I}_2}{\mathbf{I}_1}\right|_{\mathbf{V}_1=0} \tag{19.28}$$

From Eq. (19.28) and from our experience so far, it is evident that these parameters are known individually as

$$\begin{aligned} \mathbf{a} &= \text{Open-circuit voltage gain} \\ \mathbf{b} &= \text{Negative short-circuit transfer impedance} \\ \mathbf{c} &= \text{Open-circuit transfer admittance} \\ \mathbf{d} &= \text{Negative short-circuit current gain} \end{aligned} \tag{19.29}$$

While **a** and **d** are dimensionless, **b** and **c** are in ohms and siemens, respectively.

In terms of the transmission or inverse transmission parameters, a network is reciprocal if

$$\mathbf{AD} - \mathbf{BC} = 1, \qquad \mathbf{ad} - \mathbf{bc} = 1 \tag{19.30}$$

These relations can be proved in the same way as the transfer impedance relations for the *z* parameters. Alternatively, we will be able to use Table 19.1 a little later to derive Eq. (19.30) from the fact that $\mathbf{z}_{12} = \mathbf{z}_{21}$ for reciprocal networks.

Example 19.8

Find the transmission parameters for the two-port network in Fig. 19.32.

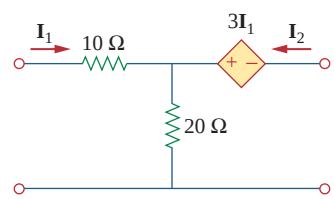


Figure 19.32
For Example 19.8.

Solution:

To determine **A** and **C**, we leave the output port open as in Fig. 19.33(a) so that $\mathbf{I}_2 = 0$ and place a voltage source $\mathbf{V}_1$ at the input port. We have

$$\mathbf{V}_1 = (10 + 20)\mathbf{I}_1 = 30\mathbf{I}_1 \qquad \text{and} \qquad \mathbf{V}_2 = 20\mathbf{I}_1 - 3\mathbf{I}_1 = 17\mathbf{I}_1$$

Thus,

$$\mathbf{A} = \frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{30\mathbf{I}_1}{17\mathbf{I}_1} = 1.765, \qquad \mathbf{C} = \frac{\mathbf{I}_1}{\mathbf{V}_2} = \frac{\mathbf{I}_1}{17\mathbf{I}_1} = 0.0588 \text{ S}$$

To obtain **B** and **D**, we short-circuit the output port so that $\mathbf{V}_2 = 0$ as shown in Fig. 19.33(b) and place a voltage source $\mathbf{V}_1$ at the input port. At node *a* in the circuit of Fig. 19.33(b), KCL gives

$$\frac{\mathbf{V}_1 - \mathbf{V}_a}{10} - \frac{\mathbf{V}_a}{20} + \mathbf{I}_2 = 0 \tag{19.8.1}$$

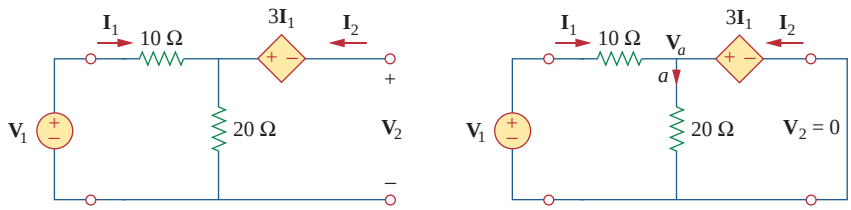


Figure 19.33
For Example 19.8: (a) finding **A** and **C**, (b) finding **B** and **D**.

But $\mathbf{V}_a = 3\mathbf{I}_1$ and $\mathbf{I}_1 = (\mathbf{V}_1 - \mathbf{V}_a)/10$. Combining these gives

$$\mathbf{V}_a = 3\mathbf{I}_1 \qquad \mathbf{V}_1 = 13\mathbf{I}_1 \tag{19.8.2}$$

Substituting $\mathbf{V}_a = 3\mathbf{I}_1$ into Eq. (19.8.1) and replacing the first term with $\mathbf{I}_1$,

$$\mathbf{I}_1 - \frac{3\mathbf{I}_1}{20} + \mathbf{I}_2 = 0 \quad \Rightarrow \quad \frac{17}{20}\mathbf{I}_1 = -\mathbf{I}_2$$

Therefore,

$$\mathbf{D} = -\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{20}{17} = 1.176, \qquad \mathbf{B} = -\frac{\mathbf{V}_1}{\mathbf{I}_2} = \frac{-13\mathbf{I}_1}{(-17/20)\mathbf{I}_1} = 15.29\ \Omega$$

Practice Problem 19.8

Find the transmission parameters for the circuit in Fig. 19.16 (see Practice Prob. 19.3).

Answer: $\mathbf{A} = 1.5$, $\mathbf{B} = 22\ \Omega$, $\mathbf{C} = 125$ mS, $\mathbf{D} = 2.5$.

Example 19.9

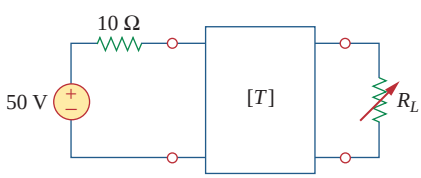


Figure 19.34
For Example 19.9.

The **ABCD** parameters of the two-port network in Fig. 19.34 are

$$\begin{bmatrix} 4 & 20\ \Omega \\ 0.1\ \text{S} & 2 \end{bmatrix}$$

The output port is connected to a variable load for maximum power transfer. Find R_L and the maximum power transferred.

Solution:

What we need is to find the Thevenin equivalent ($\mathbf{Z}_{\text{Th}}$ and $\mathbf{V}_{\text{Th}}$) at the load or output port. We find $\mathbf{Z}_{\text{Th}}$ using the circuit in Fig. 19.35(a). Our goal is to get $\mathbf{Z}_{\text{Th}} = \mathbf{V}_2/\mathbf{I}_2$. Substituting the given **ABCD** parameters into Eq. (19.22), we obtain

$$\mathbf{V}_1 = 4\mathbf{V}_2 - 20\mathbf{I}_2 \tag{19.9.1}$$

$$\mathbf{I}_1 = 0.1\mathbf{V}_2 - 2\mathbf{I}_2 \tag{19.9.2}$$

At the input port, $\mathbf{V}_1 = -10\mathbf{I}_1$. Substituting this into Eq. (19.9.1) gives

$$-10\mathbf{I}_1 = 4\mathbf{V}_2 - 20\mathbf{I}_2$$

or

$$\mathbf{I}_1 = -0.4\mathbf{V}_2 + 2\mathbf{I}_2 \tag{19.9.3}$$

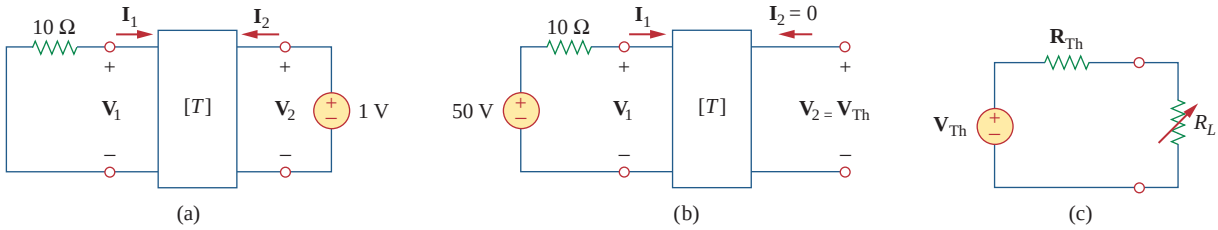


Figure 19.35
Solution of Example 19.9: (a) finding $\mathbf{Z}_{\text{Th}}$, (b) finding $\mathbf{V}_{\text{Th}}$, (c) finding R_L for maximum power transfer.

Setting the right-hand sides of Eqs. (19.9.2) and (19.9.3) equal,

$$0.1\mathbf{V}_2 - 2\mathbf{I}_2 = -0.4\mathbf{V}_2 + 2\mathbf{I}_2 \quad \Rightarrow \quad 0.5\mathbf{V}_2 = 4\mathbf{I}_2$$

Hence,

$$\mathbf{Z}_{\text{Th}} = \frac{\mathbf{V}_2}{\mathbf{I}_2} = \frac{4}{0.5} = 8\ \Omega$$

To find $\mathbf{V}_{\text{Th}}$, we use the circuit in Fig. 19.35(b). At the output port $\mathbf{I}_2 = 0$ and at the input port $\mathbf{V}_1 = 50 - 10\mathbf{I}_1$. Substituting these into Eqs. (19.9.1) and (19.9.2),

$$50 - 10\mathbf{I}_1 = 4\mathbf{V}_2 \qquad \textbf{(19.9.4)}$$

$$\mathbf{I}_1 = 0.1\mathbf{V}_2 \qquad \textbf{(19.9.5)}$$

Substituting Eq. (19.9.5) into Eq. (19.9.4),

$$50 - \mathbf{V}_2 = 4\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{V}_2 = 10$$

Thus,

$$\mathbf{V}_{\text{Th}} = \mathbf{V}_2 = 10\ \text{V}$$

The equivalent circuit is shown in Fig. 19.35(c). For maximum power transfer,

$$R_L = \mathbf{Z}_{\text{Th}} = 8\ \Omega$$

From Eq. (4.24), the maximum power is

$$P = I^2 R_L = \left(\frac{\mathbf{V}_{\text{Th}}}{2R_L}\right)^2 R_L = \frac{\mathbf{V}_{\text{Th}}^2}{4R_L} = \frac{100}{4 \times 8} = 3.125\ \text{W}$$

Practice Problem 19.9

Find $\mathbf{I}_1$ and $\mathbf{I}_2$ if the transmission parameters for the two-port in Fig. 19.36 are

$$\begin{bmatrix} 5 & 10\ \Omega \\ 0.4\ \text{S} & 1 \end{bmatrix}$$

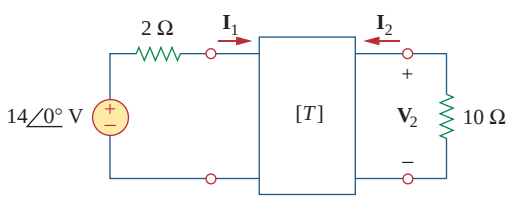


Figure 19.36
For Practice Prob. 19.9.

Answer: 1 A, −0.2 A.

19.6 †Relationships Between Parameters

Since the six sets of parameters relate the same input and output terminal variables of the same two-port network, they should be interrelated. If two sets of parameters exist, we can relate one set to the other set. Let us demonstrate the process with two examples.

Given the *z* parameters, let us obtain the *y* parameters. From Eq. (19.2),

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = [\mathbf{z}] \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} \tag{19.31}$$

or

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = [\mathbf{z}]^{-1} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} \tag{19.32}$$

Also, from Eq. (19.9),

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = [\mathbf{y}] \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} \tag{19.33}$$

Comparing Eqs. (19.32) and (19.33), we see that

$$[\mathbf{y}] = [\mathbf{z}]^{-1} \tag{19.34}$$

The adjoint of the [**z**] matrix is

$$\begin{bmatrix} \mathbf{z}_{22} & -\mathbf{z}_{12} \\ -\mathbf{z}_{21} & \mathbf{z}_{11} \end{bmatrix}$$

and its determinant is

$$\Delta_z = \mathbf{z}_{11}\mathbf{z}_{22} - \mathbf{z}_{12}\mathbf{z}_{21}$$

Substituting these into Eq. (19.34), we get

$$\begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} = \frac{\begin{bmatrix} \mathbf{z}_{22} & -\mathbf{z}_{12} \\ -\mathbf{z}_{21} & \mathbf{z}_{11} \end{bmatrix}}{\Delta_z} \tag{19.35}$$

Equating terms yields

$$\mathbf{y}_{11} = \frac{\mathbf{z}_{22}}{\Delta_z}, \quad \mathbf{y}_{12} = -\frac{\mathbf{z}_{12}}{\Delta_z}, \quad \mathbf{y}_{21} = -\frac{\mathbf{z}_{21}}{\Delta_z}, \quad \mathbf{y}_{22} = \frac{\mathbf{z}_{11}}{\Delta_z} \tag{19.36}$$

As a second example, let us determine the *h* parameters from the *z* parameters. From Eq. (19.1),

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2 \tag{19.37a}$$

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2 \tag{19.37b}$$

Making $\mathbf{I}_2$ the subject of Eq. (19.37b),

$$\mathbf{I}_2 = -\frac{\mathbf{z}_{21}}{\mathbf{z}_{22}}\mathbf{I}_1 + \frac{1}{\mathbf{z}_{22}}\mathbf{V}_2 \tag{19.38}$$

Substituting this into Eq. (19.37a),

$$\mathbf{V}_1 = \frac{\mathbf{z}_{11}\mathbf{z}_{22} - \mathbf{z}_{12}\mathbf{z}_{21}}{\mathbf{z}_{22}}\mathbf{I}_1 + \frac{\mathbf{z}_{12}}{\mathbf{z}_{22}}\mathbf{V}_2 \tag{19.39}$$

Putting Eqs. (19.38) and (19.39) in matrix form,

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \dfrac{\Delta_z}{\mathbf{z}_{22}} & \dfrac{\mathbf{z}_{12}}{\mathbf{z}_{22}} \\ -\dfrac{\mathbf{z}_{21}}{\mathbf{z}_{22}} & \dfrac{1}{\mathbf{z}_{22}} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{V}_2 \end{bmatrix} \tag{19.40}$$

From Eq. (19.15),

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{h}_{11} & \mathbf{h}_{12} \\ \mathbf{h}_{21} & \mathbf{h}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{V}_2 \end{bmatrix}$$

Comparing this with Eq. (19.40), we obtain

$$\mathbf{h}_{11} = \frac{\Delta_z}{\mathbf{z}_{22}}, \quad \mathbf{h}_{12} = \frac{\mathbf{z}_{12}}{\mathbf{z}_{22}}, \quad \mathbf{h}_{21} = -\frac{\mathbf{z}_{21}}{\mathbf{z}_{22}}, \quad \mathbf{h}_{22} = \frac{1}{\mathbf{z}_{22}} \tag{19.41}$$

Table 19.1 provides the conversion formulas for the six sets of two-port parameters. Given one set of parameters, Table 19.1 can be used to find other parameters. For example, given the *T* parameters, we find the corresponding *h* parameters in the fifth column of the third row.

TABLE 19.1

Conversion of two-port parameters.

	z		**y**		**h**		**g**		**T**		**t**	
z	$\mathbf{z}_{11}$	$\mathbf{z}_{12}$	$\dfrac{\mathbf{y}_{22}}{\Delta_y}$	$-\dfrac{\mathbf{y}_{12}}{\Delta_y}$	$\dfrac{\Delta_h}{\mathbf{h}_{22}}$	$\dfrac{\mathbf{h}_{12}}{\mathbf{h}_{22}}$	$\dfrac{1}{\mathbf{g}_{11}}$	$-\dfrac{\mathbf{g}_{12}}{\mathbf{g}_{11}}$	$\dfrac{\mathbf{A}}{\mathbf{C}}$	$\dfrac{\Delta_T}{\mathbf{C}}$	$\dfrac{\mathbf{d}}{\mathbf{c}}$	$\dfrac{1}{\mathbf{c}}$
	$\mathbf{z}_{21}$	$\mathbf{z}_{22}$	$-\dfrac{\mathbf{y}_{21}}{\Delta_y}$	$\dfrac{\mathbf{y}_{11}}{\Delta_y}$	$-\dfrac{\mathbf{h}_{21}}{\mathbf{h}_{22}}$	$\dfrac{1}{\mathbf{h}_{22}}$	$\dfrac{\mathbf{g}_{21}}{\mathbf{g}_{11}}$	$\dfrac{\Delta_g}{\mathbf{g}_{11}}$	$\dfrac{1}{\mathbf{C}}$	$\dfrac{\mathbf{D}}{\mathbf{C}}$	$\dfrac{\Delta_t}{\mathbf{c}}$	$\dfrac{\mathbf{a}}{\mathbf{c}}$
y	$\dfrac{\mathbf{z}_{22}}{\Delta_z}$	$-\dfrac{\mathbf{z}_{12}}{\Delta_z}$	$\mathbf{y}_{11}$	$\mathbf{y}_{12}$	$\dfrac{1}{\mathbf{h}_{11}}$	$-\dfrac{\mathbf{h}_{12}}{\mathbf{h}_{11}}$	$\dfrac{\Delta_g}{\mathbf{g}_{22}}$	$\dfrac{\mathbf{g}_{12}}{\mathbf{g}_{22}}$	$\dfrac{\mathbf{D}}{\mathbf{B}}$	$-\dfrac{\Delta_T}{\mathbf{B}}$	$\dfrac{\mathbf{a}}{\mathbf{b}}$	$-\dfrac{1}{\mathbf{b}}$
	$-\dfrac{\mathbf{z}_{21}}{\Delta_z}$	$\dfrac{\mathbf{z}_{11}}{\Delta_z}$	$\mathbf{y}_{21}$	$\mathbf{y}_{22}$	$\dfrac{\mathbf{h}_{21}}{\mathbf{h}_{11}}$	$\dfrac{\Delta_h}{\mathbf{h}_{11}}$	$-\dfrac{\mathbf{g}_{21}}{\mathbf{g}_{22}}$	$\dfrac{1}{\mathbf{g}_{22}}$	$-\dfrac{1}{\mathbf{B}}$	$\dfrac{\mathbf{A}}{\mathbf{B}}$	$-\dfrac{\Delta_t}{\mathbf{b}}$	$\dfrac{\mathbf{d}}{\mathbf{b}}$
h	$\dfrac{\Delta_z}{\mathbf{z}_{22}}$	$\dfrac{\mathbf{z}_{12}}{\mathbf{z}_{22}}$	$\dfrac{1}{\mathbf{y}_{11}}$	$-\dfrac{\mathbf{y}_{12}}{\mathbf{y}_{11}}$	$\mathbf{h}_{11}$	$\mathbf{h}_{12}$	$\dfrac{\mathbf{g}_{22}}{\Delta_g}$	$-\dfrac{\mathbf{g}_{12}}{\Delta_g}$	$\dfrac{\mathbf{B}}{\mathbf{D}}$	$\dfrac{\Delta_T}{\mathbf{D}}$	$\dfrac{\mathbf{b}}{\mathbf{a}}$	$\dfrac{1}{\mathbf{a}}$
	$-\dfrac{\mathbf{z}_{21}}{\mathbf{z}_{22}}$	$\dfrac{1}{\mathbf{z}_{22}}$	$\dfrac{\mathbf{y}_{21}}{\mathbf{y}_{11}}$	$\dfrac{\Delta_y}{\mathbf{y}_{11}}$	$\mathbf{h}_{21}$	$\mathbf{h}_{22}$	$-\dfrac{\mathbf{g}_{21}}{\Delta_g}$	$\dfrac{\mathbf{g}_{11}}{\Delta_g}$	$-\dfrac{1}{\mathbf{D}}$	$\dfrac{\mathbf{C}}{\mathbf{D}}$	$\dfrac{\Delta_t}{\mathbf{a}}$	$\dfrac{\mathbf{c}}{\mathbf{a}}$
g	$\dfrac{1}{\mathbf{z}_{11}}$	$-\dfrac{\mathbf{z}_{12}}{\mathbf{z}_{11}}$	$\dfrac{\Delta_y}{\mathbf{y}_{22}}$	$\dfrac{\mathbf{y}_{12}}{\mathbf{y}_{22}}$	$\dfrac{\mathbf{h}_{22}}{\Delta_h}$	$-\dfrac{\mathbf{h}_{12}}{\Delta_h}$	$\mathbf{g}_{11}$	$\mathbf{g}_{12}$	$\dfrac{\mathbf{C}}{\mathbf{A}}$	$-\dfrac{\Delta_T}{\mathbf{A}}$	$\dfrac{\mathbf{c}}{\mathbf{d}}$	$-\dfrac{1}{\mathbf{d}}$
	$\dfrac{\mathbf{z}_{21}}{\mathbf{z}_{11}}$	$\dfrac{\Delta_z}{\mathbf{z}_{11}}$	$-\dfrac{\mathbf{y}_{21}}{\mathbf{y}_{22}}$	$\dfrac{1}{\mathbf{y}_{22}}$	$-\dfrac{\mathbf{h}_{21}}{\Delta_h}$	$\dfrac{\mathbf{h}_{11}}{\Delta_h}$	$\mathbf{g}_{21}$	$\mathbf{g}_{22}$	$\dfrac{1}{\mathbf{A}}$	$\dfrac{\mathbf{B}}{\mathbf{A}}$	$\dfrac{\Delta_t}{\mathbf{d}}$	$-\dfrac{\mathbf{b}}{\mathbf{d}}$
T	$\dfrac{\mathbf{z}_{11}}{\mathbf{z}_{21}}$	$\dfrac{\Delta_z}{\mathbf{z}_{21}}$	$-\dfrac{\mathbf{y}_{22}}{\mathbf{y}_{21}}$	$-\dfrac{1}{\mathbf{y}_{21}}$	$-\dfrac{\Delta_h}{\mathbf{h}_{21}}$	$-\dfrac{\mathbf{h}_{11}}{\mathbf{h}_{21}}$	$\dfrac{1}{\mathbf{g}_{21}}$	$\dfrac{\mathbf{g}_{22}}{\mathbf{g}_{21}}$	$\mathbf{A}$	$\mathbf{B}$	$\dfrac{\mathbf{d}}{\Delta_t}$	$\dfrac{\mathbf{b}}{\Delta_t}$
	$\dfrac{1}{\mathbf{z}_{21}}$	$\dfrac{\mathbf{z}_{22}}{\mathbf{z}_{21}}$	$-\dfrac{\Delta_y}{\mathbf{y}_{21}}$	$-\dfrac{\mathbf{y}_{11}}{\mathbf{y}_{21}}$	$-\dfrac{\mathbf{h}_{22}}{\mathbf{h}_{21}}$	$-\dfrac{1}{\mathbf{h}_{21}}$	$\dfrac{\mathbf{g}_{11}}{\mathbf{g}_{21}}$	$\dfrac{\Delta_g}{\mathbf{g}_{21}}$	$\mathbf{C}$	$\mathbf{D}$	$\dfrac{\mathbf{c}}{\Delta_t}$	$\dfrac{\mathbf{a}}{\Delta_t}$
t	$\dfrac{\mathbf{z}_{22}}{\mathbf{z}_{12}}$	$\dfrac{\Delta_z}{\mathbf{z}_{12}}$	$-\dfrac{\mathbf{y}_{11}}{\mathbf{y}_{12}}$	$-\dfrac{1}{\mathbf{y}_{12}}$	$\dfrac{1}{\mathbf{h}_{12}}$	$\dfrac{\mathbf{h}_{11}}{\mathbf{h}_{12}}$	$-\dfrac{\Delta_g}{\mathbf{g}_{12}}$	$-\dfrac{\mathbf{g}_{22}}{\mathbf{g}_{12}}$	$\dfrac{\mathbf{D}}{\Delta_T}$	$\dfrac{\mathbf{B}}{\Delta_T}$	$\mathbf{a}$	$\mathbf{b}$
	$\dfrac{1}{\mathbf{z}_{12}}$	$\dfrac{\mathbf{z}_{11}}{\mathbf{z}_{12}}$	$-\dfrac{\Delta_y}{\mathbf{y}_{12}}$	$-\dfrac{\mathbf{y}_{22}}{\mathbf{y}_{12}}$	$\dfrac{\mathbf{h}_{22}}{\mathbf{h}_{12}}$	$\dfrac{\Delta_h}{\mathbf{h}_{12}}$	$-\dfrac{\mathbf{g}_{11}}{\mathbf{g}_{12}}$	$-\dfrac{1}{\mathbf{g}_{12}}$	$\dfrac{\mathbf{C}}{\Delta_T}$	$\dfrac{\mathbf{A}}{\Delta_T}$	$\mathbf{c}$	$\mathbf{d}$

$\Delta_z = \mathbf{z}_{11}\mathbf{z}_{22} - \mathbf{z}_{12}\mathbf{z}_{21}$, $\Delta_h = \mathbf{h}_{11}\mathbf{h}_{22} - \mathbf{h}_{12}\mathbf{h}_{21}$, $\Delta_T = \mathbf{AD} - \mathbf{BC}$
$\Delta_y = \mathbf{y}_{11}\mathbf{y}_{22} - \mathbf{y}_{12}\mathbf{y}_{21}$, $\Delta_g = \mathbf{g}_{11}\mathbf{g}_{22} - \mathbf{g}_{12}\mathbf{g}_{21}$, $\Delta_t = \mathbf{ad} - \mathbf{bc}$

Also, given that $\mathbf{z}_{21} = \mathbf{z}_{12}$ for a reciprocal network, we can use the table to express this condition in terms of other parameters. It can also be shown that

$$[\mathbf{g}] = [\mathbf{h}]^{-1} \tag{19.42}$$

but

$$[\mathbf{t}] \neq [\mathbf{T}]^{-1} \tag{19.43}$$

Example 19.10

Find [**z**] and [**g**] of a two-port network if

$$[\mathbf{T}] = \begin{bmatrix} 10 & 1.5\ \Omega \\ 2\ \text{S} & 4 \end{bmatrix}$$

Solution:

If $\mathbf{A} = 10, \mathbf{B} = 1.5, \mathbf{C} = 2, \mathbf{D} = 4$, the determinant of the matrix is

$$\Delta_T = \mathbf{AD} - \mathbf{BC} = 40 - 3 = 37$$

From Table 19.1,

$$\mathbf{z}_{11} = \frac{\mathbf{A}}{\mathbf{C}} = \frac{10}{2} = 5, \qquad \mathbf{z}_{12} = \frac{\Delta_T}{\mathbf{C}} = \frac{37}{2} = 18.5$$

$$\mathbf{z}_{21} = \frac{1}{\mathbf{C}} = \frac{1}{2} = 0.5, \qquad \mathbf{z}_{22} = \frac{\mathbf{D}}{\mathbf{C}} = \frac{4}{2} = 2$$

$$\mathbf{g}_{11} = \frac{\mathbf{C}}{\mathbf{A}} = \frac{2}{10} = 0.2, \qquad \mathbf{g}_{12} = -\frac{\Delta_T}{\mathbf{A}} = -\frac{37}{10} = -3.7$$

$$\mathbf{g}_{21} = \frac{1}{\mathbf{A}} = \frac{1}{10} = 0.1, \qquad \mathbf{g}_{22} = \frac{\mathbf{B}}{\mathbf{A}} = \frac{1.5}{10} = 0.15$$

Thus,

$$[\mathbf{z}] = \begin{bmatrix} 5 & 18.5 \\ 0.5 & 2 \end{bmatrix} \Omega, \qquad [\mathbf{g}] = \begin{bmatrix} 0.2\ \text{S} & -3.7 \\ 0.1 & 0.15\ \Omega \end{bmatrix}$$

Practice Problem 19.10

Determine [**y**] and [**T**] of a two-port network whose *z* parameters are

$$[\mathbf{z}] = \begin{bmatrix} 6 & 4 \\ 4 & 6 \end{bmatrix} \Omega$$

Answer: $[\mathbf{y}] = \begin{bmatrix} 0.3 & -0.2 \\ -0.2 & 0.3 \end{bmatrix}$ S, $[\mathbf{T}] = \begin{bmatrix} 1.5 & 5\ \Omega \\ 0.25\ \text{S} & 1.5 \end{bmatrix}$.

Example 19.11

Obtain the *y* parameters of the op amp circuit in Fig. 19.37. Show that the circuit has no *z* parameters.

Solution:

Since no current can enter the input terminals of the op amp, $\mathbf{I}_1 = 0$, which can be expressed in terms of $\mathbf{V}_1$ and $\mathbf{V}_2$ as

$$\mathbf{I}_1 = 0\mathbf{V}_1 + 0\mathbf{V}_2 \tag{19.11.1}$$

Comparing this with Eq. (19.8) gives

$$\mathbf{y}_{11} = 0 = \mathbf{y}_{12}$$

Also,

$$\mathbf{V}_2 = R_3\mathbf{I}_2 + \mathbf{I}_o(R_1 + R_2)$$

where $\mathbf{I}_o$ is the current through R_1 and R_2. But $\mathbf{I}_o = \mathbf{V}_1/R_1$. Hence,

$$\mathbf{V}_2 = R_3\mathbf{I}_2 + \frac{\mathbf{V}_1(R_1 + R_2)}{R_1}$$

which can be written as

$$\mathbf{I}_2 = -\frac{(R_1 + R_2)}{R_1R_3}\mathbf{V}_1 + \frac{\mathbf{V}_2}{R_3}$$

Comparing this with Eq. (19.8) shows that

$$\mathbf{y}_{21} = -\frac{(R_1 + R_2)}{R_1R_3}, \qquad \mathbf{y}_{22} = \frac{1}{R_3}$$

The determinant of the [**y**] matrix is

$$\Delta_y = \mathbf{y}_{11}\mathbf{y}_{22} - \mathbf{y}_{12}\mathbf{y}_{21} = 0$$

Since $\Delta_y = 0$, the [**y**] matrix has no inverse; therefore, the [**z**] matrix does not exist according to Eq. (19.34). Note that the circuit is not reciprocal because of the active element.

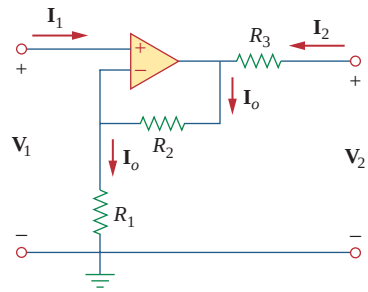


Figure 19.37
For Example 19.11.

Practice Problem 19.11

Find the *z* parameters of the op amp circuit in Fig. 19.38. Show that the circuit has no *y* parameters.

Answer: $[\mathbf{z}] = \begin{bmatrix} R_1 & 0 \\ -R_2 & 0 \end{bmatrix}$. Since $[\mathbf{z}]^{-1}$ does not exist, [**y**] does not exist.

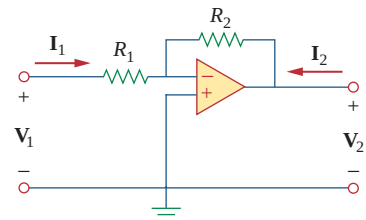


Figure 19.38
For Practice Prob. 19.11.

19.7 Interconnection of Networks

A large, complex network may be divided into subnetworks for the purposes of analysis and design. The subnetworks are modeled as two-port networks, interconnected to form the original network. The two-port networks may therefore be regarded as building blocks that can be interconnected to form a complex network. The interconnection can be in series, in parallel, or in cascade. Although the interconnected network can be described by any of the six parameter sets, a certain set of parameters may have a definite advantage. For example, when the networks are in series, their individual *z* parameters add up to give the *z* parameters of the larger network. When they are in parallel, their individual *y* parameters add up to give the *y* parameters of the larger network. When they are cascaded, their individual transmission parameters can be multiplied together to get the transmission parameters of the larger network.

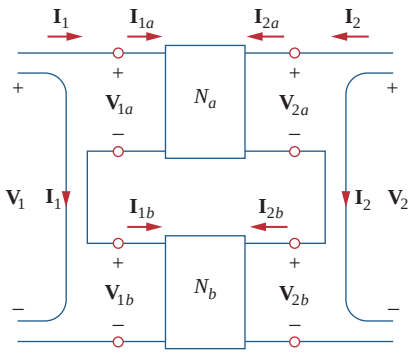


Figure 19.39
Series connection of two two-port networks.

Consider the series connection of two two-port networks shown in Fig. 19.39. The networks are regarded as being in series because their input currents are the same and their voltages add. In addition, each network has a common reference, and when the circuits are placed in series, the common reference points of each circuit are connected together. For network N_a,

$$\begin{aligned} \mathbf{V}_{1a} &= \mathbf{z}_{11a}\mathbf{I}_{1a} + \mathbf{z}_{12a}\mathbf{I}_{2a} \\ \mathbf{V}_{2a} &= \mathbf{z}_{21a}\mathbf{I}_{1a} + \mathbf{z}_{22a}\mathbf{I}_{2a} \end{aligned} \tag{19.44}$$

and for network N_b,

$$\begin{aligned} \mathbf{V}_{1b} &= \mathbf{z}_{11b}\mathbf{I}_{1b} + \mathbf{z}_{12b}\mathbf{I}_{2b} \\ \mathbf{V}_{2b} &= \mathbf{z}_{21b}\mathbf{I}_{1b} + \mathbf{z}_{22b}\mathbf{I}_{2b} \end{aligned} \tag{19.45}$$

We notice from Fig. 19.39 that

$$\mathbf{I}_1 = \mathbf{I}_{1a} = \mathbf{I}_{1b}, \qquad \mathbf{I}_2 = \mathbf{I}_{2a} = \mathbf{I}_{2b} \tag{19.46}$$

and that

$$\begin{aligned} \mathbf{V}_1 &= \mathbf{V}_{1a} + \mathbf{V}_{1b} = (\mathbf{z}_{11a} + \mathbf{z}_{11b})\mathbf{I}_1 + (\mathbf{z}_{12a} + \mathbf{z}_{12b})\mathbf{I}_2 \\ \mathbf{V}_2 &= \mathbf{V}_{2a} + \mathbf{V}_{2b} = (\mathbf{z}_{21a} + \mathbf{z}_{21b})\mathbf{I}_1 + (\mathbf{z}_{22a} + \mathbf{z}_{22b})\mathbf{I}_2 \end{aligned} \tag{19.47}$$

Thus, the z parameters for the overall network are

$$\begin{bmatrix} \mathbf{z}_{11} & \mathbf{z}_{12} \\ \mathbf{z}_{21} & \mathbf{z}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{z}_{11a} + \mathbf{z}_{11b} & \mathbf{z}_{12a} + \mathbf{z}_{12b} \\ \mathbf{z}_{21a} + \mathbf{z}_{21b} & \mathbf{z}_{22a} + \mathbf{z}_{22b} \end{bmatrix} \tag{19.48}$$

or

$$\boxed{[\mathbf{z}] = [\mathbf{z}_a] + [\mathbf{z}_b]} \tag{19.49}$$

showing that the z parameters for the overall network are the sum of the z parameters for the individual networks. This can be extended to n networks in series. If two two-port networks in the [**h**] model, for example, are connected in series, we use Table 19.1 to convert the **h** to **z** and then apply Eq. (19.49). We finally convert the result back to **h** using Table 19.1.

Two two-port networks are in parallel when their port voltages are equal and the port currents of the larger network are the sums of the individual port currents. In addition, each circuit must have a common reference and when the networks are connected together, they must all have their common references tied together. The parallel connection of two two-port networks is shown in Fig. 19.40. For the two networks,

$$\begin{aligned} \mathbf{I}_{1a} &= \mathbf{y}_{11a}\mathbf{V}_{1a} + \mathbf{y}_{12a}\mathbf{V}_{2a} \\ \mathbf{I}_{2a} &= \mathbf{y}_{21a}\mathbf{V}_{1a} + \mathbf{y}_{22a}\mathbf{V}_{2a} \end{aligned} \tag{19.50}$$

and

$$\begin{aligned} \mathbf{I}_{1b} &= \mathbf{y}_{11b}\mathbf{V}_{1b} + \mathbf{y}_{12b}\mathbf{V}_{2b} \\ \mathbf{I}_{2a} &= \mathbf{y}_{21b}\mathbf{V}_{1b} + \mathbf{y}_{22b}\mathbf{V}_{2b} \end{aligned} \tag{19.51}$$

But from Fig. 19.40,

$$\mathbf{V}_1 = \mathbf{V}_{1a} = \mathbf{V}_{1b}, \qquad \mathbf{V}_2 = \mathbf{V}_{2a} = \mathbf{V}_{2b} \tag{19.52a}$$

$$\mathbf{I}_1 = \mathbf{I}_{1a} + \mathbf{I}_{1b}, \qquad \mathbf{I}_2 = \mathbf{I}_{2a} + \mathbf{I}_{2b} \tag{19.52b}$$

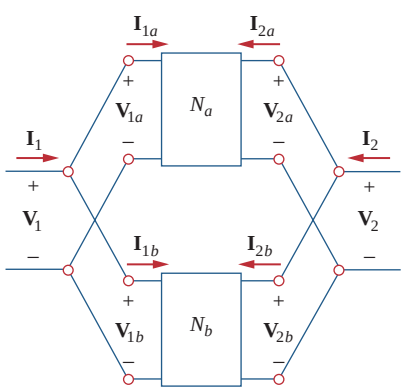


Figure 19.40
Parallel connection of two two-port networks.

Substituting Eqs. (19.50) and (19.51) into Eq. (19.52b) yields

$$\begin{aligned} \mathbf{I}_1 &= (\mathbf{y}_{11a} + \mathbf{y}_{11b})\mathbf{V}_1 + (\mathbf{y}_{12a} + \mathbf{y}_{12b})\mathbf{V}_2 \\ \mathbf{I}_2 &= (\mathbf{y}_{21a} + \mathbf{y}_{21b})\mathbf{V}_1 + (\mathbf{y}_{22a} + \mathbf{y}_{22b})\mathbf{V}_2 \end{aligned} \tag{19.53}$$

Thus, the *y* parameters for the overall network are

$$\begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{11a} + \mathbf{y}_{11b} & \mathbf{y}_{12a} + \mathbf{y}_{12b} \\ \mathbf{y}_{21a} + \mathbf{y}_{21b} & \mathbf{y}_{22a} + \mathbf{y}_{22b} \end{bmatrix} \tag{19.54}$$

or

$$\boxed{[\mathbf{y}] = [\mathbf{y}_a] + [\mathbf{y}_b]} \tag{19.55}$$

showing that the *y* parameters of the overall network are the sum of the *y* parameters of the individual networks. The result can be extended to *n* two-port networks in parallel.

Two networks are said to be *cascaded* when the output of one is the input of the other. The connection of two two-port networks in cascade is shown in Fig. 19.41. For the two networks,

$$\begin{bmatrix} \mathbf{V}_{1a} \\ \mathbf{I}_{1a} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_a & \mathbf{B}_a \\ \mathbf{C}_a & \mathbf{D}_a \end{bmatrix} \begin{bmatrix} \mathbf{V}_{2a} \\ -\mathbf{I}_{2a} \end{bmatrix} \tag{19.56}$$

$$\begin{bmatrix} \mathbf{V}_{1b} \\ \mathbf{I}_{1b} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_b & \mathbf{B}_b \\ \mathbf{C}_b & \mathbf{D}_b \end{bmatrix} \begin{bmatrix} \mathbf{V}_{2b} \\ -\mathbf{I}_{2b} \end{bmatrix} \tag{19.57}$$

From Fig. 19.41,

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{V}_{1a} \\ \mathbf{I}_{1a} \end{bmatrix}, \quad \begin{bmatrix} \mathbf{V}_{2a} \\ -\mathbf{I}_{2a} \end{bmatrix} = \begin{bmatrix} \mathbf{V}_{1b} \\ \mathbf{I}_{1b} \end{bmatrix}, \quad \begin{bmatrix} \mathbf{V}_{2b} \\ -\mathbf{I}_{2b} \end{bmatrix} = \begin{bmatrix} \mathbf{V}_2 \\ -\mathbf{I}_2 \end{bmatrix} \tag{19.58}$$

Substituting these into Eqs. (19.56) and (19.57),

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{I}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_a & \mathbf{B}_a \\ \mathbf{C}_a & \mathbf{D}_a \end{bmatrix} \begin{bmatrix} \mathbf{A}_b & \mathbf{B}_b \\ \mathbf{C}_b & \mathbf{D}_b \end{bmatrix} \begin{bmatrix} \mathbf{V}_2 \\ -\mathbf{I}_2 \end{bmatrix} \tag{19.59}$$

Thus, the transmission parameters for the overall network are the product of the transmission parameters for the individual transmission parameters:

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_a & \mathbf{B}_a \\ \mathbf{C}_a & \mathbf{D}_a \end{bmatrix} \begin{bmatrix} \mathbf{A}_b & \mathbf{B}_b \\ \mathbf{C}_b & \mathbf{D}_b \end{bmatrix} \tag{19.60}$$

or

$$\boxed{[\mathbf{T}] = [\mathbf{T}_a][\mathbf{T}_b]} \tag{19.61}$$

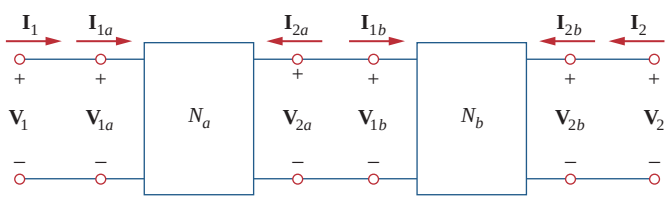


Figure 19.41
Cascade connection of two two-port networks.

It is this property that makes the transmission parameters so useful. Keep in mind that the multiplication of the matrices must be in the order in which the networks N_a and N_b are cascaded.

Example 19.12

Evaluate $\mathbf{V}_2/\mathbf{V}_s$ in the circuit in Fig. 19.42.

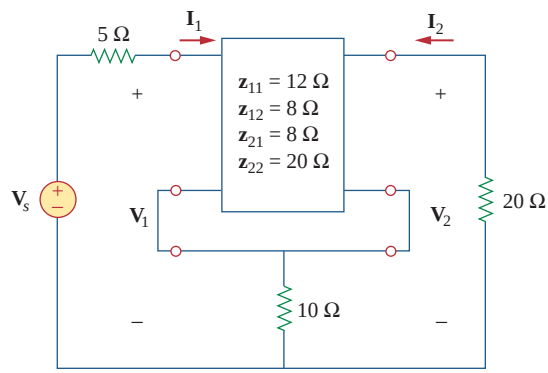


Figure 19.42
For Example 19.12.

Solution:
This may be regarded as two two-ports in series. For N_b,

$$\mathbf{z}_{12b} = \mathbf{z}_{21b} = 10 = \mathbf{z}_{11b} = \mathbf{z}_{22b}$$

Thus,

$$[\mathbf{z}] = [\mathbf{z}_a] + [\mathbf{z}_b] = \begin{bmatrix} 12 & 8 \\ 8 & 20 \end{bmatrix} + \begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix} = \begin{bmatrix} 22 & 18 \\ 18 & 30 \end{bmatrix}$$

But

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2 = 22\mathbf{I}_1 + 18\mathbf{I}_2 \qquad \textbf{(19.12.1)}$$

$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2 = 18\mathbf{I}_1 + 30\mathbf{I}_2 \qquad \textbf{(19.12.2)}$$

Also, at the input port

$$\mathbf{V}_1 = \mathbf{V}_s - 5\mathbf{I}_1 \qquad \textbf{(19.12.3)}$$

and at the output port

$$\mathbf{V}_2 = -20\mathbf{I}_2 \quad \Rightarrow \quad \mathbf{I}_2 = -\frac{\mathbf{V}_2}{20} \qquad \textbf{(19.12.4)}$$

Substituting Eqs. (19.12.3) and (19.12.4) into Eq. (19.12.1) gives

$$\mathbf{V}_s - 5\mathbf{I}_1 = 22\mathbf{I}_1 - \frac{18}{20}\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{V}_s = 27\mathbf{I}_1 - 0.9\mathbf{V}_2 \qquad \textbf{(19.12.5)}$$

while substituting Eq. (19.12.4) into Eq. (19.12.2) yields

$$\mathbf{V}_2 = 18\mathbf{I}_1 - \frac{30}{20}\mathbf{V}_2 \quad \Rightarrow \quad \mathbf{I}_1 = \frac{2.5}{18}\mathbf{V}_2 \qquad \textbf{(19.12.6)}$$

Substituting Eq. (19.12.6) into Eq. (19.12.5), we get

$$\mathbf{V}_s = 27 \times \frac{2.5}{18}\mathbf{V}_2 - 0.9\mathbf{V}_2 = 2.85\mathbf{V}_2$$

And so,

$$\frac{\mathbf{V}_2}{\mathbf{V}_s} = \frac{1}{2.85} = 0.3509$$

Practice Problem 19.12

Find $\mathbf{V}_2/\mathbf{V}_s$ in the circuit in Fig. 19.43.

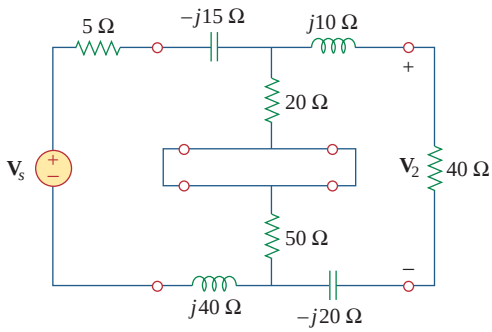


Figure 19.43
For Practice Prob. 19.12.

Answer: $0.6799\angle -29.05°$.

Example 19.13

Find the y parameters of the two-port in Fig. 19.44.

Solution:
Let us refer to the upper network as N_a and the lower one as N_b. The two networks are connected in parallel. Comparing N_a and N_b with the circuit in Fig. 19.13(a), we obtain

$$y_{12a} = -j4 = y_{21a}, \qquad y_{11a} = 2 + j4, \qquad y_{22a} = 3 + j4$$

or

$$[\mathbf{y}_a] = \begin{bmatrix} 2 + j4 & -j4 \\ -j4 & 3 + j4 \end{bmatrix} \text{S}$$

and

$$y_{12b} = -4 = y_{21b}, \qquad y_{11b} = 4 - j2, \qquad y_{22b} = 4 - j6$$

or

$$[\mathbf{y}_b] = \begin{bmatrix} 4 - j2 & -4 \\ -4 & 4 - j6 \end{bmatrix} \text{S}$$

The overall y parameters are

$$[\mathbf{y}] = [\mathbf{y}_a] + [\mathbf{y}_b] = \begin{bmatrix} 6 + j2 & -4 - j4 \\ -4 - j4 & 7 - j2 \end{bmatrix} \text{S}$$

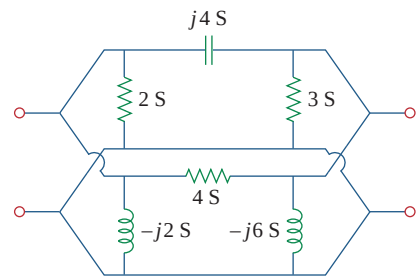


Figure 19.44
For Example 19.13.

Practice Problem 19.13

Obtain the y parameters for the network in Fig. 19.45.

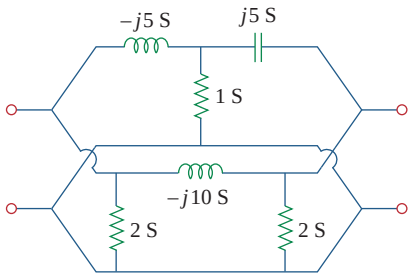


Figure 19.45
For Practice Prob. 19.13.

Answer: $\begin{bmatrix} 27 - j15 & -25 + j10 \\ -25 + j10 & 27 - j5 \end{bmatrix}$ S.

Example 19.14

Find the transmission parameters for the circuit in Fig. 19.46.

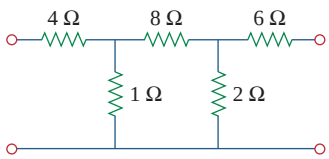


Figure 19.46
For Example 19.14.

Solution:
We can regard the given circuit in Fig. 19.46 as a cascade connection of two T networks as shown in Fig. 19.47(a). We can show that a T network, shown in Fig. 19.47(b), has the following transmission parameters [see Prob. 19.52(b)]:

$$\mathbf{A} = 1 + \frac{R_1}{R_2}, \qquad \mathbf{B} = R_3 + \frac{R_1(R_2 + R_3)}{R_2}$$

$$\mathbf{C} = \frac{1}{R_2}, \qquad \mathbf{D} = 1 + \frac{R_3}{R_2}$$

Applying this to the cascaded networks N_a and N_b in Fig. 19.47(a), we get

$$\mathbf{A}_a = 1 + 4 = 5, \qquad \mathbf{B}_a = 8 + 4 \times 9 = 44\ \Omega$$
$$\mathbf{C}_a = 1\ \text{S}, \qquad \mathbf{D}_a = 1 + 8 = 9$$

or in matrix form,

$$[\mathbf{T}_a] = \begin{bmatrix} 5 & 44\ \Omega \\ 1\ \text{S} & 9 \end{bmatrix}$$

and

$$\mathbf{A}_b = 1, \qquad \mathbf{B}_b = 6\ \Omega, \qquad \mathbf{C}_b = 0.5\ \text{S}, \qquad \mathbf{D}_b = 1 + \frac{6}{2} = 4$$

i.e.,

$$[\mathbf{T}_b] = \begin{bmatrix} 1 & 6\ \Omega \\ 0.5\ \text{S} & 4 \end{bmatrix}$$

Thus, for the total network in Fig. 19.46,

$$[\mathbf{T}] = [\mathbf{T}_a][\mathbf{T}_b] = \begin{bmatrix} 5 & 44 \\ 1 & 9 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 0.5 & 4 \end{bmatrix}$$
$$= \begin{bmatrix} 5 \times 1 + 44 \times 0.5 & 5 \times 6 + 44 \times 4 \\ 1 \times 1 + 9 \times 0.5 & 1 \times 6 + 9 \times 4 \end{bmatrix}$$
$$= \begin{bmatrix} 27 & 206\ \Omega \\ 5.5\ \text{S} & 42 \end{bmatrix}$$

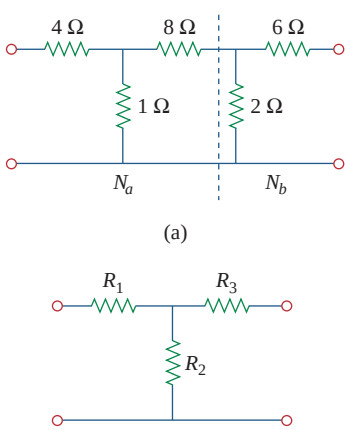


Figure 19.47
For Example 19.14: (a) Breaking the circuit in Fig. 19.46 into two two-ports, (b) a general T two-port.

Notice that

$$\Delta_{T_a} = \Delta_{T_b} = \Delta_T = 1$$

showing that the network is reciprocal.

Practice Problem 19.14

Obtain the **ABCD** parameter representation of the circuit in Fig. 19.48.

Answer: $[\mathbf{T}] = \begin{bmatrix} 29.25 & 2200\ \Omega \\ 0.425\ \text{S} & 32 \end{bmatrix}$.

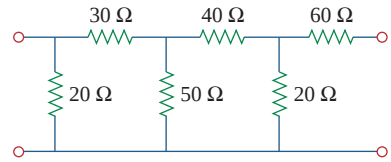


Figure 19.48
For Practice Prob. 19.14.

19.8 Computing Two-Port Parameters Using *PSpice*

Hand calculation of the two-port parameters may become difficult when the two-port is complicated. We resort to *PSpice* in such situations. If the circuit is purely resistive, *PSpice* dc analysis may be used; otherwise, *PSpice* ac analysis is required at a specific frequency. The key to using *PSpice* in computing a particular two-port parameter is to remember how that parameter is defined and to constrain the appropriate port variable with a 1-A or 1-V source while using an open or short circuit to impose the other necessary constraints. The following two examples illustrate the idea.

Example 19.15

Find the *h* parameters of the network in Fig. 19.49.

Solution:
From Eq. (19.16),

$$\mathbf{h}_{11} = \left.\frac{\mathbf{V}_1}{\mathbf{I}_1}\right|_{\mathbf{V}_2=0}, \qquad \mathbf{h}_{21} = \left.\frac{\mathbf{I}_2}{\mathbf{I}_1}\right|_{\mathbf{V}_2=0}$$

showing that $\mathbf{h}_{11}$ and $\mathbf{h}_{21}$ can be found by setting $\mathbf{V}_2 = 0$. Also by setting $\mathbf{I}_1 = 1$ A, $\mathbf{h}_{11}$ becomes $\mathbf{V}_1/1$ while $\mathbf{h}_{21}$ becomes $\mathbf{I}_2/1$. With this in mind, we draw the schematic in Fig. 19.50(a). We insert a 1-A dc current source

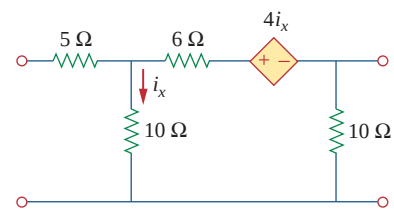
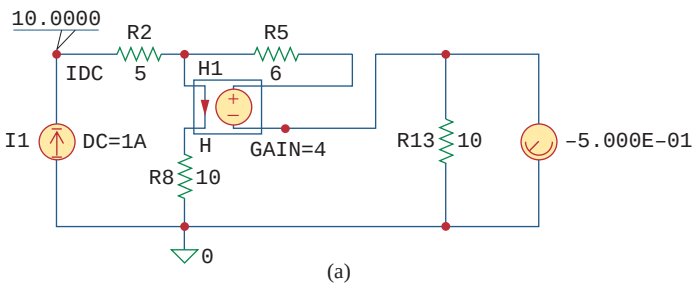
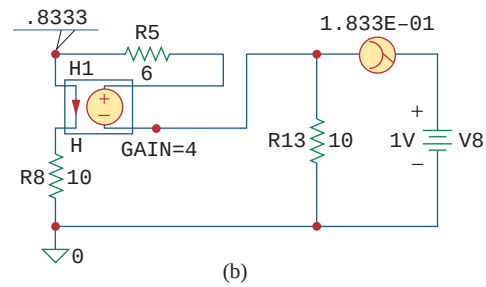


Figure 19.49
For Example 19.15.



(a)



(b)

Figure 19.50
For Example 19.15: (a) computing $\mathbf{h}_{11}$ and $\mathbf{h}_{21}$, (b) computing $\mathbf{h}_{12}$ and $\mathbf{h}_{22}$.

IDC to take care of $\mathbf{I}_1 = 1$ A, the pseudocomponent VIEWPOINT to display $\mathbf{V}_1$ and pseudocomponent IPROBE to display $\mathbf{I}_2$. After saving the schematic, we run *PSpice* by selecting **Analysis/Simulate** and note the values displayed on the pseudocomponents. We obtain

$$\mathbf{h}_{11} = \frac{\mathbf{V}_1}{1} = 10\ \Omega, \qquad \mathbf{h}_{21} = \frac{\mathbf{I}_2}{1} = -0.5$$

Similarly, from Eq. (19.16),

$$\mathbf{h}_{12} = \left.\frac{\mathbf{V}_1}{\mathbf{V}_2}\right|_{\mathbf{I}_1=0}, \qquad \mathbf{h}_{22} = \left.\frac{\mathbf{I}_2}{\mathbf{V}_2}\right|_{\mathbf{I}_1=0}$$

indicating that we obtain $\mathbf{h}_{12}$ and $\mathbf{h}_{22}$ by open-circuiting the input port ($\mathbf{I}_1 = 0$). By making $\mathbf{V}_2 = 1$ V, $\mathbf{h}_{12}$ becomes $\mathbf{V}_1/1$ while $\mathbf{h}_{22}$ becomes $\mathbf{I}_2/1$. Thus, we use the schematic in Fig. 19.50(b) with a 1-V dc voltage source VDC inserted at the output terminal to take care of $\mathbf{V}_2 = 1$ V. The pseudocomponents VIEWPOINT and IPROBE are inserted to display the values of $\mathbf{V}_1$ and $\mathbf{I}_2$, respectively. (Notice that in Fig. 19.50(b), the 5-Ω resistor is ignored because the input port is open-circuited and *PSpice* will not allow such. We may include the 5-Ω resistor if we replace the open circuit with a very large resistor, say, 10 MΩ.) After simulating the schematic, we obtain the values displayed on the pseudocomponents as shown in Fig. 19.50(b). Thus,

$$\mathbf{h}_{12} = \frac{\mathbf{V}_1}{1} = 0.8333, \qquad \mathbf{h}_{22} = \frac{\mathbf{I}_2}{1} = 0.1833\ \text{S}$$

Practice Problem 19.15

Obtain the *h* parameters for the network in Fig. 19.51 using *PSpice*.

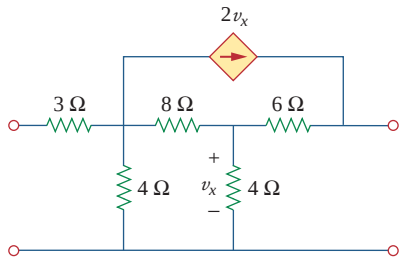


Figure 19.51
For Practice Prob. 19.15.

Answer: $h_{11} = 4.238\ \Omega$, $h_{21} = -0.6190$, $h_{12} = -0.7143$, $h_{22} = -0.1429$ S.

Example 19.16

Find the *z* parameters for the circuit in Fig. 19.52 at $\omega = 10^6$ rad/s.

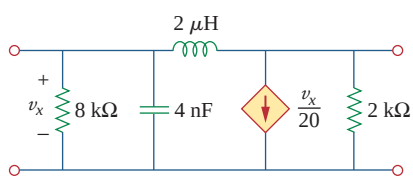


Figure 19.52
For Example 19.16.

Solution:

Notice that we used dc analysis in Example 19.15 because the circuit in Fig. 19.49 is purely resistive. Here, we use ac analysis at $f = \omega/2\pi = 0.15915$ MHz, because *L* and *C* are frequency dependent.

In Eq. (19.3), we defined the *z* parameters as

$$\mathbf{z}_{11} = \left.\frac{\mathbf{V}_1}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0}, \qquad \mathbf{z}_{21} = \left.\frac{\mathbf{V}_2}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0}$$

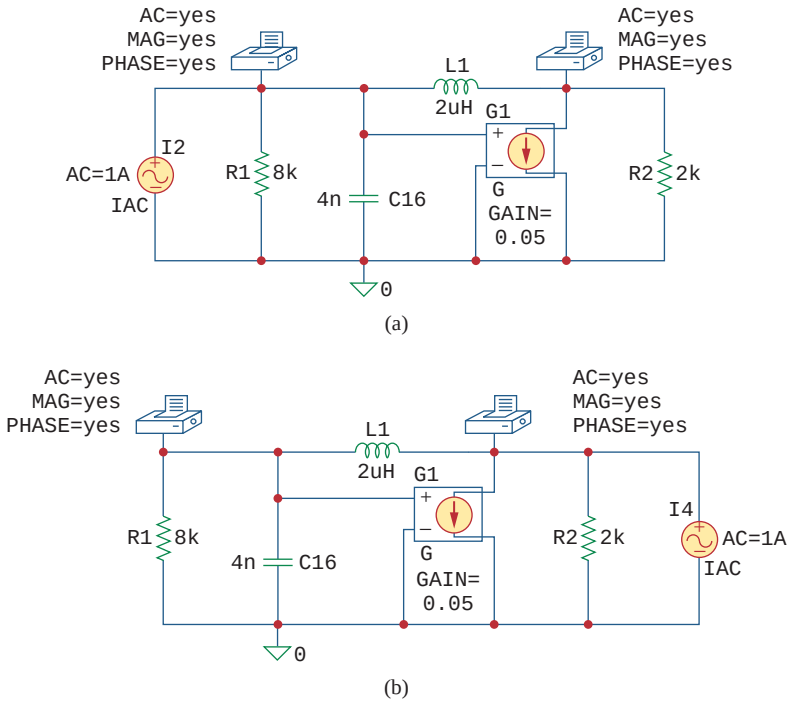


Figure 19.53
For Example 19.16: (a) circuit for determining $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$, (b) circuit for determining $\mathbf{z}_{12}$ and $\mathbf{z}_{22}$.

This suggests that if we let $\mathbf{I}_1 = 1$ A and open-circuit the output port so that $\mathbf{I}_2 = 0$, then we obtain

$$\mathbf{z}_{11} = \frac{\mathbf{V}_1}{1} \qquad \text{and} \qquad \mathbf{z}_{21} = \frac{\mathbf{V}_2}{1}$$

We realize this with the schematic in Fig. 19.53(a). We insert a 1-A ac current source IAC at the input terminal of the circuit and two VPRINT1 pseudocomponents to obtain $\mathbf{V}_1$ and $\mathbf{V}_2$. The attributes of each VPRINT1 are set as *AC = yes*, *MAG = yes*, and *PHASE = yes* to print the magnitude and phase values of the voltages. We select **Analysis/Setup/AC Sweep** and enter 1 as *Total Pts*, 0.1519MEG as *Start Freq*, and 0.1519MEG as *Final Freq* in the **AC Sweep and Noise Analysis** dialog box. After saving the schematic, we select **Analysis/Simulate** to simulate it. We obtain $\mathbf{V}_1$ and $\mathbf{V}_2$ from the output file. Thus,

$$\mathbf{z}_{11} = \frac{\mathbf{V}_1}{1} = 19.70\underline{/175.7^\circ}\ \Omega, \qquad \mathbf{z}_{21} = \frac{\mathbf{V}_2}{1} = 19.79\underline{/170.2^\circ}\ \Omega$$

In a similar manner, from Eq. (19.3),

$$\mathbf{z}_{12} = \left.\frac{\mathbf{V}_1}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0}, \qquad \mathbf{z}_{22} = \left.\frac{\mathbf{V}_2}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0}$$

suggesting that if we let $\mathbf{I}_2 = 1$ A and open-circuit the input port,

$$\mathbf{z}_{12} = \frac{\mathbf{V}_1}{1} \qquad \text{and} \qquad \mathbf{z}_{22} = \frac{\mathbf{V}_2}{1}$$

This leads to the schematic in Fig. 19.53(b). The only difference between this schematic and the one in Fig. 19.53(a) is that the 1-A ac current source IAC is now at the output terminal. We run the schematic in Fig. 19.53(b) and obtain $\mathbf{V}_1$ and $\mathbf{V}_2$ from the output file. Thus,

$$\mathbf{z}_{12} = \frac{\mathbf{V}_1}{1} = 19.70\angle 175.7^\circ\ \Omega, \qquad \mathbf{z}_{22} = \frac{\mathbf{V}_2}{1} = 19.56\angle 175.7^\circ\ \Omega$$

Practice Problem 19.16 Obtain the z parameters of the circuit in Fig. 19.54 at $f = 60$ Hz.

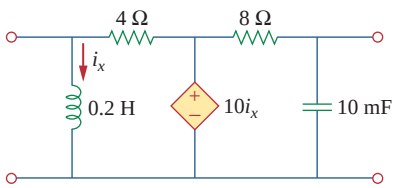


Figure 19.54
For Practice Prob. 19.16.

Answer: $z_{11} = 3.987\angle 175.5^\circ\ \Omega$, $z_{21} = 0.0175\angle -2.65^\circ\ \Omega$, $z_{12} = 0$, $z_{22} = 0.2651\angle 91.9^\circ\ \Omega$.

19.9 †Applications

We have seen how the six sets of network parameters can be used to characterize a wide range of two-port networks. Depending on the way two-ports are interconnected to form a larger network, a particular set of parameters may have advantages over others, as we noticed in Section 19.7. In this section, we will consider two important application areas of two-port parameters: transistor circuits and synthesis of ladder networks.

19.9.1 Transistor Circuits

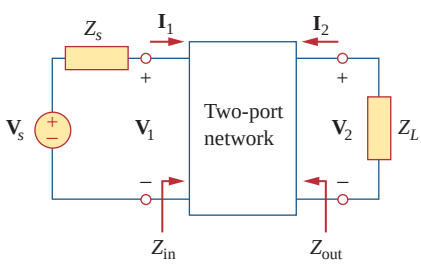


Figure 19.55
Two-port network isolating source and load.

The two-port network is often used to isolate a load from the excitation of a circuit. For example, the two-port in Fig. 19.55 may represent an amplifier, a filter, or some other network. When the two-port represents an amplifier, expressions for the voltage gain A_v, the current gain A_i, the input impedance Z_{in}, and the output impedance Z_{out} can be derived with ease. They are defined as follows:

$$A_v = \frac{V_2(s)}{V_1(s)} \tag{19.62}$$

$$A_i = \frac{I_2(s)}{I_1(s)} \tag{19.63}$$

$$Z_{\text{in}} = \frac{V_1(s)}{I_1(s)} \tag{19.64}$$

$$Z_{\text{out}} = \left.\frac{V_2(s)}{I_2(s)}\right|_{V_s=0} \tag{19.65}$$

Any of the six sets of two-port parameters can be used to derive the expressions in Eqs. (19.62) to (19.65). However, the hybrid (h) parameters are the most useful for transistors; they are easily measured and are often provided in the manufacturer's data or spec sheets for transistors. The h parameters provide a quick estimate of the performance of transistor circuits. They are used for finding the exact voltage gain, input impedance, and output impedance of a transistor.

The *h* parameters for transistors have specific meanings expressed by their subscripts. They are listed by the first subscript and related to the general *h* parameters as follows:

$$h_i = h_{11}, \qquad h_r = h_{12}, \qquad h_f = h_{21}, \qquad h_o = h_{22} \tag{19.66}$$

The subscripts *i*, *r*, *f*, and *o* stand for input, reverse, forward, and output. The second subscript specifies the type of connection used: *e* for common emitter (CE), *c* for common collector (CC), and *b* for common base (CB). Here we are mainly concerned with the common-emitter connection. Thus, the four *h* parameters for the common-emitter amplifier are:

$$\begin{aligned} h_{ie} &= \text{Base input impedance} \\ h_{re} &= \text{Reverse voltage feedback ratio} \\ h_{fe} &= \text{Base-collector current gain} \\ h_{oe} &= \text{Output admittance} \end{aligned} \tag{19.67}$$

These are calculated or measured in the same way as the general *h* parameters. Typical values are $h_{ie} = 6\ \text{k}\Omega$, $h_{re} = 1.5 \times 10^{-4}$, $h_{fe} = 200$, $h_{oe} = 8\ \mu\text{S}$. We must keep in mind that these values represent ac characteristics of the transistor, measured under specific circumstances.

Figure 19.56 shows the circuit schematic for the common-emitter amplifier and the equivalent hybrid model. From the figure, we see that

$$\mathbf{V}_b = h_{ie}\mathbf{I}_b + h_{re}\mathbf{V}_c \tag{19.68a}$$

$$\mathbf{I}_c = h_{fe}\mathbf{I}_b + h_{oe}\mathbf{V}_c \tag{19.68b}$$

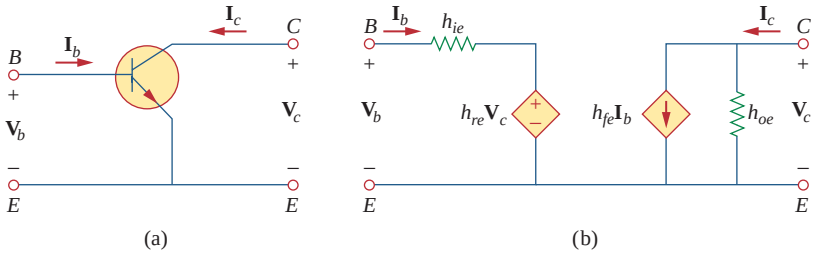


Figure 19.56
Common emitter amplifier: (a) circuit schematic, (b) hybrid model.

Consider the transistor amplifier connected to an ac source and a load as in Fig. 19.57. This is an example of a two-port network embedded within a larger network. We can analyze the hybrid equivalent circuit as usual with Eq. (19.68) in mind. (See Example 19.6.) Recognizing

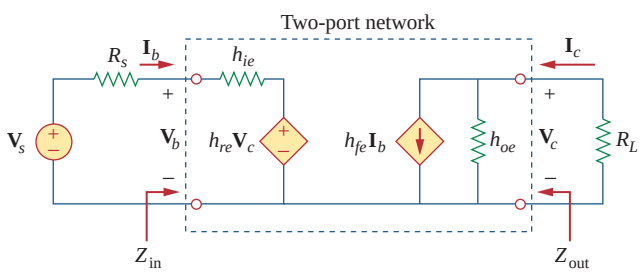


Figure 19.57
Transistor amplifier with source and load resistance.

from Fig. 19.57 that $\mathbf{V}_c = -R_L\mathbf{I}_c$ and substituting this into Eq. (19.68b) gives

$$\mathbf{I}_c = h_{fe}\mathbf{I}_b - h_{oe}R_L\mathbf{I}_c$$

or

$$(1 + h_{oe}R_L)\mathbf{I}_c = h_{fe}\mathbf{I}_b \tag{19.69}$$

From this, we obtain the current gain as

$$A_i = \frac{\mathbf{I}_c}{\mathbf{I}_b} = \frac{h_{fe}}{1 + h_{oe}R_L} \tag{19.70}$$

From Eqs. (19.68b) and (19.70), we can express $\mathbf{I}_b$ in terms of $\mathbf{V}_c$:

$$\mathbf{I}_c = \frac{h_{fe}}{1 + h_{oe}R_L}\mathbf{I}_b = h_{fe}\mathbf{I}_b + h_{oe}\mathbf{V}_c$$

or

$$\mathbf{I}_b = \frac{h_{oe}\mathbf{V}_c}{\dfrac{h_{fe}}{1 + h_{oe}R_L} - h_{fe}} \tag{19.71}$$

Substituting Eq. (19.71) into Eq. (19.68a) and dividing by $\mathbf{V}_c$ gives

$$\begin{aligned}\frac{\mathbf{V}_b}{\mathbf{V}_c} &= \frac{h_{oe}h_{ie}}{\dfrac{h_{fe}}{1 + h_{oe}R_L} - h_{fe}} + h_{re} \\ &= \frac{h_{ie} + h_{ie}h_{oe}R_L - h_{re}h_{fe}R_L}{-h_{fe}R_L}\end{aligned} \tag{19.72}$$

Thus, the voltage gain is

$$A_v = \frac{\mathbf{V}_c}{\mathbf{V}_b} = \frac{-h_{fe}R_L}{h_{ie} + (h_{ie}h_{oe} - h_{re}h_{fe})R_L} \tag{19.73}$$

Substituting $\mathbf{V}_c = -R_L\mathbf{I}_c$ into Eq. (19.68a) gives

$$\mathbf{V}_b = h_{ie}\mathbf{I}_b - h_{re}R_L\mathbf{I}_c$$

or

$$\frac{\mathbf{V}_b}{\mathbf{I}_b} = h_{ie} - h_{re}R_L\frac{\mathbf{I}_c}{\mathbf{I}_b} \tag{19.74}$$

Replacing $\mathbf{I}_c/\mathbf{I}_b$ by the current gain in Eq. (19.70) yields the input impedance as

$$Z_{\text{in}} = \frac{\mathbf{V}_b}{\mathbf{I}_b} = h_{ie} - \frac{h_{re}h_{fe}R_L}{1 + h_{oe}R_L} \tag{19.75}$$

The output impedance Z_{out} is the same as the Thevenin equivalent at the output terminals. As usual, by removing the voltage source and placing a

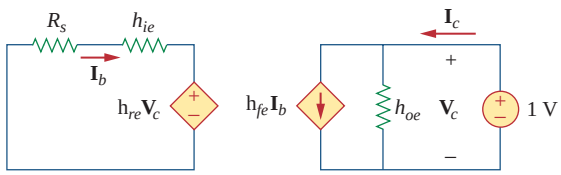


Figure 19.58
Finding the output impedance of the amplifier circuit in Fig. 19.57.

1-V source at the output terminals, we obtain the circuit in Fig. 19.58, from which Z_{out} is determined as $1/\mathbf{I}_c$. Since $\mathbf{V}_c = 1$ V, the input loop gives

$$h_{re}(1) = -\mathbf{I}_b(R_s + h_{ie}) \quad \Rightarrow \quad \mathbf{I}_b = -\frac{h_{re}}{R_s + h_{ie}} \tag{19.76}$$

For the output loop,

$$\mathbf{I}_c = h_{oe}(1) + h_{fe}\mathbf{I}_b \tag{19.77}$$

Substituting Eq. (19.76) into Eq. (19.77) gives

$$\mathbf{I}_c = \frac{(R_s + h_{ie})h_{oe} - h_{re}h_{fe}}{R_s + h_{ie}} \tag{19.78}$$

From this, we obtain the output impedance Z_{out} as $1/\mathbf{I}_c$; that is,

$$Z_{\text{out}} = \frac{R_s + h_{ie}}{(R_s + h_{ie})h_{oe} - h_{re}h_{fe}} \tag{19.79}$$

Example 19.17

Consider the common-emitter amplifier circuit of Fig. 19.59. Determine the voltage gain, current gain, input impedance, and output impedance using these h parameters:

$$h_{ie} = 1\ \text{k}\Omega, \qquad h_{re} = 2.5 \times 10^{-4}, \qquad h_{fe} = 50, \qquad h_{oe} = 20\ \mu\text{S}$$

Find the output voltage $\mathbf{V}_o$.

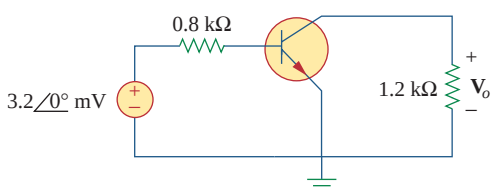


Figure 19.59
For Example 19.17.

Solution:

1. **Define.** In an initial look at this problem, it appears to be clearly stated. However, when we are asked to determine the input impedance and the voltage gain, do they refer to the transistor or the circuit? As far as the current gain and the output impedance are concerned, they are the same for both cases.

   We ask for clarification and are told that we should calculate the input impedance, the output impedance, and the voltage gain

for the circuit and not the transistor. It is interesting to note that the problem can be restated so that it becomes a simple design problem: Given the *h* parameters, design a simple amplifier that has a gain of −60.

2. **Present.** Given a simple transistor circuit, an input voltage of 3.2 mV, and the *h* parameters of the transistor, calculate the output voltage.
3. **Alternative.** There are a couple of ways we can approach the problem, the most straightforward being to use the equivalent circuit shown in Fig. 19.57. Once you have the equivalent circuit you can use circuit analysis to determine the answer. Once you have a solution, you can check it by plugging in the answer into the circuit equations to see if they are correct. Another approach is to simplify the right-hand side of the equivalent circuit and work backward to see if you obtain approximately the same answer. We will use that approach here.
4. **Attempt.** We note that $R_s = 0.8\,\text{k}\Omega$ and $R_L = 1.2\,\text{k}\Omega$. We treat the transistor of Fig. 19.59 as a two-port network and apply Eqs. (19.70) to (19.79).

$$\begin{aligned} h_{ie}h_{oe} - h_{re}h_{fe} &= 10^3 \times 20 \times 10^{-6} - 2.5 \times 10^{-4} \times 50 \\ &= 7.5 \times 10^{-3} \end{aligned}$$

$$A_v = \frac{-h_{fe}R_L}{h_{ie} + (h_{ie}h_{oe} - h_{re}h_{fe})\,R_L} = \frac{-50 \times 1200}{1000 + 7.5 \times 10^{-3} \times 1200}$$
$$= -59.46$$

A_v is the voltage gain of the amplifier $= V_o/V_b$. To calculate the gain of the circuit we need to find V_o/V_s. We can do this by using the mesh equation for the circuit on the left and Eqs. (19.71) and (19.73).

$$-\text{V}_s + R_s\text{I}_b + \text{V}_b = 0$$

or

$$\text{V}_s = 800\frac{20 \times 10^{-6}}{\dfrac{50}{1 + 20 \times 10^{-6} \times 1.2 \times 10^3} - 50} - \frac{1}{59.46}\text{V}_o$$
$$= -0.03047\ \text{V}_o.$$

Thus, the circuit gain is equal to $-\mathbf{32.82}$. Now we can calculate the output voltage.

$$V_o = \text{ gain} \times V_s = -\mathbf{105.09}\angle\mathbf{0}^\circ\ \mathbf{mV}.$$

$$A_i = \frac{h_{fe}}{1 + h_{oe}R_L} = \frac{50}{1 + 20 \times 10^{-6} \times 1200} = 48.83$$

$$Z_{\text{in}} = h_{ie} - \frac{h_{re}h_{fe}R_L}{1 + h_{oe}R_L}$$
$$= 1000 - \frac{2.5 \times 10^{-4} \times 50 \times 1200}{1 + 20 \times 10^{-6} \times 1200}$$
$$= 985.4\ \Omega$$

You can modify Z_{in} to include the 800-ohm resistor so that

$$\text{Circuit input impedance} = 800 + 985.4 = \mathbf{1785.4}\ \Omega.$$

$$(R_s + h_{ie})h_{oe} - h_{re}h_{fe}$$
$$= (800 + 1000) \times 20 \times 10^{-6} - 2.5 \times 10^{-4} \times 50 = 23.5 \times 10^{-3}$$
$$Z_{\text{out}} = \frac{R_s + h_{ie}}{(R_s + h_{ie})h_{oe} - h_{re}h_{fe}} = \frac{800 + 1000}{23.5 \times 10^{-3}} = 76.6\ \text{k}\Omega$$

5. **Evaluate.** In the equivalent circuit, h_{oe} represents a resistor of 50,000 Ω. This is in parallel with a load resistor equal to 1.2 kΩ. The size of the load resistor is so small relative to the h_{oe} resistor that h_{oe} can be neglected. This then leads to

$$\text{I}_c = h_{fe}\text{I}_b = 50I_b, \quad \text{V}_c = -1200\text{I}_c,$$

and the following loop equation from the left-hand side of the circuit:

$$-0.0032 + (800 + 1000)\text{I}_b + (0.00025)(-1200)(50)\text{I}_b = 0$$
$$\text{I}_b = 0.0032/(1785) = 1.7927\ \mu\text{A}.$$
$$\text{I}_c = 50 \times 1.7927 = 89.64\ \mu\text{A and V}_c = -1200 \times 89.64 \times 10^{-6}$$
$$= -107.57\ \text{mV}$$

This is a good approximation to −105.09 mV.

$$\text{Voltage gain} = -107.57/3.2 = -33.62$$

Again, this is a good approximation to 32.82.

$$\text{Circuit input impedance} = 0.032/1.7927 \times 10^{-6} = \mathbf{1785\ \Omega}$$

which clearly compares well with the 1785.4 Ω we obtained before.

For these calculations, we assumed that $Z_{\text{out}} = \infty\ \Omega$. Our calculations produced 72.6 kΩ. We can test our assumption by calculating the equivalent resistance of this and the load resistance.

$$72{,}600 \times 1200/(72{,}600 + 1200) = 1{,}180.5 = 1.1805\ \text{k}\Omega$$

Again, we have a good approximation.

6. **Satisfactory?** We have satisfactorily solved the problem and checked the results. We can now present our results as a solution to the problem.

Practice Problem 19.17

For the transistor amplifier of Fig. 19.60, find the voltage gain, current gain, input impedance, and output impedance. Assume that

$$h_{ie} = 6\ \text{k}\Omega, \quad h_{re} = 1.5 \times 10^{-4}, \quad h_{fe} = 200, \quad h_{oe} = 8\ \mu\text{S}$$

Answer: −123.61 for the transistor and −4.753 for the circuit, 194.17, 6 kΩ for the transistor and 156 kΩ for the circuit, 128.08 kΩ.

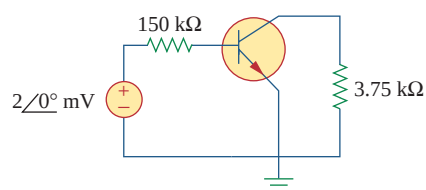


Figure 19.60
For Practice Prob. 19.17.

19.9.2 Ladder Network Synthesis

Another application of two-port parameters is the synthesis (or building) of ladder networks, which are found frequently in practice and

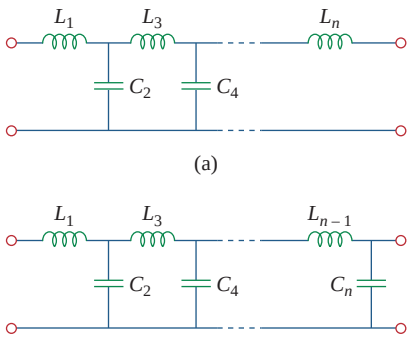


Figure 19.61
LC ladder networks for lowpass filters of: (a) odd order, (b) even order.

have particular use in designing passive lowpass filters. Based on our discussion of second-order circuits in Chapter 8, the order of the filter is the order of the characteristic equation describing the filter and is determined by the number of reactive elements that cannot be combined into single elements (e.g., through series or parallel combination). Figure 19.61(a) shows an *LC* ladder network with an odd number of elements (to realize an odd-order filter), while Fig. 19.61(b) shows one with an even number of elements (for realizing an even-order filter). When either network is terminated by the load impedance Z_L and the source impedance Z_s, we obtain the structure in Fig. 19.62. To make the design less complicated, we will assume that $Z_s = 0$. Our goal is to synthesize the transfer function of the *LC* ladder network. We begin by characterizing the ladder network by its admittance parameters, namely,

$$\mathbf{I}_1 = \mathbf{y}_{11}\mathbf{V}_1 + \mathbf{y}_{12}\mathbf{V}_2 \tag{19.80a}$$

$$\mathbf{I}_2 = \mathbf{y}_{21}\mathbf{V}_1 + \mathbf{y}_{22}\mathbf{V}_2 \tag{19.80b}$$

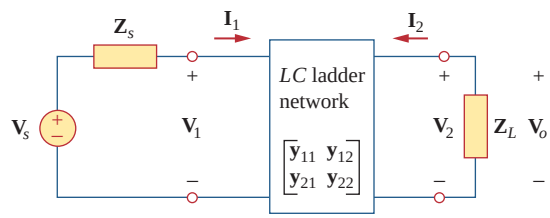


Figure 19.62
LC ladder network with terminating impedances.

(Of course, the impedance parameters could be used instead of the admittance parameters.) At the input port, $\mathbf{V}_1 = \mathbf{V}_s$ since $\mathbf{Z}_s = 0$. At the output port, $\mathbf{V}_2 = \mathbf{V}_o$ and $\mathbf{I}_2 = -\mathbf{V}_2/\mathbf{Z}_L = -\mathbf{V}_o\mathbf{Y}_L$. Thus, Eq. (19.80b) becomes

$$-\mathbf{V}_o\mathbf{Y}_L = \mathbf{y}_{21}\mathbf{V}_s + \mathbf{y}_{22}\mathbf{V}_o$$

or

$$\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{-\mathbf{y}_{21}}{\mathbf{Y}_L + \mathbf{y}_{22}} \tag{19.81}$$

We can write this as

$$\mathbf{H}(s) = -\frac{\mathbf{y}_{21}/\mathbf{Y}_L}{1 + \mathbf{y}_{22}/\mathbf{Y}_L} \tag{19.82}$$

We may ignore the negative sign in Eq. (19.82) because filter requirements are often stated in terms of the magnitude of the transfer function. The main objective in filter design is to select capacitors and inductors so that the parameters $\mathbf{y}_{21}$ and $\mathbf{y}_{22}$ are synthesized, thereby realizing the desired transfer function. To achieve this, we take advantage of an important property of the *LC* ladder network: all *z* and *y* parameters are ratios of polynomials that contain only even powers of *s* or odd powers

of s—that is, they are ratios of either Od(s)/Ev(s) or Ev(s)/Od(s), where Od and Ev are odd and even functions, respectively. Let

$$\mathbf{H}(s) = \frac{\mathbf{N}(s)}{\mathbf{D}(s)} = \frac{\mathbf{N}_o + \mathbf{N}_e}{\mathbf{D}_o + \mathbf{D}_e} \tag{19.83}$$

where $\mathbf{N}(s)$ and $\mathbf{D}(s)$ are the numerator and denominator of the transfer function $\mathbf{H}(s)$; $\mathbf{N}_o$ and $\mathbf{N}_e$ are the odd and even parts of $\mathbf{N}$; $\mathbf{D}_o$ and $\mathbf{D}_e$ are the odd and even parts of $\mathbf{D}$. Since $\mathbf{N}(s)$ must be either odd or even, we can write Eq. (19.83) as

$$\mathbf{H}(s) = \begin{cases} \dfrac{\mathbf{N}_o}{\mathbf{D}_o + \mathbf{D}_e}, & (\mathbf{N}_e = 0) \\ \dfrac{\mathbf{N}_e}{\mathbf{D}_o + \mathbf{D}_e}, & (\mathbf{N}_o = 0) \end{cases} \tag{19.84}$$

and can rewrite this as

$$\mathbf{H}(s) = \begin{cases} \dfrac{\mathbf{N}_o/\mathbf{D}_e}{1 + \mathbf{D}_o/\mathbf{D}_e}, & (\mathbf{N}_e = 0) \\ \dfrac{\mathbf{N}_e/\mathbf{D}_o}{1 + \mathbf{D}_e/\mathbf{D}_o}, & (\mathbf{N}_o = 0) \end{cases} \tag{19.85}$$

Comparing this with Eq. (19.82), we obtain the y parameters of the network as

$$\frac{\mathbf{y}_{21}}{\mathbf{Y}_L} = \begin{cases} \dfrac{\mathbf{N}_o}{\mathbf{D}_e}, & (\mathbf{N}_e = 0) \\ \dfrac{\mathbf{N}_e}{\mathbf{D}_o}, & (\mathbf{N}_o = 0) \end{cases} \tag{19.86}$$

and

$$\frac{\mathbf{y}_{22}}{\mathbf{Y}_L} = \begin{cases} \dfrac{\mathbf{D}_o}{\mathbf{D}_e}, & (\mathbf{N}_e = 0) \\ \dfrac{\mathbf{D}_e}{\mathbf{D}_o}, & (\mathbf{N}_o = 0) \end{cases} \tag{19.87}$$

The following example illustrates the procedure.

Example 19.18

Design the *LC* ladder network terminated with a 1-Ω resistor that has the normalized transfer function

$$\mathbf{H}(s) = \frac{1}{s^3 + 2s^2 + 2s + 1}$$

(This transfer function is for a Butterworth lowpass filter.)

Solution:

The denominator shows that this is a third-order network, so that the *LC* ladder network is shown in Fig. 19.63(a), with two inductors and one capacitor. Our goal is to determine the values of the inductors and

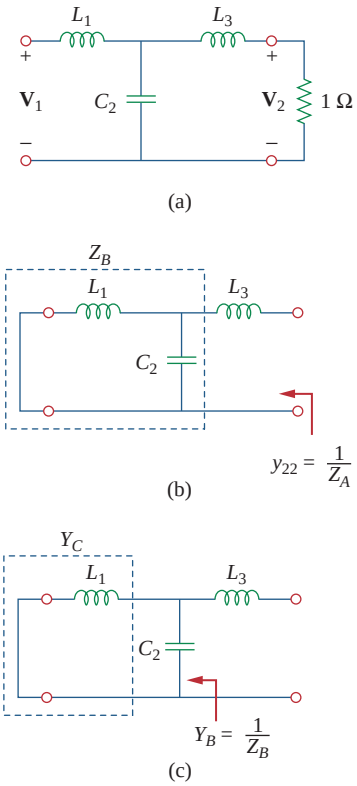


Figure 19.63
For Example 19.18.

capacitor. To achieve this, we group the terms in the denominator into odd or even parts:

$$\mathbf{D}(s) = (s^3 + 2s) + (2s^2 + 1)$$

so that

$$\mathbf{H}(s) = \frac{1}{(s^3 + 2s) + (2s^2 + 1)}$$

Divide the numerator and denominator by the odd part of the denominator to get

$$\mathbf{H}(s) = \frac{\dfrac{1}{s^3 + 2s}}{1 + \dfrac{2s^2 + 1}{s^3 + 2s}} \tag{19.18.1}$$

From Eq. (19.82), when $\mathbf{Y}_L = 1$,

$$\mathbf{H}(s) = \frac{-y_{21}}{1 + y_{22}} \tag{19.18.2}$$

Comparing Eqs. (19.19.1) and (19.19.2), we obtain

$$y_{21} = -\frac{1}{s^3 + 2s}, \qquad y_{22} = \frac{2s^2 + 1}{s^3 + 2s}$$

Any realization of y_{22} will automatically realize y_{21}, since y_{22} is the output driving-point admittance, that is, the output admittance of the network with the input port short-circuited. We determine the values of L and C in Fig. 19.63(a) that will give us y_{22}. Recall that y_{22} is the short-circuit output admittance. So we short-circuit the input port as shown in Fig. 19.63(b). First we get L_3 by letting

$$Z_A = \frac{1}{y_{22}} = \frac{s^3 + 2s}{2s^2 + 1} = sL_3 + Z_B \tag{19.18.3}$$

By long division,

$$Z_A = 0.5s + \frac{1.5s}{2s^2 + 1} \tag{19.18.4}$$

Comparing Eqs. (19.18.3) and (19.18.4) shows that

$$L_3 = 0.5\text{H}, \qquad Z_B = \frac{1.5s}{2s^2 + 1}$$

Next, we seek to get C_2 as in Fig. 19.63(c) and let

$$Y_B = \frac{1}{Z_B} = \frac{2s^2 + 1}{1.5s} = 1.333s + \frac{1}{1.5s} = sC_2 + Y_C$$

from which $C_2 = 1.33$ F and

$$Y_C = \frac{1}{1.5s} = \frac{1}{sL_1} \quad \Rightarrow \quad L_1 = 1.5\text{ H}$$

Thus, the LC ladder network in Fig. 19.63(a) with $L_1 = 1.5$ H, $C_2 = 1.333$ F, and $L_3 = 0.5$ H has been synthesized to provide the given transfer function $\mathbf{H}(s)$. This result can be confirmed by finding $\mathbf{H}(s) = \mathbf{V}_2/\mathbf{V}_1$ in Fig. 19.63(a) or by confirming the required y_{21}.

Practice Problem 19.18

Realize the following transfer function using an *LC* ladder network terminated in a 1-Ω resistor:

$$H(s) = \frac{2}{s^3 + s^2 + 4s + 2}$$

Answer: Ladder network in Fig. 19.63(a) with $L_1 = L_3 = 1.0$ H and $C_2 = 0.5$ F.

19.10 Summary

1. A two-port network is one with two ports (or two pairs of access terminals), known as input and output ports.
2. The six parameters used to model a two-port network are the impedance [**z**], admittance [**y**], hybrid [**h**], inverse hybrid [**g**], transmission [**T**], and inverse transmission [**t**] parameters.
3. The parameters relate the input and output port variables as

$$\begin{bmatrix}\mathbf{V}_1\\ \mathbf{V}_2\end{bmatrix} = [\mathbf{z}]\begin{bmatrix}\mathbf{I}_1\\ \mathbf{I}_2\end{bmatrix}, \quad \begin{bmatrix}\mathbf{I}_1\\ \mathbf{I}_2\end{bmatrix} = [\mathbf{y}]\begin{bmatrix}\mathbf{V}_1\\ \mathbf{V}_2\end{bmatrix}, \quad \begin{bmatrix}\mathbf{V}_1\\ \mathbf{I}_2\end{bmatrix} = [\mathbf{h}]\begin{bmatrix}\mathbf{I}_1\\ \mathbf{V}_2\end{bmatrix}$$

$$\begin{bmatrix}\mathbf{I}_1\\ \mathbf{V}_2\end{bmatrix} = [\mathbf{g}]\begin{bmatrix}\mathbf{V}_1\\ \mathbf{I}_2\end{bmatrix}, \quad \begin{bmatrix}\mathbf{V}_1\\ \mathbf{I}_1\end{bmatrix} = [\mathbf{T}]\begin{bmatrix}\mathbf{V}_2\\ -\mathbf{I}_2\end{bmatrix}, \quad \begin{bmatrix}\mathbf{V}_2\\ \mathbf{I}_2\end{bmatrix} = [\mathbf{t}]\begin{bmatrix}\mathbf{V}_1\\ -\mathbf{I}_1\end{bmatrix}$$

4. The parameters can be calculated or measured by short-circuiting or open-circuiting the appropriate input or output port.
5. A two-port network is reciprocal if $\mathbf{z}_{12} = \mathbf{z}_{21}$, $\mathbf{y}_{12} = \mathbf{y}_{21}$, $\mathbf{h}_{12} = -\mathbf{h}_{21}$, $\mathbf{g}_{12} = -\mathbf{g}_{21}$, $\Delta_T = 1$ or $\Delta_t = 1$. Networks that have dependent sources are not reciprocal.
6. Table 19.1 provides the relationships between the six sets of parameters. Three important relationships are

$$[\mathbf{y}] = [\mathbf{z}]^{-1}, \quad [\mathbf{g}] = [\mathbf{h}]^{-1}, \quad [\mathbf{t}] \neq [\mathbf{T}]^{-1}$$

7. Two-port networks may be connected in series, in parallel, or in cascade. In the series connection the *z* parameters are added, in the parallel connection the *y* parameters are added, and in the cascade connection the transmission parameters are multiplied in the correct order.
8. One can use *PSpice* to compute the two-port parameters by constraining the appropriate port variables with a 1-A or 1-V source while using an open or short circuit to impose the other necessary constraints.
9. The network parameters are specifically applied in the analysis of transistor circuits and the synthesis of ladder *LC* networks. Network parameters are especially useful in the analysis of transistor circuits because these circuits are easily modeled as two-port networks. *LC* ladder networks, important in the design of passive low-pass filters, resemble cascaded T networks and are therefore best analyzed as two-ports.

Review Questions

19.1 For the single-element two-port network in Fig. 19.64(a), $\mathbf{z}_{11}$ is:

(a) 0 (b) 5 (c) 10
(d) 20 (e) undefined

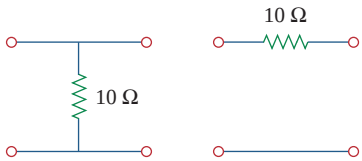


Figure 19.64
For Review Questions.

19.2 For the single-element two-port network in Fig. 19.64(b), $\mathbf{z}_{11}$ is:

(a) 0 (b) 5 (c) 10
(d) 20 (e) undefined

19.3 For the single-element two-port network in Fig. 19.64(a), $\mathbf{y}_{11}$ is:

(a) 0 (b) 5 (c) 10
(d) 20 (e) undefined

19.4 For the single-element two-port network in Fig. 19.64(b), $\mathbf{h}_{21}$ is:

(a) −0.1 (b) −1 (c) 0
(d) 10 (e) undefined

19.5 For the single-element two-port network in Fig. 19.64(a), **B** is:

(a) 0 (b) 5 (c) 10
(d) 20 (e) undefined

19.6 For the single-element two-port network in Fig. 19.64(b), **B** is:

(a) 0 (b) 5 (c) 10
(d) 20 (e) undefined

19.7 When port 1 of a two-port circuit is short-circuited, $\mathbf{I}_1 = 4\mathbf{I}_2$ and $\mathbf{V}_2 = 0.25\mathbf{I}_2$. Which of the following is true?

(a) $y_{11} = 4$ (b) $y_{12} = 16$
(c) $y_{21} = 16$ (d) $y_{22} = 0.25$

19.8 A two-port is described by the following equations:

$$\mathbf{V}_1 = 50\mathbf{I}_1 + 10\mathbf{I}_2$$
$$\mathbf{V}_2 = 30\mathbf{I}_1 + 20\mathbf{I}_2$$

Which of the following is *not* true?

(a) $\mathbf{z}_{12} = 10$ (b) $\mathbf{y}_{12} = -0.0143$
(c) $\mathbf{h}_{12} = 0.5$ (d) $\mathbf{A} = 50$

19.9 If a two-port is reciprocal, which of the following is *not* true?

(a) $\mathbf{z}_{21} = \mathbf{z}_{12}$ (b) $\mathbf{y}_{21} = \mathbf{y}_{12}$
(c) $\mathbf{h}_{21} = \mathbf{h}_{12}$ (d) $AD = BC + 1$

19.10 If the two single-element two-port networks in Fig. 19.64 are cascaded, then **D** is:

(a) 0 (b) 0.1 (c) 2
(d) 10 (e) undefined

Answers: 19.1c, 19.2e, 19.3e, 19.4b, 19.5a, 19.6c, 19.7b, 19.8d, 19.9c, 19.10c.

Problems

Section 19.2 Impedance Parameters

19.1 Obtain the *z* parameters for the network in Fig. 19.65.

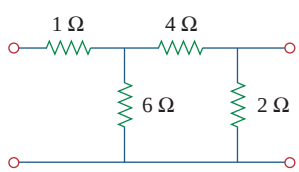


Figure 19.65
For Probs. 19.1 and 19.28

***19.2** Find the impedance parameter equivalent of the network in Fig. 19.66.

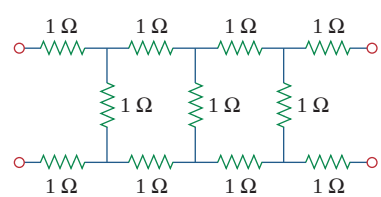


Figure 19.66
For Prob. 19.2.

* An asterisk indicates a challenging problem.

19.3 Find the *z* parameters of the circuit in Fig. 19.67.

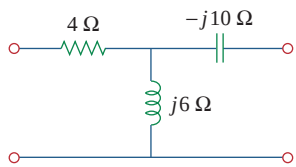


Figure 19.67
For Prob. 19.3.

19.4 Using Fig. 19.68, design a problem to help other students better understand how to determine *z* parameters from an electrical circuit.

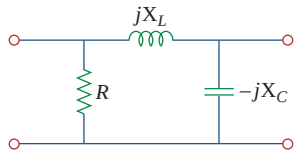


Figure 19.68
For Prob. 19.4.

19.5 Obtain the *z* parameters for the network in Fig. 19.69 as functions of *s*.

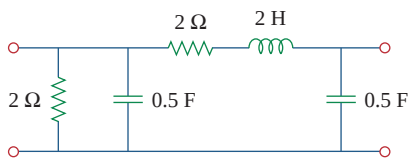


Figure 19.69
For Prob. 19.5.

19.6 Compute the *z* parameters of the circuit in Fig. 19.70.

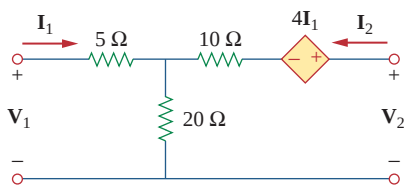


Figure 19.70
For Prob. 19.6 and 19.73.

19.7 Calculate the impedance-parameter equivalent of the circuit in Fig. 19.71.

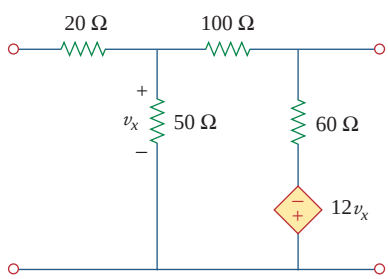


Figure 19.71
For Prob. 19.7 and 19.80.

19.8 Find the *z* parameters of the two-port in Fig. 19.72.

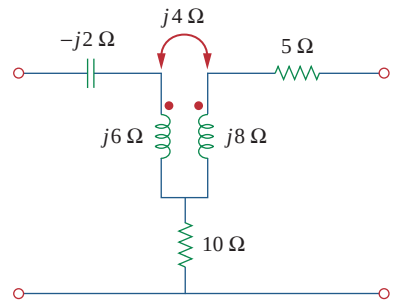


Figure 19.72
For Prob. 19.8.

19.9 The *y* parameters of a network are:

$$[\mathbf{y}] = \begin{bmatrix} 0.25 & -0.1 \\ -0.1 & 0.2 \end{bmatrix} S$$

Determine the *z* parameters for the network.

19.10 Construct a two-port that realizes each of the following *z* parameters.

(a) $[\mathbf{z}] = \begin{bmatrix} 25 & 20 \\ 5 & 10 \end{bmatrix} \Omega$

(b) $[\mathbf{z}] = \begin{bmatrix} 1 + \dfrac{3}{s} & \dfrac{1}{s} \\ \dfrac{1}{s} & 2s + \dfrac{1}{s} \end{bmatrix} \Omega$

19.11 Determine a two-port network that is represented by the following *z* parameters:

$$[\mathbf{z}] = \begin{bmatrix} 6 + j3 & 5 - j2 \\ 5 - j2 & 8 - j \end{bmatrix} \Omega$$

19.12 For the circuit shown in Fig. 19.73, let

$$[\mathbf{z}] = \begin{bmatrix} 10 & -6 \\ -4 & 12 \end{bmatrix} \Omega$$

Find I_1, I_2, V_1, and V_2.

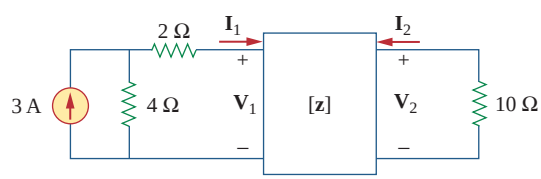


Figure 19.73
For Prob. 19.12.

19.13 Determine the average power delivered to $Z_L = 5 + j4$ in the network of Fig. 19.74. *Note:* The voltage is rms.

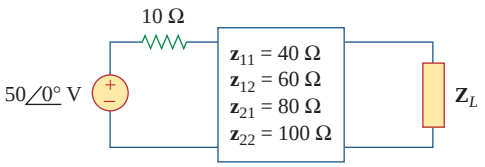


Figure 19.74
For Prob. 19.13.

19.14 For the two-port network shown in Fig. 19.75, show that at the output terminals,

$$\mathbf{Z}_{\text{Th}} = \mathbf{z}_{22} - \frac{\mathbf{z}_{12}\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}$$

and

$$\mathbf{V}_{\text{Th}} = \frac{\mathbf{z}_{21}}{\mathbf{z}_{11} + \mathbf{Z}_s}\mathbf{V}_s$$

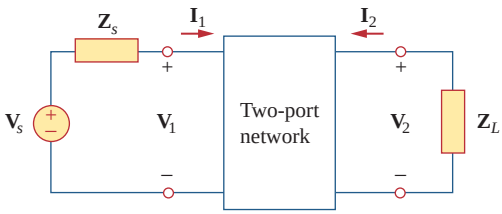


Figure 19.75
For Probs. 19.14 and 19.41.

19.15 For the two-port circuit in Fig. 19.76,

$$[\mathbf{z}] = \begin{bmatrix} 40 & 60 \\ 80 & 120 \end{bmatrix} \Omega$$

(a) Find $\mathbf{Z}_L$ for maximum power transfer to the load.

(b) Calculate the maximum power delivered to the load.

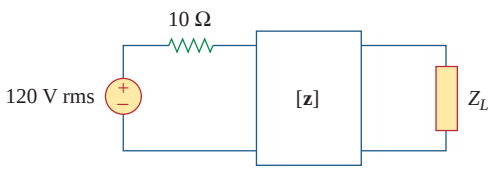


Figure 19.76
For Prob. 19.15.

19.16 For the circuit in Fig. 19.77, at $\omega = 2$ rad/s, $\mathbf{z}_{11} = 10\ \Omega$, $\mathbf{z}_{12} = \mathbf{z}_{21} = j6\ \Omega$, $\mathbf{z}_{22} = 4\ \Omega$. Obtain the Thevenin equivalent circuit at terminals *a-b* and calculate v_o.

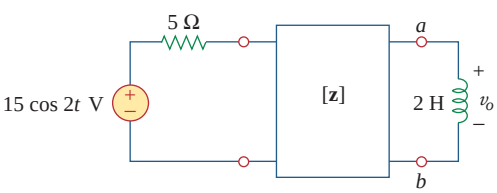


Figure 19.77
For Prob. 19.16.

Section 19.3 Admittance Parameters

***19.17** Determine the *z* and *y* parameters for the circuit in Fig. 19.78.

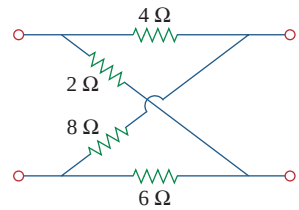


Figure 19.78
For Prob. 19.17.

19.18 Calculate the *y* parameters for the two-port in Fig. 19.79.

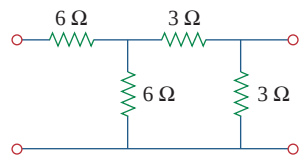


Figure 19.79
For Probs. 19.18 and 19.37.

19.19 Using Fig. 19.80, design a problem to help other students better understand how to find *y* parameters in the *s*-domain.

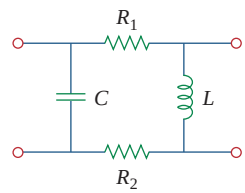


Figure 19.80
For Prob. 19.19.

19.20 Find the *y* parameters for the circuit in Fig. 19.81.

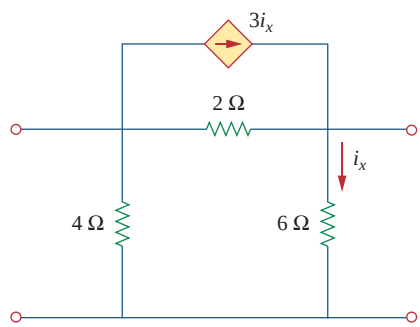


Figure 19.81
For Prob. 19.20.

19.21 Obtain the admittance parameter equivalent circuit of the two-port in Fig. 19.82.

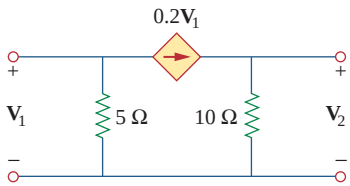


Figure 19.82
For Prob. 19.21.

19.22 Obtain the *y* parameters of the two-port network in Fig. 19.83.

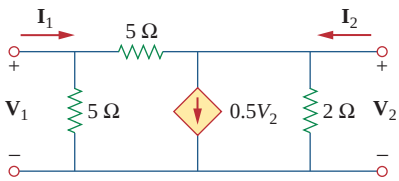


Figure 19.83
For Prob. 19.22.

19.23 (a) Find the *y* parameters of the two-port in Fig. 19.84.

(b) Determine $\mathbf{V}_2(s)$ for $v_s = 2u(t)$ V.

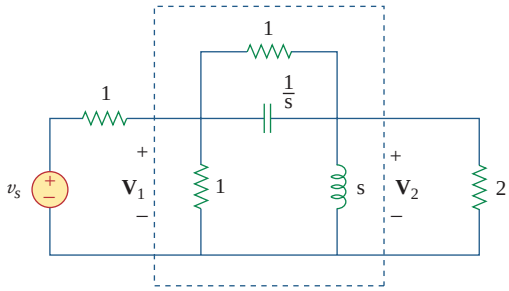


Figure 19.84
For Prob. 19.23.

19.24 Find the resistive circuit that represents these *y* parameters:

$$[\mathbf{y}] = \begin{bmatrix} \frac{1}{4} & -\frac{1}{8} \\ -\frac{1}{8} & \frac{3}{16} \end{bmatrix} \text{S}$$

19.25 Draw the two-port network that has the following *y* parameters:

$$[\mathbf{y}] = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1.5 \end{bmatrix} \text{S}$$

19.26 Calculate [**y**] for the two-port in Fig. 19.85.

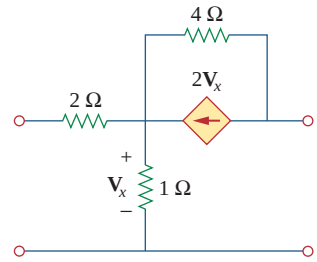


Figure 19.85
For Prob. 19.26.

19.27 Find the *y* parameters for the circuit in Fig. 19.86.

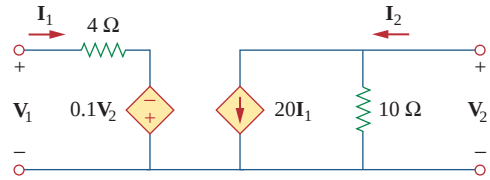


Figure 19.86
For Prob. 19.27.

19.28 In the circuit of Fig. 19.65, the input port is connected to a 1-A dc current source. Calculate the power dissipated by the 2-Ω resistor by using the *y* parameters. Confirm your result by direct circuit analysis.

19.29 In the bridge circuit of Fig. 19.87, $I_1 = 10$ A and $I_2 = -4$ A.

(a) Find V_1 and V_2 using *y* parameters.

(b) Confirm the results in part (a) by direct circuit analysis.

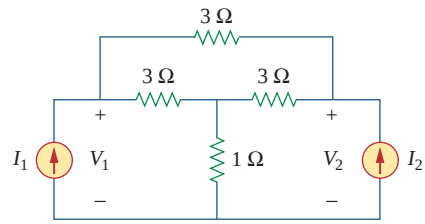


Figure 19.87
For Prob. 19.29.

Section 19.4 Hybrid Parameters

19.30 Find the *h* parameters for the networks in Fig. 19.88.

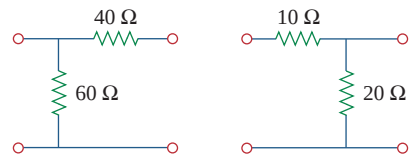


Figure 19.88
For Prob. 19.30.

19.31 Determine the hybrid parameters for the network in Fig. 19.89.

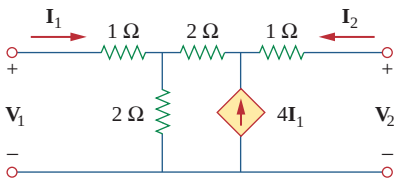


Figure 19.89
For Prob. 19.31.

19.32 Using Fig. 19.90, design a problem to help other students better understand how to find the *h* and *g* parameters for a circuit in the *s*-domain.

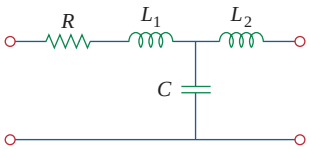


Figure 19.90
For Prob. 19.32.

19.33 Obtain the *h* parameters for the two-port of Fig. 19.91.

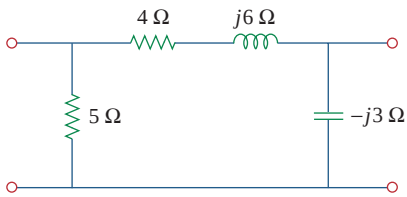


Figure 19.91
For Prob. 19.33.

19.34 Obtain the *h* and *g* parameters of the two-port in Fig. 19.92.

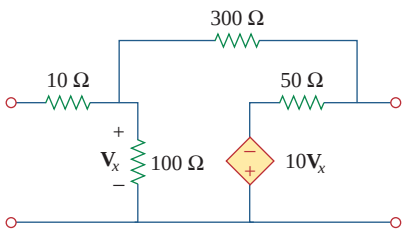


Figure 19.92
For Prob. 19.34.

19.35 Determine the *h* parameters for the network in Fig. 19.93.

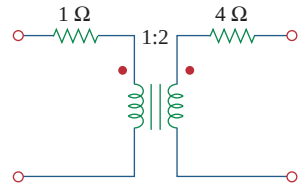


Figure 19.93
For Prob. 19.35.

19.36 For the two-port in Fig. 19.94,

$$[\mathbf{h}] = \begin{bmatrix} 16\ \Omega & 3 \\ -2 & 0.01\ \text{S} \end{bmatrix}$$

Find:

(a) V_2/V_1 (b) I_2/I_1

(c) I_1/V_1 (d) V_2/I_1

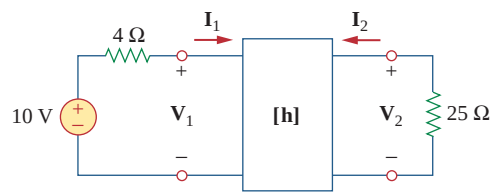


Figure 19.94
For Prob. 19.36.

19.37 The input port of the circuit in Fig. 19.79 is connected to a 10-V dc voltage source while the output port is terminated by a 5-Ω resistor. Find the voltage across the 5-Ω resistor by using *h* parameters of the circuit. Confirm your result by using direct circuit analysis.

19.38 The *h* parameters of the two-port of Fig. 19.95 are:

$$[\mathbf{h}] = \begin{bmatrix} 600\ \Omega & 0.04 \\ 30 & 2\ \text{mS} \end{bmatrix}$$

Given the $Z_s = 2\ \text{k}\Omega$ and $Z_L = 400\ \Omega$, find Z_{in} and Z_{out}.

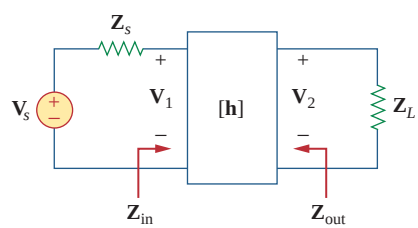


Figure 19.95
For Prob. 19.38.

19.39 Obtain the *g* parameters for the wye circuit of Fig. 19.96.

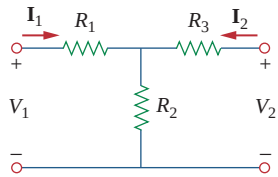


Figure 19.96
For Prob. 19.39.

19.40 Using Fig. 19.97, design a problem to help other students better understand how to find *g* parameters in an ac circuit.

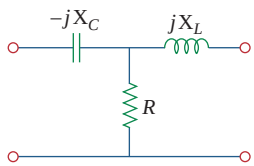


Figure 19.97
For Prob. 19.40.

19.41 For the two-port in Fig. 19.75, show that

$$\frac{\mathbf{I}_2}{\mathbf{I}_1} = \frac{-\mathbf{g}_{21}}{\mathbf{g}_{11}\mathbf{Z}_L + \Delta_g}$$

$$\frac{\mathbf{V}_2}{\mathbf{V}_s} = \frac{\mathbf{g}_{21}\mathbf{Z}_L}{(1 + \mathbf{g}_{11}\mathbf{Z}_s)(\mathbf{g}_{22} + \mathbf{Z}_L) - \mathbf{g}_{21}\mathbf{g}_{12}\mathbf{Z}_s}$$

where Δ_g is the determinant of [**g**] matrix.

19.42 The *h* parameters of a two-port device are given by

$$\mathbf{h}_{11} = 600\ \Omega, \quad \mathbf{h}_{12} = 10^{-3}, \quad \mathbf{h}_{21} = 120, \quad \mathbf{h}_{22} = 2 \times 10^{-6}\ \text{S}$$

Draw a circuit model of the device including the value of each element.

Section 19.5 Transmission Parameters

19.43 Find the transmission parameters for the single-element two-port networks in Fig. 19.98.

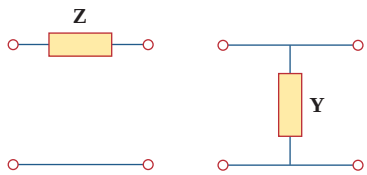


Figure 19.98
For Prob. 19.43.

19.44 Using Fig. 19.99, design a problem to help other students better understand how to find the transmission parameters of an ac circuit.

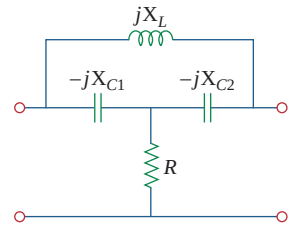


Figure 19.99
For Prob. 19.44.

19.45 Find the **ABCD** parameters for the circuit in Fig. 19.100.

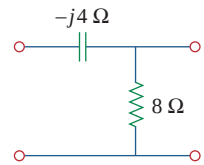


Figure 19.100
For Prob. 19.45.

19.46 Find the transmission parameters for the circuit in Fig. 19.101.

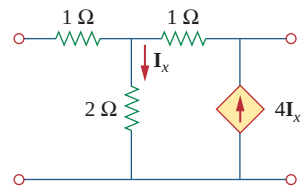


Figure 19.101
For Prob. 19.46.

19.47 Obtain the **ABCD** parameters for the network in Fig. 19.102.

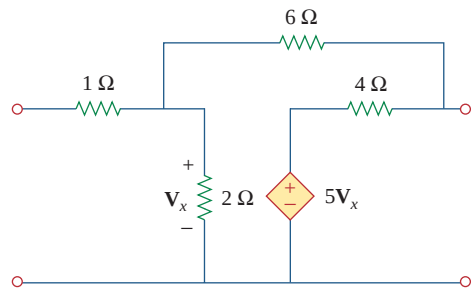


Figure 19.102
For Prob. 19.47.

19.48 For a two-port, let $\mathbf{A} = 4$, $\mathbf{B} = 30\ \Omega$, $\mathbf{C} = 0.1$ S, and $\mathbf{D} = 1.5$. Calculate the input impedance $\mathbf{Z}_{\text{in}} = \mathbf{V}_1/\mathbf{I}_1$, when:

(a) the output terminals are short-circuited,

(b) the output port is open-circuited,

(c) the output port is terminated by a 10-Ω load.

19.49 Using impedances in the *s*-domain, obtain the transmission parameters for the circuit in Fig. 19.103.

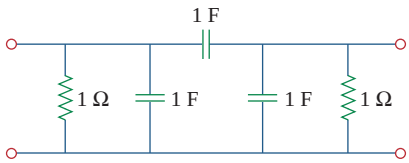


Figure 19.103
For Prob. 19.49.

19.50 Derive the *s*-domain expression for the *t* parameters of the circuit in Fig. 19.104.

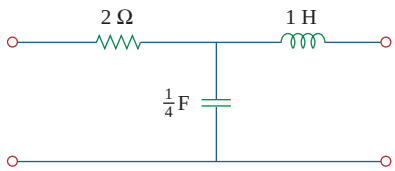


Figure 19.104
For Prob. 19.50.

19.51 Obtain the *t* parameters for the network in Fig. 19.105.

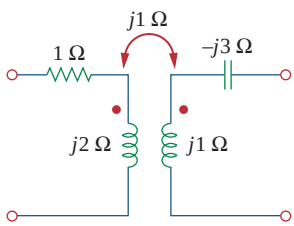


Figure 19.105
For Prob. 19.51.

Section 19.6 Relationships Between Parameters

19.52 (a) For the *T* network in Fig. 19.106, show that the *h* parameters are:

$$\mathbf{h}_{11} = R_1 + \frac{R_2R_3}{R_1 + R_3}, \qquad \mathbf{h}_{12} = \frac{R_2}{R_2 + R_3}$$

$$\mathbf{h}_{21} = -\frac{R_2}{R_2 + R_3}, \qquad \mathbf{h}_{22} = \frac{1}{R_2 + R_3}$$

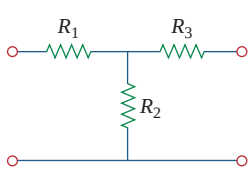


Figure 19.106
For Prob. 19.52.

(b) For the same network, show that the transmission parameters are:

$$\mathbf{A} = 1 + \frac{R_1}{R_2}, \qquad \mathbf{B} = R_3 + \frac{R_1}{R_2}(R_2 + R_3)$$

$$\mathbf{C} = \frac{1}{R_2}, \qquad \mathbf{D} = 1 + \frac{R_3}{R_2}$$

19.53 Through derivation, express the *z* parameters in terms of the **ABCD** parameters.

19.54 Show that the transmission parameters of a two-port may be obtained from the *y* parameters as:

$$\mathbf{A} = -\frac{\mathbf{y}_{22}}{\mathbf{y}_{21}}, \qquad \mathbf{B} = -\frac{1}{\mathbf{y}_{21}}$$

$$\mathbf{C} = -\frac{\Delta_y}{\mathbf{y}_{21}}, \qquad \mathbf{D} = -\frac{\mathbf{y}_{11}}{\mathbf{y}_{21}}$$

19.55 Prove that the *g* parameters can be obtained from the *z* parameters as

$$\mathbf{g}_{11} = \frac{1}{\mathbf{z}_{11}}, \qquad \mathbf{g}_{12} = -\frac{\mathbf{z}_{12}}{\mathbf{z}_{11}}$$

$$\mathbf{g}_{21} = \frac{\mathbf{z}_{21}}{\mathbf{z}_{11}}, \qquad \mathbf{g}_{22} = \frac{\Delta_z}{\mathbf{z}_{11}}$$

19.56 For the network of Fig. 19.107, obtain $\mathbf{V_o}/\mathbf{V_s}$.

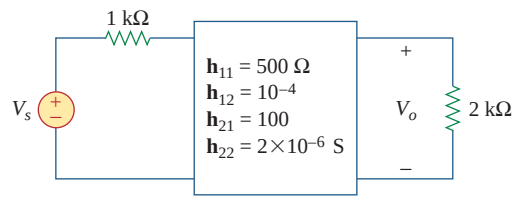


Figure 19.107
For Prob. 19.56.

19.57 Given the transmission parameters

$$[\mathbf{T}] = \begin{bmatrix} 3 & 20 \\ 1 & 7 \end{bmatrix}$$

obtain the other five two-port parameters.

19.58 Design a problem to help other students better understand how to develop the *y* parameters and transmission parameters, given equations in terms of the hybrid parameters.

19.59 Given that

$$[\mathbf{g}] = \begin{bmatrix} 0.06\ \text{S} & -0.4 \\ 0.2 & 2\ \Omega \end{bmatrix}$$

determine:

(a) [**z**] (b) [**y**] (c) [**h**] (d) [**T**]

19.60 Design a T network necessary to realize the following z parameters at $\omega = 10^6$ rad/s.

$$[\mathbf{z}] = \begin{bmatrix} 4 + j3 & 2 \\ 2 & 5 - j \end{bmatrix} \text{k}\Omega$$

19.61 For the bridge circuit in Fig. 19.108, obtain:

(a) the z parameters

(b) the h parameters

(c) the transmission parameters

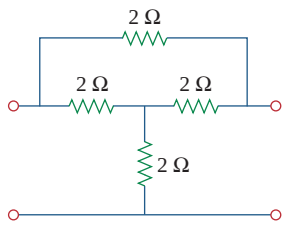


Figure 19.108
For Prob. 19.61.

19.62 Find the z parameters of the op amp circuit in Fig. 19.109. Obtain the transmission parameters.

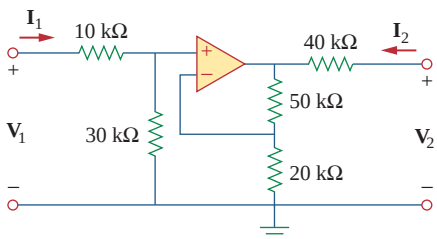


Figure 19.109
For Prob. 19.62.

19.63 Determine the z parameters of the two-port in Fig. 19.110.

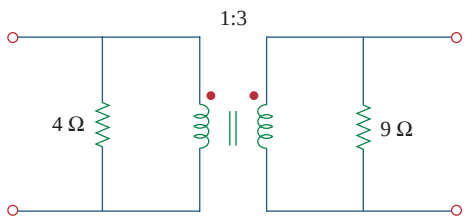


Figure 19.110
For Prob. 19.63.

19.64 Determine the y parameters at $\omega = 1{,}000$ rad/s for the op amp circuit in Fig. 19.111. Find the corresponding h parameters.

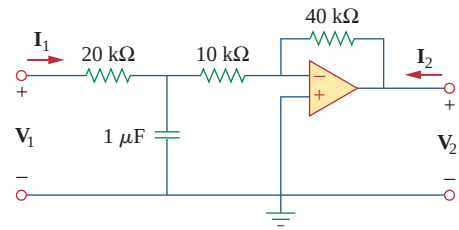


Figure 19.111
For Prob. 19.64.

Section 19.7 Interconnection of Networks

19.65 What is the y parameter presentation of the circuit in Fig. 19.112?

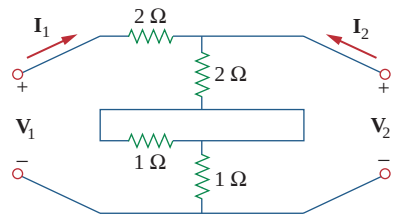


Figure 19.112
For Prob. 19.65.

19.66 In the two-port of Fig. 19.113, let $\mathbf{y}_{12} = \mathbf{y}_{21} = 0$, $\mathbf{y}_{11} = 2$ mS, and $\mathbf{y}_{22} = 10$ mS. Find $\mathbf{V}_o/\mathbf{V}_s$.

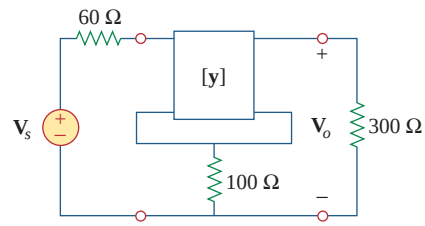


Figure 19.113
For Prob. 19.66.

19.67 If three copies of the circuit in Fig. 19.114 are connected in parallel, find the overall transmission parameters.

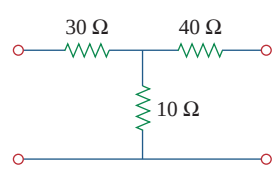


Figure 19.114
For Prob. 19.67.

19.68 Obtain the *h* parameters for the network in Fig. 19.115.

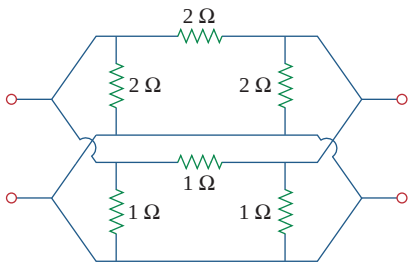


Figure 19.115
For Prob. 19.68.

***19.69** The circuit in Fig. 19.116 may be regarded as two two-ports connected in parallel. Obtain the *y* parameters as functions of *s*.

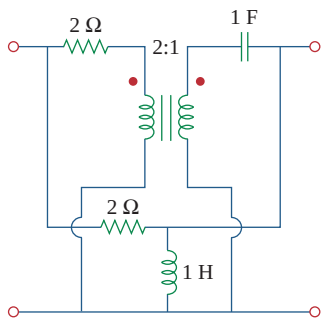


Figure 19.116
For Prob. 19.69.

***19.70** For the parallel-series connection of the two two-ports in Fig. 19.117, find the *g* parameters.

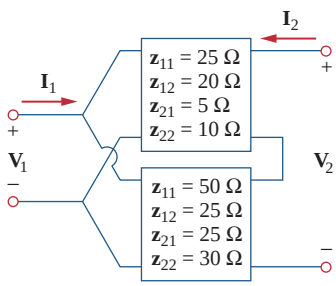


Figure 19.117
For Prob. 19.70.

***19.71** Determine the *z* parameters for the network in Fig. 19.118.

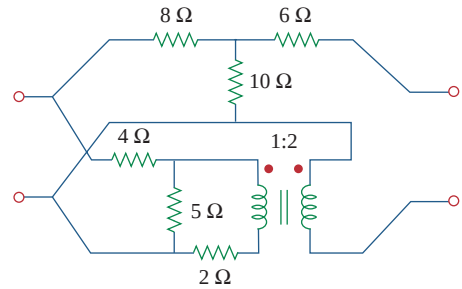


Figure 19.118
For Prob. 19.71.

***19.72** A series-parallel connection of two two-ports is shown in Fig. 19.119. Determine the *z* parameter representation of the network.

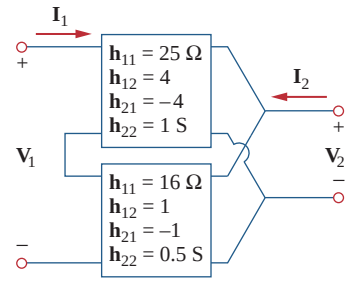


Figure 19.119
For Prob. 19.72.

19.73 Three copies of the circuit shown in Fig. 19.70 are connected in cascade. Determine the *z* parameters.

***19.74** Determine the **ABCD** parameters of the circuit in Fig. 19.120 as functions of *s*. (*Hint:* Partition the circuit into subcircuits and cascade them using the results of Prob. 19.43.)

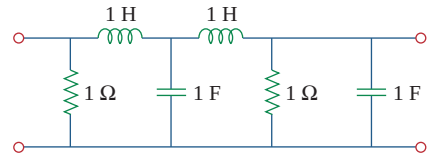


Figure 19.120
For Prob. 19.74.

***19.75** For the individual two-ports shown in Fig. 19.121 where,

$$[\mathbf{z}_a] = \begin{bmatrix} 8 & 6 \\ 4 & 5 \end{bmatrix} \Omega \quad [\mathbf{y}_b] = \begin{bmatrix} 8 & -4 \\ 2 & 10 \end{bmatrix} \text{S}$$

(a) Determine the *y* parameters of the overall two-port.

(b) Find the voltage ratio $\mathbf{V}_o/\mathbf{V}_i$ when $\mathbf{Z}_L = 2\ \Omega$.

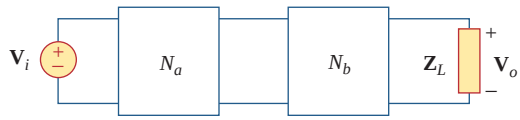


Figure 19.121
For Prob. 19.75.

Section 19.8 Computing Two-Port Parameters Using *PSpice*

19.76 Use *PSpice* to obtain the *z* parameters of the network in Fig. 19.122.

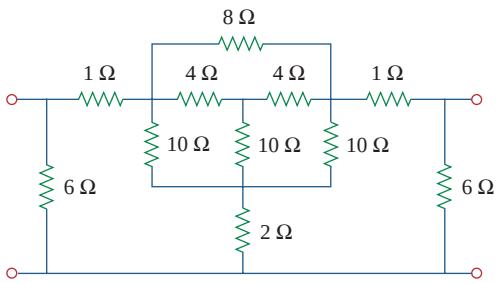


Figure 19.122
For Prob. 19.76.

19.77 Using *PSpice*, find the *h* parameters of the network in Fig. 19.123. Take $\omega = 1$ rad/s.

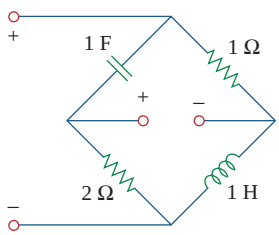


Figure 19.123
For Prob. 19.77.

19.78 Obtain the *h* parameters at $\omega = 4$ rad/s for the circuit in Fig. 19.124 using *PSpice*.

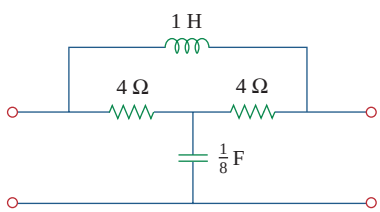


Figure 19.124
For Prob. 19.78.

19.79 Use *PSpice* to determine the *z* parameters of the circuit in Fig. 19.125. Take $\omega = 2$ rad/s.

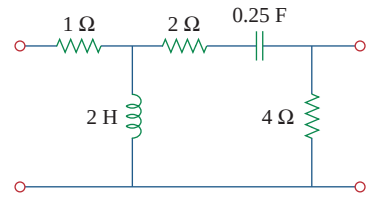


Figure 19.125
For Prob. 19.79.

19.80 Use *PSpice* to find the *z* parameters of the circuit in Fig. 19.71.

19.81 Repeat Prob. 19.26 using *PSpice*.

19.82 Use *PSpice* to rework Prob. 19.31.

19.83 Rework Prob. 19.47 using *PSpice*.

19.84 Using *PSpice*, find the transmission parameters for the network in Fig. 19.126.

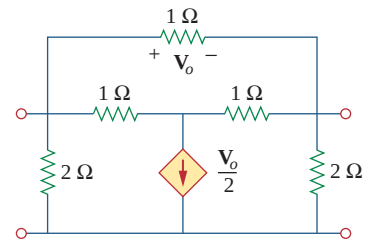


Figure 19.126
For Prob. 19.84.

19.85 At $\omega = 1$ rad/s, find the transmission parameters of the network in Fig. 19.127 using *PSpice*.

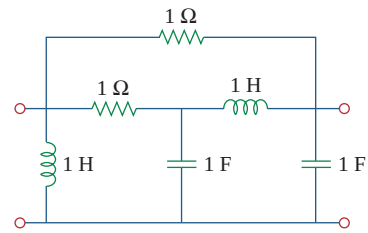


Figure 19.127
For Prob. 19.85.

19.86 Obtain the *g* parameters for the network in Fig. 19.128 using *PSpice*.

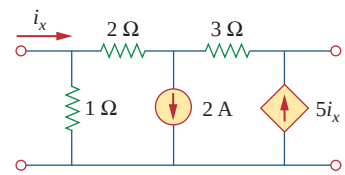


Figure 19.128
For Prob. 19.86.

19.87 For the circuit shown in Fig. 19.129, use *PSpice* to obtain the *t* parameters. Assume $\omega = 1$ rad/s.

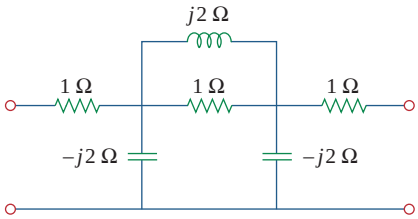


Figure 19.129
For Prob. 19.87.

Section 19.9 Applications

19.88 Using the *y* parameters, derive formulas for Z_{in}, Z_{out}, A_i, and A_v for the common-emitter transistor circuit.

19.89 A transistor has the following parameters in a common-emitter circuit:

$$h_{ie} = 2{,}640\ \Omega, \quad h_{re} = 2.6 \times 10^{-4}$$
$$h_{fe} = 72, \quad h_{oe} = 16\ \mu\text{S}, \quad R_L = 100\ \text{k}\Omega$$

What is the voltage amplification of the transistor? How many decibels gain is this?

19.90 A transistor with

$$h_{fe} = 120, \quad h_{ie} = 2\ \text{k}\Omega$$
$$h_{re} = 10^{-4}, \quad h_{oe} = 20\ \mu\text{S}$$

is used for a CE amplifier to provide an input resistance of 1.5 kΩ.

(a) Determine the necessary load resistance R_L.

(b) Calculate A_v, A_i, and Z_{out} if the amplifier is driven by a 4-mV source having an internal resistance of 600 Ω.

(c) Find the voltage across the load.

19.91 For the transistor network of Fig. 19.130,

$$h_{fe} = 80, \quad h_{ie} = 1.2\ \text{k}\Omega$$
$$h_{re} = 1.5 \times 10^{-4}, \quad h_{oe} = 20\ \mu\text{S}$$

Determine the following:

(a) voltage gain $A_v = V_o/V_s$,

(b) current gain $A_i = I_o/I_i$,

(c) input impedance Z_{in},

(d) output impedance Z_{out}.

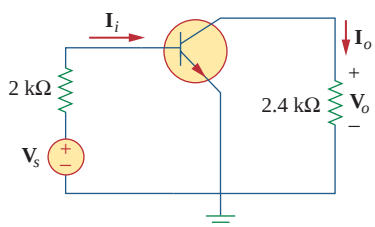


Figure 19.130
For Prob. 19.91.

***19.92** Determine A_v, A_i, Z_{in}, and Z_{out} for the amplifier shown in Fig. 19.131. Assume that

$$h_{ie} = 4\ \text{k}\Omega, \quad h_{re} = 10^{-4}$$
$$h_{fe} = 100, \quad h_{oe} = 30\ \mu\text{S}$$

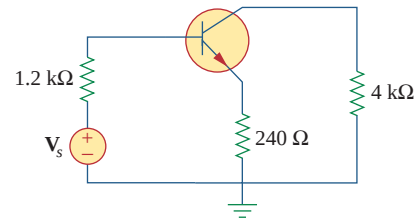


Figure 19.131
For Prob. 19.92.

***19.93** Calculate A_v, A_i, Z_{in}, and Z_{out} for the transistor network in Fig. 19.132. Assume that

$$h_{ie} = 2\ \text{k}\Omega, \quad h_{re} = 2.5 \times 10^{-4}$$
$$h_{fe} = 150, \quad h_{oe} = 10\ \mu\text{S}$$

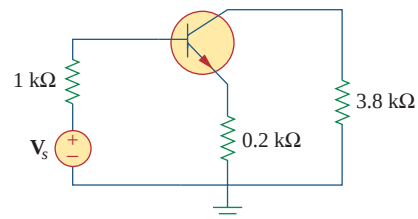


Figure 19.132
For Prob. 19.93.

19.94 A transistor in its common-emitter mode is specified by

$$[\mathbf{h}] = \begin{bmatrix} 200\ \Omega & 0 \\ 100 & 10^{-6}\ \text{S} \end{bmatrix}$$

Two such identical transistors are connected in cascade to form a two-stage amplifier used at audio frequencies. If the amplifier is terminated by a 4-kΩ resistor, calculate the overall A_v and Z_{in}.

19.95 Realize an *LC* ladder network such that

$$y_{22} = \frac{s^3 + 5s}{s^4 + 10s^2 + 8}$$

19.96 Design an *LC* ladder network to realize a lowpass filter with transfer function

$$H(s) = \frac{1}{s^4 + 2.613s^2 + 3.414s^2 + 2.613s + 1}$$

19.97 Synthesize the transfer function

$$H(s) = \frac{V_o}{V_s} = \frac{s^3}{s^3 + 6s + 12s + 24}$$

using the *LC* ladder network in Fig. 19.133.

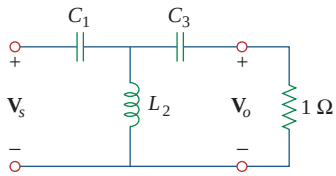


Figure 19.133
For Prob. 19.97.

19.98 A two-stage amplifier in Fig. 19.134 contains two identical stages with

$$[\mathbf{h}] = \begin{bmatrix} 2\ \text{k}\Omega & 0.004 \\ 200 & 500\ \mu\text{S} \end{bmatrix}$$

If $\mathbf{Z}_L = 20\ \text{k}\Omega$, find the required value of $\mathbf{V}_s$ to produce $\mathbf{V}_o = 16$ V.

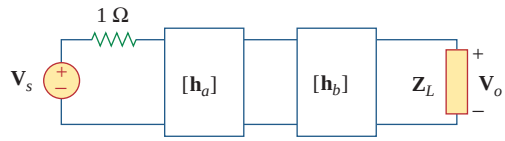
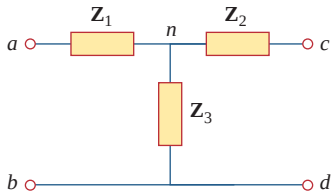


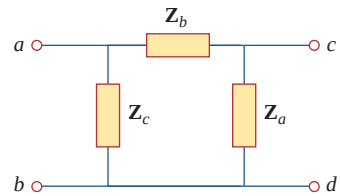
Figure 19.134
For Prob. 19.98.

Comprehensive Problem

19.99 Assume that the two circuits in Fig. 19.135 are equivalent. The parameters of the two circuits must be equal. Using this factor and the z parameters, derive Eqs. (9.67) and (9.68).



(a)



(b)

Figure 19.135
For Prob. 19.99.

Appendix A

Simultaneous Equations and Matrix Inversion

In circuit analysis, we often encounter a set of simultaneous equations having the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \quad\quad & \vdots \quad\quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned} \tag{A.1}$$

where there are n unknown $x_1, x_2, \ldots, x_n$ to be determined. Equation (A.1) can be written in matrix form as

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ldots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_2 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \tag{A.2}$$

This matrix equation can be put in a compact form as

$$\mathbf{AX} = \mathbf{B} \tag{A.3}$$

where

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ldots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \tag{A.4}$$

A is a square ($n \times n$) matrix while **X** and **B** are column matrices.

There are several methods for solving Eq. (A.1) or (A.3). These include substitution, Gaussian elimination, Cramer's rule, matrix inversion, and numerical analysis.

A.1 Cramer's Rule

In many cases, Cramer's rule can be used to solve the simultaneous equations we encounter in circuit analysis. Cramer's rule states that the solution to Eq. (A.1) or (A.3) is

$$\begin{aligned} x_1 &= \frac{\Delta_1}{\Delta} \\ x_2 &= \frac{\Delta_2}{\Delta} \\ &\vdots \\ x_n &= \frac{\Delta_n}{\Delta} \end{aligned} \tag{A.5}$$

where the Δ's are the determinants given by

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}, \quad \Delta_1 = \begin{vmatrix} b_1 & a_{12} & \cdots & a_{1n} \\ b_2 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ b_n & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$$\vdots \qquad\qquad \vdots$$

$$\Delta_2 = \begin{vmatrix} a_{11} & b_1 & \cdots & a_{1n} \\ a_{21} & b_2 & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & b_n & \cdots & a_{nn} \end{vmatrix}, \ldots, \Delta_n = \begin{vmatrix} a_{11} & a_{12} & \cdots & b_1 \\ a_{21} & a_{22} & \cdots & b_2 \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & b_n \end{vmatrix} \tag{A.6}$$

Notice that Δ is the determinant of matrix $\mathbf{A}$ and Δ_k is the determinant of the matrix formed by replacing the kth column of $\mathbf{A}$ by $\mathbf{B}$. It is evident from Eq. (A.5) that Cramer's rule applies only when $\Delta \neq 0$. When $\Delta = 0$, the set of equations has no unique solution, because the equations are linearly dependent.

The value of the determinant Δ, for example, can be obtained by expanding along the first row:

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix} \tag{A.7}$$

$$= a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} + \cdots + (-1)^{1+n}a_{1n}M_{1n}$$

where the minor M_{ij} is an $(n-1) \times (n-1)$ determinant of the matrix formed by striking out the ith row and jth column. The value of Δ may also be obtained by expanding along the first column:

$$\Delta = a_{11}M_{11} - a_{21}M_{21} + a_{31}M_{31} + \cdots + (-1)^{n+1}a_{n1}M_{n1} \tag{A.8}$$

We now specifically develop the formulas for calculating the determinants of 2×2 and 3×3 matrices, because of their frequent occurrence in this text. For a 2×2 matrix,

$$\Delta = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21} \tag{A.9}$$

For a 3×3 matrix,

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}(-1)^2\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{21}(-1)^3\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

$$+ a_{31}(-1)^4\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

$$= a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{21}(a_{12}a_{33} - a_{32}a_{13})$$

$$+ a_{31}(a_{12}a_{23} - a_{22}a_{13}) \tag{A.10}$$

An alternative method of obtaining the determinant of a 3×3 matrix is by repeating the first two rows and multiplying the terms diagonally as follows.

$$\Delta = \begin{array}{ccccc} & a_{11} & a_{12} & a_{13} & \\ & a_{21} & a_{22} & a_{23} & \\ & a_{31} & a_{32} & a_{33} & \\ - & a_{11} & a_{12} & a_{13} & + \\ - & a_{21} & a_{22} & a_{23} & + \\ - & & & & + \end{array}$$

$$= a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - a_{13}a_{22}a_{31} - a_{23}a_{32}a_{11} - a_{33}a_{12}a_{21} \quad \textbf{(A.11)}$$

In summary:

The solution of linear simultaneous equations by Cramer's rule boils down to finding

$$x_k = \frac{\Delta_k}{\Delta}, \qquad k = 1, 2, \ldots, n \qquad \textbf{(A.12)}$$

where Δ is the determinant of matrix **A** and Δ_k is the determinant of the matrix formed by replacing the kth column of **A** by **B**.

One may use other methods, such as matrix inversion and elimination. Only Cramer's method is covered here, because of its simplicity and also because of the availability of powerful calculators.

You may not find much need to use Cramer's method described in this appendix, in view of the availability of calculators, computers, and software packages such as *MATLAB*, which can be used easily to solve a set of linear equations. But in case you need to solve the equations by hand, the material covered in this appendix becomes useful. At any rate, it is important to know the mathematical basis of those calculators and software packages.

Example A.1

Solve the simultaneous equations

$$4x_1 - 3x_2 = 17, \qquad -3x_1 + 5x_2 = -21$$

Solution:

The given set of equations is cast in matrix form as

$$\begin{bmatrix} 4 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 17 \\ -21 \end{bmatrix}$$

The determinants are evaluated as

$$\Delta = \begin{vmatrix} 4 & -3 \\ -3 & 5 \end{vmatrix} = 4 \times 5 - (-3)(-3) = 11$$

$$\Delta_1 = \begin{vmatrix} 17 & -3 \\ -21 & 5 \end{vmatrix} = 17 \times 5 - (-3)(-21) = 22$$

$$\Delta_2 = \begin{vmatrix} 4 & 17 \\ -3 & -21 \end{vmatrix} = 4 \times (-21) - 17 \times (-3) = -33$$

Hence,

$$x_1 = \frac{\Delta_1}{\Delta} = \frac{22}{11} = 2, \qquad x_2 = \frac{\Delta_2}{\Delta} = \frac{-33}{11} = -3$$

Practice Problem A.1

Find the solution to the following simultaneous equations:

$$3x_1 - x_2 = 4, \qquad -6x_1 + 18x_2 = 16$$

Answer: $x_1 = 1.833, x_2 = 1.5.$

Example A.2

Determine x_1, x_2, and x_3 for this set of simultaneous equations:

$$25x_1 - 5x_2 - 20x_3 = 50$$
$$-5x_1 + 10x_2 - 4x_3 = 0$$
$$-5x_1 - 4x_2 + 9x_3 = 0$$

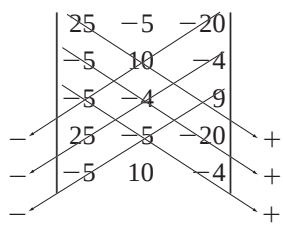
Solution:

In matrix form, the given set of equations becomes

$$\begin{bmatrix} 25 & -5 & -20 \\ -5 & 10 & -4 \\ -5 & -4 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \\ 0 \end{bmatrix}$$

We apply Eq. (A.11) to find the determinants. This requires that we repeat the first two rows of the matrix. Thus,

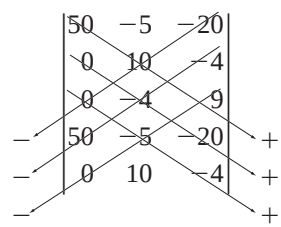
$$\Delta = \begin{vmatrix} 25 & -5 & -20 \\ -5 & 10 & -4 \\ -5 & -4 & 9 \end{vmatrix} =$$



$$= 25(10)9 + (-5)(-4)(-20) + (-5)(-5)(-4) - (-20)(10)(-5) - (-4)(-4)25 - 9(-5)(-5)$$
$$= 2250 - 400 - 100 - 1000 - 400 - 225 = 125$$

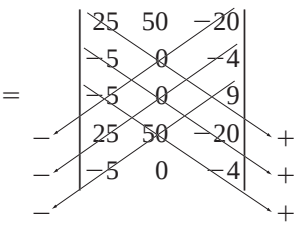
Similarly,

$$\Delta_1 = \begin{vmatrix} 50 & -5 & -20 \\ 0 & 10 & -4 \\ 0 & -4 & 9 \end{vmatrix} =$$



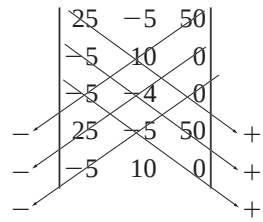
$$= 4500 + 0 + 0 - 0 - 800 - 0 = 3700$$

$$\Delta_2 = \begin{vmatrix} 25 & 50 & -20 \\ -5 & 0 & -4 \\ -5 & 0 & 9 \end{vmatrix} =$$



$$= 0 + 0 + 1000 - 0 - 0 + 2250 = 3250$$

$$\Delta_3 = \begin{vmatrix} 25 & -5 & 50 \\ -5 & 10 & 0 \\ -5 & -4 & 0 \end{vmatrix} =$$



$$= 0 + 1000 + 0 + 2500 - 0 - 0 = 3500$$

Hence, we now find

$$x_1 = \frac{\Delta_1}{\Delta} = \frac{3700}{125} = 29.6$$

$$x_2 = \frac{\Delta_2}{\Delta} = \frac{3250}{125} = 26$$

$$x_3 = \frac{\Delta_2}{\Delta} = \frac{3500}{125} = 28$$

Practice Problem A.2 Obtain the solution of this set of simultaneous equations:

$$3x_1 - x_2 - 2x_3 = 1$$
$$-x_1 + 6x_2 - 3x_3 = 0$$
$$-2x_1 - 3x_2 + 6x_3 = 6$$

Answer: $x_1 = 3 = x_3, x_2 = 2.$

A.2 Matrix Inversion

The linear system of equations in Eq. (A.3) can be solved by matrix inversion. In the matrix equation $\mathbf{AX} = \mathbf{B}$, we may invert $\mathbf{A}$ to get $\mathbf{X}$, i.e.,

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} \tag{A.13}$$

where $\mathbf{A}^{-1}$ is the inverse of $\mathbf{A}$. Matrix inversion is needed in other applications apart from using it to solve a set of equations.

By definition, the inverse of matrix $\mathbf{A}$ satisfies

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I} \tag{A.14}$$

where $\mathbf{I}$ is an identity matrix. $\mathbf{A}^{-1}$ is given by

$$\mathbf{A}^{-1} = \frac{\text{adj } \mathbf{A}}{\det \mathbf{A}} \tag{A.15}$$

where adj $\mathbf{A}$ is the adjoint of $\mathbf{A}$ and $\det \mathbf{A} = |\mathbf{A}|$ is the determinant of $\mathbf{A}$. The adjoint of $\mathbf{A}$ is the transpose of the cofactors of $\mathbf{A}$. Suppose we are given an $n \times n$ matrix $\mathbf{A}$ as

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \tag{A.16}$$

The cofactors of $\mathbf{A}$ are defined as

$$\mathbf{C} = \text{cof}\,(\mathbf{A}) = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & & & \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix} \tag{A.17}$$

where the cofactor c_{ij} is the product of $(-1)^{i+j}$ and the determinant of the $(n-1) \times (n-1)$ submatrix is obtained by deleting the ith row and jth column from $\mathbf{A}$. For example, by deleting the first row and the first column of $\mathbf{A}$ in Eq. (A.16), we obtain the cofactor c_{11} as

$$c_{11} = (-1)^2 = \begin{vmatrix} a_{22} & a_{23} & \cdots & a_{2n} \\ a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & & & \\ a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix} \tag{A.18}$$

Once the cofactors are found, the adjoint of $\mathbf{A}$ is obtained as

$$\text{adj}\,(\mathbf{A}) = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & & & \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix}^T = \mathbf{C}^T \tag{A.19}$$

where T denotes transpose.

In addition to using the cofactors to find the adjoint of $\mathbf{A}$, they are also used in finding the determinant of $\mathbf{A}$ which is given by

$$|\mathbf{A}| = \sum_{j=1}^{n} a_{ij} c_{ij} \tag{A.20}$$

where i is any value from 1 to n. By substituting Eqs. (A.19) and (A.20) into Eq. (A.15), we obtain the inverse of $\mathbf{A}$ as

$$\boxed{\mathbf{A}^{-1} = \frac{\mathbf{C}^T}{|\mathbf{A}|}} \tag{A.21}$$

For a 2×2 matrix, if

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{A.22}$$

its inverse is

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \tag{A.23}$$

For a 3×3 matrix, if

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \tag{A.24}$$

we first obtain the cofactors as

$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \tag{A.25}$$

where

$$\begin{aligned} c_{11} &= \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, & c_{12} &= -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, & c_{13} &= \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}, \\ c_{21} &= -\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}, & c_{22} &= \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}, & c_{23} &= -\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}, \\ c_{31} &= \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}, & c_{32} &= -\begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix}, & c_{33} &= \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{aligned} \tag{A.26}$$

The determinant of the 3×3 matrix can be found using Eq. (A.11). Here, we want to use Eq. (A.20), i.e.,

$$|\mathbf{A}| = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13} \tag{A.27}$$

The idea can be extended $n > 3$, but we deal mainly with 2×2 and 3×3 matrices in this book.

Example A.3

Use matrix inversion to solve the simultaneous equations

$$2x_1 + 10x_2 = 2, \qquad -x_1 + 3x_2 = 7$$

Solution:

We first express the two equations in matrix form as

$$\begin{bmatrix} 2 & 10 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$$

or

$$\mathbf{AX} = \mathbf{B} \longrightarrow \mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$$

where

$$\mathbf{A} = \begin{bmatrix} 2 & 10 \\ -1 & 3 \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$$

The determinant of $\mathbf{A}$ is $|\mathbf{A}| = 2 \times 3 - 10(-1) = 16$, so the inverse of $\mathbf{A}$ is

$$\mathbf{A}^{-1} = \frac{1}{16}\begin{bmatrix} 3 & -10 \\ 1 & 2 \end{bmatrix}$$

Hence,

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} = \frac{1}{16}\begin{bmatrix} 3 & -10 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 7 \end{bmatrix} = \frac{1}{16}\begin{bmatrix} -64 \\ 16 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$$

i.e., $x_1 = -4$ and $x_2 = 1$.

Practice Problem A.3

Solve the following two equations by matrix inversion.

$$2y_1 - y_2 = 4, \quad y_1 + 3y_2 = 9$$

Answer: $y_1 = 3, y_2 = 2$.

Example A.4

Determine x_1, x_2, and x_3 for the following simultaneous equations using matrix inversion.

$$\begin{aligned} x_1 + x_2 + x_3 &= 5 \\ -x_1 + 2x_2 &= 9 \\ 4x_1 + x_2 - x_3 &= -2 \end{aligned}$$

Solution:

In matrix form, the equations become

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 2 & 0 \\ 4 & 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \\ -2 \end{bmatrix}$$

or

$$\mathbf{AX} = \mathbf{B} \longrightarrow \mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$$

where

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 2 & 0 \\ 4 & 1 & -1 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 5 \\ 9 \\ -2 \end{bmatrix}$$

We now find the cofactors

$$c_{11} = \begin{vmatrix} 2 & 0 \\ 1 & -1 \end{vmatrix} = -2, \quad c_{12} = -\begin{vmatrix} -1 & 0 \\ 4 & -1 \end{vmatrix} = -1, \quad c_{13} = \begin{vmatrix} -1 & 2 \\ 4 & 1 \end{vmatrix} = -9$$

$$c_{21} = -\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = 2, \quad c_{22} = \begin{vmatrix} 1 & 1 \\ 4 & -1 \end{vmatrix} = -5, \quad c_{23} = -\begin{vmatrix} 1 & 1 \\ 4 & 1 \end{vmatrix} = 3$$

$$c_{31} = \begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = -2, \quad c_{32} = -\begin{vmatrix} 1 & 1 \\ -1 & 0 \end{vmatrix} = -1, \quad c_{33} = \begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix} = 3$$

The adjoint of matrix **A** is

$$\text{adj } \mathbf{A} = \begin{bmatrix} -2 & -1 & -9 \\ 2 & -5 & 3 \\ -2 & -1 & 3 \end{bmatrix}^T = \begin{bmatrix} -2 & 2 & -2 \\ -1 & -5 & -1 \\ -9 & 3 & 3 \end{bmatrix}$$

We can find the determinant of **A** using any row or column of **A**. Since one element of the second row is 0, we can take advantage of this to find the determinant as

$$|\mathbf{A}| = -1c_{21} + 2c_{22} + (0)c_{23} = -1(2) + 2(-5) = -12$$

Hence, the inverse of **A** is

$$\mathbf{A}^{-1} = \frac{1}{-12}\begin{bmatrix} -2 & 2 & -2 \\ -1 & -5 & -1 \\ -9 & 3 & 3 \end{bmatrix}$$

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} = \frac{1}{-12}\begin{bmatrix} -2 & 2 & -2 \\ -1 & -5 & -1 \\ -9 & 3 & 3 \end{bmatrix}\begin{bmatrix} 5 \\ 9 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix}$$

i.e., $x_1 = -1, x_2 = 4, x_3 = 2$.

Practice Problem A.4

Solve the following equations using matrix inversion.

$$\begin{aligned} y_1 - y_3 &= 1 \\ 2y_1 + 3y_2 - y_3 &= 1 \\ y_1 - y_2 - y_3 &= 3 \end{aligned}$$

Answer: $y_1 = 6, y_2 = -2, y_3 = 5$.

Appendix B

Complex Numbers

The ability to manipulate complex numbers is very handy in circuit analysis and in electrical engineering in general. Complex numbers are particularly useful in the analysis of ac circuits. Again, although calculators and computer software packages are now available to manipulate complex numbers, it is still advisable for a student to be familiar with how to handle them by hand.

B.1 Representations of Complex Numbers

A complex number z may be written in *rectangular form* as

$$z = x + jy \tag{B.1}$$

where $j = \sqrt{-1}$; x is the *real part* of z while y is the *imaginary part* of z; that is,

$$x = \text{Re}(z), \qquad y = \text{Im}(z) \tag{B.2}$$

The complex number z is shown plotted in the complex plane in Fig. B.1. Since $j = \sqrt{-1}$,

$$\begin{aligned} \frac{1}{j} &= -j \\ j^2 &= -1 \\ j^3 &= j \cdot j^2 = -j \\ j^4 &= j^2 \cdot j^2 = 1 \\ j^5 &= j \cdot j^4 = j \\ &\vdots \\ j^{n+4} &= j^n \end{aligned} \tag{B.3}$$

The complex plane looks like the two-dimensional curvilinear coordinate space, but it is not.

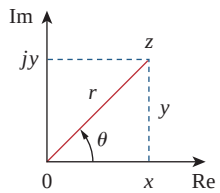


Figure B.1
Graphical representation of a complex number.

A second way of representing the complex number z is by specifying its magnitude r and the angle θ it makes with the real axis, as Fig. B.1 shows. This is known as the *polar form*. It is given by

$$z = |z| \angle \theta = r \angle \theta \tag{B.4}$$

where

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \tan^{-1}\frac{y}{x} \tag{B.5a}$$

or

$$x = r\cos\theta, \qquad y = r\sin\theta \tag{B.5b}$$

that is,

$$z = x + jy = r\angle\theta = r\cos\theta + jr\sin\theta \tag{B.6}$$

In converting from rectangular to polar form using Eq. (B.5), we must exercise care in determining the correct value of θ. These are the four possibilities:

$$\begin{aligned} z &= x + jy, & \theta &= \tan^{-1}\frac{y}{x} && \text{(1st Quadrant)} \\ z &= -x + jy, & \theta &= 180^\circ - \tan^{-1}\frac{y}{x} && \text{(2nd Quadrant)} \\ z &= -x - jy, & \theta &= 180^\circ + \tan^{-1}\frac{y}{x} && \text{(3rd Quadrant)} \\ z &= x - jy, & \theta &= 360^\circ - \tan^{-1}\frac{y}{x} && \text{(4th Quadrant)} \end{aligned} \tag{B.7}$$

assuming that x and y are positive.

In the exponential form, $z = re^{j\theta}$ so that $dz/d\theta = jre^{j\theta} = jz$.

The third way of representing the complex z is the *exponential form*:

$$z = re^{j\theta} \tag{B.8}$$

This is almost the same as the polar form, because we use the same magnitude r and the angle θ.

The three forms of representing a complex number are summarized as follows.

$z = x + jy,$	$(x = r\cos\theta, y = r\sin\theta)$	Rectangular form
$z = r\angle\theta,$	$\left(r = \sqrt{x^2 + y^2}, \theta = \tan^{-1}\frac{y}{x}\right)$	Polar form
$z = re^{j\theta},$	$\left(r = \sqrt{x^2 + y^2}, \theta = \tan^{-1}\frac{y}{x}\right)$	Exponential form

(B.9)

The first two forms are related by Eqs. (B.5) and (B.6). In Section B.3 we will derive Euler's formula, which proves that the third form is also equivalent to the first two.

Example B.1

Express the following complex numbers in polar and exponential form: (a) $z_1 = 6 + j8$, (b) $z_2 = 6 - j8$, (c) $z_3 = -6 + j8$, (d) $z_4 = -6 - j8$.

Solution:

Notice that we have deliberately chosen these complex numbers to fall in the four quadrants, as shown in Fig. B.2.

(a) For $z_1 = 6 + j8$ (1st quadrant),

$$r_1 = \sqrt{6^2 + 8^2} = 10, \qquad \theta_1 = \tan^{-1}\frac{8}{6} = 53.13^\circ$$

Hence, the polar form is $10\angle 53.13^\circ$ and the exponential form is $10e^{j53.13^\circ}$.

(b) For $z_2 = 6 - j8$ (4th quadrant),

$$r_2 = \sqrt{6^2 + (-8)^2} = 10, \qquad \theta_2 = 360^\circ - \tan^{-1}\frac{8}{6} = 306.87^\circ$$

so that the polar form is $10\angle 306.87°$ and the exponential form is $10e^{j306.87°}$. The angle θ_2 may also be taken as $-53.13°$, as shown in Fig. B.2, so that the polar form becomes $10\angle -53.13°$ and the exponential form becomes $10e^{-j53.13°}$.

(c) For $z_3 = -6 + j8$ (2nd quadrant),

$$r_3 = \sqrt{(-6)^2 + 8^2} = 10, \qquad \theta_3 = 180° - \tan^{-1}\frac{8}{6} = 126.87°$$

Hence, the polar form is $10\angle 126.87°$ and the exponential form is $10e^{j126.87°}$.

(d) For $z_4 = -6 - j8$ (3rd quadrant),

$$r_4 = \sqrt{(-6)^2 + (-8)^2} = 10, \qquad \theta_4 = 180° + \tan^{-1}\frac{8}{6} = 233.13°$$

so that the polar form is $10\angle 233.13°$ and the exponential form is $10e^{j233.13°}$.

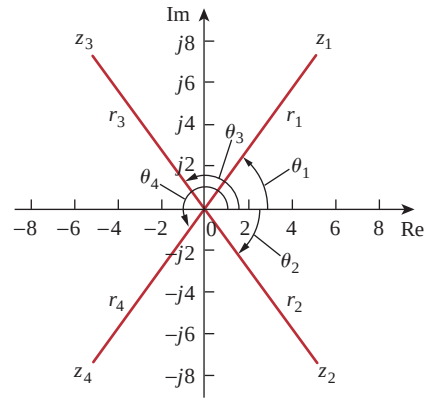


Figure B.2
For Example B.1.

Practice Problem B.1

Convert the following complex numbers to polar and exponential forms: (a) $z_1 = 3 - j4$, (b) $z_2 = 5 + j12$, (c) $z_3 = -3 - j9$, (d) $z_4 = -7 + j$.

Answer: (a) $5\angle 306.9°$, $5e^{j306.9°}$, (b) $13\angle 67.38°$, $13e^{j67.38°}$, (c) $9.487\angle 251.6°$, $9.487e^{j251.6°}$, (d) $7.071\angle 171.9°$, $7.071e^{j171.9°}$.

Example B.2

Convert the following complex numbers into rectangular form: (a) $12\angle -60°$, (b) $-50\angle 285°$, (c) $8e^{j10°}$, (d) $20e^{-j\pi/3}$.

Solution:

(a) Using Eq. (B.6),

$$12\angle -60° = 12\cos(-60°) + j12\sin(-60°) = 6 - j10.39$$

Note that $\theta = -60°$ is the same as $\theta = 360° - 60° = 300°$.

(b) We can write

$$-50\angle 285° = -50\cos 285° - j50\sin 285° = -12.94 + j48.3$$

(c) Similarly,

$$8e^{j10°} = 8\cos 10° + j8\sin 10° = 7.878 + j1.389$$

(d) Finally,

$$20e^{-j\pi/3} = 20\cos(-\pi/3) + j20\sin(-\pi/3) = 10 - j17.32$$

Practice Problem B.2

Find the rectangular form of the following complex numbers: (a) $-8\angle 210°$, (b) $40\angle 305°$, (c) $10e^{-j30°}$, (d) $50e^{j\pi/2}$.

Answer: (a) $6.928 + j4$, (b) $22.94 - j32.77$, (c) $8.66 - j5$, (d) $j50$.

We have used lightface notation for complex numbers—since they are not time- or frequency-dependent—whereas we use boldface notation for phasors.

B.2 Mathematical Operations

Two complex numbers $z_1 = x_1 + jy_1$ and $z_2 = x_2 + jy_2$ are equal if and only if their real parts are equal and their imaginary parts are equal,

$$x_1 = x_2, \qquad y_1 = y_2 \tag{B.10}$$

The *complex conjugate* of the complex number $z = x + jy$ is

$$z^* = x - jy = r\angle -\theta = re^{-j\theta} \tag{B.11}$$

Thus, the complex conjugate of a complex number is found by replacing every j by $-j$.

Given two complex numbers $z_1 = x_1 + jy_1 = r_1\angle\theta_1$ and $z_2 = x_2 + jy_2 = r_2\angle\theta_2$, their sum is

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2) \tag{B.12}$$

and their difference is

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2) \tag{B.13}$$

While it is more convenient to perform addition and subtraction of complex numbers in rectangular form, the product and quotient of the two complex numbers are best done in polar or exponential form. For their product,

$$z_1 z_2 = r_1 r_2 \angle\theta_1 + \theta_2 \tag{B.14}$$

Alternatively, using the rectangular form,

$$\begin{aligned} z_1 z_2 &= (x_1 + jy_1)(x_2 + jy_2) \\ &= (x_1 x_2 - y_1 y_2) + j(x_1 y_2 + x_2 y_1) \end{aligned} \tag{B.15}$$

For their quotient,

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\angle\theta_1 - \theta_2 \tag{B.16}$$

Alternatively, using the rectangular form,

$$\frac{z_1}{z_2} = \frac{x_1 + jy_1}{x_2 + jy_2} \tag{B.17}$$

We rationalize the denominator by multiplying both the numerator and denominator by z_2^*.

$$\frac{z_1}{z_2} = \frac{(x_1 + jy_1)(x_2 - jy_2)}{(x_2 + jy_2)(x_2 - jy_2)} = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + j\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2} \tag{B.18}$$

Example B.3

If $A = 2 + j5$, $B = 4 - j6$, find: (a) $A^*(A + B)$, (b) $(A + B)/(A - B)$.

Solution:

(a) If $A = 2 + j5$, then $A^* = 2 - j5$ and

$$A + B = (2 + 4) + j(5 - 6) = 6 - j$$

so that

$$A^*(A + B) = (2 - j5)(6 - j) = 12 - j2 - j30 - 5 = 7 - j32$$

(b) Similarly,

$$A - B = (2 - 4) + j(5 - -6) = -2 + j11$$

Hence,

$$\frac{A + B}{A - B} = \frac{6 - j}{-2 + j11} = \frac{(6 - j)(-2 - j11)}{(-2 + j11)(-2 - j11)}$$
$$= \frac{-12 - j66 + j2 - 11}{(-2)^2 + 11^2} = \frac{-23 - j64}{125} = -0.184 - j0.512$$

Practice Problem B.3

Given that $C = -3 + j7$ and $D = 8 + j$, calculate:
(a) $(C - D^*)(C + D^*)$, (b) D^2/C^*, (c) $2CD/(C + D)$.

Answer: (a) $-103 - j26$, (b) $-5.19 + j6.776$, (c) $6.045 + j11.53$.

Example B.4

Evaluate:

(a) $\dfrac{(2 + j5)(8e^{j10^\circ})}{2 + j4 + 2\angle -40^\circ}$ (b) $\dfrac{j(3 - j4)^*}{(-1 + j6)(2 + j)^2}$

Solution:

(a) Since there are terms in polar and exponential forms, it may be best to express all terms in polar form:

$$2 + j5 = \sqrt{2^2 + 5^2}\angle \tan^{-1} 5/2 = 5.385\angle 68.2^\circ$$
$$(2 + j5)(8e^{j10^\circ}) = (5.385\angle 68.2^\circ)(8\angle 10^\circ) = 43.08\angle 78.2^\circ$$
$$2 + j4 + 2\angle -40^\circ = 2 + j4 + 2\cos(-40^\circ) + j2\sin(-40^\circ)$$
$$= 3.532 + j2.714 = 4.454\angle 37.54^\circ$$

Thus,

$$\frac{(2 + j5)(8e^{j10^\circ})}{2 + j4 + 2\angle -40^\circ} = \frac{43.08\angle 78.2^\circ}{4.454\angle 37.54^\circ} = 9.672\angle 40.66^\circ$$

(b) We can evaluate this in rectangular form, since all terms are in that form. But

$$j(3 - j4)^* = j(3 + j4) = -4 + j3$$
$$(2 + j)^2 = 4 + j4 - 1 = 3 + j4$$
$$(-1 + j6)(2 + j)^2 = (-1 + j6)(3 + j4) = -3 - 4j + j18 - 24$$
$$= -27 + j14$$

Hence,

$$\frac{j(3 - j4)^*}{(-1 + j6)(2 + j)^2} = \frac{-4 + j3}{-27 + j14} = \frac{(-4 + j3)(-27 - j14)}{27^2 + 14^2}$$
$$= \frac{108 + j56 - j81 + 42}{925} = 0.1622 - j0.027$$

Practice Problem B.4

Evaluate these complex fractions:

(a) $\dfrac{6\angle 30^\circ + j5 - 3}{-1 + j + 2e^{j45^\circ}}$ (b) $\left[\dfrac{(15 - j7)(3 + j2)^*}{(4 + j6)^*(3\angle 70^\circ)}\right]^*$

Answer: (a) $3.387\angle -5.615^\circ$, (b) $2.759\angle -287.6^\circ$.

B.3 Euler's Formula

Euler's formula is an important result in complex variables. We derive it from the series expansion of e^x, $\cos\theta$, and $\sin\theta$. We know that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots \tag{B.19}$$

Replacing x by $j\theta$ gives

$$e^{j\theta} = 1 + j\theta - \frac{\theta^2}{2!} - j\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + \cdots \tag{B.20}$$

Also,

$$\begin{aligned} \cos\theta &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \cdots \\ \sin\theta &= \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots \end{aligned} \tag{B.21}$$

so that

$$\cos\theta + j\sin\theta = 1 + j\theta - \frac{\theta^2}{2!} - j\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + j\frac{\theta^5}{5!} - \cdots \tag{B.22}$$

Comparing Eqs. (B.20) and (B.22), we conclude that

$$\boxed{e^{j\theta} = \cos\theta + j\sin\theta} \tag{B.23}$$

This is known as *Euler's formula*. The exponential form of representing a complex number as in Eq. (B.8) is based on Euler's formula. From Eq. (B.23), notice that

$$\boxed{\cos\theta = \text{Re}(e^{j\theta}), \qquad \sin\theta = \text{Im}(e^{j\theta})} \tag{B.24}$$

and that

$$|e^{j\theta}| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$$

Replacing θ by $-\theta$ in Eq. (B.23) gives

$$e^{-j\theta} = \cos\theta - j\sin\theta \tag{B.25}$$

Adding Eqs. (B.23) and (B.25) yields

$$\boxed{\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})} \tag{B.26}$$

Subtracting Eq. (B.25) from Eq. (B.23) yields

$$\sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta}) \tag{B.27}$$

Useful Identities

The following identities are useful in dealing with complex numbers. If $z = x + jy = r\angle\theta$, then

$$zz^* = x^2 + y^2 = r^2 \tag{B.28}$$

$$\sqrt{z} = \sqrt{x + jy} = \sqrt{r}e^{j\theta/2} = \sqrt{r}\angle\theta/2 \tag{B.29}$$

$$z^n = (x + jy)^n = r^n\angle n\theta = r^n e^{jn\theta} = r^n(\cos n\theta + j\sin n\theta) \tag{B.30}$$

$$z^{1/n} = (x + jy)^{1/n} = r^{1/n}\angle\theta/n + 2\pi k/n \qquad k = 0, 1, 2, \ldots, n - 1 \tag{B.31}$$

$$\ln(re^{j\theta}) = \ln r + \ln e^{j\theta} = \ln r + j\theta + j2k\pi \qquad (k = \text{integer}) \tag{B.32}$$

$$\begin{aligned} \frac{1}{j} &= -j \\ e^{\pm j\pi} &= -1 \\ e^{\pm j2\pi} &= 1 \\ e^{j\pi/2} &= j \\ e^{-j\pi/2} &= -j \end{aligned} \tag{B.33}$$

$$\begin{aligned} \text{Re}(e^{(\alpha+j\omega)t}) &= \text{Re}(e^{\alpha t}e^{j\omega t}) = e^{\alpha t}\cos\omega t \\ \text{Im}(e^{(\alpha+j\omega)t}) &= \text{Im}(e^{\alpha t}e^{j\omega t}) = e^{\alpha t}\sin\omega t \end{aligned} \tag{B.34}$$

Example B.5

If $A = 6 + j8$, find: (a) $\sqrt{A}$, (b) A^4.

Solution:

(a) First, convert A to polar form:

$$r = \sqrt{6^2 + 8^2} = 10, \qquad \theta = \tan^{-1}\frac{8}{6} = 53.13°, \qquad A = 10\angle 53.13°$$

Then

$$\sqrt{A} = \sqrt{10}\angle 53.13°/2 = 3.162\angle 26.56°$$

(b) Since $A = 10\angle 53.13°$,

$$A^4 = r^4\angle 4\theta = 10^4\angle 4 \times 53.13° = 10{,}000\angle 212.52°$$

Practice Problem B.5

If $A = 3 - j4$, find: (a) $A^{1/3}$ (3 roots), and (b) $\ln A$.

Answer: (a) $1.71\angle 102.3°$, $1.71\angle 222.3°$, $1.71\angle 342.3°$,
(b) $1.609 + j5.356 + j2n\pi$ $(n = 0, 1, 2, \ldots)$.

Appendix C

Mathematical Formulas

This appendix—by no means exhaustive—serves as a handy reference. It does contain all the formulas needed to solve circuit problems in this book.

C.1 Quadratic Formula

The roots of the quadratic equation $ax^2 + bx + c = 0$ are

$$x_1, x_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

C.2 Trigonometric Identities

$$\sin(-x) = -\sin x$$
$$\cos(-x) = \cos x$$
$$\sec x = \frac{1}{\cos x}, \qquad \csc x = \frac{1}{\sin x}$$
$$\tan x = \frac{\sin x}{\cos x}, \qquad \cot x = \frac{1}{\tan x}$$
$$\sin(x \pm 90°) = \pm\cos x$$
$$\cos(x \pm 90°) = \mp\sin x$$
$$\sin(x \pm 180°) = -\sin x$$
$$\cos(x \pm 180°) = -\cos x$$
$$\cos^2 x + \sin^2 x = 1$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \qquad \text{(law of sines)}$$
$$a^2 = b^2 + c^2 - 2bc\cos A \qquad \text{(law of cosines)}$$
$$\frac{\tan\frac{1}{2}(A - B)}{\tan\frac{1}{2}(A + B)} = \frac{a - b}{a + b} \qquad \text{(law of tangents)}$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$
$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$
$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$
$$2\sin x \sin y = \cos(x - y) - \cos(x + y)$$
$$2\sin x \cos y = \sin(x + y) + \sin(x - y)$$
$$2\cos x \cos y = \cos(x + y) + \cos(x - y)$$
$$\sin 2x = 2\sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$$

$$\tan 2x = \frac{2\tan x}{1 - \tan^2 x}$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$K_1 \cos x + K_2 \sin x = \sqrt{K_1^2 + K_2^2}\cos\left(x + \tan^{-1}\frac{-K_2}{K_1}\right)$$

$$e^{jx} = \cos x + j \sin x \qquad \text{(Euler's formula)}$$

$$\cos x = \frac{e^{jx} + e^{-jx}}{2}$$

$$\sin x = \frac{e^{jx} - e^{-jx}}{2j}$$

$$1 \text{ rad} = 57.296°$$

C.3 Hyperbolic Functions

$$\sinh x = \frac{1}{2}(e^x - e^{-x})$$

$$\cosh x = \frac{1}{2}(e^x + e^{-x})$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\coth x = \frac{1}{\tanh x}$$

$$\operatorname{csch} x = \frac{1}{\sinh x}$$

$$\operatorname{sech} x = \frac{1}{\cosh x}$$

$$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$$

$$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$$

C.4 Derivatives

If $U = U(x)$, $V = V(x)$, and $a =$ constant,

$$\frac{d}{dx}(aU) = a\frac{dU}{dx}$$

$$\frac{d}{dx}(UV) = U\frac{dV}{dx} + V\frac{dU}{dx}$$

$$\frac{d}{dx}\left(\frac{U}{V}\right) = \frac{V\dfrac{dU}{dx} - U\dfrac{dV}{dx}}{V^2}$$

$$\frac{d}{dx}(aU^n) = naU^{n-1}$$

$$\frac{d}{dx}(a^U) = a^U \ln a \frac{dU}{dx}$$

$$\frac{d}{dx}(e^U) = e^U \frac{dU}{dx}$$

$$\frac{d}{dx}(\sin U) = \cos U \frac{dU}{dx}$$

$$\frac{d}{dx}(\cos U) = -\sin U \frac{dU}{dx}$$

C.5 Indefinite Integrals

If $U = U(x)$, $V = V(x)$, and a = constant,

$$\int a\,dx = ax + C$$

$$\int U\,dV = UV - \int V\,dU \qquad \text{(integration by parts)}$$

$$\int U^n\,dU = \frac{U^{n+1}}{n+1} + C, \qquad n \neq 1$$

$$\int \frac{dU}{U} = \ln U + C$$

$$\int a^U\,dU = \frac{a^U}{\ln a} + C, \qquad a > 0, a \neq 1$$

$$\int e^{ax}\,dx = \frac{1}{a}e^{ax} + C$$

$$\int xe^{ax}\,dx = \frac{e^{ax}}{a^2}(ax - 1) + C$$

$$\int x^2e^{ax}\,dx = \frac{e^{ax}}{a^3}(a^2x^2 - 2ax + 2) + C$$

$$\int \ln x\,dx = x\ln x - x + C$$

$$\int \sin ax\,dx = -\frac{1}{a}\cos ax + C$$

$$\int \cos ax\,dx = \frac{1}{a}\sin ax + C$$

$$\int \sin^2 ax\,dx = \frac{x}{2} - \frac{\sin 2ax}{4a} + C$$

$$\int \cos^2 ax\,dx = \frac{x}{2} + \frac{\sin 2ax}{4a} + C$$

$$\int x \sin ax\, dx = \frac{1}{a^2}(\sin ax - ax \cos ax) + C$$

$$\int x \cos ax\, dx = \frac{1}{a^2}(\cos ax + ax \sin ax) + C$$

$$\int x^2 \sin ax\, dx = \frac{1}{a^3}(2ax \sin ax + 2 \cos ax - a^2x^2 \cos ax) + C$$

$$\int x^2 \cos ax\, dx = \frac{1}{a^3}(2ax \cos ax - 2 \sin ax + a^2x^2 \sin ax) + C$$

$$\int e^{ax} \sin bx\, dx = \frac{e^{ax}}{a^2 + b^2}(a \sin bx - b \cos bx) + C$$

$$\int e^{ax} \cos bx\, dx = \frac{e^{ax}}{a^2 + b^2}(a \cos bx + b \sin bx) + C$$

$$\int \sin ax \sin bx\, dx = \frac{\sin(a - b)x}{2(a - b)} - \frac{\sin(a + b)x}{2(a + b)} + C, \qquad a^2 \neq b^2$$

$$\int \sin ax \cos bx\, dx = -\frac{\cos(a - b)x}{2(a - b)} - \frac{\cos(a + b)x}{2(a + b)} + C, \qquad a^2 \neq b^2$$

$$\int \cos ax \cos bx\, dx = \frac{\sin(a - b)x}{2(a - b)} + \frac{\sin(a + b)x}{2(a + b)} + C, \qquad a^2 \neq b^2$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$\int \frac{x^2\, dx}{a^2 + x^2} = x - a \tan^{-1} \frac{x}{a} + C$$

$$\int \frac{dx}{(a^2 + x^2)^2} = \frac{1}{2a^2}\left(\frac{x}{x^2 + a^2} + \frac{1}{a} \tan^{-1} \frac{x}{a}\right) + C$$

C.6 Definite Integrals

If m and n are integers,

$$\int_0^{2\pi} \sin ax\, dx = 0$$

$$\int_0^{2\pi} \cos ax\, dx = 0$$

$$\int_0^{\pi} \sin^2 ax\, dx = \int_0^{\pi} \cos^2 ax\, dx = \frac{\pi}{2}$$

$$\int_0^{\pi} \sin mx \sin nx\, dx = \int_0^{\pi} \cos mx \cos nx\, dx = 0, \qquad m \neq n$$

$$\int_0^{\pi} \sin mx \cos nx\, dx = \begin{cases} 0, & m + n = \text{even} \\ \dfrac{2m}{m^2 - n^2}, & m + n = \text{odd} \end{cases}$$

$$\int_0^{2\pi} \sin mx \sin nx\, dx = \int_{-\pi}^{\pi} \sin mx \sin nx\, dx = \begin{cases} 0, & m \neq n \\ \pi, & m = n \end{cases}$$

$$\int_0^\infty \frac{\sin ax}{x}\,dx = \begin{cases} \dfrac{\pi}{2}, & a > 0 \\ 0, & a = 0 \\ -\dfrac{\pi}{2}, & a < 0 \end{cases}$$

C.7 L'Hopital's Rule

If $f(0) = 0 = h(0)$, then

$$\lim_{x \to 0} \frac{f(x)}{h(x)} = \lim_{x \to 0} \frac{f'(x)}{h'(x)}$$

where the prime indicates differentiation.

Appendix D

PSpice for Windows

There are several computer software packages, such as *Spice*, *Mathcad*, *Quattro*, *MATLAB*, and *Maple*, which can be used for circuit analysis. The most popular is *Spice*, which stands for *Simulation Program with Integrated-Circuit Emphasis*. *Spice* was developed at the Department of Electrical and Computer Engineering at the University of California at Berkeley in the 1970s for mainframe computers. Since then about 20 versions have been developed. *PSpice*, a version of *Spice* for personal computers, was developed by MicroSim Corporation in California and made available in 1984 and later by OrCAD and cadence. *PSpice* has been made available in different operating systems (DOS, Windows, Unix, etc.).

The student version of *PSpice* can be obtained free of charge.

If you do not have access to *PSpice*, you can find out how to obtain a free student copy by going to the text website (www.mhhe.com/alexander). The instructions and examples in this appendix were developed for version 9.1, but also work for later versions.

Assuming that you are using Windows and have the *PSpice* software installed in your computer, you can access *PSpice* by clicking the Start icon on the left-hand corner of your PC; drag the cursor to Programs, to *PSpice* students and to *Schematics*, and then click as shown in Fig. D.1.

The objective of this appendix is to provide a short tutorial on using the Windows-based *PSpice* on an IBM PC or equivalent.

PSpice can analyze up to roughly 130 elements and 100 nodes. It is capable of performing three major types of circuit analysis: dc analysis, transient analysis, and ac analysis. In addition, it can also perform transfer function analysis, Fourier analysis, and operating point analysis.The circuit can contain resistors, inductors, capacitors, independent and dependent voltage and current sources, op amps, transformers, transmission lines, and semiconductor devices.

We will assume that you are familiar with using the Microsoft Windows operating system and that *PSpice for Windows* is already installed in your computer. As with any standard Windows application, *PSpice* provides an on-line help system.

If you need help on any topic at any level, click Help, click Help Topics, click Search, and type in the topic.

D.1 Design Center for Windows

In earlier versions of *PSpice* prior to Windows 95, *PSpice for Windows* is formally known as the *MicroSim Design Center*, which is a computer environment for simulating electric circuits. The *Design Center for Windows* includes the following programs:

Schematics: This program is a graphical editor used to draw the circuit to be simulated on the screen. It allows the user to enter

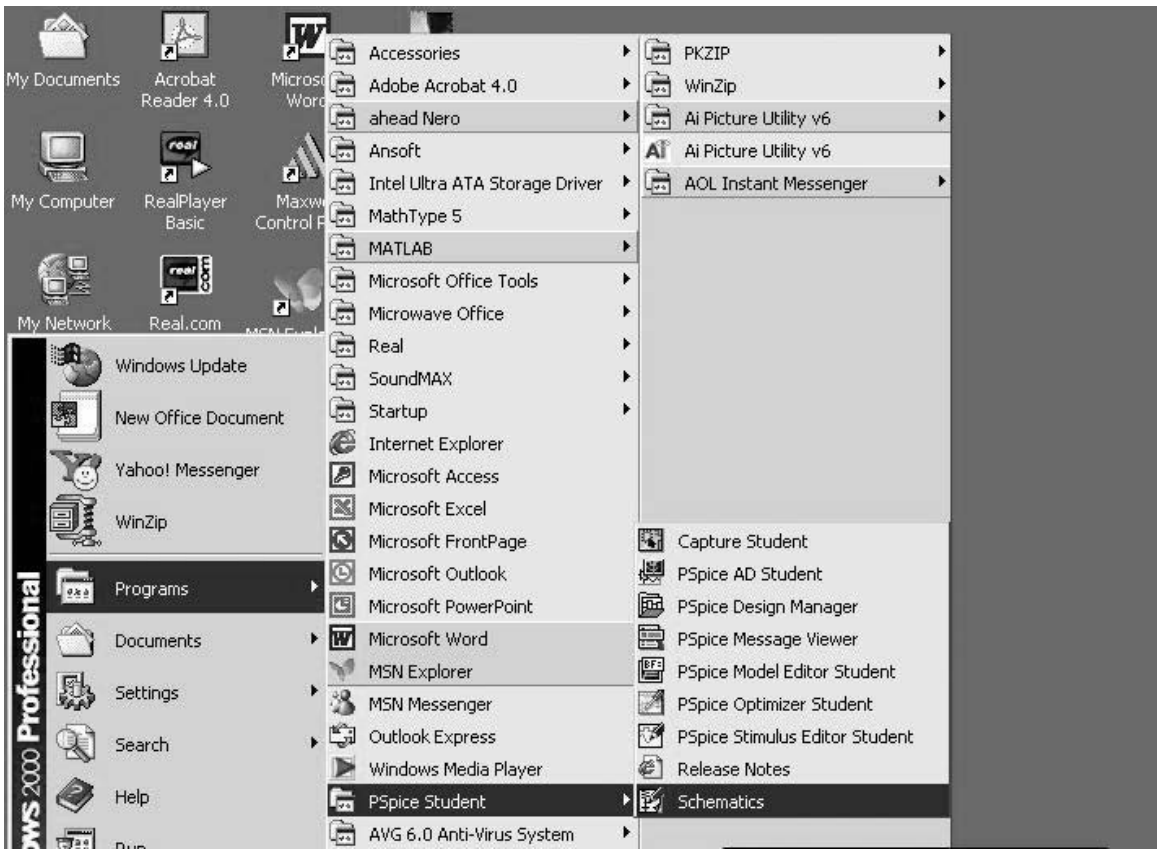


Figure D.1
Accessing *PSpice* on Windows.

the components, wire the components together to form the circuit, and specify the type of analysis to be performed.

Pspice: This program simulates the circuit created using *Schematics*. By simulation, we mean a method of analysis described in a program by which a circuit is represented by mathematical models of the components comprising the circuit.

Orcad PSpice: This program provides a graphic display of the output generated by the *PSpice* program. It can be used to observe any voltage or current in the circuit.

One may think of *Schematics* as the computer breadboard for setting up the circuit topology, *PSpice* as the simulator (performing the computation), and *Orcad PSpice* as the oscilloscope. Using the *Schematics* program is perhaps the hardest part of circuit simulation using *PSpice*. The next section covers the essential skills needed to operate the *Schematics*.

D.2 Creating a Circuit

For a circuit to be analyzed by *PSpice*, we must take three steps: (1) create the circuit, (2) simulate it, and (3) print or plot the results. In this section, we learn how to create the circuit using the Schematics program.

Before we discuss how to use the *Schematics* capture, we need to know how to use the mouse to select an object and perform an action. One uses the mouse in *Schematics* in conjunction with the keyboard to

carry out various instructions. Throughout this text, we will use the following terms to represent actions to be performed by the mouse:

- **CLICKL** : click the left button once to select an item.
- **CLICKR** : click the right button once to abort a mode.
- **DCLICKL** : double-click the left button to edit a selection or end a mode.
- **DCLICKR** : double-click the right button to repeat an action.
- **CLICKLH** : click the left button, hold down, and move the mouse to drag a selected item. Release the left button after the item has been placed.
- **DRAG** : drag the mouse (without clicking) to move an item.

When the term "click" is used, it means that you quickly press and release the **left** mouse button. To select an item requires **CLICKL**, while to perform an action requires **DCLICKL**. Also, to avoid writing "click" several times, the menu to be clicked will be highlighted in bold. For example, "click **Draw**, click **Get New Part**" will be written as **Draw/Get New Part**. Of course, we can always press the <Esc> key to abort any action.

Assuming that you are using Windows, you can access *Pspice* by clicking the Start icon on the left-hand corner of your PC, drag the cursor to Programs, *PSpice* student; and to *Schematics*, as shown in Fig. D.1. Alternatively, you have the *PSpice* icon on your screen. **DCLICK** on it. Either way, a blank screen will appear as shown in Fig. D.2. The file

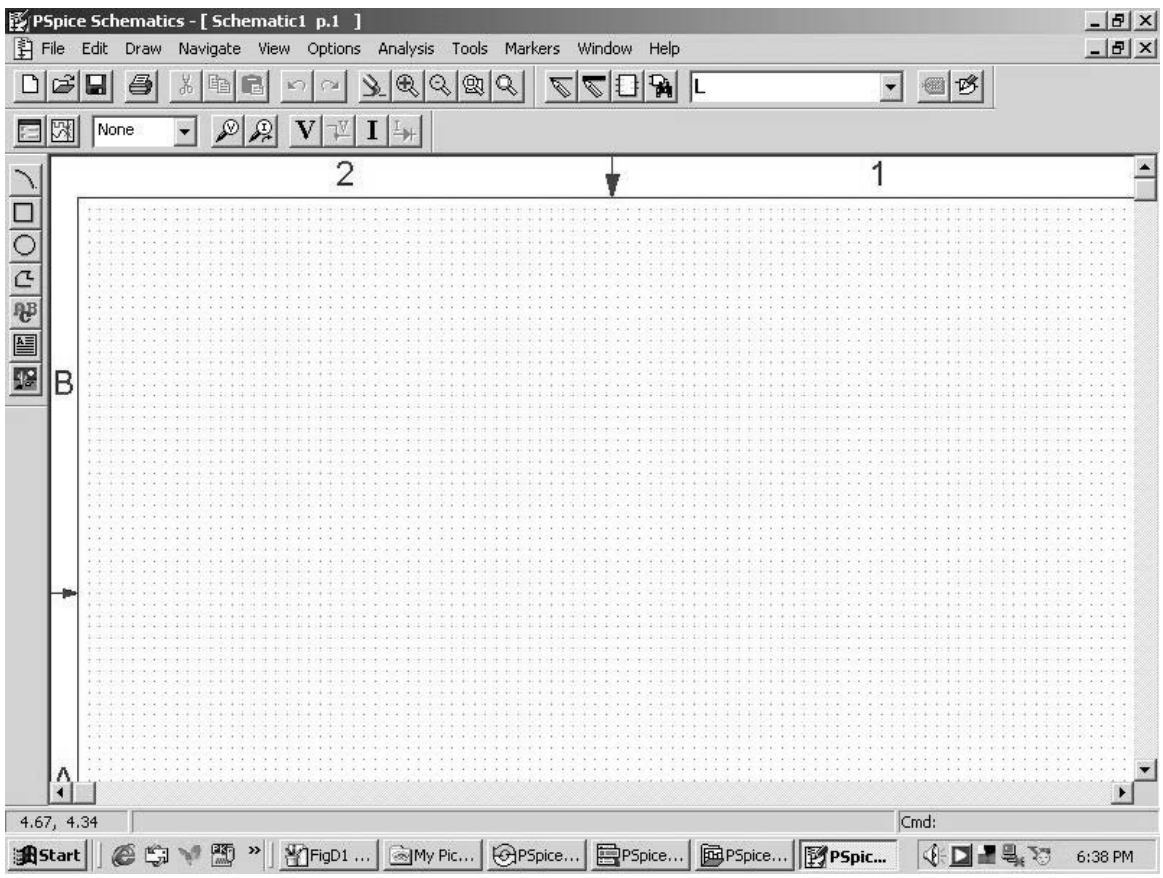


Figure D.2
Schematics window.

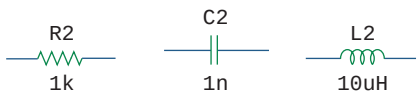
name [*Schematic1* **p.1**] next to *PSpice Schematics* is assigned to a circuit which is yet to be saved. You can change it by pulling down the *File* menu.

To create a circuit using *Schematics* requires three steps: (1) placing the parts or components of the circuit, (2) wiring the parts together to form the circuit, and (3) changing attributes of the parts.

Step 1: Placing the Parts

Each circuit part is retrieved by following this procedure:

- Select **Draw/Get New Part** to pull down the Draw menu (or type <Ctrl-G>).
- Use scroll bar to select the part (or type the part name, e.g., R for resistor, in the *PartName* box). Figures D.3 to D.5 show some part names and symbols for circuit elements and independent voltage and current sources.
- Click **Place & Close** (or press <Enter>).
- **DRAG** part to the desired location on the screen.
- **CLICKR** to terminate the placement mode.



(a) (b) (c)

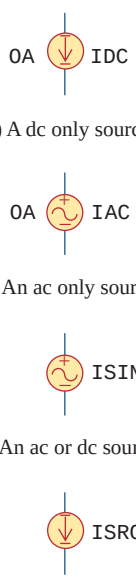
Figure D.3
Part symbols and attributes for circuit elements: (a) a resistor, (b) a capacitor, (c) an inductor.



(a) A dc only source (b) An ac only source (c) An ac or dc source (d) An ac, dc, or transient source

Figure D.4
Part symbols and attributes for independent voltage sources.

Sometimes, we want to rotate a part 90°. To rotate a resistor, for example, select the part R and click **Edit/Rotate** (or type <Ctrl R>). To delete a part, **CLICKL** to select (highlight red) the part, then click **Edit/Cut** (or press <Delete>).



(a) A dc only source

(b) An ac only source

(c) An ac or dc source

(d) An ac, dc, or transient source

Figure D.5
Part symbols and attributes for independent current sources.

Step 2: Wiring Parts Together

We complete the circuit by wiring the parts together. We first select **Draw/Wire** (or type <Ctrl-W>) to be in wiring mode. A pencil cursor will appear in place of an arrow cursor. **DRAG** the pencil cursor to the first point you want to connect and **CLICKL**. Next, **DRAG** the pencil cursor to the second point and **CLICKL** to change the dashed line to a solid line. (Only solid lines are wires.) **CLICKR** to end the wiring mode. To resume the wiring mode, press the <Space bar>. Repeat the above procedure for each connection in the circuit until all the parts are wired. The wiring is not complete without adding a ground connection (part AGND) to a schematic; *PSpice* will not operate without it. To verify that the parts are actually connected together, the *Junctions* option available in the **Options/Set Display Level** menu should be in the on position when wiring the parts. By default, the *Junctions* option is marked with a checksign (✓) in the dialog box, indicating that it is on.

Some of the connections have a black dot indicating a connection. Although it is not necessary to have a dot where a wire joins a pin,

having the dot shows the presence of a connection. To be sure a dot appears, make sure the wire overlaps the pin.

If you make a mistake, you can delete the part or wire by highlighting it (select **CLICKL**) and pressing the <Delete> key. Typing <Ctrl-L> will erase the fragments that are not really on the shematic.

Step 3: Changing Attributes of Parts

As shown in Figs. D.3 to D.5, each component has an attribute in addition to its symbol. Attributes are the labels for parts. Each attribute consists of a *name* and its designated *value.* For example, R and VSRC are the names of resistor and voltage source (dc, ac, or transient source), while 2k and DC = +10V are the designated values of the resistor and voltage source, respectively.

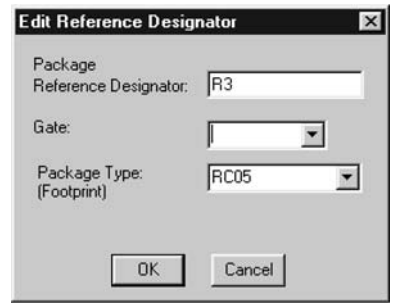
As parts are placed on the screen, they are automatically assigned names by successive numbers (R1, R2, R3, etc.). Also, some parts are assigned some predetermined values. For example, all resistors are placed horizontally and assigned a value of 1 kΩ. We may need to change the attributes (names and values) of a part. Although there are several ways of changing the attributes, the following is one simple way.

To change the name R3 to RX, for example, **DCLICKL** on the text R3 to bring up the *Edit Reference Designator* dialog box of Fig. D.6(a). Type the new name RX and click the **OK** button to accept the change. The same procedure can be used to change VDC to V1 or whatever.

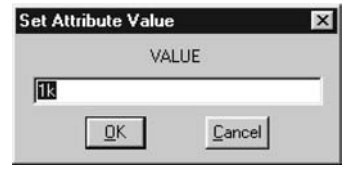
To change the value 1k to 10Meg, for example, **DCLICKL** on the 1k attribute (not the symbol) to open up the *Set Attribute Value* dialog box of Fig. D.6(b). Type the new value 10Meg (no space between 10 and Meg) and click the **OK** button to accept the change. Similarly, to change the default value 0V to 15kV for voltage source VDC, **DCLICKL** the symbol for VDC to bring up the *PartName* dialog box. **DCLICKL** on the *DC* = attribute and type 15kV in the value box. For convenience, one can express numbers with the scale factors in Table D.1. For example, 6.6×10^{-8} can be written as 66N or 0.066U.

Except for the ground, which is automatically assigned node 0, every node is either given a name (or number) or is assigned one in the netlist. A node is labeled by giving a name to a wire connected to that node. **DCLICKL** the wire to open up the *Set Attribute Value* dialog box, and type the label.

To obtain a hard copy of the screen/schematic, click **File/Print/OK**. To save the schematic created, select **File/Save As** and type Filename. Click **OK** or press <Enter>. This creates a file named "filename" and saves it with extension .sch.



(a)



(b)

Figure D.6
(a) Changing name R3 to RX,
(b) changing 1k to 10Meg.

A component may have several attributes; some are displayed by default. If need be, we may add more attributes for display, but we should hide unimportant attributes to avoid clutter.

TABLE D.1
Scale factors.

Symbol	Value	Name of suffix
T	10^{12}	tera
G	10^{9}	giga
MEG	10^{6}	mega
K	10^{3}	kilo
M	10^{-3}	milli
U	10^{-6}	micro
N	10^{-9}	nano
P	10^{-12}	pico
F	10^{-15}	femto

It is always expedient to number the nodes by numbering the wires. Otherwise, *Schematics* will label the nodes its own way, and one may not understand which node is which in the output results.

Example D.1

Draw the circuit in Fig. D.7 using *Schematics*.

Solution:
We will follow the three steps mentioned above. We begin by double-clicking the *Schematics* icon. This provides us with a blank screen as

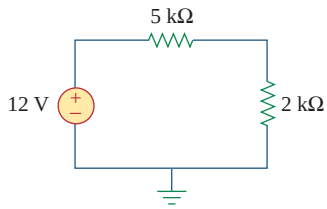


Figure D.7
For Example D.1.

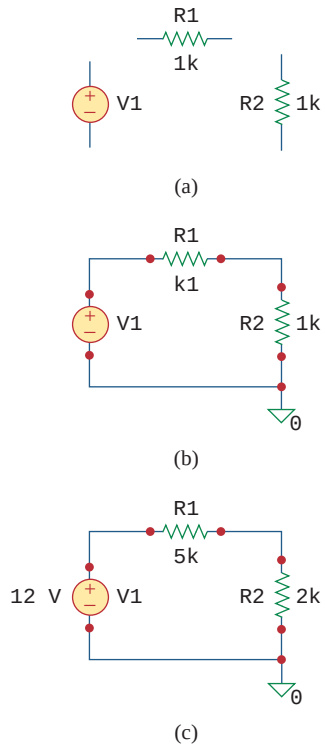


Figure D.8
Creating the circuit in Fig. D.7: (a) placing the parts, (b) wiring the parts together, (c) changing the attributes.

a worksheet to draw the circuit on. We now take the following steps to create the circuit in Fig. D.7.
To place the voltage source, we need to:

1. Click **Draw/Get New Part** (or type <Ctrl-G>).
2. Type VSRC in the *Part Browser Basic* box.
3. Click **OK** (or type <Enter>).
4. **DRAG** the part to the desired location on the screen.
5. **CLICKL** to place VSRC and **CLICKR** to terminate placement mode.

At this point, only the voltage source V1 in Fig. D.8(a) is shown on the screen, highlighted red. To place the resistors, we need to:

1. Click **Draw/Get New Part**.
2. Type R in the *Part Browser Basic* box.
3. Click **OK**.
4. **DRAG** resistor to R1's location on the screen.
5. **CLICKL** to place R1.
6. **CLICKL** to place R2 and **CLICKR** to terminate placement mode.
7. **DRAG** R2 to its location.
8. **Edit/Rotate** (or type <Ctrl-R>) to rotate R2.

At this point, the three parts have been created as shown in Fig. D.8(a). The next step is to connect the parts by wiring. To do this:

1. Click **Draw/Wire** to be in wiring mode, indicated by the pencil cursor.
2. **DRAG** the pencil cursor to the top of V1.
3. **CLICKL** to join the wire to the top of V1.
4. **DRAG** the dotted wire to the top corner.
5. **CLICKL** to turn wire segment solid, and anchor at corner.
6. **DRAG** dotted wire to left of R1.
7. **CLICKL** to turn wire segment solid and anchor to left of R1.
8. **CLICKR** to end placement mode.

Follow the same steps to connect R1 with R2 and V1 with R2. (You can resume the wiring mode by pressing <Space bar>.) At this point, we have the circuit in Fig. D.8(b), except that the ground symbol is missing. We insert the ground by taking the following steps:

1. Click **Draw/Get New Part**.
2. Type AGND in the *Part Browser Basic* box.
3. Click **OK**.
4. **DRAG** the part to the desired location on the screen.
5. **CLICKL** to place AGND and **CLICKR** to terminate placement mode.

The last thing to be done is to change or assign values to the attributes. To assign the attribute 12V to V1, we take these steps:

1. **DCLICKL** on the V1 symbol to open up the *PartName* dialog box.
2. **DCLICKL** on the *DC* = attribute.
3. Type +12V (or simply 12) in the *Value* box.
4. Click **Save Attr**.
5. Click **OK**.

To assign 5k to R1, we follow these steps:

1. **DCLICKL** on 1k attribute of R1 to bring up the *Set Attribute Value* dialog box.
2. Type 5k in the *Value* box.
3. Click **OK**.

Use the same procedure in assigning value 2k to R2. Figure D.8(c) shows the final circuit.

Practice Problem D.1

Construct the circuit in Fig. D.9 with *Schematics*.

Answer: See the schematic Fig. D.10.

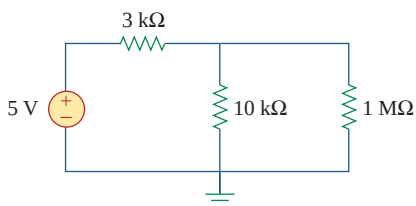


Figure D.9
For Practice Prob. D.1.

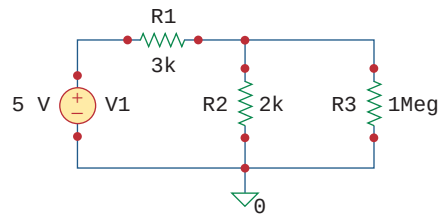


Figure D.10
For Practice Prob. D.1.

D.3 DC Analysis

DC analysis is one of the standard analyses that we can perform using *PSpice*. Other standard analyses include transient, AC, and Fourier. Under DC analysis, there are two kinds of simulation that *PSpice* can execute: DC nodal analysis and DC sweep.

1. DC Nodal Analysis

PSpice allows dc nodal analysis to be performed on sources with an attribute of the form DC = value and provides the dc voltage at each node of the circuit and dc branch currents if required. To view dc node voltages and branch currents requires adding two kinds of additional parts, shown in Fig. D.11. The symbol VIEWPOINT is connected to each node at which the voltage is to be viewed, while the symbol IPROBE is connected in the branch where the current is to be displayed. This necessitates modifying the schematic. For example, let us consider placing voltage VIEWPOINTS and current IPROBES to the schematic in Fig. D.8(c). To add VIEWPOINTS, we take the following steps:

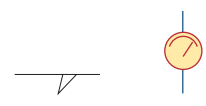


Figure D.11
Symbols for: (a) voltage VIEWPOINT, (b) current IPROBE.

1. Click **Draw/Get New Part** (or type <Ctrl-G>).
2. Type VIEWPOINT in the *Part Browser Basic* box.
3. Click **OK** (or type <Enter>).
4. **DRAG** to locate VIEWPOINT above V1 and **CLICKL**.
5. **DRAG** to locate VIEWPOINT above R2 and **CLICKL**.
6. **CLICKR** to end placement mode.

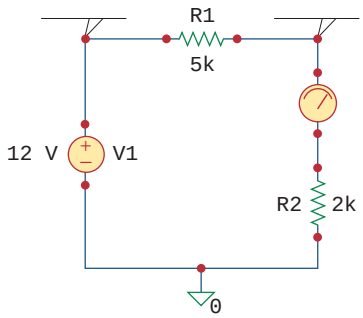


Figure D.12
Placing VIEWPOINTS and IPROBES.

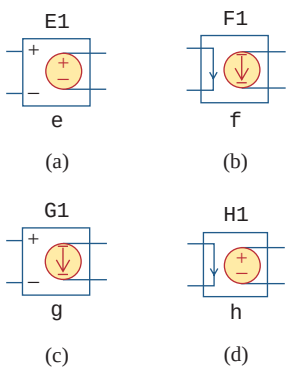


Figure D.13
Dependent sources:
(a) voltage-controlled voltage source (VCVS),
(b) current-controlled current source (CCCS),
(c) voltage-controlled current source (VCCS),
(d) current-controlled voltage source (CCVS).

A netlist can be generated manually or automatically by *Schematics*.

There are two kinds of common errors in *PSpice*: (1) errors involving wiring of the circuit, and (2) errors that occur during simulation.

Figure D.12 shows the two voltage VIEWPOINTS. Since the IPROBE symbol must be connected in series with a branch element, we need to move R2 down by clicking and dragging R2 and the wires. Once this is done, we add IPROBE as follows:

1. Click **Draw/Get New Part** (or type <Ctrl-G>).
2. Type IPROBE in the *Part Browser Basic* box.
3. Click **OK** (or type <Enter>).
4. **DRAG** to locate IPROBE above R2 and **CLICKL**.
5. **CLICKR** to end placement mode.
6. Use wiring to join all gaps.

The schematic becomes that shown in Fig. D.12. We are ready to simulate the circuit. At this point, we must save the schematic—*PSpice* will not run without first saving the schematic to be simulated. Before learning how to run *PSpice*, note the following points:

1. There must be a reference node or ground connection (part AGND) in the schematic. Any node can be used as ground, and the voltages at other nodes will be measured with respect to the selected ground.
2. Dependent sources are found in the Parts library. Obtain them by selecting **Draw/Get New Part** and typing the part name. Figure D.13 shows the part name for each type, with the gain. *E* is a voltage-controlled voltage source with gain *e*; *F* is a current-controlled current source with gain *f*; *G* is a voltage-controlled current source with a transconductance gain *g*; and *H* is a current-controlled voltage source with transresistance gain *h*.
3. By convention, we assume in dc analysis that all capacitors are open circuits and all inductors are short circuits.

We run *PSpice* by clicking **Analysis/Simulate**. This invokes the *electric rule check* (ERC), which generates the *netlist*. The ERC performs a connectivity check on the schematic before creating the netlist. The netlist is a list describing the operational behavior of each component in the circuit and its connections. Each line in the netlist represents a single component of the circuit. The netlist can be examined by clicking **Analysis/Examine Netlist** from the *Schematics* window. If there are errors in the schematic, an *error* window will appear. Click **OK** (or type <Enter>) to display the error list. After noting the errors, exit from the error list and go back to *Schematics* to correct the errors. If no errors are found, the system automatically enters *PSpice* and performs the simulation (nodal analysis). When the analysis is complete, the program displays **Bias point calculated**, and creates the result/output file with extension .out.

To examine the output file, click **Analysis/Examine Output** from the *Schematics* window (or click **File/Examine Output** from the *PSpice* window). To print the output file, click **File/Print**, and to exit the output file, click **File/Exit**.

We can also examine the results of the simulation by looking at the values displayed on the VIEWPOINTS and IPROBES parts of the *schematics* after the simulation is complete. The values displayed with VIEWPOINTS and IPROBES should be the same as those in the output file.

2. DC Sweep

DC nodal analysis allows simulation for DC sources with fixed voltages or currents. DC sweep provides more flexibility in that it allows the calculation of node voltages and branch currents of a circuit when a source is swept over a range of values. As in nodal analysis, we assume capacitors to be open circuits and inductors to be short circuits.

Suppose we desire to perform a DC sweep of voltage source V1 in Fig. D.12 from 0 to 20 volts in 1-volt increments. We proceed as follows:

1. Click **Analysis/Setup**.
2. **CLICKL DC Sweep** button.
3. Click *Name* box and type V1.
4. Click *Start Value* box and type 0.
5. Click *End Value* box and type 20.
6. Click *Increment* box and type 1.
7. Click **OK** to end the *DC Sweep* dialog box and save parameters.
8. Click **Close** to end the *Analysis Setup* menu.

Figure D.14 shows the *DC Sweep* dialog box. Notice that the default setting is *Voltage Source* for the *Swept Var. Type*, while it is *Linear* for *Sweep Type*. If needed, other options can be selected by clicking the appropriate buttons.

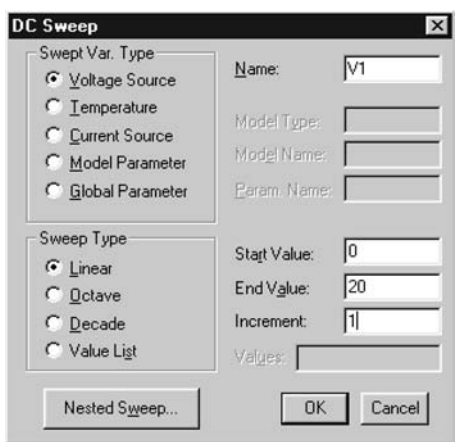


Figure D.14
DC sweep analysis dialog box.

To run DC sweep analysis, click **Analysis/Simulate**. *Schematics* will create a netlist and then run *PSpice* if no errors are found. If errors are found in the schematic, check for them in the *Error List* and correct them as usual. If no errors are found, the data generated by *PSpice* is passed to *Orcad PSpice*. The *Orcad PSpice* window will appear, displaying a graph in which the X axis is by default set to the DC sweep variable and range, and the Y axis is blank for now. To display some specific plots, click **Trace/Add** in the *Orcad PSpice* menu to open the *Add Traces* dialog box. The box contains traces, which are the output variables (node voltages and branch currents) in the data file available for display. Select the traces to be displayed by clicking or typing them, and click **OK**. The selected traces will be plotted and displayed on the

screen. As many traces as you want may be added to the same plot or on different windows. Select a new window by clicking **Window/New**. To delete a trace, click the trace name in the legend of the plot to highlight it and click **Edit/Delete** (or press <Delete>).

It is important to understand how to interpret the traces. We must interpret the voltage and current variables according to the passive sign convention. As parts are initially placed horizontally in a schematic as shown typically in Fig. D.3, the left-hand terminal is named pin 1 while the right-hand terminal is pin 2. When a component (say R1) is rotated counterclockwise once, pin 2 would be on the top, since rotation is about pin 1. Therefore, if current enters through pin 2, the current I(R1) through R1 would be negative. In other words, positive current implies that the current enters through pin 1, and negative current means that the current enters through pin 2. As for voltage variables, they are always with respect to the ground. For example, V(R1:2) is the voltage (with respect to the ground) at pin 2 of resistor R1; V(V1:+) is the voltage (with respect to the ground) at the positive terminal of voltage source V1; and V(E2:1) is the voltage at pin 1 of component E2 with respect to ground, regardless of the polarity.

Example D.2

For the circuit in Fig. D.15, find the dc node voltages and the current i_o.

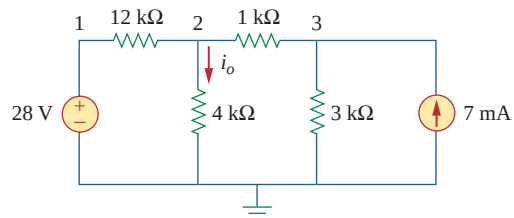


Figure D.15
For Example D.2.

Solution:

We use *Schematics* to create the circuit. After saving the circuit, click **Analysis/Simulate** to simulate the circuit. We obtain the results of the dc analysis from the output file or from the VIEWPOINT AND IPROBE parts, as shown in Fig. D.16. The netlist file is shown in Fig. D.17. Notice that the netlist contains the name, value, and

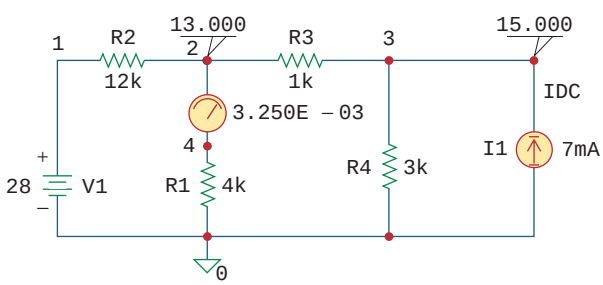


Figure D.16
For Example D.2; schematic for the circuit in Fig. D.15.

```
* Schematics Netlist *

V_V1      1 0 28
R_R1      0 4 4k
R_R2      1 2 12k
R_R3      2 3 1k
R_R4      0 3 3k
I_I1      0 3 DC 7mA
v_V2      2 4 0
```

Figure D.17
The Netlist file for Example D.2.

connection for each element in the circuit. First example, the first line shows that the voltage source V1 has a value of 28 V and is connected between nodes 0 and 1. Figure D.18 shows the edited version of the output file. The output file also contains the Netlist file, but this was removed from Fig. D.18. From IPROBE or the output file, we obtain i_o as 3.25 mA.

```
**** 11/26/99 20:56:05 ********* NT Evaluation PSpice (Nov. 1999) *********

* C:\ MSIMEV63\ examd2.sch

**** CIRCUIT DESCRIPTION

******************************************************************************

* Schematics Version 6.3 - April 1996
* Sat Jul 26 20:56:04 1997

**** INCLUDING examd2.als ****
* Schematics Aliases *

.ALIASES
V_V1       V1(+=1 -=0 )
R_R1       R1(1=0 2=4 )
R_R2       R2(1=1 2=2 )
R_R3       R3(1=2 2=3 )
R_R4       R4(1=0 2=3 )
I_I1       I1(+=0 -=3 )
v_V2       V2(+=2 -=4 )
_ _(1=1)
_ _(2=2)
_ _(3=3)
.ENDALIASES

.probe

.END

NODE VOLTAGE    NODE VOLTAGE     NODE VOLTAGE    NODE VOLTAGE

( 1) 28.0000    ( 2) 13.0000     ( 3) 15.0000    ( 4) 13.0000

VOLTAGE SOURCE CURRENTS
NAME    CURRENT

V_V1     -1.250E-03
v_V2      3.250E-03

TOTAL POWER DISSIPATION 3.50E-02 WATTS
```

Figure D.18
Output file (edited version) for Example D.2.

Practice Problem D.2

Use *PSpice* to determine the node voltages and the current i_x in the circuit of Fig. D.19.

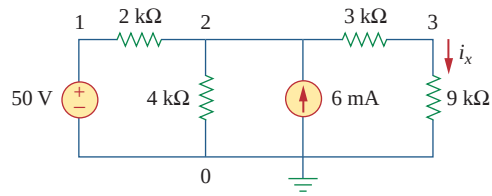


Figure D.19
For Practice Prob. D.2.

Answer: $V_1 = 50$, $V_2 = 37.2$, $V_3 = 27.9$, $i_x = 3.1$ mA.

Example D.3

Plot I_1 and I_2 if the dc voltage source in Fig. D.20 is swept from 2 V to 10 V.

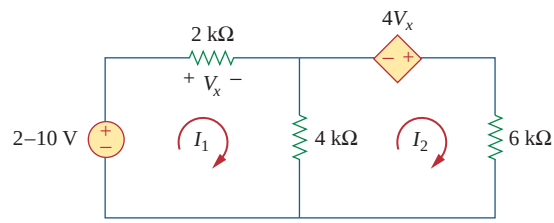


Figure D.20
For Example D.3.

Solution:
We draw the schematic of the circuit and set the attributes as shown in Fig. D.21. Notice how the voltage-controlled voltage source E1 is connected. After completing the schematic, we select **Analysis/Setup** and input the start, end, and increment values as 2, 10, and 0.5, respectively. By selecting **Analysis/Simulate**, we bring up the *Orcad PSpice* window. We select **Trace/Add** and click I(R1) and −I(R3) to

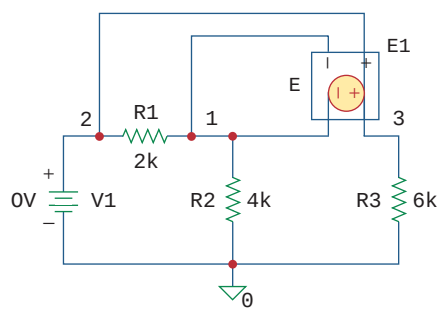


Figure D.21
The schematic for the circuit in Fig. D.20.

be displayed. (The negative sign is needed to make the current through R3 positive.) Figure D.22 shows the result.

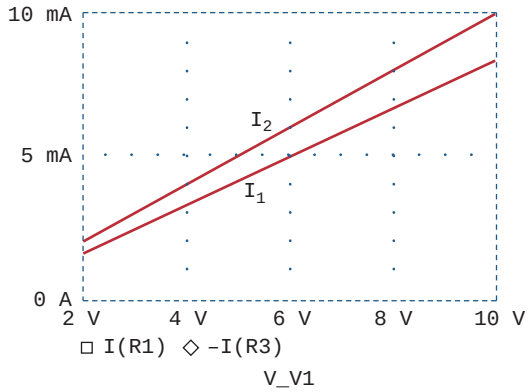


Figure D.22
Plots of I_1 and I_2 against V1.

Practice Problem D.3

Use *PSpice* to obtain the plots of i_x and i_o if the dc voltage source in Fig. D.23 is swept from 2 V to 10 V.

Answer: The plots of i_x and i_o are displayed in Fig. D.24.

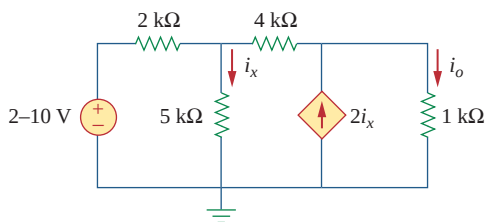


Figure D.23
For Practice Prob. D.3.

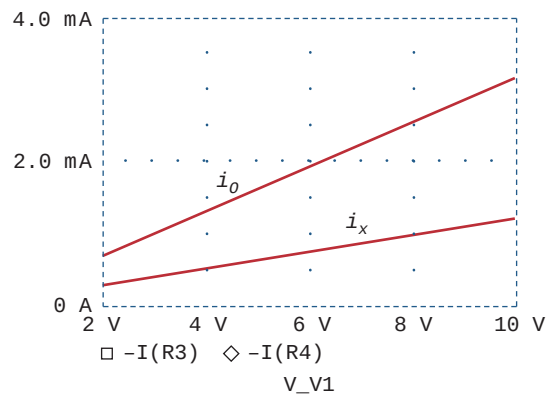


Figure D.24
Plots of i_x and i_o versus V1.

D.4 Transient Analysis

In *PSpice*, transient analysis is generally used to examine the behavior of a waveform (voltage or current) as time varies. Transient analysis solves some differential equations describing a circuit and obtains voltages and currents versus time. Transient analysis is also used to obtain Fourier analysis. To perform transient analysis on a circuit using *PSpice*

Transient analysis is used to view the transient response of inductors and capacitors.

usually involves these steps: (1) drawing the circuit, (2) providing specifications, and (3) simulating the circuit.

1. Drawing the Circuit

In order to run a transient analysis on a circuit, the circuit must first be created using *Schematics* and the source must be specified. *PSpice* has several time-varying functions or sources that enhance the performance of transient analysis. Sources used in the transient analysis include:

- VSIN, ISIN: damped sinusoidal voltage or current source, e.g., $v(t) = 10e^{-0.2t}\sin(120\pi t - 60°)$.
- VPULSE, IPULSE: voltage or current pulse.
- VEXP, IEXP: voltage or current exponential source, e.g., $i(t) = 6[1 - \exp(-0.5t)]$.
- VPWL, IPWL: piecewise linear voltage or current function, which can be used to create an arbitrary waveform.

It is expedient to take a close look at these functions.

VSIN is the exponentially damped sinusoidal voltage source, for example,

$$v(t) = V_o + V_m e^{-\alpha(t-t_d)}\sin[2\pi f(t - t_d) + \phi] \qquad \textbf{(D.1)}$$

The VSIN source has the following attributes, which are illustrated in Fig. D.25 and compared with Eq. (D.1):

$$\begin{aligned} \text{VOFF} &= \text{Offset voltage, } V_o \\ \text{VAMPL} &= \text{Amplitude, } V_m \\ \text{TD} &= \text{Time delay in seconds, } t_d \\ \text{FREQ} &= \text{Frequency in Hz, } f \\ \text{DF} &= \text{Damping factor (dimensionless), } \alpha \\ \text{PHASE} &= \text{Phase in degrees, } \phi \end{aligned} \qquad \textbf{(D.2)}$$

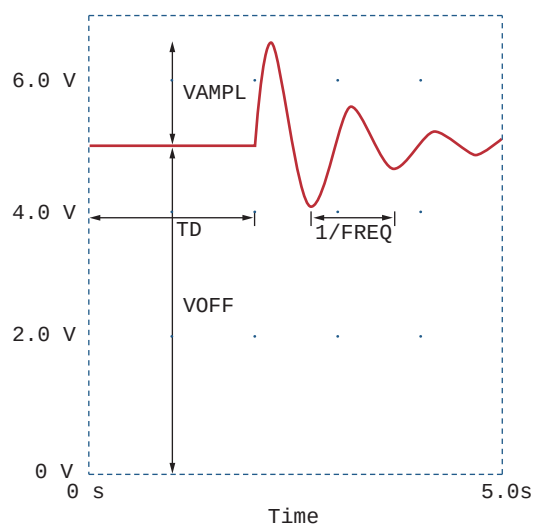


Figure D.25
Sinusoidal voltage source VSIN.

Attributes TD, DF, and PHASE are set to 0 by default but can be assigned other values if necessary. What has been said about VSIN is also true for ISIN.

The VPULSE source has the following attributes, which are portrayed in Fig. D.26:

$$\begin{aligned} \text{V1} &= \text{Low voltage} \\ \text{V2} &= \text{High voltage} \\ \text{TD} &= \text{Initial time delay in seconds} \\ \text{TR} &= \text{Rise time in seconds} \\ \text{TF} &= \text{Fall time in seconds} \\ \text{PW} &= \text{Pulse width in seconds} \\ \text{PER} &= \text{Period in seconds} \end{aligned} \tag{D.3}$$

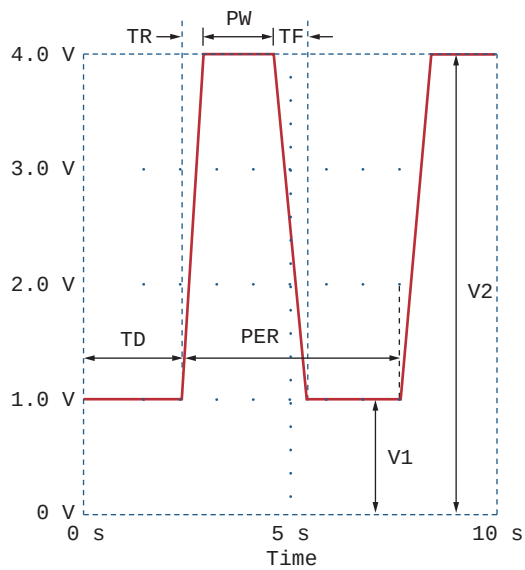


Figure D.26
Pulse voltage source VPULSE.

Attributes V1 and V2 must be assigned values. By default, attribute TD is assigned 0; TR and TF are assigned the *print step* value; and PW and PER are assigned the *final time* value. The values of the *print time* and *final time* are obtained as default values from the specifications provided by the user in the **Transient Analysis/Setup**, to be discussed a little later.

The exponential voltage source VEXP has the following attributes, typically illustrated in Fig. D.27:

$$\begin{aligned} \text{V1} &= \text{Initial voltage} \\ \text{V2} &= \text{Final voltage} \\ \text{TD1} &= \text{Rise delay in seconds} \\ \text{TC1} &= \text{Rise time constant in seconds} \\ \text{TD2} &= \text{Fall delay in seconds} \\ \text{TC2} &= \text{Fall time in seconds} \end{aligned} \tag{D.4}$$

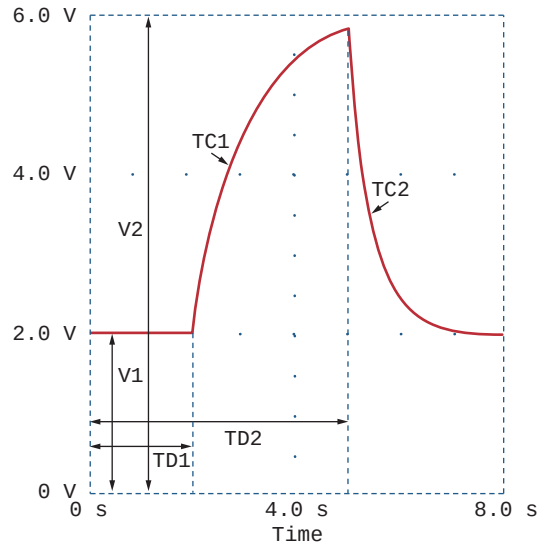


Figure D.27
Exponential voltage source VEXP.

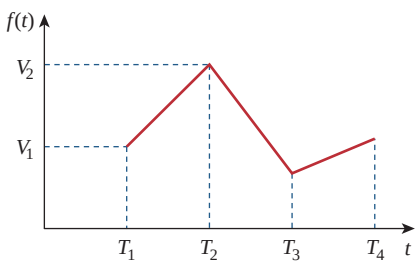


Figure D.28
Piecewise linear voltage source VPWL.

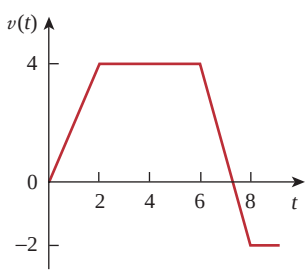


Figure D.29
An example of a piecewise linear voltage source VPWL.

The piecewise linear voltage source VPWL, such as shown in Fig. D.28, requires specifying pairs of TN, VN, where VN is the voltage at time TN for N = 1, 2, . . . , 10. For example, for the function in Fig. D.29, we will need to specify the attributes T1 = 0, V1 = 0, T2 = 2, V2 = 4, T3 = 6, V3 = 4, and T4 = 8, V4 = −2.

To obtain information about other sources, click **Help/Search for Help on** . . . and type in the name of the source. To add a source to the schematic, take the following steps:

1. Select **Draw/Get New Part**.
2. Type the name of the source.
3. Click **OK** and **DRAG** the symbol to the desired location.
4. **DCLICKL** the symbol of the source to open up the *PartName* dialog box.
5. For each attribute, **DCLICKL** on the attribute, enter the value, and click **Save Attr** to accept changes.
6. Click **OK** to accept new attributes.

In step 5, the attributes may not be shown on the schematic after entering their values. To display an attribute, select **Change Display/Both Name and Value** in the *PartName* dialog box.

In addition to specifying the source to be used in transient analysis, there may be need to set initial conditions on capacitors and inductors in the circuit. To do so, **DCLICKL** the part symbol to bring up the *PartName* dialog box, click **IC** = and type in the initial condition. The IC attribute allows for setting the initial conditions on a capacitor or inductor. The default value of IC is 0. The attributes of open/close switches (with part names Sw_tClose and Sw_tOpen) can be changed in a similar manner.

2. Providing Specifications

After the circuit is drawn and the source is specified with its attributes, we need to add some specifications for the transient analysis.

For example, suppose we want the analysis to run the simulation from 0 to 10 ms with a print interval of 2 ns; we enter these specifications as follows:

1. Select **Analysis/Setup/Transient** to open up the *Transient Analysis* dialog box.
2. **CLICKL** *Print Step* and type 2 ns.
3. **CLICKL** *Final Time* and type 10 ms.
4. **CLICKL** *Step Ceiling* and type 5 μs.
5. **CLICKL OK/Close** to accept specifications.

These specifications control the simulation and the display of output variables. *Final Time* specifies how long the simulation should run. In other words, the simulation runs from $t = 0$ to $t =$ Final Time. *Print Step* refers to the time interval the print part will print out; it controls how often simulation results are written to the output file. The value of *Print Step* can be any value less than the *Final Time*, but it cannot be zero. *Step Ceiling* is the maximum time between simulation points; specifying its value is optional. By selecting 10 ms as *Final Time* and 5 μs as *Step Ceiling*, the simulation will have a minimum of 10 ms/5 μs = 2000 points. When *Step Ceiling* is unspecified, *PSpice* selects its own internal time step—the time between simulation points. The time step is selected as large as possible to reduce simulation time. If the user has no idea of what the plot may look like, it is recommended that the value of *Step Ceiling* be unspecified. If the plot is jagged as a result of a large time step assumed by *PSpice*, the user may now specify a *Step Ceiling* that will smooth the plot. Keep in mind that a smaller value gives more points in the simulation but takes more time.

To obtain the Fourier component of a signal, we enable the Fourier option in the *Transient Analysis* dialog box. Chapter 17 has more on this.

3. Simulating the Circuit

After the circuit is drawn, the specifications for the transient analysis are given, and the circuit is saved, we are ready to simulate it. To perform transient analysis, we select **Analysis/Simulate**. If there are no errors, the *Orcad PSpice* window will automatically appear. As usual, the time axis (or X axis) is drawn but no curves are drawn yet. Select **Trace/Add** and click on the variables to be displayed.

An alternative way of displaying the results is to use *markers*. Although there are many types of markers, we will discuss only voltage and current markers. A voltage marker is used to display voltage at a node relative to ground; a current marker is for displaying current through a component pin. To place a voltage marker at a node, take the following steps while in the *Schematics* window:

1. Select **Markers/Mark Voltage/Level**.
2. **DRAG** the voltage marker to the desired node.
3. **CLICKL** to place the marker and **CLICKR** to end the placement mode.

This will cause two things to happen immediately. The voltage marker becomes part of the circuit and the appropriate node voltage is automatically displayed by the *Orcad PSpice* window when the *Schematics* is

run. To place a current marker at a component pin, take the following steps in the *Schematics* window:

1. Select **Markers/Mark current into pin**.
2. **DRAG** the current marker to the desired pin.
3. **CLICKL** to place the marker and **CLICKR** to end the placement mode.

This will automatically add the current through the pin to your graph. It is important that the current marker be placed at the pin of the component; otherwise the system would reject the marker. You can place as many voltage and current markers as you want on a circuit. To remove the markers from the circuit as well as the plots from the *Orcad PSpice* window, select **Markers/Clear All** from *Schematics* window.

Example D.4

Assuming that $i(0) = 10$A, plot the zero-input response $i(t)$ in the circuit of Fig. D.30 for $0 < t < 4$ s using *PSpice*.

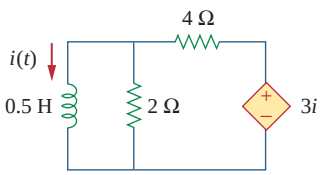


Figure D.30
For Example D.4.

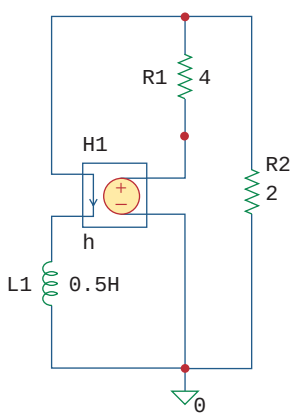


Figure D.31
The schematic of the circuit in Fig. D.30.

Solution:
The circuit is the same as the one for Example 7.3, where we obtained the solution as

$$i(t) = 10e^{(-2/3)t}$$

For *PSpice* analysis, the schematic is in Fig. D.31, where the current-controlled source H1 has been wired to agree with the circuit in Fig. D.30. The voltage of H1 is 3 times the current through inductor L1. Therefore, for H1, we set GAIN = 3 and for the inductor L1, we set the initial condition IC = 10. Using the **Analysis/Setup/Transient** dialog box, we set *Print Step* = 0.25 s and *Final Time* = 4 s. After simulating the circuit, the output is taken as the inductor current $i(t)$, which is plotted in Fig. D.32.

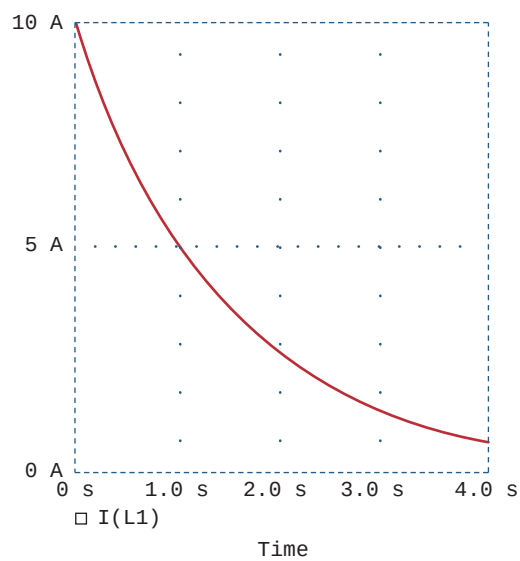


Figure D.32
Output plot for Example D.4.

Practice Problem D.4

Using *PSpice*, plot the source-free response $v(t)$ in the circuit of Fig. D.33, assuming that $v(0) = 10$ V.

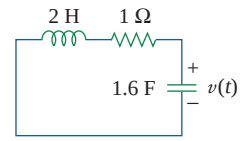


Figure D.33
For Practice Prob. D.4.

Answer: Figure D.34 shows the plot. Note that $v(t) = 10e^{-0.25t}\cos 0.5t + 5e^{-0.25t}\sin 0.5t$ V.

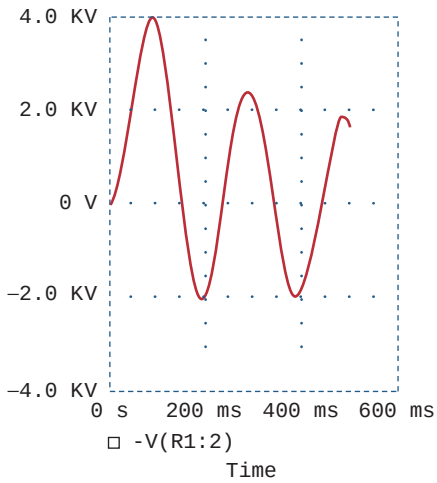


Figure D.34
Output plot for Practice Prob. D.4.

Example D.5

Plot the forced response $v_o(t)$ in the circuit of Fig. D.35(a) for $0 < t < 5$ s if the source voltage is as shown in Fig. D.35(b).

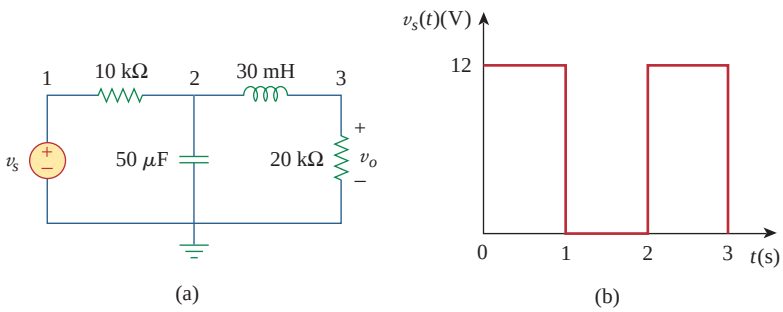


Figure D.35
For Example D.5.

Solution:
We draw the circuit and set the attributes as shown in Fig. D.36. We enter in the data in Fig. D.35(b) by double-clicking the symbol of the voltage source V1 and typing in T1 = 0, V1 = 0, T2 = 1 ns, V2 = 12, T3 = 1 s, V3 = 12, T4 = 1.001 s, V4 = 0, T5 = 2 s, V5 = 0, T6 = 2.001 s, V6 = 12, T7 = 3 s, V7 = 12, T8 = 3.001 s, V8 = 0. In the **Analysis/Setup/Transient** dialog box, we set *Print Step* = 0.2 s and *Final Time* = 5 s. When the circuit is simulated and we are in the *Orcad PSpice* window, we close or minimize the window to go back to the

Schematics window. We place two voltage markers as shown in Fig. D.36 to get the plots of input v_s and output v_o. We press <Alt-Esc> to get into the *Orcad PSpice* window and obtain the plots shown in Fig. D.37.

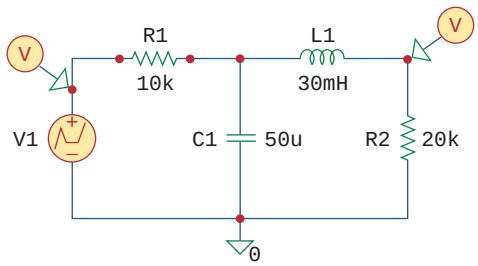


Figure D.36
The schematic of the circuit in Fig. D.35.

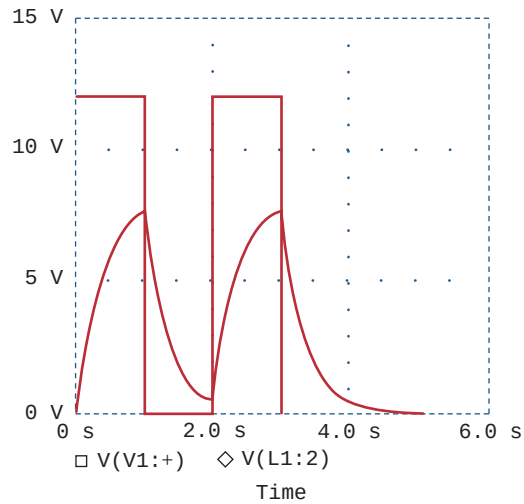


Figure D.37
Output plot for Example D.5.

Practice Problem D.5

Obtain the plot of $v(t)$ in the circuit in Fig. D.38 for $0 < t < 0.5$ s if $i_s = 2e^{-t} \sin 2\pi(5)t$ A.

Answer: See Fig. D.39.

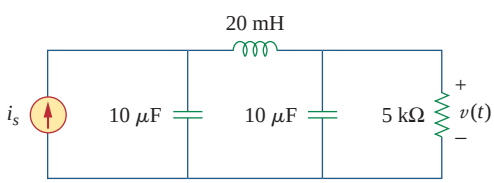


Figure D.38
For Practice Prob. D.5.

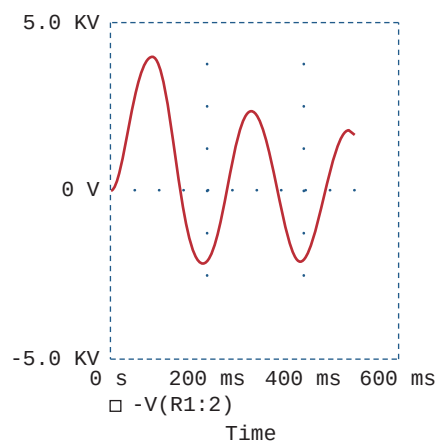


Figure D.39
Output plot for Practice Prob. D.5.

D.5 AC Analysis/Frequency Response

Using AC sweep, *PSpice* can perform AC analysis of a circuit for a single frequency or over a range of frequencies in increments that can vary linearly, by decade, or by octave. In AC sweep, one or more sources

are swept over a range of frequencies while the voltages and currents of the circuit are calculated. Thus, we use AC sweep both for phasor analysis and for frequency response analysis: it will output Bode gain and phase plots. (Keep in mind that a phasor is a complex quantity with real and imaginary parts or with magnitude and phase.)

While transient analysis is done in the time domain, AC analysis is performed in the frequency domain. For example, if $v_s = 10\cos(377t + 40^\circ)$, transient analysis can be used to display v_s as a function of time, whereas AC sweep will give the magnitude as 10 and phase as 40°. To perform AC sweep requires taking three steps similar to those for transient analysis: (1) drawing the circuit, (2) providing specifications, and (3) simulating the circuit.

1. Drawing the Circuit

We first draw the circuit using *Schematics* and specify the source(s). Sources used in AC sweep are AC sources VAC and IAC. The sources and attributes are entered into the *Schematics* as stated in the previous section. For each independent source, we must specify its magnitude and phase.

2. Providing Specifications

Before simulating the circuit, we need to add some specifications for AC sweep. For example, suppose we want a linear sweep at frequencies 50, 100, and 150 Hz. We enter these parameters as follows:

1. Select **Analysis/Setup/AC Sweep** to open up the dialog box for AC Sweep.
2. **CLICKL** *Linear* for the *X* axis to have a linear scale.
3. Type 3 in the *Total Pts* box.
4. Type 50 in the *Start Freq* box.
5. Type 150 in the *End Freq* box.
6. **CLICKL OK/Close** to accept specifications.

A linear sweep implies that simulation points are spread uniformly between the starting and ending frequencies. Note that the *Start Freq* cannot be zero because 0 Hz corresponds to DC analysis. If we want the simulation to be done at a single frequency, we enter 1 in step 3 and the same frequency in steps 4 and 5. If we want the AC sweep to simulate the circuit from 1 Hz to 10 MHz at 10 points per decade, we **CLICKL** on *Decade* in step 2 to make the *X* axis logarithmic, enter 10 in the *Total Pts* box in step 3, enter 1 in the *Start Freq* box, and enter 10Meg in the *End Freq* box. Keep in mind that a decade is a factor of 10. In this case, a decade is from 1 Hz to 10 Hz, from 10 Hz to 100, from 100 to 1 kHz, and so forth.

3. Simulating the Circuit

After providing the necessary specifications and saving the circuit, we perform the AC sweep by selecting **Analysis/Simulate**. If no errors are encountered, the circuit is simulated. At the end of the simulation, the system displays **AC analysis finished** and creates an output file with extension .out. Also, the *Orcad PSpice* program will automatically run if there are no errors. The frequency axis (or *X* axis) is drawn but no curves are shown yet. Select **Trace/Add** from the *Orcad PSpice* menu bar and click on the variables to be displayed. We may also use

current or voltage markers to display the traces as explained in the previous section. To use advanced markers such as *vdb*, *idb*, *vphase*, *iphase*, *vreal*, and *ireal*, select **Markers/Mark Advanced**.

In case the resolution of the trace is not good enough, we may need to check the data points to see if they are enough. To do so, select **Tools/Options/Mark Data Points/OK** in the *Orcad PSpice* menu and the data points will be displayed. If necessary, we can improve the resolution by increasing the value of the entry in the **Total Pts** box in the **Analysis/Setup/AC Sweep and Noise Analysis** dialog box for AC sweep.

To generate Bode plots involves using the AC sweep and the dB command in *Orcad PSpice*.

Bode plots are separate plots of magnitude and phase versus frequency. To obtain Bode plots, it is common to use an AC source, say V1, with 1 volt magnitude and zero phase. After we have selected **Analysis/Simulate** and have the *Orcad PSpice* program running, we can display the magnitude and phase plots as mentioned above. Suppose we want to display a Bode magnitude plot of V(4). We select **Trace/Add** and type **dB(V(4))** in **Trace Command** box. dB(V(4)) is equivalent to 20log(V(4)), and because the magnitude of V1 or V(R1:1) is unity, dB(V(4)) actually corresponds to dB(V(4)/V(R1:1)), which is the gain. Adding the trace dB(V(4)) will give a Bode magnitude/gain plot with the *Y* axis in dB.

Once a plot is obtained in the *Orcad PSpice* window, we can add labels to it for documentation purposes. To add a title to the plot, select **Edit/Modify Title** in the *Orcad PSpice* menu and type the title in the dialog box. To add a label to the *Y* axis, select **Plot/Y Axis Settings**, type the label, and **CLICKL OK**. Add a label to the *X* axis in the same manner.

As an alternative approach, we can avoid running the *Orcad PSpice* program by using *pseudocomponents* to send results to the output file. Pseudocomponents are like parts that can be inserted into a schematic as if they were circuit elements, but they do not correspond to circuit elements. We can add them to the circuit for specifying initial conditions or for output control. In fact, we have already used two pseudocomponents, VIEWPOINT and IPROBE, for DC analysis. Other important pseudocomponents and their usage are shown in Fig. D.40 and listed in Table D.2. The pseudocomponents are added to the schematic. To add a pseudocomponent, select **Draw/Get New Parts** in the *Schematics* window, select the pseudocomponent, place it at the desired location, and add the appropriate attributes as usual. Once the pseudocomponents are added to the schematic, we select **Analysis/Setup/AC Sweep** and enter the specifications for the AC sweep, and finally, select **Analysis/Simulate** to perform AC sweep. If no errors are encountered, the voltages and currents specified in the pseudocomponents will be saved in the output file. We obtain the output file by selecting **Analysis/Examine Output** in the *Schematics* window.



Figure D.40
Print and plot pseudocomponents.

TABLE D.2

Print and plot pseudocomponents.

Symbol	Description
IPLOT	Plot showing branch current; symbol must be placed in series
IPRINT	Table showing branch current; symbol must be placed in series
VPLOT1	Plot showing voltages at the node to which the symbol is connected
VPLOT2	Plot showing voltage differentials between two points to which the symbol is connected
VPRINT1	Table showing voltages at the node to which the symbol is connected
VPRINT2	Table showing voltage differentials between two points to which the symbol is connected

Example D.6

Find current i in the circuit in Fig. D.41.

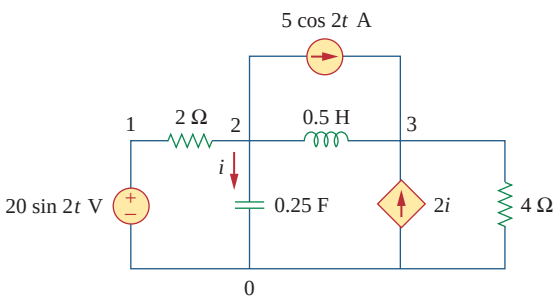


Figure D.41
For Example D.6.

Solution:

Recall that $20 \sin 2t = 20 \cos(2t - 90°)$ and that $f = \omega/2\pi = 2/2\pi = 0.31831$. The schematic is shown in Fig. D.42. The attributes of V1 are set as *ACMAG* = 20, *ACPHASE* = −90; while the attributes of IAC are set as AC = 5. The current-controlled current source is connected in such a way as to conform with the original circuit in Fig. D.41; its gain is set equal to 2. The attributes of the pseudocomponent IPRINT are set as *AC* = yes, *MAG* = yes, *PHASE* = ok, *REAL* =, and *IMAG* =. Since this is a single-frequency ac analysis, we select **Analysis/Setup/AC Sweep** and enter *Total Pts* = 1, *Start Freq* = 0.31831, and *Final Freq* = 0.31831. We save the circuit and select **Analysis/Simulate** for simulation. The output file includes

```
FREQ          IM(V_PRINT3)     IP(V_PRINT3)

3.183E-01     7.906E+00        4.349E+01
```

From the output file, we obtain $I = 7.906\angle 43.49°$ A or $i(t) = 7.906 \cos(2t + 43.49°)$ A. This example is for a single-frequency ac analysis; Example D.7 is for AC sweep over a range of frequencies.

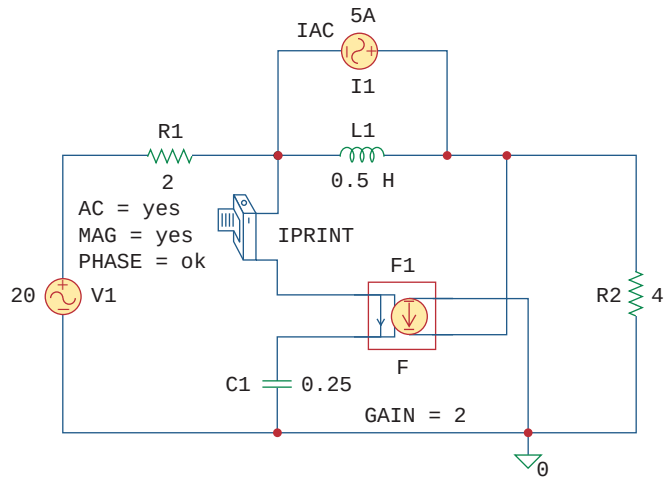


Figure D.42
The schematic of the circuit in Fig. D.41.

Practice Problem D.6

Find $i_x(t)$ in the circuit in Fig. D.43.

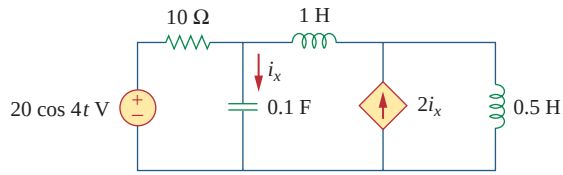


Figure D.43
For Practice Prob. D.6.

Answer: From the output file, $I_x = 7.59\angle 108.43°$ or $i_x = 7.59\cos(4t + 108.43°)$ A.

Example D.7

For the *RC* circuit shown in Fig. D.44, obtain the magnitude plot of the output voltage v_o for frequencies from 1 Hz to 10 kHz. Let $R = 1\ \text{k}\Omega$ and $C = 4\ \mu\text{F}$.

Solution:
The schematic is shown in Fig. D.45. We assume that the magnitude of V1 is 1 and its phase is zero; we enter these as the attributes of V1. We

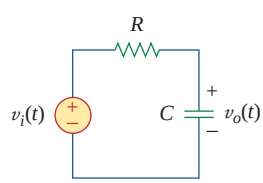


Figure D.44
For Example D.7.

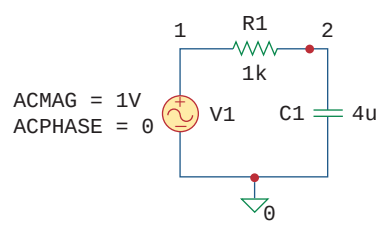


Figure D.45
The schematic of the circuit in Fig. D.44.

also assume 10 points per decade. For the AC sweep specifications, we select **Analysis/Setup/AC Sweep** and enter 10 in the *Total Pts* box, 1 in the *Start Freq* box, and 10k in the *Final Freq* box. After saving the circuit, we select **Analysis/Simulate**. From the *Orcad PSpice* menu, we obtain the plot in Fig. D.46(a) by selecting **Traces/Add** and clicking V(2). Also, by selecting **Trace/Add** and typing dB(V(2)) in the *Trace Command* box, we obtain the Bode plot in Fig. D.46(b). The two plots in Fig. D.46 indicate that the circuit is a lowpass filter: low frequencies are passed while high frequencies are blocked by the circuit.

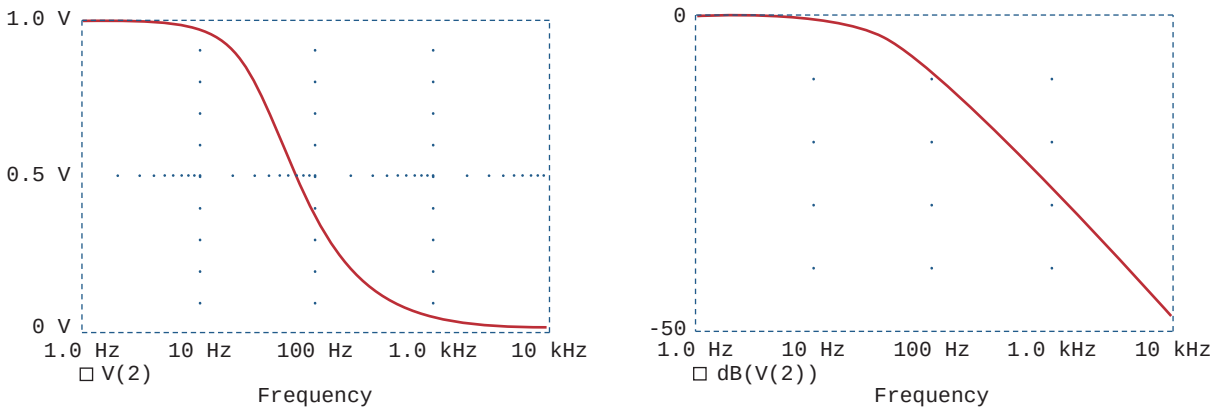


Figure D.46
Result of Example D.7: (a) linear, (b) Bode plot.

Practice Problem D.7

For the circuit in Fig. D.44, replace the capacitor C with an inductor $L = 4$ mH and obtain the magnitude plot (both linear and Bode) for v_o for $10 < f < 100$ MHz.

Answer: See the plots in Fig. D.47.

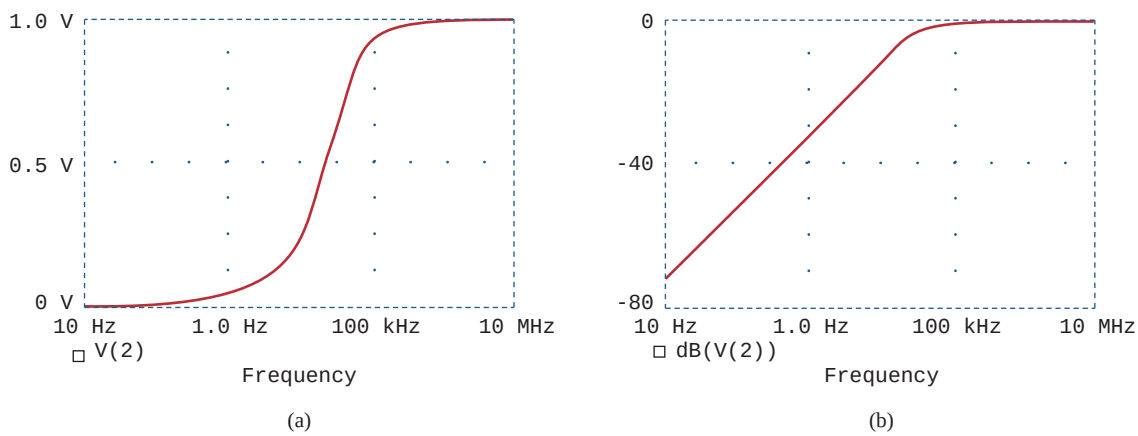


Figure D.47
Result of Practice Prob. D.7: (a) linear plot, (b) Bode plot.

Appendix E

MATLAB

MATLAB has become a powerful tool of technical professionals worldwide. The term *MATLAB* is an abbreviation for *MAT*rix *LAB*oratory, implying that *MATLAB* is a computational tool that uses matrices and vectors (or arrays) to carry out numerical analysis, signal processing, and scientific visualization tasks. Because *MATLAB* uses matrices as its fundamental building blocks, one can write mathematical expressions involving matrices just as easily as one would on paper. *MATLAB* is available for Macintosh, Unix, and Windows operating systems. A student version of *MATLAB* is available for personal computers (PCs). A copy of *MATLAB* can be obtained from

The Mathworks, Inc.
3 Apple Hill Drive
Natick, MA 01760-2098
Phone:(508) 647-7000
Website: http://www.mathworks.com

A brief introduction to *MATLAB* is presented in this appendix. What is presented is sufficient for solving problems in this book. More about *MATLAB* can be found in *MATLAB* books and from on-line help. The best way to learn *MATLAB* is to work with it after having learned the basics.

E.1 *MATLAB* Fundamentals

The Command window is the primary area where you interact with *MATLAB*. A little later, we will learn how to use the text editor to create M-files, which allow for execution of sequences of commands. For now, we focus on how to work in the Command window. We will first learn how to use *MATLAB* as a calculator.

Using *MATLAB* as a Calculator

The following are algebraic operators used in *MATLAB*:

`+`	Addition
`-`	Subtraction
`*`	Multiplication
`^`	Exponentiation
`/`	Right division (`a/b` means $a \div b$)
`\`	Left division (`a\b` means $b \div a$)

To begin to use *MATLAB*, we use these operators. Type commands to the *MATLAB* prompt "`>>`" in the Command window

(correct any mistakes by backspacing) and press the Enter key. For example,

```
>>  a = 2; b = 4;c = -6;
>>  dat = b^2 - 4*a*c
dat =
    64
>>  e = sqrt(dat)/10
e =
    0.8000
```

The first command assigns the values 2, 4, and −6 to the variables `a`, `b`, and `c`, respectively. *MATLAB* does not respond because this line ends with a colon. The second command sets `dat` to $b^2 - 4ac$ and *MATLAB* returns the answer as 64. Finally, the third line sets `e` equal to the square root of `dat` and divides by 10. *MATLAB* prints the answer as 0.8. Other mathematical functions, listed in Table E.1, can be used similarly to how the function `sqrt` is used here. Table E.1 provides just a tiny sample of *MATLAB* functions. Others can be obtained from the on-line help. To get help, type

```
>> help
```

A long list of topics will come up. For a specific topic, type the command name. For example, to get help on "log to base 2," type

```
>> help log2
```

A help message on the log function will be displayed. Note that *MATLAB* is case sensitive, so `sin(a)` is not the same as `sin(A)`.

TABLE E.1

Typical elementary math functions.

Function	Remark
`abs(x)`	Absolute value or complex magnitude of `x`
`acos, acosh(x)`	Inverse cosine and inverse hyperbolic cosine of `x` in radians
`acot, acoth(x)`	Inverse cotangent and inverse hyperbolic cotangent of `x` in radians
`angle(x)`	Phase angle (in radian) of a complex number `x`
`asin, asinh(x)`	Inverse sine and inverse hyperbolic sine of `x` in radians
`atan, atanh(x)`	Inverse tangent and inverse hyperbolic tangent of `x` in radians
`conj(x)`	Complex conjugate of `x`
`cos, cosh(x)`	Cosine and hyperbolic cosine of `x` in radians
`cot, coth(x)`	Cotangent and hyperbolic cotangent of `x` in radians
`exp(x)`	Exponential of `x`
`fix`	Round toward zero
`imag(x)`	Imaginary part of a complex number `x`
`log(x)`	Natural logarithm of `x`
`log2(x)`	Logarithm of `x` to base 2
`log10(x)`	Common logarithms (base 10) of `x`
`real(x)`	Real part of a complex number `x`
`sin, sinh(x)`	Sine and hyperbolic sine of `x` in radians
`sqrt(x)`	Square root of `x`
`tan, tanh`	Tangent and hyperbolic tangent of `x` in radians

Try the following examples:

```
>> 3^(log10(25.6))
>> y = 2* sin(pi/3)
>> exp(y+4-1)
```

In addition to operating on mathematical functions, *MATLAB* allows one to work easily with vectors and matrices. A vector (or array) is a special matrix with one row or one column. For example,

```
>> a = [1 -3 6 10 -8 11 14];
```

is a row vector. Defining a matrix is similar to defining a vector. For example, a 3×3 matrix can be entered as

```
>> A = [1 2 3; 4 5 6; 7 8 9]
```

or as

```
>> A = [ 1 2 3
         4 5 6
         7 8 9]
```

In addition to the arithmetic operations that can be performed on a matrix, the operations in Table E.2 can be implemented.

Using the operations in Table E.2, we can manipulate matrices as follows:

```
>> B = A'
B =
       1   4   7
       2   5   8
       3   6   9
>> C = A + B
C =
       2   6  10
       6  10  14
      10  14  18
>> D = A^3 - B*C
D =
     372   432   492
     948  1131  1314
    1524  1830  2136
>> e = [1 2; 3 4]
e =
   1 2
   3 4
>> f = det(e)
f =
   -2
>> g = inv(e)
g =
   -2.0000   1.0000
    1.5000  -0.5000
```

TABLE E.2

Matrix operations.

Operation	Remark
`A'`	Finds the transpose of matrix A
`det(A)`	Evaluates the determinant of matrix A
`inv(A)`	Calculates the inverse of matrix A
`eig(A)`	Determines the eigenvalues of matrix A
`diag(A)`	Finds the diagonal elements of matrix A

TABLE E.3

Special matrices, variables, and constants.

Matrix, Variable, Constant	Remark
`eye`	Identity matrix
`ones`	An array of 1s
`zeros`	An array of 0s
`i or j`	Imaginary unit or `sqrt(-1)`
`pi`	3.142
`NaN`	Not a number
`inf`	Infinity
`eps`	A very small number, `2.2e - 16`
`rand`	Random element

```
>> H = eig(g)
H =
   -2.6861
    0.1861
```

Note that not all matrices can be inverted. A matrix can be inverted if and only if its determinant is nonzero. Special matrices, variables, and constants are listed in Table E.3. For example, type

```
>> eye(3)
ans =
     1  0  0
     0  1  0
     0  0  1
```

to get a 3×3 identity matrix.

Plotting

To plot using *MATLAB* is easy. For a two-dimensional plot, use the plot command with two arguments as follows:

```
>> plot(xdata,ydata)
```

where `xdata` and `ydata` are vectors of the same length containing the data to be plotted.

For example, suppose we want to plot `y = 10*sin(2*pi*x)` from `0` to `5*pi`. We will proceed with the following commands:

```
% x is a vector, 0 <= x <= 5*pi, increments of pi/100
% creates a vector y
% creates the plot
```

```
>> x = 0:pi/100:5*pi;
>> y = 10*sin(2*pi*x);
>> plot(x,y);
```

With this, *MATLAB* responds with the plot in Fig. E.1.

MATLAB will let you graph multiple plots together and distinguish them with different colors. This is obtained with the format `plot(xdata, ydata, 'color')`, where the color is indicated by using a character string from the options listed in Table E.4.

For example,

```
>> plot(x1,y1,'r',x2,y2,'b',x3,y3,'--');
```

will graph data `(x1, y1)` in red, data `(x2, y2)` in blue, and data `(x3, y3)` in dashed line all on the same plot.

TABLE E.4

Various color and line types.

y	Yellow	.	Point
m	Magenta	o	Circle
c	Cyan	x	x mark
r	Red	+	Plus
g	Green	-	Solid
b	Blue	*	Star
w	White	:	Dotted
k	Black	-.	Dashdot
		--	Dashed

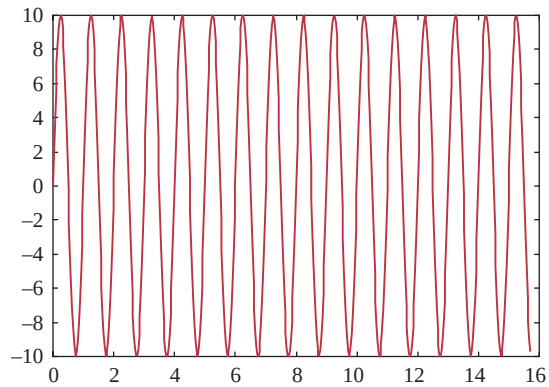


Figure E.1
MATLAB plot of `y = 10*sin(2*pi*x)`.

MATLAB also allows for logarithm scaling. Rather than using the plot command, we use

`loglog log(y)` versus `log(x)`
`semilogx y` versus `log(x)`
`semilogy log(y)` versus `x`

Three-dimensional plots are drawn using the functions mesh and meshdom (mesh domain). For example, to draw the graph of `z = x*exp( - x^2 - y^2)` over the domain `-1 < x, y < 1`, we type the following commands:

```
>> xx = -1:.1:1;
>> yy = xx;
>> [x,y] = meshgrid(xx,yy);
>> z = x.*exp(-x.^2 -y.^2);
>> mesh(z);
```

(The dot symbol used in `x.` and `y.` allows element-by-element multiplication.) The result is shown in Fig. E.2.

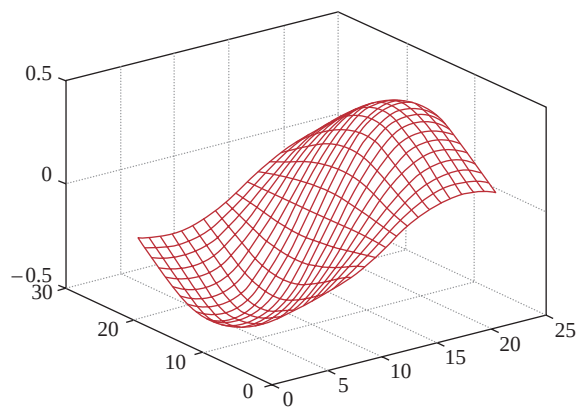


Figure E.2
A three-dimensional plot.

Programming *MATLAB*

So far we have used *MATLAB* as a calculator. You can also use *MATLAB* to create your own program. The command line editing in *MATLAB* can be inconvenient if one has several lines to execute. To avoid this problem, you can create a program that is a sequence of statements to be executed. If you are in the Command window, click **File/New/M**-files to open a new file in the *MATLAB* Editor/Debugger or simple text editor. Type the program and save it in a file with an extension `.m`, say `filename.m`; it is for this reason that it is called an M-file. Once the program is saved as an M-file, exit the Debugger window. You are now back in the Command window. Type the file without the extension `.m` to get results. For example, the plot that was made in Fig. E.2 can be improved by adding title and labels and being typed as an M-file called `example1.m`.

```
% x is a vector, 0 <= x <= 5*pi, increments of pi/100
% creates a vector y
% create the plot
% label the x axis
% label the y axis
% title the plot
% add grid
```

```
x = 0:pi/100:5*pi;
y = 10*sin(2*pi*x);
plot(x,y);
xlabel('x (in radians)');
ylabel('10*sin(2*pi*x)');
title('A sine functions');
grid
```

Once the file is saved as `example1.m` and you exit the text editor, type

```
>> example1
```

in the Command window and hit **Enter** to obtain the result shown in Fig. E.3.

To allow flow control in a program, certain relational and logical operators are necessary. They are shown in Table E.5. Perhaps the most commonly used flow control statements are `for` and `if`. The `for`

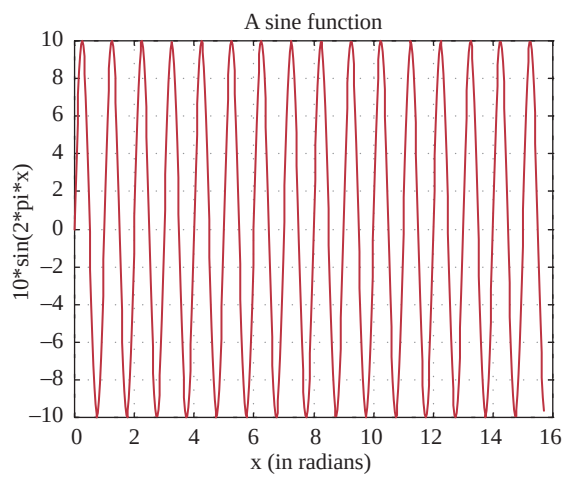


Figure E.3
MATLAB plot of `y = 10*sin(2*pi*x)` with title and labels.

TABLE E.5
Relational and logical operators.

Operator	Remark
<	less than
<=	less than or equal
>	greater than
>=	greater than or equal
==	equal
~=	not equal
&	and
|	or
~	not

statement is used to create a loop or a repetitive procedure and has the general form

```
for x = array
   [commands]
end
```

The `if` statement is used when certain conditions need to be met before an expression is executed. It has the general form

```
if expression
   [commands if expression is True]
else
   [commands if expression is False]
end
```

For example, suppose we have an array `y(x)` and we want to determine the minimum value of `y` and its corresponding index `x`. This can be done by creating an M-file as shown here.

```
% example2.m
% This program finds the minimum y value and
  its corresponding x index
x = [1 2 3 4 5 6 7 8 9 10]; %the nth term in y
y = [3 9 15 8 1 0 -2 4 12 5];
min1 = y(1); for k = 1:10
   min2 = y(k);
   if(min2 < min1)
      min1 = min2;
      xo = x(k);
   else
      min1 = min1;
   end
end
diary
min1, xo
diary off
```

Note the use of the `for` and `if` statements. When this program is saved as `example2.m`, we execute it in the Command window and obtain the minimum value of `y` as `-2` and the corresponding value of `x` as 7, as expected.

```
>> example2
min1 =
  -2
xo =
   7
```

If we are not interested in the corresponding index, we could do the same thing using the command

```
>> min(y)
```

The following tips are helpful in working effectively with *MATLAB*:

- Comment your M-file by adding lines beginning with a % character.
- To suppress output, end each command with a semicolon (;); you may remove the semicolon when debugging the file.
- Press the up and down arrow keys to retrieve previously executed commands.
- If your expression does not fit on one line, use an ellipse (. . .) at the end of the line and continue on the next line. For example, *MATLAB* considers

```
y = sin(x + log10(2x + 3)) + cos(x + ...
log10(2x + 3));
```

  as one line of expression.
- Keep in mind that variable and function names are case sensitive.

Solving Equations

Consider the general system of n simultaneous equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

or in matrix form

$$\mathbf{AX} = \mathbf{B}$$

where

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & a_{n3} & a_{n4} \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \cdots \\ b_n \end{bmatrix}$$

A is a square matrix and is known as the coefficient matrix, while **X** and **B** are vectors. **X** is the solution vector we are seeking to get. There are two ways to solve for **X** in *MATLAB*. First, we can use the back-slash operator(\) so that

```
X = A\B
```

Second, we can solve for **X** as

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$$

which in *MATLAB* is the same as

```
X = inv(A)*B
```

Example E.1

Use *MATLAB* to solve Example A.2.

Solution:
From Example A.2, we obtain matrix **A** and vector **B** and enter them in *MATLAB* as follows.

```
>> A = [25 -5 -20; -5 10 -4; -5 -4 9]
A =
   25  -5  -20
   -5  10   -4
   -5  -4    9
>> B = [50 0 0]'
B =
   50
    0
    0
>> X = inv(A)*B
X =
   29.6000
   26.0000
   28.0000
>> X = A\B
X =
   29.6000
   26.0000
   28.0000
```

Thus, $x_1 = 29.6$, $x_2 = 26$, and $x_3 = 28$.

Practice Problem E.1

Solve the problem in Practice Prob. A.2 using *MATLAB*.

Answer: $x_1 = 3 = x_3$, $x_2 = 2$.

E.2 DC Circuit Analysis

There is nothing special in applying *MATLAB* to resistive dc circuits. We apply mesh and nodal analysis as usual and solve the resulting simultaneous equations using *MATLAB* as is described in Section E.1. Examples E.2 to E.5 illustrate.

Example E.2

Use nodal analysis to solve for the nodal voltages in the circuit of Fig. E.4.

Solution:
At node 1,

$$2 = \frac{V_1 - V_2}{4} + \frac{V_1 - 0}{8} \rightarrow 16 = 3V_1 - 2V_2 \qquad \textbf{(E.2.1)}$$

At node 2,

$$3I_x = \frac{V_2 - V_1}{4} + \frac{V_2 - V_3}{2} + \frac{V_2 - V_4}{2}$$

But

$$I_x = \frac{V_4 - V_3}{4}$$

so that

$$3\left(\frac{V_4 - V_3}{4}\right) = \frac{V_2 - V_1}{4} + \frac{V_2 - V_3}{2} + \frac{V_2 - V_4}{2} \rightarrow \quad \textbf{(E.2.2)}$$

$$0 = -V_1 + 5V_2 + V_3 - 5V_4$$

At node 3,

$$3 = \frac{V_3 - V_2}{2} + \frac{V_3 - V_4}{4} \rightarrow 12 = -2V_2 + 3V_3 - V_4 \quad \textbf{(E.2.3)}$$

At node 4,

$$0 = 2 + \frac{V_4 - V_2}{2} + \frac{V_4 - V_3}{4} \rightarrow -8 = -2V_2 - V_3 + 3V_4 \quad \textbf{(E.2.4)}$$

Combining Eqs. (E.2.1) to (E.2.4) gives

$$\begin{bmatrix} 3 & -2 & 0 & 0 \\ -1 & 5 & 1 & -5 \\ 0 & -2 & 3 & -1 \\ 0 & -2 & -1 & 3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 16 \\ 0 \\ 12 \\ -8 \end{bmatrix}$$

or

$$\mathbf{AV} = \mathbf{B}$$

We now use *MATLAB* to determine the nodal voltages contained in vector **V**.

```
>> A = [ 3 -2  0  0;
        -1  5  1 -5;
         0 -2  3 -1;
         0 -2 -1  3];
>> B = [16  0 12 -8]';
>> V = inv(A)*B
V =
    -6.0000
   -17.0000
   -13.5000
   -18.5000
```

Hence $V_1 = -6.0$, $V_2 = -17$, $V_3 = -13.5$, and $V_4 = -18.5$ V.

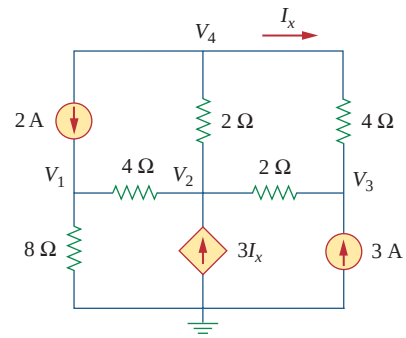


Figure E.4
For Example E.2.

Practice Problem E.2

Find the nodal voltages in the circuit in Fig. E.5 using *MATLAB*.

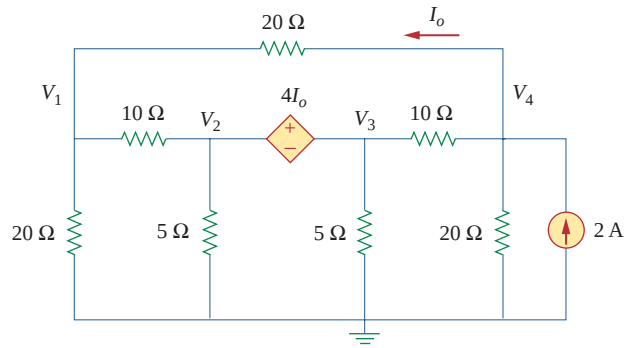


Figure E.5
For Practice Prob. E.2.

Answer:
$V_1 = 14.55$, $V_2 = 38.18$, $V_3 = -34.55$, and $V_4 = -3.636$ V.

Example E.3

Use *MATLAB* to solve for the mesh currents in the circuit in Fig. E.6.

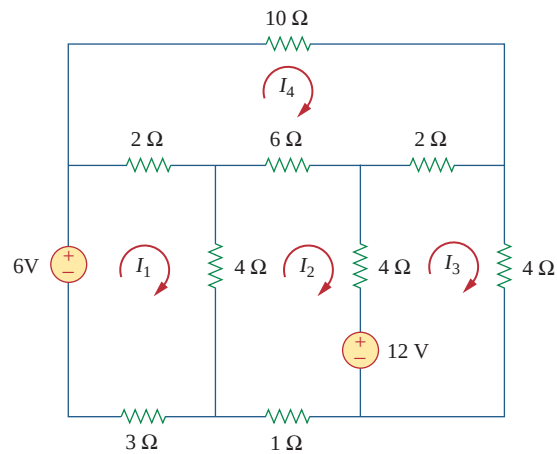


Figure E.6
For Example E.3.

Solution:
For the four meshes,

$$-6 + 9I_1 - 4I_2 - 2I_4 = 0 \longrightarrow 6 = 9I_1 - 4I_2 - 2I_4 \quad \textbf{(E.3.1)}$$

$$12 + 15I_2 - 4I_1 - 4I_3 - 6I_4 = 0 \longrightarrow$$

$$-12 = -4I_1 + 15I_2 - 4I_3 - 6I_4 \quad \textbf{(E.3.2)}$$

$$-12 + 10I_3 - 4I_2 - 2I_4 = 0 \longrightarrow 12 = -4I_2 + 10I_3 - 2I_4 \quad \textbf{(E.3.3)}$$

$$20I_4 - 2I_1 - 6I_2 - 2I_3 = 0 \longrightarrow 0 = -2I_1 - 6I_2 - 2I_3 + 20I_4 \quad \textbf{(E.3.4)}$$

Putting Eqs. (E.3.1) to (E.3.4) together in matrix form, we have

$$\begin{bmatrix} 9 & -4 & 0 & -2 \\ -4 & 15 & -4 & -6 \\ 0 & -4 & 10 & -2 \\ -2 & -6 & -2 & 20 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 6 \\ -12 \\ 12 \\ 0 \end{bmatrix}$$

or **AI** = **B**, where the vector **I** contains the unknown mesh currents. We now use *MATLAB* to determine **I** as follows:

```
>> A = [9 -4  0 -2; -4 15 -4 -6;
        0 -4 10 -2;  -2 -6 -2 20]
A =
     9  -4   0  -2
    -4  15  -4  -6
     0  -4  10  -2
    -2  -6  -2  20
>> B = [6 -12 12 0]'
B =
     6
   -12
    12
     0
>> I = inv(A)*B
I =
    0.5203
   -0.3555
    1.0682
    0.0522
```

Thus, $I_1 = 0.5203$, $I_2 = -0.3555$, $I_3 = 1.0682$, and $I_4 = 0.0522$ A.

Practice Problem E.3

Find the mesh currents in the circuit in Fig. E.7 using *MATLAB*.

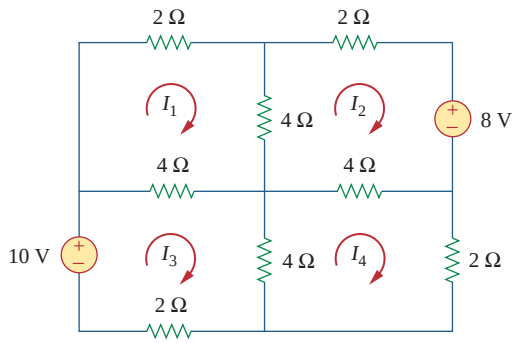


Figure E.7
For Practice Prob. E.3.

Answer: $I_1 = 0.2222$, $I_2 = -0.6222$, $I_3 = 1.1778$, and $I_4 = 0.2222$ A.

E.3 AC Circuit Analysis

Using *MATLAB* in ac circuit analysis is similar to how *MATLAB* is used for dc circuit analysis. We must first apply nodal or mesh analysis to the circuit and then use *MATLAB* to solve the resulting system of equations. However, the circuit is in the frequency domain, and we are dealing with phasors or complex numbers. So in addition to what we learned in Section E.2, we need to understand how *MATLAB* handles complex numbers.

MATLAB expresses complex numbers in the usual manner, except that the imaginary part can be either j or i representing $\sqrt{-1}$. Thus, $3 - j4$ can be written in *MATLAB* as `3 - j4`, `3 - j*4`, `3 - i4`, or `3 - I*4`. Here are the other complex functions:

`abs(A)`	Absolute value of magnitude of `A`
`angle(A)`	Angle of `A` in radians
`conj(A)`	Complex conjugate of `A`
`imag(A)`	Imaginary part of `A`
`real(A)`	Real part of `A`

Keep in mind that an angle in radians must be multiplied by $180/\pi$ to convert it to degrees, and vice versa. Also, the transpose operator (`'`) gives the complex conjugate transpose, whereas the dot-transpose (`.'`) transposes an array without conjugating it.

Example E. 4

In the circuit of Fig. E.8, let $v = 4\cos(5t - 30°)$ V and $i = 0.8\cos 5t$ A. Find v_1 and v_2.

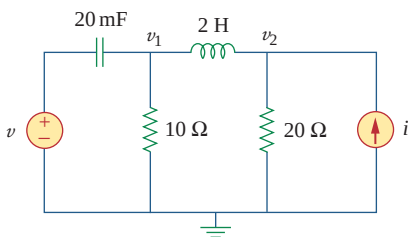


Figure E.8
For Example E.4.

Solution:
As usual, we convert the circuit in the time-domain to its frequency-domain equivalent.

$$v = 4\cos(5t - 30°) \longrightarrow \mathbf{V} = 4\angle -30°, \quad \omega = 5$$
$$i = 0.8\cos 5t \longrightarrow \mathbf{I} = 8\angle 0°$$
$$2\text{ H} \longrightarrow j\omega L = j5 \times 2 = j10$$
$$20\text{ mF} \longrightarrow \frac{1}{j\omega C} = \frac{1}{\text{j10 }\Omega \times 10^{-3}} = -j10$$

Thus, the frequency-domain equivalent circuit is shown in Fig. E.9. We now apply nodal analysis to this.

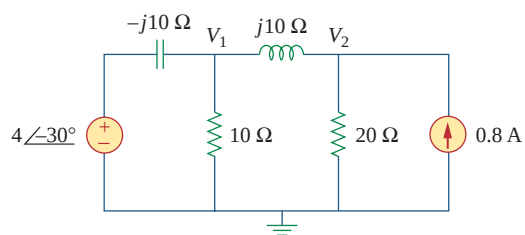


Figure E.9
The frequency-domain equivalent circuit of the circuit in Fig. E.8.

At node 1,

$$\frac{4\angle -30^\circ - V_1}{-j10} = \frac{V_1}{10} + \frac{V_1 - V_2}{j10} \longrightarrow 4\angle -30^\circ = 3.468 - j2 = -jV_1 + V_2 \quad \textbf{(E.4.1)}$$

At node 2,

$$0.8 = \frac{V_2}{20} + \frac{V_2 - V_1}{j10} \longrightarrow j16 = -2V_1 + (2 + j)V_2 \quad \textbf{(E.4.2)}$$

Equations (E.4.1) and (E.4.2) can be cast in matrix form as

$$\begin{bmatrix} -j & 1 \\ -2 & (2+j) \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 3.468 - j2 \\ j16 \end{bmatrix}$$

or $\mathbf{AV} = \mathbf{B}$. We use *MATLAB* to invert $\mathbf{A}$ and multiply the inverse by $\mathbf{B}$ to get $\mathbf{V}$.

```
>> A = [-j 1; -2 (2+j)]
A =
   0 - 1.0000i  1.000
   -2.0000  2.0000 + 1.000 i
>> B = [(3.468 - 2j) 16j].' %note the dot-transpose
B =
   3.4680 - 2.0000i
   0 + 16.0000i
>> V = inv(A)*B
V =
   4.6055 - 2.4403i
   5.9083 + 2.6055i
>> abs(V(1))
ans =
   5.2121
>> angle(V(1))*180/pi %converts angle from
   radians to degrees
ans =
  -27.9175
>> abs(V(2))
ans =
  6.4573
>> angle(V(2))*180/pi
ans =
  23.7973
```

Thus,

$$V_1 = 4.6055 - j2.4403 = 5.212\angle -27.92^\circ$$
$$V_2 = 5.908 + j2.605 = 6.457\angle 23.8^\circ$$

In the time domain,

$$v_1 = 4.605\cos(5t - 27.92°)\text{ V}, \qquad v_2 = 6.457\cos(5t + 23.8°)\text{ V}$$

Practice Problem E.4

Calculate v_1 and v_2 in the circuit in Fig. E.10 given $i = 4\cos(10t + 40°)$ A and $v = 12\cos 10t$ V.

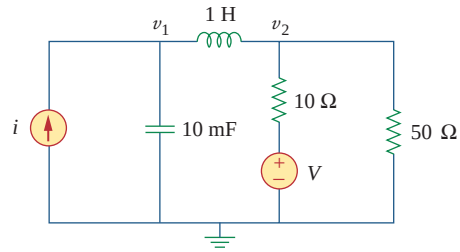


Figure E.10
For Practice Prob. E.4.

Answer: $63.58\cos(10t - 10.68°)$ V, $40\cos(10t - 50°)$ V.

Example E.5

In the unbalanced three-phase system shown in Fig. E.11, find currents I_1, I_2, I_3, and I_{Bb}. Let

$$Z_A = 12 + j10\ \Omega, \qquad Z_B = 10 - j8\ \Omega, \qquad Z_C = 15 + j6\ \Omega$$

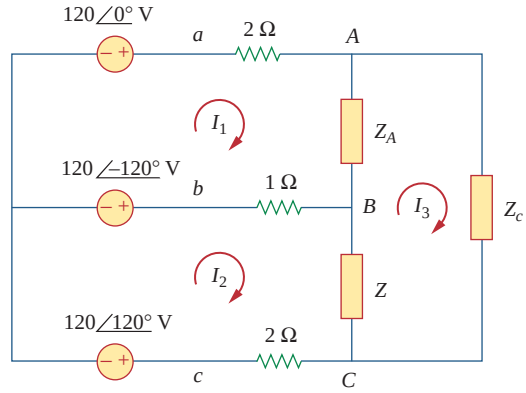


Figure E.11
For Example E.5.

Solution:
For mesh 1,

$$120\angle -120° - 120\angle 0° + I_1(2 + 1 + 12 + j10) - I_2 - I_3(12 + j10) = 0$$

or

$$I_1(15 + j10) - I_2 - I_3(12 + j10) = 120\angle 0° - 120\angle -120° \qquad \textbf{(E.5.1)}$$

For mesh 2,

$$120\angle 120^\circ - 120\angle -120^\circ + I_2(2 + 1 + 10 - j8) - I_1 - I_3(10 - j8) = 0$$

or

$$-I_1 + I_2(13 - j8) - I_3(10 - j8) = 120\angle -120^\circ - 120\angle 120^\circ \qquad \textbf{(E.5.2)}$$

For mesh 3,

$$I_3(12 + j10 + 10 - j8 + 15 + j6) - I_1(12 + j10) - I_2(10 - j8) = 0$$

or

$$-I_1(12 + j10) - I_2(10 - j8) - I_3(37 + j8) = 0 \qquad \textbf{(E.5.3)}$$

In matrix form, we can express Eqs. (E.5.1) to (E.5.3) as

$$\begin{bmatrix} 15 + j10 & -1 & -12 - j10 \\ -1 & 13 - j8 & -10 + j8 \\ -12 - j10 & -10 + j8 & 37 + j8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 120\angle 0^\circ - 120\angle -120^\circ \\ 120\angle -120^\circ - 120\angle 120^\circ \\ 0 \end{bmatrix}$$

or

$$\mathbf{ZI} = \mathbf{V}$$

We input matrices **Z** and **V** into *MATLAB* to get I.

```
>> z = [(15 + 10j) -1 (-12 - 10j);
        -1 (13 - 8j) (-10 + 8j);
        (-12 - 10j) (-10 + 8j) (37 + 8j)];
>> c1=120*exp(j*pi*(-120)/180);
>> c2=120*exp(j*pi*(-120)/180);
>> a1=120 - c1; a2=c1 - c2;
>> V = [a1; a2; 0]
>> I = inv(z)*V
I=
   16.9910 - 6.5953i
   12.4023 - 16.9993i
    5.6621 - 6.0471i
>> IbB = I(2) - I(1)
IbB =
   -4.5887 - 10.4039i
>> abs(I(1))
ans =
    18.2261
>> angle(I(1))*180/pi
ans =
   -21.2146
```

```
>> abs (I(2))
ans =
    21.0426
>> angle(I(2))*180/pi
ans =
   -53.8864
>> abs(I(3))
ans =
    8.2841
>> angle(I(3))*180/pi
ans =
   -46.8833
>> abs(IbB)
ans =
   11.3709
>> angle(IbB)*180/pi
ans =
   -113.8001
```

Thus, $I_1 = 18.23\angle -21.21°$, $I_2 = 21.04\angle -58.89°$, $I_3 = 8.284\angle -46.88°$, and $I_{bB} = 11.37\angle -113.8°$A.

Practice Problem E.5

In the unbalanced wye-wye three-phase system in Fig. E.12, find the line currents I_1, I_2, and I_3 and the phase voltage V_{CN}.

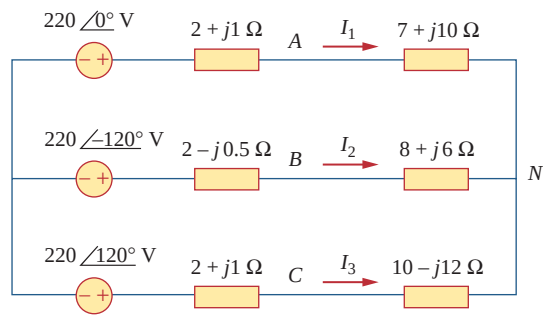


Figure E.12
For Practice Prob. E.5.

Answer: $22.66\angle -26.54°$ *A*, $6.036\angle -150.48°$ *A*, $19.93\angle 138.9°$ *A*, $94.29\angle 159.3°$ *V*.

E.4 Frequency Response

Frequency response involves plotting the magnitude and phase of the transfer function $H(s) = D(s)/N(s)$ or obtaining the Bode magnitude and phase plots of $H(s)$. One hard way to obtain the plots is to generate

data using the `for` loop for each value of $s = j\omega$ for a given range of ω and then plot the data as we did in Section E.1. However, there is an easy way that allows us to use one of two *MATLAB* commands: `freqs` and `bode`. For each command, we must first specify `H(s)` as `num` and `den`, where `num` and `den` are the vectors of coefficients of the numerator `N(s)` and denominator `D(s)` in descending powers of `s`, i.e., from the highest power to the constant term. The general form of the `bode` function is

```
bode(num, den, range);
```

where `range` is a specified frequency interval for the plot. If `range` is omitted, *MATLAB* automatically selects the frequency range. The `range` could be linear or logarithmic. For example, for $1 < \omega < 1000$ rad/s with 50 plot points, we can specify a linear `range` as

```
range = linspace(1,1000,50);
```

For a logarithmic `range` with $10^{-2} < \omega < 10^{4}$ rad/s and 100 plot points in between, we specify `range` as

```
range = logspace(-2,4,100);
```

For the `freqs` function, the general form is

```
hs = freqs(num, den, range);
```

where `hs` is the frequency response (generally complex). We still need to calculate the magnitude in decibels as

```
mag = 20*log 10(abs(hs))
```

and phase in degrees as

```
phase = angle(hs)*180/pi
```

and plot them, whereas the `bode` function does it all at once. We illustrate with an example.

Example E.6

Use *MATLAB* to obtain the Bode plots of

$$G(s) = \frac{s^3}{s^3 + 14.8s^2 + 38.1s + 2554}$$

Solution:

With the explanation previously given, we develop the *MATLAB* code as shown here.

```
% for example e.6
num=[1 0 0 0];
den = [1 14.8 38.1 2554];
w = logspace(-1,3);
bode(num, den, w);
title('Bode plot for a highpass filter')
```

Running the program produces the Bode plots in Fig. E.13. It is evident from the magnitude plot that $G(s)$ represents a highpass filter.

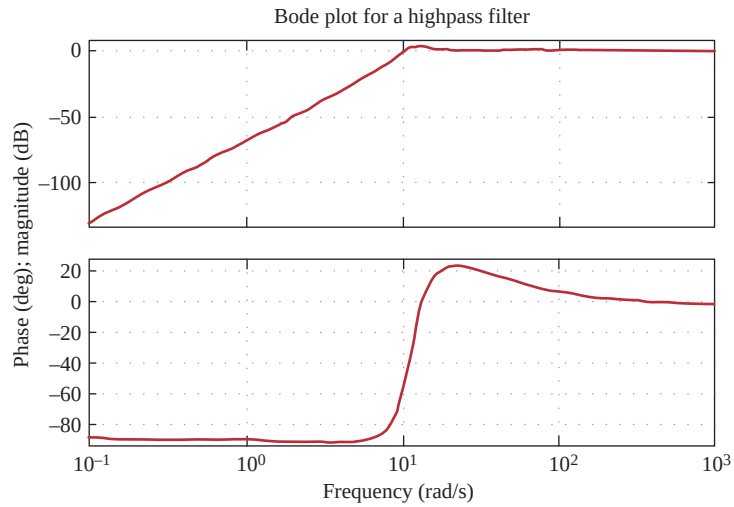


Figure E.13
For Example E.6.

Practice Problem E.6

Use *MATLAB* to determine the frequency response of

$$H(s) = \frac{10(s + 1)}{s^2 + 6s + 100}$$

Answer: See Fig. E.14.

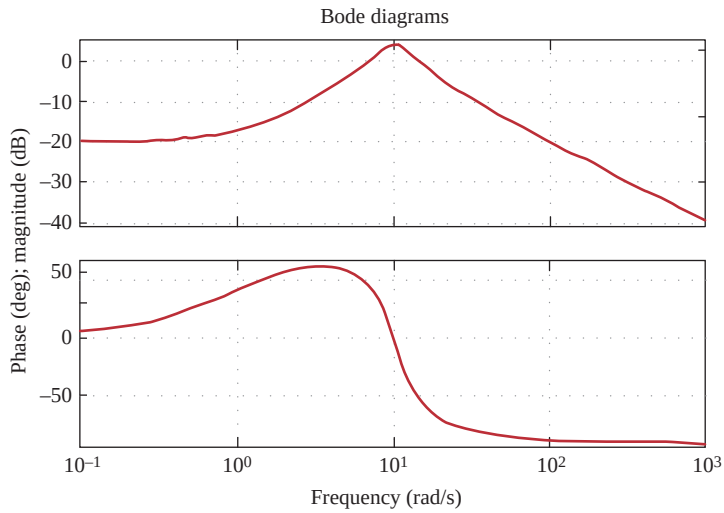


Figure E.14
For Practice Prob. E.6.

Appendix F

KCIDE for Circuits

Engineers of the twenty-first century will need to be able to work in a "knowledge capturing integrated design environment" (known as KCIDE). Essentially, engineers will go to their computer where they will do their work on a platform, much like Windows, where their various software packages, laboratory work, and other support software packages (such as *Word*) will all come together and interact with each other to help them with their work. These platforms will capture the work being done by the engineer and make the data available to be used in any manner the engineer chooses (such as preliminary design reports, user manuals, papers, books, proposals, or requests for proposals).

A detailed presentation of all the elements associated with learning how to work in such an environment is beyond the scope of this book. However, a platform to begin the process of training engineers to work in this environment is included in this textbook. *KCIDE for Circuits* was designed to assist the circuits student to learn how to work in a simplified KCIDE environment designed especially for electrical circuits students. The software that is used in the platform includes *PSpice*, *MATLAB*, *Excel*, *Word*, and *PowerPoint*.

In this appendix, we will help you to understand the *KCIDE for Circuits* platform and how to use it. The software can be obtained, free of any charges, from the website http://KCIDE.FennResearch.org. More details and examples are also included at the website. In addition, we will also have support services available at the website.

F.1 How to Work with *KCIDE for Circuits*

The structure of the platform and how it is effectively used follows the problem-solving process used throughout the text. This is essentially a systems approach to problem solving that uses a structured process to capture your work and present it in two different formats. It will be helpful to work through an example to see how to use the platform.

Use the *KCIDE for Circuits* platform to solve Example 3.2. **Example F.1**

Solution:

Opening the software, we see the screen shown in Fig. F.1, where we define a new project. Although we can name the project anyway we



Figure F.1
Creating a new project in *KCIDE for Circuits.*

wish to, we name it KCIDE Example F-1 050626 (see Fig. F.2). Note that the last six digits are: year/month/day. The reason for this is that if we create different files corresponding to different dates, the files will always appear in chronological order.

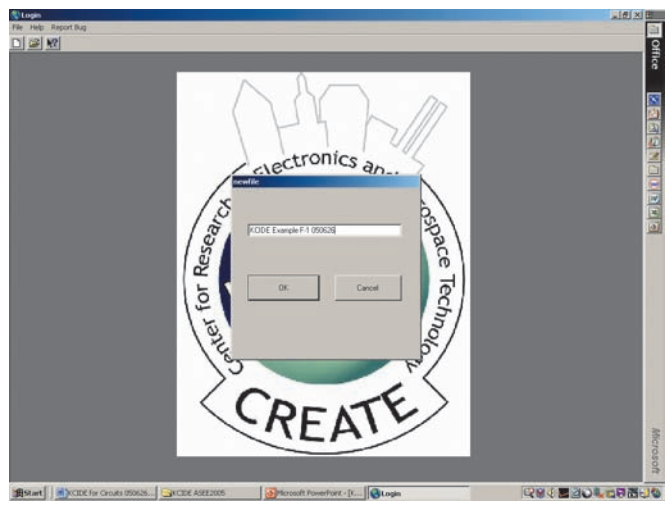


Figure F.2
Naming the project.

We now enter the problem statement into the screen shown in Fig. F.3. After we have entered the problem statement, we can click on the button to Open *PSpice*. The next screen, Fig. F.4, shows what is seen when the Open *PSpice* button is clicked. To open the *PSpice* schematic capture, we need to click on the page 1 icon. In Schematic, we create the circuit representing our problem. This is shown in Fig. F.5.

We now need to enter all we know about the problem by entering our problem analysis into the text box and then identifying the number

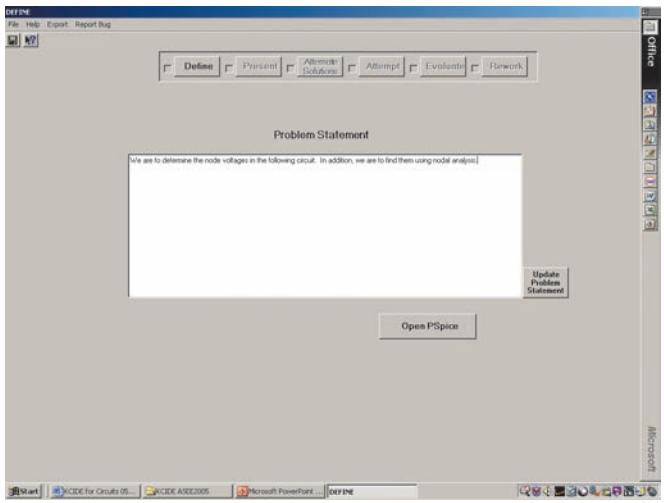


Figure F.3
Entering the problem statement.

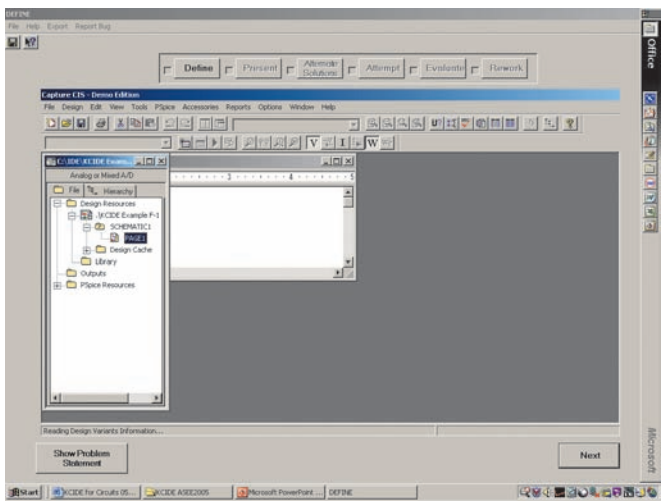


Figure F.4
How to open the schematic capture of *PSpice*.

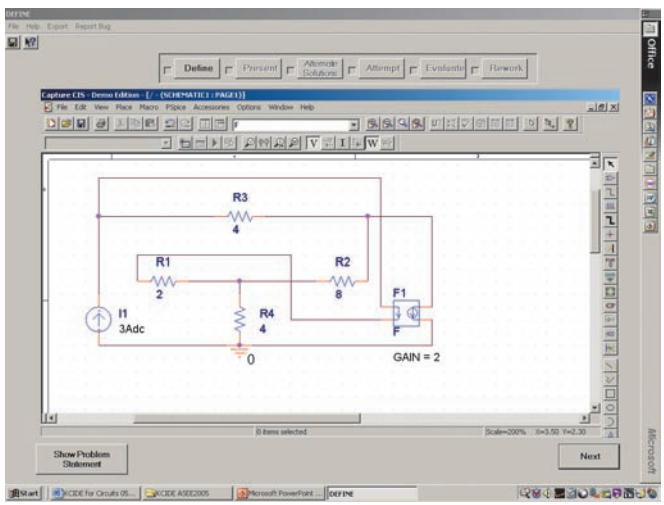


Figure F.5
Circuit for Example F.1.

of unknown nodes and loops for the circuit (see Fig. F.6). We continue this process by going to the next screen and entering the requested information (see Fig. F.7).

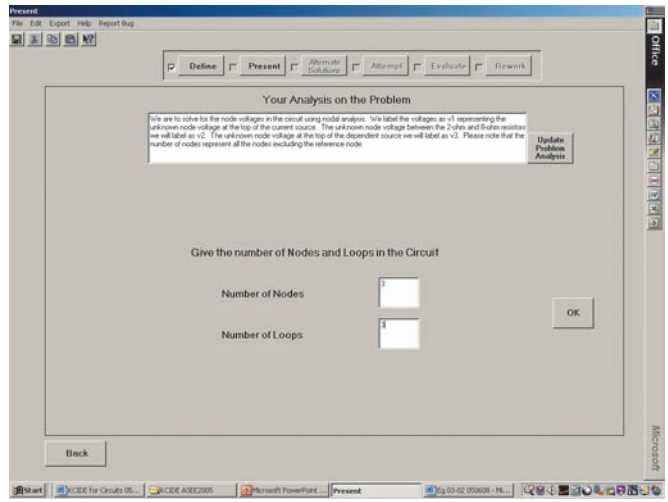


Figure F.6
Presenting what we know about the problem, part 1.

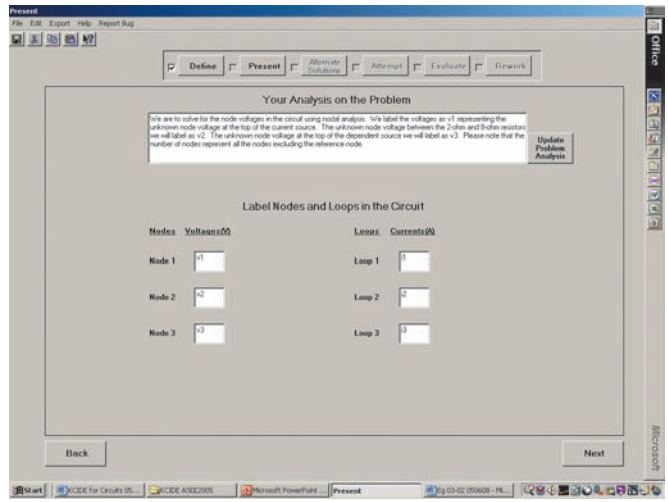


Figure F.7
Identifying the unknown node voltages and unknown loop currents, part 2.

We now proceed to selecting the method of solution. We do this by entering the requested information into the screen shown in Fig. F.8. Now we can develop the equations that will generate a solution for the problem. Since nodal analysis is required for the solution for the node voltages, all we need to do is to write the nodal equations. Once we have the appropriate equations, as shown in Fig. F.9, we can select a solution technique. In this case we chose Excel to solve our simultaneous equations.

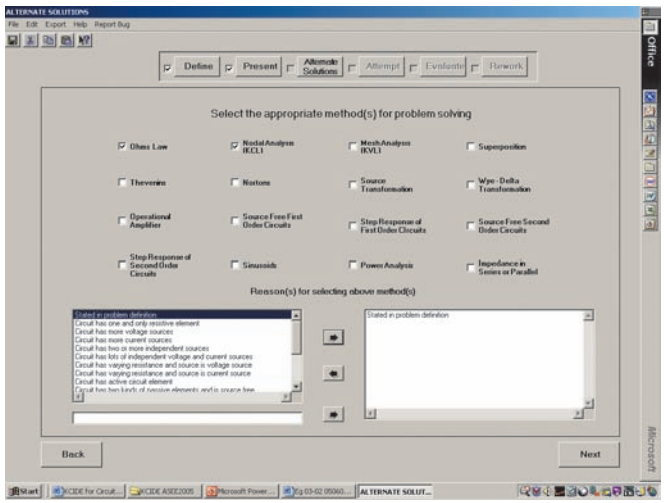


Figure F.8
Selecting the method of solving the problem.

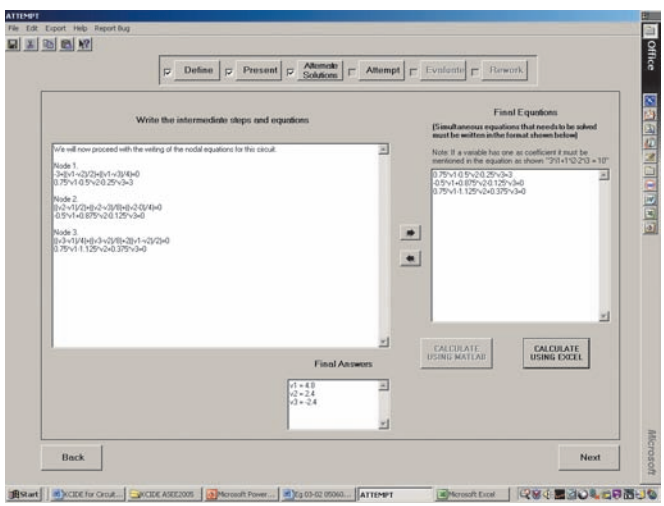


Figure F.9
Solving for the unknown node voltages.

Now, when we go to the next screen, the Evaluate portion of the solution, we actually open *PSpice* again. We need to open page 1 to retrieve our original circuit (see Fig. F.10). Once we have our original *PSpice* circuit, we need to prepare it for solving for our unknowns. The first step in this task is to go to the *PSpice* button and select New Simulation Profile (see Fig. F.11).

We next need to assign a name to the new simulation profile (Fig. F.12). Clicking on the Create button produces the screen shown in Fig. F.13. For this problem, we select Bias Point for the Analysis type.

Clicking on OK returns the screen to the original condition. Now we go to the *PSpice* button and select Run from the dropdown menu,

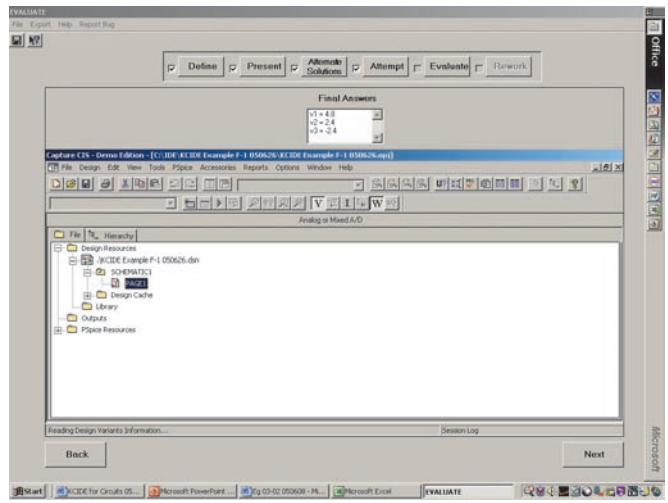


Figure F.10
Opening *PSpice* again.

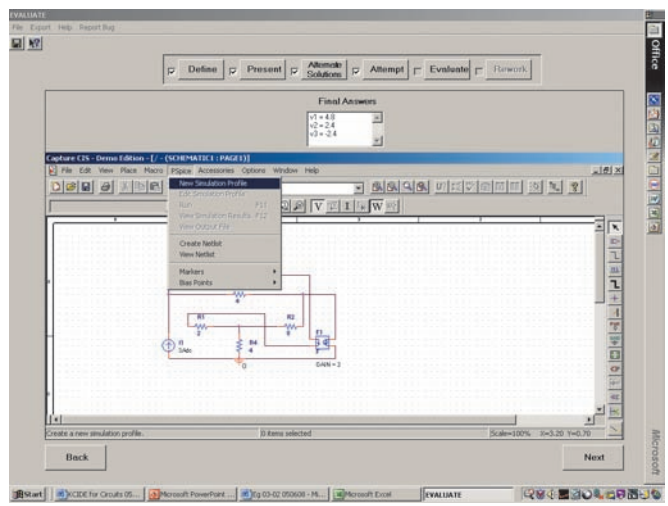


Figure F.11
Setting up our circuit for solution by *PSpice*.

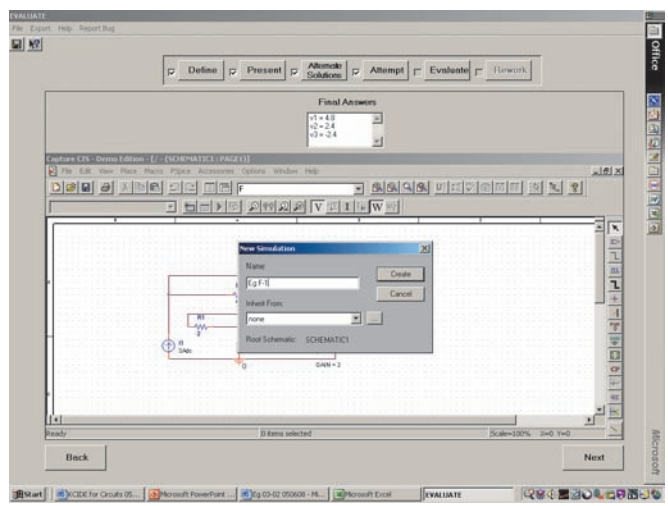


Figure F.12
Setting up our circuit for solution by *PSpice*.

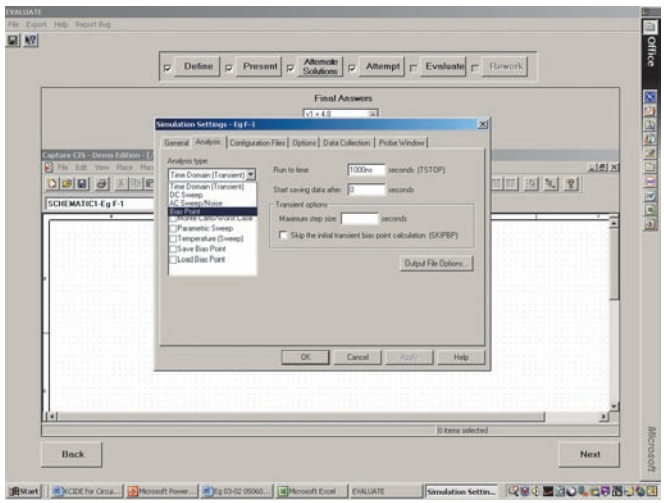


Figure F.13
Setting up our circuit for solution by *PSpice*.

Fig. F.14. Running *PSpice* produces the screen shown in Fig. F.15. We can immediately see that the voltages agree with the solution we obtained by using nodal analysis. Clicking on Next leads to our being asked, as in Fig. F.16, if we have any graphics to export. For this problem, we have no graphs.

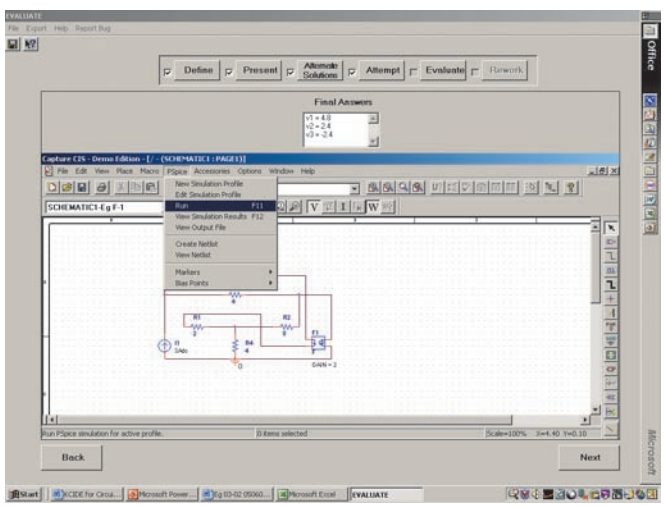


Figure F.14
Setting up our circuit for solution by *PSpice*.

We are now approaching the end of the process. We are asked to comment about the solution, Fig. F.17. And, we are asked if the answers agree with the *PSpice* solution. The answers do agree and we can proceed to determining what we want to export, Fig. F.18. We can generate *Word* and/or *PowerPoint* files, Fig. F.19. In this case, we select both but will only show the output of the *Word* file, Fig. F.20. Note: This output was modified so that it could be presented on two pages.

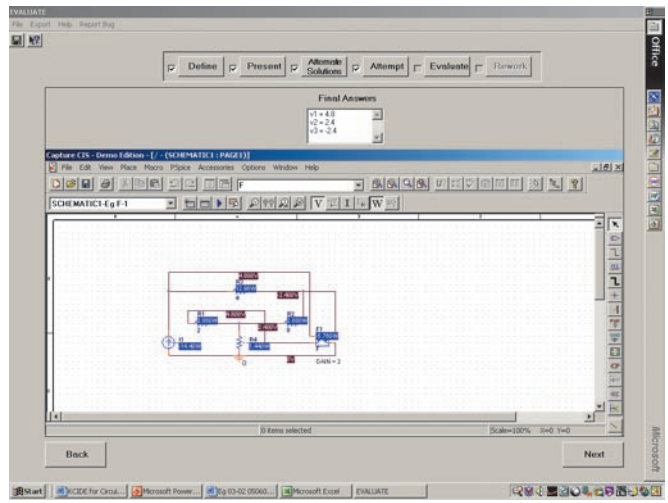


Figure F.15
Problem solution using *PSpice.*

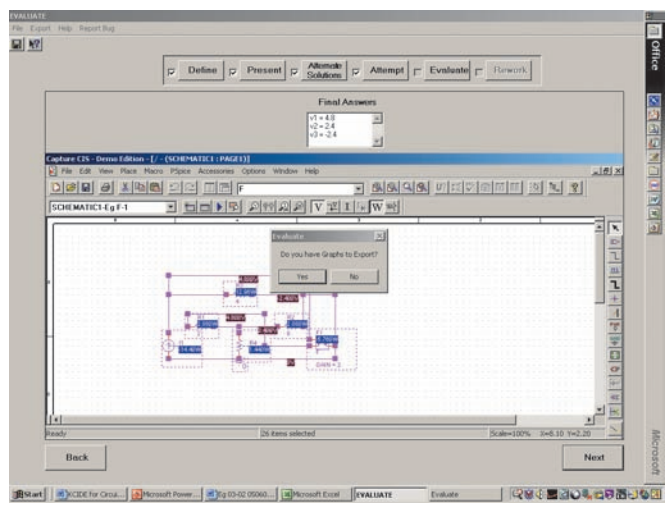


Figure F.16
Screen for exporting graphs.

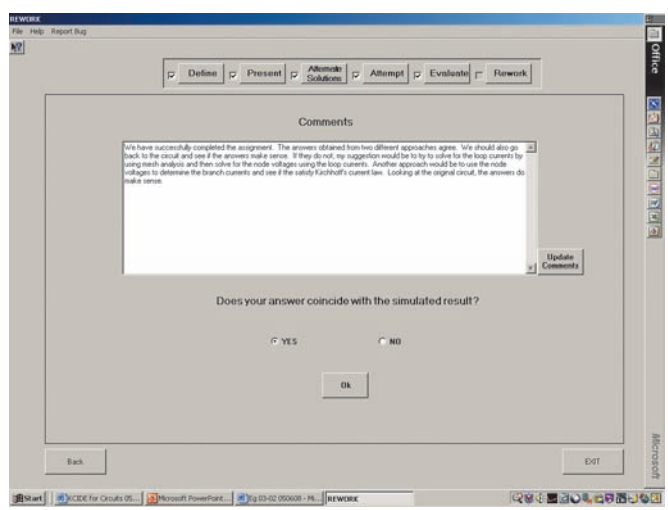


Figure F.17
Determining if the problem has been solved correctly.

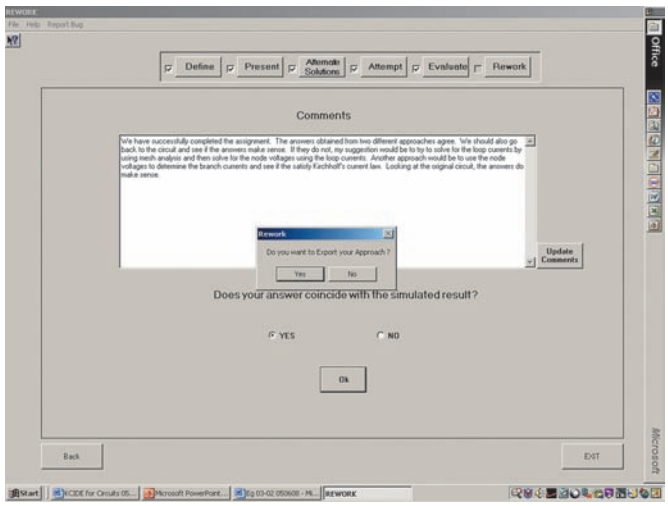


Figure F.18
Determining if you want to generate *Word* and/or *PowerPoint* documents.

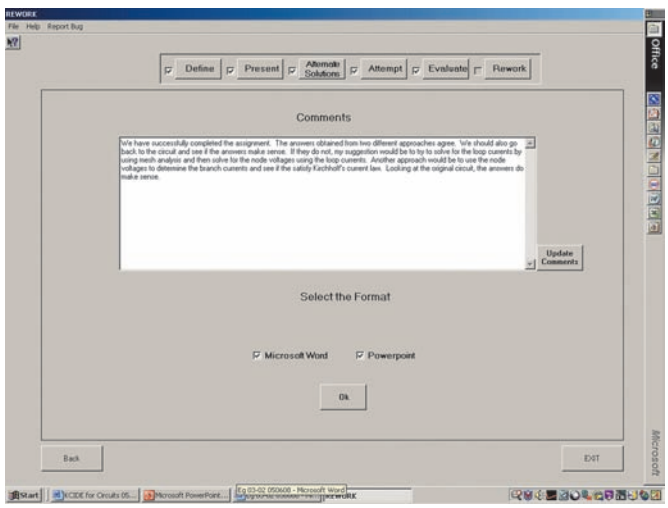


Figure F.19
Generating *Word* and *PowerPoint* files for Example F.1.

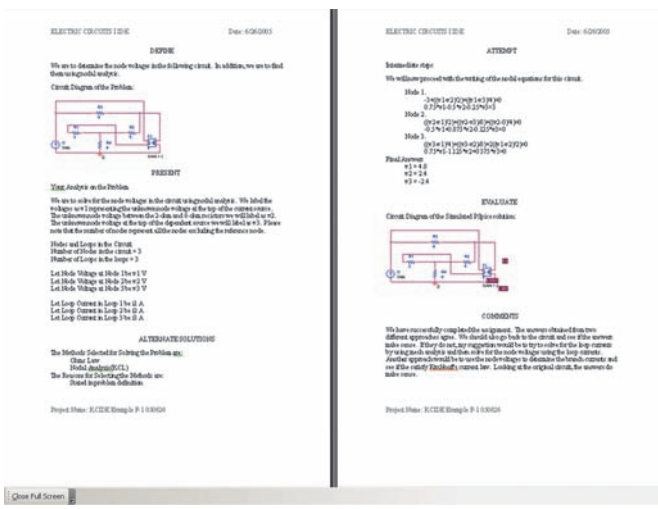


Figure F.20
Word file output.

We have now completed a detailed example. We suggest that you first try to do this with the platform and look at your output in both *Word* and *PowerPoint.* To help you to continue to develop your facility with the platform, try working the following practice problem, using mesh analysis. For more examples, please go to the website.

Practice Problem F.1

Use the *KCIDE for Circuits* platform to solve Practice Prob. 3.2.

Answer: $v_1 = 80$ V, $v_2 = -64$ V, and $v_3 = 156$ V.

Appendix G

Answers to Odd-Numbered Problems

Chapter 1

1.1 (a) −0.1038 C, (b) −0.19865 C, (c) −3.941 C, (d) −26.08 C

1.3 (a) $3t + 1$ C, (b) $t^2 + 5t$ mC, (c) $2\sin(10t + \pi/6) + 1\ \mu$C, (d) $-e^{-30t}[0.16\cos 40t + 0.12\sin 40t]$ C

1.5 25 C

1.7 $i = \begin{cases} 25\text{ A}, & 0 < t < 2 \\ -25\text{ A}, & 2 < t < 6 \\ 25\text{ A}, & 6 < t < 8 \end{cases}$

See the sketch in Fig. G.1.

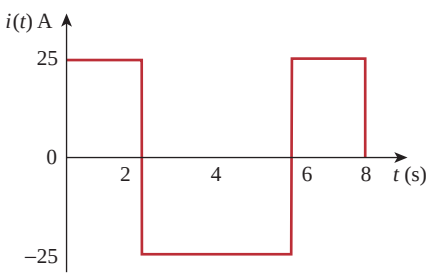


Figure G.1
For Prob. 1.7.

1.9 (a) 10 C, (b) 22.5 C, (c) 30 C

1.11 3.672 kC, 4.406 kJ

1.13 164.5 mW, 78.34 mJ

1.15 (a) 1.297 C, (b) $-90e^{-4t}$ W, (c) −22.5 J

1.17 70 W

1.19 3 A

1.21 2.696×10^{23} electrons, 43,200 C

1.23 $1.35

1.25 21.6 cents

1.27 (a) 43.2 kC, (b) 475.2 kJ, (c) 1.188 cents

1.29 39.6 cents

1.31 $42.05

1.33 6 C

1.35 2.333 MWh

1.37 29.84 kWh

1.39 24 cents

Chapter 2

2.1 This is a design problem with several answers.

2.3 184.3 mm

2.5 $n = 9, b = 15, l = 7$

2.7 (a) 6 branches and 5 nodes, and (b) 7 branches and 5 nodes.

2.9 14 A, −2 A, 10 A

2.11 6 V, 3 V

2.13 12 A, −10 A, 5 A, −2 A

2.15 10 V, −2 A

2.17 2 V, −22 V, 10 V

2.19 2 A, 12 W, 8 W, −40 W, 20 W

2.21 4.167 W

2.23 2 V, 1.92 W

2.25 0.1 A, 2 kV, 0.2 kW

2.27 6.4 V

2.29 1.625 Ω

2.31 11.2 A, 1.6 A, 9.6 A, 6.4 A, 3.2 A

2.33 3 V, 6 A

2.35 8 V, 0.2 A

2.37 2.5 Ω

2.39 (a) 727.3 Ω, (b) 3 kΩ

2.41 16 Ω

2.43 (a) 12 Ω, (b) 16 Ω

2.45 (a) 59.8 Ω, (b) 32.5 Ω

2.47 24 Ω

2.49 (a) 4 Ω, (b) $R_1 = 18\ \Omega$, $R_2 = 6\ \Omega$, $R_3 = 3\ \Omega$

2.51 (a) 9.231 Ω, (b) 36.25 Ω

2.53 (a) 142.32 Ω, (b) 33.33 Ω

2.55 997.4 mA

2.57 12.21 Ω, 1.64 A

2.59 1.2 A

2.61 Use R_1 and R_3 bulbs

2.63 0.4 Ω, ≅ 1 W

2.65 4 kΩ

2.67 (a) 4 V, (b) 2.857 V, (c) 28.57%, (d) 6.25%

2.69 (a) 1.278 V (with), 1.29 V (without)
(b) 9.30 V (with), 10 V (without)
(c) 25 V (with), 30.77 V (without)

2.71 10 Ω

2.73 45 Ω

2.75 (a) 19.9 kΩ, (b) 20 kΩ

2.77 (a) Four 20-Ω resistors in parallel
(b) One 300-Ω resistor in series with a 1.8-Ω resistor and a parallel combination of two 20-Ω resistors
(c) Two 24-kΩ resistors in parallel connected in series with two 56-kΩ resistors in parallel
(d) A series combination of a 20-Ω resistor, 300-Ω resistor, 24-kΩ resistor, and a parallel combination of two 56-kΩ resistors

2.79 75 Ω

2.81 38 kΩ, 3.333 kΩ

2.83 3 kΩ, ∞ Ω (best answer)

Chapter 3

3.1 This is a design problem with several answers.

3.3 4 A, 2 A, 1.3333 A, 0.667 A, 40 V

3.5 20 V

3.7 5.714 V

3.9 39.67 mA

3.11 293.9 W, 177.79 W, 238 W

3.13 8 V, 8 V

3.15 29.45 A, 144.6 W, 129.6 W, 12 W

3.17 1.73 A

3.19 10 V, 4.933 V, 12.267 V

3.21 1 V, 3 V

3.23 22.34 V

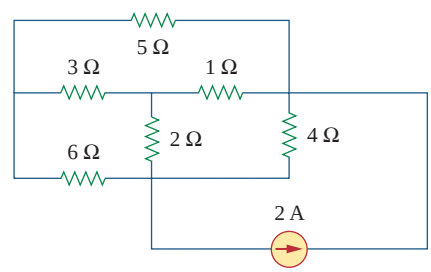
3.25 25.52 V, 22.05 V, 14.842 V, 15.055 V

3.27 625 mV, 375 mV, 1.625 V

3.29 −0.7708 V, 1.209 V, 2.309 V, 0.7076 V

3.31 4.97 V, 4.85 V, −0.12 V

3.33 (a) and (b) are both planar and can be redrawn as shown in Fig. G.2.



(a)

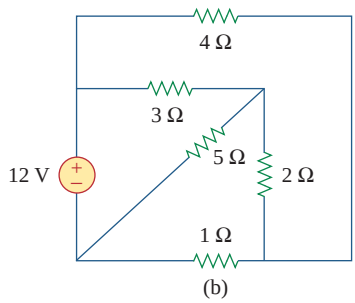


Figure G.2
For Prob. 3.33.

3.35 20 V

3.37 1.1111 V

3.39 0.8 A, −0.9 A

3.41 1.188 A

3.43 1.7778 A, 53.33 V

3.45 8.561 A

3.47 10 V, 4.933 V, 12.267 V

3.49 33.78 V, 10.67 A

3.51 20 V

3.53 1.6196 mA, −1.0202 mA, −2.461 mA, 3 mA, −2.423 mA

3.55 −1 A, 0 A, 2 A

3.57 3.23 kΩ, 28 V, 72 V

3.59 −1.344 kV, −5.6 A

3.61 −0.3

3.63 −4 V, 2.105 A

3.65 2.17 A, 1.9912 A, 1.8119 A, 2.094 A, 2.249 A

3.67 −12 V

3.69
$$\begin{bmatrix} 1.75 & -0.25 & -1 \\ -0.25 & 1 & -0.25 \\ -1 & -0.25 & 1.25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 5 \\ 5 \end{bmatrix}$$

3.71 2.085 A, 653.3 mA, 1.2312 A

3.73
$$\begin{bmatrix} 9 & -3 & -4 & 0 \\ -3 & 8 & 0 & 0 \\ -4 & 0 & 6 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \\ -3 \end{bmatrix}$$

3.75 −3 A, 0 A, 3 A

3.77 3.111 V, 1.4444 V

3.79 −5.278 V, 10.28 V, 694.4 mV, −26.88 V

3.81 26.67 V, 6.667 V, 173.33 V, −46.67 V

3.83 See Fig. G.3; −12.5 V

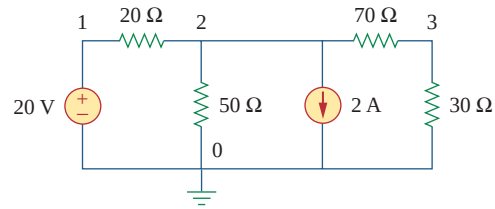


Figure G.3
For Prob. 3.83.

3.85 9 Ω

3.87 −8

3.89 30 μA, 12 V

3.91 0.61 μA, 8.641 V, 49 mV

Chapter 4

4.1 0.1 A, 1 A

4.3 (a) 0.5 V, 0.5 A, (b) 5 V, 5 A, (c) 5 V, 500 mA

4.5 4.5 V

4.7 888.9 mV

4.9 7 V

4.11 17.99 V, 1.799 A

4.13 8.696 V

4.15 1.875 A, 10.55 W

4.17 −8.571 V

4.19 −26.67 V

4.21 This is a design problem with multiple answers.

4.23 2 A, 32 W

4.25 −6.6 V

4.27 −48 V

4.29 3 V

4.31 3.652 V

4.33 (a) 8 Ω, 16 V, (b) 20 Ω, 50 V

4.35 −125 mV

4.37 10 Ω, 1 A

4.39 20 Ω, −16.4 V

4.41 4 Ω, −8 V, −2 A

4.43 10 Ω, 0 V

4.45 3 Ω, 3 A

4.47 476.2 mΩ, 1.9841 V, 4.176 A

4.49 28 Ω, 3.286 A

4.51 (a) 2 Ω, 7 A, (b) 1.5 Ω, 12.667 A

4.53 3 Ω, 1 A

4.55 100 kΩ, −20 mA

4.57 10 Ω, 166.67 V, 16.667 A

4.59 22.5 Ω, 40 V, 1.7778 A

4.61 1.2 Ω, 9.6 V, 8 A

4.63 −3.333 Ω, 0 A

4.65 $V_0 = (48 - 5I_0)$ V

4.67 25 Ω, 7.84 W

4.69 ∞ (theoretically)

4.71 8 kΩ, 1.152 W

4.73 20.77 W

4.75 $R_L = 10$ Ω, P_L tends toward infinity.

4.77 (a) 3.8 Ω, 4 V, (b) 3.2 Ω, 15 V

4.79 10 Ω, 167 V

4.81 3.3 Ω, 10 V (Note, values obtained graphically)

4.83 8 Ω, 12 V

4.85 (a) 24 V, 30 kΩ, (b) 9.6 V

4.87 (a) 10 mA, 8 kΩ, (b) 9.926 mA

4.89 (a) 99.99 μA, (b) 99.99 μA

4.91 (a) 100 Ω, 20 Ω, (b) 100 Ω, 200 Ω

4.93 $\dfrac{V_s}{R_s + (1 + \beta)R_o}$

4.95 5.333 V, 66.67 kΩ

4.97 2.4 kΩ, 4.8 V

Chapter 5

5.1 (a) 1.5 MΩ, (b) 60 Ω, (c) 98.06 dB

5.3 10 V

5.5 0.9999990

5.7 −100 nV, −10 mV

5.9 (a) 2 V, (b) 3 V

5.11 This is a design problem with multiple answers.

5.13 2.7 V, 288 μA

5.15 (a) $-\left(R_1 + R_3 + \dfrac{R_1R_3}{R_2}\right)$, (b) −92 kΩ

5.17 (a) −1.2, (b) −8, (c) −200

5.19 −0.375 mA

5.21 −4 V

5.23 $-\dfrac{R_f}{R_1}$

5.25 1.25 V

5.27 1.8 V

5.29 $\dfrac{R_2}{R_1}$

5.31 727.2 μA

5.33 −6 mA, 108 mW

5.35 If $R_1 = 10\ \text{k}\Omega$, then $R_f = 90\ \text{k}\Omega$.

5.37 -3 V

5.39 3 V

5.41 See Fig. G.4.

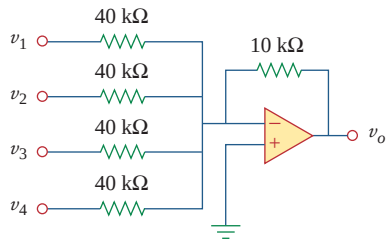


Figure G.4
For Prob. 5.41.

5.43 3 kΩ

5.45 See Fig. G.5, where $R \leq 100\ \text{k}\Omega$.

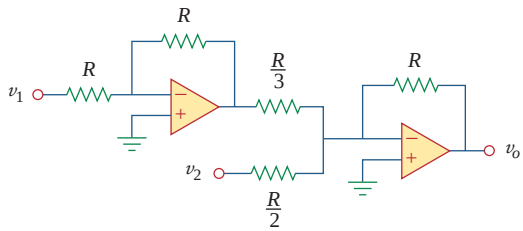


Figure G.5
For Prob. 5.45.

5.47 14.09 V

5.49 $R_1 = R_3 = 10\ \text{k}\Omega$, $R_2 = R_4 = 20\ \text{k}\Omega$

5.51 See Fig. G.6.

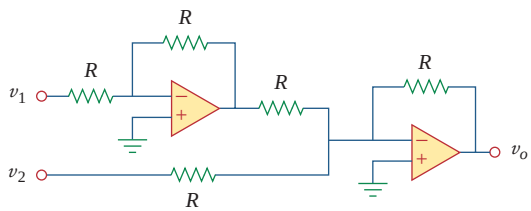


Figure G.6
For Prob. 5.51.

5.53 Proof.

5.55 7.956, 7.956, 1.989

5.57 $6v_{s1} - 6v_{s2}$

5.59 -16

5.61 -4.8 V

5.63 $\dfrac{R_2R_4/R_1R_5 - R_4/R_6}{1 - R_2R_4/R_3R_5}$

5.65 -21.6 mV

5.67 2.4 V

5.69 -17.143 mV

5.71 10 V

5.73 18 V

5.75 -2, 200 μA

5.77 -3.343 mV

5.79 -14.61 V

5.81 343.4 mV, 24.51 μA

5.83 The result depends on your design. Hence, let
$R_G = 10$ k ohms, $R_1 = 10$ k ohms,
$R_2 = 20$ k ohms, $R_3 = 40$ k ohms,
$R_4 = 80$ k ohms, $R_5 = 160$ k ohms,
$R_6 = 320$ k ohms, then,

$$-v_o = (R_f/R_1)v_1 + \text{———} + (R_f/R_6)v_6$$
$$= v_1 + 0.5v_2 + 0.25v_3 + 0.125v_4 + 0.0625v_5 + 0.03125v_6$$

(a) $|v_o| = 1.1875 = 1 + 0.125 + 0.0625 = 1 + (1/8) + (1/16)$, which implies, $[v_1\ v_2\ v_3\ v_4\ v_5\ v_6] = [\mathbf{100110}]$

(b) $|v_o| = 0 + (1/2) + (1/4) + 0 + (1/16) + (1/32) = (27/32) =$ **843.75 mV**

(c) This corresponds to [111111].
$|v_o| = 1 + (1/2) + (1/4) + (1/8) + (1/16) + (1/32)$
$= 63/32 =$ **1.96875 V**

5.85 160 kΩ

5.87 $\left(1 + \dfrac{R_4}{R_3}\right)v_2 - \left[\dfrac{R_4}{R_3} + \left(\dfrac{R_2R_4}{R_1R_3}\right)\right]v_1$

Let $R_4 = R_1$ and $R_3 = R_2$;

then $v_0 = \left(1 + \dfrac{R_4}{R_3}\right)(v_2 - v_1)$

a subtractor with a gain of $\left(1 + \dfrac{R_4}{R_3}\right)$.

5.89 A summer with $v_0 = -v_1 - (5/3)v_2$ where $v_2 = 6$ V battery and an inverting amplifier with $v_1 = -12\ v_2$.

5.91 9

5.93 $A = \dfrac{1}{(1 + \frac{R_1}{R_3})R_L - R_1(\frac{R_2 + R_L}{R_2R_3})(R_4 + \frac{R_2R_L}{R_2 + R_L})}$

Chapter 6

6.1 $10(1 - 3t)e^{-3t}$ A, $20t(1 - 3t)e^{-6t}$ W

6.3 This is a design problem with multiple answers.

6.5 $v = \begin{cases} 20 \text{ mA}, & 0 < t < 2 \text{ ms} \\ -20 \text{ mA}, & 2 < t < 6 \text{ ms} \\ 20 \text{ mA}, & 6 < t < 8 \text{ ms} \end{cases}$

6.7 $0.04t^2 + 10$ V

6.9 13.624 V, 70.66 W

6.11 $v(t) = \begin{cases} 10 + 3.75t \text{ V}, & 0 < t < 2\text{s} \\ 22.5 - 2.5t \text{ V}, & 2 < t < 4\text{s} \\ 12.5 \text{ V}, & 4 < t < 6\text{s} \\ 2.5t - 2.5 \text{ V}, & 6 < t < 8\text{s} \end{cases}$

6.13 30 V, 40 V

6.15 (a) 100 mJ, 150 mJ, (b) 36 mJ, 24 mJ

6.17 (a) 3 F, (b) 8 F, (c) 1 F

6.19 10 μF

6.21 2.5 μF

6.23 This is a design problem with multiple answers.

6.25 (a) For the capacitors in series,

$$Q_1 = Q_2 \rightarrow C_1 v_1 = C_2 v_2 \rightarrow \frac{v_1}{v_2} = \frac{C_2}{C_1}$$

$$v_s = v_1 + v_2 = \frac{C_2}{C_1} v_2 + v_2 = \frac{C_1 + C_2}{C_1} v_2$$

$$\rightarrow v_2 = \frac{C_1}{C_1 + C_2} v_s$$

Similarly, $v_1 = \dfrac{C_2}{C_1 + C_2} v_s$

(b) For capacitors in parallel,

$$v_1 = v_2 = \frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

$$Q_s = Q_1 + Q_2 = \frac{C_1}{C_2} Q_2 + Q_2 = \frac{C_1 + C_2}{C_2} Q_2$$

or

$$Q_2 = \frac{C_2}{C_1 + C_2}$$

$$Q_1 = \frac{C_1}{C_1 + C_2} Q_s$$

$$i = \frac{dQ}{dt} \rightarrow i_1 = \frac{C_1}{C_1 + C_2} i_s,$$

$$i_2 = \frac{C_2}{C_1 + C_2} i_s$$

6.27 1 μF, 16 μF

6.29 (a) 1.6 C, (b) 1 C

6.31 $v(t) = \begin{cases} t^2 \text{ kV}, & 0 < t < 1\text{s} \\ 2t - 1 \text{ kV}, & 1 < t < 3\text{s} \\ 0.5t^2 - 5t + 15.5 \text{ kV}, & 3 < t < 5\text{s} \end{cases}$

$i_1(t) = \begin{cases} 12t \text{ mA}, & 0 < t < 1\text{s} \\ 12 \text{ mA}, & 1 < t < 3\text{s} \\ 6t - 30 \text{ mA}, & 3 < t < 5\text{s} \end{cases}$

$i_2(t) = \begin{cases} 8t \text{ mA}, & 0 < t < 1\text{s} \\ 8 \text{ mA}, & 1 < t < 3\text{s} \\ 4t - 20 \text{ mA}, & 3 < t < 5\text{s} \end{cases}$

6.33 10 F, 7.5 V

6.35 6.4 mH

6.37 $4.8 \cos 100t$ V, 96 mJ

6.39 $(5t^3 + 5t^2 + 20t + 1)$ A

6.41 5.977 A, 35.72 J

6.43 144 μJ

6.45 $i(t) = \begin{cases} 100t^2 \text{ A}, & 0 < t < 1\text{s} \\ [100t^2 - 400t + 400] \text{ A}, & 1 < t < 2\text{s} \end{cases}$

6.47 5 Ω

6.49 3.75 mH

6.51 7.778 mH

6.53 20 mH

6.55 (a) 1.4 L, (b) 0.5 L

6.57 6.625 H

6.59 Proof.

6.61 (a) 6.667 mH, e^{-t} mA, $2e^{-t}$ mA
(b) $-20e^{-t}$ μV (c) 1.3534 nJ

6.63 See Fig. G.7.

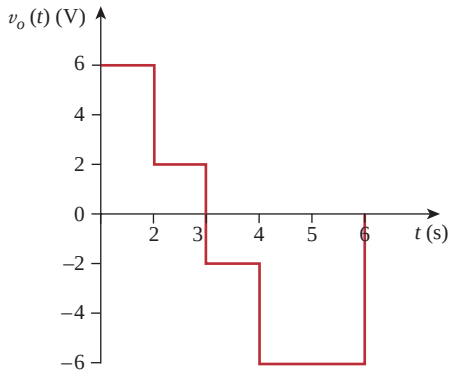


Figure G.7
For Prob. 6.63.

6.65 (a) 40 J, 40 J, (b) 80 J, (c) $5 \times 10^{-5}(e^{-200t} - 1) + 4$ A, $1.25 \times 10^{-5}(e^{-200t} - 1) - 2$ A
(d) $6.25 \times 10^{-5}(e^{-200t} - 1) + 2$ A

6.67 200 cos(50*t*) mV

6.69 See Fig. G.8.

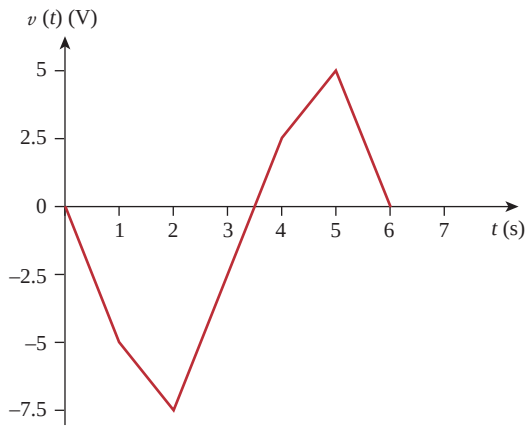


Figure G.8
For Prob. 6.69.

6.71 By combining a summer with an integrator, we have the circuit shown in Fig. G.9.

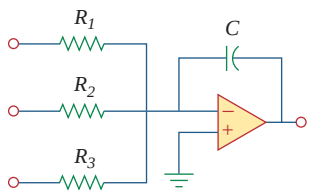


Figure G.9
For Prob. 6.71.

$$v_o = -\frac{1}{R_1 C}\int v_1\,dt - \frac{1}{R_2 C}\int v_2\,dt - \frac{1}{R_2 C}\int v_2\,dt$$

For the given problem, $C = 2\mu\text{F} : R_1 = 500\ \text{k}\Omega$, $R_2 = 125\ \text{k}\Omega$, $R_3 = 50\ \text{k}\Omega$.

6.73 Consider the op amp as shown in Fig. G.10.

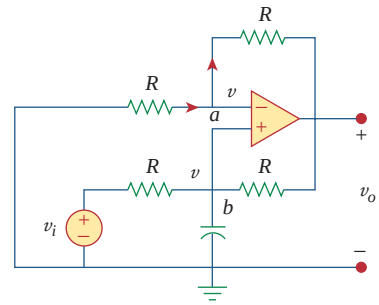


Figure G.10
For Prob. 6.73.

Let $v_a = v_b = v$. At node *a*,

$$\frac{0 - v}{R} = \frac{v - v_0}{R} \rightarrow 2v - v_0 = 0 \qquad \textbf{(1)}$$

At node *b*, $\dfrac{v_i - v}{R} = \dfrac{v - v_0}{R} + C\dfrac{dv}{dt}$

$$v_i = 2v - v_o + RC\frac{dv}{dt} \qquad \textbf{(2)}$$

Combining Eqs. (1) and (2),

$$v_i = v_o - v_o + \frac{RC}{2}\frac{dv_o}{dt} \quad \text{or} \quad v_o = \frac{2}{RC}\int v_i\,dt$$

showing that the circuit is a noninverting integrator.

6.75 −30 mV

6.77 See Fig. G.11.

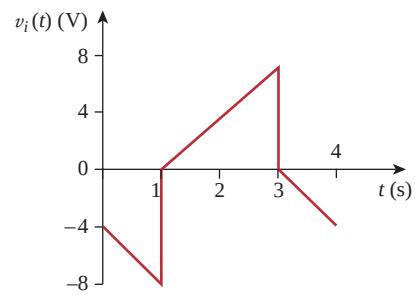


Figure G.11
For Prob. 6.77.

6.79 See Fig. G.12.

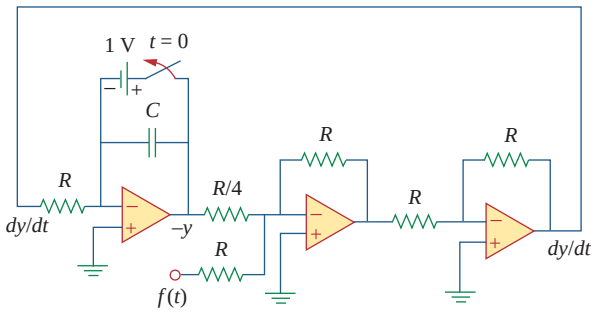


Figure G.12
For Prob. 6.79.

6.81 See Fig. G.13.

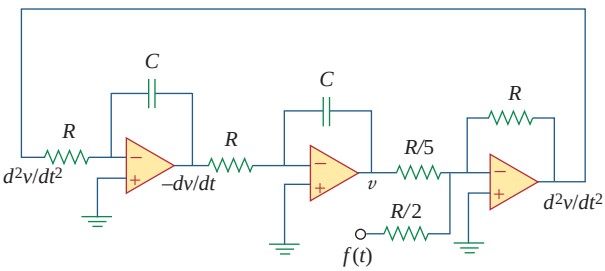


Figure G.13
For Prob. 6.81.

6.83 Eight groups in parallel with each group made up of two capacitors in series

6.85 1.25 mH inductor

Chapter 7

7.1 (a) 0.7143 μF, (b) 5 ms, (c) 3.466 ms

7.3 3.222 μs

7.5 This is a design problem with multiple answers.

7.7 $v_o(t) = [6 + 2e^{-t/20}]$ V for all $t > 0$.

7.9 $v_o(t) = 4e^{-5t}$ V for all $t > 0$.

7.11 $1.4118e^{-3t}$ A

7.13 (a) 5 kΩ, 5 H, 1 ms, (b) 25.28 μJ

7.15 (a) 0.25 s, (b) 0.5 ms

7.17 $-2e^{-16t}u(t)$ V

7.19 $2e^{-5t}u(t)$ A

7.21 13.333 Ω

7.23 $2e^{-4t}$ V, $t > 0$, $0.5e^{-4t}$ V, $t > 0$

7.25 This is a design problem with multiple answers.

7.27 $[10u(t + 1) + 20u(t) - 50u(t - 1) + 30u(t - 2)]$ V

7.29 (a) See Fig. G.14(a). (b) See Fig. G.14(b). (c) $z(t) = 5\cos 4t\,\delta(t - 1) = 5\cos 4\delta(t - 1) = -3.268\delta(t - 1)$, which is sketched in Fig. G.14(c).

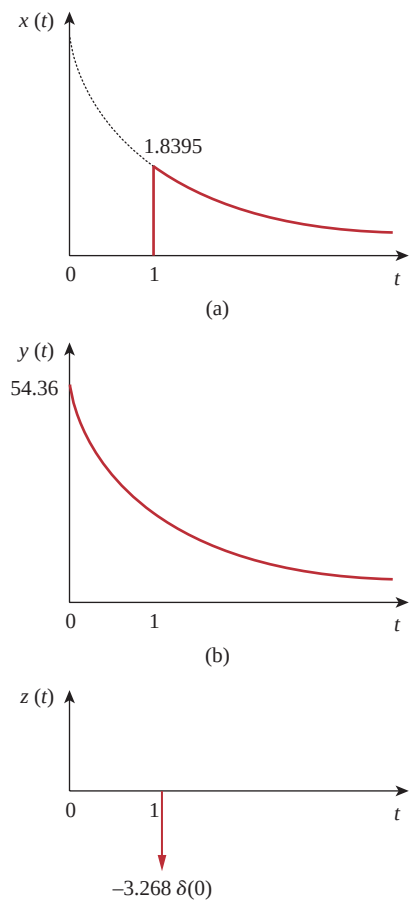


Figure G.14
For Prob. 7.29.

7.31 (a) 112×10^{-9}, (b) 7

7.33 $2u(t-2)$ A

7.35 (a) $-e^{-2t}u(t)$ V, (b) $2e^{1.5t}u(t)$ A

7.37 (a) 4 s, (b) 10 V, (c) $(10 - 8e^{-t/4})\,u(t)$ V

7.39 (a) 4 V, $t < 0$, $20 - 16e^{-t/8}$, $t > 0$,
(b) 4 V, $t < 0$, $12 - 8e^{-t/6}$ V, $t > 0$.

7.41 This is a design problem with multiple answers.

7.43 0.8 A, $0.8e^{-t/480}u(t)$ A

7.45 $(4 - 3e^{-14.286t})\,u(t)$ V

7.47 $\begin{cases} 48(1 - e^{-t})\text{ V}, & 0 < t < 1 \\ (60 - 29.66e^{-(t-1)})\text{ V}, & t > 1 \end{cases}$

7.49 $\begin{cases} 8(1 - e^{-t/5})\text{ V}, & 0 < t < 1 \\ 1.45e^{-(t-1)/5}\text{ V}, & t > 1 \end{cases}$

7.51 $V_S = Ri + L\dfrac{di}{dt}$

or $L\dfrac{di}{dt} = -R\left(i - \dfrac{V_S}{R}\right)$

$$\frac{di}{i - V_S/R} = \frac{-R}{L}dt$$

Integrating both sides,

$$\ln\left(i - \frac{V_S}{R}\right)\Big|_{I_0}^{i(t)} = \frac{-R}{L}t$$

$$\ln\left(\frac{i - V_S/R}{I_0 - V_S/R}\right) = \frac{-t}{\tau}$$

or $\dfrac{i - V_S/R}{I_0 - V_S/R} = e^{-t/\tau}$

$$i(t) = \frac{V_S}{R} + \left(I_0 - \frac{V_S}{R}\right)e^{-t/\tau}$$

which is the same as Eq. (7.60).

7.53 (a) 5 A, $5e^{-t/2}u(t)$ A, (b) 6 A, $6e^{-2t/3}u(t)$ A

7.55 96 V, $96e^{-4t}u(t)$ V

7.57 $4.8e^{-2t}u(t)$ A, $1.2e^{-5t}u(t)$ A

7.59 $3e^{-4t}u(t)$ V

7.61 $20e^{-80t}u(t)$ V, $(5 + (5 - 5e^{-80t})\,u(t))$ A

7.63 $-16e^{-8t}u(t)$ V, $(4 + (-4 + 4e^{-8t})u(t))$ A

7.65 $\begin{cases} 4(1 - e^{-2t})\text{ A} & 0 < t < 1 \\ 3.458e^{-2(t-1)}\text{ A} & t > 1 \end{cases}$

7.67 $10e^{-100t/3}u(t)$ A

7.69 $48(e^{-t/3000} - 1)\,u(t)$ V

7.71 $6(1 - e^{-5t})u(t)$ V

7.73 $-50u(t)$ mA

7.75 $(6 - 3e^{-50t})u(t)$ V, -0.2 mA

7.77 See Fig. G.15.

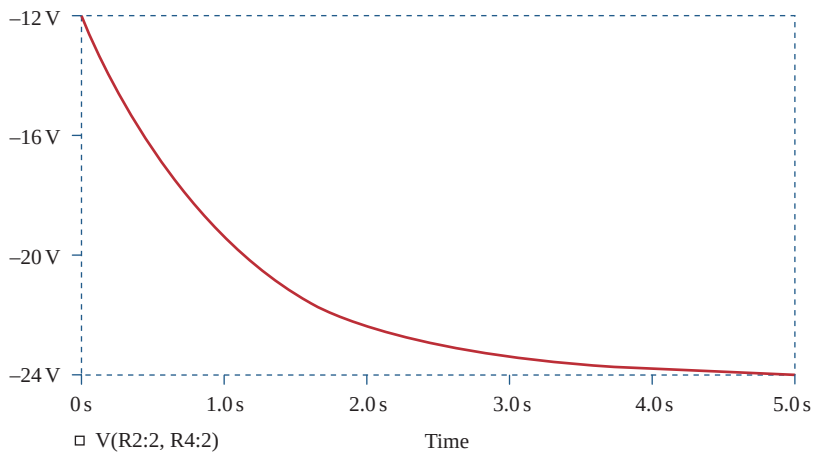


Figure G.15
For Prob. 7.77.

7.79 $(-0.5 + 4.5e^{-80t/3})u(t)$ A

7.81 See Fig. G.16.

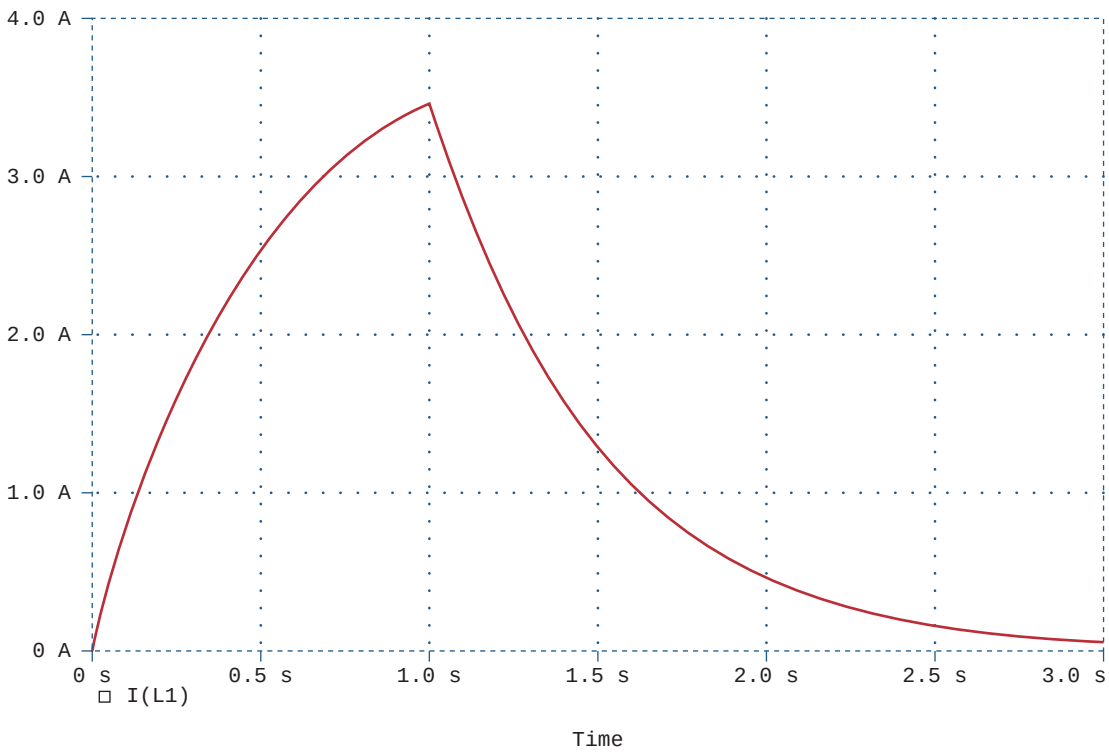


Figure G.16
For Prob. 7.81.

7.83 6.278 m/s

7.85 (a) 659.7 μs, (b) 16.636 s

7.87 441 mA

7.89 $L < 200$ mH

7.91 1.271 Ω

Chapter 8

8.1 (a) 2 A, 12 V, (b) −4 A/s, −5 V/s, (c) 0 A, 0 V

8.3 (a) 0 A, −10 V, 0 V, (b) 0 A/s, 8 V/s, 8 V/s, (c) 400 mA, 6 V, 16 V

8.5 (a) 0 A, 0 V, (b) 4 A/s, 0 V/s, (c) 2.4 A, 9.6 V

8.7 overdamped

8.9 $(2 + 10t)e^{-5t}u(t)$ A

8.11 $20(1 + t)e^{-t}$ V for $t > 0$.

8.13 120 Ω

8.15 750 Ω, 200 μF, 25 H

8.17 $(64.65e^{-2.679t} - 4.641e^{-37.32t})$ V

8.19 18 sin(0.5t) V for $t > 0$.

8.21 $18e^{-t} - 2e^{-9t}$ V

8.23 40 mF

8.25 This is a design problem with multiple answers.

8.27 $(6 - 6(\cos(2t) + \sin(2t)e^{-2t})u(t))$ V

8.29 (a) $3 - 3\cos 2t + \sin 2t$ V,
(b) $2 - 4e^{-t} + e^{-4t}$ A,
(c) $3 + (2 + 3t)e^{-t}$ V,
(d) $2 + 2\cos 2te^{-t}$ A

8.31 80 V, 40 V

8.33 $[20 + 0.001125e^{-4.95t} - 10.001e^{-0.05t}]$ V

8.35 This is a design problem with multiple answers.

8.37 $5e^{-4t}$ A

8.39 $[30 + (0.021e^{-47.33t} - 6.021e^{-0.167t})]$ V

8.41 $(0.3638\sin(4.583t)e^{-2t})$ A for $t > 0$.

8.43 8 Ω, 2.392 mF

8.45 $[4 - [3\cos(1.3229t) + 1.1339\sin(1.3229t)]e^{-t/2}]$ A,
$[4.536\sin(1.3229t)e^{-t/2}]$ V

8.47 $(200te^{-10t})$ V

8.49 $[3 + (3 + 6t)e^{-2t}]$ A

8.51 $\left[-\dfrac{i_0}{\omega_o C}\sin(\omega_o t)\right]$ V where $\omega_o = 1/\sqrt{LC}$

8.53 $(d^2i/dt^2) + 0.125(di/dt) + 400i = 600$

8.55 $7.448 - 3.448e^{-7.25t}$ V, $t > 0$

8.57 (a) $s^2 + 20s + 36 = 0$,
(b) $-\dfrac{3}{4}e^{-2t} - \dfrac{5}{4}e^{-18t}$ A, $6e^{-2t} + 10e^{-18t}$ V

8.59 $-32te^{-t}$ V

8.61 $2.4 - 2.667e^{-2t} + 0.2667e^{-5t}$ A,
$9.6 - 16e^{-2t} + 6.4e^{-5t}$ V

8.63 $\dfrac{d^2i(t)}{dt^2} = -\dfrac{v_s}{RCL}$

8.65 $\dfrac{d^2v_o}{dt^2} - \dfrac{v_o}{R^2C^2} = 0$, $e^{10t} - e^{-10t}$ V
Note, circuit is unstable.

8.67 $-te^{-t}u(t)$ V

8.69 See Fig. G.17.

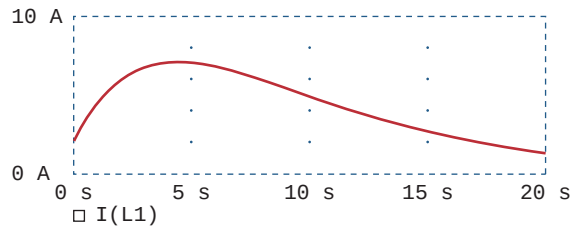


Figure G.17
For Prob. 8.69.

8.71 See Fig. G.18.

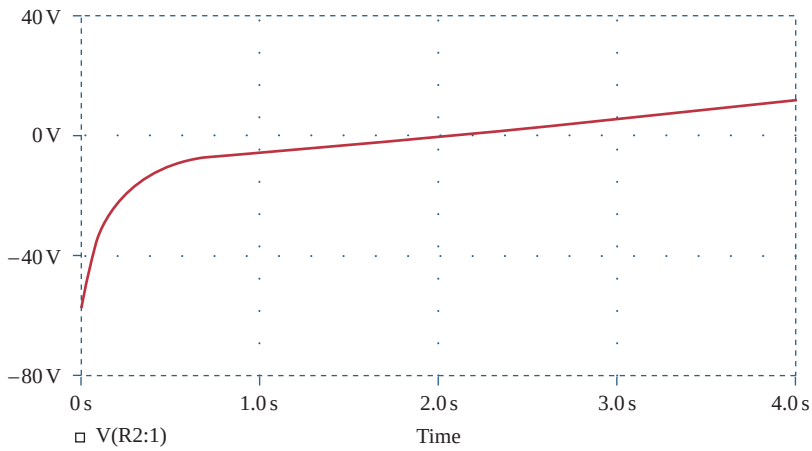


Figure G.18
For Prob. 8.71.

8.73 This is a design problem with multiple answers.

8.75 See Fig. G.19.

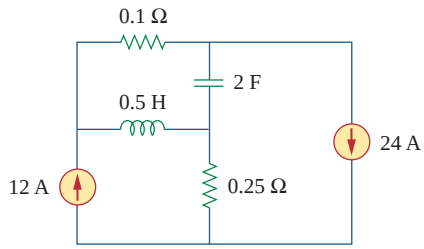


Figure G.19
For Prob. 8.75.

8.77 See Fig. G.20.

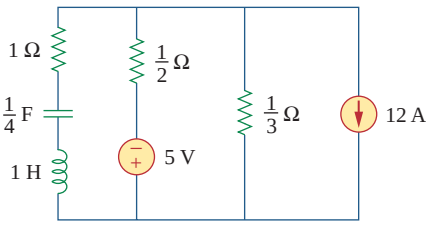


Figure G.20
For Prob. 8.77.

8.79 434 μF

8.81 2.533 μH, 625 μF

8.83 $\dfrac{d^2v}{dt^2} + \dfrac{R}{L}\dfrac{dv}{dt} + \dfrac{R}{LC}i_D + \dfrac{1}{C}\dfrac{di_D}{dt} = \dfrac{v_s}{LC}$

Chapter 9

9.1 (a) 50 V, (b) 209.4 ms, (c) 4.775 Hz, (d) 44.48 V, 0.3 rad

9.3 (a) $4\cos(\omega t - 120°)$, (b) $2\cos(6t + 90°)$, (c) $10\cos(\omega t + 110°)$

9.5 20°, v_1 lags v_2

9.7 Proof

9.9 (a) $50.88\angle -15.52°$, (b) $60.02\angle -110.96°$

9.11 (a) $21\angle -15°$ V, (b) $8\angle 160°$ mA,
(c) $120\angle -140°$ V, (d) $60\angle 190°$ mA

9.13 (a) $-1.2749 + j0.1520$, (b) -2.083, (c) $35 + j14$

9.15 (a) $-6 - j11$, (b) $120.99 + j4.415$, (c) -1

9.17 $15.62\cos(50t - 9.8°)$ V

9.19 (a) $3.32\cos(20t + 114.49°)$,
(b) $64.78\cos(50t - 70.89°)$,
(c) $9.44\cos(400t - 44.7°)$

9.21 (a) $f(t) = 8.324\cos(30t + 34.86°)$,
(b) $g(t) = 5.565\cos(t - 62.49°)$,
(c) $h(t) = 1.2748\cos(40t - 168.69°)$

9.23 (a) $43.49\cos(\omega t - 6.59°)$ V,
(b) $18.028\cos(\omega t + 78.69°)$ A

9.25 (a) $0.8\cos(2t - 98.13°)$ A,
(b) $0.745\cos(5t - 4.56°)$ A

9.27 $0.289\cos(377t - 92.45°)$ V

9.29 $2\sin(10^6 t - 65°)$

9.31 $78.3\cos(2t + 51.21°)$ mA

9.33 69.82 V

9.35 $4.789\cos(200t - 16.7°)$ A

9.37 $(500 - j50)$ mS

9.39 $9.135 + j27.47\ \Omega$,
$414.5\cos(10t - 71.6°)$ mA

9.41 $15.812\cos(t - 18.43°)$ V

9.43 $499.7\angle -28.85°$ mA

9.45 -5 A

9.47 $1.8428\cos(2{,}000t + 52.63°)$ A

9.49 $1.4142\sin(200t - 45°)$ V

9.51 $-12.5\cos(2t - 53.13°)$ A

9.53 $8.873\angle -21.67°$ A

9.55 $2.798 - j16.403\ \Omega$

9.57 $0.3171 - j0.1463$ S

9.59 $2.707 + j2.509$

9.61 $1 + j0.5\ \Omega$

9.63 $34.69 - j6.93\ \Omega$

9.65 $17.35\angle 0.9°$ A, $6.83 + j1.094\ \Omega$

9.67 (a) $14.8\angle -20.22°$ mS, (b) $19.7\angle 74.57°$ mS

9.69 $1.661 + j0.6647$ S

9.71 $1.058 - j2.235\ \Omega$

9.73 $0.3796 + j1.46\ \Omega$

9.75 Can be achieved by the RL circuit shown in Fig. G.21.

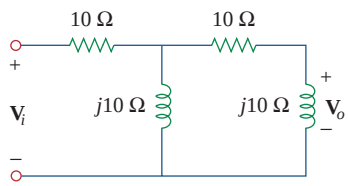


Figure G.21
For Prob. 9.75.

9.77 (a) 51.49° lagging, (b) 1.5915 MHz

9.79 (a) 140.2°, (b) leading, (c) 18.43 V

9.81 1.8 kΩ, 0.1 μF

9.83 104.17 mH

9.85 Proof

9.87 $38.21\angle -8.97°\ \Omega$

9.89 8.05 mH

9.91 235 pF

9.93 $1.7958\angle -38.66°$ A

Chapter 10

10.1 $1.9704\cos(10t + 5.65°)$ A

10.3 $7.67\cos(4t - 35.02°)$ V

10.5 $12.398\cos(4 \times 10^3 t + 4.06°)$ mA

10.7 $124.08\angle -154°$ V

10.9 $6.154\cos(10^3 t + 70.26°)$ V

10.11 $498.7\angle 86.87°$ mA

10.13 $29.36\angle 62.88°$ V

10.15 $15.812\angle 43.49°$ A

10.17 $13.875\angle -162.12°$ A

10.19 $7.682\angle 50.19°$ V

10.21 (a) $1, 0, -\frac{j}{R}\sqrt{\frac{L}{C}}$, (b) $0, 1, \frac{j}{R}\sqrt{\frac{L}{C}}$

10.23 $\dfrac{(1 - \omega^2 LC)V_s}{1 - \omega^2 LC + j\omega RC(2 - \omega^2 LC)}$

10.25 $2.828\cos(2t + 45°)$ A

10.27 $4.698\angle 95.24°$ A, $0.9928\angle 37.71°$ A

10.29 This is a design problem with multiple answers.

10.31 $2.179\angle 61.44°$ A

10.33 $15.92\angle 43.49°$ A

10.35 $985.5\angle -2.1°$ mA

10.37 $2.38\angle -96.37°$ A, $2.38\angle 143.63°$ A, $2.38\angle 23.63°$ A

10.39 $0.6357\angle 109.6°$ A, $0.5738\angle 124.4°$ A,
$0.2425\angle -60.42°$ A, $0.1675\angle 48.5°$ A

10.41 $2.122\cos(2t + 45°) + 7.156\sin(4t + 25.56°)$ V

10.43 $9.902\cos(2t - 129.17°)$ A

10.45 $[989.1\cos(10t + 21.47°)$
$+ 499\sin(4t + 176.57°)]$ mA

10.47 $[4 + 0.504\sin(t + 19.1°)$
$+ 0.3352\cos(3t - 76.43°)]$ A

10.49 $8.944\sin(200t + 56.56°)$ A

10.51 $109.3\angle 30°$ mA

10.53 $(3.529 - j5.883)$ V

10.55 (a) $\mathbf{Z}_N = \mathbf{Z}_{\text{Th}} = 22.63\angle -63.43°\ \Omega$,
$\mathbf{V}_{\text{Th}} = -50\angle 30°$ V, $\mathbf{I}_N = 2.236\angle 273.4°$ A,
(b) $\mathbf{Z}_N = \mathbf{Z}_{\text{Th}} = 10\angle 26°\ \Omega$,
$\mathbf{V}_{\text{Th}} = 33.92\angle 58°$ V, $\mathbf{I}_N = 3.392\angle 32°$ A

10.57 This is a design problem with multiple answers.

10.59 $-6 + j38\ \Omega$

10.61 $-24 + j12$ V, $-8 + j6\ \Omega$

10.63 1 kΩ, $5.657\cos(200t + 75°)$ A

10.65 This is a design problem with multiple answers.

10.67 $4.945\angle -69.76°$ V, $0.4378\angle -75.24°$ A, $11.243 + j1.079\ \Omega$

10.69 $-j\omega RC,\ -V_m \cos\omega t$

10.71 $96\cos(2t + 44.52°)$ V

10.73 $21.21\angle -45°$ kΩ

10.75 $0.12499\angle 180°$

10.77 $\dfrac{R_2 + R_3 + j\omega C_2R_2R_3}{(1 + j\omega R_1C_1)(R_3 + j\omega C_2R_2R_3)}$

10.79 $3.578\cos(1{,}000t + 26.56°)$ V

10.81 $11.27\angle 128.1°$ V

10.83 $6.611\cos(1{,}000t - 159.2°)$ V

10.85 This is a design problem with multiple answers.

10.87 $15.91\angle 169.6°$ V, $5.172\angle -138.6°$ V, $2.27\angle -152.4°$ V

10.89 Proof

10.91 (a) 180 kHz,
(b) 40 kΩ

10.93 Proof

10.95 Proof

Chapter 11

(Assume all values of currents and voltages are rms unless otherwise specified.)

11.1 $800 + 1{,}600\cos(100t + 60°)$ W, 800 W

11.3 13.333 W

11.5 $P_{1\Omega} = 11.33$ W, $P_{2\Omega} = 40.79$ W, $P_{3H} = P_{0.25F} = 0$

11.7 160 W

11.9 1.794 mW

11.11 12.751 mW

11.13 (a) $120 - j60\ \Omega$, (b) 12.605 W

11.15 $0.5 - j0.5\ \Omega$, 9 kW

11.17 20 Ω, 5 W

11.19 2.576 Ω, 3.798 W

11.21 19.58 Ω

11.23 This is a design problem with multiple answers.

11.25 8.165

11.27 2.887 A

11.29 5.773 A, 400 W

11.31 2.944 V

11.33 6.665

11.35 21.6 V

11.37 This is a design problem with multiple answers.

11.39 (a) 0.7592, 6.643 kW, 5.695 kVAR,
(b) 312 μF

11.41 (a) 0.5547 (leading), (b) 0.9304 (lagging)

11.43 This is a design problem with multiple answers.

11.45 (a) 46.9 V, 1.061 A, (b) 20 W

11.47 (a) $S = 112 + j194$ VA,
average power = 112 W,
reactive power = 194 VAR
(b) $S = 226.3 - j226.3$ VA,
average power = 226.3 W,
reactive power = −226.3 VAR
(c) $S = 110.85 + j64$ VA, average power = 110.85 W, reactive power = 64 VAR
(d) $S = 7.071 + j7.071$ kVA, average power = 7.071 kW, reactive power = 7.071 kVAR

11.49 (a) $4 + j2.373$ kVA,
(b) $1.6 - j1.2$ kVA,
(c) $0.4624 + j1.2705$ kVA,
(d) $110.77 + j166.16$ VA

11.51 (a) 0.9956 (lagging),
(b) 1.751 kW,
(c) 164.9 VAR,
(d) 1.7587 kVA,
(e) $(1{,}751 + j164.9)$ VA

11.53 (a) $93.97\angle 29.8°$ A, (b) 1.0 (lagging)

11.55 This is a design problem with multiple answers.

11.57 (50.45 − *j*33.64) VA

11.59 *j*339.3 VAR, −*j*1.4146 kVAR

11.61 33.1$\angle$92.4° A, 6.62$\angle$−2.4° kVA

11.63 443.3$\angle$−28.13° A

11.65 80 μW

11.67 36$\angle$36.86° mVA, 12.042 mW

11.69 (a) 0.6402 (lagging),
(b) 295.1 W,
(c) 130.4 μF

11.71 (a) 50.14 + *j*1.7509 mΩ,
(b) 0.9994 lagging,
(c) 2.392$\angle$−2° kA

11.73 (a) 12.21 kVA, (b) 50.86$\angle$−35° A,
(c) 4.083 kVAR, 188.03 μF, (d) 43.4$\angle$−16.26° A

11.75 (a) 1,835.9 − *j*114.68 VA, (b) 0.998 (leading),
(c) no correction is necessary

11.77 157.69 W

11.79 50 mW

11.81 This is a design problem with multiple answers.

11.83 (a) 688.1 W, (b) 840 VA,
(c) 481.8 VAR, (d) 0.8191 (lagging)

11.85 (a) 20 A, 17.85$\angle$163.26° A, 5.907$\angle$−119.5° A,
(b) 4,451 + *j*617 VA, (c) 0.9904 (lagging)

11.87 0.5333

11.89 (a) 12 kVA, 9.36 + *j*7.51 kVA,
(b) 2.866 + *j*2.3 Ω

11.91 0.9775, 104 μF

11.93 (a) 7.328 kW, 1.196 kVAR, (b) 0.987

11.95 (a) 2.814 kHz,
(b) 431.8 mW

11.97 547.3 W

Chapter 12

(Assume all values of currents and voltages are rms unless otherwise specified.)

12.1 (a) 231$\angle$−30°, 231$\angle$−150°, 231$\angle$90° V,
(b) 231$\angle$30°, 231$\angle$150°, 231$\angle$−90° V

12.3 *abc* sequence, 208$\angle$250° V

12.5 260 cos(ωt + 62°) V, 260 cos(ωt − 58°) V,
260 cos(ωt + 182°) V

12.7 44$\angle$53.13° A, 44$\angle$−66.87° A, 44$\angle$173.13° A

12.9 4.8$\angle$−36.87° A, 4.8$\angle$−156.87° A, 4.8$\angle$83.13° A

12.11 207.8 V, 99.85 A

12.13 40.85 A,15.02 kW

12.15 13.66 A

12.17 5.773$\angle$5° A, 5.773 $\angle$−115° A,
5.773 $\angle$125° A

12.19 5.47$\angle$−18.43° A, 5.47$\angle$−138.43° A, 5.47$\angle$101.57° A,
9.474$\angle$−48.43° A, 9.474$\angle$−168.43° A,
9.474$\angle$71.57° A

12.21 34.36$\angle$−98.66° A, 59.51$\angle$171.34° A

12.23 (a) 13.995 A,
(b) 2.448 kW

12.25 8.87$\angle$4.78°, 8.87$\angle$−115.22°, 8.87$\angle$124.78° A

12.27 91.79 V

12.29 1.3 + *j*1.1465 kVA

12.31 (a) 6.144 + *j*4.608 Ω,
(b) 18.04 A, (c) 207.2 μF

12.33 15.385 A, 360.3 V

12.35 (a) 14.61 − *j*5.953 A,
(b) 3.361 + *j*1.368 kVA,
(c) 0.9261

12.37 55.51 A, 1.298 − *j*1.731 Ω

12.39 431.1 W

12.41 9.021 A

12.43 4.373 − *j*1.145 kVA

12.45 2.109$\angle$24.83° kV

12.47 39.19 A (rms), 0.9982 (lagging)

12.49 (a) 5.808 kW, (b) 1.9356 kW

12.51 (a) $19.2 - j14.4$ A, $-42.76 + j27.09$ A, $-12 - j20.78$ A,
(b) $31.2 + j6.38$ A, $-61.96 + j41.48$ A, $30.76 - j47.86$ A

12.53 This is a design problem with multiple answers.

12.55 $9.6\angle -90°$ A, $6\angle 120°$ A, $8\angle -150°$ A, $3.103 + j3.264$ kVA

12.57 $I_a = 1.9585\angle -18.1°$ A, $I_b = 1.4656\angle -130.55°$ A, $I_c = 1.947\angle 117.8°$ A

12.59 $220.6\angle -34.56°$, $214.1\angle -81.49°$, $49.91\angle -50.59°$ V, assuming that N is grounded.

12.61 $11.15\angle 37°$ A, $230.8\angle -133.4°$ V, assuming N is grounded.

12.63 $18.67\angle 158.9°$ A, $12.38\angle 144.1°$ A

12.65 $11.02\angle 12°$ A, $11.02\angle -108°$ A, $11.02\angle 132°$ A

12.67 (a) 97.67 kW, 88.67 kW, 82.67 kW,
(b) 108.97 A

12.69 $I_a = 94.32\angle -62.05°$ A, $I_b = 94.32\angle 177.95°$ A, $I_c = 94.32\angle 57.95°$ A, $28.8 + j18.03$ kVA

12.71 (a) 2,590 W, 4,808 W,
(b) 8,335 VA

12.73 2,360 W, −632.8 W

12.75 (a) 20 mA,
(b) 200 mA

12.77 320 W

12.79 $17.15\angle -19.65°$, $17.15\angle -139.65°$, $17.15\angle 100.35°$ A, $223\angle 2.97°$, $223\angle -117.03°$, $223\angle 122.97°$ V

12.81 516 V

12.83 183.42 A

12.85 $Z_Y = 2.133\ \Omega$

12.87 $1.448\angle -176.6°$ A, $1{,}252 + j711.6$ VA, $1{,}085 + j721.2$ VA

Chapter 13

(Assume all values of currents and voltages are rms unless otherwise specified.)

13.1 10 H

13.3 150 mH, 50 mH, 25 mH, 0.2887

13.5 (a) 123.7 mH, (b) 24.31 mH

13.7 $540.5\angle 144.16°$ mV

13.9 $4.148\angle 21.12°$ V

13.11 $412.3\cos(600t - 140.43°)$ mA

13.13 $4.308 + j6.538\ \Omega$

13.15 $1 + j19.5\ \Omega$, $1.404\angle 9.44°$ A

13.17 $13.073 + j25.86\ \Omega$

13.19 See Fig. G.22.

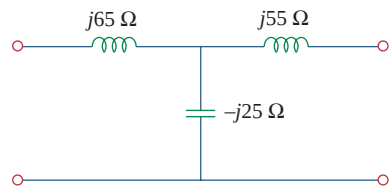


Figure G.22
For Prob. 13.19.

13.21 This is a design problem with multiple answers.

13.23 $50.68\cos(10t + 52.54°)$ A, $27.19\cos(10t - 100.89°)$ A, 1.5 kJ

13.25 $2.2\sin(2t - 4.88°)$ A, $1.5085\angle 17.9°\ \Omega$

13.27 1.567 W

13.29 0.984, 130.5 mJ

13.31 This is a design problem with multiple answers.

13.33 $12.769 + j7.154\ \Omega$

13.35 $10.143\angle -21.4°$ A, $532.8\angle -134.85°$ mA, $529.4\angle -110.41°$ mA

13.37 (a) 5, (b) 104.17 A, (c) 20.83 A

13.39 $15.7\angle 20.31°$ A, $78.5\angle 20.31°$ A

13.41 7.857 A, −23.57 A

13.43 4.186 V, 16.744 V

13.45 58.72 W

13.47 $118.03\cos(3t + 59.93°)$ V

13.49 $0.937\cos(2t + 51.34°)$ A

13.51 $8 - j1.5\ \Omega$, $29.49\angle 10.62°$ A

13.53 (a) 5, (b) 8 W

13.55 1.6669 Ω

13.57 (a) $25.9\angle 69.96°$, $12.95\angle 69.96°$ A (rms), (b) $21.06\angle 147.4°$, $42.12\angle 147.4°$, $42.12\angle 147.4°$ V(rms), (c) $1554\angle 20.04°$ VA

13.59 $P_{10\Omega} = 395$ W, $P_{12\Omega} = 266.6$ W, $P_{20\Omega} = 49.39$ W

13.61 6 A, 0.36 A, −60 V

13.63 $3.795\angle 18.43°$ A, $1.8975\angle 18.43°$ A, $0.6325\angle 161.6°$ A

13.65 11.05 W

13.67 (a) 160 V, (b) 31.25 A, (c) 12.5 A

13.69 $(1.2 - j2)$ kΩ, 5.333 W

13.71 $[1 + (N_1/N_2)]^2 Z_L$

13.73 (a) three-phase Δ-Y transformer, (b) $8.66\angle 156.87°$ A, $5\angle -83.13°$ A, (c) 1.8 kW

13.75 (a) 0.11547, (b) 76.98 A, 15.395 A

13.77 (a) a single-phase transformer, $1:n$, $n = 1/110$, (b) 7.576 mA

13.79 $7.836\angle -68.01°$ A, $2.441\angle -77.86°$ A, $8.016\angle -54.92°$ A

13.81 $104.5\angle 13.96°$ mA, $29.54\angle -143.8°$ mA, $208.8\angle 24.4°$ mA

13.83 $21.6\angle 33.91°$ A, $302.8\angle -34.21°$ V

13.85 100 turns

13.87 0.5

13.89 0.5, 41.67 A, 83.33 A

13.91 (a) 1,875 kVA, (b) 7,812 A

13.93 (a) See Fig. G.23(a). (b) See Fig. G.23(b).

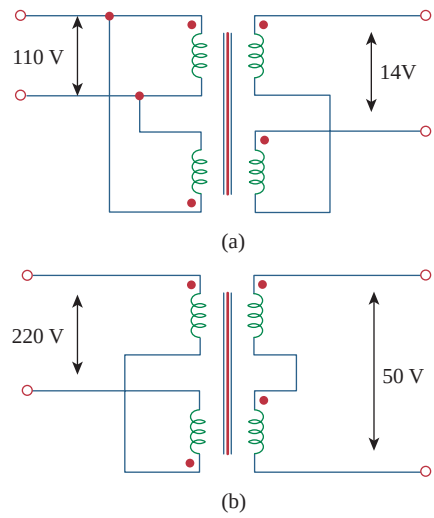


Figure G.23
For Prob. 13.93.

13.95 (a) 1/60, (b) 139 mA

Chapter 14

14.1 $\dfrac{j\omega/\omega_o}{1 + j\omega/\omega_o}$, $\omega_o = \dfrac{1}{RC}$

14.3 $\dfrac{5}{s^2 + 8s + 5}$

14.5 (a) $\dfrac{sRL}{(R + R_s)Ls + RR_s}$

(b) $\dfrac{R}{LRCs^2 + Ls + R}$

14.7 (a) 1.005773, (b) 0.4898, (c) 1.718×10^5

14.9 See Fig. G.24.

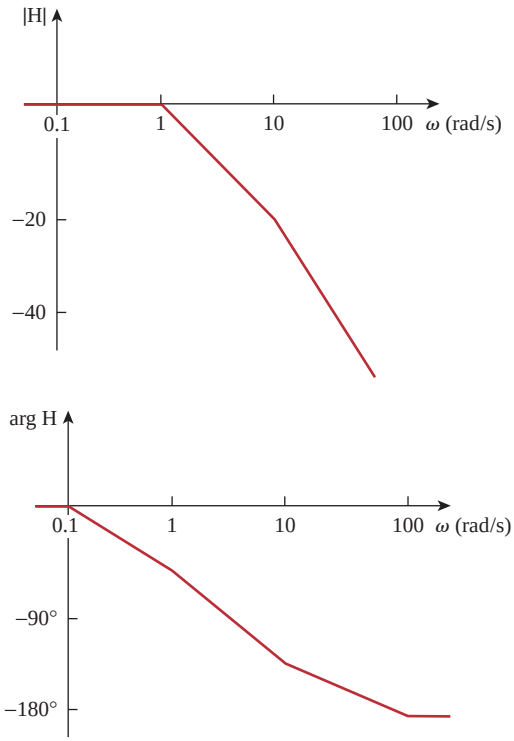


Figure G.24
For Prob. 14.9.

14.11 See Fig. G.25.

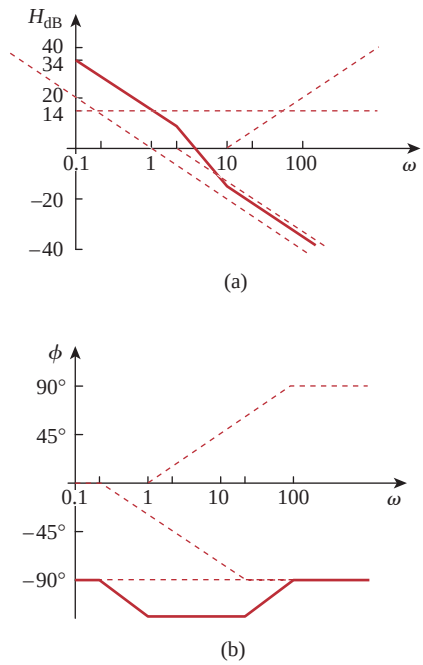


Figure G.25
For Prob. 14.11.

14.13 See Fig. G.26.

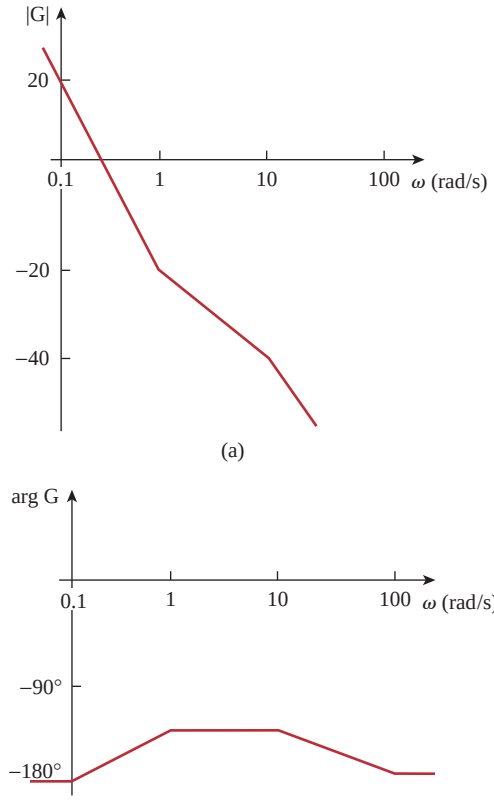


Figure G.26
For Prob. 14.13.

14.15 See Fig. G.27.

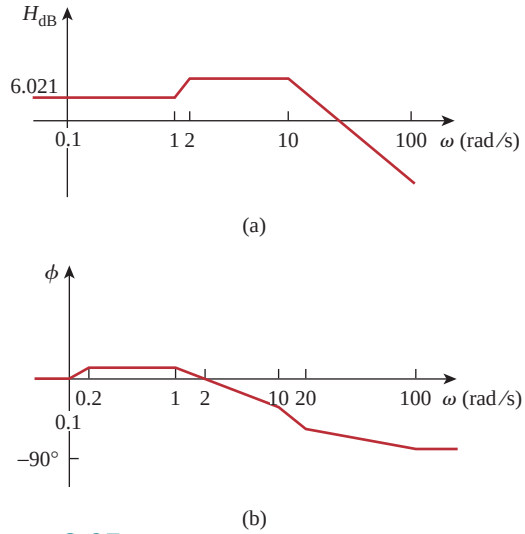


Figure G.27
For Prob. 14.15: (a) magnitude plot, (b) phase plot.

14.17 See Fig. G.28.

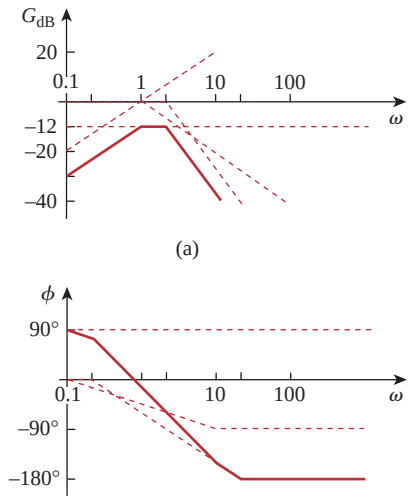


Figure G.28
For Prob. 14.17.

14.19 See Fig. G.29.

14.21 See Fig. G.30.

14.23 $\dfrac{10^4\, j\omega}{(10 + j\omega)(100 + j\omega)^2}$

14.25 2 kΩ, $2 - j0.75$ kΩ, $2 - j0.3$ kΩ, $2 + j0.3$ kΩ, $2 + j0.75$ kΩ

14.27 $R = 1\ \Omega$, $L = 0.1$ H, $C = 25$ mF

14.29 4.082 krad/s, 38.67, 105.55 rad/s

14.31 8.796×10^6 rad/s,

14.33 14.21 μH, 56.84 pF

14.35 40 Ω, 2.5 μH, 10 μF, 2.5 krad/s, 198.75 krad/s, 202.25 krad/s

14.37 $\dfrac{1}{\sqrt{LC - R^2C^2}}$

14.39 (a) 19.89 nF (b) 164.45 μH, (c) 552.9 krad/s, (d) 25.13 krad/s, (e) 22

14.41 This is a design problem with multiple answers.

14.43 (a) 2.357 krad/s, (b) 1 Ω

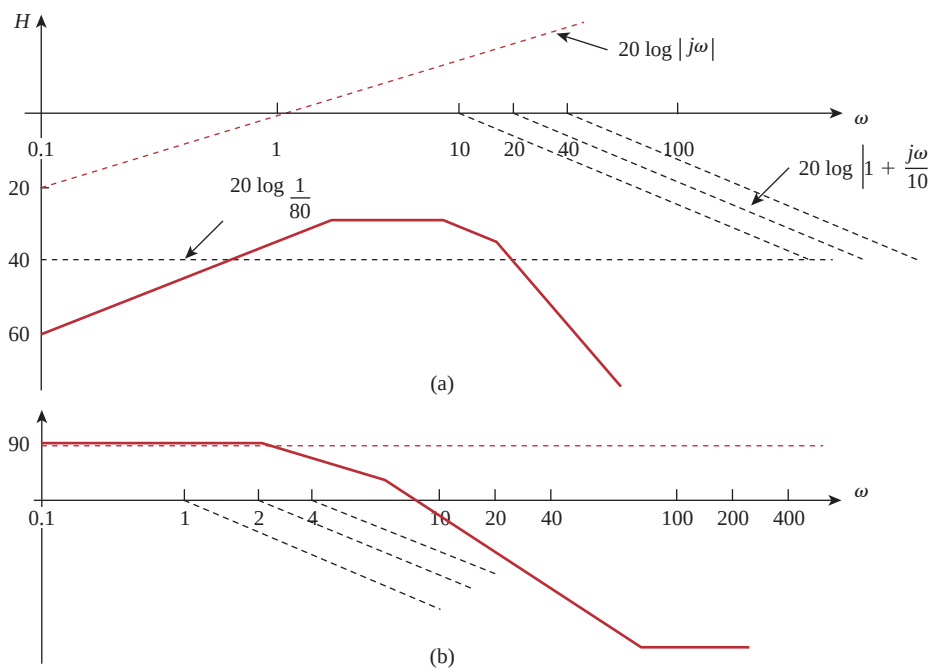


Figure G.29
For Prob. 14.19.

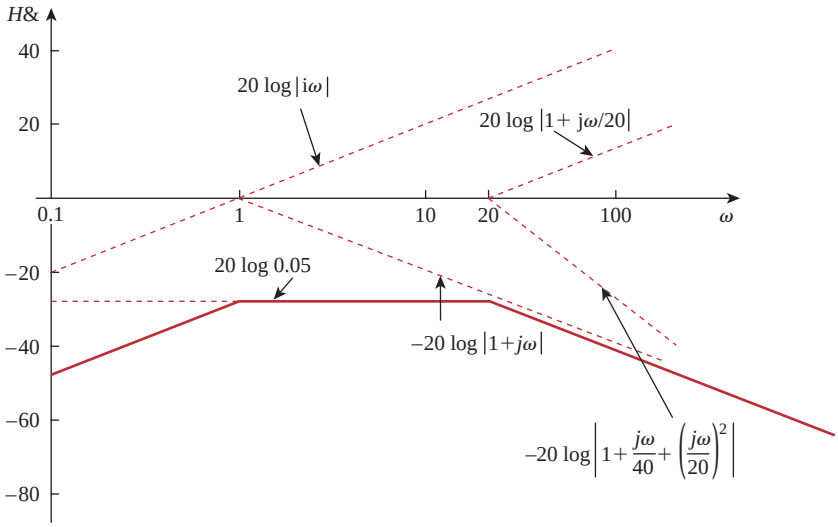


Figure G.30
For Prob. 14.21.

14.45 (a) $\dfrac{j\omega}{2(1+j\omega)^2}$, (b) 0.25

14.47 796 kHz

14.49 This is a design problem with multiple answers.

14.51 1.256 kΩ

14.53 18.045 kΩ. 2.872 H, 10.5

14.55 1.56 kHz $< f <$ 1.62 kHz, 25

14.57 (a) 1 rad/s, 3 rad/s, (b) 1 rad/s, 3 rad/s

14.59 2.408 krad/s, 15.811 krad/s

14.61 (a) $\dfrac{1}{1+j\omega RC}$,
(b) $\dfrac{j\omega RC}{1+j\omega RC}$

14.63 10 MΩ, 100 kΩ

14.65 Proof

14.67 If $R_f = 20$ kΩ, then $R_i = 80$ kΩ and $C = 15.915$ nF.

14.69 Let $R = 10$ kΩ, then $R_f = 25$ kΩ, $C = 7.96$ nF.

14.71 $K_f = 2 \times 10^{-4}$, $K_m = 5 \times 10^{-3}$

14.73 9.6 MΩ, 32 μH, 0.375 pF

14.75 200 Ω, 400 μH, 1 μF

14.77 (a) 1,200 H, 0.5208 μF, (b) 2 mH, 312.5 nF, (c) 8 mH, 7.81 pF

14.79 (a) $8s + 5 + \dfrac{10}{s}$,
(b) $0.8s + 50 + \dfrac{10^4}{s}$, 111.8 rad/s

14.81 (a) 0.4 Ω, 0.4 H, 1 mF, 1 mS,
(b) 0.4 Ω, 0.4 mH, 1 μF, 1 mS

14.83 0.1 pF, 0.5 pF, 1 MΩ, 2 MΩ

14.85 See Fig. G.31.

14.87 See Fig. G.32; highpass filter, $f_0 = 1.2$ Hz.

14.89 See Fig. G.33.

14.91 See Fig. G.34; $f_o = 800$ Hz.

14.93 $\dfrac{-RCs+1}{RCs+1}$

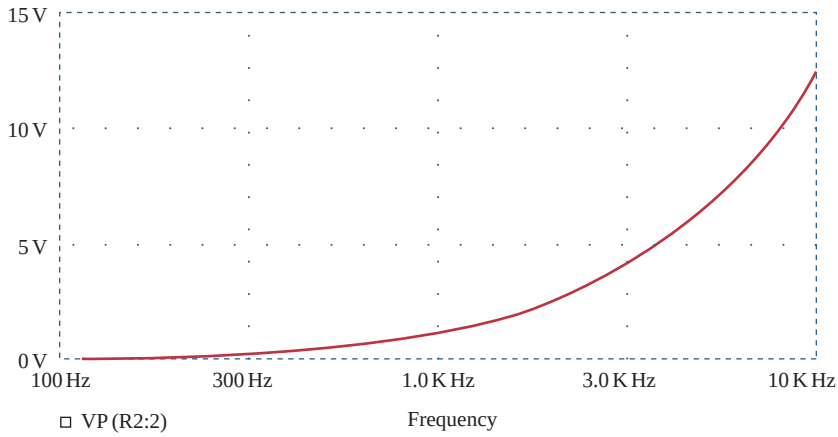
14.95 (a) 0.541 MHz $< f_o <$ 1.624 MHz,
(b) 67.98, 204.1

14.97 $\dfrac{s^3LR_LC_1C_2}{(sR_iC_1+1)(s^2LC_2+sR_LC_2+1)+s^2LC_1(sR_LC_2+1)}$

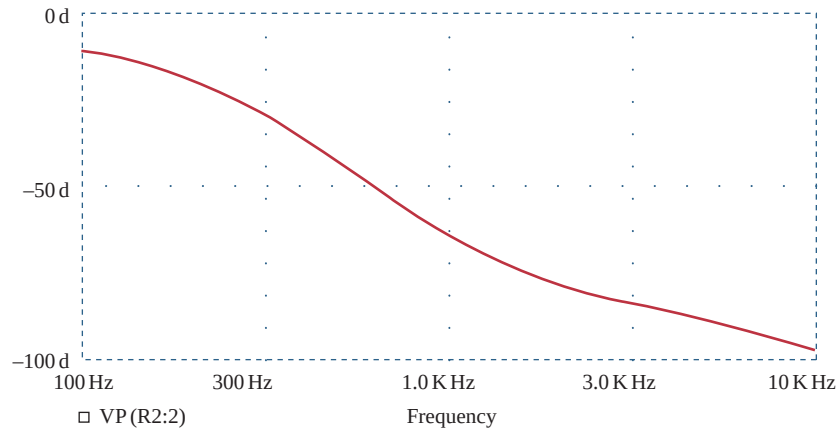
14.99 8.165 MHz, 4.188×10^6 rad/s

14.101 1.061 kΩ

14.103 $\dfrac{R_2(1+sCR_1)}{R_1+R_2+sCR_1R_2}$



(a)



(b)

Figure G.31
For Prob. 14.85.

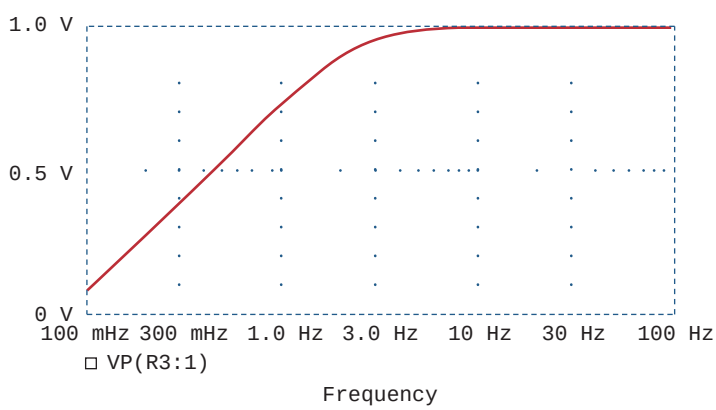


Figure G.32
For Prob. 14.87.

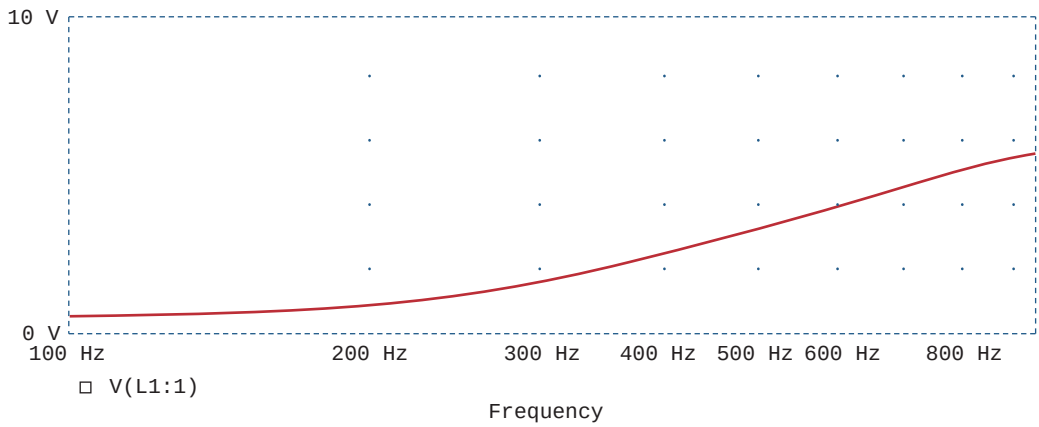


Figure G.33
For Prob. 14.89.

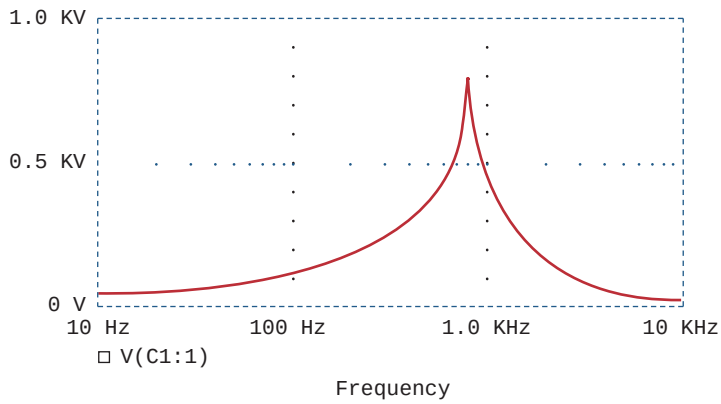


Figure G.34
For Prob. 14.91.

Chapter 15

15.1 (a) $\dfrac{s}{s^2 - a^2}$,
(b) $\dfrac{a}{s^2 - a^2}$

15.3 (a) $\dfrac{s+2}{(s+2)^2 + 9}$, (b) $\dfrac{4}{(s+2)^2 + 16}$,
(c) $\dfrac{s+3}{(s+3)^2 - 4}$ (d) $\dfrac{1}{(s+4)^2 - 1}$,
(e) $\dfrac{4(s+1)}{[(s+1)^2 + 4]^2}$

15.5 (a) $\dfrac{8 - 12\sqrt{3}s - 6s^2 + \sqrt{3}s^3}{(s^2+4)^3}$,
(b) $\dfrac{72}{(s+2)^5}$, (c) $\dfrac{2}{s^2} - 4s$,
(d) $\dfrac{2e}{s+1}$, (e) $\dfrac{5}{s}$, (f) $\dfrac{18}{3s+1}$, (g) s^n

15.7 (a) $\dfrac{2}{s^2} + \dfrac{4}{s}$, (b) $\dfrac{4}{s} + \dfrac{3}{s+2}$,
(c) $\dfrac{8s+18}{s^2+9}$, (d) $\dfrac{s+2}{s^2+4s-12}$

15.9 (a) $\dfrac{e^{-2s}}{s^2} - \dfrac{2e^{-2s}}{s^2}$, (b) $\dfrac{2e^{-s}}{e^4(s+4)}$,
(c) $\dfrac{2.702s}{s^2+4} + \dfrac{8.415}{s^2+4}$,
(d) $\dfrac{6}{s}e^{-2s} - \dfrac{6}{s}e^{-4s}$

15.11 (a) $\dfrac{6(s+1)}{s^2+2s-3}$,
(b) $\dfrac{24(s+2)}{(s^2+4s-12)^2}$,
(c) $\dfrac{e^{-(2s+6)}[(4e^2 + 4e^{-2})s + (16e^2 + 8e^{-2})]}{s^2+6s+8}$

15.13 (a) $\dfrac{s^2-1}{(s^2+1)^2}$,
(b) $\dfrac{2(s+1)}{(s^2+2s+2)^2}$,
(c) $\tan^{-1}\left(\dfrac{\beta}{s}\right)$

15.15 $5\dfrac{1-e^{-s}-se^{-s}}{s^2(1-e^{-3s})}$

15.17 This is a design problem with multiple answers.

15.19 $\dfrac{1}{1-e^{-s}}$

15.21 $\dfrac{(2\pi s-1+e^{-2\pi s})}{2\pi s^2(1-e^{-2\pi s})}$

15.23 (a) $\dfrac{(1-e^{-s})^2}{s(1-e^{-2s})}$,
(b) $\dfrac{2(1-e^{-2s})-4se^{-2s}(s+s^2)}{s^3(1-e^{-2s})}$

15.25 (a) 5 and 0, (b) 5 and 0

15.27 (a) $u(t)+2e^{-t}u(t)$, (b) $3\delta(t)-11e^{-4t}u(t)$,
(c) $(2e^{-t}-2e^{-3t})u(t)$,
(d) $(3e^{-4t}-3e^{-2t}+6te^{-2t})u(t)$

15.29 $(1-e^{-2t}\cos(3t)-(1/3)e^{-2t}\sin(3t))u(t)$

15.31 (a) $(-5e^{-t}+20e^{-2t}-15e^{-3t})u(t)$,
(b) $\left(-e^{-t}+\left(1+3t-\dfrac{t^2}{2}\right)e^{-2t}\right)u(t)$,
(c) $(-0.2e^{-2t}+0.2e^{-t}\cos(2t)$
$+0.4e^{-t}\sin(2t))u(t)$

15.33 (a) $(3e^{-t}-3\cos(t)+3\sin(t))u(t)$,
(b) $\sin(t-\pi)u(t-\pi)$,
(c) $3u(t)[1-e^{-t}-te^{-t}-0.5t^2e^{-t}]$

15.35 (a) $[2e^{-(t-6)}-e^{-2(t-6)}]u(t-6)$,
(b) $\dfrac{4}{3}u(t)[e^{-t}-e^{-4t}]-\dfrac{1}{3}u(t-2)[e^{-(t-2)}-e^{-4(t-2)}]$,
(c) $\dfrac{1}{13}u(t-1)[-3e^{-3(t-1)}+3\cos 2(t-1)$
$+2\sin 2(t-1)]$

15.37 (a) $(2-e^{-2t})u(t)$,
(b) $[0.4e^{-3t}+0.6e^{-t}\cos t+0.8e^{-t}\sin t]u(t)$,
(c) $e^{-2(t-4)}u(t-4)$,
(d) $\left(\dfrac{10}{3}\cos t-\dfrac{10}{3}\cos 2t\right)u(t)$

15.39 (a) $(-1.6e^{-t}\cos 4t-4.05e^{-t}\sin 4t$
$+3.6e^{-2t}\cos 4t+(3.45e^{-2t}\sin 4t)u(t)$,
(b) $[0.08333\cos 3t+0.02778\sin 3t$
$+0.0944e^{-0.551t}-0.1778e^{-5.449t}]u(t)$

15.41 $z(t)=\begin{cases}8t, & 0<t<2\\ 16-8t, & 2<t<6\\ -16, & 6<t<8\\ 8t-80, & 8<t<12\\ 112-8t, & 12<t<14\\ 0, & \text{otherwise}\end{cases}$

15.43 (a) $y(t)=\begin{cases}\dfrac{1}{2}t^2, & 0<t<1\\ -\dfrac{1}{2}t^2+2t-1, & 1<t<2\\ 1, & t>2\\ 0, & \text{otherwise}\end{cases}$
(b) $y(t)=2(1-e^{-t}),\ t>0$,
(c) $y(t)=\begin{cases}\dfrac{1}{2}t^2+t+\dfrac{1}{2}, & -1<t<0\\ -\dfrac{1}{2}t^2+t+\dfrac{1}{2}, & 0<t<2\\ \dfrac{1}{2}t^2-3t+\dfrac{9}{2}, & 2<t<3\\ 0, & \text{otherwise}\end{cases}$

15.45 $(4e^{-2t}-8te^{-2t})u(t)$

15.47 (a) $(-e^{-t}+2e^{-2t})u(t)$, (b) $(e^{-t}-e^{-2t})u(t)$

15.49 (a) $\left(\dfrac{t}{a}(e^{at}-1)-\dfrac{1}{a^2}-\dfrac{e^{at}}{a^2}(at-1)\right)u(t)$,
(b) $[0.5\cos(t)(t+0.5\sin(2t))$
$-0.5\sin(t)(\cos(t)-1)]u(t)$

15.51 $(3e^{-t}+4e^{-2t}-5e^{-3t})u(t)V$

15.53 $\cos(t)+\sin(t)$ or $1.4142\cos(t-45°)$

15.55 $\left(\dfrac{1}{40}+\dfrac{1}{20}e^{-2t}-\dfrac{3}{104}e^{-4t}-\dfrac{3}{65}e^{-t}\cos(2t)\right.$
$\left.-\dfrac{2}{65}e^{-t}\sin(2t)\right)u(t)$

15.57 This is a design problem with multiple answers.

15.59 $[-2.5e^{-t}+12e^{-2t}-10.5e^{-3t}]u(t)$

Chapter 16

16.1 $1.155e^{-0.5t}\sin(0.866t)u(t)$ A

16.3 $[16 + 104e^{-15t}]u(t)$ mA

16.5 $\left(e^{-2t} - \frac{2}{\sqrt{7}}\sin\left(\frac{\sqrt{7}}{2}t\right)\right)u(t)$ A

16.7 $\left(1 - e^{-3t/4}\cos\frac{\sqrt{7}}{4}t + 4.9135e^{-3t/4}\sin\frac{\sqrt{7}}{4}t\right)u(t)$ V

16.9 (a) $\frac{2(s^2+1)}{s^2+2s+1}$,
(b) $\frac{s(5s+6)}{3s^2+7s+6}$

16.11 $\frac{50s+160}{s(s^2+9s+16)}$

16.13 $5(e^{-t} - e^{-2t})u(t)$ A

16.15 $-\frac{5s(s^2+20)}{s(s+2)(s^2+0.5s+40)}$

16.17 $[4 - e^{-t} + 1.5811e^{-jt+161.57^\circ} + 1.5811e^{jt-161.57^\circ}]u(t)$ A

16.19 This is a design problem with multiple answers.

16.21 $v_o(t) = \frac{20}{3}[1 - e^{-t}\cos 0.7071t - 1.414e^{-t}\sin 0.7071t]u(t)$ V

16.23 $(5e^{-4t}\cos 2t + 230e^{-4t}\sin 2t)u(t)$ V,
$(6 - 6e^{-4t}\cos 2t - 11.37e^{-4t}\sin 2t)u(t)$ A

16.25 $[2.202e^{-3t} + 3.84te^{-3t} - 0.202\cos(4t) + 0.6915\sin(4t)]u(t)$ V

16.27 $\frac{20(s+1)}{(s+3)(3s^2+4s+1)}$, $\frac{10(s+1)}{(s+3)(3s^2+4s+1)}$

16.29 $10[2e^{-1.5t} - e^{-t}]u(t)$ A

16.31 $\frac{10s^2}{s^2+4}$

16.33 $4 + \frac{s}{2(s+3)} - \frac{2s(s+2)}{s^2+4s+20} - \frac{12s}{s^2+4s+20}$

16.35 $\frac{9s}{3s^2+9s+2}$

16.37 (a) $\frac{s^2-3}{3s^2+2s-9}$, (b) $\frac{-3}{2s}$

16.39 $sRC + 1$

16.41 $\left(2 + \frac{8}{3}e^{-t} - \frac{14}{3}e^{-4t}\right)u(t)$ A

16.43 $\begin{bmatrix} v_C' \\ i' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}\begin{bmatrix} v_C \\ i \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}u(t)$;
$i(t) = [0 \quad 1]\begin{bmatrix} v_C \\ i \end{bmatrix} + [0]u(t)$

16.45 $\begin{bmatrix} i_L' \\ v_C' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 4 & -2 \end{bmatrix}\begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$;
$v_o(t) = [0 \quad -1]\begin{bmatrix} i_L \\ v_C \end{bmatrix} + [1 \quad 0]\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$

16.47 $\begin{bmatrix} i_L' \\ v_C' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 4 & -2 \end{bmatrix}\begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$;
$\begin{bmatrix} i_1(t) \\ i_2(t) \end{bmatrix} = \begin{bmatrix} 1 & -0.5 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} i_L \\ v_C \end{bmatrix} + [0.5 \quad 0]\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$

16.49 $\begin{bmatrix} 0 & 1 \\ -9 & -7 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -5 \end{bmatrix}$, $[1 \quad 0]$, $[0]$

16.51 $[1 - e^{-2t}(\cos 2t + \sin 2t)]u(t)$

16.53 $s_{1,2} = \frac{-1}{2RC} \pm \sqrt{\frac{1}{(2RC)^2} - \frac{1}{LC}}$. Note that both roots lie in the left half-plane since R, L, and C are positive quantities; thus the circuit is stable.

16.55 The circuit is unstable.

16.57 100 Ω, 12.8 Ω, 20 μF

16.59 We have three equations and four unknowns. Thus, there is a family of solutions. One such solution is

$$R_2 = 1\ \text{k}\Omega,\ C_1 = 50\ \text{nF},\ C_2 = 20\ \mu\text{F}$$

16.61 See Fig. G.35.

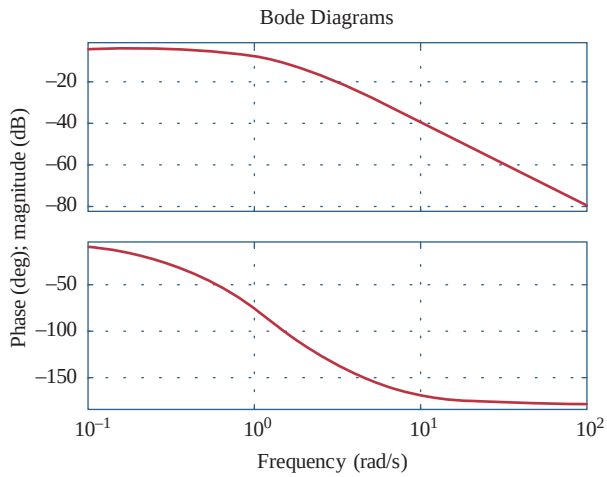


Figure G.35
For Prob. 16.61.

16.63 See Fig. G.36.

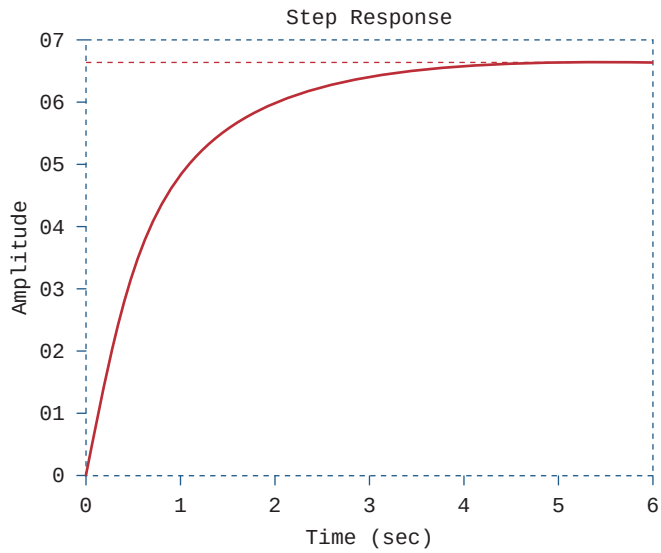


Figure G.36
For Prob. 16.63.

16.65 See Fig. G.37.

16.67 $a = -100$, $b = 400$, $c = 20{,}000$

16.69 Proof

Chapter 17

17.1 (a) periodic, 2, (b) not periodic, (c) periodic, 2π, (d) periodic, π, (e) periodic, 10, (f) not periodic, (g) not periodic

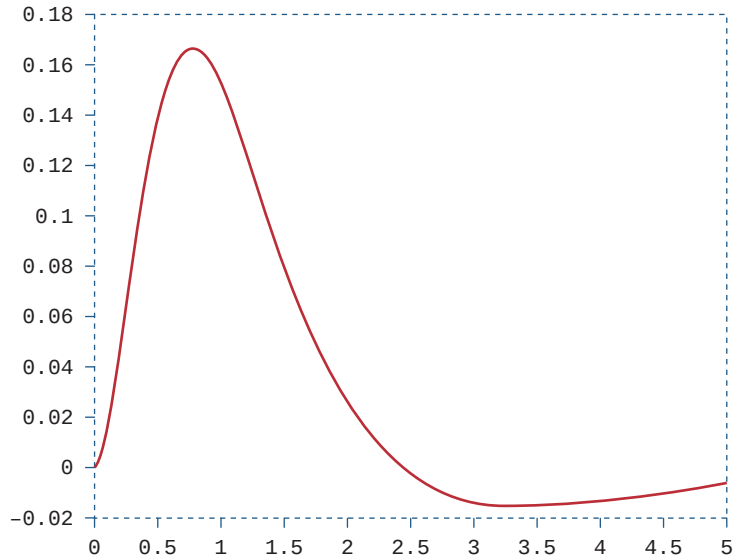


Figure G.37
For Prob. 16.65.

17.3 $a_0 = 7.5$

$$a_n = \begin{cases} \dfrac{10}{n\pi}(-1)^{(n+1)/2}, & n = \text{odd}, \\ 0, & n = \text{even} \end{cases}$$

$$b_n = \frac{10}{n\pi}\left[3 - 2\cos n\pi - \cos\frac{n\pi}{2}\right]$$

17.5 $-0.5 + \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \frac{6}{n\pi}\sin nt$

17.7 $3 + \sum_{n=0}^{\infty}\left[\frac{9}{n\pi}\sin\frac{4n\pi}{3}\cos\frac{2n\pi t}{3} + \frac{9}{n\pi}\left(1 - \cos\frac{4n\pi}{3}\right)\sin\frac{2n\pi t}{3}\right]$

17.9 $a_0 = 3.183$, $a_1 = 10$, $a_2 = 6.362$, $a_3 = 0$, $b_1 = 0 = b_2 = b_3$

17.11 $\sum_{n=-\infty}^{\infty} \frac{1}{n^2\pi^2}[1 + j(jn\pi/2 - 1)\sin n\pi/2 + n\pi \sin n\pi/2]e^{jn\pi t/2}$

17.13 This is a design problem with multiple answers.

17.15 (a) $10 + \sum_{n=1}^{\infty}\sqrt{\frac{16}{(n^2+1)^2} + \frac{1}{n^6}}\cos\left(10nt - \tan^{-1}\frac{n^2+1}{4\pi^3}\right)$,

(b) $10 + \sum_{n=1}^{\infty}\sqrt{\frac{16}{(n^2+1)} + \frac{1}{n^6}}\sin\left(10nt + \tan^{-1}\frac{4n^3}{n^2+1}\right)$

17.17 (a) neither odd nor even, (b) even, (c) odd, (d) even, (e) neither odd nor even

17.19 $\frac{5}{n^2\omega_o^2}\sin n\pi/2 - \frac{10}{n\omega_o}(\cos \pi n - \cos n\pi/2) - \frac{5}{n^2\omega_o^2}(\sin \pi n - \sin n\pi/2) - \frac{2}{n\omega_o}\cos n\pi - \frac{\cos \pi n/2}{n\omega_o}$

17.21 $\frac{1}{2} + \sum_{n=1}^{\infty}\frac{8}{n^2\pi^2}\left[1 - \cos\left(\frac{n\pi}{2}\right)\right]\cos\left(\frac{n\pi t}{2}\right)$

17.23 This is a design problem with multiple answers.

17.25
$$\sum_{\substack{n=1\\ n=\text{odd}}}^{\infty}\left\{\left[\frac{3}{\pi^2n^2}\left(\cos\left(\frac{2\pi n}{3}\right)-1\right)+\frac{2}{\pi n}\sin\left(\frac{2\pi n}{3}\right)\right]\cos\left(\frac{2\pi n}{3}\right)\right.$$
$$\left.+\left[\frac{3}{\pi^2n^2}\sin\left(\frac{2\pi n}{3}\right)-\frac{2}{n\pi}\cos\left(\frac{2\pi n}{3}\right)\right]\sin\left(\frac{2\pi n}{3}\right)\right\}$$

17.27 (a) odd, (b) -0.045, (c) 0.3829

17.29 $2\sum_{k=1}^{\infty}\left[\frac{2}{n^2\pi}\cos(nt)-\frac{1}{n}\sin(nt)\right]$, $n=2k-1$

17.31 $\omega_o'=\frac{2\pi}{T'}=\frac{2\pi}{T/\alpha}=\alpha\omega_o$

$$a_n'=\frac{2}{T'}\int_0^{T'}f(\alpha t)\cos n\omega_o' t\,dt$$

Let $\alpha t=\lambda$, $dt=d\lambda/\alpha$, and $\alpha T'=T$. Then

$$a_n'=\frac{2\alpha}{T}\int_0^{T}f(\lambda)\cos n\omega_o\lambda\,d\lambda/\alpha=a_n$$

Similarly, $b_n'=b_n$

17.33 $v_o(t)=\sum_{n=1}^{\infty}A_n\sin(n\pi t-\theta_n)$ V,

$$A_n=\frac{8(4-2n^2\pi^2)}{\sqrt{(20-10n^2\pi^2)^2-64n^2\pi^2}},$$

$$\theta_n=90^\circ-\tan^{-1}\left(\frac{8n\pi}{20-10n^2\pi^2}\right)$$

17.35 $\frac{3}{8}+\sum_{n=\text{odd}}^{\infty}A_n\cos\left(\frac{2\pi n}{3}+\theta_n\right)$, where

$$A_n=\frac{\frac{6}{n\pi}\sin\frac{2n\pi}{3}}{\sqrt{9\pi^2n^2+(2\pi^2n^2/3-3)^2}},$$

$$\theta_n=\frac{\pi}{2}-\tan^{-1}\left(\frac{2n\pi}{9}-\frac{1}{n\pi}\right)$$

17.37 $\sum_{n=1}^{\infty}\frac{8(1-\cos\pi n)}{\sqrt{1+n^2\pi^2}}\cos(n\pi t-\tan^{-1}n\pi)$ V

17.39 $\frac{1}{20}+\frac{200}{\pi}\sum_{k=1}^{\infty}I_n\sin(n\pi t-\theta_n)$, $n=2k-1$,

$$\theta_n=90^\circ+\tan^{-1}\frac{2n^2\pi^2-1{,}200}{802n\pi},$$

$$I_n=\frac{1}{n\sqrt{(804n\pi)^2+(2n^2\pi^2-1{,}200)}}$$

17.41 $\frac{10}{\pi}+\sum_{n=1}^{\infty}A_n\cos(2nt+\theta_n)$,

$$A_n=\frac{100}{\pi(4n^2-1)\sqrt{16n^2-40n+29}},$$

$$\theta_n=90^\circ-\tan^{-1}(2n-2.5)$$

17.43 (a) 33.91 V,
(b) 6.782 A,
(c) 203.1 W

17.45 4.263 A, 181.7 W

17.47 10%

17.49 (a) 1.5326,
(b) 1.7086,
(c) 3.061%

17.51 This is a design problem with multiple answers.

17.53 $\sum_{n=-\infty}^{\infty}\frac{0.6321e^{j2n\pi t}}{1+j2n\pi}$

17.55 $\sum_{n=-\infty}^{\infty}\frac{1+e^{-jn\pi}}{2\pi(1-n^2)}e^{jnt}$

17.57 $-3+\sum_{n=\infty,\,n\neq0}^{\infty}\frac{3}{n^3-2}e^{j50nt}$

17.59 $-\sum_{\substack{n=-\infty\\ n\neq0}}^{\infty}\frac{j4e^{-j(2n+1)\pi t}}{(2n+1)\pi}$

17.61 (a) $6+2.571\cos t-3.83\sin t+1.638\cos 2t-1.147\sin 2t+0.906\cos 3t-0.423\sin 3t+0.47\cos 4t-0.171\sin 4t$, (b) 6.828

17.63 See Fig. G.38.

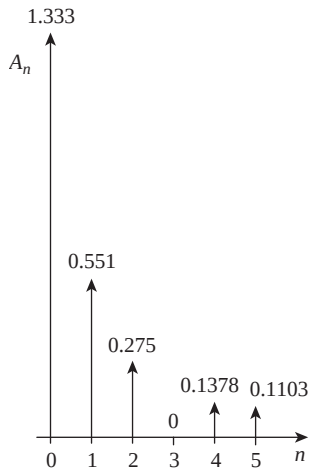


Figure G.38
For Prob. 17.63.

17.65 See Fig. G.39.

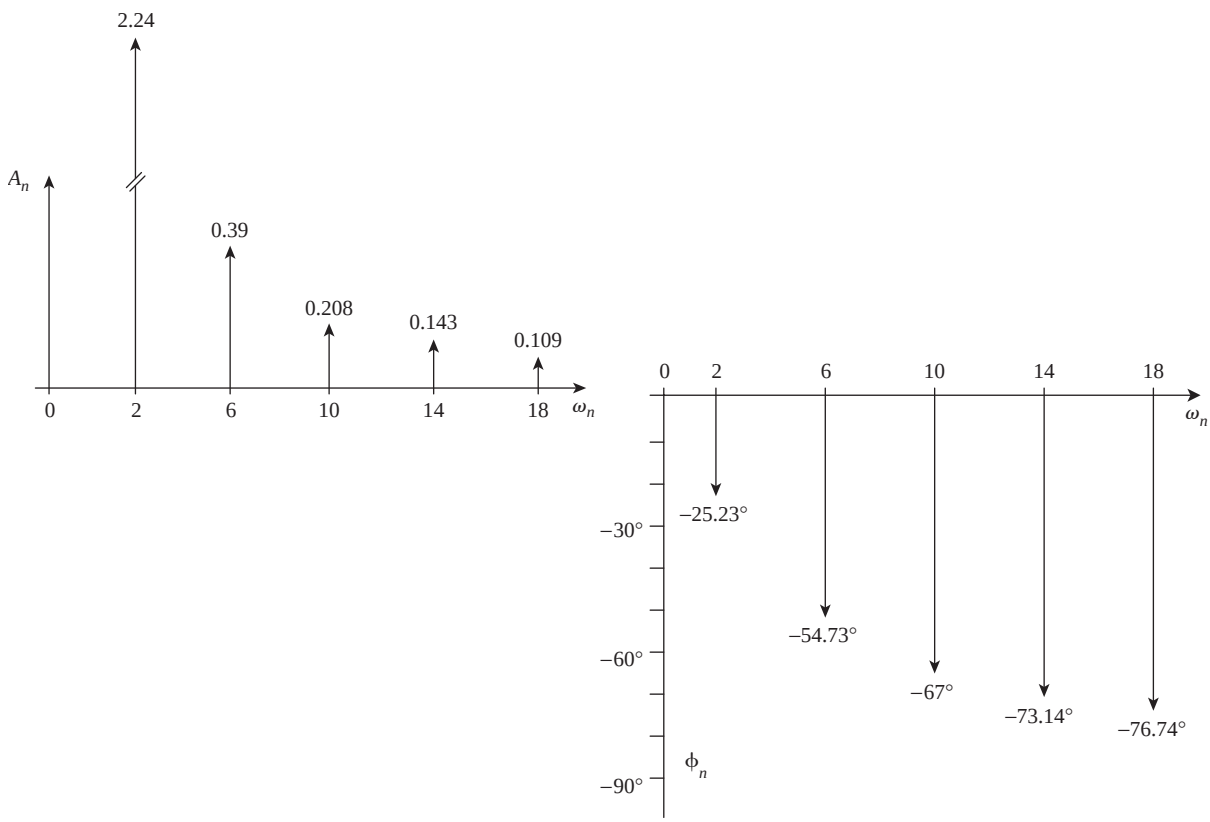


Figure G.39
For Prob. 17.65.

17.67 DC COMPONENT = 4.950000E-01

HARMONIC NO	FREQUENCY (HZ)	FOURIER COMPONENT	NORMALIZED COMPONENT	PHASE (DEG)	NORMALIZED PHASE (DEG)
1	1.667E-01	2.432E+00	1.000E+00	-8.996E+01	0.000E+00
2	3.334E-01	6.576E-04	2.705E-04	-8.932E+01	6.467E-01
3	5.001E-01	5.403E-01	2.222E-01	9.011E+01	1.801E+02
4	6.668E+00	3.343E-04	1.375E-04	9.134E+01	1.813E+02
5	8.335E-01	9.716E-02	3.996E-02	-8.982E+01	1.433E-01
6	1.000E+00	7.481E-06	3.076E-06	-9.000E+01	-3.581E-02
7	1.167E+00	4.968E-02	2.043E-01	-8.975E+01	2.173E-01
8	1.334E+00	1.613E-04	6.634E-05	-8.722E+01	2.748E+00
9	1.500E+00	6.002E-02	2.468E-02	-9.032E+01	1.803E+02

17.69

HARMONIC NO	FREQUENCY (HZ)	FOURIER COMPONENT	NORMALIZED COMPONENT	PHASE (DEG)	NORMALIZED PHASE (DEG)
1	5.000E-01	4.056E-01	1.000E+00	-9.090E+01	0.000E+00
2	1.000E+00	2.977E-04	7.341E-04	-8.707E+01	3.833E+00
3	1.500E+00	4.531E-02	1.117E-01	-9.266E+01	-1.761E+00
4	2.000E+00	2.969E-04	7.320E-04	-8.414E+01	6.757E+00
5	2.500E+00	1.648E-02	4.064E-02	-9.432E+01	-3.417E+00
6	3.000E+00	2.955E-04	7.285E-04	-8.124E+01	9.659E+00
7	3.500E+00	8.535E-03	2.104E-02	-9.581E+01	-4.911E+00
8	4.000E+00	2.935E-04	7.238E-04	-7.836E+01	1.254E+01
9	4.500E+00	5.258E-03	1.296E-02	-9.710E+01	-6.197E+00

TOTAL HARMONIC DISTORTION = 1.214285+01 PERCENT

17.71 See Fig. G.40.

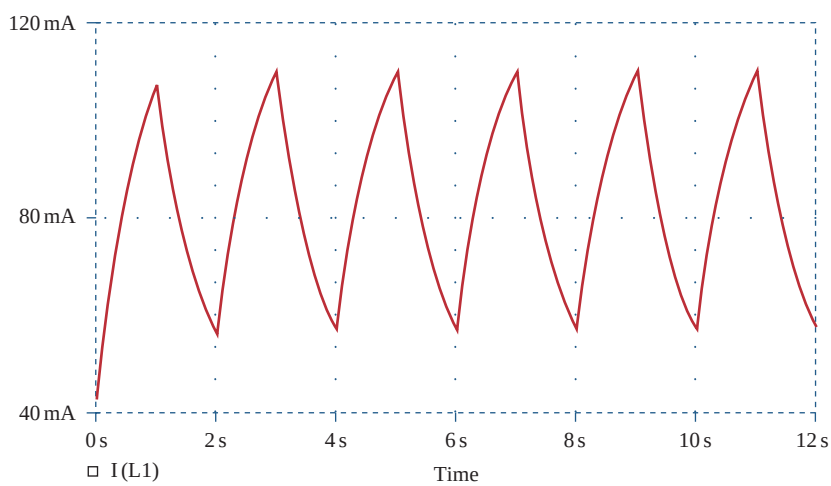


Figure G.40
For Prob. 17.71.

17.73 300 mW

17.75 24.59 mF

17.77 (a) π, (b) -2 V, (c) 11.02 V

17.79 See below for the program in *MATLAB* and the results.

```
% for problem 17.79
a = 10;
c = 4.*a/pi
for n = 1:10
  b(n)=c/(2*n-1);
end
diary
n, b
diary off
```

n	b_n
1	12.7307
2	4.2430
3	2.5461
4	1.8187
5	1.414
6	1.1573
7	0.9793
8	0.8487
9	0.7488
10	0.6700

17.81 (a) $\frac{A^2}{2}$, (b) $|c_1| = 2A/(3\pi)$, $|c_2| = 2A/(15\pi)$, $|c_3| = 2A/(35\pi)$, $|c_4| = 2A/(63\pi)$ (c) 81.1% (d) 0.72%

Chapter 18

18.1 $\frac{2(\cos 2\omega - \cos\omega)}{j\omega}$

18.3 $\frac{j}{\omega^2}(2\omega\cos 2\omega - \sin 2\omega)$

18.5 $\frac{2j}{\omega} - \frac{2j}{\omega^2}\sin\omega$

18.7 (a) $\frac{2 - e^{-j\omega} - e^{-j2\omega}}{j\omega}$, (b) $\frac{5e^{-j2\omega}}{\omega^2}(1 + j\omega 2) - \frac{5}{\omega^2}$

18.9 (a) $\frac{2}{\omega}\sin 2\omega + \frac{4}{\omega}\sin\omega$,
(b) $\frac{2}{\omega^2} - \frac{2e^{-j\omega}}{\omega^2}(1 + j\omega)$

18.11 $\frac{5\pi}{\omega^2 - \pi^2}(e^{-j\omega 2} - 1)$

18.13 (a) $\pi e^{-j\pi/3}\delta(\omega - a) + \pi e^{j\pi/3}\delta(\omega + a)$,
(b) $\frac{e^{j\omega}}{\omega^2 - 1}$, (c) $\pi[\delta(\omega + b) + \delta(\omega - b)]$
$+ \frac{j\pi A}{2}[\delta(\omega + a + b) - \delta(\omega - a + b)$
$+ \delta(\omega + a - b) - \delta(\omega - a - b)]$,
(d) $\frac{1}{\omega^2} - \frac{e^{-j4\omega}}{j\omega} - \frac{e^{-j4\omega}}{\omega^2}(j4\omega + 1)$

18.15 (a) $2j\sin 3\omega$, (b) $\frac{2e^{-j\omega}}{j\omega}$, (c) $\frac{1}{3} - \frac{j\omega}{2}$

18.17 (a) $\frac{\pi}{2}[\delta(\omega + 2) + \delta(\omega - 2)] - \frac{j\omega}{\omega^2 - 4}$,
(b) $\frac{j\pi}{2}[\delta(\omega + 10) - \delta(\omega - 10)] - \frac{10}{\omega^2 - 100}$

18.19 $\frac{j\omega}{\omega^2 - 4\pi^2}(e^{-j\omega} - 1)$

18.21 Proof

18.23 (a) $\frac{30}{(6 - j\omega)(15 - j\omega)}$,
(b) $\frac{20e^{-j\omega/2}}{(4 + j\omega)(10 + j\omega)}$,
(c) $\frac{5}{[2 + j(\omega + 2)][5 + j(\omega + 2)]} +$
$\frac{5}{[2 + j(\omega - 2)][5 + j(\omega - 2)]}$,
(d) $\frac{j\omega 10}{(2 + j\omega)(5 + j\omega)}$,
(e) $\frac{10}{j\omega(2 + j\omega)(5 + j\omega)} + \pi\delta(\omega)$

18.25 (a) $10e^{2t}u(t)$,
(b) $1.5e^{-2|t|}$,
(c) $-5e^{t}u(t) + 5e^{2t}u(t)$

18.27 (a) $5\text{sgn}(t) - 10e^{-10t}u(t)$,
(b) $4e^{2t}u(-t) - 6e^{-3t}u(t)$,
(c) $2e^{-20t}\sin(30t)u(t)$, (d) $\frac{1}{4}\pi$

18.29 (a) $\frac{1}{2\pi}(1 + 8\cos 3t)$, (b) $\frac{4\sin 2t}{\pi t}$,
(c) $3\delta(t + 2) + 3\delta(t - 2)$

18.31 (a) $x(t) = e^{-at}u(t)$,
(b) $x(t) = u(t + 1) - u(t - 1)$,
(c) $x(t) = \frac{1}{2}\delta(t) - \frac{a}{2}e^{-at}u(t)$

18.33 (a) $\frac{2j\sin t}{t^2 - \pi^2}$, (b) $u(t - 1) - u(t - 2)$

18.35 (a) $\frac{e^{-j\omega/3}}{6 + j\omega}$, (b) $\frac{1}{2}\left[\frac{1}{2 + j(\omega + 5)} + \frac{1}{2 + j(\omega - 5)}\right]$,
(c) $\frac{j\omega}{2 + j\omega}$, (d) $\frac{1}{(2 + j\omega)^2}$, (e) $\frac{1}{(2 + j\omega)^2}$

18.37 $\frac{j\omega}{4 + j3\omega}$

18.39 $\frac{2x}{10^6 + j\omega}\left(\frac{1}{j\omega} + \frac{1}{\omega^2} - \frac{1}{\omega^2}e^{-j\omega}\right)$

18.41 $\frac{2j\omega(4.5 + j2\omega)}{(2 + j\omega)(4 - 2\omega^2 + j\omega)}$

18.43 $1000(e^{-1t} - e^{-1.25t})u(t)$ V

18.45 $5(e^{-t} - e^{-2t})u(t)$ A

18.47 $16(e^{-t} - e^{-2t})u(t)$ V

18.49 $0.542\cos(t + 13.64°)$ V

18.51 16.667 J

18.53 π

18.55 682.5 J

18.57 12.5 J, 87.41%

18.59 $(16e^{-t} - 20e^{-2t} + 4e^{-4t})u(t)$ V

18.61 $2X(\omega) + 0.5X(\omega + \omega_0) + 0.5X(\omega - \omega_0)$

18.63 106 stations

18.65 6.8 kHz

18.67 200 Hz, 5 ms

18.69 35.24%

Chapter 19

19.1 $\begin{bmatrix} 4 & 1 \\ 1 & 1.667 \end{bmatrix}\Omega$

19.3 $\begin{bmatrix} 4 + j6 & j6 \\ j6 & -j4 \end{bmatrix}\Omega$

19.5 $\begin{bmatrix} \frac{2(s^2 + s + 1)}{s^3 + 2s^2 + 3s + 1} & \frac{2}{s^3 + 2s^2 + 3s + 1} \\ \frac{2}{s^3 + 2s^2 + 3s + 1} & \frac{2(s^2 + 2s + 2)}{s^3 + 2s^2 + 3s + 1} \end{bmatrix}\Omega$

19.7 $\begin{bmatrix} 29.88 & 3.704 \\ -70.37 & 11.11 \end{bmatrix}\Omega$

19.9 $\begin{bmatrix} 5 & 2.5 \\ 2.5 & 6.25 \end{bmatrix}\Omega$

19.11 See Fig. G.41.

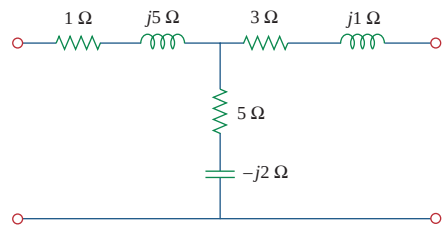


Figure G.41
For Prob. 19.11.

19.13 329.9 W

19.15 24 Ω, 384 W

19.17 $\begin{bmatrix} 4.8 & -0.4 \\ -0.4 & 4.2 \end{bmatrix}\Omega$, $\begin{bmatrix} 0.21 & 0.02 \\ 0.02 & 0.24 \end{bmatrix}$ S

19.19 This is a design problem with multiple answers.

19.21 See Fig. G.42.

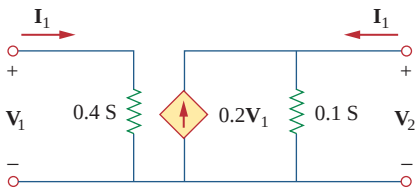


Figure G.42
For Prob. 19.21.

19.23 $\begin{bmatrix} s+2 & -(s+1) \\ -(s+1) & \dfrac{s^2+s+1}{s} \end{bmatrix}, \dfrac{0.8(s+1)}{s^2+1.8s+1.2}$

19.25 See Fig. G.43.

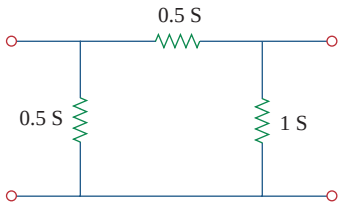


Figure G.43
For Prob. 19.25.

19.27 $\begin{bmatrix} 0.25 & 0.025 \\ 5 & 0.6 \end{bmatrix}$ S

19.29 (a) 22 V, 8 V, (b) same

19.31 $\begin{bmatrix} 3.8\,\Omega & 0.4 \\ -3.6 & 0.2\text{ S} \end{bmatrix}$

19.33 $\begin{bmatrix} 3.077 + j1.2821 & 0.3846 - j0.2564 \\ -0.3846 + j0.2564 & 0.0769 + j0.2821 \end{bmatrix}$

19.35 $\begin{bmatrix} 2\,\Omega & 0.5 \\ -0.5 & 0 \end{bmatrix}$

19.37 1.1905 V

19.39 $g_{11} = \dfrac{1}{R_1+R_2}, g_{12} = -\dfrac{R_2}{R_1+R_2}$

$g_{21} = \dfrac{R_2}{R_1+R_2}, g_{22} = R_3 + \dfrac{R_1R_2}{R_1+R_2}$

19.41 Proof

19.43 (a) $\begin{bmatrix} 1 & \mathbf{Z} \\ 0 & 1 \end{bmatrix}$, (b) $\begin{bmatrix} 1 & 0 \\ \mathbf{Y} & 1 \end{bmatrix}$

19.45 $\begin{bmatrix} 1 - j0.5 & -j4\,\Omega \\ 0.125\text{ S} & 1 \end{bmatrix}$

19.47 $\begin{bmatrix} 0.3235 & 1.176 \\ 0.02941 & 0.4706 \end{bmatrix}$

19.49 $\begin{bmatrix} \dfrac{2s+1}{s} & \dfrac{1}{s}\,\Omega \\ \dfrac{(s+1)(3s+1)}{s}S & 2+\dfrac{1}{s} \end{bmatrix}$

19.51 $\begin{bmatrix} 2 & 2+j5 \\ j & -2+j \end{bmatrix}$

19.53 $z_{11} = \dfrac{A}{C}, z_{12} = \dfrac{AD-BC}{C}, z_{21} = \dfrac{1}{C}, z_{22} = \dfrac{D}{C}$

19.55 Proof

19.57 $\begin{bmatrix} 3 & 1 \\ 1 & 7 \end{bmatrix}\Omega, \begin{bmatrix} \dfrac{7}{20} & \dfrac{-1}{20} \\ \dfrac{-1}{20} & \dfrac{3}{20} \end{bmatrix}$ S, $\begin{bmatrix} \dfrac{20}{7}\,\Omega & \dfrac{1}{7} \\ \dfrac{-1}{7} & \dfrac{1}{7}\text{ S} \end{bmatrix}$,
$\begin{bmatrix} \dfrac{1}{3}\text{ S} & \dfrac{-1}{3} \\ \dfrac{1}{3} & \dfrac{20}{3}\,\Omega \end{bmatrix}, \begin{bmatrix} 7 & 20\,\Omega \\ 1\text{ S} & 3 \end{bmatrix}$

19.59 $\begin{bmatrix} 16.667 & 6.667 \\ 3.333 & 3.333 \end{bmatrix}\Omega, \begin{bmatrix} 0.1 & -0.2 \\ -0.1 & 0.5 \end{bmatrix}$ S,
$\begin{bmatrix} 10\,\Omega & 2 \\ -1 & 0.3\text{ S} \end{bmatrix}, \begin{bmatrix} 5 & 10\,\Omega \\ 0.3\text{ S} & 1 \end{bmatrix}$

19.61 $\begin{bmatrix} \dfrac{10}{3} & \dfrac{8}{3} \\ \dfrac{8}{3} & \dfrac{10}{3} \end{bmatrix}\Omega, \begin{bmatrix} \dfrac{6}{5}\,\Omega & \dfrac{4}{5} \\ \dfrac{-4}{5} & \dfrac{3}{10}\text{ S} \end{bmatrix}, \begin{bmatrix} \dfrac{5}{4} & \dfrac{3}{2}\,\Omega \\ \dfrac{3}{8}\text{ S} & \dfrac{5}{4} \end{bmatrix}$

19.63 $\begin{bmatrix} 0.8 & 2.4 \\ 2.4 & 7.2 \end{bmatrix}\Omega$

19.65 $\begin{bmatrix} \dfrac{1}{3} & -\dfrac{1}{3} \\ -\dfrac{1}{3} & \dfrac{2}{3} \end{bmatrix}$ S

19.67 $\begin{bmatrix} 4 & 63.29 \\ 0.1576 & 4.994 \end{bmatrix}$

19.69 $\begin{bmatrix} \dfrac{s+1}{s+2} & \dfrac{-(3s+2)}{2(s+2)} \\ \dfrac{-(3s+2)}{2(s+2)} & \dfrac{5s^2+4s+4}{2s(s+2)} \end{bmatrix}$

19.71 $\begin{bmatrix} 2 & -3.334 \\ 3.334 & 20.22 \end{bmatrix}\ \Omega$

19.73 $\begin{bmatrix} 14.628 & 3.141 \\ 5.432 & 19.625 \end{bmatrix}$

19.75 (a) $\begin{bmatrix} 0.3015 & -0.1765 \\ 0.0588 & 10.94 \end{bmatrix}$, (b) -0.0051

19.77 $\begin{bmatrix} 0.9488\angle -161.6^\circ & 0.3163\angle 18.42^\circ \\ 0.3163\angle -161.6^\circ & 0.9488\angle -161.6^\circ \end{bmatrix}$

19.79 $\begin{bmatrix} 4.669\angle -136.7^\circ & 2.53\angle -108.4^\circ \\ 2.53\angle -108.4^\circ & 1.789\angle -153.4^\circ \end{bmatrix}\ \Omega$

19.81 $\begin{bmatrix} 1.5 & -0.5 \\ 3.5 & 1.5 \end{bmatrix}$ S

19.83 $\begin{bmatrix} 0.3235 & 1.1765 \\ 0.02941 & 0.4706 \end{bmatrix}$

19.85 $\begin{bmatrix} 1.581\angle 71.59^\circ & -j \\ j & 5.661 \times 10^{-4} \end{bmatrix}$

19.87 $\begin{bmatrix} -j1{,}765 & -j1{,}765 \\ j888.2 & j888.2 \end{bmatrix}$

19.89 $-1{,}613$, 64.15 dB

19.91 (a) -25.64 for the transistor and -9.615 for the circuit.

19.93 -17.74, 144.5, 31.17 Ω, $-$ 6.148 MΩ

19.95 See Fig. G.44.

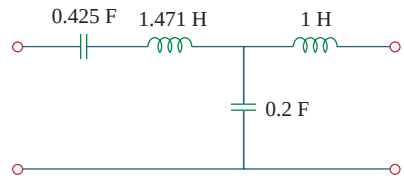


Figure G.44
For Prob. 19.95.

19.97 0.25 F, 0.3333 H, 0.5 F

19.99 Proof

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